AZ-900 Exam Prep 2026

Microsoft Azure Fundamentals - 424 Questions & Answers

Revised edition - verified answers, per-question explanations and diagrams

What this edition contains

Every one of the 424 questions now has its own content, written for that question. Nothing is copied from question to question:

Verified answer	Each printed answer was re-checked against current Microsoft Learn facts. 17 wrong answers were corrected and are marked in red at the question.
2026 update notes	31 questions use retired or renamed services (Azure AD, Security Center, Blueprints, Premier support...). The expected answer is kept and an amber note explains what is true today.
Answer explained	A brief, to-the-point reason for the correct answer and a one-line reason for every wrong option.
What / Where / When / Why / How	Five short, question-specific facts about the service or concept tested.
High-resolution diagram	A 3040 x 1702 px (400 DPI) concept diagram chosen for the question's topic, placed directly under the explanation. 33 diagrams in all.
Exam strategy + impact	How to spot and eliminate under pressure, and why the concept matters in real Azure work.
Exam tip + memory trick	A trap or keyword specific to that question, and a mnemonic for that fact.
Example, use case, scenario	Three different real-world illustrations for every question.

Exam outline (current)

Describe cloud concepts 25-30% · Describe Azure architecture and services 35-40% · Describe Azure management and governance 30-35%. Passing score 700 / 1000.

How to read each question

✔ **Green row** = correct option. **Red box** = answer corrected in this edition. **Amber box** = premise outdated; expected exam answer kept. Use the PDF bookmarks panel to jump to any question.

What's New in v7

424 questions: the v6 set minus 3 more duplicates found across domains (v6 Q222, Q390, Q399). Every explanation is rewritten in **simple English**: short sentences, everyday words, and technical terms explained in brackets. **350** questions that are easy to misread now start with a yellow **Clarification** box that says exactly what is being asked, the word to watch, and any look-alike question with a different answer.

Example: Q283 asks only about **compute** charges of a deallocated VM (answer No). Storage still costs money - that is tested separately in Q236 (answer: Storage).

Answer Corrections (17)

These answers were wrong in the previous PDF. Each one is explained at the question.

Q	Question (abridged)	Previously printed	Correct answer
17	Which of the following can be used to allow an organization to manage the compliance of resources across multiple subscriptions?	Azure Policies.	**Management Groups.**
39	You have a virtual machine named VM1 that runs Windows Server 2016. VM1 is in the East US Azure region. Which Azure service should you us...	Azure virtual machines.	**Azure Monitor.**
47	You have an Azure subscription named Subscription1. You sign in to the Azure portal and create a resource group named RG1. From Azure doc...	No.	**Yes.**
80	[Resource groups] provide organizations with the ability to manage the compliance of Azure resources across multiple subscriptions.	Azure policies.	**Management groups.**
116	You are planning to migrate a company to Azure. Each of the company has numerous divisions will have an administrator in place to manage ...	No.	**Yes.**
134	Every Azure region has multiple datacenters.	Yes.	**No.**
143	An Azure subscription can be managed by using a Microsoft account only.	Yes.	**No.**
150	Azure Monitor can send alerts to Microsoft Entra ID security groups.	Yes.	**No.**
162	An Azure administrator plans to run a PowerShell script that creates Azure resources. You need to recommend which computer configuration ...	A computer that runs macOS and has PowerShell Core 6.0 installed.; A computer that runs Windows 10 and has the Azure PowerShell module installed.; A computer that runs Chrome OS and uses Azure Cloud Shell.	**A computer that runs Windows 10 and has the Azure PowerShell module installed.; A computer that runs Linux and has the Azure PowerShell module installed.; A computer that runs Chrome OS and uses Azure Cloud Shell.**
165	Azure virtual machines that run Windows Server 2016 can encrypt network traffic sent to the Internet.	No.	**Yes.**
181	You can create Group Polices in Microsoft Entra ID.	Yes.	**No.**
209	Which task can you perform by using Azure Advisor?	Estimate the costs of an Azure solution.	**Confirm that Azure subscription security follows best practices.**
210	If your company uses an Azure free account, you will only be able to use a subset of Azure services.	Yes.	**No.**
227	An Azure Reservation is used to reserve server capacity at a specific data center.	Yes.	**No.**
248	Azure DevOps Services can be used to build and host web apps.	Yes.	**No.**
263	Which cloud deployment solution is used for Azure virtual machines and Azure SQL databases?	Infrastructure as a service (IaaS).	**Infrastructure as a service (IaaS).; Platform as a service (PaaS).**
265	Azure Reserved VM Instances are an example of OpEx.	No.	**Yes.**

Renamed and Retired Services

Older study material may use previous names. Use the new name when you answer:

Old name	Name today
Azure Active Directory (Azure AD)	Microsoft Entra ID
Azure Security Center / Azure Defender	Microsoft Defender for Cloud
Azure Sentinel	Microsoft Sentinel
Cognitive Services	Azure AI services
Windows Virtual Desktop	Azure Virtual Desktop
Azure Cost Management	Microsoft Cost Management
Compliance Manager	Microsoft Purview Compliance Manager
Azure Information Protection	Microsoft Purview Information Protection
Azure Machine Learning Studio (classic)	Azure Machine Learning studio (designer, AutoML)
Premier support	Unified support
Azure MFA	Microsoft Entra multifactor authentication
Azure Blueprints	Retired July 2026: template specs + deployment stacks
Azure Germany	Closed 2021: Germany West Central / North regions
Unmanaged (classic) disks	Retired 2025: managed disks
MSDN forums	Microsoft Q&A;

Questions by Topic

Topic	#	Questions
Identity	48	8, 9, 16, 53, 81, 106, 107, 108, 111, 112, 145, 146, 147, 169, 175, 176, 177, 181, 182, 183, 193, 194, 195, 196, 199, 200, 201, 206, 207, 251, 261, 354, 355, 356, 357, 358, 362, 363, 365, 366, 367, 368, 369, 372, 374, 375, 377, 378
Cost	39	2, 31, 58, 64, 83, 84, 93, 94, 102, 135, 140, 160, 216, 219, 225, 226, 227, 234, 235, 236, 240, 241, 259, 264, 282, 283, 284, 285, 286, 287, 379, 380, 381, 382, 383, 384, 387, 388, 396
Governance	36	17, 59, 80, 98, 114, 116, 119, 128, 129, 144, 180, 185, 190, 191, 192, 198, 204, 205, 250, 252, 257, 258, 271, 272, 273, 293, 385, 386, 390, 391, 392, 393, 395, 397, 398, 400
Monitoring	29	6, 12, 14, 39, 136, 149, 150, 151, 152, 153, 155, 157, 158, 202, 203, 209, 255, 403, 404, 405, 406, 407, 410, 412, 414, 416, 419, 420, 424
Networking	29	40, 44, 51, 52, 54, 78, 104, 109, 110, 163, 164, 170, 172, 173, 174, 214, 249, 254, 327, 328, 329, 339, 340, 341, 342, 343, 344, 349, 350
Subscriptions	27	22, 42, 61, 62, 115, 118, 120, 121, 122, 123, 124, 125, 143, 210, 211, 212, 215, 222, 223, 224, 306, 307, 308, 309, 310, 311, 326
Storage	24	26, 41, 117, 130, 131, 132, 237, 242, 330, 331, 332, 333, 334, 335, 336, 337, 338, 345, 346, 347, 348, 351, 352, 353
Security	23	50, 55, 56, 77, 79, 105, 156, 165, 166, 167, 168, 171, 178, 197, 256, 359, 360, 361, 364, 370, 371, 373, 376
Compute	22	1, 4, 43, 72, 137, 138, 139, 253, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325
Infrastructure	18	36, 75, 126, 127, 133, 134, 141, 142, 148, 189, 262, 270, 300, 301, 302, 303, 304, 305
Service models	15	10, 28, 29, 32, 34, 35, 89, 90, 263, 274, 277, 278, 295, 296, 297
Deployment models	15	11, 27, 30, 65, 66, 67, 69, 91, 92, 246, 267, 268, 269, 280, 281

Topic	#	Questions
DevOps & IaC	15	33, 48, 49, 97, 101, 103, 247, 248, 402, 408, 411, 413, 417, 418, 422
Cloud concepts	15	68, 70, 74, 87, 88, 265, 266, 275, 276, 279, 288, 289, 290, 291, 292
SLA	14	5, 23, 24, 60, 99, 100, 208, 217, 218, 229, 238, 239, 298, 299
Management tools	14	25, 37, 46, 47, 154, 159, 161, 162, 294, 401, 409, 415, 421, 423
Compliance	13	7, 76, 82, 179, 184, 186, 187, 188, 244, 245, 389, 394, 399
Lifecycle	12	18, 19, 20, 21, 63, 86, 213, 220, 221, 231, 232, 233
Data & AI	8	3, 13, 38, 71, 73, 95, 96, 260
Databases	3	15, 228, 243
Support	3	57, 85, 113
IoT	2	45, 230

Question 1

AZURE ARCHITECTURE & SERVICES

Compute
v5 Q1

If you plan to host a web application in the Azure platform as a service solution of Azure Web Apps, then the platform will have the ability to scale automatically?

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
App Service (Web Apps) is PaaS (Azure runs the platform for you). It can add instances by itself. Autoscale rules need the Standard tier or higher. Premium v2/v3 also offers automatic scaling based on HTTP traffic.

✖ **No.**
App Service adds instances for you. Only rule-based autoscale needs Standard tier or higher, but the platform can still scale by itself.

EXAM STRATEGY
Look for two clues: 'platform as a service' and 'scale automatically'. In PaaS, Azure owns the scaling. So the answer is Yes. You do not need to build VM scale sets yourself.

IMPACT
App Service saves you from writing your own scaling scripts. A site that cannot scale out during a traffic spike loses sales and may go down.

What · Where · When · Why · How

What	Azure App Service (Web Apps) is a PaaS host for web apps. It can scale out by itself.
Where	You set it on the 'Scale out' page of the App Service plan in the Azure portal.
When	Use it when web traffic goes up and down and you do not want to manage VMs.
Why	Azure adds and removes instances for you, so the app scales automatically.
How	Set autoscale rules (for example CPU above 70%) on Standard or higher. Or turn on automatic scaling on Premium v2/v3.

Exam tip
PaaS + 'scale automatically' = Yes. The trap is thinking you must build a VM scale set. That is the IaaS (you manage the VMs) way. Autoscale needs Standard tier or higher.

Memory trick
App Service: you bring the code, Azure handles the crowd.

Real-world example, use case and scenario

EXAMPLE
CPU rule
A Standard S1 plan grows from 2 to 6 instances when average CPU is above 70 percent for 10 minutes. It shrinks again when CPU falls below 30 percent.

USE CASE
Ticket sales site
A concert company runs its booking site on App Service. The site handles the rush when tickets go on sale at 10 a.m. The company does not buy extra servers.

SCENARIO
Exam-season portal
A university results portal was slow every results day. IT moved it from one IIS VM to App Service Premium v3 with automatic scaling. On the next results day it grew to 8 instances for two hours, then went back to 2. No admin had to act.

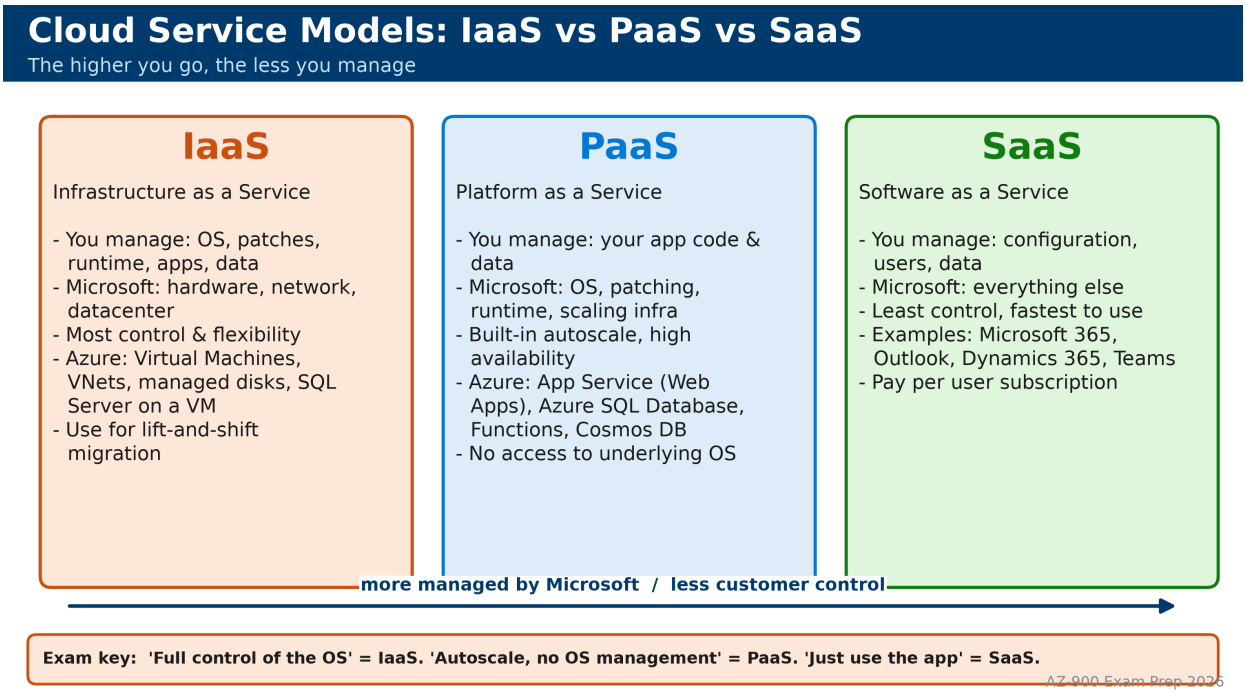


Figure Q1a: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

• Serverless options such as Azure Functions are treated as PaaS-style: no OS access, automatic scaling and per-execution billing.

• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

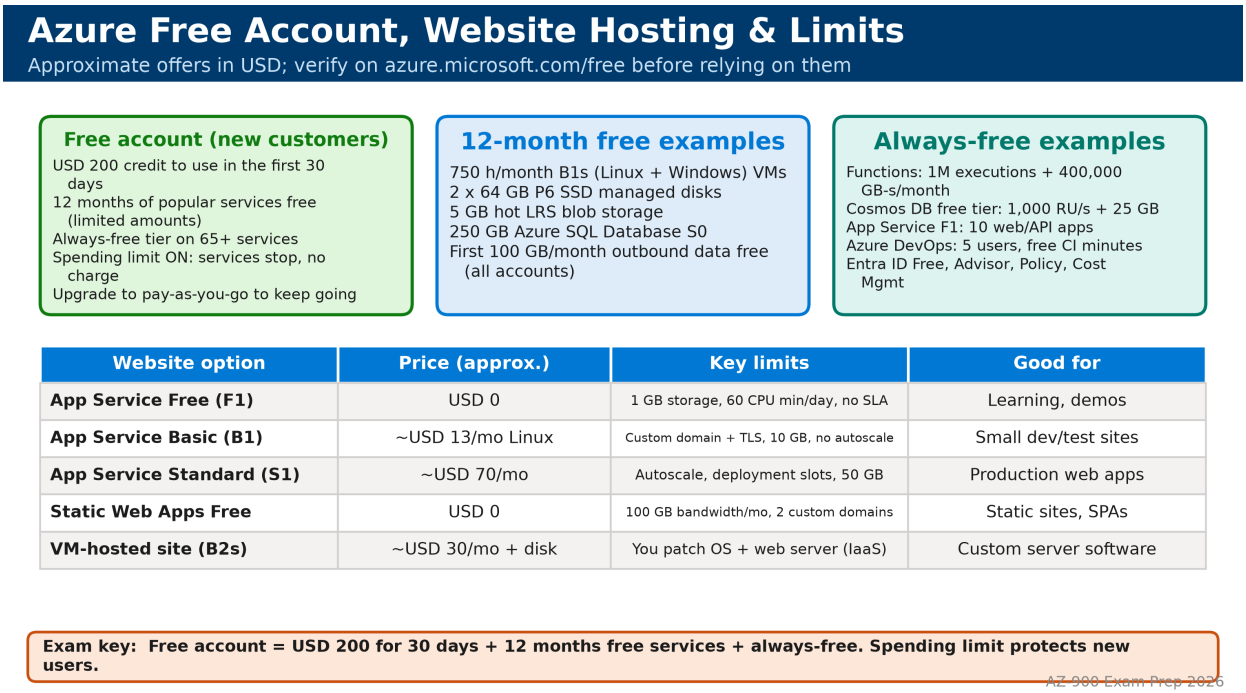


Figure Q1b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 2

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q2, Q316

You decide to create 2 Virtual machines. Each virtual machine is of the size D2s v3. Would these machines always generate the same monthly cost?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The word to watch is 'always'. The question asks if two VMs of the same size must cost exactly the same each month. One different factor (region, OS licence, running hours) is enough to make the answer No.

Answer explained

✔ **No.**
VM size only sets the compute rate. The monthly bill also changes with region, OS licence, disks, egress (data leaving Azure), running hours and purchase option (reservation, Spot, Hybrid Benefit).

✖ **Yes.**
Same size does not mean same cost. Region, OS licence, disks, egress (data leaving Azure) and running time can differ between the VMs.

EXAM STRATEGY
The word 'always' decides it. Any one factor (region, licence, running hours, reservation) can change the cost. So answer No.

IMPACT
If you plan budgets by VM size alone, you will overspend. Region and shutdown schedules can change the bills of identical VMs a lot.

What · Where · When · Why · How

What	VM cost = compute + OS licence + disks + networking + purchase option.
Where	You see it per resource in Cost Management. You can estimate it in the Pricing calculator.
When	When you plan budgets for VMs of the same size but different region, OS or running hours.
Why	Other factors besides size change the bill. So the same size does not mean the same cost.
How	Compute is billed per running hour at a rate that depends on the region. Disks and egress (data leaving Azure) are billed separately.

Exam tip
'Always' is the trap. Region alone breaks it: East US and Switzerland North have different prices. Running hours also break it: a VM that is deallocated (stopped, hardware released) at night pays much less for compute.

Memory trick
Same shirt size, different shop, different price. Size is not the price tag.

Real-world example, use case and scenario

EXAMPLE
Linux vs Windows
A D2s v3 VM running Ubuntu costs less per hour than one running Windows Server. The Windows licence is included in the price unless you use Hybrid Benefit.

USE CASE
Dev box schedules
A software company shuts down its developer VMs at 7 p.m. So these D2s v3 VMs pay for about half the compute hours of the same-size production VMs.

SCENARIO
Budget surprise
A retailer planned the same budget for two D2s v3 VMs. One was in West Europe with Premium SSD and served downloads. The other was in East US with Standard SSD. The first bills differed by over 40 percent. Cost Management showed the cause: region, disk type and egress.

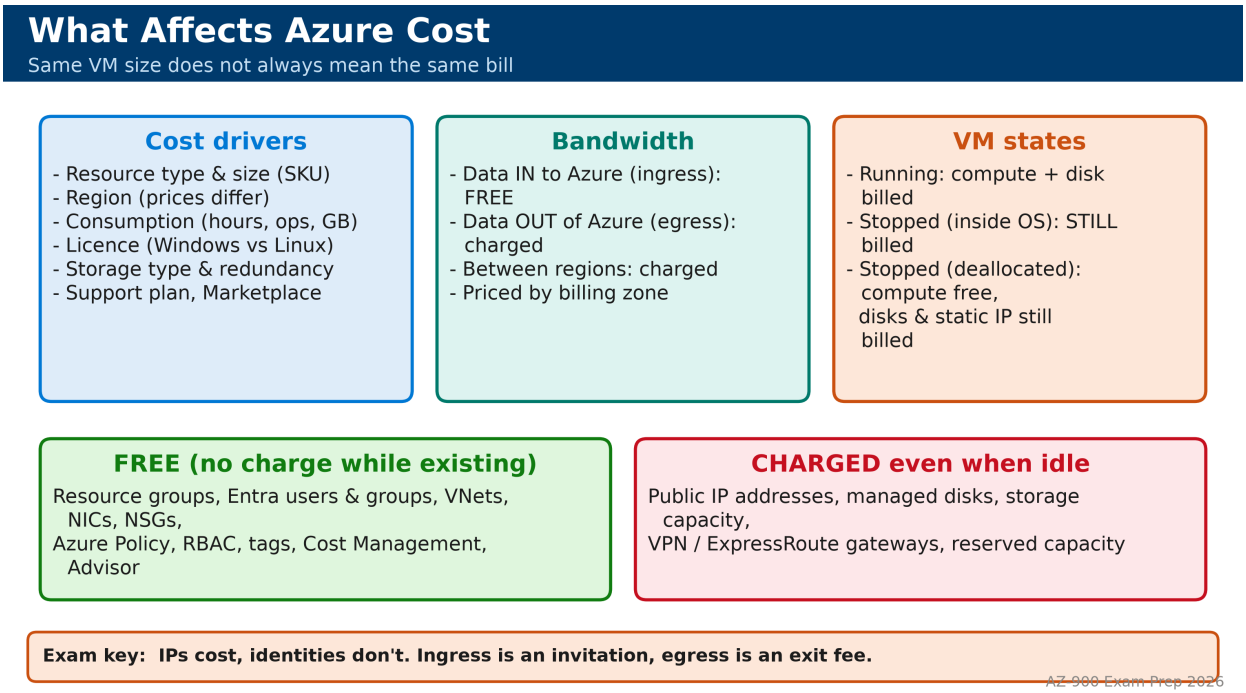


Figure Q2a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

• Bandwidth: ingress is free; the first 100 GB/month of Internet egress is free, then approx. USD 0.087/GB in North America and Europe.

• Inter-region transfer is approx. USD 0.02/GB within a continent; VNet peering is approx. USD 0.01/GB each way in the same region.

• A VM stopped inside the guest OS is still allocated and billed for compute; only Stopped (deallocated) from the portal, CLI or API stops compute charges.

• A Standard static public IP costs approx. USD 3.65/month even unattached; a 128 GB Premium SSD costs approx. USD 20/month even when its VM is deallocated.

• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q2b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 3

AZURE ARCHITECTURE & SERVICES

Data & AI
v5 Q3

Your team needs to have a tool that provides a digital online assistant that can provide speech support. Which of the following service can be used for this purpose?

✖ A. Azure Machine Learning.

✖ B. Azure loT Hub.

✔ **C.** Azure Al bot.

✖ D. Azure Functions.

Correct answer: Azure Al bot.

Clarification (read this first): The option 'Azure Al bot' means Azure AI Bot Service (the 'Al' is a typo for 'AI'). The question asks for a digital assistant that can talk. A chat or voice assistant is a bot.

2026 update: The option says 'Azure Al bot'. The real name is Azure AI Bot Service. For voice, it is usually used with Azure AI Speech (part of Azure AI services).

Answer explained

✔ **Azure AI bot.**
Azure AI Bot Service builds and runs chat assistants. Azure AI Speech adds voice through the Direct Line Speech channel.

✖ **Azure Machine Learning.**
It trains your own prediction models from data. It does not give you a chat assistant.

✖ **Azure IoT Hub.**
It connects IoT (internet-connected) devices and collects their data. It has no chat or speech features.

✖ **Azure Functions.**
It runs small pieces of code when an event happens. A bot can call it, but it is not a bot platform.

EXAM STRATEGY
The words 'digital assistant' and 'speech' mean a conversation. Remove Machine Learning (training models), IoT Hub (devices) and Functions (code). The bot service is left.

IMPACT
With the managed bot service and Speech, you can build a voice assistant in days. You do not need to build language tools yourself. This 'service for scenario' matching appears often on AZ-900.

What · Where · When · Why · How

What	Azure AI Bot Service is a managed platform to build and run chat bots.
Where	You create an Azure Bot resource and connect it to channels such as Teams, web chat or phone.
When	When users need a digital assistant that answers by chat or voice.
Why	It gives the chat assistant. Azure AI Speech gives the speech support.
How	Write the bot logic, register it in Bot Service, and add Direct Line Speech for voice.

Exam tip
'Digital assistant', 'chatbot', 'virtual agent' and 'answer questions in a conversation' all point to Bot Service. 'Speech' just adds Azure AI Speech.

Memory trick
Talks back? That is a bot.

Real-world example, use case and scenario

EXAMPLE
Voice FAQ
A bot answers 'What are your opening hours?' It works when typed in web chat or spoken on a phone line using Azure AI Speech.

USE CASE
Airline rebooking
An airline puts a voice bot on its phone line. The bot rebooks cancelled flights, so callers do not wait for an agent.

SCENARIO
Clinic receptionist
A chain of dental clinics missed calls at lunch time. They built an assistant with Azure AI Bot Service and Speech that books and cancels appointments by voice. Complaints about missed calls fell. Staff now handle only difficult cases.

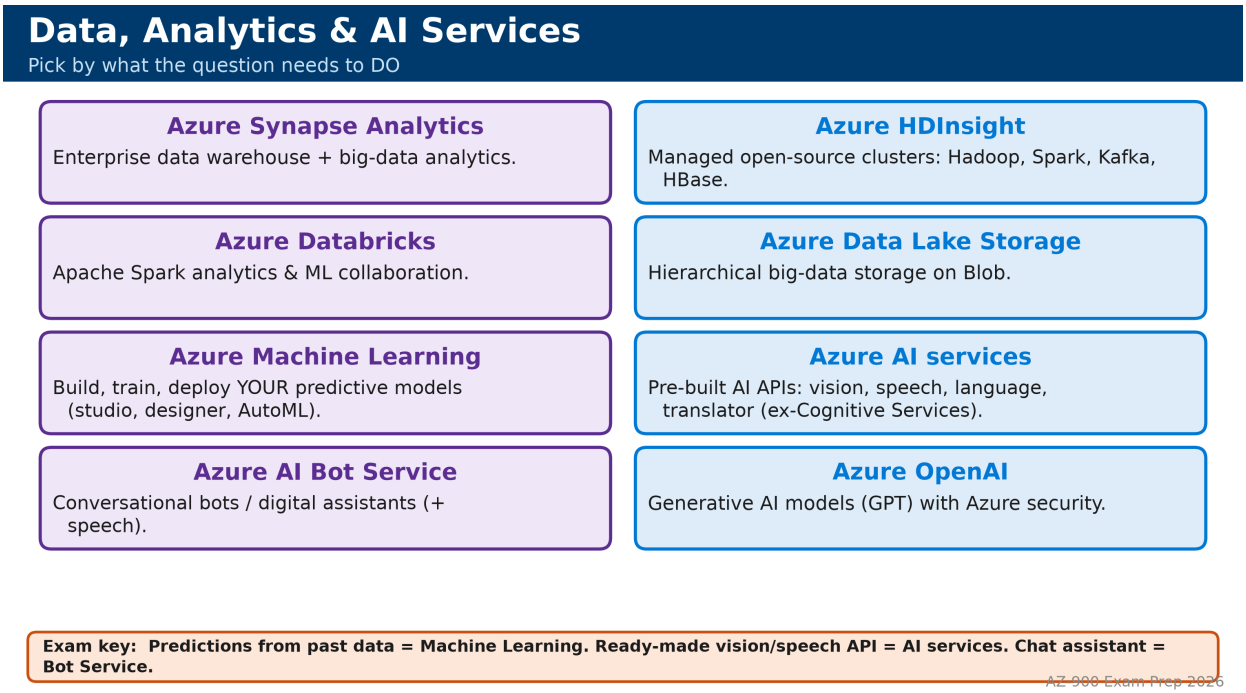


Figure Q3: Data, analytics and AI services

About this figure

- Left column covers data and analytics platforms; right column covers AI. Decide whether the scenario needs storage, processing, custom models or ready-made AI.
- Azure Machine Learning trains your own models from your data; Azure AI services are prebuilt APIs needing no data science, billed per transaction.
- Azure AI services include Vision, Speech, Language, Translator and Document Intelligence; many offer a free F0 tier with limited monthly calls.
- Azure OpenAI models are now offered through Microsoft Foundry (formerly Azure AI Foundry), billed per 1,000 or million tokens, varying widely by model.
- Microsoft Fabric is the newer unified analytics SaaS combining data engineering, warehousing and Power BI, and overlaps much of Synapse.

Question 4

AZURE ARCHITECTURE & SERVICES

Compute
v5 Q5

Your team needs to have a tool that can provide serverless computing capabilities. Which of the following service can be used for this purpose?

✖ A. Azure Machine Learning.

✖ B. Azure IoT Hub.

✖ C. Azure AI bot.

✔ **D.** Azure Functions.

Correct answer: Azure Functions.

Clarification (read this first): The key word is 'serverless'. Look-alike: Q43 also asks for a serverless solution, but for rule-based email with no code. Its answer is a logic app. Functions = your own code; Logic Apps = workflow with ready-made connectors.

Answer explained

✔ **Azure Functions.**
Azure Functions is Azure's serverless compute (you manage no servers). Code runs when an event happens and scales by itself. On Consumption plans you pay per run.

✖ **Azure Machine Learning.**
A platform to train and deploy ML models on compute you choose. It is not general serverless compute.

✖ **Azure IoT Hub.**
A messaging hub for IoT device data. It does not run your code.

✖ **Azure AI bot.**
A platform to build chat bots. It is not general serverless compute.

EXAM STRATEGY
The word 'serverless' points straight to Functions. The other three are AI or device services. One keyword match answers the question.

IMPACT
On the Consumption plan you pay per run. Event-based jobs cost cents instead of an always-on VM. Spotting serverless appears in almost every exam.

What · Where · When · Why · How

What	Azure Functions is event-driven serverless compute. It runs code when a trigger fires.
Where	A Function App in a region, on a Consumption, Flex Consumption or Premium plan.
When	When small pieces of code must run on events such as HTTP calls, timers or queue messages.
Why	No servers to manage, automatic scaling, and pay-per-run billing on Consumption plans.
How	Write a function and link it to a trigger. Azure runs it when the event happens.

Exam tip
'Serverless', 'event-driven', 'pay per execution', 'run code without managing servers' = Functions. If it is a workflow with connectors and no code, the answer is Logic Apps.

Memory trick
Functions run when poked, and you pay per poke.

Real-world example, use case and scenario

EXAMPLE
Thumbnail on upload
A blob trigger runs a function each time an image arrives in a container. The function saves a 200 px small copy.

USE CASE
Webhook handler
An online shop uses an HTTP-triggered function to handle messages from its payment provider. At low volume it costs cents per month.

SCENARIO
Nightly report
A delivery company ran a VM all day just to email a 2 a.m. report. They replaced it with a timer-triggered function. The report still runs every night, but they pay only for a few seconds of run time.

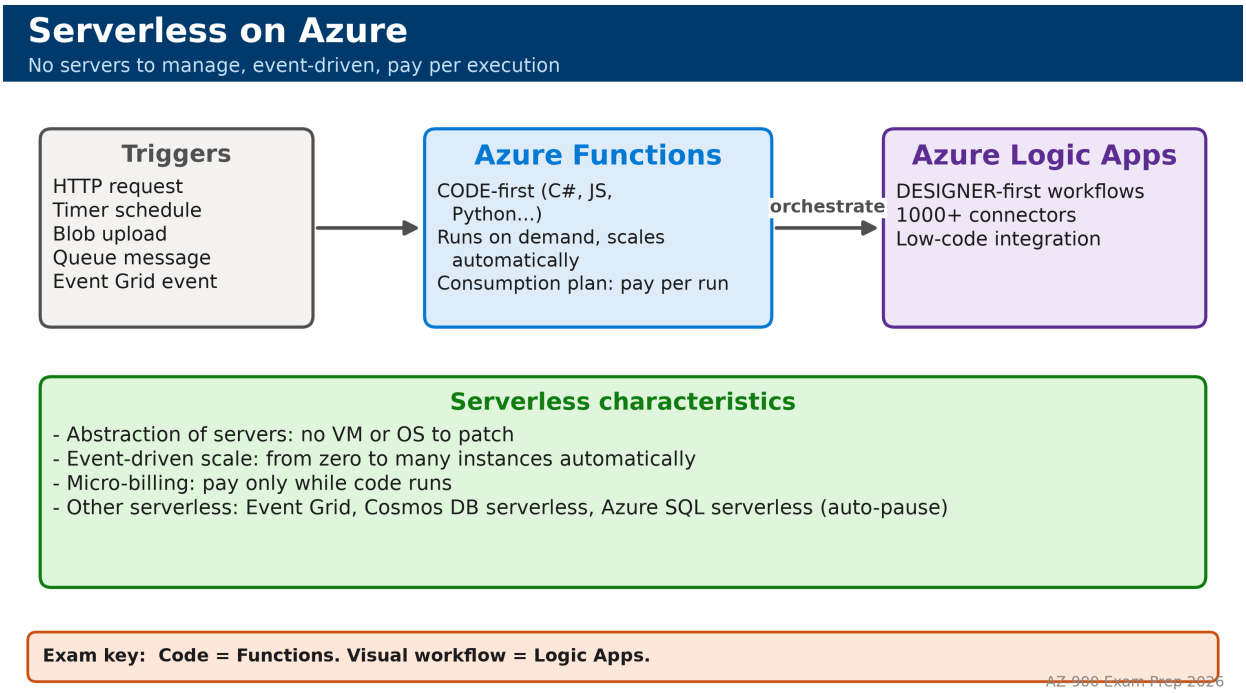


Figure Q4: Serverless on Azure

About this figure

- Flow left to right: an event trigger starts code (Functions) or a workflow (Logic Apps); the green box lists traits shared by serverless services.
- Functions Consumption: approx. USD 0.20 per million executions plus GB-seconds, with 1 million executions and 400,000 GB-s free monthly.
- Consumption-plan functions can cold start and have a default 5-minute, maximum 10-minute timeout; Flex Consumption and Premium plans reduce cold starts.
- Logic Apps Consumption bills per action and connector call, roughly fractions of a cent each; Standard plan bills per hosting plan instead.
- Serverless still runs on servers; Microsoft manages and scales them, and you pay nothing while nothing runs, except storage or plan minimums.

Question 5

CLOUD CONCEPTS

SLA
v5 Q8, Q172, Q211, Q212

Your company needs to deploy an application to virtual machines hosted in Azure. The solution must ensure an SLA of 99.99%. What is the minimum number of virtual machines and availability zones that you need to recommend for the deployment?

✖ A. One virtual machine and One availability zone.

✖ B. Two virtual machines and availability zone.

✖ C. One virtual machine and Two availability zones.

✔ **D.** Two virtual machines and Two availability zones.

Correct answer: Two virtual machines and Two availability zones.

Clarification (read this first): The key words are 'minimum' and '99.99%'. You need the smallest layout that still reaches 99.99%. Look-alikes: Q99 (2 VMs in 1 zone) and Q100 (1 VM in 2 zones) both answer No.

Answer explained

✔ **Two virtual machines and Two availability zones.**
At least two VMs, placed in at least two availability zones, is the smallest layout that gets the 99.99% VM SLA (uptime promise).

✖ **One virtual machine and One availability zone.**
One VM gets at most 99.9%, and only with Premium SSD or Ultra disks.

✖ **Two virtual machines and availability zone.**
Two VMs in one zone cannot survive the loss of that datacenter. Even in an availability set they get only 99.95%.

✖ **One virtual machine and Two availability zones.**
A VM lives in exactly one zone. So one VM cannot use two zones.

EXAM STRATEGY
Look for 'minimum' and '99.99%'. Four nines needs zones. Remove every option with one VM or one zone. Only two VMs in two zones is left.

IMPACT
If you build too small, you get no SLA credit, and one datacenter failure takes the app down. SLA levels by VM layout are tested often.

What · Where · When · Why · How

What	Availability zones are physically separate datacenters inside one Azure region.
Where	In regions that support zones. You pick the zone (1, 2 or 3) for each VM when you create it.
When	When a workload must survive the failure of a whole datacenter.
Why	Two VMs in two zones is the smallest layout for the 99.99% VM SLA.
How	Put at least two VMs in different zones behind a zone-redundant load balancer.

Exam tip
Ladder: 1 VM = 99.9%. 2 VMs in an availability set = 99.95%. 2 VMs across zones = 99.99%.
'Minimum' + 'four nines' = two VMs, two zones.

Memory trick
Four nines needs two-by-two: two VMs, two zones.

Real-world example, use case and scenario

EXAMPLE
Zone placement
VM-A is in zone 1 and VM-B in zone 2 of West Europe, behind a Standard Load Balancer. This layout qualifies for 99.99 percent.

USE CASE
Payment API
A finance company runs its payment API on VMs in zones 1, 2 and 3. One datacenter outage cannot take it down.

SCENARIO
Contract requirement
An insurer's contract required 99.99 percent uptime. The architect first suggested two VMs in an availability set, but that gives only 99.95 percent. Moving the second VM to a different zone met the contract with no extra VM cost.

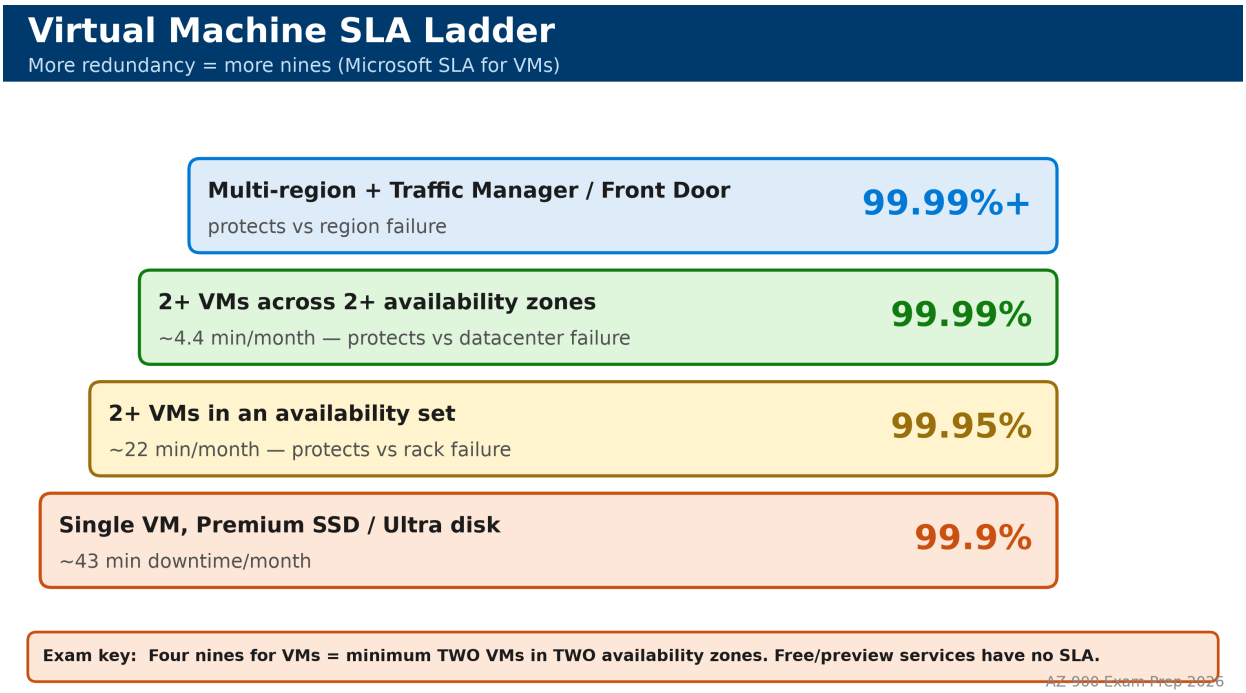


Figure Q5a: Virtual machine SLA ladder

About this figure

- Climb the ladder: each step adds redundancy and a higher VM SLA; the downtime figures assume a 30-day month.
- A single VM with Standard SSD gets about 99.5% and with Standard HDD about 95%; all its OS and data disks must meet that tier.
- Availability sets spread VMs across fault domains (racks) and update domains (maintenance batches) in one datacenter; zones spread across datacenters.
- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

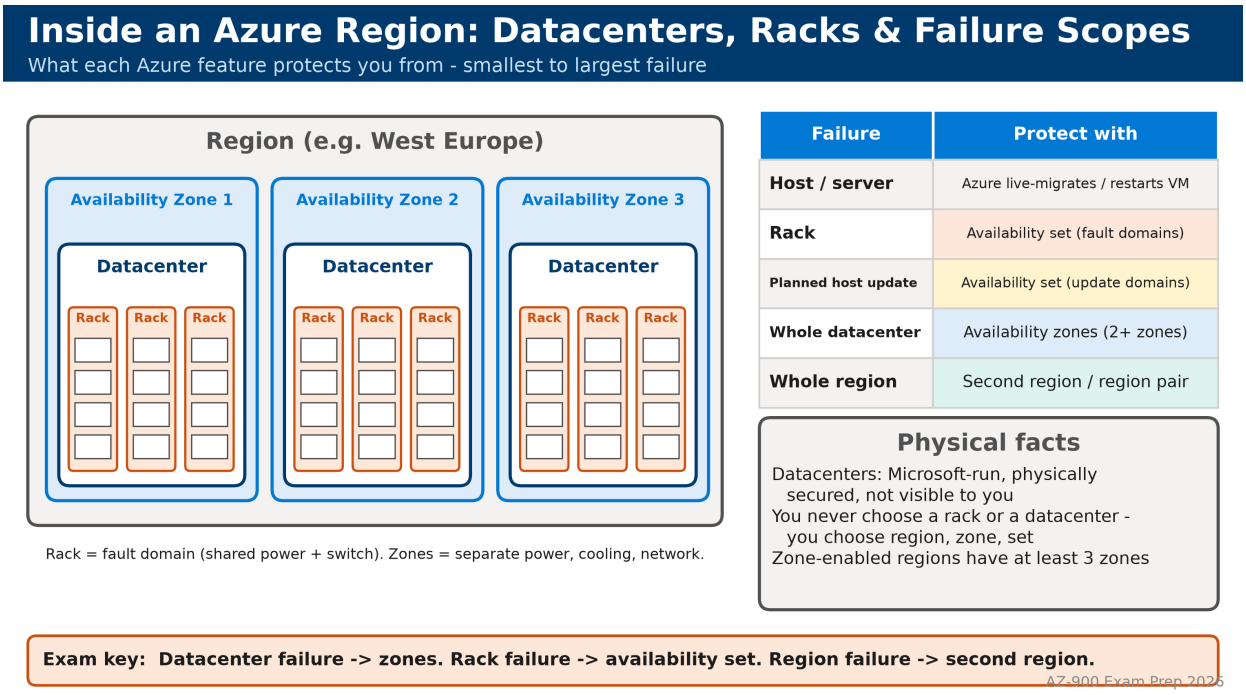


Figure Q5b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 6

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q9, Q92

Your team needs a tool that can be used to correlate events from multiple resources into a central repository. Which of the following can be used for this purpose?

✖ A. Azure Event Hubs.

✖ B. Microsoft Defender for Cloud.

✖ C. Microsoft Entra ID.

✔ **D.** Azure Log Analytics.

Correct answer: Azure Log Analytics.

Clarification (read this first): Watch the words 'correlate' (link events together) and 'central repository' (one place that stores them). Event Hubs moves events but does not keep them for searching. You need a store you can query.

Answer explained

✔ **Azure Log Analytics.**
A Log Analytics workspace (Azure Monitor Logs) collects logs from many resources into one store. KQL queries then link events together.

✖ **Azure Event Hubs.**
It streams events to other services. It does not store them for searching and linking.

✖ **Microsoft Defender for Cloud.**
It is a security tool that checks your setup and finds threats. It uses a workspace, but it is not the central log store.

✖ **Microsoft Entra ID.**
It is the identity service. It creates sign-in and audit logs, but it does not collect logs from other resources.

EXAM STRATEGY
'Correlate' and 'central repository' mean store logs and query them. Event Hubs streams but does not store for queries. Defender and Entra are distractors.

IMPACT
With one central workspace, responders can trace a failure across resources in minutes. They do not need to check each service one by one.

What · Where · When · Why · How

What	A Log Analytics workspace is the Azure Monitor Logs store for log and event data.
Where	A workspace resource in a region. Resources send data to it through diagnostic settings or agents.
When	When events from many resources must be collected and linked in one place.
Why	It keeps all logs together, so one query can join events from many sources.
How	Send diagnostics to the workspace. Then link events with KQL (Kusto Query Language).

Exam tip
'Central repository', 'correlate', 'query logs' = Log Analytics. Event Hubs moves events. Log Analytics keeps and searches them.

Memory trick
Log Analytics is the library; KQL is the librarian.

Real-world example, use case and scenario

EXAMPLE
KQL join
A query joins failed sign-ins from Entra ID with network flow logs. It shows which IP addresses appear in both.

USE CASE
Retail ops center
A retailer sends logs from 300 store VMs and its App Service apps into one workspace. The team uses it to fix checkout failures.

SCENARIO
Slow-down mystery
A software company had random slowdowns, but logs were spread across VMs, SQL and the load balancer. They sent all diagnostics to one workspace and matched times with KQL. The cause was a nightly backup that overloaded the disks.

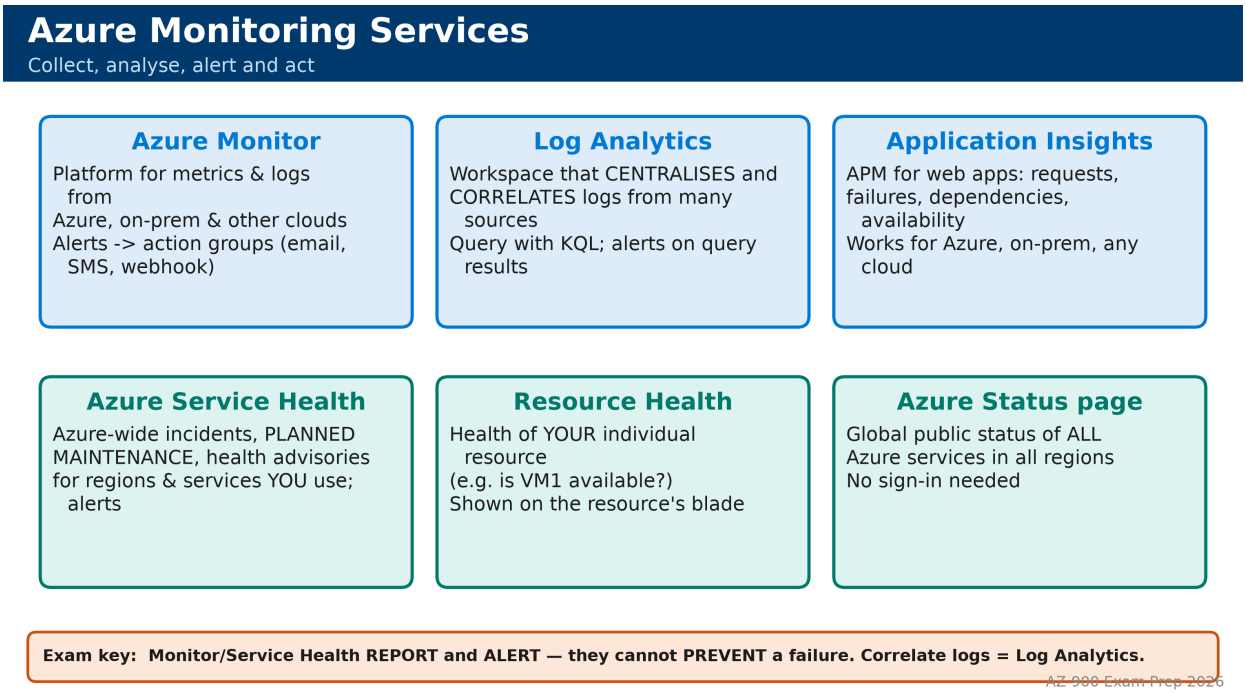


Figure Q6: Azure monitoring services

About this figure

- Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).
- Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.
- Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.
- Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.
- Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 7

AZURE ARCHITECTURE & SERVICES

Compliance
v5 Q12, Q284

Which of the following customers are eligible to use Azure Government to develop a cloud solution? Choose 2 answers from the options given below.

- ✔ **A.** United states government Entity.
- ✔ **B.** A United states government contractor.
- ✖ C. A European government Entity.
- ✖ D. A European government contractor.

Correct answer: **United states government Entity.; A United states government contractor.**

Clarification (read this first): Choose TWO answers. Azure Government is for the United States only. Both US options are correct; both European options are wrong. See also Q302 (US agency needs an isolated cloud -> Azure Government).

Answer explained

✔ **United states government Entity.; A United states government contractor.**
Azure Government is a US sovereign cloud (a separate cloud for one country). Only US federal, state, local and tribal government bodies and their approved partners or contractors can use it.

✖ **A European government Entity.**
Non-US government bodies cannot use it. European public bodies use public EU regions or EU sovereign offers.

✖ **A European government contractor.**
A contractor qualifies only when it works for US government agencies. A European contractor fails the check.

EXAM STRATEGY
'Azure Government' means United States only. Remove both European options at once. The two US options are the pair.

IMPACT
Putting regulated US government workloads in normal regions can break FedRAMP or CJIS rules (US government security standards) and break contracts.

What · Where · When · Why · How

What	Azure Government is a separate, isolated US cloud for government workloads.
Where	Special US Gov regions such as US Gov Virginia and US Gov Arizona, with their own portal.
When	When a US government agency or its contractor needs standards such as FedRAMP High or DoD.
Why	Only US government bodies and their approved partners or contractors can use it.
How	The organization applies, passes an eligibility check, and then gets an Azure Government subscription.

Exam tip
Azure Government = US government AND its partners/contractors. Anything non-US is out. The other separate cloud to remember is Azure China (run by 21Vianet).

Memory trick
The Gov cloud wears a US flag: agencies and their contractors only.

Real-world example, use case and scenario

EXAMPLE
State agency
A US state's motor vehicle department runs its licensing system in US Gov Virginia.

USE CASE
Defense contractor
A US defense supplier builds a logistics portal for the Department of Defense in Azure Government. This meets DoD security rules.

SCENARIO
Rejected sign-up
An IT supplier for a French ministry tried to sign up for Azure Government to keep data in-country. It failed the check. The architect used France Central with the EU Data Boundary instead. The project went ahead in public Azure.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

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Figure Q7: Compliance, privacy and trust

About this figure

- Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.
- Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.
- Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.
- Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.
- The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 8

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q13

A team is currently planning on using Azure for hosting resources. They are going to create users which would have access to the Azure resources. They want to implement Multi-Factor authentication for the users. Is it required to deploy a federated solution to implement Multi-Factor authentication?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The word to watch is 'required'. The question asks if you MUST set up federation (sign-in through another system, such as AD FS) before you can use MFA. You do not. MFA is built into Entra ID.

Answer explained

✔ **No.**
Microsoft Entra MFA (multifactor authentication: a second check at sign-in) is built into Entra ID. You turn it on with security defaults or Conditional Access. Cloud-only users need no federation.

✖ **Yes.**
Federation (for example AD FS) is an optional way to sign in, not a must. Entra MFA works directly in the cloud.

EXAM STRATEGY
The deciding word is 'required'. MFA is part of Entra ID through security defaults or Conditional Access. So any 'you need federation first' claim is false.

IMPACT
If you think MFA needs federation, you delay it. MFA stops most account-takeover attacks, so turn it on from day one.

What · Where · When · Why · How

What	Microsoft Entra MFA asks for a second proof of identity at sign-in.
Where	Turned on in the Entra admin center with security defaults or Conditional Access policies.
When	When cloud users need stronger sign-in security than a password alone.
Why	It is built into Entra ID, so you need no federation and no on-premises servers.
How	Turn on security defaults or a Conditional Access policy. Users register a method such as Microsoft Authenticator.

Exam tip
Watch for 'required'. MFA needs no federation, no Entra Connect and no on-premises AD. Security defaults turn it on for free.

Memory trick
MFA is born in the cloud; it needs no passport from on-prem.

Real-world example, use case and scenario

EXAMPLE
Security defaults
A new tenant has security defaults on. Every user must register Microsoft Authenticator within 14 days.

USE CASE
Startup with no servers
A 20-person design company has no on-premises AD. It requires MFA for all staff using Conditional Access.

SCENARIO
Avoided AD FS project
A consultant offered a charity a project to build AD FS servers before MFA. The board asked for a second opinion. It showed Entra MFA could be turned on directly. MFA was live the same afternoon with no new servers.

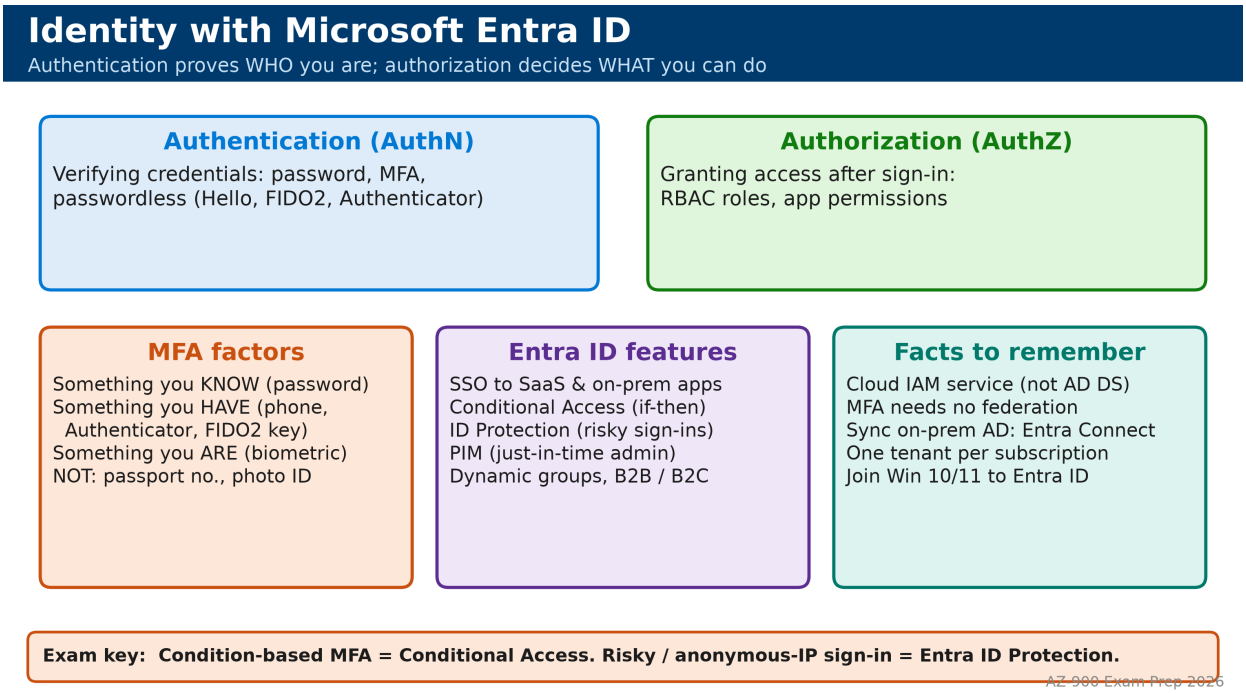


Figure Q8: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 9

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q14

A team is currently planning on using Azure for hosting resources. They are going to create users which would have access to the Azure resources. They want to implement Multi-Factor authentication for the users. Are two valid methods for Azure Multi-Factor authentication picture identification and entering a passport number.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The question asks if BOTH picture ID and passport number are valid MFA methods. Neither is. Entra MFA uses things like an app, a key, a phone call or a text, not document details.

Answer explained

✔ **No.**
Entra MFA methods are Authenticator, passkeys/FIDO2 keys, Windows Hello, OATH tokens, SMS, voice call and certificates. Photo ID and passport numbers are not supported.

✖ **Yes.**
Neither picture ID nor a passport number is an Entra MFA method. Azure cannot check them at sign-in.

EXAM STRATEGY
Entra MFA has a fixed list of methods (app, passkey, SMS, voice, token). Details typed from a paper document are not on it. So the answer is No.

IMPACT
If you know the valid methods, you can plan strong sign-in with passkeys. Weak factors fail audits.

What · Where · When · Why · How

What	Entra MFA methods are the ways users can prove who they are at sign-in.
Where	Managed in the Entra admin center under Authentication methods.
When	When you choose which second factors users may register.
Why	Only supported methods work. Photo ID and passport numbers are not on the list.
How	Users register Authenticator, FIDO2 keys, OATH tokens, SMS or voice, and use them at sign-in.

Exam tip
MFA factors are something you know (password), something you have (phone, key) or something you are (face, fingerprint). A typed document number is none of these in Entra.

Memory trick
MFA wants your phone, key or face, not your passport.

Real-world example, use case and scenario

EXAMPLE
Number matching
A user signs in and types the two-digit number shown on screen into the Authenticator app.

USE CASE
Factory floor keys
A factory gives FIDO2 security keys to shift workers who cannot carry phones on the production line.

SCENARIO
Auditor question
An auditor asked if staff could verify by typing their passport number. The security lead said Entra supports Authenticator, passkeys/FIDO2, Windows Hello, tokens, SMS and voice, but no document numbers. The audit report then required Authenticator for admins.

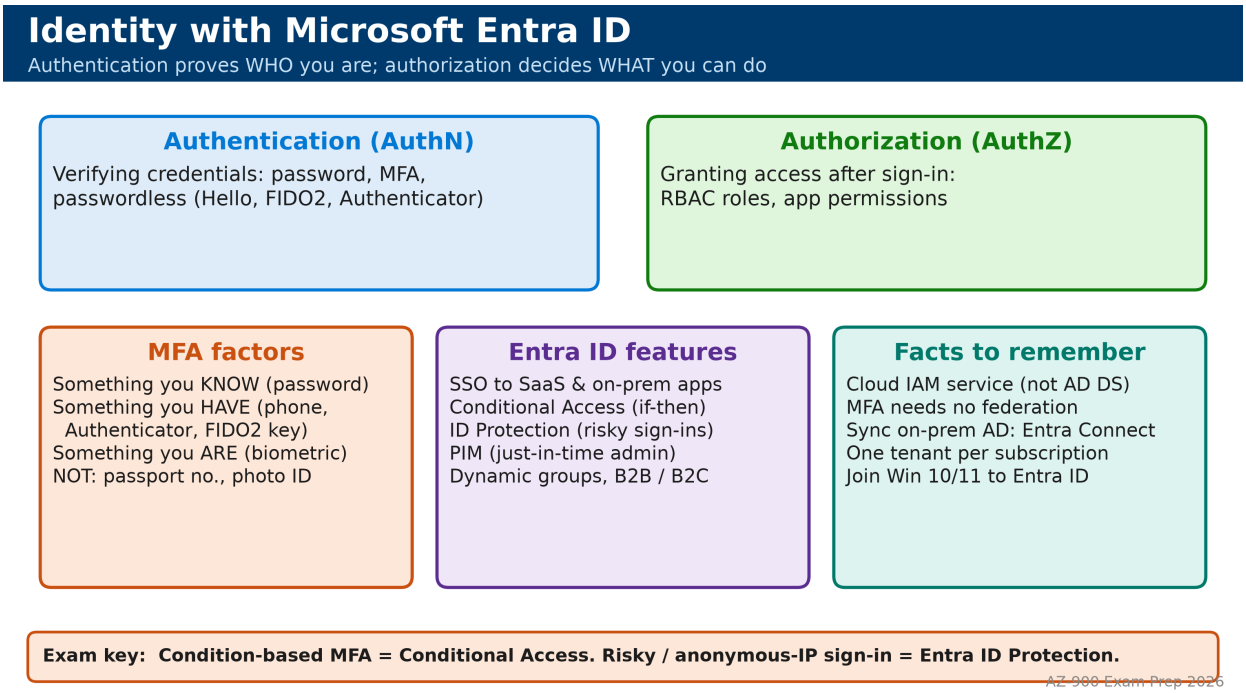


Figure Q9: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 10

CLOUD CONCEPTS

Service models
v5 Q18, Q380

You are looking at using a Software as a Service in Azure. Which of the following would you be responsible for?

- ✖ A. High availability of the solution.
- ✔ **B.** Configuration of the solution.
- ✖ C. Installing the solution.
- ✖ D. Scalability of the solution.

Correct answer: Configuration of the solution.

Clarification (read this first): The question asks what YOU (the customer) still do in SaaS (ready-made software). The provider does install, uptime and scaling. You still set up the app. Related: Q276 asks what is ALWAYS yours in every model: data and accounts/identities.

Answer explained

✔ **Configuration of the solution.**
In SaaS (ready-made software) the provider installs, runs, scales and keeps the app available. The customer still owns the settings, data, users and access.

✖ **High availability of the solution.**
Keeping the app available is the SaaS provider's job, backed by its SLA (uptime promise).

✖ **Installing the solution.**
SaaS comes ready to use. The provider does the installing.

✖ **Scalability of the solution.**
The SaaS provider scales the service for all customers.

EXAM STRATEGY
In SaaS, remove everything the vendor runs: install, availability, scaling. The only customer task in the list is configuration.

IMPACT
If you misunderstand shared responsibility, you may leave SaaS settings open and leak data. Who-does-what per model is tested again and again.

What · Where · When · Why · How

What	SaaS is complete software that the provider delivers and runs over the internet.
Where	Your tasks are in the app's admin settings, for example the Microsoft 365 admin center.
When	When you use ready-made software such as Microsoft 365 or Dynamics 365.
Why	Under shared responsibility, you still own settings, data, identities and access.
How	The provider runs everything else. You set up settings, users and policies.

Exam tip
In every model the customer always keeps data, identities, devices and configuration. SaaS removes everything else from you.

Memory trick
SaaS: they build the house, you arrange the furniture.

Real-world example, use case and scenario

EXAMPLE
Mailbox policies
In Microsoft 365 the admin sets retention policies and MFA. Microsoft runs the Exchange servers.

USE CASE
CRM rollout
A real-estate agency starts using Dynamics 365. It spends its time setting up sales stages and user roles, not servers.

SCENARIO
Shared link leak
A law firm's SaaS file-sharing app exposed documents because external sharing was left on. The vendor's platform worked as designed. The firm learned that settings were its job and turned sharing off.

Shared Responsibility Model

Who manages each layer: on-premises vs IaaS vs PaaS vs SaaS

Responsibility	On-premises	IaaS	PaaS	SaaS
Information & data	Customer	Customer	Customer	Customer
Devices (mobile, PCs)	Customer	Customer	Customer	Customer
Accounts & identities	Customer	Customer	Customer	Customer
Identity & directory infra	Customer	Customer	Shared	Shared
Applications	Customer	Customer	Shared	Microsoft
Network controls	Customer	Customer	Shared	Microsoft
Operating system	Customer	Customer	Microsoft	Microsoft
Physical hosts	Customer	Microsoft	Microsoft	Microsoft
Physical network	Customer	Microsoft	Microsoft	Microsoft
Physical datacenter	Customer	Microsoft	Microsoft	Microsoft

Exam key: Data, devices, accounts & identities are ALWAYS the customer's. Physical datacenter is NEVER yours in cloud.

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Figure Q10: Shared responsibility model

About this figure

- Read each row across: orange cells are yours, blue are Microsoft's, yellow are shared. Your share shrinks moving from on-premises to IaaS, PaaS and SaaS.
- Information and data, devices, and accounts and identities stay with the customer in every model, even in SaaS such as Microsoft 365.
- Physical hosts, network and datacenter move to Microsoft in every cloud model, so hardware failures, power and building security are never the customer's task.
- With IaaS you still patch the guest OS and configure network controls; with PaaS the OS becomes Microsoft's job, but your code and access settings remain yours.
- Responsibility does not equal liability: misconfigured access, weak passwords or leaked keys are customer failures even when the service itself is fully managed.

Question 11

CLOUD CONCEPTS

Deployment models
v5 Q20, Q122

Under which of the following circumstances could a company look at decommissioning its data center?

✖ A. If they have their infrastructure hosted in a private cloud.

✔ **B.** If they have their infrastructure hosted in the public cloud.

✖ C. If they have their infrastructure hosted in a hybrid cloud.

Correct answer: If they have their infrastructure hosted in the public cloud.

Clarification (read this first): 'Decommission its data center' means close it completely, with no own servers left. Only public cloud allows that. Hybrid keeps some on-premises servers (see Q66), and private cloud usually runs on your own hardware.

Answer explained

✔ **If they have their infrastructure hosted in the public cloud.**
When everything is in the public cloud, the provider owns and runs all the hardware. The company no longer needs its own datacenter.

✖ **If they have their infrastructure hosted in a private cloud.**
A private cloud usually runs on the company's own hardware. So it still needs its datacenter.

✖ **If they have their infrastructure hosted in a hybrid cloud.**
Hybrid cloud always keeps some servers on-premises. So a datacenter stays.

EXAM STRATEGY
Decommission means no hardware left. Private and hybrid both keep your own hardware, so remove them. Only public cloud is left.

IMPACT
Closing a datacenter removes power, cooling, rent and hardware replacement costs. This is a main reason companies move to the cloud.

What · Where · When · Why · How

What	Public cloud is hardware owned and run by a provider and shared over the internet.
Where	In the provider's datacenters, such as Azure regions around the world.
When	When a company wants to stop owning and running its own hardware.
Why	With all workloads in public cloud, the company needs no datacenter, so it can close it.
How	Move every workload to the provider, then retire the on-premises servers and building.

Exam tip
Close the datacenter = everything in public cloud. Hybrid and private both mean hardware you still run.

Memory trick
Public cloud: hand back the building keys.

Real-world example, use case and scenario

EXAMPLE
Lease ends
A company moves all servers to Azure before its datacenter rental ends, and does not renew it.

USE CASE
Media company exit
A video streaming company closed both its datacenters after moving video processing and web workloads to Azure.

SCENARIO
Half-way plan
An insurer planned to close its datacenter but kept a mainframe on-premises, so it was hybrid. The datacenter could not close until the mainframe work was replaced. It moved to a small rented space instead.

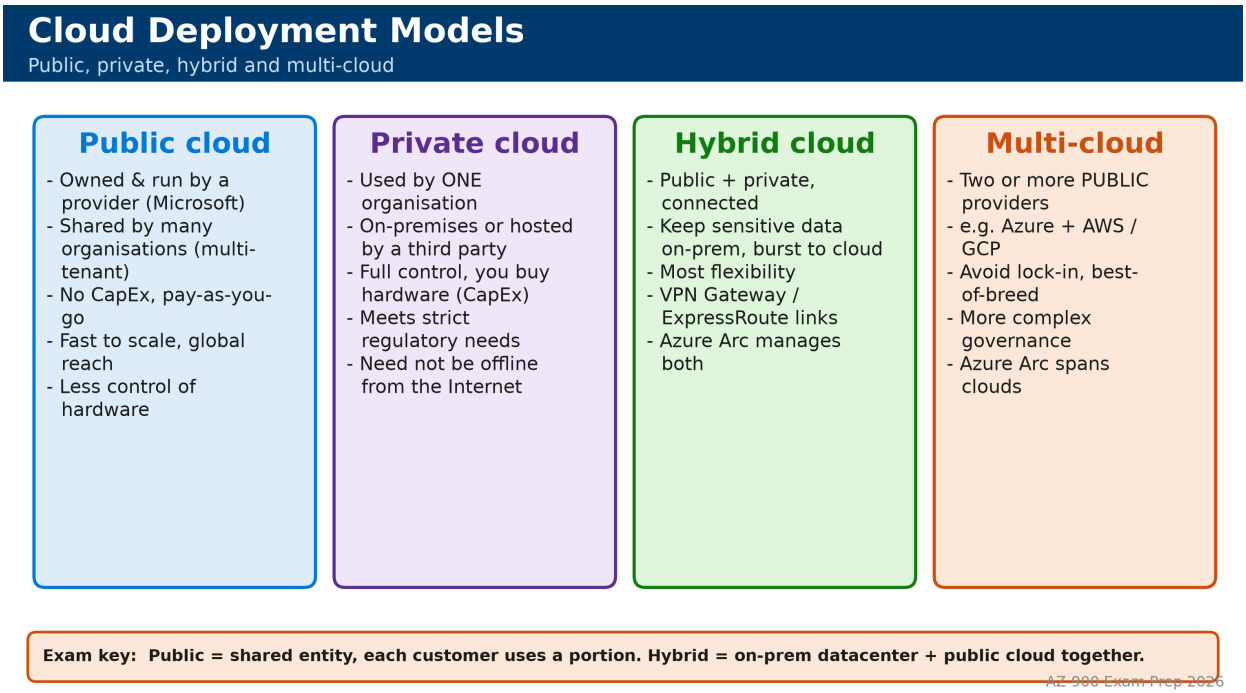


Figure Q11a: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

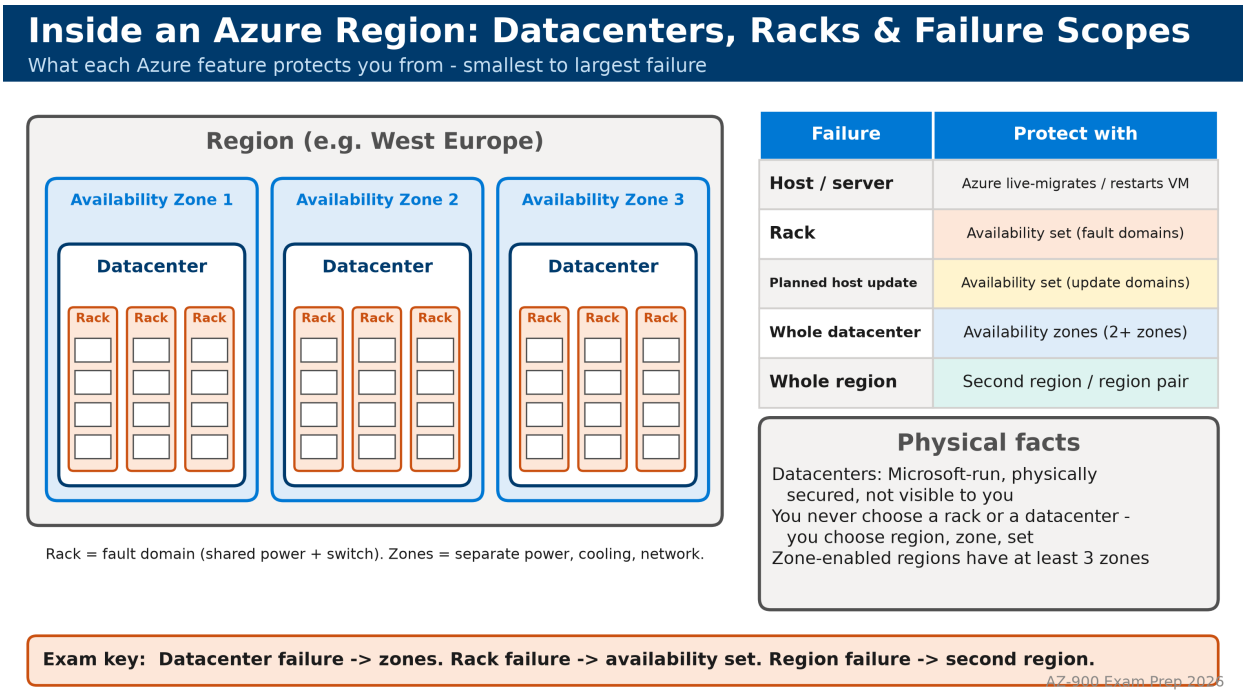


Figure Q11b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 12

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q27

Your team needs a tool that can provide guidance and recommendations that can be used to improve an Azure environment. Which of the following service can be used for this purpose?

✔ **A.** Azure Advisor.

✖ B. Azure Cognitive Services.

✖ C. Azure Application Insights.

✖ D. Azure DevOps.

Correct answer: Azure Advisor.

Answer explained

✔ **Azure Advisor.**
Azure Advisor checks your resources and gives personal recommendations for reliability, security, performance, cost and operational excellence (running things well).

✖ **Azure Cognitive Services.**
Ready-made AI APIs for apps. They do not check your Azure environment.

✖ **Azure Application Insights.**
It collects data about how an app runs. It does not give best-practice advice for your whole environment.

✖ **Azure DevOps.**
Tools to build and release code. Not a recommendation service.

EXAM STRATEGY
The words 'guidance and recommendations' to 'improve' point to Advisor. The other options deploy code, monitor apps or provide AI.

IMPACT
Advisor is free. It often finds unused resources and weak spots, which saves money and avoids outages with no extra tools.

What · Where · When · Why · How

What	Azure Advisor is a free service that gives personal best-practice recommendations.
Where	In the Azure portal under Advisor, for your subscriptions and resources.
When	When you want advice to improve cost, security, reliability, performance or operations.
Why	It looks at your setup and usage and tells you how to improve.
How	Read recommendations by category, check the Advisor score and apply the fixes.

Exam tip
'Recommendations', 'guidance', 'best practices' = Advisor. Five categories: Reliability, Security, Performance, Cost, Operational Excellence.

Memory trick
Advisor advises: RSPCO, 'Real Smart People Cut Overhead'.

Real-world example, use case and scenario

EXAMPLE
Right-size
Advisor sees a VM that averages 3 percent CPU. It suggests a smaller size to save money.

USE CASE
Monthly review
A nonprofit's only IT admin reads Advisor each month as a free list of improvements.

SCENARIO
Backup gap
A dental software company ran Advisor for the first time. It found five production VMs with no backup. The Reliability recommendation linked directly to turning on Azure Backup. When a disk failed a month later, they recovered in minutes.

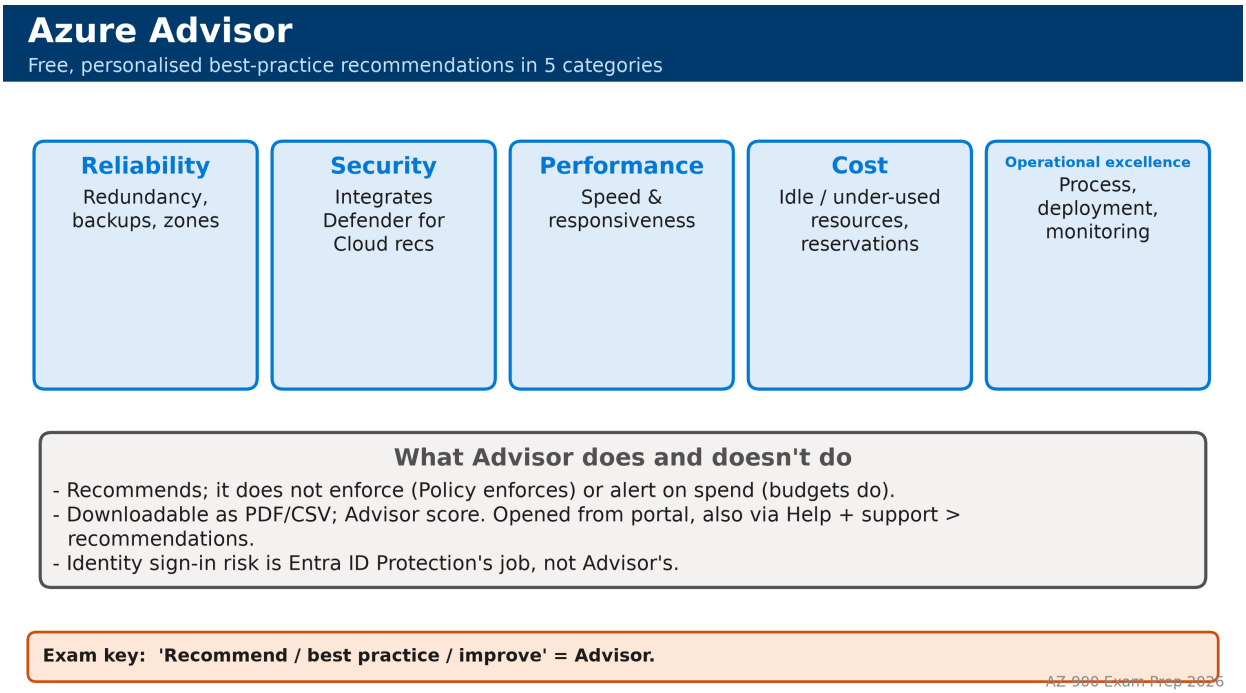


Figure Q12: Azure Advisor

About this figure

- The five boxes are Advisor's categories; each recommendation shows impact (high, medium, low) and suggested actions you can apply or postpone.
- Advisor is free and analyses your deployed resources' configuration and usage telemetry, giving subscription- and resource-group-level recommendations.
- Advisor score rates how well you follow best practices, overall and per category, helping prioritise fixes with the largest impact.
- Recommendations can trigger alerts and digest emails; security recommendations come from Defender for Cloud, so fixing them there also updates Advisor.
- Advisor is aligned with the Azure Well-Architected Framework pillars: reliability, security, cost optimisation, operational excellence and performance efficiency.

Question 13

AZURE ARCHITECTURE & SERVICES

Data & AI
v5 Q28

Your team needs to use a tool that can be used to build intelligent Artificial Intelligence based applications. Which of the following service can be used for this purpose?

✖ A. Azure Advisor.

✔ **B.** Azure Cognitive Services.

✖ C. Azure Application Insights.

✖ D. Azure DevOps.

Correct answer: Azure Cognitive Services.

Clarification (read this first): Renamed service: Azure Cognitive Services is now called Azure AI services. Pick it even with the old name. Look-alike: Q260 asks to build, test and deploy your OWN predictive models; that answer is Azure Machine Learning.

2026 update: Azure Cognitive Services is now called Azure AI services. This includes Azure AI Vision, Speech, Language, Translator and Azure OpenAI. They are now offered through Microsoft Foundry (before: Azure AI Foundry).

Answer explained

✔ **Azure Cognitive Services.**
Cognitive Services (now Azure AI services) give ready-trained AI APIs for vision, speech, language and decisions. You use them to build smart apps.

✖ **Azure Advisor.**
Advisor gives advice about your resources. It gives no AI features.

✖ **Azure Application Insights.**
Application Insights monitors apps. It adds no AI features.

✖ **Azure DevOps.**
Azure DevOps builds and releases apps. It gives no AI features.

EXAM STRATEGY
The phrase 'build intelligent AI applications' with ready-made AI points here. The old name Cognitive Services = today's Azure AI services. Pick it whatever the label.

IMPACT
Ready-made AI APIs add vision, speech or language to an app in hours, with no data scientists. Knowing old and new names matters on the exam.

What · Where · When · Why · How

What	Azure Cognitive Services (now Azure AI services) are ready-trained AI models you call as APIs.
Where	Created as Azure AI resources in a region. Apps call them with REST or SDKs.
When	When an app needs vision, speech, language or translation without training a model.
Why	They let developers add AI to apps quickly with ready-made models.
How	Create the resource, get its endpoint and key, and call the API from your code.

Exam tip
'Build intelligent apps' with ready-made AI = Cognitive Services / Azure AI services. 'Train your own model from data' = Azure Machine Learning.

Memory trick
Cognitive = ready-made brains on tap.

Real-world example, use case and scenario

EXAMPLE
Image tagging
An app sends a photo to Azure AI Vision. It gets back tags like 'dog', 'beach', 'sunset'.

USE CASE
Multilingual support
An online shop uses Azure AI Translator to translate product reviews into 12 languages automatically.

SCENARIO
Invoice processing
An accounts team typed invoice data by hand. Developers added Azure AI Document Intelligence to read the supplier, amount and date. Each invoice now takes seconds instead of minutes.

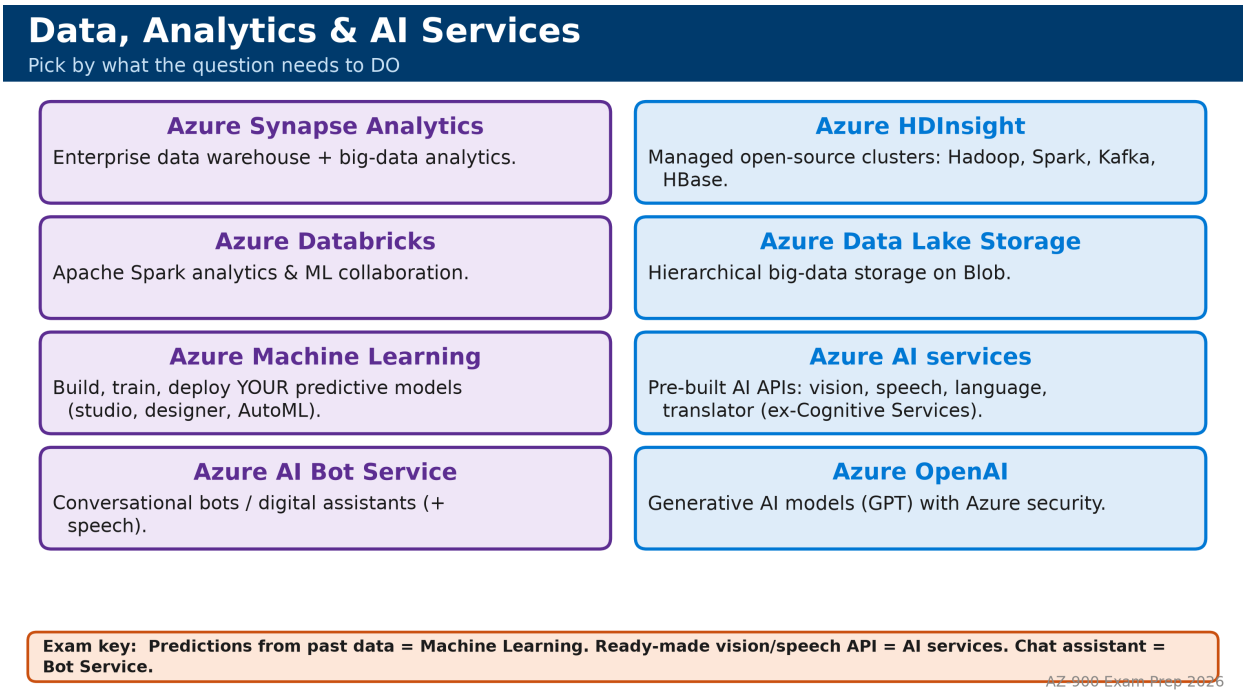


Figure Q13: Data, analytics and AI services

About this figure

- Left column covers data and analytics platforms; right column covers AI. Decide whether the scenario needs storage, processing, custom models or ready-made AI.
- Azure Machine Learning trains your own models from your data; Azure AI services are prebuilt APIs needing no data science, billed per transaction.
- Azure AI services include Vision, Speech, Language, Translator and Document Intelligence; many offer a free F0 tier with limited monthly calls.
- Azure OpenAI models are now offered through Microsoft Foundry (formerly Azure AI Foundry), billed per 1,000 or million tokens, varying widely by model.
- Microsoft Fabric is the newer unified analytics SaaS combining data engineering, warehousing and Power BI, and overlaps much of Synapse.

Question 14

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q29

Your team needs to have a tool that can be used to monitor web applications deployed to Azure. Which of the following service can be used for this purpose?

✖ A. Azure Advisor.

✖ B. Azure Cognitive Services.

✔ **C.** Azure Application Insights.

✖ D. Azure DevOps.

Correct answer: Azure Application Insights.

Answer explained

✔ **Azure Application Insights.**
Application Insights is part of Azure Monitor. It tracks a web app's requests, response times, failures, dependencies (services the app calls) and user behavior.

✖ **Azure Advisor.**
Advisor gives best-practice advice, not live data about how an app runs.

✖ **Azure Cognitive Services.**
Cognitive Services are AI APIs, not a monitoring tool.

✖ **Azure DevOps.**
Azure DevOps releases apps. Monitoring them afterwards is the job of Application Insights.

EXAM STRATEGY
'Monitor web applications' means APM (application performance monitoring). Remove Advisor (advice), DevOps (release) and Cognitive Services (AI). Application Insights is left.

IMPACT
Application Insights finds failing requests and slow dependencies before customers complain. Problems get fixed faster.

What · Where · When · Why · How

What	Application Insights is the app performance monitoring feature of Azure Monitor.
Where	An Application Insights resource linked to the web app, often stored in a Log Analytics workspace.
When	When you need to watch a web app's speed, failures and usage.
Why	It collects request, dependency and error data built for web apps.
How	Turn it on for the App Service or add the SDK. Then view live metrics, the app map and alerts.

Exam tip
'Monitor web applications', 'APM', 'find performance problems', 'page load times' = Application Insights.

Memory trick
Insights looks inside the app.

Real-world example, use case and scenario

EXAMPLE
Slow dependency
The application map shows a SQL call that takes 4 seconds. This makes pages load slowly.

USE CASE
Food delivery app
A food delivery service tracks failed checkouts with Application Insights. It sends an alert when they go above 2 percent.

SCENARIO
Hidden exception
An online bookstore saw many abandoned carts but no server errors. Application Insights found a JavaScript error in one browser version. A fix brought sales back within a day.

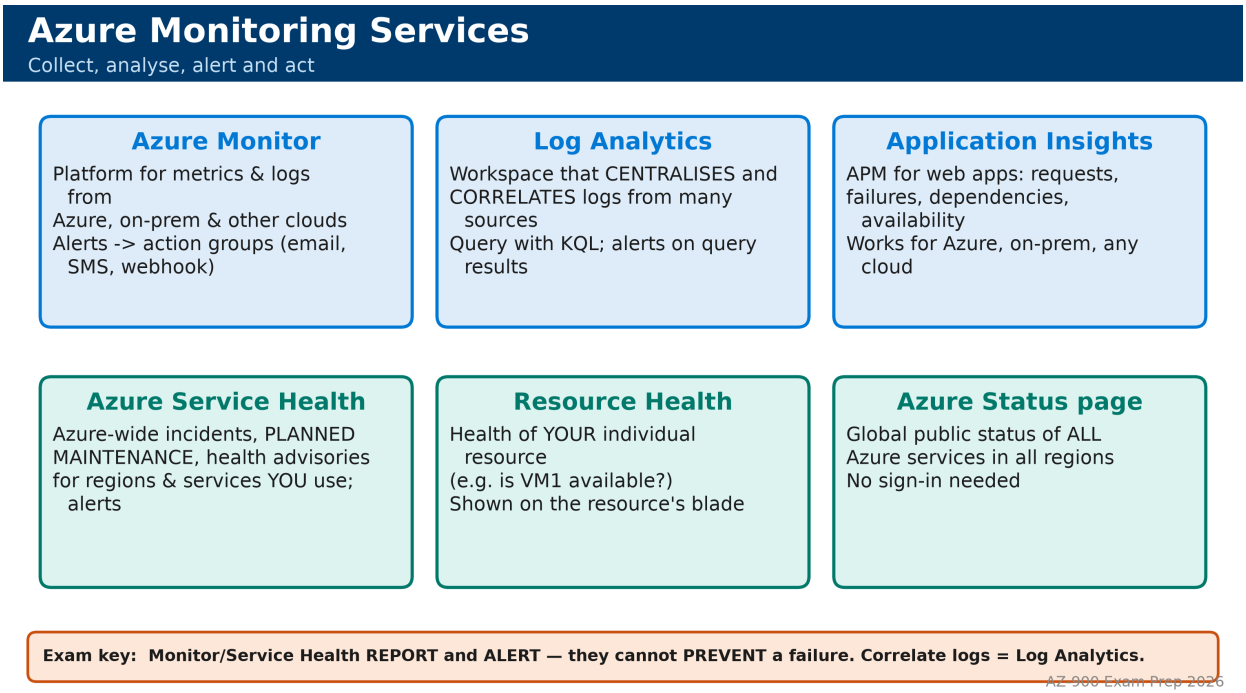


Figure Q14a: Azure monitoring services

About this figure

- Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).
- Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.
- Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.
- Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.
- Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

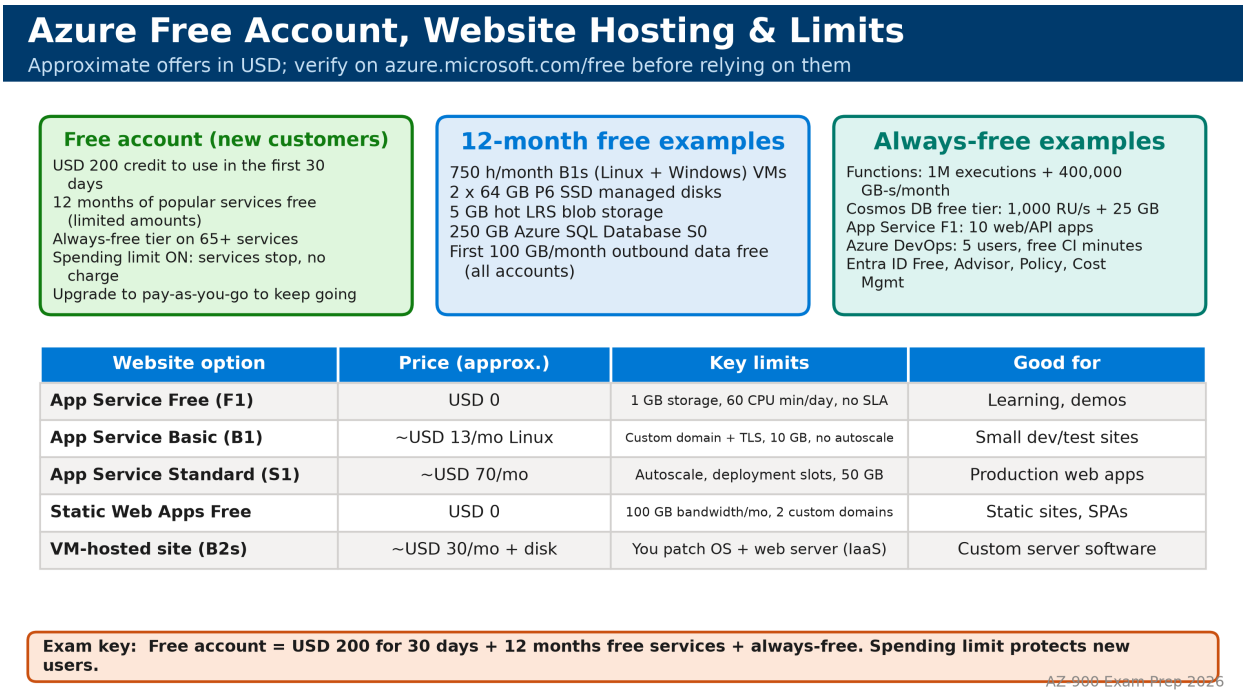


Figure Q14b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 15

AZURE ARCHITECTURE & SERVICES — **Databases** v5 Q31, Q207

A company wants to have a data store hosted in Azure. Below are the key requirements for the data store: The data store must be able to store JSON documents. The data store must be able to handle data writes from multiple regions. Which of the following would be the ideal data store for this requirement?

✖ A. Azure SQL Database.

✔ **B.** Azure Cosmos DB.

✖ C. Azure Redis Cache.

✖ D. Azure Synapse Analytics.

Correct answer: Azure Cosmos DB.

Answer explained

✔ **Azure Cosmos DB.**
Cosmos DB is a NoSQL database (stores data without fixed tables) spread across the world. It stores JSON documents directly and can accept writes in many regions.

✖ **Azure SQL Database.**
A relational (table-based) database. Its copies in other regions are read-only, so it cannot write in many regions and does not store documents natively.

✖ **Azure Redis Cache.**
An in-memory cache for fast data. It is not a lasting document store across regions.

✖ **Azure Synapse Analytics.**
An analytics and data warehouse service. It is not a live store for JSON.

EXAM STRATEGY
Two clues: 'JSON documents' and 'writes from multiple regions'. Only Cosmos DB meets both. SQL is relational, Redis is a cache, Synapse is analytics.

IMPACT
Writes in many regions keep global apps fast. They also keep the app working during a regional outage, so no transactions are lost.

What · Where · When · Why · How

What	Azure Cosmos DB is a global, multi-model NoSQL database service.
Where	A Cosmos DB account copied to any Azure regions you add.
When	When users in several regions must write JSON documents with low delay.
Why	It stores JSON directly and supports writes in many regions. The others do not.
How	Turn on multi-region writes in the account and add regions. Each region then accepts writes.

Exam tip
JSON + global + writes in many regions = Cosmos DB, every time.

Memory trick
Cosmos writes anywhere in the cosmos.

Real-world example, use case and scenario

EXAMPLE
Profile document
A user profile is saved as a JSON document with nested settings. It is written in the nearest region.

USE CASE
Multiplayer leaderboard
A game company stores player data in Cosmos DB with writes turned on in the US, Europe and Asia.

SCENARIO
Global shopping cart
A fashion shop's carts were slow for buyers in Asia because all writes went to a US database. Moving carts to Cosmos DB with multi-region writes let each region write locally. Cart delay in Asia dropped a lot.

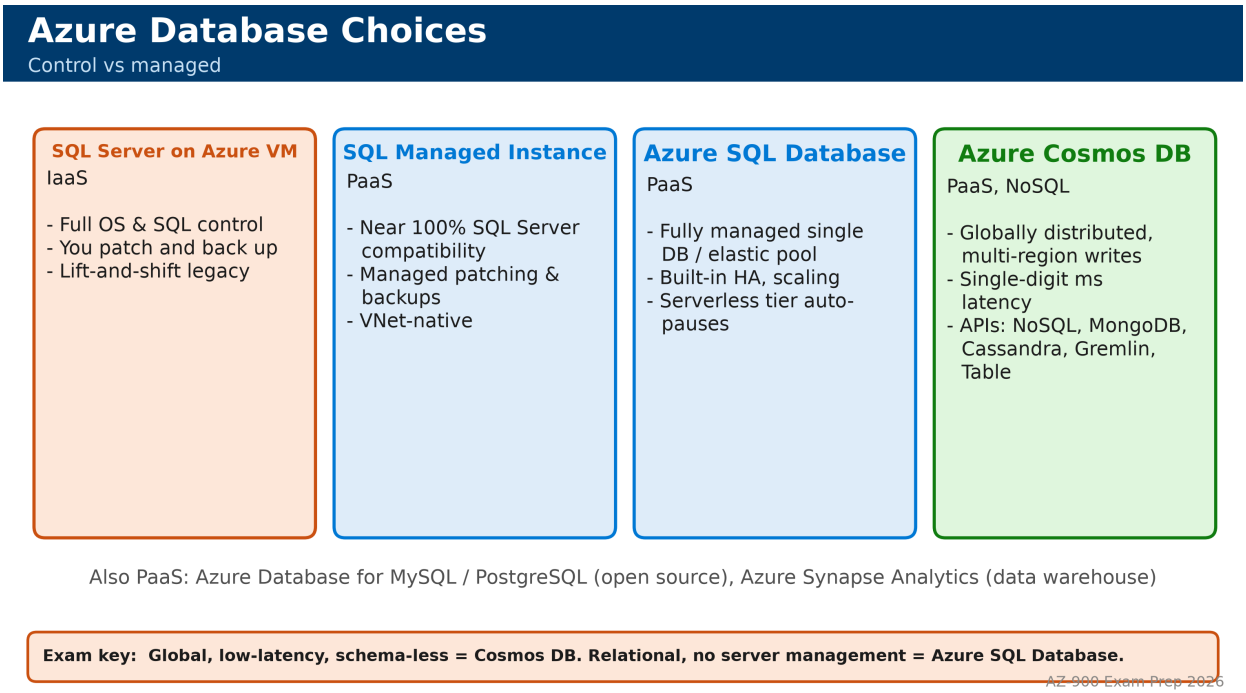


Figure Q15: Azure database choices

About this figure

• Columns run from most control (SQL Server on a VM, IaaS) to fully managed relational PaaS and globally distributed NoSQL Cosmos DB.

• Azure SQL Database Basic costs approx. USD 5/month and S0 approx. USD 15/month; serverless bills per vCore-second and storage only while paused.

• Azure SQL Database SLA is up to 99.99%, and up to 99.995% with zone redundancy in Business Critical; automated backups are kept 7 days by default.

• Cosmos DB free tier gives 1,000 RU/s and 25 GB free per account; it offers five consistency levels and up to 99.999% SLA for multi-region accounts.

• SQL Managed Instance suits migrations needing SQL Agent, cross-database queries or CLR; Azure Hybrid Benefit can reuse SQL Server licences for any of them.

Question 16

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q40, Q181

Your company has just setup an Azure account and Microsoft Entra ID. They need to ensure that when users connect to Microsoft Entra ID from the Internet from an Anonymous IP address, they are prompted to change their password automatically. Which of the following Azure service can help them achieve this requirement?

✖ A. Microsoft Entra Connect.

✔ **B.** Microsoft Entra ID Protection.

✖ C. Microsoft Entra Privileged Identity Management.

✖ D. Microsoft Defender for Identity.

Correct answer: Microsoft Entra ID Protection.

Clarification (read this first): The clue is 'anonymous IP address' plus 'change their password automatically'. That is a risky sign-in rule, which is Entra ID Protection. Look-alikes: Q107 (Key Vault) and Q108 (PIM) for the same goal both answer No.

Answer explained

✔ **Microsoft Entra ID Protection.**
Entra ID Protection finds risky sign-ins, such as ones from anonymous IP addresses. Its risk rules can force MFA or a safe password change. It needs Entra ID P2.

✖ **Microsoft Entra Connect.**
Entra Connect copies on-premises AD users to Entra ID. It does not check sign-in risk.

✖ **Microsoft Entra Privileged Identity Management.**
PIM (Privileged Identity Management) gives admin roles for a short, set time. It does not force password changes based on risk.

✖ **Microsoft Defender for Identity.**
Defender for Identity watches on-premises AD domain controllers for attacks. It does not check Entra sign-in risk.

EXAM STRATEGY
'Anonymous IP address' + forced password change = a sign-in risk rule. Connect copies users, PIM controls admin roles, Defender for Identity watches on-premises AD.

IMPACT
Risk rules stop password-guessing and Tor-based attacks automatically. They need Entra ID P2 licences, a cost to plan for.

What · Where · When · Why · How

What	Microsoft Entra ID Protection finds and reacts to identity risks in sign-ins and users.
Where	In the Entra admin center under ID Protection. Risk rules are set in Conditional Access (P2).
When	When sign-ins from anonymous IPs or leaked passwords must trigger an automatic fix.
Why	It marks anonymous-IP sign-ins as risky and can force a safe password change.
How	Create a risk-based Conditional Access policy that needs MFA or a password change at chosen risk levels.

Exam tip
'Anonymous IP', 'leaked credentials', 'risky sign-in', 'force password change' = Entra ID Protection.

Memory trick
Protection smells risk and makes you change the lock.

Real-world example, use case and scenario

EXAMPLE
Tor sign-in
A sign-in through Tor (a tool that hides your location) gets medium risk. The policy makes the user complete MFA and reset the password.

USE CASE
Bank workforce
A bank uses a user-risk policy. It forces a password change when a user's password appears in a leak.

SCENARIO
Compromised account
An online shop saw a staff account signing in through an anonymous proxy. ID Protection marked it high risk and forced a password change at the next sign-in. The attacker was locked out before reaching any data.

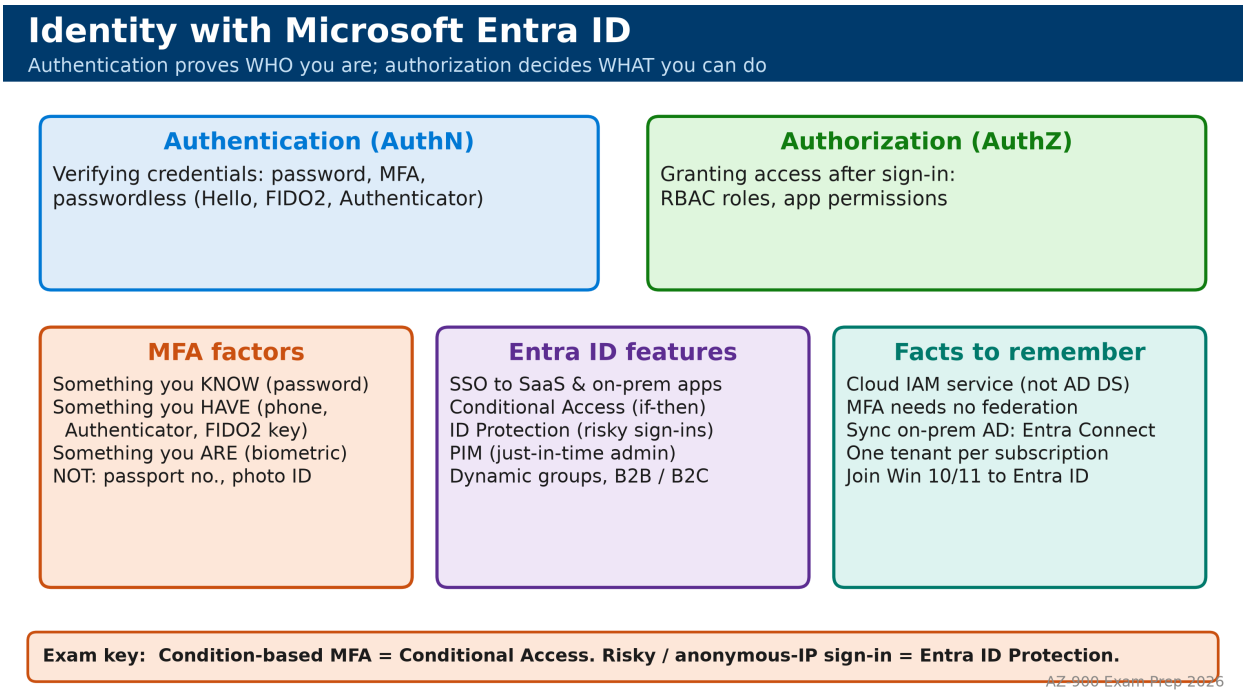


Figure Q16: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 17

AZURE ARCHITECTURE & SERVICES — **Governance** v5 Q42

Which of the following can be used to allow an organization to manage the compliance of resources across multiple subscriptions?

- ✖ A. Resource Groups.
- ✔ **B.** Management Groups.
- ✖ C. Azure Policies.
- ✖ D. Azure Resource Manager templates.

Correct answer: Management Groups.

Clarification (read this first): The key phrase is 'across multiple subscriptions'. Azure Policy is the rule, but the place you assign it to cover many subscriptions is a management group. Look-alike: Q80 says resource groups for the same goal; the answer is still management groups.

Answer corrected in this edition. The previous PDF printed: *Azure Policies.*. The PDF printed Azure Policies. Microsoft Learn says management groups are the way to manage access, policies and compliance across many subscriptions. The answer key and the standard exam answer are Management Groups. Policy is the tool, but the scope across subscriptions is the management group.

Answer explained

✔ **Management Groups.**
Management groups sit above subscriptions. Policies and RBAC (role-based access) set on a management group pass down to every subscription below it. This manages compliance across all of them.

✖ **Resource Groups.**
A resource group lives inside one subscription. It cannot cover many subscriptions.

✖ **Azure Policies.**
Policy defines the rules. But the scope (level) that covers many subscriptions is the management group.

✖ **Azure Resource Manager templates.**
ARM templates create resources. They do not keep checking compliance over time.

EXAM STRATEGY
Key phrase: 'across multiple subscriptions'. Resource groups sit inside one subscription. Policy is the rule, not the scope. Management groups cover many subscriptions.

IMPACT
Assign a policy once at a management group, and it applies to every current and future subscription. New subscriptions do not slip through.

What · Where · When · Why · How

What	Management groups are containers above subscriptions that organize governance (rules and control).
Where	At the top of the structure, under the tenant root group and above subscriptions.
When	When policy and access must be managed the same way across many subscriptions.
Why	Policies and RBAC set on a management group apply to every subscription under it.
How	Put subscriptions under a management group, then assign a policy initiative (a group of policies) there.

Exam tip
'Across multiple subscriptions' = management groups. Policy is the rule; the management group is where you attach it for many subscriptions.

Memory trick
Management groups are the umbrella over many subscriptions.

Real-world example, use case and scenario

EXAMPLE
Allowed locations
A policy 'allowed locations: EU only' set on the Corp management group applies to all 25 subscriptions under it.

USE CASE
Holding company
A parent company has each daughter company in its own subscription. It applies its ISO 27001 policy initiative once, at a top management group.

SCENARIO
Subscription sprawl
A university had 40 research subscriptions, each with its own drifting policies. IT created a Research management group, moved the subscriptions under it and assigned one policy initiative. Compliance reports became one dashboard.

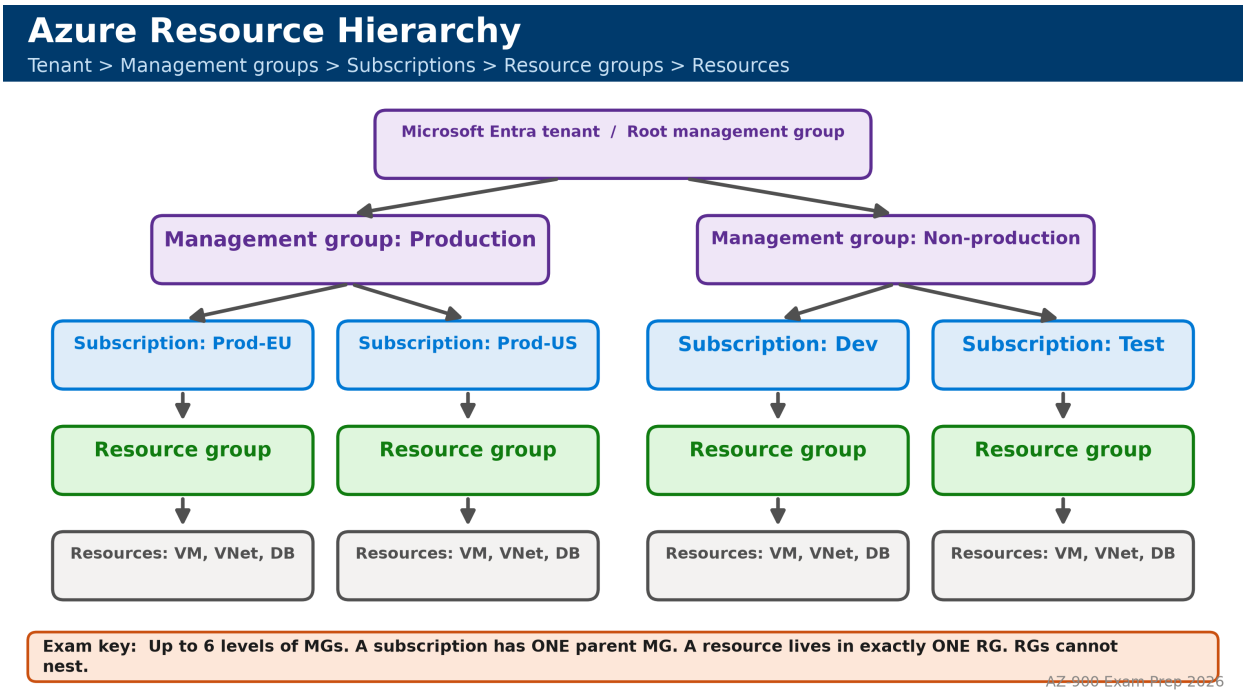


Figure Q17: Azure resource hierarchy

About this figure

- Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.
- A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.
- Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.
- Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.
- Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 18

CLOUD CONCEPTS

Lifecycle
v5 Q43

Does Microsoft release most of the Azure services in private preview before being introduced in public preview?

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The question asks about the ORDER: private preview first, then public preview. 'Most' allows a few exceptions, so Yes. Related: Q20 asks if private preview is open to ALL customers; that is No.

Answer explained

✔ **Yes.**
Most Azure features go through private preview (invited customers only), then public preview (all customers), then general availability (GA: SLA and full support).

✖ **No.**
The normal lifecycle puts private preview before public preview for most features.

EXAM STRATEGY
Lifecycle order: private preview, then public preview, then GA. 'Private before public' matches, so Yes.

IMPACT
Knowing the preview stages helps you decide when a feature is safe for production. Previews have no SLA (uptime promise).

What · Where · When · Why · How

What	The Azure lifecycle moves features from private preview to public preview to GA.
Where	Announced on Azure Updates. Preview features have a 'Preview' label in the portal.
When	When planning to use a newly announced Azure feature.
Why	Most features are first tested privately with invited customers before opening to all.
How	Private preview by invitation, then public preview for everyone, then GA with an SLA.

Exam tip
Order: private preview (invite only) > public preview (anyone, no SLA) > GA (SLA, full support).

Memory trick
Private party, then public opening, then grand opening.

Real-world example, use case and scenario

EXAMPLE
Invite form
A new database feature first appears only for customers who filled in a sign-up form and were accepted into private preview.

USE CASE
Early adopter
A media data company joins private previews to shape features it depends on.

SCENARIO
Roadmap planning
An architect at a bank saw a feature announced in private preview. She knew public preview and GA come next. She planned production use only after GA, so the bank did not depend on an unsupported feature.

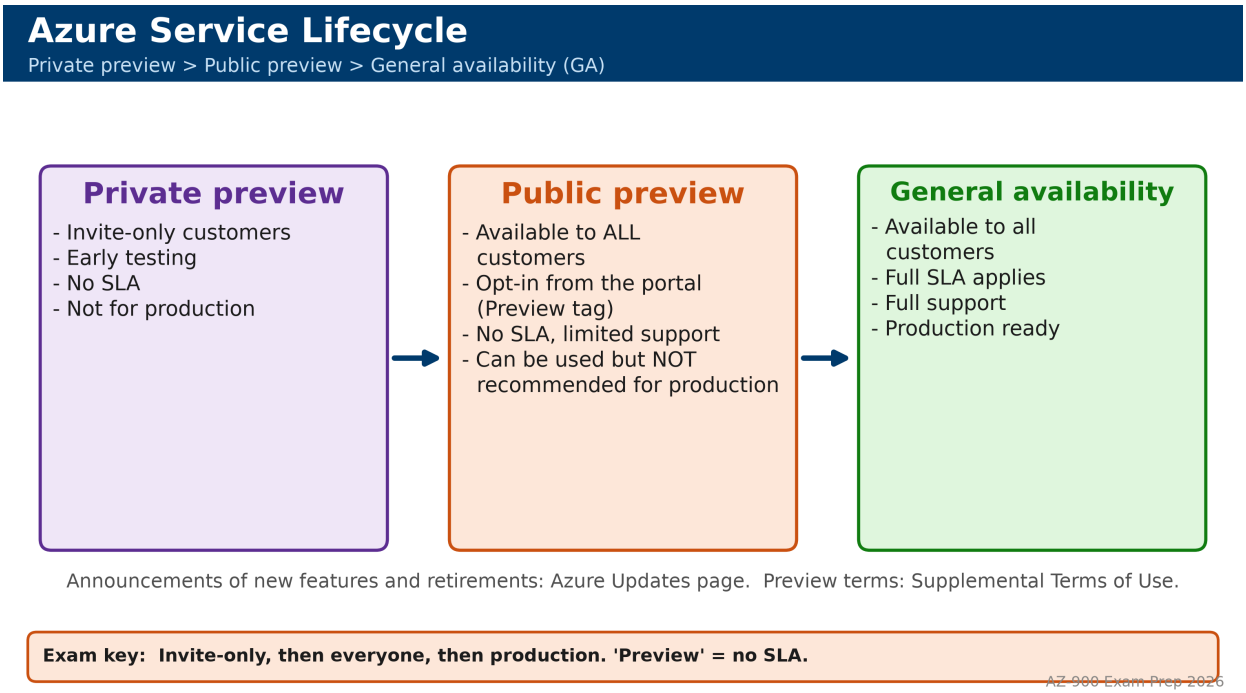


Figure Q18: Service lifecycle: preview to GA

About this figure

- Read the arrows left to right: invite-only private preview, open public preview, then general availability with SLA and full support.
- Previews are offered as-is under the Supplemental Terms of Use for Microsoft Azure Previews; features, pricing and availability may change or be withdrawn.
- Some previews are discounted or free, but many bill at reduced or standard rates; check the pricing page before relying on preview resources.
- Follow Azure Updates for previews, GA announcements and retirements; Service Health also sends retirement notices for services you actually use.
- Microsoft usually gives at least 12 months' notice before retiring a GA service or feature that customers depend on, with migration guidance.

Question 19

CLOUD CONCEPTS

Lifecycle
v5 Q44

Is it true that Azure services released in public preview can only be managed via the Azure command line interface?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is 'only'. The question says preview services can be managed ONLY with the CLI (command-line tool). That is false; the portal and other tools also work. Related: Q213 says private preview needs a separate portal; also No.

Answer explained

✔ **No.**
Public preview features are usually in the portal (with a 'Preview' label). They also work with the CLI, PowerShell, REST and templates.

✖ **Yes.**
No rule limits public preview to the CLI. Preview features often appear in the Azure portal.

EXAM STRATEGY
Trap word: 'only'. Public preview features appear in the portal and all usual tools. So 'only by CLI' is No.

IMPACT
Teams can try preview features with tools they know. But no SLA (uptime promise) applies, so avoid production use.

What · Where · When · Why · How

What	Public preview is the stage where a feature is open to all customers to try.
Where	Usually in the Azure portal with a 'Preview' label, plus CLI, PowerShell and APIs.
When	When customers want to try a new feature before general availability.
Why	Preview features are not limited to the CLI. The portal usually supports them too.
How	Sign up if needed, then manage the feature with any normal Azure tool.

Exam tip
'Only' is the trap. Public preview = all customers, normal tools, no SLA.

Memory trick
Preview wears a badge in the portal, not a CLI-only ticket.

Real-world example, use case and scenario

EXAMPLE
Preview label
A portal page shows 'Preview' next to a new monitoring feature. You set it up by clicking.

USE CASE
Feature evaluation
A delivery startup tests a preview AI feature directly in the portal before deciding to use it.

SCENARIO
Assumed CLI-only
A junior admin thought she had to learn the CLI to try a public preview feature. Her lead showed her the same feature in the portal with a Preview tag. She finished testing that afternoon.

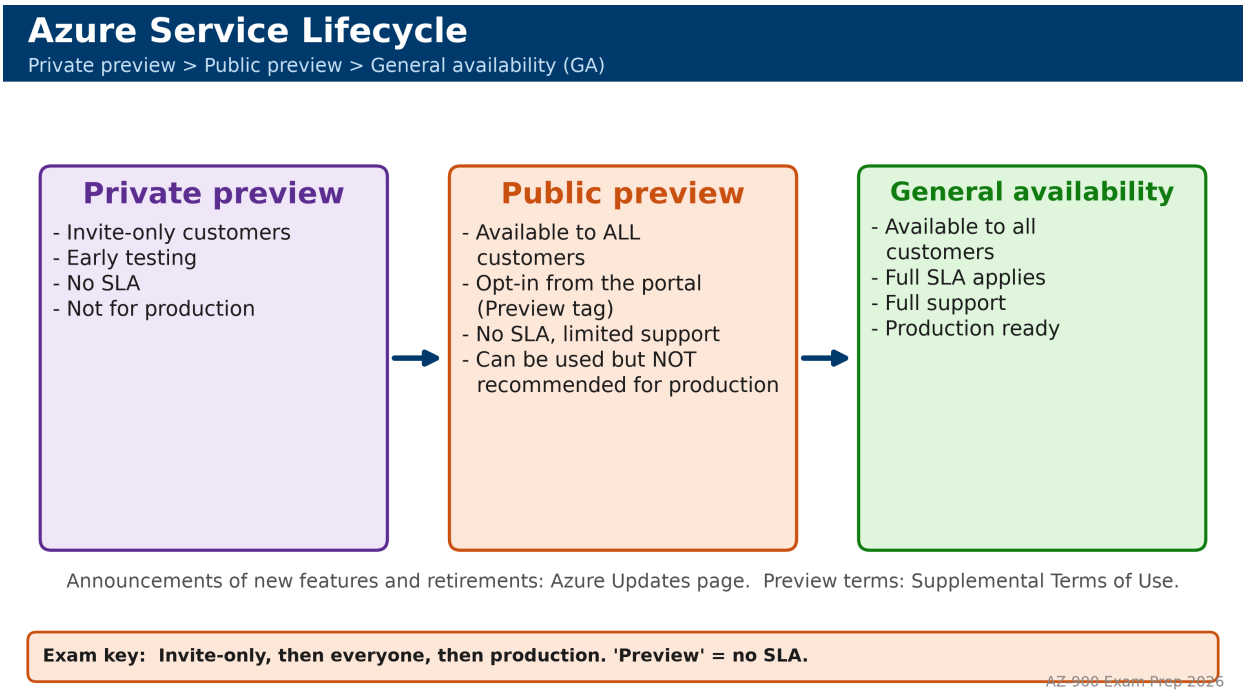


Figure Q19: Service lifecycle: preview to GA

About this figure

- Read the arrows left to right: invite-only private preview, open public preview, then general availability with SLA and full support.
- Previews are offered as-is under the Supplemental Terms of Use for Microsoft Azure Previews; features, pricing and availability may change or be withdrawn.
- Some previews are discounted or free, but many bill at reduced or standard rates; check the pricing page before relying on preview resources.
- Follow Azure Updates for previews, GA announcements and retirements; Service Health also sends retirement notices for services you actually use.
- Microsoft usually gives at least 12 months' notice before retiring a GA service or feature that customers depend on, with migration guidance.

Question 20

CLOUD CONCEPTS

Lifecycle
v5 Q45, Q327

Is it true that an Azure service that is released in private preview is made available to all Azure customers?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The word to watch is 'all'. Private preview is for invited customers only, so No. Look-alikes: Q21 asks the same about general availability (GA) -> Yes. Q220 asks which stage is open to all customers -> public preview.

Answer explained

✔ **No.**
Private preview is only for invited or approved customers. Access for all customers starts at public preview.

✖ **Yes.**
'Available to all customers' describes public preview. Private preview is limited to chosen customers.

EXAM STRATEGY
Private preview is invite-only. 'All Azure customers' describes public preview and GA. So No.

IMPACT
If you plan a project around a private preview feature you cannot access, the project stalls. Check the release stage first.

What · Where · When · Why · How

What	Private preview is an early stage open only to chosen customers.
Where	Turned on per subscription by Microsoft after an invite or an approved sign-up.
When	When Microsoft wants feedback from a small group before public release.
Why	Access is limited, so it is not open to all customers.
How	Customers apply or are invited. Microsoft then adds their subscriptions to an allow-list.

Exam tip
Private = invited few. Public = everyone with a subscription. GA = everyone, with an SLA.

Memory trick
Private means guest list only.

Real-world example, use case and scenario

EXAMPLE
Allow-listed subscription
Microsoft adds a customer's subscription ID to a list, so the private preview feature appears only for them.

USE CASE
Partner testing
A large healthcare software company is invited to test a private preview compliance feature and give feedback.

SCENARIO
Feature not visible
A developer read a blog about a new feature but could not find it in his portal. It was in private preview and his subscription was not enrolled. He applied through the sign-up form and waited for public preview.

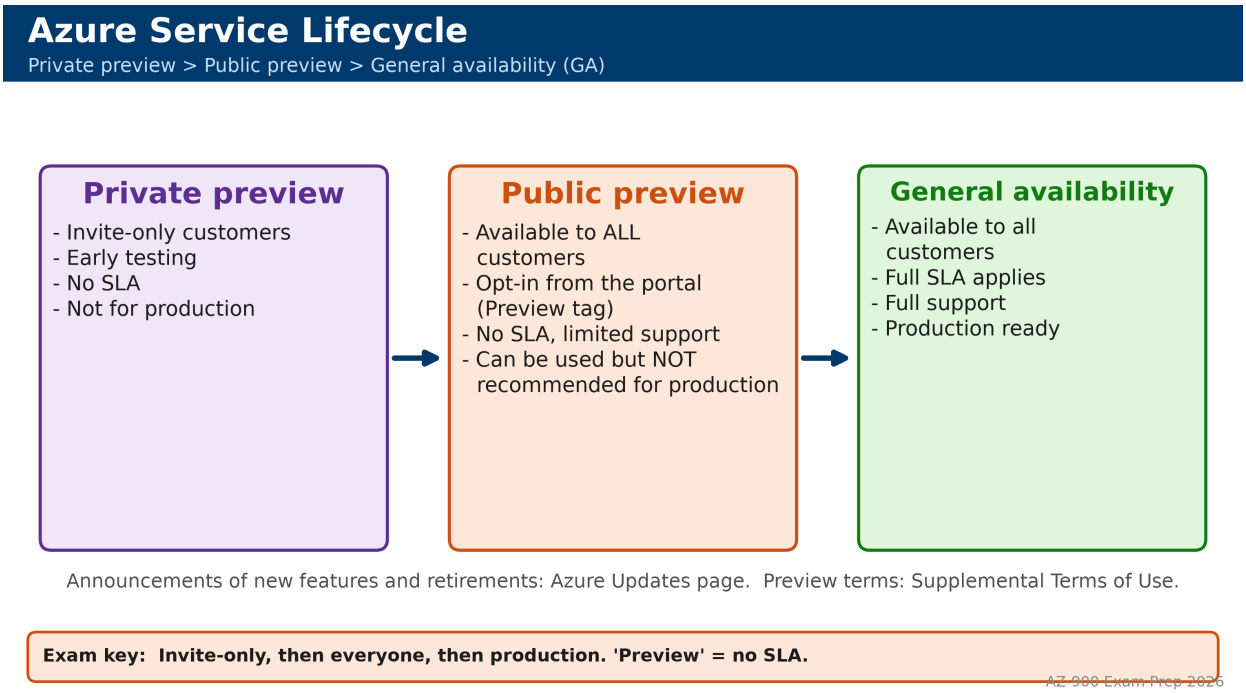


Figure Q20: Service lifecycle: preview to GA

About this figure

- Read the arrows left to right: invite-only private preview, open public preview, then general availability with SLA and full support.
- Previews are offered as-is under the Supplemental Terms of Use for Microsoft Azure Previews; features, pricing and availability may change or be withdrawn.
- Some previews are discounted or free, but many bill at reduced or standard rates; check the pricing page before relying on preview resources.
- Follow Azure Updates for previews, GA announcements and retirements; Service Health also sends retirement notices for services you actually use.
- Microsoft usually gives at least 12 months' notice before retiring a GA service or feature that customers depend on, with migration guidance.

Question 21

CLOUD CONCEPTS

Lifecycle
v5 Q47, Q328

Is it true that an Azure service that is released as generally available is made available to all Azure customers?

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): GA (general availability) means the final, full release. It is open to all customers, so Yes. Look-alike: Q20 asks the same about private preview -> No. Note: a GA service may reach some regions later, but every customer is allowed to use it.

Answer explained

✔ **Yes.**
General availability is the production release. Every Azure customer can use the service in the regions where it runs. It has a published SLA (uptime promise) and full support.

✖ **No.**
Only private preview is limited to invited customers. GA opens the service to all customers.

EXAM STRATEGY
GA means production release for every customer, with an SLA. Different regional rollout does not change who may use it. So Yes.

IMPACT
Only GA services have a financially backed SLA (you get credit if it fails). Contracts and compliance rules usually require this.

What · Where · When · Why · How

What	General availability (GA) is the final, production-ready stage of the Azure lifecycle.
Where	Announced on Azure Updates. It applies per service in each region where it has arrived.
When	When you need a service for production with an SLA and full support.
Why	GA opens the service to all customers and adds a published SLA. Previews do not.
How	Microsoft moves a feature from private preview to public preview to GA. Customers just turn it on.

Exam tip
Three stages, two questions. Who can use it: private = invited, public = everyone, GA = everyone. Is there an SLA: only GA. Do not confuse regional rollout with who may use it.

Memory trick
GA = Gates All open: every customer walks in, with the SLA.

Real-world example, use case and scenario

EXAMPLE
Feature goes GA
A new Azure Container Apps feature moves from public preview to GA. From that day any subscription can turn it on, and the Container Apps SLA covers it.

USE CASE
Production gating
An insurer's cloud rule allows only GA services in production. GA status shows that the SLA and full support apply.

SCENARIO
Waiting for GA
A delivery startup tested a preview AI feature, but its auditor required SLA-backed services. The team waited for the GA announcement. Then they turned it on in production with no special sign-up, because GA opened it to all customers.

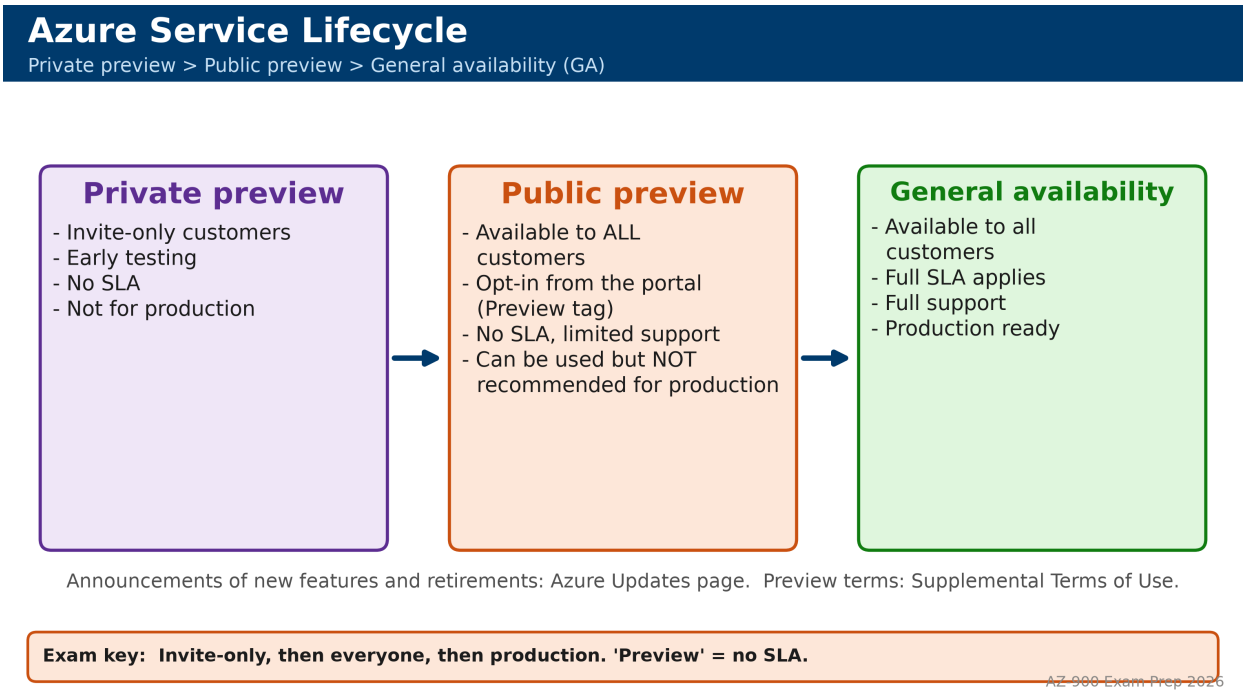


Figure Q21: Service lifecycle: preview to GA

About this figure

- Read the arrows left to right: invite-only private preview, open public preview, then general availability with SLA and full support.
- Previews are offered as-is under the Supplemental Terms of Use for Microsoft Azure Previews; features, pricing and availability may change or be withdrawn.
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- Microsoft usually gives at least 12 months' notice before retiring a GA service or feature that customers depend on, with migration guidance.

Question 22

AZURE ARCHITECTURE & SERVICES — **Subscriptions** v5 Q48

Can a single Microsoft Account be used to manage multiple Azure subscriptions?

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): An account is a sign-in (who you are). A subscription is a billing container. One account can manage many subscriptions, so Yes. Look-alike: Q121 asks if one subscription can link to many Entra tenants -> No.

Answer explained

✔ **Yes.**
An account is an identity (a sign-in), not a subscription. One account can own or have roles on many subscriptions and switch between them in the portal.

✖ **No.**
There is no 'one account, one subscription' limit. Managing dev, test and prod subscriptions from one sign-in is normal.

EXAM STRATEGY
An account is an identity that can have roles on many subscriptions. One-to-many means Yes.

IMPACT
One sign-in for many subscriptions lets you keep prod, dev and billing separate without many logins.

What · Where · When · Why · How

What	An Azure account is a sign-in identity. A subscription is a billing and management container.
Where	Subscriptions appear in the portal under 'Directories + subscriptions' for the signed-in account.
When	When one admin must run dev, test and prod subscriptions from a single login.
Why	One account can link to many subscriptions, so one identity can manage many.
How	The account owns subscriptions or gets RBAC (role-based access) roles on them. It then switches between them in the portal.

Exam tip
Account to subscription is one-to-many. Subscription to tenant is many-to-one. A subscription trusts exactly one directory, but an account can reach many subscriptions.

Memory trick
One key ring, many doors: one sign-in, many subscriptions.

Real-world example, use case and scenario

EXAMPLE
Subscription switcher
An admin signs in once to the portal. She uses the 'Directories + subscriptions' filter to work in Dev, Test and Prod.

USE CASE
MSP management
An engineer at an IT service company uses one work account with roles on several subscriptions to run daily operations.

SCENARIO
Separate billing
A university IT lead wanted separate bills for research and teaching. She created a second subscription under her existing account. She now manages both from one login, and each department sees its own bill.

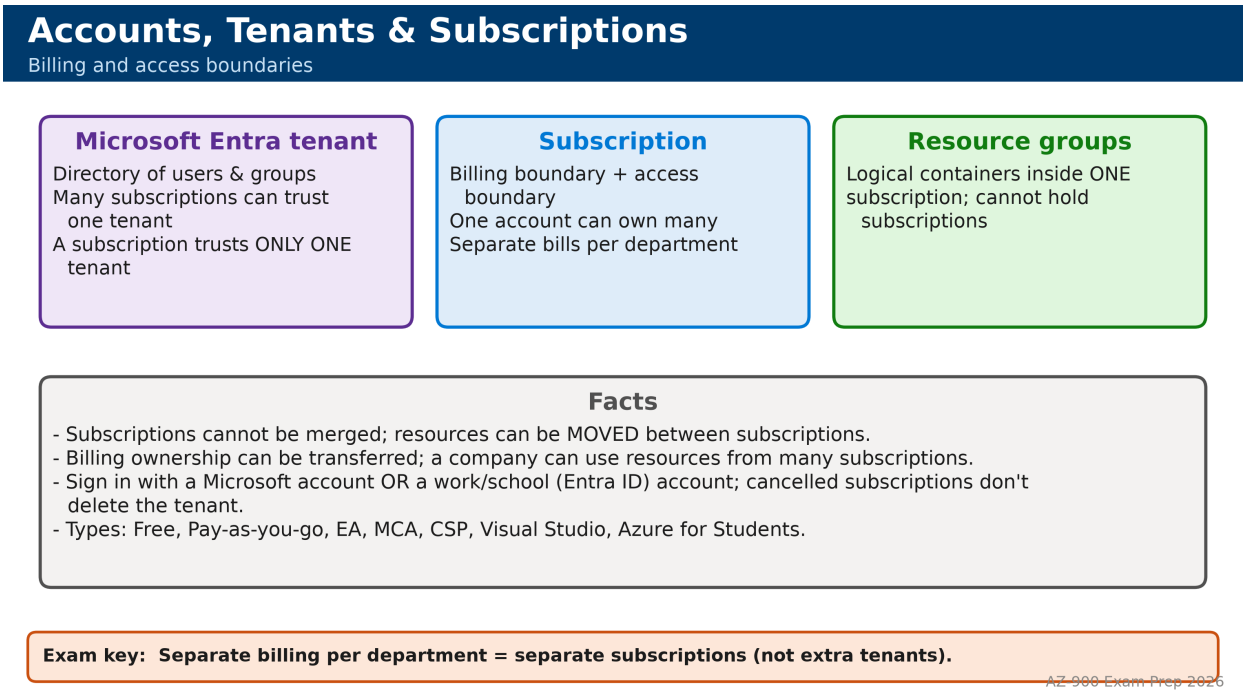


Figure Q22a: Accounts, tenants and subscriptions

About this figure

- Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.
- Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.
- Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.
- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

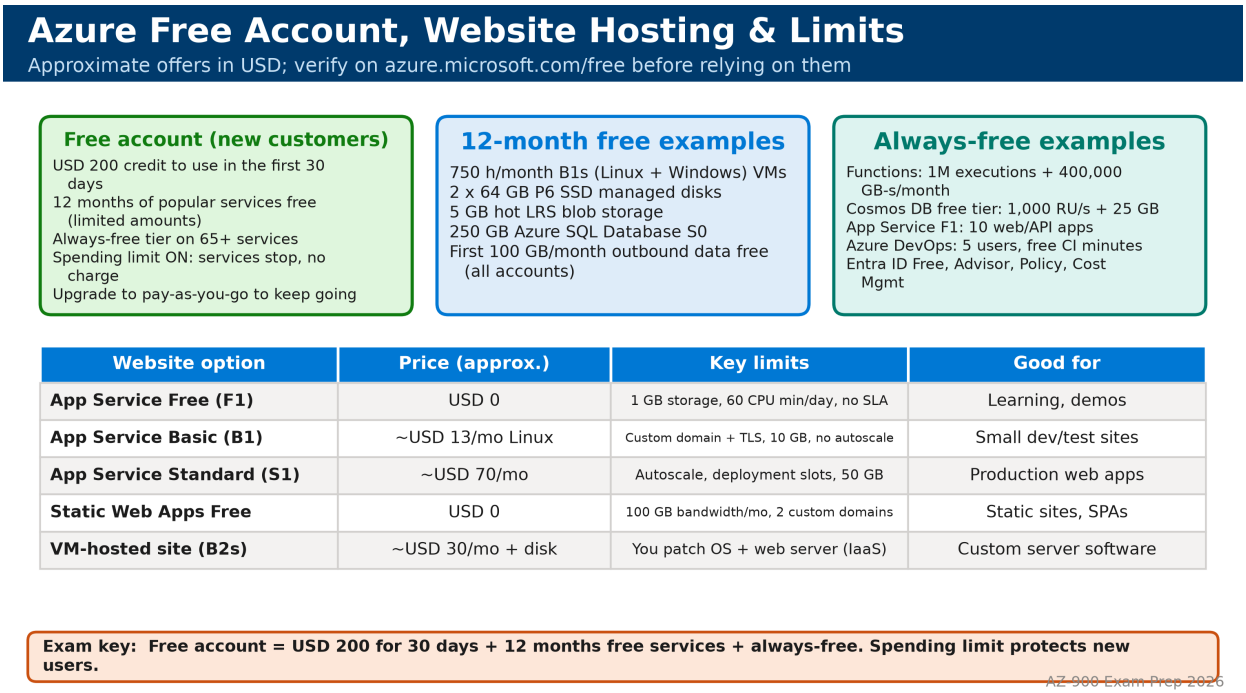


Figure Q22b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 23

CLOUD CONCEPTS

SLA
v5 Q50

A company has just started using Azure. They have setup a subscription and started deploying resources to Azure. Could the company increase the SLA for their resources by deploying resources across multiple regions?

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The question asks if spreading resources over several REGIONS raises the SLA (uptime). Yes, because both regions must fail before you go down. Look-alike: Q24 asks the same about buying more SUBSCRIPTIONS -> No.

Answer explained

✔ **Yes.**
With copies running in two regions, an outage needs both regions to fail. So the combined uptime is much higher than one region's SLA (uptime promise).

✖ **No.**
Copies running side by side in several regions add backup, which raises combined uptime. Only parts that depend on each other in a chain lower it.

EXAM STRATEGY
Copies in several regions run in parallel, which raises combined uptime. Your design changes the SLA, so Yes.

IMPACT
A multi-region design survives a full regional outage, and these do happen. It is the difference between hours offline and a smooth switch-over.

What · Where · When · Why · How

What	Multi-region deployment runs copies of a workload in two or more Azure regions.
Where	At the region level, with Front Door or Traffic Manager sending users to a working region.
When	When a workload must survive a full regional outage.
Why	Parallel copies fail only if all regions fail, so combined uptime is higher than one region.
How	Deploy the same setup in two regions, copy the data, and route users with failover (automatic switch).

Exam tip
Parts in a chain multiply and lower the SLA. Parallel copies across regions raise it. Multiple regions = Yes. Multiple subscriptions = No.

Memory trick
Two regions, two lifeboats: both must sink before you drown.

Real-world example, use case and scenario

EXAMPLE
Parallel math
Two web setups at 99.95% each, in East US and West Europe, behind Front Door: 1 - 0.0005 x 0.0005 = 99.99997% in theory.

USE CASE
Global checkout
An airline runs its booking API in two paired regions with Azure SQL geo-replication (copying to another region). A regional outage does not stop ticket sales.

SCENARIO
Regional outage drill
A healthcare portal had everything in one region and went offline during a regional incident. The architect added a second region with Traffic Manager priority routing. In the next test, traffic moved over in minutes.

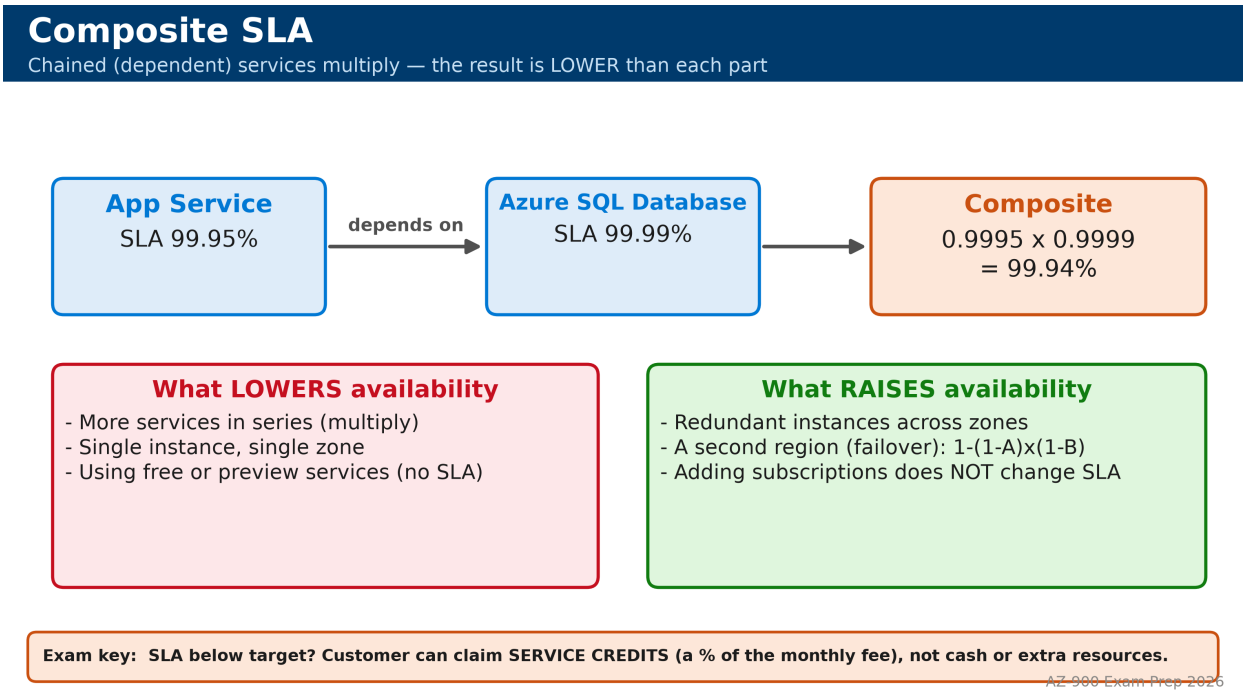


Figure Q23: Composite SLA

About this figure

• Follow the arrows: when one service depends on another, multiply their SLAs; the result is always lower than the weakest single component.

• Example: 99.95% x 99.99% = 99.94%, which allows roughly 26 minutes of downtime per month instead of about 22 for App Service alone.

• Parallel redundancy raises availability: two independent 99.9% deployments in failover give 1 - (0.001 x 0.001) = 99.9999% in theory.

• Preview services and free tiers (for example App Service F1) carry no SLA, so including them removes any guaranteed availability.

• Credits must be claimed through a support request, usually by the end of the following billing month, and are capped at the service's monthly fee.

Question 24

CLOUD CONCEPTS

SLA
v5 Q51

A company has just started using Azure. They have setup a subscription and started deploying resources to Azure. Could the company increase the SLA for their resources by purchasing multiple subscriptions?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The question asks if buying more SUBSCRIPTIONS raises the SLA (uptime). No: a subscription is only for billing and access, not hardware. Look-alike: Q23 asks the same about multiple REGIONS -> Yes.

Answer explained

✔ **No.**
A subscription is a billing and access boundary. It adds no backup hardware. The SLA depends only on the service tier, number of instances, zones and regions.

✖ **Yes.**
More subscriptions add no backup. Resources run on the same hardware with the same SLA.

EXAM STRATEGY
Subscriptions are billing and access boundaries with no physical backup. Buying more changes nothing: No.

IMPACT
Paying for extra subscriptions to get resilience wastes money. The real single point of failure stays.

What · Where · When · Why · How

What	A subscription is a billing, quota and access boundary, not a hardware layer.
Where	Above resource groups and below management groups in the structure.
When	When you separate billing, environments or admin work, never to raise uptime.
Why	Subscriptions add no backup, so buying more cannot raise any SLA.
How	Raise the SLA with availability sets, availability zones, more regions or higher tiers instead.

Exam tip
SLA levers are about design (zones, sets, regions, premium tiers). Admin items (subscriptions, resource groups, tags, support plans) do not change the SLA.

Memory trick
More wallets do not make the car more reliable.

Real-world example, use case and scenario

EXAMPLE
Same SLA, different sub
One VM with Premium SSD in Sub-A and the same VM in Sub-B each have 99.9%. Neither changes the other.

USE CASE
Why add subscriptions
A bank uses separate subscriptions for production and development to keep billing and access apart, not to raise uptime.

SCENARIO
Misguided purchase
A manager wanted to buy three subscriptions to raise uptime for a shop website. The architect showed the SLA would not change. Instead he put two VMs in different availability zones, raising the VM SLA to 99.99%.

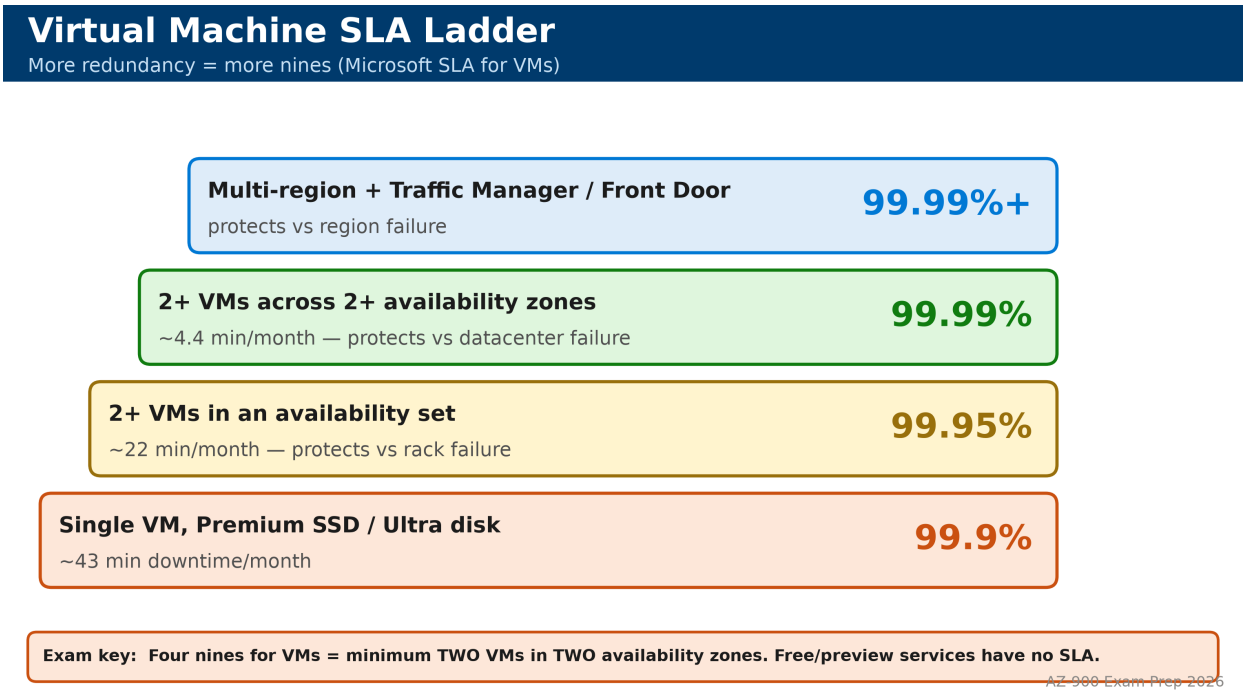


Figure Q24: Virtual machine SLA ladder

About this figure

- Climb the ladder: each step adds redundancy and a higher VM SLA; the downtime figures assume a 30-day month.
- A single VM with Standard SSD gets about 99.5% and with Standard HDD about 95%; all its OS and data disks must meet that tier.
- Availability sets spread VMs across fault domains (racks) and update domains (maintenance batches) in one datacenter; zones spread across datacenters.
- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

Question 25

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q58, Q383, Q384

Which of the following URL is used to manage the creation of Azure resources?

✖ A. `https://admin.azure.com`.

✖ B. `https://admin.azurewebsites.com`.

✖ C. `https://portal.microsoft.com`.

✖ D. `https://portal.azurewebsites.com`.

✔ **E.** `https://portal.azure.com`.

Correct answer: `https://portal.azure.com`.

Answer explained

✔ **`https://portal.azure.com`.**
https://portal.azure.com is the Azure portal. It is the website for creating and managing Azure resources.

✖ **`https://admin.azure.com`.**
Not an Azure management address. Microsoft 365 admin is at admin.microsoft.com.

✖ **`https://admin.azurewebsites.com`.**
Not a real console. App Service apps use the azurewebsites.net name, not a management portal.

✖ **`https://portal.microsoft.com`.**
Not the Azure portal. Microsoft 365 uses admin.microsoft.com and microsoft365.com.

✖ **`https://portal.azurewebsites.com`.**
A made-up address that mixes 'portal' with the App Service domain. It manages nothing.

EXAM STRATEGY
Match the exact text portal.azure.com. Remove anything with admin, microsoft.com (Microsoft 365) or azurewebsites (App Service app names).

IMPACT
Knowing the real portal address helps staff spot fake look-alike sites that steal admin passwords.

What · Where · When · Why · How

What	The Azure portal is the website for creating, setting up and watching Azure resources.
Where	Global Azure uses portal.azure.com. Azure Government uses portal.azure.us.
When	When you manage Azure resources with a graphical interface.
Why	portal.azure.com is the only listed address that is the real Azure portal.
How	Open the address, sign in with your Entra or Microsoft account, and choose resources.

Exam tip
Exact text: portal.azure.com. Anything with azurewebsites is an App Service app name. Anything with microsoft.com is Microsoft 365.

Memory trick
Portal dot Azure dot com: the door is named after the house.

Real-world example, use case and scenario

EXAMPLE
Bookmark
New engineers save portal.azure.com on day one. It opens the dashboard, resource groups and Cloud Shell.

USE CASE
Government cloud
A US federal contractor uses portal.azure.us because its resources are in Azure Government, a separate cloud.

SCENARIO
Phishing check
An employee got an email with a link to portal-azure-login.com. Training had taught him the real portal is portal.azure.com. He reported the email instead of signing in.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

AZ-900 Exam Prep 2026

Figure Q25: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 26

AZURE ARCHITECTURE & SERVICES

Storage
v5 Q59

You need to create a virtual machine in Azure. Which of the following storage account service is used to store the data disks for the virtual machine?

✔ **A.** Blobs.

✖ B. Files.

✖ C. Tables.

✖ D. Queues.

Correct answer: Blobs.

Clarification (read this first): The question asks which of the four storage account services holds VM disks. VM disks are stored as page blobs, so Blobs. Look-alike: Q41 asks for a mapped network drive; that answer is Files. If 'Disks' is an option in another version, pick Disks.

2026 update: VM disks are stored as page blobs. Today all new VM disks are managed disks. Azure stores and manages them for you, with no visible storage account. Unmanaged disks (VHDs in your own storage account) were retired in 2025. The technology underneath is still page blobs.

Answer explained

✔ **Blobs.**
VM OS and data disks are VHD files (virtual hard disks) stored as page blobs. Managed disks still use page blobs in the background.

✖ **Files.**
Azure Files gives SMB/NFS file shares (shared network folders), not VM disks.

✖ **Tables.**
Table storage is a simple NoSQL key-value store. It cannot hold a disk image.

✖ **Queues.**
Queue storage holds messages between apps, not disk data.

EXAM STRATEGY
'Data disks' means VHD page blobs. Files, Tables and Queues never hold VM disks, so Blobs. If Disks is offered, choose Disks.

IMPACT
Knowing that disks are managed page blobs explains disk billing and speed tiers when you size VMs.

What · Where · When · Why · How

What	Page blobs are blob storage built for random reads and writes in 512-byte pages.
Where	Inside Azure Storage. Managed disks use page blobs in storage accounts that Microsoft manages.
When	Every time a VM gets an OS or data disk.
Why	VM disks are VHD files stored as page blobs, so Blobs is the service.
How	Create a managed disk and attach it. Azure stores the VHD as a page blob.

Exam tip
Disks = page blobs. Files and backups = block blobs. Logs = append blobs. If 'Disks' is in the list, pick Disks for modern questions.

Memory trick
A disk is a book of pages: page blobs.

Real-world example, use case and scenario

EXAMPLE
Page blob VHD
A 128 GiB Premium SSD data disk on a VM is a managed disk stored as a page blob.

USE CASE
Database VM
A factory company attaches several Premium SSD v2 data disks to a SQL Server VM for data and log files.

SCENARIO
Legacy VHD migration
An engineering company still had unmanaged VHDs in a storage account. Before the 2025 retirement, they changed them to managed disks. Now Azure handles the page blob storage for them.

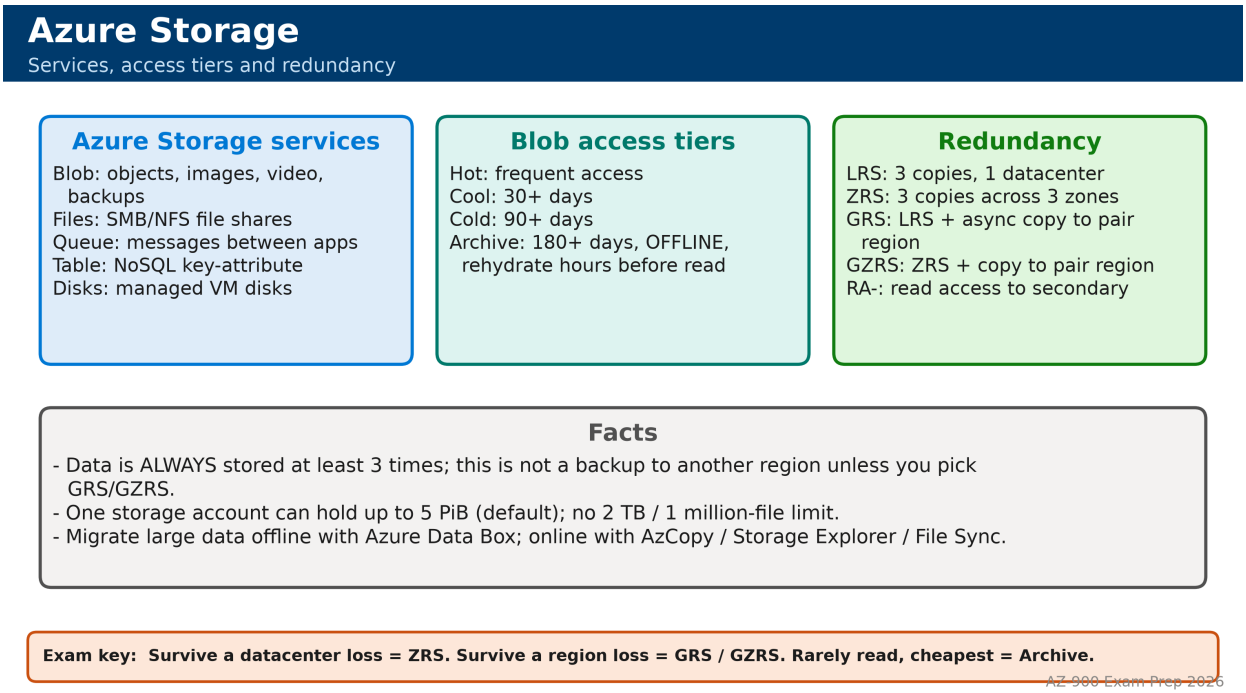


Figure Q26: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 27

CLOUD CONCEPTS

Deployment models
v5 Q63

What are two key characteristics of the public cloud?

✖ A. Dedicated hardware resources.

✔ **B.** Shared infrastructure.

✔ **C.** Pay-as-you-go model.

✖ D. On-premises deployment.

✖ E. Limited geographic availability.

Correct answer: Shared infrastructure.; Pay-as-you-go model.

Clarification (read this first): Choose TWO answers. Look-alike: Q69 asks the same thing with different options; there the answers are metered pricing and self-service management. Both describe the same idea: shared, pay for what you use.

Answer explained

✔ **Shared infrastructure.; Pay-as-you-go model.**
A public cloud runs on hardware owned by the provider and shared by many customers. It bills by use (pay-as-you-go).

✖ **Dedicated hardware resources.**
Dedicated hardware is a mark of private cloud or on-premises, not the shared public cloud.

✖ **On-premises deployment.**
On-premises means your own servers in your own datacenter.

✖ **Limited geographic availability.**
Public clouds reach the whole world through many regions, not limited areas.

EXAM STRATEGY
Remove anything about owning hardware or one location: dedicated hardware, on-premises and limited area describe private setups. Shared infrastructure and pay-as-you-go are left.

IMPACT
These two traits drive the cloud cost model. Public vs private trait pairs appear in many exam versions.

What · Where · When · Why · How

What	Public cloud is capacity owned by a provider and offered to any customer.
Where	Azure regions around the world, reached over the internet or private links.
When	When you want capacity without buying hardware and want to pay only for use.
Why	Its main traits are shared multi-tenant infrastructure (many customers on the same hardware) and pay-per-use billing.
How	Customers create resources when needed and pay per second, hour or unit used.

Exam tip
Public = shared + pay per use + global. Private = dedicated + CapEx (upfront buying) + your location.

Memory trick
Share the ride, pay the fare.

Real-world example, use case and scenario

EXAMPLE
Per-second billing
An Azure VM is billed per second while it runs. Compute charges stop when it is deallocated (stopped, hardware released).

USE CASE
Seasonal tax app
A tax company grows on shared Azure capacity during tax season and pays only for those weeks.

SCENARIO
Choosing a model
A school district compared buying servers with renting public cloud. Shared hardware and pay-as-you-go meant no hardware budget request. They moved their learning platform to Azure.

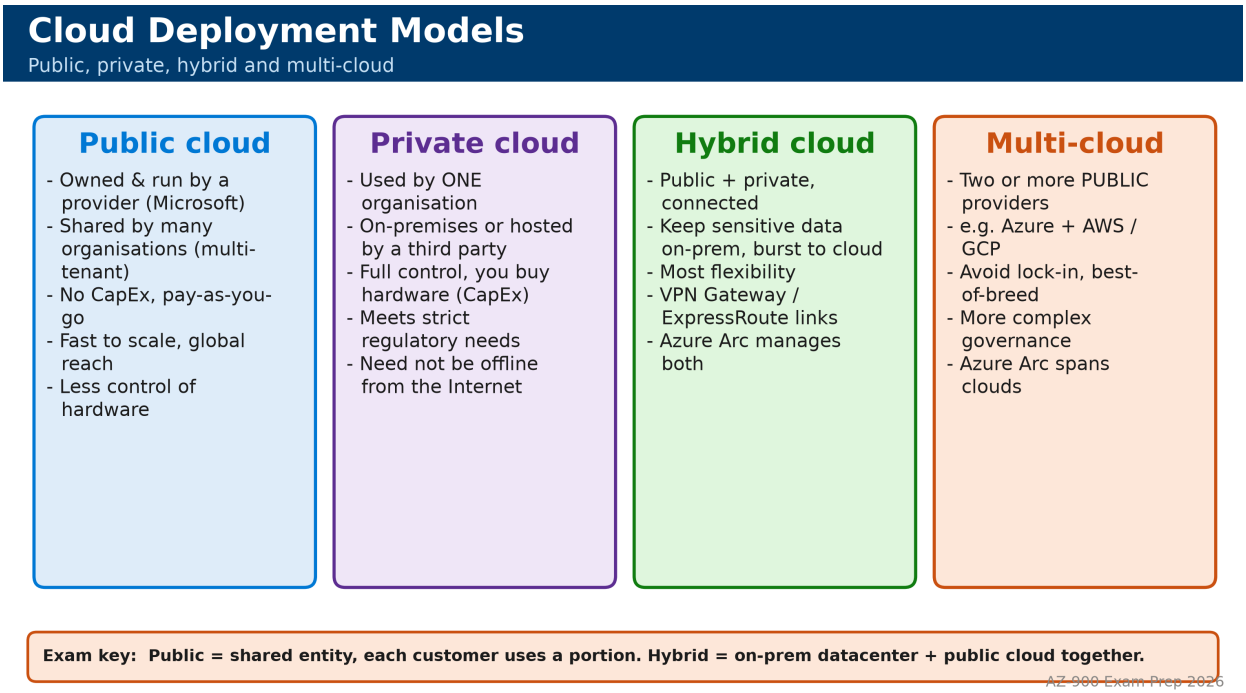


Figure Q27: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 28

CLOUD CONCEPTS

Service models
v5 Q64, Q360

Your company plans to migrate all its data and resources to Azure. The company migration plan states that only Platform as a Service (PaaS) solutions must be used in Azure. You need to deploy an Azure environment that meets the company migration plan. Solution: You create an Azure App Service and Azure SQL databases. Does this meet the goal?

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): This is a 'Does this meet the goal?' series. The rule is PaaS only. App Service and Azure SQL Database are both PaaS, so Yes. Look-alike: Q34 adds virtual machines (IaaS), so its answer is No.

Answer explained

✔ **Yes.**
Azure App Service and Azure SQL Database are both PaaS (Microsoft manages the OS and platform). So a PaaS-only plan is met.

✖ **No.**
Neither service gives you a VM or OS to manage. Both are PaaS, so the PaaS-only rule is met.

EXAM STRATEGY
Check: does the solution include any VM? App Service and Azure SQL Database are both PaaS, so Yes.

IMPACT
A PaaS-only setup removes OS patching from the team. This cuts work and security risks.

What · Where · When · Why · How

What	App Service and Azure SQL Database are PaaS services. Microsoft manages the OS and platform.
Where	Both are Azure resources placed in a resource group through the portal or templates.
When	When a migration plan allows only PaaS services.
Why	Neither gives you a VM or OS to manage, so the PaaS-only rule is met.
How	Put the web app on App Service and connect it to an Azure SQL database.

Exam tip
PaaS-only series: App Service, Azure SQL Database, Functions, Logic Apps = PaaS. Any option with virtual machines breaks the rule.

Memory trick
Apps and SQL, no servers to fiddle.

Real-world example, use case and scenario

EXAMPLE
Classic PaaS pair
A .NET web app on App Service connects to Azure SQL Database. Neither needs OS patching.

USE CASE
Booking site
A small hotel group runs its booking site on App Service with Azure SQL Database. Its small IT team no longer looks after servers.

SCENARIO
Policy compliance
A charity's migration rule was PaaS only. The architect proposed App Service plus Azure SQL Database. The review board approved it because neither needed a VM.

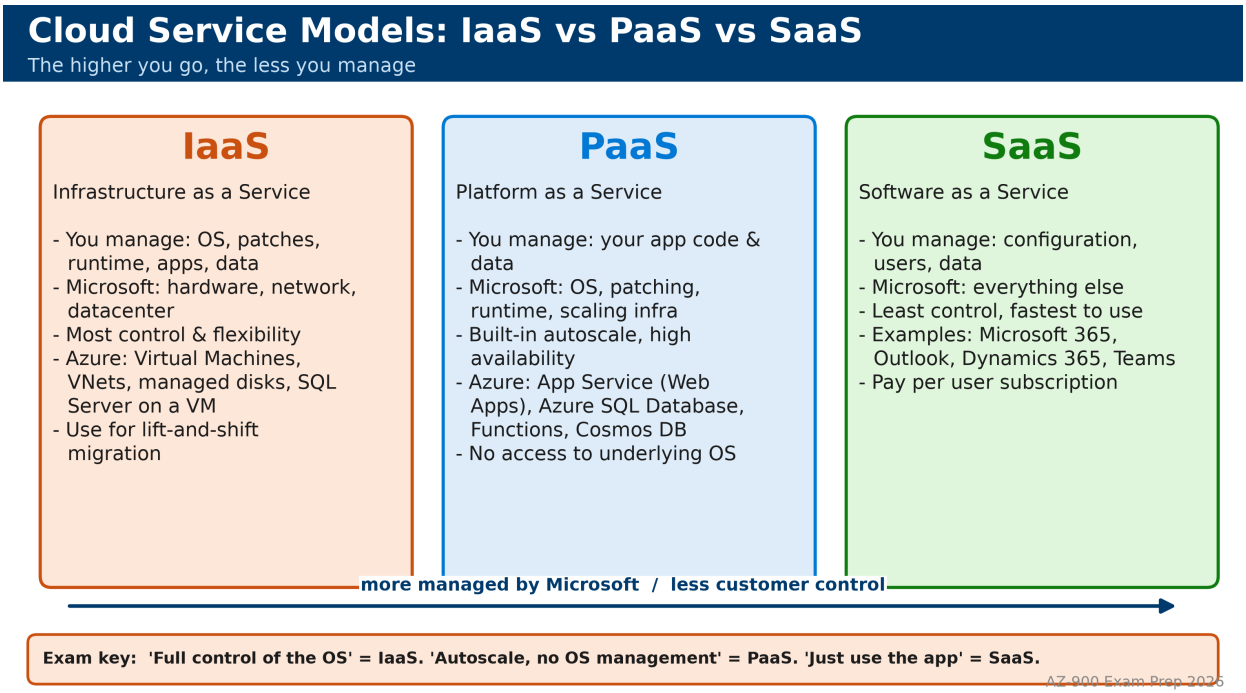


Figure Q28a: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

• Serverless options such as Azure Functions are treated as PaaS-style: no OS access, automatic scaling and per-execution billing.

• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

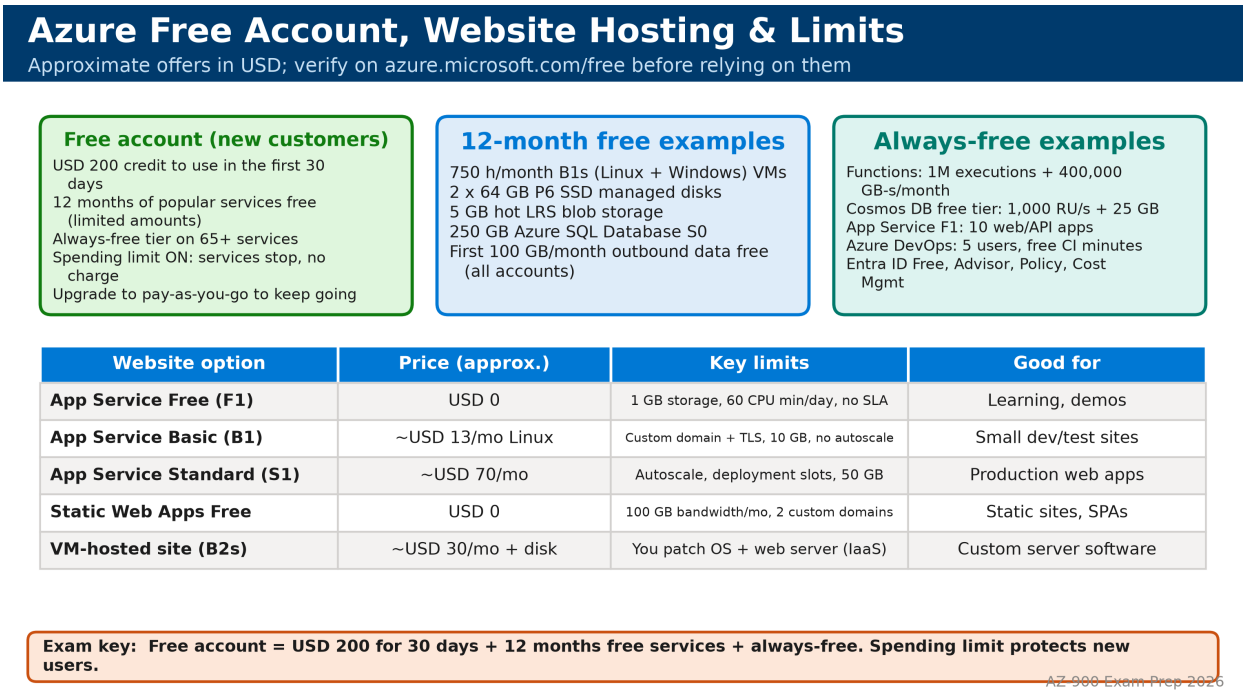


Figure Q28b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 29

CLOUD CONCEPTS

Service models
v5 Q65

You plan to migrate a web application to Azure. The web application is accessed by external users. You need to recommend a cloud deployment solution to minimize the amount of administrative effort used to manage the web application. What should you include in the recommendation?

✖ A. Software as a Service (SaaS).

✔ **B.** Platform as a Service (PaaS).

✖ C. Infrastructure as a Service (IaaS).

✖ D. Database as a Service (DaaS).

Correct answer: Platform as a Service (PaaS).

Clarification (read this first): It is YOUR OWN web app, and you want the LEAST admin work. SaaS is a trap: SaaS is someone else's finished app. Look-alike: Q35 also moves custom apps, but they need extra software installed, so its answer is IaaS.

Answer explained

✔ **Platform as a Service (PaaS).**
PaaS, such as Azure App Service, runs your own web app. Microsoft manages the OS, patching and scaling. This gives the least admin work.

✖ **Software as a Service (SaaS).**
SaaS is a finished app from a vendor. You cannot host your own custom web app as SaaS.

✖ **Infrastructure as a Service (IaaS).**
With IaaS (you manage the OS) you handle the OS, patches, web server and scaling. That is the most admin work.

✖ **Database as a Service (DaaS).**
It covers only managed databases, not web app hosting. It is not one of the three main service models.

EXAM STRATEGY
'Minimize administrative effort' for your own web app means PaaS. SaaS is a vendor's app. IaaS adds OS management.

IMPACT
PaaS frees the team from patching and scaling. They can spend time on features instead of servers.

What · Where · When · Why · How

What	PaaS gives a managed platform that runs your code. You do not manage the OS.
Where	Azure App Service is the main PaaS host for web apps and APIs.
When	When you move your own web app and want the least admin work.
Why	Microsoft patches the OS and runtime and handles scaling, so admin work is lowest.
How	Deploy code from Git or a pipeline to an App Service plan and set up scaling.

Exam tip
Your own app + least management = PaaS. Choose SaaS only when the app itself is bought from a vendor.

Memory trick
Bring your code, borrow the platform.

Real-world example, use case and scenario

EXAMPLE
App Service deploy
A team deploys a Node.js site to App Service from GitHub Actions. Scaling out is a slider, not new servers.

USE CASE
Customer portal
An insurance broker moves its quote website to App Service. Its two-person IT team no longer patches IIS (the Windows web server).

SCENARIO
VM fatigue
A nonprofit ran its donation site on a VM and spent weekends on Windows updates. Moving to PaaS ended the patching. It also added built-in autoscale for fundraising events.

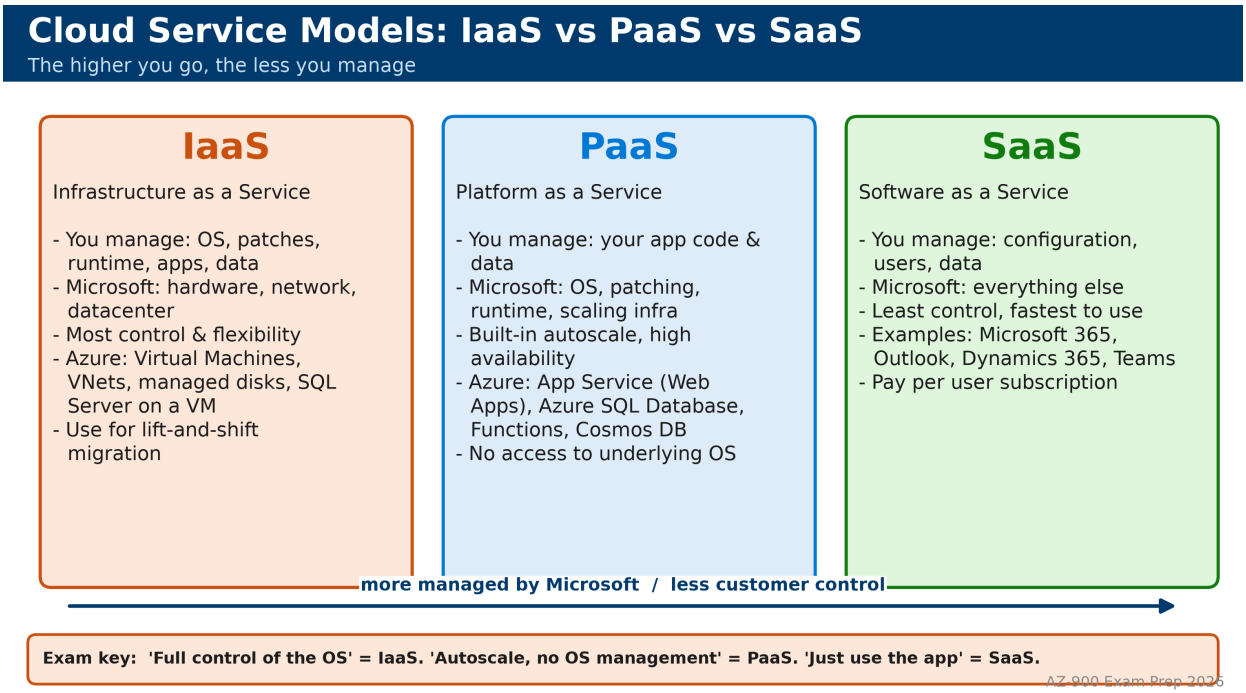


Figure Q29a: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

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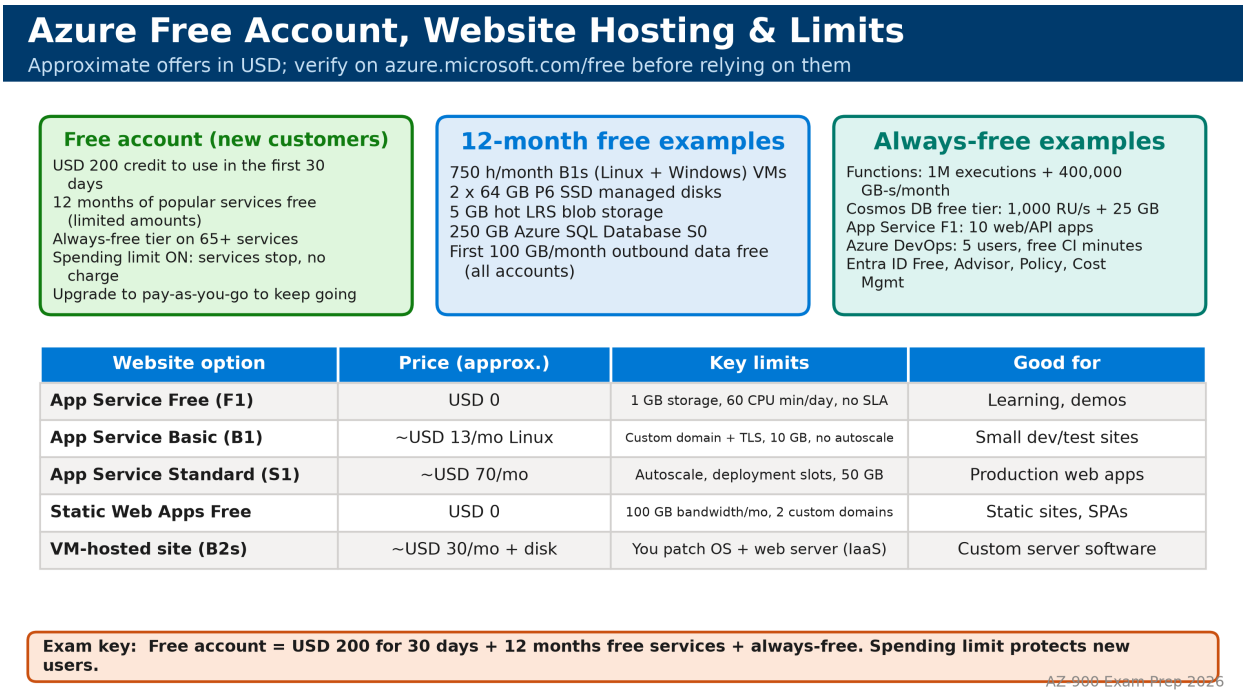


Figure Q29b: Free account, website hosting and limits

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- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
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- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 30

CLOUD CONCEPTS

Deployment models
v5 Q67, Q62, Q405

You plan to migrate several servers from an on-premises network to Azure. What is an advantage of using a public cloud service for the servers over an on-premises network?

✖ A. The public cloud is owned by the public, NOT a private corporation.

✖ B. The public cloud is a crowd-sourcing solution that provides corporations with the ability to enhance the cloud.

✖ C. All public cloud resources can be freely accessed by every member of the public.

✔ **D.** The public cloud is a shared entity whereby multiple corporations each use a portion of the resources in the cloud.

Correct answer: The public cloud is a shared entity whereby multiple corporations each use a portion of the resources in the cloud.

Clarification (read this first): 'Public' cloud does NOT mean owned by the public or open to everyone. It means many companies share the provider's hardware, each using a part. Pick the 'shared entity' option.

Answer explained

✔ **The public cloud is a shared entity whereby multiple corporations each use a portion of the resources in the cloud.**
Public cloud resources are pooled and shared by many organizations. Each uses and pays for a part. This gives economies of scale (lower cost because of large size).

✖ **The public cloud is owned by the public, NOT a private corporation.**
Public clouds are owned and run by companies such as Microsoft. 'Public' means any customer can sign up.

✖ **The public cloud is a crowd-sourcing solution that provides corporations with the ability to enhance the cloud.**
Customers use the provider's services. They do not crowd-source or improve the platform.

✖ **All public cloud resources can be freely accessed by every member of the public.**
Each customer's resources are kept apart by identity and network controls. Only endpoints you choose to open are public.

EXAM STRATEGY
Remove the false meanings of 'public': owned by the public, crowd-sourced, open to everyone. Only the shared, part-of-the-resources option is correct.

IMPACT
Explaining public cloud correctly calms the common fear that 'public' means exposed data. This fear often blocks migrations.

What · Where · When · Why · How

What	Public cloud multi-tenancy means many organizations share one pool of hardware.
Where	In the provider's datacenters in Azure regions, kept separate per customer.
When	When comparing cloud costs with running your own servers.
Why	Shared hardware spreads the cost across customers, so each pays only for its part.
How	Each customer takes a slice of the shared compute, storage and network when needed.

Exam tip
'Shared entity / portion of resources' is always the right choice.

Memory trick
A shared office tower: many companies, each renting a floor.

Real-world example, use case and scenario

EXAMPLE
Shared pool
Azure's capacity in a region is pooled. When one customer releases VMs, that capacity serves other customers.

USE CASE
Biotech compute
A biotech startup rents large-memory VMs for gene analysis only during experiments. It uses a slice of Azure's shared capacity.

SCENARIO
Board presentation
A CIO was asked why cloud was cheaper. She said Azure spreads hardware costs across thousands of customers. So the company pays for a part instead of buying whole servers.

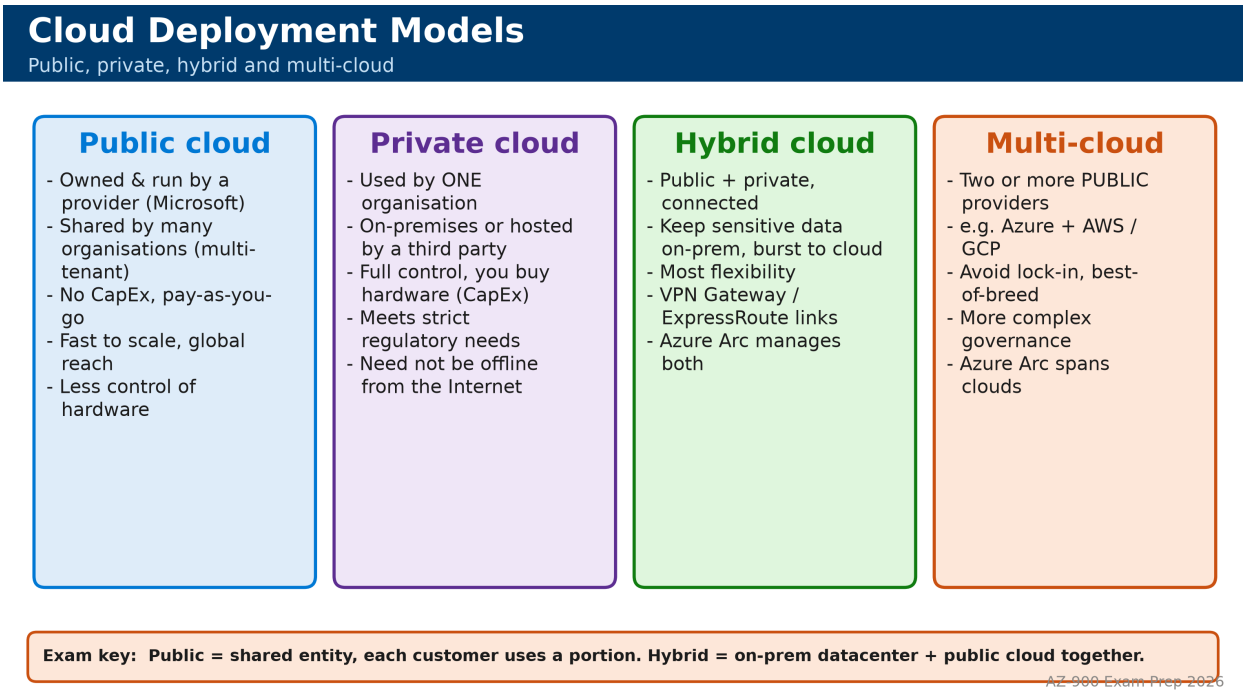


Figure Q30: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 31

CLOUD CONCEPTS

Cost
v5 Q68, Q163, Q391

You have 1,000 virtual machines hosted on the Hyper-V hosts in a data center. You plan to migrate all the virtual machines to an Azure pay-as-you-go subscription. You need to identify which expenditure model to use for the planned Azure solution. Which expenditure model should you identify?

✔ **A.** Operational.

✖ B. Elastic.

✖ C. Capital.

✖ D. Scalable.

Correct answer: Operational.

Clarification (read this first): There are only two expenditure (spending) models: CapEx (buy up front) and OpEx (pay as you use). 'Elastic' and 'Scalable' are cloud benefits, not spending models. Look-alikes: Q93 (elastic) and Q94 (scalable) both answer No.

Answer explained

✔ **Operational.**
Pay-as-you-go means regular payments for services you use, with no upfront purchase of assets. That is OpEx (operational expenditure).

✖ **Elastic.**
Elasticity is a cloud benefit (growing and shrinking automatically), not a spending model.

✖ **Capital.**
CapEx (capital expenditure) is upfront spending on physical assets, such as the current Hyper-V servers.

✖ **Scalable.**
Scalability is the ability to add resources, not a spending model.

EXAM STRATEGY
'Pay-as-you-go' means OpEx. Elastic and scalable are cloud benefits. Only Operational and Capital are real spending models.

IMPACT
Moving 1,000 VMs to OpEx changes accounting and cash flow. Finance teams weigh this in migration decisions.

What · Where · When · Why · How

What	Operational expenditure (OpEx) is ongoing spending on services you use over time.
Where	It shows as monthly Azure bills tracked in Cost Management + Billing.
When	When workloads move to a pay-as-you-go subscription instead of owned hardware.
Why	Pay-as-you-go has no upfront asset purchase, so it is OpEx, not CapEx.
How	Azure measures use and bills monthly. Costs are recorded in the month they happen.

Exam tip
Only two spending models exist: CapEx and OpEx. Elastic and scalable are benefits, never spending models.

Memory trick
Pay as you go = pay as you operate = OpEx.

Real-world example, use case and scenario

EXAMPLE
Monthly invoice
Running 1,000 Azure VMs gives a monthly bill based on hours used. It is recorded as an operating cost.

USE CASE
Budget shift
A delivery company moves from 5-year server replacement budgets to monthly cloud budgets tracked in Cost Management.

SCENARIO
CFO question
After moving 1,000 Hyper-V VMs to pay-as-you-go, the finance director asked how to record the cost. The cloud team said it is OpEx, so no new assets go on the balance sheet.

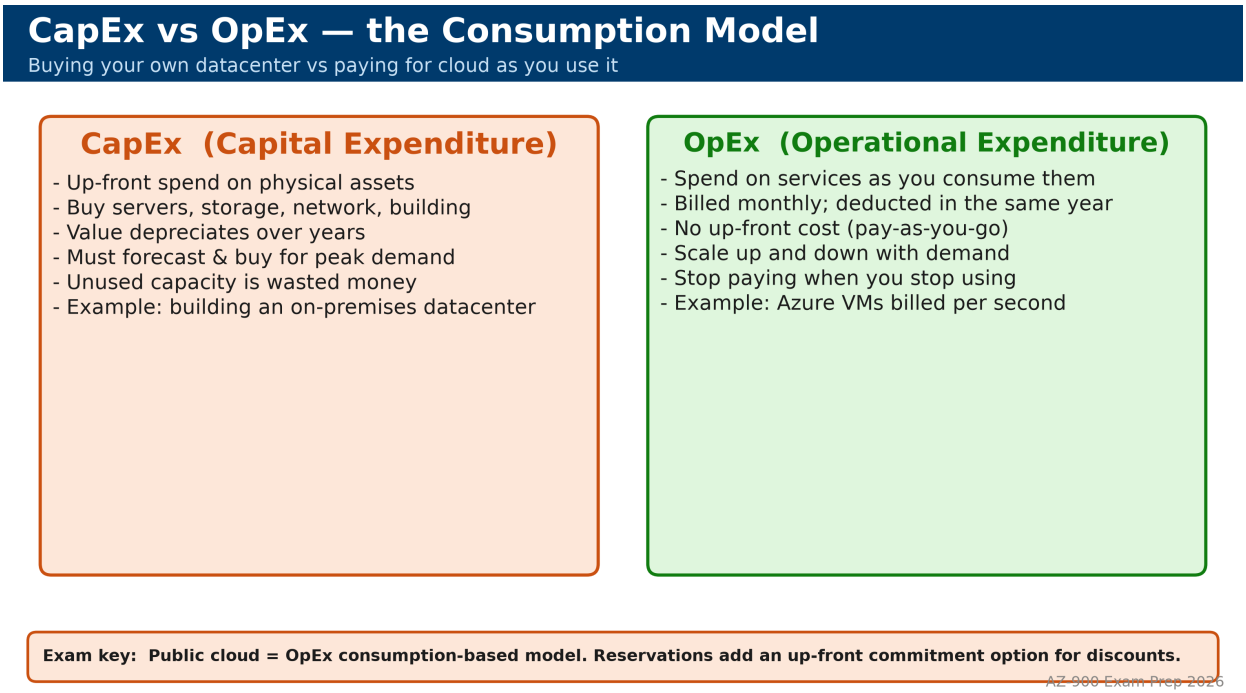


Figure Q31a: CapEx vs OpEx

About this figure

• Left box is buying assets up front (CapEx); right box is paying for consumption (OpEx). Cloud moves spending from left to right.

• CapEx assets are depreciated over several years; OpEx is expensed in the period it is incurred, which suits variable or uncertain demand.

• The consumption model means no up-front cost, pay only for what you use, and stop paying when resources are deleted or deallocated.

• Reservations and savings plans add a 1- or 3-year commitment for discounts, but they remain operational spending, not purchased hardware.

• Use the TCO calculator to compare on-premises CapEx (hardware, power, labour) with expected Azure OpEx before migrating.

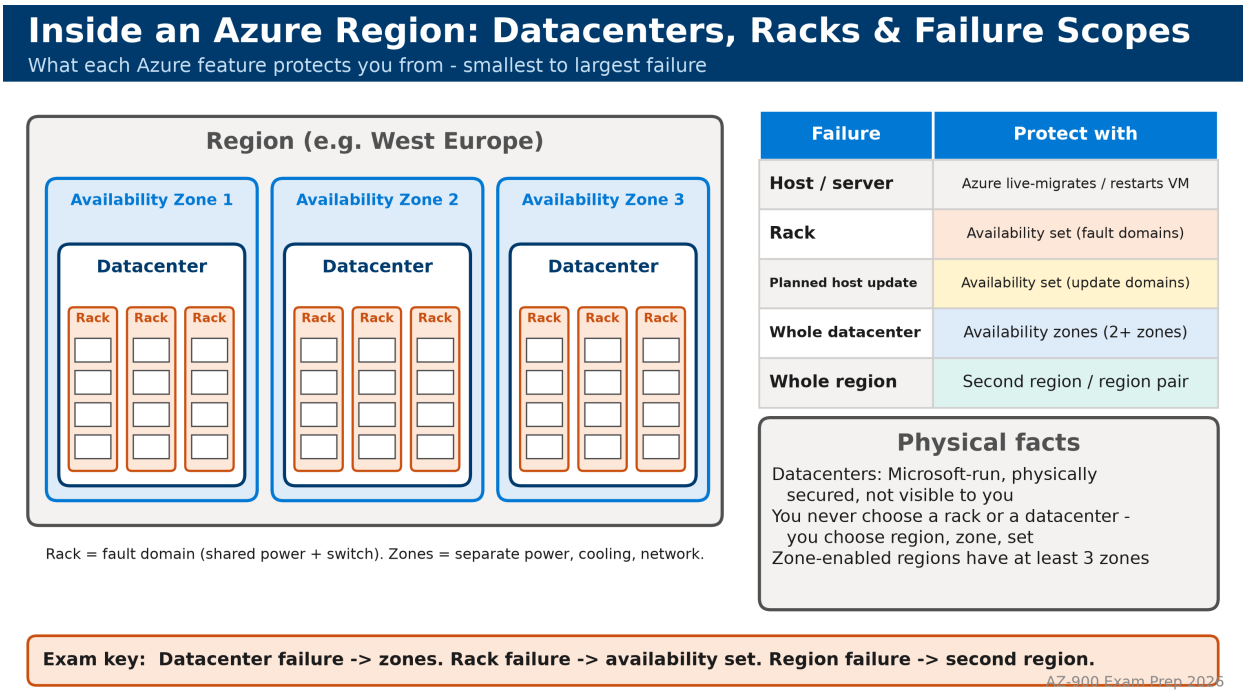


Figure Q31b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 32

CLOUD CONCEPTS

Service models
v5 Q69, Q157

You plan to provision Infrastructure as a Service (IaaS) resources in Azure. Which resource is an example of IaaS?

✖ A. An Azure web app.

✔ **B.** An Azure virtual machine.

✖ C. An Azure logic app.

✖ D. An Azure SQL database.

Correct answer: An Azure virtual machine.

Answer explained

✔ **An Azure virtual machine.**
An Azure VM is IaaS. Microsoft runs the hardware and hypervisor (the layer that runs VMs). You manage the OS, middleware and apps.

✖ **An Azure web app.**
App Service web apps are PaaS. You never manage the OS.

✖ **An Azure logic app.**
Logic Apps is a serverless workflow service, not infrastructure.

✖ **An Azure SQL database.**
Azure SQL Database is a fully managed PaaS database.

EXAM STRATEGY
Ask: who manages the OS? Only the virtual machine leaves the OS to you. Web apps, logic apps and SQL Database are PaaS.

IMPACT
Choosing IaaS means you own patching, backups and hardening (securing). If you ignore this work, VMs stay open to attack.

What · Where · When · Why · How

What	IaaS gives virtual compute, storage and networking. You manage from the OS upward.
Where	Azure Virtual Machines, placed in a virtual network in a region.
When	When you need full OS control or custom software installed.
Why	With a VM you install and patch the OS yourself. That is what makes it IaaS.
How	Create a VM from an image, connect by RDP or SSH (remote login), and manage it like a server.

Exam tip
If you patch the OS, it is IaaS. VMs, VNets and managed disks are IaaS. Web apps, SQL Database, Functions and Logic Apps are PaaS.

Memory trick
IaaS = I administer a server.

Real-world example, use case and scenario

EXAMPLE
Full OS control
An admin connects to a Windows Server VM by RDP to install an old accounting program.

USE CASE
Lift and shift
A factory company moves its on-premises production server to an Azure VM without changes. It keeps full OS control.

SCENARIO
Choosing IaaS
A research lab needed a custom Linux kernel module for lab machines. PaaS hides the OS, so they chose an Azure Linux VM. They accepted patching work in return for control.

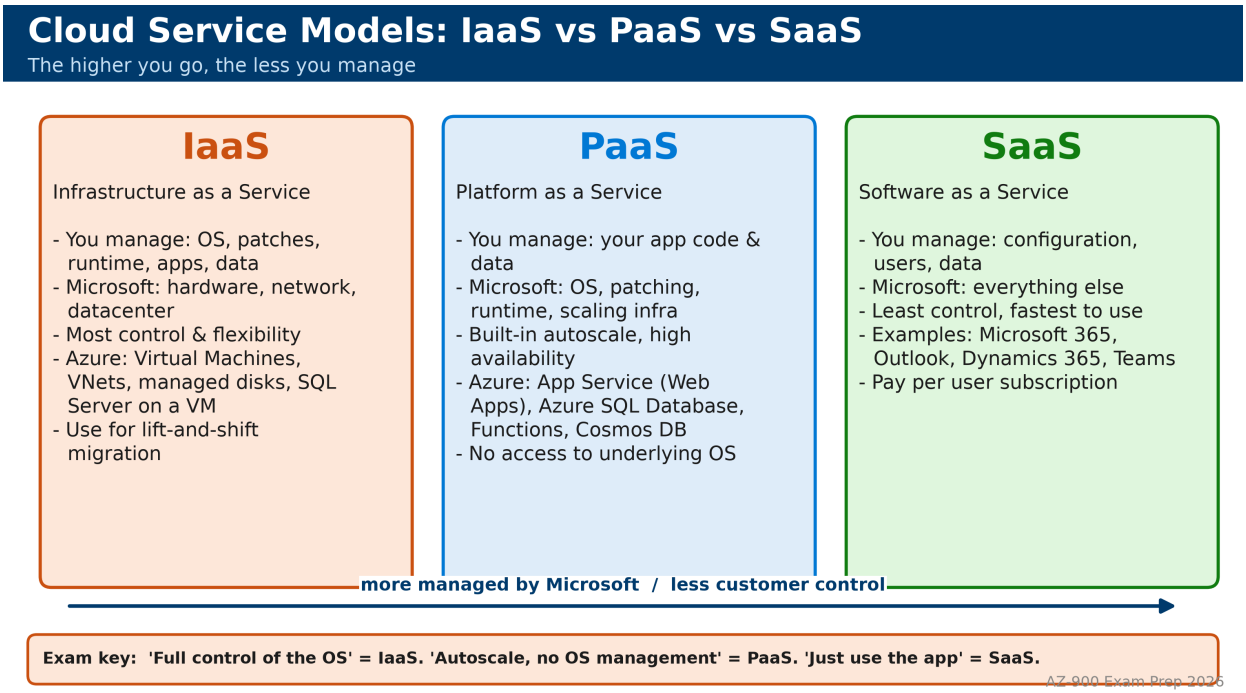


Figure Q32: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

• Serverless options such as Azure Functions are treated as PaaS-style: no OS access, automatic scaling and per-execution billing.

• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

Question 33

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q70

A team of developers at your company plans to deploy, and then remove, 50 virtual machines each week. All the virtual machines are configured by using Azure Resource Manager templates. You need to recommend which Azure service will minimize the administrative effort required to deploy and remove the virtual machines. What should you recommend?

✖ A. Azure Reserved Virtual Machine (VM) Instances.

✔ **B.** Azure DevTest Labs.

✖ C. Azure virtual machine scale sets.

✖ D. Microsoft Managed Desktop.

Correct answer: Azure DevTest Labs.

Clarification (read this first): Clues: developers, create AND delete VMs every week, ARM templates, least effort. Look-alikes in the same series: Q103 (DevTest Labs) -> Yes, Q102 (Reserved VM Instances) -> No, Q101 (Microsoft Managed Desktop) -> No.

Answer explained

✔ **Azure DevTest Labs.**
Azure DevTest Labs creates VMs from ARM templates, formulas and images. It has auto-shutdown, expiry dates and quotas (limits). It suits weekly create-and-delete cycles.

✖ **Azure Reserved Virtual Machine (VM) Instances.**
Reservations are 1- or 3-year price commitments for VMs that run all the time. They are not a tool to deploy short-lived VMs.

✖ **Azure virtual machine scale sets.**
Scale sets add and remove identical copies of one workload. They do not manage developer VM lifecycles.

✖ **Microsoft Managed Desktop.**
A managed service for company Windows PCs. It has nothing to do with deploying Azure VMs.

EXAM STRATEGY
Clues: developers, deploy and remove weekly, templates, least effort. DevTest Labs fits. Reservations are long-term deals. Scale sets are for scaling production.

IMPACT
DevTest Labs auto-shutdown, expiry and quotas stop forgotten test VMs from running all week. Forgotten VMs waste a lot of dev budget.

What · Where · When · Why · How

What	Azure DevTest Labs manages self-service dev/test VMs with templates and cost controls.
Where	A lab resource in a subscription, set up in the Azure portal.
When	When developers create and delete VMs from ARM templates again and again.
Why	Templates, auto-shutdown, expiry and quotas keep admin work low for frequent VM changes.
How	Create a lab, add ARM templates or formulas, and set expiry and shutdown rules.

Exam tip
Developers + deploy and remove + templates + least effort/cost = DevTest Labs.

Memory trick
DevTest Labs: build it Monday, bin it Friday.

Real-world example, use case and scenario

EXAMPLE
Expiration dates
A lab rule makes each VM expire after 5 days and shut down every night at 7 p.m.

USE CASE
QA environments
A game studio's test team creates test machines from a lab formula for each build. They delete them when testing ends.

SCENARIO
Weekly churn
A software consultancy's developers made 50 VMs a week by hand and often forgot to delete them. Moving to DevTest Labs with ARM templates and auto-expiry cut admin time. Forgotten VMs no longer ran up bills.

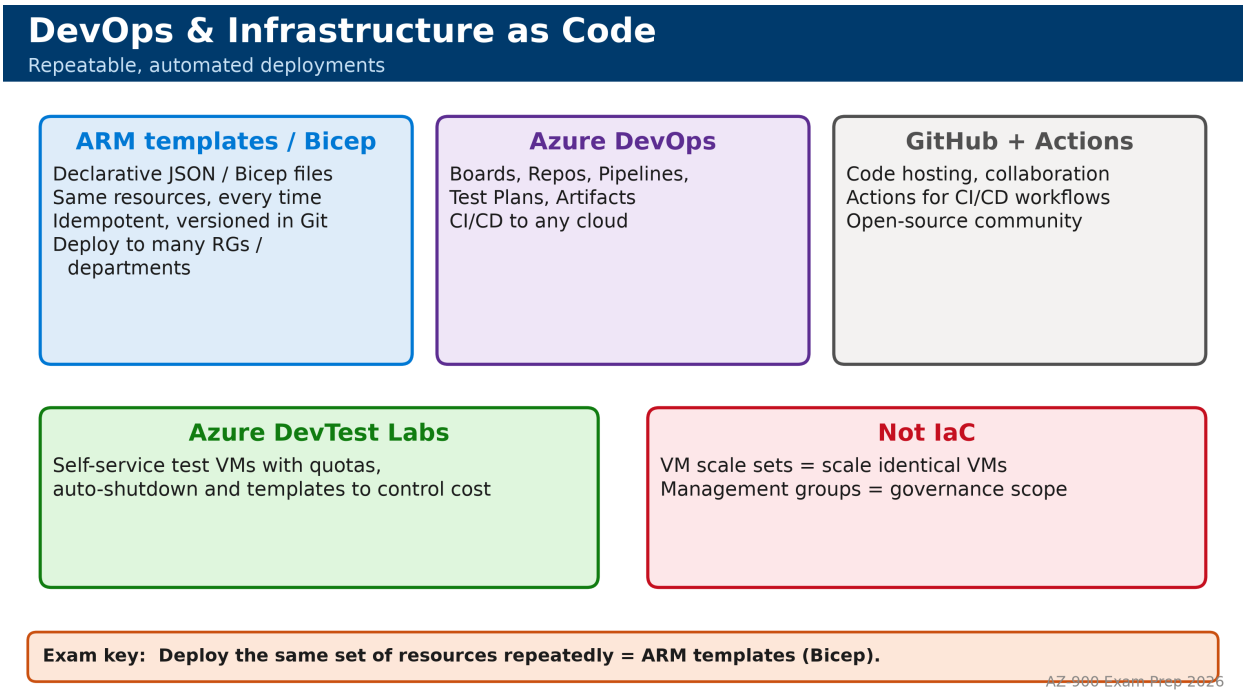


Figure Q33a: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

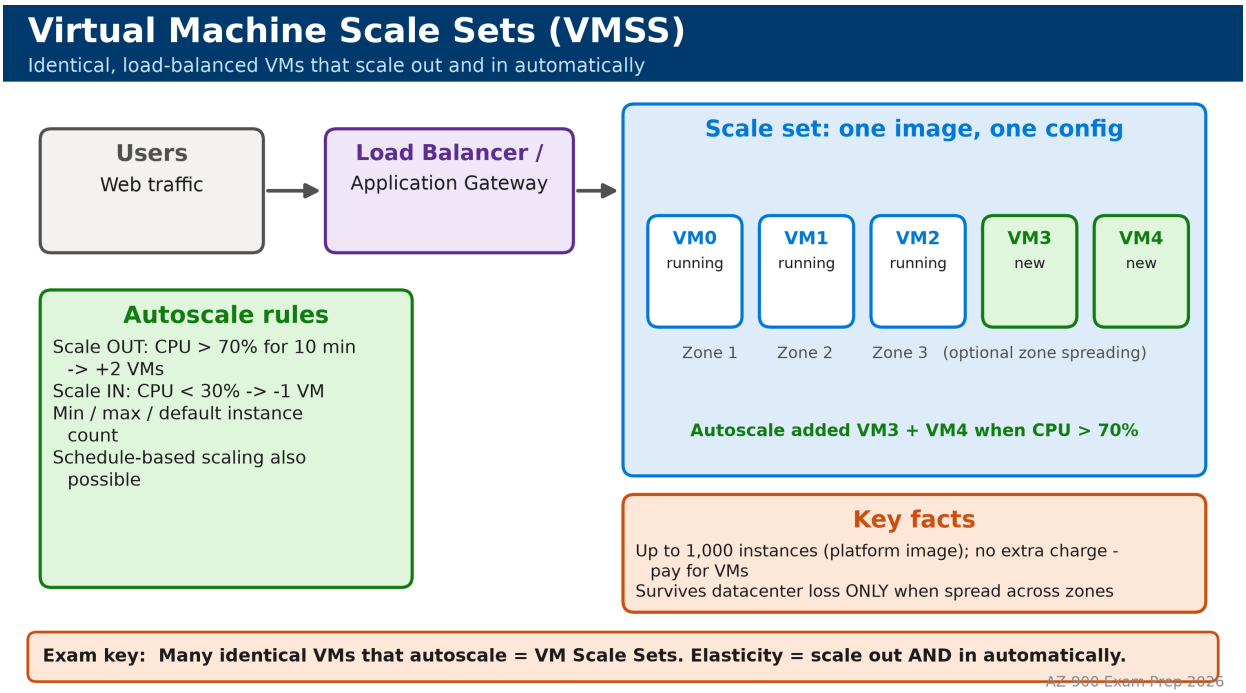


Figure Q33b: Virtual Machine Scale Sets

About this figure

- A scale set runs identical VMs from one image and configuration behind a load balancer or Application Gateway.
- Autoscale adds instances when a metric (e.g. CPU > 70%) is high and removes them when low, within min/max instance counts - this is elasticity.
- Scale sets support up to 1,000 VM instances (platform images); the scale set itself is free - you pay only for the VMs, disks and networking.
- Orchestration modes: Uniform (identical VMs) and Flexible (mixed sizes, recommended for new deployments).
- A scale set survives a datacenter failure only when its instances are spread across availability zones; a non-zonal set does not.
- Scaling out adds more VMs (horizontal); scaling up means a bigger VM size (vertical) and usually needs a restart.

Question 34

CLOUD CONCEPTS

Service models
v5 Q71, Q124

Your company plans to migrate all its data and resources to Azure. The company migration plan states that only Platform as a Service (PaaS) solutions must be used in Azure. You need to deploy an Azure environment that meets the company migration plan. Solution: You create an Azure virtual machines, Azure SQL databases, and Azure Storage accounts. Does this meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same series as Q28, but this solution includes virtual machines. VMs are IaaS (you manage the OS), so the PaaS-only rule is broken: No. Q28 (App Service + SQL Database, no VMs) answers Yes.

Answer explained

✔ **No.**
Azure virtual machines are IaaS. Including them breaks a PaaS-only rule, even though Azure SQL Database is PaaS.

✖ **Yes.**
Azure VMs are IaaS (you manage the OS). So the solution is not PaaS only.

EXAM STRATEGY
Scan for 'virtual machines'. Any VM is IaaS, so the PaaS-only goal fails whatever else is there: No.

IMPACT
Adding a VM to a PaaS-only plan brings back OS patching and a rule break that auditors will report.

What · Where · When · Why · How

What	Azure Virtual Machines are IaaS compute where you manage the OS.
Where	VMs run in a virtual network next to the PaaS SQL database and storage.
When	When you check a design against a PaaS-only rule.
Why	Including VMs adds IaaS, so the PaaS-only goal is not met.
How	Replace the VMs with PaaS hosting, such as App Service, to follow the rule.

Exam tip
In PaaS-only series, scan the solution for 'virtual machine'. If it is there, answer No.

Memory trick
One VM spoils the PaaS soup.

Real-world example, use case and scenario

EXAMPLE
Mixed stack
VMs plus Azure SQL Database is a mix of IaaS and PaaS, not pure PaaS.

USE CASE
Replacing the VM
A marketing agency replaces its web VM with App Service. Its setup now fully meets a PaaS-only standard.

SCENARIO
Review rejection
An architect sent a design with VMs, SQL Database and Storage for a PaaS-only migration. The review board rejected it because of the VMs. Replacing them with App Service got approval.

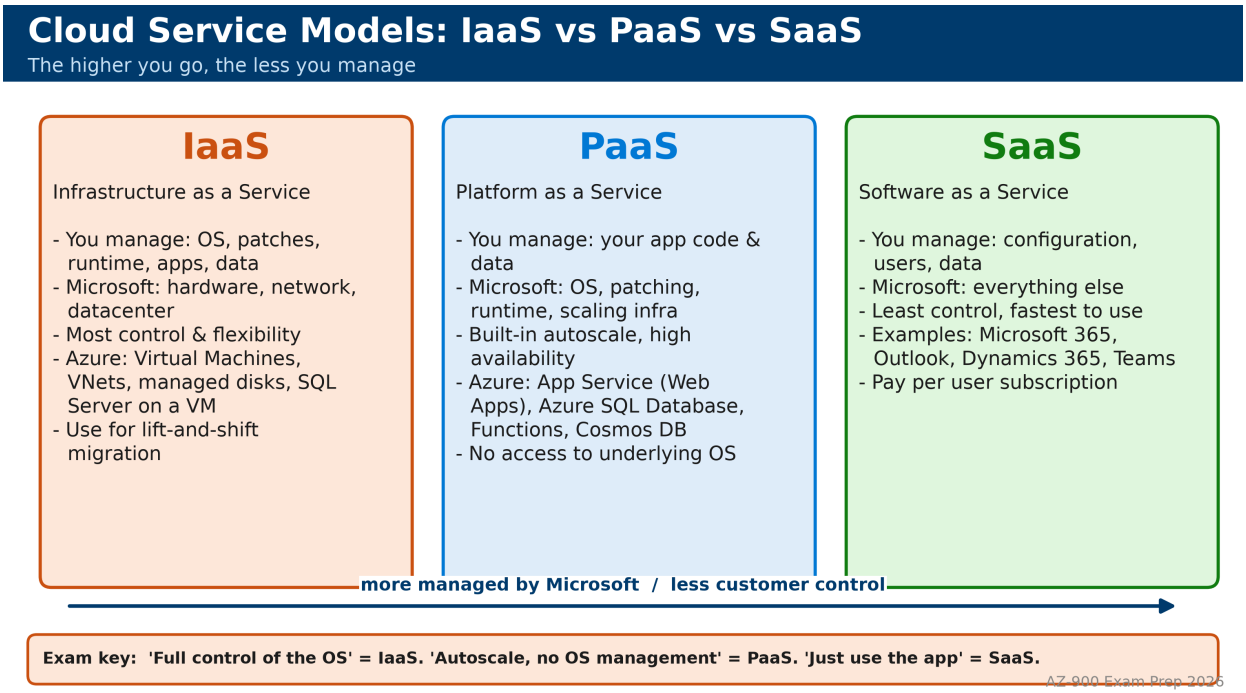


Figure Q34: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

• Serverless options such as Azure Functions are treated as PaaS-style: no OS access, automatic scaling and per-execution billing.

• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

Question 35

CLOUD CONCEPTS

Service models
v5 Q72

Your company plans to deploy several custom applications to Azure. The applications will provide invoicing services to the customers of the company. Each application will have several prerequisite applications and services installed. You need to recommend a cloud deployment solution for all the applications. What should you recommend?

✖ A. Software as a Service (SaaS).

✖ B. Platform as a Service (PaaS).

✔ **C.** Infrastructure as a Service (Iaas).

Correct answer: Infrastructure as a Service (Iaas).

Clarification (read this first): The key words are 'prerequisite applications and services installed'. Installing your own software needs OS access, which only IaaS gives. Look-alike: Q29 asks for the least admin work for your own web app; that answer is PaaS.

Answer explained

✔ **Infrastructure as a Service (Iaas).**
Installing several required apps and services needs control of the OS. Only IaaS (Azure VMs) gives that.

✖ **Software as a Service (SaaS).**
SaaS is a finished vendor app. You cannot deploy your own custom apps on it.

✖ **Platform as a Service (PaaS).**
PaaS manages the OS for you. It does not let you install any software or services you want.

EXAM STRATEGY
'Prerequisite applications installed' means you must work on the OS. Remove SaaS (nothing to install) and PaaS (no OS access). Only IaaS is left.

IMPACT
Choosing PaaS for software that needs custom installs stops a migration halfway. 'Install = IaaS' appears in many service-model questions.

What · Where · When · Why · How

What	IaaS gives you virtual machines with full admin access to the OS.
Where	Azure Virtual Machines in a virtual network.
When	When apps need extra software and services installed on the server.
Why	Only IaaS lets you install any required apps and services.
How	Create VMs, install the required software and the invoicing apps, and patch them yourself.

Exam tip
'Prerequisite applications installed' or 'OS-level control' = IaaS. 'Minimize administrative effort' = PaaS.

Memory trick
Need to install it? IaaS lets you in.

Real-world example, use case and scenario

EXAMPLE
Custom installs
An invoicing app needs a PDF service and a licence service installed on Windows Server. So it runs on Azure VMs.

USE CASE
ISV hosting
A billing software company hosts each customer's app on VMs. The product depends on Windows services and third-party drivers.

SCENARIO
PaaS attempt fails
A utility tried to put its invoicing app on App Service. The app needed a component installed as a Windows service. They moved it to VMs, where the installer worked normally.

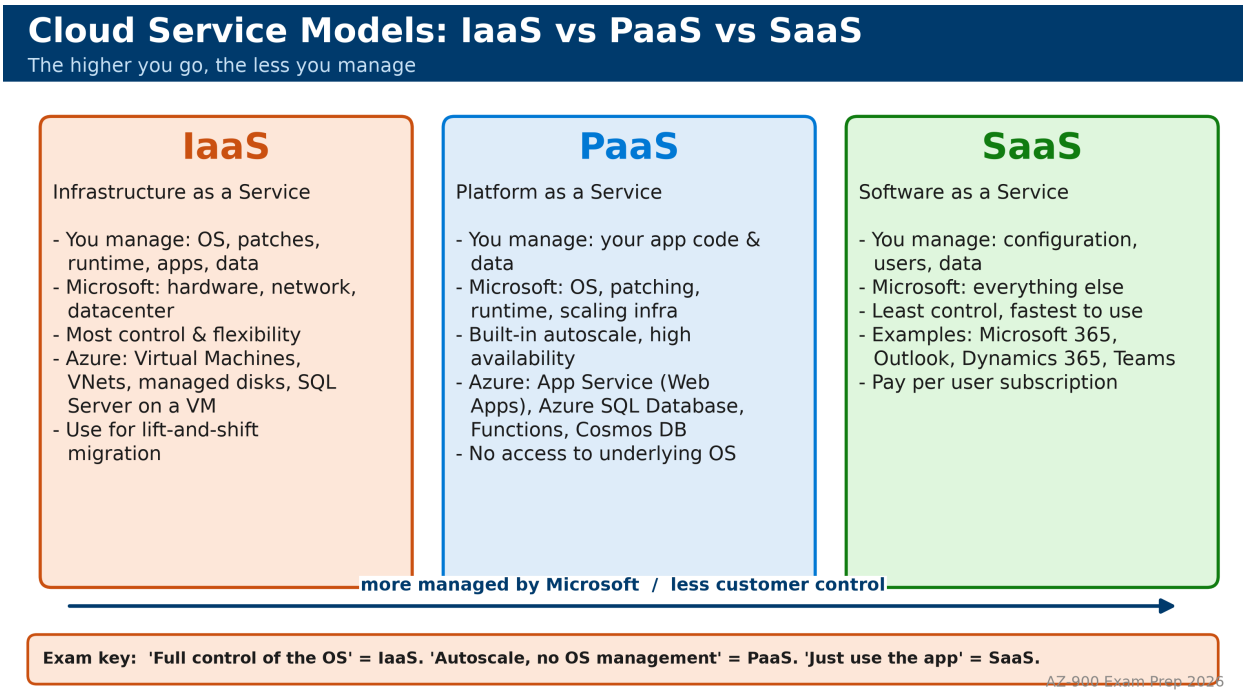


Figure Q35: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

• Serverless options such as Azure Functions are treated as PaaS-style: no OS access, automatic scaling and per-execution billing.

• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

Question 36

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q73

[An Azure region] contains one or more data centers that are connected by using a low-latency network.

✔ **A.** No change is needed.

✖ B. Is found in each country where Microsoft has a subsidiary office.

✖ C. Can be found in every country in Europe and the Americas only.

✖ D. Contains one or more data centers that are connected by using a high-latency network.

Correct answer: No change is needed.

Clarification (read this first): This is a 'fix the bracketed text' question. If the sentence is already true, choose 'No change is needed'. Here it is true: a region is datacenters linked by a LOW-latency (fast) network. The trap option says 'high-latency'.

Answer explained

✔ **No change is needed.**
An Azure region is a group of datacenters inside a latency-defined perimeter (an area small enough for fast links). They are linked by a fast, low-latency network. So the sentence is correct.

✖ **Is found in each country where Microsoft has a subsidiary office.**
Regions are placed by customer demand, laws and infrastructure, not by where Microsoft has offices.

✖ **Can be found in every country in Europe and the Americas only.**
Regions also exist in Asia, Africa, the Middle East and Australia. And not every European country has one.

✖ **Contains one or more data centers that are connected by using a high-latency network.**
Regions are defined by a low-latency network. High latency (slow links) would stop instant data copying.

EXAM STRATEGY
Look for 'low-latency' in the bracketed sentence. Regions are defined by fast links, so the 'high-latency' option is the trap. The country options are false claims.

IMPACT
Region choice affects speed, where data is stored and price. Thinking every country has a region can lead to storing data in the wrong place.

What · Where · When · Why · How

What	An Azure region is a group of datacenters within a latency-defined perimeter.
Where	A geographic location you choose when you deploy, such as West Europe or East US.
When	When choosing where to deploy for speed, data location or service availability.
Why	Its datacenters are linked by a fast, low-latency network, so the sentence is true.
How	Pick the region when you create a resource. Resources and data stay in that region.

Exam tip
Region = datacenters + low-latency network. Availability zone = separate datacenters inside a region. Region pair = two regions in the same geography, usually hundreds of miles apart (some newer regions have no pair).

Memory trick
Region: close together, quick together.

Real-world example, use case and scenario

EXAMPLE
West Europe
West Europe (Netherlands) has several datacenters linked by a fast regional network. It offers three availability zones.

USE CASE
Data residency
A German retailer picks Germany West Central so data stays in Germany and users get fast responses.

SCENARIO
Sync replication
A trading platform needed instant copying between database servers. Putting them in different zones of one region worked because zones share the region's fast network. Between regions it would have been too slow.

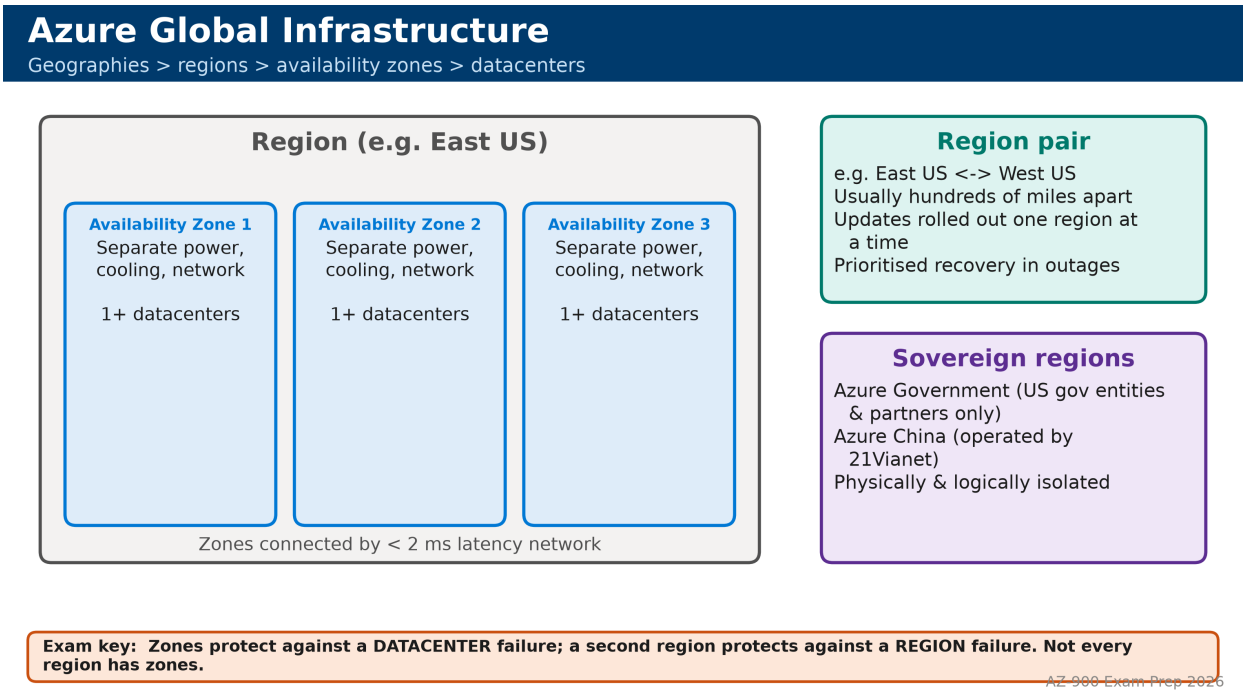


Figure Q36a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

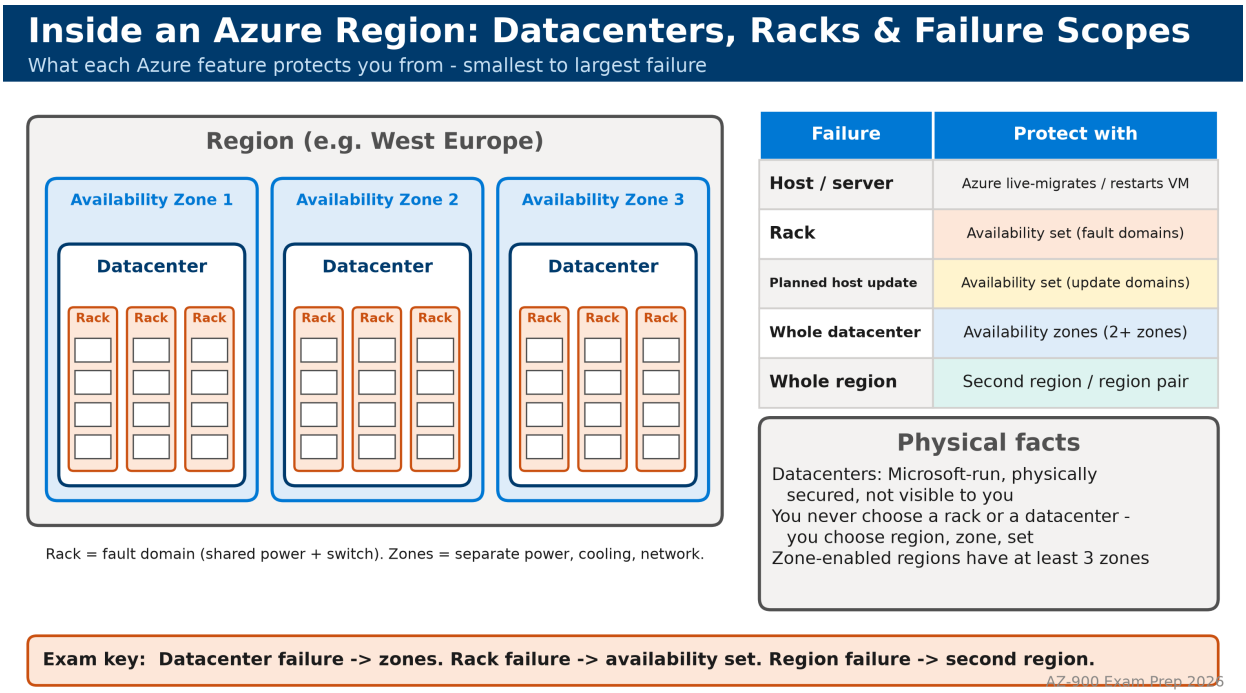


Figure Q36b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 37

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q74, Q130, Q131

A support engineer plans to perform several Azure management tasks by using the Azure CLI. You install the CLI on a computer. You need to tell the support engineer which tools to use to run the CLI. Which two tools should you instruct the support engineer to use?

- ✔ **A.** Command Prompt.
- ✖ B. Azure Resource Explorer.
- ✔ **C.** Windows PowerShell.
- ✖ D. Windows Defender Firewall.
- ✖ E. Network and Sharing Center.

Correct answer: **Command Prompt.; Windows PowerShell.**

Clarification (read this first): Choose TWO answers. The CLI was installed on the computer, so the question asks where you type its commands. Any command shell works. Look-alike: Q46 asks about an iPhone, where nothing can be installed; that answer is the portal and Cloud Shell.

Answer explained

✔ **Command Prompt.; Windows PowerShell.**
The Azure CLI (az) is a command-line tool. On Windows it runs from any shell, including Command Prompt and Windows PowerShell.

✖ **Azure Resource Explorer.**
A web tool to browse resource JSON (code view). It is not a command shell.

✖ **Windows Defender Firewall.**
A firewall on the computer that filters traffic. It cannot run commands.

✖ **Network and Sharing Center.**
A Control Panel page for local network settings. It cannot run commands.

EXAM STRATEGY
'Run the CLI' after a local install means any OS shell. Pick the two shells. Resource Explorer, Firewall and Network and Sharing Center are not shells.

IMPACT
Knowing az runs in plain Command Prompt or PowerShell lets admins script from any Windows computer. Shell questions appear several times in the exam.

What · Where · When · Why · How

What	The Azure CLI (az) is a command-line tool for managing Azure on any OS.
Where	Installed on Windows, macOS or Linux, and already included in Cloud Shell.
When	When admins want to script or run Azure tasks from a command line.
Why	az runs in any shell, so Command Prompt and Windows PowerShell both work.
How	Open the shell, run az login, then run commands such as az group list.

Exam tip
az runs in any shell: Command Prompt, PowerShell, bash, zsh, Cloud Shell. Pick the options that are command shells.

Memory trick
az needs a prompt, any prompt.

Real-world example, use case and scenario

EXAMPLE
Cmd.exe
An engineer types az login and then az group list in Command Prompt.

USE CASE
Scripted checks
A support desk runs a PowerShell script that calls az vm list to check VM states each morning.

SCENARIO
Which window?
A new support engineer installed the CLI and asked where to type commands. His lead said Command Prompt or PowerShell. He ran az account show in both, and it worked.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

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Figure Q37: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 38

AZURE ARCHITECTURE & SERVICES

Data & AI
v5 Q75, Q60

You plan to store 20 TB of data in Azure. The data will be accessed infrequently and visualized by using Microsoft Power BI. You need to recommend a storage solution for the data Which two solutions should you recommend?

✔ **A.** Azure Data Lake.

✖ B. Azure Cosmos DB.

✔ **C.** Azure Synapse Analytics.

✖ D. Azure SQL Database.

✖ E. Azure Database for PostgreSQL.

Correct answer: Azure Data Lake.; Azure Synapse Analytics.

Clarification (read this first): Choose TWO answers. Clues: very large data (20 TB), 'accessed infrequently' (rarely used, so cheap storage), and Power BI reports. This points to analytics storage, not everyday app databases. Q64 is the same idea.

Answer explained

✔ **Azure Data Lake.; Azure Synapse Analytics.**
Data Lake Storage holds 20 TB cheaply with cool, cold and archive tiers (lower-cost levels for rarely used data). Synapse Analytics queries it. Both connect directly to Power BI.

✖ **Azure Cosmos DB.**
Fast NoSQL database for live apps, priced by throughput. Expensive for rarely used analytics data.

✖ **Azure SQL Database.**
A transactional database for daily app use. Costly and a poor fit for 20 TB of rarely used analytics data.

✖ **Azure Database for PostgreSQL.**
A relational database for live apps, not low-cost analytics storage.

EXAM STRATEGY
'20 TB, infrequent access, Power BI' means analytics-size storage. Keep Data Lake and Synapse. Remove the app databases (SQL Database, PostgreSQL, Cosmos DB) built for daily transactions.

IMPACT
Putting 20 TB of rarely used analytics data in an app database raises cost and slows queries. The analytics option keeps BI reports cheap.

What · Where · When · Why · How

What	Data Lake Storage is large analytics storage. Synapse Analytics is a data warehouse and query service.
Where	Data Lake is a storage account with folders (hierarchical namespace). Synapse is a workspace on top of it.
When	When large, rarely used data must be shown in Power BI.
Why	Both store or query 20 TB at low cost and connect directly to Power BI.
How	Store data in the lake, query it with Synapse, and connect Power BI to Synapse.

Exam tip
Size + Power BI point to Data Lake + Synapse, whatever the number.

Memory trick
Big, cold, charted: lake and warehouse.

Real-world example, use case and scenario

EXAMPLE
Tiered lake
Website click data older than 90 days moves to the cool tier in Data Lake Storage Gen2. Synapse views feed Power BI.

USE CASE
Clinical archive
A hospital group stores 20 TB of past admissions data in a data lake. It builds quarterly Power BI reports through Synapse.

SCENARIO
Analyst wishlist
A sports league's analysts wanted ten seasons of tracking data in Power BI. Cosmos DB was quoted as costly. The data team chose Data Lake plus Synapse serverless SQL and pays mostly for storage.

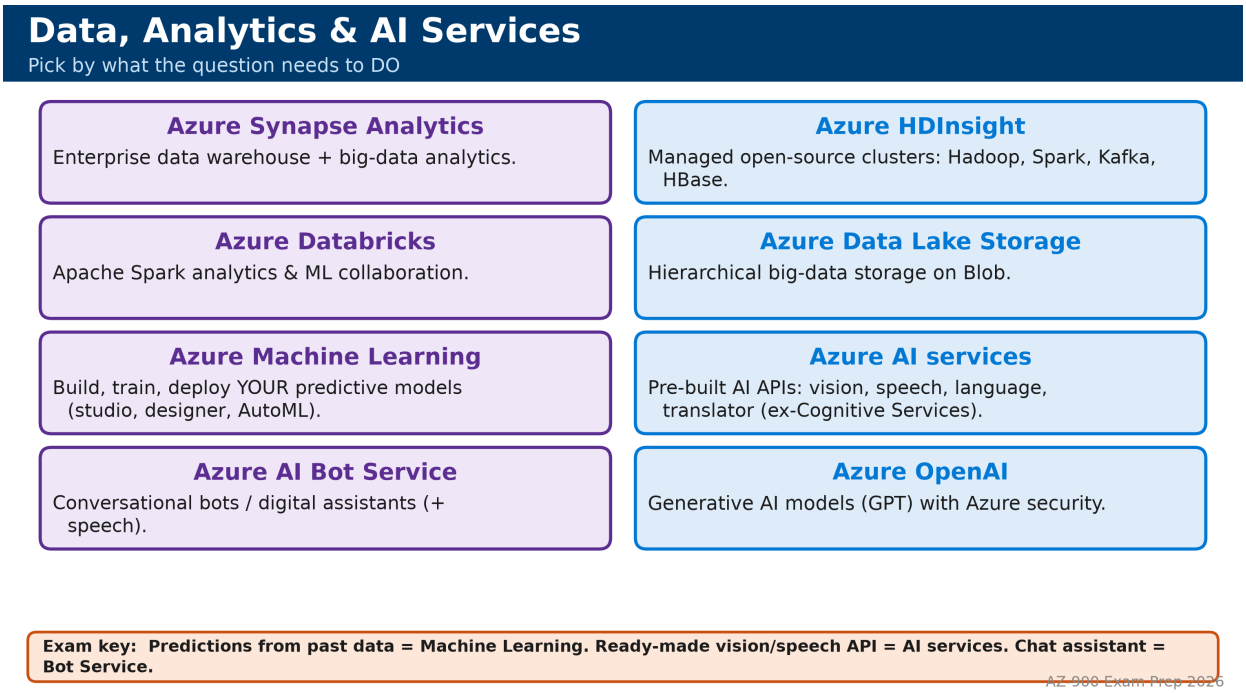


Figure Q38: Data, analytics and AI services

About this figure

- Left column covers data and analytics platforms; right column covers AI. Decide whether the scenario needs storage, processing, custom models or ready-made AI.
- Azure Machine Learning trains your own models from your data; Azure AI services are prebuilt APIs needing no data science, billed per transaction.
- Azure AI services include Vision, Speech, Language, Translator and Document Intelligence; many offer a free F0 tier with limited monthly calls.
- Azure OpenAI models are now offered through Microsoft Foundry (formerly Azure AI Foundry), billed per 1,000 or million tokens, varying widely by model.
- Microsoft Fabric is the newer unified analytics SaaS combining data engineering, warehousing and Power BI, and overlaps much of Synapse.

Question 39

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q77, Q240

You have a virtual machine named VM1 that runs Windows Server 2016. VM1 is in the East US Azure region. Which Azure service should you use from the Azure portal to view service failure notifications that can affect the availability of VM1?

✖ A. Azure Service Fabric.

✔ **B.** Azure Monitor.

✖ C. Azure virtual machines.

✖ D. Azure Advisor.

Correct answer: Azure Monitor.

Clarification (read this first): The exact tool is Azure Service Health, but it is not in the list. Service Health sits inside Azure Monitor in the portal, so pick Azure Monitor. Related: Q136 asks the same with Service Health in the list.

Answer corrected in this edition. The previous PDF printed: *Azure virtual machines.*. The PDF printed 'Azure virtual machines'. Azure Service Health shows service failure notices that can affect a VM. In the portal it is inside Azure Monitor (Monitor > Service Health). Of the listed options, Azure Monitor is correct. This matches the checked answer key.

Answer explained

✔ **Azure Monitor.**
Azure Service Health shows Azure service problems and planned maintenance that affect your resources. In the portal you open it under Azure Monitor.

✖ **Azure Service Fabric.**
A platform to host microservices (small app parts), not a health notice service.

✖ **Azure virtual machines.**
The VM page shows only that VM's Resource Health, not notices about Azure service failures.

✖ **Azure Advisor.**
Advisor gives best-practice advice, not outage notices.

EXAM STRATEGY
'View service failure notifications' points to Service Health, found inside Azure Monitor. Remove Advisor (advice only), Service Fabric (app platform) and the VM page.

IMPACT
Service Health alerts tell you an outage is Microsoft's, not yours. This saves hours of fault-finding and lets you switch over before customers notice.

What · Where · When · Why · How

What	Azure Service Health reports Azure service problems and planned maintenance that affect you.
Where	In the portal under Azure Monitor > Service Health.
When	When checking if an Azure incident could affect VM1 in East US.
Why	Service Health is inside Monitor, the only listed service that shows these notices.
How	Open Monitor > Service Health, filter by region and service, and create alerts.

Exam tip
Service failure or planned maintenance notices = Service Health. If it is not listed, choose Azure Monitor. Advisor = advice, not incidents.

Memory trick
Monitor is the hospital; Service Health is its emergency room.

Real-world example, use case and scenario

EXAMPLE
Service issue alert
Monitor > Service Health shows an active Virtual Machines incident in East US and lists the affected subscriptions.

USE CASE
Ops alerting
An online learning company sets Service Health alerts to email and Teams when maintenance is planned in its regions.

SCENARIO
Is it us or Azure?
An admin saw VM1 in East US stop responding. Service Health in Azure Monitor showed a regional network incident. The team waited for Microsoft's fix instead of rebuilding the VM.

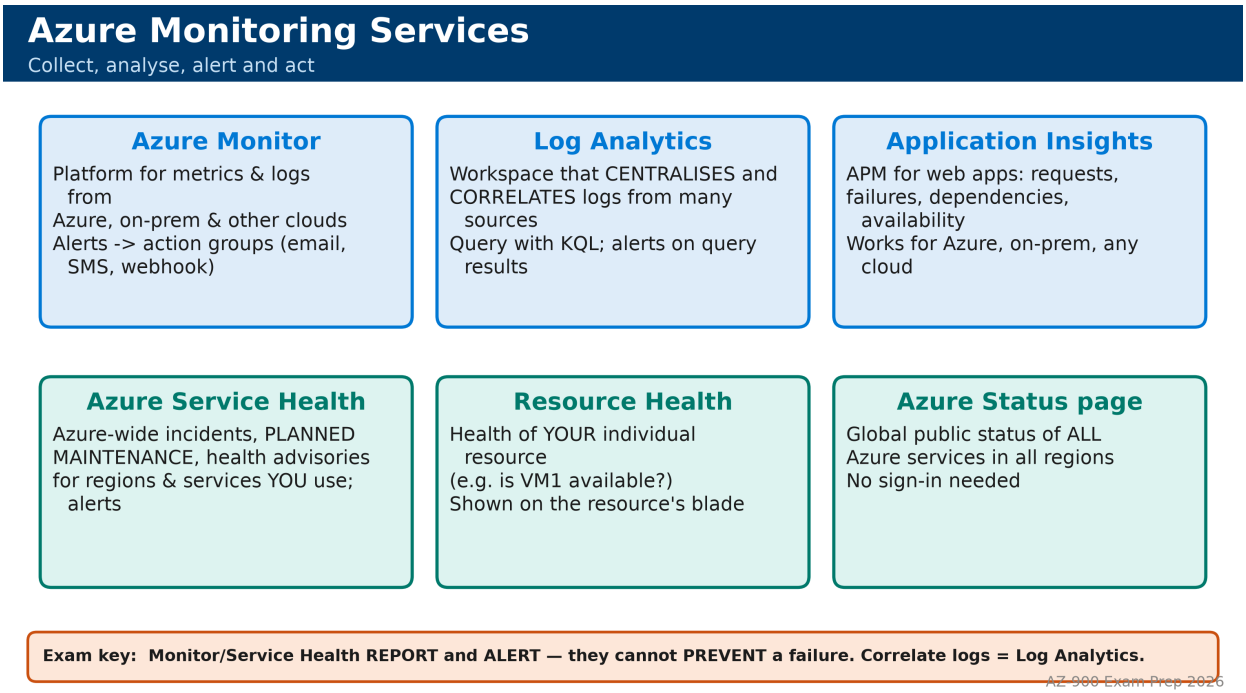


Figure Q39: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 40

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q79

You have an Azure environment that contains multiple Azure virtual machines. You plan to implement a solution that enables the client computers on your on-premises network to communicate to the Azure virtual machines. You need to recommend which Azure resources must be created for the planned solution. Which two Azure resources should you include in the recommendation?

✔ **A.** A virtual network gateway.

✖ B. A load balancer.

✖ C. An application gateway.

✖ D. A virtual network.

✔ **E.** A gateway subnet.

Correct answer: A virtual network gateway.; A gateway subnet.

Clarification (read this first): Choose TWO answers. Watch 'must be created': the VMs already exist, so their virtual network already exists. You only need to add the VPN gateway and the special subnet it lives in (GatewaySubnet).

Answer explained

✔ **A virtual network gateway.; A gateway subnet.**
Connecting on-premises computers needs a VPN gateway (a virtual network gateway). It must be placed in its own subnet named GatewaySubnet.

✖ **A load balancer.**
It spreads traffic across VMs. It does not connect on-premises networks.

✖ **An application gateway.**
A Layer-7 (web traffic) load balancer with WAF (web app firewall). It does not connect sites.

✖ **A virtual network.**
The VMs are already in a virtual network, so you do not need to create one.

EXAM STRATEGY
'On-premises clients to Azure VMs' means a site-to-site VPN. You need a virtual network gateway in a GatewaySubnet. Load balancers and application gateways spread traffic; they do not link sites.

IMPACT
If you forget the GatewaySubnet, the gateway cannot be deployed and the connection is delayed. The gateway + gateway subnet pair appears often on the exam.

What · Where · When · Why · How

What	A virtual network gateway (VPN gateway) connects an Azure VNet to on-premises networks.
Where	Placed in a special subnet named GatewaySubnet inside the VMs' VNet.
When	When on-premises computers must reach Azure VMs privately.
Why	The VPN needs the gateway, and the gateway must sit in a GatewaySubnet.
How	Add GatewaySubnet, deploy the gateway, then create a local network gateway and a connection.

Exam tip
A VPN gateway always sits in a subnet named exactly GatewaySubnet. The VMs mean a VNet already exists, so do not pick 'virtual network'.

Memory trick
The gateway needs its own room: GatewaySubnet.

Real-world example, use case and scenario

EXAMPLE
S2S setup
An admin adds GatewaySubnet (/27) to VNet1, deploys a VpnGw1 gateway, and connects it to the office firewall over IPsec (an encrypted tunnel).

USE CASE
Branch access
A law firm connects its office network to Azure VMs running its document system through a site-to-site VPN.

SCENARIO
Deployment error
An engineer tried to deploy a VPN gateway and got an error. She had not created a subnet named GatewaySubnet. After adding it, the gateway deployed and office PCs reached the VMs.

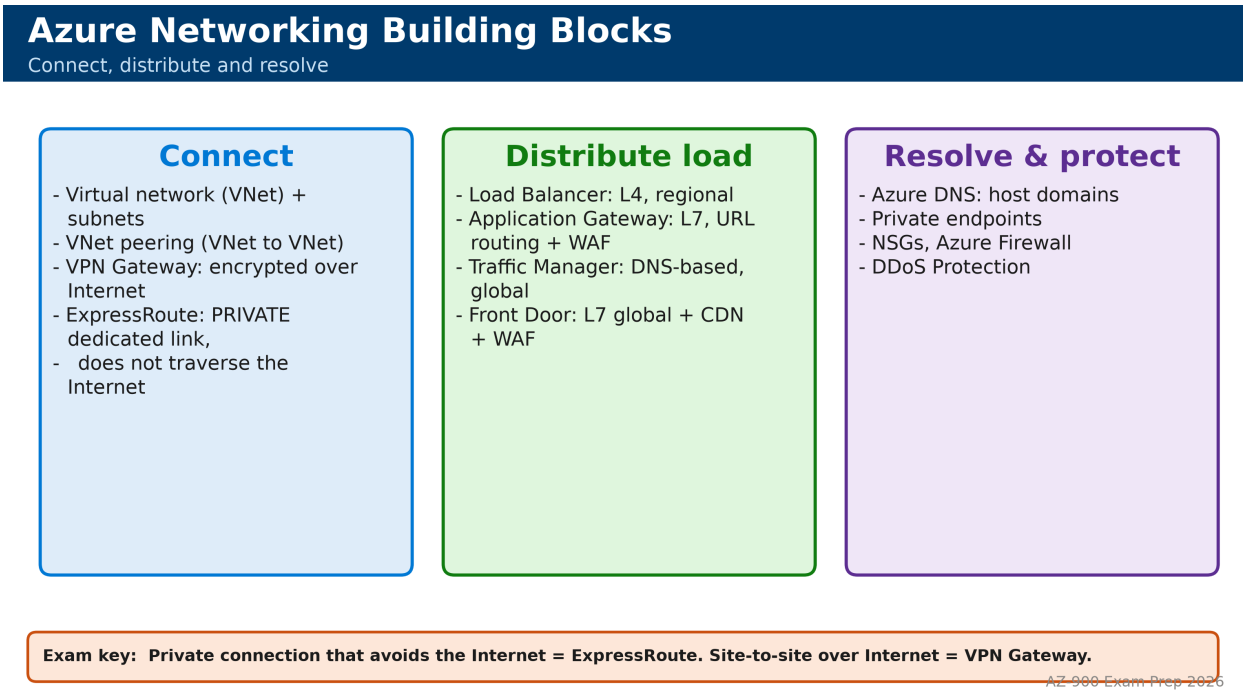


Figure Q40: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 41

AZURE ARCHITECTURE & SERVICES

Storage
v5 Q82, Q30

You plan to map a network drive from several computers that run Windows 10 to Azure Storage. You need to create a storage solution in Azure for the planned mapped drive. What should you create?

✖ A. An Azure SQL database.

✖ B. A virtual machine data disk.

✔ **C.** A Files service in a storage account.

✖ D. A Blobs service in a storage account.

Correct answer: A Files service in a storage account.

Clarification (read this first): 'Map a network drive' means a shared folder that shows as a drive letter (like Z:). That is Azure Files. Look-alike: Q26 asks where VM disks are stored; that answer is Blobs.

Answer explained

✔ **A Files service in a storage account.**
Azure Files gives managed SMB file shares (shared network folders). Many Windows 10 computers can map one as a network drive.

✖ **An Azure SQL database.**
A relational database, not a file share you can map as a drive.

✖ **A virtual machine data disk.**
A data disk attaches to one VM only. It cannot be mapped as a drive on many Windows 10 computers.

✖ **A Blobs service in a storage account.**
Blob storage is object storage reached over HTTP. It cannot be mapped with SMB.

EXAM STRATEGY
'Map a network drive' means an SMB file share, which is Azure Files. Blob is object storage, data disks attach to one VM, SQL is a database.

IMPACT
Azure Files replaces office file servers with no VM to look after. Picking Blob for an SMB need breaks every mapped drive in the office.

What · Where · When · Why · How

What	Azure Files is a managed file share service using SMB or NFS.
Where	A file share inside a storage account, reached at <account>.file.core.windows.net.
When	When many computers need a shared, mapped network drive.
Why	Windows 10 can map an Azure file share as a drive letter over SMB.
How	Create a file share, then map it with net use or File Explorer over port 445.

Exam tip
Mapped drive, SMB, shared folder, moving a file server = Azure Files.

Memory trick
Map a drive? File it under Files.

Real-world example, use case and scenario

EXAMPLE
Net use
The net use Z: command, pointed at the share on contoso.file.core.windows.net, maps an Azure file share as drive Z on Windows 10.

USE CASE
Shared templates
An accounting firm keeps shared spreadsheets on an Azure file share mapped as S: on every staff PC.

SCENARIO
Retiring a NAS
A small architecture firm's old storage box was failing. They created an Azure Files share, mapped it on all Windows 10 PCs and copied the data. Now there is no server to look after.

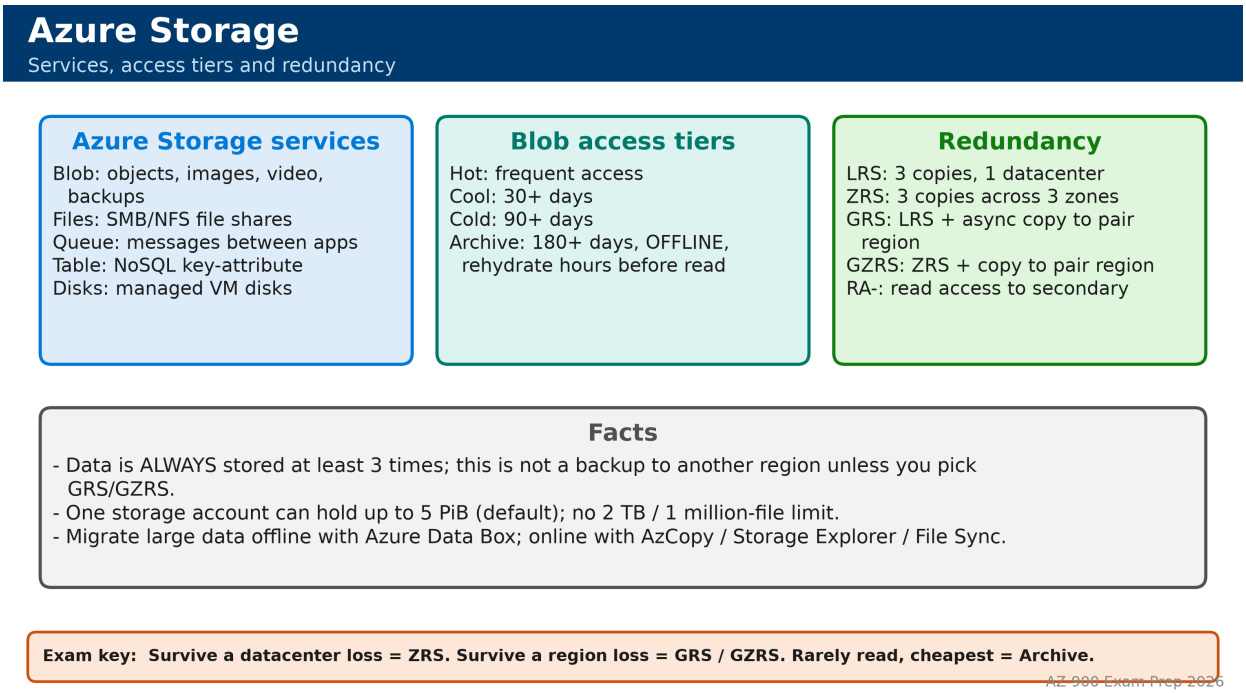


Figure Q41: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 42

AZURE ARCHITECTURE & SERVICES — **Subscriptions** v5 Q83

Your company plans to migrate all its network resources to Azure. You need to start the planning process by exploring Azure. What should you create first?

✔ **A.** A subscription.

✖ B. A resource group.

✖ C. A virtual network.

✖ D. A management group.

Correct answer: A subscription.

Clarification (read this first): The word to watch is 'first'. Resource groups and VNets can only be made inside a subscription. Management groups are optional and come later. So the subscription comes first.

Answer explained

✔ **A subscription.**
Every Azure resource must belong to a subscription. So a subscription is the first thing to create before you explore or deploy.

✖ **A resource group.**
Resource groups are created inside a subscription, so the subscription must exist first.

✖ **A virtual network.**
A VNet is a resource inside a resource group, inside a subscription.

✖ **A management group.**
Management groups organize subscriptions that already exist. They are not a starting point.

EXAM STRATEGY
The 'first' step to explore Azure is the billing and access container that holds everything. Resource groups and VNets need a subscription. Management groups are optional and come later.

IMPACT
Nothing can be deployed without a subscription. It is the base for billing and access. Questions about creation order appear often in the governance area.

What · Where · When · Why · How

What	A subscription is the billing and access container needed for any Azure resource.
Where	At the top of the deployment structure, under optional management groups.
When	When an organization first starts using or exploring Azure.
Why	Resource groups, VNets and resources all need a subscription, so it comes first.
How	Sign up for a free or pay-as-you-go subscription, then create resource groups.

Exam tip
Creation order: account/tenant, then subscription, then resource group, then resources.
Management groups are optional and sit above subscriptions.

Memory trick
No subscription, no admission.

Real-world example, use case and scenario

EXAMPLE
Free account
A team signs up for an Azure free account. This creates a subscription with credit for the first 30 days.

USE CASE
Migration pilot
A manufacturer opens a pay-as-you-go subscription to run a migration check with Azure Migrate before committing.

SCENARIO
Where to start
A city council IT manager wanted to build a VNet but found nothing to click. A partner explained that a subscription comes first. Once it existed, she could build resource groups and networks.

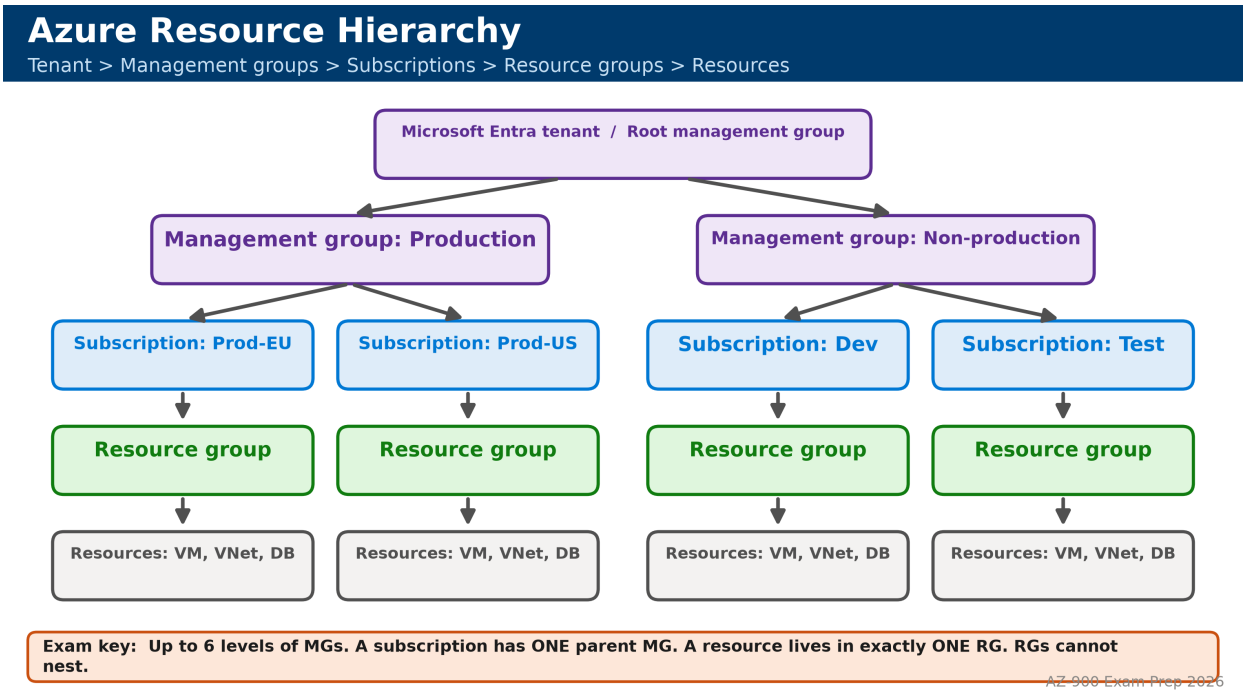


Figure Q42: Azure resource hierarchy

About this figure

• Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.

• A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.

• Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.

• Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.

• Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 43

AZURE ARCHITECTURE & SERVICES — Compute v5 Q84

You have an on-premises application that sends email notifications automatically based on a rule, You plan to migrate the application to Azure. You need to recommend a serverless computing solution for the application. What should you include in the recommendation?

✖ A. A web app.

✖ B. A server image in Azure Marketplace.

✔ **C.** A logic app.

✖ D. An API app.

Correct answer: A logic app.

Clarification (read this first): Both Functions and Logic Apps are serverless. Here the app sends email 'based on a rule' and Functions is not even listed, so pick Logic Apps (workflow with ready-made connectors). Look-alike: Q4 asks for general serverless computing; that answer is Azure Functions.

Answer explained

✔ **A logic app.**
Azure Logic Apps is a serverless workflow service. A trigger or rule runs actions, such as sending email, through ready-made connectors. There are no servers to manage.

✖ **A web app.**
Web apps run on an App Service plan you pay for all the time. They are PaaS but not serverless.

✖ **A server image in Azure Marketplace.**
A Marketplace server image creates a VM. That is IaaS, the opposite of serverless.

✖ **An API app.**
API apps host REST APIs on an App Service plan you pay for all the time. Not serverless.

EXAM STRATEGY
'Serverless' + 'based on a rule' + sending email = workflow automation: Logic Apps. Web and API apps run on App Service plans. Marketplace images are VMs.

IMPACT
Logic Apps has built-in email connectors and you pay per run. It replaces an always-on server and cuts cost for a small number of automatic emails.

What · Where · When · Why · How

What	Azure Logic Apps is a serverless workflow service with ready-made connectors.
Where	A logic app resource built in the portal's workflow designer.
When	When a process, such as rule-based email notices, must run without servers.
Why	It is serverless and has ready-made email connectors, so you need no server or code.
How	Set a trigger or schedule, add a condition, then add a 'Send an email' action.

Exam tip
Serverless + workflow/rule/connectors/no code = Logic Apps. Serverless + your own code run by events = Functions.

Memory trick
Logic Apps: if this, then email that.

Real-world example, use case and scenario

EXAMPLE
Threshold email
A logic app starts when a SharePoint list item's status changes to Overdue. It sends an Outlook email to the owner.

USE CASE
Order alerts
An online flower shop uses a logic app to email the manager whenever a large order arrives.

SCENARIO
Retiring a VM
A property manager's on-premises app sent rent reminder emails based on due-date rules. They rebuilt it as a Logic Apps workflow and removed the server. Now they pay only when reminders run.

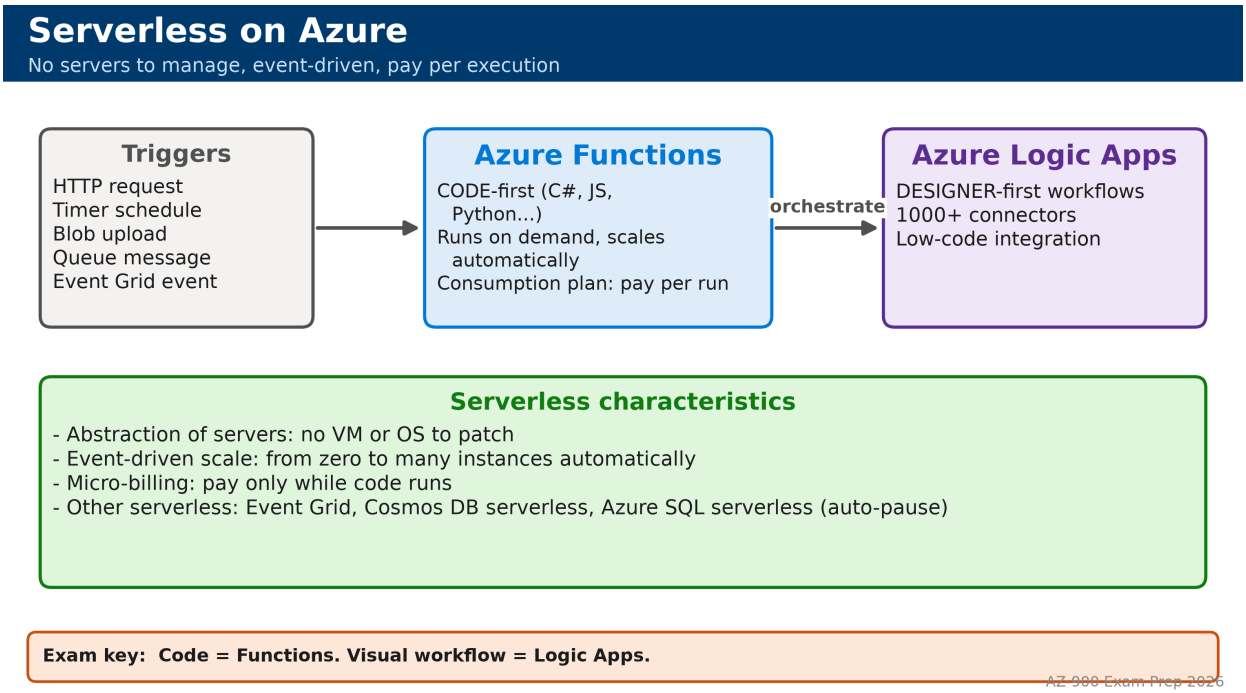


Figure Q43: Serverless on Azure

About this figure

• Flow left to right: an event trigger starts code (Functions) or a workflow (Logic Apps); the green box lists traits shared by serverless services.

• Functions Consumption: approx. USD 0.20 per million executions plus GB-seconds, with 1 million executions and 400,000 GB-s free monthly.

• Consumption-plan functions can cold start and have a default 5-minute, maximum 10-minute timeout; Flex Consumption and Premium plans reduce cold starts.

• Logic Apps Consumption bills per action and connector call, roughly fractions of a cent each; Standard plan bills per hosting plan instead.

• Serverless still runs on servers; Microsoft manages and scales them, and you pay nothing while nothing runs, except storage or plan minimums.

Question 44

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q85

You plan to deploy a website to Azure. The website will be accessed by users worldwide and will host large video files. You need to recommend which Azure feature must be used to provide the best video playback experience. What should you recommend?

✖ A. An application gateway.

✖ B. An Azure ExpressRoute circuit.

✔ **C.** A content delivery network (CDN).

✖ D. An Azure Traffic Manager profile.

Correct answer: A content delivery network (CDN).

Answer explained

✔ **A content delivery network (CDN).**
A CDN (content delivery network) keeps copies of large files, such as videos, at edge locations (servers near users) worldwide. This cuts delay and buffering.

✖ **An application gateway.**
A regional web load balancer and WAF (web app firewall). It does not keep copies of content around the world.

✖ **An Azure ExpressRoute circuit.**
A private connection from your office to Azure. It does not help internet viewers.

✖ **An Azure Traffic Manager profile.**
DNS-based routing to the best endpoint. It does not store or deliver content.

EXAM STRATEGY
'Users worldwide' + 'large video files' = keep copies near users: CDN. Traffic Manager only routes by DNS, Application Gateway is regional, ExpressRoute is a private link.

IMPACT
CDN copies near users stop buffering for viewers around the world. They also lower egress (data leaving Azure) costs from the main server. This is a common exam pattern.

What · Where · When · Why · How

What	A content delivery network keeps copies of content at edge locations close to users.
Where	Edge locations worldwide, through Azure Front Door or Azure CDN, in front of the main server.
When	When users around the world download large static files such as videos.
Why	Users get video from a nearby edge, so there is less delay and buffering.
How	Create a CDN or Front Door endpoint pointing to the main server. Edges keep copies of content.

Exam tip
Worldwide users + large static files (video, images) = CDN. Global routing without copies = Traffic Manager or Front Door.

Memory trick
CDN: Copies Delivered Nearby.

Real-world example, use case and scenario

EXAMPLE
Edge caching
A user in Sydney opens an MP4 file once, and the Sydney edge keeps a copy. The next viewer in Australia gets it from there.

USE CASE
Training videos
A global cooking-course website sends lesson videos through Azure Front Door caching to students on five continents.

SCENARIO
Buffering complaints
An independent film site in West Europe had buffering complaints from Asia. Adding a CDN in front of blob storage made videos start much faster, because they now came from nearby edges.

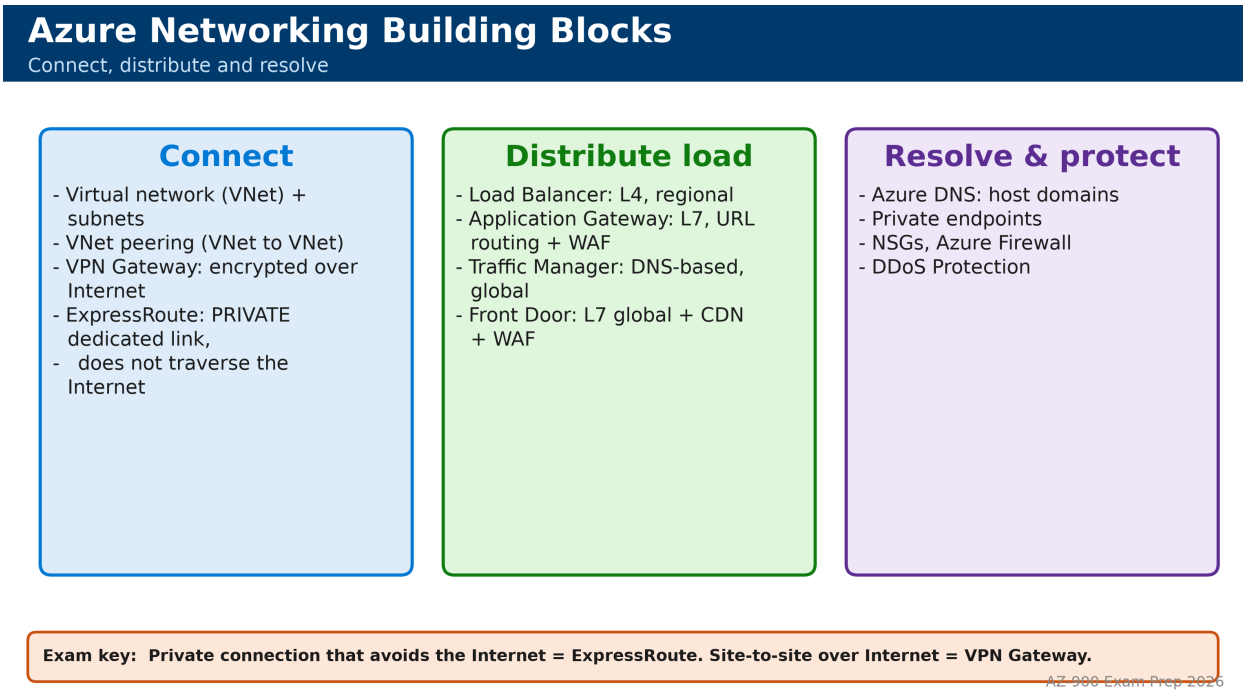


Figure Q44a: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

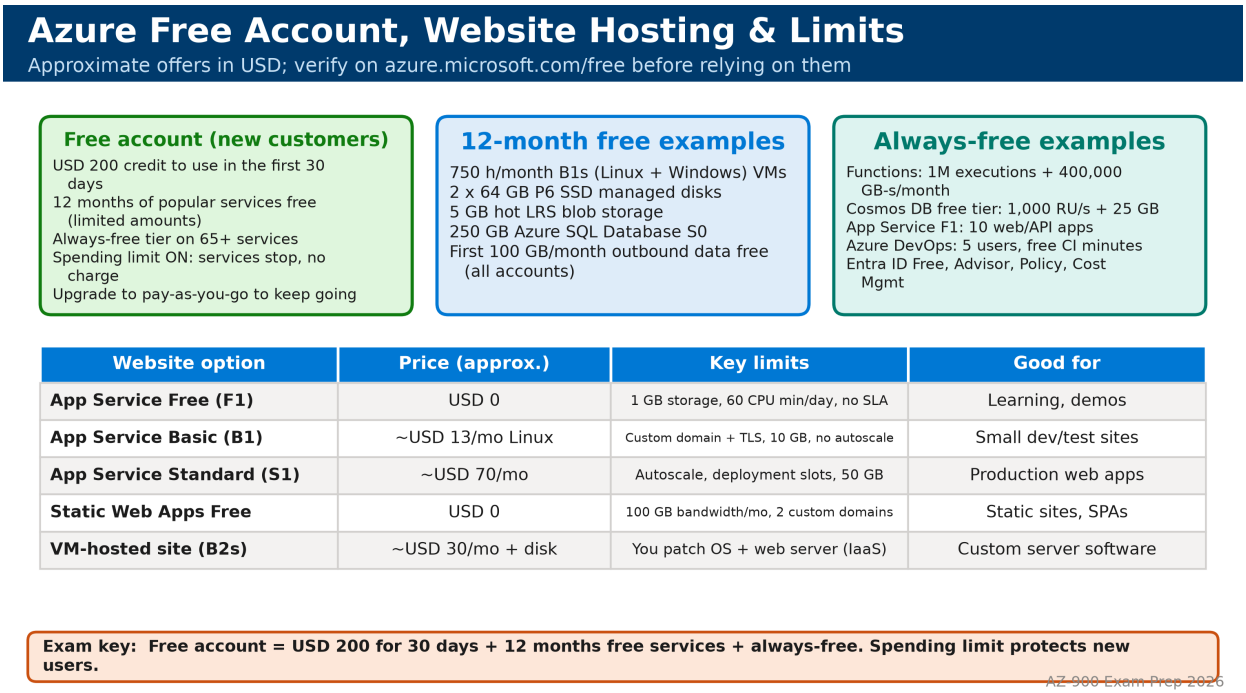


Figure Q44b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 45

AZURE ARCHITECTURE & SERVICES

IoT
v5 Q86, Q6

Your company plans to deploy several million sensors that will upload data to Azure. You need to identify which Azure resources must be created to support the planned solution. Which two Azure resources should you identify?

✔ **A.** Azure Data Lake.

✖ B. Azure Queue storage.

✖ C. Azure File Storage.

✔ **D.** Azure IoT Hub.

✖ E. Azure Notification Hubs.

Correct answer: Azure Data Lake.; Azure IoT Hub.

Clarification (read this first): Choose TWO answers: one service to RECEIVE data from millions of sensors (IoT Hub) and one place to STORE it (Data Lake). Notification Hubs is a trap: it sends messages OUT to phones, it does not receive sensor data.

Answer explained

✔ **Azure Data Lake.; Azure IoT Hub.**
IoT Hub receives data from millions of devices, each with its own identity. Data Lake Storage stores the huge amount of data at low cost.

✖ **Azure Queue storage.**
A simple message queue. It has no identity per device and is not built for IoT scale.

✖ **Azure File Storage.**
SMB file shares are not built to receive streams of sensor data or store analytics data.

✖ **Azure Notification Hubs.**
It sends push notifications out to mobile apps. It does not receive sensor data.

EXAM STRATEGY
'Millions of sensors' means receiving device data at scale (IoT Hub) plus a big place to store it (Data Lake). Notification Hubs sends to phones, a common trap.

IMPACT
IoT Hub gives each device an identity and secure two-way messaging. A simple queue loses device management and opens a security gap at scale.

What · Where · When · Why · How

What	IoT Hub receives device data. Data Lake Storage stores it at very large scale.
Where	IoT Hub is the cloud entry point for devices. Data Lake is a storage account target.
When	When millions of sensors must send data to Azure for storage and analysis.
Why	IoT Hub scales to millions of signed-in devices, and Data Lake stores the volume cheaply.
How	Register devices in IoT Hub and use message routing to write data to Data Lake.

Exam tip
Sensors upload = IoT Hub (in). Push to phones = Notification Hubs (out). Store huge amounts = Data Lake.

Memory trick
Hub to catch it, Lake to keep it.

Real-world example, use case and scenario

EXAMPLE
Routing to the lake
IoT Hub sends all temperature messages to a Data Lake Storage Gen2 container, split into folders by date.

USE CASE
Smart agriculture
A farming tech company sends soil readings from millions of field sensors into IoT Hub. It analyzes years of history in the lake.

SCENARIO
Wrong hub
A smart-meter startup first tried Notification Hubs for meter readings and found it only sends outward. Switching to IoT Hub for receiving and Data Lake for storage handled all the meters.

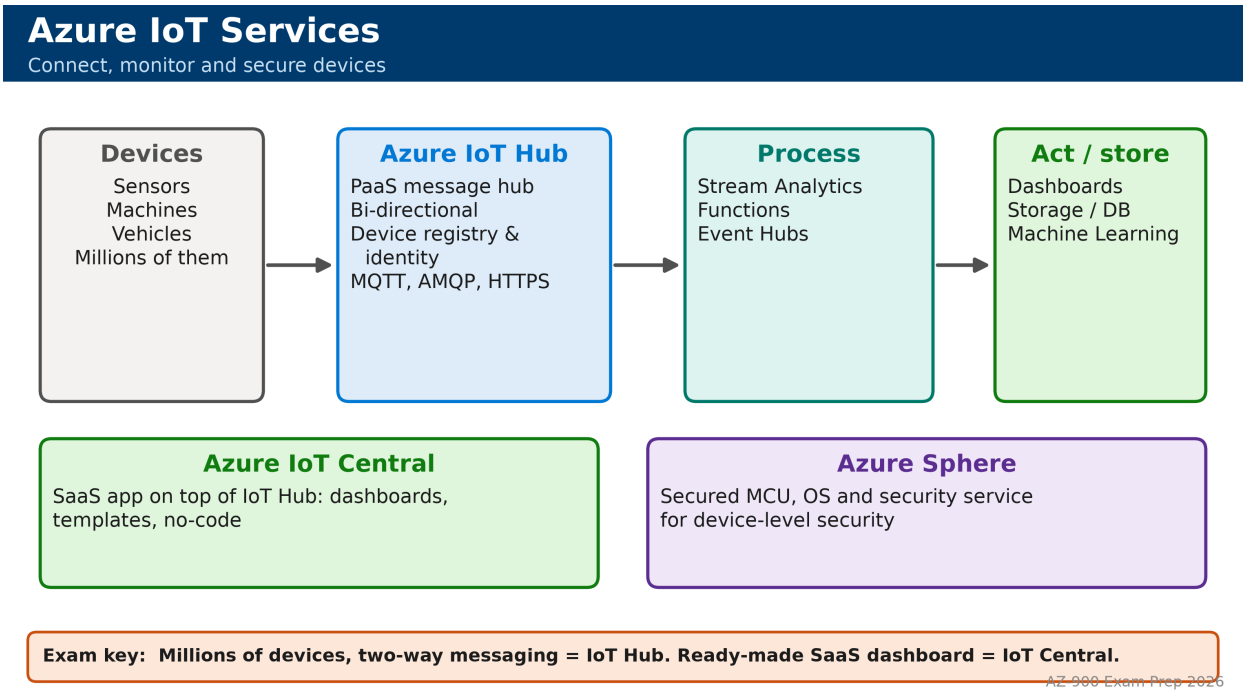


Figure Q45: Azure IoT services

About this figure

- Flow left to right: devices send telemetry to IoT Hub, data is processed, then stored or acted on. Bottom boxes are the SaaS and device-security options.
- IoT Hub has a free tier of 8,000 messages/day; Standard S1 costs approx. USD 25/month per unit for 400,000 messages/day.
- IoT Hub supports cloud-to-device commands and device twins; Event Hubs only ingests events one way, at very high throughput, without per-device identity.
- Azure Sphere combines a certified microcontroller, a Linux-based OS and a cloud security service that delivers updates and certificate-based authentication.
- Use IoT Central for quick, managed dashboards with device templates; use IoT Hub when you build a custom solution and need full control.

Question 46

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q87

You have an Azure web app. You need to manage the settings of the web app from an iPhone. What are two Azure management tools that you can use?

- ✖ A. Azure CLI.
- ✔ **B.** The Azure portal.
- ✔ **C.** Azure Cloud Shell.
- ✖ D. Windows PowerShell.
- ✖ E. Azure Storage Explorer.

Correct answer: **The Azure portal.; Azure Cloud Shell.**

Clarification (read this first): Choose TWO answers. On an iPhone you cannot install desktop tools. Only tools that run in a web browser work: the Azure portal and Cloud Shell. Look-alikes: Q37 (CLI installed on a Windows PC -> Command Prompt + PowerShell) and Q159 (Android tablet).

Answer explained

✔ **The Azure portal.; Azure Cloud Shell.**
The Azure portal and Cloud Shell both run in a phone's web browser (and in the Azure mobile app). So they work from an iPhone.

✖ **Azure CLI.**
The installed CLI is not available on iOS. On a phone it runs only inside Cloud Shell.

✖ **Windows PowerShell.**
Windows PowerShell works only on Windows. It cannot be installed on iOS.

✖ **Azure Storage Explorer.**
A desktop app for Windows, macOS and Linux that manages storage, not web app settings.

EXAM STRATEGY
'iPhone' means browser or mobile app only. The portal and Cloud Shell run in Safari. The CLI, Windows PowerShell and Storage Explorer need a desktop install.

IMPACT
Changing web app settings from a phone helps you fix problems when you are away from a laptop. Device-to-tool matching appears in several questions.

What · Where · When · Why · How

What	The Azure portal and Cloud Shell are Azure management tools that run in a browser.
Where	portal.azure.com or shell.azure.com in Safari, or in the Azure mobile app.
When	When managing Azure from a phone without desktop tools.
Why	Both need only a browser, so they work on an iPhone.
How	Open the web app's Configuration page in the portal, or run az webapp commands in Cloud Shell.

Exam tip
Phone or tablet = portal, Cloud Shell, Azure mobile app. Installed tools (CLI, PowerShell, Storage Explorer) = desktop only.

Memory trick
The iPhone has Safari, Safari has the portal, the portal has Cloud Shell.

Real-world example, use case and scenario

EXAMPLE
App settings in Safari
An admin opens portal.azure.com in Safari and edits a web app's Configuration > Application settings.

USE CASE
Weekend tweak
A restaurant chain's developer switches a feature on in App Service settings from her iPhone on a Saturday.

SCENARIO
Quick restart
A web app froze during a concert ticket sale while the engineer was away from his desk. He used Cloud Shell on his iPhone to run az webapp restart, and the site came back.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

AZ-900 Exam Prep 2026

Figure Q46a: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

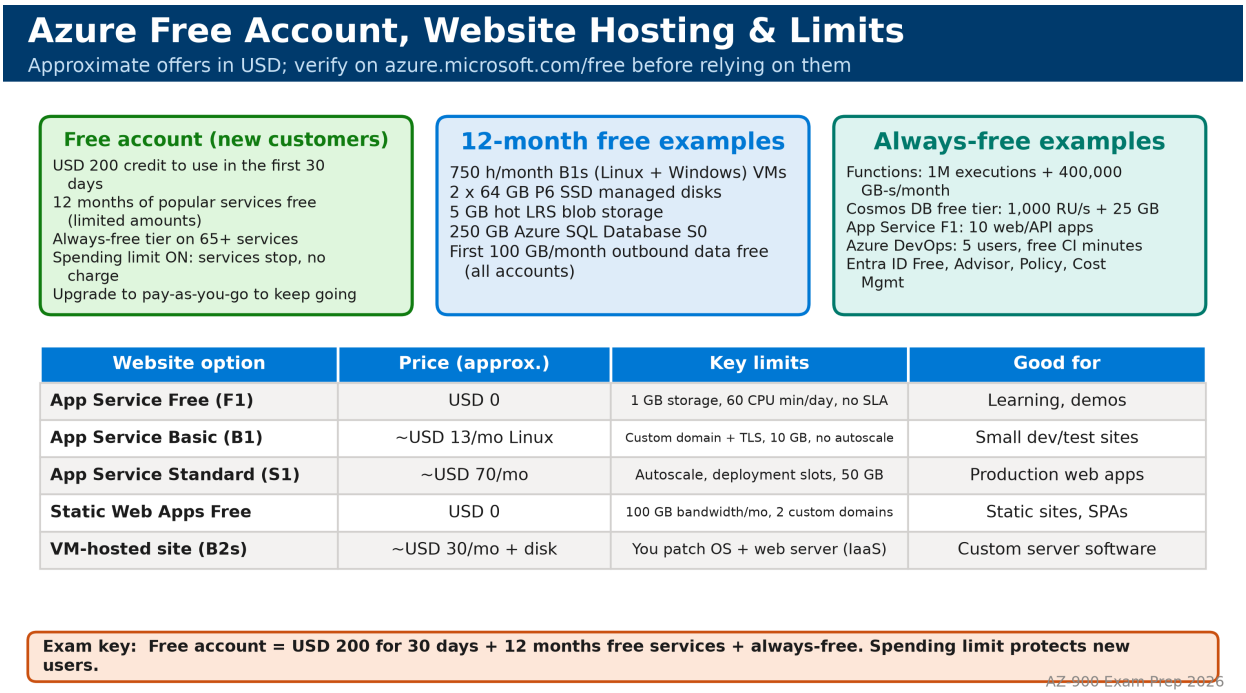


Figure Q46b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 47

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q88, Q90

You have an Azure subscription named Subscription1. You sign in to the Azure portal and create a resource group named RG1. From Azure documentation, you have the following command that creates a virtual machine named VM1. `az vm create --resource-group RG1 --name VM1 --image UbuntuLTS --generate-ssh-keys`. You need to create VM1 in Subscription1 by using the command. Solution: From the Azure portal, launch Azure Cloud Shell and select PowerShell. Run the command in Cloud Shell. Does this meet the goal?

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The question asks if an az (Azure CLI) command runs when Cloud Shell is in PowerShell mode, not Bash. Yes: Cloud Shell has the Azure CLI installed in both modes.

Answer corrected in this edition. The previous PDF printed: *No.*. The PDF printed No. Azure Cloud Shell's PowerShell mode also has the Azure CLI installed and signed in, so az vm create works there. The checked answer key gives Yes.

2026 update: The UbuntuLTS image name in the command is outdated in current Azure CLI versions. Today you would use a name such as Ubuntu2204 or Ubuntu2404. This does not change what is being tested.

Answer explained

✔ **Yes.**
Cloud Shell's PowerShell mode also has the Azure CLI installed and signed in. So az vm create runs there and creates VM1.

✖ **No.**
The Azure CLI is installed in Cloud Shell's PowerShell session too. So the az command runs successfully.

EXAM STRATEGY
One fact decides it: Cloud Shell's PowerShell session also has az installed. Ignore the old UbuntuLTS name; the exam tests the shell, not the image name.

IMPACT
Knowing az works in either Cloud Shell mode saves switching back and forth. This four-part Yes/No series costs several marks if misunderstood.

What · Where · When · Why · How

What	Cloud Shell PowerShell is a PowerShell session in the browser with Azure tools already installed.
Where	Opened from the Azure portal toolbar by choosing PowerShell.
When	When running Azure CLI commands from the docs in the PowerShell mode.
Why	Both Cloud Shell modes include the Azure CLI, so az vm create runs. Answer Yes.
How	Open Cloud Shell, choose PowerShell, paste the az command and run it.

Exam tip
Cloud Shell PowerShell and Bash both have az and the Az PowerShell commands. Current Microsoft behavior is Yes.

Memory trick
Cloud Shell speaks both languages in both rooms.

Real-world example, use case and scenario

EXAMPLE
Az in PowerShell
In Cloud Shell PowerShell, typing az account show returns the current subscription, just as in Bash.

USE CASE
Mixed scripting
An operations team at a shipping company writes PowerShell scripts that call az commands they know from the CLI docs.

SCENARIO
Doc command, PowerShell shell
An admin copied az vm create from the docs, but Cloud Shell opened in PowerShell. He ran it without switching, and VM1 was created in RG1.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

AZ-900 Exam Prep 2026

Figure Q47: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 48

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q91, Q7, Q169

Your company has several business units. Each business unit requires 20 different Azure resources for daily operation. All the business units require the same type of Azure resources. You need to recommend a solution to automate the creation of the Azure resources. What should you include in the recommendations?

✔ **A.** Azure Resource Manager templates.

✖ B. Virtual machine scale sets.

✖ C. The Azure API Management service.

✖ D. Management groups.

Correct answer: Azure Resource Manager templates.

Clarification (read this first): The key words are 'automate the creation' of the SAME set of resources again and again. That is infrastructure as code: ARM templates. Look-alikes: Q97 (API Management) and Q98 (management groups) for the same goal both answer No.

Answer explained

✔ **Azure Resource Manager templates.**
ARM templates (or Bicep) describe a whole set of resources as code. One template creates the same 20 resources in the same way for every business unit.

✖ **Virtual machine scale sets.**
Scale sets create and scale identical VMs only. They cannot create 20 different resource types.

✖ **The Azure API Management service.**
API Management publishes and protects APIs. It creates no infrastructure.

✖ **Management groups.**
Management groups organize subscriptions for policy and RBAC (role-based access). They create nothing.

EXAM STRATEGY
'Automate creation' of the 'same' resources many times = infrastructure as code: ARM templates (or Bicep). Scale sets copy VMs only. Management groups govern; they do not deploy.

IMPACT
Templates give identical, repeatable setups for each business unit. They remove manual differences and setup mistakes that cause outages.

What · Where · When · Why · How

What	ARM templates are JSON files that describe Azure resources as code. Bicep files turn into them.
Where	Deployed through Azure Resource Manager at resource group, subscription, management group or tenant level.
When	When the same set of resources must be deployed again and again in the same way.
Why	One template gives every business unit the same result each time, with no manual clicking.
How	Write the template once, add parameters for values that differ per unit, and deploy it for each unit with the CLI, PowerShell or a pipeline.

Exam tip
'Automate', 'same resources', 'each department/business unit' always point to ARM templates (or Bicep). Scale sets are a trap: they repeat only VMs.

Memory trick
Template = cookie cutter: one shape, stamped out for each business unit.

Real-world example, use case and scenario

EXAMPLE
One file, many units
A template defines a VNet, two subnets, a storage account, an App Service plan and a SQL database. Deploying it with unit=Finance or unit=HR creates two identical setups.

USE CASE
Franchise restaurant chain
A restaurant franchise company deploys a standard ordering system for every new region from one Bicep file stored in Git.

SCENARIO
Drift between units
An insurer's three business units built their setups by hand, and each had different firewall settings. The architect replaced this with an ARM template pipeline. New units are now ready in minutes, and audits find identical settings.

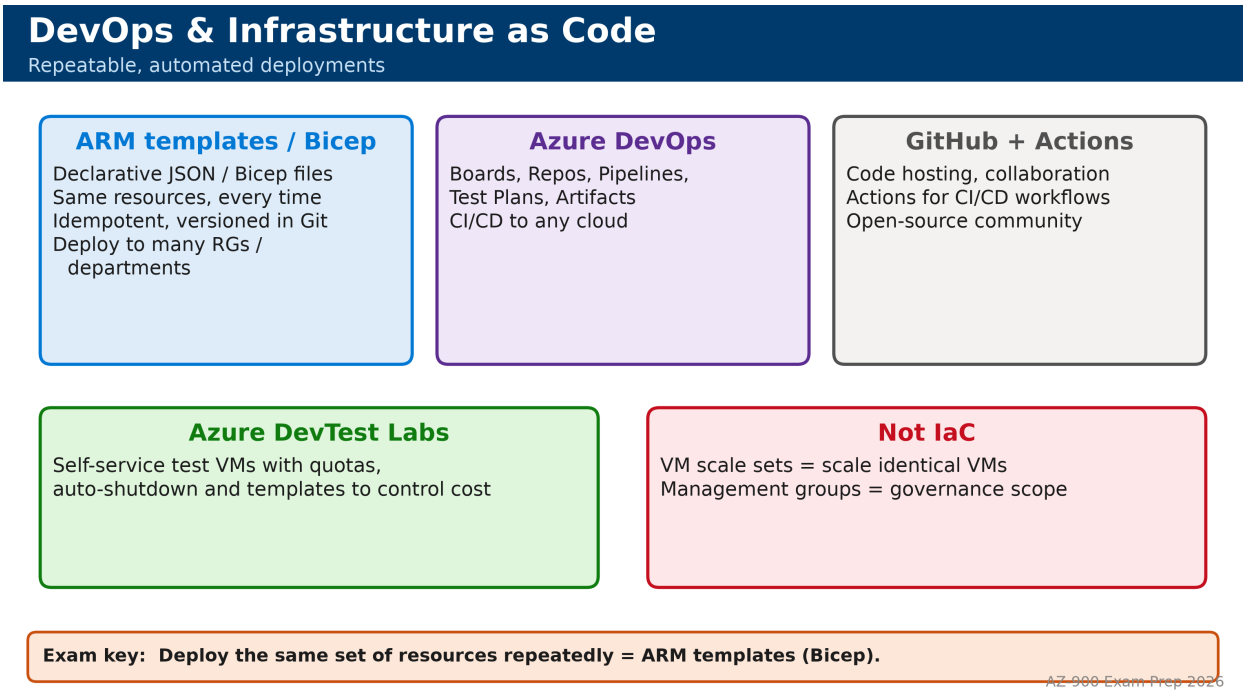


Figure Q48a: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

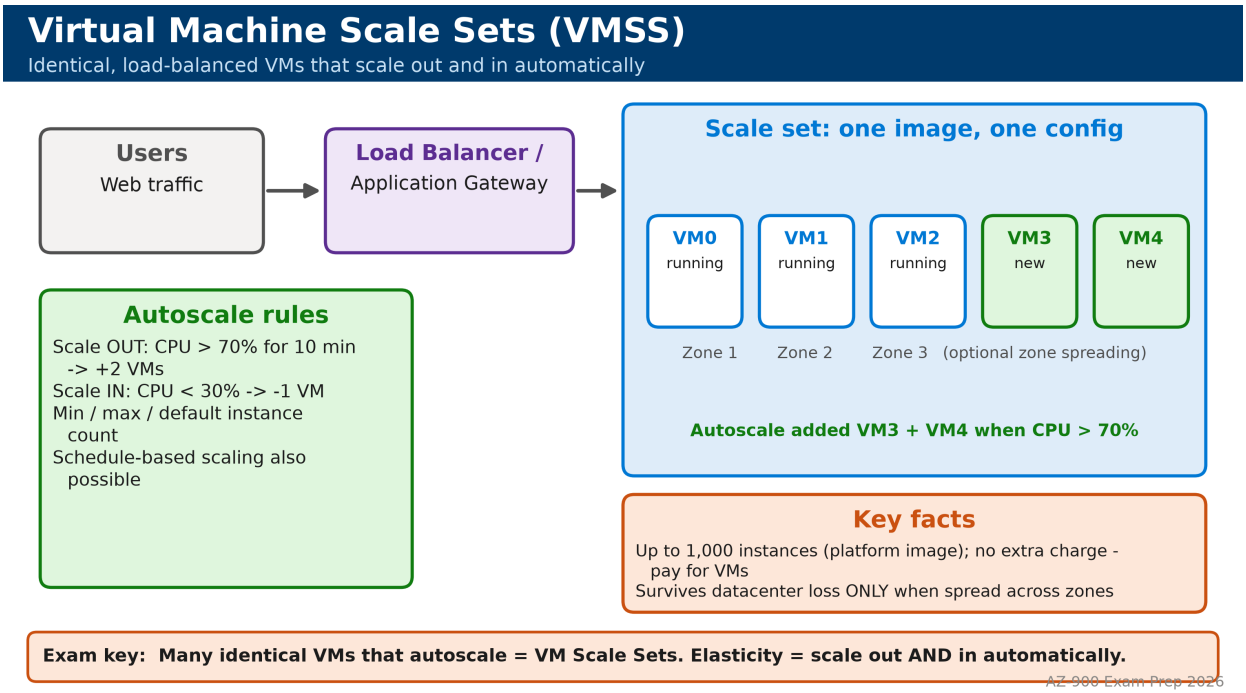


Figure Q48b: Virtual Machine Scale Sets

About this figure

- A scale set runs identical VMs from one image and configuration behind a load balancer or Application Gateway.
- Autoscale adds instances when a metric (e.g. CPU > 70%) is high and removes them when low, within min/max instance counts - this is elasticity.
- Scale sets support up to 1,000 VM instances (platform images); the scale set itself is free - you pay only for the VMs, disks and networking.
- Orchestration modes: Uniform (identical VMs) and Flexible (mixed sizes, recommended for new deployments).
- A scale set survives a datacenter failure only when its instances are spread across availability zones; a non-zonal set does not.
- Scaling out adds more VMs (horizontal); scaling up means a bigger VM size (vertical) and usually needs a restart.

Question 49

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q94, Q364

Which Azure service provides a set of version control tools to manage code?

✔ **A.** Azure Repos.

✖ B. Azure DevTest Labs.

✖ C. Azure Storage.

✖ D. Azure Cosmos DB.

Correct answer: Azure Repos.

Answer explained

✔ **Azure Repos.**
Azure Repos is the Azure DevOps service that gives Git (and TFVC) repositories for version control (tracking every change to code).

✖ **Azure DevTest Labs.**
DevTest Labs creates cost-controlled dev/test VMs. It keeps no code history.

✖ **Azure Storage.**
Storage holds blobs and files. It has no commit, branch or merge features.

✖ **Azure Cosmos DB.**
Cosmos DB is a NoSQL database, not a source control system.

EXAM STRATEGY
'Version control' points straight to Repos in Azure DevOps. DevTest Labs manages test VMs. Storage and Cosmos DB store data, not code history.

IMPACT
Good source control lets you roll back after a bad release and see who changed what. It is the base of any CI/CD (automatic build and release) pipeline.

What · Where · When · Why · How

What	Azure Repos is the Azure DevOps service that hosts Git repositories and TFVC for version control.
Where	Inside an Azure DevOps organization and project, under Repos.
When	When a team needs code history, branches and pull requests (reviewed merges).
Why	It is the only listed service built to manage versions of code.
How	Push commits to a Git repo, work in branches, and merge through pull requests with branch rules.

Exam tip
Azure DevOps services: Boards (work tracking), Repos (version control), Pipelines (CI/CD), Test Plans, Artifacts (packages). 'Version control' = Repos.

Memory trick
Repos = repository = where code versions live.

Real-world example, use case and scenario

EXAMPLE
Pull request flow
A developer pushes a feature branch to Azure Repos and opens a pull request. Two reviewers must approve it before it merges to main.

USE CASE
Bank code governance
A bank keeps its payment services code in Azure Repos. Branch rules block merges unless the build passes.

SCENARIO
Shared drive chaos
A game studio kept scripts on a file share, and people kept overwriting each other's work. They moved to Azure Repos Git. Now every change has history, an author and an easy rollback.

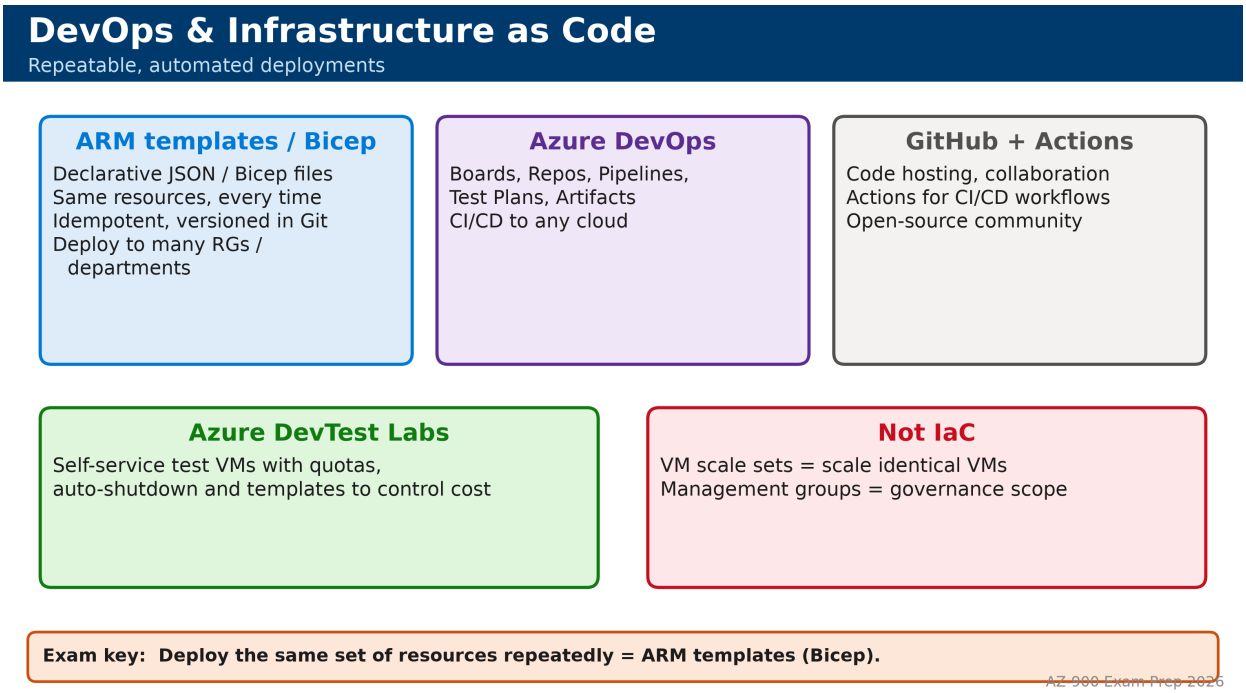


Figure Q49: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 50

AZURE ARCHITECTURE & SERVICES

Security
v5 Q95, Q178

Your company plans to automate the deployment of servers to Azure. Your manager is concerned that you may expose administrative credentials during the deployment. You need to recommend an Azure solution that encrypts the administrative credentials during the deployment. What should you include in the recommendation?

✔ **A.** Azure Key Vault.

✖ B. Microsoft Purview Information Protection.

✖ C. Microsoft Defender for Cloud.

✖ D. Azure Multi-Factor Authentication (MFA).

Correct answer: Azure Key Vault.

Clarification (read this first): The trap word is 'encrypts'. Purview Information Protection encrypts documents and emails, not deployment passwords. Passwords and secrets used by automation belong in Key Vault. Look-alikes: Q105 (Purview) and Q106 (MFA) for the same goal both answer No.

Answer explained

✔ **Azure Key Vault.**
Key Vault stores secrets, such as admin passwords, in encrypted form. Templates read them during deployment, so passwords never appear in code or logs.

✖ **Microsoft Purview Information Protection.**
Purview Information Protection labels and encrypts documents and emails, not deployment passwords.

✖ **Microsoft Defender for Cloud.**
Defender for Cloud checks security settings and finds threats. It does not store secrets.

✖ **Azure Multi-Factor Authentication (MFA).**
MFA makes interactive sign-in stronger. It does not protect passwords given to an automated deployment.

EXAM STRATEGY
'Credentials' or 'secrets' during deployment = Key Vault. Purview labels documents, Defender for Cloud checks security, MFA protects sign-in, not stored secrets.

IMPACT
Passwords written into scripts are a top cause of cloud breaches. Key Vault references keep them out of code and pipelines.

What · Where · When · Why · How

What	Azure Key Vault is a managed service that stores secrets, keys and certificates encrypted.
Where	A Key Vault resource in a region, protected with Azure RBAC or access policies.
When	When automation needs passwords or keys without showing them in scripts or templates.
Why	Passwords stay encrypted in the vault and never appear in template files, pipelines or logs.
How	Save the admin password as a secret and point to it from the ARM/Bicep parameter file during deployment.

Exam tip
Secrets, passwords, connection strings, certificates, encryption keys = Key Vault. Do not let the word 'encrypts' pull you to Information Protection.

Memory trick
Keys and secrets go in the Vault, never in the script.

Real-world example, use case and scenario

EXAMPLE
Template reference
A Bicep parameter file uses a Key Vault reference for adminPassword. The deployment reads the secret while running without showing it.

USE CASE
Retail CI/CD
A retailer's pipelines get SQL admin passwords from Key Vault through a managed identity when they build store inventory servers.

SCENARIO
Leaked in Git
A software company found an admin password inside a script in a public code repository. They changed it, stored the new one in Key Vault, and made the deployment read the secret from there. The leak path was closed.

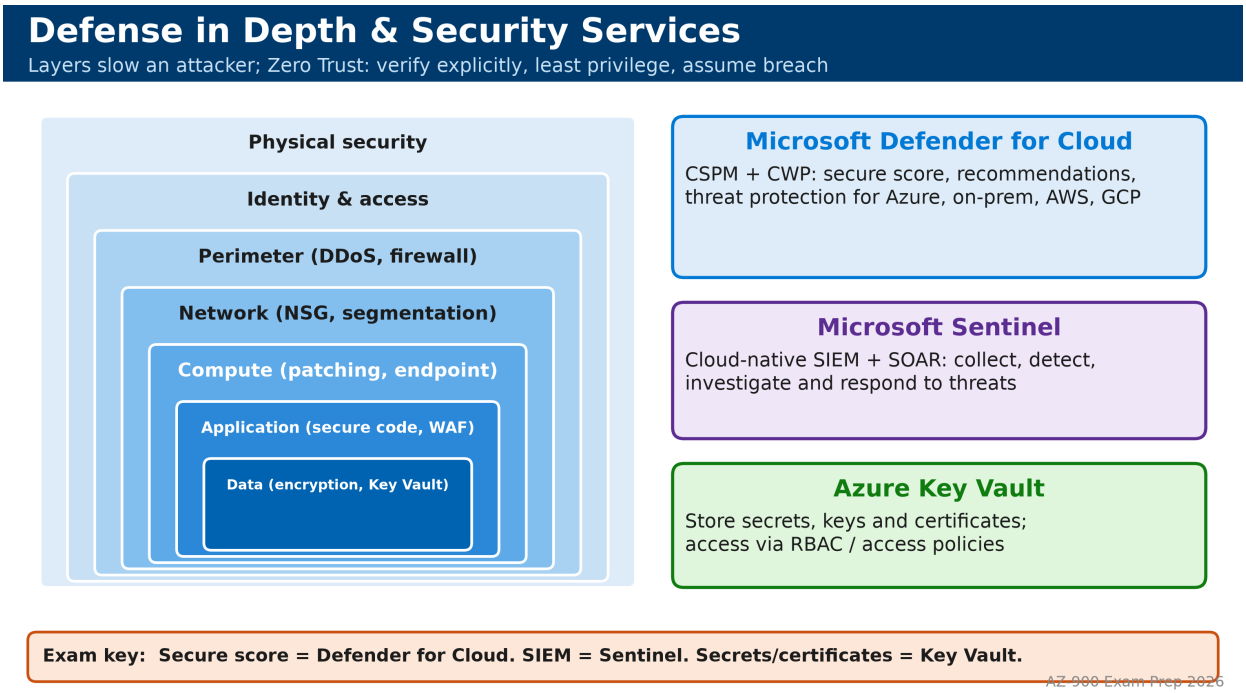


Figure Q50: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 51

AZURE ARCHITECTURE & SERVICES | **Networking** v5 Q99, Q11

Your Azure environment contains multiple Azure virtual machines. You need to ensure that a virtual machine named VM1 is accessible from the Internet over HTTP. Solution: You modify an Azure Traffic Manager profile. Does this meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The question asks if changing Traffic Manager makes VM1 reachable over HTTP. No: Traffic Manager only answers DNS (name lookups) and never opens ports. Look-alikes: Q79 (DDoS plan) -> No; Q170 says the working fixes are changing the NSG or Azure Firewall.

Answer explained

✔ **No.**
Traffic Manager is DNS-based routing (it only answers name lookups). It never opens or filters ports, so a blocked port stays blocked.

✖ **Yes.**
Changing a Traffic Manager profile changes only DNS answers. HTTP access needs an NSG or Azure Firewall rule.

EXAM STRATEGY
Traffic Manager is DNS routing and never opens ports. If the solution cannot filter traffic or open ports, answer No.

IMPACT
If you expect DNS routing to give access, the site stays unreachable. Telling routing apart from filtering is a common AZ-900 trap.

What · Where · When · Why · How

What	Azure Traffic Manager is a DNS-based load balancer across endpoints and regions.
Where	A global Traffic Manager profile with a trafficmanager.net DNS name.
When	When sending users to the best or healthy endpoint by priority, weight, speed or location.
Why	It only answers DNS lookups and cannot open ports, so the answer is No.
How	It returns an endpoint's DNS name based on the routing method. Clients then connect straight to that endpoint.

Exam tip
Traffic Manager = DNS load balancing across regions. It opens nothing; it only points clients somewhere.

Memory trick
Traffic Manager gives directions, not keys.

Real-world example, use case and scenario

EXAMPLE
Priority routing
A profile normally returns the East US endpoint. If East US fails its health checks, DNS answers switch to West Europe.

USE CASE
Global e-learning
An online learning company uses performance routing, so students reach the closest regional copy.

SCENARIO
Still unreachable
An engineer at a ticket company added VM1 to Traffic Manager to fix HTTP access. DNS now worked, but the browser still timed out because the NSG blocked port 80. Only the NSG change fixed it.

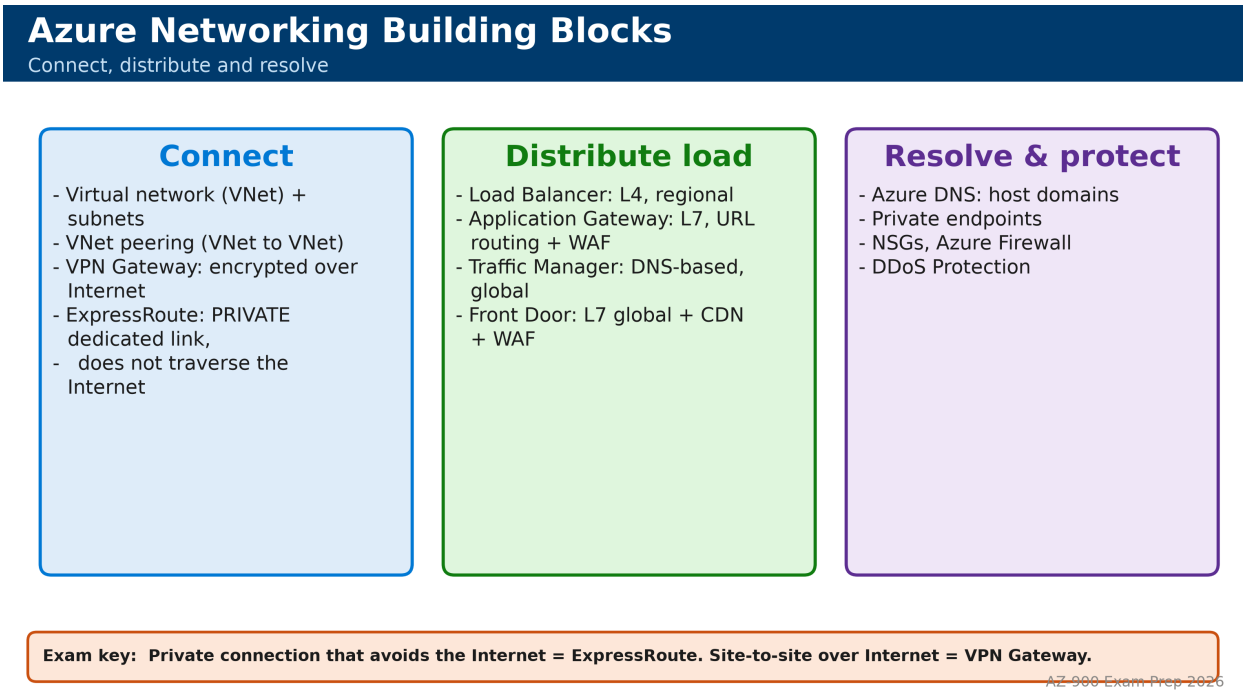


Figure Q51: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 52

AZURE ARCHITECTURE & SERVICES | **Networking** v5 Q100, Q41, Q184

Your company plans to deploy several web servers and several database servers to Azure. You need to recommend an Azure solution to limit the types of connections from the web servers to the database servers. What should you include in the recommendation?

✔ **A.** Network Security Groups (NSGs).

✖ B. Azure Service Bus.

✖ C. A local network gateway.

✖ D. A route filter.

Correct answer: Network Security Groups (NSGs).

Clarification (read this first): 'Limit the types of connections' means allow only certain ports and protocols between the two server groups. That is filtering, done by NSGs. Look-alikes: Q109 (Service Bus) and Q110 (local network gateway) for the same goal both answer No.

Answer explained

✔ **Network Security Groups (NSGs).**
An NSG (network security group) on the database subnet can allow only the needed port, for example TCP 1433, from the web subnet. It blocks other kinds of connections.

✖ **Azure Service Bus.**
Service Bus passes messages between apps. It does not filter network connections.

✖ **A local network gateway.**
A local network gateway describes the on-premises VPN device and its addresses. It does not filter traffic between Azure servers.

✖ **A route filter.**
Route filters choose which ExpressRoute Microsoft routes are shared. They do not limit connection types.

EXAM STRATEGY
'Limit the types of connections' between server groups = port and protocol filtering between subnets: NSGs. Service Bus is messaging. Local gateways and route filters handle connections, not filtering.

IMPACT
Allowing only the SQL port from web to database contains a hacked web server. Auditors check this basic layered defense.

What · Where · When · Why · How

What	NSGs filter traffic between subnets in a virtual network, and from the internet.
Where	Applied to the database subnet or to the database servers' network cards.
When	When web and database servers must talk only on certain ports.
Why	They limit which kinds of connections the web servers can open to the database servers.
How	Allow SQL port 1433 from the web subnet to the database subnet, and block other incoming traffic.

Exam tip
'Limit the types of connections between servers' = NSG (optionally with application security groups). Route filters and gateways are about routing, not filtering.

Memory trick
Web talks to DB only on 1433: the NSG keeps the conversation short.

Real-world example, use case and scenario

EXAMPLE
Tiered rules
NSG on the DB subnet: allow 1433 from WebSubnet, block all other incoming traffic from the virtual network.

USE CASE
E-commerce PCI zone
An online shop isolates its card data database subnet so only the payment API servers can connect. This supports PCI DSS (card security standard) separation.

SCENARIO
Compromised web server
A ticket reseller's web VM was hacked. The attacker could not scan or RDP to the database servers, because the NSG allowed only SQL from the web servers. The breach stayed contained.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

AZ-900 Exam Prep 2026

Figure Q52: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 53

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q101

To what should an application connect to retrieve security tokens?

✖ A. An Azure Storage account.

✔ **B.** Microsoft Entra ID.

✖ C. A certificate store.

✖ D. An Azure key vault.

Correct answer: Microsoft Entra ID.

Clarification (read this first): A security token is a digital pass that proves who an app or user is. It is issued by the identity provider, Microsoft Entra ID. Key Vault is a trap: it STORES secrets and certificates but does not issue tokens.

Answer explained

✔ **Microsoft Entra ID.**
Microsoft Entra ID is the identity provider and token service. Apps sign in with OAuth 2.0/OpenID Connect (standard sign-in protocols) and get access and ID tokens.

✖ **An Azure Storage account.**
Storage accepts tokens or SAS (shared access signatures) for access. It does not issue identity tokens.

✖ **A certificate store.**
A certificate store keeps certificates on a computer. It issues no OAuth tokens.

✖ **An Azure key vault.**
Key Vault stores secrets and certificates. It needs an Entra token itself and does not issue security tokens.

EXAM STRATEGY
'Security tokens' means an identity provider issuing OAuth/OpenID tokens: Microsoft Entra ID. Key Vault stores secrets and certificates but does not issue access tokens.

IMPACT
Using Entra ID tokens instead of stored passwords enables modern sign-in, Conditional Access and MFA. It removes the risk of leaked passwords in apps.

What · Where · When · Why · How

What	Microsoft Entra ID is Microsoft's cloud identity provider and security token service.
Where	The tenant's OAuth 2.0 / OpenID Connect endpoints at login.microsoftonline.com.
When	When an app must prove who a user or the app is, and call protected APIs.
Why	It issues the access and ID tokens that apps show to APIs.
How	The app asks Entra ID for a token, then sends it as a bearer token (a pass in the request) to the API.

Exam tip
Tokens come from the identity provider. Key Vault stores secrets; it does not hand out access tokens.

Memory trick
Entra ID is the ticket booth at the entrance.

Real-world example, use case and scenario

EXAMPLE
Managed identity
A web app with a managed identity (an identity Azure manages for it) asks Entra ID for a token. It uses the token to read secrets from Key Vault.

USE CASE
Mobile banking app
A bank's mobile app signs users in through Entra ID. It calls the bank's APIs with the access token it gets.

SCENARIO
Hard-coded keys removed
A delivery company's integration service used a stored API key for every call. They changed it to get short-lived tokens from Entra ID. Long-lived secrets are no longer in config files.

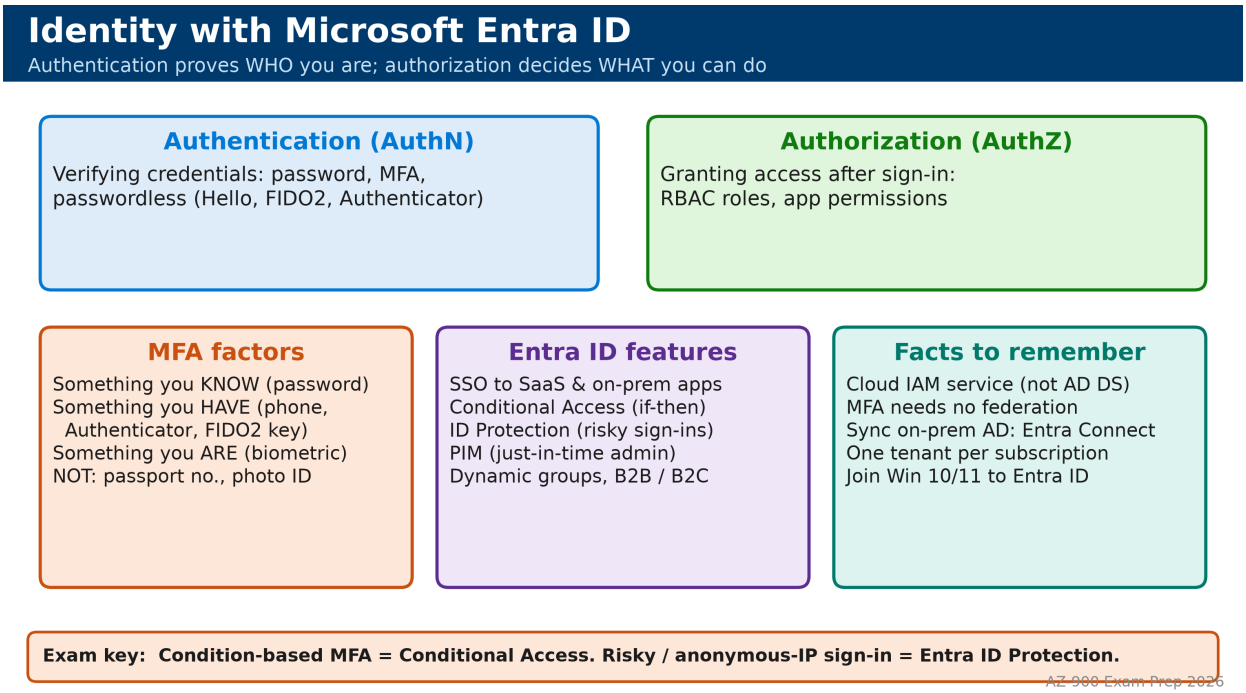


Figure Q53: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 54

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q102, Q78

Which service provides network traffic filtering across multiple Azure subscriptions and virtual networks?

✔ **A.** Azure Firewall.

✖ B. An application security group.

✖ C. Azure DDoS protection.

✖ D. A Network Security Group (NSG).

Correct answer: Azure Firewall.

Clarification (read this first): Key words: 'across multiple subscriptions and virtual networks'. That means one central filter for many networks: Azure Firewall. An NSG also filters, but only for one subnet or network card.

Answer explained

✔ **Azure Firewall.**
Azure Firewall is a central stateful firewall (it tracks each connection). It usually sits in a hub VNet. Its rules filter traffic for connected VNets across many subscriptions.

✖ **An application security group.**
An ASG only groups VM network cards for use in NSG rules. It is not a central filter.

✖ **Azure DDoS protection.**
DDoS Protection blocks flood attacks on public IP addresses. You cannot write your own filter rules in it.

✖ **A Network Security Group (NSG).**
An NSG filters at one subnet or network card. It is not a central filter across subscriptions and VNets.

EXAM STRATEGY
'Across multiple subscriptions and virtual networks' = central, stateful filtering: Azure Firewall. NSGs and ASGs work on one VNet's subnets or network cards. DDoS stops floods.

IMPACT
One firewall policy across subscriptions makes audits easier. It prevents gaps caused by different rules in each VNet.

What · Where · When · Why · How

What	Azure Firewall is a managed, cloud-based stateful firewall service.
Where	A hub virtual network that spoke VNets in any subscription connect and route to.
When	When one set of filter rules must cover many VNets and subscriptions.
Why	It filters traffic in one central place, not per subnet like an NSG.
How	Send spoke traffic to the firewall with user-defined routes. Manage rules with Firewall Policy or Firewall Manager.

Exam tip
'Across multiple subscriptions and virtual networks', 'centrally' = Azure Firewall. Per subnet = NSG.

Memory trick
One Firewall at the hub guards every spoke.

Real-world example, use case and scenario

EXAMPLE
Hub-and-spoke
Ten spoke VNets in three subscriptions connect to a hub VNet. The hub's Azure Firewall filters all outgoing internet traffic.

USE CASE
Conglomerate subsidiaries
A large group gives each daughter company its own subscription. It enforces one list of allowed websites through a shared Azure Firewall.

SCENARIO
Rule sprawl
A hospital group had 60 NSGs with different outgoing rules. They added Azure Firewall in a hub and routed all spokes through it. Now they manage outgoing rules in one place.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

AZ-900 Exam Prep 2026

Figure Q54: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 55

AZURE ARCHITECTURE & SERVICES

Security v5 Q103

Which Azure service should you use to store certificates?

✖ A. Microsoft Defender for Cloud.

✖ B. An Azure Storage account.

✔ **C.** Azure Key Vault.

✖ D. Microsoft Purview Information Protection.

Correct answer: Azure Key Vault.

Answer explained

✔ **Azure Key Vault.**
Key Vault (a secure store for secrets) keeps keys, secrets and certificates. It can import, create and automatically renew certificates with partner CAs (certificate authorities).

✖ **Microsoft Defender for Cloud.**
Defender for Cloud checks your security setup and finds threats. It does not store certificates.

✖ **An Azure Storage account.**
A storage account can hold a .pfx file. But it cannot renew certificates or protect them with an HSM (hardware security module).

✖ **Microsoft Purview Information Protection.**
Purview Information Protection labels and encrypts documents and emails. It is not a certificate store.

EXAM STRATEGY
'Store certificates' means Key Vault. Key Vault holds keys, secrets and certificates. A storage account holds files but does not manage certificates. Defender and Purview do not store certificates.

IMPACT
Key Vault can renew certificates automatically. This stops websites going offline because a TLS (website security) certificate expired.

What · Where · When · Why · How

What	Azure Key Vault safely stores and manages keys, secrets and certificates.
Where	In a Key Vault resource in a region, under Objects > Certificates.
When	When you must store, renew and use TLS or signing certificates safely.
Why	It is the Azure service built to hold certificates, with renewal and access control.
How	Import or create a certificate, turn on auto-renewal if needed, and give apps access.

Exam tip
Key Vault stores K-S-C: Keys, Secrets, Certificates. If a question names any of these three, pick Key Vault.

Memory trick
Key Vault keeps the KSC trio: Keys, Secrets, Certs.

Real-world example, use case and scenario

EXAMPLE
Auto-renewal
A TLS certificate from Key Vault's DigiCert link is set to renew by itself 30 days before it expires.

USE CASE
Online gaming portal
A gaming site uses its TLS certificate from Key Vault in App Service. Web teams never touch the private key.

SCENARIO
Expired certificate outage
A courier company's tracking site went down when a certificate, managed by hand, expired. They moved it into Key Vault with auto-renew and expiry alerts. The outage has not happened again.

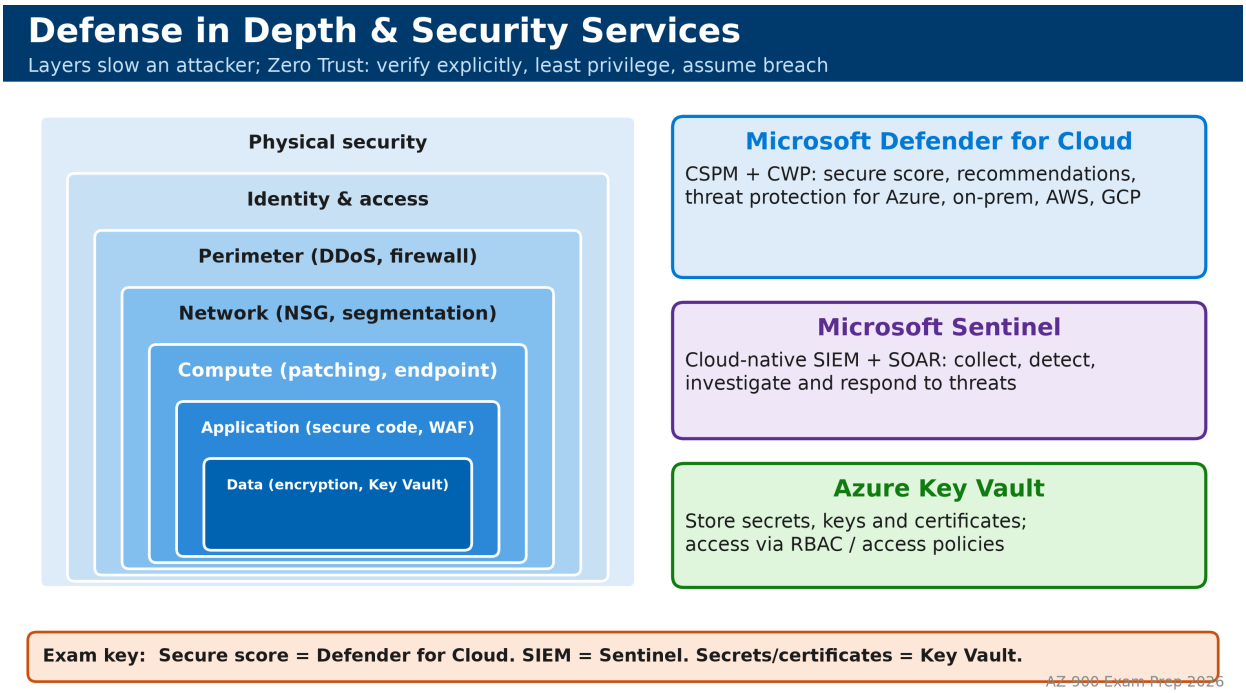


Figure Q55: Defense in depth and security services

About this figure

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- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 56

AZURE MANAGEMENT & GOVERNANCE

Security v5 Q105

What can Microsoft Purview Information Protection encrypt?

- ✖ A. Network traffic.
- ✔ **B.** Documents and email messages.
- ✖ C. An Azure Storage account.
- ✖ D. An Azure SQL database.

Correct answer: Documents and email messages.

Clarification (read this first): The question asks what this service encrypts. It works on content (files and emails), not on infrastructure like networks, storage accounts or databases. Q179 also uses Purview Information Protection, for watermarking Word files.

2026 update: Its old name was Azure Information Protection (AIP). The labelling client and service are now Microsoft Purview Information Protection.

Answer explained

✔ **Documents and email messages.**
Purview Information Protection adds sensitivity labels (tags like 'Confidential') to content. A label can encrypt documents and emails. The protection stays with the file wherever it goes.

✖ **Network traffic.**
Network traffic is encrypted by TLS, VPN Gateway or MACsec, not by Information Protection.

✖ **An Azure Storage account.**
Storage accounts are encrypted at rest (while stored) by Storage Service Encryption, not by labels.

✖ **An Azure SQL database.**
Azure SQL uses Transparent Data Encryption and Always Encrypted, not label encryption.

EXAM STRATEGY
'Encrypt documents and email' points to Purview Information Protection labels. Remove network, storage account and SQL. They use TLS or built-in platform encryption.

IMPACT
Label encryption stays with the file. So a leaked document still cannot be read outside the company. This matters for data loss and compliance.

What · Where · When · Why · How

What	Microsoft Purview Information Protection classifies and protects content with sensitivity labels.
Where	Set up in the Microsoft Purview portal. Used in Office apps, Outlook and supported files.
When	When documents and emails need protection that stays with them.
Why	Labels can encrypt documents and emails so only allowed users can open them.
How	Publish label policies. A label adds encryption through Azure Rights Management to the file or email.

Exam tip
Information Protection = content (documents, emails). Storage, SQL and network use other encryption.

Memory trick
Labels stick to the letter: docs and mail.

Real-world example, use case and scenario

EXAMPLE
Confidential label
A Word file labelled Confidential is encrypted. Only employees can open it, even if it is emailed outside the company.

USE CASE
M&A deal room
An investment firm labels merger (company buy-out) documents Highly Confidential. People cannot print or forward them.

SCENARIO
Misdirected email
An HR officer emailed a salary spreadsheet to a supplier by mistake. The file had an encrypted label, so the supplier could not open it. HR then removed access.

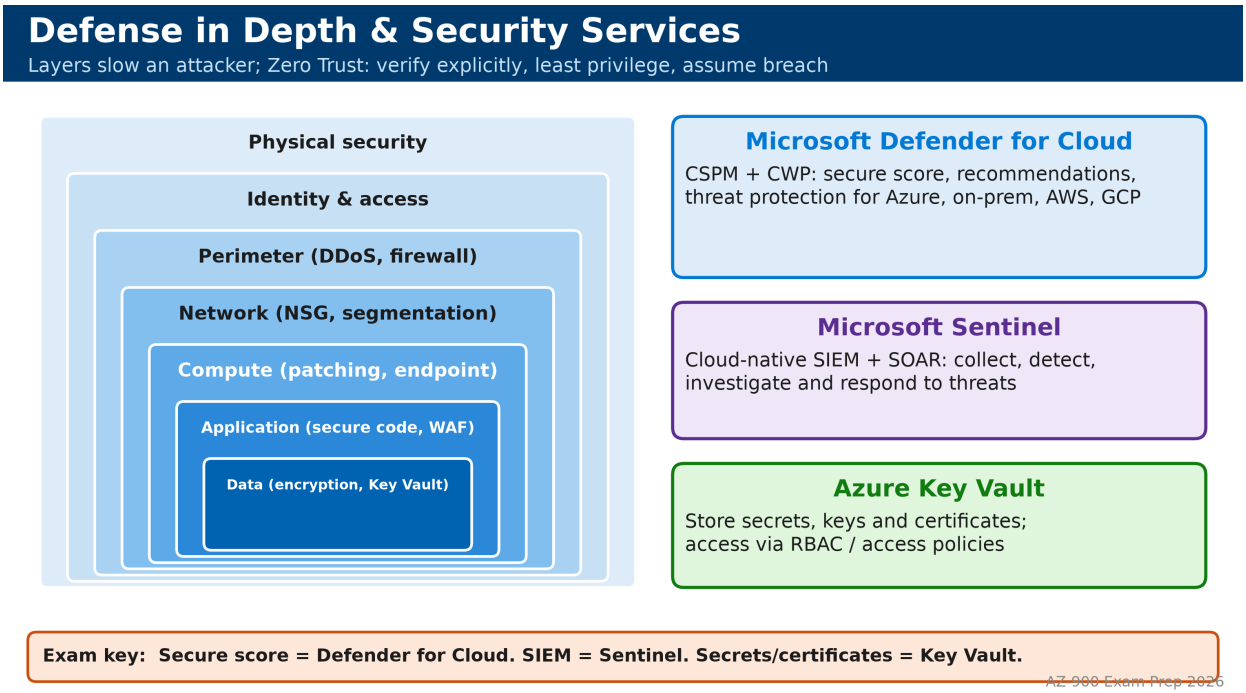


Figure Q56: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 57

AZURE MANAGEMENT & GOVERNMENT

Support

v5 Q110, Q191, Q337

You attempt to create several managed Microsoft SQL Server instances in an Azure environment and receive a message that you must increase your Azure subscription limits. What should you do to increase the limits?

✖ A. Create a service health alert.

✖ B. Upgrade your support plan.

✖ C. Modify an Azure policy.

✔ **D.** Create a new support request.

Correct answer: Create a new support request.

Clarification (read this first): The question asks how to raise a quota (a limit on how many resources you can create). Watch the trap 'Upgrade your support plan': quota requests are free on every plan. Q113 (Service Health alert) and Q114 (Azure Policy) are the same scenario - both No.

Answer explained

✔ **Create a new support request.**
You raise a subscription quota (resource limit) by creating a 'Service and subscription limits (quotas)' support request. It is free on every support plan, including Basic.

✖ **Create a service health alert.**
A Service Health alert tells you about Azure problems. It does not change any limit.

✖ **Upgrade your support plan.**
You do not need a paid plan for a quota request. Basic can open one.

✖ **Modify an Azure policy.**
Azure Policy limits what you can deploy. Microsoft sets quotas and raises them only on request.

EXAM STRATEGY
'Increase subscription limits' means a quota support request. Remove the plan upgrade (quota requests are free on every plan), policy (rules) and Service Health alerts.

IMPACT
Quota increases are free and quick. Knowing this saves money on support plans and unblocks deployments that hit vCPU (virtual CPU) or instance limits.

What · Where · When · Why · How

What	Azure quotas are limits on resources, such as vCores (virtual CPU cores), per subscription and region.
Where	Help + support > Create a support request, or the Quotas page in the portal.
When	When a deployment fails because it reached a subscription limit.
Why	Only Microsoft can raise quotas, and quota requests are free on every support plan.
How	Open a 'Service and subscription limits (quotas)' request. Name the service, region and new limit.

Exam tip
'Increase limits/quota' = new support request. It is free on Basic. Do not upgrade the plan.

Memory trick
Hit a limit? Ask support to raise it - free of charge.

Real-world example, use case and scenario

EXAMPLE
Quota request
Help + support > Create a support request > Issue type: Service and subscription limits > SQL Managed Instance > ask for more vCores in West Europe.

USE CASE
Seasonal analytics
A tax company asks for extra vCPU quota each January, before its busy filing season.

SCENARIO
Launch day blocker
A gaming studio's deployment of extra servers failed with a quota error hours before a launch. The engineer used the link in the error to file a quota request. It was approved automatically within minutes.

Azure Support Plans

Plan names appear in older dumps; know the ladder

Feature	Basic	Developer	Standard	Prof. Direct	Unified
Price	Free	Low	Mid	Higher	Enterprise
Billing & subscription	Yes	Yes	Yes	Yes	Yes
Advisor, Service Health, docs	Yes	Yes	Yes	Yes	Yes
Technical support	No	Email, bus. hrs	24x7 phone/email	24x7 phone/email	24x7
Sev A (< 1 h)	No	No	Yes	Yes	Yes
Architecture guidance	No	General	General	Reviews (ProDirect)	Customer-specific
Proactive / account mgr	No	No	No	Delivery mgr pool	Technical account mgr

Exam key: Basic = billing only (free). Developer = test/non-prod, email in business hours. Standard = cheapest 24x7 production.

Premier support has been replaced by Unified support. Support plans are no longer on the current skills outline.

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Figure Q57: Azure support plans

About this figure

- Columns climb from free Basic to enterprise Unified; read down a column to see what each plan adds, especially 24x7 support and Sev A.
- Approximate list prices: Developer ~USD 29/month, Standard ~USD 100/month, Professional Direct ~USD 1,000/month; Unified is a custom enterprise contract.
- Initial response targets: Developer under 8 business hours; Standard Sev A under 1 hour; Professional Direct Sev A also under 1 hour with escalation management.
- Premier was replaced by Unified support. Every plan includes billing and subscription help, documentation, Azure Advisor, Service Health and community forums.
- Support plans are purchased per billing scope; free accounts get Basic only, so technical issues require upgrading or using community support.

Question 58

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q111

Your company has 10 offices. You plan to generate several billing reports from the Azure portal. Each report will contain the Azure resource utilization of each office. Which Azure Resource Manager feature should you use before you generate the reports?

✔ **A.** Tags.

✖ B. Templates.

✖ C. Locks.

✖ D. Policies.

Correct answer: Tags.

Clarification (read this first): The question asks what to set up BEFORE making the reports. You need a label on each resource that says which office owns it. That label is a tag.

Answer explained

✔ **Tags.**
Tags (name/value labels) such as Office=Leeds are added to resources. Cost Management can then group and filter costs by office.

✖ **Templates.**
Templates deploy resources. They do not sort costs, though they can add tags.

✖ **Locks.**
Locks stop deletion or changes. They add no reporting labels.

✖ **Policies.**
A policy can require a tag. But the tag itself is what groups costs in reports.

EXAM STRATEGY
'Report per office' means sorting resources with labels: tags. Templates deploy, locks protect, policies enforce rules. Only tags group costs in Cost Management.

IMPACT
Tagging every resource lets you charge each office or department for its own use (chargeback). Without tags, sharing costs is guesswork.

What · Where · When · Why · How

What	Tags are name/value labels added to resources and resource groups.
Where	Set on resources or resource groups. Used in Cost Management cost analysis.
When	When you must report costs by office, department, project or cost center.
Why	Tagged resources can be grouped and filtered by office in billing reports.
How	Add an Office tag to each resource. Then group cost analysis by that tag.

Exam tip
Cost reports by department, office, project or cost center = Tags. Tags are not inherited by default.

Memory trick
Tag it to track it.

Real-world example, use case and scenario

EXAMPLE
Cost by tag
Cost analysis grouped by the tag 'Office' shows Leeds 3,200, Bristol 2,100 and so on for the month.

USE CASE
Architecture practice
An architecture firm with 10 studios tags every resource with Studio and CostCenter. Each month it charges each studio for its cloud use.

SCENARIO
Untagged spend
A recruitment agency's first office report showed 40 percent 'untagged'. They added a policy to require the Office tag and copy it from resource groups. The next report placed nearly all costs.

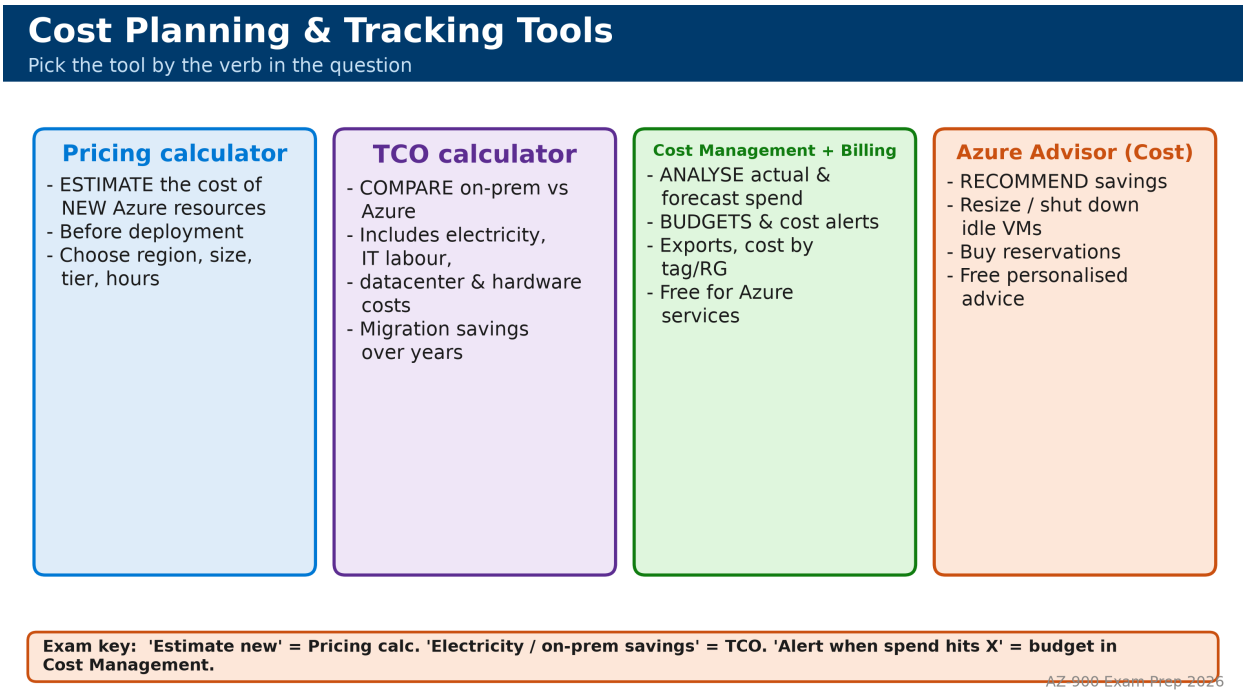


Figure Q58a: Cost planning and tracking tools

About this figure

• Each column is one tool; pick it by the question's verb: estimate (Pricing calculator), compare (TCO), analyse or alert (Cost Management), recommend (Advisor).

• Pricing and TCO calculators are free websites that need no Azure subscription; estimates use list prices and can be saved or exported to Excel.

• Cost Management is free for Azure resources; budgets can be set monthly, quarterly or yearly with alerts at thresholds such as 50%, 80% or 100%.

• Budgets alert but do not stop spending; only the free account and some credit offers have a spending limit that disables resources when exhausted.

• Advisor cost recommendations include resizing or shutting down VMs with low CPU and buying reservations or savings plans based on 30 days of usage.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q58b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 59

AZURE ARCHITECTURE & SERVICES

Governance
v5 Q112, Q194

Your company plans to migrate to Azure. The company has several departments. All the Azure resources used by each department will be managed by a department administrator. What are two possible techniques to segment Azure for the departments?

✔ **A.** Multiple subscriptions.

✖ B. Multiple Microsoft Entra ID directories.

✖ C. Multiple regions.

✔ **D.** Multiple resource groups.

Correct answer: Multiple subscriptions.; Multiple resource groups.

Clarification (read this first): This question has TWO answers. 'Segment' here means split who manages what. Watch 'regions': a region is a place, not a permission boundary. Q115 and Q116 are look-alikes: multiple directories is No, multiple resource groups is Yes.

Answer explained

✔ **Multiple subscriptions.; Multiple resource groups.**
Subscriptions and resource groups are both RBAC scopes (levels where you give permissions). So each department admin can get rights only on their own subscription or resource group.

✖ **Multiple Microsoft Entra ID directories.**
Separate Entra tenants (directories) split user identities and add a lot of admin work. They are not the recommended way to segment.

✖ **Multiple regions.**
Regions are physical places, not management or permission boundaries.

EXAM STRATEGY
'Segment Azure' for department admins means scopes where RBAC applies: subscriptions and resource groups. Regions are geography. Many Entra tenants split identity into pieces.

IMPACT
Giving each department its own subscription or resource group limits the damage of admin mistakes. It also keeps billing clear.

What · Where · When · Why · How

What	Subscriptions and resource groups are RBAC scopes that separate the management of resources.
Where	Inside one Entra tenant: subscriptions under management groups, resource groups inside subscriptions.
When	When each department admin must manage only that department's resources.
Why	Both let you give department admins rights on just their own scope.
How	Create one subscription or resource group per department. Give its admin Owner or Contributor there.

Exam tip
Segment for admins = subscriptions and/or resource groups (RBAC scopes). Regions never segment management.

Memory trick
Subs and RGs divide the house; regions only say where it stands.

Real-world example, use case and scenario

EXAMPLE
RG per department
RG-Sales and RG-Legal each get a department admin with Contributor rights only on that group.

USE CASE
University faculties
A university gives each faculty its own subscription. Each faculty's IT lead manages its resources and sees only its own bill.

SCENARIO
Too many tenants
A food distributor first made a separate Entra tenant for each department. Each employee then had four user accounts. They moved to one tenant with a subscription per department, which cut identity admin work a lot.

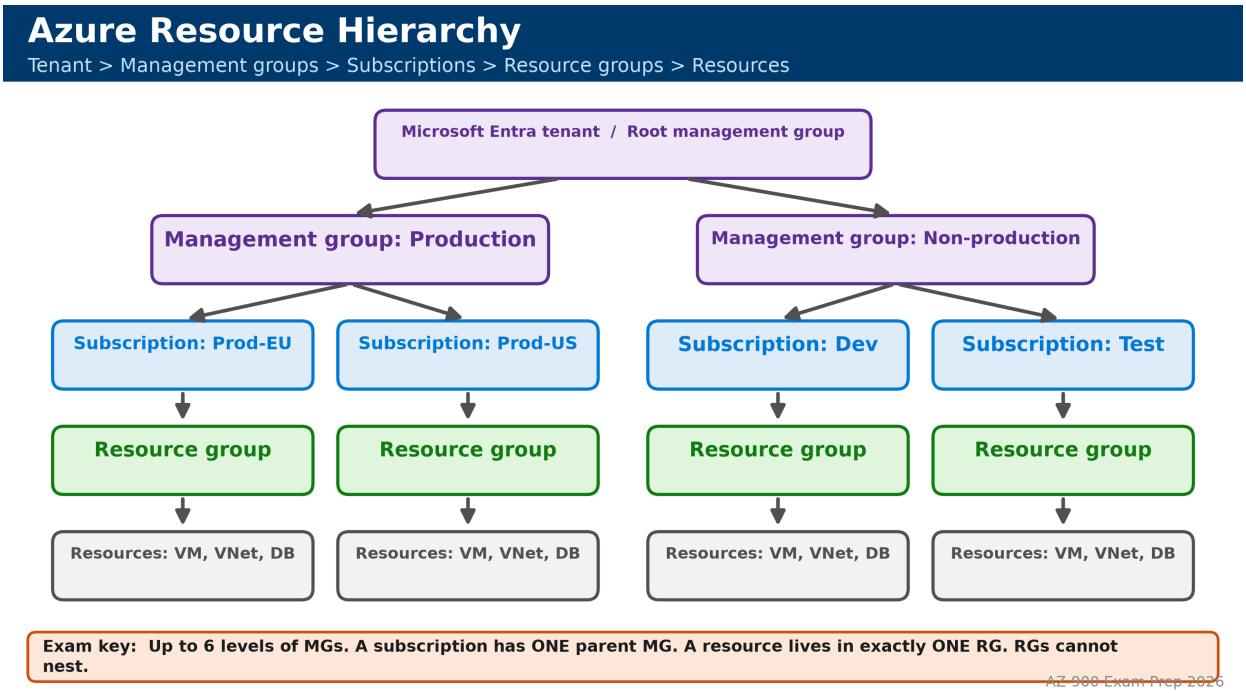


Figure Q59: Azure resource hierarchy

About this figure

• Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.

• A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.

• Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.

• Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.

• Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 60

CLOUD CONCEPTS

SLA
v5 Q113

What is guaranteed in an Azure Service Level Agreement (SLA) for virtual machines?

✔ **A.** Uptime.

✖ B. Feature availability.

✖ C. Bandwidth.

✖ D. Performance.

Correct answer: Uptime.

Clarification (read this first): The question asks what the SLA (Service Level Agreement - Microsoft's written uptime promise) guarantees. It only promises uptime. A slow VM that is still reachable does not break the SLA.

Answer explained

✔ **Uptime.**
The VM SLA guarantees a monthly uptime (connectivity) percentage, for example 99.99% across availability zones. If Microsoft misses it, you get service credits (money back on your bill).

✖ **Feature availability.**
SLAs do not promise that certain features exist or stay the same.

✖ **Bandwidth.**
Network speed depends on VM size. The SLA does not guarantee it.

✖ **Performance.**
SLAs do not guarantee CPU, disk or app performance.

EXAM STRATEGY
SLAs are always about an uptime percentage. Remove performance, bandwidth and features. Microsoft guarantees uptime and gives credits when it falls short.

IMPACT
SLA uptime sets your service credits and design targets. SLA questions, including combined (composite) SLA maths, appear often.

What · Where · When · Why · How

What	An Azure SLA is Microsoft's promise of a monthly uptime or connectivity percentage.
Where	Published per service on Microsoft's SLA website. The VM SLA depends on how you place the VMs.
When	When you design for uptime targets or work out combined availability.
Why	It guarantees uptime, not performance, bandwidth or features.
How	If monthly uptime falls below the promised percentage, you claim service credits.

Exam tip
Azure SLAs = uptime/connectivity percentages and service credits. Never performance or bandwidth.

Memory trick
SLA = Stays Live Agreement.

Real-world example, use case and scenario

EXAMPLE
Downtime budget
99.95% monthly uptime allows about 21.6 minutes of downtime in a 30-day month before credits apply.

USE CASE
Logistics planning
A shipping company runs its route-planning VMs across two availability zones to get the 99.99% uptime SLA.

SCENARIO
Slow but up
A design agency asked for SLA credits because its VM was slow all week. Microsoft said no, because the VM was reachable the whole time. The fix was a larger VM size.

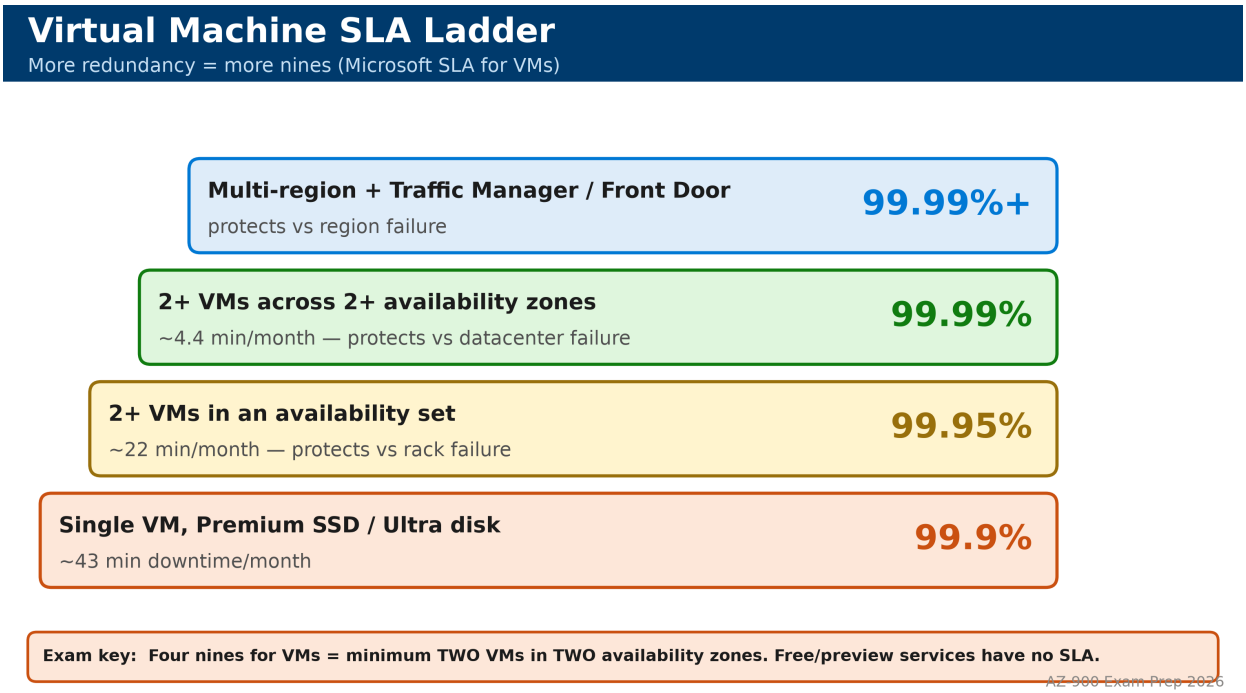


Figure Q60: Virtual machine SLA ladder

About this figure

- Climb the ladder: each step adds redundancy and a higher VM SLA; the downtime figures assume a 30-day month.
- A single VM with Standard SSD gets about 99.5% and with Standard HDD about 95%; all its OS and data disks must meet that tier.
- Availability sets spread VMs across fault domains (racks) and update domains (maintenance batches) in one datacenter; zones spread across datacenters.
- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

Question 61

AZURE ARCHITECTURE & SERVICES

Subscriptions
v5 Q114

Your Azure trial account expired last week. You are now unable to [create additional Microsoft Entra ID user accounts].

✖ A. No change is needed.

✔ **B.** Start an existing Azure virtual machine.

✖ C. Access your data stored in Azure.

✖ D. Access the Azure portal.

Correct answer: Start an existing Azure virtual machine.

Clarification (read this first): The sentence asks what you can NO LONGER do after the trial ends. Only things that need running compute stop. You can still sign in to the portal, see your data and create Entra ID users.

Answer explained

✔ **Start an existing Azure virtual machine.**
When a trial ends, the subscription is disabled. Existing VMs are deallocated (stopped, with their hardware released) and cannot start until you upgrade.

✖ **No change is needed.**
Entra ID Free is not tied to the trial subscription. You can still create users.

✖ **Access your data stored in Azure.**
Your data is kept for a time after expiry. You can reach it again once you upgrade.

✖ **Access the Azure portal.**
You can still sign in to the portal. That is where you upgrade the expired subscription.

EXAM STRATEGY
An expired trial disables the subscription, so compute stops. Creating Entra ID users, using the portal and reaching data still work. Pick the option that needs running compute.

IMPACT
A disabled subscription stops VMs and halts production. Upgrading to pay-as-you-go before the trial ends avoids a surprise outage.

What · Where · When · Why · How

What	An Azure free account gives credit and free services for a limited time.
Where	Check subscription status in Subscriptions or Cost Management + Billing in the portal.
When	When the trial ends without an upgrade, the subscription becomes disabled.
Why	A disabled subscription cannot start VMs. Portal access, data and Entra ID remain.
How	Upgrade to pay-as-you-go in the portal. This turns the subscription back on so VMs can start.

Exam tip
Expired or disabled subscription: compute stops and cannot start. Portal access and Entra ID remain.

Memory trick
Trial over: the engines stop, but the doors stay open.

Real-world example, use case and scenario

EXAMPLE
Disabled subscription
After 30 days, clicking Start on a VM shows an error that the subscription is disabled.

USE CASE
Student project
A student's final-year project VM stops when her free credit ends. She upgrades to pay-as-you-go in the portal and starts it again.

SCENARIO
Demo day surprise
A startup built a prototype on a free account. The trial ended the week before a pitch, and the VM would not start. The founder could still sign in, upgrade to pay-as-you-go and have the demo running within the hour.

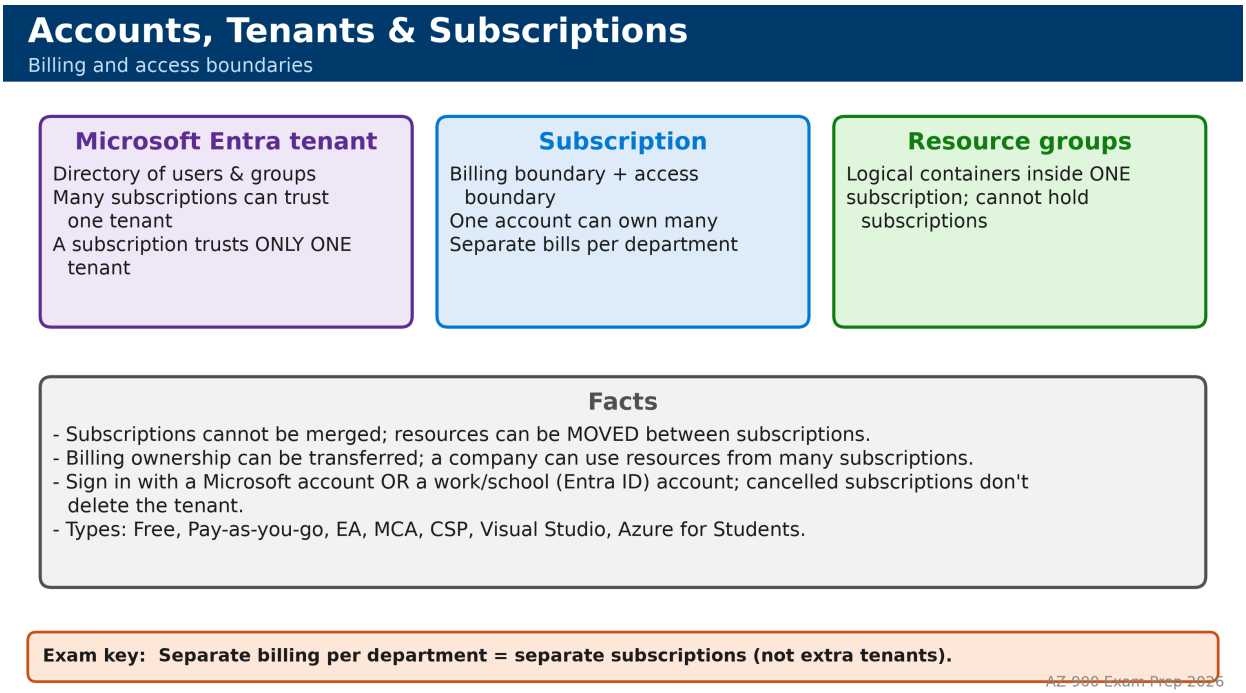


Figure Q61a: Accounts, tenants and subscriptions

About this figure

- Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.
- Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.
- Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.
- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

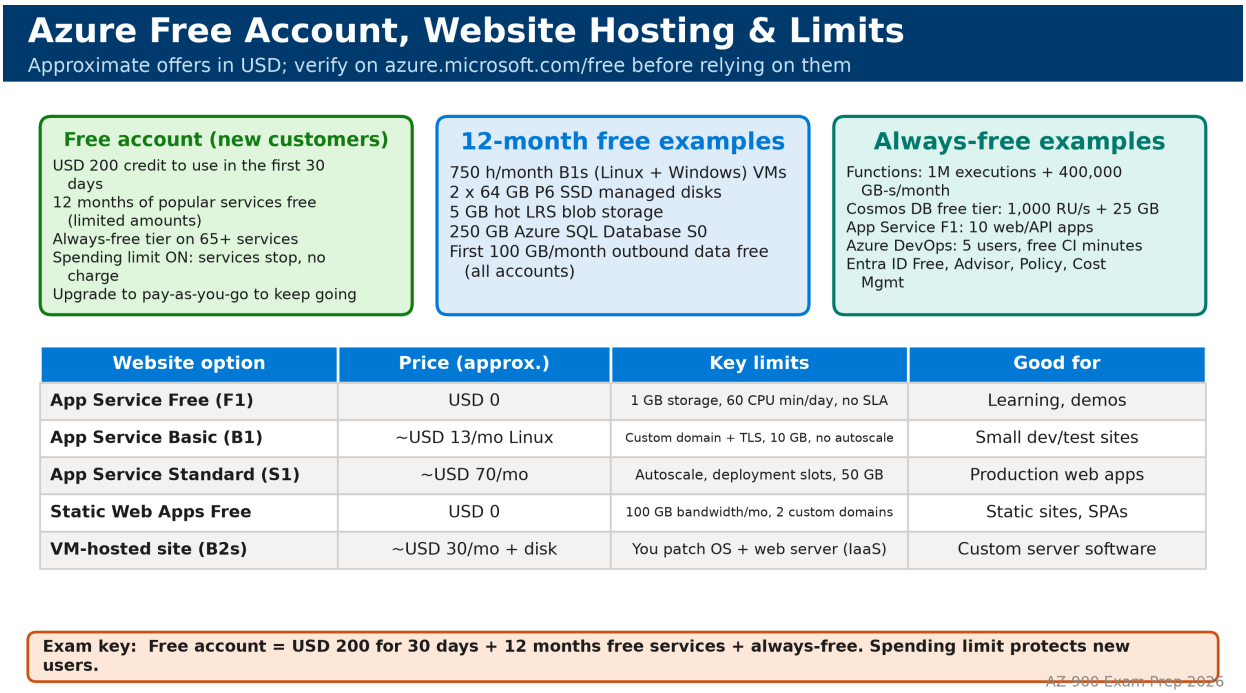


Figure Q61b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 62

AZURE ARCHITECTURE & SERVICES — **Subscriptions** v5 Q115

Your company has 10 departments. The company plans to implement Azure environment. You need to ensure that each department can use a different payment option for the Azure services it consumes. What should you create for each department?

✖ A. A reservation.

✔ **B.** A subscription.

✖ C. A resource group.

✖ D. A container instance.

Correct answer: A subscription.

Clarification (read this first): Key words: 'different payment option'. You need something that gets its own bill. A resource group shares its subscription's bill. Q118-Q120 are look-alikes: role No, policy No, subscription Yes.

Answer explained

✔ **A subscription.**
Each subscription is a billing boundary. It has its own billing profile, invoice and payment method. So one subscription per department allows different payment options.

✖ **A reservation.**
A reservation is a discount you buy with a 1- or 3-year commitment. It is not a separate billing container.

✖ **A resource group.**
Resource groups use their subscription's billing and payment method.

✖ **A container instance.**
Container Instances is a service that runs containers. It is not a billing unit.

EXAM STRATEGY
'Different payment option' per department means a separate billing unit: a subscription. Resource groups share their subscription's billing. Reservations are discounts.

IMPACT
Separate subscriptions give each department its own invoice and budget. This makes charging departments simple and stops one team using another's money.

What · Where · When · Why · How

What	A subscription is a container for resources and a billing boundary.
Where	Under a billing account or profile. It can be grouped in management groups.
When	When departments need separate invoices, payment methods or offers.
Why	Each subscription is billed separately, so departments can pay in different ways.
How	Create one subscription per department. Link each to its own billing profile or payment method.

Exam tip
'Different payment option', 'separate invoice', 'billing boundary' = subscription.

Memory trick
One wallet per subscription.

Real-world example, use case and scenario

EXAMPLE
Two payment methods
Marketing's subscription is charged to a company card. R&D's subscription is paid by invoice.

USE CASE
Holding company
A holding company gives each of its 10 brands its own subscription. Each brand's finance team pays its own Azure bill.

SCENARIO
Chargeback disputes
A media group shared one subscription across departments and argued every month over who owed what. They split into one subscription per department. Each got its own invoice and payment method, and the arguments ended.

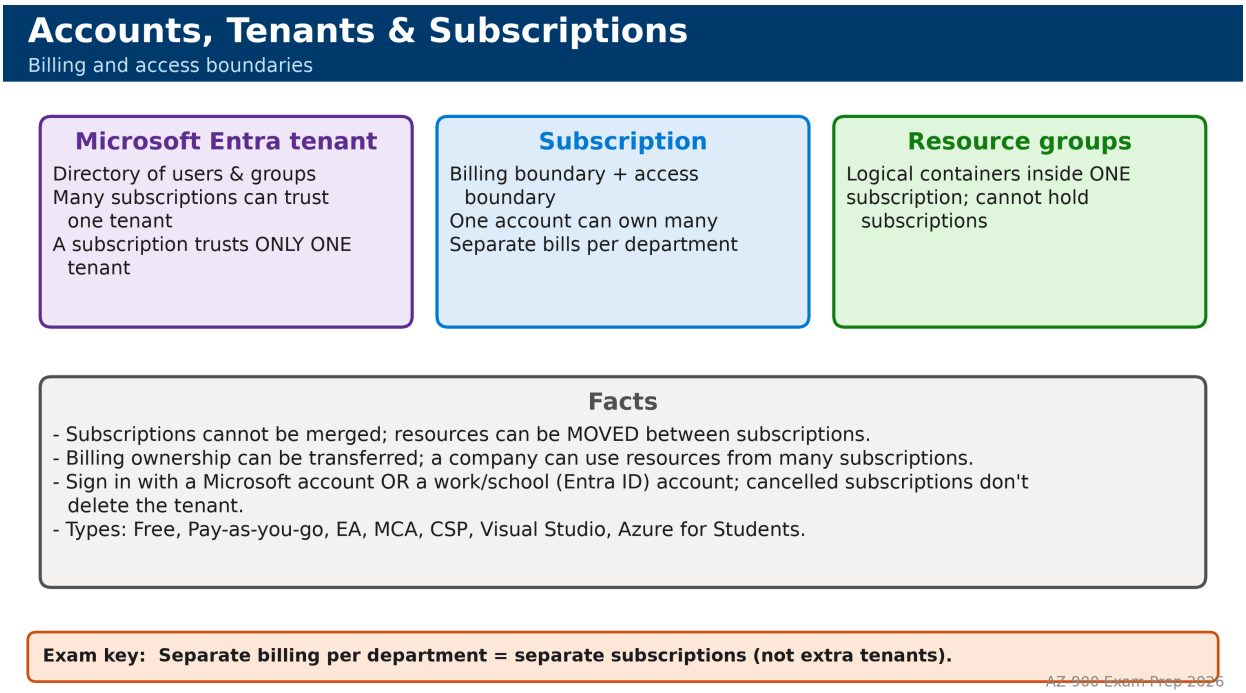


Figure Q62a: Accounts, tenants and subscriptions

About this figure

- Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.
- Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.
- Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.
- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

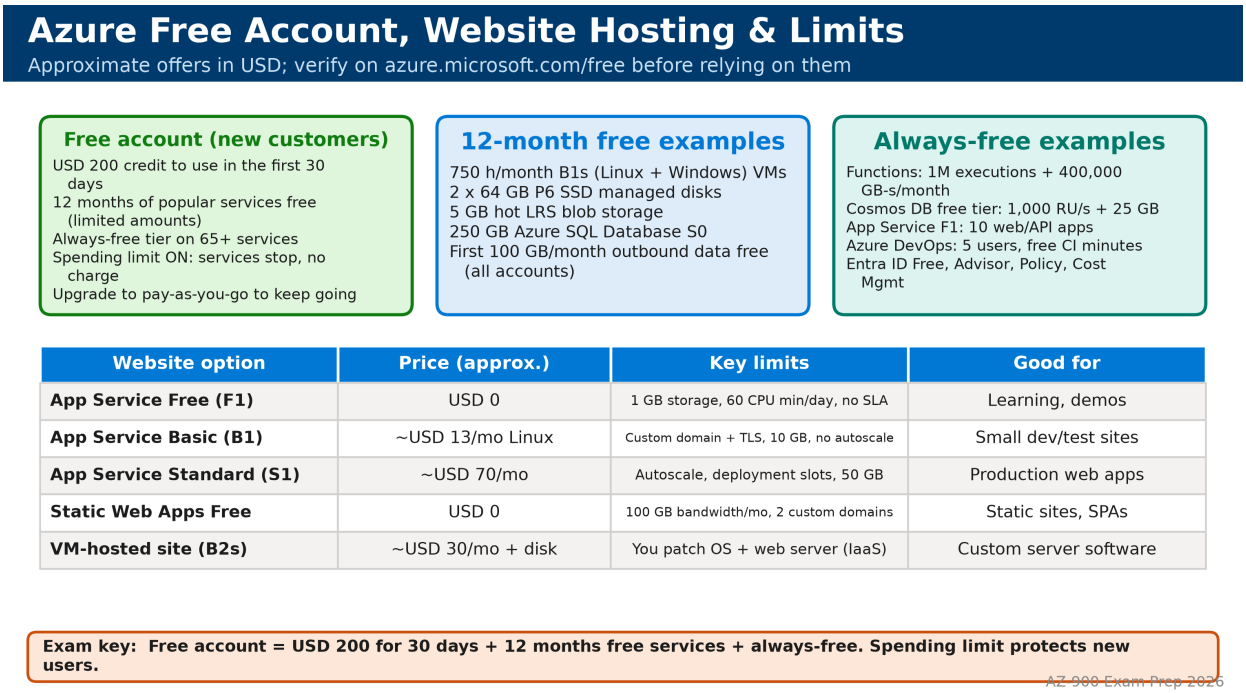


Figure Q62b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 63

CLOUD CONCEPTS

Lifecycle
v5 Q116, Q144

Which statement accurately describes the Modern Lifecycle Policy for Azure services?

✖ A. Microsoft provides mainstream support for a service for five years.

✔ **B.** Microsoft provides a minimum of 12 months notice before ending support for a service.

✖ C. After a service is made generally available, Microsoft provides support for the service for a minimum of four years.

✖ D. When a service is retired, you can purchase extended support for the service for up to five years.

Correct answer: Microsoft provides a minimum of 12 months notice before ending support for a service.

Clarification (read this first): Do not mix up the two Microsoft policies. Modern Lifecycle (for cloud services like Azure) = at least 12 months' notice. Fixed Lifecycle (for boxed products) = 5 years mainstream + 5 years extended support.

Answer explained

✔ **Microsoft provides a minimum of 12 months notice before ending support for a service.**
Under the Modern Lifecycle Policy, Microsoft gives at least 12 months' notice before support ends for a service with no replacement.

✖ **Microsoft provides mainstream support for a service for five years.**
Five years of mainstream support is part of the Fixed Lifecycle Policy for traditional products, not Azure services.

✖ **After a service is made generally available, Microsoft provides support for the service for a minimum of four years.**
The Modern Lifecycle Policy has no four-year minimum. Support continues while customers stay up to date.

✖ **When a service is retired, you can purchase extended support for the service for up to five years.**
Paid extended support (such as ESUs - Extended Security Updates) is for fixed-lifecycle products, not retired Azure services.

EXAM STRATEGY
The key Modern Lifecycle number is 12 months' notice. Remove fixed-year support terms (five or four years). They belong to the Fixed Lifecycle Policy.

IMPACT
Twelve months' notice gives teams time to plan a move before a service retires. Missing these notices leads to rushed migrations.

What · Where · When · Why · How

What	The Modern Lifecycle Policy covers services that are updated all the time, such as Azure.
Where	On the Microsoft Lifecycle site. It applies to Azure and other cloud products.
When	When you plan for a service that might be retired.
Why	It promises at least 12 months' notice before support ends for a service with no replacement.
How	Customers keep up with updates. Microsoft announces retirements at least 12 months ahead.

Exam tip
Modern Lifecycle = 'stay current' + 12 months' notice. Fixed Lifecycle = 5 years mainstream + 5 years extended.

Memory trick
Modern gives a year's warning: 12 months.

Real-world example, use case and scenario

EXAMPLE
Retirement notice
Microsoft announces a service will retire in September of the next year, more than 12 months ahead, with migration guidance.

USE CASE
Platform roadmap
A SaaS company follows Azure Updates retirement notices and plans moves within the 12-month window.

SCENARIO
Planned migration
A hospital IT team learned that an old Azure service they used was retiring. With over a year's notice, they moved it outside flu season. Patients saw no downtime.

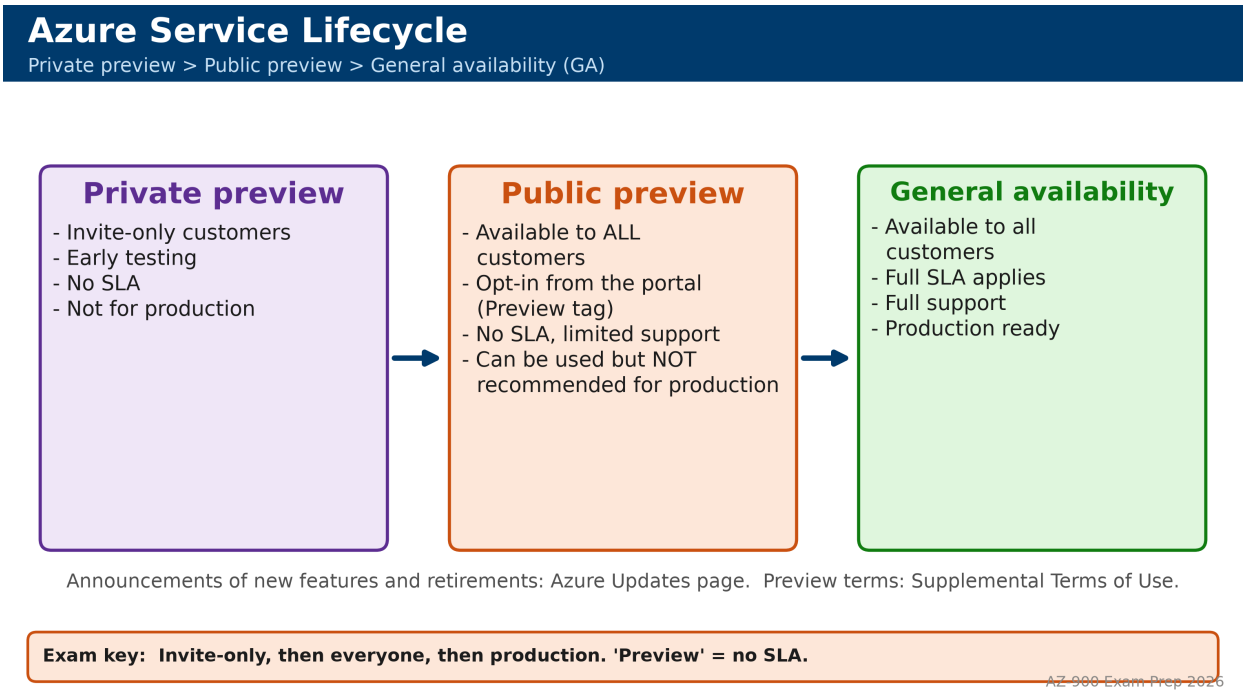


Figure Q63: Service lifecycle: preview to GA

About this figure

- Read the arrows left to right: invite-only private preview, open public preview, then general availability with SLA and full support.
- Previews are offered as-is under the Supplemental Terms of Use for Microsoft Azure Previews; features, pricing and availability may change or be withdrawn.
- Some previews are discounted or free, but many bill at reduced or standard rates; check the pricing page before relying on preview resources.
- Follow Azure Updates for previews, GA announcements and retirements; Service Health also sends retirement notices for services you actually use.
- Microsoft usually gives at least 12 months' notice before retiring a GA service or feature that customers depend on, with migration guidance.

Question 64

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q117

You can use Advisor recommendations in Azure to send email alerts when the cost of the current billing Azure subscription exceeds a specified limit.

- ✖ A. No change is needed.
- ✖ B. Access control (LAM).
- ✔ **C.** Budget alerts.
- ✖ D. Compliance.

Correct answer: Budget alerts.

Clarification (read this first): The underlined tool (Advisor recommendations) is wrong. Advisor only gives tips; it does not watch your spending. 'Email me when cost goes over X' is a budget alert. Q382: a budget alert does not stop resources by itself.

Answer explained

✔ **Budget alerts.**
Cost Management budgets have thresholds (trigger levels) on actual or forecast spending. Budget alerts email people when cost crosses them.

✖ **No change is needed.**
Advisor gives cost-saving tips. It does not alert you when spending goes over a limit.

✖ **Access control (LAM).**
Access control (IAM) manages RBAC role assignments, not cost alerts.

✖ **Compliance.**
Compliance views track policy or regulation status, not spending.

EXAM STRATEGY
'Email alerts when cost exceeds a limit' means budgets in Cost Management. Advisor only suggests savings. IAM and compliance do not watch spending.

IMPACT
Budget alerts catch runaway spending early, such as a forgotten GPU VM, before it becomes a very large bill.

What · Where · When · Why · How

What	Budget alerts are Cost Management messages sent when spending crosses budget thresholds.
Where	Cost Management > Budgets, at subscription, resource group or management group level.
When	When people must be emailed as cost nears or passes a limit.
Why	They email people automatically when actual or forecast spending crosses the threshold.
How	Create a budget amount with percentage thresholds and email recipients. Action groups are optional.

Exam tip
'Email when cost exceeds X' = budget alert. Budgets alert but do not stop resources by themselves.

Memory trick
Budgets bark when you overspend; Advisor only advises.

Real-world example, use case and scenario

EXAMPLE
Threshold set
A 1,000 USD monthly budget with alerts at 75% actual and 100% forecast emails the finance lead.

USE CASE
Research grant
A university research group sets a budget equal to its grant. The lead researcher is warned before the money runs out.

SCENARIO
Runaway GPU VM
An AI startup left a GPU VM running over a long weekend. Their 90% budget alert emailed the CTO on Saturday. He shut it down before the month's budget was used up.

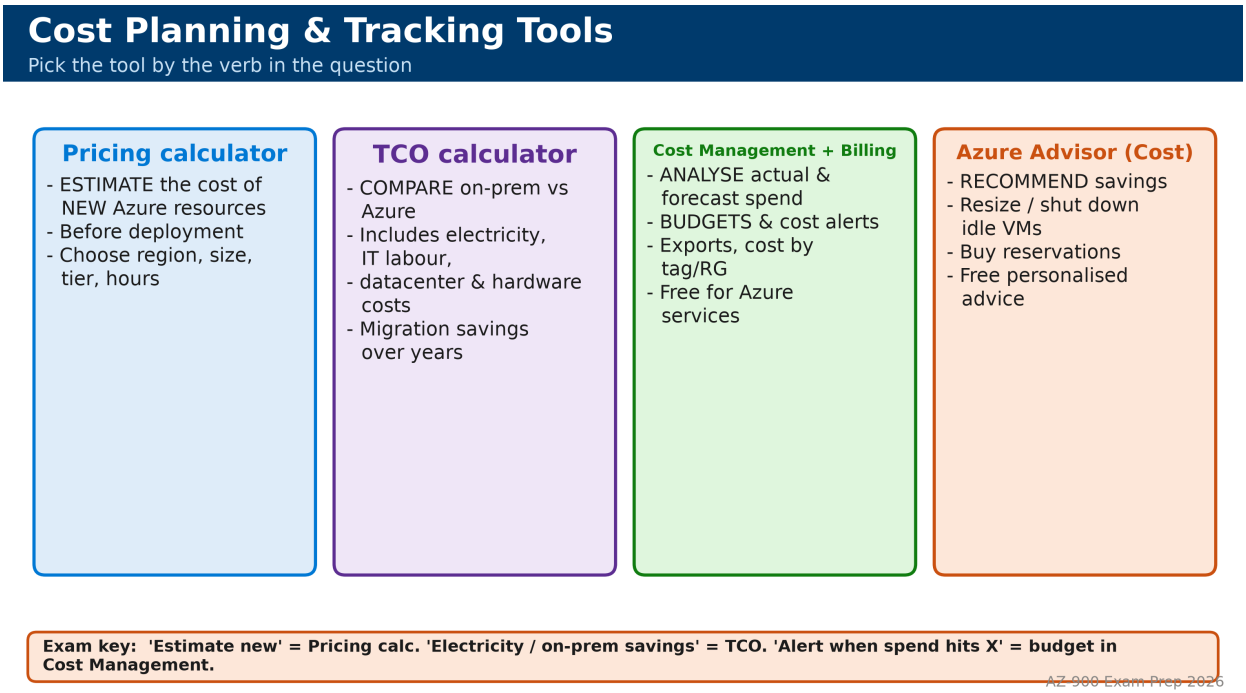


Figure Q64a: Cost planning and tracking tools

About this figure

• Each column is one tool; pick it by the question's verb: estimate (Pricing calculator), compare (TCO), analyse or alert (Cost Management), recommend (Advisor).

• Pricing and TCO calculators are free websites that need no Azure subscription; estimates use list prices and can be saved or exported to Excel.

• Cost Management is free for Azure resources; budgets can be set monthly, quarterly or yearly with alerts at thresholds such as 50%, 80% or 100%.

• Budgets alert but do not stop spending; only the free account and some credit offers have a spending limit that disables resources when exhausted.

• Advisor cost recommendations include resizing or shutting down VMs with low CPU and buying reservations or savings plans based on 30 days of usage.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

Figure Q64b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 65

CLOUD CONCEPTS

Deployment models
v5 Q118

To implement a hybrid cloud model, a company must first have a private cloud.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch the words 'must first'. Hybrid cloud needs on-premises servers plus public cloud. The on-premises part can be ordinary servers - it does not have to be a private cloud. Q268 (hybrid needs public cloud resources) is Yes.

Answer explained

✔ **No.**
Hybrid cloud joins public cloud with any on-premises setup, such as normal servers linked by VPN. A private cloud is not required first.

✖ **Yes.**
There is no such requirement. Ordinary on-premises servers connected to Azure already make a hybrid cloud.

EXAM STRATEGY
Key fact: hybrid means on-premises plus public cloud, and on-premises need not be a private cloud. Words like 'must first' often signal a false absolute (an always/never claim).

IMPACT
Companies can start hybrid with the datacenters they already have. They do not need to build a private cloud before using Azure.

What · Where · When · Why · How

What	Hybrid cloud combines on-premises infrastructure with public cloud services.
Where	Across the company's own datacenter and Azure, joined by VPN or ExpressRoute (a private link).
When	When some workloads stay on-premises and others use the public cloud.
Why	Any on-premises setup counts, so no private cloud is needed first.
How	Connect existing on-premises servers to Azure with a site-to-site VPN or ExpressRoute.

Exam tip
Hybrid = on-premises (any form) + public cloud. 'Must first' statements about deployment models are usually false.

Memory trick
Hybrid needs a home and the cloud, not a private cloud.

Real-world example, use case and scenario

EXAMPLE
Simple hybrid
A file server in the office plus Azure VMs, connected by site-to-site VPN, is a hybrid cloud.

USE CASE
Manufacturing plant
A factory keeps machine-control servers on-premises and sends production data to Azure for analytics.

SCENARIO
No private cloud project
A regional credit union thought it had to buy a private-cloud platform before using Azure. Its consultant connected the existing datacenter to Azure with a VPN. They were running hybrid within a month.

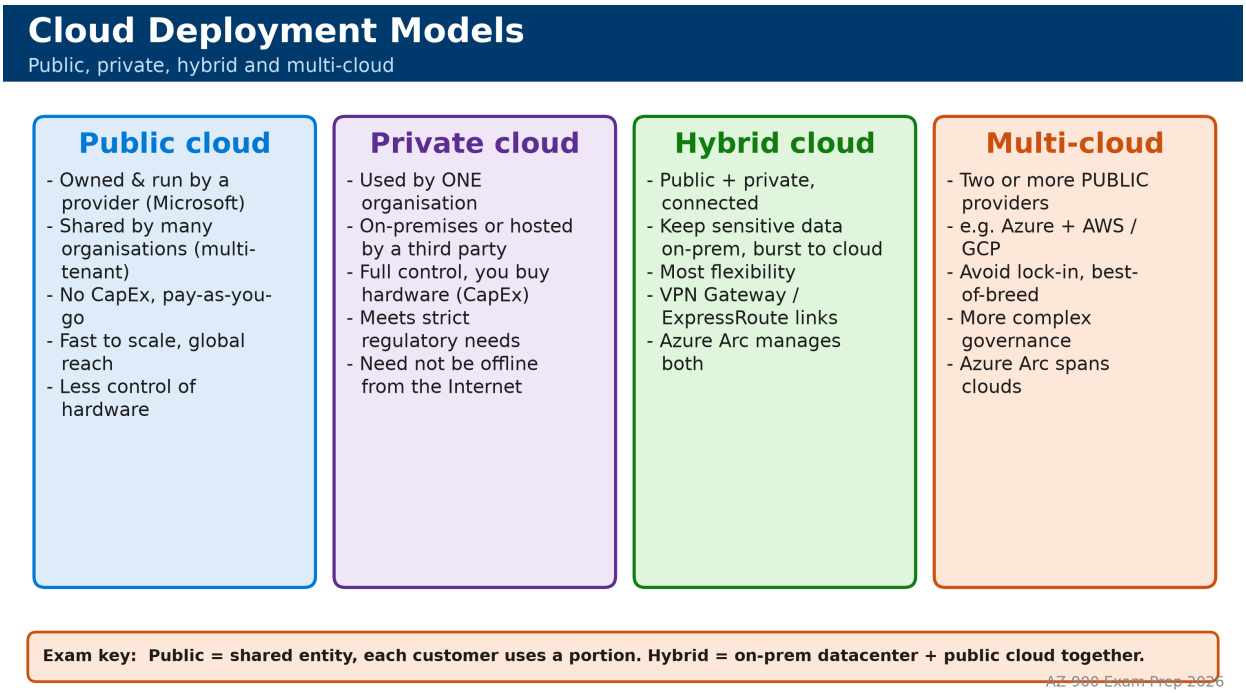


Figure Q65: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 66

CLOUD CONCEPTS

Deployment models
v5 Q119, Q159

A company can extend the computing resources of its internal network by using a hybrid cloud.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): 'Extend' means add capacity to your own network by using the cloud. That is exactly what hybrid cloud does. Q246 asks the same with 'public cloud' - also Yes. Q267 is different: you cannot add your own physical servers to the public cloud (No).

Answer explained

✔ **Yes.**
Hybrid cloud connects the internal network to the public cloud over VPN or ExpressRoute. This adds capacity to on-premises without new hardware.

✖ **No.**
Extending on-premises compute into the public cloud, for example bursting (overflowing to the cloud) at peaks, is the main use of hybrid cloud.

EXAM STRATEGY
Key fact: hybrid cloud lets on-premises networks burst or extend into public cloud resources. Extending capacity is the definition, so Yes.

IMPACT
Hybrid bursting handles peaks without buying hardware, which saves capital spending. This core idea is tested often.

What · Where · When · Why · How

What	Hybrid cloud links a company's internal network with public cloud resources.
Where	On-premises datacenter plus Azure virtual networks, connected by VPN or ExpressRoute.
When	When internal capacity is not enough, for example at peak demand.
Why	It adds compute capacity without buying more on-premises hardware.
How	Connect the networks. Run extra workloads on Azure VMs, or burst into Azure at peaks.

Exam tip
'Extend', 'burst', 'keep some on-premises' = hybrid cloud.

Memory trick
Hybrid = home plus an extension built in the cloud.

Real-world example, use case and scenario

EXAMPLE
Cloud bursting
An on-premises render farm sends extra jobs to Azure VMs near deadlines.

USE CASE
Tax office peak
A tax-filing service extends its on-premises web servers into Azure every April.

SCENARIO
Full datacenter
A biotech firm's datacenter had no space for new genome-analysis servers. They connected it to Azure with ExpressRoute and ran new jobs in the cloud. This avoided a costly datacenter expansion.

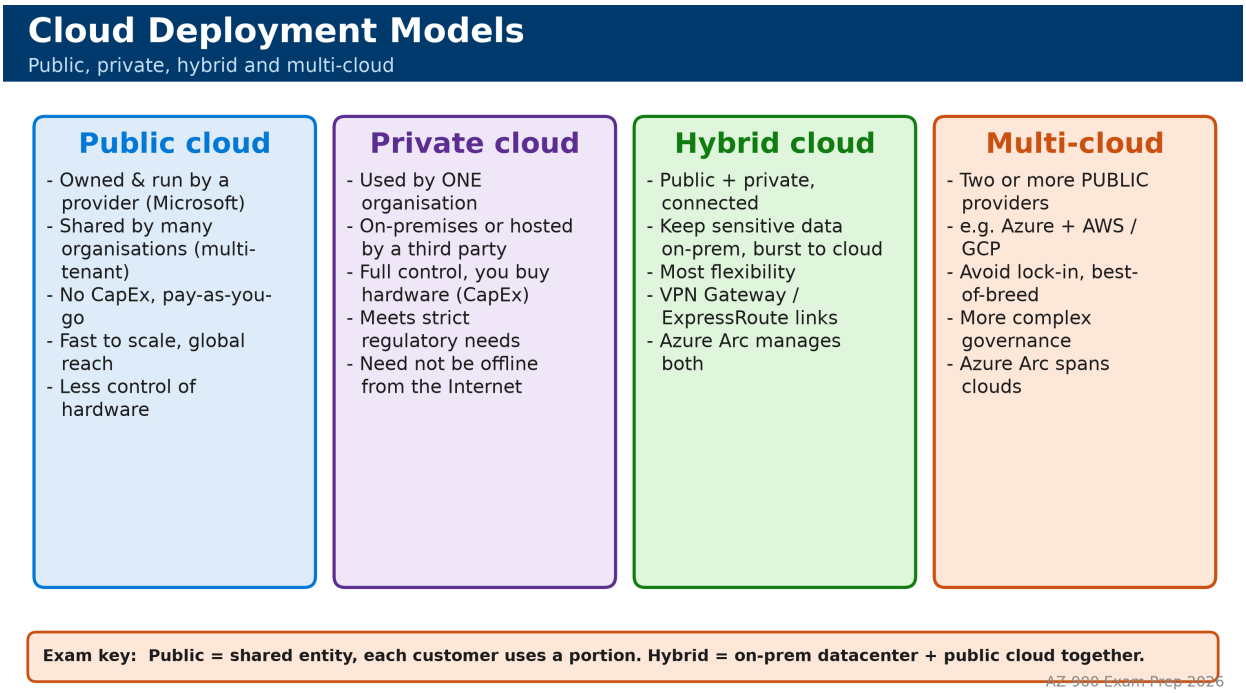


Figure Q66: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 67

CLOUD CONCEPTS

Deployment models
v5 Q120

In a public cloud model, only guest users at your company can access the resources in the cloud.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch the word 'only'. 'Public' describes who owns the hardware (shared provider hardware). It does not mean that only guests, or anyone, can reach your resources. You decide who gets access.

Answer explained

✔ **No.**
Public cloud means shared provider hardware. You decide who can reach your resources, using your own Entra ID, RBAC (role-based access) and network controls.

✖ **Yes.**
Nothing limits public cloud access to guest users. Members, guests or anonymous web users can have access, as you set it up.

EXAM STRATEGY
Key fact: any user you allow can reach public cloud resources, not only guests. 'Only guest users' is a false limit, so No.

IMPACT
Misreading 'public' leads to wrong identity designs. Public cloud means shared provider hardware; your own identity system controls access.

What · Where · When · Why · How

What	Public cloud is provider-owned hardware shared by many customers. Anyone can sign up.
Where	Provider datacenters such as Azure regions. Access to your resources is set in Entra ID and RBAC.
When	When you want resources on demand without owning hardware.
Why	Access is whatever you set up, not only guests, so the statement is false.
How	Give members, guests or anonymous users access through Entra ID, RBAC and network settings.

Exam tip
Public cloud describes who owns and shares the hardware, not who in your company may sign in.

Memory trick
Public infrastructure, private keys.

Real-world example, use case and scenario

EXAMPLE
Mixed access
An app in Azure is used by employees (members), a partner (guest) and anonymous website visitors.

USE CASE
Charity portal
A charity runs a volunteer portal in Azure. Staff, outside volunteers and the public all use it with different permissions.

SCENARIO
Misunderstood term
A board member at a dental group worried 'public cloud' meant anyone could see patient records. The IT lead explained that Entra ID and RBAC protect the records. Only allowed staff can open them.

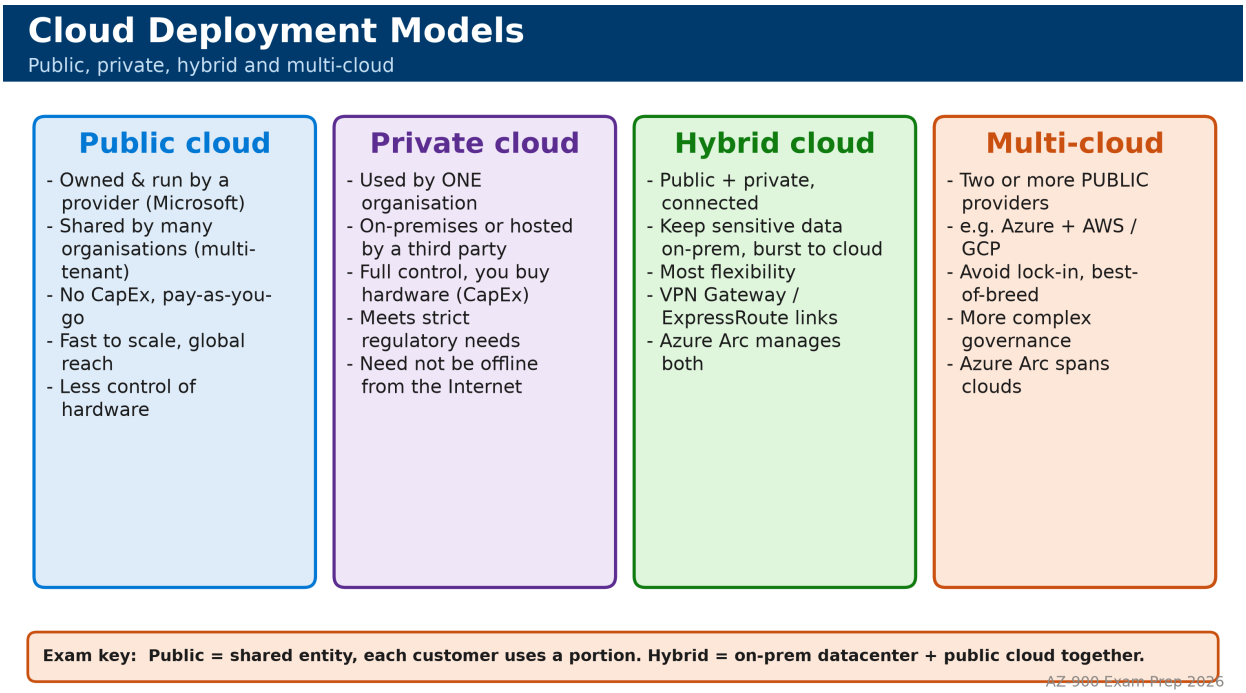


Figure Q67: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 68

CLOUD CONCEPTS

Cloud concepts
v5 Q121, Q19, Q154

You have an on-premises network that contains several servers. You plan to migrate all the servers to Azure. You need to recommend a solution to ensure that some of the servers are available if a single Azure data center goes offline for an extended period. What should you include in the recommendation?

✔ **A.** Fault tolerance.

✖ B. Elasticity.

✖ C. Scalability.

✖ D. Low latency.

Correct answer: **Fault tolerance.**

Clarification (read this first): The key phrase is 'a single Azure data center goes offline'. That is about surviving failure, not about handling more users. Q87 (elasticity) and Q88 (scalability) are the same scenario - both No.

Answer explained

✔ **Fault tolerance.**
Fault tolerance keeps a system running when a part fails. Spreading servers across availability zones or regions lets them survive a datacenter outage.

✖ **Elasticity.**
Elasticity adds and removes resources as demand changes. It does not deal with datacenter failure.

✖ **Scalability.**
Scalability is the ability to add capacity, not to survive outages.

✖ **Low latency.**
Low latency (fast response) is about speed, not staying available during failures.

EXAM STRATEGY
'Available if a data center goes offline' means continuing despite failure: fault tolerance. Elasticity and scalability are about load. Low latency is about speed.

IMPACT
Fault tolerance across zones keeps money-making systems online during datacenter outages. Questions defining cloud benefits appear often.

What · Where · When · Why · How

What	Fault tolerance is a system's ability to keep running when parts fail.
Where	Built with availability zones in a region, or with paired regions.
When	When workloads must survive a datacenter outage.
Why	It keeps some servers available if one datacenter is offline for a long time.
How	Spread servers across two or more availability zones or regions, so one failure does not stop them all.

Exam tip
'Data center goes offline', 'failure', 'remain available' = fault tolerance / high availability, built with availability zones.

Memory trick
Fault tolerance: one datacenter falls, the app stays up.

Real-world example, use case and scenario

EXAMPLE
Zone spread
Two app servers in Zone 1 and Zone 2 of West Europe keep the app running when Zone 1's datacenter loses power.

USE CASE
Emergency dispatch
A 911 dispatch software company spreads its servers across zones. Calls still get routed during a datacenter outage.

SCENARIO
Cooling failure
A payroll company moved its servers into a single datacenter. It had a day-long outage when cooling failed. They redeployed across availability zones, and customers did not notice a later zone problem.

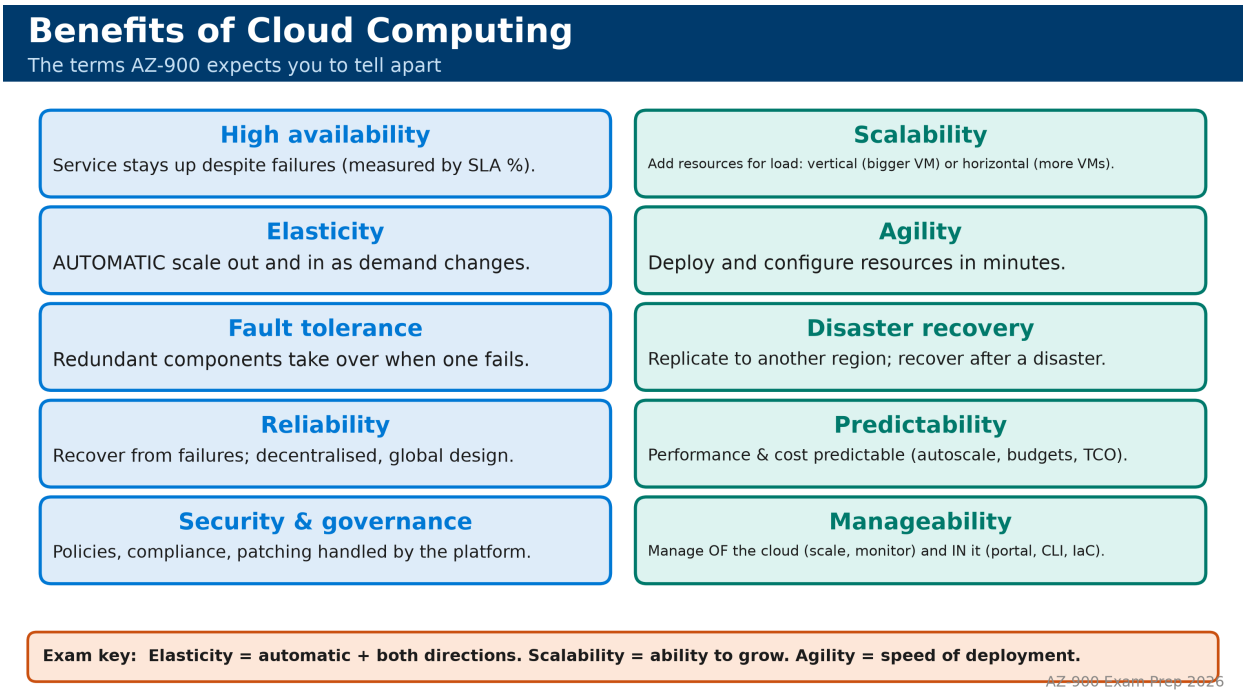


Figure Q68a: Benefits of cloud computing

About this figure

- Each box pairs a term with its meaning; the exam often gives a scenario and asks which benefit it describes, so focus on keywords.
- Scalability is the capacity to grow, manual or automatic; elasticity is automatic scaling out and back in with demand, for example VM Scale Sets autoscale.
- High availability is expressed as an SLA percentage such as 99.9% or 99.99%; fault tolerance and disaster recovery are ways of achieving it.
- Agility means provisioning in minutes instead of weeks of hardware procurement; predictability covers both performance and cost forecasting.
- Governance tools such as Azure Policy, RBAC and locks deliver the security and governance benefit; manageability covers portal, CLI, PowerShell and IaC.

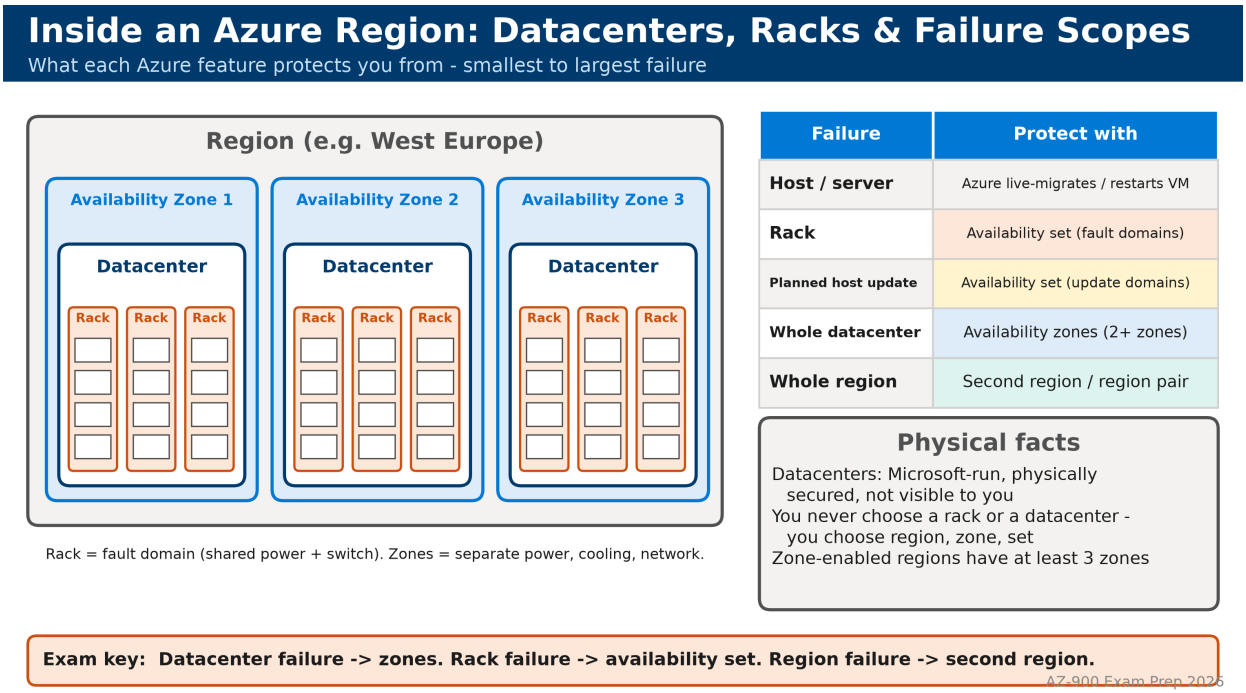


Figure Q68b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 69

CLOUD CONCEPTS

Deployment models
v5 Q123

What are two characteristics of the public cloud?

- ✖ A. Dedicated hardware.
- ✖ B. Unsecured connections.
- ✖ C. Limited storage.
- ✔ **D.** Metered pricing.
- ✔ **E.** Self-service management.

Correct answer: Metered pricing.; Self-service management.

Clarification (read this first): This question has TWO answers. 'Metered pricing' means pay-as-you-go. Q27 is a look-alike with different words: shared infrastructure + pay-as-you-go.

Answer explained

✔ **Metered pricing.; Self-service management.**
Public cloud is metered (you pay only for what you use) and self-service (you create resources yourself through the portal, CLI or API, when you need them).

✖ **Dedicated hardware.**
Public cloud runs on shared hardware used by many customers. Dedicated hardware is a private cloud trait, or the special Dedicated Host option.

✖ **Unsecured connections.**
Public cloud connections use TLS (encryption) and can be private over VPN or ExpressRoute. They are not unsecured by nature.

✖ **Limited storage.**
Public cloud storage is almost unlimited and can grow. Limited capacity describes on-premises hardware.

EXAM STRATEGY
Public cloud traits: pay for what you use, and create resources yourself. Remove dedicated hardware (private cloud), unsecured connections and limited storage, which are false.

IMPACT
Metered pricing plus self-service lets teams deploy in minutes and pay only for use. This is the main business reason to adopt cloud.

What · Where · When · Why · How

What	Public cloud is shared hardware offered as a metered, on-demand service.
Where	Provider regions, reached through the Azure portal, CLI and APIs.
When	When you want to pay per use and create resources yourself at once.
Why	Metered pricing and self-service management are defining public cloud traits.
How	Usage is measured per second, hour or GB and billed. You create resources without help from the provider.

Exam tip
Public cloud keywords: shared, metered/pay-as-you-go, self-service, no CapEx (up-front buying). Dedicated hardware = private.

Memory trick
Public cloud is like a taxi: you call it yourself and pay by the meter.

Real-world example, use case and scenario

EXAMPLE
Per-second billing
A developer creates a VM in the portal, uses it for 3 hours and pays only for those hours.

USE CASE
Market research spike
A market research firm creates analytics clusters for each survey and deletes them after. It pays only for use.

SCENARIO
Ticket queue removed
A university's researchers used to wait two weeks for IT to set up servers. With Azure self-service and per-use billing charged to grants, they now deploy in minutes.

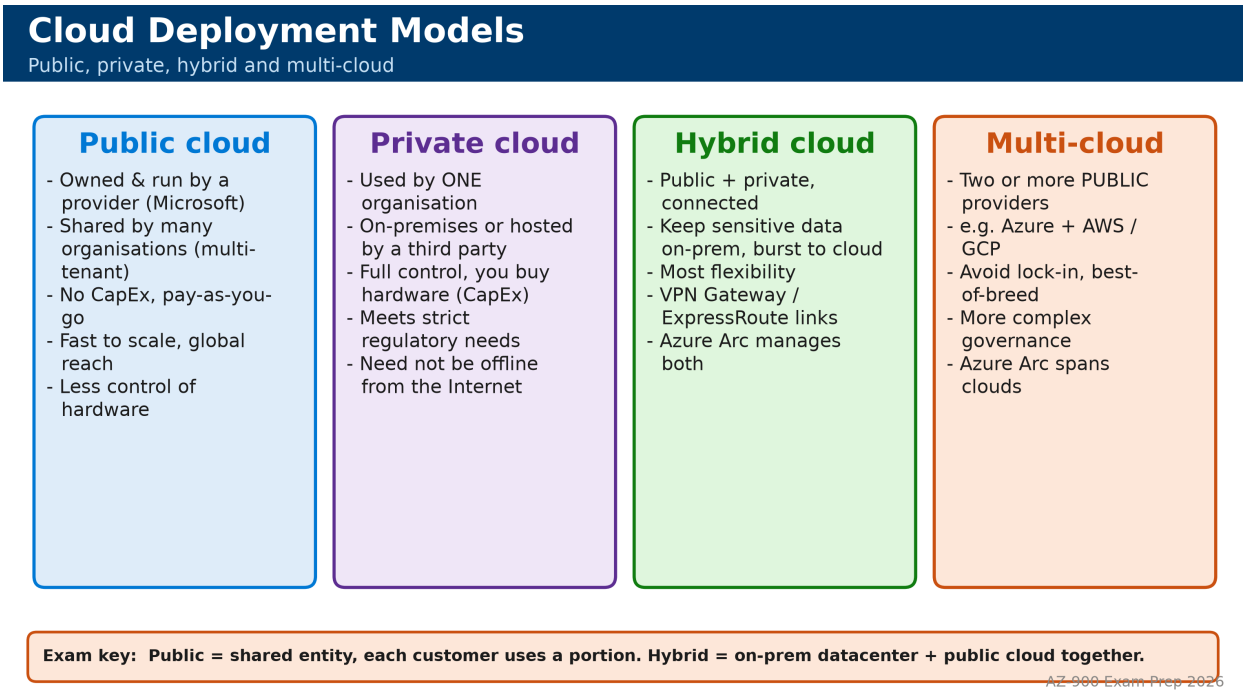


Figure Q69: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 70

CLOUD CONCEPTS

Cloud concepts
v5 Q125

Your company hosts an accounting application named App1 that is used by all the customers of the company. App1 has low usage during the first three weeks of each month and very high usage during the last week of each month. Which benefit of Azure Cloud Services supports cost management for this type of usage pattern?

✖ A. High availability.

✖ B. High latency.

✔ **C.** Elasticity.

✖ D. Load balancing.

Correct answer: Elasticity.

Clarification (read this first): The usage goes up and down in a cycle, and the question asks about COST. Elasticity grows and shrinks resources automatically, so you stop paying for idle capacity. Load balancing only spreads traffic over servers you already have.

Answer explained

✔ **Elasticity.**
Elasticity automatically adds resources for the month-end peak and removes them afterwards. You pay for extra capacity only when you use it.

✖ **High availability.**
High availability keeps the app running through failures. It does not lower cost for changing usage.

✖ **High latency.**
High latency means slow responses. It is a problem, not a benefit.

✖ **Load balancing.**
Load balancing spreads traffic across servers you already have. It does not add or remove capacity to save money.

EXAM STRATEGY
'Low usage, then very high usage' in a repeating cycle signals elasticity: automatic scale out and in. High availability and load balancing do not change capacity.

IMPACT
Elastic scaling pays for peak capacity only in the last week. This costs much less than running peak capacity all month.

What · Where · When · Why · How

What	Elasticity means automatically adding and removing resources as demand changes.
Where	Set up through autoscale settings on App Service plans or VM scale sets.
When	When usage has peaks, expected or not, such as month-end.
Why	You pay for extra capacity only during the peak week, which cuts cost.
How	Autoscale rules add instances on high load or on a schedule, then remove them after.

Exam tip
Peaks (expected or not) + 'cost' = elasticity. Scalability alone does not mean automatic scale-in (shrinking).

Memory trick
Elastic band: stretches for the month-end rush, goes back after.

Real-world example, use case and scenario

EXAMPLE
Autoscale rule
App Service autoscale adds instances when CPU goes above 70% and removes them when it drops below 30%.

USE CASE
Payroll processing
A payroll company runs 2 instances most of the month and 12 during the last week's pay runs.

SCENARIO
Over-provisioned all month
An accounting software company sized its servers for month-end peaks. It paid for idle capacity for three weeks. After it turned on autoscale, its monthly compute bill fell by more than half.

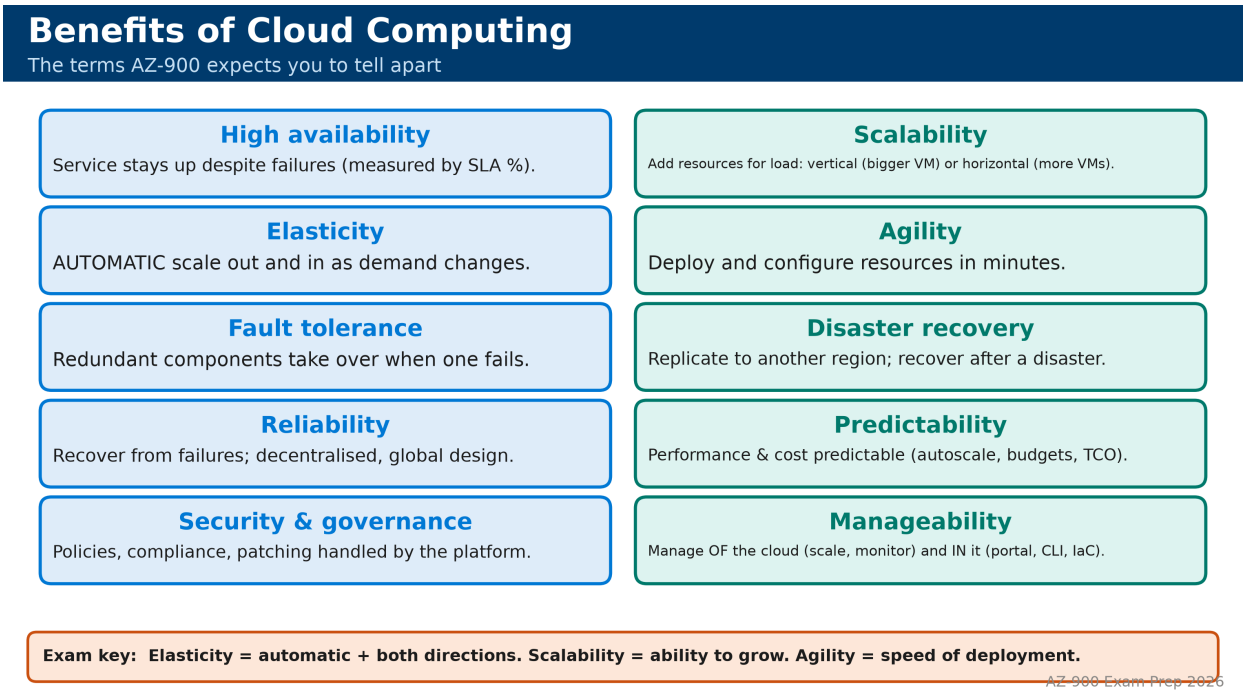


Figure Q70: Benefits of cloud computing

About this figure

- Each box pairs a term with its meaning; the exam often gives a scenario and asks which benefit it describes, so focus on keywords.
- Scalability is the capacity to grow, manual or automatic; elasticity is automatic scaling out and back in with demand, for example VM Scale Sets autoscale.
- High availability is expressed as an SLA percentage such as 99.9% or 99.99%; fault tolerance and disaster recovery are ways of achieving it.
- Agility means provisioning in minutes instead of weeks of hardware procurement; predictability covers both performance and cost forecasting.
- Governance tools such as Azure Policy, RBAC and locks deliver the security and governance benefit; manageability covers portal, CLI, PowerShell and IaC.

Question 71

AZURE ARCHITECTURE & SERVICES

Data & AI
v5 Q127

One of the benefits of Azure Synapse Analytics is that [high availability] is built into the platform.

✔ **A.** No change is needed.

✖ B. Automatic scaling.

✖ C. Data compression.

✖ D. Versioning.

Correct answer: No change is needed.

Clarification (read this first): 'No change is needed' means the sentence is already true. Synapse (a PaaS service) has high availability built in, so keep the bracketed words.

2026 update: Azure Synapse Analytics (formerly SQL Data Warehouse) is still available. Microsoft also promotes Microsoft Fabric as its newer, all-in-one analytics platform.

Answer explained

✔ **No change is needed.**
Synapse is a managed PaaS service. Its storage is copied (replicated) and failed compute nodes are replaced automatically. So high availability is built in (99.9% SLA).

✖ **Automatic scaling.**
Dedicated SQL pools are scaled by changing DWU (performance unit) levels. Automatic scaling is not the built-in platform benefit.

✖ **Data compression.**
Columnstore compression is a storage feature, not the built-in platform benefit the sentence means.

✖ **Versioning.**
Synapse has no built-in data versioning. That is a Blob Storage or source-control feature.

EXAM STRATEGY
Built-in high availability is a typical PaaS feature, so the Synapse sentence is correct. Versioning and compression are not the main platform benefits.

IMPACT
PaaS analytics with built-in HA (high availability) removes cluster management work. Know Synapse, and know Microsoft Fabric is the newer analytics option.

What · Where · When · Why · How

What	Azure Synapse Analytics is a PaaS analytics service. It combines data warehousing and big data.
Where	A Synapse workspace with dedicated or serverless SQL pools in an Azure region.
When	When you need large-scale analytics without managing warehouse servers.
Why	As PaaS it has built-in high availability, with a 99.9% SLA for dedicated SQL pools.
How	Storage is copied separately from compute. Failed compute nodes are replaced automatically.

Exam tip
PaaS data services (SQL Database, Synapse, Cosmos DB) include built-in high availability. That is a key PaaS advantage over IaaS.

Memory trick
Synapse never naps: HA comes in the box.

Real-world example, use case and scenario

EXAMPLE
Node replacement
If a compute node in a Synapse dedicated SQL pool fails, the service connects the storage to a new node automatically.

USE CASE
Retail analytics
A supermarket chain runs nightly sales analytics on Synapse without building its own failover clusters.

SCENARIO
Cluster maintenance avoided
An energy company's on-premises data warehouse needed a weekend-long failover cluster upgrade every year. After moving to Synapse, the platform handles HA and those weekends ended.

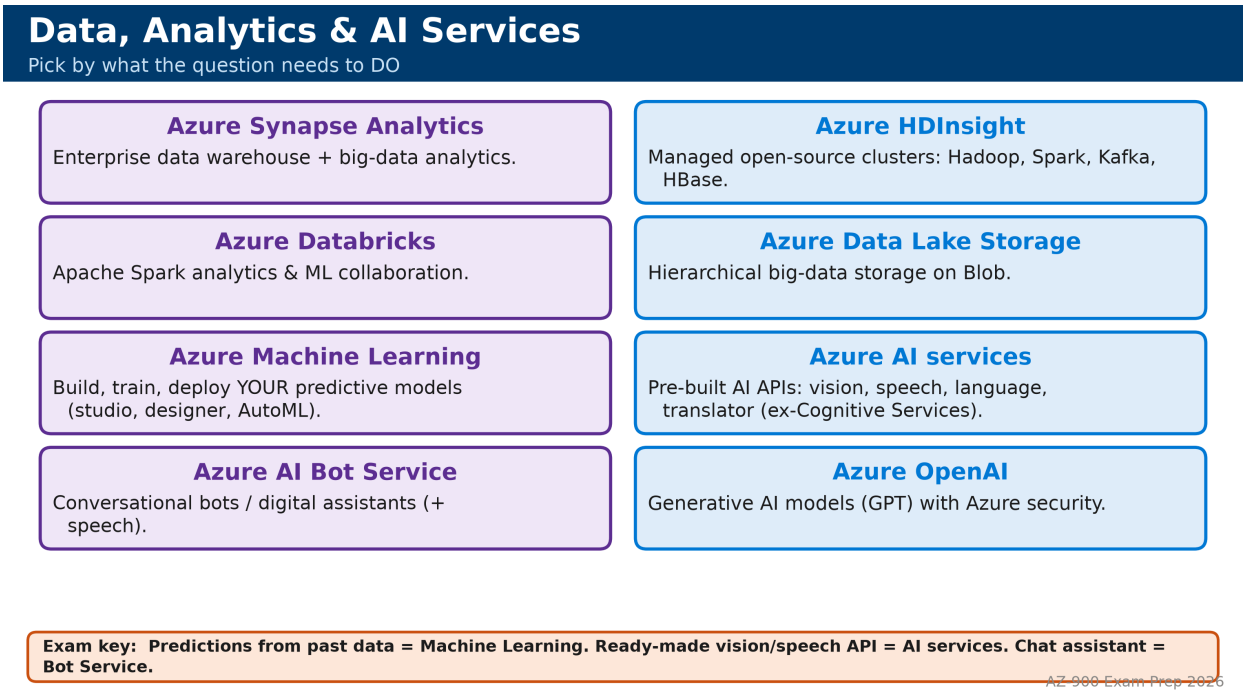


Figure Q71: Data, analytics and AI services

About this figure

• Left column covers data and analytics platforms; right column covers AI. Decide whether the scenario needs storage, processing, custom models or ready-made AI.

• Azure Machine Learning trains your own models from your data; Azure AI services are prebuilt APIs needing no data science, billed per transaction.

• Azure AI services include Vision, Speech, Language, Translator and Document Intelligence; many offer a free F0 tier with limited monthly calls.

• Azure OpenAI models are now offered through Microsoft Foundry (formerly Azure AI Foundry), billed per 1,000 or million tokens, varying widely by model.

• Microsoft Fabric is the newer unified analytics SaaS combining data engineering, warehousing and Power BI, and overlaps much of Synapse.

Question 72

AZURE ARCHITECTURE & SERVICES | **Compute** v5 Q128

Match the Azure services to the correct descriptions.

✖ A. Provide operating system virtualization: Azure App Service. Provide portable environment for virtualized applications: Azure virtual machines. Used to build, deploy, and scale web apps: Azure Container Instances. Provide a platform for serverless code: Azure Functions.

✖ B. Provide operating system virtualization: Azure Functions. Provide portable environment for virtualized applications: Azure virtual machines. Used to build, deploy, and scale web apps: Azure Container Instances. Provide a platform for serverless code: Azure App Service.

✔ **C.** Provide operating system virtualization: Azure virtual machines. Provide portable environment for virtualized applications: Azure Container Instances. Used to build, deploy, and scale web apps: Azure App Service. Provide a platform for serverless code: Azure Functions.

✖ D. Provide operating system virtualization: Azure virtual machines. Provide portable environment for virtualized applications: Azure Functions. Used to build, deploy, and scale web apps: Azure App Service. Provide a platform for serverless code: Azure Container Instances.

Correct answer: Provide operating system virtualization: Azure virtual machines. Provide portable environment for virtualized applications: Azure Container Instances. Used to build, deploy, and scale web apps: Azure App Service. Provide a platform for serverless code: Azure Functions.

Clarification (read this first): Match each description to one service. Clue words: 'operating system' = VMs, 'portable' = containers, 'web apps' = App Service, 'serverless' = Functions.

Answer explained

✔ **Provide operating system virtualization: Azure virtual machines. Provide portable environment for virtualized applications: Azure Container Instances. Used to build, deploy, and scale web apps: Azure App Service. Provide a platform for serverless code: Azure Functions.**
VMs virtualise (simulate) the operating system. Container Instances package apps so they move easily. App Service hosts web apps. Functions runs serverless code that starts on events.

✖ **Provide operating system virtualization: Azure App Service. Provide portable environment for virtualized applications: Azure virtual machines. Used to build, deploy, and scale web apps: Azure Container Instances. Provide a platform for serverless code: Azure Functions.**
App Service is PaaS, not OS virtualisation. VMs and ACI are also swapped. Only the Functions match is right.

✖ **Provide operating system virtualization: Azure Functions. Provide portable environment for virtualized applications: Azure virtual machines. Used to build, deploy, and scale web apps: Azure Container Instances. Provide a platform for serverless code: Azure App Service.**
Every match is wrong. Functions is serverless code, VMs are not portable app packages, and App Service is not serverless.

✖ **Provide operating system virtualization: Azure virtual machines. Provide portable environment for virtualized applications: Azure Functions. Used to build, deploy, and scale web apps: Azure App Service. Provide a platform for serverless code: Azure Container Instances.**
ACI and Functions are swapped. Containers are the portable environment and Functions runs serverless code.

EXAM STRATEGY
Clue words: 'operating system virtualization' = VMs, 'portable environment' = containers, 'web apps' = App Service, 'serverless code' = Functions. Match one to one, then check no service repeats.

IMPACT
The right compute model sets how much you manage and pay. Compute-matching questions appear in almost every AZ-900 exam.

What · Where · When · Why · How

What	Four compute models: VMs, Container Instances, App Service and Functions.
Where	All are Azure compute services, from IaaS VMs to serverless Functions.
When	When matching a workload to the right level of control.
Why	VMs virtualise the OS, ACI is portable, App Service runs web apps, Functions is serverless.
How	Pick VMs for OS control, containers for portability, App Service for web apps, Functions for event code.

Exam tip
VM = OS virtualisation; container = app virtualisation (portable); App Service = web apps; Functions = serverless code.

Memory trick
VMs virtualise the OS, containers carry the app, App Service hosts the web, Functions fire on events.

Real-world example, use case and scenario

EXAMPLE
Four layers
A company runs an old app on a VM, a batch tool in ACI, its website on App Service and image-resize code in Functions.

USE CASE
Photo-sharing startup
A photo app uses Functions to make thumbnails each time a picture is uploaded to Blob Storage. It pays only per run.

SCENARIO
Works on my machine
A data-science team's scripts kept failing on colleagues' laptops because of different library versions. They packaged them as containers and ran them in ACI. The scripts now behave the same everywhere.

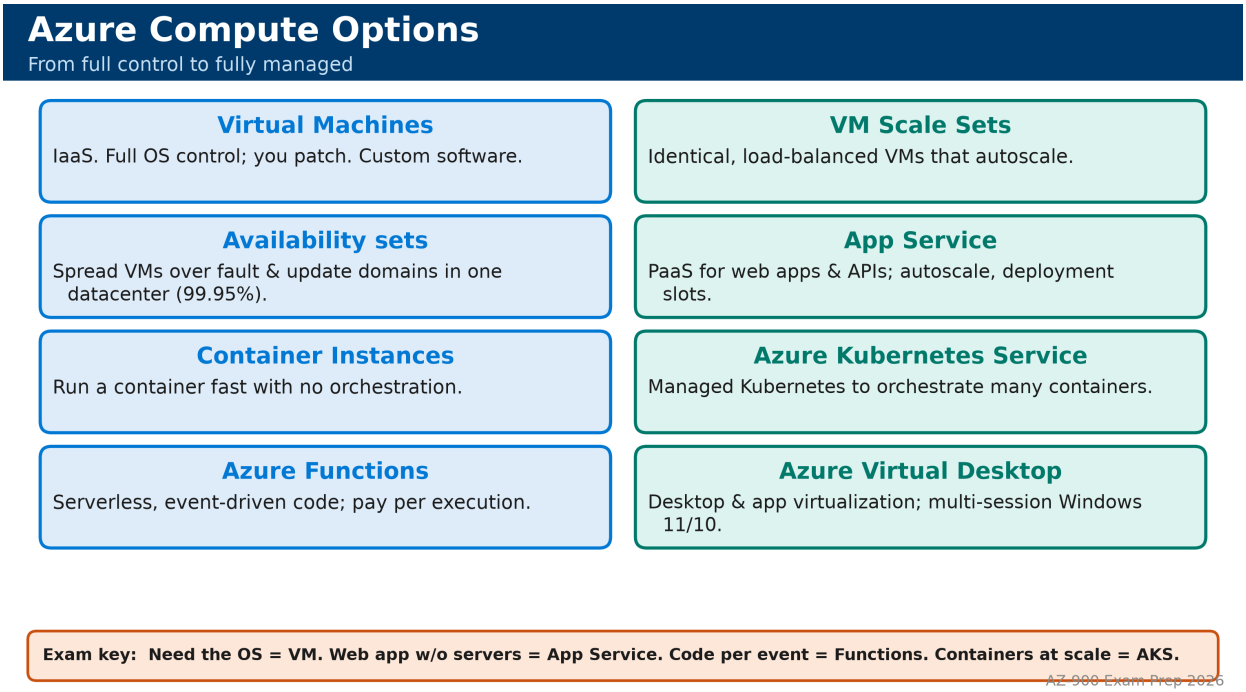


Figure Q72: Azure compute options

About this figure

- Read left column then right: options run from full control (VMs) to fully managed or serverless (Functions); pick by how much you must manage.
- VM sizes range from tiny B-series (B1s approx. USD 8/month Linux) to large memory and GPU VMs; per-second billing applies while running.
- VM Scale Sets support up to 1,000 instances (600 with custom images); availability sets use up to 3 fault domains and 20 update domains.
- App Service plans set the price: Free F1 USD 0, Basic B1 approx. USD 13/month Linux, Standard S1 approx. USD 70/month with autoscale and slots.
- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

Question 73

AZURE ARCHITECTURE & SERVICES

Data & AI
v5 Q133

[Azure Databricks] is an Apache Spark-based analytics service.

✔ **A.** No change is needed.

✖ B. Azure Data Factory.

✖ C. Azure DevOps.

✖ D. Azure HDInsight.

Correct answer: No change is needed.

Clarification (read this first): 'No change is needed' means the sentence is already true. HDInsight can also run Spark, but the sentence is correct as written, so do not change it.

Answer explained

✔ **No change is needed.**
Azure Databricks is Microsoft's Apache Spark-based analytics platform (Spark is a big-data engine). It has shared notebooks and Spark clusters that scale automatically.

✖ **Azure Data Factory.**
Data Factory moves and transforms data (ETL - extract, transform, load). It is not a Spark analytics engine.

✖ **Azure DevOps.**
Azure DevOps gives code repos, pipelines and boards for software delivery, not data analytics.

✖ **Azure HDInsight.**
HDInsight runs managed open-source clusters (Hadoop, Kafka, Spark). But the standard answer for Spark-based analytics is Databricks.

EXAM STRATEGY
'Apache Spark-based analytics' is the phrase that defines Databricks. HDInsight also offers Spark, but the sentence is already true. Data Factory runs ETL; DevOps is CI/CD (build and release).

IMPACT
Databricks gives managed Spark for data engineering and ML without constant cluster care. Matching analytics services comes up across the exam.

What · Where · When · Why · How

What	Azure Databricks is an Apache Spark-based analytics platform with shared notebooks.
Where	Deployed as an Azure Databricks workspace resource in a subscription.
When	When teams need Spark-based big data processing, data engineering or machine learning.
Why	Microsoft describes it as the Apache Spark-based analytics service, so no change is needed.
How	Create a workspace, start Spark clusters that scale automatically, and run notebooks in Python, SQL or Scala.

Exam tip
Spark-based + shared notebooks = Databricks.
Hadoop family / open-source clusters = HDInsight.
ETL = Data Factory.

Memory trick
Databricks lights the Spark.

Real-world example, use case and scenario

EXAMPLE
Notebook job
A data engineer opens a Databricks notebook, reads Parquet files from ADLS Gen2 with PySpark and writes a Delta table.

USE CASE
Retail demand forecasting
A grocery chain uses Databricks Spark clusters to process billions of sales and train demand forecast models.

SCENARIO
Picking the engine
A telecom analytics team compares HDInsight and Databricks for a new Spark workload. They pick Databricks for auto-scaling clusters and shared notebooks. Jobs that took hours on their old Hadoop now finish in minutes.

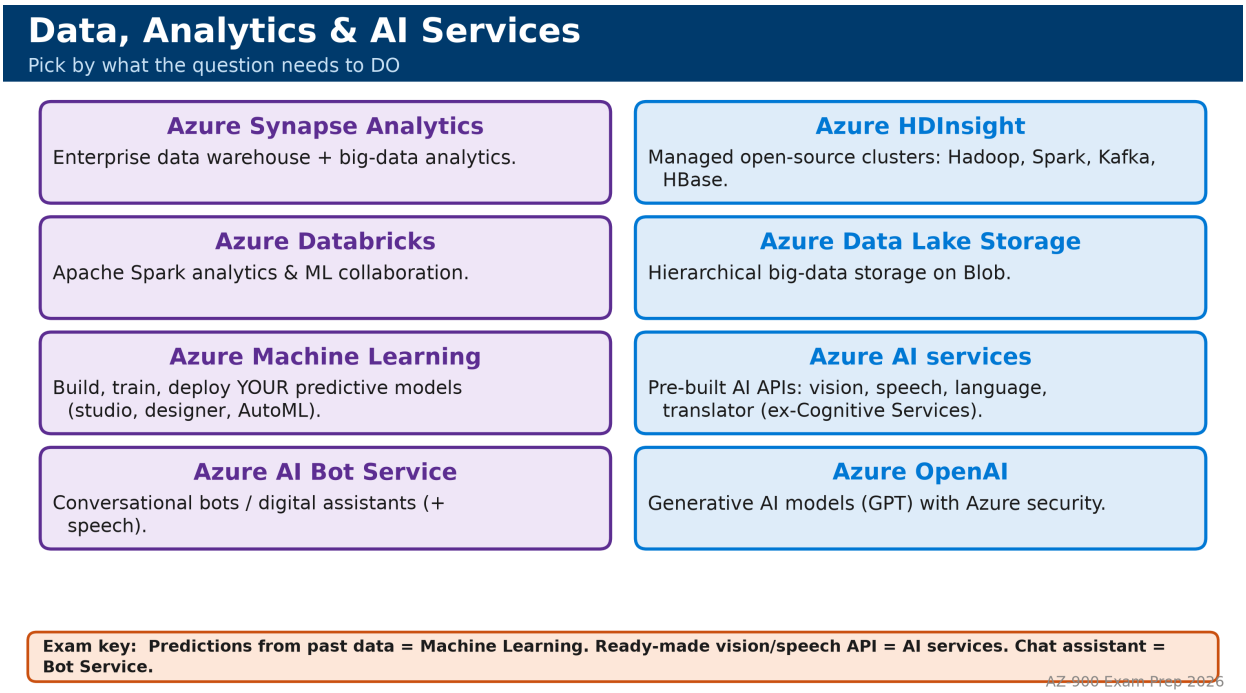


Figure Q73: Data, analytics and AI services

About this figure

- Left column covers data and analytics platforms; right column covers AI. Decide whether the scenario needs storage, processing, custom models or ready-made AI.
- Azure Machine Learning trains your own models from your data; Azure AI services are prebuilt APIs needing no data science, billed per transaction.
- Azure AI services include Vision, Speech, Language, Translator and Document Intelligence; many offer a free F0 tier with limited monthly calls.
- Azure OpenAI models are now offered through Microsoft Foundry (formerly Azure AI Foundry), billed per 1,000 or million tokens, varying widely by model.
- Microsoft Fabric is the newer unified analytics SaaS combining data engineering, warehousing and Power BI, and overlaps much of Synapse.

Question 74

AZURE ARCHITECTURE & SERVICES

Cloud concepts
v5 Q134

Azure Site Recovery provides [fault tolerance] for virtual machines.

✖ A. No change is needed.

✔ **B.** Disaster recovery.

✖ C. Elasticity.

✖ D. High availability.

Correct answer: Disaster recovery.

Clarification (read this first): Watch the difference: fault tolerance means no interruption at all. Site Recovery copies VMs to another place and switches over after a disaster, with a short outage. That is disaster recovery.

Answer explained

✔ **Disaster recovery.**
Site Recovery copies (replicates) VMs to a second region or to Azure. It runs failover (switching to the copy) and failback. This is disaster recovery, with set RPO/RTO (data-loss and recovery-time targets).

✖ **No change is needed.**
Fault tolerance means no interruption when a part fails. Site Recovery uses a failover, with some downtime and possible data loss.

✖ **Elasticity.**
Elasticity is automatic scaling with demand. It has nothing to do with copying or failover.

✖ **High availability.**
High availability keeps a service up inside a region with extra copies (sets, zones). It does not fail over to another site.

EXAM STRATEGY
'Site Recovery' means copying to another region so you can fail over: disaster recovery. Fault tolerance and high availability are designs inside a region. Elasticity is scaling.

IMPACT
Site Recovery with tested failover cuts recovery time from days to minutes after a regional disaster. It also helps pass business continuity audits.

What · Where · When · Why · How

What	Azure Site Recovery is a disaster recovery service that copies machines and fails them over.
Where	Set up in a Recovery Services vault. It copies to a second Azure region, or from on-premises to Azure.
When	When workloads must be recovered after a site or region outage.
Why	It gives failover and failback with set RPO/RTO, which is disaster recovery.
How	Turn on replication per VM, build a recovery plan, then run test or real failovers.

Exam tip
Site Recovery = DR (copy + fail over). Backup = data protection. Availability zones/sets = HA/fault tolerance.

Memory trick
Site Recovery: when the site dies, recovery moves to another region.

Real-world example, use case and scenario

EXAMPLE
Region failover
VMs in West Europe are copied to North Europe. A recovery plan starts them there in the right order.

USE CASE
Hospital EHR DR
A regional hospital copies its on-premises VMware servers that run patient records to Azure with Site Recovery. It does not need a second datacenter.

SCENARIO
Flood at the primary site
A manufacturer's main datacenter floods. Site Recovery has been copying its ERP VMs to Azure, so IT runs the recovery plan. ERP is back online in Azure within an hour, with only minutes of data lost.

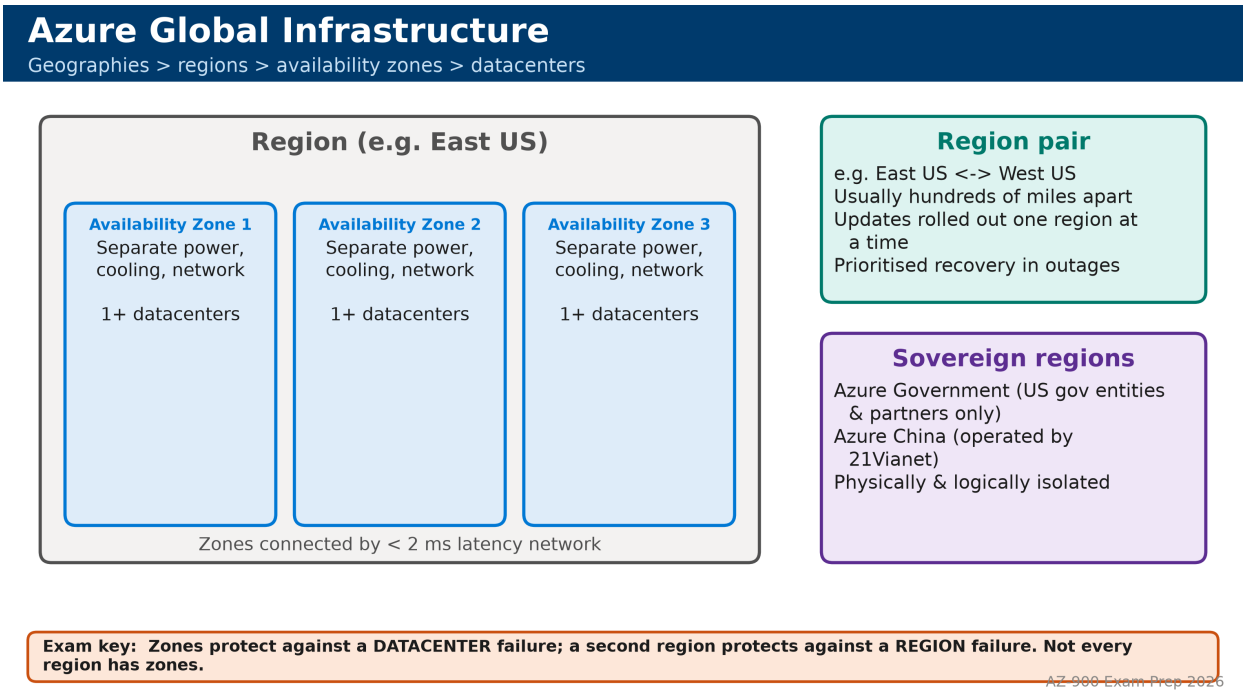


Figure Q74: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

Question 75

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q135, Q206

An Availability Zone in Azure has physically separate locations [across two continents].

✖ A. No change is needed.

✔ **B.** Within a single Azure region.

✖ C. Within multiple Azure regions.

✖ D. Within a single Azure datacenter.

Correct answer: Within a single Azure region.

Clarification (read this first): Think in sizes: datacenter < availability zone < region < geography. Zones are separate datacenters inside ONE region, never across continents. Q126: not every region has zones.

Answer explained

✔ **Within a single Azure region.**
Availability zones are physically separate datacenters inside one Azure region. Each has its own power, cooling and networking.

✖ **No change is needed.**
Zones are kilometres apart inside one region, not on different continents. Continental distance needs more than one region.

✖ **Within multiple Azure regions.**
Each zone belongs to exactly one region. Spanning regions is what region pairs and multi-region designs do.

✖ **Within a single Azure datacenter.**
A zone is itself one or more datacenters, and zones are separate. So they cannot share one datacenter.

EXAM STRATEGY
Availability zones are inside one region. Remove 'two continents' and 'multiple regions' (region pairs). Remove 'single datacenter', since zones are separate datacenters.

IMPACT
Putting zones in the wrong place leads to wrong DR (disaster recovery) plans. Zones protect against datacenter loss; region pairs protect against regional disasters.

What · Where · When · Why · How

What	Availability zones are physically separate datacenters with their own power, cooling and networking.
Where	Inside a single zone-enabled Azure region, usually three zones.
When	When a workload must survive the loss of a datacenter without leaving the region.
Why	Zones are close enough for fast links but far enough apart to fail separately.
How	Deploy resources into different zones, or use zone-redundant (spread across zones) services.

Exam tip
Ladder: availability set = racks in a datacenter; availability zone = datacenters in a region; region pair = regions in a geography.

Memory trick
Zones are areas of one city (region), not countries on different continents.

Real-world example, use case and scenario

EXAMPLE
Three zones
East US 2 has zones 1, 2 and 3. A VM in zone 1 and one in zone 3 survive a fire in either building.

USE CASE
Payment API resilience
A fintech spreads its payment API VMs and zone-redundant SQL across three zones in UK South to meet a 99.99% target.

SCENARIO
Misreading 'zone'
A startup thinks zone 2 is in another country and skips DR planning. A review shows all zones are in the same region. They add a second region for disaster recovery, as well as zones.

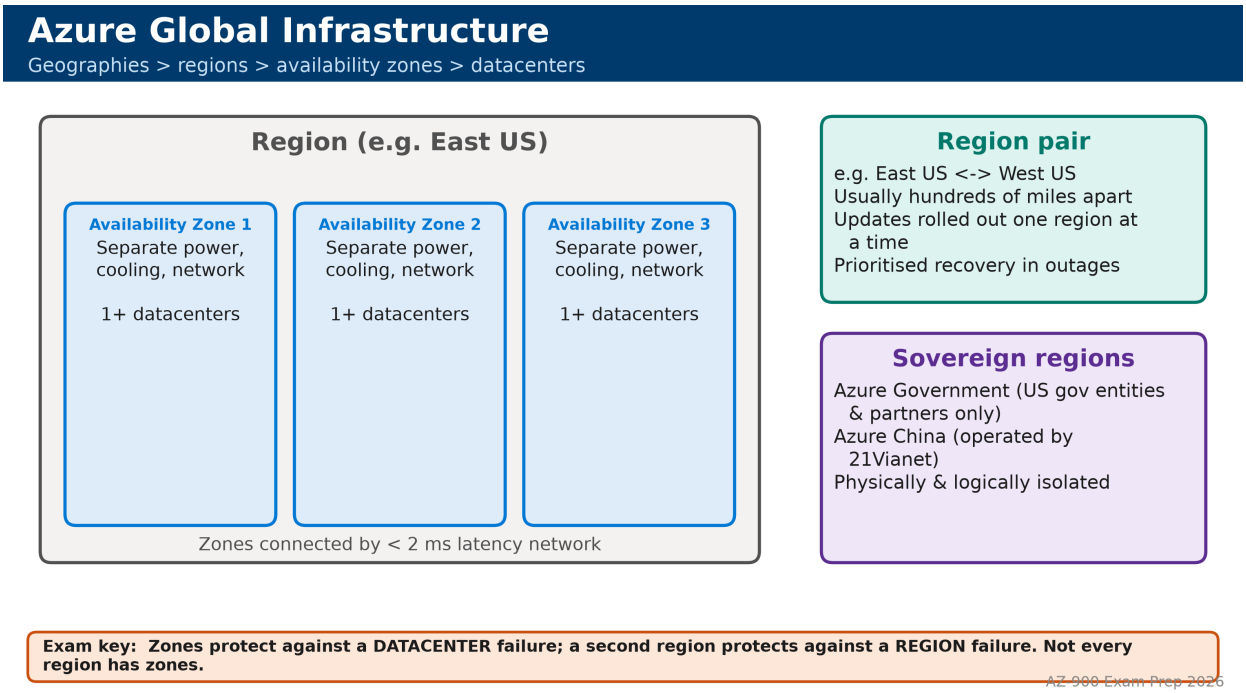


Figure Q75a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

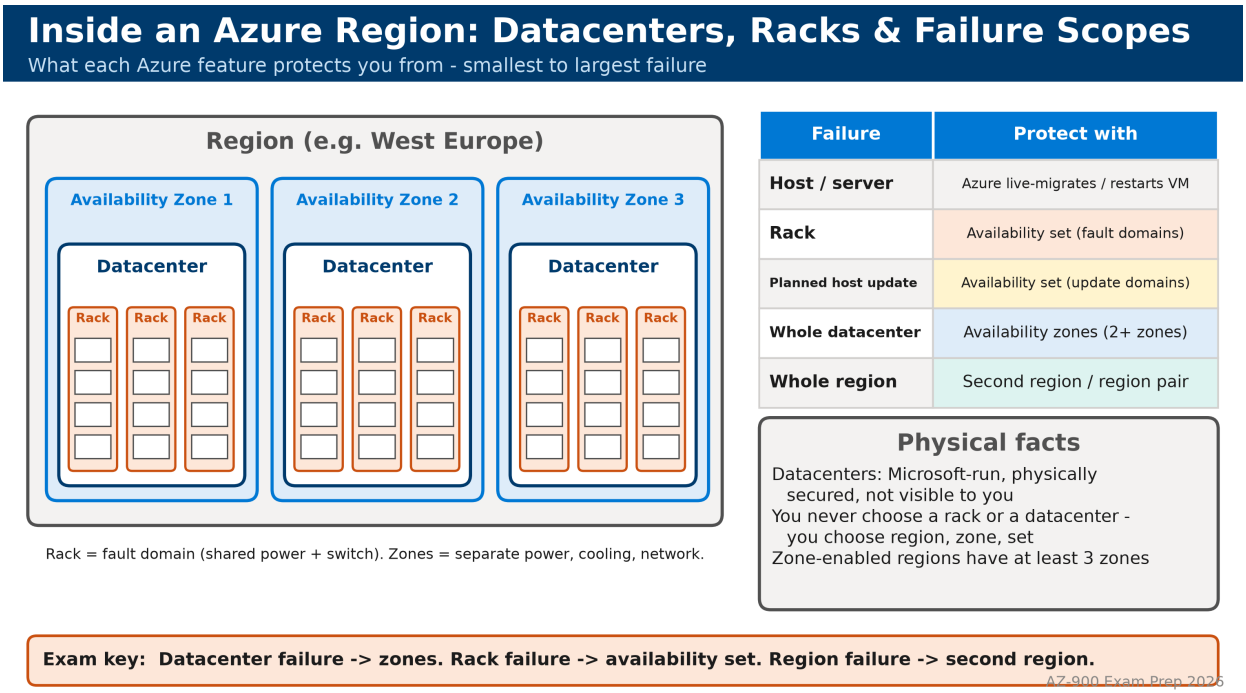


Figure Q75b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 76

AZURE MANAGEMENT & GOVERNANCE — **Compliance** v5 Q136

Your company plans to migrate all on-premises data to Azure. You need to identify whether Azure complies with the company regional requirements. What should you use?

✖ A. The Knowledge Center.

✖ B. Azure Marketplace.

✖ C. The MyApps portal.

✔ **D.** The Trust Center.

Correct answer: The Trust Center.

Clarification (read this first): The question asks whether AZURE (Microsoft) meets the rules. That is Microsoft's published compliance information: the Trust Center. Q82 asks whether YOUR environment meets the rules - that is Compliance Manager.

2026 update: The Trust Center content now sits next to the Service Trust Portal and the 'Azure compliance offerings' pages on Microsoft Learn. Today, for regional rules, you browse that compliance list or download audit reports from the Service Trust Portal.

Answer explained

✔ **The Trust Center.**
The Trust Center (with the Service Trust Portal) shows how Azure meets regional and national standards, such as GDPR or IRAP.

✖ **The Knowledge Center.**
Knowledge Center is self-help troubleshooting content, not compliance proof.

✖ **Azure Marketplace.**
Azure Marketplace is a catalogue of ready-made solutions to deploy, not compliance documents.

✖ **The MyApps portal.**
MyApps is the Entra ID user portal for opening assigned SSO (single sign-on) apps.

EXAM STRATEGY
'Whether Azure complies' with regional rules is about Microsoft's own certificates: Trust Center or Service Trust Portal. Marketplace, MyApps and Knowledge Center are distractors (wrong answers made to look right).

IMPACT
Checking Microsoft's regional certifications before migrating stops you moving regulated data into a service or region that does not comply.

What · Where · When · Why · How

What	The Microsoft Trust Center publishes Microsoft's security, privacy and compliance promises and offerings.
Where	Online at Microsoft's Trust Center, next to the Service Trust Portal and Azure compliance documentation.
When	When checking if Azure meets regional or national rules before migrating.
Why	It lists regional compliance offerings such as GDPR, IRAP and C5.
How	Browse the compliance offerings for your region. Download audit reports from the Service Trust Portal.

Exam tip
Compliance documents = Trust Center / Service Trust Portal. Checking your own controls = Compliance Manager. MyApps and Marketplace are distractors.

Memory trick
Trust Center tells you whether you can trust Azure in your region.

Real-world example, use case and scenario

EXAMPLE
Checking IRAP
An Australian agency checks Microsoft's compliance offerings to confirm Azure holds an IRAP assessment for its planned regions.

USE CASE
EU data residency check
A German insurer reads Trust Center pages on GDPR and the EU Data Boundary before approving a migration.

SCENARIO
Legal sign-off
A Canadian credit union's legal team asks if Azure meets local privacy rules. IT points them to the Trust Center and downloads audit reports from the Service Trust Portal. Legal approves without paying for a separate audit.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

AZ-900 Exam Prep 2026

Figure Q76: Compliance, privacy and trust

About this figure

- Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.
- Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.
- Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.
- Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.
- The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 77

AZURE ARCHITECTURE & SERVICES

Security v5 Q137

Azure Key Vault is used to store secrets for [Microsoft Entra ID user accounts].

✖ A. No change is needed.

✖ B. Microsoft Entra ID administrative accounts.

✖ C. Personally Identifiable Information (PII).

✔ **D.** Server applications.

Correct answer: Server applications.

Clarification (read this first): Key Vault holds secrets that APPLICATIONS use (connection strings, API keys). It does not hold user or admin passwords - Entra ID protects those.

Answer explained

✔ **Server applications.**
Key Vault stores secrets, keys and certificates that server applications read while running. Ideally they use a managed identity (an automatic Azure identity for the app). This keeps secrets out of code.

✖ **No change is needed.**
Entra ID stores and protects user passwords itself. They are not put in Key Vault.

✖ **Microsoft Entra ID administrative accounts.**
Admin accounts are protected with PIM, MFA and Conditional Access, not by storing their passwords in Key Vault.

✖ **Personally Identifiable Information (PII).**
Key Vault is not a store for business records. PII (personal data) belongs in encrypted databases, and Key Vault may hold their keys.

EXAM STRATEGY
Key Vault stores secrets that applications use, such as connection strings and keys. User and admin passwords live in Entra ID. PII is not its job.

IMPACT
Moving app secrets into Key Vault with managed identities removes passwords from config files. Leaked config files are a top cause of credential leaks.

What · Where · When · Why · How

What	Azure Key Vault is a managed store for secrets, encryption keys and certificates.
Where	A Key Vault resource in a resource group. Access is through RBAC or access policies.
When	When apps need connection strings, API keys or certificates without writing them into code.
Why	Server applications read secrets while running, which keeps them out of code and config.
How	Store the secret in the vault. Give the app's managed identity permission to read it.

Exam tip
Key Vault = secrets, keys, certificates for apps and services. If the option says users, admins or business data, it is a trap.

Memory trick
Key Vault is the app's keyring, not the people's password book.

Real-world example, use case and scenario

EXAMPLE
Connection string
An App Service app reads its SQL connection string from Key Vault, using a Key Vault reference and its managed identity.

USE CASE
E-commerce payment key
An online shop keeps its payment gateway API key in Key Vault. Developers never see the production value.

SCENARIO
Leaked secret in GitHub
A media company finds a database password in a public code repository. They change it, move all secrets to Key Vault and let the web servers read them through managed identities. New code holds only the vault address.

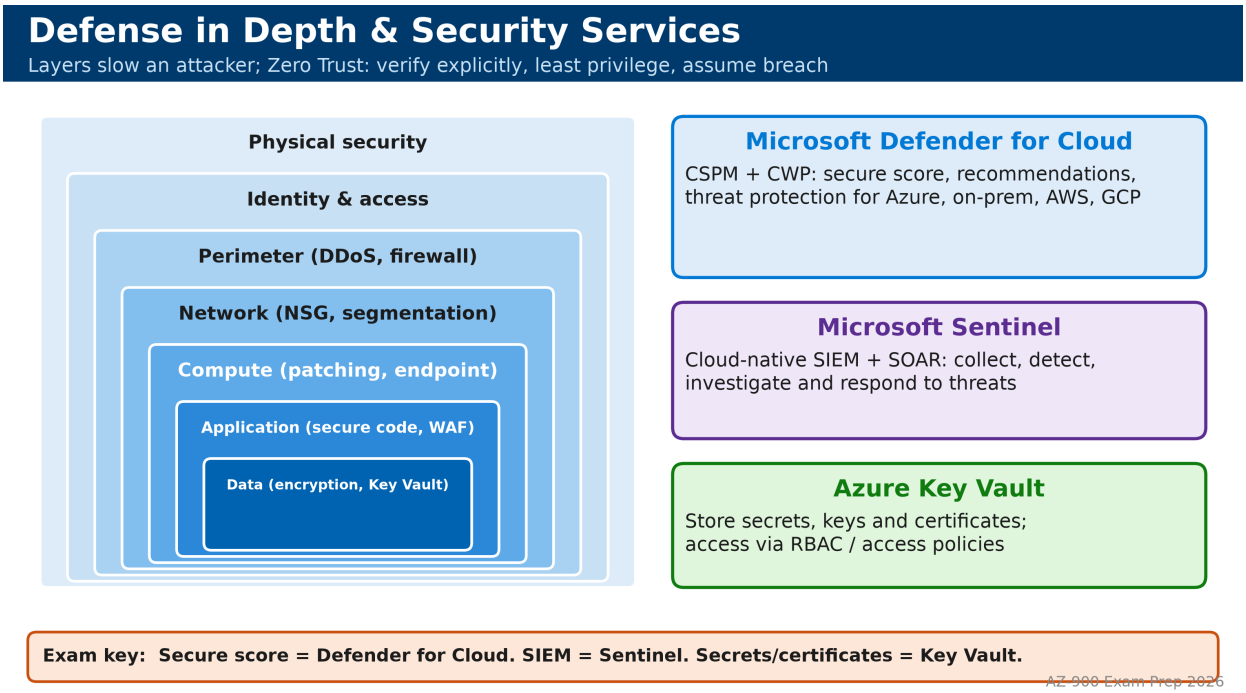


Figure Q77: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 78

AZURE ARCHITECTURE & SERVICES | **Networking** | v5 Q138, Q96, Q266

After you create a virtual machine, you need to modify the [Network Security Group (NSG)] to allow connections to TCP port 8080 on the virtual machine.

✔ **A.** No change is needed.

✖ B. Virtual network gateway.

✖ C. Virtual network.

✖ D. Route table.

Correct answer: No change is needed.

Clarification (read this first): 'No change is needed' means the sentence is already true. An NSG (Network Security Group - a list of allow/deny rules) is the right thing to change to open port 8080.

Answer explained

✔ **No change is needed.**
An NSG filters traffic by port and protocol. Adding an inbound allow rule for TCP 8080 opens that port to the VM.

✖ **Virtual network gateway.**
A VNet gateway connects networks over VPN or ExpressRoute. It has no port rules for each VM.

✖ **Virtual network.**
A virtual network sets the address space and subnets. Port filtering is done by NSGs attached to it.

✖ **Route table.**
A route table sets where traffic goes next. It does not allow or block ports.

EXAM STRATEGY
'Allow connections to TCP port' means a filter rule, so the NSG sentence is correct. Gateways, VNets and route tables do not allow or deny ports.

IMPACT
Forgetting to open the app port in the NSG is the classic reason a new service times out. This pattern comes up often.

What · Where · When · Why · How

What	A network security group (NSG) is a set of allow/deny rules that filter traffic by IP, port and protocol.
Where	Attached to a VM's network interface or to its subnet.
When	When a port such as TCP 8080 must be opened or blocked on a VM.
Why	Default rules block inbound internet traffic, so only an NSG rule can allow port 8080.
How	Add an inbound security rule for TCP 8080 with Allow and a suitable priority.

Exam tip
Open or close a port on a VM = NSG rule. Route tables steer traffic; gateways connect networks.

Memory trick
NSG = Never Skip the Gatekeeper for ports.

Real-world example, use case and scenario

EXAMPLE
Port 8080 rule
Inbound rule: priority 300, source Any, destination port 8080, protocol TCP, action Allow.

USE CASE
Tomcat app server
A logistics firm runs a Java app on Tomcat on port 8080. An NSG rule opens it only to its office IP range.

SCENARIO
App installed but unreachable
A developer installs Jenkins on a VM, but the browser times out on port 8080. The NSG only allows 22 and 3389. Adding a TCP 8080 allow rule fixes it at once, with no change to routes or the VNet.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

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Figure Q78: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 79

AZURE ARCHITECTURE & SERVICES

Security
v5 Q139, Q10

Your Azure environment contains multiple Azure virtual machines. You need to ensure that a virtual machine named VM1 is accessible from the Internet over HTTP. Solution: You modify a DDoS protection plan. Does this meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Asked: does this solution make VM1 reachable? DDoS protection only blocks attack floods; it never opens access. Q170 gives the right fixes: change the NSG or Azure Firewall. Q51 (Traffic Manager) is also No.

Answer explained

✔ **No.**
A DDoS protection plan only blocks flood attacks (huge amounts of fake traffic) on public IPs. It never creates allow rules, so HTTP stays blocked until an NSG rule is added.

✖ **Yes.**
DDoS Protection cleans out attack traffic but gives no access. Inbound HTTP is still controlled by NSG rules.

EXAM STRATEGY
Key fact: DDoS protection only reduces floods and never opens ports. If the solution cannot allow port 80 traffic, answer No.

IMPACT
A DDoS protection plan costs a lot. Buying it to fix connectivity wastes money and does not solve the access problem.

What · Where · When · Why · How

What	Azure DDoS Protection reduces distributed denial-of-service (flood) attacks against public IPs.
Where	Turned on per VNet with a DDoS protection plan, or per public IP with IP Protection.
When	When public endpoints need protection from flood or protocol attacks.
Why	It only drops attack traffic and creates no allow rules, so it cannot open HTTP.
How	It watches traffic to public IPs and removes harmful packets automatically.

Exam tip
DDoS Protection = protection from floods, never 'allow access'. Access questions are answered by NSG rules.

Memory trick
DDoS blocks the mob, it doesn't unlock the door.

Real-world example, use case and scenario

EXAMPLE
Attack mitigation
During a 200 Gbps UDP flood, DDoS Network Protection cleans the traffic to the public IP and sends a report.

USE CASE
Online gaming
A multiplayer game studio turns on DDoS Network Protection on its VNets because rivals often flood its game servers.

SCENARIO
Wrong setting
An intern tries to fix an unreachable web VM by turning on a DDoS protection plan. The site stays unreachable and the plan's monthly fee is charged. The senior engineer adds an NSG rule for port 80 instead.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
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Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

AZ-900 Exam Prep 2026

Figure Q79: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 80

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q140

[Resource groups] provide organizations with the ability to manage the compliance of Azure resources across multiple subscriptions.

✖ A. No change is needed.

✔ **B.** Management groups.

✖ C. Azure policies.

✖ D. Azure App Service plans.

Correct answer: Management groups.

Clarification (read this first): Key phrase: 'across multiple subscriptions'. A resource group sits inside one subscription. Management groups sit above subscriptions. Q17 asks the same thing - Management Groups.

Answer corrected in this edition. The previous PDF printed: *Azure policies.*. The PDF prints 'Azure policies', but the answer key and usual exam answer is 'Management groups'. A management group lets you manage access, policy and compliance across many subscriptions. A policy is assigned once at the management group, and every subscription below inherits (receives) it.

Answer explained

✔ **Management groups.**
Management groups hold many subscriptions. Azure Policy and RBAC assigned to a management group are inherited by (passed down to) every subscription below it.

✖ **No change is needed.**
A resource group sits inside one subscription. So it cannot manage compliance across many subscriptions.

✖ **Azure policies.**
Azure Policy defines the rules, but it needs a scope (a level to apply to). The cross-subscription scope here is the management group.

✖ **Azure App Service plans.**
An App Service plan sets compute for web apps. It has nothing to do with governance or compliance.

EXAM STRATEGY
'Across multiple subscriptions' points to management groups, which sit above subscriptions. Resource groups live inside one subscription. Policies are applied at scopes.

IMPACT
Applying policy at management group level enforces compliance on every subscription, including new ones. This avoids gaps in governance.

What · Where · When · Why · How

What	Management groups are containers that organise many subscriptions into a hierarchy (tree).
Where	Above subscriptions, under the tenant root management group.
When	When policy, RBAC or compliance must apply across many subscriptions.
Why	Assignments at a management group are inherited by all subscriptions below it.
How	Build the hierarchy, move subscriptions in, then assign Azure Policy initiatives (policy sets) at that level.

Exam tip
Key phrase 'across multiple subscriptions' = management groups. Policy is the rule; the management group is the cross-subscription scope it is assigned to.

Memory trick
Many subscriptions? Manage them with a Management group.

Real-world example, use case and scenario

EXAMPLE
Allowed regions everywhere
An 'Allowed locations = EU only' policy on the Production management group applies to all 12 subscriptions under it.

USE CASE
Pharma GxP estate
A drug company puts its regulated subscriptions under a GxP management group and assigns its compliance initiative once.

SCENARIO
Subscription sprawl
A university has 30 department subscriptions, each with hand-assigned policies that drift apart. The cloud team builds a management group hierarchy and moves the assignments to the top. New subscriptions get the rules automatically, and reports cover everything.

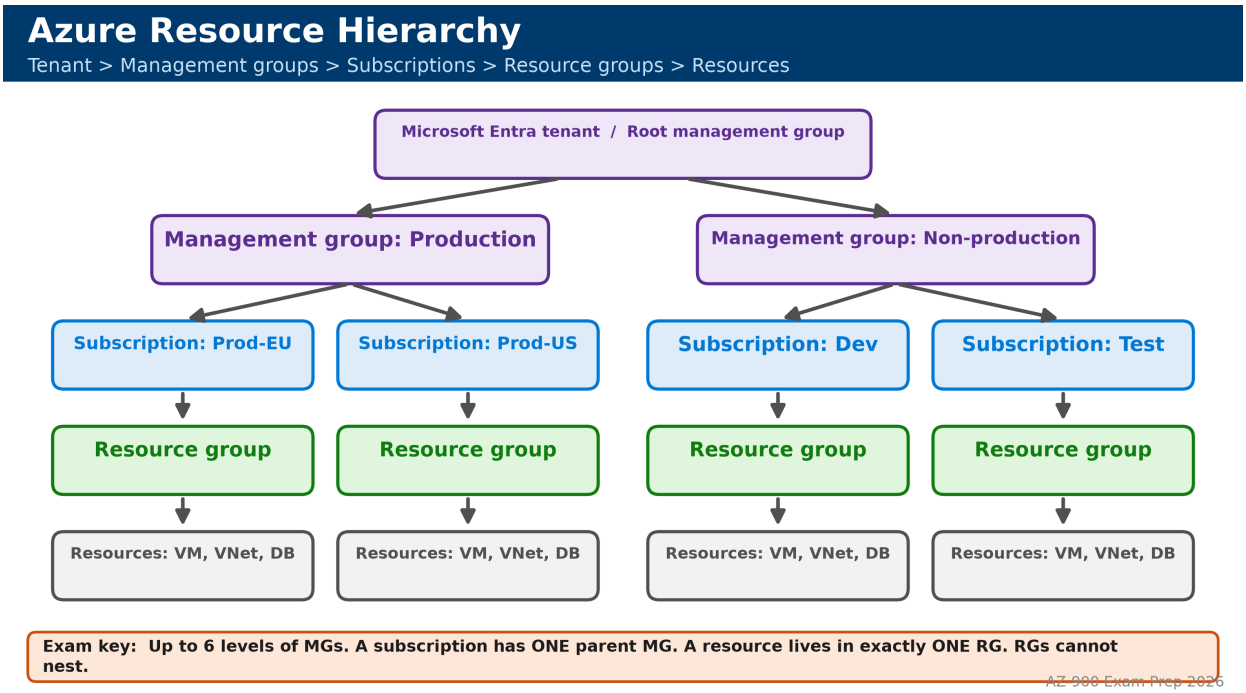


Figure Q80: Azure resource hierarchy

About this figure

• Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.

• A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.

• Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.

• Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.

• Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 81

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q141, Q187

Your network contains an Active Directory forest. The forest contains 5,000 user accounts. Your company plans to migrate all network resources to Azure and to decommission the on-premises data center. You need to recommend a solution to minimize the impact on users after the planned migration. What should you recommend?

✖ A. Implement Azure Multi-Factor Authentication (MFA).

✔ **B.** Sync all the Active Directory user accounts to Microsoft Entra ID.

✖ C. Instruct all users to change their password.

✖ D. Create a guest user account in Microsoft Entra ID for each user.

Correct answer: Sync all the Active Directory user accounts to Microsoft Entra ID.

Clarification (read this first): The goal is 'minimize the impact on users' - users should notice as little change as possible. Syncing keeps the same username and password. Q111 (MFA) and Q112 (password change) are the same scenario - both No.

Answer explained

✔ **Sync all the Active Directory user accounts to Microsoft Entra ID.**
Syncing AD accounts to Entra ID lets users keep the same username and password after the datacenter closes. Use Entra Connect with password hash sync (it copies a scrambled form of the password).

✖ **Implement Azure Multi-Factor Authentication (MFA).**
MFA adds a sign-in step and gives users no cloud accounts. It adds impact instead of reducing it.

✖ **Instruct all users to change their password.**
Forcing password changes adds disruption and moves no accounts to the cloud.

✖ **Create a guest user account in Microsoft Entra ID for each user.**
Guest (B2B) accounts are for outside partners. Employees would get new accounts with limited access.

EXAM STRATEGY
'Minimize impact on users' after closing the datacenter means users keep the same sign-in details. Sync AD to Entra ID with Entra Connect. Password changes and guest accounts add friction.

IMPACT
Synced accounts avoid thousands of help-desk calls and password resets after migration. Users can work normally on day one.

What · Where · When · Why · How

What	Microsoft Entra Connect copies on-premises AD users and groups to Microsoft Entra ID.
Where	Installed on a server in the on-premises domain, syncing to the Entra tenant.
When	When moving to Azure while keeping users' existing accounts.
Why	Users keep the same usernames and passwords, so there is little disruption.
How	Run Entra Connect with password hash sync so accounts and password hashes copy to Entra ID.

Exam tip
'Minimize impact on users' in an AD migration = sync identities (Entra Connect). Users keep one username and password.

Memory trick
Same user, same password, new home: sync, don't rebuild.

Real-world example, use case and scenario

EXAMPLE
Password hash sync
Entra Connect syncs jane@contoso.com and her password hash every 2 minutes. She signs in to the Azure-hosted apps with her usual details.

USE CASE
Law firm cloud move
A law firm with 600 staff syncs its AD to Entra ID before moving file shares and business apps to Azure.

SCENARIO
Avoiding a helpdesk flood
A retail group plans to close its datacenter and thinks about creating new cloud accounts for 5,000 staff. The architect warns of password resets and lost group memberships. They use Entra Connect instead, and on switch-over day the helpdesk gets almost no sign-in tickets.

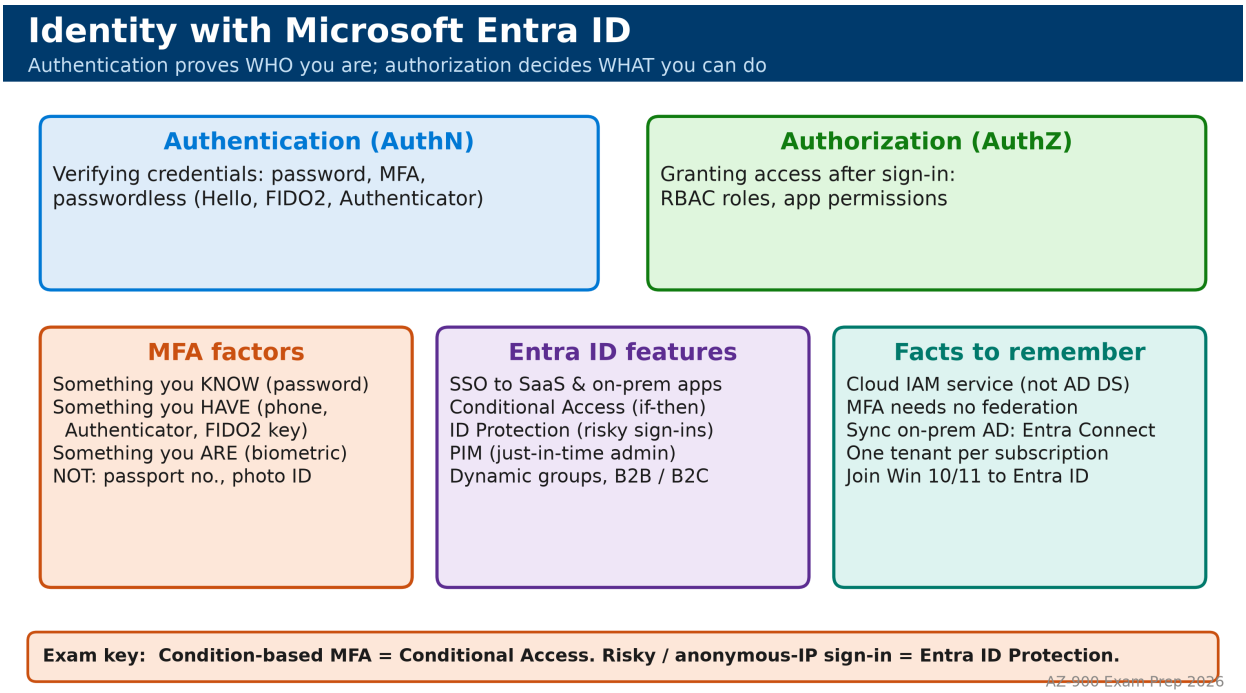


Figure Q81a: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

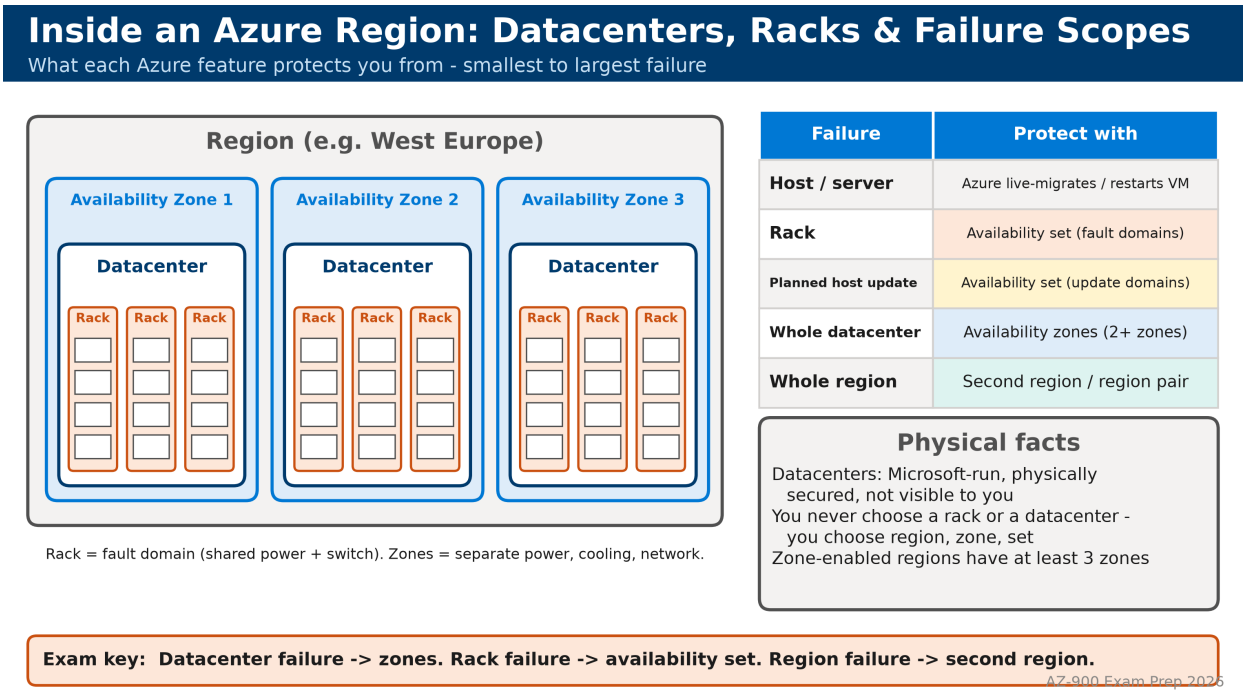


Figure Q81b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 82

AZURE MANAGEMENT & GOVERNANCE | **Compliance** | v5 Q142, Q108, Q358

What should you use to evaluate whether your company Azure environment meets regulatory requirements?

✖ A. The Knowledge Center website.

✖ B. The Advisor blade from the Azure portal.

✔ **C.** Compliance Manager from the Service Trust Portal.

✖ D. The Security Center blade from the Azure portal.

Correct answer: Compliance Manager from the Service Trust Portal.

Clarification (read this first): The question asks whether YOUR company's environment meets regulations. That is Compliance Manager (it gives you a score). Q76 asks whether AZURE itself complies - Trust Center. 'Security Center' is the old name for Defender for Cloud.

2026 update: Compliance Manager has moved out of the Service Trust Portal. It is now Microsoft Purview Compliance Manager, in the Microsoft Purview portal. Security Center is now Microsoft Defender for Cloud. Its regulatory compliance dashboard also checks your Azure resources against standards.

Answer explained

✔ **Compliance Manager from the Service Trust Portal.**
Compliance Manager checks your environment against regulations (GDPR, ISO 27001, NIST). It gives a compliance score and actions to improve. It is now part of Microsoft Purview.

✖ **The Knowledge Center website.**
Knowledge Center is general help content. It does not check your environment.

✖ **The Advisor blade from the Azure portal.**
Advisor gives best-practice tips (cost, security, reliability, performance, operations), not regulatory checks.

✖ **The Security Center blade from the Azure portal.**
Security Center (now Defender for Cloud) focuses on security posture (your overall security state). The expected answer for regulation checks is Compliance Manager.

EXAM STRATEGY
'Evaluate your environment against regulatory requirements' points to Compliance Manager, now in Microsoft Purview. Advisor gives best practice; Knowledge Center is documentation.

IMPACT
Ongoing compliance checks give audit evidence and show gaps early. This lowers the risk of fines and cuts audit preparation time.

What · Where · When · Why · How

What	Compliance Manager checks compliance against regulations. It gives a score and actions to improve.
Where	First reached through the Service Trust Portal; now in the Microsoft Purview portal.
When	When checking if your environment meets standards such as GDPR or ISO 27001.
Why	It maps Microsoft's controls and your own, and scores your progress.
How	Create an assessment from a regulation template, then complete the suggested improvement actions.

Exam tip
Regulatory assessment with a score = Compliance Manager. Audit reports to download = Service Trust Portal. Security posture = Defender for Cloud.

Memory trick
Compliance Manager manages your compliance homework and grades it.

Real-world example, use case and scenario

EXAMPLE
GDPR assessment
An admin creates a GDPR assessment in Compliance Manager. It shows a score of 62% and a list of actions the customer must do.

USE CASE
Healthcare HIPAA readiness
A telehealth startup uses Compliance Manager's HIPAA template. It tracks which safeguards Microsoft covers and which it must add itself.

SCENARIO
Auditor's question
An auditor asks a payments company how it knows it meets PCI DSS. The compliance officer opens Compliance Manager and exports the PCI assessment with evidence. She gives the remaining actions to owners, and the audit closes with no major findings.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

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Figure Q82: Compliance, privacy and trust

About this figure

- Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.
- Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.
- Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.
- Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.
- The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 83

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q145

Your company has an Azure subscription that contains the following unused resources: 20 user accounts in Microsoft Entra ID. Five groups in Microsoft Entra ID. 10 public IP addresses. 10 network interfaces. You need to reduce the Azure costs for the company. Solution: You remove the unused network interfaces. Does this meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Asked: does deleting the unused network interfaces (NICs) lower the bill? NICs are free, so No. Q235 uses the same list: only the public IP addresses cost money. Q84 (user accounts) is also No.

Answer explained

✔ **No.**
Network interfaces (NICs - the virtual network card that links a VM to a network) have no Azure charge. Deleting unused NICs saves nothing. Only the public IP addresses in the list cost money.

✖ **Yes.**
There is no price for NICs on the Azure price list. Removing them leaves the bill the same.

EXAM STRATEGY
Decide with one fact: does the item cost money on its own? A NIC is free. Items that use paid capacity, like public IPs, have a charge. So No.

IMPACT
Cleaning up free items wastes time while the real costs keep growing. This four-part 'unused resources' series is common on the exam.

What · Where · When · Why · How

What	A network interface (NIC) connects a VM to a virtual network.
Where	A separate resource in a resource group, attached to a VM.
When	When reviewing unused resources to cut Azure costs.
Why	NICs have no charge, so removing them does not lower the bill.
How	Cost Management shows no charge for NICs. Only paid resources like public IPs cut costs.

Exam tip
Free: Entra users/groups, NICs, VNets, resource groups. Charged: public IPs, disks, running VMs, storage.

Memory trick
A NIC costs nothing.

Real-world example, use case and scenario

EXAMPLE
Orphaned NIC
After a VM is deleted, its NIC stays in the resource group. Cost Management shows no charges for it.

USE CASE
Cleanup priorities
A SaaS company's cost team deletes unattached disks and public IPs before NICs, because only those cost money.

SCENARIO
Expected savings that never came
An admin deletes 10 unused NICs and promises lower costs. The next invoice is the same. Cost Management shows the 10 idle public IPs were the real cost, and deleting them brings the saving.

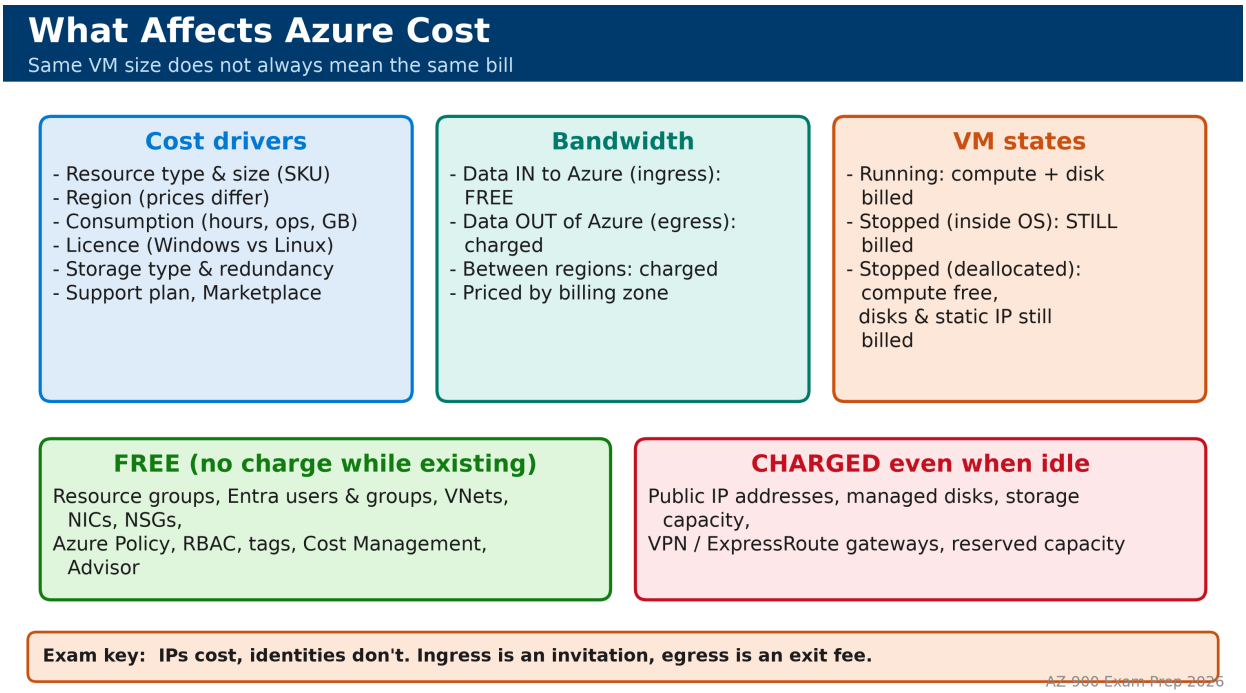


Figure Q83a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

• Bandwidth: ingress is free; the first 100 GB/month of Internet egress is free, then approx. USD 0.087/GB in North America and Europe.

• Inter-region transfer is approx. USD 0.02/GB within a continent; VNet peering is approx. USD 0.01/GB each way in the same region.

• A VM stopped inside the guest OS is still allocated and billed for compute; only Stopped (deallocated) from the portal, CLI or API stops compute charges.

• A Standard static public IP costs approx. USD 3.65/month even unattached; a 128 GB Premium SSD costs approx. USD 20/month even when its VM is deallocated.

• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q83b: Approximate Azure charges (USD)

About this figure

• Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.

• Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.

• Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.

• Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.

• Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 84

AZURE MANAGEMENT & GOVERNANCE

Cost

v5 Q147, Q53, Q54, Q149

Your company has an Azure subscription that contains the following unused resources: 20 user accounts in Microsoft Entra ID. Five groups in Microsoft Entra ID. 10 public IP addresses 10. network interfaces. You need to reduce the Azure costs for the company. Solution: You remove the unused user accounts. Does this meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Asked: does deleting unused user accounts lower the AZURE bill? User accounts are free. Licences can cost money, but no licences are mentioned. Q235 uses the same list: only the public IPs cost money.

Answer explained

✔ **No.**
Entra ID user accounts have no Azure usage charge. So deleting unused accounts does not lower Azure costs.

✖ **Yes.**
Directory user accounts are free. No licences are mentioned, so deleting them saves no Azure money.

EXAM STRATEGY
Directory objects (users, groups) never have an Azure usage charge. If the item is an Entra ID object, the answer to 'reduces Azure cost' is No.

IMPACT
Licences cost money, user accounts do not. Mixing them up leads to deleting accounts for nothing while paid network resources keep charging.

What · Where · When · Why · How

What	Microsoft Entra ID user accounts are identities in the tenant directory.
Where	In the Entra tenant linked to the subscription, not in a resource group.
When	When cleaning up old accounts, usually for security reasons.
Why	User accounts have no Azure usage charge, so deleting them saves nothing.
How	Delete or disable unused users in Entra ID. The Azure bill stays the same.

Exam tip
Directory objects (users, groups) are free. In this series, only removing the public IPs is Yes.

Memory trick
Users ride free on the Azure bill.

Real-world example, use case and scenario

EXAMPLE
No user charge
Adding or removing 20 Entra users changes nothing in Cost Management's Azure usage view.

USE CASE
Security cleanup
A charity removes old accounts for security, not cost, using Entra access reviews.

SCENARIO
Wrong lever
A finance manager asks IT to delete 20 unused accounts to cut Azure spending. IT explains users are free, though old accounts are a security risk. They delete them for security and remove idle public IPs for the real saving.

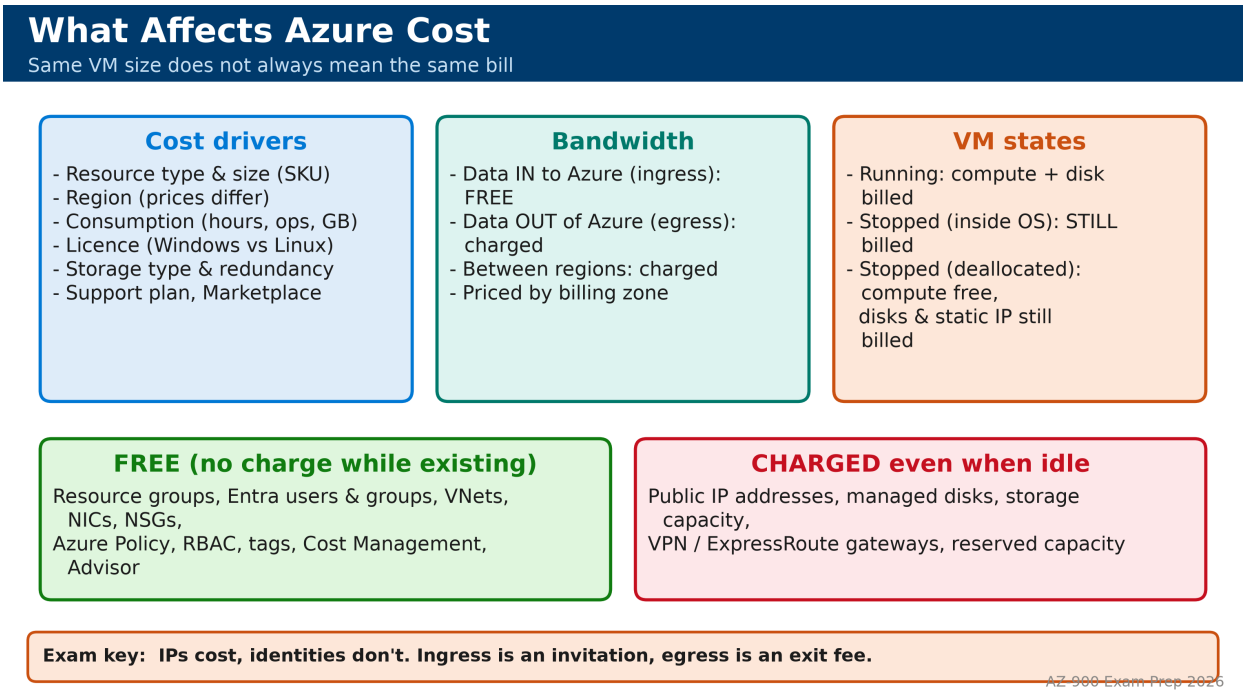


Figure Q84a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

• Bandwidth: ingress is free; the first 100 GB/month of Internet egress is free, then approx. USD 0.087/GB in North America and Europe.

• Inter-region transfer is approx. USD 0.02/GB within a continent; VNet peering is approx. USD 0.01/GB each way in the same region.

• A VM stopped inside the guest OS is still allocated and billed for compute; only Stopped (deallocated) from the portal, CLI or API stops compute charges.

• A Standard static public IP costs approx. USD 3.65/month even unattached; a 128 GB Premium SSD costs approx. USD 20/month even when its VM is deallocated.

• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

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Figure Q84b: Approximate Azure charges (USD)

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• Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.

• Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.

• Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 85

AZURE MANAGEMENT & GOVERNMENT

Support
v5 Q148

You can create an Azure support request from [support.microsoft.com].

✖ A. No change is needed.

✔ **B.** The Azure portal.

✖ C. The Knowledge Center.

✖ D. The Security & Compliance admin center.

Correct answer: The Azure portal.

Clarification (read this first): The bracketed website is wrong. support.microsoft.com is for consumer and Microsoft 365 help. Azure support tickets are created in the Azure portal.

Answer explained

✔ **The Azure portal.**
You create Azure support requests in the Azure portal, under Help + support > Create a support request.

✖ **No change is needed.**
support.microsoft.com is for consumer and Microsoft 365 product help, not Azure subscription tickets.

✖ **The Knowledge Center.**
Knowledge Center has self-help content. It cannot open support tickets.

✖ **The Security & Compliance admin center.**
The Security & Compliance center managed Microsoft 365 compliance settings, not Azure support tickets.

EXAM STRATEGY
Signal: 'Azure support request'. Help + support is in the Azure portal. Remove support.microsoft.com, Knowledge Center and the compliance admin center; none of them open Azure tickets.

IMPACT
Quota increases and outage tickets all go through the portal's Help + support page. Knowing this saves time during a live problem.

What · Where · When · Why · How

What	An Azure support request is a ticket for technical, billing, subscription or quota issues.
Where	Azure portal, under Help + support.
When	When you need Microsoft's help with an Azure subscription or resource problem.
Why	The portal is the official place to create Azure support tickets.
How	Select Help + support > Create a support request. Choose the issue type, subscription and severity (how urgent).

Exam tip
Create an Azure support ticket = Azure portal, Help + support. Quota increases are raised the same way.

Memory trick
Trouble in the portal? Ticket from the portal.

Real-world example, use case and scenario

EXAMPLE
Quota increase
An admin selects Help + support > Create a support request > Service and subscription limits (quotas) to raise a vCPU limit.

USE CASE
Retailer severity A ticket
A retailer on the Professional Direct plan opens a Severity A (most urgent) ticket from the portal during a database outage.

SCENARIO
Wrong website
An engineer at a hotel chain spends 20 minutes on support.microsoft.com looking for an Azure ticket form. A colleague shows him Help + support in the portal. He files the ticket for the right subscription in two minutes.

Azure Support Plans

Plan names appear in older dumps; know the ladder

Feature	Basic	Developer	Standard	Prof. Direct	Unified
Price	Free	Low	Mid	Higher	Enterprise
Billing & subscription	Yes	Yes	Yes	Yes	Yes
Advisor, Service Health, docs	Yes	Yes	Yes	Yes	Yes
Technical support	No	Email, bus. hrs	24x7 phone/email	24x7 phone/email	24x7
Sev A (< 1 h)	No	No	Yes	Yes	Yes
Architecture guidance	No	General	General	Reviews (ProDirect)	Customer-specific
Proactive / account mgr	No	No	No	Delivery mgr pool	Technical account mgr

Exam key: Basic = billing only (free). Developer = test/non-prod, email in business hours. Standard = cheapest 24x7 production.

Premier support has been replaced by Unified support. Support plans are no longer on the current skills outline.

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Figure Q85: Azure support plans

About this figure

- Columns climb from free Basic to enterprise Unified; read down a column to see what each plan adds, especially 24x7 support and Sev A.
- Approximate list prices: Developer ~USD 29/month, Standard ~USD 100/month, Professional Direct ~USD 1,000/month; Unified is a custom enterprise contract.
- Initial response targets: Developer under 8 business hours; Standard Sev A under 1 hour; Professional Direct Sev A also under 1 hour with escalation management.
- Premier was replaced by Unified support. Every plan includes billing and subscription help, documentation, Azure Advisor, Service Health and community forums.
- Support plans are purchased per billing scope; free accounts get Basic only, so technical issues require upgrading or using community support.

Question 86

CLOUD CONCEPTS

Lifecycle
v5 Q150, Q311

All Azure services that are in public preview are [provided without any documentation].

✖ A. No change is needed.

✖ B. Only configurable from Azure CLI.

✔ **C.** Excluded from the Service Level Agreements.

✖ D. Only configurable from the Azure portal.

Correct answer: Excluded from the Service Level Agreements.

Clarification (read this first): Watch the word 'All' and the bracketed words. Preview services DO have documentation. What they lack is an SLA (uptime promise). Q19: previews are not limited to the CLI either.

Answer explained

✔ **Excluded from the Service Level Agreements.**
Public preview services are given 'as is' under the Supplemental Terms of Use for Previews. They are not covered by SLAs (uptime promises).

✖ **No change is needed.**
Preview services are documented on Microsoft Learn so customers can try them.

✖ **Only configurable from Azure CLI.**
Previews are usually available in the portal, PowerShell and APIs, not only the CLI.

✖ **Only configurable from the Azure portal.**
Previews are usually also available in the CLI, PowerShell and REST APIs, not only the portal.

EXAM STRATEGY
Public preview = no SLA. Remove the tool claims (CLI-only, portal-only) and 'no documentation'. Previews are documented and work from the portal and CLI.

IMPACT
Running production on a preview feature means no uptime promise and no service credits. This is a common risk when teams adopt new features early.

What · Where · When · Why · How

What	Public preview is a release stage for testing features before general availability (GA - full release).
Where	Shown as '(Preview)' in the portal. Covered by the Supplemental Terms of Use for Previews.
When	When you test new features outside production.
Why	Preview services are given as-is and are not covered by SLAs.
How	Opt in or register the feature and test it. Keep production on GA services.

Exam tip
Preview = no SLA (and limited support); GA = SLA. Previews still have documentation.

Memory trick
Preview = Promise-free: no SLA promise.

Real-world example, use case and scenario

EXAMPLE
Preview label
A portal page marked '(Preview)' links to the Supplemental Terms of Use, which say no SLA applies.

USE CASE
Innovation lab
A car maker's lab tests a preview AI feature in a test subscription. Production stays on GA services.

SCENARIO
Outage with no credit
A startup runs its checkout on a preview database feature. A four-hour outage happens and they ask for a service credit. Microsoft says no because previews have no SLA, and the team moves to the GA tier.

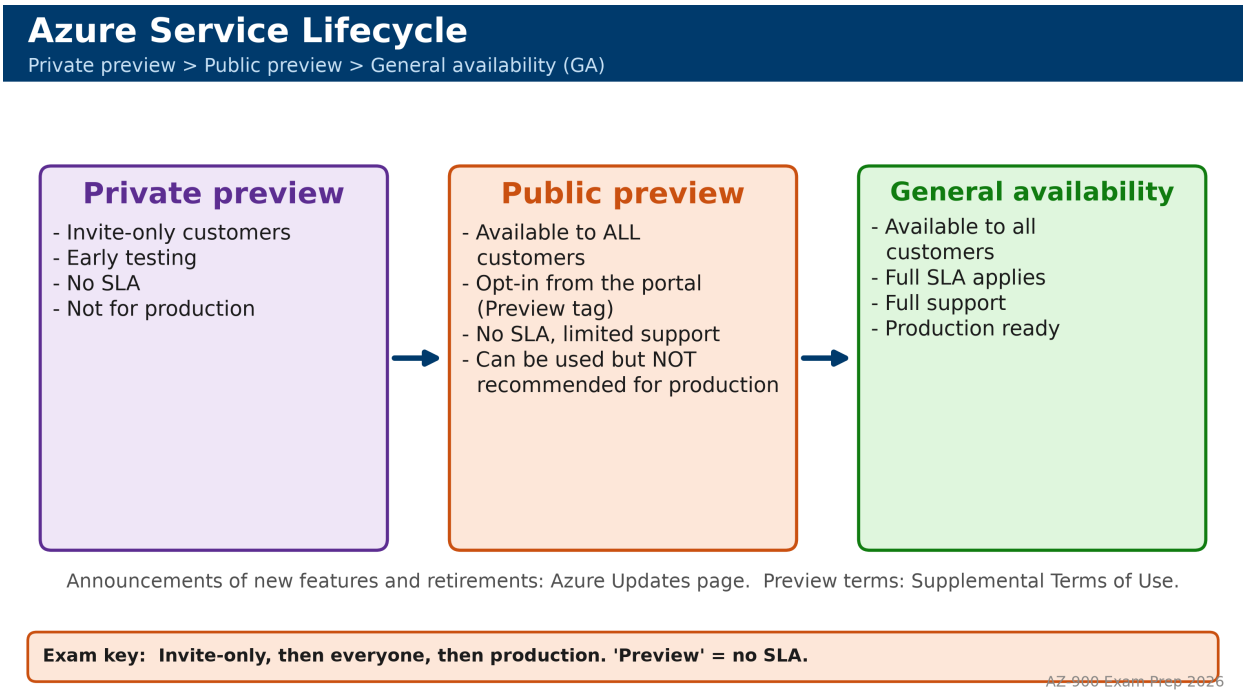


Figure Q86: Service lifecycle: preview to GA

About this figure

- Read the arrows left to right: invite-only private preview, open public preview, then general availability with SLA and full support.
- Previews are offered as-is under the Supplemental Terms of Use for Microsoft Azure Previews; features, pricing and availability may change or be withdrawn.
- Some previews are discounted or free, but many bill at reduced or standard rates; check the pricing page before relying on preview resources.
- Follow Azure Updates for previews, GA announcements and retirements; Service Health also sends retirement notices for services you actually use.
- Microsoft usually gives at least 12 months' notice before retiring a GA service or feature that customers depend on, with migration guidance.

Question 87

CLOUD CONCEPTS

Cloud concepts
v5 Q152

Your company has an on-premises network with numerous servers, which they intend to migrate to Azure. You have been tasked with devising a plan that allows for the availability of a few of the servers, in the event that one of the Azure data centers becomes unavailable for a lengthy interval. Solution: You should include elasticity in your plan. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same scenario as Q68 and Q88. The goal is to survive a datacenter going offline. Elasticity handles changes in demand, not failures, so No. Q68 gives the right answer: fault tolerance.

Answer explained

✔ **No.**
Elasticity grows and shrinks resources with demand. It does not keep servers available when a datacenter fails. That needs copies in more than one datacenter.

✖ **Yes.**
Extra instances in the failed datacenter fail too. Surviving a datacenter outage needs fault tolerance across zones or regions.

EXAM STRATEGY
'Datacenter unavailable' needs copies in other places. Elasticity is about scaling with demand, not surviving site failure, so No.

IMPACT
Autoscaling inside a failed datacenter scales nothing. A design with only elasticity stays down during a site outage.

What · Where · When · Why · How

What	Elasticity is the ability to add and remove resources automatically as demand changes.
Where	Built with autoscale in VM Scale Sets, App Service and similar services.
When	When demand goes up and down without warning.
Why	It deals with load, not datacenter failure, so it does not meet this goal.
How	Autoscale rules add instances on high load and remove them when load drops.

Exam tip
Datacenter outage = fault tolerance/HA/DR.
Demand spikes = elasticity/scalability.

Memory trick
Elastic stretches with load, not across disasters.

Real-world example, use case and scenario

EXAMPLE
Autoscale rule
A VM scale set grows from 2 to 10 instances when CPU goes above 70%, then shrinks at night.

USE CASE
Tax season spikes
A tax-filing service uses elasticity for April peaks. It does not pay for idle capacity the rest of the year.

SCENARIO
Wrong design goal
An architect at a utility company suggests autoscaling to protect against datacenter loss. The review board points out all new instances would be in the same failed site. The plan is changed to spread servers across availability zones.

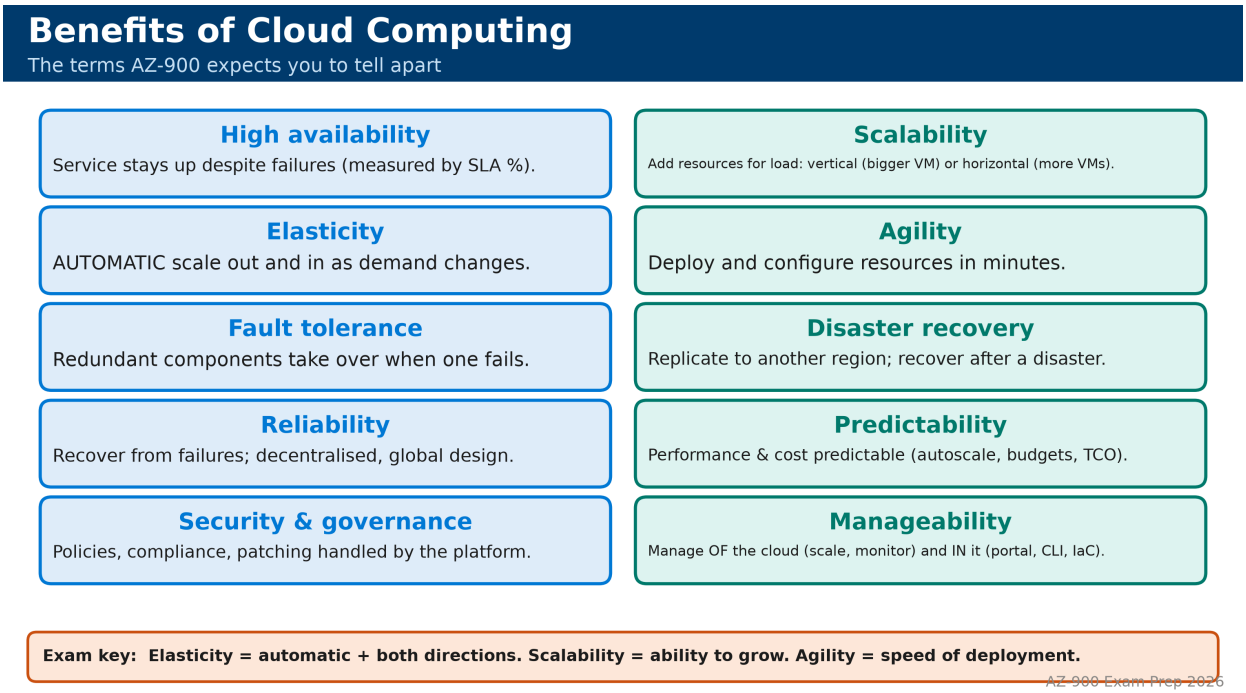


Figure Q87a: Benefits of cloud computing

About this figure

- Each box pairs a term with its meaning; the exam often gives a scenario and asks which benefit it describes, so focus on keywords.
- Scalability is the capacity to grow, manual or automatic; elasticity is automatic scaling out and back in with demand, for example VM Scale Sets autoscale.
- High availability is expressed as an SLA percentage such as 99.9% or 99.99%; fault tolerance and disaster recovery are ways of achieving it.
- Agility means provisioning in minutes instead of weeks of hardware procurement; predictability covers both performance and cost forecasting.
- Governance tools such as Azure Policy, RBAC and locks deliver the security and governance benefit; manageability covers portal, CLI, PowerShell and IaC.

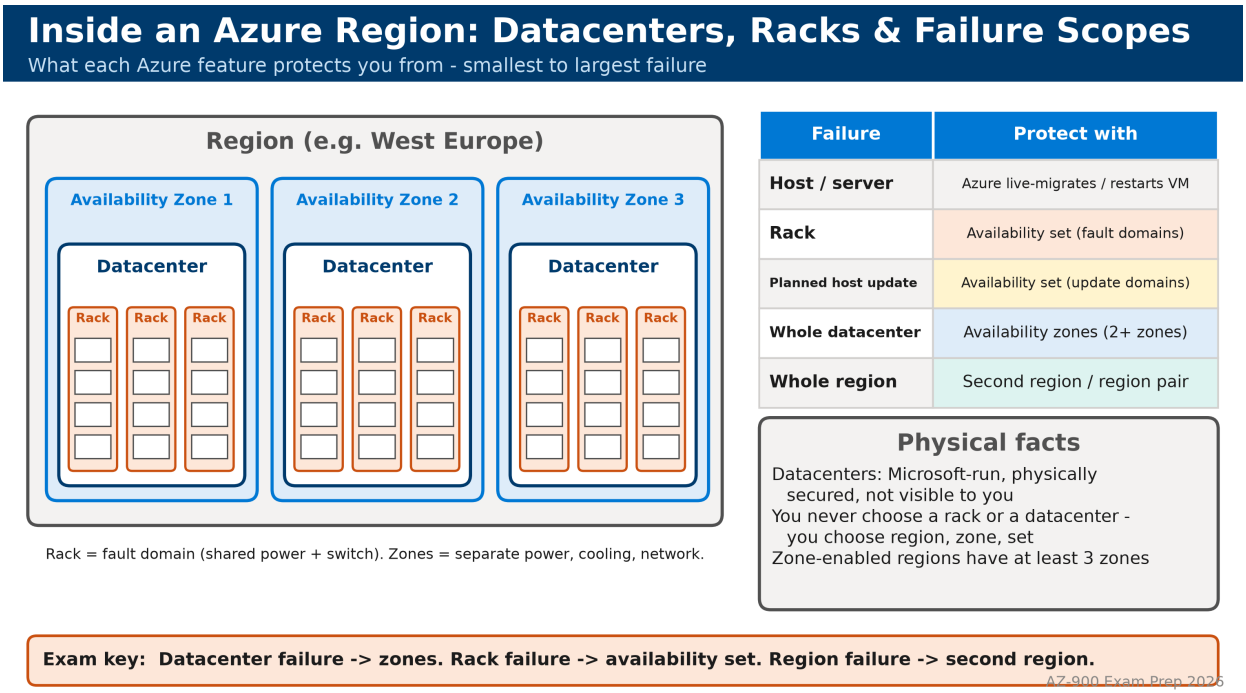


Figure Q87b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 88

CLOUD CONCEPTS

Cloud concepts
v5 Q153

Your company has an on-premises network with numerous servers, which they intend to migrate to Azure. You have been tasked with devising a plan that allows for the availability of a few of the servers, in the event that one of the Azure data centers becomes unavailable for a lengthy interval. Solution: You should include scalability in your plan. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same scenario as Q68 and Q87. Scalability adds capacity (more or bigger servers), but they stay in the same place. It does not help when a datacenter fails, so No.

Answer explained

✔ **No.**
Scalability adds capacity to meet demand but says nothing about location. So it does not keep servers available during a datacenter outage.

✖ **Yes.**
More or bigger servers in the same failed datacenter still go down together. Scalability is about capacity, not resilience (surviving failure).

EXAM STRATEGY
Scalability adds capacity but no spread across locations. 'Datacenter unavailable' needs fault tolerance, so any scale-related answer is No.

IMPACT
Confusing scale with resilience is a common design mistake. It turns one datacenter outage into a full business outage.

What · Where · When · Why · How

What	Scalability is the ability to add capacity by scaling up (bigger) or out (more).
Where	Done by resizing VMs or adding instances inside a deployment.
When	When demand grows and more capacity is needed.
Why	It adds capacity but not copies in other places, so a datacenter outage still stops the servers.
How	Scale up to larger VM sizes, or scale out by adding instances.

Exam tip
In this three-part series only fault tolerance is Yes. Elasticity and scalability are about capacity.

Memory trick
Scale for size, tolerate for failure.

Real-world example, use case and scenario

EXAMPLE
Scale up
Resizing a VM from D2s_v5 to D8s_v5 gives four times the vCPUs, but it stays in the same datacenter.

USE CASE
Growing e-learning platform
An e-learning company adds web servers as student numbers grow each term.

SCENARIO
Bigger is not safer
A clinic network moves to the largest VM sizes, thinking this makes them more resilient. A datacenter power failure still takes them all offline. They spread across zones to get fault tolerance.

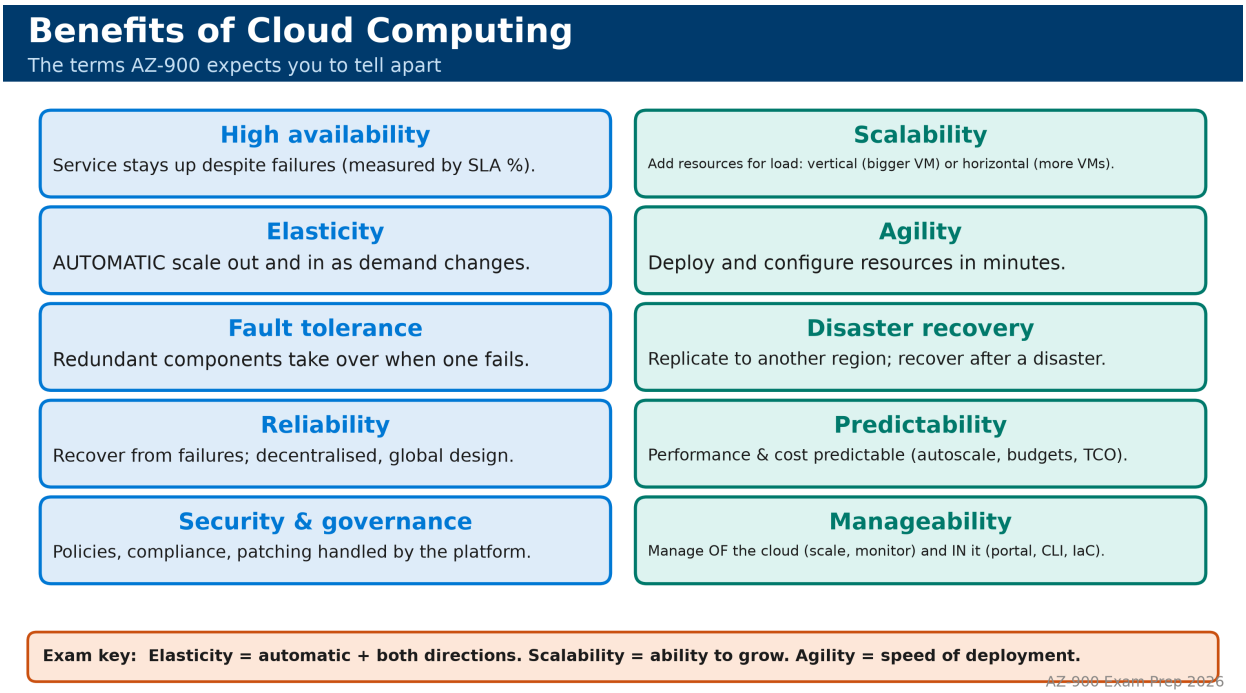


Figure Q88a: Benefits of cloud computing

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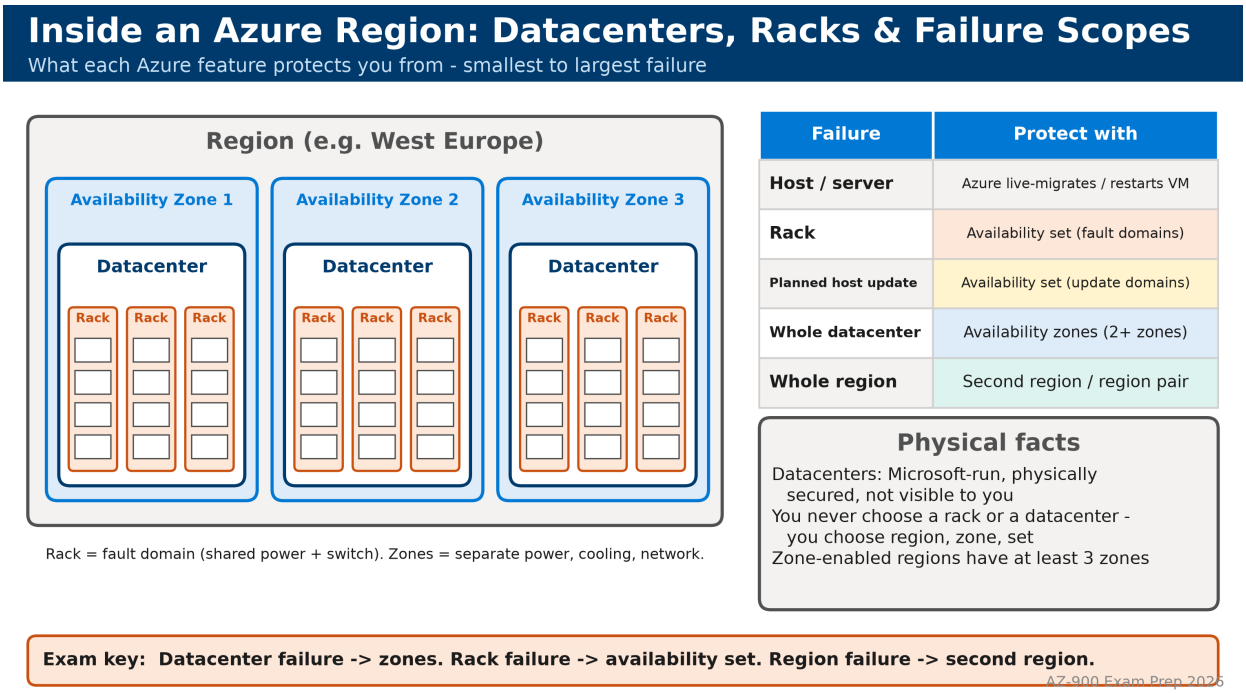


Figure Q88b: Inside a region: datacenters, racks and failure scopes

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- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 89

CLOUD CONCEPTS

Service models
v5 Q155

You are tasked with deploying Azure virtual machines for your company. You need to make use of the appropriate cloud deployment solution. Solution: You should make use of Software as a Service (SaaS). Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The question asks which service model fits deploying VMs. VMs where you manage the operating system are IaaS. So SaaS (Q89) and PaaS (Q90) are both No. Q32 confirms a VM is IaaS.

Answer explained

✔ **No.**
Deploying VMs where you manage the OS and apps is IaaS (Infrastructure as a Service). SaaS (Software as a Service) gives finished apps such as Microsoft 365.

✖ **Yes.**
With SaaS you only use the software. You never deploy or manage virtual machines.

EXAM STRATEGY
'Deploy virtual machines' = you manage the OS = IaaS. SaaS gives finished apps with no VMs to deploy, so No.

IMPACT
The wrong service model confuses who patches and secures the OS. In the shared responsibility model, this directly causes security gaps.

What · Where · When · Why · How

What	SaaS gives complete apps, such as Microsoft 365, run by the provider.
Where	The top of the cloud service model stack. The customer manages only data and users.
When	When users need ready-made software with nothing to manage underneath.
Why	It does not involve deploying VMs, so it fails the goal.
How	Subscribe to the app and use it in a browser or app.

Exam tip
VM = IaaS every time. SaaS = ready-to-use apps; PaaS = platform without OS management.

Memory trick
If you patch the OS, you're in IaaS.

Real-world example, use case and scenario

EXAMPLE
SaaS vs VM
Using Outlook in Microsoft 365 is SaaS. Creating a Windows Server VM in Azure is IaaS.

USE CASE
CRM decision
A sales team uses Dynamics 365 (SaaS) for customer management. IT deploys VMs (IaaS) for an old custom app.

SCENARIO
Model mismatch
A project manager at a museum writes 'SaaS' in the plan for deploying VMs. The cloud architect changes it to IaaS, because the team will patch the OS. The responsibility table in the plan is updated.

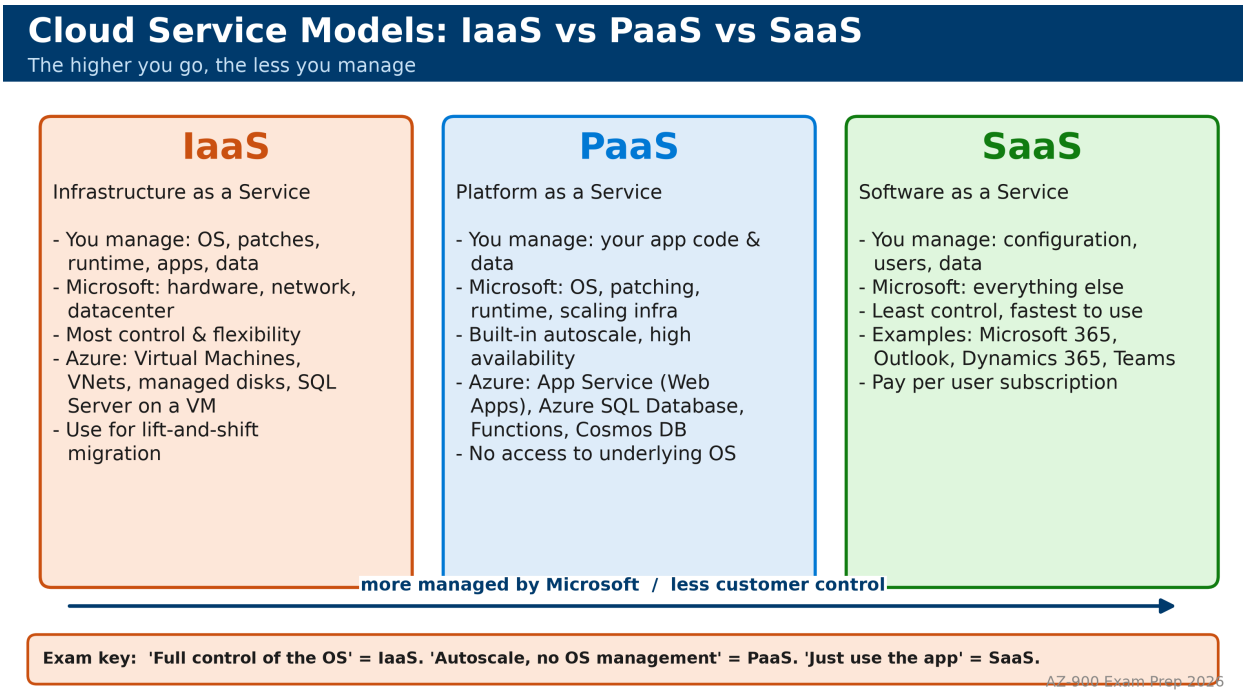


Figure Q89: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

• Serverless options such as Azure Functions are treated as PaaS-style: no OS access, automatic scaling and per-execution billing.

• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

Question 90

CLOUD CONCEPTS

Service models
v5 Q156

You are tasked with deploying Azure virtual machines for your company. You need to make use of the appropriate cloud deployment solution. Solution: You should make use of Platform as a Service (PaaS). Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same question as Q89, now with PaaS. PaaS hides the VMs and the OS, so you cannot 'deploy VMs' with it. The answer is No; VMs are IaaS.

Answer explained

✔ **No.**
Azure VMs give you control of the OS, which is IaaS. PaaS (Platform as a Service) hides the OS and runtime, so you deploy only code or data.

✖ **Yes.**
PaaS customers do not deploy or manage VMs. The platform hides them.

EXAM STRATEGY
PaaS hides the VM and OS, so you cannot deploy plain VMs as PaaS. Any question saying 'virtual machines' points only to IaaS.

IMPACT
Calling VMs PaaS makes teams think Microsoft patches the guest OS. The servers are then left unpatched and open to attack.

What · Where · When · Why · How

What	PaaS gives a managed platform for deploying code without managing the OS.
Where	Middle of the service model stack, for example App Service and Azure SQL Database.
When	When developers want to focus on code, not servers.
Why	PaaS hides the VMs, so it is not the model for deploying Azure VMs.
How	Deploy code or data to the managed service. Microsoft patches the OS and runtime.

Exam tip
Deploy VMs = IaaS. PaaS appears when the question says 'without managing the OS' or 'focus on code'.

Memory trick
PaaS hides the box; IaaS hands you the box.

Real-world example, use case and scenario

EXAMPLE
Same app, two models
Running IIS on a Windows VM is IaaS. Publishing the same site to App Service is PaaS.

USE CASE
Developer platform
A recruitment website team moves from VMs to App Service so they stop spending weekends on OS patching.

SCENARIO
Custom driver needed
A lab-equipment company needs a VM with a custom kernel driver (low-level OS software). A teammate suggests PaaS. PaaS gives no OS access, so they choose IaaS VMs.

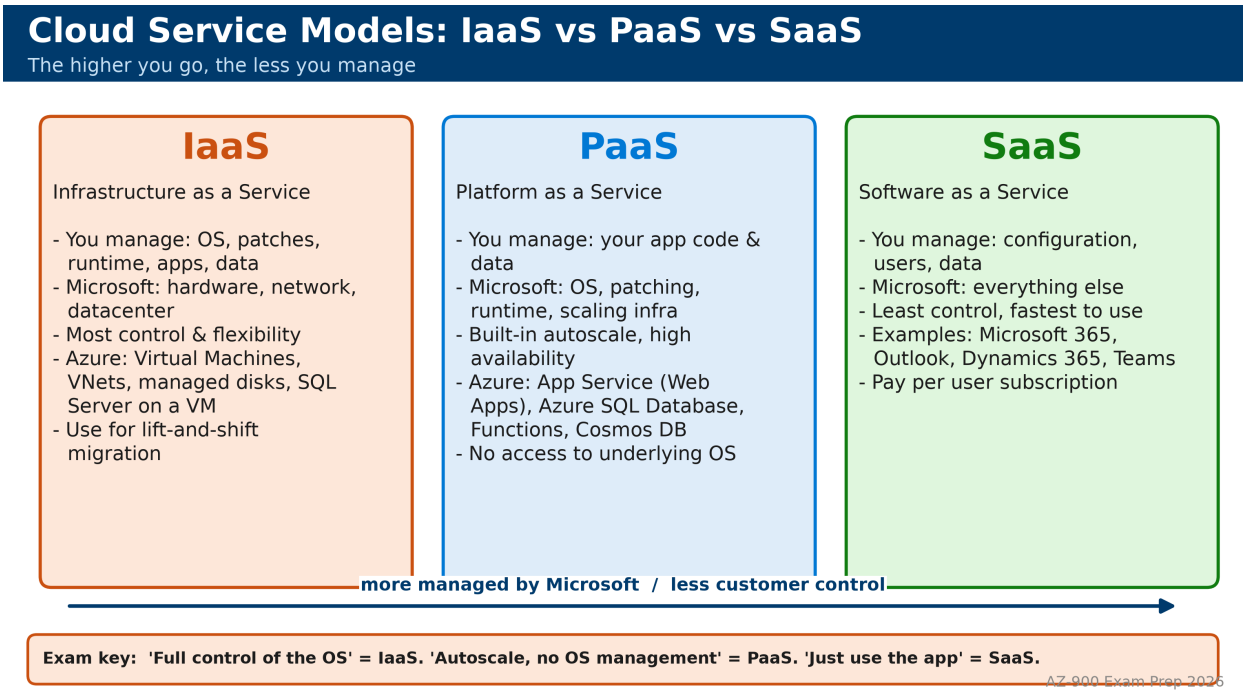


Figure Q90: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

• Serverless options such as Azure Functions are treated as PaaS-style: no OS access, automatic scaling and per-execution billing.

• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

Question 91

CLOUD CONCEPTS

Deployment models
v5 Q158

Your company has on-premises network includes a large number of servers. They would like to make extra resources available to their users, while keeping capital and operational overheads to a minimum. You are required to make recommendations that should be included in the overall solution. Solution: You should indorse the use of an added data center as part of the solution. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is extra resources with LOW CapEx (money spent up front on buying) and LOW OpEx (running costs). Anything you build or buy yourself fails. Q92 (private cloud) is also No; Q246 (extend with public cloud) is Yes.

Answer explained

✔ **No.**
A new datacenter needs a lot of money up front, plus ongoing staff and maintenance. That is the opposite of keeping CapEx and OpEx low.

✖ **Yes.**
An extra datacenter means the most capital spending and adds running costs. It fails the cost goal.

EXAM STRATEGY
'Minimise capital and operational overheads' rules out anything you build or buy. A new datacenter is the highest CapEx and OpEx, so No.

IMPACT
Building a datacenter costs millions and takes years. Cloud capacity gives the same resources in minutes.

What · Where · When · Why · How

What	An extra datacenter is a new building, owned by the company, for servers.
Where	On-premises, built and run by the company.
When	Only when the company must own all its hardware and has the capital budget.
Why	It brings the highest capital and running costs, so it fails the goal.
How	You must buy building space, hardware, power, cooling and staff.

Exam tip
'Minimise capital expenditure' rules out anything you build or buy (datacenter, private cloud hardware). Look for cloud or hybrid.

Memory trick
New datacenter = new debt.

Real-world example, use case and scenario

EXAMPLE
Build cost
A second server room needs racks, UPS (backup power), cooling and staff before any workload runs.

USE CASE
Burst to cloud
A video post-production company rents Azure GPU VMs for busy rendering periods instead of building more server rooms.

SCENARIO
Board rejects building
A regional bank's IT director asks for a new datacenter for growth. The CFO refuses the multi-million up-front cost. The team extends into Azure by VPN and adds capacity within weeks, paid monthly.

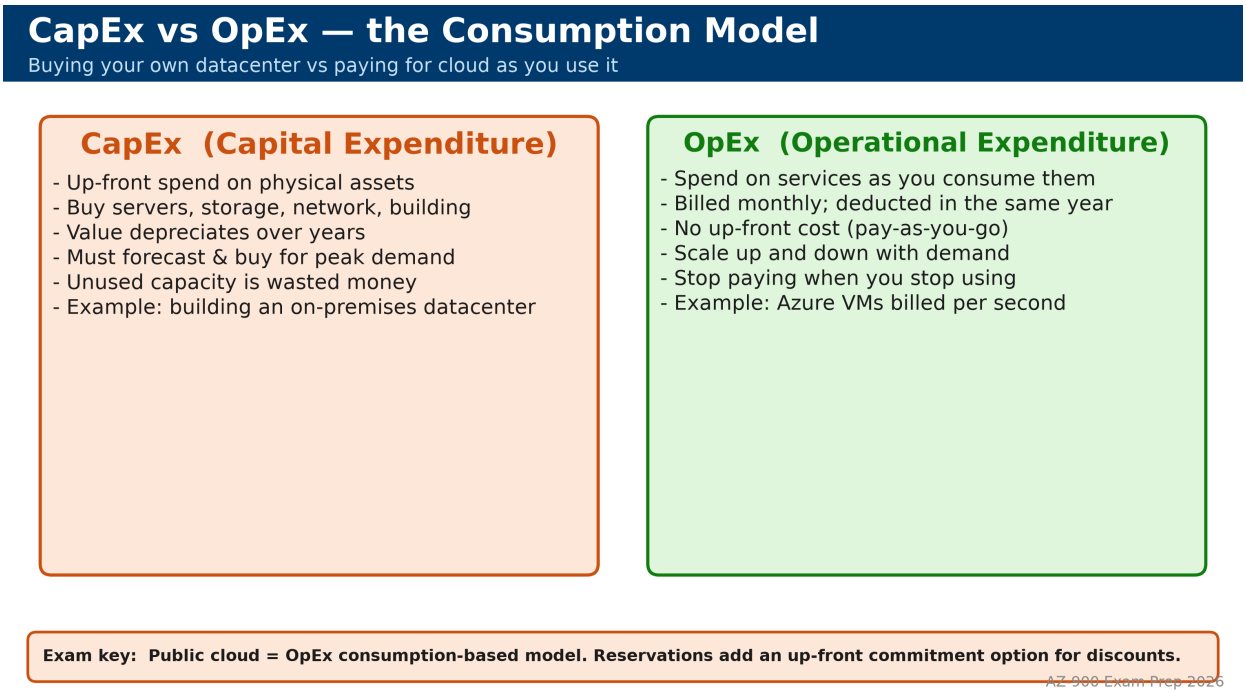


Figure Q91a: CapEx vs OpEx

About this figure

• Left box is buying assets up front (CapEx); right box is paying for consumption (OpEx). Cloud moves spending from left to right.

• CapEx assets are depreciated over several years; OpEx is expensed in the period it is incurred, which suits variable or uncertain demand.

• The consumption model means no up-front cost, pay only for what you use, and stop paying when resources are deleted or deallocated.

• Reservations and savings plans add a 1- or 3-year commitment for discounts, but they remain operational spending, not purchased hardware.

• Use the TCO calculator to compare on-premises CapEx (hardware, power, labour) with expected Azure OpEx before migrating.

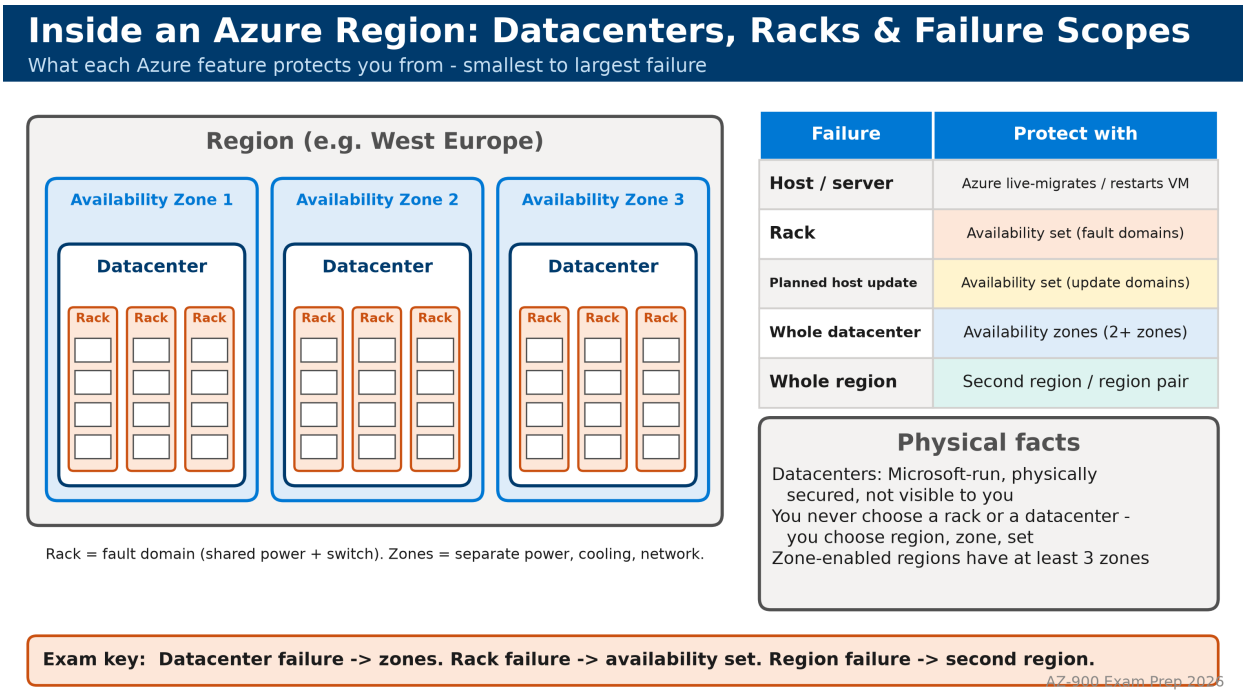


Figure Q91b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 92

CLOUD CONCEPTS

Deployment models
v5 Q160

Your company has on-premises network includes a large number of servers. They would like to make extra resources available to their users, while keeping capital and operational overheads to a minimum. You are required to make recommendations that should be included in the overall solution. Solution: You should indorse the use of a private cloud as part of the solution. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same goal as Q91. A private cloud runs on hardware you buy and run yourself, so it raises both CapEx and OpEx. The answer is No.

Answer explained

✔ **No.**
A private cloud runs on hardware the company buys and runs. Adding capacity raises both capital and running costs.

✖ **Yes.**
The company would pay the full cost of new hardware, datacenter space and staff. That fails the cost goal.

EXAM STRATEGY
Private cloud = company-owned hardware, so new capacity means new purchases and staff. That conflicts with 'minimal overhead', so No.

IMPACT
A private cloud keeps all the up-front cost on you. Recommending it to cut costs would fail a business case review.

What · Where · When · Why · How

What	A private cloud is cloud infrastructure used by only one organisation.
Where	Usually in the company's own datacenter on its own hardware.
When	When full control or regulation needs dedicated hardware, whatever the cost.
Why	Adding capacity means buying hardware and hiring staff, so costs stay high.
How	Buy and virtualise servers, for example with Azure Stack Hub or VMware.

Exam tip
Private cloud = your hardware = CapEx. To keep costs low, use public or hybrid cloud.

Memory trick
Private cloud, private bill.

Real-world example, use case and scenario

EXAMPLE
On-prem private cloud
Running Azure Stack Hub or VMware in your own datacenter gives cloud features, but you buy the servers.

USE CASE
Defence contractor
A defence contractor picks a private cloud because rules require full physical control. It accepts the higher cost.

SCENARIO
Cost goal missed
A retail company suggests a private cloud to add capacity. The finance review shows a big hardware purchase plus new staff. They switch to a hybrid plan that uses Azure for the extra capacity.

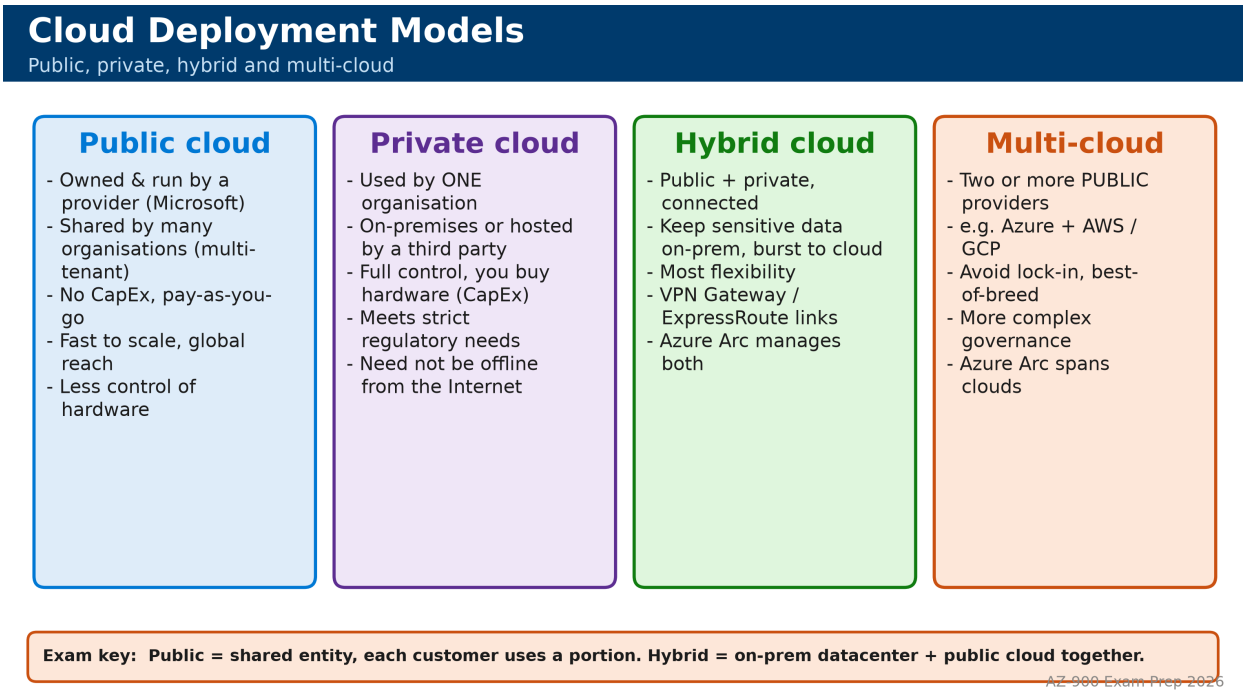


Figure Q92: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 93

CLOUD CONCEPTS

Cost
v5 Q161

Your company is planning to migrate all their virtual machines to an Azure pay-as-you-go subscription. The virtual machines are currently hosted on the Hyper-V hosts in a data center. You are required make sure that the intended Azure solution uses the correct expenditure model. Solution: You should recommend the use of the elastic expenditure model. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Only two expenditure models exist: CapEx (buy up front) and OpEx (pay as you go). 'Elastic' and 'scalable' (Q94) are cloud features, not expenditure models, so No. Q31 gives the right answer: OpEx.

Answer explained

✔ **No.**
The expenditure models are CapEx (up-front buying) and OpEx (ongoing spending). Elasticity is a cloud feature, not an expenditure model. Pay-as-you-go is OpEx.

✖ **Yes.**
There is no 'elastic expenditure model'. Elasticity describes scaling with demand, not how spending is classified.

EXAM STRATEGY
Only two expenditure models exist: CapEx and OpEx. 'Elastic' is a cloud feature, so any label other than CapEx/OpEx is No.

IMPACT
Finance teams budget by CapEx versus OpEx. The right term decides how cloud spending is approved and recorded.

What · Where · When · Why · How

What	Elasticity is a cloud feature for scaling with demand. It is not an expenditure model.
Where	It applies to scaling resources, not to accounting.
When	When matching resources to changing load, not when classifying spending.
Why	Only CapEx and OpEx are expenditure models, so 'elastic' fails.
How	Pay-as-you-go spending is classified as OpEx instead.

Exam tip
Only two expenditure models exist: CapEx and OpEx. Pay-as-you-go = OpEx.

Memory trick
Money comes in two types: Cap and Op. Elastic is a band, not a budget.

Real-world example, use case and scenario

EXAMPLE
Monthly bill
Azure VM charges appear as a monthly running cost, not as an asset (something owned) on the balance sheet.

USE CASE
Budget reclassification
A hospital's finance team moves server spending from the capital budget to the operating budget after moving to Azure.

SCENARIO
Invented term
A consultant's proposal for a law firm mentions an 'elastic expenditure model'. The finance director asks which accounting line that is. The proposal is corrected to say the migration moves costs to OpEx.

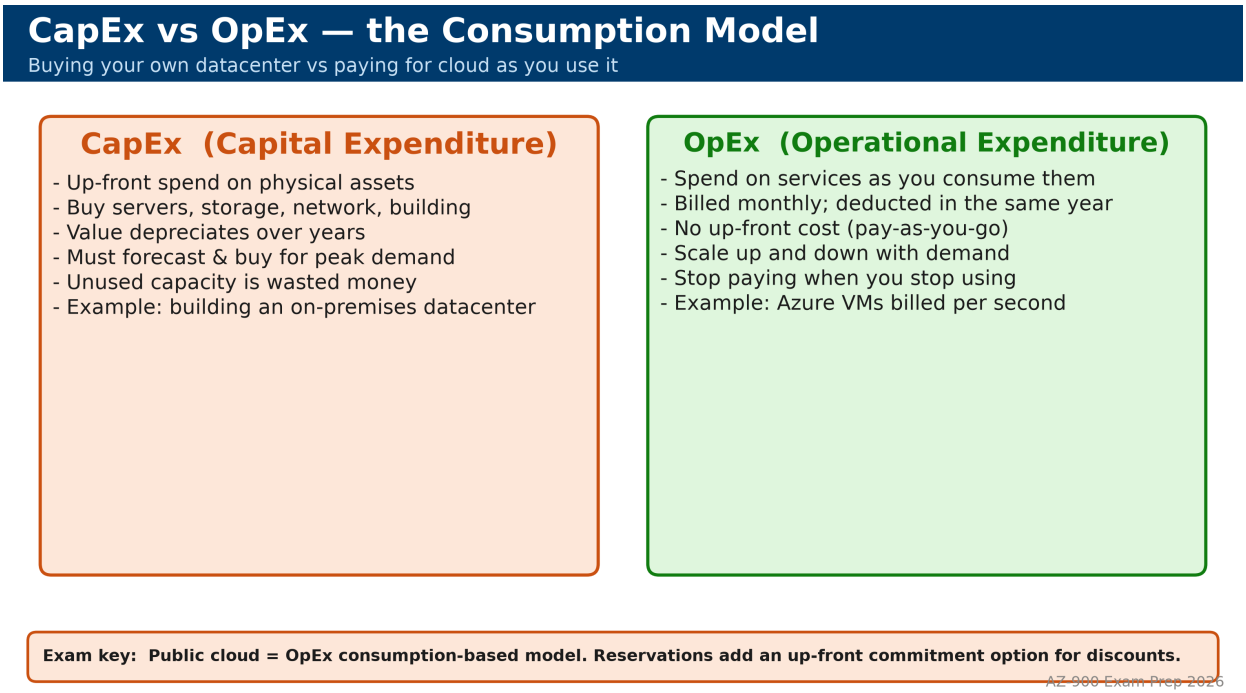


Figure Q93a: CapEx vs OpEx

About this figure

• Left box is buying assets up front (CapEx); right box is paying for consumption (OpEx). Cloud moves spending from left to right.

• CapEx assets are depreciated over several years; OpEx is expensed in the period it is incurred, which suits variable or uncertain demand.

• The consumption model means no up-front cost, pay only for what you use, and stop paying when resources are deleted or deallocated.

• Reservations and savings plans add a 1- or 3-year commitment for discounts, but they remain operational spending, not purchased hardware.

• Use the TCO calculator to compare on-premises CapEx (hardware, power, labour) with expected Azure OpEx before migrating.

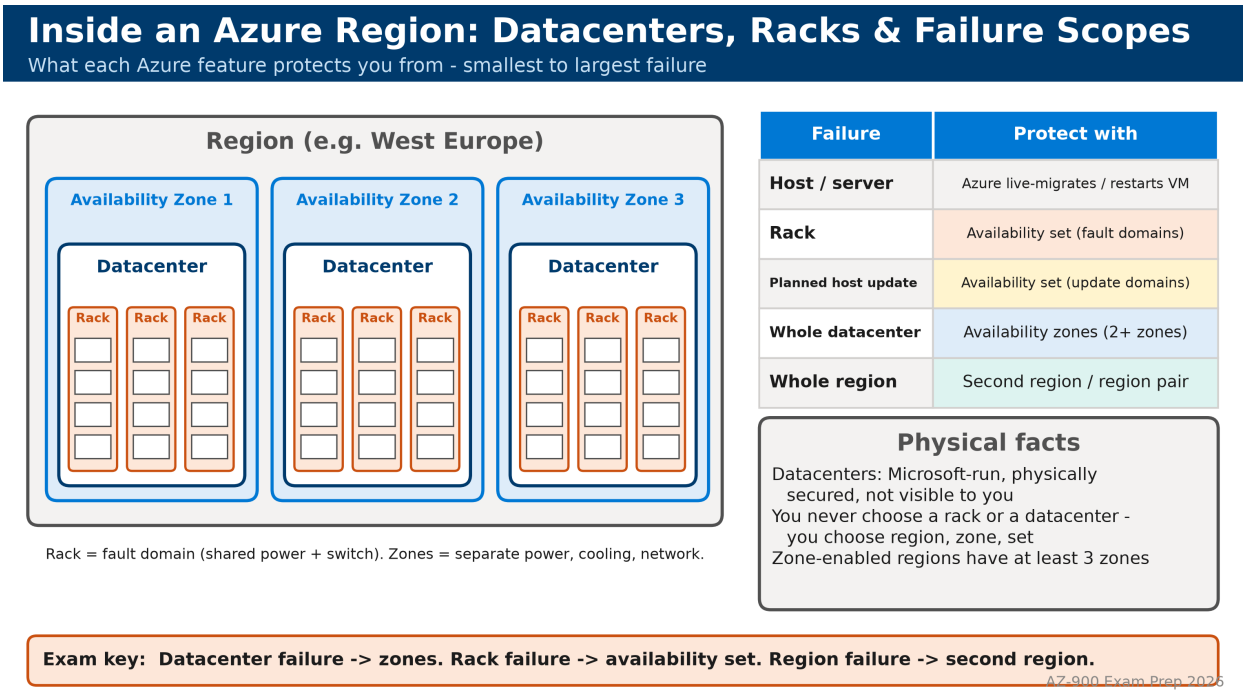


Figure Q93b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 94

CLOUD CONCEPTS

Cost
v5 Q162

Your company is planning to migrate all their virtual machines to an Azure pay-as-you-go subscription. The virtual machines are currently hosted on the Hyper-V hosts in a data center. You are required make sure that the intended Azure solution uses the correct expenditure model. Solution: You should recommend the use of the scalable expenditure model. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same as Q93. Only CapEx and OpEx are expenditure models. 'Scalable' is a cloud feature, so No. Pay-as-you-go is OpEx (Q31).

Answer explained

✔ **No.**
Scalability is a cloud feature, not an expenditure model. Pay-as-you-go Azure spending is OpEx (operational expenditure - ongoing spending).

✖ **Yes.**
There is no 'scalable expenditure model'. The correct class for pay-as-you-go spending is OpEx.

EXAM STRATEGY
Same trap as 'elastic': scalability is a feature, not an expenditure model. Accept only CapEx or OpEx here.

IMPACT
Mixing technical and money words confuses budget talks. This CapEx/OpEx series comes up often on AZ-900.

What · Where · When · Why · How

What	Scalability is a cloud feature for adding capacity. It is not an expenditure model.
Where	It applies to resource capacity, not to accounting.
When	When handling more demand, not when classifying spending.
Why	Only CapEx and OpEx are expenditure models, so 'scalable' fails.
How	Pay-as-you-go spending is classified as OpEx instead.

Exam tip
Elastic and scalable are cloud benefits, never expenditure models. The right answer in this series is OpEx.

Memory trick
Scale is for load, OpEx is for ledgers.

Real-world example, use case and scenario

EXAMPLE
Capacity vs cost
Scaling a VM scale set to 20 instances raises the OpEx bill. It does not change the expenditure model.

USE CASE
Startup finance
A food-delivery startup tells investors its Azure spending is OpEx that grows with orders.

SCENARIO
Exam-style trap
A trainee studying for AZ-900 picks 'scalable expenditure' because Azure is scalable. The teacher points out the question is about accounting. The trainee changes to OpEx and remembers the difference.

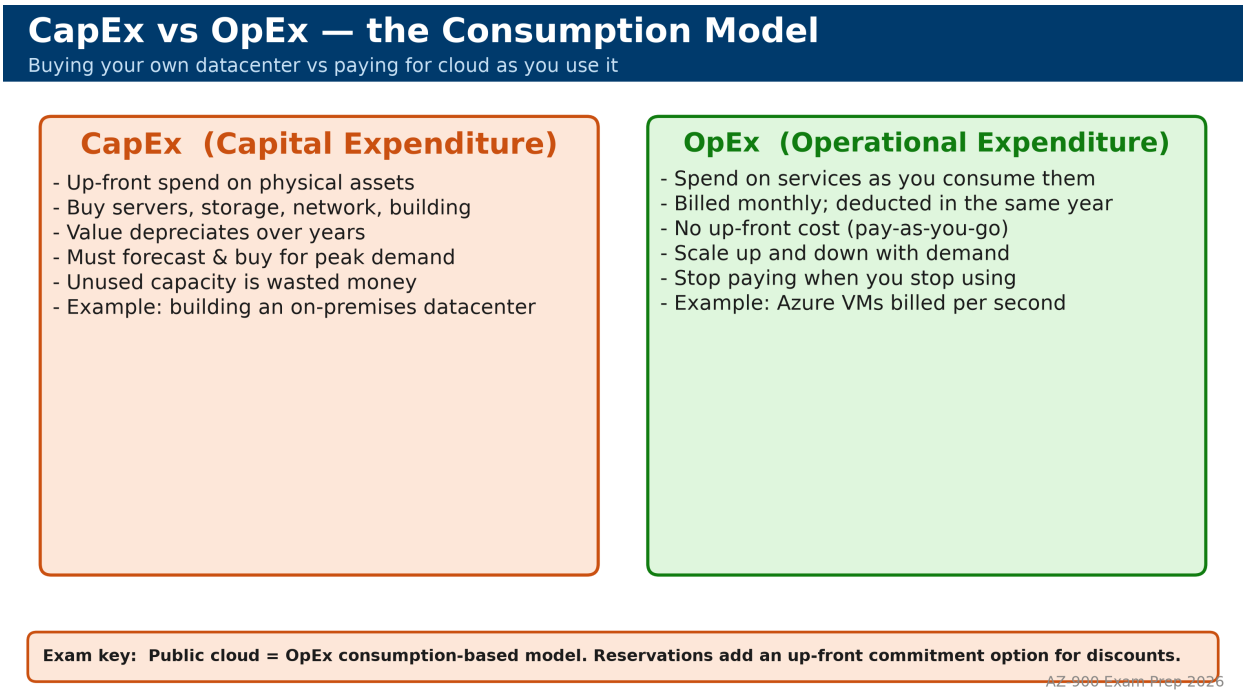


Figure Q94a: CapEx vs OpEx

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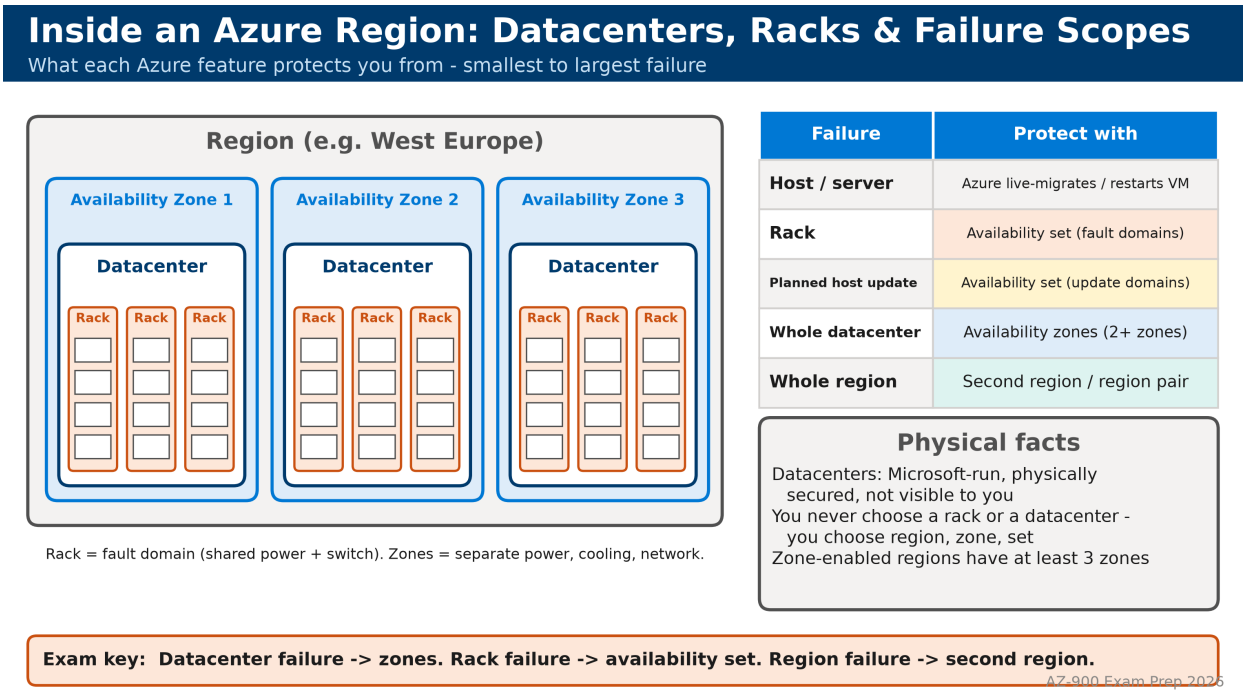


Figure Q94b: Inside a region: datacenters, racks and failure scopes

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- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 95

AZURE ARCHITECTURE & SERVICES

Data & AI
v5 Q164

You are required to deploy an Artificial Intelligence (AI) solution in Azure. You want to make sure that you are able to build, test, and deploy predictive analytics for the solution. Solution: You should make use of Azure Cosmos DB. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): 'Build, test, and deploy predictive analytics' means a machine learning tool. Cosmos DB is only a database, so No. Q96 (Logic Apps) is also No. Q260 gives the right answer: Azure Machine Learning designer.

Answer explained

✔ **No.**
Cosmos DB is a NoSQL (non-table) database spread across the world. It stores data but cannot build, train, test or deploy prediction models.

✖ **Yes.**
A database cannot train, test or deploy models. That is Azure Machine Learning.

EXAM STRATEGY
'Build, test, deploy predictive analytics' = a machine learning platform. Cosmos DB is a database, and storage services never train models, so No.

IMPACT
Choosing a database for ML (machine learning) work leaves no way to train or deploy models. The AI project then stalls.

What · Where · When · Why · How

What	Azure Cosmos DB is a NoSQL database service spread across the world.
Where	Deployed as a database account copied to the Azure regions you choose.
When	When apps need fast data storage spread across the world.
Why	It stores data but cannot build, test or deploy prediction models.
How	Create an account, choose an API, and read or write data in a few milliseconds.

Exam tip
Build, train, test, deploy prediction models = Azure Machine Learning. Databases and workflow tools are distractors.

Memory trick
Cosmos stores the stars; it doesn't predict them.

Real-world example, use case and scenario

EXAMPLE
Cosmos role
A shopping cart is stored in Cosmos DB, with reads in a few milliseconds across regions.

USE CASE
Gaming leaderboards
A mobile game uses Cosmos DB for player profiles and leaderboards worldwide.

SCENARIO
Feature store mix-up
A loyalty-program team plans to 'use Cosmos DB for predictions'. The data scientist explains Cosmos can hold the customer data, but the churn model must be built in Azure Machine Learning. They keep Cosmos as the data source and deploy the model as an ML endpoint.

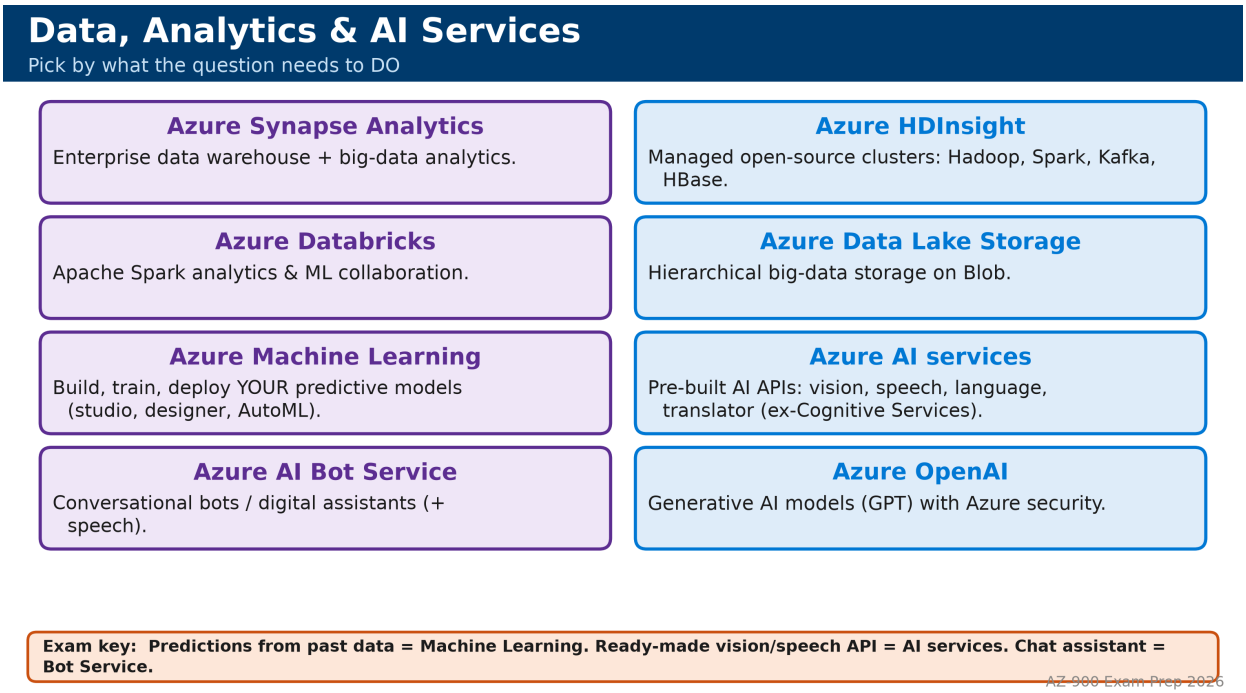


Figure Q95: Data, analytics and AI services

About this figure

- Left column covers data and analytics platforms; right column covers AI. Decide whether the scenario needs storage, processing, custom models or ready-made AI.
- Azure Machine Learning trains your own models from your data; Azure AI services are prebuilt APIs needing no data science, billed per transaction.
- Azure AI services include Vision, Speech, Language, Translator and Document Intelligence; many offer a free F0 tier with limited monthly calls.
- Azure OpenAI models are now offered through Microsoft Foundry (formerly Azure AI Foundry), billed per 1,000 or million tokens, varying widely by model.
- Microsoft Fabric is the newer unified analytics SaaS combining data engineering, warehousing and Power BI, and overlaps much of Synapse.

Question 96

AZURE ARCHITECTURE & SERVICES

Data & AI
v5 Q165

You are required to deploy an Artificial Intelligence (AI) solution in Azure. You want to make sure that you are able to build, test, and deploy predictive analytics for the solution. Solution: You should make use of Azure Logic Apps. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same goal as Q95. Logic Apps runs workflows (automated steps). It can call a model, but cannot build one, so No. Q260: Azure Machine Learning designer is the right tool.

Answer explained

✔ **No.**
Logic Apps runs workflows with connectors. It can call a model endpoint, but it cannot build, train, test or deploy prediction models.

✖ **Yes.**
Logic Apps cannot train models. It is a tool for linking systems in workflows.

EXAM STRATEGY
Logic Apps automates workflows. It can call a model but cannot build one. If the service is about linking systems or workflows, answer No.

IMPACT
Workflow tools call ML endpoints; they do not replace them. Mixing up these roles creates weak, unsupported AI pipelines.

What · Where · When · Why · How

What	Azure Logic Apps is a workflow service with connectors.
Where	Deployed as a Logic App resource and designed in the portal designer.
When	When automating links between systems and business processes.
Why	It can call models but cannot train or deploy predictive analytics.
How	Triggers start workflows that run actions through connectors.

Exam tip
Logic Apps = workflow glue. Predictive analytics = Azure Machine Learning.

Memory trick
Logic Apps follow logic; they don't learn it.

Real-world example, use case and scenario

EXAMPLE
Workflow, not model
A logic app starts when a file lands in Blob Storage and posts a Teams message.

USE CASE
Invoice routing
An accounts team uses Logic Apps to send scanned invoices to approvers after Document Intelligence reads the fields.

SCENARIO
Orchestration vs training
A retail analyst wants Logic Apps to forecast sales. Her architect shows that Logic Apps can only call a deployed model. They train the forecast in Azure Machine Learning and use Logic Apps to email the weekly results.

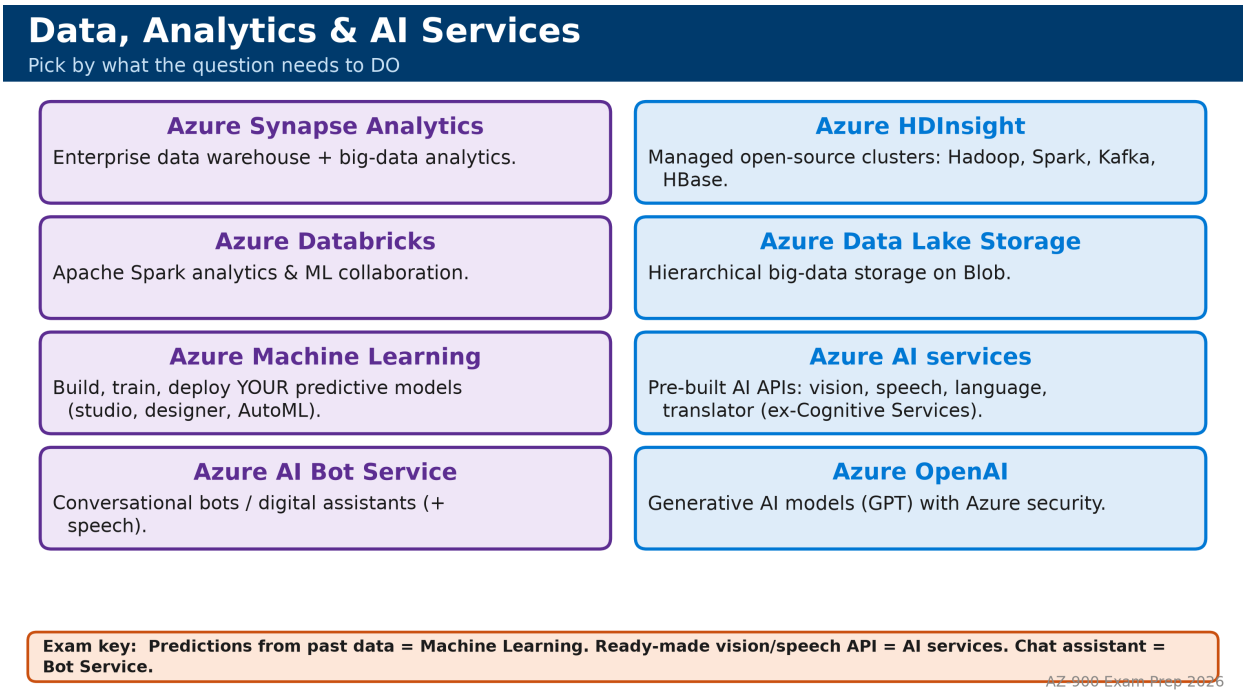


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- Microsoft Fabric is the newer unified analytics SaaS combining data engineering, warehousing and Power BI, and overlaps much of Synapse.

Question 97

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q167

Your company has infrastructure includes a number of business units that each need a large number of various Azure resources for everyday operation. The resources required by each business unit are identical. You are required to sanction a strategy to create Azure resources automatically. Solution: You recommend that the Azure API Management service be included in the strategy. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): 'Create Azure resources automatically' means infrastructure as code. API Management only manages APIs, so No. Q98 (management groups) is also No. Q48 gives the right answer: ARM templates.

Answer explained

✔ **No.**
API Management publishes, secures and limits APIs. It does not create Azure resources. Repeated creation needs ARM templates or Bicep (infrastructure as code).

✖ **Yes.**
API Management manages API traffic and access, not the automatic creation of Azure resources.

EXAM STRATEGY
'Create resources automatically' = infrastructure as code (resources described in files). API Management publishes and secures APIs and creates nothing, so No.

IMPACT
Mistaking an API gateway for a deployment tool wastes money. Environments are still built by hand and differ between units.

What · Where · When · Why · How

What	Azure API Management is a gateway for publishing, securing and monitoring APIs.
Where	Deployed as an API Management instance in front of back-end services.
When	When sharing APIs with developers or partners using keys, limits and analytics.
Why	It manages API traffic and cannot create Azure resources automatically.
How	Import APIs, apply policies such as rate limits, and publish them through a developer portal.

Exam tip
Create identical resources automatically = ARM templates/Bicep. API Management, management groups and policies do not deploy anything.

Memory trick
APIM manages calls, not creation.

Real-world example, use case and scenario

EXAMPLE
APIM role
APIM sets a limit of 100 calls per minute on a weather API and gives subscription keys to partners.

USE CASE
Bank open APIs
A bank shares account-information APIs with fintech partners through API Management to meet open-banking rules.

SCENARIO
Wrong 'API' idea
A team lead at a university thinks API Management can 'call the Azure API' to build department environments. The architect explains APIM sits in front of their own APIs. They write a Bicep template instead and deploy it for each department.

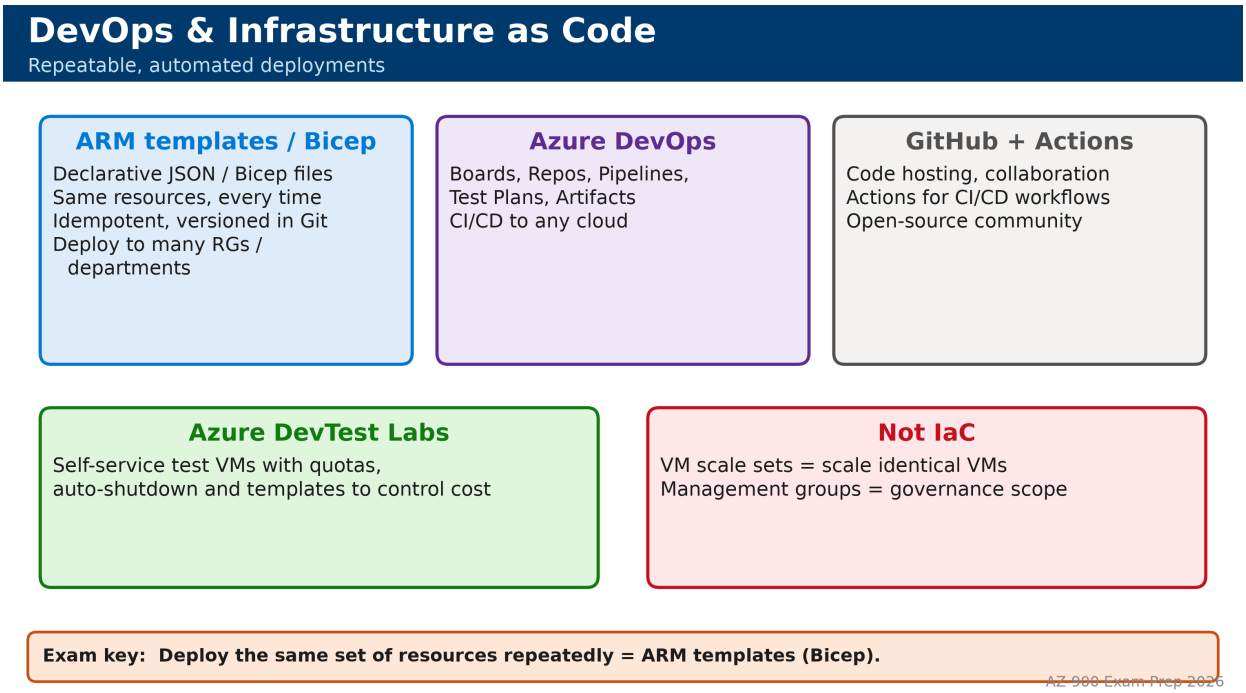


Figure Q97: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 98

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q168

Your company has infrastructure includes a number of business units that each need a large number of various Azure resources for everyday operation. The resources required by each business unit are identical. You are required to sanction a strategy to create Azure resources automatically. Solution: You recommend that management groups be included in the strategy. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same goal as Q97. Management groups organise subscriptions for rules and permissions. They create nothing, so No. Q48: ARM templates is the right answer. Q80 shows what management groups ARE for: compliance across subscriptions.

Answer explained

✔ **No.**
Management groups organise subscriptions for policy and RBAC. They deploy no resources, so they cannot create identical resources automatically.

✖ **Yes.**
Management groups only set the scope for governance across subscriptions. They create no VMs, storage or networks.

EXAM STRATEGY
Management groups organise subscriptions for policy and RBAC; they deploy nothing. Any governance container is No for 'create resources'.

IMPACT
Governance structures do not replace deployment automation. Relying on them leaves each business unit building resources by hand, in different ways.

What · Where · When · Why · How

What	Management groups are containers for organising subscriptions.
Where	Above subscriptions, under the tenant root management group.
When	When applying policy or RBAC across many subscriptions.
Why	They govern but deploy no resources, so they do not meet the goal.
How	Assign policies and roles at the management group; subscriptions inherit them.

Exam tip
Organise/govern = management groups; deploy/repeat = ARM templates/Bicep.

Memory trick
Management groups manage, they don't manufacture.

Real-world example, use case and scenario

EXAMPLE
MG hierarchy
Root > Corp > Finance and HR subscriptions, with a tagging policy assigned at Corp.

USE CASE
Franchise governance
A fast-food franchise groups each country's subscriptions under regional management groups to apply data-location policies.

SCENARIO
Expecting automation
A manufacturer creates a management group for each business unit and expects resources to appear. Nothing is deployed. They add an ARM template pipeline to create the standard resources and keep the management groups for governance.

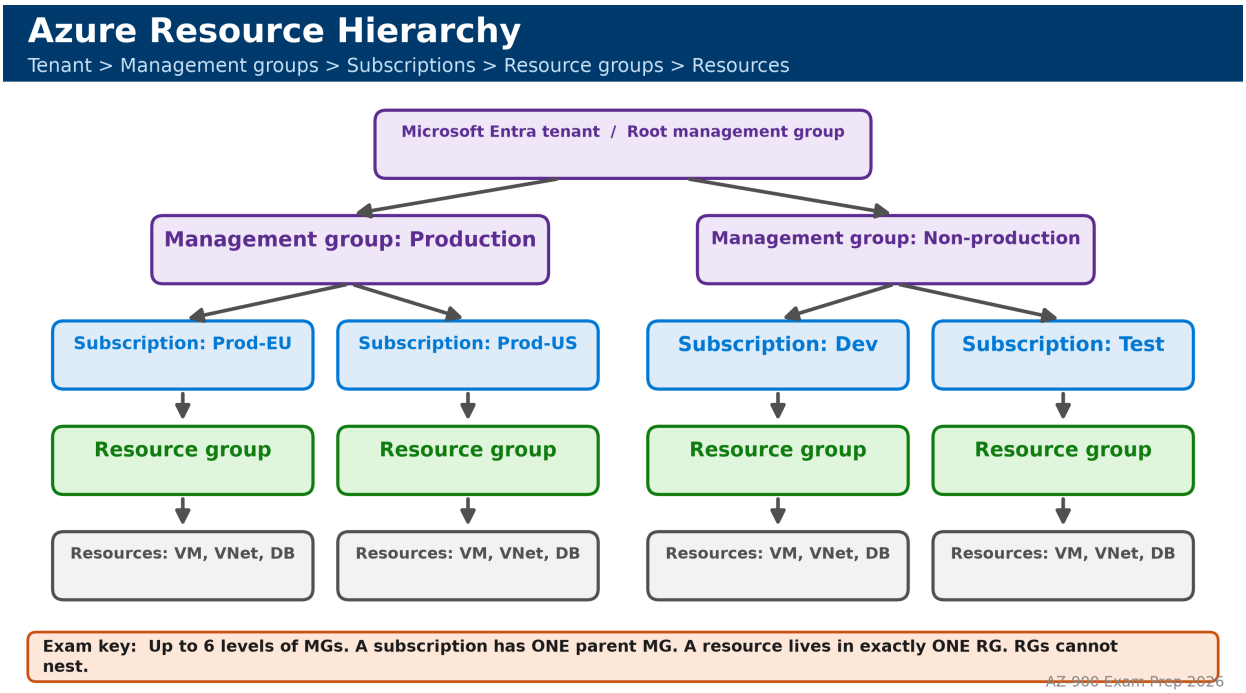


Figure Q98: Azure resource hierarchy

About this figure

• Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.

• A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.

• Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.

• Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.

• Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 99

CLOUD CONCEPTS

SLA
v5 Q170

You are tasked with deploying a critical LOB application, which will be installed on a virtual machine, to Azure. You are informed that the application deployment strategy should allow for a guaranteed availability of 99.99 percent. You need to make sure that the strategy requires as little virtual machines and availability zones as possible. Solution: You include two virtual machines and one availability zone in your strategy. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): You need BOTH at least two VMs AND at least two zones. Here there is only one zone, so No. Q100 (one VM, two zones) is also No. Q5 gives the right answer: 2 VMs in 2 zones.

Answer explained

✔ **No.**
The 99.99% VM SLA needs two or more VMs across two or more availability zones. Two VMs in one zone get at most 99.95%.

✖ **Yes.**
Two VMs in one zone fail together if that zone fails. They cannot reach the 99.99% multi-zone SLA.

EXAM STRATEGY
99.99% for VMs needs at least two VMs in at least two zones. One zone gives at most 99.95%, so any single-zone answer is No.

IMPACT
With both VMs in one zone, one building failure takes the critical LOB (line-of-business) app down, even though you pay for two VMs.

What · Where · When · Why · How

What	The VM SLA sets the guaranteed uptime for different VM layouts.
Where	Published in Microsoft's VM SLA. You choose a zone for each VM when you create it.
When	When a workload must meet a 99.99% uptime target.
Why	Two VMs in one zone get at most 99.95%, below the needed 99.99%.
How	Place at least two VMs in at least two availability zones in the same region.

Exam tip
SLA ladder: single VM with premium SSD 99.9%, availability set 99.95%, two zones 99.99%. Minimum for four nines = 2 VMs + 2 zones.

Memory trick
Four nines needs two and two: two VMs, two zones.

Real-world example, use case and scenario

EXAMPLE
Zone layout
VM-A and VM-B both in zone 1 = at most 99.95%. Move VM-B to zone 2 = 99.99%.

USE CASE
Pharmacy ordering app
A pharmacy chain places two LOB VMs in zones 1 and 2 of Central US to support 99.99% uptime in its contracts.

SCENARIO
Contract at risk
A service provider promises a client 99.99% but puts both VMs in zone 1. In review, the architect points out the SLA is at most 99.95%. One VM is moved to zone 2 before go-live.

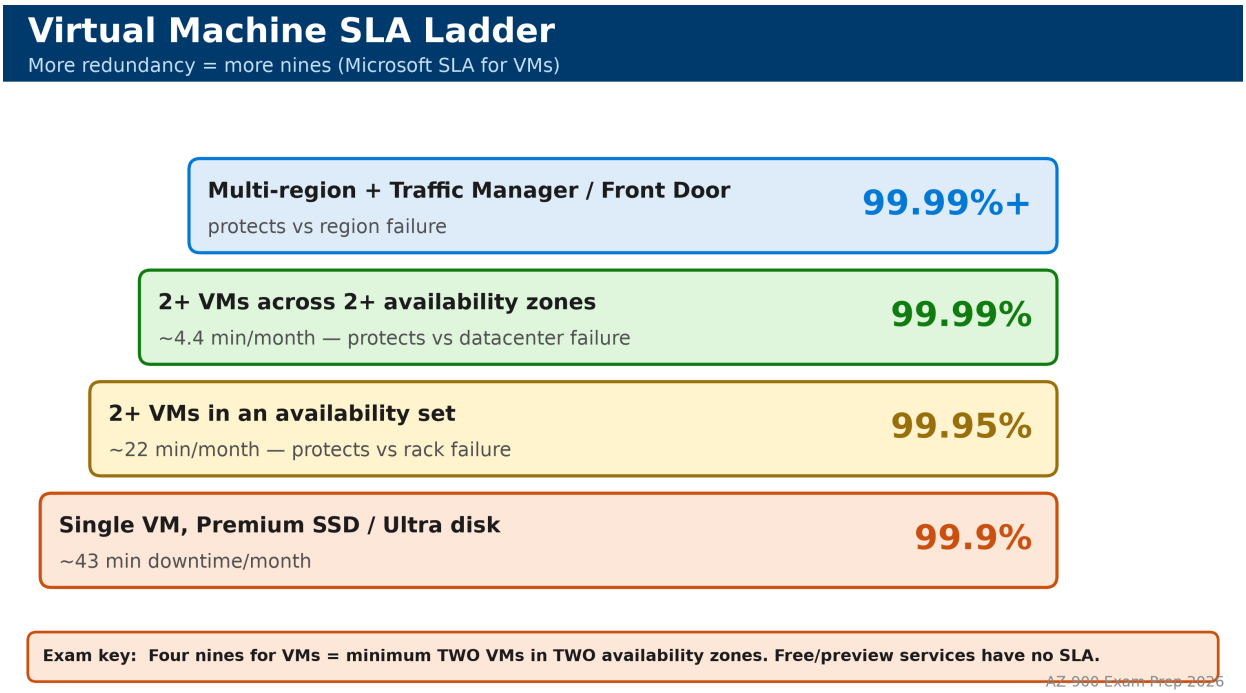


Figure Q99a: Virtual machine SLA ladder

About this figure

- Climb the ladder: each step adds redundancy and a higher VM SLA; the downtime figures assume a 30-day month.
- A single VM with Standard SSD gets about 99.5% and with Standard HDD about 95%; all its OS and data disks must meet that tier.
- Availability sets spread VMs across fault domains (racks) and update domains (maintenance batches) in one datacenter; zones spread across datacenters.
- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

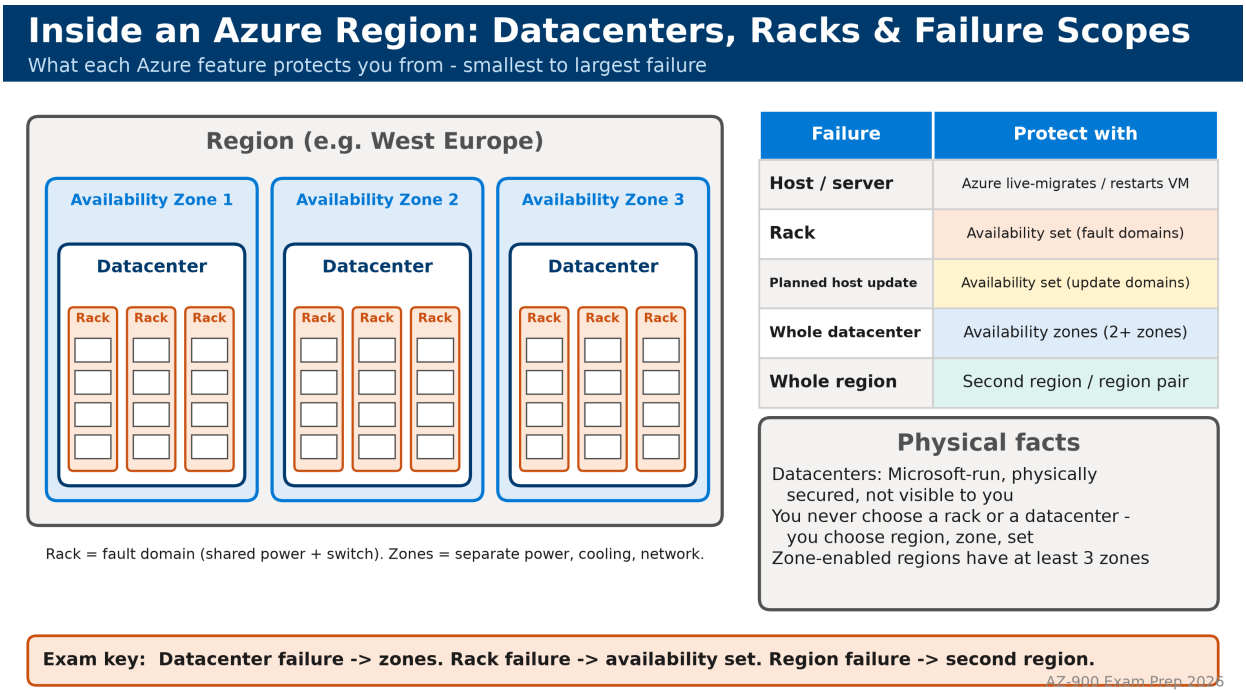


Figure Q99b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 100

CLOUD CONCEPTS

SLA
v5 Q171

You are tasked with deploying a critical LOB application, which will be installed on a virtual machine, to Azure. You are informed that the application deployment strategy should allow for a guaranteed availability of 99.99 percent. You need to make sure that the strategy requires as little virtual machines and availability zones as possible. Solution: You include one virtual machine and two availability zones in your strategy. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same goal as Q99. Here there is only ONE VM. A VM lives in one zone and cannot cover two. One VM gets at most 99.9%, so No. The right answer (Q5) is 2 VMs in 2 zones.

Answer explained

✔ **No.**
A VM lives in a single zone, so one VM cannot cover two zones. A single VM gets at most 99.9%. 99.99% needs two VMs in two zones.

✖ **Yes.**
One VM has no backup copy. It gets at most 99.9% (with premium SSD or Ultra disks), below 99.99%.

EXAM STRATEGY
A single VM sits in one zone and cannot cover two. With one VM you get 99.9% at best, so No.

IMPACT
Overestimating a single VM's SLA leaves critical apps unprotected and breaks uptime promises in contracts.

What · Where · When · Why · How

What	In an availability zone deployment, each VM is placed in a single zone.
Where	In a zone-enabled region. You set the zone for each VM when you create it.
When	When designing for a 99.99% VM uptime target.
Why	One VM has no backup copy and gets at most 99.9%, so it fails.
How	Deploy two or more VMs across two or more zones to get 99.99%.

Exam tip
Count the VMs first: one VM gives at most 99.9%. 99.99% = two VMs in two zones.

Memory trick
One VM can't stand in two zones at once.

Real-world example, use case and scenario

EXAMPLE
Single-instance SLA
One VM with premium SSD in zone 2 has a 99.9% SLA, about 43 minutes of allowed downtime a month.

USE CASE
Kiosk back end
A museum's ticket-kiosk system needs 99.99%. It runs two VMs in zones 1 and 3, not one large VM.

SCENARIO
Budget shortcut
A small airline's IT manager tries to save money with one VM 'covering two zones'. The portal allows only one zone per VM. He adds a second, smaller VM in another zone to meet 99.99%.

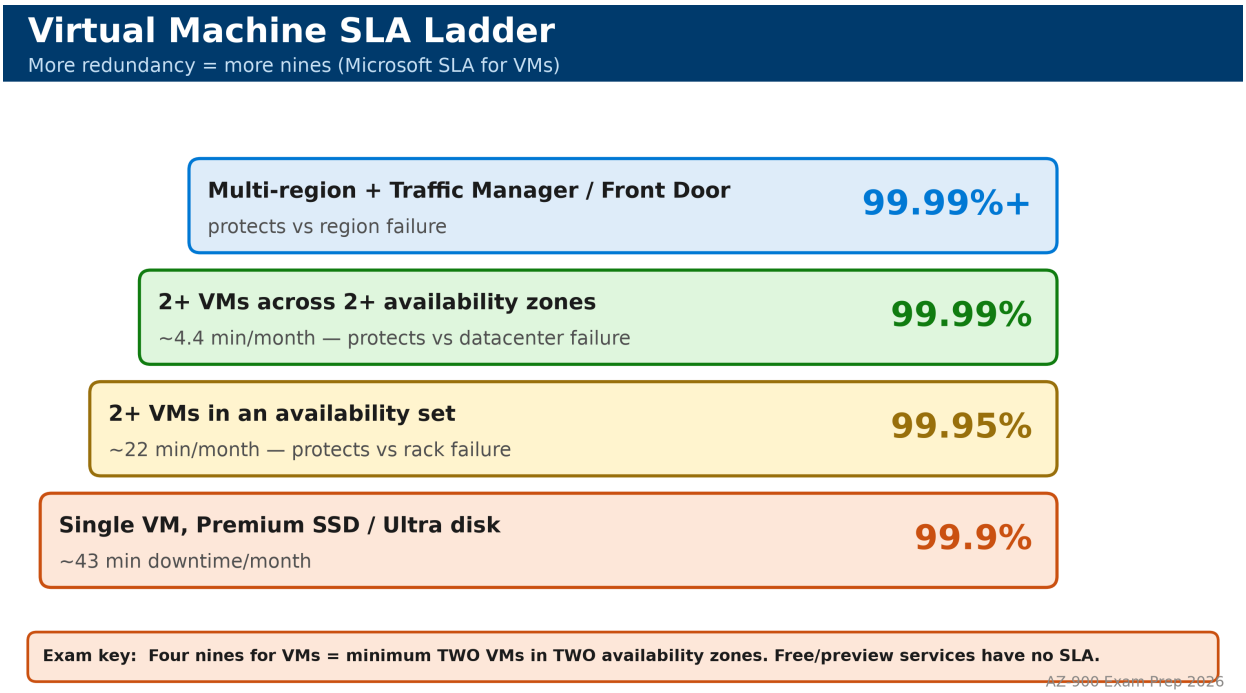


Figure Q100a: Virtual machine SLA ladder

About this figure

- Climb the ladder: each step adds redundancy and a higher VM SLA; the downtime figures assume a 30-day month.
- A single VM with Standard SSD gets about 99.5% and with Standard HDD about 95%; all its OS and data disks must meet that tier.
- Availability sets spread VMs across fault domains (racks) and update domains (maintenance batches) in one datacenter; zones spread across datacenters.
- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

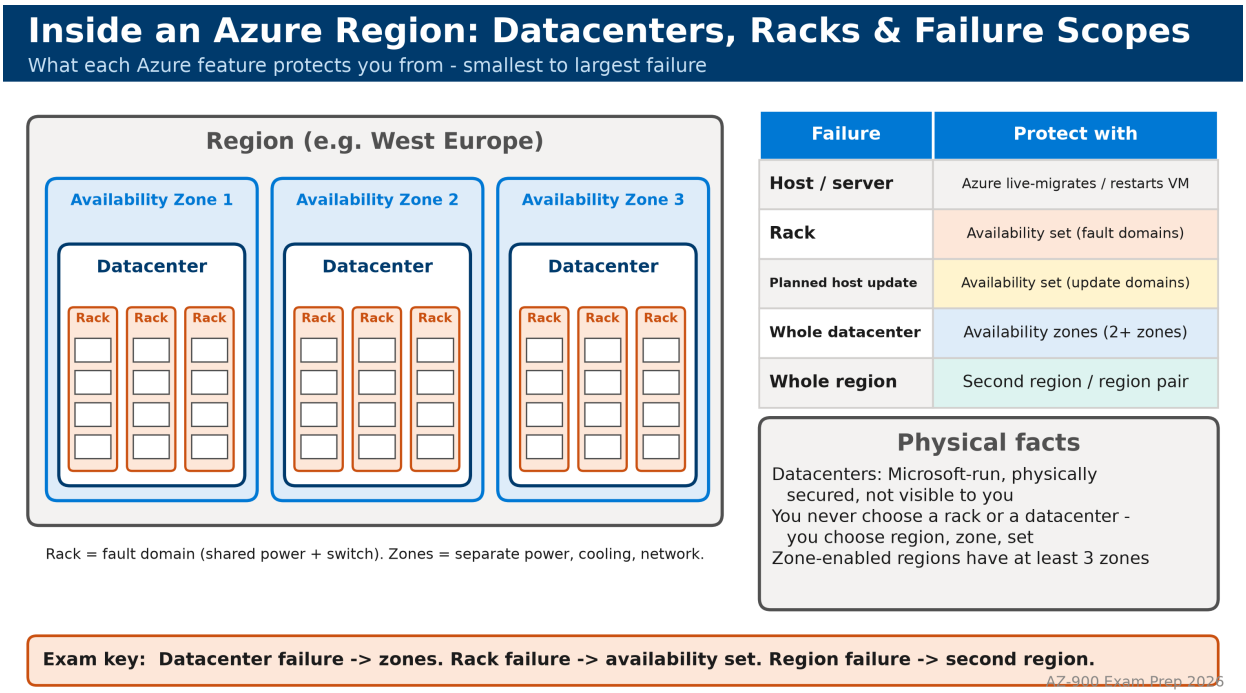


Figure Q100b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 101

AZURE ARCHITECTURE & SERVICES

DevOps & IaC
v5 Q173

Your company has developers intend to deploy a large number of custom virtual machines on a weekly basis. They will also be removing these virtual machines during the same week it was deployed. Sixty percent of the virtual machines have Windows Server 2016 installed, while the other forty percent has Ubuntu Linux installed. You are required to make sure that the administrative effort, needed for this process, is reduced by employing a suitable Azure service. Solution: you recommend the use of Microsoft Managed Desktop. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is less admin work for many short-lived Windows and Linux VMs. Only DevTest Labs (Q103) is Yes. Managed Desktop (Q101) and Reserved Instances (Q102) are No. Q33 has the same idea.

Answer explained

✔ **No.**
Microsoft Managed Desktop is a service where Microsoft manages employees' Windows PCs. It cannot create or delete Azure VMs, and it does nothing for Ubuntu servers.

✖ **Yes.**
Managed Desktop manages user Windows devices, not weekly Azure VM churn (frequent create and delete). Azure DevTest Labs is the service that reduces that work.

EXAM STRATEGY
Microsoft Managed Desktop manages user Windows PCs, not Azure VMs. The mix of Windows Server and Ubuntu rules it out, so No.

IMPACT
Buying a desktop management service for server labs wastes money. Developers still create and delete VMs by hand.

What · Where · When · Why · How

What	Microsoft Managed Desktop is a service where Microsoft manages a company's Windows user PCs.
Where	It covers employee laptops and desktops enrolled through Intune and Autopilot, not Azure subscriptions.
When	When a company wants Microsoft to handle enrolment, updates and security for its PCs.
Why	It cannot create or delete Azure VMs or manage Linux, so it does not reduce lab VM work.
How	Devices are enrolled and Microsoft runs their policies and updates. Weekly lab VMs need DevTest Labs instead.

Exam tip
Anything with 'Desktop' in the name (Managed Desktop, Virtual Desktop) is about user workspaces, not lab VMs you create and delete weekly.

Memory trick
Managed Desktop manages DESKS (people's PCs), not LABS.

Real-world example, use case and scenario

EXAMPLE
Laptop fleet
Microsoft Managed Desktop enrols 800 new Windows laptops for a law firm and keeps them updated and within policy.

USE CASE
What the dev team really needs
A game studio's build engineers need 40 new Windows and Ubuntu VMs each Monday from custom images, gone by Friday. That is a DevTest Labs job.

SCENARIO
Wrong product, wrong problem
An IT manager hears 'managed' and suggests Managed Desktop for the weekly VMs. The architect explains it cannot create Azure VMs or handle Linux. The team uses DevTest Labs with auto-shutdown and expiry dates instead.

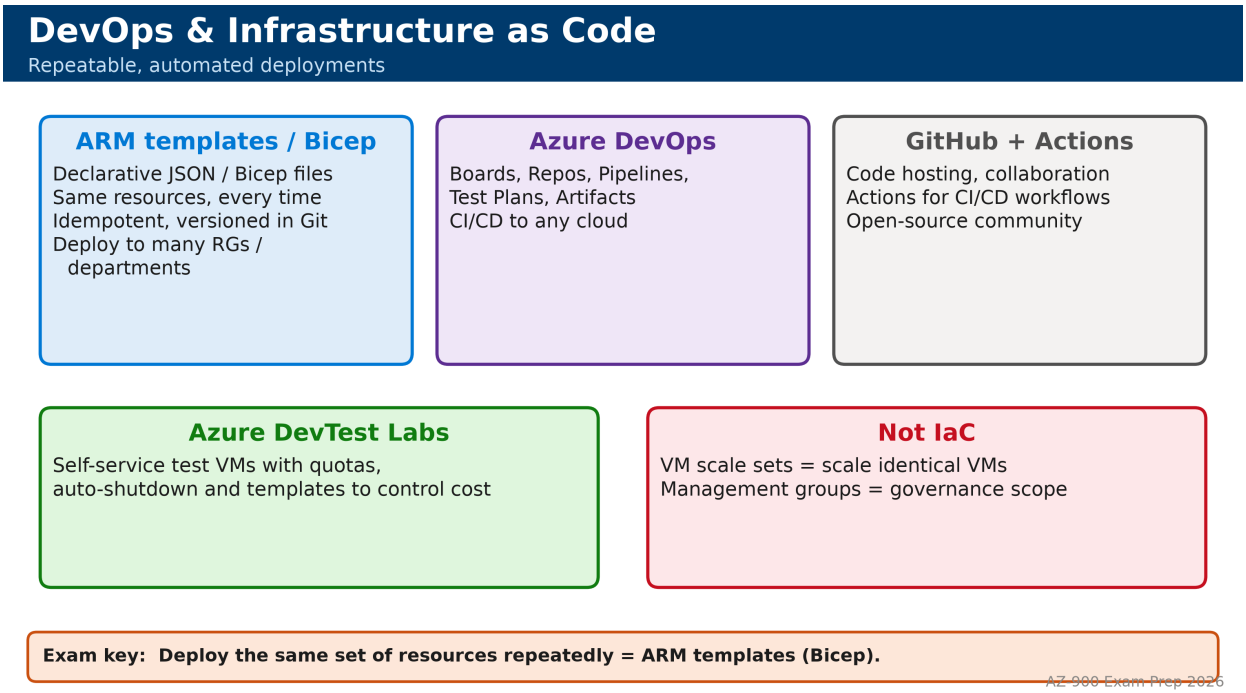


Figure Q101: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 102

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q174

Your company has developers intend to deploy a large number of custom virtual machines on a weekly basis. They will also be removing these virtual machines during the same week it was deployed. Sixty percent of the virtual machines have Windows Server 2016 installed, while the other forty percent has Ubuntu Linux installed. You are required to make sure that the administrative effort, needed for this process, is reduced by employing a suitable Azure service. Solution: you recommend the use of Azure Reserved Virtual Machines (VM) Instances.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same scenario as Q101. Watch the goal: less ADMIN WORK, not a lower price. A reservation only lowers the price of VMs that run for 1-3 years. It automates nothing, so No.

2026 update: Reserved VM Instances are now sold under Azure Reservations, next to the more flexible Azure savings plan for compute.

Answer explained

✔ **No.**
Reserved VM Instances are a 1- or 3-year price commitment. They automate nothing and suit VMs that are always on, not VMs deleted within a week.

✖ **Yes.**
A reservation only lowers the compute price. It does not create, delete or manage VMs, so admin work stays the same.

EXAM STRATEGY
Reservations are a 1- or 3-year price commitment, not automation. Short-lived weekly VMs do not fit any long-term commitment, so No.

IMPACT
Reserving capacity for VMs deleted within a week means paying for unused commitments. Reservations suit steady, always-on workloads.

What · Where · When · Why · How

What	Azure Reserved VM Instances are a 1- or 3-year commitment that lowers VM compute prices.
Where	Bought under Reservations in the Azure portal, for a subscription, resource group or billing account.
When	When VMs run all the time, predictably, for one to three years.
Why	It changes only the price, so it does not reduce the work of creating and deleting VMs.
How	The discount is applied automatically on the bill to matching running VMs.

Exam tip
When the goal says 'reduce administrative effort', remove every pricing option (reservations, savings plan, Hybrid Benefit) at once.

Memory trick
Reserve = a parking spot you pay for all year; weekly visitors don't need one.

Real-world example, use case and scenario

EXAMPLE
Steady workload discount
A 3-year reservation on a D4s v5 VM that runs 24x7 can cut its compute rate by about 60 percent compared with pay-as-you-go.

USE CASE
Bank core VM
A regional bank reserves the VMs behind its always-on core banking software, because their use never drops.

SCENARIO
Paying for idle commitment
A QA team buys reservations for VMs they delete every Thursday. Two months later, Cost Management shows low reservation use. They switch to DevTest Labs with auto-shutdown and keep the reservation only for a long-running build server.

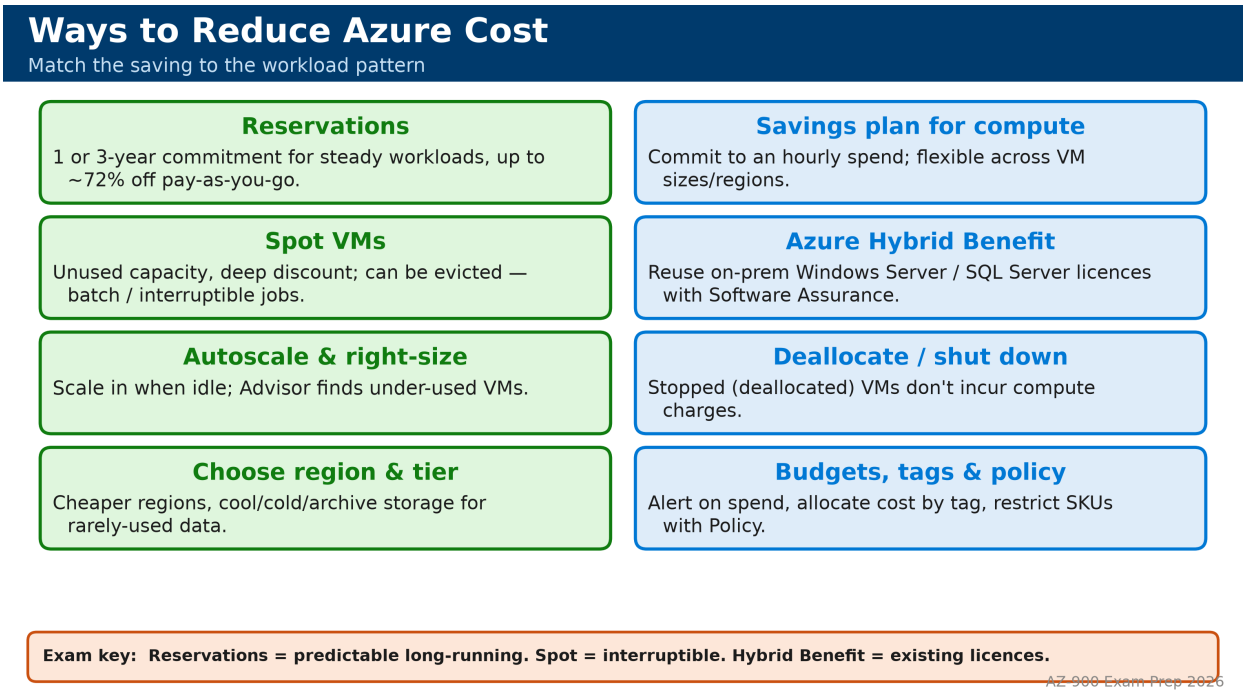


Figure Q102a: Ways to reduce cost

About this figure

• Each box matches a saving to a workload pattern: steady (reservations), flexible (savings plan), interruptible (Spot), existing licences (Hybrid Benefit).

• Reservations save up to approx. 72% versus pay-as-you-go; savings plans for compute save up to approx. 65% and apply across VM families and regions.

• Spot VMs can be up to approx. 90% cheaper, but Azure can evict them with 30 seconds' notice when it needs capacity back.

• Combining Azure Hybrid Benefit with reservations can cut Windows VM costs by up to approx. 80%; Linux subscriptions such as RHEL and SUSE also qualify.

• Example: moving rarely read data from Hot (~USD 0.02/GB-month) to Archive (~USD 0.002 or less) saves storage cost, but reads and early deletion cost more.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q102b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 103

AZURE ARCHITECTURE & SERVICES

DevOps & IaC
v5 Q175

Your company has developers intend to deploy a large number of custom virtual machines on a weekly basis. They will also be removing these virtual machines during the same week it was deployed. Sixty percent of the virtual machines have Windows Server 2016 installed, while the other forty percent has Ubuntu Linux installed. You are required to make sure that the administrative effort, needed for this process, is reduced by employing a suitable Azure service. Solution: you recommend the use of Azure DevTest Labs. Does the solution meet the goal?

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): Same scenario as Q101 and Q102, but this one is Yes. Key words: developers, custom VMs, created and deleted each week, less admin work. That is DevTest Labs. Q33 has the same answer.

Answer explained

✔ **Yes.**
Azure DevTest Labs creates Windows and Linux VMs from custom images or formulas (saved VM recipes). It sets auto-shutdown, quotas and expiry dates that delete VMs automatically.

✖ **No.**
DevTest Labs is built for short-lived dev/test VMs with automatic cleanup. So it does reduce the weekly create-and-delete work.

EXAM STRATEGY
'Often create and delete custom Windows and Linux VMs, with less admin work' = Azure DevTest Labs. Its formulas, auto-shutdown and expiry fit exactly.

IMPACT
DevTest Labs' expiry dates and auto-shutdown stop forgotten developer VMs from billing for weeks. Forgotten VMs are a common cause of Azure overspending.

What · Where · When · Why · How

What	Azure DevTest Labs is a service for quickly creating and managing dev/test VMs.
Where	Created as a lab resource in an Azure subscription, with policies set on the lab.
When	When developers often create and delete Windows or Linux VMs from custom images.
Why	Formulas, quotas, auto-shutdown and expiry dates cut the admin work of weekly VM churn.
How	Users claim or create VMs from custom images or formulas. Expiry policies delete them automatically.

Exam tip
Keywords 'developers', 'custom VMs', 'deploy and remove weekly' = DevTest Labs. The OS mix (Windows plus Ubuntu) is a distractor; Labs supports both.

Memory trick
DevTest Labs: Deploy, Test, Expire.

Real-world example, use case and scenario

EXAMPLE
Expiration date
A lab formula builds an Ubuntu 22.04 VM with Docker installed and sets it to expire after five days. It is deleted without anyone remembering to do it.

USE CASE
Insurance app testing
An insurer's test team takes ready-made Windows Server images from a lab each sprint to test its claims website. Each tester can have at most five VMs.

SCENARIO
Forgotten VMs
A startup found 120 forgotten test VMs costing thousands per month. Moving to DevTest Labs with nightly auto-shutdown and 7-day expiry ended the cleanup work and the surprise bills.

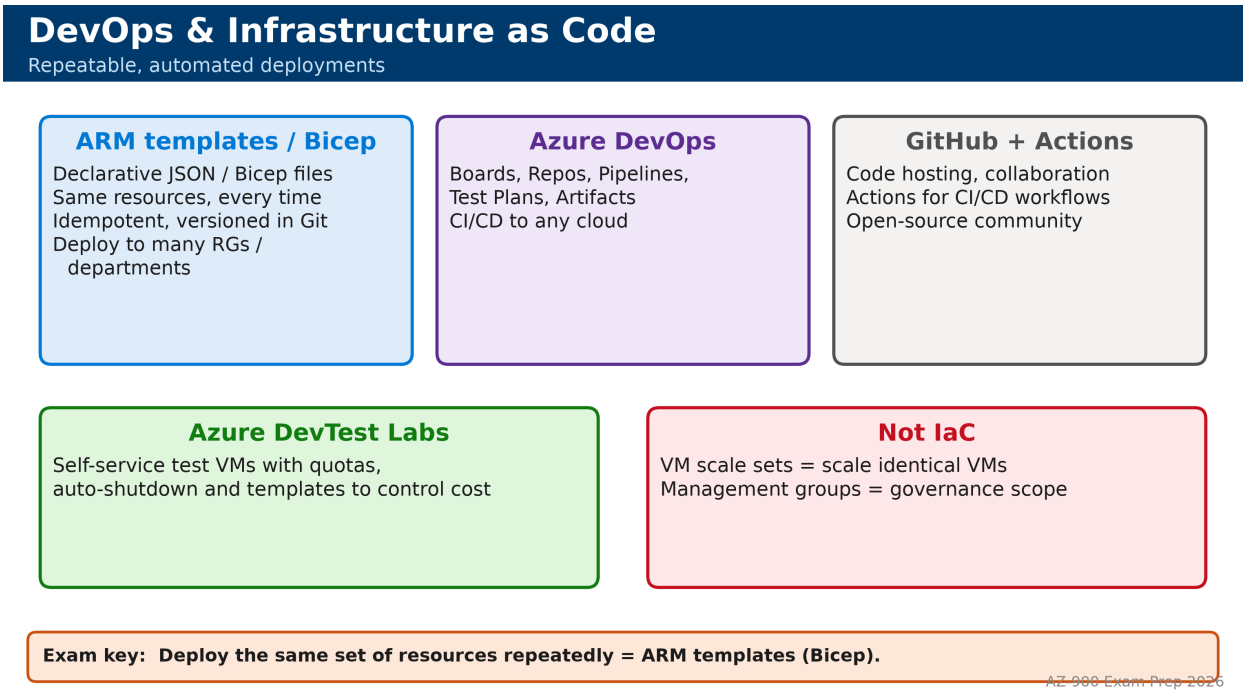


Figure Q103: DevOps and infrastructure as code

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- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 104

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q176

Your company has virtual machines (VMs) hosted in Microsoft Azure. The VMs are located in a single Azure virtual network named VNet1. The company has users that work remotely. The remote workers require access to the VMs on VNet1. You need to provide access for the remote workers. What should you do?

- ✖ A. Configure a Site-to-Site (S2S) VPN.
- ✖ B. Configure a VNet-toVNet VPN.
- ✔ **C.** Configure a Point-to-Site (P2S) VPN.
- ✖ D. Configure DirectAccess on a Windows Server 2012 server VM.
- ✖ E. Configure a Multi-Site VPN.

Correct answer: Configure a Point-to-Site (P2S) VPN.

Clarification (read this first): Key words: 'remote workers' (single people, not an office). Point-to-Site connects one device. Site-to-Site connects a whole office network.

Answer explained

✔ **Configure a Point-to-Site (P2S) VPN.**
Point-to-Site VPN connects single remote devices to an Azure VNet through the VPN gateway. Each laptop only needs a VPN client.

✖ **Configure a Site-to-Site (S2S) VPN.**
Site-to-Site links a whole on-premises network through a VPN device with a public IP. It is not for single remote workers.

✖ **Configure a VNet-toVNet VPN.**
VNet-to-VNet connects two Azure virtual networks to each other, not users to a VNet.

✖ **Configure DirectAccess on a Windows Server 2012 server VM.**
DirectAccess is an old on-premises Windows remote-access feature. It is not supported on Azure VMs.

✖ **Configure a Multi-Site VPN.**
Multi-Site is several Site-to-Site tunnels from many offices to one gateway. It still needs on-premises VPN devices.

EXAM STRATEGY
'Individual remote workers' = Point-to-Site.
Site-to-Site and Multi-Site link networks.
VNet-to-VNet links Azure networks. DirectAccess is an on-premises Windows feature.

IMPACT
P2S gives laptops safe VNet access without putting VMs on public IPs. This avoids open RDP or SSH ports that attackers scan all the time.

What · Where · When · Why · How

What	A Point-to-Site VPN connects one client device to an Azure virtual network.
Where	Set up on the VNet's VPN gateway, with a client address pool and a sign-in type.
When	When remote or home users need private access to VMs in a VNet.
Why	Each worker connects from their own device, with no on-premises VPN device needed.
How	Users install a VPN client and sign in by certificate, Entra ID or RADIUS, over OpenVPN, IKEv2 or SSTP.

Exam tip
'Remote workers / individual users' = Point-to-Site.
'Office / on-premises network' = Site-to-Site.
'Private, not over internet' = ExpressRoute.

Memory trick
P2S = Person to Site.

Real-world example, use case and scenario

EXAMPLE
Azure VPN Client
Employees install the Azure VPN Client and sign in with Entra ID. They get a private IP from the P2S pool and can RDP into VMs on VNet1.

USE CASE
Field auditors
An accounting firm's travelling auditors use P2S from client offices to reach a document server in Azure.

SCENARIO
No router at home
An admin first suggests S2S, then sees that 60 home-based staff have no VPN devices. Switching to P2S with Entra sign-in connects everyone in a day.

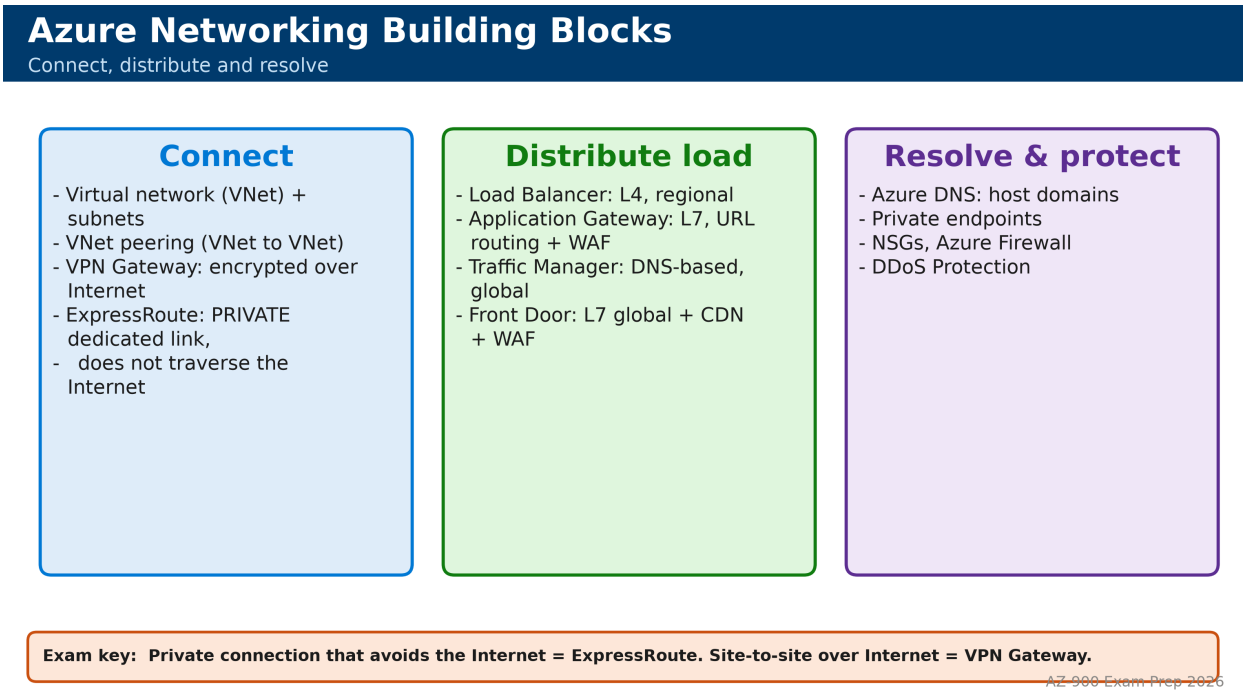


Figure Q104: Azure networking building blocks

About this figure

- Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.
- VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.
- VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.
- Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.
- Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 105

AZURE ARCHITECTURE & SERVICES

Security v5 Q177

You have been informed by your superiors of the company has intentions to automate server deployment to Azure. There is, however, some concern that administrative credentials could be uncovered during this process. You are required to make sure that during the deployment, the administrative credentials are encrypted using a suitable Azure solution. Solution: You recommend the use of Microsoft Purview Information Protection. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is to protect admin passwords used by a deployment script. Purview protects documents and emails, so No. Q106 (MFA) is also No. Q50 gives the right answer: Azure Key Vault.

2026 update: Microsoft Purview Information Protection is the current name for the old Azure Information Protection.

Answer explained

✔ **No.**
Microsoft Purview Information Protection labels and encrypts documents and emails. It is not a secret store that deployment templates can read passwords from.

✖ **Yes.**
Purview protects files and messages. Deployment passwords must be stored in Azure Key Vault and referenced by the template.

EXAM STRATEGY
Purview Information Protection labels documents and emails; it is not a secret store. Deployment passwords need Key Vault, so No.

IMPACT
Using a document-labelling tool for secrets leaves admin passwords inside scripts and code. That is a top cause of cloud breaches.

What · Where · When · Why · How

What	Microsoft Purview Information Protection classifies, labels and encrypts documents and emails.
Where	Managed in the Microsoft Purview portal. Applied in Office apps, email and files.
When	When sensitive documents must stay protected wherever they are shared.
Why	It is not a secret store, so deployment scripts cannot use it to protect admin passwords.
How	Sensitivity labels add encryption and access rights to content. Key Vault is used for deployment secrets.

Exam tip
Credentials, secrets, keys, certificates in deployments = Key Vault. Purview Information Protection = documents and emails.

Memory trick
Purview protects PAGES; Key Vault protects PASSWORDS.

Real-world example, use case and scenario

EXAMPLE
Sensitivity label
A Purview label marks a spreadsheet Confidential, so only employees can open it.

USE CASE
Legal contracts
A law firm uses Purview Information Protection to encrypt client contracts shared with outside lawyers.

SCENARIO
Wrong vault
A cloud engineer suggests Purview to protect the VM admin password in a pipeline. A reviewer notes the template cannot read a Purview label. The password is moved to a Key Vault secret that the Bicep file references.

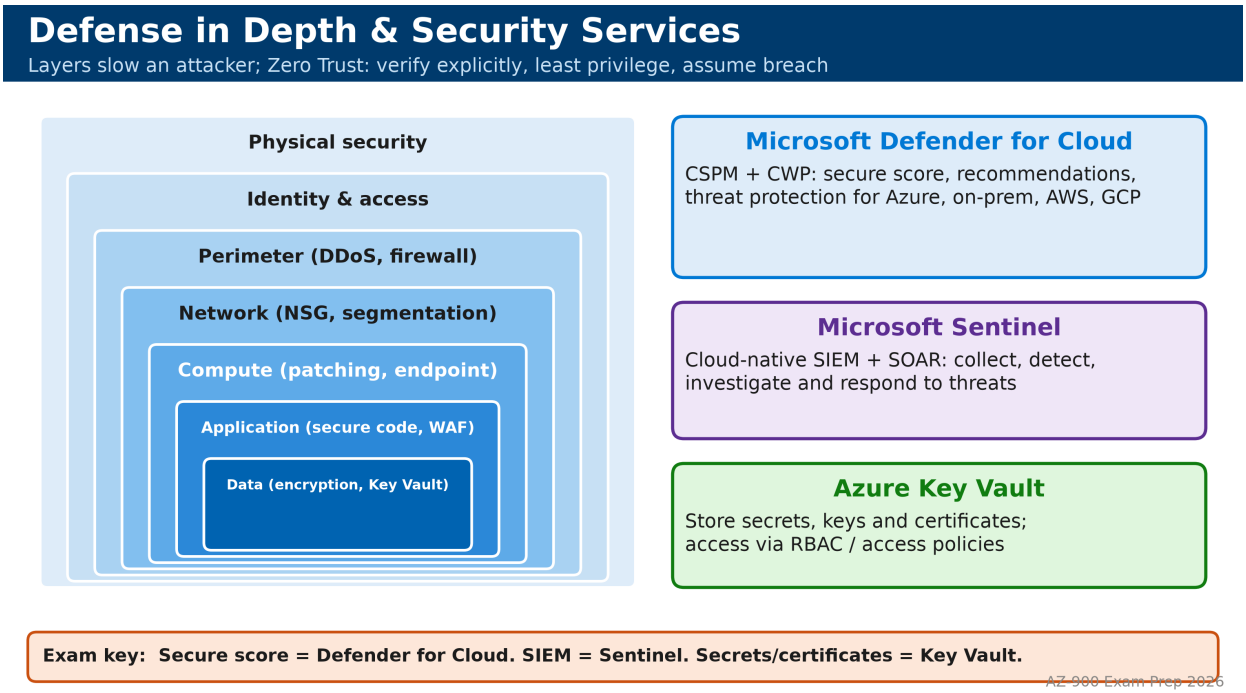


Figure Q105: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 106

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q179

You have been informed by your superiors of the company has intentions to automate server deployment to Azure. There is, however, some concern that administrative credentials could be uncovered during this process. You are required to make sure that during the deployment, the administrative credentials are encrypted using a suitable Azure solution. Solution: You recommend the use of Azure Multi-Factor Authentication (MFA). Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same goal as Q105. MFA checks who is signing in. It does not store or encrypt passwords, so No. Q50 gives the right answer: Azure Key Vault.

2026 update: Azure Multi-Factor Authentication is now called Microsoft Entra multifactor authentication.

Answer explained

✔ **No.**
MFA adds a second check when a person signs in. It does not store or encrypt passwords used by a deployment.

✖ **Yes.**
MFA checks who is signing in. It has no store for admin passwords, so Key Vault is needed instead.

EXAM STRATEGY
MFA protects sign-ins by people; it does not store or encrypt passwords. A storage need rules out any sign-in method, so No.

IMPACT
MFA is vital for users, but an automated pipeline needs a secret store. Mixing them up leaves deployment passwords unprotected.

What · Where · When · Why · How

What	Microsoft Entra multifactor authentication asks for a second proof at sign-in.
Where	Enforced on user sign-ins through Conditional Access or security defaults in Entra ID.
When	When you need to protect sign-ins against stolen passwords.
Why	It checks people and stores no passwords, so it cannot encrypt deployment secrets.
How	After the password, the user approves an Authenticator prompt, enters a code or uses a FIDO2 key.

Exam tip
MFA = proving who you are at sign-in. It never encrypts or stores anything. Reject it whenever the goal is protecting secrets in code.

Memory trick
MFA asks 'Is it really you?'; Key Vault asks 'Where's it kept?'

Real-world example, use case and scenario

EXAMPLE
Authenticator push
An admin enters a password and then approves a number-matching prompt in Microsoft Authenticator.

USE CASE
Hospital admins
A hospital requires MFA for every account with Contributor rights on its Azure subscription.

SCENARIO
Pipeline stuck
A team turns on MFA for the user account their deployment script used. The nightly build then waits forever for a prompt. They replace the user with a managed identity that reads secrets from Key Vault.

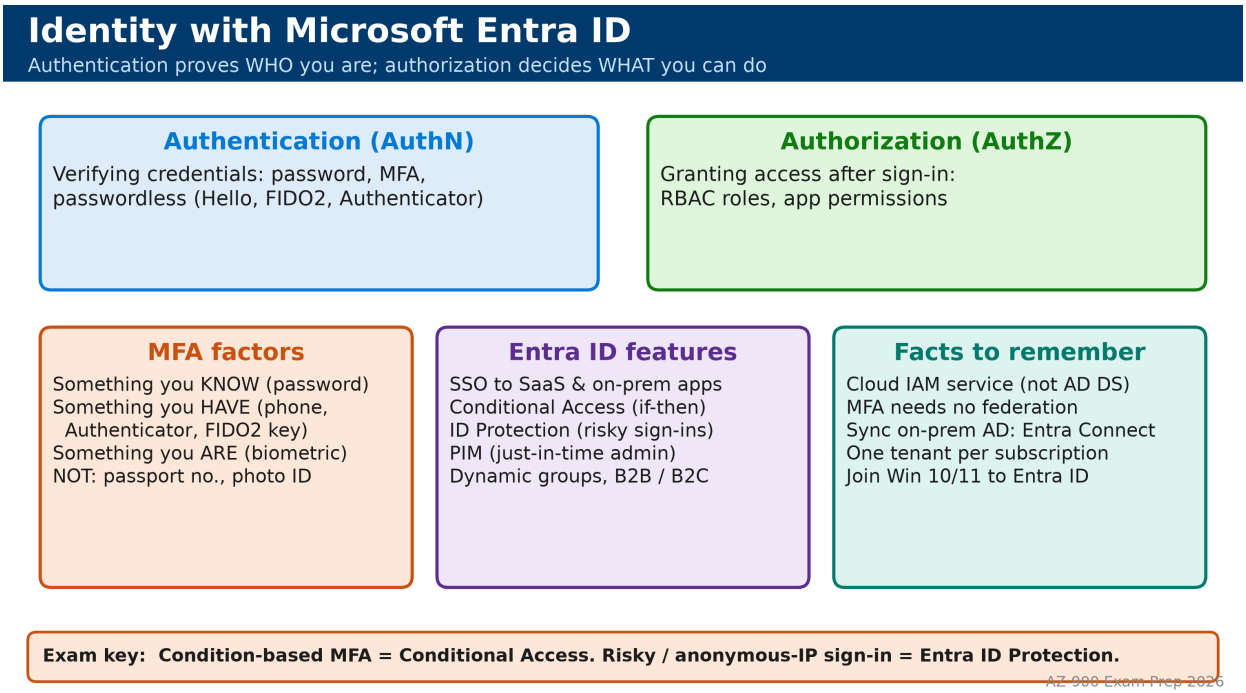


Figure Q106: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 107

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q180

Your company has a Microsoft Entra ID environment. Users occasionally connect to Microsoft Entra ID via the Internet. You have been tasked with making sure that users who connect to Microsoft Entra ID via the internet from an unidentified IP address, are automatically encouraged to change passwords. Solution: You configure the use of Azure Key Vault. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The question says 'encouraged', but it means users are prompted to change their password after a risky sign-in. Key Vault cannot see sign-ins, so No. Q108 (PIM) is also No. Q16 gives the right answer: Microsoft Entra ID Protection.

Answer explained

✔ **No.**
Azure Key Vault stores keys, secrets and certificates for apps. It cannot see user sign-ins or ask anyone to change a password.

✖ **Yes.**
Finding sign-ins from anonymous IPs and forcing a password change is done by Microsoft Entra ID Protection, not Key Vault.

EXAM STRATEGY
Key Vault stores secrets for apps; it never checks user sign-ins. 'Risky sign-in leads to password change' is an identity-protection task, so No.

IMPACT
Wrong controls leave hacked accounts active. Real risk response must come from sign-in analysis, not a secret store.

What · Where · When · Why · How

What	Azure Key Vault is a secure store for app keys, secrets and certificates.
Where	A vault resource in an Azure subscription, used by apps and identities.
When	When apps need connection strings, API keys or certificates kept out of code.
Why	It cannot see user sign-ins, so it cannot ask risky users to change passwords.
How	Apps sign in to the vault and read secrets. Sign-in risk is handled by Entra ID Protection.

Exam tip
'Anonymous IP', 'unfamiliar location', 'leaked credentials', 'risky sign-in' = Microsoft Entra ID Protection.

Memory trick
Key Vault holds keys; it never knocks on a user's door.

Real-world example, use case and scenario

EXAMPLE
Vault scope
Key Vault holds the API key a web app uses for a payment gateway. It knows nothing about who signs in to Entra ID.

USE CASE
Proper Key Vault use
A logistics firm keeps its shipping-label API certificates in Key Vault with automatic renewal.

SCENARIO
Mixed-up meeting
In a security meeting, someone suggests Key Vault to handle sign-ins from the Tor network (an anonymous network). The identity lead explains Key Vault cannot see sign-ins. She turns on an Entra ID Protection user-risk policy instead.

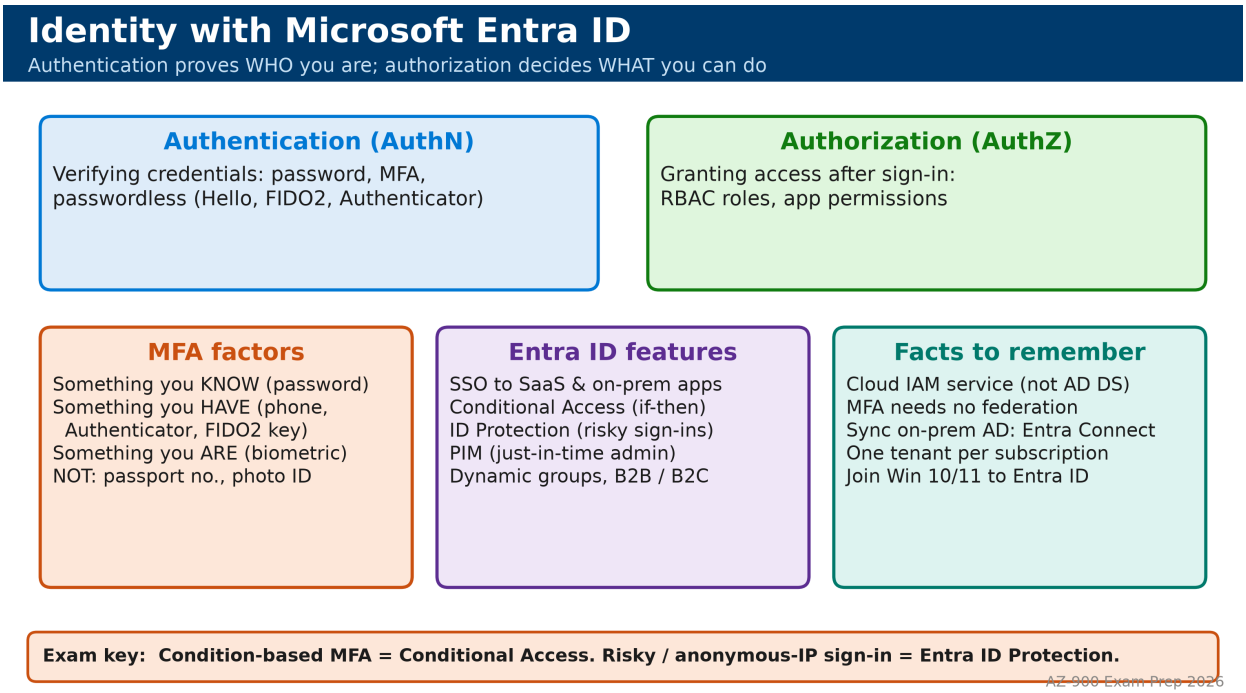


Figure Q107: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 108

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q182

Your company has a Microsoft Entra ID environment. Users occasionally connect to Microsoft Entra ID via the Internet. You have been tasked with making sure that users who connect to Microsoft Entra ID via the internet from an unidentified IP address, are automatically encouraged to change passwords. Solution: You configure the use of Microsoft Entra Privileged Identity Management. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Same goal as Q107. PIM (Privileged Identity Management) gives admins temporary admin rights. It does not check sign-in IP addresses, so No. Q16 gives the right answer: Microsoft Entra ID Protection.

Answer explained

✔ **No.**
Privileged Identity Management (PIM) gives admin roles just-in-time (only when needed), for a limited time, with approval. It does not check sign-in IP addresses or force password changes.

✖ **Yes.**
PIM controls when admin roles are turned on. Finding risky IPs and forcing a password change come from Entra ID Protection.

EXAM STRATEGY
PIM controls just-in-time admin role activation. It never checks sign-in IPs or forces password resets, so No here.

IMPACT
PIM and ID Protection are both P2 features and easy to confuse. Picking PIM would leave risky sign-ins unhandled.

What · Where · When · Why · How

What	Microsoft Entra Privileged Identity Management gives just-in-time, time-limited access to admin roles.
Where	In the Microsoft Entra admin center, for Entra roles and Azure resource roles.
When	When admins should turn on higher roles only when needed, with approval and logging.
Why	It manages role activation and does not detect risky IPs or force password changes.
How	Users are made eligible for a role. They activate it for a limited time and give a reason.

Exam tip
PIM keywords: just-in-time, eligible vs active, approval, time-limited, admin roles. None mention IP addresses.

Memory trick
PIM = Promotion In Minutes (temporary admin), not a sign-in guard.

Real-world example, use case and scenario

EXAMPLE
Eligible role
A database admin is eligible, not permanently assigned, for the SQL Server Contributor role. He activates it for one hour when needed.

USE CASE
Bank admin rights
A bank uses PIM so no one holds permanent Owner rights on production subscriptions.

SCENARIO
Right tool, wrong problem
An admin turns on PIM hoping it will stop suspicious sign-ins from abroad. Nothing changes for normal users. The team adds an ID Protection sign-in risk policy and keeps PIM for admin roles.

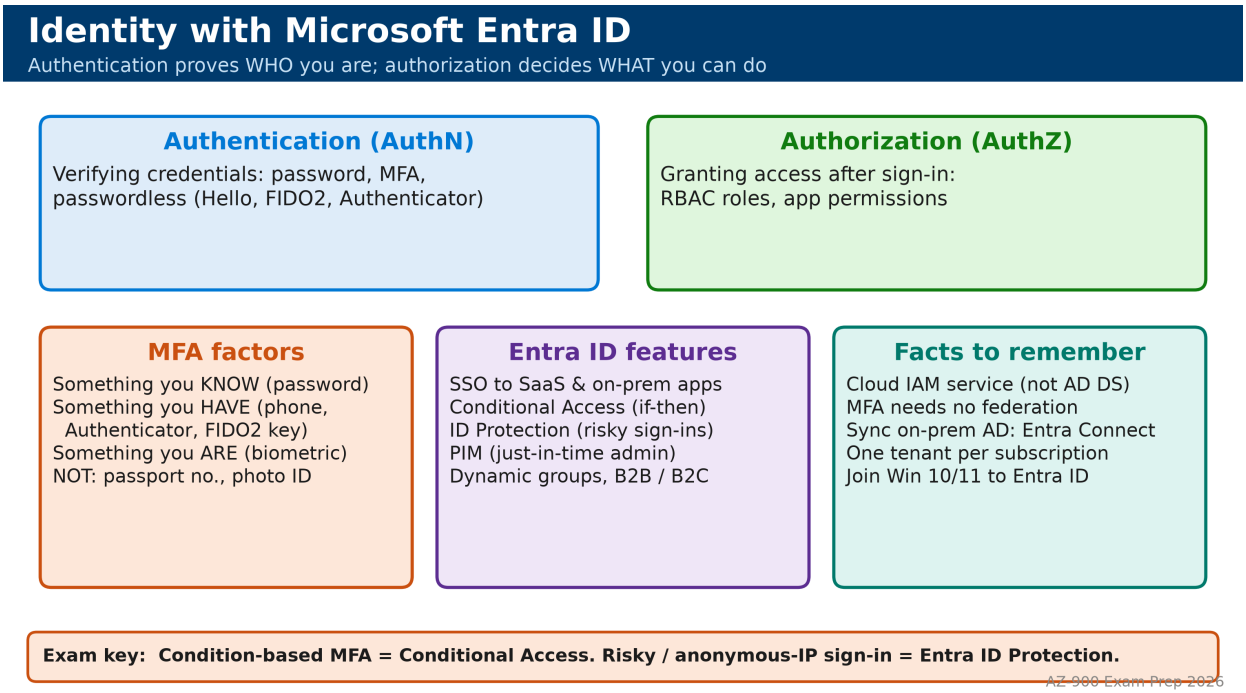


Figure Q108: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 109

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q183

You are planning a strategy to deploy numerous web servers and database servers to Azure. This strategy should allow for connection types between the web servers and database servers to be controlled. Solution: You include the Azure Service Bus in your strategy. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is to control which kinds of network traffic (ports and protocols) can go between web servers and database servers. This is one of a series of 'Solution' questions. Q52 asks the same goal directly and the answer is network security groups (NSGs). Q110 tries a local network gateway, also No.

Answer explained

✔ **No.**
Azure Service Bus is a message broker (a service that passes messages between apps using queues and topics). It cannot allow or block network connections by port or protocol.

✖ **Yes.**
Service Bus moves messages between apps. Network security groups (NSGs) filter traffic between web and database servers.

EXAM STRATEGY
Service Bus is for messages (queues, topics). It cannot filter ports or protocols. 'Control connection types' between servers means network filtering, so the answer is No.

IMPACT
If you use a message service as a firewall, the database ports stay open. You must block traffic at the network layer.

What · Where · When · Why · How

What	Azure Service Bus is a managed message broker with queues and topics.
Where	A Service Bus namespace (a container for queues and topics) in an Azure subscription, used by app code.
When	When apps must send messages to each other reliably and work independently.
Why	It moves messages. It cannot allow or block network traffic between web and database servers.
How	A sender puts a message on a queue or topic. A receiver reads and handles it later.

Exam tip
'Control connection types / ports / protocols between subnets or VMs' = NSG. Service Bus = messages. Event Grid = events.

Memory trick
Service Bus carries passengers (messages). It does not guard the road.

Real-world example, use case and scenario

EXAMPLE
Queue, not firewall
A Service Bus queue holds 'order placed' messages until a worker reads them. It has no rules about TCP ports.

USE CASE
Airline bookings
An airline uses Service Bus topics so its booking site and ticket system can work separately.

SCENARIO
Design review catch
A junior architect adds Service Bus to 'secure' web-to-database traffic. The reviewer sees that the web servers still talk SQL directly to the database VMs. He adds an NSG on the database subnet that allows only port 1433 from the web subnet.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

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Figure Q109: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 110

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q185

You are planning a strategy to deploy numerous web servers and database servers to Azure. This strategy should allow for connection types between the web servers and database servers to be controlled. Solution: You include a local network gateway in your strategy. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is to control which kinds of traffic can go between web servers and database servers. Do not be misled by the word 'gateway' - it is not a firewall. Q52 gives the real answer: NSGs. Q249 is a look-alike where a local network gateway IS the answer, because there you must represent an on-premises VPN device.

Answer explained

✔ **No.**
A local network gateway only describes your on-premises VPN device (its public IP and address ranges) for a Site-to-Site VPN (a link from your office network to Azure). It has no rules to filter traffic.

✖ **Yes.**
A local network gateway describes the remote VPN site. To control web-to-database traffic you need NSGs.

EXAM STRATEGY
A local network gateway only describes the on-premises VPN end of a Site-to-Site link. It has no rules, so it cannot control traffic. The answer is No.

IMPACT
If you think a gateway filters traffic, you may leave connections between servers fully open.

What · Where · When · Why · How

What	A local network gateway is an Azure object that stands for an on-premises VPN device.
Where	Created in a resource group and used by a Site-to-Site connection on a VPN gateway.
When	When you set up a Site-to-Site VPN to an office or datacenter.
Why	It only holds the remote site's IP and address ranges. It cannot filter traffic between Azure servers.
How	Enter the on-premises device's public IP and address ranges, then link it to a VPN connection.

Exam tip
'Local network gateway' = the on-premises side of a Site-to-Site VPN. 'Virtual network gateway' = the Azure side. Neither filters traffic like an NSG.

Memory trick
LOCAL network gateway = a business card for your LOCAL (on-premises) router.

Real-world example, use case and scenario

EXAMPLE
What it stores
The local network gateway 'HQ-LNG' holds IP 203.0.113.10 and range 192.168.0.0/16 for the head office firewall.

USE CASE
Factory link
A manufacturer creates one local network gateway for each plant's VPN device to connect the plants to Azure.

SCENARIO
Misread name
An engineer thinks 'gateway' means firewall. He plans to filter web-to-database traffic with it. Tests show all ports are still open. The team adds NSGs on the subnets to really enforce the rules.

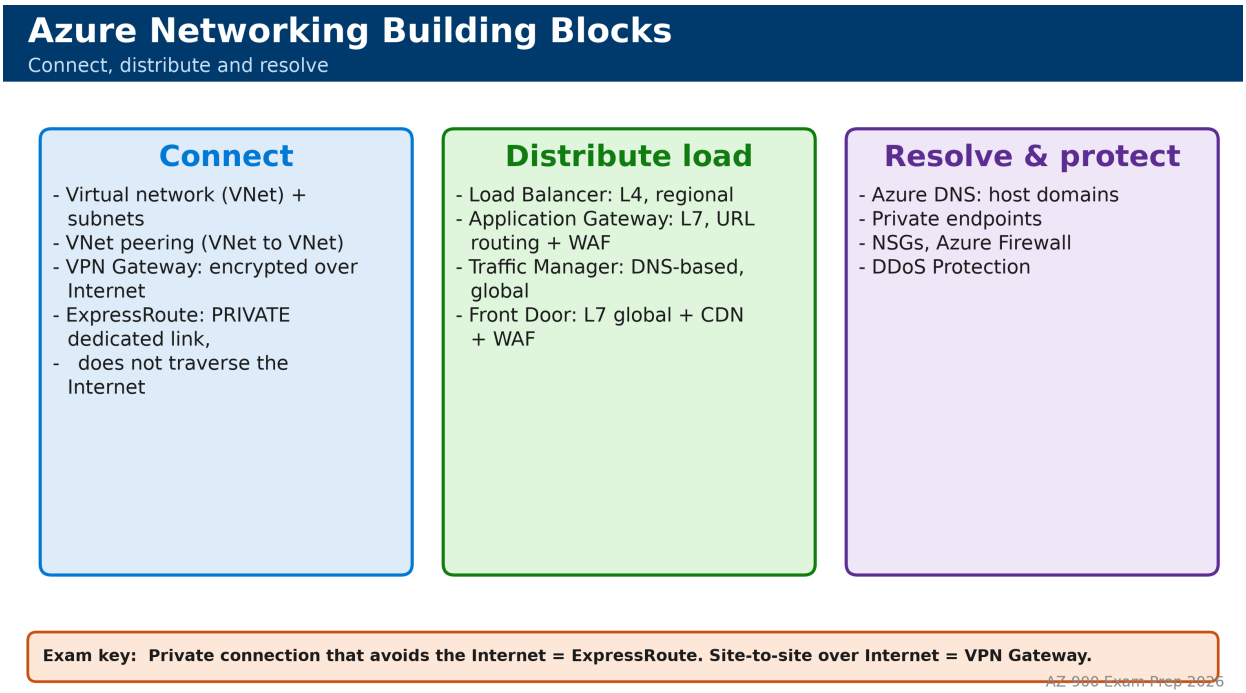


Figure Q110: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 111

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q186

Your company has Active Directory forest includes thousands of user accounts. You have been informed that all network resources will be migrated to Azure. Thereafter, the on-premises data center will be retired. You are required to employ a strategy that reduces the effect on users, once the planned migration has been completed. Solution: You plan to require Azure Multi-Factor Authentication (MFA). Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is to REDUCE the effect on users (fewer changes for them), not to improve security. MFA is good security, but it adds a new step for every user, so here it is No. Q81 shows the right answer: sync the existing AD accounts to Microsoft Entra ID.

2026 update: Azure Multi-Factor Authentication is now called Microsoft Entra multifactor authentication.

Answer explained

✔ **No.**
MFA (multifactor authentication - a second check at sign-in) makes every user register and do a new sign-in step. This increases the effect on users after the migration.

✖ **Yes.**
MFA improves security, but it changes every user's sign-in. Syncing the existing accounts is what keeps the effect on users small.

EXAM STRATEGY
'Reduce the effect on users' means fewer changes for users. MFA adds registration and new sign-in steps, so the answer is No.

IMPACT
MFA is still important for security. But starting it in the middle of a migration, without warning users, floods the help desk. This question tests user effect, not security.

What · Where · When · Why · How

What	Multifactor authentication adds a second check when users sign in.
Where	Turned on in Microsoft Entra ID through Conditional Access (rules for sign-in) or security defaults.
When	When account security must be stronger, usually as a separate planned project.
Why	It adds registration and prompts for every user, so it increases the effect on users.
How	Users register a method such as the Authenticator app and approve a prompt after typing their password.

Exam tip
'Reduce the effect on users' after migration = keep the same usernames and passwords (Entra Connect sync). Any option that makes users do something new is wrong.

Memory trick
Small effect = make it feel the same. MFA makes it feel different.

Real-world example, use case and scenario

EXAMPLE
Extra step
After MFA starts, every user must install Authenticator and approve prompts. The help desk gets hundreds of calls in the first week.

USE CASE
Where MFA fits
A credit union turns on MFA after the migration is stable, as a separate security project.

SCENARIO
Day-one chaos avoided
A CIO wants MFA on the day of the switch. The project lead says the goal is no disruption for users. So accounts are synced first, and MFA is added a month later with training.

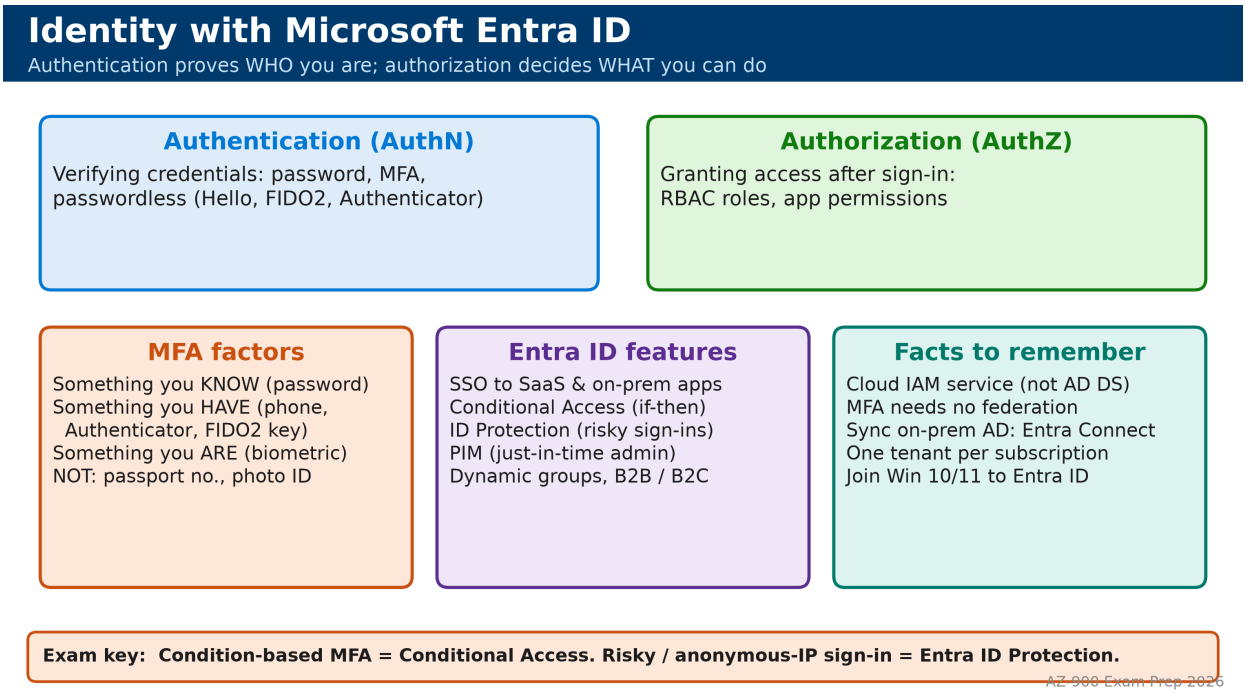


Figure Q111a: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

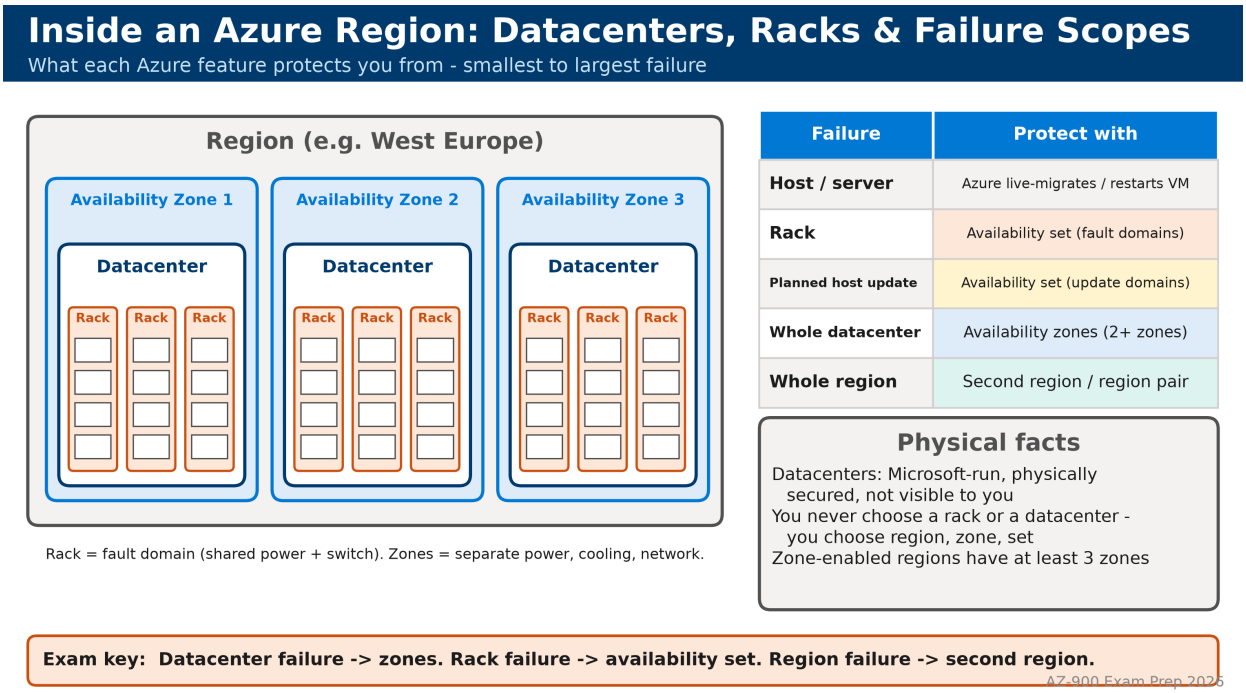


Figure Q111b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 112

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q188

Your company has Active Directory forest includes thousands of user accounts. You have been informed that all network resources will be migrated to Azure. Thereafter, the on-premises data center will be retired. You are required to employ a strategy that reduces the effect on users, once the planned migration has been completed. Solution: You plan to enforce password change. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is to REDUCE the effect on users. A forced password change makes every user do something new, so it is No. Same series as Q111 (MFA, also No). Q81 gives the right answer: sync the existing accounts to Microsoft Entra ID.

Answer explained

✔ **No.**
Forcing a password change makes thousands of users create new passwords. This causes lockouts. It also does not make their accounts available in Azure.

✖ **Yes.**
A mass password reset adds disruption. It also does not move user accounts to Entra ID, so it fails the goal.

EXAM STRATEGY
A forced password change makes every user do something new. That is the opposite of reducing the effect. It also does not move accounts to Azure. The answer is No.

IMPACT
Mass forced resets lock people out and waste work time for thousands of users. They give no migration benefit.

What · Where · When · Why · How

What	A forced password change makes users set a new password at their next sign-in.
Where	Applied to user accounts in Active Directory or Microsoft Entra ID.
When	When passwords are known or suspected to be stolen.
Why	It disrupts thousands of users and does not move their accounts to Azure.
How	Admins mark accounts to need a change. Users must choose a new password when they sign in.

Exam tip
In the 'reduce the effect on users' series, only the option that keeps existing passwords (sync) is Yes. MFA and password resets are both No.

Memory trick
New password = new pain.

Real-world example, use case and scenario

EXAMPLE
Reset fallout
Forcing 5,000 resets on one morning locks out shift workers who forget the new password by lunch.

USE CASE
When to force it
A media company forces a reset only for 40 accounts that appeared in a public data leak.

SCENARIO
Undoing a bad idea
A migration plan said 'reset all passwords on the day of the switch'. The service desk manager expected 2,000 extra tickets. The step was removed, and accounts were synced with their existing passwords.

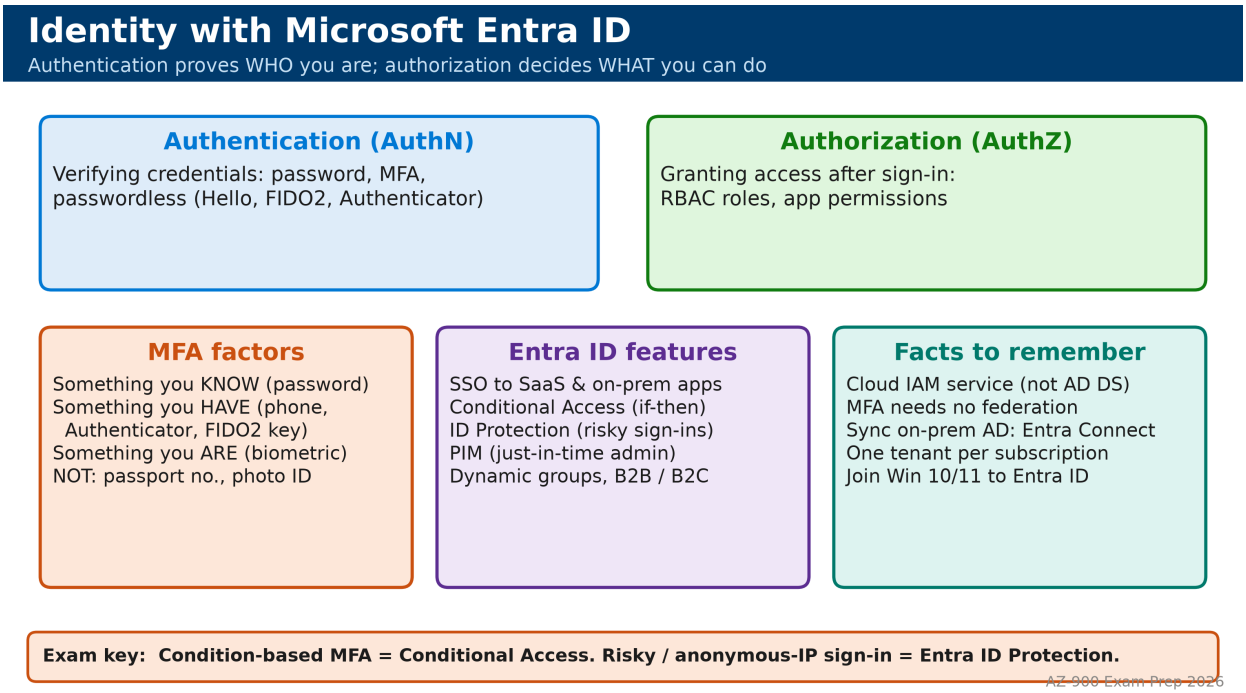


Figure Q112a: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
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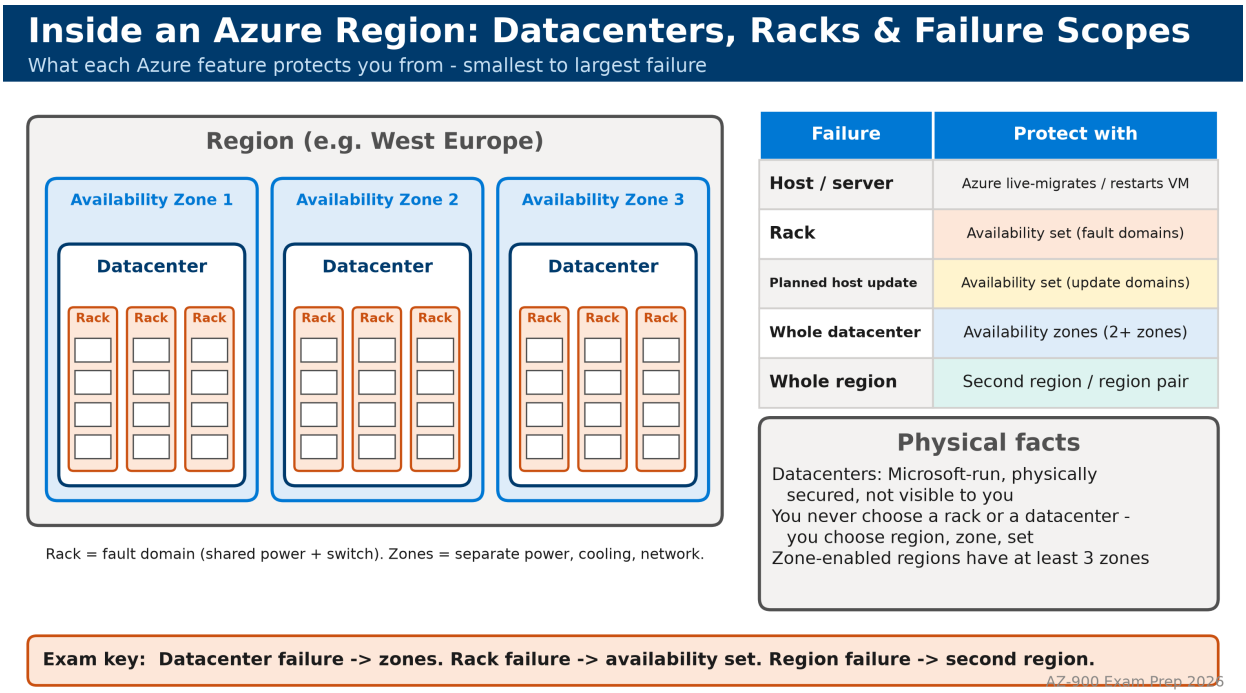


Figure Q112b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 113

AZURE MANAGEMENT & GOVERNANCE

Support
v5 Q189

You are currently trying to generate a number of managed Microsoft SQL Server instances in an Azure environment. During the process, you are presented with a notification informing you that the Azure subscription limits must be increased. You want to make sure that you can complete your task. Solution: You generate a service health alert. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is to RAISE a subscription limit (quota - the maximum amount of a resource you may use). An alert only tells you something; it changes nothing. Q57 gives the right answer: create a new support request. Q114 tries Azure Policy, also No.

Answer explained

✔ **No.**
A Service Health alert only tells you about Azure problems, planned maintenance and advisories. It cannot raise subscription limits.

✖ **Yes.**
Limits are raised with a quota or support request. A Service Health alert only sends messages.

EXAM STRATEGY
Service Health alerts only tell you about Azure problems and maintenance. They change nothing. 'Increase limits' needs a support request, so the answer is No.

IMPACT
Waiting for an alert to fix a quota stops your deployment. Know the one path that really raises limits.

What · Where · When · Why · How

What	A Service Health alert tells you about Azure service problems, maintenance and health advisories.
Where	Set up in Service Health in the Azure portal, sending to action groups (lists of who to notify).
When	When teams need to know about Azure platform problems in their regions or services.
Why	It only informs you. It cannot change subscription limits.
How	Create an alert rule for chosen services and regions, linked to email, SMS or webhook actions.

Exam tip
Increase limits or quotas = 'New support request' (quota type, free on every plan, including Basic). Alerts and policies never change quotas.

Memory trick
Health alerts TELL you. Support requests ASK Microsoft.

Real-world example, use case and scenario

EXAMPLE
Outage notice
A Service Health alert emails the ops team when SQL Database in West Europe has a slow-performance problem.

USE CASE
Broadcaster
A TV streaming company sends Service Health alerts to Teams so on-call staff know about maintenance times.

SCENARIO
Waiting for nothing
An engineer sets up a Service Health alert, hoping it will tell Microsoft about the quota block. Days pass with no change. She files a quota support request, and the limit is raised within hours.

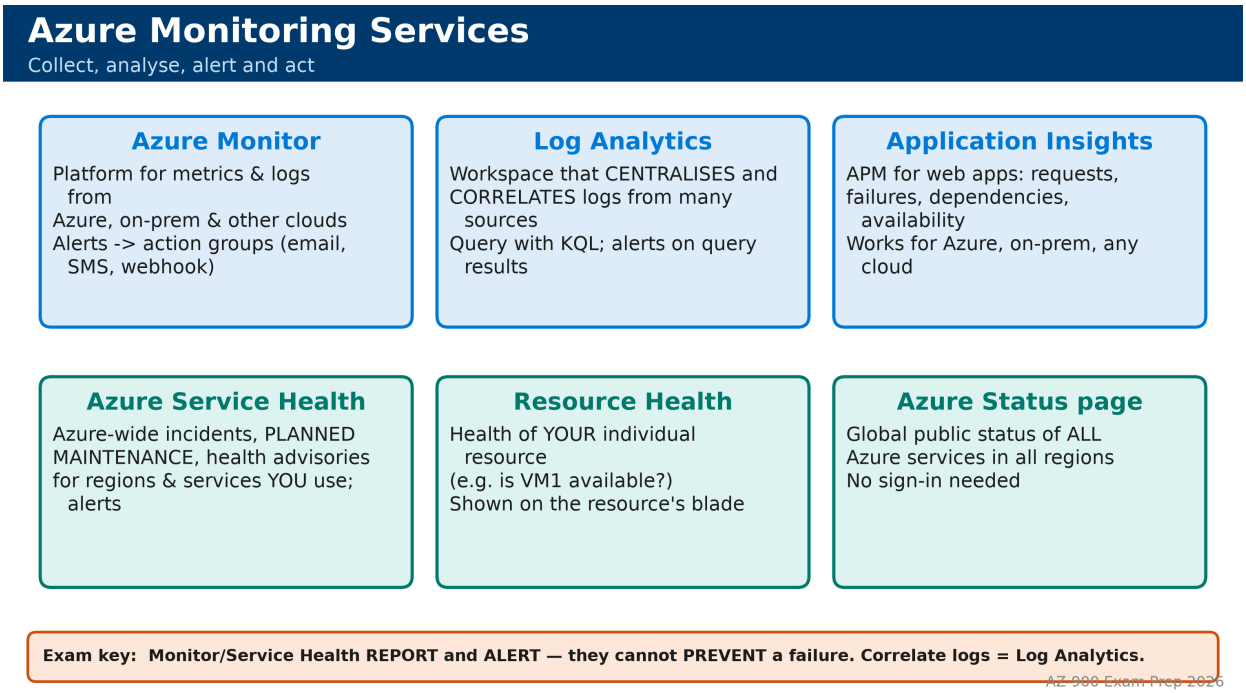


Figure Q113: Azure monitoring services

About this figure

- Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).
- Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.
- Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.
- Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.
- Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 114

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q190

You are currently trying to generate a number of managed Microsoft SQL Server instances in an Azure environment. During the process, you are presented with a notification informing you that the Azure subscription limits must be increased. You want to make sure that you can complete your task. Solution: You alter an Azure policy. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is to RAISE a subscription limit (quota). Azure Policy is your own rules; quotas are Microsoft's limits. Q57 gives the right answer: a new support request. Q113 tries a Service Health alert, also No.

Answer explained

✔ **No.**
Azure Policy enforces your company's rules. Subscription limits are quotas set by Microsoft, and no policy change can raise them.

✖ **Yes.**
Quotas are not policies. Only a quota or support request to Microsoft raises a subscription limit.

EXAM STRATEGY
Azure Policy limits what you deploy. It cannot raise Microsoft's quotas. Policy can only make things stricter, never add capacity. The answer is No.

IMPACT
Mixing up policy and quotas wastes time when a rollout is blocked. Microsoft owns quotas and changes them only on request.

What · Where · When · Why · How

What	Azure Policy enforces company rules on resource settings.
Where	Assigned at management group, subscription or resource group level.
When	When you must limit regions or sizes, or require tags, for compliance.
Why	Microsoft sets quotas, so editing a policy cannot raise a subscription limit.
How	Assign a policy definition with an effect such as deny, audit or modify at a chosen level.

Exam tip
Error says 'subscription limit/quota' = support request. Error says 'RequestDisallowedByPolicy' = Azure Policy.

Memory trick
Policy = YOUR rules. Quota = MICROSOFT's ceiling.

Real-world example, use case and scenario

EXAMPLE
Policy versus quota
A policy that blocks VM sizes above D8 is your rule. The regional limit 'Total Regional vCPUs = 20' is Microsoft's quota.

USE CASE
Proper policy use
A government contractor uses Azure Policy to allow deployments only in US regions.

SCENARIO
Editing the wrong thing
An admin turns off several policy assignments to clear the 'limit exceeded' error. Now the subscription breaks company rules. The error stays until a quota increase request is approved. Then the policies are turned back on.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

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Figure Q114: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 115

AZURE ARCHITECTURE & SERVICES

Subscriptions
v5 Q192

You are planning to migrate a company to Azure. Each of the company has numerous divisions will have an administrator in place to manage the Azure resources used by their respective division. You want to make sure that the Azure deployment you employ allows for Azure to be segmented for the divisions, while keeping administrative effort to a minimum. Solution: You plan to make use of several Microsoft Entra ID directories. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Two goals: split Azure by division AND keep admin work small. Several directories (tenants) split things, but they add a lot of admin work, so No. Q116 (resource groups) is Yes, and Q59 says subscriptions and resource groups are the two right ways.

Answer explained

✔ **No.**
Each Entra ID directory (tenant - a separate store of users) has its own users and its own admins. So several directories increase admin work.

✖ **Yes.**
Several tenants copy users and admin work. Divisions are split with subscriptions or resource groups inside one tenant.

EXAM STRATEGY
'Minimal admin effort' rules out several directories, because each Entra tenant repeats users and admins. Split with resource groups or subscriptions instead. The answer is No.

IMPACT
Several tenants split identity, make single sign-on harder and multiply admin work. This mistake is costly to undo later.

What · Where · When · Why · How

What	A Microsoft Entra ID directory (tenant) is a separate store of users and apps.
Where	Each tenant stands alone. Each subscription trusts exactly one tenant.
When	Only when a company needs full identity separation, such as separate legal companies.
Why	Several directories repeat users and admin tasks, so they increase effort.
How	Instead, split divisions with subscriptions or resource groups inside one tenant.

Exam tip
'Minimal admin effort' + split by division = subscriptions and/or resource groups in ONE tenant. Several tenants is the trap.

Memory trick
One company, one tenant. Split the house into rooms, not into new houses.

Real-world example, use case and scenario

EXAMPLE
Duplicate users
With three tenants, a manager who works with two divisions needs accounts or guest access in each one.

USE CASE
When separate tenants fit
A large group keeps a separate tenant for a smaller company it plans to sell next year.

SCENARIO
Tenant sprawl
A manufacturer made one tenant per division. Soon staff spent hours adding and removing users in five directories. They moved to one tenant with one subscription per division, and identity admin time dropped a lot.

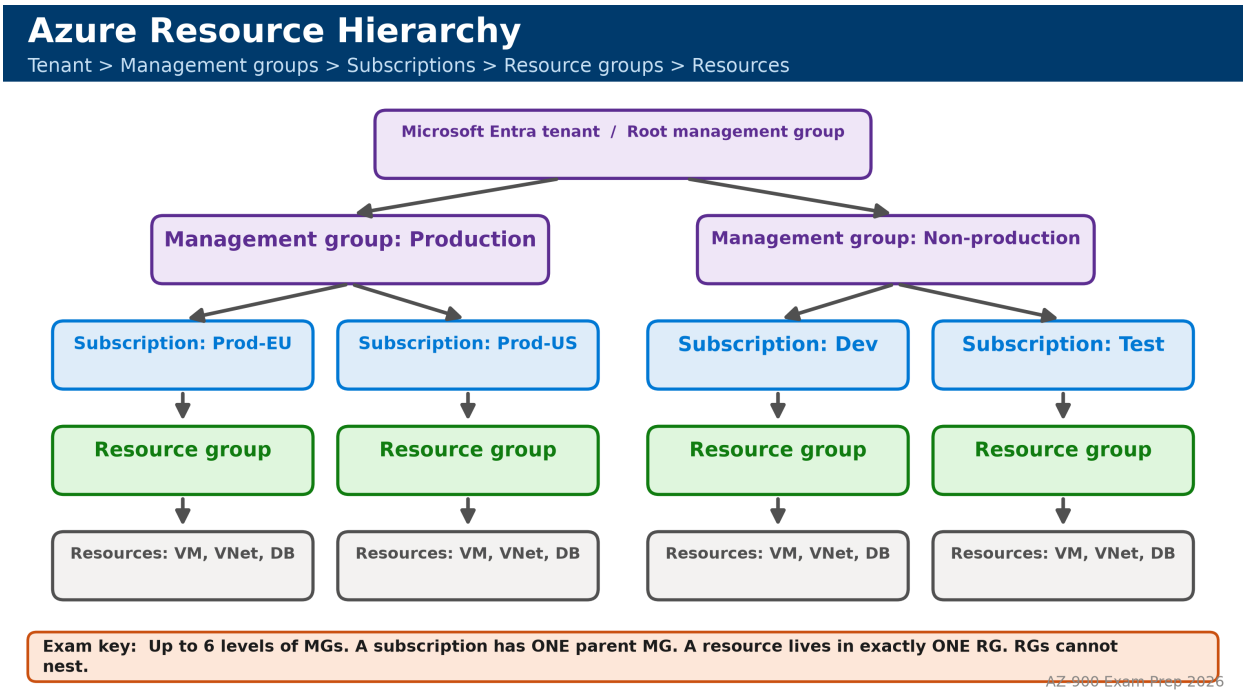


Figure Q115: Azure resource hierarchy

About this figure

• Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.

• A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.

• Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.

• Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.

• Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 116

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q193

You are planning to migrate a company to Azure. Each of the company has numerous divisions will have an administrator in place to manage the Azure resources used by their respective division. You want to make sure that the Azure deployment you employ allows for Azure to be segmented for the divisions, while keeping administrative effort to a minimum. Solution: You plan to make use of several resource groups. Does the solution meet the goal?

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): Two goals: split Azure by division AND keep admin work small. One resource group per division meets both, so Yes. Compare Q115 (several Entra directories) = No. Q59 says subscriptions and resource groups are both correct. Note: separate BILLING per division needs subscriptions (Q120).

Answer corrected in this edition. The previous PDF printed: *No.*. The printed answer No does not match the source set. The companion question (Q59) says 'multiple subscriptions + multiple resource groups' are the two correct ways to split divisions, and the answer key marks this item Yes. Resource groups in one subscription, with an RBAC role for each division admin, split Azure with very little extra work.

Answer explained

✔ **Yes.**
Give each division its own resource group, and give that division's admin a role on it with RBAC (role-based access control - who can do what). This splits management inside one subscription with little effort.

✖ **No.**
Resource groups are an accepted, low-effort way to split divisions in this series. Only several Entra directories is No.

EXAM STRATEGY
One resource group per division, plus RBAC for each division's admin, splits resources inside one tenant with little effort. The answer is Yes.

IMPACT
Limiting each division's admin to their own resource group gives least privilege (only the access needed). One division cannot break another division's live resources.

What · Where · When · Why · How

What	A resource group is a logical container for Azure resources that share a lifecycle and access.
Where	Inside a subscription. RBAC, policies and locks set on it apply to its resources.
When	When divisions need their own management area inside one subscription.
Why	One group per division with its own RBAC splits admin work with very little setup.
How	Create a resource group per division and give that division's admin group the Contributor role on it.

Exam tip
In this repeated series, both subscriptions and resource groups are accepted for 'split divisions, minimal effort'. Only several Entra directories is No. Separate billing, however, needs subscriptions.

Memory trick
Resource Group = a Room for each division, with its own key (RBAC).

Real-world example, use case and scenario

EXAMPLE
RG per division
rg-marketing, rg-finance and rg-hr each give the Contributor role to that division's admin group.

USE CASE
Mid-size publisher
A publisher with four divisions uses one subscription and four resource groups. Each team manages its own web apps.

SCENARIO
Keeping it simple
A consultant suggests five subscriptions for a 200-person firm. The IT lead says billing is central anyway. Resource groups with their own RBAC give each division control with much less setup.

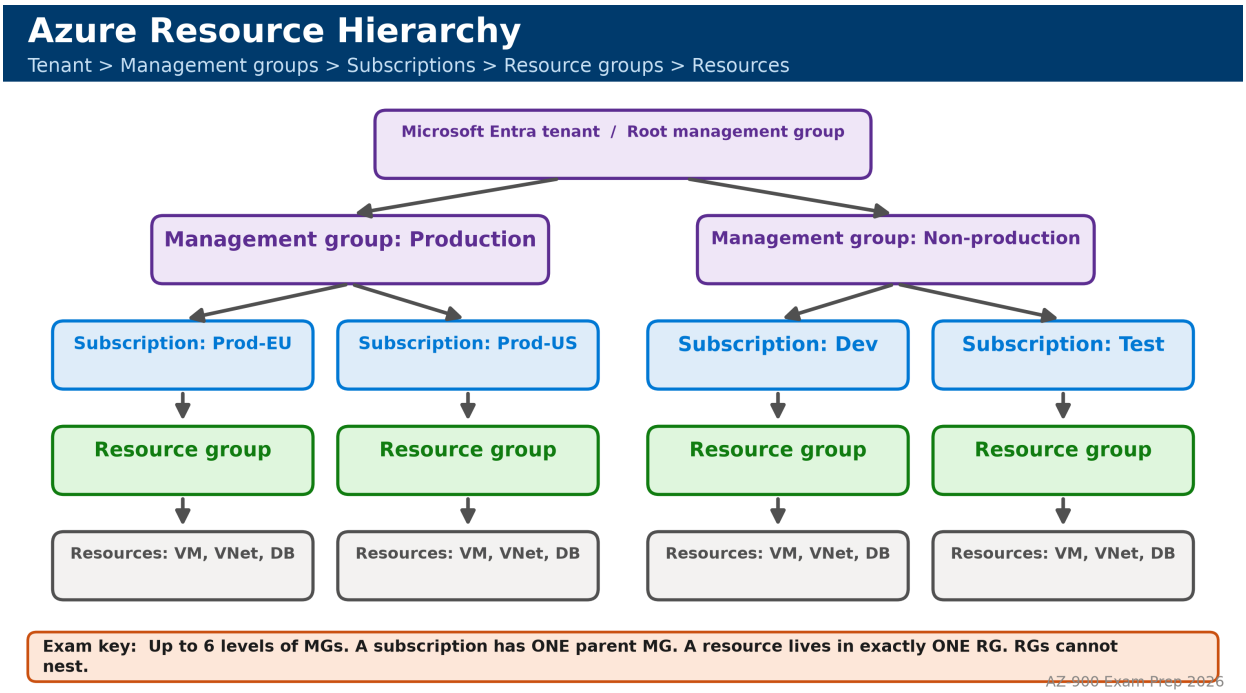


Figure Q116: Azure resource hierarchy

About this figure

• Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.

• A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.

• Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.

• Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.

• Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 117

AZURE ARCHITECTURE & SERVICES — Storage v5 Q195

Your company has datacenters in Los Angeles and New York. The company has a Microsoft Azure subscription. You are configuring the two datacenters as geo-clustered sites for site resiliency. You need to recommend an Azure storage redundancy option. You have the following data storage requirements: Data must be stored on multiple nodes. Data must be stored on nodes in separate geographic locations. Data can be read from the secondary location as well as from the primary location Which of the following Azure stored redundancy options should you recommend?

✖ A. Geo-redundant storage.

✔ **B.** Read-only geo-redundant storage.

✖ C. Zone-redundant storage.

✖ D. Locally redundant storage.

Correct answer: Read-only geo-redundant storage.

Clarification (read this first): Three needs: many copies, copies in different geographic places, and you can READ the second copy at any time. 'Read-only geo-redundant storage' in the options means RA-GRS (read-access geo-redundant storage). Plain GRS has a second copy, but you cannot read it until failover. Q331 is a look-alike that adds 'three zones' - that one is RA-GZRS.

2026 update: Microsoft's official name is read-access geo-redundant storage (RA-GRS). The option's 'read-only' wording means the same thing. RA-GZRS is the version that also uses zones.

Answer explained

✓ **Read-only geo-redundant storage.**
RA-GRS keeps three copies in the main region and three in a distant second region. You can read the second copy at any time.

✖ **Geo-redundant storage.**
GRS copies data to a second region, but you cannot read that copy until a failover (switch to the second region) happens.

✖ **Zone-redundant storage.**
ZRS copies data across three zones in one region only. It has no copy in a different geographic place.

✖ **Locally redundant storage.**
LRS keeps three copies in one datacenter. It has no geographic separation and no reading from a second place.

EXAM STRATEGY
Look for three signals: many copies, different places, read from the second copy. 'Read' removes plain GRS. 'Geographic' removes ZRS and LRS.

IMPACT
A readable second copy lets reports and read traffic keep working during a regional outage, without waiting for a failover decision.

What · Where · When · Why · How

What	Read-access geo-redundant storage (RA-GRS) copies data to a second region that you can read.
Where	Chosen as the redundancy setting of an Azure Storage account. The second region is the paired region.
When	When data must survive a regional outage and be readable from the second place.
Why	It gives many copies in two geographic places, plus read access to the second copy.
How	Three copies are written at once in the main region and three later in the second region. You read through the '-secondary' address.

Exam tip
'Read from the second location' is the clue for the RA- prefix. GRS alone = the second copy exists but is offline until failover.

Memory trick
RA = Read Anytime from the faraway copy.

Real-world example, use case and scenario

EXAMPLE
Second address
An app reads files from contoso-secondary.blob.core.windows.net in West US when reads from East US are slow.

USE CASE
Medical imaging
A radiology network stores scans on RA-GRS. A West Coast clinic can read them from the second region during a main-region outage.

SCENARIO
Two-coast resilience
A media firm with sites in LA and New York first picks GRS. Then it sees New York staff could not read the copy during an outage without a failover. Moving to RA-GRS lets New York read at once while the main region recovers.

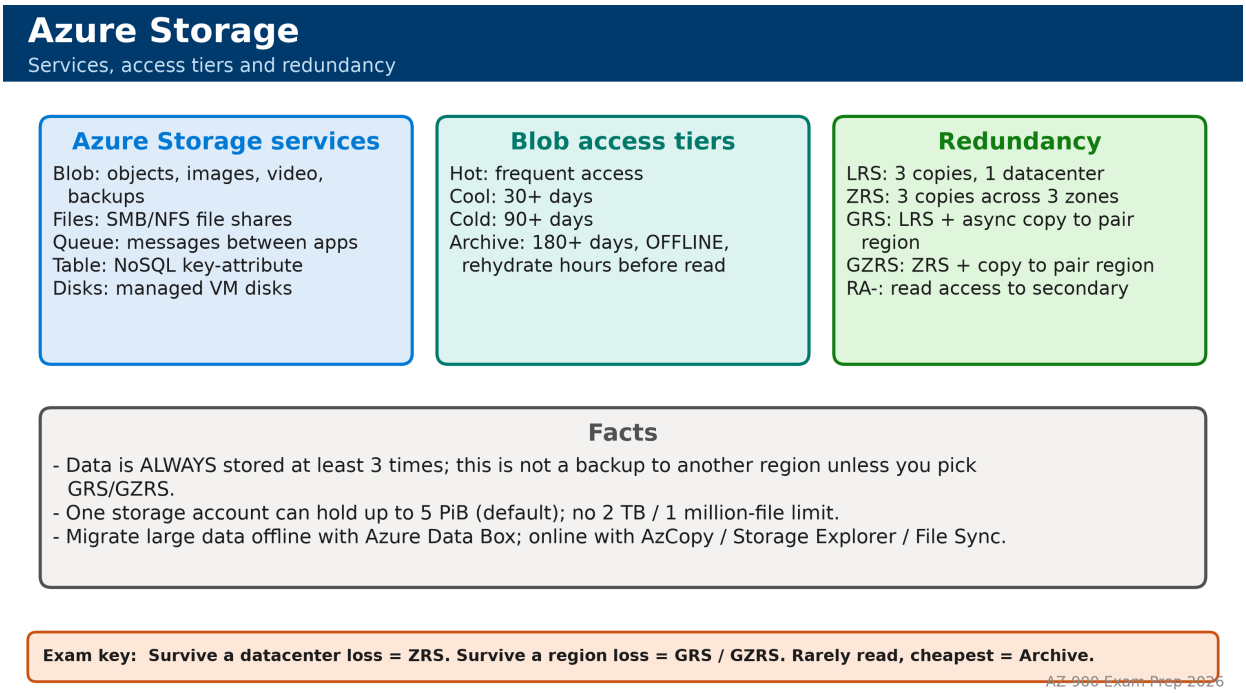


Figure Q117a: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

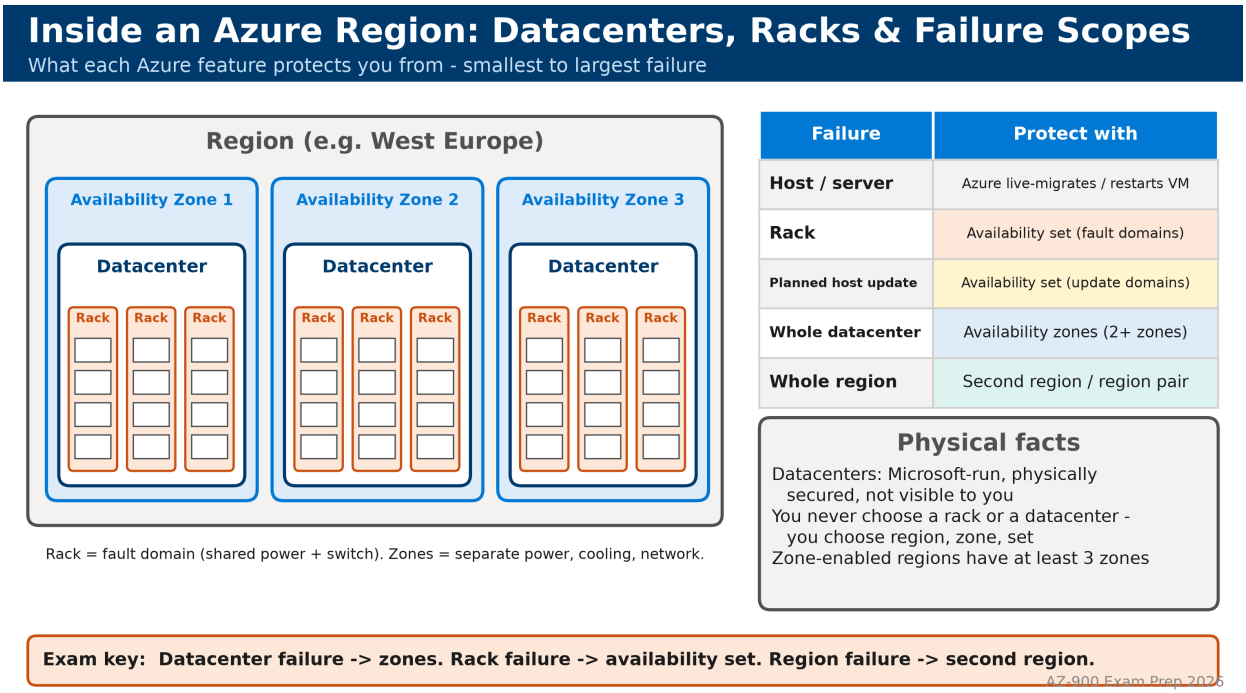


Figure Q117b: Inside a region: datacenters, racks and failure scopes

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- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 118

AZURE MANAGEMENT & GOVERNANCE

Subscriptions
v5 Q196

A company that has numerous divisions is planning to deploy an Azure environment. The company would like each division has option to pay for the Azure services it utilizes, not be the same. You have been asked to recommend a solution to meet the requirements. Solution: You recommend that an Azure role be created for each division. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The need is that each division PAYS separately (billing). A role controls permissions (who can do what), not who pays. Same series: Q119 (policy) = No, Q120 (subscription) = Yes. Q62 also says create a subscription.

Answer explained

✔ **No.**
An Azure RBAC role only says what actions a person or app can do. It has no billing or payment setting.

✖ **Yes.**
Roles give permissions. Separate payment for each division needs separate subscriptions, because the subscription is the billing boundary (the unit that gets a bill).

EXAM STRATEGY
Roles set permissions, not payment. 'Each division pays separately' means a billing boundary. RBAC roles are never billing, so the answer is No.

IMPACT
Billing based on permissions fails at month end. You get cost disputes between teams and spending nobody owns.

What · Where · When · Why · How

What	An Azure RBAC role is a set of permissions given to a user, group or app identity.
Where	Assigned at management group, subscription, resource group or resource level.
When	When you need to control who can do what on Azure resources.
Why	Roles have no billing setting, so they cannot give divisions separate payment.
How	Give a role such as Contributor to a person or group at a chosen level.

Exam tip
Different billing or payment per division = separate subscriptions. Roles = who can do what. Policies = what is allowed.

Memory trick
Roles hand out keys, not credit cards.

Real-world example, use case and scenario

EXAMPLE
Role scope
The Billing Reader role lets someone see invoices. It does not change which card pays them.

USE CASE
Correct role use
A hotel chain gives each hotel's IT lead the Virtual Machine Contributor role on that hotel's resource group.

SCENARIO
Shared invoice
A company creates a custom role per division and expects split bills. The next invoice still comes as one charge on the head-office card. So they create one subscription per division, each with its own payment method.

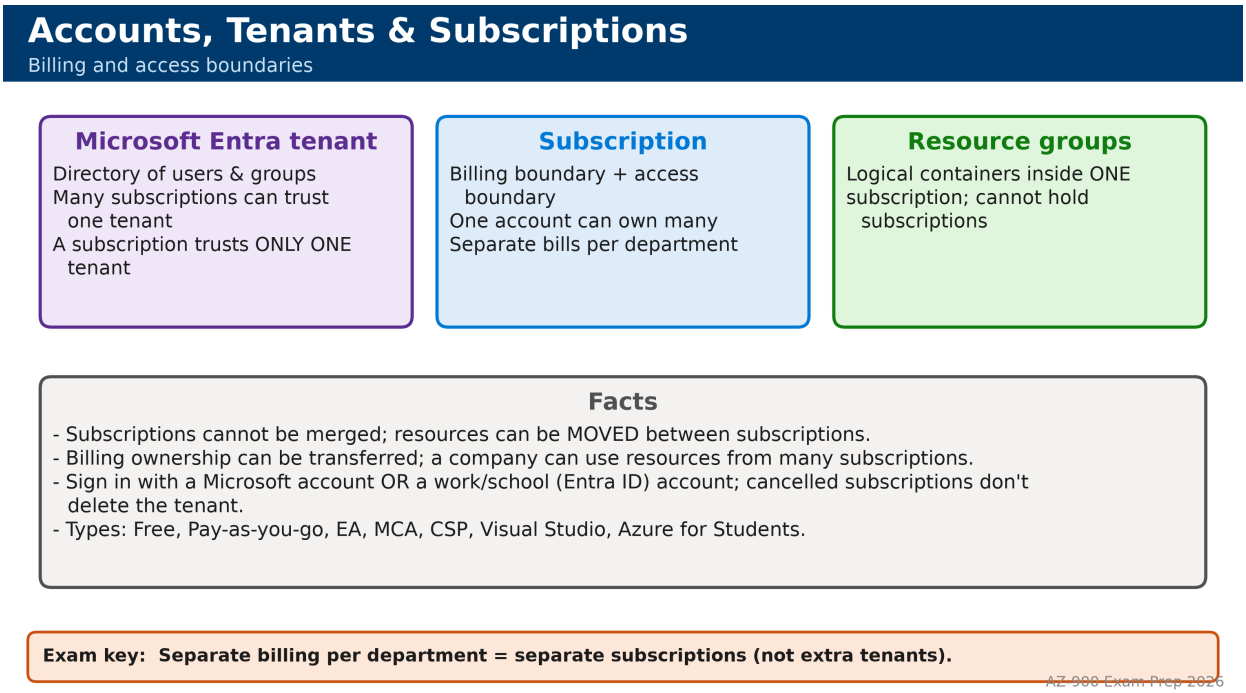


Figure Q118a: Accounts, tenants and subscriptions

About this figure

- Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.
- Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.
- Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.
- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

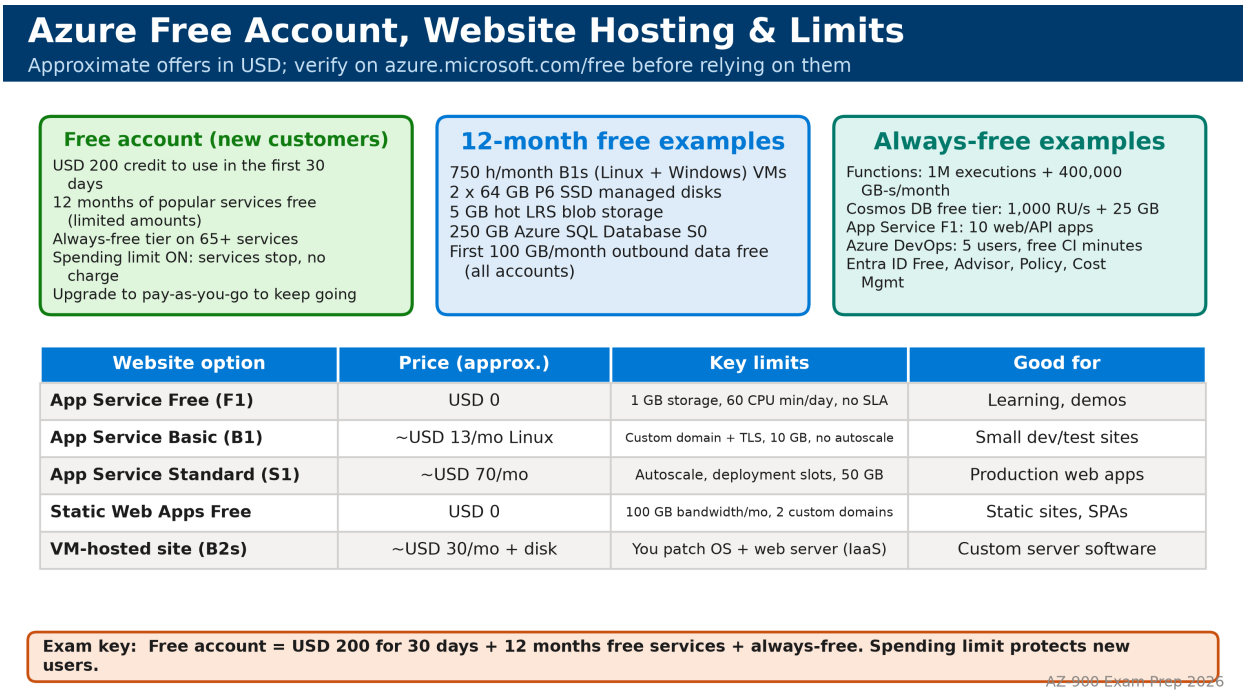


Figure Q118b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 119

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q197

A company that has numerous divisions is planning to deploy an Azure environment. The company would like each division has option to pay for the Azure services it utilizes, not be the same. You have been asked to recommend a solution to meet the requirements. Solution: You recommend that an Azure policy be created for each division. Does the solution meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The need is that each division PAYS separately. A policy controls what can be deployed, not who pays. Same series: Q118 (role) = No, Q120 (subscription) = Yes. Do not confuse with Q58: tags help with cost REPORTS, but they do not split the bill.

Answer explained

✔ **No.**
Azure Policy enforces rules on resource settings, such as location, size and tags. It cannot give a division its own payment method or invoice.

✖ **Yes.**
Policies control what may be deployed, not who pays. Separate billing for each division needs separate subscriptions.

EXAM STRATEGY
Policy controls resource settings like region, size and tags, not invoices. Any rule-based answer to a billing need is No.

IMPACT
Tags set by policy can help cost reports. But only subscriptions give each division its own invoice or payment method.

What · Where · When · Why · How

What	Azure Policy checks and enforces rules on resource settings.
Where	Assigned at management group, subscription or resource group level.
When	When resources must follow rules, such as allowed regions or required tags.
Why	Policy cannot change invoices or payment methods, so it cannot split billing.
How	Assign a definition with a deny, audit or modify effect, for example to require a CostCenter tag.

Exam tip
Policy can REQUIRE a cost-center tag for reports, but it cannot SPLIT payment. Splitting payment = subscriptions.

Memory trick
Policy sets the rules of the game, not who pays for the tickets.

Real-world example, use case and scenario

EXAMPLE
Allowed locations
A policy blocks any resource created outside North Europe and West Europe.

USE CASE
Tag enforcement
A university requires a 'Department' tag through policy. The single central bill can then be split in reports by faculty.

SCENARIO
Reporting is not billing
A CFO is told a tag policy will let divisions pay separately. Tags give cost reports, but each division still cannot be charged to its own card. So separate subscriptions are created.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q119: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 120

AZURE MANAGEMENT & GOVERNANCE

Subscriptions
v5 Q198

A company that has numerous divisions is planning to deploy an Azure environment. The company would like each division has option to pay for the Azure services it utilizes, not be the same. You have been asked to recommend a solution to meet the requirements. Solution: You recommend that a subscription be created for each division. Does the solution meet the goal?

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The need is that each division PAYS separately. The subscription is the billing boundary, so Yes. Same series: Q118 (role) and Q119 (policy) are both No. Q62 asks the same thing directly: a subscription.

Answer explained

✔ **Yes.**
A subscription is Azure's billing boundary. Each one can be billed separately, with its own payment method, invoice or invoice section, while all share one Entra tenant.

✖ **No.**
Separate subscriptions do give each division separate billing, so the solution meets the goal.

EXAM STRATEGY
'Each division pays for its own services' = subscription, Azure's billing boundary. Each subscription can have its own invoice or payment. The answer is Yes.

IMPACT
One subscription per division makes chargeback (charging each team for its use) and budgets simple. No one has to split one bill by hand.

What · Where · When · Why · How

What	An Azure subscription is the billing boundary for Azure use.
Where	Linked to a billing account, with its own payment method or invoice section / billing profile.
When	When each division must pay for the Azure services it uses.
Why	Separate subscriptions give separate billing for each division under one tenant.
How	Create a subscription per division and link each to its own payment method or invoice section.

Exam tip
Billing boundary = subscription. Resource groups and tags help with cost reports, but they do not create separate invoices.

Memory trick
Subscription = separate bill. A Sub for each Sub-division.

Real-world example, use case and scenario

EXAMPLE
Separate invoices
sub-research is billed to the R&D card and sub-sales to the sales cost center. Each gets its own invoice.

USE CASE
Franchise model
A fitness franchise gives each franchise owner their own subscription, so they pay Microsoft directly.

SCENARIO
Chargeback ends
A holding company spent days each month splitting one Azure bill. After moving to one subscription per division, each division gets its own invoice. The manual splitting stops.

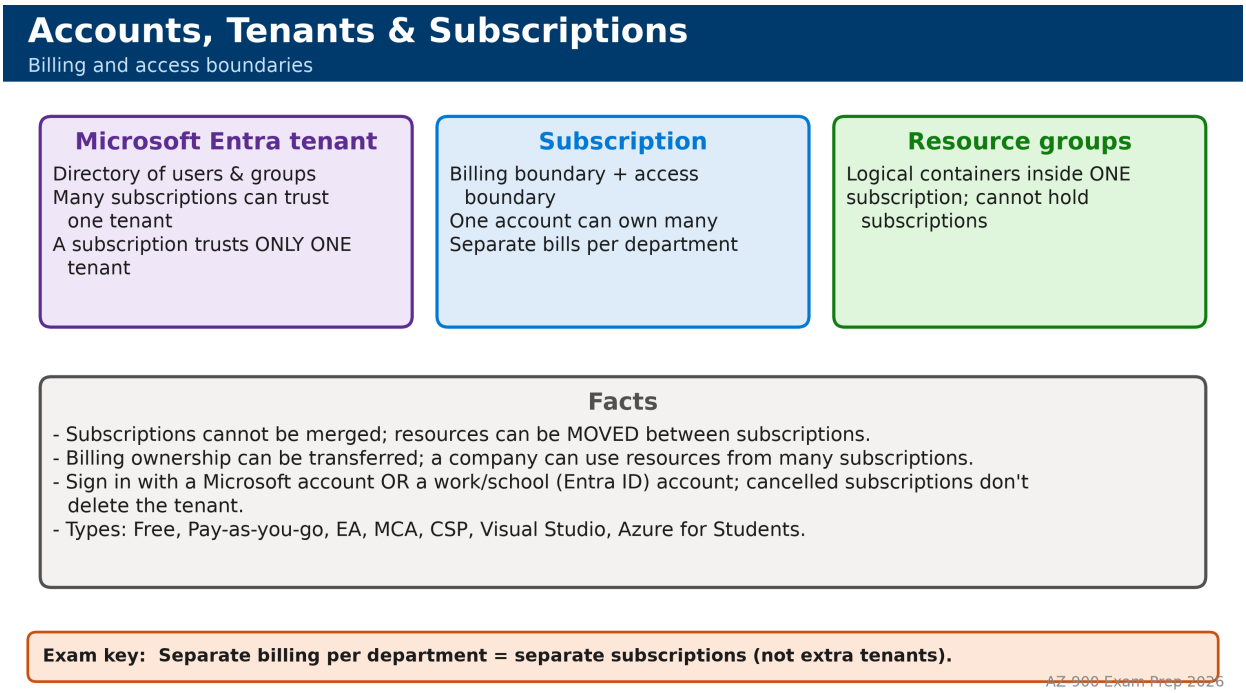


Figure Q120a: Accounts, tenants and subscriptions

About this figure

- Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.
- Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.
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- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

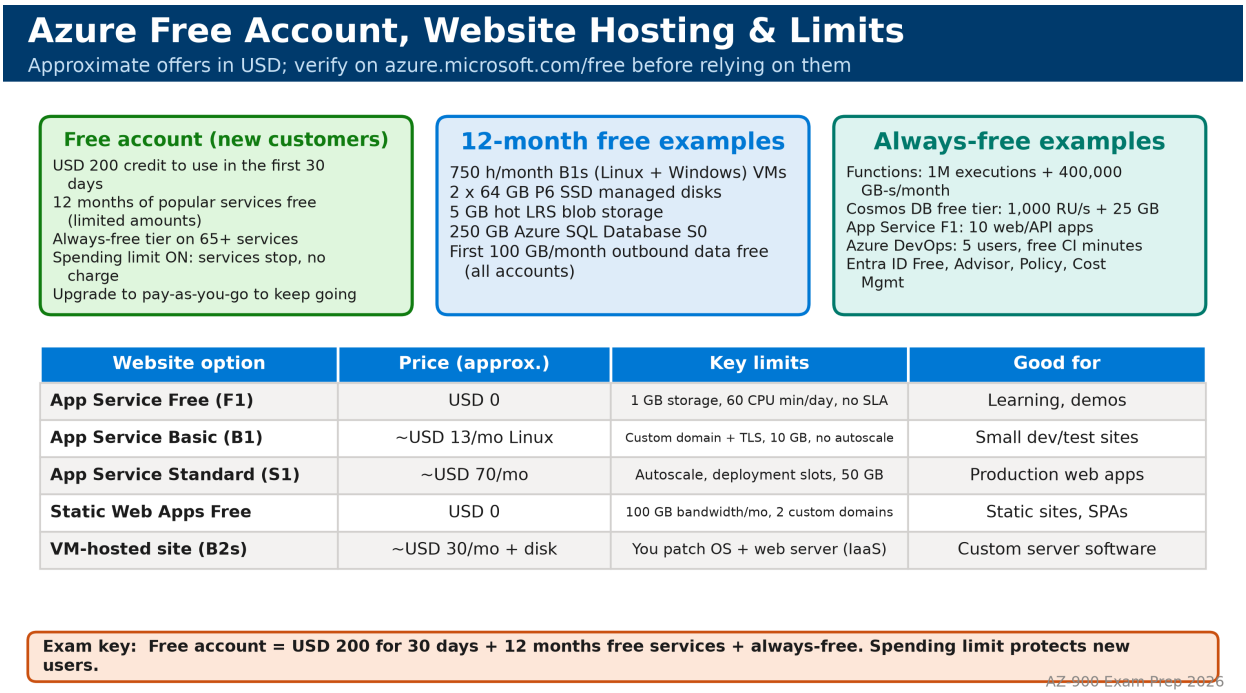


Figure Q120b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
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- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 121

AZURE ARCHITECTURE & SERVICES — **Subscriptions** v5 Q199

An Azure subscription can be associated to multiple Microsoft Entra ID tenants.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch the direction. One subscription trusts only ONE tenant at a time (so No). But one tenant can have MANY subscriptions. Q122 is a look-alike: you CAN change which tenant a subscription trusts (Yes) - changing is not the same as having several at once.

Answer explained

✔ **No.**
Each Azure subscription trusts exactly one Microsoft Entra tenant at a time. One tenant, however, can have many subscriptions.

✖ **Yes.**
A subscription cannot trust several tenants. Users from other tenants get access only as B2B guests (invited outside users) in the trusted tenant.

EXAM STRATEGY
Rule: one subscription trusts exactly one tenant. One tenant can have many subscriptions. 'Several tenants' for one subscription is No.

IMPACT
Knowing about the one-tenant link shapes identity design. A plan built on a subscription with several tenants would fail when you build it.

What · Where · When · Why · How

What	The tenant-subscription trust decides which Entra directory signs in a subscription's users.
Where	Shown as the 'Directory' on the subscription's overview page in the portal.
When	When deciding who can sign in and get roles on a subscription.
Why	A subscription trusts exactly one tenant, so it cannot be linked to several.
How	Give outside users access by inviting them as B2B guests into the trusted tenant.

Exam tip
Tenant to subscription is one-to-many. Subscription to tenant is always exactly one.

Memory trick
A subscription has one parent tenant. A tenant can have many child subscriptions.

Real-world example, use case and scenario

EXAMPLE
Directory field
The subscription overview shows one 'Directory' value, for example contoso.onmicrosoft.com.

USE CASE
Partner access
An agency's staff use a client's subscription as B2B guests in the client's tenant. The subscription cannot also trust the agency's tenant.

SCENARIO
Merger assumption
After a merger, IT thinks a subscription can trust both companies' tenants. They learn it cannot. They invite the other company's staff as guests until the tenants are joined.

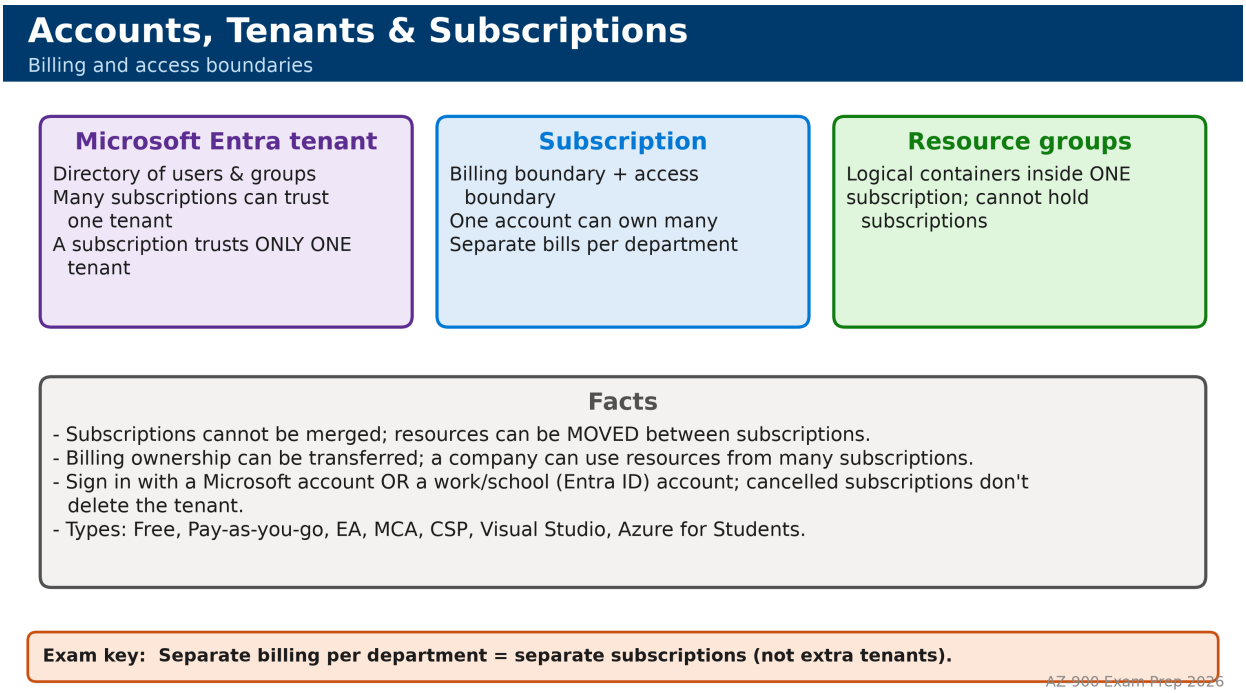


Figure Q121a: Accounts, tenants and subscriptions

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- Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.
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- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

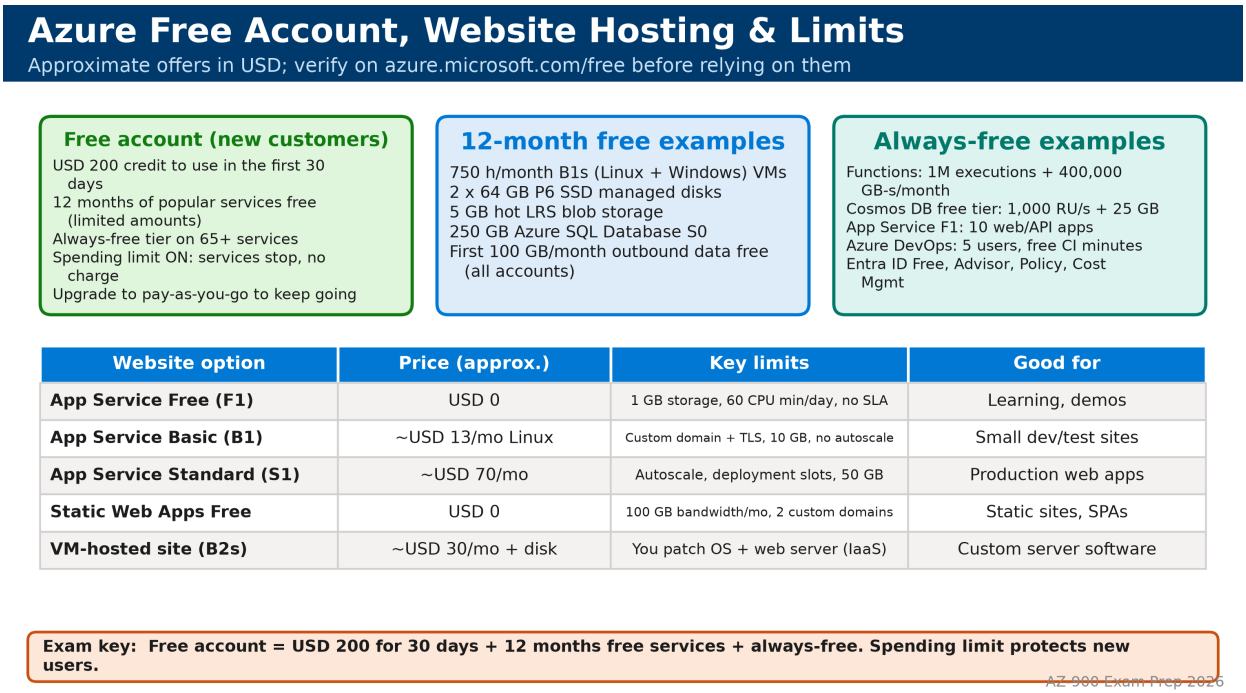


Figure Q121b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
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- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 122

AZURE ARCHITECTURE & SERVICES — **Subscriptions** v5 Q200

You can change the Microsoft Entra ID tenant to which an Azure subscription is associated.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The word is CHANGE. You can move a subscription to a different tenant (Yes). Q121 is the look-alike: a subscription cannot trust SEVERAL tenants at the same time (No). Q124: merging subscriptions is not possible (No).

Answer explained

✔ **Yes.**
The 'Change directory' action moves a subscription to a different Entra tenant. Existing RBAC role assignments are removed and must be added again.

✖ **No.**
Changing a subscription's directory is a supported action in the Azure portal.

EXAM STRATEGY
'Change directory' is supported: a subscription can move to another tenant. 'Can change' is Yes. 'Linked to several at once' is No.

IMPACT
Moving a subscription to another tenant removes existing RBAC assignments and can break managed identities (app identities). Mergers need careful planning.

What · Where · When · Why · How

What	'Change directory' moves a subscription's trust to another Microsoft Entra tenant.
Where	On the subscription's overview page in the Azure portal.
When	When a subscription must move to another company's tenant or a corporate tenant.
Why	It shows that a subscription's tenant can be changed.
How	An allowed user selects Change directory, picks the new tenant, then adds the RBAC assignments again.

Exam tip
Change directory = supported, but RBAC assignments are lost. Merge subscriptions = not supported.

Memory trick
Subscriptions can move house (tenant), but they leave the keys (RBAC) behind.

Real-world example, use case and scenario

EXAMPLE
Change directory
An Owner selects Change directory on a subscription and moves it from a personal tenant to fabrikam.onmicrosoft.com.

USE CASE
Startup purchase
After buying a startup, a company moves the startup's subscription into its own corporate tenant.

SCENARIO
After the move
An IT team moves a subscription to the parent company's directory. Developers lose access because role assignments did not move with it. The team adds the roles again from a script they saved before.

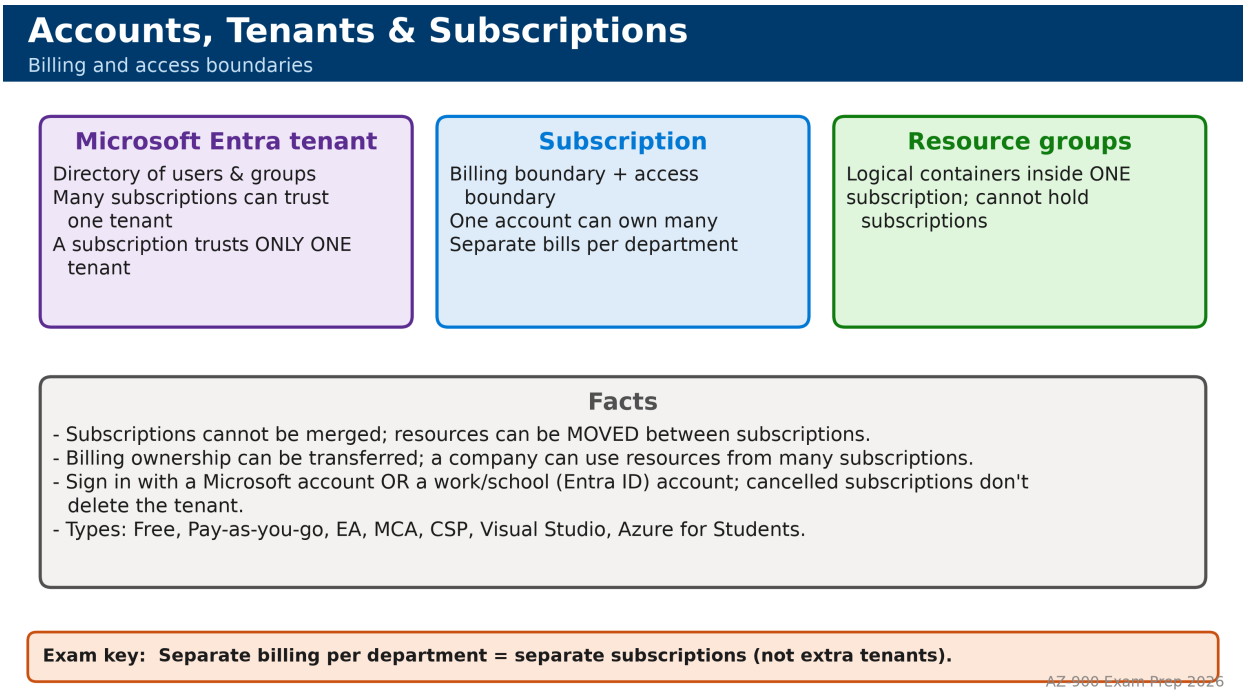


Figure Q122a: Accounts, tenants and subscriptions

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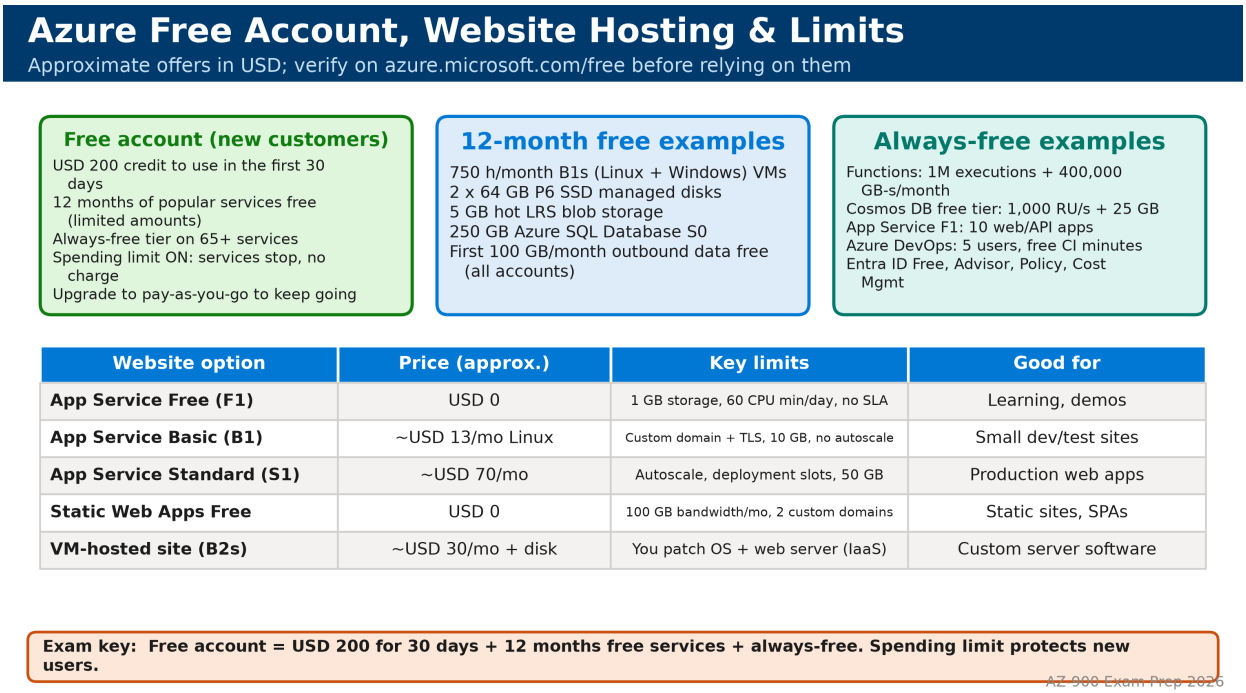


Figure Q122b: Free account, website hosting and limits

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- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 123

AZURE ARCHITECTURE & SERVICES — **Subscriptions** v5 Q201

When an Azure subscription expires, the associated Microsoft Entra ID tenant is deleted automatically.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The question says the tenant is deleted AUTOMATICALLY. It is not. When a subscription ends, only its Azure resources are turned off and later deleted. The tenant and its users stay.

Answer explained

✔ **No.**
An Entra tenant exists on its own, separate from any subscription. When a subscription expires, only its resources are turned off and later deleted. The tenant and users stay.

✖ **Yes.**
Deleting a tenant is a manual action by an admin. A subscription ending never deletes the tenant.

EXAM STRATEGY
Subscriptions and tenants have separate lifecycles. Expiry turns off resources but never deletes the directory. Any 'tenant deleted automatically' claim is No.

IMPACT
Your tenant, users and other subscriptions survive an expired subscription. Sign-in for Microsoft 365 and other services keeps working.

What · Where · When · Why · How

What	A Microsoft Entra tenant is a user directory separate from any Azure subscription.
Where	It exists at company level and may also serve Microsoft 365.
When	Matters when a subscription expires, is cancelled or is turned off.
Why	Expiry removes only Azure resources. The tenant is never deleted automatically.
How	Only a Global Administrator can delete a tenant, by hand, after removing what depends on it.

Exam tip
Subscription ends = resources go away; directory stays. Tenants are only deleted by hand.

Memory trick
The lease (subscription) ends; the building (tenant) still stands.

Real-world example, use case and scenario

EXAMPLE
Free trial end
A 30-day free trial ends. The user can still sign in to the tenant and create a new pay-as-you-go subscription.

USE CASE
Microsoft 365 keeps working
A charity's Azure sponsorship ends. Its Microsoft 365 mailboxes in the same tenant keep working.

SCENARIO
Panic avoided
A school lets its Azure subscription end and fears staff logins are gone. The admin checks: all users still exist in Entra ID. Only Azure resources were affected.

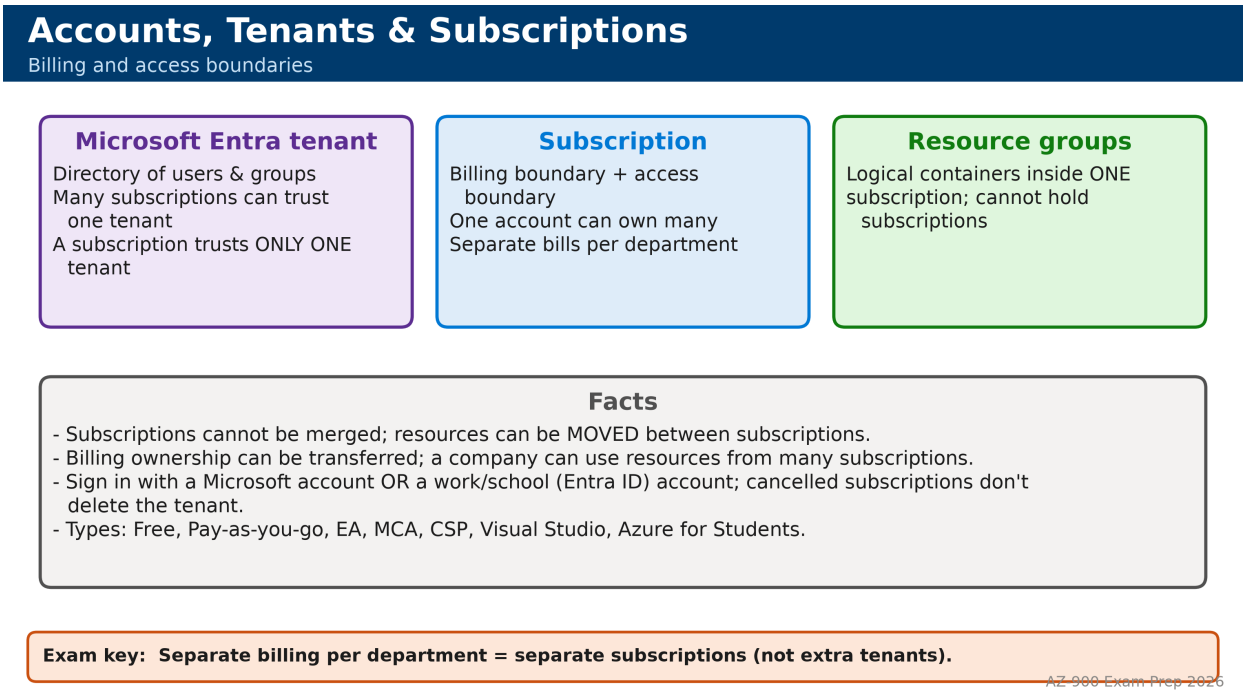


Figure Q123a: Accounts, tenants and subscriptions

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• Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.

• Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.

• Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.

• Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

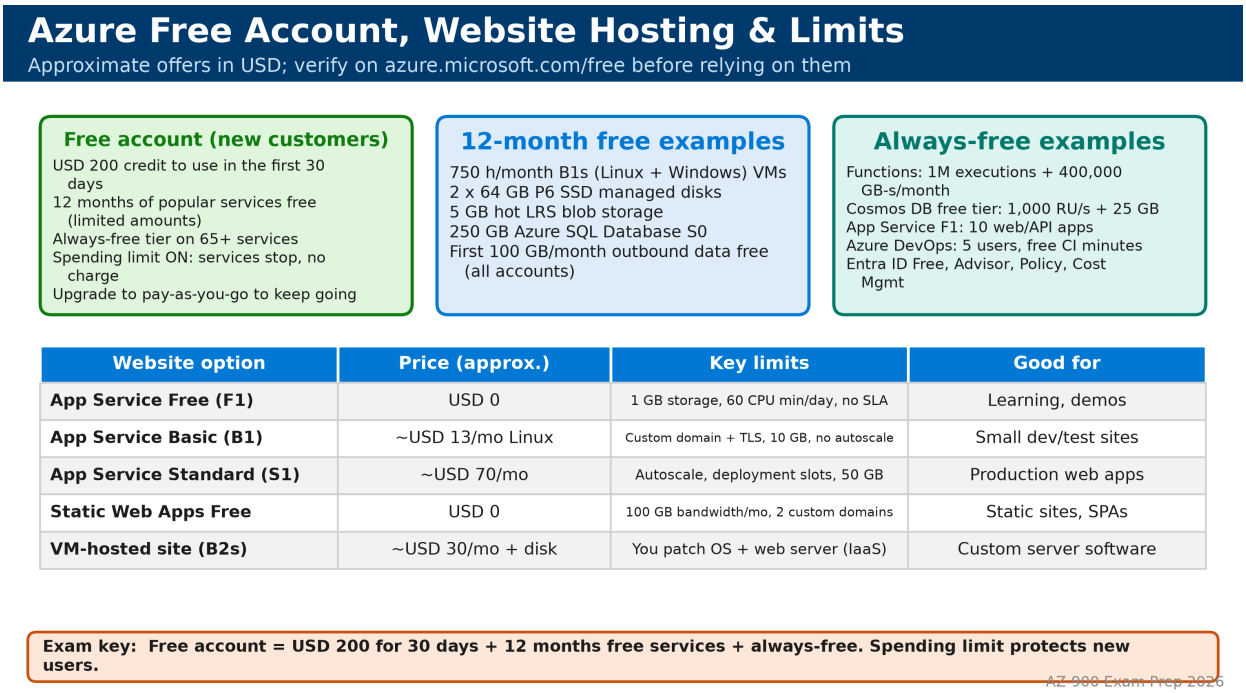


Figure Q123b: Free account, website hosting and limits

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- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
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- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 124

AZURE ARCHITECTURE & SERVICES — **Subscriptions** v5 Q202, Q49

Two Azure subscriptions can be merged into a single subscription.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): MERGE is the key word. There is no merge feature. You can MOVE resources between subscriptions (Yes), and you can CHANGE a subscription's tenant (Q122, Yes). But you cannot merge two subscriptions into one.

Answer explained

✔ **No.**
Azure has no feature to merge subscriptions. You move resources from one subscription to the other, then cancel the empty one.

✖ **Yes.**
There is no merge action. You can only move supported resources between subscriptions.

EXAM STRATEGY
No merge feature exists for subscriptions. You move resources between them instead. 'Merge subscriptions' is always No.

IMPACT
Joining subscriptions means planning resource moves. Some resource types cannot move. So a merger project needs a migration plan, not a button.

What · Where · When · Why · How

What	Joining subscriptions means putting resources from two Azure subscriptions into one.
Where	Done with the Move action on resources or resource groups in the portal.
When	When a company purchase or reorganisation leaves extra subscriptions.
Why	Azure has no merge feature, so two subscriptions cannot be merged.
How	Move supported resources to the target subscription, rebuild the ones that cannot move, then cancel the empty subscription.

Exam tip
Merge subscriptions = No. Move resources between subscriptions = Yes. Change a subscription's directory = Yes.

Memory trick
Subscriptions do not marry; their belongings move in.

Real-world example, use case and scenario

EXAMPLE
Move action
An admin selects a resource group's VMs, disks and network cards and uses Move to another subscription. Then the admin cancels the empty source subscription.

USE CASE
After a purchase
A bakery chain buys a competitor and moves the competitor's Azure resources into its own subscription.

SCENARIO
Resource that cannot move
A team tries to join two subscriptions. One service type cannot be moved. They rebuild it from an ARM template (a deployment file) in the target subscription, then close the old one.

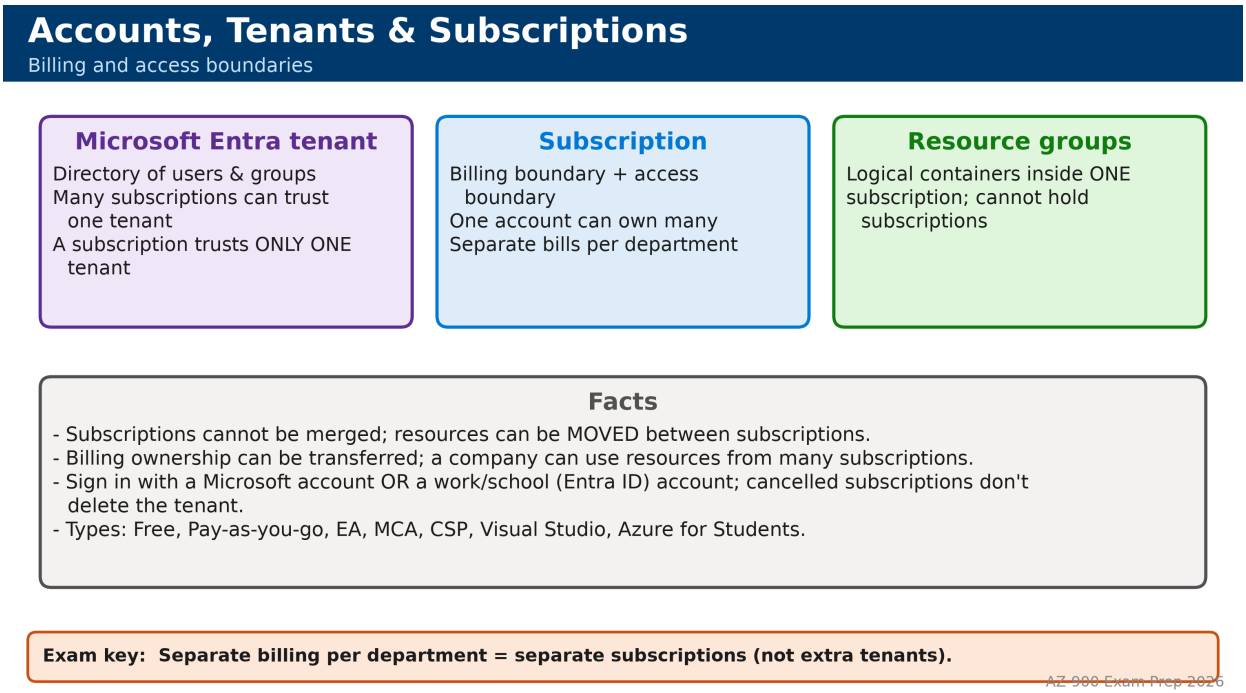


Figure Q124a: Accounts, tenants and subscriptions

About this figure

• Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.

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• Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.

• Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.

• Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

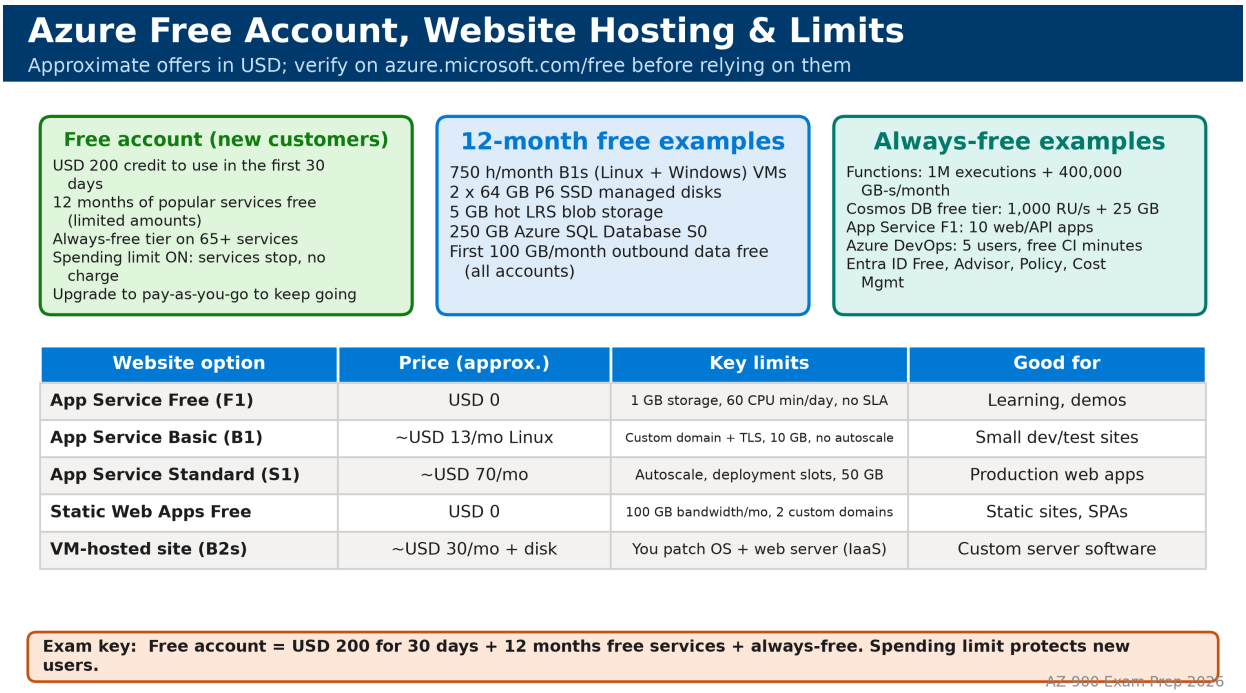


Figure Q124b: Free account, website hosting and limits

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Question 125

AZURE ARCHITECTURE & SERVICES — **Subscriptions** v5 Q203

A company can use resources from multiple subscriptions.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Companies often run many subscriptions under one tenant. They manage them with management groups (containers for subscriptions) and connect them, for example with VNet peering across subscriptions.

✖ **No.**
Using many subscriptions is normal and recommended by the Cloud Adoption Framework (Microsoft's cloud guidance). Nothing limits a company to one.

EXAM STRATEGY
A tenant can hold many subscriptions, and a company uses them all. 'Several subscriptions per company' is Yes.

IMPACT
Several subscriptions are normal for separating dev, test and production, each with its own budget, quotas and policies.

What · Where · When · Why · How

What	A multi-subscription design spreads a company's resources across several subscriptions.
Where	Several subscriptions under one Entra tenant, often grouped in management groups.
When	When separating environments, departments, billing or subscription limits.
Why	Nothing limits a company to one subscription, and resources can connect across them.
How	Group subscriptions in management groups and connect them, for example with VNet peering.

Exam tip
Several subscriptions per company is normal. Management groups exist to manage many subscriptions together.

Memory trick
One company, many subscriptions, one management-group umbrella.

Real-world example, use case and scenario

EXAMPLE
Peering across subscriptions
A VNet in the Prod subscription connects (peers) with a hub VNet in the Connectivity subscription.

USE CASE
Landing zones
A telecom company uses separate subscriptions for identity, connectivity, management and each app team.

SCENARIO
Limits outgrown
A gaming company hits per-subscription limits as it grows. It splits workloads into several subscriptions under one management group and keeps them connected through a hub network.

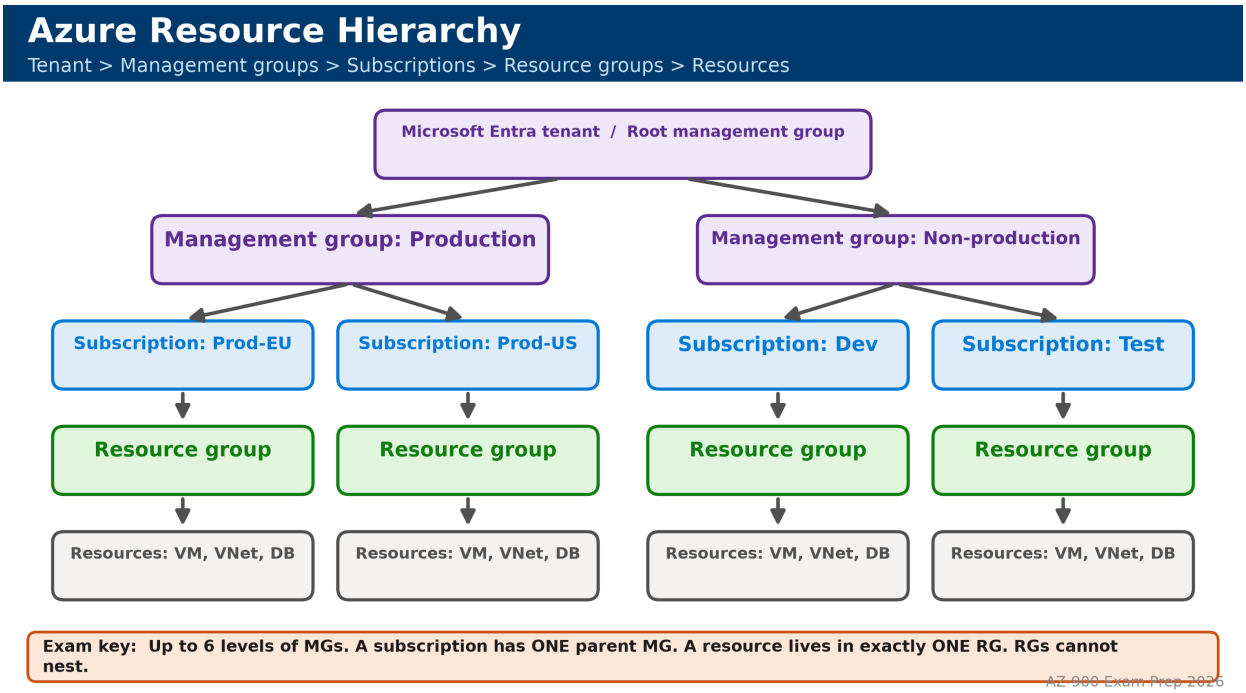


Figure Q125: Azure resource hierarchy

About this figure

• Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.

• A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.

• Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.

• Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.

• Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 126

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q204, Q216

Availability zones can be implemented in all Azure regions.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is ALL. Availability zones (separate datacenters inside one region) exist only in some regions. Q304 is a look-alike: a region that DOES have zones has at least three - that one is Yes.

Answer explained

✔ **No.**
Availability zones exist only in zone-enabled regions. Some Azure regions still have no zones.

✖ **Yes.**
Not every region supports availability zones, so you cannot use them in all regions.

EXAM STRATEGY
'All regions' is the trap. Zones exist only in selected regions. 'Always/all' claims about zones are No.

IMPACT
A region without zones cannot give the 99.99% zonal VM SLA (uptime promise). Check zone support before you pick a region.

What · Where · When · Why · How

What	Availability zones are physically separate datacenters inside one Azure region.
Where	Only in zone-enabled regions, usually three zones per region.
When	When a workload must survive a datacenter failure inside one region.
Why	Some regions have no zones, so zones cannot be used in every region.
How	Check Microsoft's list of regions with zones, then choose a zone or a zone-redundant option.

Exam tip
Watch for absolute words like 'all regions'. Zones = only in zone-enabled regions, usually three per region.

Memory trick
Zones are a feature of some regions, not all of them.

Real-world example, use case and scenario

EXAMPLE
Region choice
East US 2 offers zones 1, 2 and 3. A team using a region without zones sees no zone list when creating a VM.

USE CASE
Region selection
A stock-trading platform picks a zone-enabled region so it can deploy zone-redundant services.

SCENARIO
Data residency clash
A company must use one region for legal data reasons, but it has no zones. They use availability sets for a 99.95% VM SLA and plan disaster recovery to another region.

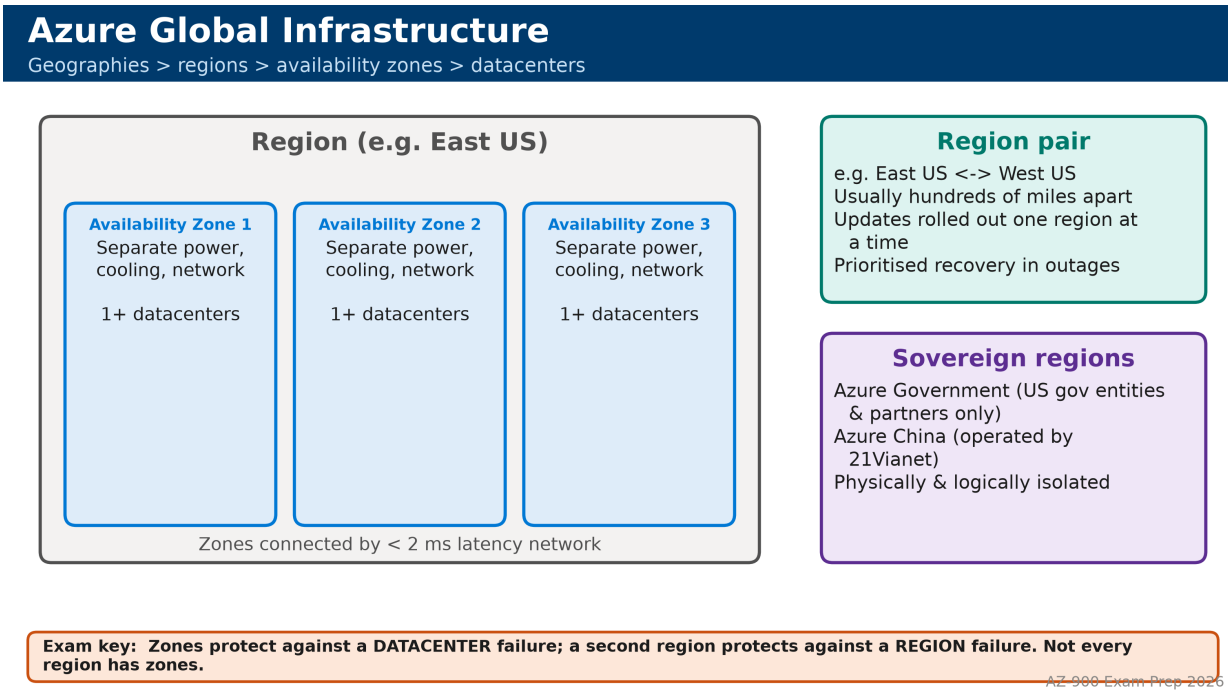


Figure Q126a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

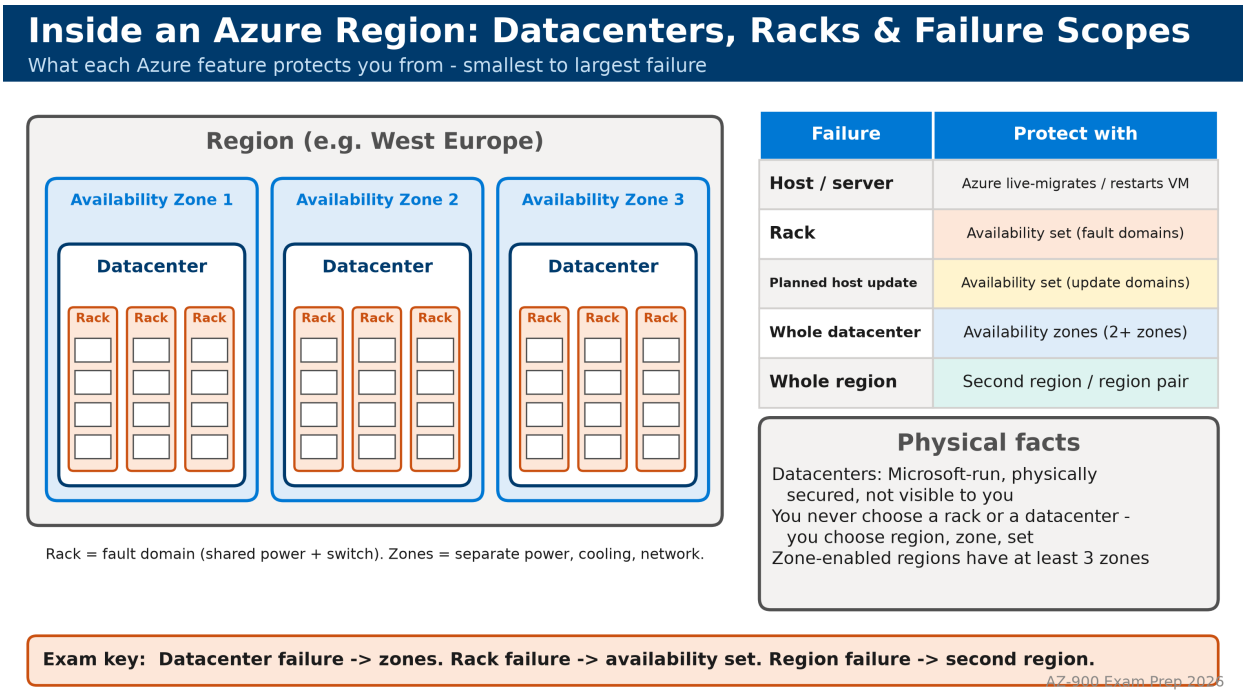


Figure Q126b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 127

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q205

Only virtual machines that run Windows Server can be created in availability zones.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is ONLY. Zones limit by region (and sometimes VM size), never by operating system. Linux VMs can use zones too.

Answer explained

✔ **No.**
Availability zones are a choice of physical location. Both Windows and Linux VMs (and many PaaS services - services where Azure manages the OS) can be placed in zones.

✖ **Yes.**
Linux VMs are fully supported in availability zones. Zone support depends on region and VM size, not the operating system.

EXAM STRATEGY
'Only Windows' is an absolute limit. Zones do not care about the OS; Linux VMs work too. The answer is No.

IMPACT
Linux workloads get the same zone protection. Thinking otherwise could push teams to the wrong region or design.

What · Where · When · Why · How

What	Availability zones are a physical location choice that does not depend on the OS.
Where	Chosen for each VM when you create it, in zone-enabled regions.
When	When Windows or Linux VMs need protection from a datacenter failure.
Why	Linux VMs work in zones too, so 'Windows only' is false.
How	Set the availability zone field (1, 2 or 3) when creating any supported VM.

Exam tip
Zones are limited by REGION and sometimes VM SIZE, never by operating system.

Memory trick
Zones care about WHERE, not WHICH OS.

Real-world example, use case and scenario

EXAMPLE
Ubuntu in zone 3
An Ubuntu 24.04 VM is deployed with availability zone 3 in North Europe.

USE CASE
Linux web farm
A news site runs Nginx on Linux VMs spread across three zones, behind a zone-redundant load balancer.

SCENARIO
Myth busted
A Linux team thinks only Windows can use zones and plans a costly second region. The architect shows that Linux VMs work in zones. They get 99.99% uptime inside one region.

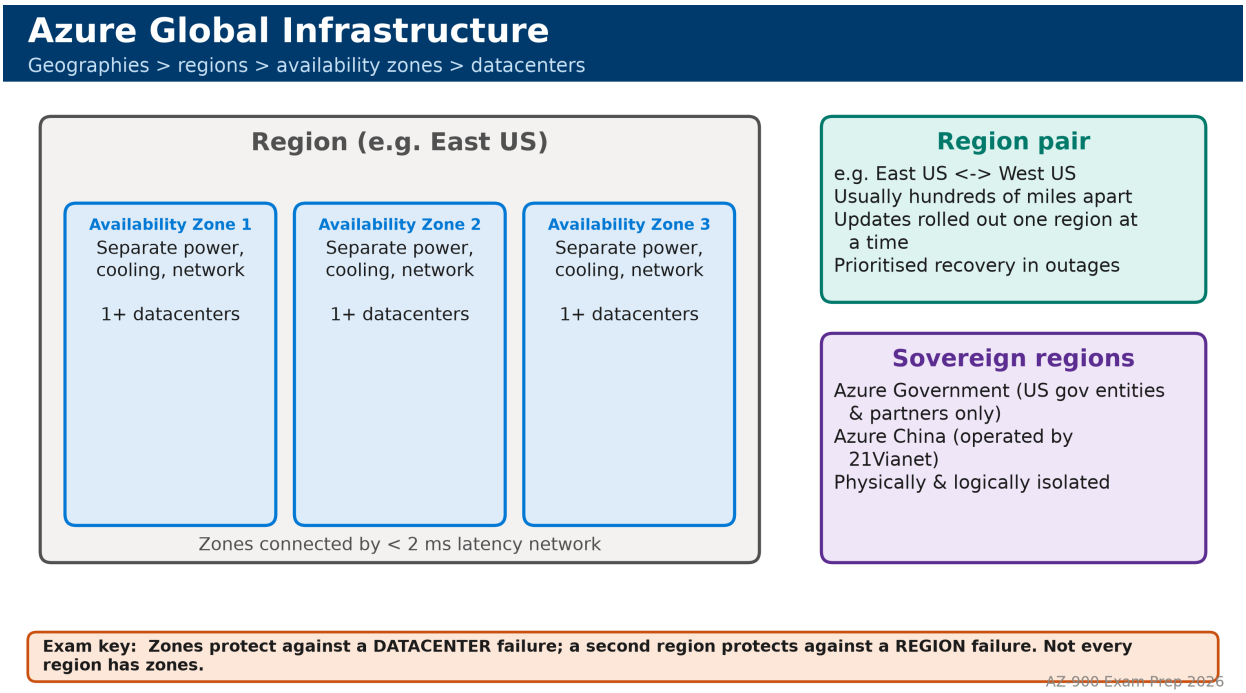


Figure Q127a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
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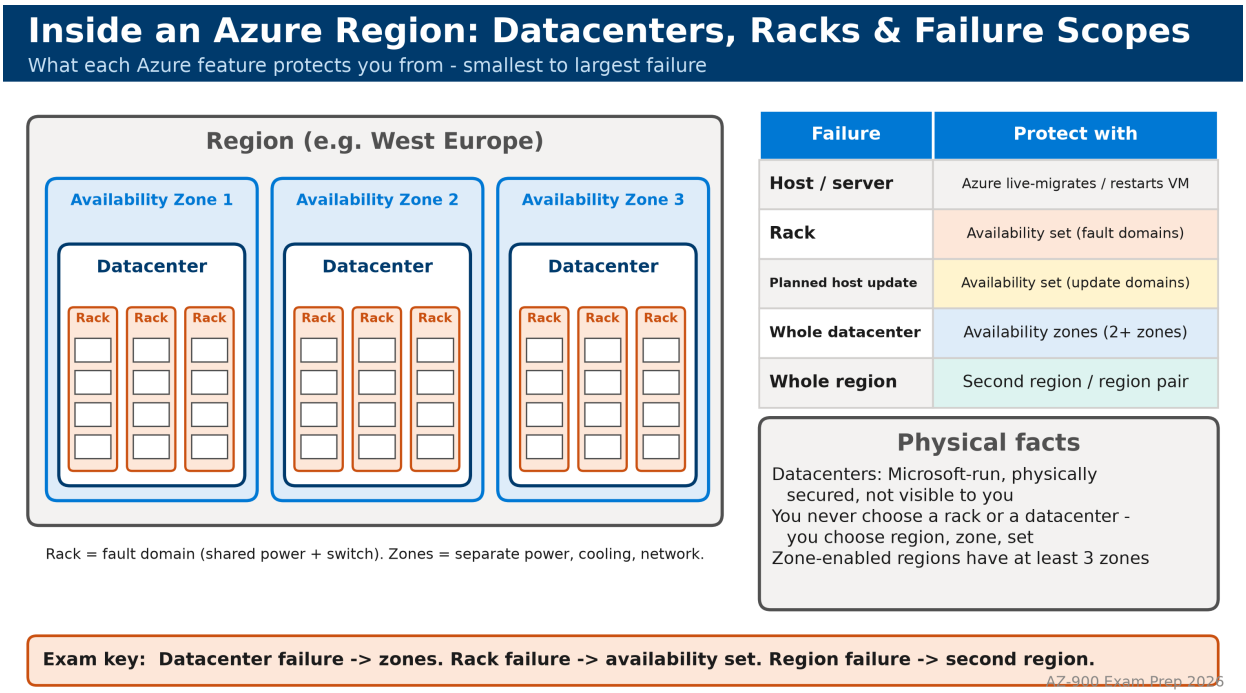


Figure Q127b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 128

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q208

All the Azure resources deployed to a resource group must use the same Azure region.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is MUST. A resource group's own region only says where its metadata (information about the resources) is stored. Q273 asks the same idea the other way: a resource group CAN contain resources from several regions (Yes).

Answer explained

✔ **No.**
A resource group's location only stores its metadata. Resources inside it can be in any region.

✖ **Yes.**
Resources in one resource group can be in different regions. The group's own region does not limit them.

EXAM STRATEGY
A resource group's location only stores metadata. Resources inside can be in any region. 'Must use the same region' is No.

IMPACT
Grouping by lifecycle, not by region, lets one app's resources in several regions be managed and deleted together.

What · Where · When · Why · How

What	A resource group is a logical container. Its own location only stores its metadata.
Where	Created in a subscription, with a location setting for the metadata.
When	When grouping resources that share a lifecycle, even across regions.
Why	Resources in a group can be in any region, so they do not need to match.
How	Create the group in one region, and deploy each resource to the region it needs.

Exam tip
Resource group region = where its metadata lives. Resources inside can be in any region.

Memory trick
The folder lives in one city; its files can travel the world.

Real-world example, use case and scenario

EXAMPLE
Mixed regions
rg-webapp (location: East US) holds an App Service in East US and a storage account in Japan East.

USE CASE
Multi-region app
A shipping company keeps its main and disaster-recovery resources from two regions in one resource group, so they share lifecycle and RBAC.

SCENARIO
Extra work for nothing
A team made one resource group per region, thinking it was required. This doubled their permission settings. After learning groups can span regions, they reorganised by app instead.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

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Figure Q128: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 129

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q209, Q33

If you assign a tag to a resource group, all the Azure resources in that resource group are assigned to the same tag.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Tags (name-value labels such as Department=Finance) are NOT passed down to resources by default. Look-alikes with a different answer: RBAC permissions (Q258) and locks (Q191) ARE inherited - both Yes. Q385: a policy with the Modify effect can add missing tags.

Answer explained

✔ **No.**
Tags on a resource group are not inherited by its resources by default. You need an Azure Policy such as 'Inherit a tag from the resource group'.

✖ **Yes.**
Unlike RBAC, policies and locks, tags do not pass down to the resources inside automatically.

EXAM STRATEGY
Tags are not inherited by default. To pass them down you need an Azure Policy such as 'inherit a tag from the resource group'. The answer is No.

IMPACT
If you assume tags are inherited, cost reports break. Untagged resources show up as unassigned spending in Cost Management.

What · Where · When · Why · How

What	Tags are name-value pairs that label Azure resources for organising and cost reports.
Where	Added to subscriptions, resource groups and single resources.
When	When costs or resources must be sorted by department, environment or owner.
Why	Tags are not inherited by default, so resources do not get the group's tag automatically.
How	Tag each resource, or assign the 'Inherit a tag from the resource group' policy with remediation (a fix task for existing resources).

Exam tip
Inherited: RBAC, Policy, Locks. NOT inherited: Tags (unless a policy copies them).

Memory trick
Tags stick only where you stick them.

Real-world example, use case and scenario

EXAMPLE
Missing tag
rg-finance has Department=Finance, but a VM inside shows no tags.

USE CASE
Policy-based inheritance
A healthcare provider assigns the built-in policy 'Inherit a tag from the resource group if missing'. Every resource then gets its CostCenter tag.

SCENARIO
Blank cost report
A cost analyst groups costs by Department and finds most spending 'untagged', even though the resource groups have tags. Adding a tag-inheritance policy with remediation fixes next month's report.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

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Figure Q129: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 130

AZURE ARCHITECTURE & SERVICES

Storage
v5 Q213

Data that is stored in an Azure Storage account automatically has at least three copies.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): 'At least three copies' is about the minimum. Even the cheapest option, LRS, keeps three copies, so Yes. Q131 is a look-alike: copies in ANOTHER datacenter are NOT automatic (No), because LRS keeps all three in one building.

Answer explained

✔ **Yes.**
Even the lowest option, LRS (locally redundant storage), keeps three copies of the data written at the same time. Geo-redundant options keep six.

✖ **No.**
Every redundancy option stores at least three copies (LRS is the minimum), so the statement is true.

EXAM STRATEGY
Even the cheapest option, LRS, keeps three copies written at the same time. So 'at least three copies' is always Yes.

IMPACT
Three copies protect against disk or rack failure, but not the loss of a region. Knowing LRS limits helps you pick GRS for disaster recovery.

What · Where · When · Why · How

What	Azure Storage redundancy copies data automatically to protect against hardware failure.
Where	Chosen per storage account as LRS, ZRS, GRS, RA-GRS, GZRS or RA-GZRS.
When	Applies to every storage account from the moment data is written.
Why	Even LRS, the minimum, keeps three copies, so there are always at least three.
How	Each write is saved to three copies at the same time. Geo options add three more in another region.

Exam tip
Minimum copies = 3 (LRS). Geo options = 6. Copies protect against hardware loss, not accidental deletion, so they are not a backup.

Memory trick
Three is the floor: LRS starts at three, geo doubles to six.

Real-world example, use case and scenario

EXAMPLE
LRS copies
A file uploaded to an LRS account is written to three separate fault domains (separate power and network groups) in one datacenter before the upload is confirmed.

USE CASE
Log archive
A security firm stores logs it can recreate on LRS. Three copies still protect them against disk failure.

SCENARIO
Copies are not backup
A developer deletes a container, thinking the three copies will save it. All three copies are deleted too. The team turns on soft delete and versioning so future mistakes can be undone.

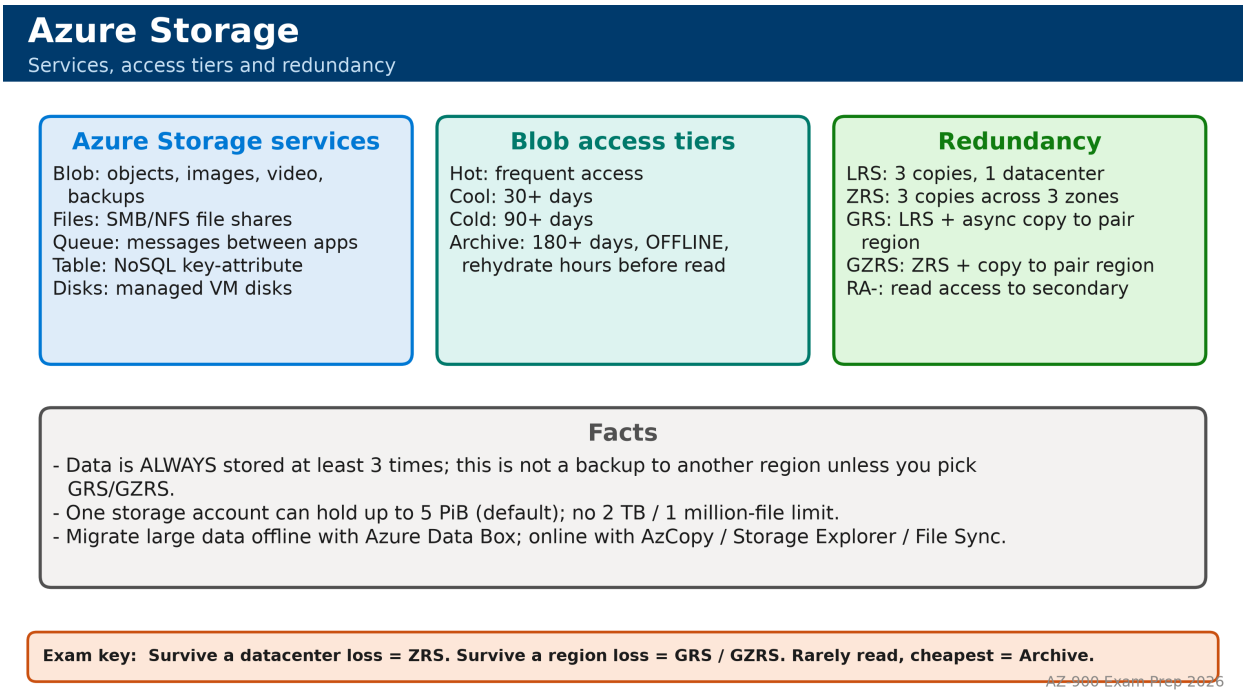


Figure Q130: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 131

AZURE ARCHITECTURE & SERVICES — Storage v5 Q214

All data that is copied to an Azure Storage account is backed up automatically to another Azure data center.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch 'ALL data', 'AUTOMATICALLY' and 'ANOTHER datacenter'. LRS keeps all three copies in ONE datacenter, so copies elsewhere are not automatic. Also, copying (replication) is not backup. Compare Q130: at least three copies always exist (Yes).

Answer explained

✔ **No.**
You choose redundancy for each account. LRS keeps all three copies in one datacenter. Also, replication (copying, as in ZRS or GRS) is not a backup.

✖ **Yes.**
Only ZRS, GRS or GZRS copy data to other datacenters. The LRS option keeps every copy in one building.

EXAM STRATEGY
'Automatically' plus 'another datacenter' decides it. LRS never leaves one datacenter, so answer No. You do not need to think about GRS.

IMPACT
Teams that think backup is built in skip Azure Backup or soft delete. One accidental delete or ransomware attack is copied at once, and the data is gone.

What · Where · When · Why · How

What	Azure Storage redundancy (LRS, ZRS, GRS, GZRS) keeps several copies of account data.
Where	Chosen per storage account when it is created, or later under Redundancy.
When	When choosing how far apart copies must be: one datacenter, three zones or a second region.
Why	LRS keeps all copies in one datacenter, so data is not automatically copied to another datacenter.
How	Pick ZRS or GRS for copies in other datacenters. Add soft delete, versioning or Azure Backup for real backups.

Exam tip
'All data' and 'automatically' are the trap. Redundancy is a choice per account, LRS stays in one datacenter, and replication is never the same as backup.

Memory trick
LRS = Local, one building: three copies, one site.

Real-world example, use case and scenario

EXAMPLE
LRS choice
A dev team creates a storage account with LRS to save money. All three copies are in one West Europe datacenter, so a fire there would affect every copy.

USE CASE
Insurance archive on GRS
An insurer stores claim scans in GRS. A second copy is kept hundreds of kilometres away in the paired region, to meet regulator rules.

SCENARIO
Deleted invoices
A bookkeeping startup thought GRS was a backup. A script deleted a container, and the delete was copied to the second region at once. They then turned on soft delete and versioning to get real restore points.

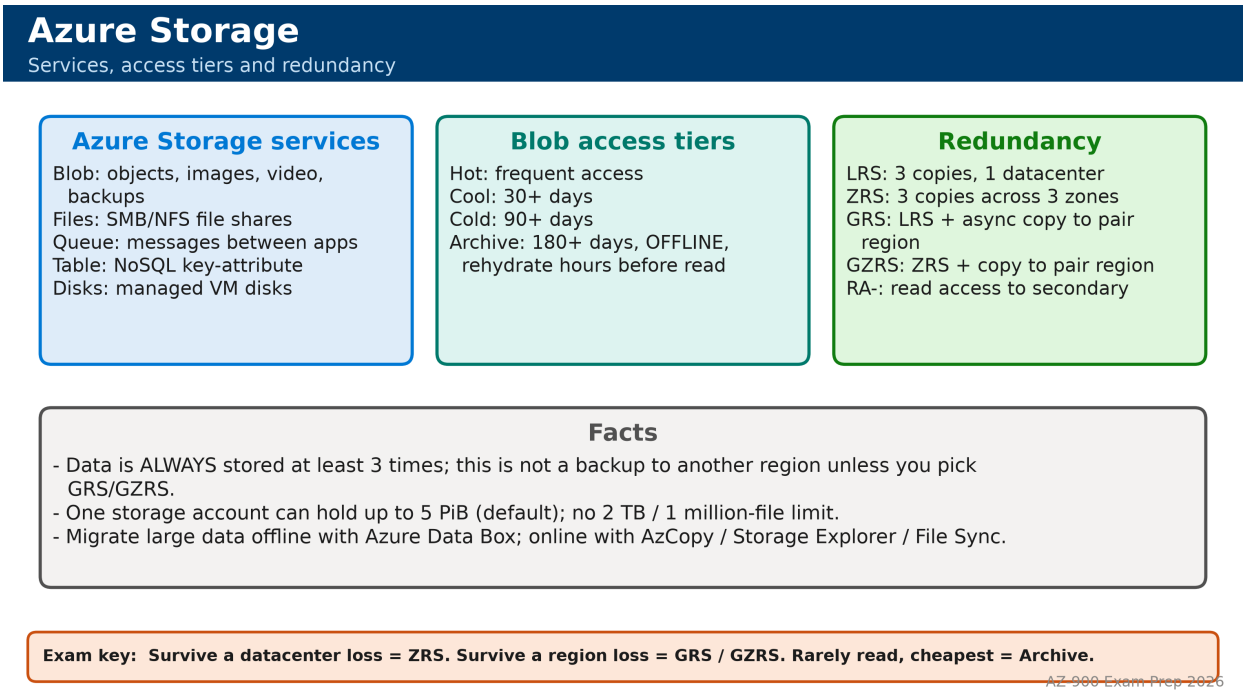


Figure Q131a: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

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• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

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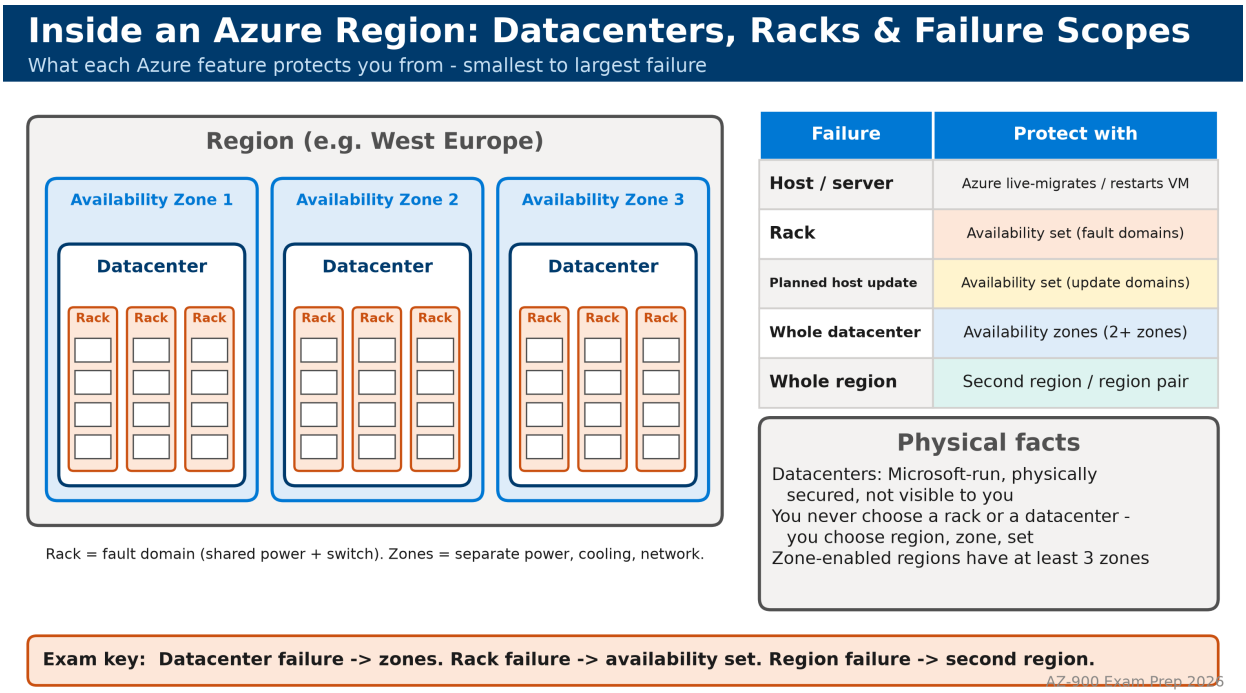


Figure Q131b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 132

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q215

An Azure Storage account can contain up to 2 TB of data and up to one million files.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The numbers are made up. A storage account holds petabytes (thousands of terabytes), not 2 TB. Q223 is a similar trap: a free account is not limited to 2 TB of upload either (No).

Answer explained

✔ **No.**
A general-purpose v2 (GPv2) storage account holds up to 5 PiB (about 5,000 TB) by default. It has no practical limit on the number of files.

✖ **Yes.**
2 TB and one million files are made-up limits. Storage accounts grow to petabytes and billions of files.

EXAM STRATEGY
Any small fixed number (2 TB, one million files) for a whole storage account means No. Azure gives limits in petabytes.

IMPACT
If you think limits are small, you create too many accounts and a complex design. Knowing the 5 PiB default keeps designs simple and cheaper.

What · Where · When · Why · How

What	Storage account limits set the maximum size and number of objects in one account.
Where	Listed per account type. A GPv2 account allows 5 PiB by default in most regions.
When	When deciding how many storage accounts a large amount of data needs.
Why	Accounts hold petabytes and billions of files, so a 2 TB / one million file limit is false.
How	Store data in one GPv2 account. Ask support for more if 5 PiB is not enough.

Exam tip
Azure Storage is built to grow very large. Any tiny size number for a whole storage account means No.

Memory trick
Storage accounts think in Petabytes, not Terabytes: P beats T.

Real-world example, use case and scenario

EXAMPLE
Video library
A streaming service keeps several hundred terabytes of master videos in one GPv2 account.

USE CASE
Genomics data lake
A research lab stores billions of gene files in Data Lake Storage Gen2 on one storage account, with no file-count limit.

SCENARIO
Wrong size guess
An architect planned 40 storage accounts for a 60 TB photo archive, thinking each held only 2 TB. A reviewer pointed out the 5 PiB default. The design shrank to one account with storage tiers.

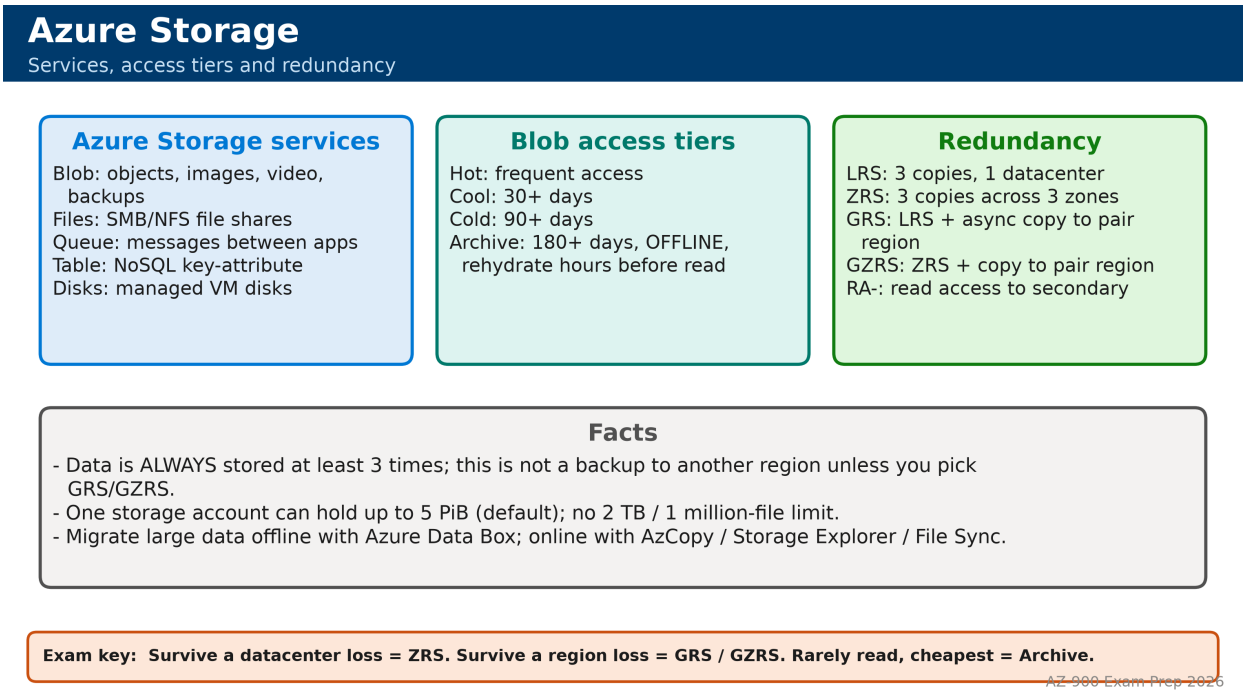


Figure Q132: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 133

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q217

North America is represented by a single Azure region.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Answer explained

✔ **No.**
North America has many Azure regions, such as East US, Central US, West US 2 and Canada Central, in several geographies (groups of regions).

✖ **Yes.**
A continent has many regions and geographies. A region is one area, like West US 3 in Arizona.

EXAM STRATEGY
'Single region' for a whole continent is wrong. Think of East US, West US, Central US and Canada Central. Answer No.

IMPACT
Region choice affects speed, data location rules and price. Treating a continent as one region leads to wrong data-location and recovery decisions.

What · Where · When · Why · How

What	An Azure region is a set of datacenters in one area. A geography groups several regions.
Where	North America includes regions like East US, Central US, West US 2 and Canada Central.
When	When choosing where to deploy for speed, data location rules or disaster recovery.
Why	A continent has many regions, never just one.
How	Pick the region nearest to your users in the portal's region list, and add more regions when needed.

Exam tip
Order to remember: geography > region > availability zone > datacenter. A continent is never one region.

Memory trick
Continent holds countries, country holds regions, region holds zones.

Real-world example, use case and scenario

EXAMPLE
US regions
In the portal region list you can pick East US, South Central US, North Central US and West US 2, all in North America.

USE CASE
Canadian data rules
A Canadian health provider uses Canada Central and Canada East to keep patient data inside Canada.

SCENARIO
Delay for gamers
A gaming studio thought one North American region would serve all players. Players in Vancouver had high delay from East US. The studio added West US 2 and sent players to the nearest region with Front Door.

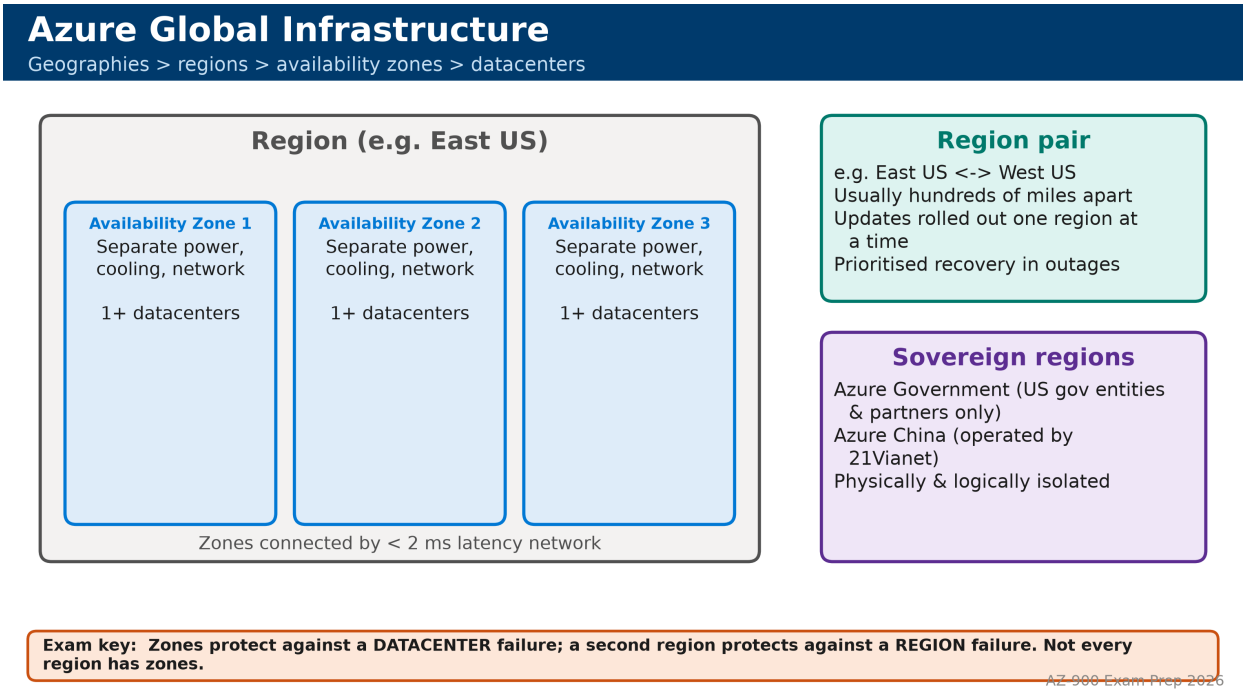


Figure Q133: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

Question 134

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q218

Every Azure region has multiple datacenters.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap words are EVERY and MULTIPLE. Microsoft Learn says a region has 'at least one, but potentially multiple' datacenters. So not every region has several. Q304 is different: zone-ENABLED regions have at least three zones (Yes).

Answer corrected in this edition. The previous PDF printed: *Yes.*. The printed answer was Yes. Microsoft Learn says a region has at least one, and possibly several, datacenters. So not every region is sure to have several datacenters.

2026 update: The current Microsoft Learn wording (one or more datacenters) supports No.

Answer explained

✔ **No.**
Microsoft Learn says a region has at least one, and possibly several, datacenters. So 'every region has multiple' is not guaranteed.

✖ **Yes.**
Only zone-enabled regions are sure to have several datacenters. The official definition says one or more.

EXAM STRATEGY
'Every' plus 'multiple' decides it. Learn says at least one datacenter. Choose No on today's exam.

IMPACT
If you think every region has several datacenters, you may miss a single-datacenter risk. Zone-based designs, not just picking a region, give datacenter-level protection.

What · Where · When · Why · How

What	An Azure region is an area with at least one, and possibly several, datacenters.
Where	Definition from the Microsoft Learn module on Azure global infrastructure.
When	When judging how resilient a region is, or reading 'every region' statements.
Why	Only zone-enabled regions are sure to have several datacenters, so 'every region has multiple' is false.
How	Choose a zone-enabled region when you need several separate datacenters for sure.

Exam tip
Learn the exact Learn phrase: at least one, but potentially multiple, datacenters. Absolute words like 'every' are the trap.

Memory trick
Region = one or more. Zone-enabled region = three or more.

Real-world example, use case and scenario

EXAMPLE
Small region
A newer region that opened without availability zones may use only one datacenter building.

USE CASE
Choosing a strong region
A delivery firm picks a zone-enabled region because that guarantees several separate datacenters for its tracking system.

SCENARIO
Study group debate
Two students argued about this statement. One quoted an old answer key that said Yes. The other showed the Learn definition of 'at least one datacenter'. The group agreed on No.

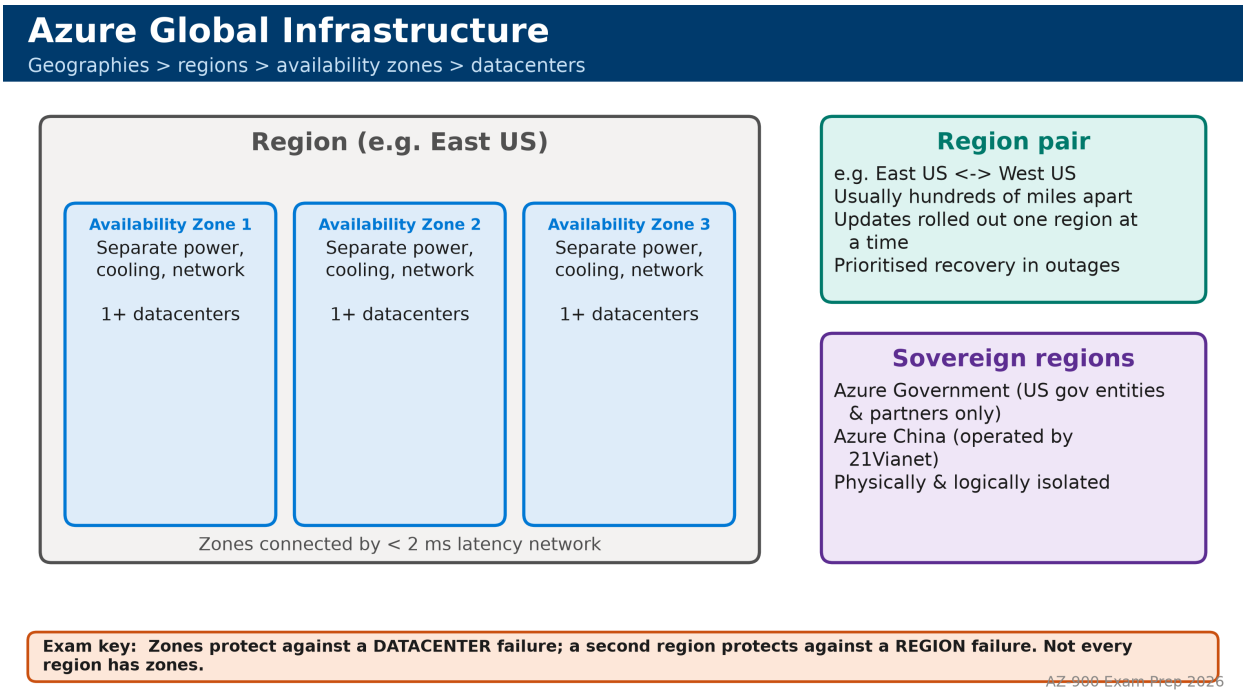


Figure Q134a: Regions, availability zones and region pairs

About this figure

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- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

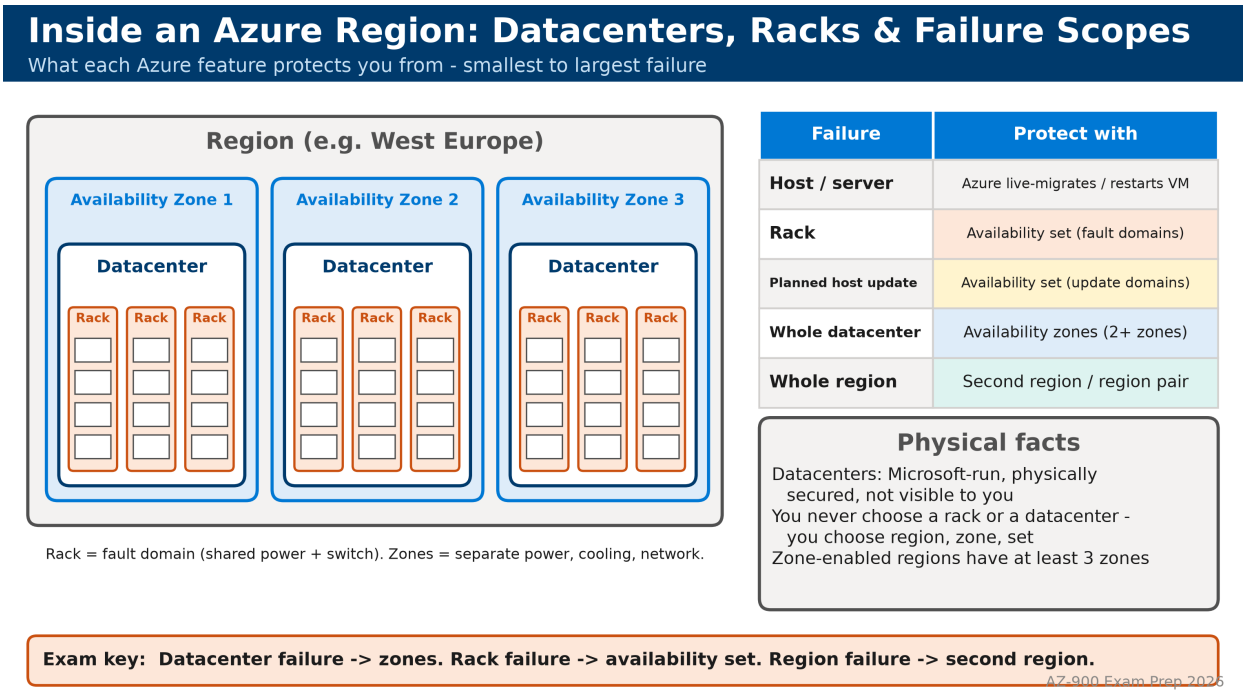


Figure Q134b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 135

AZURE MANAGEMENT & GOVERNANCE

Cost

v5 Q219, Q348

Data transfers between Azure services located in different Azure regions are always free.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is ALWAYS. Data going INTO Azure is free, but data moving BETWEEN regions is charged per GB. Look-alikes: Q219 copying data INTO Azure over VPN has no transfer cost (No); Q241 copying data OUT of Azure to on-premises does cost (Yes).

Answer explained

✔ **No.**
Data moving between Azure regions is charged per GB. Only data coming in (ingress) and most traffic inside one region are free.

✖ **Yes.**
Transfer between regions, such as VNet peering across regions or geo-replication, is charged per GB based on source and destination.

EXAM STRATEGY
'Always free' is the giveaway. Only incoming data is free. Any claim that transfer between regions is free is No.

IMPACT
Data leaving a region for GRS, geo-replicated databases and global VNet peering slowly adds to monthly bills. Architects must budget for it.

What · Where · When · Why · How

What	Bandwidth pricing charges for data leaving a region, including between Azure regions.
Where	Billed on the subscription as data transfer, with prices by source and destination.
When	When designing replication across regions, global VNet peering or multi-region apps.
Why	Transfer between regions is charged per GB, so it is not always free.
How	Estimate the GB moved between regions in the pricing calculator, and keep services that talk a lot in the same region.

Exam tip
Data in (ingress) is free. Data out (egress) is charged. Copying across regions (GRS, geo-replicated SQL, VNet peering across regions) adds bandwidth cost.

Memory trick
In is free, out costs a fee, crossing regions costs a toll.

Real-world example, use case and scenario

EXAMPLE
Global VNet peering
Traffic between a VNet in East US and a peered VNet in West Europe is charged per GB in each direction.

USE CASE
DR copy budget
A software company adds region-to-region bandwidth to its cost plan, because nightly database copying to the second region moves 2 TB.

SCENARIO
Surprise bill
A media agency put its rendering computers in one region and its storage in another. The monthly bill showed thousands in data transfer charges, so they moved both into the same region.

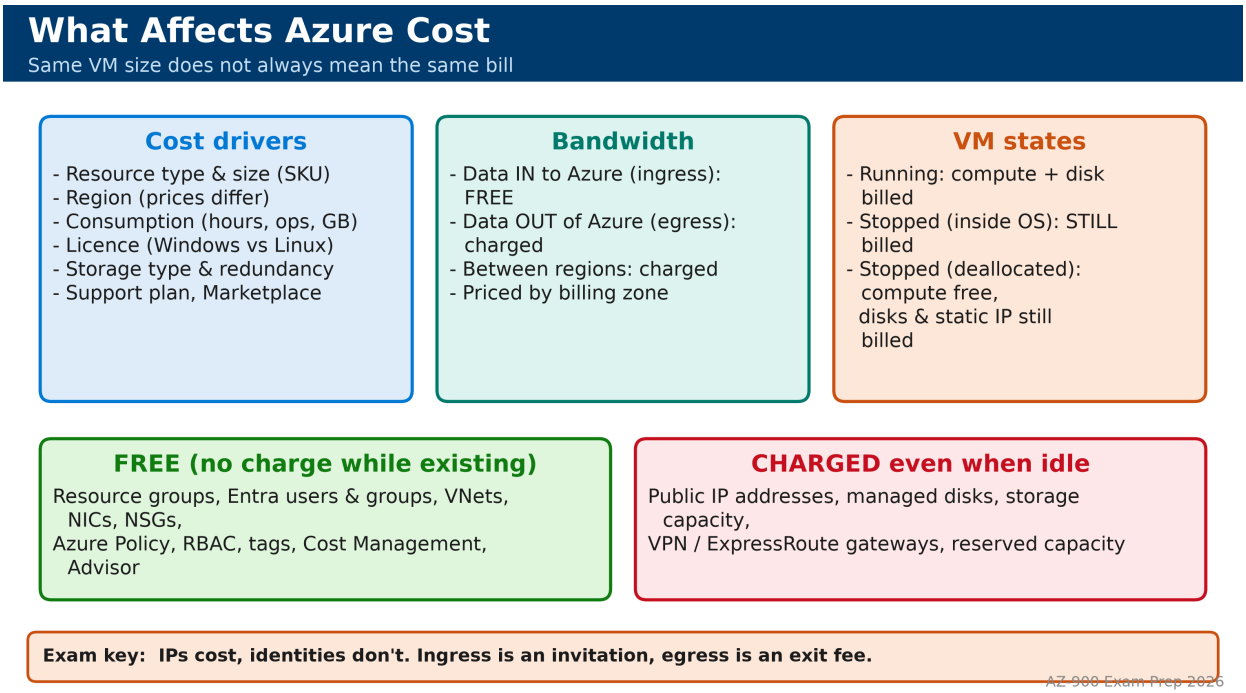


Figure Q135a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

• Bandwidth: ingress is free; the first 100 GB/month of Internet egress is free, then approx. USD 0.087/GB in North America and Europe.

• Inter-region transfer is approx. USD 0.02/GB within a continent; VNet peering is approx. USD 0.01/GB each way in the same region.

• A VM stopped inside the guest OS is still allocated and billed for compute; only Stopped (deallocated) from the portal, CLI or API stops compute charges.

• A Standard static public IP costs approx. USD 3.65/month even unattached; a 128 GB Premium SSD costs approx. USD 20/month even when its VM is deallocated.

• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q135b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 136

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q220

You need to be notified when Microsoft plans to perform maintenance that can affect the resources deployed to an Azure subscription. What should you use?

✖ A. Azure Monitor.

✔ **B.** Azure Service Health.

✖ C. Azure Advisor.

✖ D. Microsoft Trust Center.

Correct answer: Azure Service Health.

Clarification (read this first): Note who does the maintenance: MICROSOFT. Notices about Microsoft's own outages and planned maintenance come from Service Health. Q405 is different: with NO Azure account, you check the public Azure status page.

Answer explained

✔ **Azure Service Health.**
Azure Service Health reports planned maintenance, service problems and advisories for your subscriptions. It can also send you alerts about them.

✖ **Azure Monitor.**
Azure Monitor tracks metrics and logs of your own resources. Notices about Microsoft maintenance come from Service Health.

✖ **Azure Advisor.**
Advisor gives best-practice advice on cost, security, reliability and performance. It does not send maintenance notices.

✖ **Microsoft Trust Center.**
The Trust Center is a public website about Microsoft security, privacy and compliance. It sends no notices for your subscription.

EXAM STRATEGY
'Microsoft plans maintenance' points to Service Health. Remove Advisor (advice for you), Trust Center (compliance documents) and general Monitor.

IMPACT
Missing a maintenance notice can mean unplanned downtime for live systems. Service Health alerts give you time to fail over or change plans.

What · Where · When · Why · How

What	Azure Service Health is a personal view of Azure service problems, planned maintenance and advisories.
Where	Azure portal > Monitor > Service Health (or search for Service Health).
When	When you must know about Microsoft maintenance or outages that affect your subscriptions.
Why	It is the service that announces planned maintenance and can alert you.
How	Create a Service Health alert for 'Planned maintenance' events, linked to an action group (email, SMS).

Exam tip
Microsoft's side (outages, planned maintenance, advisories) = Service Health. Your own resource's side = Resource Health / Azure Monitor.

Memory trick
Service Health is Microsoft's notice board for you.

Real-world example, use case and scenario

EXAMPLE
Maintenance notice
Service Health lists a planned host update for VMs in North Europe next Tuesday and shows which of your resources are affected.

USE CASE
Airline ops team
An airline creates a Service Health alert so its on-call team gets SMS warnings about maintenance in the regions that run its booking system.

SCENARIO
Avoiding a clash
An online learning company saw in Service Health that maintenance fell in exam week. They used the self-service maintenance window to apply updates early, so there were no restarts during exams.

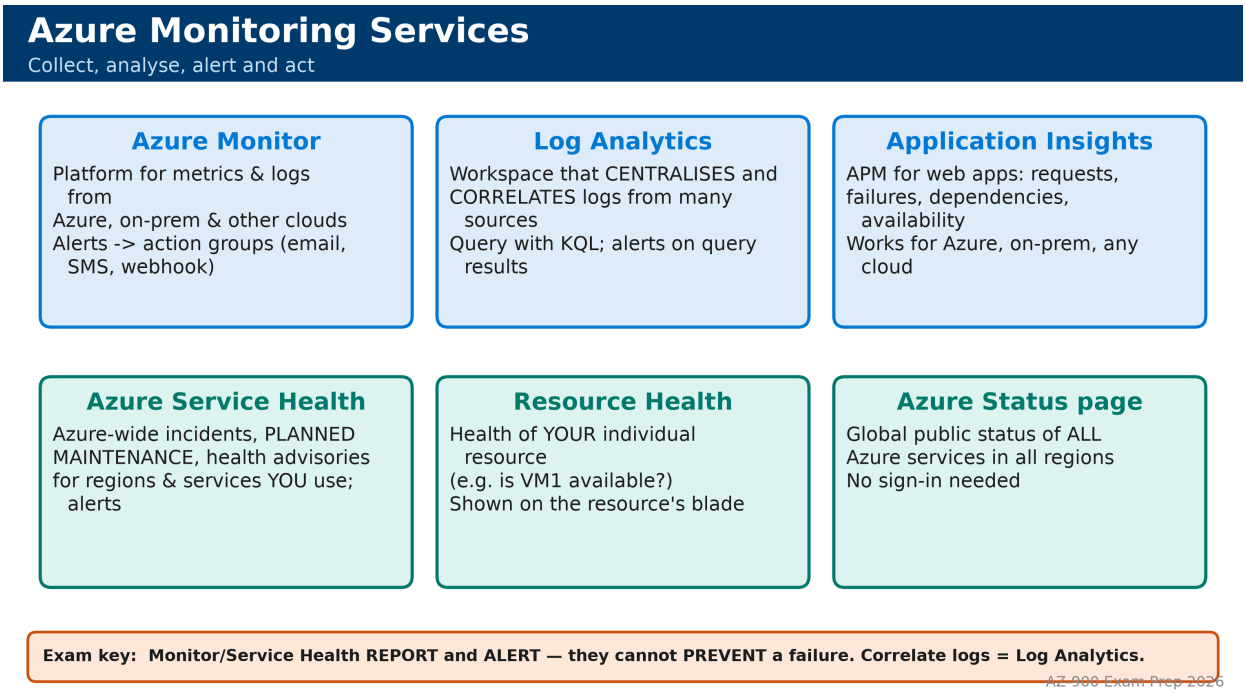


Figure Q136: Azure monitoring services

About this figure

- Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).
- Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.
- Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.
- Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.
- Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 137

AZURE ARCHITECTURE & SERVICES | **Compute** v5 Q221

A Windows Virtual Desktop session host can run Windows 10 only.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is ONLY. Also, 'Windows Virtual Desktop' is the old name of Azure Virtual Desktop (AVD). Session hosts (the VMs users connect to) can run Windows 11, Windows 10 or Windows Server.

2026 update: Windows Virtual Desktop was renamed Azure Virtual Desktop in 2021. Session hosts can run Windows 11 and Windows 10 Enterprise (including multi-session) and Windows Server 2016 or later.

Answer explained

✔ **No.**
Azure Virtual Desktop session hosts can run Windows 11 and Windows 10 Enterprise (including multi-session - many users on one VM) and Windows Server 2016 or later.

✖ **Yes.**
Windows 11 and Windows Server images are also supported, so Windows 10 is not the only one.

EXAM STRATEGY
The word 'only' decides it. AVD supports Windows 11, Windows 10 and Windows Server. Read the old WVD name as Azure Virtual Desktop.

IMPACT
Windows 11 multi-session is only available in AVD. Putting several users on one VM cuts licence and compute costs.

What · Where · When · Why · How

What	Azure Virtual Desktop (formerly Windows Virtual Desktop) session hosts are Azure VMs that run user sessions.
Where	Set up in an AVD host pool (a group of session host VMs), built from Marketplace or custom images.
When	When choosing the OS image for virtual desktops or published apps.
Why	Windows 11, Windows 10 and Windows Server 2016+ are supported, so it is not Windows 10 only.
How	Select a Windows 11, Windows 10 or Windows Server image when adding session hosts to the host pool.

Exam tip
Any 'only' statement about AVD operating systems is false. Remember multi-session Windows 10/11 exists only in AVD.

Memory trick
AVD hosts: 10, 11 and Server - three families, not one.

Real-world example, use case and scenario

EXAMPLE
Server host pool
A team builds a host pool from a Windows Server 2022 Marketplace image to publish an old business app.

USE CASE
Call centre desktops
A call centre uses Windows 11 Enterprise multi-session hosts so many agents share each VM.

SCENARIO
App needs Server
A law firm needed an app that is approved only on Windows Server. They thought AVD supported only Windows 10 and nearly bought on-premises servers. Instead, they deployed a Windows Server host pool in AVD.

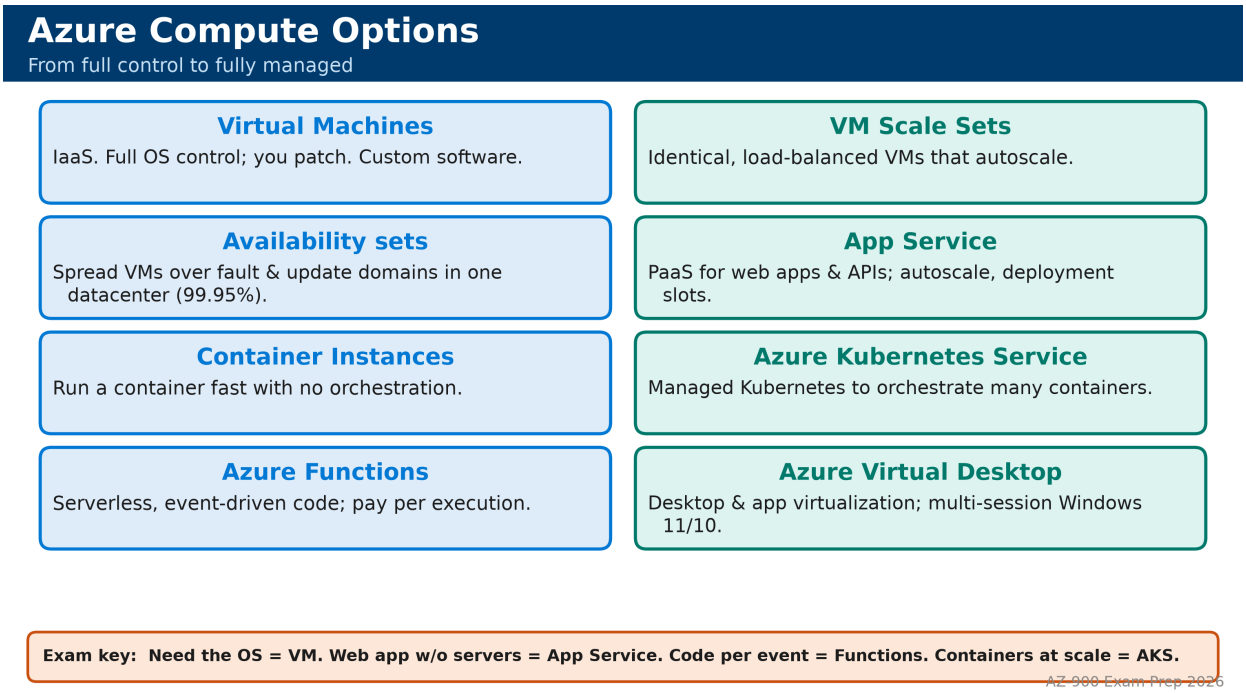


Figure Q137: Azure compute options

About this figure

- Read left column then right: options run from full control (VMs) to fully managed or serverless (Functions); pick by how much you must manage.
- VM sizes range from tiny B-series (B1s approx. USD 8/month Linux) to large memory and GPU VMs; per-second billing applies while running.
- VM Scale Sets support up to 1,000 instances (600 with custom images); availability sets use up to 3 fault domains and 20 update domains.
- App Service plans set the price: Free F1 USD 0, Basic B1 approx. USD 13/month Linux, Standard S1 approx. USD 70/month with autoscale and slots.
- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

Question 138

AZURE ARCHITECTURE & SERVICES — Compute v5 Q222

A Windows Virtual Desktop host pool that includes 20 session hosts supports a maximum of 20 simultaneous user connections.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap is thinking one host = one user. In a pooled host pool, each multi-session host serves MANY users, so 20 hosts can serve far more than 20. One user per host applies only to personal host pools. Windows Virtual Desktop is now Azure Virtual Desktop.

2026 update: Windows Virtual Desktop is now Azure Virtual Desktop.

Answer explained

✔ **No.**
Pooled host pools use multi-session Windows or Windows Server. Each session host serves many users, up to a session limit you set.

✖ **Yes.**
One user per host applies only to personal host pools. Pooled multi-session hosts serve many users at once each.

EXAM STRATEGY
The trap is hosts = users. Pooled multi-session hosts serve many users each, so 20 hosts allow far more than 20 users.

IMPACT
Choosing how many users share each host is the main AVD cost control. Thinking one user per host multiplies VM costs.

What · Where · When · Why · How

What	An AVD host pool is a group of session host VMs. Pooled pools share hosts between users.
Where	Set up in the Azure Virtual Desktop service, with a maximum session limit per pooled host pool.
When	When working out how many VMs you need for a number of users at the same time.
Why	Multi-session hosts serve many users each, so 20 hosts can serve more than 20 connections.
How	Create a pooled host pool with multi-session images, then set the max session limit and load-balancing method.

Exam tip
Pooled + multi-session = many users per host.
Personal = one user per host.

Memory trick
Pooled hosts are buses; personal hosts are taxis.

Real-world example, use case and scenario

EXAMPLE
Session limit
A pooled host pool with 20 hosts and a max session limit of 12 can serve 240 users at the same time.

USE CASE
University labs
A university runs pooled multi-session hosts so thousands of students can use engineering software from home.

SCENARIO
Too many VMs planned
A retailer planned 300 VMs for 300 store staff. After learning about pooled multi-session hosts, they deployed 25 hosts with breadth-first load balancing (spreading users evenly). Compute cost dropped a lot.

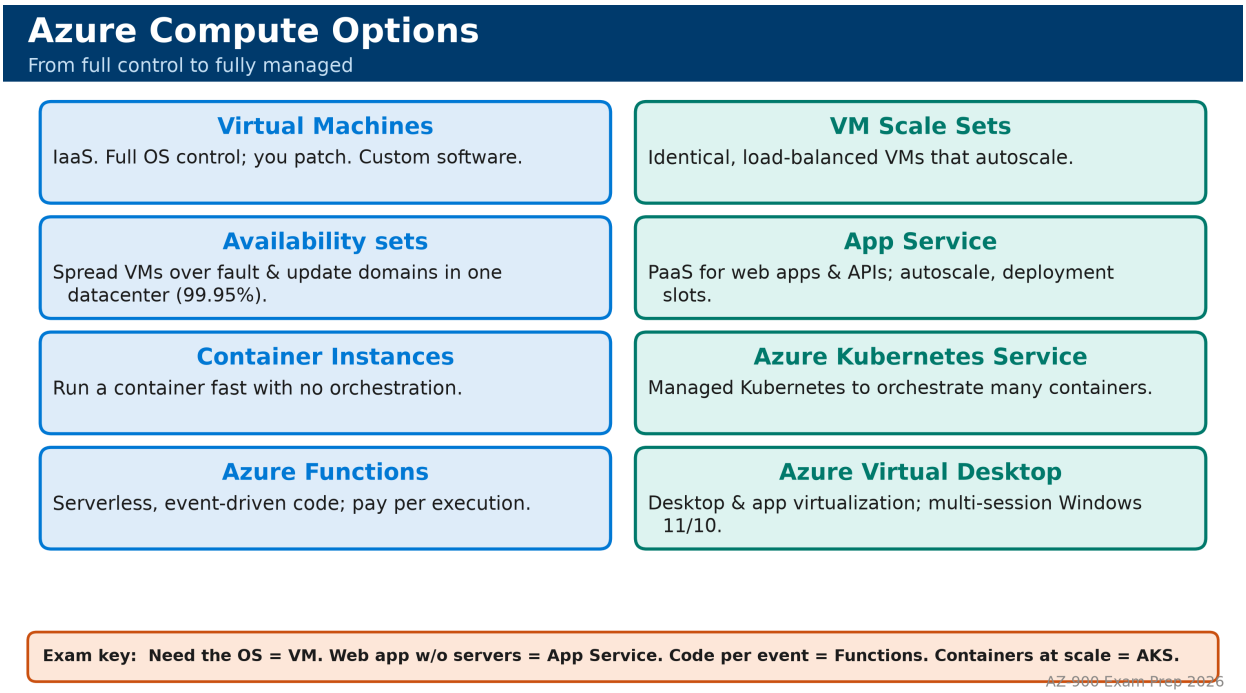


Figure Q138: Azure compute options

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- VM sizes range from tiny B-series (B1s approx. USD 8/month Linux) to large memory and GPU VMs; per-second billing applies while running.
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- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

Question 139

AZURE ARCHITECTURE & SERVICES — Compute v5 Q223

Windows Virtual Desktop supports desktop and app virtualization.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): 'Windows Virtual Desktop' is the old name of Azure Virtual Desktop. It delivers both full desktops and single apps, so Yes.

2026 update: Windows Virtual Desktop is now Azure Virtual Desktop.

Answer explained

✔ **Yes.**
Azure Virtual Desktop publishes full desktops (desktop application groups) and single apps (RemoteApp application groups).

✖ **No.**
AVD supports both desktop and RemoteApp application groups, so it delivers both desktop and app virtualization.

EXAM STRATEGY
Remember the two application group types: desktop and RemoteApp. Both exist, so Yes. The old name WVD means AVD.

IMPACT
Publishing single RemoteApps instead of full desktops confuses users less and leaves fewer things to attack. This is a common AVD design choice.

What · Where · When · Why · How

What	Azure Virtual Desktop delivers virtual Windows desktops and single remote apps from Azure.
Where	Set up through desktop and RemoteApp application groups linked to a host pool and workspace.
When	When users need a full desktop, or only certain Windows apps, from any device.
Why	AVD supports both desktop and app virtualization, so the statement is true.
How	Publish a desktop application group for full desktops, or a RemoteApp group for single apps.

Exam tip
Desktop application group = full desktop. RemoteApp application group = single apps. AVD has both.

Memory trick
AVD serves the whole meal (desktop) or single dishes (apps).

Real-world example, use case and scenario

EXAMPLE
RemoteApp Excel
An accountant on a Mac opens only Excel as a RemoteApp from AVD. It looks like a local window.

USE CASE
Contractor desktops
An engineering firm gives contractors full AVD desktops, so source data never leaves Azure.

SCENARIO
Old app on Chromebooks
A school needed a Windows-only grading tool on Chromebooks. They published just that tool as a RemoteApp through AVD, and teachers used it from the browser.

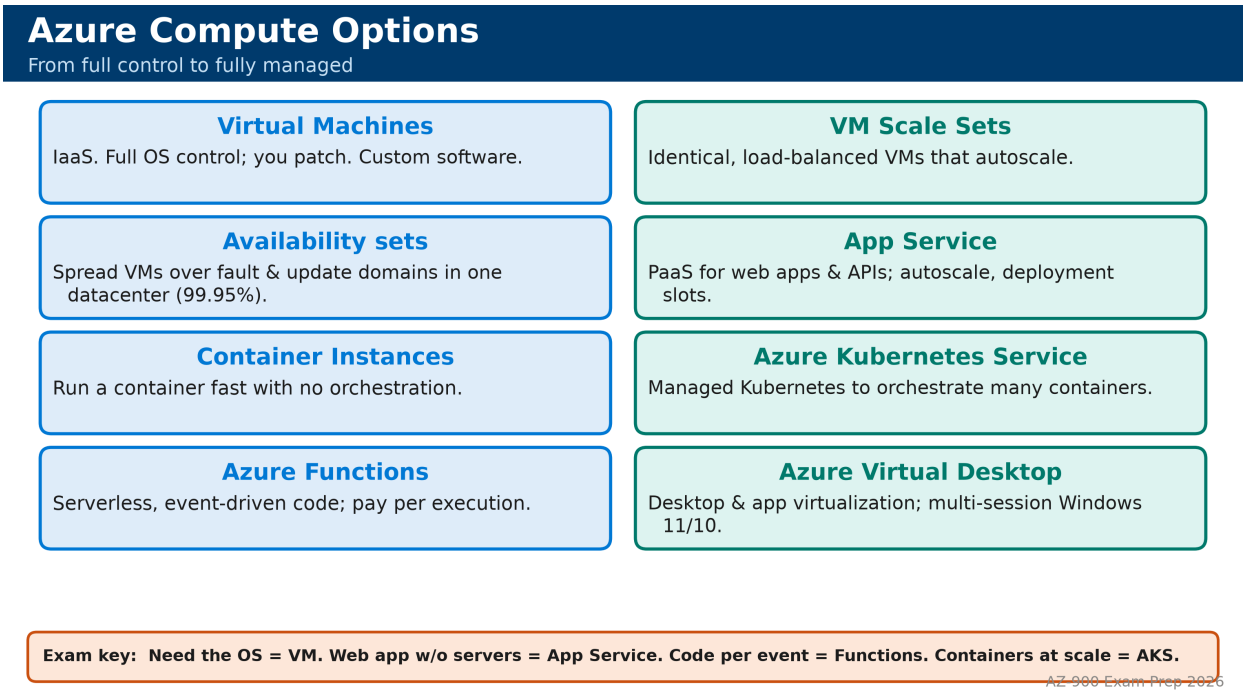


Figure Q139: Azure compute options

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- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

Question 140

CLOUD CONCEPTS

Cost
v5 Q224

[...] can calculate cost savings due to reduced electricity consumption as a result of migrating on-premises Microsoft SQL servers to Azure.

- ✖ A. The Azure Migrate: Server Assessment tool.
- ✔ **B.** The Azure Total Cost of Ownership (TCO) calculator.
- ✖ C. The Database Migration Assistant.
- ✖ D. The pricing calculator in Azure.

Correct answer: The Azure Total Cost of Ownership (TCO) calculator.

Clarification (read this first): Key words: ELECTRICITY and ON-PREMISES. Only the TCO (Total Cost of Ownership) calculator compares your own datacenter costs with Azure. Q396 is a look-alike: estimating the monthly cost of planned Azure services uses the PRICING calculator, not TCO.

Answer explained

✔ **The Azure Total Cost of Ownership (TCO) calculator.**
The TCO calculator compares on-premises costs, including electricity, datacenter space and staff, with the same setup's cost in Azure.

✖ **The Azure Migrate: Server Assessment tool.**
Azure Migrate checks if servers are ready, their size and their Azure cost. It does not model on-premises electricity savings.

✖ **The Database Migration Assistant.**
Data Migration Assistant checks SQL Server compatibility and features. It has no cost model.

✖ **The pricing calculator in Azure.**
The pricing calculator estimates only Azure service costs. It has no on-premises costs to compare.

EXAM STRATEGY
'Electricity' and 'on-premises' are the TCO clue words. Remove the pricing calculator (Azure costs only) and the migration check tools.

IMPACT
TCO estimates build the business case that managers approve. Leaving out power, cooling and space makes savings look smaller and can stop a migration.

What · Where · When · Why · How

What	The Azure TCO calculator compares on-premises costs with the same setup's costs in Azure.
Where	A free web tool on the Azure website, separate from the Azure portal.
When	When building a business case to move on-premises servers, such as SQL Server, to Azure.
Why	Only TCO includes on-premises electricity, datacenter space and staff costs, so it shows power savings.
How	Enter server, database, storage and network details, change values such as electricity cost, then view the savings.

Exam tip
Anything about on-premises vs Azure, electricity, cooling or datacenter costs points to TCO.

Memory trick
TCO counts the electric bill. The pricing calculator only reads the Azure menu.

Real-world example, use case and scenario

EXAMPLE
Power line item
The TCO report shows the estimated electricity cost for 12 SQL servers over five years, next to Azure SQL costs.

USE CASE
CFO business case
A manufacturing CFO uses the TCO report to show the board why the company can close a server room.

SCENARIO
Wrong tool first
A database admin ran DMA hoping for savings numbers and got only compatibility warnings. The architect then ran the TCO calculator, which showed five-year savings including power and cooling.

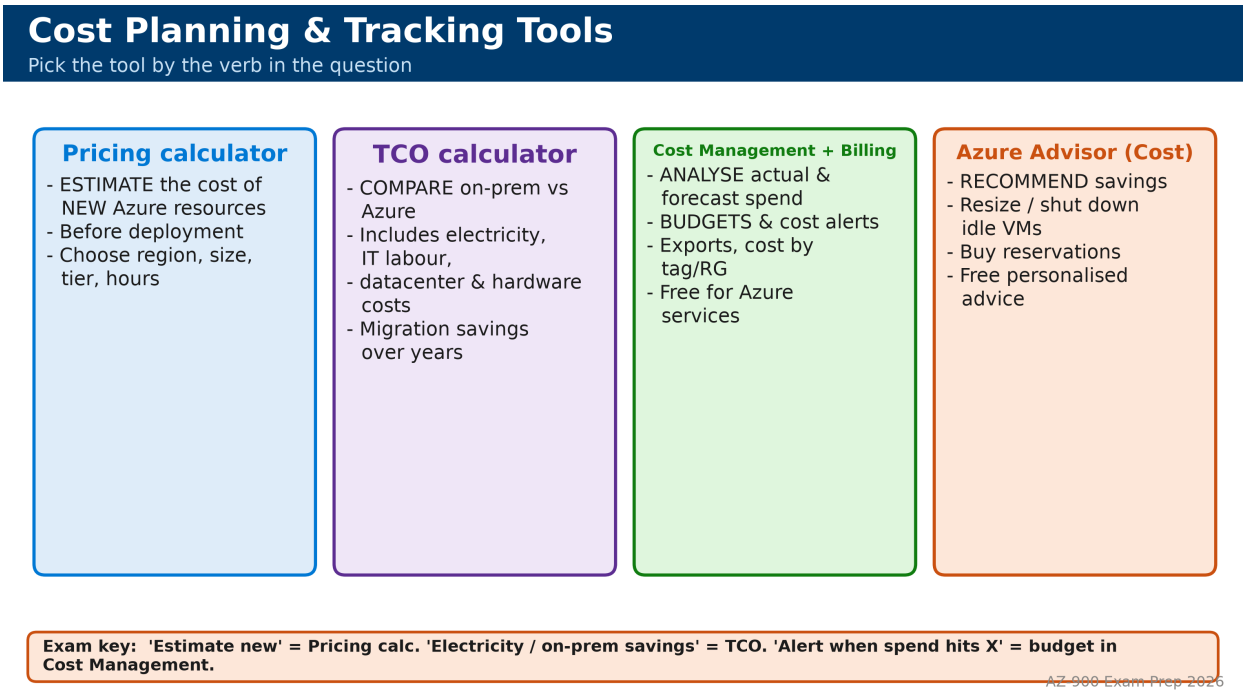


Figure Q140a: Cost planning and tracking tools

About this figure

• Each column is one tool; pick it by the question's verb: estimate (Pricing calculator), compare (TCO), analyse or alert (Cost Management), recommend (Advisor).

• Pricing and TCO calculators are free websites that need no Azure subscription; estimates use list prices and can be saved or exported to Excel.

• Cost Management is free for Azure resources; budgets can be set monthly, quarterly or yearly with alerts at thresholds such as 50%, 80% or 100%.

• Budgets alert but do not stop spending; only the free account and some credit offers have a spending limit that disables resources when exhausted.

• Advisor cost recommendations include resizing or shutting down VMs with low CPU and buying reservations or savings plans based on 30 days of usage.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q140b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 141

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q225, Q76, Q227

You can use Availability Zones in Azure to protect Azure virtual machines from a datacenter failure.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The failure here is a DATACENTER, and zones are separate datacenters, so Yes. Q142 is the look-alike: zones do NOT protect against a whole REGION failure (No), because all zones are inside one region.

Answer explained

✔ **Yes.**
Each availability zone has its own power, cooling and networking. VMs spread across zones survive a datacenter failure.

✖ **No.**
Datacenter failure is exactly what zones protect against. 'No' would be right only for a whole-region failure.

EXAM STRATEGY
'Datacenter failure' plus zones = Yes. Zones are physically separate datacenters with their own power and cooling.

IMPACT
Spreading VMs across zones gives the 99.99% VM SLA (uptime promise) and avoids an outage when one building loses power.

What · Where · When · Why · How

What	Availability zones are physically separate datacenters in one region, with their own power and cooling.
Where	In zone-enabled regions. You choose a zone for each VM when you create it.
When	When VMs must keep running if one datacenter fails.
Why	VMs in the other zones keep running. Two or more VMs in two or more zones get a 99.99% SLA.
How	Deploy VMs into zones 1, 2 and 3 behind a zone-redundant Standard Load Balancer.

Exam tip
Zones = datacenter failure. Region pairs = region failure. Availability sets = rack or host failure.

Memory trick
Set saves the rack, Zone saves the building, Region pair saves the city.

Real-world example, use case and scenario

EXAMPLE
VMs in zones
Three web VMs in zones 1, 2 and 3, behind a Standard Load Balancer, keep working when zone 2 loses power.

USE CASE
Ticket website
A concert ticket site deploys across zones to get the 99.99% VM SLA during big sales.

SCENARIO
Cooling failure
A cooling fault took one datacenter offline. A hospital app on VMs in several zones kept running within seconds. A nearby app on a single VM was down for hours.

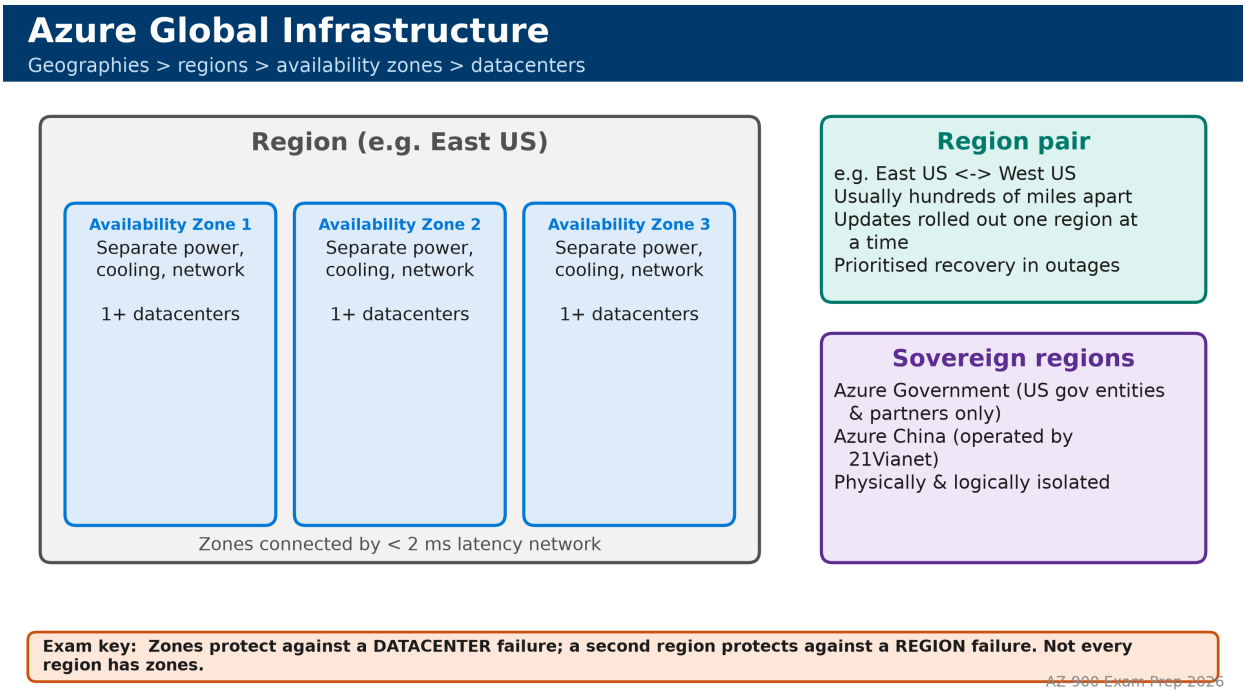


Figure Q141a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

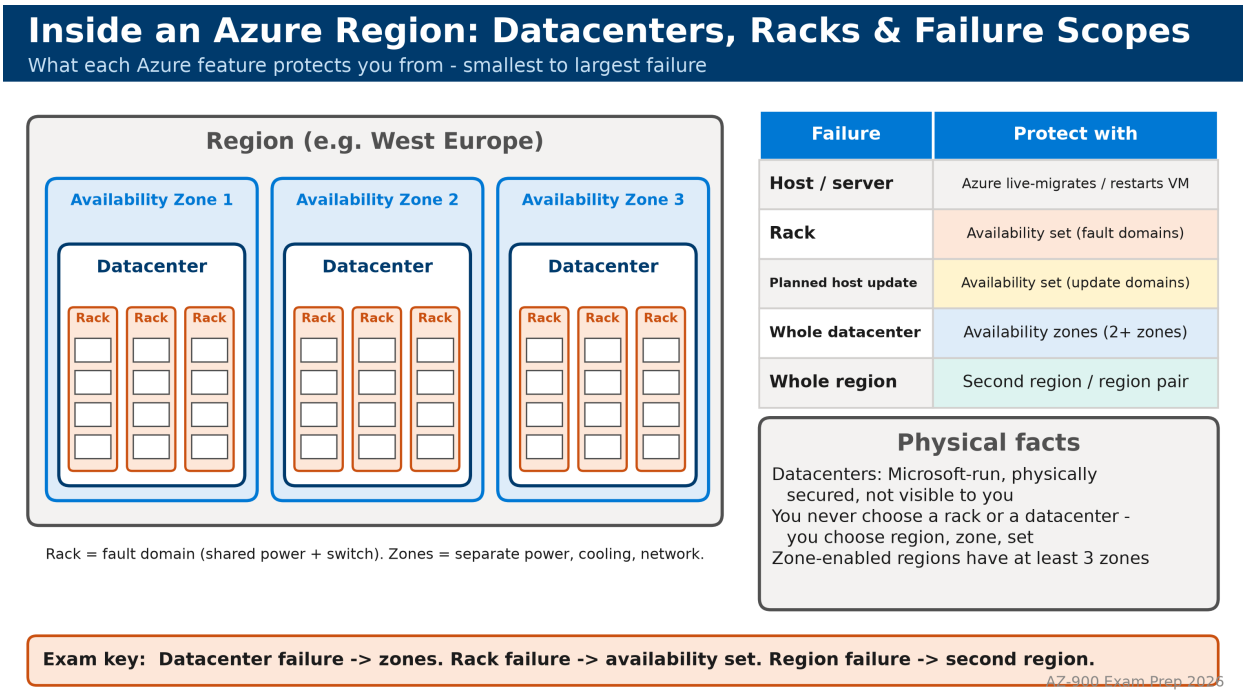


Figure Q141b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 142

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q226

You can use Availability Zones in Azure to protect Azure virtual machines from a region failure.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The failure here is a whole REGION. All zones are inside one region, so zones alone do not help (No). Q141 is the look-alike: zones DO protect against a single DATACENTER failure (Yes).

Answer explained

✔ **No.**
All availability zones are inside one region. A region-wide outage affects every zone, so region failure needs a second region.

✖ **Yes.**
Zones are inside one region. To survive a region failure, you must deploy or copy to another region.

EXAM STRATEGY
'Region failure' means you need a second region. All zones are inside one region, so zones alone = No.

IMPACT
Relying on zones for disaster recovery leaves apps down during a regional outage. Real disaster recovery needs Site Recovery or geo-redundancy.

What · Where · When · Why · How

What	Availability zones protect against a datacenter failure inside a single region.
Where	All zones of a region belong to that one region.
When	When checking protection against a complete regional outage.
Why	A region-wide outage affects every zone, so zones cannot protect against region failure.
How	Add a second region with Azure Site Recovery, geo-replication or an active-active setup.

Exam tip
Region failure in the question = the answer uses another region (region pairs, Site Recovery, geo-redundancy).

Memory trick
Zones live under one regional roof. When the roof falls, they all get wet.

Real-world example, use case and scenario

EXAMPLE
Regional outage
A regional network problem makes all three zones in a region unreachable at the same time.

USE CASE
Trading platform recovery
A trading platform copies to a second region with Site Recovery to meet a regulator's rule about losing a region.

SCENARIO
False sense of safety
An insurer used zones and said disaster recovery was done. An audit asked what happens if the region is lost. They added a second region and geo-replicated SQL to pass.

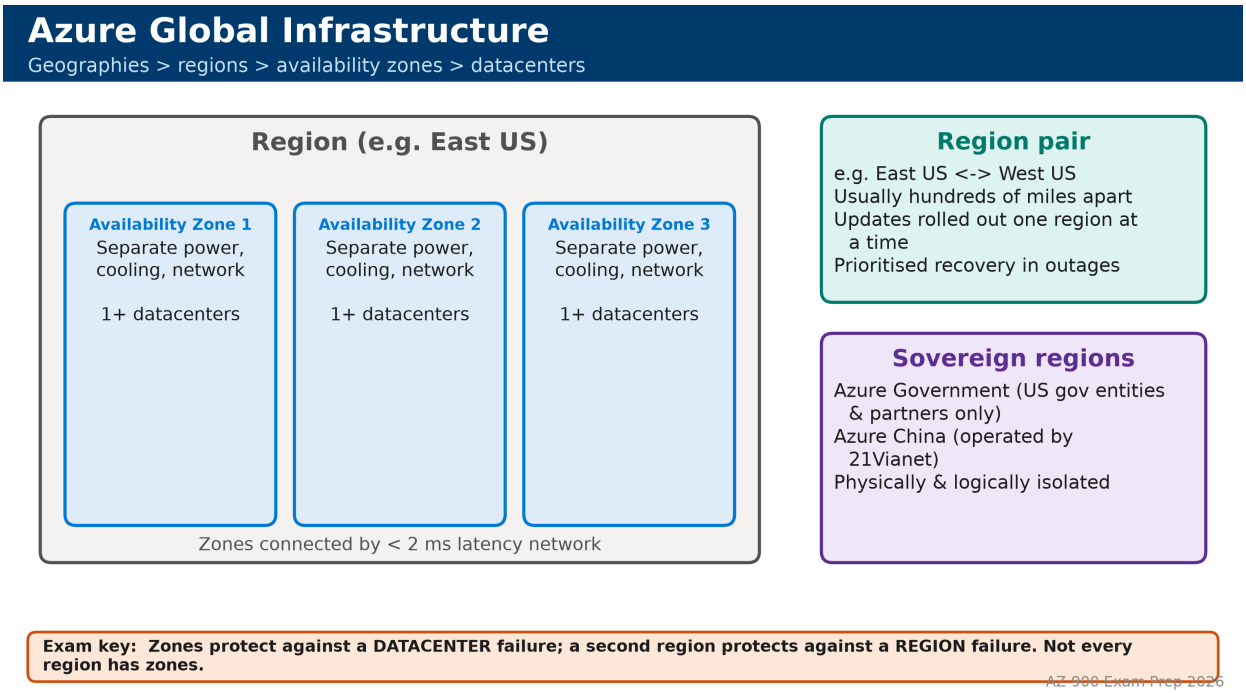


Figure Q142a: Regions, availability zones and region pairs

About this figure

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- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

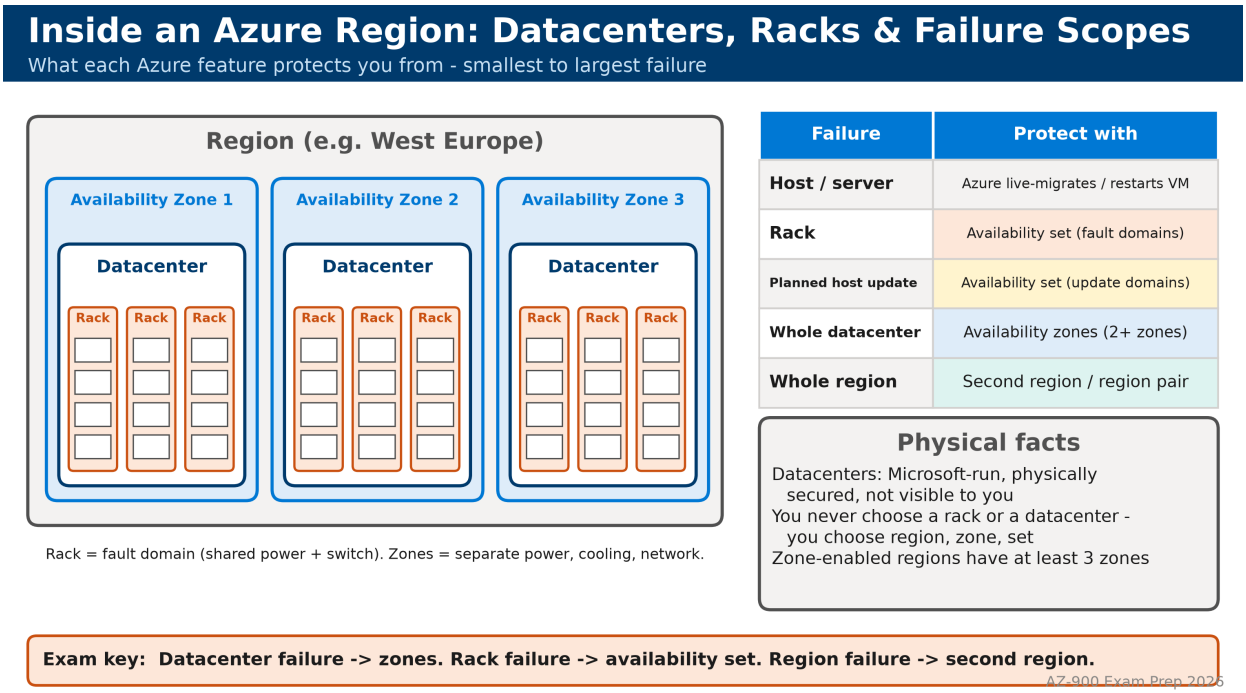


Figure Q142b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 143

AZURE ARCHITECTURE & SERVICES — **Subscriptions** v5 Q228

An Azure subscription can be managed by using a Microsoft account only.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is ONLY. Microsoft accounts CAN manage subscriptions (Q22: one Microsoft account can manage several subscriptions - Yes), but Entra work or school accounts and guests can too. So 'Microsoft account only' is No.

Answer corrected in this edition. The previous PDF printed: *Yes.*. The printed answer was Yes. Subscriptions can be managed with Microsoft Entra ID work or school accounts as well as Microsoft accounts, so 'Microsoft account only' is false.

Answer explained

✔ **No.**
Subscriptions trust a Microsoft Entra ID tenant. So Entra work or school accounts and guests can manage them, not just Microsoft accounts (personal accounts like outlook.com).

✖ **Yes.**
Entra ID work or school accounts are the normal way to manage subscriptions, so 'Microsoft account only' is false.

EXAM STRATEGY
'Only' decides it. Subscriptions trust an Entra tenant, so work, school and guest accounts can also manage them. The answer is No.

IMPACT
Companies should manage subscriptions with Entra work accounts so they can enforce MFA, Conditional Access and removal when staff leave. Personal accounts escape those controls.

What · Where · When · Why · How

What	Azure subscriptions are managed by identities from the Microsoft Entra ID tenant they trust.
Where	Access is given with Azure RBAC on the subscription's Access control (IAM) page.
When	When deciding which accounts can manage a subscription.
Why	Entra work or school accounts and guests can manage subscriptions, not only Microsoft accounts.
How	Give RBAC roles such as Owner or Contributor to Entra users, groups or guests.

Exam tip
The word 'only' is the trap. Subscriptions trust an Entra tenant, so work, school and guest accounts work too.

Memory trick
The subscription trusts the tenant, and the tenant accepts many account types.

Real-world example, use case and scenario

EXAMPLE
Work account login
An admin signs in to the portal with admin@contoso.com, an Entra ID work account, and manages the production subscription.

USE CASE
Guest consultant
A partner company's employee is invited as an Entra B2B guest and given Contributor on a project subscription.

SCENARIO
Moving off personal accounts
A growing agency signed up with an outlook.com account. They created an Entra tenant, linked the subscription to it, and gave staff work accounts with RBAC roles. They gained MFA and control when staff leave.

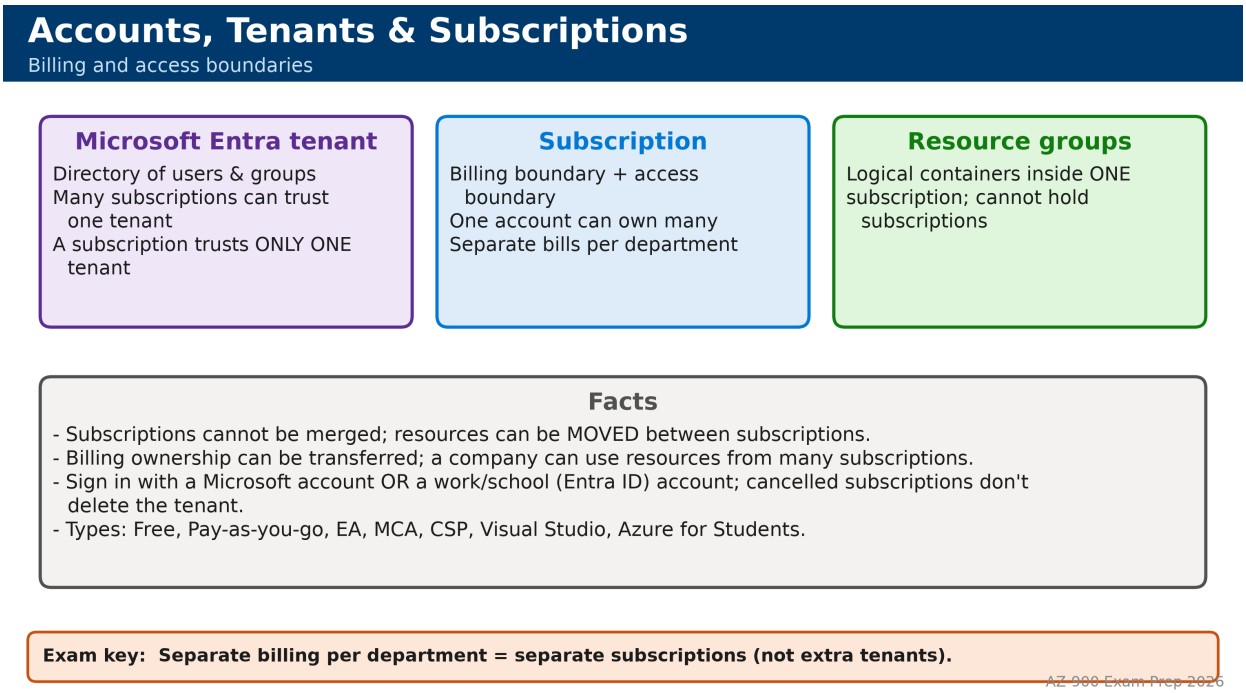


Figure Q143a: Accounts, tenants and subscriptions

About this figure

- Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.
- Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.
- Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.
- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

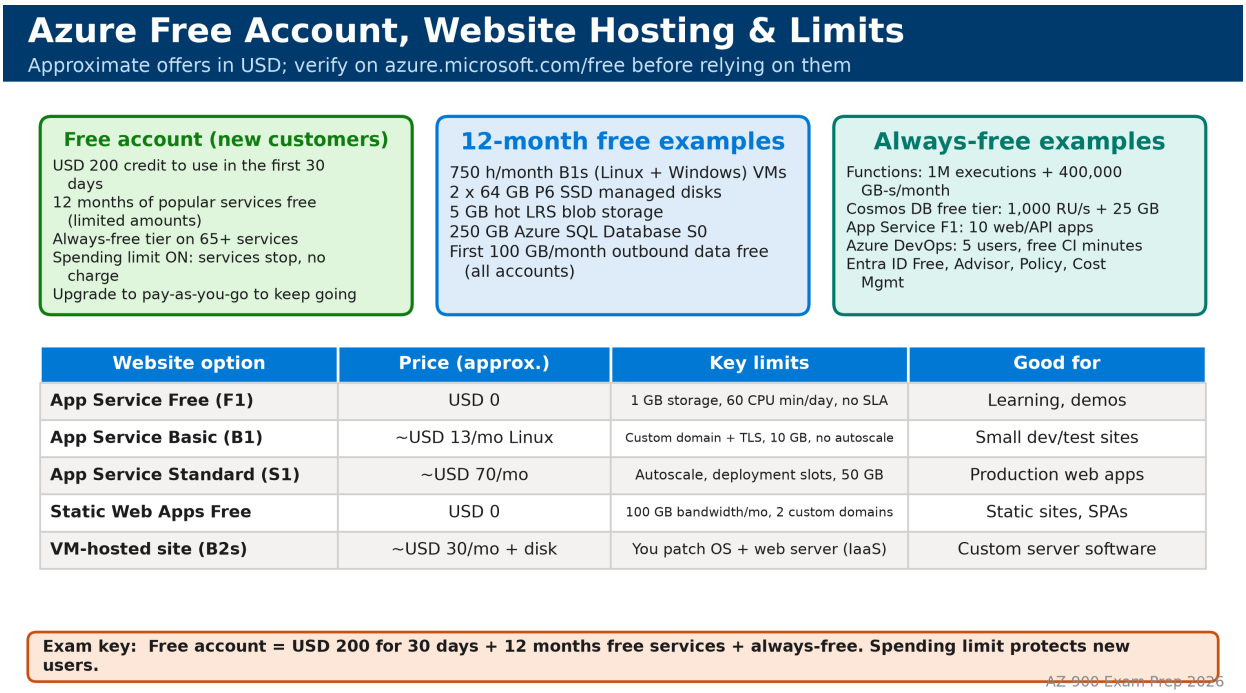


Figure Q143b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 144

AZURE ARCHITECTURE & SERVICES

Governance
v5 Q229

An Azure resource group can contain multiple Azure subscriptions.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Check the direction of the hierarchy. A resource group sits INSIDE one subscription, so it cannot hold subscriptions. To group subscriptions, use a management group (Q80). Q309: resource groups also cannot be nested (No).

Answer explained

✔ **No.**
The order is management group > subscription > resource group > resource. A resource group sits inside exactly one subscription.

✖ **Yes.**
It turns the hierarchy upside down. Several subscriptions are grouped with management groups, not resource groups.

EXAM STRATEGY
Check the hierarchy direction: resource groups live inside one subscription. Any container that holds its own parent is No.

IMPACT
Misreading the hierarchy breaks RBAC, policy and billing design. AZ-900 tests these levels in many governance questions.

What · Where · When · Why · How

What	A resource group is a logical container for Azure resources inside one subscription.
Where	Order: management group > subscription > resource group > resource.
When	When organising resources by app, environment or lifecycle.
Why	Resource groups sit inside subscriptions, so they cannot contain subscriptions.
How	Group several subscriptions with a management group, and group resources with resource groups.

Exam tip
Need to group subscriptions? Management group. Need to group resources? Resource group. Never the other way round.

Memory trick
MG > Sub > RG > Res: My Small Red Rabbit.

Real-world example, use case and scenario

EXAMPLE
Hierarchy view
In the portal, RG-Web appears under the Production subscription, which is under the Corp management group.

USE CASE
Policy at scale
A drug company groups its 15 subscriptions in a management group to apply one allowed-regions policy.

SCENARIO
Wrong container
A new admin tried to put the Dev and Test subscriptions into one resource group to share a budget. He learned to create a management group instead and set budgets for each subscription.

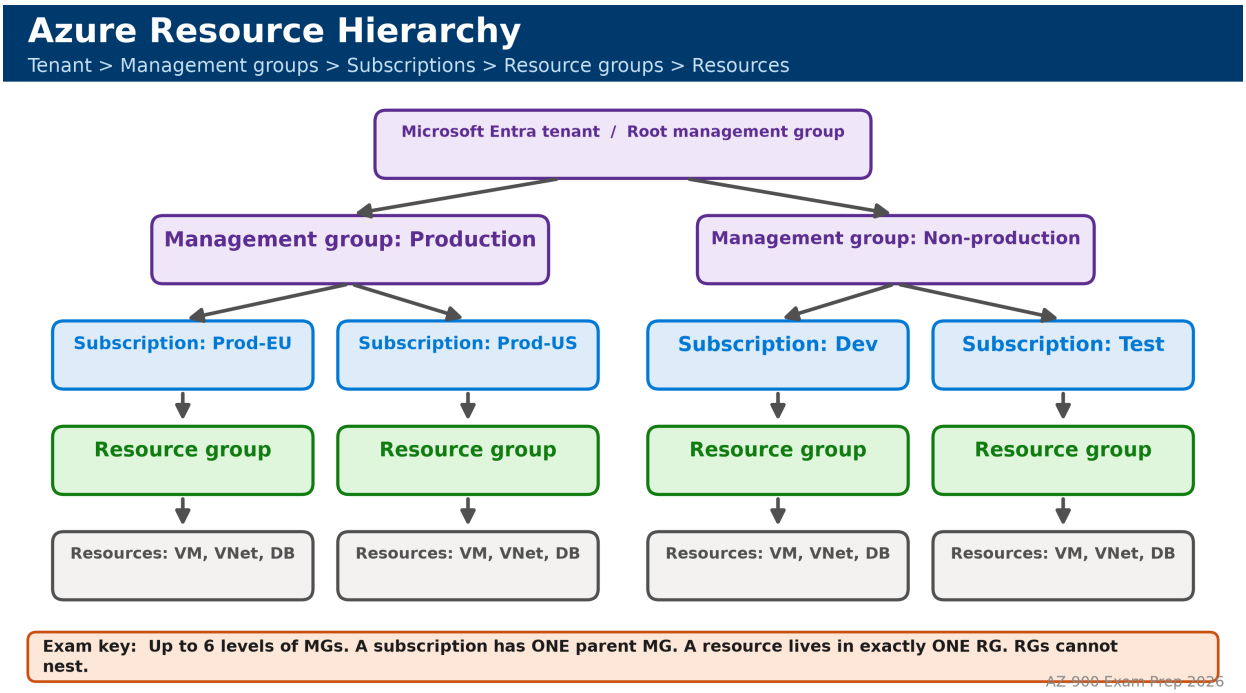


Figure Q144: Azure resource hierarchy

About this figure

- Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.
- A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.
- Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.
- Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.
- Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 145

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q230

To use Microsoft Entra ID credentials to sign in to a computer that runs Windows 10, the computer must be joined to Microsoft Entra ID.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The word to watch is JOINED vs REGISTERED. To sign in to Windows itself with Entra credentials, the PC must be Microsoft Entra joined (or hybrid joined). A registered (personal) device cannot. Q183 is different: Android devices cannot be JOINED to Entra ID (No).

2026 update: Azure AD join is now called Microsoft Entra join. Microsoft Entra hybrid join (domain-joined devices synced to Entra ID) also lets users sign in with Entra credentials.

Answer explained

✔ **Yes.**
To sign in to Windows 10 with Entra ID credentials, the device must be Microsoft Entra joined or hybrid joined (joined to both on-premises AD and Entra ID).

✖ **No.**
A PC in a workgroup, or one that is only Entra registered, cannot use Entra ID credentials at the Windows sign-in screen.

EXAM STRATEGY
'Sign in to a computer with Entra credentials' means joined (or hybrid joined). Registered devices do not count. The answer is Yes.

IMPACT
Entra join removes the need for on-premises domain controllers for cloud-first devices. It also allows Conditional Access when users sign in to the device.

What · Where · When · Why · How

What	Microsoft Entra join links a Windows device to Entra ID so users sign in with work accounts.
Where	Set up during Windows setup, through Autopilot, or in Settings > Accounts > Access work or school.
When	When users must sign in to Windows 10 or 11 with Entra ID credentials.
Why	Only Entra joined or hybrid joined devices accept Entra credentials at the Windows sign-in screen.
How	Join the device to Microsoft Entra ID, or hybrid join a domain-joined device through Entra Connect.

Exam tip
Joined = sign in to the device with Entra.
Registered = a personal device that only reaches company resources.

Memory trick
Join to log in; register to reach in.

Real-world example, use case and scenario

EXAMPLE
Join at first setup
During Windows setup a user picks 'Set up for an organization'. The laptop becomes Entra joined, and the user signs in with name@company.com.

USE CASE
Cloud-only staff
A remote-first marketing agency Entra-joins all laptops through Autopilot. Staff sign in with company credentials and MFA.

SCENARIO
Login fails at home
A new worker's PC was only Entra registered, so his work password failed at the lock screen. IT reset it through Autopilot as Entra joined, and sign-in worked.

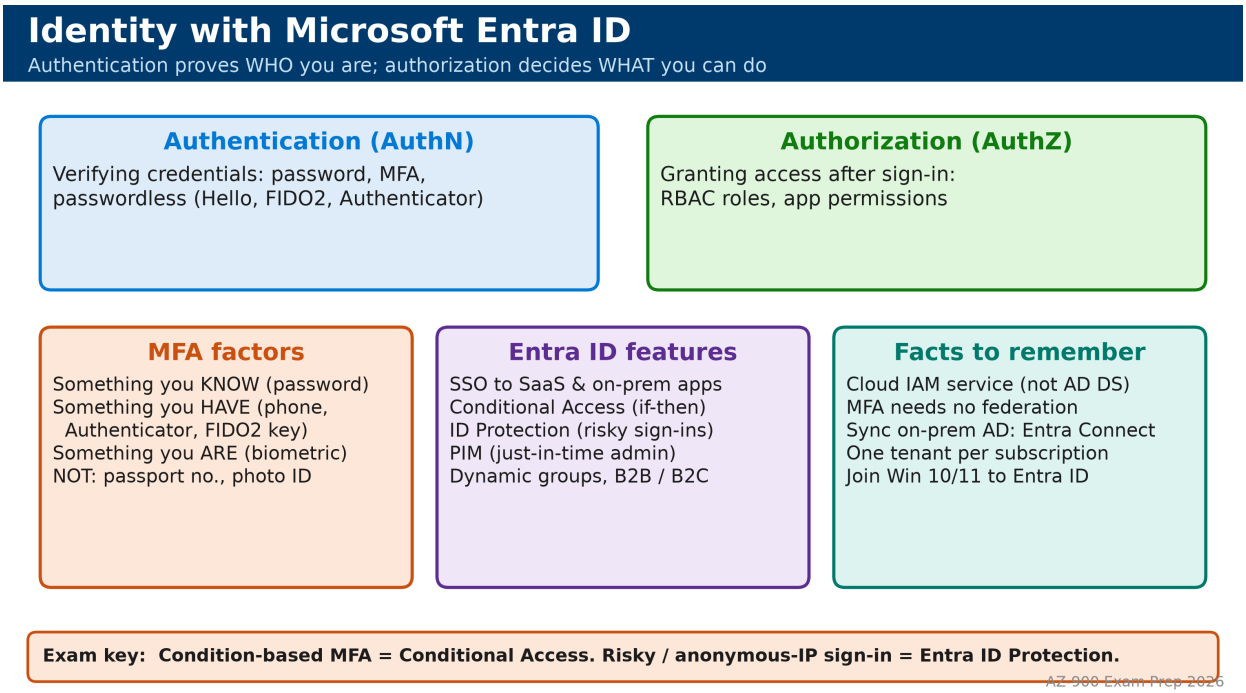


Figure Q145: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 146

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q231

Users in Microsoft Entra ID are organized by using resource groups.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Do not mix up GROUPS and RESOURCE GROUPS. Users are organised in Entra groups (and administrative units). Resource groups hold only Azure resources, such as VMs and storage.

Answer explained

✔ **No.**
Entra ID users are organised with groups and administrative units (sets of users with their own admins). Resource groups only hold Azure resources.

✖ **Yes.**
Users are identity objects in the Entra tenant, not Azure resources. They cannot be put in resource groups.

EXAM STRATEGY
Resource groups hold only Azure resources. If you see users and resource groups together, the answer is No.

IMPACT
Mixing up identity groups and resource groups leads to wrong permissions. Users belong in Entra groups, and those groups then get RBAC roles.

What · Where · When · Why · How

What	Entra ID organises users with security groups, Microsoft 365 groups and administrative units.
Where	In the Microsoft Entra admin center, separate from Azure resource groups.
When	When grouping users for access, licences or shared admin tasks.
Why	Resource groups hold only Azure resources, so users are not organised with them.
How	Create Entra groups or administrative units, and add users by hand or by dynamic rules.

Exam tip
Identity objects (users, groups) live in the Entra tenant. Resource groups only hold Azure resources.

Memory trick
People go in groups; machines go in resource groups.

Real-world example, use case and scenario

EXAMPLE
Security group
Admins add 40 finance users to the Finance-Team security group in Entra ID.

USE CASE
Local admin rights
A global company uses administrative units so the Germany help desk can reset passwords only for German users.

SCENARIO
Users in the wrong place
A trainee looked for an 'Add user' button inside RG-HR. A colleague explained that users are managed in Entra ID. Then they gave the HR group a role on RG-HR.

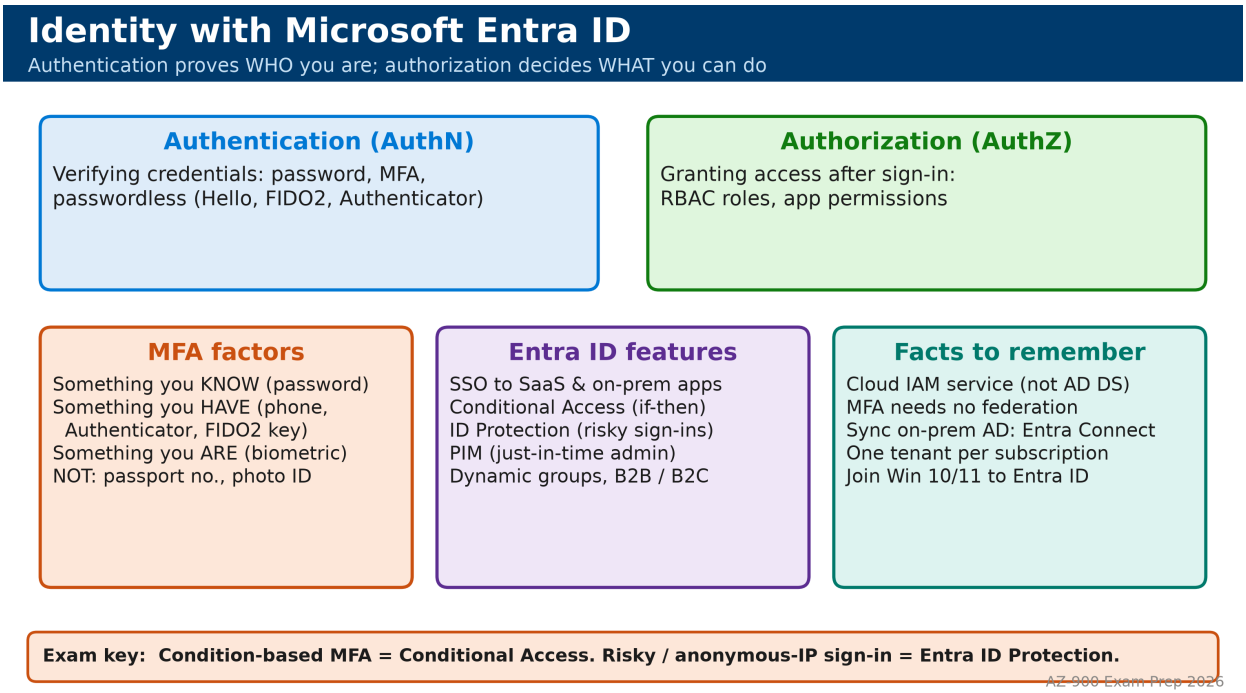


Figure Q146: Identity with Microsoft Entra ID

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- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 147

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q232

Microsoft Entra ID groups support dynamic membership rules.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Entra ID security groups and Microsoft 365 groups support dynamic membership rules based on user or device attributes (details like department). This needs Entra ID P1.

✖ **No.**
Dynamic user and device groups are a documented Entra ID feature, available with a P1 licence or higher.

EXAM STRATEGY
'Dynamic membership' is a real Entra feature based on attributes. The answer is Yes. Remember it needs Entra ID P1.

IMPACT
Dynamic groups update access automatically when people join, move or leave. This cuts mistakes that leave former staff with old permissions.

What · Where · When · Why · How

What	Dynamic membership groups add and remove members automatically based on attribute rules.
Where	Entra ID security or Microsoft 365 groups with membership type Dynamic User or Dynamic Device.
When	When membership should follow attributes like department or job title, without manual updates.
Why	Entra ID supports dynamic rules, so the statement is true. It needs Entra ID P1.
How	Write a rule such as user.department -eq "Sales". Entra updates the members when attributes change.

Exam tip
Dynamic membership = rules based on attributes. Needs Entra ID P1.

Memory trick
Dynamic groups update themselves when HR updates the attribute.

Real-world example, use case and scenario

EXAMPLE
Department rule
The rule user.department -eq Sales automatically puts every sales employee into the Sales-All group.

USE CASE
Licence assignment
A retailer gives Microsoft 365 licences to a dynamic group of store staff, so new workers get licences on day one.

SCENARIO
Moving between teams
An engineer moved from Support to Development, and HR changed her department. Within minutes she left the Support group and joined the Dev group. She got the right app access without a ticket.

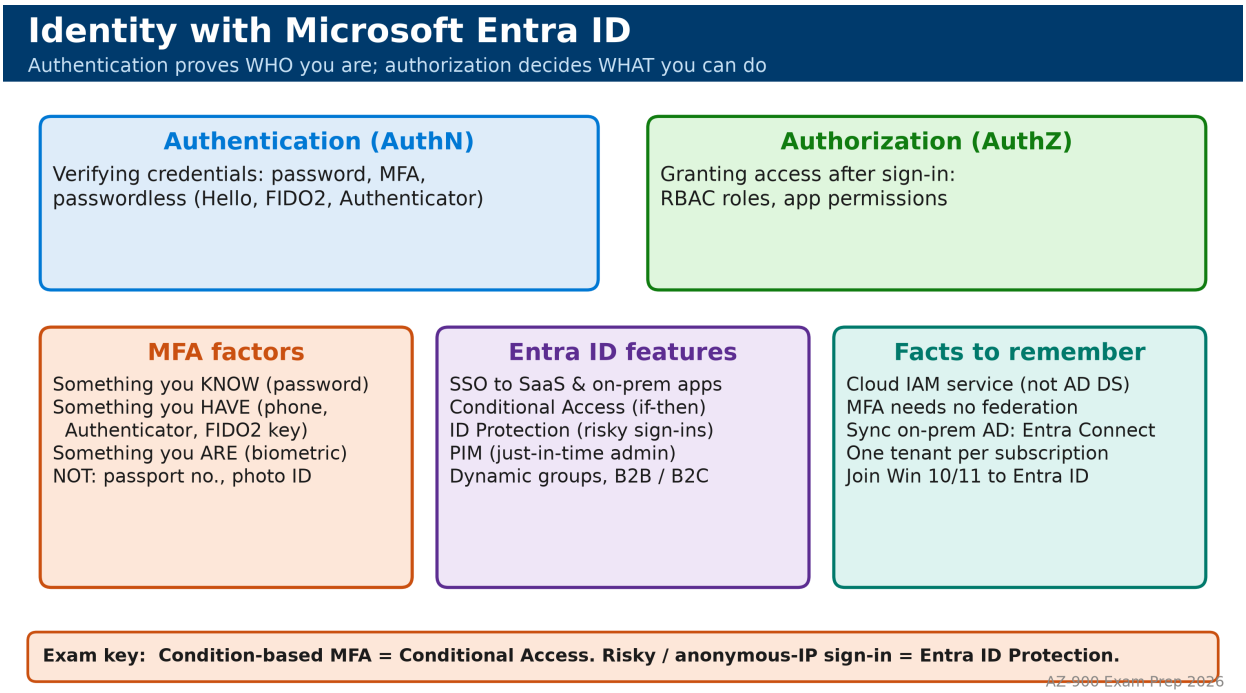


Figure Q147: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 148

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q233, Q398, Q399

You plan to deploy several Azure virtual machines. You need to ensure that the services running on the virtual machines remain available if a single data center fails. What are two possible solutions?

- ✔ **A.** Deploy the virtual machines to two or more availability zones.
- ✖ B. Deploy the virtual machines to two or more resource groups.
- ✖ C. Deploy the virtual machines to a scale set.
- ✔ **D.** Deploy the virtual machines to two or more regions.

Correct answer: Deploy the virtual machines to two or more availability zones.; Deploy the virtual machines to two or more regions.

Clarification (read this first): Pick TWO answers. The failure is ONE DATACENTER, so you need VMs in physically separate buildings: zones or regions. Q262 is a look-alike: putting VMs in two or more resource groups does NOT help (No), because resource groups are not physical places.

Answer explained

✔ **Deploy the virtual machines to two or more availability zones.; Deploy the virtual machines to two or more regions.**
Availability zones and multiple regions both put VMs in physically separate datacenters. If one datacenter is lost, the others keep running.

✖ **Deploy the virtual machines to two or more resource groups.**
Resource groups are logical containers. They say nothing about physical location, and both groups can be in one datacenter.

✖ **Deploy the virtual machines to a scale set.**
A scale set adds or removes identical VMs automatically. It spreads them across datacenters only if you set it up with zones.

EXAM STRATEGY
'Datacenter fails' needs physical separation, so pick zones and regions. Remove resource groups (logical only) and scale sets (no guaranteed zone spread).

IMPACT
Using logical containers for resilience is a classic mistake. Production goes down when a datacenter loses power.

What · Where · When · Why · How

What	Availability zones and multiple regions put VM copies in physically separate datacenters.
Where	Zones are inside one zone-enabled region. Regions are spread across Azure geographies.
When	When services on VMs must stay available if one datacenter fails.
Why	Physical separation survives the loss of a datacenter. Resource groups and plain scale sets do not guarantee it.
How	Deploy VMs across two or more zones, or two or more regions, behind a load balancer or Front Door.

Exam tip
Answers for datacenter failure are physical: zones or regions. Logical containers like resource groups never count.

Memory trick
Physical problems need physical separation.

Real-world example, use case and scenario

EXAMPLE
Web tier in zones
Two API VMs are deployed to zone 1 and zone 2 in Australia East, behind a zone-redundant load balancer.

USE CASE
Payment service
A payment company runs active-active in two regions, so one datacenter, or even one region, can fail.

SCENARIO
Resource groups are not resilience
A startup split its VMs into RG-A and RG-B, thinking this added protection. A datacenter power problem took both down. Afterwards they deployed across zones.

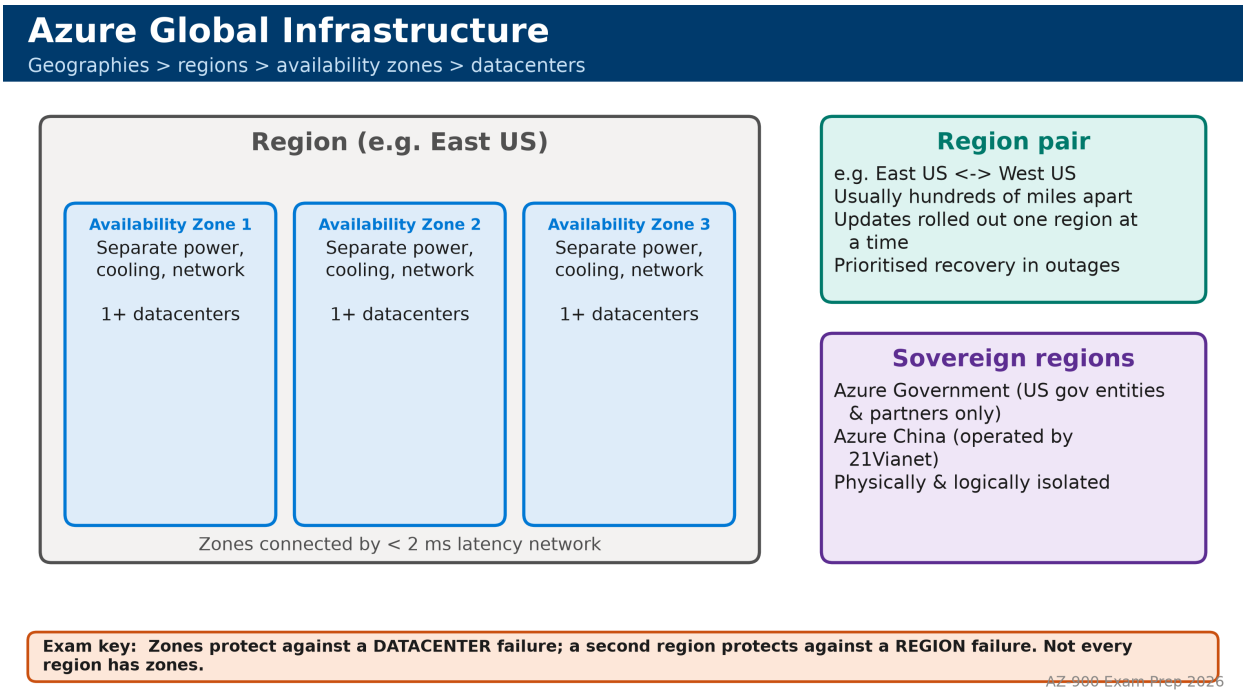


Figure Q148a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

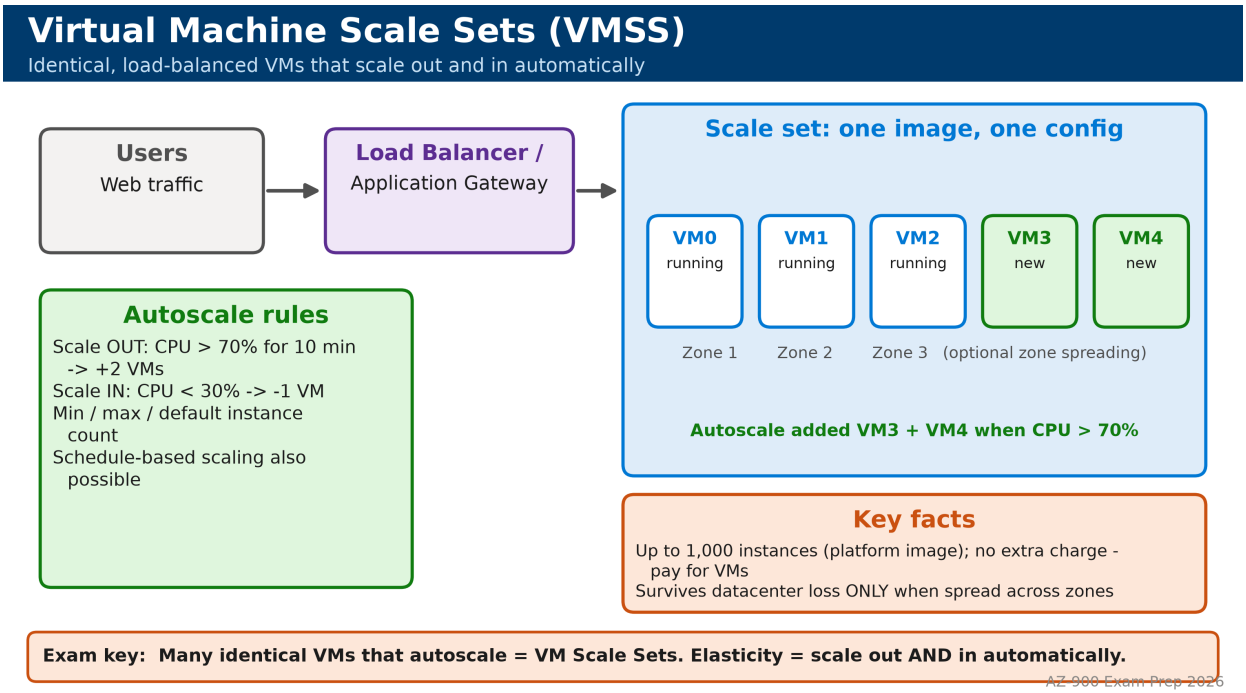


Figure Q148b: Virtual Machine Scale Sets

About this figure

- A scale set runs identical VMs from one image and configuration behind a load balancer or Application Gateway.
- Autoscale adds instances when a metric (e.g. CPU > 70%) is high and removes them when low, within min/max instance counts - this is elasticity.
- Scale sets support up to 1,000 VM instances (platform images); the scale set itself is free - you pay only for the VMs, disks and networking.
- Orchestration modes: Uniform (identical VMs) and Flexible (mixed sizes, recommended for new deployments).
- A scale set survives a datacenter failure only when its instances are spread across availability zones; a non-zonal set does not.
- Scaling out adds more VMs (horizontal); scaling up means a bigger VM size (vertical) and usually needs a restart.

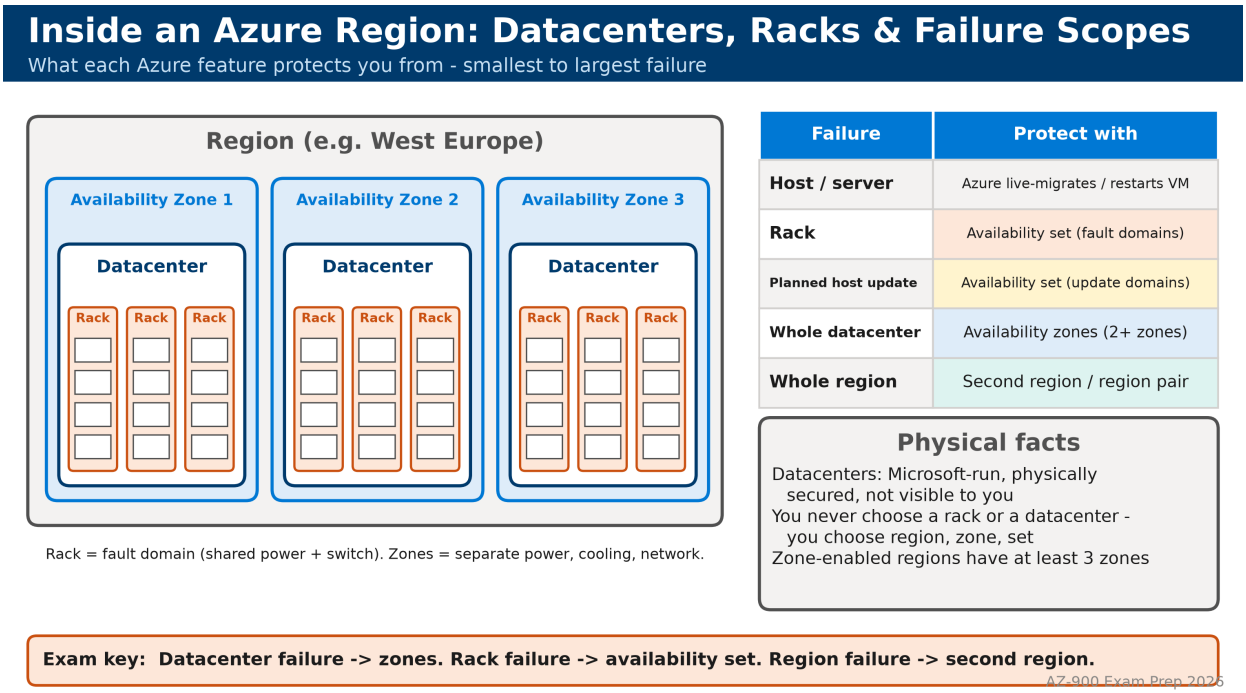


Figure Q148c: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 149

AZURE MANAGEMENT & GOVERNMENT | **Monitoring** v5 Q234

Azure Monitor can monitor the performance of on-premises computers.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): Azure Monitor is not only for Azure. With an agent installed, it can watch on-premises computers too, so Yes. Similar Yes answers: Q166 (Defender for Cloud covers on-premises) and Q414 (Application Insights for on-premises apps).

Answer explained

✔ **Yes.**
Azure Monitor collects performance data and logs from on-premises servers through the Azure Monitor Agent, usually with Azure Arc (a service that lets Azure manage non-Azure servers).

✖ **No.**
Azure Monitor works for hybrid setups. Agents on non-Azure machines send data to Log Analytics for analysis and alerts.

EXAM STRATEGY
'On-premises' plus Azure Monitor = Yes, through the Azure Monitor Agent and Azure Arc.

IMPACT
One monitoring tool for cloud and on-premises avoids paying for two tools and avoids blind spots during migration.

What · Where · When · Why · How

What	Azure Monitor is Azure's service for collecting and analysing metrics and logs from any place.
Where	On-premises machines send data to a Log Analytics workspace (a log store) through the Azure Monitor Agent.
When	When on-premises and Azure servers should be watched in one place.
Why	Agents on non-Azure machines let Azure Monitor track their performance, so the statement is true.
How	Connect the server with Azure Arc, install the Azure Monitor Agent, and add a data collection rule.

Exam tip
Azure Monitor + Azure Monitor Agent (through Azure Arc) = on-premises servers watched in the same place as Azure VMs.

Memory trick
Monitor watches anywhere an agent can live.

Real-world example, use case and scenario

EXAMPLE
CPU charts
An on-premises Windows file server sends CPU and disk data to Log Analytics and appears in VM insights.

USE CASE
Factory servers
A car-parts maker watches its factory Linux servers next to its Azure VMs in one Azure Monitor workbook.

SCENARIO
Disk full at night
A hospital's on-premises imaging server kept filling its disk at night. After connecting it with Arc and adding a log alert, the on-call team was paged at 80 percent and cleared space before any outage.

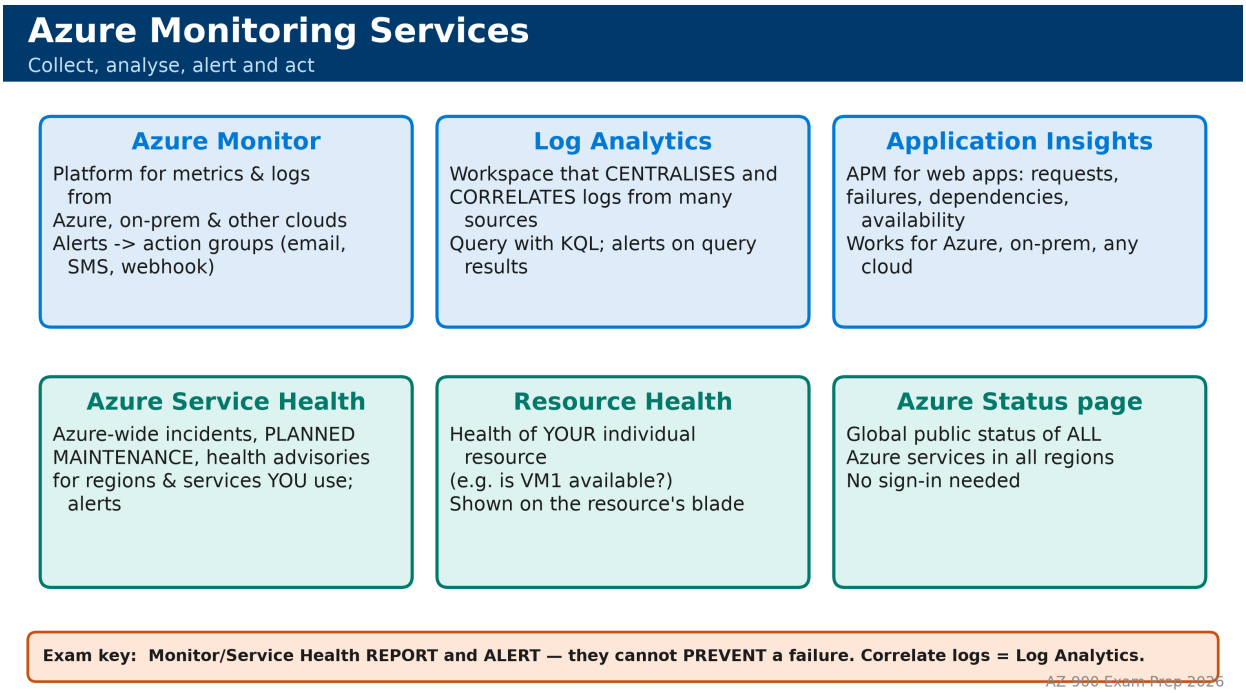


Figure Q149: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 150

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q235

Azure Monitor can send alerts to Microsoft Entra ID security groups.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Alert recipients are set in ACTION GROUPS (lists of who to notify and what to run). An Entra ID security group is not one of the choices, so No. Q419 lists what action groups CAN do: SMS/email and Logic App/webhook.

Answer corrected in this edition. The previous PDF printed: *Yes.*. The printed answer was Yes. Azure Monitor alerts notify through action groups (email, SMS, push, voice, webhook, or members of an Azure Resource Manager role). An Entra ID security group cannot be chosen as a recipient.

Answer explained

✔ **No.**
Azure Monitor alerts notify through action groups (email, SMS, push, voice, ARM role, webhook). An Entra ID security group cannot be chosen as a target.

✖ **Yes.**
Action groups cannot target Entra security groups. Use email addresses, ARM role notifications or webhooks instead.

EXAM STRATEGY
Ask where alert recipients are set: action groups. An Entra security group is not a target option, so the answer is No.

IMPACT
Wrong action groups mean alerts reach nobody during an outage. Testing who receives alerts is a basic task.

What · Where · When · Why · How

What	Action groups set who is notified and what runs when an Azure Monitor alert fires.
Where	Azure portal > Monitor > Alerts > Action groups.
When	When choosing who receives alert messages.
Why	Action groups offer email, SMS, push, voice, ARM role and webhooks, but no Entra security group target.
How	Add a team mailbox, an ARM role notification or a webhook to the action group used by the alert rule.

Exam tip
Alert recipients are set in action groups, not in Entra ID groups.

Memory trick
Alerts call Action groups, not Entra groups: A for Action.

Real-world example, use case and scenario

EXAMPLE
Action group setup
An action group named Ops-Pager sends SMS to two phone numbers and email to ops@company.com.

USE CASE
Owner notice
A fintech sets alerts to email all users who hold the Owner role on the subscription.

SCENARIO
Group not found
An admin looked for the SecOps Entra group in the action group and could not find it. She used the team's shared mailbox address instead, and alerts started arriving.

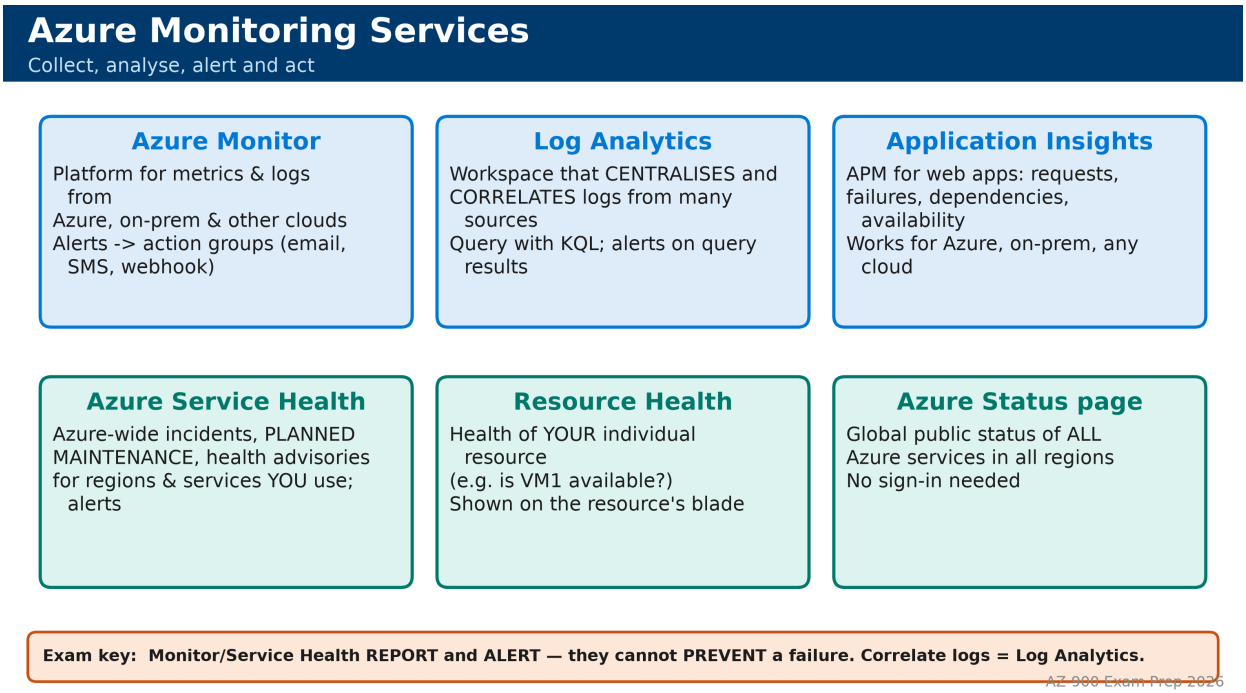


Figure Q150: Azure monitoring services

About this figure

- Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).
- Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.
- Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.
- Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.
- Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 151

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q236

Azure Monitor can trigger alerts based on data in an Azure Log Analytics workspace.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Log search alert rules run KQL queries (Kusto Query Language - the query language for logs) on a Log Analytics workspace on a schedule. They fire when results pass a limit you set.

✖ **No.**
Log alerts are a main Azure Monitor alert type, along with metric alerts and activity log alerts.

EXAM STRATEGY
'Log Analytics workspace' points to log search alerts that run KQL. The answer is Yes.

IMPACT
Log-based alerts catch patterns that metrics miss, such as repeated failed sign-ins or app errors, before customers report them.

What · Where · When · Why · How

What	Log search alerts are Azure Monitor alert rules based on KQL queries over log data.
Where	Created in Monitor > Alerts, with a Log Analytics workspace as the scope.
When	When you need alerts on events in collected logs, such as failed logons or error codes.
Why	Azure Monitor can alert on Log Analytics data, so the statement is true.
How	Write a KQL query, set how often it runs and a limit, and attach an action group.

Exam tip
Three alert sources to remember: metric, log (KQL on Log Analytics), activity log.

Memory trick
Query the logs, ring the bell.

Real-world example, use case and scenario

EXAMPLE
KQL rule
A rule runs every 5 minutes, counts Event ID 4625 failed logons, and fires above 20.

USE CASE
Online shop errors
An online shop gets an alert when HTTP 500 errors in app logs go above 50 in 10 minutes.

SCENARIO
Silent batch failures
A payroll batch job wrote failures to logs that nobody read. The team created a log alert on the error text. The next failure was caught before employees were paid late.

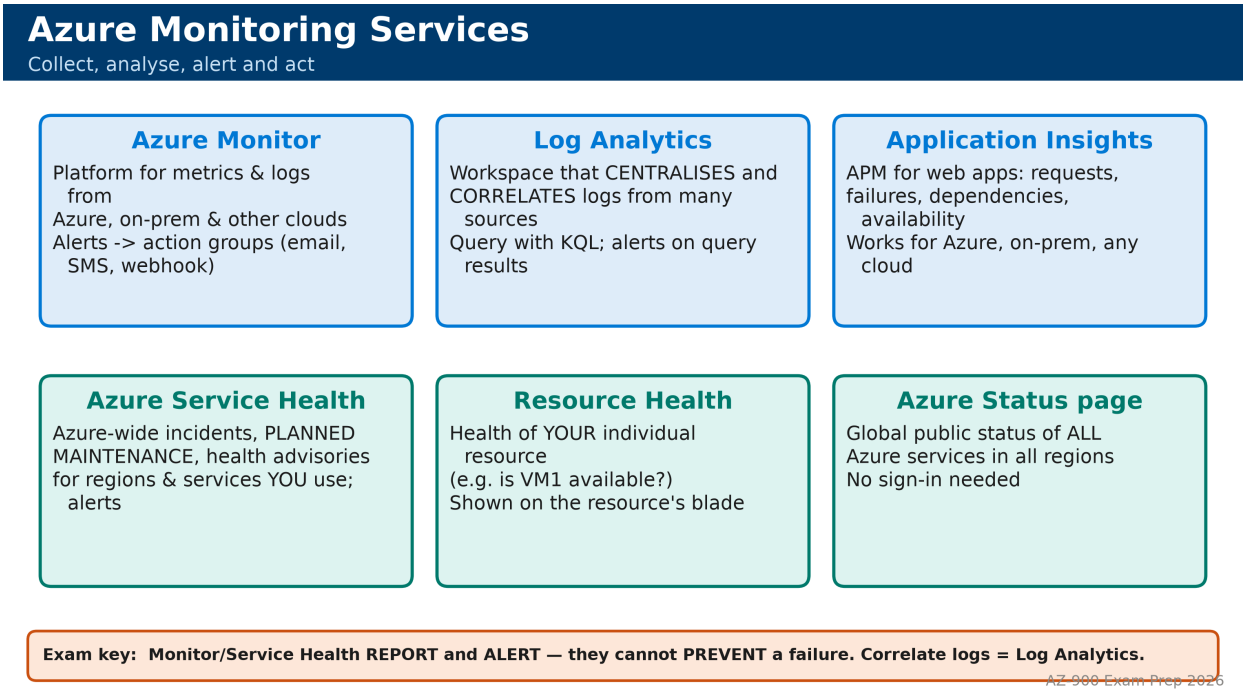


Figure Q151: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

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• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 152

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q238

From Azure Service Health, an administrator can create a rule to be alerted if an Azure service fails.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The word is ALERT. Service Health CAN alert you when an Azure service fails (Yes). Q153 is the look-alike: it CANNOT PREVENT a failure (No).

Answer explained

✔ **Yes.**
Service Health alerts, set by subscription, service, region and event type, notify you through action groups when an Azure service problem affects you.

✖ **No.**
Creating Service Health alert rules is a built-in feature, sent through action groups.

EXAM STRATEGY
Service Health supports alerts through action groups. 'Create a rule to be alerted' = Yes.

IMPACT
Service Health alerts give early warning of Azure problems. You can fail over and inform people before support tickets arrive.

What · Where · When · Why · How

What	Service Health alerts tell you about Azure service problems, maintenance and advisories.
Where	Azure portal > Service Health > Health alerts, using action groups.
When	When the team must know quickly about Azure problems in its regions and services.
Why	Creating alert rules is built into Service Health, so the statement is true.
How	Create a Service Health alert for chosen subscriptions, services, regions and event types, with an action group.

Exam tip
Service Health = view AND alert. It cannot fix or prevent the problem.

Memory trick
Service Health shouts; it does not repair.

Real-world example, use case and scenario

EXAMPLE
Alert rule
An alert fires for problems with Cosmos DB in East US and emails the database team.

USE CASE
Broadcaster
A TV broadcaster alerts its on-call engineers when media-related services report problems in its regions.

SCENARIO
Faster customer updates
A CRM software company sent Service Health alerts to Teams. During an outage, its status page was updated in minutes, and support tickets fell by half.

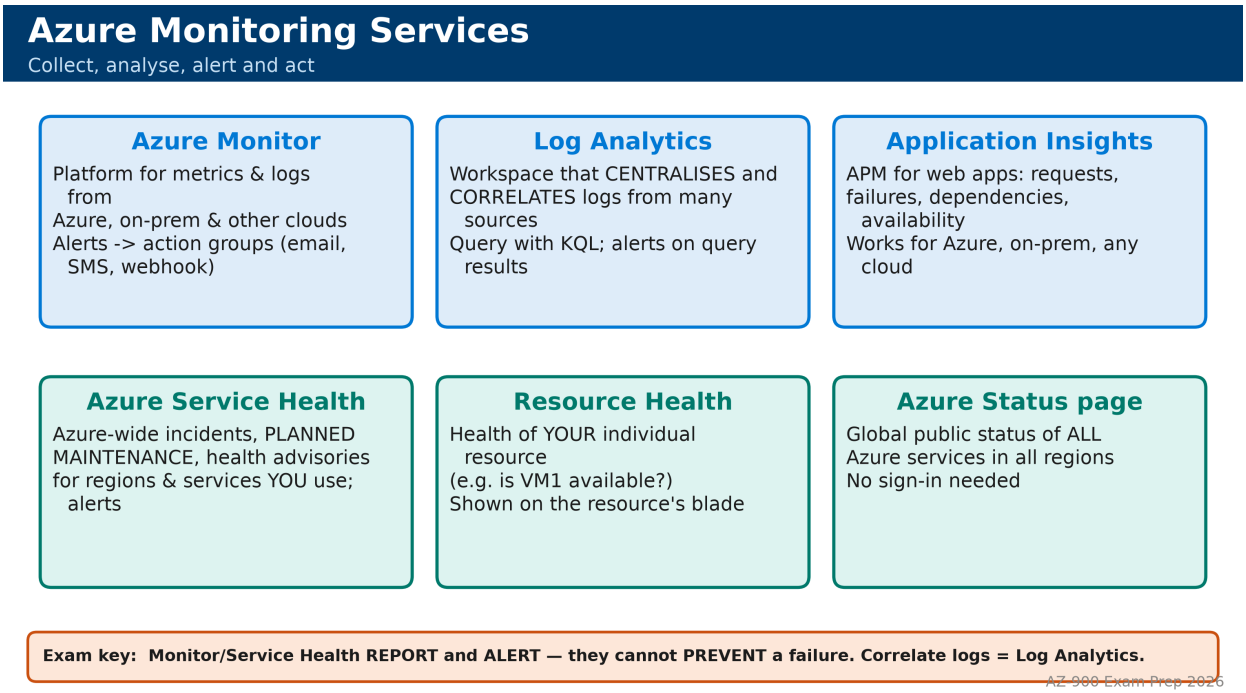


Figure Q152: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

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• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 153

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q239

From Azure Service Health, an administrator can prevent a service failure.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is PREVENT. Service Health can show and ALERT (Q152, Yes), but it cannot stop a Microsoft outage. Prevention comes from your design, such as zones and regions.

Answer explained

✔ **No.**
Service Health only informs and alerts. Microsoft runs the platform, so you reduce the impact with a strong design, such as zones and regions.

✖ **Yes.**
Service Health has no control over Azure's infrastructure. It reports and notifies, but it cannot stop a platform failure.

EXAM STRATEGY
'Prevent' decides it. Service Health shows and alerts, but it cannot stop a Microsoft outage. The answer is No.

IMPACT
Protection comes from a resilient design, not from monitoring. Expecting a dashboard to stop outages leaves live systems unprotected.

What · Where · When · Why · How

What	Azure Service Health is an information service that reports Azure platform problems.
Where	Azure portal > Monitor > Service Health.
When	When reviewing problems and maintenance, not controlling the platform.
Why	Microsoft runs the infrastructure, so customers cannot prevent service failures from Service Health.
How	Reduce the impact with a resilient design, such as availability zones and multi-region deployment.

Exam tip
View, alert, get guidance: yes. Prevent or fix a Microsoft outage: no.

Memory trick
A weather forecast cannot stop the rain.

Real-world example, use case and scenario

EXAMPLE
Problem timeline
Service Health shows a problem's timeline and root cause report after it happens.

USE CASE
Planning for resilience
A travel agency reviews past Service Health problems to decide which apps need a multi-region design.

SCENARIO
Manager's request
A manager asked IT to use Service Health to stop outages. The architect explained it only warns, and suggested a zone-redundant deployment to really reduce the impact.

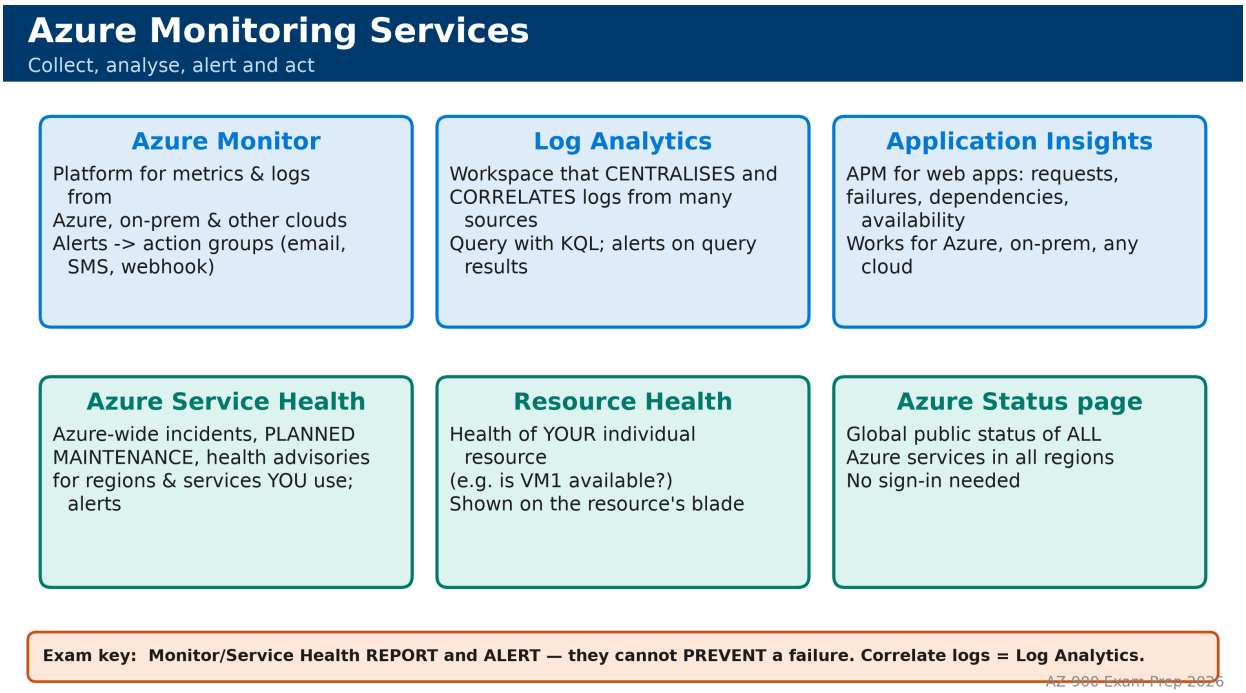


Figure Q153: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 154

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q241

You need to identify which blades in the Azure portal must be used to perform the following task: Browse available virtual machine images.

✖ A. Monitor.

✖ B. Subscriptions.

✔ **C.** Marketplace.

✖ D. Advisor.

Correct answer: Marketplace.

Answer explained

✔ **Marketplace.**
Azure Marketplace lists Microsoft and partner VM images, such as Windows Server and Ubuntu, and deploys them directly.

✖ **Monitor.**
Monitor tracks metrics, logs, alerts and service health, not VM images.

✖ **Subscriptions.**
Subscriptions manages billing, access and quotas, not a list of images.

✖ **Advisor.**
Advisor shows recommendations, not images you can deploy.

EXAM STRATEGY
'Browse images' points to Marketplace. Remove Monitor, Advisor and Subscriptions, which never list images to deploy.

IMPACT
Marketplace images, including paid partner products, can add extra software charges. Check the offer to avoid surprise bills.

What · Where · When · Why · How

What	Azure Marketplace is a catalogue of Microsoft and partner images and solutions.
Where	Azure portal > Marketplace (also through Create a resource).
When	When browsing VM images such as Windows Server or Ubuntu.
Why	Marketplace is the blade (portal page) that lists VM images to deploy.
How	Search Marketplace for an image, select it, and click Create to start the VM wizard.

Exam tip
Images, templates and third-party appliances -> Marketplace.

Memory trick
The Market is where you shop for images.

Real-world example, use case and scenario

EXAMPLE
Ubuntu image
An admin searches Marketplace for Ubuntu Server 24.04 LTS and clicks Create to start the VM wizard.

USE CASE
Firewall appliance
A bank deploys a partner firewall image from Marketplace.

SCENARIO
Ready-made WordPress
A small design studio wanted a website fast. In Marketplace they found a ready-made WordPress image and started it in minutes.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

AZ-900 Exam Prep 2026

Figure Q154: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 155

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q243

Azure Advisor can generate a list of Azure virtual machines that are protected by Azure Backup.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The word to watch is PROTECTED. Advisor lists VMs that are NOT protected by backup (the gaps). A list of protected VMs comes from Backup center. So the answer is No.

Answer explained

✔ **No.**
Advisor flags VMs that are NOT protected by Azure Backup. The list of protected VMs comes from Backup center or the Recovery Services vault (the store for backups).

✖ **Yes.**
Advisor points out gaps (unprotected VMs). It does not make a list of protected ones.

EXAM STRATEGY
The word 'protected' decides it. Advisor lists only unprotected VMs. The list of protected VMs comes from Backup center. The answer is No.

IMPACT
VMs without backup that Advisor flags are a real risk of data loss. Clearing that recommendation often protects live data.

What · Where · When · Why · How

What	Azure Advisor gives best-practice advice, including turning on backup for unprotected VMs.
Where	Azure portal > Advisor > Reliability recommendations.
When	When looking for VMs that have no Azure Backup protection.
Why	Advisor lists unprotected VMs, not protected ones, so the statement is false.
How	Use Backup center or the Recovery Services vault to list protected VMs.

Exam tip
Advisor points at problems. It lists the unprotected VMs, not the protected ones.

Memory trick
Advisor shows the holes, not the patches.

Real-world example, use case and scenario

EXAMPLE
Reliability advice
Advisor shows 'Enable backups on your virtual machines' for six VMs.

USE CASE
Backup list
A clinic uses Backup center to report every protected VM to an auditor.

SCENARIO
Audit proof
An auditor asked for a list of backed-up servers. The admin first opened Advisor but saw only unprotected VMs. Then she exported the protected list from Backup center.

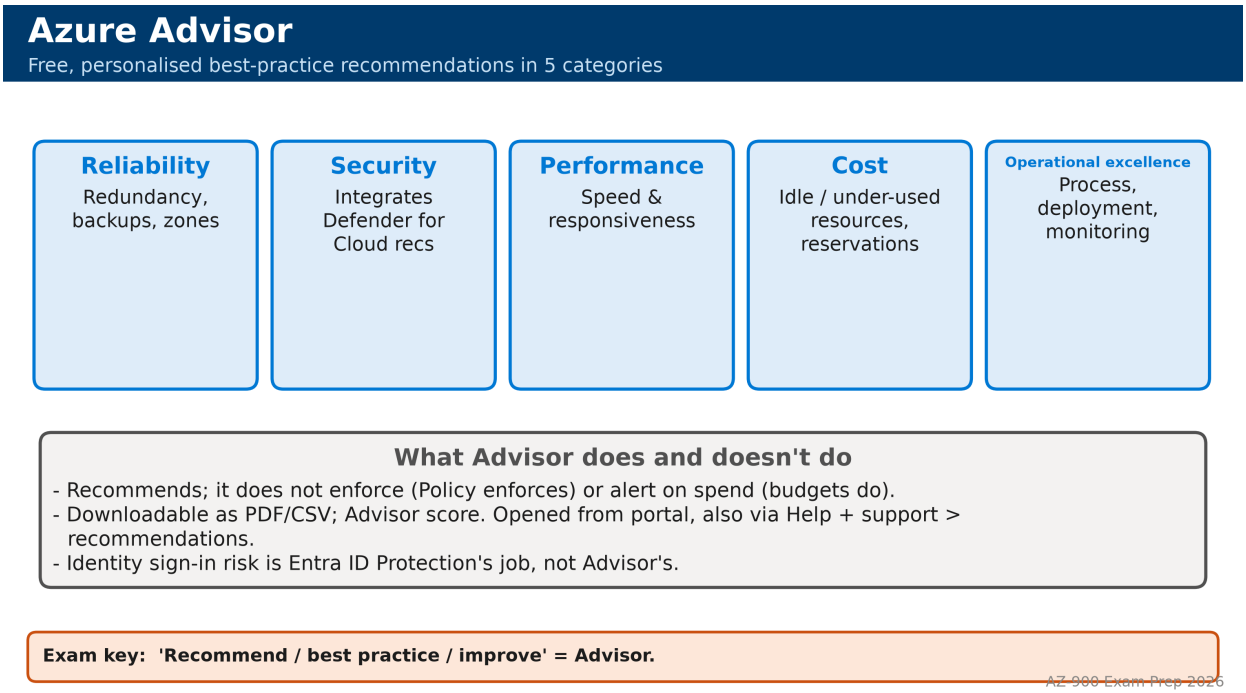


Figure Q155: Azure Advisor

About this figure

- The five boxes are Advisor's categories; each recommendation shows impact (high, medium, low) and suggested actions you can apply or postpone.
- Advisor is free and analyses your deployed resources' configuration and usage telemetry, giving subscription- and resource-group-level recommendations.
- Advisor score rates how well you follow best practices, overall and per category, helping prioritise fixes with the largest impact.
- Recommendations can trigger alerts and digest emails; security recommendations come from Defender for Cloud, so fixing them there also updates Advisor.
- Advisor is aligned with the Azure Well-Architected Framework pillars: reliability, security, cost optimisation, operational excellence and performance efficiency.

Question 156

AZURE ARCHITECTURE & SERVICES

Security v5 Q244

If you implement the security recommendations provided by Azure Advisor, your company secure score will decrease.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is DECREASE. Fixing security recommendations makes your secure score (a percentage showing how secure you are) go UP, not down.

Answer explained

✔ **No.**
Secure score goes up when you fix security recommendations, because each fixed control adds points.

✖ **Yes.**
Fixing recommendations improves security, so the secure score goes up. It drops only when new problems appear.

EXAM STRATEGY
'Decrease' is the trap. Fixing recommendations always raises the secure score. The answer is No.

IMPACT
Secure score is a security number that leaders watch. Raising it shows real improvement and lowers the chance of a breach.

What · Where · When · Why · How

What	Secure score measures your security level in Microsoft Defender for Cloud as a percentage.
Where	On the Defender for Cloud overview. Advisor's security items come from Defender for Cloud.
When	When tracking how fixes change your security level.
Why	Fixing recommendations earns control points, so the score goes up, not down.
How	Fix all recommendations in a security control to get that control's points.

Exam tip
Higher secure score = better security. Fixing recommendations always moves it up.

Memory trick
Fix it, score it - like points in a game.

Real-world example, use case and scenario

EXAMPLE
Score jump
Turning on MFA for all subscription owners raises the secure score from 52 percent to 61 percent.

USE CASE
Tracking progress
A cyber insurance company asks its clients to report their secure score every three months.

SCENARIO
Fear of change
A careful ops manager delayed fixes, worried the score would drop. After fixing encryption and patching findings, the score went up. The team set a target of 80 percent.

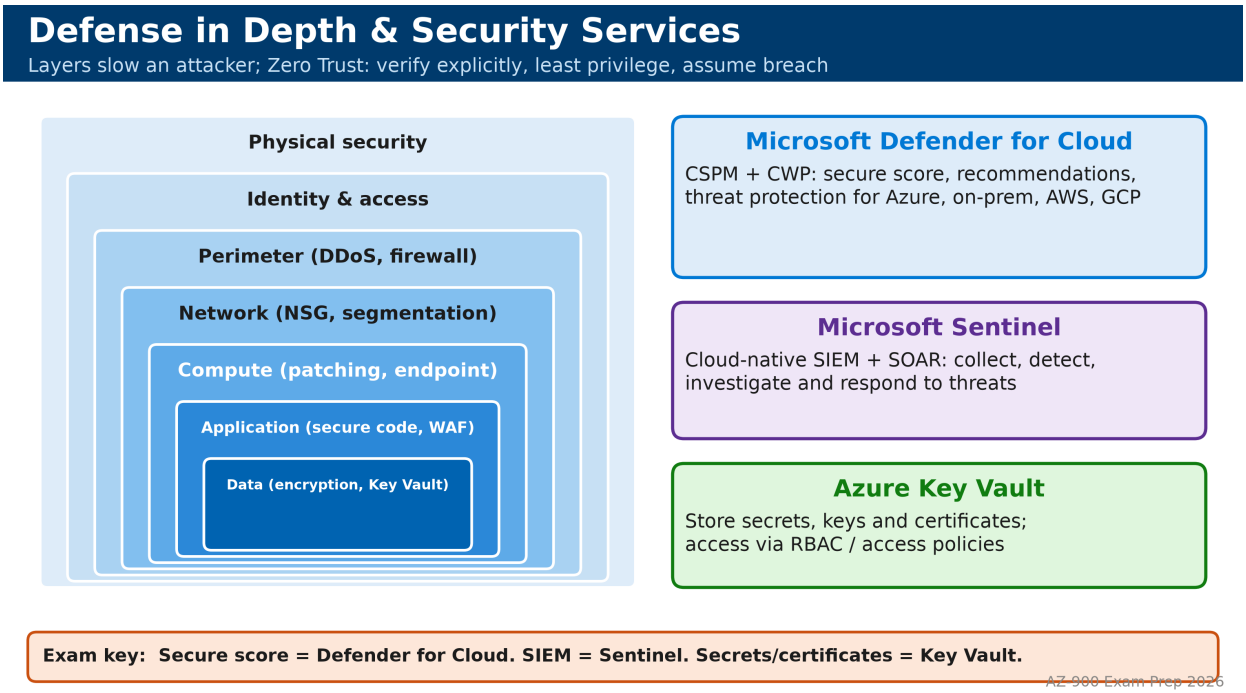


Figure Q156: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 157

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q245

To maintain Microsoft support, you must implement the security recommendations provided by Azure Advisor within a period of 30 days.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap words are MUST and '30 days'. Advisor only gives advice. Nothing is required, and your support depends only on your support plan. Q416: viewing Advisor recommendations is free (Yes).

Answer explained

✔ **No.**
Advisor recommendations are optional. You can postpone or dismiss them. Your support depends only on your support plan.

✖ **Yes.**
No Microsoft rule links support to applying Advisor recommendations by any deadline.

EXAM STRATEGY
'Must' and 'to keep support' are the trap. Advisor only advises, and support depends on your plan. The answer is No.

IMPACT
Knowing Advisor is optional lets teams fix the most important items first, based on risk and cost, instead of rushing every item for a false deadline.

What · Where · When · Why · How

What	Azure Advisor recommendations are free, optional best-practice suggestions.
Where	Azure portal > Advisor, in five categories.
When	When deciding whether Advisor advice is required.
Why	No rule links Microsoft support to applying Advisor advice within 30 days.
How	Apply, postpone or dismiss each recommendation. Use Azure Policy when you need to enforce rules.

Exam tip
Advisor suggests; Policy enforces. Nothing in Advisor is required.

Memory trick
An advisor gives advice, not orders.

Real-world example, use case and scenario

EXAMPLE
Postpone
An admin postpones a reserved instance recommendation for 90 days while budgets are set.

USE CASE
Choosing what to apply
A charity applies only Advisor cost recommendations and dismisses performance ones that do not fit its apps.

SCENARIO
Rumour in the team
A new admin rushed to apply every Advisor security fix, afraid of losing support. The team lead explained it was optional, and they ordered the changes by risk instead.

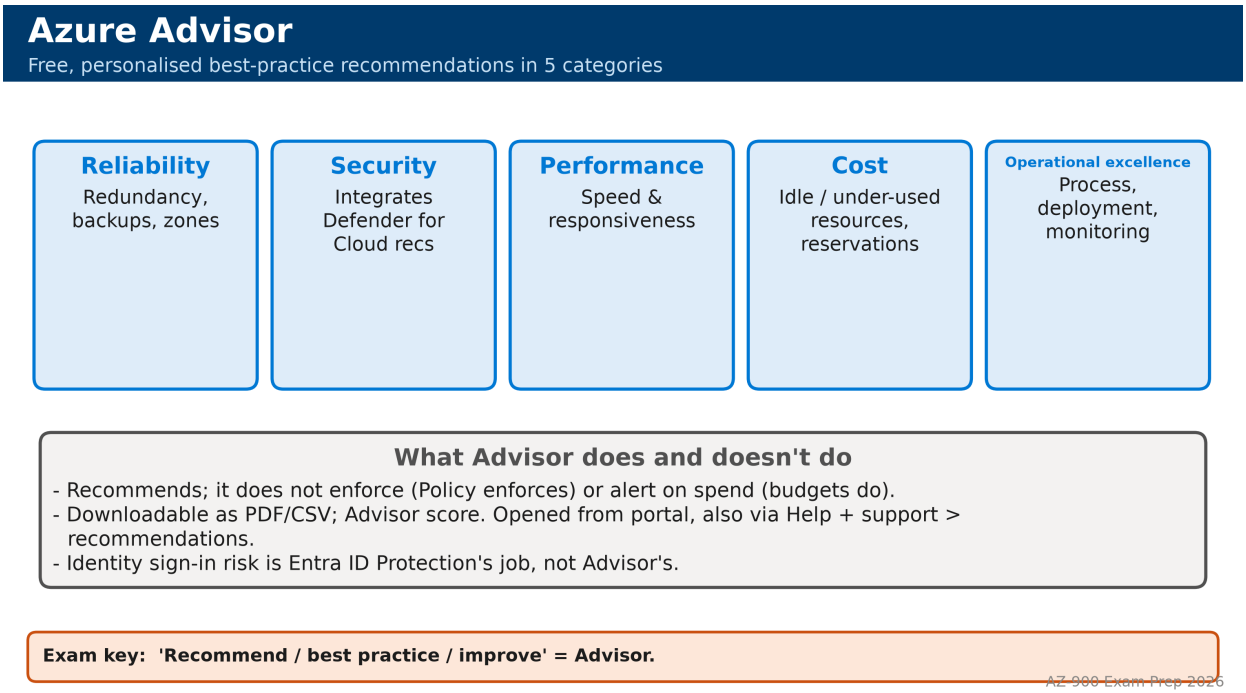


Figure Q157: Azure Advisor

About this figure

• The five boxes are Advisor's categories; each recommendation shows impact (high, medium, low) and suggested actions you can apply or postpone.

• Advisor is free and analyses your deployed resources' configuration and usage telemetry, giving subscription- and resource-group-level recommendations.

• Advisor score rates how well you follow best practices, overall and per category, helping prioritise fixes with the largest impact.

• Recommendations can trigger alerts and digest emails; security recommendations come from Defender for Cloud, so fixing them there also updates Advisor.

• Advisor is aligned with the Azure Well-Architected Framework pillars: reliability, security, cost optimisation, operational excellence and performance efficiency.

Question 158

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q246, Q299

What can you use to automatically send an alert if an administrator stops an Azure virtual machine?

✖ A. Azure Advisor.

✖ B. Azure Service Health.

✔ **C.** Azure Monitor.

✖ D. Azure Network Watcher.

Correct answer: Azure Monitor.

Clarification (read this first): The stop is done by YOUR administrator, not by Microsoft. Actions in your subscription are written to the Activity log, and Azure Monitor alerts on them. Service Health is only for Microsoft-side problems.

Answer explained

✔ **Azure Monitor.**
Stopping a VM writes an entry in the Activity log (the record of actions in your subscription). An Azure Monitor activity log alert can then notify an action group automatically.

✖ **Azure Advisor.**
Advisor gives recommendations. It cannot alert on actions such as stopping a VM.

✖ **Azure Service Health.**
Service Health reports Microsoft platform problems, not actions done by your own admins.

✖ **Azure Network Watcher.**
Network Watcher checks network connections and traffic, not VM power actions.

EXAM STRATEGY
An admin action like stopping a VM goes into the Activity log. Alert on it with Azure Monitor. Remove Service Health (Microsoft problems) and Advisor.

IMPACT
Activity log alerts catch unplanned or wrong shutdowns of live VMs within minutes. This shortens outages and helps audits.

What · Where · When · Why · How

What	An activity log alert in Azure Monitor fires on management actions, such as stopping a VM.
Where	Azure portal > Monitor > Alerts, with the Activity log as the signal.
When	When you must be told automatically if someone stops or deallocates a VM.
Why	Stop actions are written to the Activity log, and Azure Monitor alerts on it.
How	Create an alert rule on the 'Deallocate' or 'Power Off Virtual Machine' action, with an action group.

Exam tip
An admin action (stop, delete, create) -> Activity log -> Azure Monitor activity log alert.

Memory trick
Someone pressed Stop? The Activity log saw it, and Monitor tells you.

Real-world example, use case and scenario

EXAMPLE
Activity log alert
An alert rule on Microsoft.Compute/virtualMachines/deallocate/action emails the platform team.

USE CASE
Production guard
A stock trading firm gets an alert when anyone stops a live trading VM during trading hours.

SCENARIO
Mystery downtime
A website kept going offline at random times. After adding an activity log alert, the team found a contractor's script that stopped the VM every night, and fixed it.

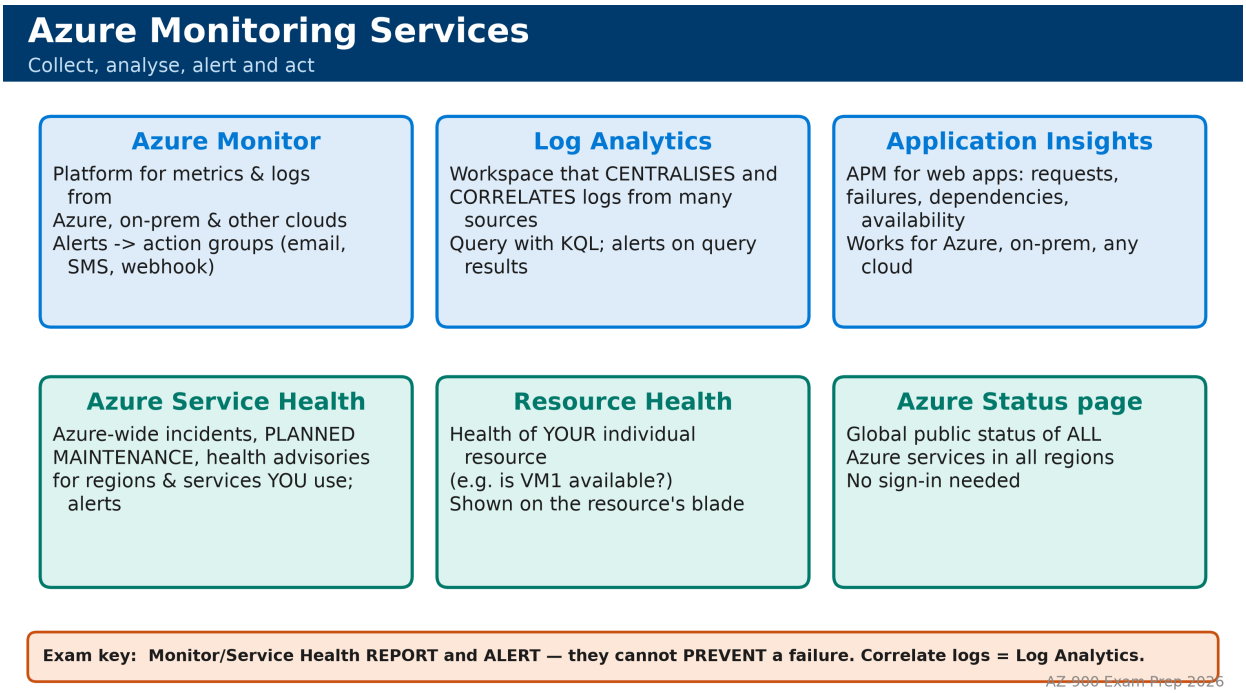


Figure Q158: Azure monitoring services

About this figure

- Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).
- Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.
- Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.
- Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.
- Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 159

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q247, Q56, Q57, Q80, Q93, Q132

You have an Azure environment. You need to create a new Azure virtual machine from a tablet that runs the Android operating system. What are three possible solutions?

- ✔ **A.** Use Bash in Azure Cloud Shell.
- ✔ **B.** Use PowerShell in Azure Cloud Shell.
- ✖ C. Use the PowerApps portal.
- ✖ D. Use the Security & Compliance admin center.
- ✔ **E.** Use the Azure portal.

Correct answer: Use Bash in Azure Cloud Shell.; Use PowerShell in Azure Cloud Shell.; Use the Azure portal.

Clarification (read this first): Pick THREE answers. Anything that runs in a browser works on an Android tablet: the Azure portal and Cloud Shell (both Bash and PowerShell). PowerApps and the Security & Compliance center are not tools for creating Azure VMs.

Answer explained

✔ **Use Bash in Azure Cloud Shell.; Use PowerShell in Azure Cloud Shell.; Use the Azure portal.**
The Azure portal and Cloud Shell (Bash or PowerShell) run in any browser. So an Android tablet can create a VM with either.

✖ **Use the PowerApps portal.**
Power Apps builds low-code business apps. It cannot create Azure virtual machines.

✖ **Use the Security & Compliance admin center.**
That Microsoft 365 compliance center (now the Microsoft Purview portal) manages compliance features, not Azure resources.

EXAM STRATEGY
Any device with a browser works. Pick the portal and both Cloud Shell options. Remove PowerApps and the Security & Compliance center, which are not Azure tools.

IMPACT
Cloud Shell lets engineers fix problems from any device without installing tools. This keeps on-call response fast.

What · Where · When · Why · How

What	The Azure portal and Azure Cloud Shell are browser-based Azure management tools.
Where	Any modern browser, including on Android, or the Azure mobile app.
When	When you must create Azure resources from a device with no local tools installed.
Why	The portal and Cloud Shell (Bash or PowerShell) need only a browser, so they work on a tablet.
How	Use the portal VM wizard, or run az vm create or New-AzVM in Cloud Shell.

Exam tip
Any device with a browser -> portal and Cloud Shell (Bash or PowerShell). Office and Power Platform portals never deploy Azure VMs.

Memory trick
Browser in hand, Azure on command.

Real-world example, use case and scenario

EXAMPLE
Cloud Shell command
From a tablet browser, an admin opens Cloud Shell Bash and runs az vm create to build a test VM.

USE CASE
Field engineer
A wind-farm engineer uses Cloud Shell in the Azure mobile app on an Android tablet to restart and rebuild monitoring VMs on site.

SCENARIO
Admin on holiday
An admin on holiday had only her Android tablet when a VM needed rebuilding. She used the portal wizard, then checked it with Get-AzVM in Cloud Shell PowerShell.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

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Figure Q159: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 160

AZURE MANAGEMENT & GOVERNANCE

Cost

v5 Q248, Q37

Azure Advisor provides recommendations on how to reduce the cost of running Azure virtual machines.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Advisor's Cost category flags VMs that are barely used, so you can resize or shut them down. It also suggests reservations or savings plans (discounts for a 1- or 3-year commitment).

✖ **No.**
Cost is one of Advisor's five categories. Resizing VMs is one of its most common recommendations.

EXAM STRATEGY
'Reduce cost' plus Advisor = Yes. Its Cost category covers resizing and reservations.

IMPACT
Following Advisor cost advice regularly cuts spending on idle or too-large VMs. This is often one of the biggest cloud savings.

What · Where · When · Why · How

What	Advisor Cost recommendations show ways to lower Azure spending.
Where	Azure portal > Advisor > Cost category.
When	When VMs are barely used, or run steadily enough for reservations.
Why	Advisor suggests resizing, shutting down or reserving VMs, so the statement is true.
How	Review the low-use VM findings, then resize, deallocate (stop and release the hardware), or buy reservations or savings plans.

Exam tip
Resize or shut down low-use VMs, buy reservations -> Advisor Cost recommendations.

Memory trick
Advisor spots the idle engine still burning fuel.

Real-world example, use case and scenario

EXAMPLE
Resize
Advisor suggests changing a D8s v5 VM that averages 3 percent CPU to a D2s v5, and shows the monthly saving.

USE CASE
Reservation advice
An HR software company buys three-year reservations after Advisor shows steady 24x7 use.

SCENARIO
Forgotten test VMs
A game studio's bill kept rising. Advisor flagged eight idle test VMs from an old release. Shutting them down cut the bill by a fifth.

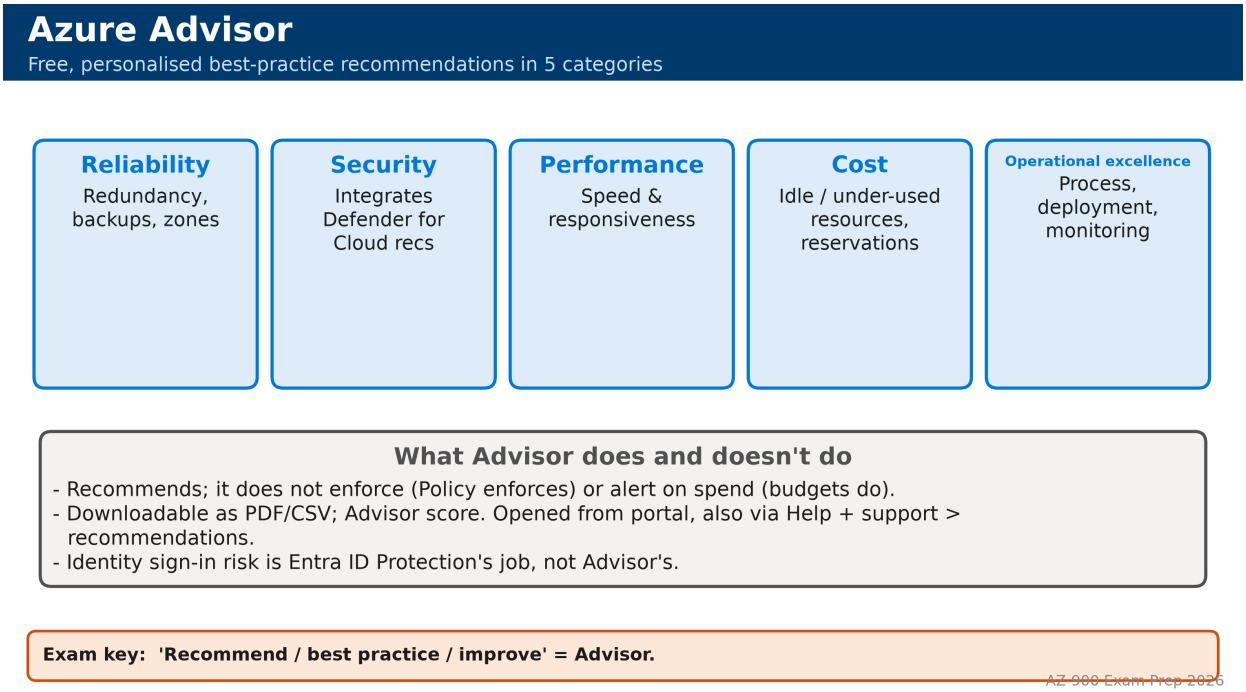


Figure Q160: Azure Advisor

About this figure

• The five boxes are Advisor's categories; each recommendation shows impact (high, medium, low) and suggested actions you can apply or postpone.

• Advisor is free and analyses your deployed resources' configuration and usage telemetry, giving subscription- and resource-group-level recommendations.

• Advisor score rates how well you follow best practices, overall and per category, helping prioritise fixes with the largest impact.

• Recommendations can trigger alerts and digest emails; security recommendations come from Defender for Cloud, so fixing them there also updates Advisor.

• Advisor is aligned with the Azure Well-Architected Framework pillars: reliability, security, cost optimisation, operational excellence and performance efficiency.

Question 161

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q250, Q39, Q249, Q251

Several support engineers plan to manage Azure by using the Computer2-Ubuntu. You need to identify which Azure management tools can be used.

- ✖ A. The Azure CLI and the Azure portal.
- ✖ B. The Azure portal and Azure PowerShell.
- ✖ C. The Azure CLI and Azure PowerShell.
- ✔ **D.** The Azure CLI, the Azure portal, and Azure PowerShell.

Correct answer: **The Azure CLI, the Azure portal, and Azure PowerShell.**

Clarification (read this first): The trap is thinking PowerShell runs only on Windows. PowerShell 7 runs on Linux too, so Ubuntu can use all three tools: CLI, portal and PowerShell.

Answer explained

✔ **The Azure CLI, the Azure portal, and Azure PowerShell.**
On Ubuntu you can use the portal in a browser, install the Azure CLI with apt, and run Azure PowerShell on PowerShell 7, which works on any OS.

✖ **The Azure CLI and the Azure portal.**
It leaves out Azure PowerShell, which runs on Ubuntu through PowerShell 7 and the Az module.

✖ **The Azure portal and Azure PowerShell.**
It leaves out the Azure CLI, which installs directly on Linux.

✖ **The Azure CLI and Azure PowerShell.**
It leaves out the Azure portal, which runs in any modern browser on Ubuntu.

EXAM STRATEGY
Linux is the PowerShell trap. PowerShell 7 runs on Ubuntu, so choose the option with all three.

IMPACT
Teams can use the same scripts on Linux build servers and admin laptops. They do not need separate tools for each OS.

What · Where · When · Why · How

What	Azure management tools: the browser-based portal, the Azure CLI and Azure PowerShell.
Where	Installed or run on an Ubuntu Linux computer.
When	When choosing which tools support engineers can use on Ubuntu.
Why	PowerShell 7 works on many OSs, so Ubuntu supports the portal, the CLI and Azure PowerShell.
How	Install the CLI with apt, install PowerShell 7 and the Az module, and use the portal in a browser.

Exam tip
PowerShell on Linux is the trap. PowerShell 7 works on Linux, so Ubuntu gets all three tools.

Memory trick
Penguins speak PowerShell too.

Real-world example, use case and scenario

EXAMPLE
Apt install
An engineer runs sudo apt install powershell, then Install-Module Az, and connects with Connect-AzAccount.

USE CASE
Linux DevOps team
A fintech DevOps team on Ubuntu laptops uses their Windows colleagues' Az PowerShell scripts without changes.

SCENARIO
Script sharing
A Linux admin was told to rewrite a PowerShell deployment script in Bash. Instead, he installed PowerShell 7 on Ubuntu and ran it as it was.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

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Figure Q161: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 162

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q252, Q22, Q23, Q89

An Azure administrator plans to run a PowerShell script that creates Azure resources. You need to recommend which computer configuration to use to run the script. Which three computers can run the script?

✖ A. A computer that runs macOS and has PowerShell Core 6.0 installed.

✔ **B.** A computer that runs Windows 10 and has the Azure PowerShell module installed.

✔ **C.** A computer that runs Linux and has the Azure PowerShell module installed.

✖ D. A computer that runs Linux and has the Azure CLI tools installed.

✔ **E.** A computer that runs Chrome OS and uses Azure Cloud Shell.

Correct answer: A computer that runs Windows 10 and has the Azure PowerShell module installed.; A computer that runs Linux and has the Azure PowerShell module installed.; A computer that runs Chrome OS and uses Azure Cloud Shell.

Clarification (read this first): Pick THREE. A PowerShell script needs two things: PowerShell AND the Az module (or Cloud Shell, which has both). The Azure CLI is a different tool and cannot run PowerShell cmdlets (commands). The printed answer key differs - see the answer note.

Answer corrected in this edition. The previous PDF printed: *A computer that runs macOS and has PowerShell Core 6.0 installed.; A computer that runs Windows 10 and has the Azure PowerShell module installed.; A computer that runs Chrome OS and uses Azure Cloud Shell.*. The printed answer was macOS + PowerShell Core 6.0; Windows 10 + Az module; Chrome OS + Cloud Shell. This version adds Linux + Azure PowerShell module, which can clearly run the script. macOS with only PowerShell Core 6.0 does not have the Az module (and Az no longer supports PowerShell 6). The original four-option version's answer was macOS, Windows and Chrome OS.

2026 update: PowerShell Core 6.0 is no longer supported. Current Az modules need PowerShell 7 (a currently supported 7.x version) or Windows PowerShell 5.1 on Windows.

Answer explained

✔ **A computer that runs Windows 10 and has the Azure PowerShell module installed.; A computer that runs Linux and has the Azure PowerShell module installed.; A computer that runs Chrome OS and uses Azure Cloud Shell.**
An Azure PowerShell script needs PowerShell with the Az module. That means Windows or Linux with Az installed, or Cloud Shell on Chrome OS.

✖ **A computer that runs macOS and has PowerShell Core 6.0 installed.**
PowerShell Core 6.0 alone does not have the Az module, and current Az versions do not support it.

✖ **A computer that runs Linux and has the Azure CLI tools installed.**
The Azure CLI runs az commands, not PowerShell scripts that use Az cmdlets.

EXAM STRATEGY
You need PowerShell plus the Az module, or Cloud Shell. Remove the Azure CLI (a different tool) and macOS with PowerShell Core 6.0 but no Az.

IMPACT
Running scripts on computers without the Az module fails in the middle of a deployment. Checking the requirements first prevents half-built environments.

What · Where · When · Why · How

What	Azure PowerShell is the Az module, which runs Azure cmdlets inside PowerShell.
Where	Any OS with PowerShell and Az installed, or Azure Cloud Shell in a browser.
When	When choosing a computer to run a script that creates Azure resources with Az cmdlets.
Why	Windows or Linux with Az, or Cloud Shell on Chrome OS, have both PowerShell and the module.
How	Install PowerShell 7 and run Install-Module Az, or open PowerShell in Cloud Shell, where Az is already installed.

Exam tip
PowerShell script = PowerShell + Az module (or Cloud Shell). The Azure CLI is a different tool and never runs PowerShell cmdlets.

Memory trick
PowerShell plus module, or Cloud Shell: no module, no Az.

Real-world example, use case and scenario

EXAMPLE
Missing module error
On a computer with PowerShell but no Az module, New-AzResourceGroup fails with a 'term not recognised' error.

USE CASE
Chromebook admins
A school district's IT staff on Chromebooks run deployment scripts in Cloud Shell PowerShell, which already has Az.

SCENARIO
CLI is not PowerShell
A Linux admin with only the Azure CLI tried to run a .ps1 deployment script, and it failed. Installing PowerShell 7 and the Az module fixed it.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

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Figure Q162: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 163

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q253

Azure Firewall will encrypt all the network traffic sent from Azure to the Internet.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The question asks if Azure Firewall itself encrypts (scrambles) all traffic going to the internet. Watch the word 'encrypt': a firewall only allows or blocks traffic. Q164 asks the same about an NSG (also No). Q165 asks about a Windows Server VM, and that answer is Yes.

Answer explained

✔ **No.**
Azure Firewall filters traffic with rules (network, application and NAT rules). It does not encrypt traffic. Encryption comes from TLS (secure web protocol) or a VPN (encrypted tunnel).

✖ **Yes.**
Azure Firewall allows or blocks traffic. The Premium tier can inspect TLS traffic, but it never encrypts all outgoing traffic.

EXAM STRATEGY
The key word is 'encrypt'. Firewalls filter. They do not encrypt. Answer No.

IMPACT
If you think a firewall encrypts, your data may travel unprotected. Audits need proof of TLS or VPN for sensitive traffic.

What · Where · When · Why · How

What	Azure Firewall is a managed, stateful (remembers open connections) network firewall for Azure virtual networks.
Where	It sits in a virtual network, often a hub VNet (central network), and checks traffic to and from the internet.
When	Use it when you must filter traffic in one central place, by IP address or FQDN (full domain name) rules.
Why	It filters traffic but does not encrypt it, so the statement is false.
How	Encrypt with TLS in your apps, or IPsec (encrypted IP traffic) through VPN Gateway. Use Firewall only for filtering.

Exam tip
Firewall = allow or block traffic. Encrypt = TLS or VPN. Do not mix them up.

Memory trick
A firewall is a door guard, not a lock maker.

Real-world example, use case and scenario

EXAMPLE
FQDN rule
An application rule lets a spoke VNet send traffic only to *.windowsupdate.com.

USE CASE
Hub-and-spoke egress
A logistics company sends all internet traffic from its spoke VNets through Azure Firewall in the hub. This gives one central log.

SCENARIO
Compliance misunderstanding
An auditor asked if data sent to the internet was encrypted. The team pointed to Azure Firewall, but the auditor needed proof of TLS. The team turned on HTTPS for the apps.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

AZ-900 Exam Prep 2026

Figure Q163: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 164

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q254

A Network Security Group (NSG) will encrypt all the network traffic sent from Azure to the Internet.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The question asks if an NSG encrypts traffic to the internet. Watch the word 'encrypt': an NSG only allows or blocks traffic. Q163 asks the same about Azure Firewall (also No). Q165 asks about a Windows Server VM, and that answer is Yes.

Answer explained

✔ **No.**
A network security group (NSG) only allows or blocks traffic by source, destination, port and protocol. It cannot encrypt anything.

✖ **Yes.**
NSGs filter traffic by IP address and port. They never encrypt the data. Encryption needs TLS or IPsec.

EXAM STRATEGY
An NSG is an allow/block list. Any claim that it encrypts is No.

IMPACT
If you rely on NSGs to keep data secret, you fail audits such as PCI DSS (card payment rules). Encryption must be set up on the server or VPN.

What · Where · When · Why · How

What	A network security group is a list of allow or block rules for network traffic.
Where	You attach it to subnets or network interfaces in a virtual network.
When	Use it to limit traffic by source, destination, port and protocol.
Why	NSGs only filter traffic and cannot encrypt, so the statement is false.
How	Use NSG rules to filter. Use TLS or IPsec to encrypt.

Exam tip
NSG = allow/block list by IP, port and protocol. Any statement that says an NSG encrypts is false.

Memory trick
NSG: No Secret Generation, just a gatekeeper.

Real-world example, use case and scenario

EXAMPLE
Allow HTTPS
An NSG rule allows TCP port 443 from the internet to a web subnet and blocks everything else.

USE CASE
Database isolation
A hospital uses NSGs so only the app subnet can reach the SQL subnet on port 1433.

SCENARIO
Plaintext API
A developer exposed an HTTP API and thought the NSG made it secure. A security test captured readable tokens. The team switched to HTTPS with a certificate.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

AZ-900 Exam Prep 2026

Figure Q164: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 165

AZURE ARCHITECTURE & SERVICES

Security v5 Q255

Azure virtual machines that run Windows Server 2016 can encrypt network traffic sent to the Internet.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): This time the question is about the VM itself, not a network filter. Software inside the VM (Windows Server) can encrypt traffic, so the answer is Yes. Compare Q163 (Azure Firewall) and Q164 (NSG): both are No, because filters cannot encrypt.

Answer corrected in this edition. The previous PDF printed: *No.*. The printed answer was No. A Windows Server 2016 VM can encrypt the traffic it sends by using TLS (HTTPS, SMB 3 encryption) or IPsec. So the statement is true.

Answer explained

✔ **Yes.**
The two ends of a connection do the encryption. Windows Server 2016 supports TLS, SMB 3 encryption and IPsec, so the VM can encrypt its traffic.

✖ **No.**
This ignores the TLS, SMB encryption and IPsec built into Windows Server 2016. Only NSGs and Azure Firewall cannot encrypt.

EXAM STRATEGY
Only the endpoints (the computers at each end) encrypt. The VM's operating system supports TLS and IPsec, so answer Yes.

IMPACT
Under shared responsibility (Microsoft and you each own some security tasks), encrypting app traffic is your job. Turning on TLS in VMs protects customer data.

What · Where · When · Why · How

What	Encryption in transit protects data while it moves on the network. The computers at each end do it.
Where	Inside the VM's Windows Server 2016 operating system and its applications.
When	When traffic from a VM to the internet must be encrypted.
Why	Windows Server 2016 supports TLS, SMB 3 encryption and IPsec, so the VM can encrypt traffic.
How	Add a TLS certificate to IIS (the Windows web server) or the app, or set IPsec rules in the operating system.

Exam tip
In this set of three questions, only the VM (the endpoint) can encrypt. NSG and Firewall cannot.

Memory trick
Endpoints encrypt, middle boxes filter.

Real-world example, use case and scenario

EXAMPLE
IIS with TLS
A Windows Server 2016 VM runs IIS with a TLS certificate. All replies to browsers are encrypted.

USE CASE
Secure file transfer
A law firm's VM sends documents to partners over SFTP, so files are encrypted the whole way.

SCENARIO
Choosing the right control
An engineer looked for an encryption setting on the NSG. A colleague explained that the VM's web server must use HTTPS. After adding a certificate, the traffic was encrypted.

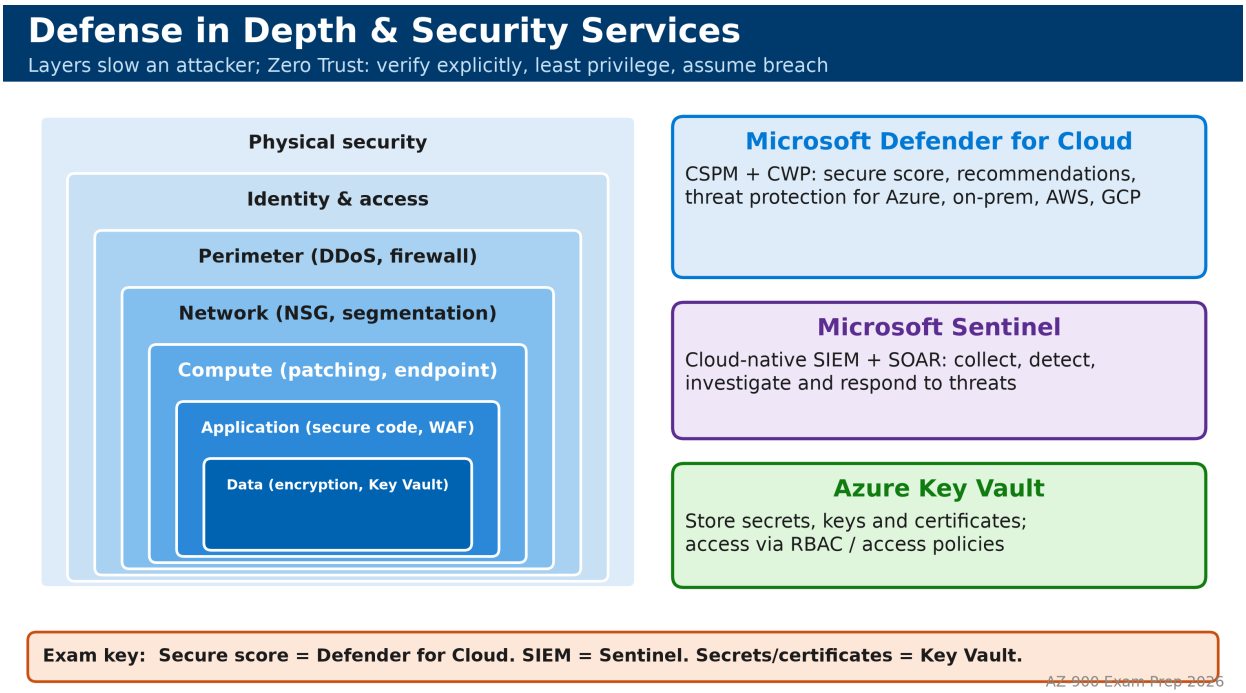


Figure Q165: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 166

AZURE ARCHITECTURE & SERVICES

Security
v5 Q256

Microsoft Defender for Cloud can monitor Azure resources and on-premises resources.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The question asks if Defender for Cloud can watch both Azure resources AND your own on-premises (in your building) servers. The answer is Yes, through Azure Arc. Q364 confirms it also covers AWS and Google Cloud.

Answer explained

✔ **Yes.**
Defender for Cloud protects Azure resources, on-premises servers through Azure Arc, and AWS and GCP workloads through connectors.

✖ **No.**
Covering on-premises servers (through Azure Arc) and other clouds is a core Defender for Cloud feature.

EXAM STRATEGY
'On-premises' with Defender for Cloud = Yes, through Azure Arc. It also covers other clouds.

IMPACT
One security view across Azure, on-premises and other clouds removes blind spots that attackers often use.

What · Where · When · Why · How

What	Microsoft Defender for Cloud is a CNAPP (cloud-native application protection platform: one tool for cloud security posture and threat protection).
Where	It covers Azure, on-premises servers through Azure Arc, and AWS and GCP through connectors.
When	When you need one security view for Azure and on-premises resources.
Why	It monitors both Azure and on-premises resources, so the statement is true.
How	Connect on-premises servers with Azure Arc, then turn on Defender plans for the subscription.

Exam tip
Defender for Cloud = Azure + on-premises (through Arc) + AWS + GCP.

Memory trick
Defender guards every castle, not just the Azure one.

Real-world example, use case and scenario

EXAMPLE
Arc-enabled server
An on-premises Windows server is connected with Azure Arc. It now gets vulnerability findings from Defender for Servers.

USE CASE
Multicloud retailer
A retailer connects its AWS account. Its EC2 and S3 security status now shows next to Azure in one secure score.

SCENARIO
Unpatched branch server
A bank's branch server was invisible to the security team. After connecting it with Arc, Defender for Cloud found missing patches and an open port within a day.

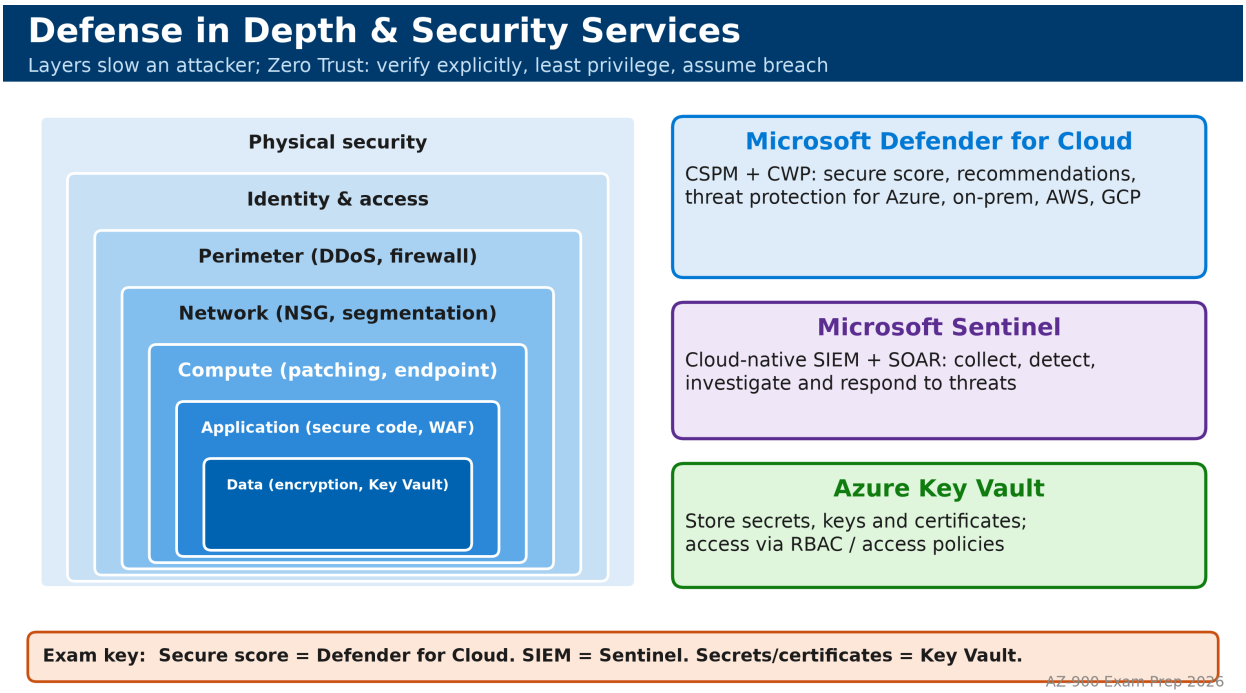


Figure Q166: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 167

AZURE ARCHITECTURE & SERVICES

Security
v5 Q257

All Microsoft Defender for Cloud features are free.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch the word 'All'. Some Defender for Cloud features are free, but not all, so the answer is No. Q370 asks only about Secure Score and recommendations, and those ARE free (Yes).

Answer explained

✔ **No.**
Only Foundational CSPM (basic security posture: secure score and recommendations) is free. Defender CSPM and workload plans such as Defender for Servers cost money.

✖ **Yes.**
Defender plans for servers, storage, SQL, containers, and Defender CSPM are charged per resource. So not all features are free.

EXAM STRATEGY
'All' plus 'free' decides it. Only Foundational CSPM is free; Defender plans are paid. Answer No.

IMPACT
Turning on Defender plans without a budget surprises finance. Skipping them leaves workloads without threat detection.

What · Where · When · Why · How

What	Defender for Cloud has free Foundational CSPM plus paid Defender plans.
Where	Defender for Cloud > Environment settings > Defender plans, for each subscription.
When	When you estimate Defender for Cloud costs or choose protection plans.
Why	Plans such as Defender for Servers and Defender CSPM are charged, so not all features are free.
How	Keep the free secure score and recommendations. Turn on paid plans only where you need workload protection.

Exam tip
Free = Foundational CSPM (secure score, recommendations). Paid = Defender plans for threat protection.

Memory trick
Posture is free, protection costs.

Real-world example, use case and scenario

EXAMPLE
Plans page
In Environment settings, Defender for Servers Plan 2 shows a price per server per hour. Foundational CSPM shows free.

USE CASE
Selective plans
An online shop turns on paid Defender for Storage only for accounts that hold customer uploads.

SCENARIO
Unexpected charges
A startup turned on all Defender plans in a trial and got charges after 30 days. They kept Defender for Servers on production and turned off plans for dev subscriptions.

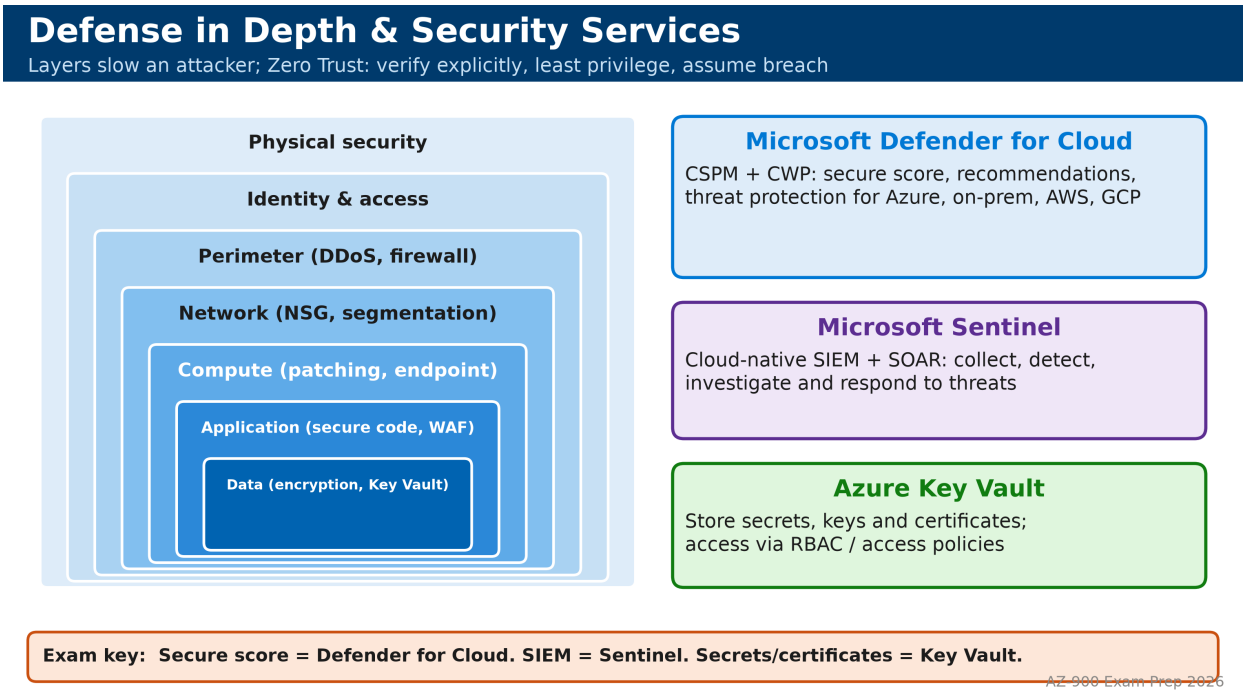


Figure Q167: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 168

AZURE ARCHITECTURE & SERVICES

Security
v5 Q259

You plan to implement several security services for an Azure environment. You need to identify which Azure services must be used to meet the following security requirements: Monitor threats by using sensors.

✖ A. Azure Monitor.

✖ B. Microsoft Defender for Cloud.

✖ C. Microsoft Entra ID Identity Protection.

✔ **D.** Microsoft Defender for Identity.

Correct answer: Microsoft Defender for Identity.

Clarification (read this first): The key word is 'sensors' (small programs installed on servers). Defender for Identity installs sensors on domain controllers (servers that run Active Directory). Do not confuse it with Entra ID Protection, which is the answer to Q169 (MFA based on a condition).

Answer explained

✔ **Microsoft Defender for Identity.**
Microsoft Defender for Identity installs sensors on on-premises domain controllers, AD FS and AD CS servers. They detect identity attacks such as pass-the-hash (reusing a stolen password hash) and lateral movement (moving from one computer to the next).

✖ **Azure Monitor.**
Azure Monitor collects metrics and logs about health and performance. It has no threat-detection sensors.

✖ **Microsoft Defender for Cloud.**
Defender for Cloud protects workloads with agents or agentless scans. It does not put identity sensors on domain controllers.

✖ **Microsoft Entra ID Identity Protection.**
Entra ID Protection calculates the risk of cloud sign-ins and users. It installs no sensors.

EXAM STRATEGY
'Sensors' is the signal word. Defender for Identity puts sensors on domain controllers. Cross out Entra ID Protection (cloud sign-in risk).

IMPACT
Finding pass-the-hash and lateral movement on domain controllers stops ransomware attackers before they get domain admin rights.

What · Where · When · Why · How

What	Microsoft Defender for Identity is a cloud service that detects attacks on on-premises Active Directory.
Where	Sensors run on domain controllers, AD FS and AD CS servers. Alerts show in the Microsoft Defender portal.
When	When you must detect identity attacks, such as pass-the-hash or reconnaissance (attackers looking around), in Active Directory.
Why	It is the only listed service that watches for threats with installed sensors.
How	Sensors read network traffic and Windows events on each domain controller and send signals to the cloud for analysis.

Exam tip
Sensors on domain controllers = Defender for Identity. Risky cloud sign-ins = Entra ID Protection. Do not mix them up.

Memory trick
Defender for IDENTITY sits on the Domain Controllers: I-D for Identity, D-C for its sensors.

Real-world example, use case and scenario

EXAMPLE
Pass-the-hash alert
A sensor on DC01 sees a sign-in that uses a password hash stolen from a helpdesk PC. It raises a pass-the-hash alert in the Defender portal.

USE CASE
Regional bank AD
A regional bank with 40 domain controllers installs Defender for Identity sensors to detect attackers moving inside its network.

SCENARIO
Recon caught early
A university saw nothing odd until a Defender for Identity sensor flagged a lab PC listing all admin accounts. The team isolated the PC within an hour. Azure Monitor alone would only have shown CPU use on the server.

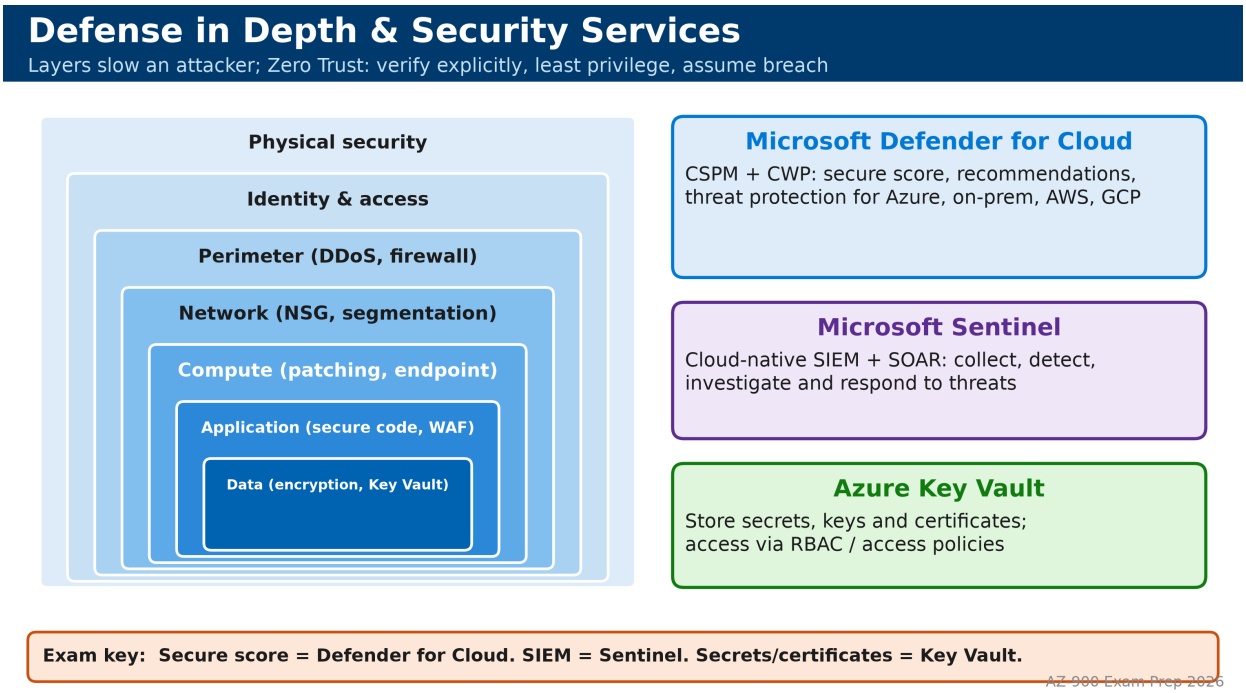


Figure Q168: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 169

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q260

You plan to implement several security services for an Azure environment. You need to identify which Azure services must be used to meet the following security requirements: Enforce Azure Multi-Factor Authentication (MFA) based on a condition.

✖ A. Azure Monitor.

✖ B. Microsoft Defender for Cloud.

✔ **C.** Microsoft Entra ID Identity Protection.

✖ D. Microsoft Defender for Identity.

Correct answer: Microsoft Entra ID Identity Protection.

Clarification (read this first): The key words are 'MFA based on a condition' (for example, only when a sign-in looks risky). Today this is done by Conditional Access, but Conditional Access is not in the list, so pick Entra ID Identity Protection. Q16 is similar (force a password change on a risky sign-in) and also points to Entra ID Protection. Do not confuse it with Defender for Identity (Q168, sensors).

2026 update: Today the product is called Microsoft Entra ID Protection. Risk-based MFA is set up in Conditional Access policies that use sign-in risk or user risk as the condition. Conditional Access is the strictly correct engine, but it is not an option here, so Identity Protection is the expected answer.

Answer explained

✔ **Microsoft Entra ID Identity Protection.**
Entra ID Protection (old name: Identity Protection) uses sign-in risk or user risk as a condition. It can require MFA when the risk is medium or high.

✖ **Azure Monitor.**
Azure Monitor collects metrics, logs and alerts. It cannot ask a user for MFA.

✖ **Microsoft Defender for Cloud.**
Defender for Cloud can recommend MFA, but it does not enforce MFA at sign-in.

✖ **Microsoft Defender for Identity.**
Defender for Identity uses sensors to detect attacks on on-premises AD. It does not control sign-ins.

EXAM STRATEGY
'Enforce MFA based on a condition' means Conditional Access. If it is not listed, pick Identity Protection. Know that the old answer and today's Conditional Access engine do the same job.

IMPACT
Risk-based MFA blocks stolen passwords without asking for MFA at every sign-in. This cuts account takeover and keeps users productive.

What · Where · When · Why · How

What	Microsoft Entra ID Protection finds risky sign-ins and users and drives risk-based access rules.
Where	Set up in the Microsoft Entra admin center. It needs Entra ID P2 licenses.
When	When MFA should be asked only for risky sign-ins, such as impossible travel or leaked passwords.
Why	It enforces MFA based on a condition (risk level). The other options cannot do this.
How	It scores sign-in risk and user risk. A risk-based Conditional Access policy then requires MFA above a set level.

Exam tip
If Conditional Access is an option, pick it for conditional MFA. If only Identity Protection is listed, pick it: its risk rules are the condition that triggers MFA.

Memory trick
Risky? Prove it: Identity Protection says if risk, then MFA.

Real-world example, use case and scenario

EXAMPLE
Impossible travel
A user signs in from Toronto and 10 minutes later from Jakarta. Sign-in risk is high, so MFA is required.

USE CASE
Insurance call center
An insurer requires MFA only for medium or high risk sign-ins. Its 2,000 agents are not asked at every office login.

SCENARIO
Leaked password
A dispatcher reused a password that appeared in a leak on the dark web. Identity Protection marked the user as high risk and forced MFA and a password change. The attacker did not have the phone, so they were blocked.

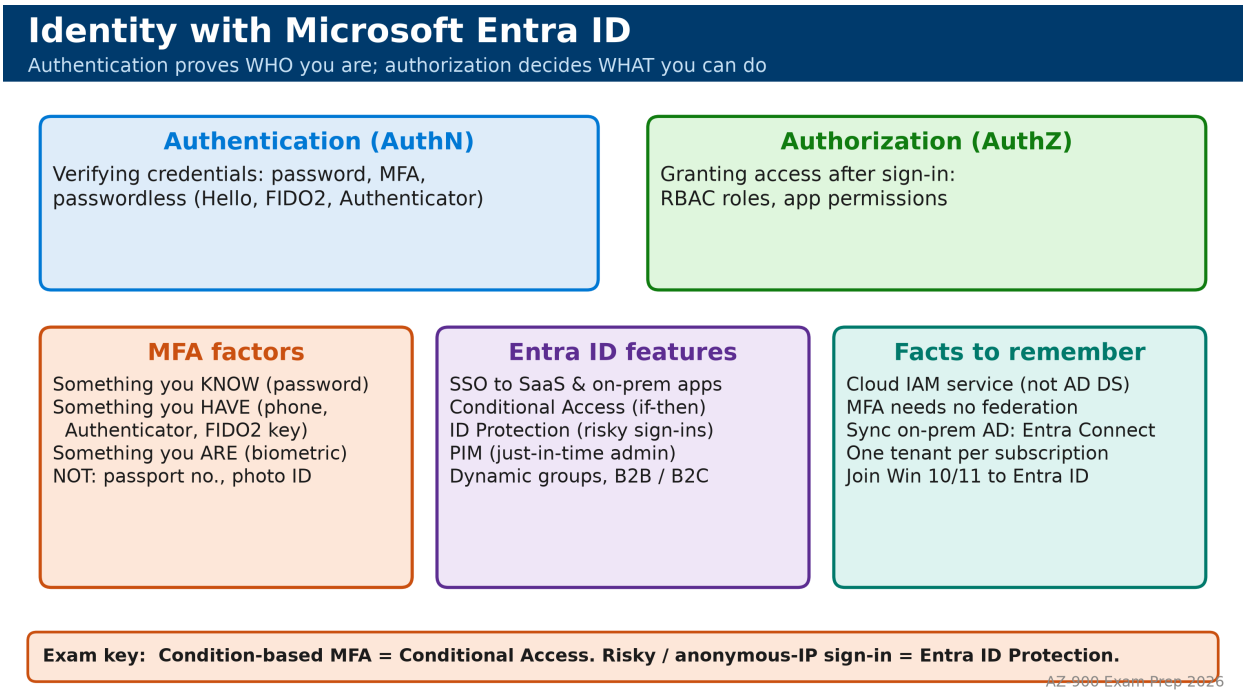


Figure Q169: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 170

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q261, Q97, Q98

Your Azure environment contains multiple Azure virtual machines. You need to ensure that a virtual machine named VM1 is accessible from the Internet over HTTP. What are two possible solutions?

✖ A. Modify an Azure Traffic Manager profile.

✔ **B.** Modify a Network Security Group (NSG).

✖ C. Modify a DDoS protection plan.

✔ **D.** Modify an Azure firewall.

Correct answer: Modify a Network Security Group (NSG).; Modify an Azure firewall.

Clarification (read this first): This question has TWO answers. 'Accessible over HTTP' means you must open port 80, so pick the two tools that allow traffic: NSG and Azure Firewall. Q51 (Traffic Manager) and Q79 (DDoS protection plan) ask about the wrong options one by one, and both are No.

Answer explained

✔ **Modify a Network Security Group (NSG).; Modify an Azure firewall.**
An NSG inbound allow rule for TCP port 80, or an Azure Firewall DNAT rule (forwards a public address to a private VM), each lets HTTP traffic from the internet reach VM1. Both are traffic filters.

✖ **Modify an Azure Traffic Manager profile.**
Traffic Manager routes users with DNS (name lookup). It opens no ports and carries no traffic.

✖ **Modify a DDoS protection plan.**
DDoS Protection defends public IPs against flood attacks. It has no allow rules to open a port.

EXAM STRATEGY
'Accessible over HTTP' means opening a port: an NSG rule or a Firewall DNAT rule. Cross out Traffic Manager (DNS routing) and DDoS (protection only).

IMPACT
Opening ports with the wrong tool, or too wide a rule, exposes VMs to the internet. Exact port 80 rules keep the risk small.

What · Where · When · Why · How

What	NSGs and Azure Firewall are network filters that allow or block traffic to Azure resources.
Where	NSGs attach to the VM's network interface or subnet. Azure Firewall sits in the VNet, often in a hub.
When	When a VM must accept traffic from the internet, such as HTTP on port 80.
Why	Each one can open port 80 to VM1, so either is a full solution.
How	Add an NSG inbound allow rule for TCP 80, or an Azure Firewall DNAT rule that maps public IP:80 to VM1.

Exam tip
Opening a port = a filtering tool (NSG or Firewall). Traffic Manager and DDoS never grant access.

Memory trick
NSG and Firewall are the gatekeepers; Traffic Manager is only the signpost.

Real-world example, use case and scenario

EXAMPLE
Port 80 rule
Add an NSG inbound rule with priority 100: source Any, destination port 80, TCP, Allow, on VM1's network interface.

USE CASE
Hub firewall publishing
A retailer with a hub-and-spoke network publishes its web VM through an Azure Firewall DNAT rule, so all incoming traffic is logged in one place.

SCENARIO
Site still unreachable
A startup developer created a Traffic Manager profile for VM1, but the site still timed out. The NSG still blocked port 80. One new NSG rule fixed it in seconds.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

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Figure Q170: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 171

AZURE ARCHITECTURE & SERVICES

Security v5 Q262

You can enable Just In Time (JIT) VM access by using [...].

✖ A. Azure Bastion.

✖ B. Azure Firewall.

✖ C. Azure Front Door.

✔ **D.** Microsoft Defender for Cloud.

Correct answer: Microsoft Defender for Cloud.

Clarification (read this first): JIT (just-in-time) access keeps VM admin ports closed until someone asks for access. The trap is Azure Bastion: it also secures RDP and SSH (remote login), but it has no JIT feature.

Answer explained

✔ **Microsoft Defender for Cloud.**
Just-in-time VM access is a feature of Defender for Servers Plan 2 in Microsoft Defender for Cloud. It opens admin ports only on approved requests, for a limited time.

✖ **Azure Bastion.**
Bastion gives RDP and SSH in the browser without public IPs. It does not offer JIT port requests.

✖ **Azure Firewall.**
JIT can change Azure Firewall rules, but JIT is set up and run from Defender for Cloud.

✖ **Azure Front Door.**
Front Door is a global web load balancer, CDN and web firewall. It has nothing to do with VM admin ports.

EXAM STRATEGY
'JIT' = Defender for Cloud. Bastion is the trap because it also secures RDP and SSH but has no JIT feature.

IMPACT
JIT keeps admin ports closed by default. This blocks password-guessing attacks on RDP that often break into exposed VMs within hours.

What · Where · When · Why · How

What	Just-in-time VM access keeps VM admin ports closed until someone requests access and it is approved.
Where	In Microsoft Defender for Cloud, with Defender for Servers Plan 2 turned on.
When	When RDP or SSH ports must not stay open to the internet all the time.
Why	JIT is a Defender for Cloud feature, so you turn it on there, not in Bastion or Firewall.
How	Defender for Cloud adds block rules, then allows the requester's IP address for a set time only.

Exam tip
JIT = Defender for Cloud (Defender for Servers). Bastion is the trap because it also secures RDP/SSH.

Memory trick
Just In Time: Defender keeps the door shut until you ring the bell.

Real-world example, use case and scenario

EXAMPLE
Three-hour window
An admin asks for RDP to VM-SQL01 for 3 hours from her office IP. Defender for Cloud adds a temporary NSG allow rule, then removes it.

USE CASE
Game studio servers
A game studio keeps SSH closed on 200 build servers. Engineers use JIT to open port 22 only during maintenance.

SCENARIO
Brute force stopped
A clinic's VM had thousands of failed RDP logins each day. With JIT, port 3389 was closed by default. The failed logins dropped to zero, and staff still got access on request.

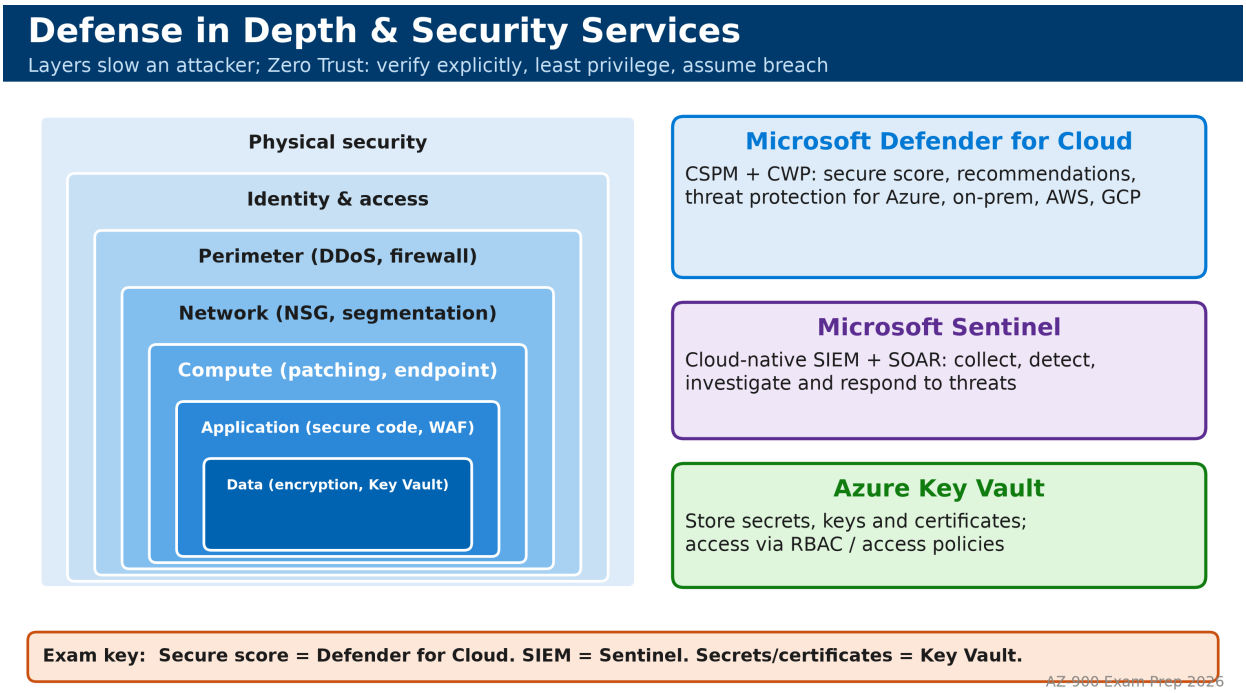


Figure Q171: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 172

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q263

You can associate a Network Security Group (NSG) to a virtual network subnet.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): This is one of three look-alike questions about where you can attach an NSG. Subnet = Yes (this one). Whole virtual network = No (Q173). Network interface = Yes (Q174).

Answer explained

✔ **Yes.**
You can attach an NSG to a subnet. Its rules then apply to every network interface in that subnet.

✖ **No.**
A subnet is one of the two places you can attach an NSG. The other is a network interface.

EXAM STRATEGY
Remember the NSG targets: subnet and network interface. Subnet is valid, so Yes.

IMPACT
Subnet NSGs give one set of rules to a whole tier. This avoids mistakes on single VMs that open production ports.

What · Where · When · Why · How

What	A network security group is a set of allow/block rules for traffic to Azure network resources.
Where	Attached to a subnet or a network interface in the same region as the NSG.
When	When every VM in a subnet should follow the same traffic rules.
Why	A subnet is a supported NSG target, so the statement is true.
How	Open the subnet settings, or the NSG's Subnets page, and attach the NSG. Its rules apply to all network interfaces there.

Exam tip
NSG targets: subnet and network interface only. A whole VNet is the trap.

Memory trick
NSGs go on Subnets and NICs: S and N, never the whole Net.

Real-world example, use case and scenario

EXAMPLE
Web subnet
NSG-Web is attached to subnet 10.0.1.0/24. It allows port 443 from the internet to all web VMs.

USE CASE
Tiered app
A payroll company puts separate NSGs on its web, app and data subnets. Only the app tier can reach SQL on port 1433.

SCENARIO
New VM auto-protected
A media agency added a fifth VM to its web subnet. The NSG was on the subnet, so the new VM got the rules at once, with no extra work.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

AZ-900 Exam Prep 2026

Figure Q172: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 173

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q264

You can associate a Network Security Group (NSG) to a virtual network.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): This asks about the WHOLE virtual network, not a subnet. You cannot attach an NSG to a whole VNet, so the answer is No. The look-alikes Q172 (subnet) and Q174 (network interface) are both Yes.

Answer explained

✔ **No.**
NSGs attach only to subnets and network interfaces. A virtual network as a whole cannot have an NSG.

✖ **Yes.**
The VNet has no NSG setting. You must attach NSGs to each subnet instead.

EXAM STRATEGY
A whole VNet is not an NSG target; only subnets and network interfaces are. Answer No.

IMPACT
If you expect one NSG for the whole VNet, new subnets stay unprotected. Every subnet needs its own NSG or a policy that requires one.

What · Where · When · Why · How

What	A network security group filters traffic but cannot be attached to a whole virtual network.
Where	NSGs attach only to subnets and network interfaces, never to the VNet itself.
When	For VNet-wide filtering, attach an NSG to each subnet or use Azure Firewall.
Why	The VNet has no NSG setting, so the statement is false.
How	Attach one NSG to each subnet. Azure Policy can require that every subnet has one.

Exam tip
Three-part NSG questions (subnet / VNet / network interface) usually go Yes / No / Yes.

Memory trick
The VNet is the Very big box; NSGs only fit the smaller boxes inside it.

Real-world example, use case and scenario

EXAMPLE
Portal check
When you open a VNet in the portal, there is no network security group setting. It appears only on each subnet and network interface.

USE CASE
Policy enforcement
A bank uses Azure Policy to require an NSG on every subnet, because none can be set at VNet level.

SCENARIO
Gap in coverage
A SaaS team thought one NSG covered their VNet. They added a new subnet without one. A review found it open to the other tiers, so they attached an NSG and added a policy.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

AZ-900 Exam Prep 2026

Figure Q173: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 174

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q265

You can associate a Network Security Group (NSG) to a network interface.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): A network interface (NIC) is the VM's network card. You can attach an NSG to it, so Yes. Look-alikes: Q172 (subnet) is Yes; Q173 (whole VNet) is No.

Answer explained

✔ **Yes.**
You can attach an NSG directly to a VM's network interface for rules for that one VM. It can be used together with a subnet NSG or instead of one.

✖ **No.**
A network interface is a supported NSG target. Only a whole VNet is not supported.

EXAM STRATEGY
A network interface is a valid NSG target. Answer Yes.

IMPACT
If you use subnet and NIC NSGs together, traffic must pass both. One forgotten rule can quietly block a production deployment.

What · Where · When · Why · How

What	An NSG on a network interface applies rules to the traffic of one VM.
Where	Attached to the VM's network interface, with or without a subnet NSG.
When	When one VM in a shared subnet needs different rules from the others.
Why	A network interface is a supported NSG target, so the statement is true.
How	Incoming traffic is checked by the subnet NSG, then by the NIC NSG. Both must allow it.

Exam tip
You can use both levels together. Traffic must pass both NSGs.

Memory trick
The NIC is the VM's own front door, and it can have its own lock (NSG).

Real-world example, use case and scenario

EXAMPLE
Jump box only
Only the jump box's network interface gets an NSG that allows SSH from the admin IP range.

USE CASE
Mixed subnet
A research lab has VMs with different needs in one subnet, so it puts NSGs on single network interfaces.

SCENARIO
Double deny
An engineer opened port 8443 on a VM's NIC NSG, but traffic still failed. The subnet NSG blocked it first. Adding the rule to the subnet NSG fixed it.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

AZ-900 Exam Prep 2026

Figure Q174: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 175

AZURE MANAGEMENT & GOVERNANCE — **Identity** v5 Q267

You can create custom Azure roles to control access to resources.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Azure RBAC (role-based access control) supports custom roles when no built-in role fits. You define them with Actions, NotActions, DataActions and AssignableScopes.

✖ **No.**
Custom roles are a documented Azure RBAC feature. You can have up to 5,000 per Microsoft Entra tenant.

EXAM STRATEGY
Custom roles exist in Azure RBAC. Answer Yes.

IMPACT
Custom roles give only the needed rights (least privilege) when built-in roles give too much. A stolen account can then do less damage.

What · Where · When · Why · How

What	An Azure custom role is a role you define yourself, with the permissions you choose.
Where	Created at management group, subscription or resource group scope. Up to 5,000 per Entra tenant.
When	When no built-in role matches the small set of permissions you need.
Why	Azure RBAC clearly supports custom roles, so the statement is true.
How	Define Actions, NotActions, DataActions and AssignableScopes in JSON or the portal, then assign the role.

Exam tip
Least privilege when no built-in role fits = custom RBAC role.

Memory trick
A built-in role is a ready-made suit; a custom role is made to fit.

Real-world example, use case and scenario

EXAMPLE
VM operator
A custom role allows starting and restarting VMs (Microsoft.Compute/virtualMachines/start and restart) but not deleting them.

USE CASE
NOC staff
A telecom network team gets a custom role to restart VMs and read metrics, but not change settings.

SCENARIO
Too much Contributor
A hotel chain gave night staff the Contributor role just to restart VMs. One person resized a database by mistake. They replaced it with a custom restart-only role.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

AZ-900 Exam Prep 2026

Figure Q175: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 176

AZURE MANAGEMENT & GOVERNANCE

Identity
v5 Q268

A user account can be assigned to multiple Azure roles.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Azure RBAC is additive (permissions add up). A user can have many role assignments, and their total permissions are all of them combined.

✖ **No.**
Azure RBAC has no one-role-per-user limit. Role assignments stack.

EXAM STRATEGY
RBAC adds permissions together. A user can have many role assignments. Answer Yes.

IMPACT
Because permissions add up, extra assignments can give unwanted write access. Regular access reviews stop users from collecting too many rights.

What · Where · When · Why · How

What	An Azure role assignment gives a role to a user, group or app at a scope (level).
Where	At management group, subscription, resource group or resource scope.
When	When a user needs permissions from more than one role, such as Reader plus a data role.
Why	RBAC adds permissions and has no one-role limit, so the statement is true.
How	Total permissions = all assignments combined, minus any deny assignments.

Exam tip
RBAC permissions add up. Reader on a subscription plus Contributor on a resource group = write access on that resource group.

Memory trick
Roles stack like badges on a lanyard.

Real-world example, use case and scenario

EXAMPLE
Two roles
Priya has Reader on the subscription and Storage Blob Data Contributor on one storage account.

USE CASE
DBA with network view
A database admin at a drug company has SQL DB Contributor plus Network Reader, to fix connection problems.

SCENARIO
Unexpected write access
An auditor asked why a Reader could edit a web app. The check found an extra Website Contributor assignment on that resource group. Permissions add up, so this was correct.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

AZ-900 Exam Prep 2026

Figure Q176: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 177

AZURE MANAGEMENT & GOVERNANCE

Identity
v5 Q269

A resource group can have the Owner role assigned to multiple users.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Any number of users, groups or service principals (app identities) can have the Owner role on the same resource group. The only limit is the number of role assignments allowed per subscription.

✖ **No.**
Owner is not a one-person role. Many users can have it at the same scope.

EXAM STRATEGY
Nothing limits Owner to one person. Answer Yes.

IMPACT
Several owners avoid lockout if one admin leaves. But too many owners add risk, so use groups and PIM (Privileged Identity Management: time-limited admin rights).

What · Where · When · Why · How

What	The Owner role gives full access to resources, including the right to give access to others.
Where	Assigned on the resource group's Access control (IAM) page.
When	When several admins must share full control of a resource group.
Why	Any number of users can have the same role at one scope, so the statement is true.
How	Add several Owner assignments, ideally to one group, within the subscription's assignment limit.

Exam tip
Many owners are allowed. Best practice is one group plus PIM, not many single-user assignments.

Memory trick
A resource group can have many co-owners, like a house owned by several people.

Real-world example, use case and scenario

EXAMPLE
Two leads
Both team leads are Owners of RG-Analytics, so either one can manage access.

USE CASE
Group-based ownership
A logistics firm gives Owner to the group RG-Fin-Owners, which has three members, instead of making three separate assignments.

SCENARIO
Holiday coverage
A museum's only resource group Owner went on leave, and nobody could give access. Adding a second Owner (through PIM) solved this for next time.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
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Resource group rules

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- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

AZ-900 Exam Prep 2026

Figure Q177: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 178	AZURE ARCHITECTURE & SERVICES	Security v5 Q271, Q270

Which Azure service can you use as a security information and event management (SIEM) solution?

- ✖ A. Azure Analysis Services.
- ✔ **B.** Azure Sentinel.
- ✖ C. Microsoft Purview Information Protection.
- ✖ D. Azure Cognitive Services.

Correct answer: Azure Sentinel.

Clarification (read this first): SIEM (security information and event management) collects security logs and finds threats. The exam option says 'Azure Sentinel', which is the old name of Microsoft Sentinel. Do not confuse it with Defender for Cloud: Q376 says Defender for Cloud is a CNAPP, not a SIEM.

2026 update: Now called Microsoft Sentinel. Azure Cognitive Services is now Azure AI services. The exam uses the current names.

Answer explained

✔ **Azure Sentinel.**
Microsoft Sentinel (old name: Azure Sentinel) is Azure's cloud SIEM and SOAR (tool that automates responses to threats).

✖ **Azure Analysis Services.**
Analysis Services hosts data models for business reports, not security events.

✖ **Microsoft Purview Information Protection.**
Purview Information Protection classifies and labels data. It is not a SIEM.

✖ **Azure Cognitive Services.**
Cognitive Services (now Azure AI services) are ready-made AI APIs. They have nothing to do with SIEM.

EXAM STRATEGY
'SIEM' = Sentinel, whether it says Azure Sentinel or Microsoft Sentinel. Cross out Analysis Services and Cognitive Services at once.

IMPACT
SIEM questions come up often on AZ-900. Knowing the answer at once saves time for harder questions.

What · Where · When · Why · How

What	A SIEM collects and links security events to find and investigate threats.
Where	Microsoft Sentinel runs on a Log Analytics workspace (log store) in Azure.
When	When an organization needs one place to watch security across cloud and on-premises sources.
Why	Microsoft Sentinel is Azure's SIEM. The other options are for reports, data labels and AI.
How	Connect data sources, turn on analytics rules, investigate incidents, and automate responses with playbooks.

Exam tip
SIEM or SOAR in the question = Sentinel, every time.

Memory trick
S for SIEM, S for Sentinel.

Real-world example, use case and scenario

EXAMPLE
Analytics rule
A scheduled rule raises an alert when someone gets an admin role outside business hours.

USE CASE
MSSP
A managed security company runs Sentinel for 30 customer tenants by using Azure Lighthouse.

SCENARIO
Replacing on-prem SIEM
A manufacturer's on-premises SIEM needed costly new hardware. Moving to Sentinel removed the servers and added ready-made connectors for Microsoft 365 and Entra ID.

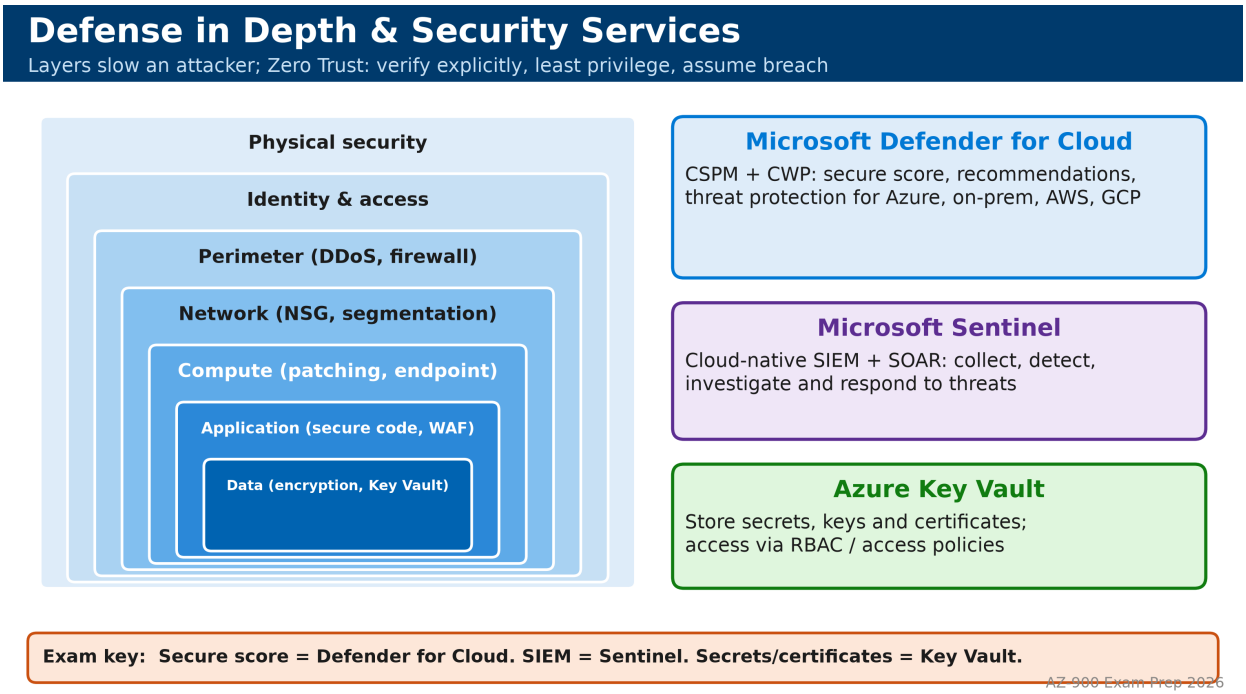


Figure Q178: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 179

AZURE MANAGEMENT & GOVERNANCE | **Compliance** v5 Q272

Your company implements [Azure policies] to automatically add a watermark to Microsoft Word documents that contain credit card information.

✖ A. No change is needed.

✖ B. DDoS protection.

✔ **C.** Microsoft Purview Information Protection.

✖ D. Microsoft Entra ID Identity Protection.

Correct answer: Microsoft Purview Information Protection.

Clarification (read this first): The part in brackets [Azure policies] may be wrong. Choose 'No change is needed' only if it is already right. Azure Policy controls Azure resources (like VMs and storage), not what is written inside a Word file. Adding a watermark to documents is a Purview Information Protection job.

Answer explained

✔ **Microsoft Purview Information Protection.**
Purview Information Protection sensitivity labels can find credit card numbers automatically and add watermarks to Word documents. Azure Policy only controls the settings of Azure resources.

✖ **No change is needed.**
Azure Policy checks Azure resource settings, not document content. It cannot add a watermark to a file.

✖ **DDoS protection.**
DDoS Protection stops network flood attacks on public IPs. It has nothing to do with documents.

✖ **Microsoft Entra ID Identity Protection.**
Entra ID Protection checks sign-in risk and user risk, not document content.

EXAM STRATEGY
'Watermark' and 'documents' point to Purview Information Protection. Azure Policy controls resources, not file content, so a change is needed.

IMPACT
Labelling credit card data automatically prevents leaks and PCI DSS fines. Resource policies cannot protect document content.

What · Where · When · Why · How

What	Microsoft Purview Information Protection classifies and protects documents with sensitivity labels.
Where	Labels are managed in the Microsoft Purview portal and applied in Office apps such as Word.
When	When documents with sensitive data, such as credit card numbers, need a mark or encryption.
Why	Its auto-labelling can add a watermark when it finds a credit card number. Azure Policy cannot.
How	Create a sensitivity label with a watermark, and set it to apply automatically when a credit card number is found.

Exam tip
Watermark, label, classify or encrypt documents = Purview Information Protection (old name: Azure Information Protection).

Memory trick
Purview stamps the page; Policy stamps rules on resources.

Real-world example, use case and scenario

EXAMPLE
Auto-label
A Word file with a Visa card number is labelled Confidential automatically and gets a CONFIDENTIAL watermark.

USE CASE
Payments company
A payment company auto-labels any document with card numbers, to meet PCI DSS rules.

SCENARIO
Wrong tool
A travel agency tried to write an Azure Policy for documents and found no such setting. They switched to Purview sensitivity labels, and files in Word, Excel and Outlook got watermarks within a week.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

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Figure Q179: Compliance, privacy and trust

About this figure

- Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.
- Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.
- Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.
- Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.
- The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 180

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q273

You have an Azure virtual network named VNET1 in a resource group named RG1. You assign the Azure Policy definition of Not Allowed Resource Type and specify that virtual networks are not an allowed resource type in RG1. VNET1 [...].

✖ A. is deleted automatically.

✖ B. is moved automatically to another resource group.

✔ **C.** continues to function normally.

✖ D. is now a read-only object.

Correct answer: continues to function normally.

Clarification (read this first): The question asks what happens to a VNet that ALREADY exists when a new policy blocks VNets. A deny policy only stops new creates and updates. The old VNET1 keeps working; it is only marked non-compliant (breaks the rule).

Answer explained

✔ **continues to function normally.**
Azure Policy deny rules block only new creates and updates. The existing VNET1 keeps working and is only reported as non-compliant.

✖ **Is deleted automatically.**
Azure Policy never deletes resources. It only blocks requests or reports that a resource breaks the rule.

✖ **Is moved automatically to another resource group.**
Azure Policy never moves resources. Moves are done by hand.

✖ **Is now a read-only object.**
Read-only comes from a ReadOnly lock, not from this policy.

EXAM STRATEGY
A deny policy acts only on new requests. Cross out delete, move and read-only. VNET1 keeps working.

IMPACT
Knowing that policies do not touch existing resources avoids panic. It also shows you must fix non-compliant resources yourself.

What · Where · When · Why · How

What	The 'Not allowed resource types' policy blocks creating certain resource types in a scope.
Where	Assigned through Azure Policy at a scope, such as resource group RG1.
When	When you want to stop new resources of a certain type from being deployed.
Why	Deny policies affect only new requests. VNET1 keeps working and is only marked non-compliant.
How	Policy checks create and update requests, and shows existing resources in the compliance view.

Exam tip
Deny policies affect future requests. Existing resources are marked non-compliant, not removed.

Memory trick
Policy is a guard at the door, not a cleaner inside the room.

Real-world example, use case and scenario

EXAMPLE
Compliance view
After the assignment, VNET1 shows as Non-compliant on the Policy compliance page, but it still carries traffic.

USE CASE
Sandbox clean-up
An agency assigns a deny policy, then uses the compliance report to find old resources and remove them by hand.

SCENARIO
Panic averted
A hospital architect feared a new policy would break the production VNet. A test showed the VNet kept working. Only new VNets in RG1 were blocked.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q180: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 181

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q274

You can create Group Polices in Microsoft Entra ID.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Group Policy (GPO: central settings for Windows PCs) belongs to on-premises Active Directory (AD DS), not to Microsoft Entra ID. Do not confuse the two: Entra ID is the cloud identity service and uses Intune for device settings.

Answer corrected in this edition. The previous PDF printed: *Yes.*. The printed answer Yes is wrong. Group Policy Objects are an Active Directory Domain Services feature. Microsoft Entra ID is a cloud identity service with no GPOs; device settings are managed with Intune profiles. Only Microsoft Entra Domain Services (a separate managed AD DS) supports GPOs. The answer key also says No.

Answer explained

✔ **No.**
Group Policy is an AD DS feature. Microsoft Entra ID has no GPOs. Cloud device settings are managed with Intune.

✖ **Yes.**
GPOs exist only in AD DS or Microsoft Entra Domain Services, not in Entra ID.

EXAM STRATEGY
GPOs belong to AD DS; Entra ID has none. Answer No, and think of Intune for cloud device settings.

IMPACT
Moving to Entra-only devices means rebuilding GPO settings in Intune. Missing this slows down cloud-first projects.

What · Where · When · Why · How

What	Group Policy Objects are an AD DS feature for setting up domain-joined computers from one place.
Where	Available in AD DS or Microsoft Entra Domain Services, not in Microsoft Entra ID.
When	For cloud-only Entra-joined devices, use Intune instead.
Why	Entra ID has no GPOs, no OUs (organizational units) and no Kerberos domain, so the statement is false.
How	Use Intune configuration profiles for Entra-joined devices, or Entra Domain Services if you must have GPOs.

Exam tip
GPOs, OUs, Kerberos, LDAP = AD DS or Entra Domain Services, not Entra ID.

Memory trick
Entra ID is flat in the cloud: no OU, no GPO.

Real-world example, use case and scenario

EXAMPLE
Intune instead
To turn on BitLocker (disk encryption) on Entra-joined laptops, an admin creates an Intune profile, not a GPO.

USE CASE
Legacy app lift
An accounting firm uses Entra Domain Services so its moved servers can still get GPOs.

SCENARIO
Missing GPO editor
A school admin looked for Group Policy in the Entra admin center and found none. Moving password and Wi-Fi settings into Intune profiles gave the same result for cloud-joined devices.

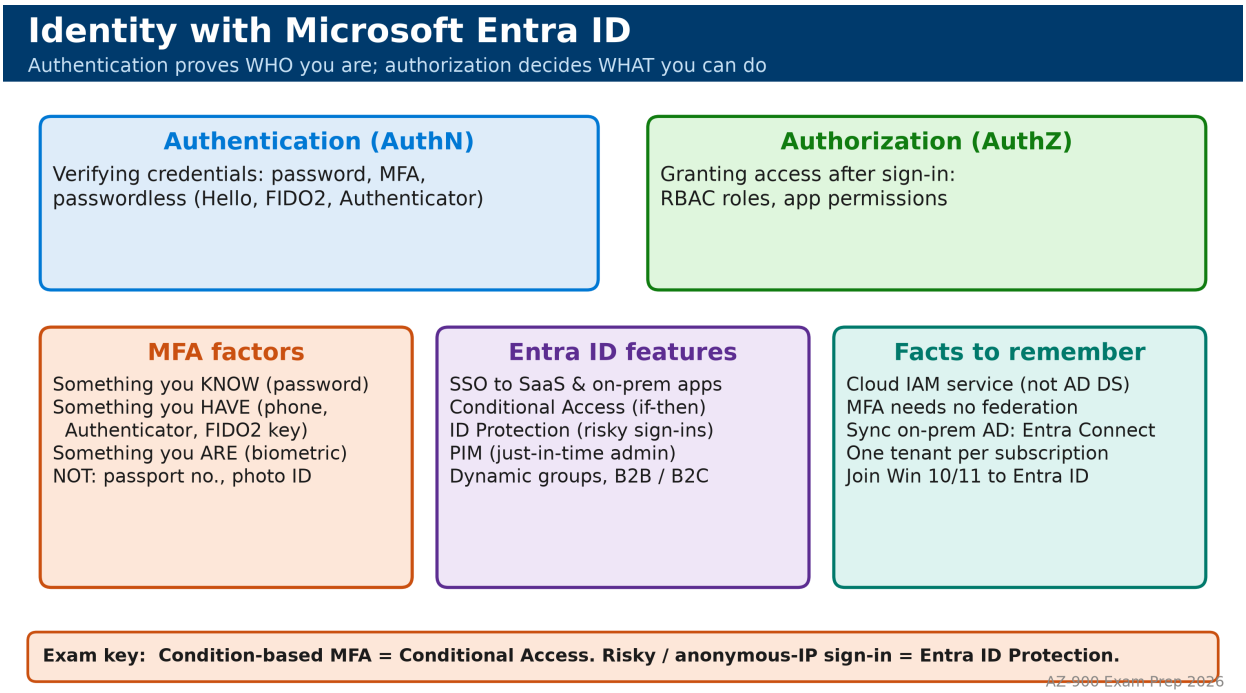


Figure Q181: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 182

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q275

You can join Windows 10 devices to Microsoft Entra ID.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): Watch the verb 'join'. Windows 10/11 devices CAN be joined to Entra ID, so Yes. Q183 asks the same about Android, and that answer is No, because Android can only be 'registered'.

Answer explained

✔ **Yes.**
Windows 10 and Windows 11 devices can be Microsoft Entra joined. Users then sign in with work accounts, with no on-premises domain.

✖ **No.**
Entra join is made for Windows 10/11. Only mobile devices are limited to registration.

EXAM STRATEGY
Windows 10/11 can be Entra joined. Answer Yes.

IMPACT
Entra join allows passwordless sign-in and Conditional Access without on-premises domain controllers. This cuts server costs.

What · Where · When · Why · How

What	Microsoft Entra join makes a Windows device a member of the Entra tenant, with no on-premises AD.
Where	Supported on Windows 10 and 11 Pro, Enterprise and Education.
When	When a company wants work laptops managed from the cloud, with no domain controllers.
Why	Windows 10 supports Entra join, so the statement is true.
How	Choose 'Set up for work' during Windows setup, or use Autopilot, then sign in with a work account.

Exam tip
Joined = Windows 10/11 (and Windows Server 2019+ Azure VMs). Registered = personal and mobile devices.

Memory trick
Windows joins the club; phones just sign in at the door.

Real-world example, use case and scenario

EXAMPLE
OOBE join
During Windows 11 setup, a user picks 'Set up for work' and signs in with user@fabrikam.com. The laptop is now Entra joined.

USE CASE
Remote-first startup
A remote-first startup ships laptops straight to staff. Autopilot joins them to Entra ID at first start.

SCENARIO
No domain controllers
A design studio closed its server room. Joining its Windows laptops to Entra ID kept single sign-on and policies with no domain controllers.

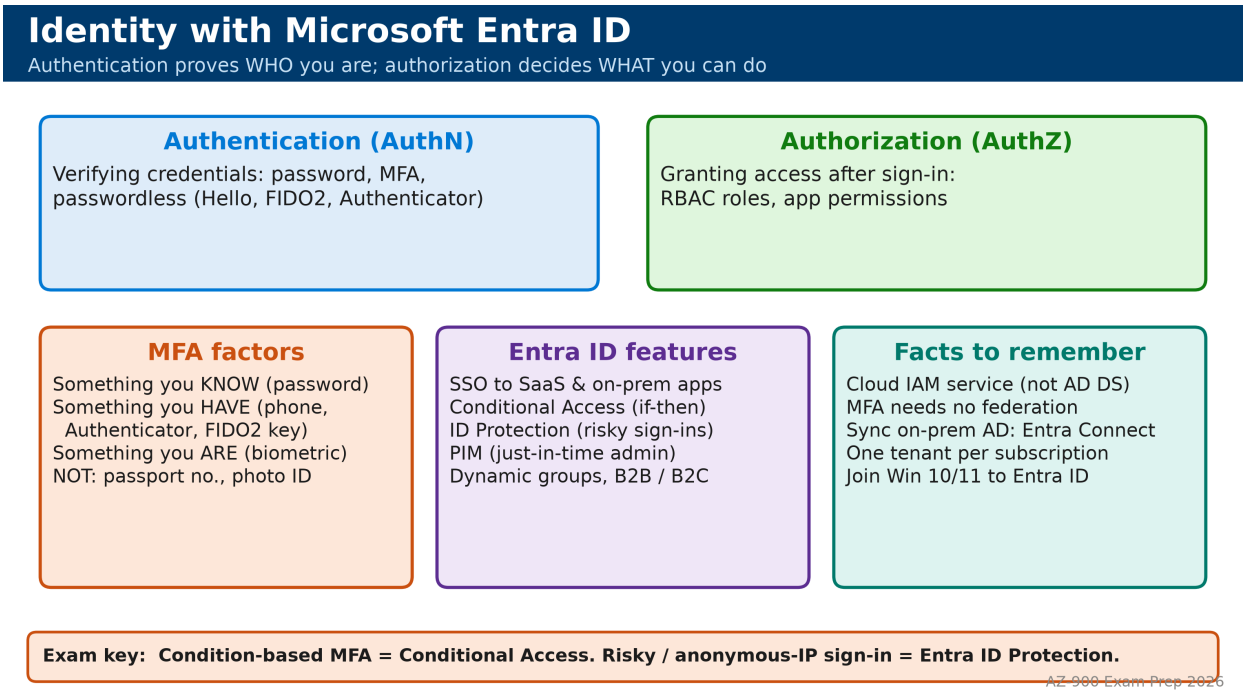


Figure Q182: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 183

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q276

You can join Android devices to Microsoft Entra ID.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch the verb 'join'. Only Windows 10/11 can be joined. Android can only be 'registered' (linked to Entra ID as a personal or mobile device). So the answer is No. Q182 (Windows 10 join) is Yes.

2026 update: Android devices can be Microsoft Entra registered (and enrolled in Intune), but not joined. Join is for Windows 10/11, so No is still the expected answer for Android.

Answer explained

✔ **No.**
Microsoft Entra join supports only Windows 10/11. Android devices can be Entra registered, not joined.

✖ **Yes.**
Android has no Entra join option. It uses Entra registration through the Company Portal or Authenticator app.

EXAM STRATEGY
Watch the verb: join is Windows only; Android registers. Answer No.

IMPACT
Using registration plus Intune for Android secures mobile access to company data. Trying an unsupported join wastes time.

What · Where · When · Why · How

What	Microsoft Entra registration links a personal or mobile device to Entra ID without joining it.
Where	Android and iOS devices register through the Company Portal or Microsoft Authenticator app.
When	When mobile devices need single sign-on and Conditional Access checks.
Why	Join is only for Windows; Android can only be registered, so the statement is false.
How	Install Company Portal or Authenticator and sign in. The device shows as Entra registered.

Exam tip
Watch the verb: join vs register. Android can register, not join.

Memory trick
Android registers like a guest; Windows joins like a member.

Real-world example, use case and scenario

EXAMPLE
Company Portal
A sales rep installs Company Portal on an Android phone. The device shows as Microsoft Entra registered.

USE CASE
Field technicians
A utility company registers its technicians' Android tablets, so Conditional Access can require safe devices.

SCENARIO
Join option missing
An IT lead looked for a Join option in Android settings for 300 phones. There is none. Enrolling them with Intune and registration met the goal.

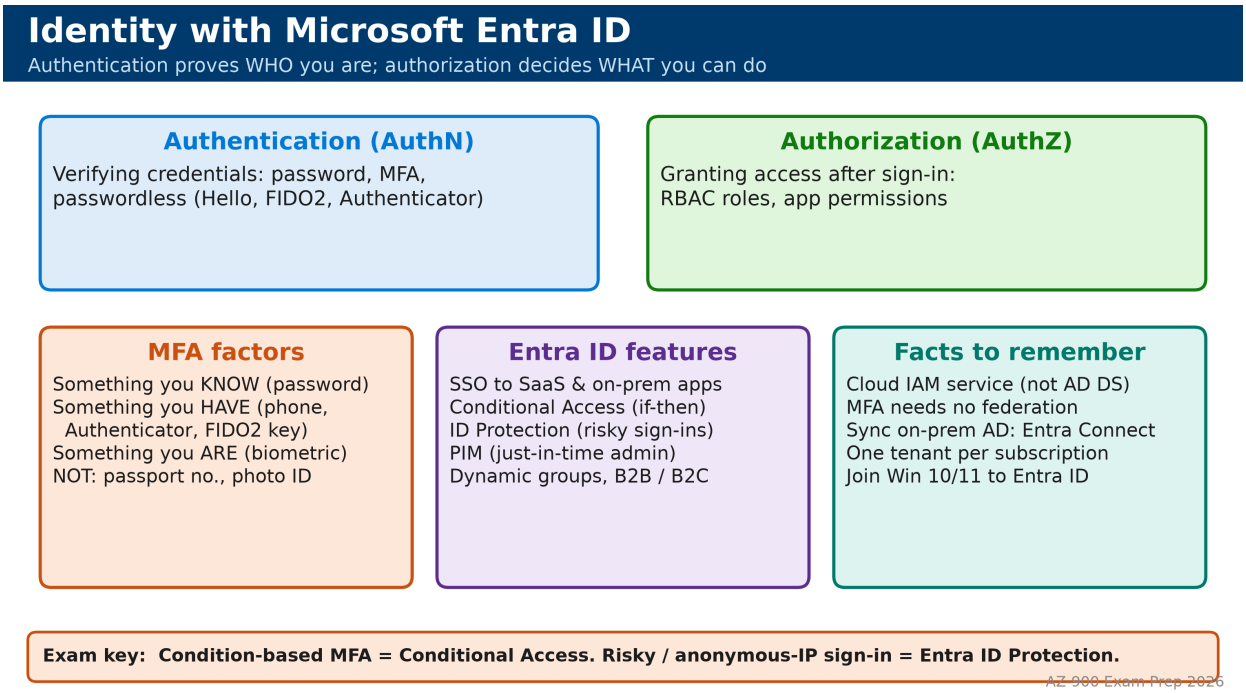


Figure Q183: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 184

AZURE MANAGEMENT & GOVERNMENT | **Compliance** v5 Q277, Q143

The [...] explains what data Microsoft processes, how Microsoft processes the data, and the purpose of processing the data.

✔ **A.** Microsoft Online Services Privacy Statement.

✖ B. Microsoft Online Services Terms.

✖ C. Microsoft Online Service Level Agreement.

✖ D. Online Subscription Agreement for Microsoft Azure.

Correct answer: Microsoft Online Services Privacy Statement.

Clarification (read this first): The question describes a document that says WHAT data Microsoft uses, HOW, and WHY. That is the Privacy Statement. Its current name is the Microsoft Privacy Statement. Do not pick the SLA, which is about uptime.

2026 update: Today this is published as the Microsoft Privacy Statement.

Answer explained

✔ **Microsoft Online Services Privacy Statement.**
The Microsoft Privacy Statement (old name: Online Services Privacy Statement) explains what personal data Microsoft processes, how, and for what purpose.

✖ **Microsoft Online Services Terms.**
The Online Services Terms (now Product Terms and DPA) are legal rules for use, not a description of why data is processed.

✖ **Microsoft Online Service Level Agreement.**
The SLA (uptime promise) sets uptime targets and service credits, not data processing.

✖ **Online Subscription Agreement for Microsoft Azure.**
The Online Subscription Agreement is the purchase contract for an Azure subscription.

EXAM STRATEGY
'What data, how, purpose' points to the Privacy Statement. Cross out SLA (uptime), Terms (rules of use) and Subscription Agreement (contract).

IMPACT
Privacy teams quote this document in GDPR records. Choosing the wrong document weakens compliance proof.

What · Where · When · Why · How

What	The Microsoft Privacy Statement describes how Microsoft collects and uses personal data.
Where	Published on Microsoft's privacy website and linked from the Microsoft Trust Center.
When	When you need to know what data Microsoft processes, how and why.
Why	It covers the data types, how they are used and why. Terms, SLAs and purchase agreements do not.
How	Read its sections on what data is collected, how it is used, and user choices.

Exam tip
What data, how, why = Privacy Statement. Uptime = SLA. Rules of use = Terms / Product Terms.

Memory trick
The Privacy Statement answers the three Ws: What, hoW, Why.

Real-world example, use case and scenario

EXAMPLE
Diagnostic data
The Privacy Statement explains that Microsoft collects diagnostic data to keep products secure and up to date.

USE CASE
Vendor assessment
A law firm's privacy officer reads the Privacy Statement before approving a move to the cloud.

SCENARIO
Customer question
A charity donor asked what Microsoft does with their data. The data protection officer pointed to the Privacy Statement, not the SLA, which only covers uptime.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

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Figure Q184: Compliance, privacy and trust

About this figure

- Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.
- Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.
- Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.
- Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.
- The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 185

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q278

An Azure Policy initiative definition is a [...].

✔ **A.** collection of policy definitions.

✖ B. collection of Azure Policy definition assignments.

✖ C. group of Azure Blueprints definitions.

✖ D. group of role-based access control (RBAC) role assignments.

Correct answer: collection of policy definitions.

Clarification (read this first): Watch 'definitions' vs 'assignments'. An initiative groups policy DEFINITIONS (the rules). You then assign the whole initiative to a scope. It does not group assignments.

Answer explained

✔ **collection of policy definitions.**
An initiative (policy set definition) groups several policy definitions toward one goal. They are then assigned and tracked as one unit.

✖ **Collection of Azure Policy definition assignments.**
Initiatives group definitions, not assignments. The initiative itself is then assigned to a scope.

✖ **Group of Azure Blueprints definitions.**
Blueprints was a separate service and is now retired. Initiatives contain policy definitions, not blueprints.

✖ **Group of role-based access control (RBAC) role assignments.**
RBAC role assignments give permissions. Initiatives contain policy rules, not role assignments.

EXAM STRATEGY
Initiatives group definitions, not assignments. Cross out the Blueprints and RBAC options at once.

IMPACT
With initiatives, you assign many controls at once, such as a set of legal rules, and track them as one compliance score.

What · Where · When · Why · How

What	An Azure Policy initiative is a named set of policy definitions grouped for one goal.
Where	Created in Azure Policy under Definitions. Assigned at management group, subscription or resource group scope.
When	When several related policies, such as the controls of a compliance standard, should be managed together.
Why	Grouping definitions lets one assignment and one compliance score cover them all.
How	Add policy definitions to an initiative definition, then assign the initiative to a scope.

Exam tip
Definition -> grouped into an initiative -> assigned at a scope. Definitions are grouped, not assignments.

Memory trick
An initiative is a playlist of policy songs.

Real-world example, use case and scenario

EXAMPLE
Tagging initiative
Three policies (require cost-center tag, copy tag from resource group, require environment tag) are put into one Tagging initiative.

USE CASE
PCI baseline
An online shop assigns the PCI DSS initiative at a management group to track many controls together.

SCENARIO
Assignment sprawl
An energy firm had 40 separate policy assignments that were hard to track. Putting them into two initiatives cut them to two assignments and gave one compliance score.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

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Figure Q185: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 186

AZURE MANAGEMENT & GOVERNANCE — **Compliance** v5 Q279

General Data Protection Regulation (GDPR) defines data protection and privacy rules.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
GDPR is the EU law that sets rules for using personal data. It also gives people privacy rights, such as the right to see or delete their data.

✖ **No.**
Setting data protection and privacy rules is the main purpose of GDPR.

EXAM STRATEGY
GDPR is a privacy law by definition. Answer Yes.

IMPACT
GDPR fines can reach 4 percent of a company's global yearly revenue. Architects must understand where data is stored and how it is used.

What · Where · When · Why · How

What	GDPR is the EU General Data Protection Regulation. It controls how personal data is used.
Where	It applies in the EU and to companies worldwide that use the data of people in the EU.
When	When a company handles personal data of people in the EU.
Why	It sets data protection rules and privacy rights, so the statement is true.
How	It sets legal reasons for using data and gives rights such as access, deletion and moving data.

Exam tip
GDPR = EU privacy law. HIPAA = US health data. FedRAMP = US government cloud.

Memory trick
GDPR: Guarding Data, Protecting Residents of the EU.

Real-world example, use case and scenario

EXAMPLE
Right to erasure
A customer asks an online shop to delete their account data under GDPR Article 17.

USE CASE
Airline loyalty
An airline maps how its loyalty program data moves, to meet GDPR rules.

SCENARIO
Subject access request
A gym chain got a request from a member to see their data, with one month to reply. Using Purview data maps, it found the data in three systems and replied on time.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

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Figure Q186: Compliance, privacy and trust

About this figure

- Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.
- Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.
- Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.
- Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.
- The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 187

AZURE MANAGEMENT & GOVERNANCE

Compliance
v5 Q280

General Data Protection Regulation (GDPR) applies to companies that offer goods or services to individuals in the EU.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The question asks if GDPR applies to a company (anywhere in the world) that sells to people in the EU. It does. What matters is where the PEOPLE are, not where the company is.

Answer explained

✔ **Yes.**
GDPR Article 3 applies to any company in the world that offers goods or services to people in the EU, or tracks them.

✖ **No.**
GDPR reaches beyond the EU. Selling to people in the EU brings a company under GDPR, wherever the company is.

EXAM STRATEGY
The location of the people decides it, not the company. Answer Yes.

IMPACT
Companies outside the EU that sell to EU residents must follow GDPR and can be fined. Ignoring this is a costly mistake.

What · Where · When · Why · How

What	GDPR's territorial scope says which companies must follow the law.
Where	Article 3 extends GDPR to companies outside the EU that target people in the EU.
When	When a company outside the EU sells to, or tracks, people in the EU.
Why	Offering goods or services to people in the EU brings a company under GDPR, so the statement is true.
How	It depends on whose data is used, not where the company is based.

Exam tip
The company's location does not matter. The location of the people whose data is used does.

Memory trick
Sell to Europe, follow Europe's privacy rules.

Real-world example, use case and scenario

EXAMPLE
US web shop
A US store that ships to France with prices in euros must follow GDPR.

USE CASE
Indian SaaS
An Indian HR software company with German clients signs data protection agreements and names an EU representative.

SCENARIO
Assumed exempt
A Canadian app maker thought GDPR did not apply. After launching in Spain, a regulator asked questions, and the company added consent and data-export features.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

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Figure Q187: Compliance, privacy and trust

About this figure

- Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.
- Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.
- Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.
- Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.
- The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 188

AZURE MANAGEMENT & GOVERNANCE

Compliance
v5 Q281

Azure can be used to build a General Data Protection Regulation (GDPR)-compliant infrastructure.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): Watch the words 'can be used to build'. Azure gives you the tools to meet GDPR, so Yes. It does not make you compliant automatically; you still have to set it up correctly.

Answer explained

✔ **Yes.**
Microsoft commits to GDPR as a processor (handles data for you). It provides EU regions, encryption and compliance tools, so customers can build GDPR-compliant solutions on Azure.

✖ **No.**
Azure offers GDPR promises in its contracts and compliance tools. Customers still set up their own compliance.

EXAM STRATEGY
'Can be used to build' decides it. Azure gives the tools; the customer reaches compliance. Answer Yes.

IMPACT
Under shared responsibility, Azure gives compliant building blocks. Customers must set them up correctly to pass GDPR audits.

What · Where · When · Why · How

What	Azure provides services and contract promises that support GDPR compliance.
Where	The promises are in Microsoft's Data Protection Addendum. Tools include Purview Compliance Manager.
When	When a company must build systems that meet GDPR duties.
Why	Microsoft acts as a GDPR processor and supplies controls, so the statement is true.
How	Use EU regions, encryption, access controls and Compliance Manager checks, under shared responsibility.

Exam tip
Azure can be used to build compliant systems. It does not make you compliant automatically.

Memory trick
Azure gives the bricks; you build the compliant house.

Real-world example, use case and scenario

EXAMPLE
EU region choice
Putting storage in West Europe, with your own encryption keys and private endpoints, supports GDPR controls.

USE CASE
Telehealth platform
A Dutch online health service uses Compliance Manager's GDPR template to track its tasks.

SCENARIO
Audit readiness
A Polish insurer had to prove its GDPR controls. Microsoft's audit reports from the Service Trust Portal, plus its own Compliance Manager tasks, satisfied the auditor.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
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AZ-900 Exam Prep 2026

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Question 189

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q283

Azure Government is operated by Microsoft.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): Pair this with Q303. Azure Government (US) is run by Microsoft, so Yes. Azure in China is run by a local partner company (21Vianet), so Q303 is No.

Answer explained

✔ **Yes.**
Azure Government is a physically separate cloud run by Microsoft, with screened US citizens as staff.

✖ **No.**
Microsoft itself runs Azure Government. Only the China cloud is run by a partner.

EXAM STRATEGY
Azure Government = run by Microsoft with screened US staff. Answer Yes.

IMPACT
US public sector workloads that need FedRAMP High or ITAR rely on Azure Government's separation and screened staff.

What · Where · When · Why · How

What	Azure Government is a physically separate sovereign cloud (cloud for one country) for US government workloads.
Where	Dedicated US regions such as US Gov Virginia and US Gov Arizona.
When	When US agencies need FedRAMP High, DoD IL5, CJIS or ITAR compliance.
Why	Microsoft runs it with screened US staff, so the statement is true.
How	Approved customers get separate subscriptions in the separate Government cloud.

Exam tip
Pair this with the China question: Government = Yes, China = No.

Memory trick
US Gov cloud, US Microsoft staff.

Real-world example, use case and scenario

EXAMPLE
Dedicated regions
US Gov Virginia and US Gov Arizona are Azure Government regions run by Microsoft.

USE CASE
State police
A state police agency stores criminal justice (CJIS) data in Azure Government.

SCENARIO
Vendor vetting
A county asked who would handle its data. Azure Government, run by Microsoft with screened US staff, met its policy. A cloud run by another company would not.

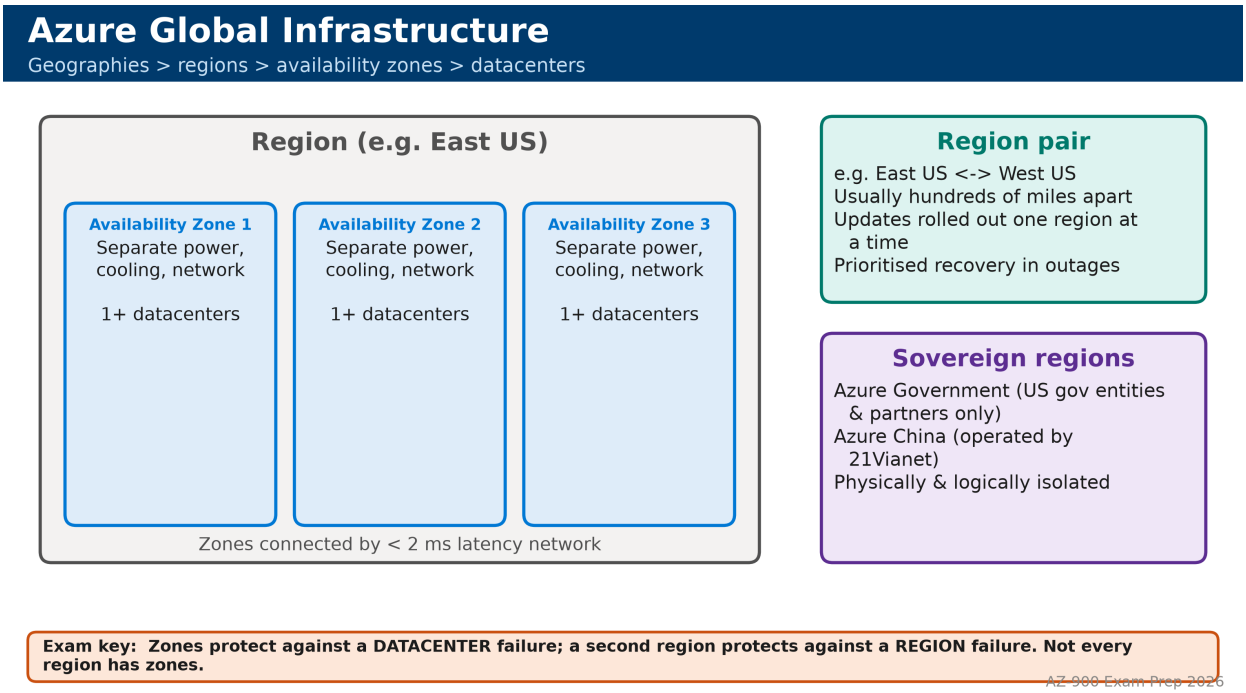


Figure Q189: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

Question 190

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q285

An Azure resource can have multiple Delete locks.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The question asks if ONE resource can have MORE THAN ONE Delete lock. It can, because each lock is a separate object with its own name. Related lock questions: Q191 (locks are inherited, Yes) and Q192 (add a Delete lock on top of a Read-only lock, Yes).

Answer explained

✔ **Yes.**
Each lock is a separate object with its own name. So a resource can have several Delete (CanNotDelete) locks at the same time.

✖ **No.**
There is no limit of one lock per type. Each lock just needs a unique name.

EXAM STRATEGY
Locks are separate named objects, not one on/off switch. So 'multiple' is fine. Only two locks with the same name at one scope would clash.

IMPACT
Teams often add their own Delete lock next to an existing one. Knowing this avoids failed automated deployments that expect only one lock per resource.

What · Where · When · Why · How

What	A resource lock stops Azure resources from being deleted (CanNotDelete) or changed (ReadOnly).
Where	Set at subscription, resource group or resource scope, on the Locks page.
When	When different teams each want to protect the same resource on their own.
Why	Each lock is a separate named object, so many Delete locks are allowed.
How	Create locks with unique names. All of them must be removed before the resource can be deleted.

Exam tip
Many locks are allowed. The strictest one applies.

Memory trick
You can put many padlocks on the same gate.

Real-world example, use case and scenario

EXAMPLE
Two locks
A storage account has two Delete locks, named Lock-App and Lock-Finance.

USE CASE
Joint ownership
A TV company's media team and security team each put a lock on the archive storage account.

SCENARIO
Delete still fails
An engineer removed one Delete lock, but the delete still failed. Security had added a second lock. After removing both (with approval), the resource could be deleted.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

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Figure Q190: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 191

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q286

An Azure resource inherits locks from its resource group.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): Inherit means a resource gets the lock from the resource group it is in. Locks ARE inherited, so Yes. Do not confuse this with tags: Q129 asks if resources inherit tags, and that answer is No.

Answer explained

✔ **Yes.**
Locks set at subscription or resource group scope apply to all resources in that scope. This includes resources added later.

✖ **No.**
Locks flow down from the parent to the resources inside it.

EXAM STRATEGY
Inheritance flows down: locks, RBAC and Policy pass from management group to subscription to resource group to resource. Only tags do not. 'Inherits' + lock = Yes.

IMPACT
One resource group lock quietly protects every resource added later. A forgotten lock can block a planned teardown or a scale change.

What · Where · When · Why · How

What	Lock inheritance means resources inside a scope are protected by the lock on that scope.
Where	Locks at subscription or resource group scope pass to all resources inside them.
When	When a whole resource group of production resources must be protected.
Why	Resources inherit locks from their resource group, so the statement is true.
How	Add one lock on the resource group. Old and new resources inside it are covered.

Exam tip
Inherited: RBAC, Policy, locks. Not inherited: tags.

Memory trick
Lock the house and every room is locked.

Real-world example, use case and scenario

EXAMPLE
RG lock
A Delete lock on RG-Prod protects all 40 resources in it.

USE CASE
Core networking
An airline locks RG-Hub-Network, so no gateway or firewall can be deleted.

SCENARIO
New VM protected
A VM was added to a locked resource group. Later a script tried to delete it. The inherited lock blocked the delete and saved production.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

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Figure Q191: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 192

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q287

If an Azure resource has a Read-only lock, you can add a Delete lock to the resource.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The question asks if you can ADD a second lock (Delete) to a resource that already has a Read-only lock. You can. A Read-only lock blocks changes to the resource itself, not adding or removing locks.

Answer explained

✔ **Yes.**
A ReadOnly lock does not block managing locks, so you can add a Delete lock. The strictest lock applies.

✖ **No.**
Adding a lock is a separate locks operation (Microsoft.Authorization/locks). A ReadOnly lock does not block it.

EXAM STRATEGY
ReadOnly blocks changes to the resource, not to the lock objects. If the question is about adding a lock, answer Yes.

IMPACT
Stacked locks are common in regulated companies. If admins misread which lock wins, they may remove the wrong one and leave production open to deletion.

What · Where · When · Why · How

What	ReadOnly and Delete are the two Azure lock types. Both can be on one resource.
Where	Both are managed on the resource's Locks page.
When	When a Delete lock should stay after a temporary ReadOnly lock is removed.
Why	A ReadOnly lock does not block lock management, so a Delete lock can be added.
How	Add the Delete lock next to the ReadOnly lock. The strictest lock applies.

Exam tip
Different lock types can stack. The strictest one wins.

Memory trick
ReadOnly is the main door lock; you can still add a chain.

Real-world example, use case and scenario

EXAMPLE
Stacked locks
A Key Vault has Lock-RO (ReadOnly) and Lock-ND (CanNotDelete).

USE CASE
Freeze period
A tax office adds a temporary ReadOnly lock during tax season, on top of its permanent Delete lock.

SCENARIO
Freeze lifted
After year-end, an accounting firm removed the ReadOnly lock. A Delete lock had also been added, so the resources stayed safe from deletion.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
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Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

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Figure Q192: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 193

AZURE MANAGEMENT & GOVERNANCE — Identity v5 Q288

Authorization to access Azure resources can be provided only to Microsoft Entra ID users.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch the word 'only'. Access can also be given to groups, apps (service principals) and managed identities, not only to users. So the answer is No. Q194 is a similar question with no 'only', and that one is Yes.

Answer explained

✔ **No.**
Azure RBAC roles can be given to users, groups, service principals (app identities) and managed identities (identities Azure manages for a resource). Not only users.

✖ **Yes.**
Groups, service principals and managed identities can also get Azure RBAC access.

EXAM STRATEGY
Signal word: 'only'. One counter-example breaks it: managed identities and service principals get RBAC roles every day. Absolute words in identity questions usually mean No.

IMPACT
Giving apps a managed identity instead of a user account removes stored passwords. If you think only users get access, teams end up sharing passwords, and they leak.

What · Where · When · Why · How

What	Azure RBAC gives roles to security principals (anything that can be given access).
Where	Assigned at management group, subscription, resource group or resource scope.
When	When apps, automation or teams need access, not only single users.
Why	Groups, service principals and managed identities can also get access, so the statement is false.
How	Create a role assignment: choose a role, a principal of any type, and a scope.

Exam tip
Watch the word 'only'. Principals = user, group, service principal, managed identity.

Memory trick
U-G-S-M: Users, Groups, Service principals, Managed identities.

Real-world example, use case and scenario

EXAMPLE
Managed identity
A Function App's managed identity gets the Storage Blob Data Reader role on a container.

USE CASE
CI/CD pipeline
A gaming company's GitHub Actions workflow uses a service principal with Contributor on RG-Build.

SCENARIO
No user needed
A weather company stored a user's password in a script to read files. Switching to a managed identity removed the password and kept the access.

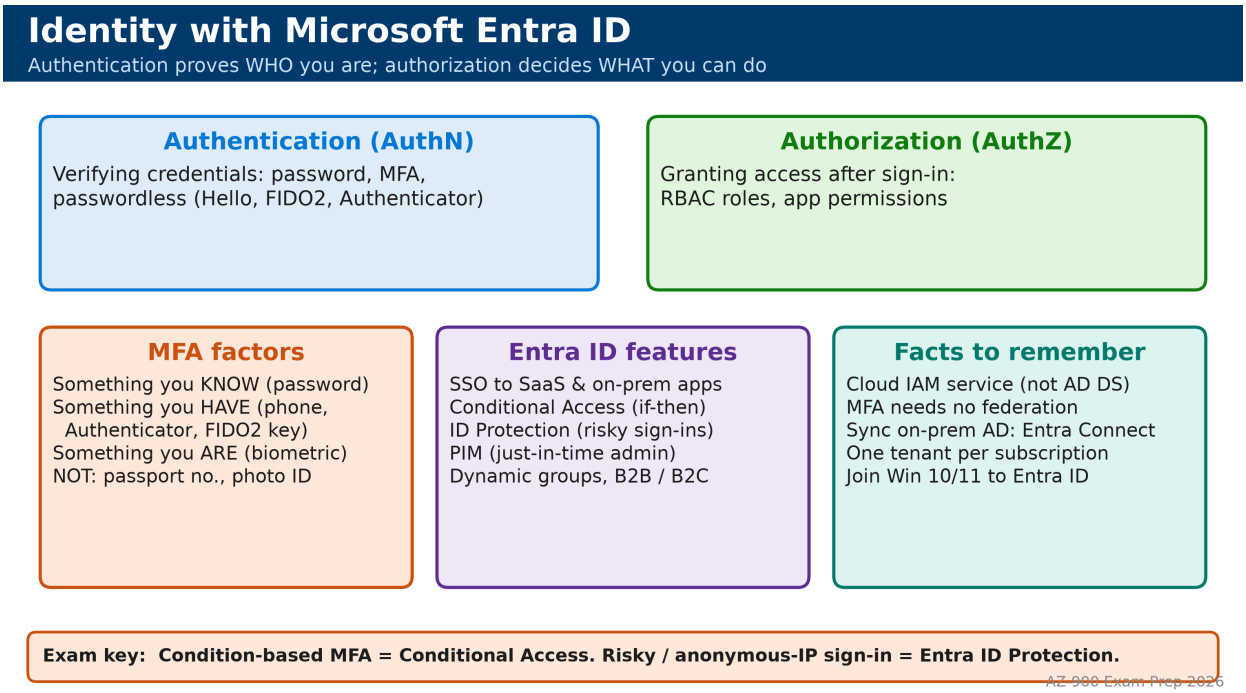


Figure Q193: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 194

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q289

Identities stored in Microsoft Entra ID, third-party cloud services, and on-premises Active Directory can be used to access Azure resources.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The question lists three sources of identities: Entra ID (cloud), third-party cloud services (such as Google accounts), and on-premises Active Directory. All three can reach Azure through Entra ID, so Yes. Compare Q193, which says 'only Entra ID users' and is No.

Answer explained

✔ **Yes.**
Entra ID cloud accounts, on-premises AD accounts that are synced or federated (linked), and external B2B (guest) accounts can all be given access to Azure.

✖ **No.**
Entra ID supports hybrid sync, federation and B2B guests from other identity providers.

EXAM STRATEGY
Entra ID syncs and federates. On-premises AD syncs through Entra Connect. Other clouds federate or join as B2B guests. All three paths lead to Entra ID.

IMPACT
Most large companies use hybrid identity. Knowing about sync and federation gives staff one sign-in, instead of extra cloud accounts to manage.

What · Where · When · Why · How

What	Azure accepts identities from Entra ID, on-premises AD and outside providers.
Where	On-premises identities come through Entra Connect. Outside ones come through B2B collaboration.
When	When partners or on-premises users need access to Azure resources.
Why	Hybrid sync, federation and B2B make these identities usable, so the statement is true.
How	Sync or federate AD, invite outside users as guests, then give RBAC roles.

Exam tip
Entra ID is the hub. Other identity sources reach Azure through it.

Memory trick
All roads (cloud, on-premises, partner) lead through Entra ID.

Real-world example, use case and scenario

EXAMPLE
Google guest
A consultant with a Gmail account is invited as a B2B guest and gets Reader on a resource group.

USE CASE
Hybrid manufacturer
A manufacturer syncs 8,000 AD users, so they use the same sign-in for Azure.

SCENARIO
Partner access
An architecture firm needed a partner's engineers in its Azure project. B2B invitations let them use their own company accounts, with no new passwords to manage.

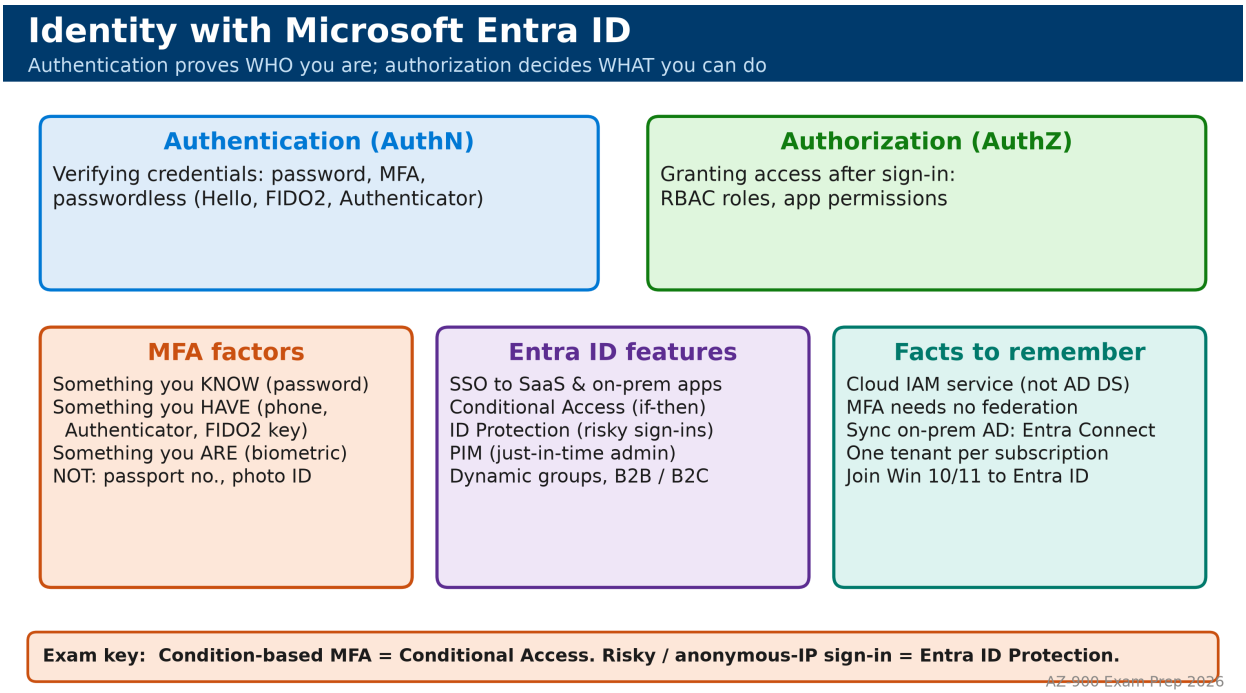


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- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 195

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q290

Azure has built-in authentication and authorization services that provide secure access to Azure resources.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Microsoft Entra ID provides authentication (checks who you are). Azure RBAC provides authorization (decides what you can do). Both are built into Azure.

✖ **No.**
Entra ID and Azure RBAC are built-in Azure services. No third-party product is needed.

EXAM STRATEGY
Entra ID (authentication) plus Azure RBAC (authorization) are built in. 'Built-in' and 'secure access' together = Yes.

IMPACT
Using built-in Entra ID and RBAC avoids adding a third-party identity product. This cuts cost and avoids mixed access rules across subscriptions.

What · Where · When · Why · How

What	Azure's built-in identity services are Microsoft Entra ID for authentication and Azure RBAC for authorization.
Where	Both work with Azure Resource Manager and every Azure subscription.
When	Every time users or apps sign in and work with Azure resources.
Why	They are built-in Azure services, so the statement is true.
How	Entra ID checks identity with passwords and MFA. RBAC role assignments decide what actions are allowed.

Exam tip
AuthN (authentication) = who you are (Entra ID).
AuthZ (authorization) = what you can do (RBAC).

Memory trick
Entra checks the ID card; RBAC checks the access list.

Real-world example, use case and scenario

EXAMPLE
Sign-in then act
Sam signs in with MFA (authentication). His role then lets him only restart VMs (authorization).

USE CASE
University tenant
A university uses Entra ID for 40,000 students and RBAC to give each lab limited access.

SCENARIO
No extra IdP
A small accounting startup planned to buy a third-party identity product. They found that Entra ID Free plus RBAC already met their Azure needs.

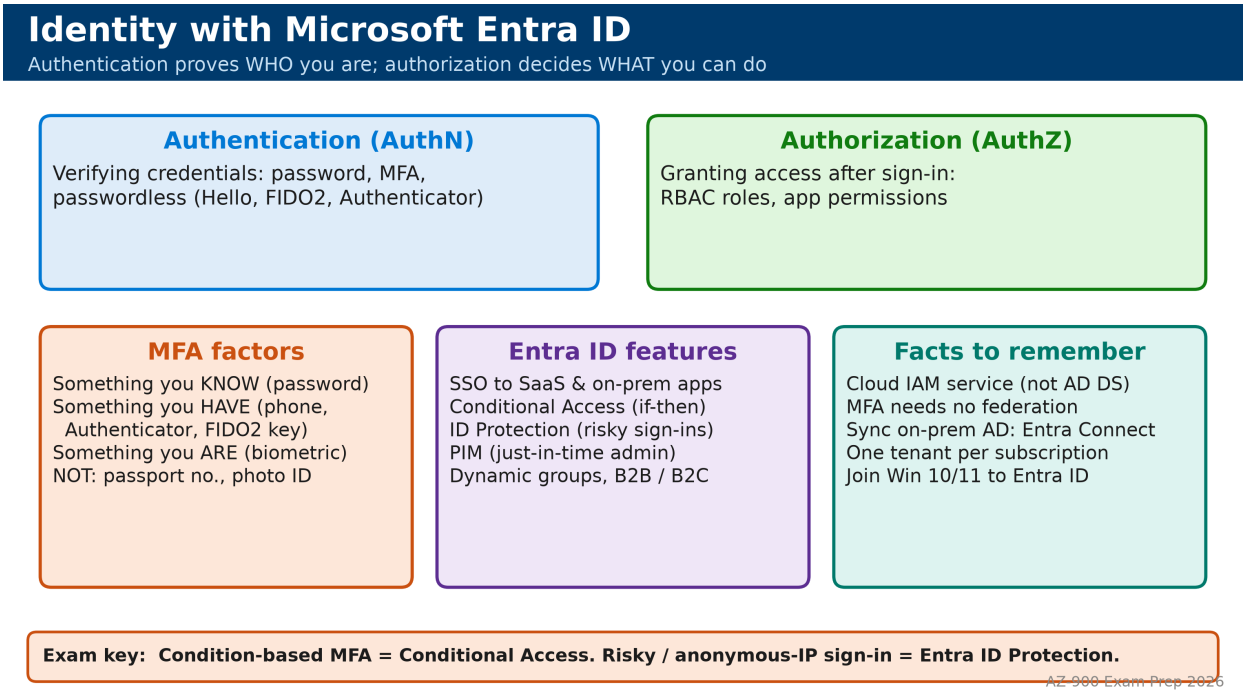


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- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Identities stored in an on-premises Active Directory can be synchronized to Microsoft Entra ID.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Microsoft Entra Connect Sync or Cloud Sync copies on-premises AD users, groups and password hashes (scrambled passwords) to Entra ID.

✖ **No.**
Directory sync is the standard way to set up hybrid identity.

EXAM STRATEGY
Microsoft Entra Connect Sync or Cloud Sync exists for exactly this. 'Synchronized' + on-premises AD = Yes. Sync goes from on-premises to the cloud.

IMPACT
Sync gives users one identity and password for on-premises and cloud. This cuts password-reset calls and lets you disable a leaver everywhere at once.

What · Where · When · Why · How

What	Directory sync copies on-premises AD objects to Microsoft Entra ID.
Where	Microsoft Entra Connect Sync on a server, or small Entra Cloud Sync agents.
When	When users should have one identity for on-premises and cloud resources.
Why	Sync is the standard hybrid identity method, so the statement is true.
How	The sync engine copies users, groups and password hashes on a schedule, every 30 minutes by default.

Exam tip
Sync tool = Microsoft Entra Connect (old name: Azure AD Connect) or Cloud Sync.

Memory trick
Connect connects on-premises AD to the cloud.

Real-world example, use case and scenario

EXAMPLE
Password hash sync
Entra Connect syncs every 30 minutes, so a new employee created in AD soon appears in Entra ID.

USE CASE
Hospital group
A hospital group syncs AD, so doctors use one login for Microsoft 365 and Azure apps.

SCENARIO
Duplicate accounts
A builder had separate cloud and AD accounts, and staff mixed up passwords. Entra Connect matched the accounts and gave everyone one identity.

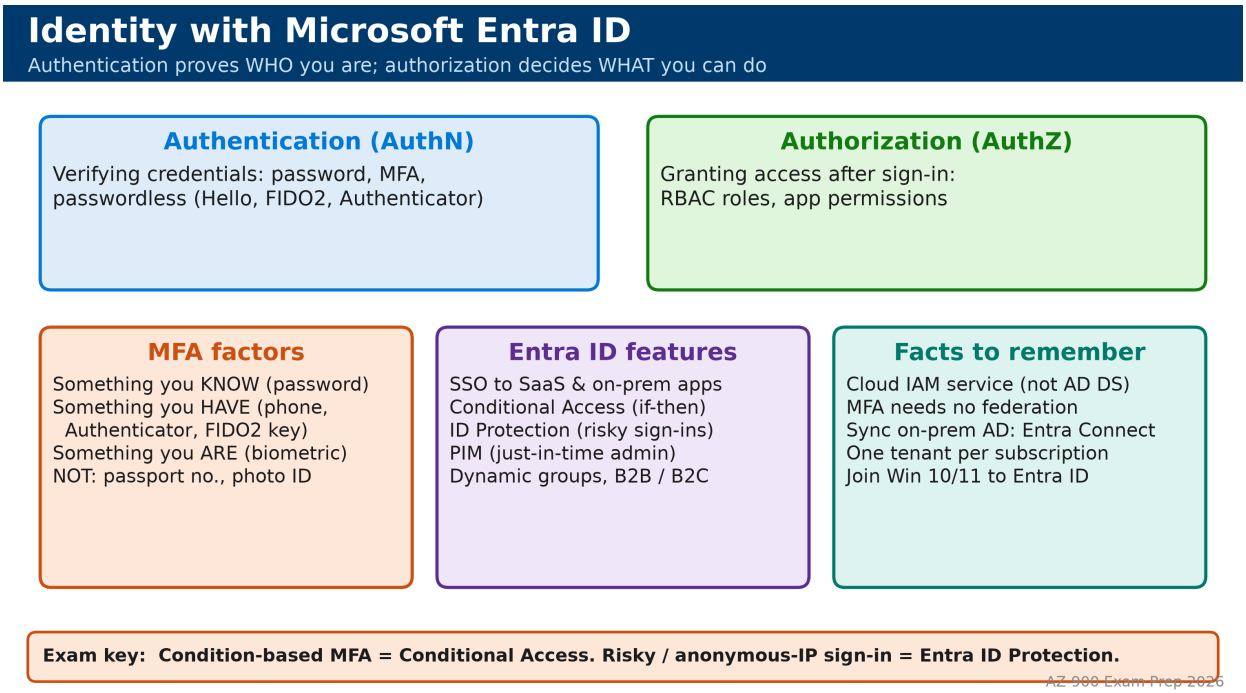


Figure Q196: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 197

AZURE MANAGEMENT & GOVERNANCE

Security
v5 Q292, Q258

You can view your company regulatory compliance report from [...].

✖ A. Azure Advisor.

✖ B. Azure Analysis Services.

✖ C. Azure Monitor.

✔ **D.** Microsoft Defender for Cloud.

Correct answer: Microsoft Defender for Cloud.

Clarification (read this first): The question asks where to see how YOUR OWN Azure resources meet rules such as PCI DSS or ISO 27001. That is Defender for Cloud. Look-alikes: Q82 (evaluate if your environment meets regulations) answers Compliance Manager, and Q394 (download Microsoft's own audit reports) answers Service Trust Portal.

Answer explained

✔ **Microsoft Defender for Cloud.**
Defender for Cloud's regulatory compliance dashboard checks your resources against standards such as PCI DSS and ISO 27001, and can export reports.

✖ **Azure Advisor.**
Advisor gives best-practice tips, not compliance reports against legal standards.

✖ **Azure Analysis Services.**
Analysis Services is a business reporting tool. It has nothing to do with compliance.

✖ **Azure Monitor.**
Azure Monitor tracks health and performance metrics and logs, not legal controls.

EXAM STRATEGY
'Regulatory compliance' = Defender for Cloud's compliance dashboard. Cross out Advisor (tips), Monitor (health data) and Analysis Services (reports) at once.

IMPACT
Auditors ask for ongoing proof against standards such as ISO 27001 or PCI DSS. This dashboard saves weeks of collecting proof by hand.

What · Where · When · Why · How

What	The regulatory compliance dashboard shows how your resources meet standards.
Where	In Microsoft Defender for Cloud, under Regulatory compliance.
When	When you need a report of your Azure resources against PCI DSS, ISO 27001 or similar.
Why	Defender for Cloud maps its checks to the controls in each standard and exports reports. The others do not.
How	Turn on standards in Defender for Cloud, review passing and failing controls, then download the report.

Exam tip
Your resources vs a standard = Defender for Cloud. Microsoft's own audit reports = Service Trust Portal.

Memory trick
Defender marks your homework against the rule book.

Real-world example, use case and scenario

EXAMPLE
PCI view
The dashboard shows that 18 of 22 PCI DSS controls pass for the subscription.

USE CASE
Payment gateway
A payments startup exports the PCI report every month for its auditor.

SCENARIO
Board request
A pension fund's board asked for its ISO 27001 status. The security chief added the standard in Defender for Cloud and exported a report the same afternoon.

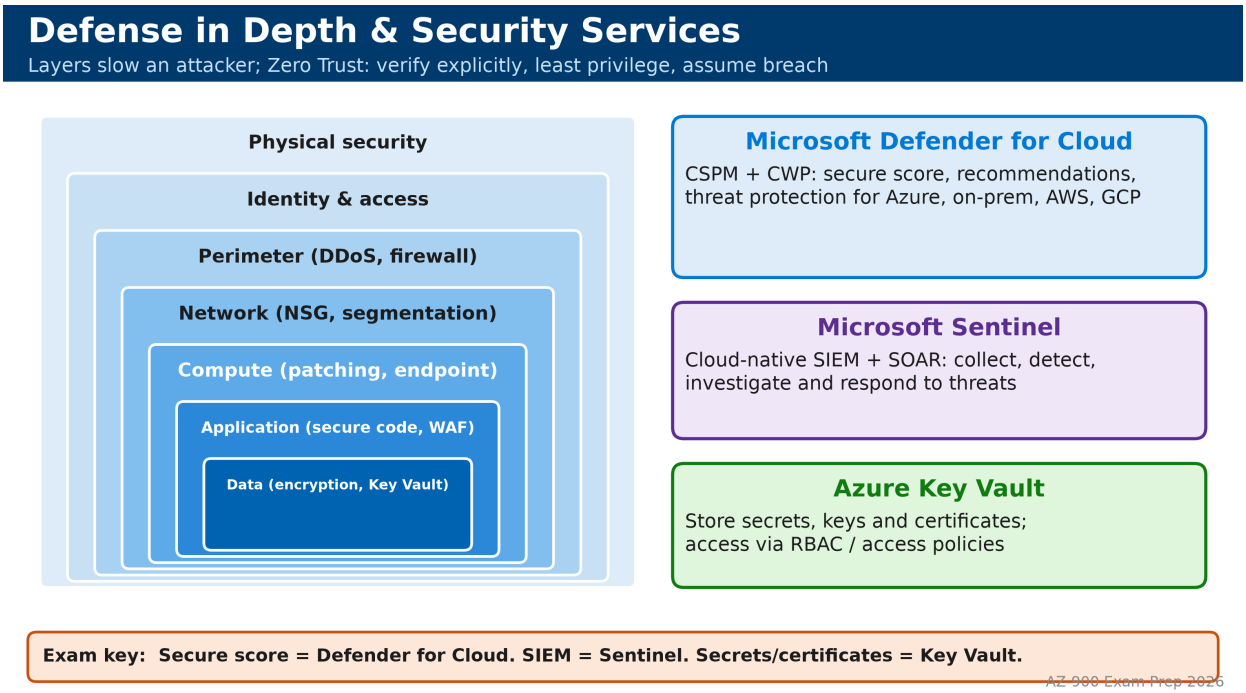


Figure Q197: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 198

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q293, Q107

Your company has an Azure subscription that contains resources in several regions. You need to ensure that administrators can only create resources in those regions. What should you use?

✖ A. A read-only lock.

✔ **B.** An Azure policy.

✖ C. A management group.

✖ D. A reservation.

Correct answer: An Azure policy.

Clarification (read this first): The question is about limiting WHERE (which regions) resources can be created. That is Azure Policy (the 'Allowed locations' rule). RBAC controls WHO can create. Q205 is similar (block only VM creation) and is also an Azure policy.

Answer explained

✔ **An Azure policy.**
The built-in 'Allowed locations' Azure Policy blocks creating resources outside the listed regions.

✖ **A read-only lock.**
A ReadOnly lock blocks all changes to existing resources. It cannot filter by region.

✖ **A management group.**
A management group is only a container. It limits nothing unless a policy is assigned to it.

✖ **A reservation.**
A reservation is a 1-year or 3-year price commitment, not a control on what you can create.

EXAM STRATEGY
'Can only create in those regions' = limit what is allowed = Azure Policy (Allowed locations). Locks protect existing resources, reservations are billing, management groups are containers.

IMPACT
Allowed locations policies enforce data residency rules (where data must be stored), such as GDPR. Without it, someone could put customer data in a region that breaks the rules.

What · Where · When · Why · How

What	The built-in 'Allowed locations' Azure Policy limits which regions resources can use.
Where	Assigned at management group, subscription or resource group scope.
When	When resources must stay in approved regions for data residency or governance.
Why	Policy blocks deployments to regions not on the list, whatever role the user has.
How	Assign Allowed locations with the allowed regions. Other deployments fail with the error RequestDisallowedByPolicy.

Exam tip
Limit what or where resources are created = Azure Policy. Who can create = RBAC.

Memory trick
Allowed locations: Policy pins the map.

Real-world example, use case and scenario

EXAMPLE
Two regions only
Allowed locations is set to UK South and UK West. A deployment to East US fails with RequestDisallowedByPolicy.

USE CASE
Data residency
An Australian health insurer limits all resources to Australia East and Australia Southeast.

SCENARIO
Stray deployment
A developer put a database in Brazil by mistake, which broke a German bank's data residency rules. After an Allowed locations policy was assigned, such mistakes were blocked at deployment.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q198: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 199

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q294

Microsoft Entra ID requires the implementation of domain controllers on Azure virtual machines.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Do not confuse Microsoft Entra ID (a cloud service Microsoft runs for you) with the old Active Directory (AD DS), which needs domain controller servers. Entra ID needs no servers, so the answer is No.

Answer explained

✔ **No.**
Microsoft Entra ID is a SaaS (ready-to-use cloud software) identity service managed by Microsoft. You do not deploy any domain controllers.

✖ **Yes.**
Entra ID needs no servers from the customer. Domain controller VMs are only needed to run AD DS in Azure.

EXAM STRATEGY
Entra ID is a fully managed SaaS identity service. 'Requires domain controllers' or 'VMs' describes old AD DS, so answer No.

IMPACT
You avoid paying for and patching domain controller VMs just to get cloud sign-in. Those VMs would waste money and add risk.

What · Where · When · Why · How

What	Microsoft Entra ID is a cloud identity and access service run by Microsoft.
Where	Delivered as a shared SaaS service. Every Azure subscription trusts one Entra tenant.
When	When you need cloud sign-in without managing directory servers.
Why	It needs no customer domain controllers, so the statement is false.
How	Microsoft hosts and patches the service. You only manage users, groups and settings.

Exam tip
Entra ID = SaaS. AD DS on VMs = IaaS. Entra Domain Services = managed domain controllers (PaaS).

Memory trick
Entra ID: no servers, no patching, no domain controllers.

Real-world example, use case and scenario

EXAMPLE
Instant tenant
When you create an Azure subscription, you get an Entra tenant automatically, with no VMs.

USE CASE
Cloud-native startup
A cloud-only startup runs all identity in Entra ID with zero servers.

SCENARIO
Unneeded VMs
A consulting firm built two domain controller VMs because it thought Entra ID needed them. A review showed their apps only used Entra sign-in, so they deleted the VMs and saved monthly costs.

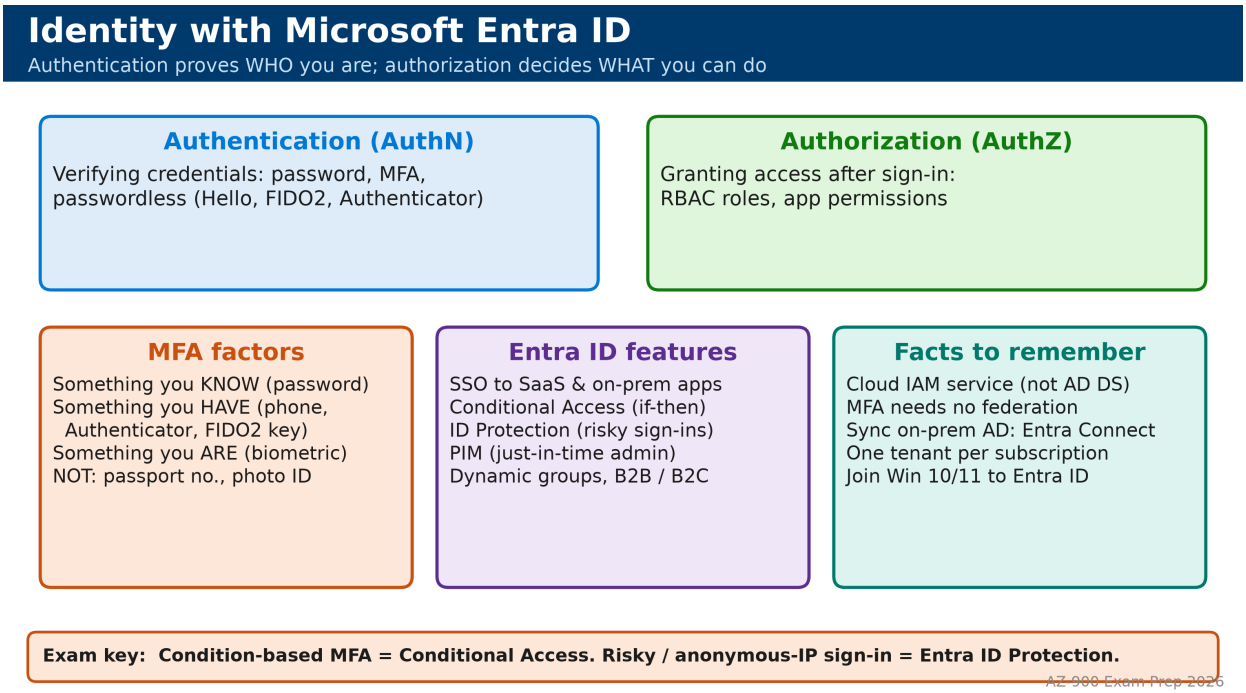


Figure Q199: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 200

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q295

Microsoft Entra ID provides authentication services for resources hosted in Azure and Microsoft 365.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Microsoft Entra ID is the identity provider that signs users in to Azure, Microsoft 365 and Dynamics 365.

✖ **No.**
Azure and Microsoft 365 both use Entra ID as their sign-in directory.

EXAM STRATEGY
Entra ID is behind both the Azure portal sign-in and the Microsoft 365 sign-in. Any statement that it signs in users for both = Yes.

IMPACT
One identity system for Azure and Microsoft 365 gives one place to enforce MFA and Conditional Access. This closes gaps that attackers use.

What · Where · When · Why · How

What	Microsoft Entra ID is the identity provider for Microsoft cloud services.
Where	One Entra tenant serves Azure, Microsoft 365, Dynamics 365 and connected SaaS apps.
When	When users need one sign-in for Azure resources and Microsoft 365 apps.
Why	Entra ID signs in users for both Azure and Microsoft 365, so the statement is true.
How	Users sign in once with their Entra account and get tokens (digital passes) for each service.

Exam tip
One Entra tenant can serve Azure, Microsoft 365 and third-party SaaS apps.

Memory trick
One key ring for Azure and Office: Entra ID.

Real-world example, use case and scenario

EXAMPLE
Same login
A user signs in to Outlook and the Azure portal with the same account.

USE CASE
Retail chain
A retail chain gives store managers single sign-on to Teams, Azure dashboards and a third-party till app.

SCENARIO
Tenant discovery
A company already on Microsoft 365 opened Azure and found its users ready to go, because both use the same Entra tenant.

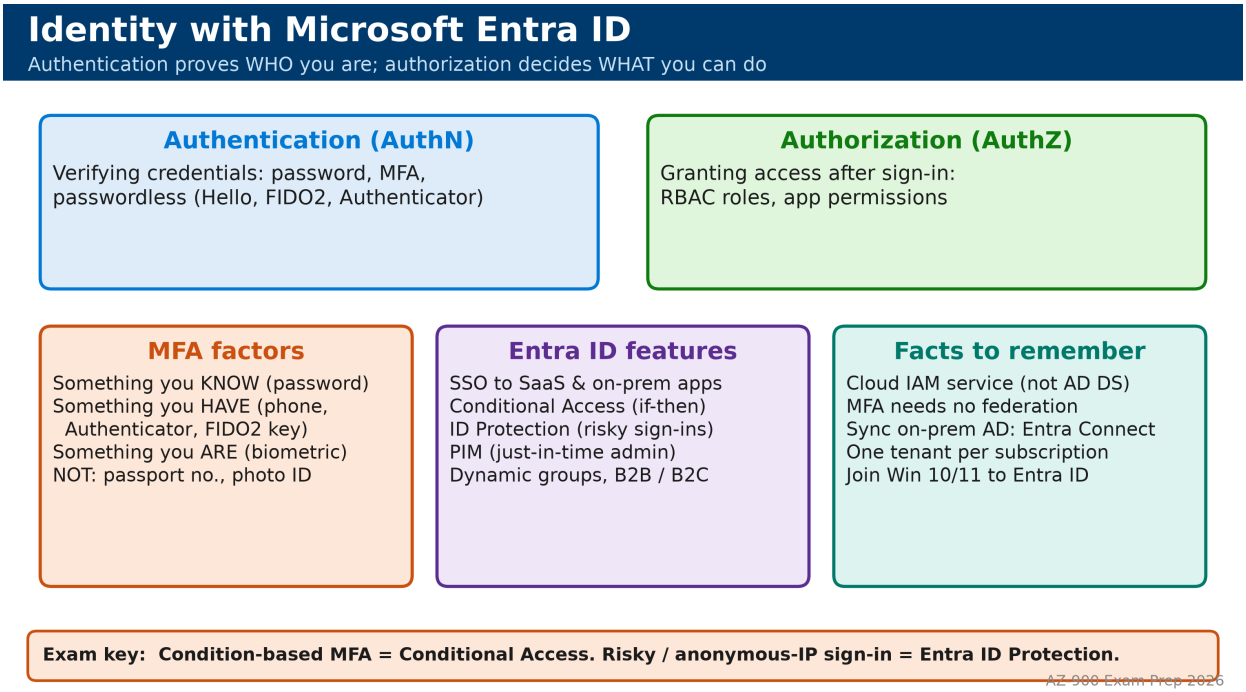


Figure Q200: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 201	AZURE ARCHITECTURE & SERVICES	Identity v5 Q296

Each user account in Microsoft Entra ID can be assigned only one license.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch the words 'only one'. A user can have many licenses at the same time, so the answer is No.

Answer explained

✔ **No.**
A user can have several licenses at once, such as Microsoft 365 E3 plus Entra ID P2. They can be assigned directly or through groups.

✖ **Yes.**
Entra ID has no one-license-per-user limit. Licenses stack.

EXAM STRATEGY
Signal words: 'only one'. Users often have several licenses, such as Microsoft 365 E3 plus Entra ID P2. Absolute limits in license questions are usually false.

IMPACT
Adding licenses on top of each other lets companies give premium security features only to some users. This controls cost instead of upgrading everyone.

What · Where · When · Why · How

What	Licenses in Entra ID give users access to paid product features.
Where	Assigned in the Microsoft 365 or Entra admin center, directly or through group-based licensing.
When	When a user needs several products, such as Microsoft 365 E3 plus Entra ID P2.
Why	A user can have many licenses, so the statement is false.
How	Assign each license to the user, or to a group the user belongs to.

Exam tip
Licenses stack for each user. Group-based licensing assigns them automatically.

Memory trick
Licenses are app tickets: one user can hold many.

Real-world example, use case and scenario

EXAMPLE
Three licenses
An analyst has Microsoft 365 E3, Power BI Pro and Visio Plan 2.

USE CASE
Group-based licensing
A university gives Entra ID P1 to all staff, and extra Project licenses to the project office group.

SCENARIO
PIM requires P2
A bank's security admin needed PIM but only had an E3 license. Adding an Entra ID P2 license next to E3 turned it on.

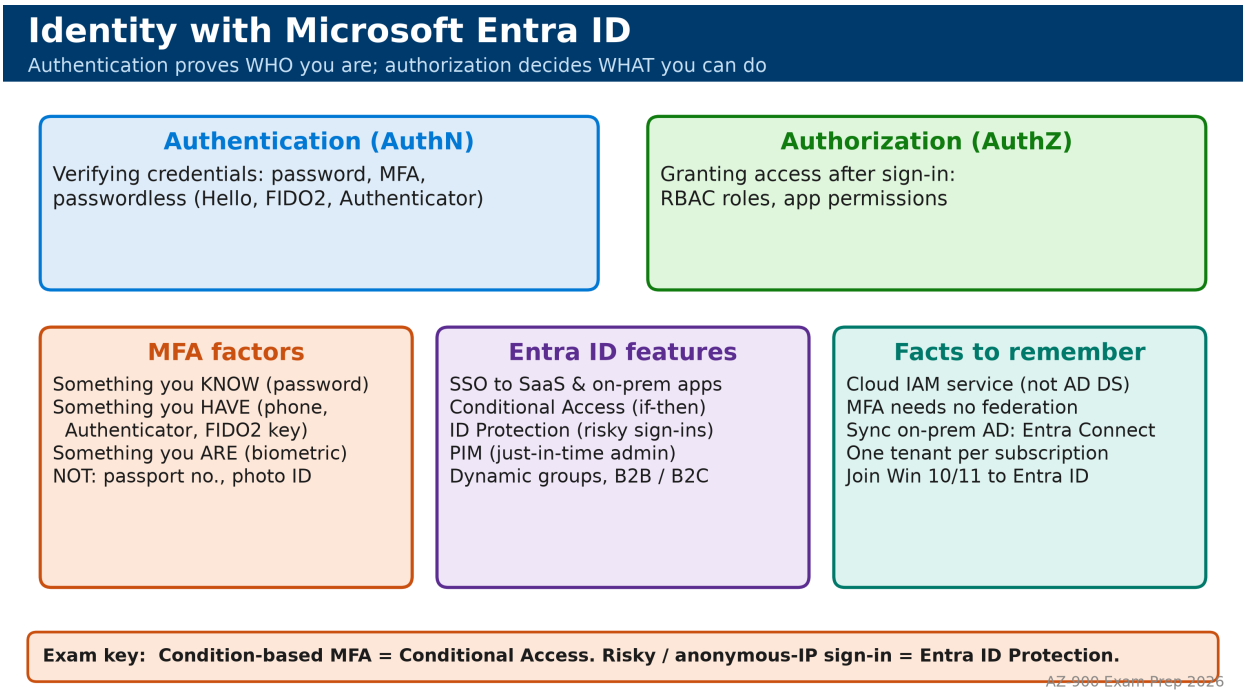


Figure Q201: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 202

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q297

You can configure the Microsoft Entra ID activity logs to appear in Azure Monitor.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Entra ID diagnostic settings send sign-in and audit logs to a Log Analytics workspace (log store). Azure Monitor can then search them and raise alerts.

✖ **No.**
Entra ID diagnostic settings can send activity logs to Azure Monitor Logs.

EXAM STRATEGY
Entra ID diagnostic settings send sign-in and audit logs to a Log Analytics workspace. 'Appear in Azure Monitor' = Yes.

IMPACT
Sending sign-in logs to Azure Monitor lets you get alerts on suspicious logins. It also keeps proof longer than the default 30 days for investigations.

What · Where · When · Why · How

What	Entra ID activity logs record sign-ins, audit events and account setup actions.
Where	Entra ID diagnostic settings send them to a Log Analytics workspace that Azure Monitor uses.
When	When you need longer storage, searches or alerts on Entra sign-in and audit data.
Why	Diagnostic settings can send these logs to Azure Monitor Logs, so the statement is true.
How	Create a diagnostic setting in Entra ID. Choose the log types and a Log Analytics workspace.

Exam tip
Send Entra logs through Diagnostic settings to Log Analytics, for longer storage and alerts.

Memory trick
Diagnostic settings is the pipe from Entra to Monitor.

Real-world example, use case and scenario

EXAMPLE
KQL query
The query SigninLogs | where ResultType != 0 lists failed sign-ins in Log Analytics.

USE CASE
Compliance retention
A stock trading firm keeps Entra sign-in logs in Log Analytics for 2 years, for regulators.

SCENARIO
Logs aged out
An investigation at a utility needed sign-ins from 60 days ago, but Entra kept only 30 days. They then set up diagnostic settings to Log Analytics, so future logs are kept.

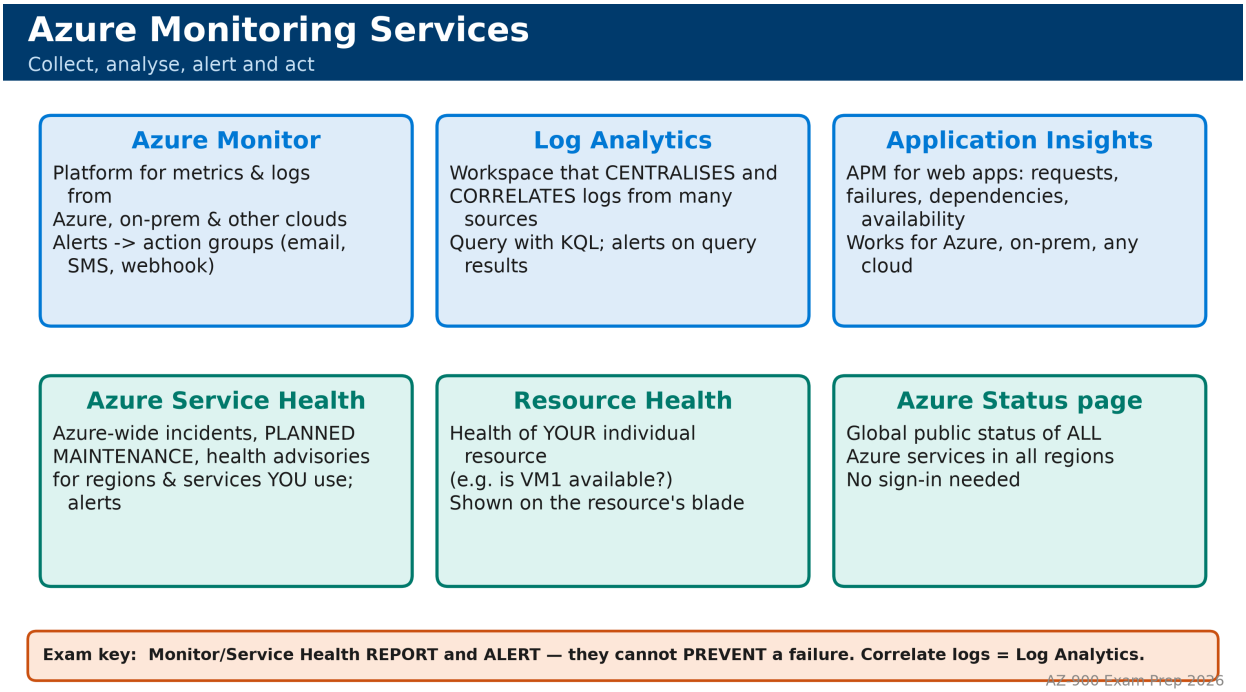


Figure Q202: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 203	AZURE MANAGEMENT & GOVERNANCE	Monitoring v5 Q298

From Azure Monitor, you can monitor resources across multiple Azure subscriptions.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Azure Monitor can show metrics, logs and alerts for resources in every subscription you have access to.

✖ **No.**
Azure Monitor is not limited to one subscription. Its views and workspaces can cover many.

EXAM STRATEGY
Azure Monitor is limited by your access, not by one subscription. 'Across multiple subscriptions' = Yes.

IMPACT
Central operations teams watch many subscriptions from one screen. Believing otherwise leads to duplicate tools and missed alerts in forgotten subscriptions.

What · Where · When · Why · How

What	Azure Monitor collects and analyzes metrics and logs for Azure resources.
Where	Metrics explorer, workbooks (report pages) and Log Analytics can cover many subscriptions.
When	When operations teams need one view of resources in many subscriptions.
Why	Monitor is not limited to one subscription, so the statement is true.
How	Select several subscriptions in Monitor views, or send data from all of them to one workspace.

Exam tip
Central monitoring across subscriptions is a core Monitor feature.

Memory trick
Monitor has a wide-angle lens across every subscription.

Real-world example, use case and scenario

EXAMPLE
Subscription filter
In Monitor, the subscription filter selects Prod, Dev and Test together.

USE CASE
Central ops team
A large company's operations center watches 60 subscriptions from one workbook.

SCENARIO
Single pane
A game publisher had one dashboard per subscription. Sending all data to one workspace gave one view and cut the time to sort incidents in half.

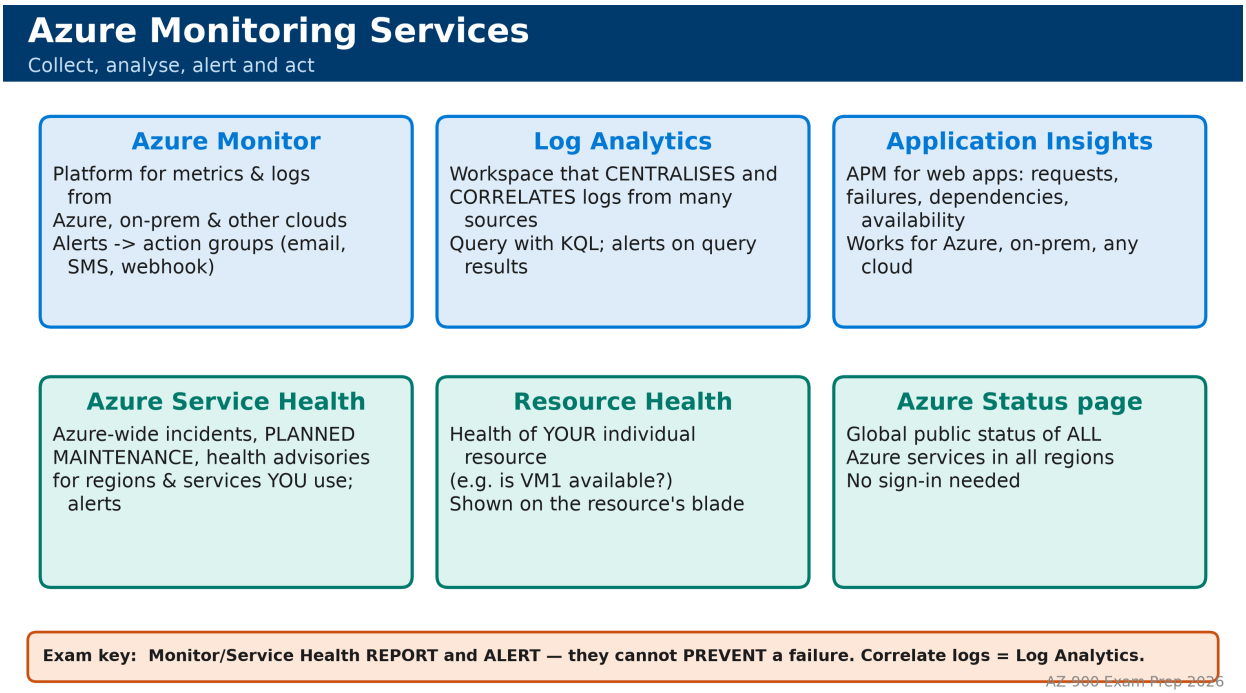


Figure Q203: Azure monitoring services

About this figure

- Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).
- Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.
- Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.
- Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.
- Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 204

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q300

You create a resource group named RG1 in Azure Resource Manager. You need to prevent the accidental deletion of the resources in RG1. Which setting should you use?

- ✖ A. Quickstart.
- ✖ B. Resource costs.
- ✖ C. Deployments.
- ✖ D. Policies.
- ✖ E. Properties.
- ✔ **F.** Locks.
- ✖ G. Automation script.

Correct answer: Locks.

Clarification (read this first): The key words are 'prevent accidental deletion'. That always means Locks (a Delete lock). Do not pick Policies: policies control what can be created or how it is set up. Q252 adds that even an Owner must remove the lock before deleting.

Answer explained

✔ **Locks.**
A Delete (CanNotDelete) lock, set on the resource group's Locks page, blocks deleting the group and its resources. It even blocks Owners.

✖ **Quickstart.**
Quickstart shows getting-started help. It protects nothing.

✖ **Resource costs.**
Resource costs shows spending data, not delete protection.

✖ **Deployments.**
Deployments lists the history of template deployments. It does not stop deletion.

✖ **Policies.**
Policies control resource settings. A denyAction policy can block deletes, but the standard tool for accidental deletion is a lock.

✖ **Properties.**
Properties shows resource group details, such as location and ID.

✖ **Automation script.**
Automation script (now Export template) exports a template. It protects nothing.

EXAM STRATEGY
'Prevent accidental deletion' always means Locks (CanNotDelete). Cross out Policies, which control what can be created or set up, not deletion.

IMPACT
One Delete lock on a production resource group stops a wrong 'delete resource group' click from wiping databases and VMs.

What · Where · When · Why · How

What	A Delete (CanNotDelete) resource lock stops a resource group and its resources from being deleted.
Where	On the resource group's Locks page in the Azure portal.
When	When production resources must be protected from deletion by mistake.
Why	Locks block deletion, even for Owners, until the lock is removed. The other settings protect nothing.
How	Open RG1, select Locks, and add a lock of type Delete with a name.

Exam tip
Accidental deletion = Lock (CanNotDelete).
Accidental change = ReadOnly lock.

Memory trick
Lock it so nobody knocks it.

Real-world example, use case and scenario

EXAMPLE
Delete lock
Add a lock named Protect-RG1, type Delete, on RG1.

USE CASE
Production database RG
A taxi app company locks its production database resource group against deletion.

SCENARIO
Cleanup script
A cleanup script at an ad agency deleted all resource groups tagged temp, including a production group tagged by mistake. After restoring, they put Delete locks on every production group, so the script now fails safely.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

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Figure Q204: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 205

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q301, Q104

You have a resource group named RG1. You need to prevent the creation of virtual machines only in RG1. The solution must ensure that other objects can be created in RG1. What should you use?

✖ A. A lock.

✖ B. An Azure role.

✖ C. A tag.

✔ **D.** An Azure policy.

Correct answer: An Azure policy.

Clarification (read this first): The key word is 'only': block VMs but still allow every other resource type. That needs a rule about resource TYPE, which is Azure Policy. A lock cannot pick one type; it applies to everything. Q198 (limit regions) is also an Azure policy.

Answer explained

✔ **An Azure policy.**
Assign Azure Policy 'Not allowed resource types' with Microsoft.Compute/virtualMachines to RG1. It blocks only VM creation there. All other resource types can still be deployed.

✖ **A lock.**
A lock applies to all resources in RG1 and cannot target one resource type. A ReadOnly lock would block every change.

✖ **An Azure role.**
RBAC roles control WHO can act, not WHICH resource types are allowed. An Owner could still create VMs.

✖ **A tag.**
Tags are name/value labels for sorting and cost reports. They enforce nothing.

EXAM STRATEGY
Block one resource type and allow others = Azure Policy (Not allowed resource types). Locks are all-or-nothing, tags are labels, roles limit people, not types.

IMPACT
The policy keeps costly VMs out of a resource group without blocking normal deployments. This is a common cost control.

What · Where · When · Why · How

What	Azure Policy is a governance service that checks and enforces rules on resource settings.
Where	Assigned at management group, subscription or resource group scope. Here, only RG1.
When	When some resource types, sizes or regions must be blocked while others stay allowed.
Why	It can block one resource type (VMs) in RG1 without blocking any other deployment.
How	Assign the built-in 'Not allowed resource types' policy to RG1, listing Microsoft.Compute/virtualMachines.

Exam tip
'Prevent creation of X but allow other objects' = Azure Policy (Not allowed resource types). Locks are all-or-nothing and never type-specific.

Memory trick
Policy controls the WHAT (type, size, region); RBAC controls the WHO.

Real-world example, use case and scenario

EXAMPLE
Not allowed resource types
An admin assigns the built-in policy 'Not allowed resource types' with Microsoft.Compute/virtualMachines to RG1. Storage accounts and web apps can still be deployed there.

USE CASE
PaaS-only sandbox
A bank's innovation team gets a resource group limited to PaaS services, so nobody creates unpatched IaaS VMs in it.

SCENARIO
The lock that blocked everything
A junior engineer put a ReadOnly lock on RG1 to stop VMs. Developers then could not even add a storage account. Replacing it with a policy that blocks only VMs fixed their work and still blocked VMs.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q205: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 206

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q302

You have an Azure subscription and 100 Windows 10 devices. You need to ensure that only users whose devices have the latest security patches installed can access Microsoft Entra ID-integrated applications. What should you implement?

✔ **A.** A conditional access policy.

✖ B. Azure Bastion.

✖ C. Azure Firewall.

✖ D. Azure Policy.

Correct answer: A conditional access policy.

Clarification (read this first): The question is about letting users in ONLY IF their device is fully patched. A rule that checks a condition at sign-in is Conditional Access. Do not pick Azure Policy: it controls Azure resource settings, not user devices.

Answer explained

✔ **A conditional access policy.**
Conditional Access, with Intune device compliance policies, lets users reach Entra ID-connected apps only from devices marked compliant, such as fully patched ones.

✖ **Azure Bastion.**
Azure Bastion gives RDP/SSH to Azure VMs in the browser. It does not check device patches at sign-in.

✖ **Azure Firewall.**
Azure Firewall filters network traffic by IP, port and domain name, not by device compliance at sign-in.

✖ **Azure Policy.**
Azure Policy controls Azure resource settings, not user device access to apps.

EXAM STRATEGY
Access depends on the device's state = Conditional Access with a compliant-device rule. Cross out Bastion and Firewall (network) and Azure Policy (resource settings).

IMPACT
Blocking unpatched devices stops attackers from using known security holes to steal sign-in tokens. This is a core Zero Trust control.

What · Where · When · Why · How

What	Conditional Access is an Entra ID feature that applies if-then access rules at sign-in.
Where	Set up in the Microsoft Entra admin center under Protection > Conditional Access. Needs Entra ID P1.
When	When app access must depend on device state, location, user risk or similar signals.
Why	It blocks sign-in from devices not marked compliant, such as unpatched Windows 10 PCs.
How	An Intune compliance policy checks the patch level. A Conditional Access policy then requires 'device marked as compliant'.

Exam tip
Words like 'only users whose devices...' or 'if/then at sign-in' point to Conditional Access, not a network or resource-control service.

Memory trick
Conditional Access = IF device patched THEN let in.

Real-world example, use case and scenario

EXAMPLE
Compliant device grant
A policy covers all cloud apps and requires 'device marked as compliant'. A laptop missing last month's update is blocked at sign-in.

USE CASE
Clinic records portal
A clinic lets staff open its patient portal only from fully patched PCs that Intune marks as compliant.

SCENARIO
Contractor laptop blocked
A contractor tried to open SharePoint from a laptop with delayed updates. Conditional Access blocked it and showed how to fix it. After patching, the device was compliant and sign-in worked.

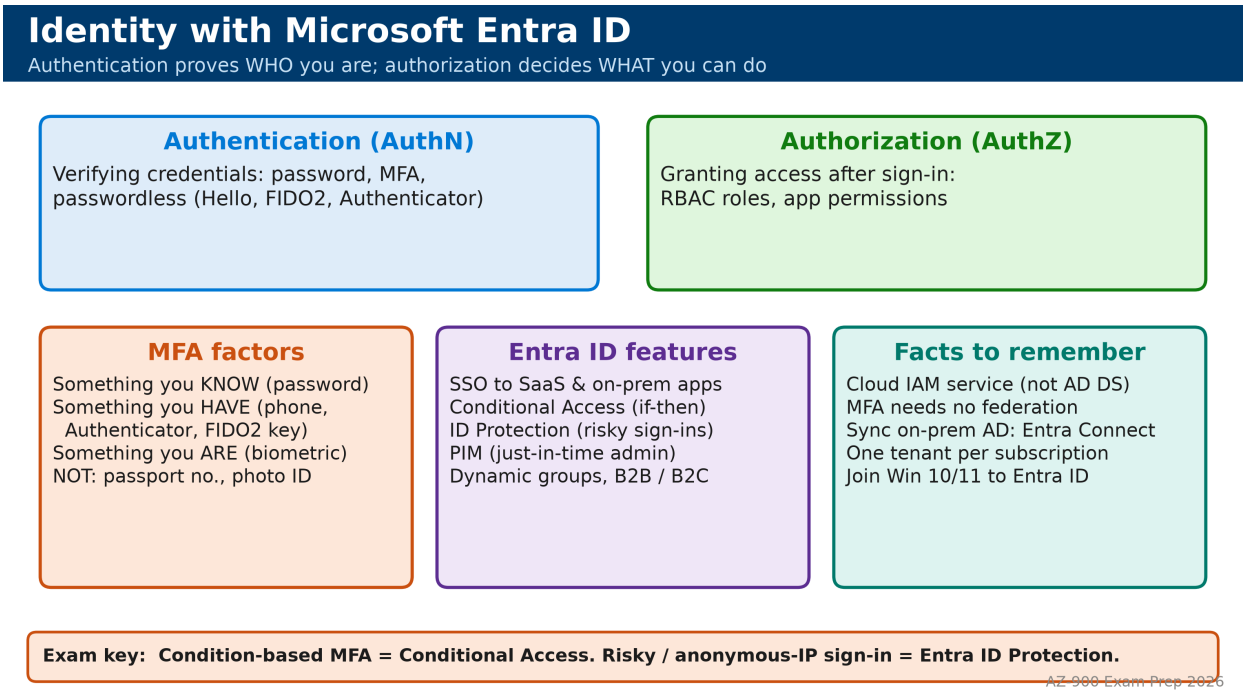


Figure Q206: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 207

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q303

[...] enables users to authenticate to multiple applications by using single sign-on (SSO).

✖ A. Application security groups in Azure.

✔ **B.** Microsoft Entra ID.

✖ C. Azure Key Vault.

✖ D. Microsoft Defender for Cloud.

Correct answer: Microsoft Entra ID.

Clarification (read this first): SSO (single sign-on) means you sign in once and can open many apps. It is an identity feature, so the answer is the identity service, Microsoft Entra ID. Older versions of the question say 'Azure Active Directory', which is the old name for the same thing.

2026 update: Older versions of this question say Azure Active Directory. The service was renamed Microsoft Entra ID in 2023.

Answer explained

✔ **Microsoft Entra ID.**
Microsoft Entra ID (old name: Azure AD) is the identity provider that gives single sign-on to Microsoft 365, the Azure portal and thousands of SaaS and custom apps.

✖ **Application security groups in Azure.**
Application security groups group VM network cards for NSG rules. They play no part in user sign-in.

✖ **Azure Key Vault.**
Key Vault stores secrets, keys and certificates. It does not sign users in.

✖ **Microsoft Defender for Cloud.**
Defender for Cloud manages security posture and workload protection. It is not an identity provider.

EXAM STRATEGY
Signal word: SSO. Single sign-on is an identity job, so pick the identity service, Entra ID. Old questions that say Azure Active Directory mean the same answer.

IMPACT
SSO means fewer passwords and less phishing risk. IT can also remove access to every connected app from one account.

What · Where · When · Why · How

What	Microsoft Entra ID is Microsoft's cloud identity and access service.
Where	A directory for the whole tenant, managed in the Microsoft Entra admin center and the Azure portal.
When	When users need one identity to sign in to many cloud and on-premises apps.
Why	One sign-in gives SSO across Microsoft 365, Azure and connected SaaS apps.
How	Apps trust Entra ID through SAML, OpenID Connect or OAuth (sign-in standards) and accept the tokens it issues.

Exam tip
SSO, MFA, Conditional Access and B2B/B2C all live in Microsoft Entra ID. If the list uses the old name, Azure Active Directory is the same thing.

Memory trick
Enter once with Entra - every app opens.

Real-world example, use case and scenario

EXAMPLE
Gallery app SSO
A user signs in to Entra ID in the morning, then opens Salesforce, ServiceNow and Outlook with no other password prompt.

USE CASE
University app portal
A university puts its library system, learning system and email in the My Apps portal, so students use one Entra identity.

SCENARIO
Password fatigue fixed
Staff at a retail chain had 9 different passwords, and password reset calls kept growing. Connecting each app to Entra ID SSO cut reset calls a lot and let IT disable leavers in one place.

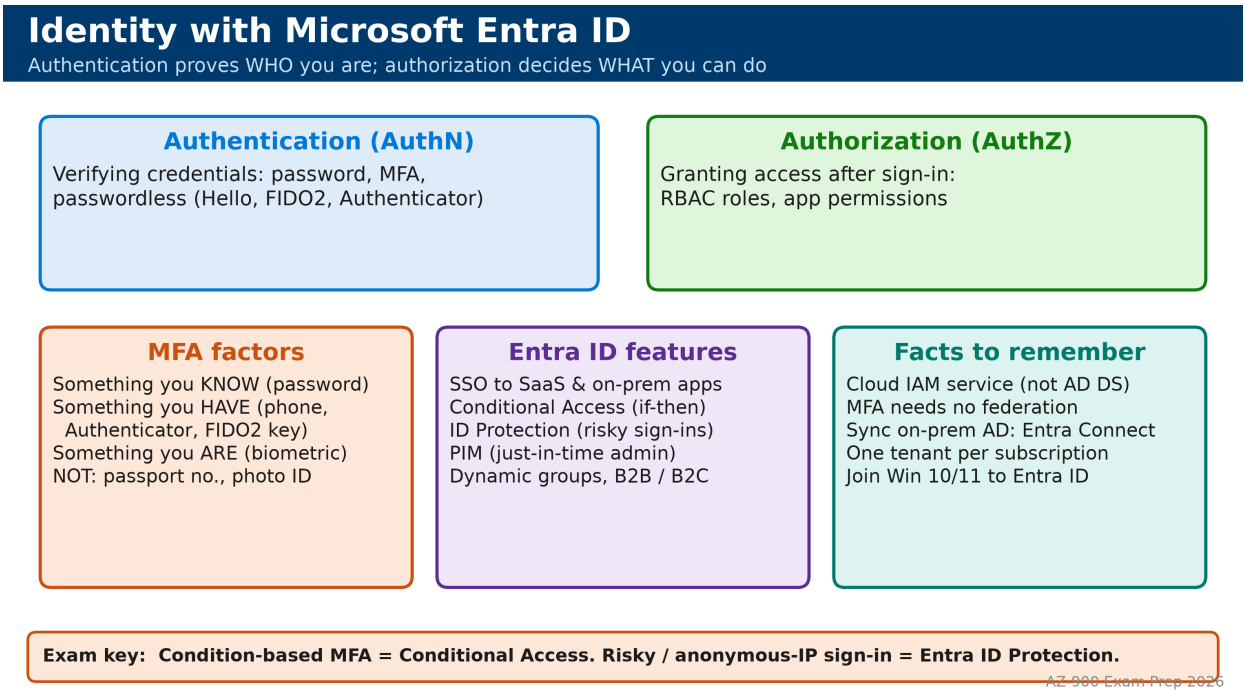


Figure Q207: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 208

CLOUD CONCEPTS

SLA
v5 Q304, Q355

You deploy an Azure resource. The resource becomes unavailable for an extended period due to a service outage. Microsoft will [...].

✖ A. refund your bank account.

✖ B. migrate the resource to another subscription.

✔ **C.** credit your Azure account.

✖ D. send you a coupon code that you can redeem for Azure credits.

Correct answer: credit your Azure account.

Clarification (read this first): When Microsoft misses its SLA (uptime promise), you get a service credit (a discount) on your Azure bill. You do not get cash back to your bank. You usually have to file a claim to get it.

Answer explained

✔ **credit your Azure account.**
When an SLA is missed, Microsoft gives a service credit (a percentage of that service's monthly fee) to your Azure account, after you file a claim.

✖ **Refund your bank account.**
SLA payments are credits against future bills, never cash refunds to a bank account.

✖ **Migrate the resource to another subscription.**
Microsoft does not move resources between subscriptions as payment. Designing for resilience is the customer's job.

✖ **Send you a coupon code that you can redeem for Azure credits.**
No coupon code is sent. The credit goes straight to the account after a checked claim.

EXAM STRATEGY
Payment for an outage under an SLA = service credit on the Azure account. Cross out refunds, coupons and moves. Microsoft never pays cash back.

IMPACT
Credits need a claim and are limited to a percentage of the service fee. They never cover your real business losses from downtime.

What · Where · When · Why · How

What	A service credit is the SLA payment: a percentage of the service's monthly fee.
Where	Added to your Azure billing account and taken off future invoices.
When	When a service's measured uptime in a month falls below its SLA promise.
Why	The SLA sets credits, not cash refunds or moves, as the payment.
How	File an SLA claim through a support request before the deadline. Microsoft checks it and credits the account.

Exam tip
SLA missed = service credit on the account. You usually must file a claim; it is not automatic.

Memory trick
Outage? Credit, not cash.

Real-world example, use case and scenario

EXAMPLE
Tiered credit
A VM's SLA of 99.9% was missed with 99.5% uptime. The SLA table gives a 10% credit on that VM's monthly charge.

USE CASE
Claim workflow
A media company's cost team records outage times and files SLA claims through Help + support, so credits lower next month's invoice.

SCENARIO
Waiting for a cheque
An online shop startup expected money back after a regional outage. Their partner explained they had to file a claim and would get a credit on the Azure bill. They filed it before the deadline.

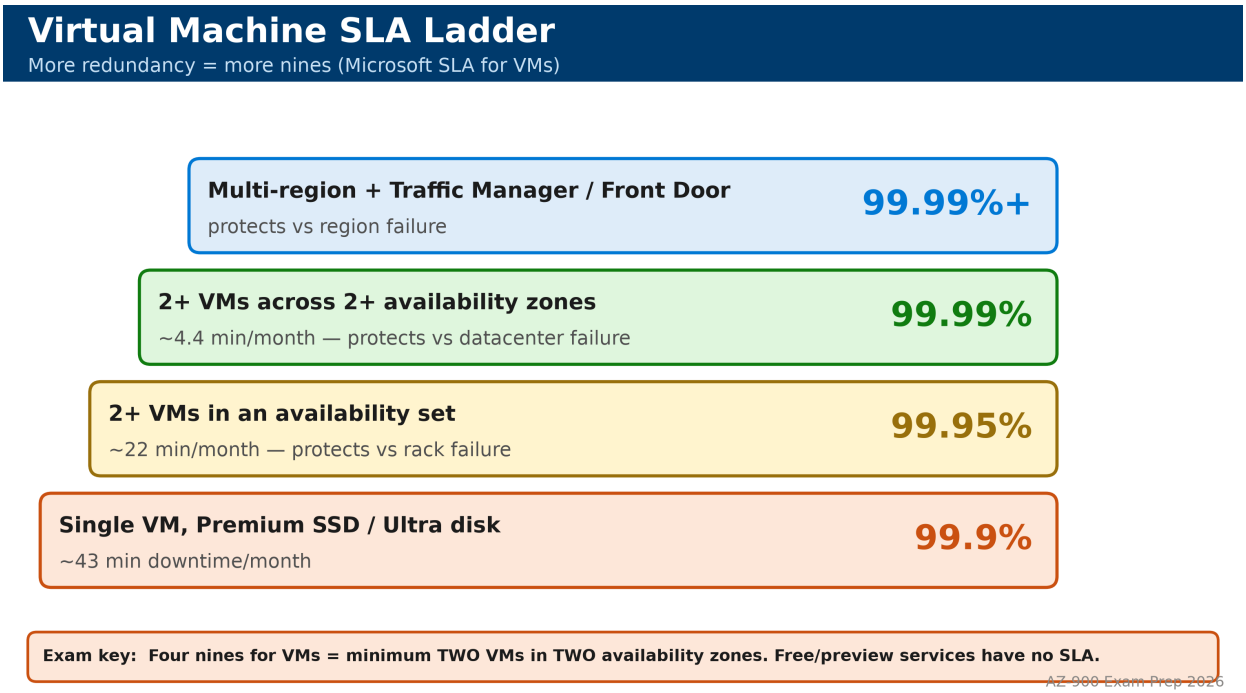


Figure Q208: Virtual machine SLA ladder

About this figure

- Climb the ladder: each step adds redundancy and a higher VM SLA; the downtime figures assume a 30-day month.
- A single VM with Standard SSD gets about 99.5% and with Standard HDD about 95%; all its OS and data disks must meet that tier.
- Availability sets spread VMs across fault domains (racks) and update domains (maintenance batches) in one datacenter; zones spread across datacenters.
- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

Question 209

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q305, Q242

Which task can you perform by using Azure Advisor?

- ✖ A. Integrate Active Directory and Microsoft Entra ID.
- ✖ B. Estimate the costs of an Azure solution.
- ✔ **C.** Confirm that Azure subscription security follows best practices.
- ✖ D. Evaluate which on-premises resources can be migrated to Azure.

Correct answer: Confirm that Azure subscription security follows best practices.

Clarification (read this first): Azure Advisor gives tips on resources you ALREADY run, including security tips. Do not confuse it with tools for future plans: estimating cost is the Pricing calculator (Q396), and checking servers for migration is Azure Migrate (Q353).

Answer corrected in this edition. The previous PDF printed: *Estimate the costs of an Azure solution.*. The printed answer 'Estimate the costs of an Azure solution' is the job of the Pricing calculator. Azure Advisor looks at deployed resources and gives best-practice tips, including a Security area, so confirming security best practices is the correct task.

Answer explained

✔ **Confirm that Azure subscription security follows best practices.**
Azure Advisor gives best-practice tips in five areas, including Security (taken from Defender for Cloud). So it can confirm that subscription security follows best practices.

✖ **Integrate Active Directory and Microsoft Entra ID.**
Syncing on-premises Active Directory to Entra ID is done by Microsoft Entra Connect, not Advisor.

✖ **Estimate the costs of an Azure solution.**
Estimating the cost of a planned solution is the Pricing calculator's job. Advisor only looks at resources already deployed.

✖ **Evaluate which on-premises resources can be migrated to Azure.**
Checking on-premises resources for migration is done by Azure Migrate.

EXAM STRATEGY
Advisor = tips on cost, security, reliability, performance and operational excellence. Cross out cost estimates (Pricing calculator) and migration checks (Azure Migrate).

IMPACT
Advisor is free and shows unused resources and security gaps. Ignoring it usually wastes money and leaves bad settings in place.

What · Where · When · Why · How

What	Azure Advisor is a free service that gives personal best-practice tips.
Where	In the Azure portal, for the subscriptions and resource groups you can access.
When	When you review deployed resources for reliability, security, performance, cost or operations problems.
Why	Its Security area, fed by Defender for Cloud, shows whether security follows best practices.
How	It looks at resource settings and usage data, then lists tips with steps to fix them.

Exam tip
Advisor = tips on what you already run (five areas). Future cost estimate = Pricing calculator. Migration check = Azure Migrate.

Memory trick
Advisor RSPCO: Reliability, Security, Performance, Cost, Operational excellence - 'Really Smart People Cut Overhead'.

Real-world example, use case and scenario

EXAMPLE
Security recommendation
Advisor warns that MFA is not turned on for subscription owners, and that storage accounts allow public network access.

USE CASE
Monthly posture review
An insurer's cloud team reviews the Advisor Security and Reliability tabs every month and copies the list to its ticket system.

SCENARIO
Wrong tool for the quote
A consulting firm opened Advisor to price a planned AKS platform for a client and found nothing to estimate. They used the Pricing calculator for the quote, and later used Advisor to tune the running cluster.

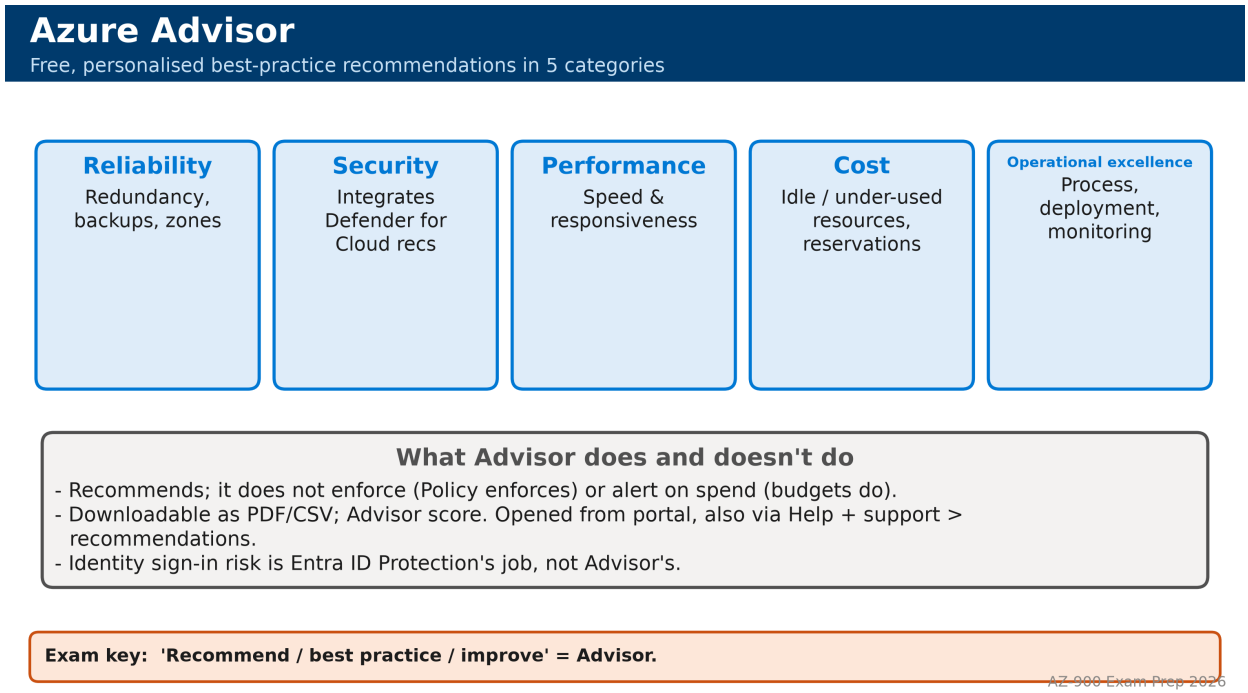


Figure Q209: Azure Advisor

About this figure

• The five boxes are Advisor's categories; each recommendation shows impact (high, medium, low) and suggested actions you can apply or postpone.

• Advisor is free and analyses your deployed resources' configuration and usage telemetry, giving subscription- and resource-group-level recommendations.

• Advisor score rates how well you follow best practices, overall and per category, helping prioritise fixes with the largest impact.

• Recommendations can trigger alerts and digest emails; security recommendations come from Defender for Cloud, so fixing them there also updates Advisor.

• Advisor is aligned with the Azure Well-Architected Framework pillars: reliability, security, cost optimisation, operational excellence and performance efficiency.

Question 210

AZURE MANAGEMENT & GOVERNANCE

Subscriptions
v5 Q306, Q378

If your company uses an Azure free account, you will only be able to use a subset of Azure services.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Separate 'limited money' from 'limited services'. The free account limits how much you can spend (credit) and the free amounts, but you can use almost any Azure service. So 'only a subset of services' is No. Q222 asks if the free account has a spending limit, and that is Yes.

Answer corrected in this edition. The previous PDF printed: *Yes.*. The printed answer was Yes. The Azure free account gives a 200 USD credit (for 30 days) that can be spent on any Azure service, plus some services free for 12 months and some always free. So it is not limited to a subset of services. The answer key and Microsoft's free account FAQ both support No.

Answer explained

✔ **No.**
The free account's 200 USD credit can be spent on almost any Azure service. Only the free amounts are capped, not the list of services you can use.

✖ **Yes.**
The credit works across the Azure catalogue. Limits apply to free amounts and credit, not to a small set of services.

EXAM STRATEGY
Separate money limits from service access. The free account limits credit and free amounts, not the service list. 'Only a subset of services' = No.

IMPACT
Teams can test with almost any service during a trial, so test results match the real production design.

What · Where · When · Why · How

What	The Azure free account is a trial offer with credit plus free amounts of services.
Where	Sign up at azure.microsoft.com/free with a Microsoft account, plus phone and card checks.
When	When people or teams want to try Azure services before paying.
Why	Its 200 USD credit works on almost all services, so services are not limited to a subset.
How	Spend the credit on any service within 30 days. Use the 12-month free and always-free amounts as well.

Exam tip
Separate 'limited free amount' from 'limited service list'. The free account limits money and amounts, not which services you can try.

Memory trick
Free account = full menu, small wallet.

Real-world example, use case and scenario

EXAMPLE
Credit on other services
A student spends part of the free credit on Azure Kubernetes Service and Azure AI services, not just on the 12-month free offers.

USE CASE
Proof of concept
A small architecture firm tests Azure Virtual Desktop on a free account before buying a paid subscription.

SCENARIO
Assumed restriction
A developer thought his free account could only run 'free tier' services, so he built a demo on another cloud. A colleague showed him the credit covers almost every Azure service, so the next test stayed on Azure.

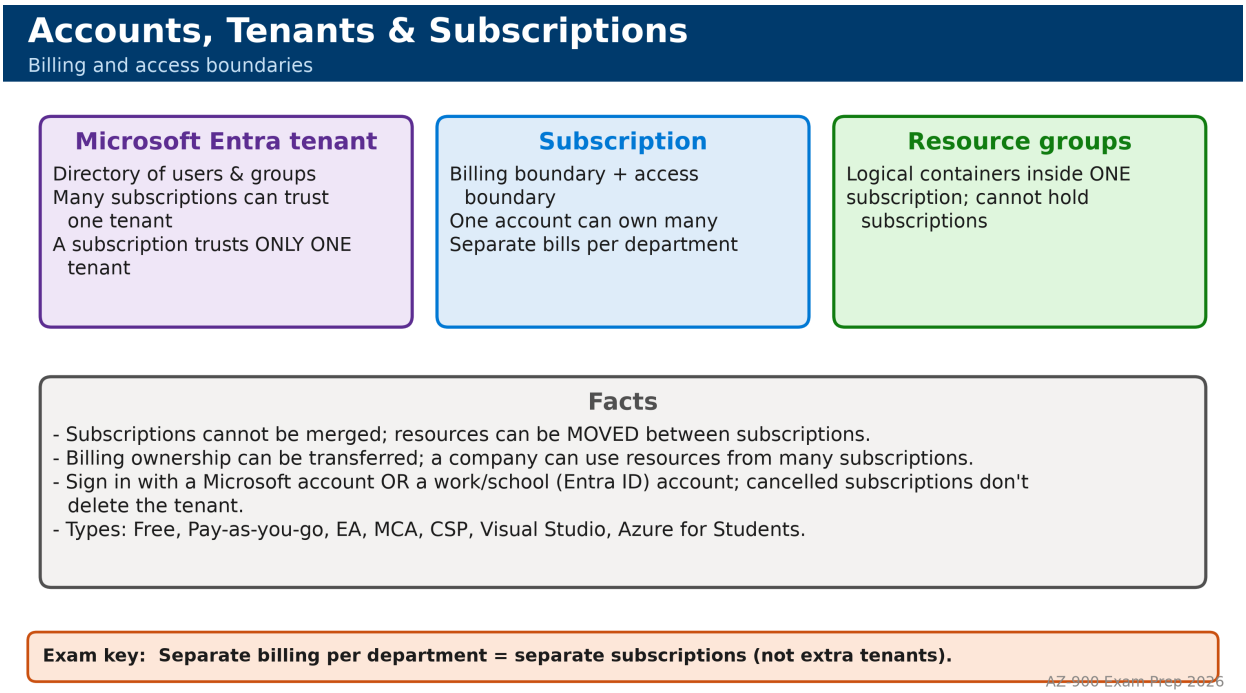


Figure Q210a: Accounts, tenants and subscriptions

About this figure

- Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.
- Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.
- Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.
- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

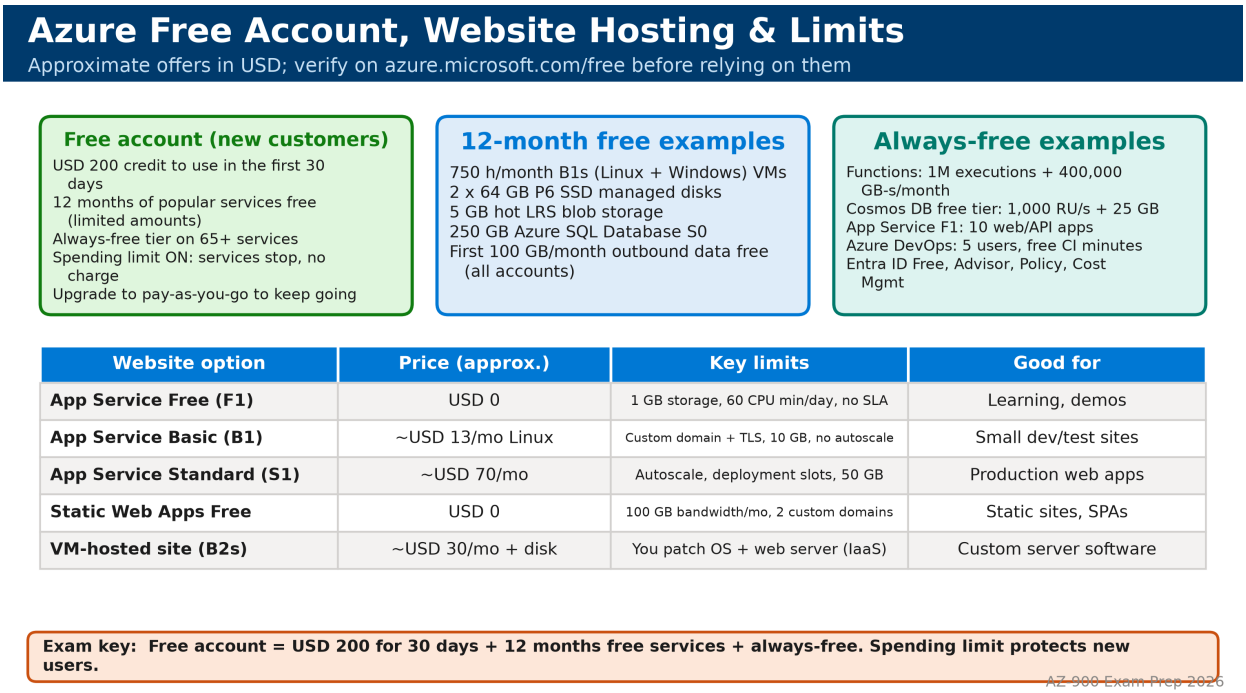


Figure Q210b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 211

AZURE MANAGEMENT & GOVERNANCE

Subscriptions
v5 Q307

All Azure free accounts expire after a specific period.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The question asks if every free account has an end date. It does: the credit ends after 30 days, and the 12-month free services end after 12 months. Q215 shows you can upgrade to pay-as-you-go to keep going.

Answer explained

✔ **Yes.**
Every free account offer has a time limit. The 200 USD credit lasts 30 days, and the 12-month free services end after 12 months unless you upgrade.

✖ **No.**
The credit ends after 30 days and the free services after 12 months. No free account runs forever without an upgrade.

EXAM STRATEGY
Every free offer has a clock: 30 days for credit, 12 months for free services. 'Expire after a specific period' = Yes.

IMPACT
If you forget the end date, resources stop in the middle of a demo. Plan the upgrade to pay-as-you-go before time runs out.

What · Where · When · Why · How

What	The free account has two time limits: a 30-day credit and 12-month free services.
Where	It applies to the free subscription created at sign-up.
When	When you plan how long a free trial will last before charges or shutdown.
Why	Every free offer ends, so the statement that all free accounts expire is true.
How	After 30 days the credit ends. Without an upgrade, the subscription is disabled.

Exam tip
Remember the two clocks: 30 days for the credit, 12 months for the free services.

Memory trick
30 for the cash, 12 for the stash.

Real-world example, use case and scenario

EXAMPLE
Day 31
On day 31, any unused part of the 200 USD credit is gone, even if only 40 USD was spent.

USE CASE
Training calendar
A training school runs its Azure labs in the first 30 days of students' free accounts, so the credit covers everything.

SCENARIO
Demo went dark
A freelancer hosted a client demo on a free account and forgot the 30-day limit. When the credit ended, the resources were disabled. Upgrading to pay-as-you-go brought them back.

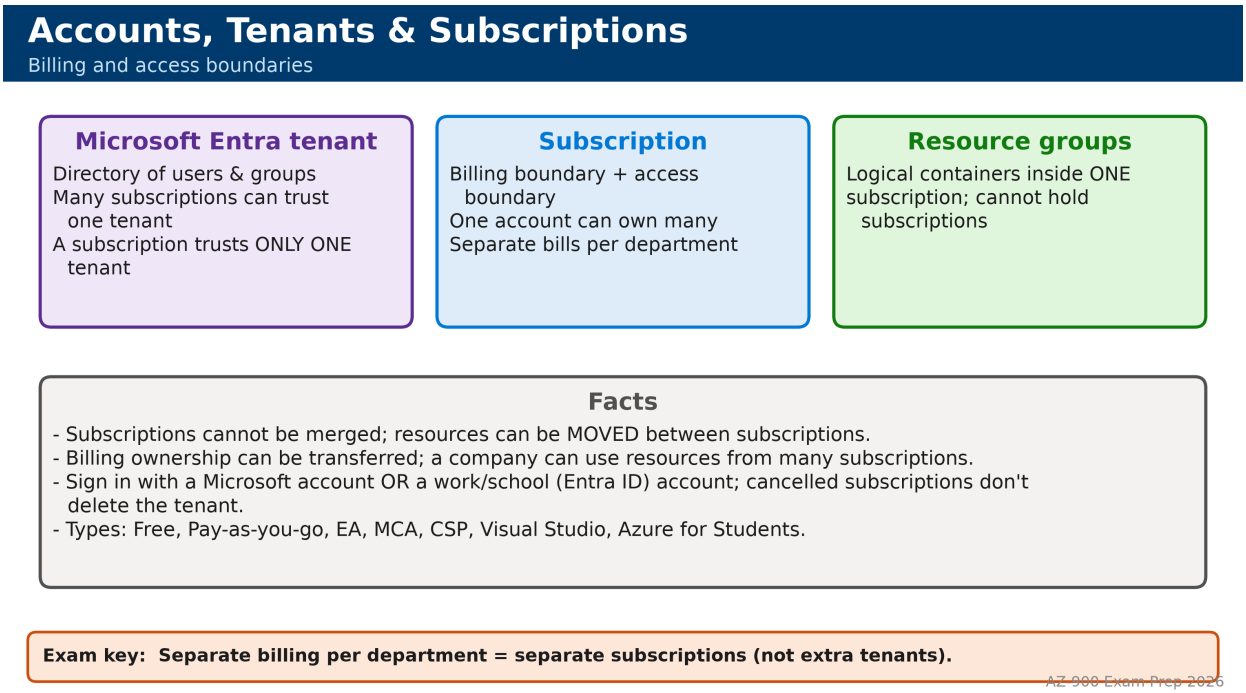


Figure Q211a: Accounts, tenants and subscriptions

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- Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.
- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

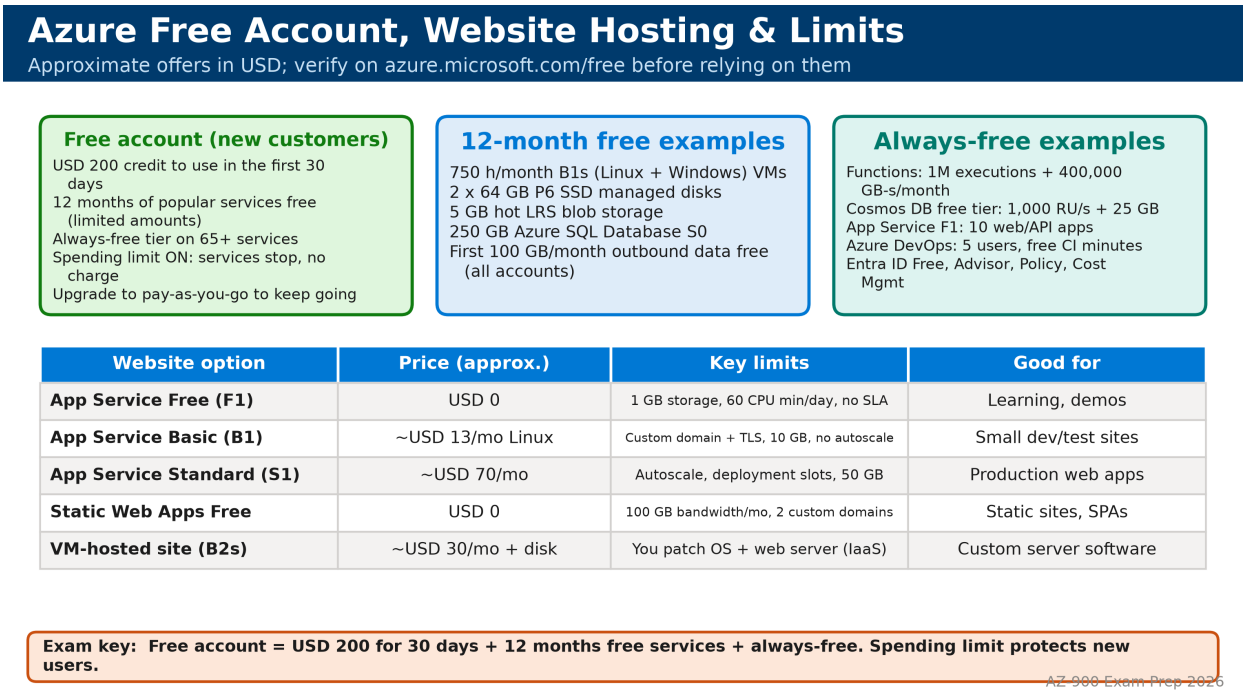


Figure Q211b: Free account, website hosting and limits

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- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 212

AZURE MANAGEMENT & GOVERNANCE

Subscriptions
v5 Q308

You can create up to 10 Azure free accounts by using the same Microsoft account.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The number 10 is a trap. You get only ONE free account per person, so the answer is No.

Answer explained

✔ **No.**
The Azure free account is offered once per customer. Sign-up checks identity by phone and card, so a second free account is refused.

✖ **Yes.**
The rule is one free account per person. Ten free accounts from one Microsoft account are not allowed.

EXAM STRATEGY
Made-up numbers are a signal. Free accounts are one per person, checked at sign-up. Any count like 10 is a trap, so answer No.

IMPACT
Trying to chain free accounts breaks the offer terms. The account can be suspended, and work in progress can be lost.

What · Where · When · Why · How

What	Free account rules allow only one free account per customer.
Where	Checked at sign-up by verifying identity with a phone and a card.
When	When someone tries to create more free accounts to get more credit.
Why	Microsoft blocks repeat free offers, so ten free accounts are impossible.
How	Create extra subscriptions as paid pay-as-you-go instead of new free accounts.

Exam tip
Any number greater than one with 'free account' = the answer is No.

Memory trick
One person, one free ride.

Real-world example, use case and scenario

EXAMPLE
Second sign-up refused
A user who had a free account last year tries again with the same Microsoft account and is told they are not eligible.

USE CASE
Team labs
A training company uses Azure Pass or sandbox subscriptions for each learner, instead of many free accounts.

SCENARIO
Ten projects, one account
A startup founder wanted 10 free accounts, one per microservice. He learned the rule is one per person. He created one pay-as-you-go subscription with separate resource groups and budgets instead.

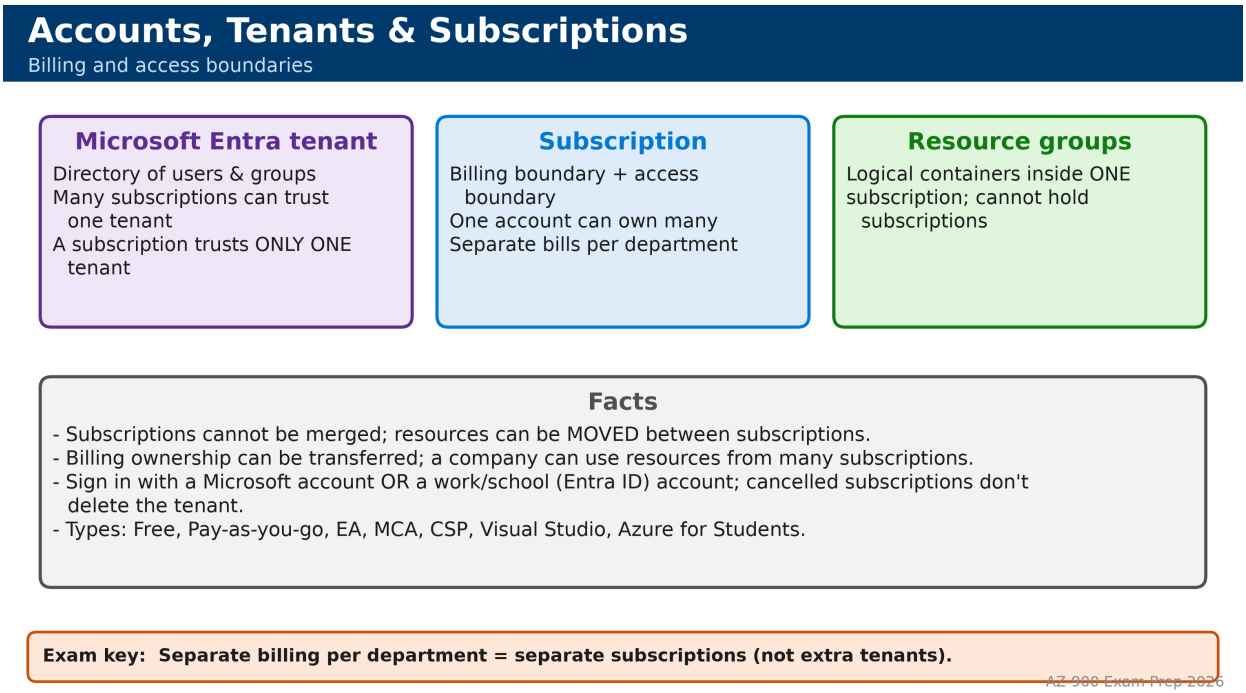


Figure Q212a: Accounts, tenants and subscriptions

About this figure

• Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.

• Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.

• Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.

• Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.

• Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

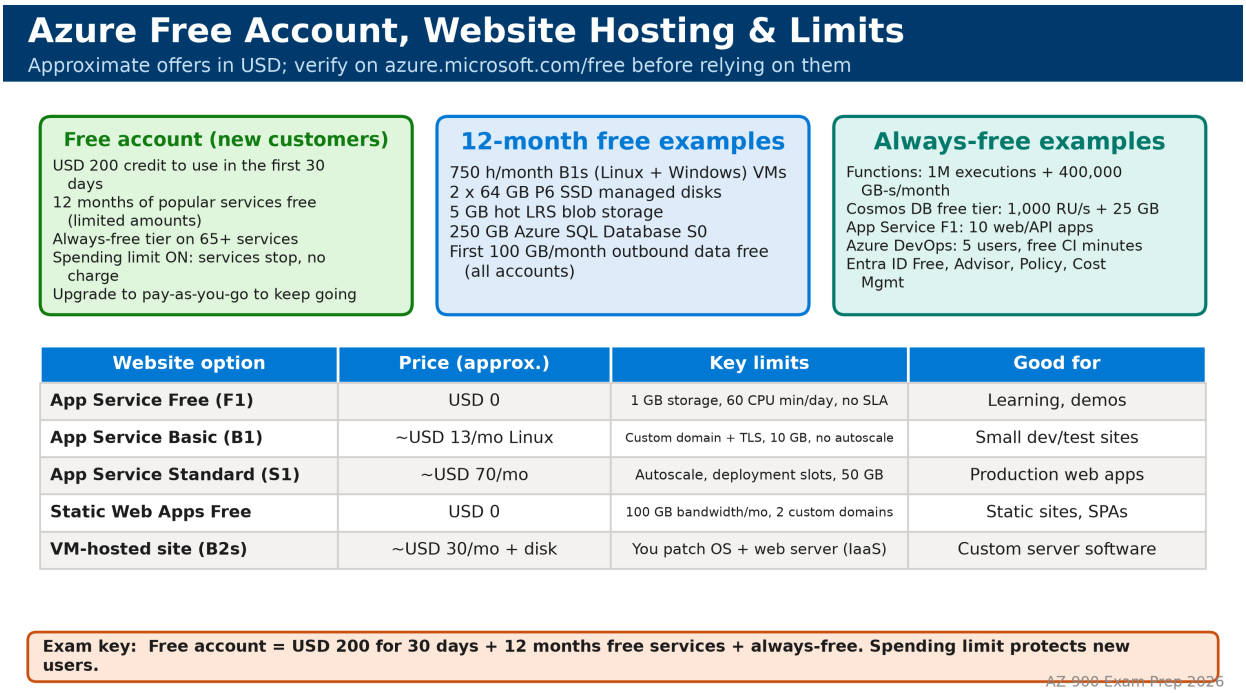


Figure Q212b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 213

CLOUD CONCEPTS

Lifecycle
v5 Q309

All Azure services in private preview must be accessed by using a separate Azure portal.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch the word 'All'. Private preview (early test version for invited customers only) features usually appear in the normal Azure portal once Microsoft turns them on for you. No separate portal is needed for all of them, so the answer is No. Q20 confirms private preview is not open to all customers.

Answer explained

✔ **No.**
Private preview access is given by adding your subscription to an allow list or by registering a feature. The feature then usually shows in the normal portal, CLI or APIs. Not every private preview needs a separate portal.

✖ **Yes.**
The word 'all' makes it false. Most private previews use the normal portal or APIs; only a few use a special link.

EXAM STRATEGY
Private previews are turned on for your subscription by registration or a feature flag, and used in the normal portal. 'Separate portal' requirement = No.

IMPACT
Knowing how preview access works lets teams test new features in their current subscription, without building a second environment.

What · Where · When · Why · How

What	Private preview is an invite-only early release of a feature for selected customers.
Where	Turned on in the customer's own subscription, and usually visible in the normal portal, CLI or APIs.
When	When Microsoft wants early feedback from a small group before public preview.
Why	Access does not need a separate portal for every private preview.
How	Microsoft adds the subscription to an allow list, or the customer registers a feature flag, for example with az feature register.

Exam tip
Absolute words like 'all' and 'only' in preview questions usually mean No.

Memory trick
Private preview = private guest list, same front door.

Real-world example, use case and scenario

EXAMPLE
Feature flag
A customer runs az feature register for a private preview. The new option appears on their normal portal page.

USE CASE
Design partner
An airline invited as a design partner tests a new networking feature in its current subscription before public preview.

SCENARIO
Hunting for a portal
An architect looked for a 'preview portal' to try an invite-only feature. Microsoft simply added the subscription to the allow list, and the feature appeared in the normal portal the next day.

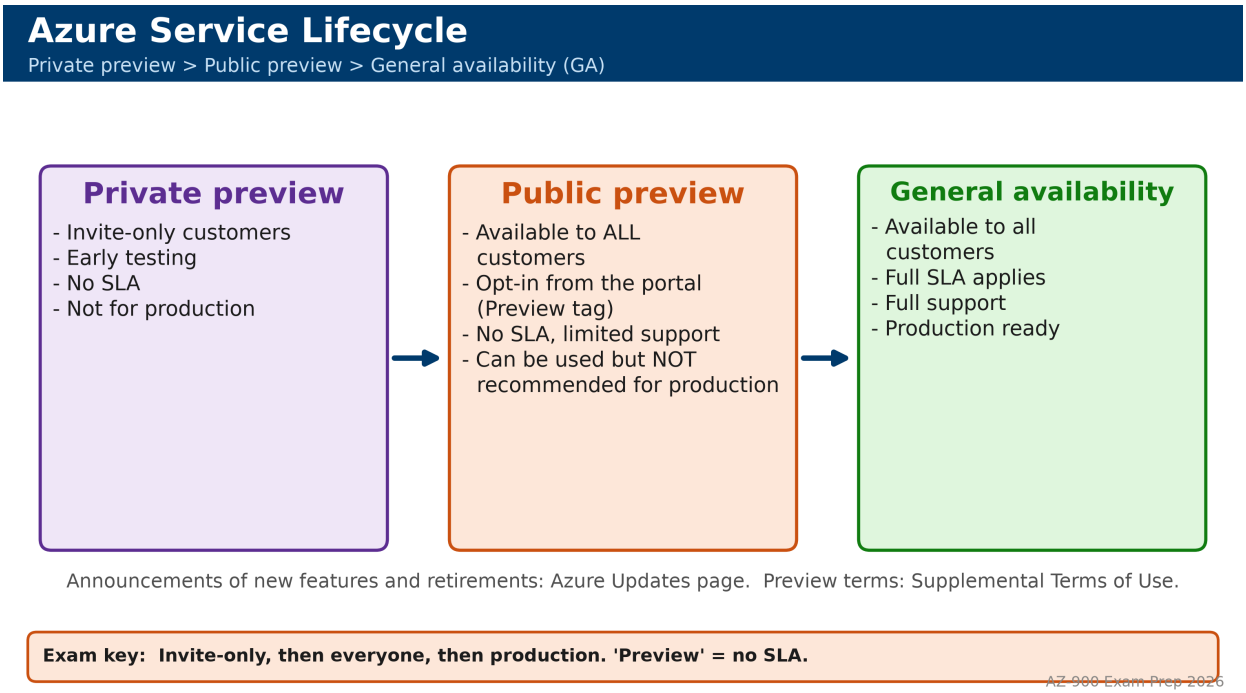


Figure Q213: Service lifecycle: preview to GA

About this figure

- Read the arrows left to right: invite-only private preview, open public preview, then general availability with SLA and full support.
- Previews are offered as-is under the Supplemental Terms of Use for Microsoft Azure Previews; features, pricing and availability may change or be withdrawn.
- Some previews are discounted or free, but many bill at reduced or standard rates; check the pricing page before relying on preview resources.
- Follow Azure Updates for previews, GA announcements and retirements; Service Health also sends retirement notices for services you actually use.
- Microsoft usually gives at least 12 months' notice before retiring a GA service or feature that customers depend on, with migration guidance.

Question 214

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q312

Match the Azure services to the appropriate descriptions.

✖ A. Extends on-premises networks to the Microsoft cloud via a private connection: VPN gateway. Combines two or more Azure virtual networks into a single logical virtual network: ExpressRoute. Provides an encrypted connection from on-premises networks to Azure via a public network: Virtual network peering.

✖ B. Extends on-premises networks to the Microsoft cloud via a private connection: ExpressRoute. Combines two or more Azure virtual networks into a single logical virtual network: VPN gateway. Provides an encrypted connection from on-premises networks to Azure via a public network: Virtual network peering.

✖ C. Extends on-premises networks to the Microsoft cloud via a private connection: VPN gateway. Combines two or more Azure virtual networks into a single logical virtual network: Virtual network peering. Provides an encrypted connection from on-premises networks to Azure via a public network: ExpressRoute.

✔ **D.** Extends on-premises networks to the Microsoft cloud via a private connection: ExpressRoute. Combines two or more Azure virtual networks into a single logical virtual network: Virtual network peering. Provides an encrypted connection from on-premises networks to Azure via a public network: VPN gateway.

Correct answer: Extends on-premises networks to the Microsoft cloud via a private connection: ExpressRoute. Combines two or more Azure virtual networks into a single logical virtual network: Virtual network peering. Provides an encrypted connection from on-premises networks to Azure via a public network: VPN gateway.

Clarification (read this first): Match each description by its key words. 'Private connection' (not over the internet) = ExpressRoute. 'Combines virtual networks' = VNet peering. 'Encrypted over a public network' = VPN gateway. Q349 also tests ExpressRoute: it does not use the public internet.

Answer explained

✔ **Extends on-premises networks to the Microsoft cloud via a private connection: ExpressRoute. Combines two or more Azure virtual networks into a single logical virtual network: Virtual network peering. Provides an encrypted connection from on-premises networks to Azure via a public network: VPN gateway.**
ExpressRoute is the private circuit that does not use the internet. VNet peering joins VNets over Microsoft's own network. VPN Gateway builds encrypted IPsec tunnels over the public internet.

✖ **Extends on-premises networks to the Microsoft cloud via a private connection: VPN gateway. Combines two or more Azure virtual networks into a single logical virtual network: ExpressRoute. Provides an encrypted connection from on-premises networks to Azure via a public network: Virtual network peering.**
All three are swapped. VPN Gateway uses the public internet, ExpressRoute does not join VNets, and peering does not connect on-premises networks.

✖ **Extends on-premises networks to the Microsoft cloud via a private connection: ExpressRoute. Combines two or more Azure virtual networks into a single logical virtual network: VPN gateway. Provides an encrypted connection from on-premises networks to Azure via a public network: Virtual network peering.**
Joining VNets is peering's job. Peering cannot give an encrypted on-premises link over the internet; that is VPN Gateway.

✖ **Extends on-premises networks to the Microsoft cloud via a private connection: VPN gateway. Combines two or more Azure virtual networks into a single logical virtual network: Virtual network peering. Provides an encrypted connection from on-premises networks to Azure via a public network: ExpressRoute.**
ExpressRoute and VPN are reversed. ExpressRoute is the private link; VPN Gateway is the encrypted link over the public internet.

EXAM STRATEGY
Key words: 'private connection' = ExpressRoute, 'combines virtual networks' = peering, 'encrypted over public network' = VPN gateway. Match the clearest word first.

IMPACT
Using VPN instead of ExpressRoute for high-volume or delay-sensitive traffic causes slow performance. Network matching questions come up often on the exam.

What · Where · When · Why · How

What	ExpressRoute, VNet peering and VPN Gateway are Azure's three main network connection options.
Where	ExpressRoute and VPN Gateway connect to a VNet's gateway subnet. Peering is set up between two VNets.
When	Private dedicated link: ExpressRoute. VNet to VNet: peering. Encrypted over the internet: VPN Gateway.
Why	Each one matches its description: private circuit, one combined network, encrypted internet tunnel.
How	ExpressRoute uses a circuit from a network provider. Peering uses Microsoft's own network. VPN uses IPsec/IKE (encryption standards) tunnels.

Exam tip
Private connection, not over the internet = ExpressRoute. Encrypted over the public internet = VPN Gateway. VNet to VNet = peering.

Memory trick
Express = Exclusive lane; VPN = Very Public Network (but encrypted); Peering = Pairing VNets.

Real-world example, use case and scenario

EXAMPLE
Three links
A company uses ExpressRoute for its head office, a site-to-site VPN for a small branch, and peering between its hub and spoke VNets.

USE CASE
Trading latency
A stock trading firm uses ExpressRoute for steady low delay and private routing to Azure for its trading analytics.

SCENARIO
Branch on a budget
A chain of dental clinics needed a cheap link from each clinic to Azure. ExpressRoute was too costly, so they used VPN Gateway tunnels over their normal broadband. They also peered the clinic VNet with the shared services VNet.

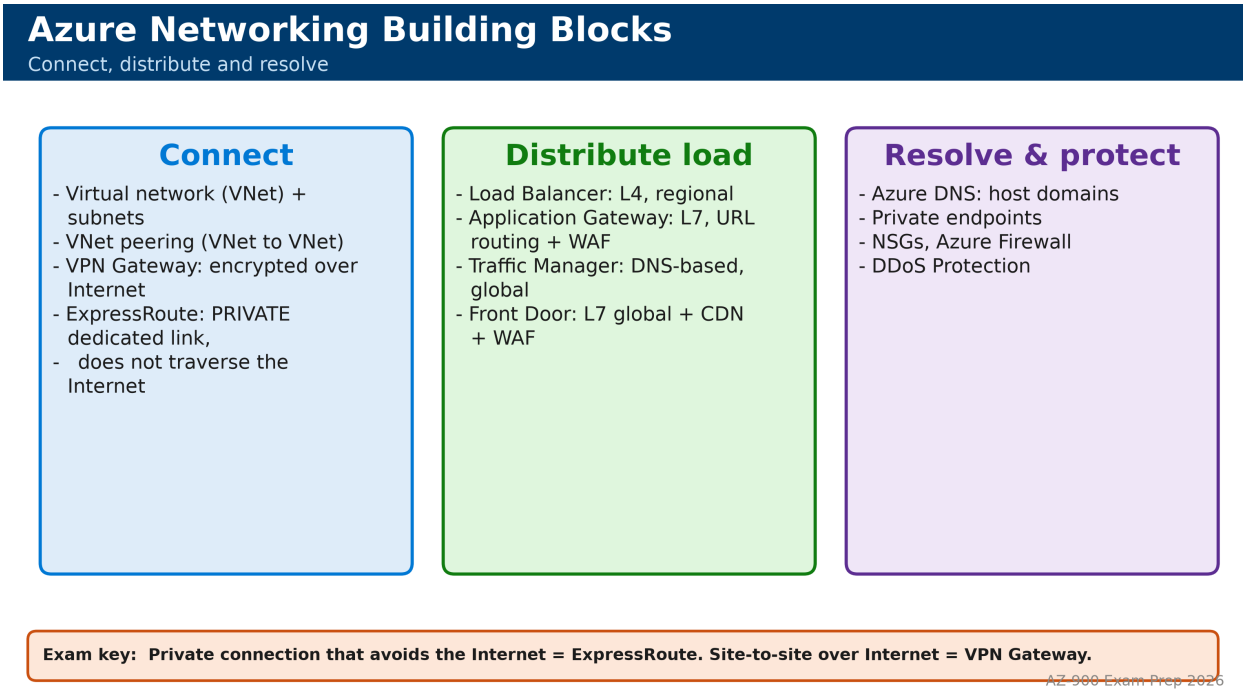


Figure Q214: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 215

AZURE ARCHITECTURE & SERVICES

Subscriptions
v5 Q314

You can convert the Azure subscription of your company from Free Trial to Pay-As-You-Go.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
You can upgrade a Free Trial / free account to pay-as-you-go in the portal at any time. You keep the same subscription, resources and data.

✖ **No.**
Microsoft supports upgrading the free subscription in place to pay-as-you-go. No new subscription or redeployment is needed.

EXAM STRATEGY
Upgrade is a built-in portal action that keeps existing resources. 'Convert Free Trial to Pay-As-You-Go' = Yes.

IMPACT
Upgrading keeps the resources and data built during the trial. You avoid costly redeployment when a test moves to production.

What · Where · When · Why · How

What	Upgrading changes a Free Trial / free account subscription to pay-as-you-go.
Where	Done in the Azure portal, from the Subscriptions page or the upgrade banner.
When	When the credit runs out, or workloads must keep running after the trial.
Why	Upgrade in place is supported, so changing to pay-as-you-go is possible.
How	Click Upgrade and confirm the payment method. Resources, data and subscription ID are kept.

Exam tip
Upgrade keeps resources and any free amounts left. You only start paying for use above them.

Memory trick
Trial to PAYG is a one-click Upgrade, not a rebuild.

Real-world example, use case and scenario

EXAMPLE
Upgrade button
A user clicks Upgrade in the portal banner and confirms the payment method. The subscription becomes Pay-As-You-Go.

USE CASE
Pilot to production
A bakery's online-ordering test, built on a free account, is upgraded so the site keeps running after the trial.

SCENARIO
No migration needed
A charity feared it would need to rebuild its donor site after the trial. Upgrading kept the same subscription ID, web app and database, so nothing had to move.

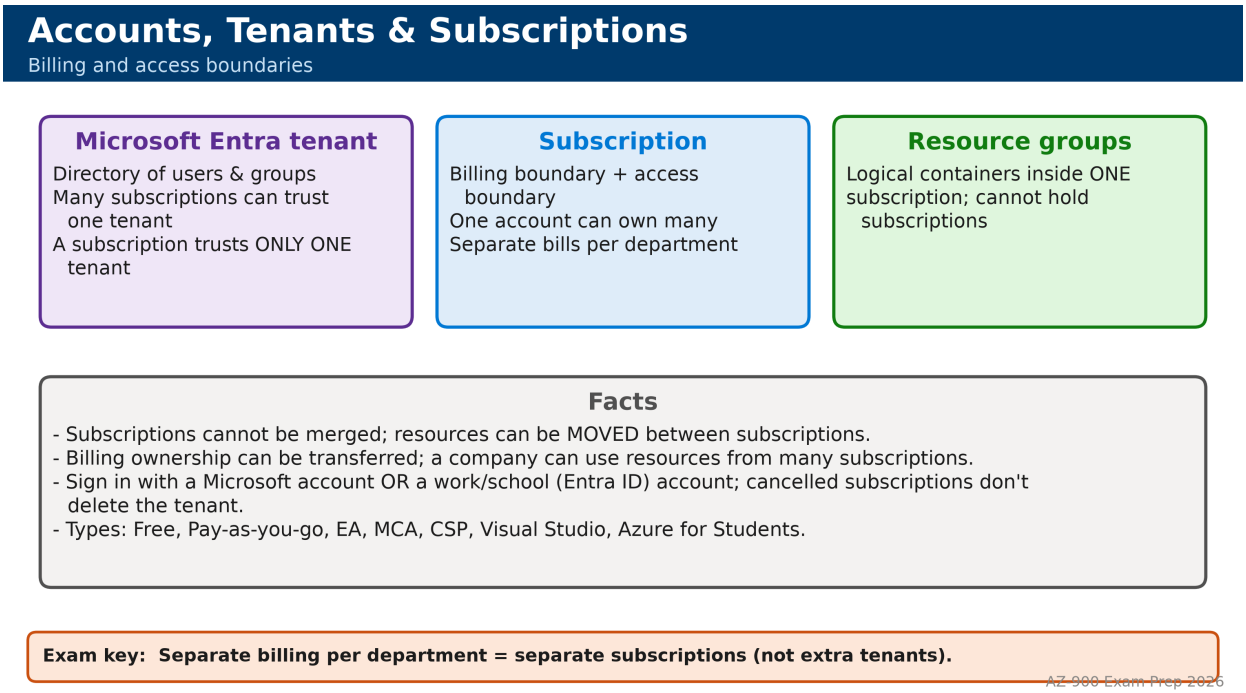


Figure Q215a: Accounts, tenants and subscriptions

About this figure

- Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.
- Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.
- Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.
- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

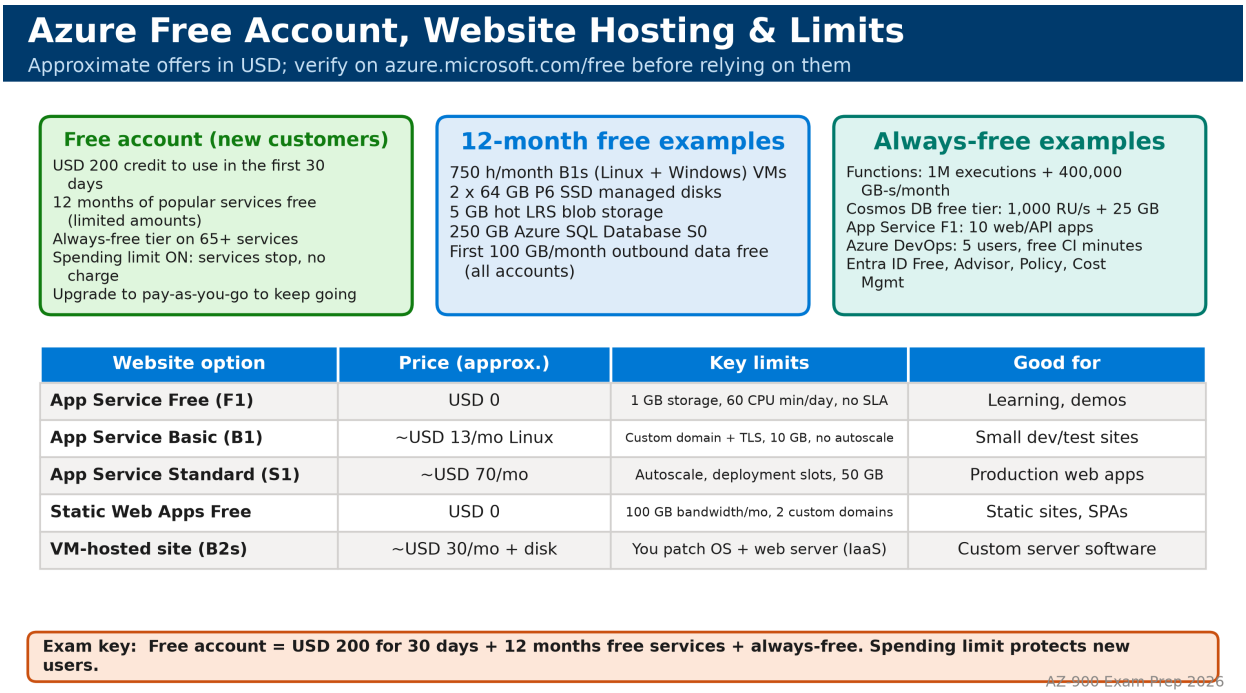


Figure Q215b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 216

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q315

With Azure Reservations, you pay less fer virtual machines than with pay as-you-go pricing.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Reservations give up to about 72% off pay-as-you-go VM prices, in return for a 1-year or 3-year commitment.

✖ **No.**
Reservations exist to give a lower price than pay-as-you-go in return for a commitment.

EXAM STRATEGY
Reservations trade a 1-year or 3-year commitment for lower prices than pay-as-you-go. 'Pay less' = Yes.

IMPACT
VMs that run all the time can save up to about 72 percent with reservations. Leaving them on pay-as-you-go wastes a lot of budget.

What · Where · When · Why · How

What	Azure Reservations are 1-year or 3-year commitments that lower the price of resource use.
Where	Bought on the portal's Reservations page and applied to a subscription, resource group or shared scope.
When	When VMs run steadily for a long time, such as production servers that run 24/7.
Why	They give up to about 72% off pay-as-you-go VM prices.
How	Buy a reservation for a VM size and region. The discount applies automatically to matching VMs.

Exam tip
Steady 24/7 workloads = Reservations (1 or 3 years). Can be stopped at any time = Spot. Flexible across services = Savings plan.

Memory trick
Commit to save: reserve and the price goes down.

Real-world example, use case and scenario

EXAMPLE
3-year D-series
A D4s v5 VM reserved for 3 years costs far less per hour than the same VM on pay-as-you-go.

USE CASE
ERP servers
A manufacturer runs its SAP servers 24/7 and buys 3-year reservations for them.

SCENARIO
Bill shock fixed
A software company's always-on API VMs cost too much on pay-as-you-go. Cost Management's reservation tips showed savings of over 50%, so they bought 1-year reservations and cut the monthly bill.

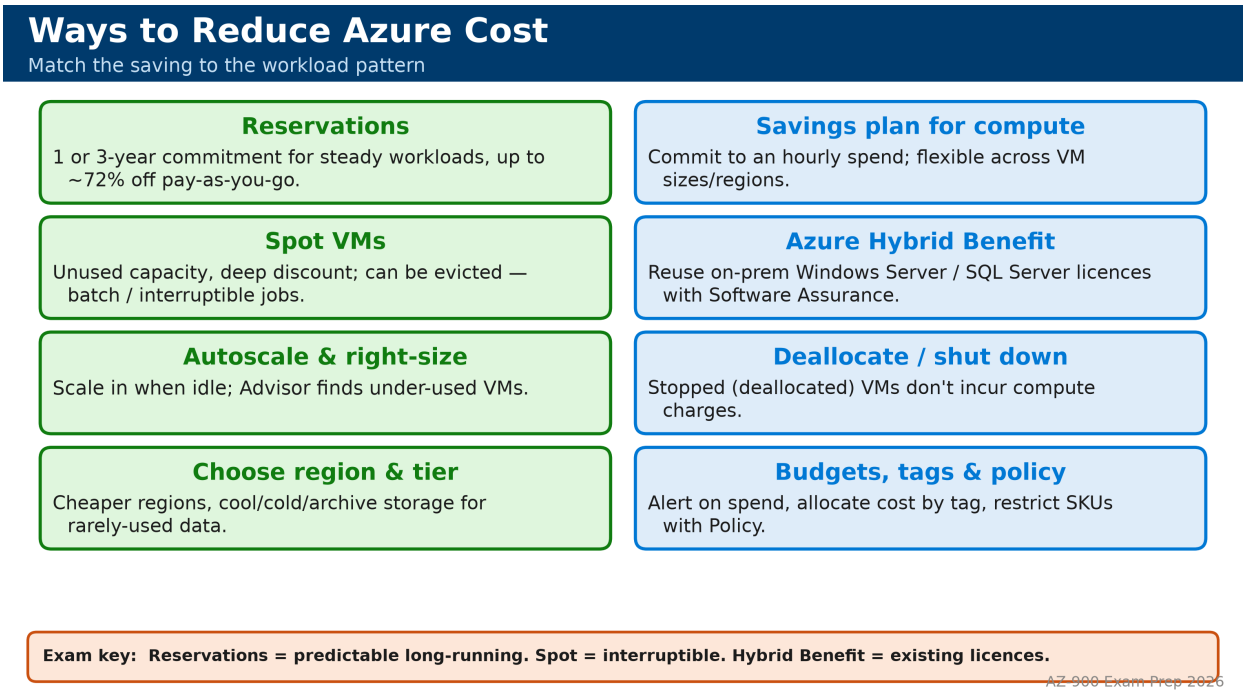


Figure Q216a: Ways to reduce cost

About this figure

• Each box matches a saving to a workload pattern: steady (reservations), flexible (savings plan), interruptible (Spot), existing licences (Hybrid Benefit).

• Reservations save up to approx. 72% versus pay-as-you-go; savings plans for compute save up to approx. 65% and apply across VM families and regions.

• Spot VMs can be up to approx. 90% cheaper, but Azure can evict them with 30 seconds' notice when it needs capacity back.

• Combining Azure Hybrid Benefit with reservations can cut Windows VM costs by up to approx. 80%; Linux subscriptions such as RHEL and SUSE also qualify.

• Example: moving rarely read data from Hot (~USD 0.02/GB-month) to Archive (~USD 0.002 or less) saves storage cost, but reads and early deletion cost more.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q216b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 217

CLOUD CONCEPTS

SLA
v5 Q317

How should you calculate the monthly uptime percentage? [...] ÷ Maximum Available Minutes × 100.

✖ A. Downtime in Minutes.

✖ B. Maximum Available Minutes.

✔ **C.** (Maximum Available Minutes-Downtime in Minutes).

Correct answer: (Maximum Available Minutes-Downtime in Minutes).

Clarification (read this first): The formula has two parts. This question asks for the TOP part (numerator). Q218 uses the same formula but asks for the BOTTOM part (denominator) - that answer is Maximum Available Minutes.

Answer explained

✔ **(Maximum Available Minutes-Downtime in Minutes).**
Monthly Uptime % = (Maximum Available Minutes - Downtime in Minutes) / Maximum Available Minutes x 100. So the top part is the minutes the service was really up.

✖ **Downtime in Minutes.**
Downtime / Maximum minutes x 100 gives the downtime percentage, not the uptime.

✖ **Maximum Available Minutes.**
Maximum / Maximum x 100 is always 100% and ignores the outage. This value is the bottom part.

EXAM STRATEGY
Formula: (Maximum Available Minutes minus Downtime) divided by Maximum Available Minutes, times 100. The blank is the top part, so pick the subtraction.

IMPACT
If you calculate uptime correctly, you can check if the SLA (uptime promise) was broken and if you can claim a credit.

What · Where · When · Why · How

What	Monthly Uptime % is the SLA (uptime promise) formula that measures how available a service was.
Where	It is written in each service's section of the Microsoft SLA for Online Services.
When	When you check if an outage broke the SLA and a credit is due.
Why	The top part must be the minutes the service was up: maximum minutes minus downtime.
How	Calculate (Maximum Available Minutes - Downtime) / Maximum Available Minutes x 100.

Exam tip
Uptime % = (Max - Down) / Max x 100. To fill the blank, ask: which part was really up?

Memory trick
Up = Max minus Down, over Max.

Real-world example, use case and scenario

EXAMPLE
30-day month
A month has 43,200 minutes. With 43 minutes of downtime: (43,200 - 43) / 43,200 x 100 = 99.90%.

USE CASE
SLA claim evidence
A logistics company uses its monitoring data to calculate uptime. Then it checks if an SLA credit is due.

SCENARIO
Wrong numerator
An analyst at a ticketing company put downtime on top. She reported 0.1% uptime and her managers panicked. With (Max - Down) on top, the real result was 99.9%, exactly at the SLA line.

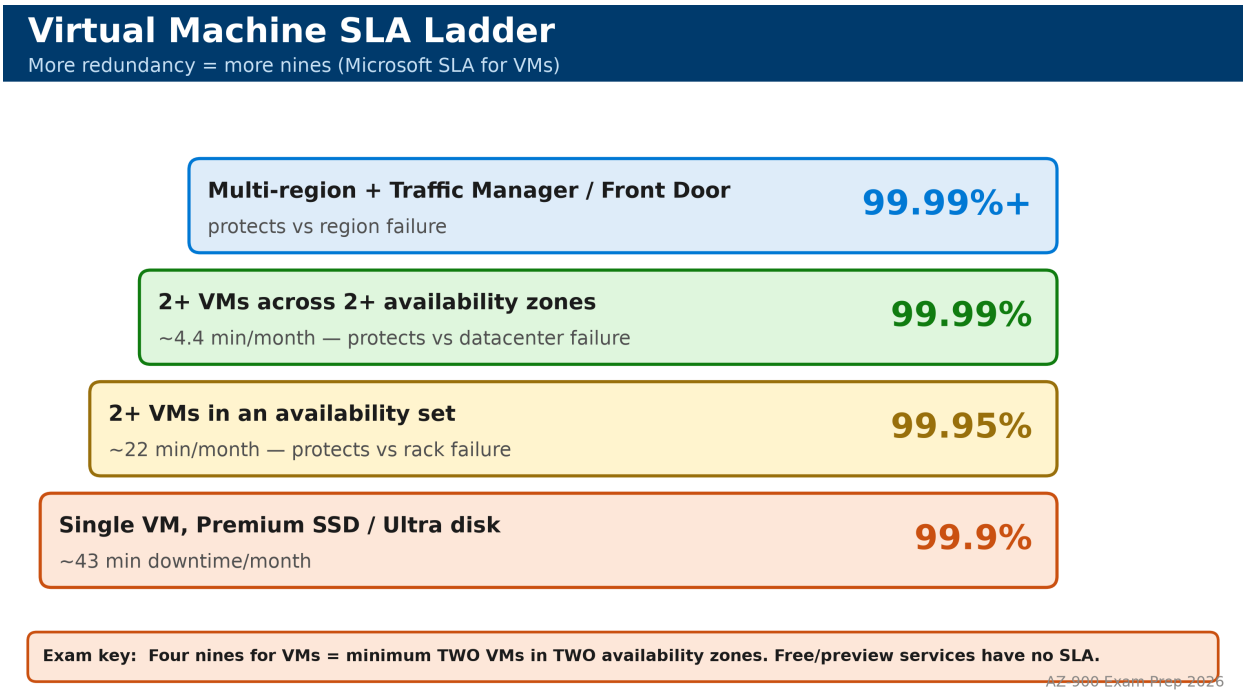


Figure Q217: Virtual machine SLA ladder

About this figure

- Climb the ladder: each step adds redundancy and a higher VM SLA; the downtime figures assume a 30-day month.
- A single VM with Standard SSD gets about 99.5% and with Standard HDD about 95%; all its OS and data disks must meet that tier.
- Availability sets spread VMs across fault domains (racks) and update domains (maintenance batches) in one datacenter; zones spread across datacenters.
- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

Question 218

CLOUD CONCEPTS

SLA
v5 Q318

How should you calculate the monthly uptime percentage? (Maximum Available Minutes-Downtime in Minutes) ÷ [...] × 100.

✖ A. 60.

✖ B. 1,440.

✔ **C.** Maximum Available Minutes.

Correct answer: Maximum Available Minutes.

Clarification (read this first): Same formula as Q217, but this question asks for the BOTTOM part (denominator). Q217 asks for the top part - that answer is (Maximum Available Minutes - Downtime).

Answer explained

✔ **Maximum Available Minutes.**
The bottom part is Maximum Available Minutes. This is the total number of minutes in the billing month. So the result is uptime as a percentage of the whole month.

✖ **60.**
60 is the minutes in one hour. Dividing by it gives hours, not a monthly percentage.

✖ **1,440.**
1,440 is the minutes in one day. Dividing by it gives days, not a monthly percentage.

EXAM STRATEGY
The bottom part is Maximum Available Minutes. Remove fixed numbers like 60 or 1,440. Months have different lengths, so the bottom part must change with the month.

IMPACT
A wrong bottom part gives a wrong uptime. You may then make a wrong claim, or miss a real service credit.

What · Where · When · Why · How

What	Maximum Available Minutes is the total number of minutes in the billing month for a service.
Where	It is the bottom part of the Monthly Uptime % formula in the Microsoft SLA (uptime promise).
When	When you turn up-minutes into a monthly uptime percentage.
Why	Dividing by all the minutes in the month gives a percentage of the whole month.
How	Example: a 30-day month has 43,200 minutes. Divide the up-minutes by 43,200 and multiply by 100.

Exam tip
The same term is in the top and the bottom of the formula: Maximum Available Minutes.

Memory trick
Divide by the whole pie - Maximum Available Minutes.

Real-world example, use case and scenario

EXAMPLE
31-day month
A 31-day month has 44,640 minutes. With 10 minutes down: 44,630 / 44,640 x 100 = 99.978%.

USE CASE
Monthly SLA report
A managed service provider writes a monthly report. For each customer, it divides the up-minutes by the maximum available minutes of the month.

SCENARIO
Divided by a day
A new engineer at a utility company divided by 1,440 and got 3,000% uptime. The team lead changed the bottom part to the month's maximum available minutes.

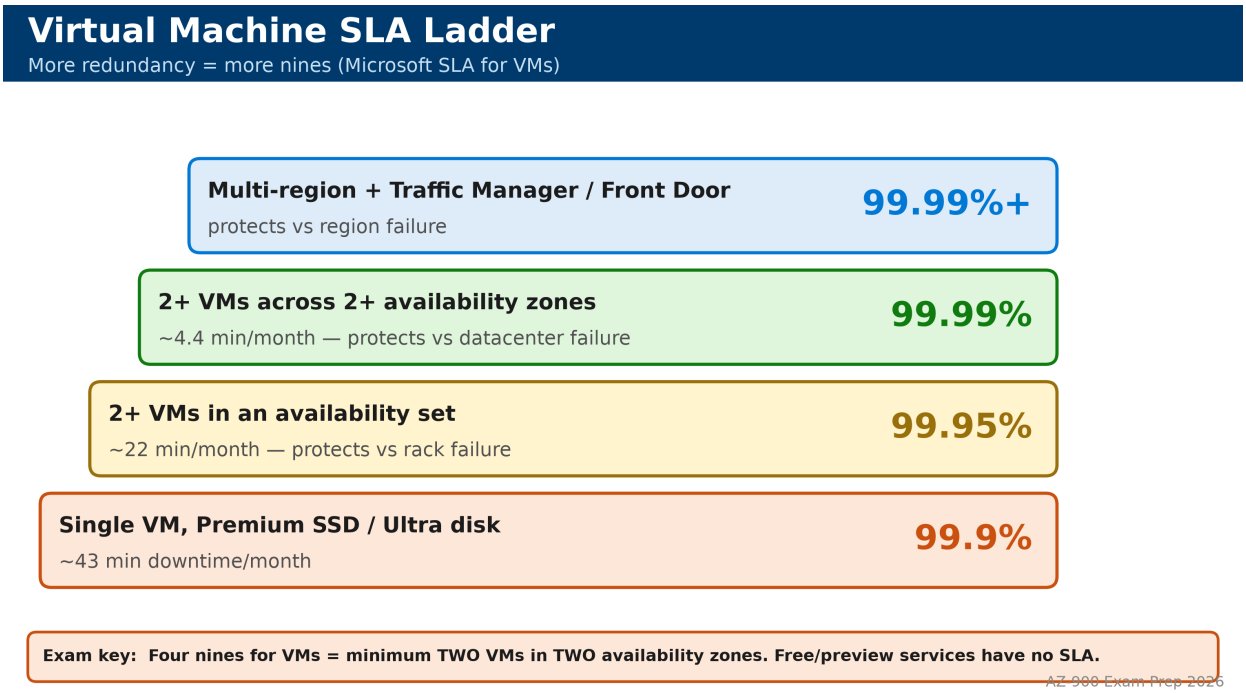


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- Availability sets spread VMs across fault domains (racks) and update domains (maintenance batches) in one datacenter; zones spread across datacenters.
- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

Question 219

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q320, Q341

By copying several gigabits of data to Azure from an on-premises network over a VPN, additional data transfer costs are incurred.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch the direction: data goes INTO Azure (from on-premises to Azure). Data going in is free. Q241 asks about data going OUT of Azure to on-premises - that answer is Yes (it costs money).

Answer explained

✔ **No.**
Data going into Azure (ingress) is free. You pay for VPN Gateway hours and for storage. You do not pay for the incoming data itself.

✖ **Yes.**
Uploading to Azure is ingress (data coming in), and it is free. Only data going out (egress) and data between regions are charged.

EXAM STRATEGY
Look at the direction. Copying 'to Azure' = ingress = free. Do not mix up the VPN gateway hourly cost with the cost per GB of data.

IMPACT
Because ingress is free, moving a lot of data into Azure is cheap. Plan your budget for gateway hours and for later egress (data leaving Azure).

What · Where · When · Why · How

What	Ingress is data that moves into Azure datacenters.
Where	It covers all incoming traffic, also traffic over a site-to-site VPN Gateway.
When	When you upload or copy on-premises data into Azure.
Why	Incoming data is free. Only egress (data leaving Azure) and traffic between regions are charged.
How	Copy the data over the VPN. You pay for gateway hours and storage, not for incoming data.

Exam tip
Direction matters: data into Azure is free, data out of Azure is charged. The gateway hourly cost is not a data transfer cost.

Memory trick
Data checks in free, pays to check out.

Real-world example, use case and scenario

EXAMPLE
Backup upload
You upload 500 GB of archive files over a site-to-site VPN. There is no data transfer charge. You pay only for storage and the gateway.

USE CASE
Log ingestion
A telecom company sends router logs into Azure. The incoming traffic is free. It plans its budget for storage and Log Analytics instead.

SCENARIO
Restore surprise
A law firm uploaded its backups for free. Later it copied 2 TB back to its office and got a bill. Egress (data leaving Azure) is charged, so now the firm restores only the files it needs.

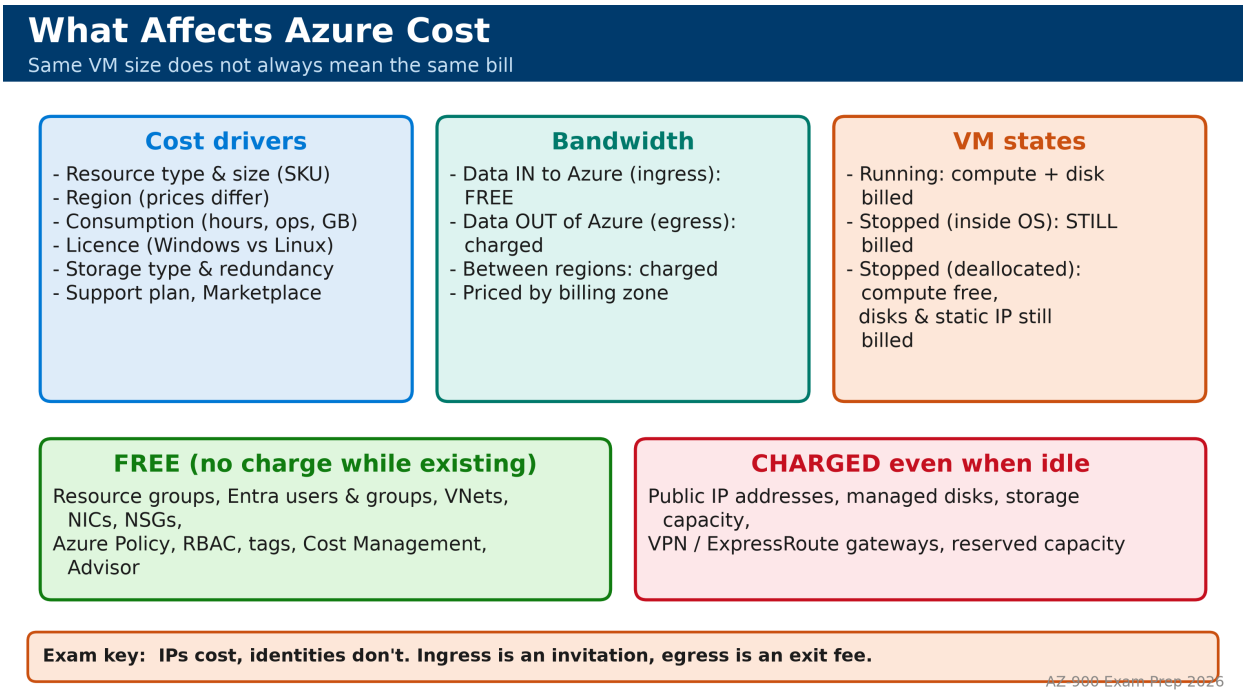


Figure Q219a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

• Bandwidth: ingress is free; the first 100 GB/month of Internet egress is free, then approx. USD 0.087/GB in North America and Europe.

• Inter-region transfer is approx. USD 0.02/GB within a continent; VNet peering is approx. USD 0.01/GB each way in the same region.

• A VM stopped inside the guest OS is still allocated and billed for compute; only Stopped (deallocated) from the portal, CLI or API stops compute charges.

• A Standard static public IP costs approx. USD 3.65/month even unattached; a 128 GB Premium SSD costs approx. USD 20/month even when its VM is deallocated.

• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q219b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 220

CLOUD CONCEPTS

Lifecycle
v5 Q321, Q46, Q151

An Azure service is available to all Azure customers when it is in [...].

✔ **A.** Public preview.

✖ B. Private preview.

✖ C. Development.

✖ D. Development an Enterprise Agreement (EA) subscription.

Correct answer: Public preview.

Clarification (read this first): The key words are 'ALL Azure customers'. Public preview is open to everyone. Private preview is only for invited customers. General availability (GA) is also open to all, but it is not an option here.

Answer explained

✔ **Public preview.**
Public preview opens a feature to all Azure customers to test. It has no SLA (uptime promise).

✖ **Private preview.**
Private preview is only for customers that Microsoft invites or approves.

✖ **Development.**
Features in development are used only inside Microsoft. Customers cannot use them.

✖ **Development an Enterprise Agreement (EA) subscription.**
This is not a release stage. An Enterprise Agreement is a licensing contract. It does not decide who can use a preview.

EXAM STRATEGY
Public preview = open to all customers. Private preview = only invited customers. Remove 'Development' and the EA option, because they are not release stages.

IMPACT
If you know the release stages, you know when a feature is open and supported. Then you can plan project dates safely.

What · Where · When · Why · How

What	Public preview is the release stage that opens a feature to all customers.
Where	It is open in any subscription. The portal often shows '(Preview)' next to it.
When	When Microsoft wants feedback from many customers before general availability.
Why	Of the stages listed, only public preview is open to all customers.
How	Customers turn on the feature and use it under the preview terms, without an SLA (uptime promise).

Exam tip
Private preview = some chosen customers. Public preview = all customers, no SLA. GA = all customers, with SLA.

Memory trick
Public preview is a public beta.

Real-world example, use case and scenario

EXAMPLE
Preview tag
A new portal feature has the label '(Preview)'. On the first day of public preview, it shows in every subscription in the world.

USE CASE
Early adopter testing
A retail analytics team tests a public preview AI feature in a test subscription. It plans to use the feature fully when it reaches GA.

SCENARIO
Waiting for an invite
A consultant thought he needed an invitation to try a new service. The service was in public preview, so he turned it on at once in his own subscription.

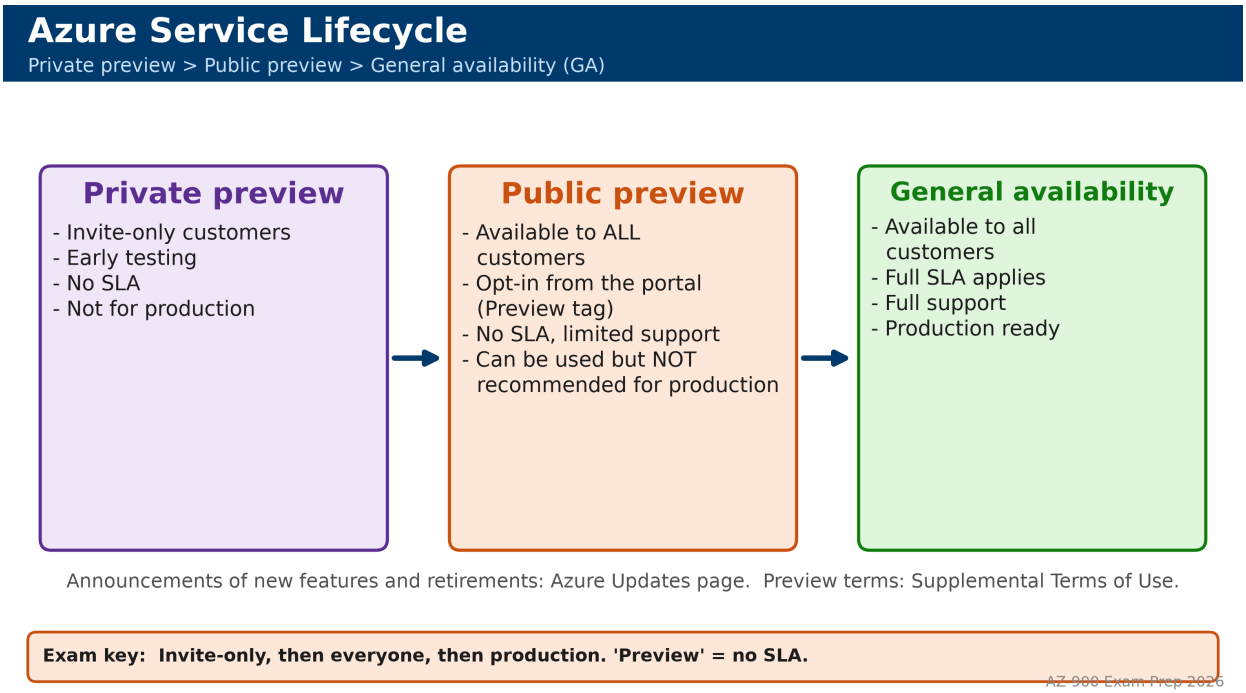


Figure Q220: Service lifecycle: preview to GA

About this figure

- Read the arrows left to right: invite-only private preview, open public preview, then general availability with SLA and full support.
- Previews are offered as-is under the Supplemental Terms of Use for Microsoft Azure Previews; features, pricing and availability may change or be withdrawn.
- Some previews are discounted or free, but many bill at reduced or standard rates; check the pricing page before relying on preview resources.
- Follow Azure Updates for previews, GA announcements and retirements; Service Health also sends retirement notices for services you actually use.
- Microsoft usually gives at least 12 months' notice before retiring a GA service or feature that customers depend on, with migration guidance.

Question 221

CLOUD CONCEPTS

Lifecycle
v5 Q322

The cost of an Azure service in private preview decreases when the service becomes generally available.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is 'decreases'. The price usually does NOT go down at GA (general availability). Preview is often free or cheaper, so the price usually goes up or stays the same.

Answer explained

✔ **No.**
Previews are often free or cheaper. Full pricing usually starts at general availability (GA). So the cost does not go down at GA.

✖ **Yes.**
GA usually brings the full price and an SLA (uptime promise). The preview discount ends; the price does not drop.

EXAM STRATEGY
No rule says prices go down at GA. Preview prices are often lower, so GA prices usually go up or stay the same. 'Decreases' = No.

IMPACT
If you plan your budget on preview prices, you may spend more than planned when the full GA price starts.

What · Where · When · Why · How

What	Preview pricing is often free or cheaper until general availability (GA).
Where	The service's Azure pricing page shows the preview price or the GA price.
When	When you plan costs for a service that moves from preview to GA.
Why	GA usually brings the full price, so the cost does not go down.
How	Check the pricing page and Azure Updates for the GA price before you plan the budget.

Exam tip
Preview = often free or cheaper. GA = full price plus SLA.

Memory trick
The free sample ends when the shop officially opens.

Real-world example, use case and scenario

EXAMPLE
Preview discount ends
A feature costs 50% of the price during preview. From the GA date, it costs the full price.

USE CASE
Budget forecasting
A media company's cost team adds extra money to the budget for every preview service that will reach GA.

SCENARIO
Waiting for a cheaper GA
A startup waited for GA, hoping the service would get cheaper. Instead, the free preview ended and GA added charges. The team had to change its budget.

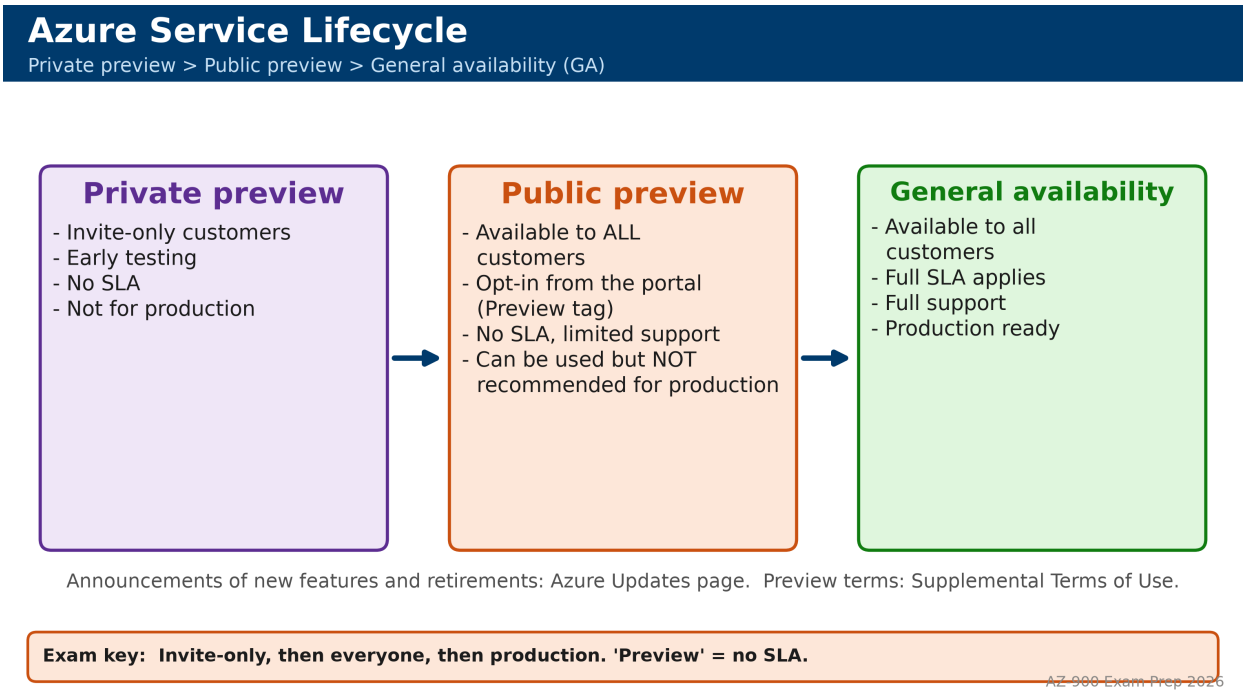


Figure Q221: Service lifecycle: preview to GA

About this figure

- Read the arrows left to right: invite-only private preview, open public preview, then general availability with SLA and full support.
- Previews are offered as-is under the Supplemental Terms of Use for Microsoft Azure Previews; features, pricing and availability may change or be withdrawn.
- Some previews are discounted or free, but many bill at reduced or standard rates; check the pricing page before relying on preview resources.
- Follow Azure Updates for previews, GA announcements and retirements; Service Health also sends retirement notices for services you actually use.
- Microsoft usually gives at least 12 months' notice before retiring a GA service or feature that customers depend on, with migration guidance.

Question 222

AZURE MANAGEMENT & GOVERNANCE

Subscriptions
v5 Q324

An Azure free account has a spending limit.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
The free account has a spending limit equal to its 200 USD credit. When the credit is used up, Azure turns off your resources. It does not charge your card.

✖ **No.**
The free account does have a spending limit. To remove it, you must upgrade to pay-as-you-go, which has no spending limit.

EXAM STRATEGY
Offers with credit have a spending limit equal to the credit. Free account = Yes. Pay-as-you-go has no spending limit; it uses budgets instead.

IMPACT
The spending limit protects learners from surprise card charges. But resources turn off as soon as the credit runs out.

What · Where · When · Why · How

What	A spending limit stops charges at the amount of credit that comes with an offer.
Where	It applies to credit offers like the free account. It does not apply to pay-as-you-go or EA.
When	When users want to avoid surprise charges while they try Azure.
Why	The free account's 200 USD credit works as a spending limit.
How	When the credit is used up, resources are turned off. You remove the limit by upgrading.

Exam tip
Credit offers have a spending limit (free account, Visual Studio, Azure for Students). Pay-as-you-go uses budgets and alerts instead.

Memory trick
Free account has a fuse; it trips at 200.

Real-world example, use case and scenario

EXAMPLE
Credit used up
A user spends the 200 USD credit in 10 days. Azure deallocates (stops and releases) the VMs and shows a banner to remove the limit.

USE CASE
Safe experimentation
A teenager learns cloud on a free account. She knows the spending limit stops surprise charges on her parents' card.

SCENARIO
Runaway GPU
A researcher forgot to stop a GPU VM on a free account. There was no large bill. The spending limit stopped the subscription when the credit ran out.

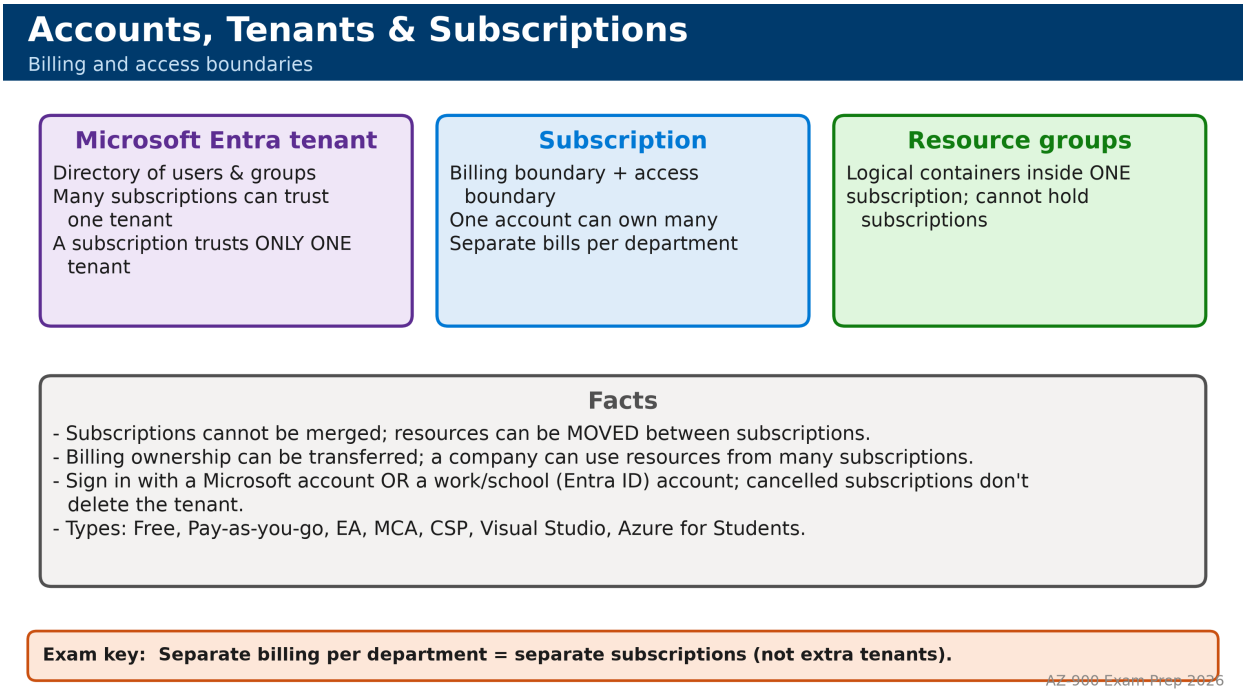


Figure Q222a: Accounts, tenants and subscriptions

About this figure

- Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.
- Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.
- Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.
- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

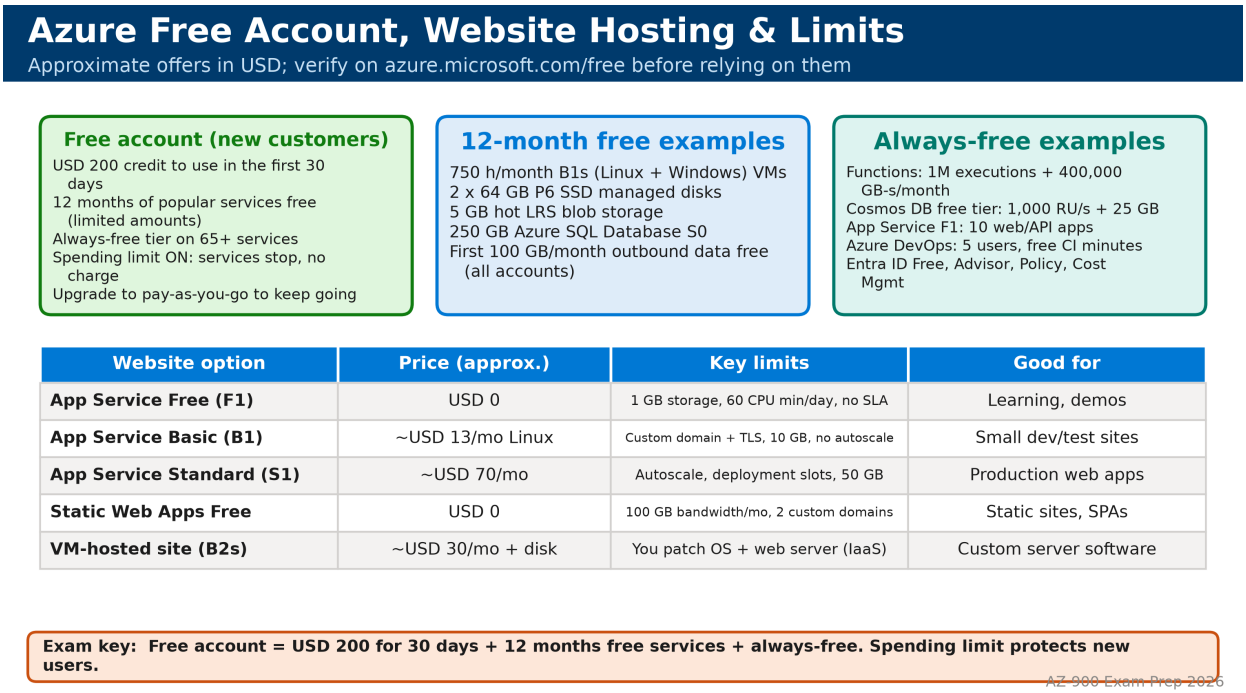


Figure Q222b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 223

AZURE MANAGEMENT & GOVERNANCE

Subscriptions
v5 Q325

An Azure free account has a limit of 2TB of data that can be uploaded to Azure.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap is the very exact number '2TB'. No such upload limit exists. A free account is limited by its credit and free amounts, not by how much data you upload.

Answer explained

✔ **No.**
The free account has no 2 TB upload limit. Ingress (data coming into Azure) is free. Storage is limited only by the free amounts and the credit.

✖ **Yes.**
The free account terms have no 2 TB upload limit. Uploads are free ingress. Only the credit and storage costs limit you.

EXAM STRATEGY
Look out for oddly exact numbers like 2 TB. Ingress is free and free accounts have no upload limit. Made-up number limits are false.

IMPACT
This stops you from wrongly rejecting Azure for data-heavy tests because of a limit that does not exist.

What · Where · When · Why · How

What	Free account use is limited by the credit and free amounts, not by upload size.
Where	It applies to incoming data and storage services in the free subscription.
When	When you move data into Azure during a free trial.
Why	There is no 2 TB upload limit, and ingress (data coming in) is free.
How	Upload freely. Storage above the free amount (for example 5 GB LRS blob) is paid from the credit.

Exam tip
Made-up number limits (2 TB upload, 10 accounts) in free account questions are usually false.

Memory trick
Free account caps the wallet, not the upload pipe.

Real-world example, use case and scenario

EXAMPLE
Free storage amount
The 12-month free offer includes 5 GB of hot LRS blob storage. Extra storage is paid from the credit.

USE CASE
Dataset trial
A data science student uploads a 50 GB dataset. The upload is free, and the storage uses the credit.

SCENARIO
Splitting uploads for no reason
A photographer split her archives into small batches because she feared a 2 TB limit. There is no upload limit, so she watched her storage cost against her credit instead.

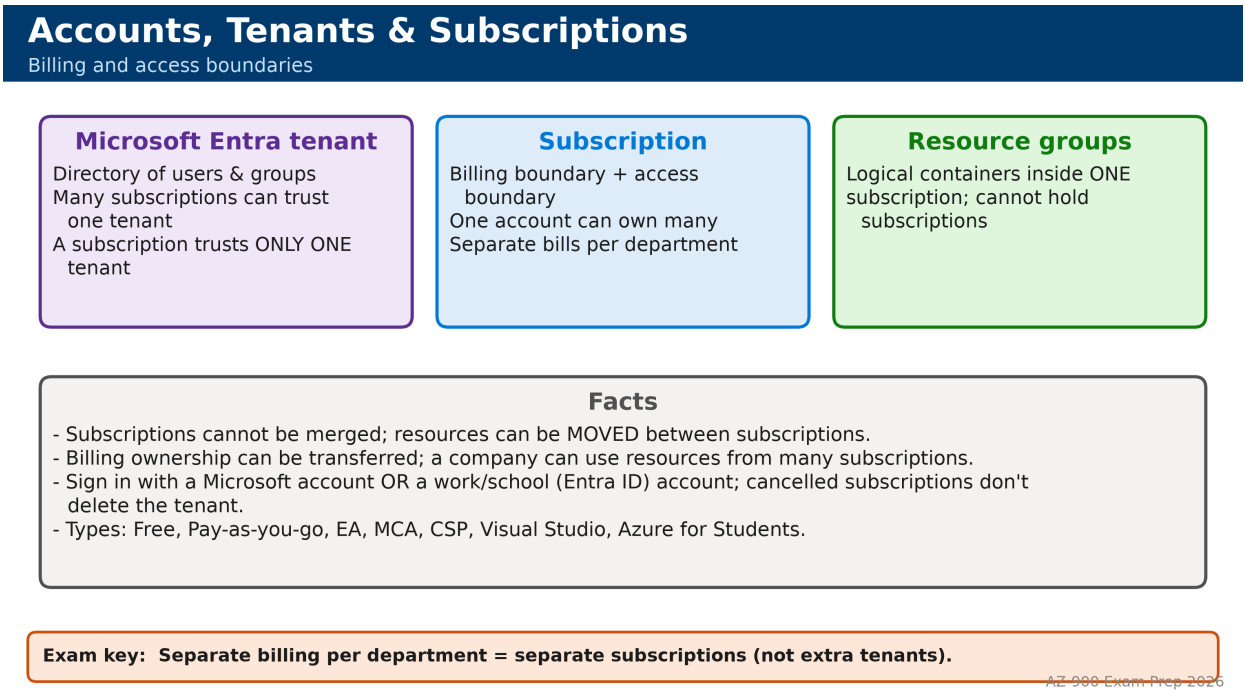


Figure Q223a: Accounts, tenants and subscriptions

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- Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.
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- Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.
- Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

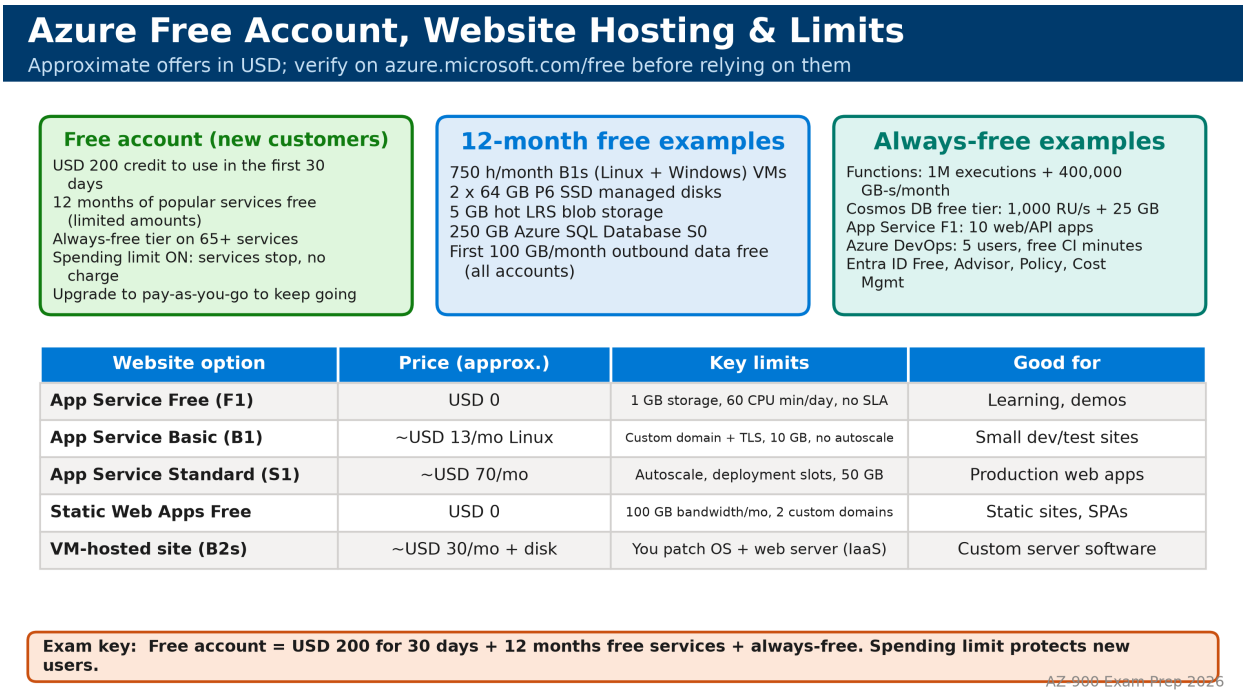


Figure Q223b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 224

AZURE MANAGEMENT & GOVERNANCE

Subscriptions
v5 Q326

An Azure free account can contain an unlimited number of web apps.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is 'unlimited'. Nothing on a free account is unlimited. The App Service Free (F1) tier allows only 10 web apps per plan.

Answer explained

✔ **No.**
Free account use is limited by free amounts, the credit and quotas (set limits). For example, the App Service Free (F1) tier allows only 10 web apps per plan.

✖ **Yes.**
Nothing is unlimited on a free account. The App Service Free tier allows at most 10 web apps per plan, and the credit and quotas also limit you.

EXAM STRATEGY
Look for the word 'unlimited'. Free accounts are limited by credit, free amounts and quotas. Any 'unlimited' claim on a free account = No.

IMPACT
Free tier limits stop large tests. Know them before you plan load testing on a trial subscription.

What · Where · When · Why · How

What	The App Service Free (F1) tier hosts web apps at no cost, with limits.
Where	The limit is set per App Service plan. The Free tier allows up to 10 apps per plan.
When	When you host small test or learning web apps on a free account.
Why	Free offers have limits, so web apps are not unlimited.
How	Create apps in an F1 plan. Move to a paid tier when you reach the limits.

Exam tip
'Unlimited' on a free account is a warning sign - the answer is No.

Memory trick
Free tier F1 = Few (10) apps.

Real-world example, use case and scenario

EXAMPLE
F1 limit
You try to create an 11th web app on a Free App Service plan. Azure does not allow it; you need a paid plan.

USE CASE
Student portfolio
A design student hosts a portfolio and two small projects on the Free tier at no cost.

SCENARIO
Agency plans
A web agency planned to host 40 client sites on a free account. It quickly reached the free tier limits and moved to a Basic plan with pay-as-you-go.

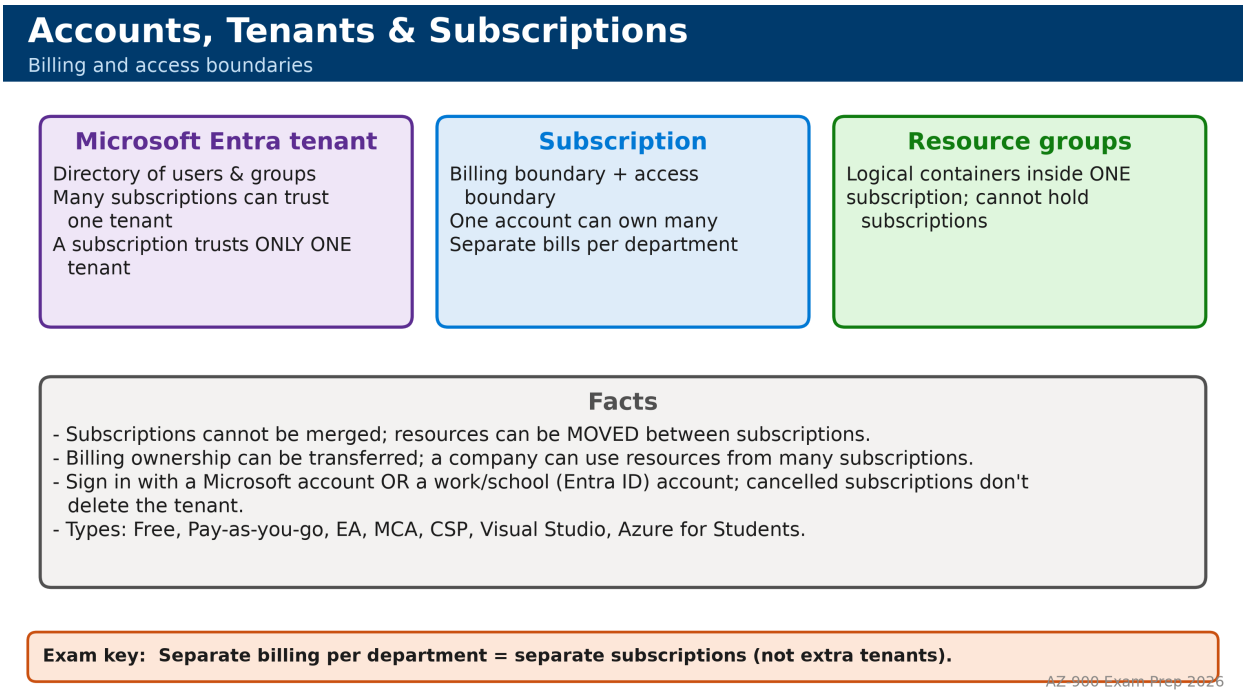


Figure Q224a: Accounts, tenants and subscriptions

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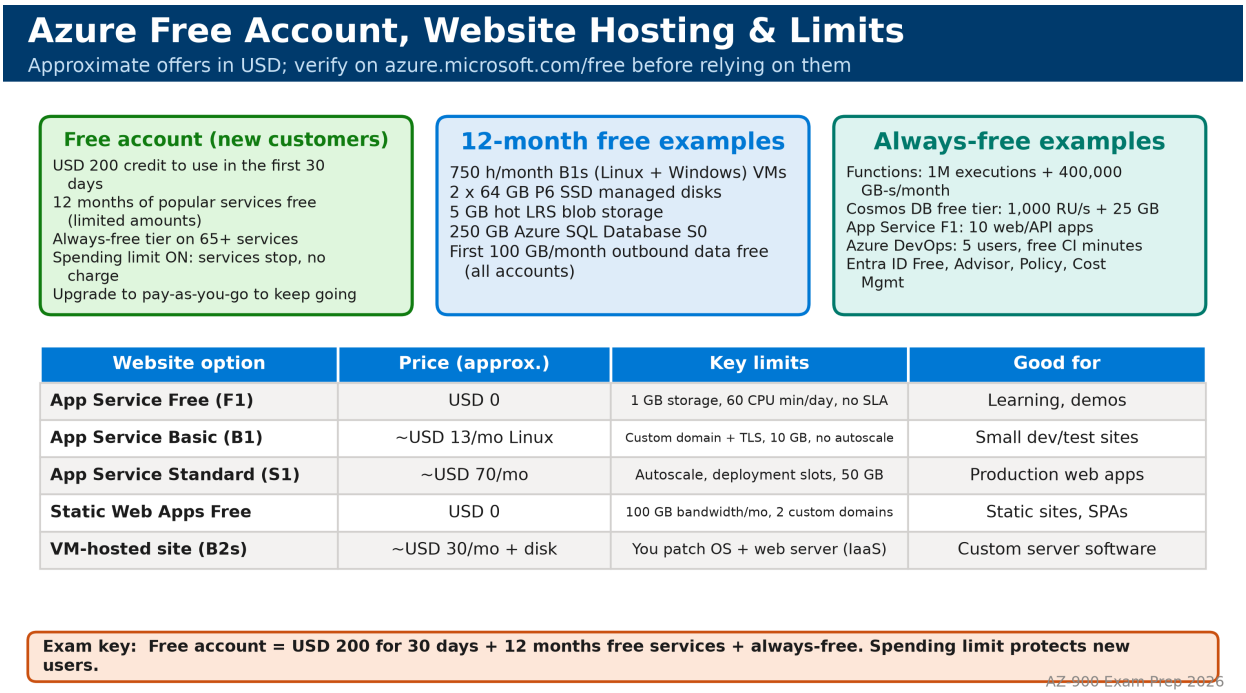


Figure Q224b: Free account, website hosting and limits

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- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 225

CLOUD CONCEPTS

Cost
v5 Q329

With a consumption-based plan, you pay a fixed rate for all data sent to or from virtual machines hosted in the cloud.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap words are 'fixed rate' and 'all data'. Consumption-based means you pay per GB you use. Data coming in is free, and data going out is charged per GB.

Answer explained

✔ **No.**
Consumption-based billing measures data per GB. Incoming data is free. Outgoing data is charged per GB after a free monthly amount. So there is no fixed rate.

✖ **Yes.**
A fixed rate does not match consumption pricing. Data is measured per GB, and only egress (data leaving Azure) is charged.

EXAM STRATEGY
Look for the phrase 'fixed rate'. Consumption means you pay for each unit you use, not a flat fee. 'Fixed rate' with 'consumption-based' = No.

IMPACT
Egress (data leaving Azure) charges grow with traffic. So budgets must estimate data amounts, not assume a flat network cost.

What · Where · When · Why · How

What	Consumption-based billing charges for measured use, including data transfer.
Where	Azure measures, per GB, the VM network traffic that leaves Azure.
When	When you estimate network costs for VMs in a pay-as-you-go model.
Why	Data is measured per GB, not charged at a fixed rate. Ingress (data coming in) is free.
How	Egress is charged per GB after a free monthly amount. Track it in Cost Management.

Exam tip
Consumption = measured and changing. 'Fixed rate' together with 'consumption-based' cannot be true.

Memory trick
Consumption is a taxi meter, not a flat fare.

Real-world example, use case and scenario

EXAMPLE
Egress meter
A VM sends 300 GB out in a month. It pays per GB above the free amount. The 2 TB it received is free.

USE CASE
Video downloads
A podcast platform tracks egress as its main changing cost. It uses Azure CDN to reduce it.

SCENARIO
Flat-fee assumption
A software company planned a flat network fee for its download servers. A product launch tripled the downloads, and egress charges tripled too. The company added a CDN and a budget alert.

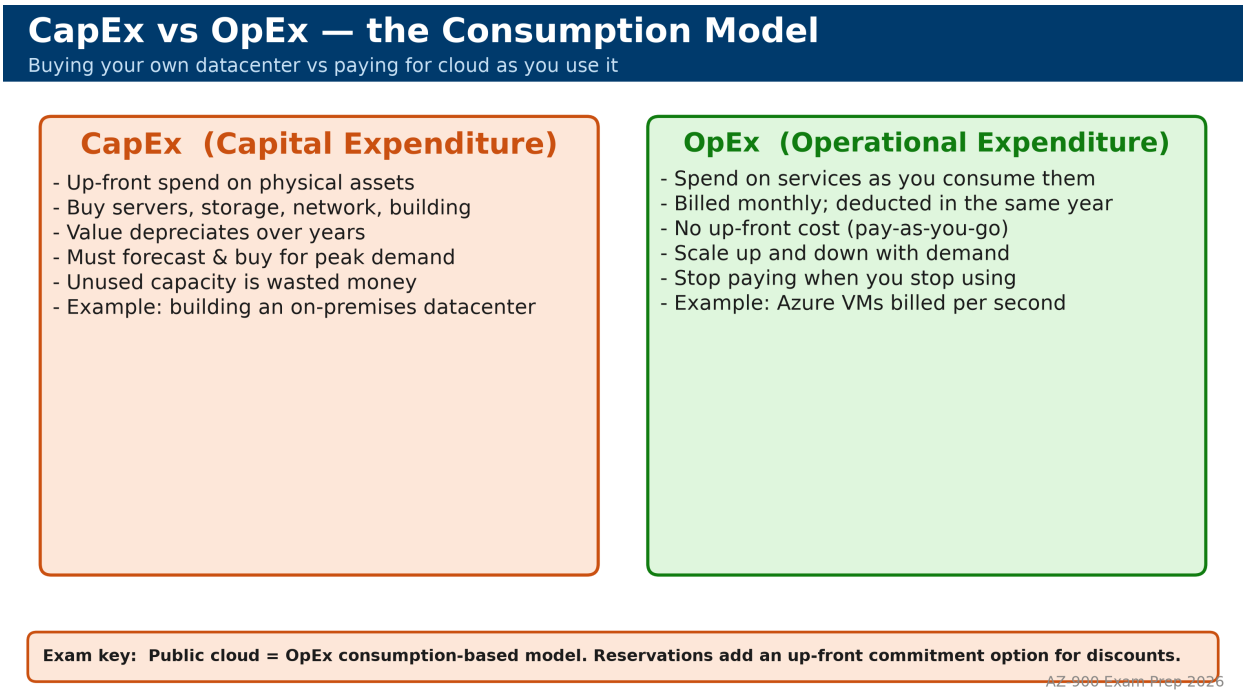


Figure Q225: CapEx vs OpEx

About this figure

- Left box is buying assets up front (CapEx); right box is paying for consumption (OpEx). Cloud moves spending from left to right.
- CapEx assets are depreciated over several years; OpEx is expensed in the period it is incurred, which suits variable or uncertain demand.
- The consumption model means no up-front cost, pay only for what you use, and stop paying when resources are deleted or deallocated.
- Reservations and savings plans add a 1- or 3-year commitment for discounts, but they remain operational spending, not purchased hardware.
- Use the TCO calculator to compare on-premises CapEx (hardware, power, labour) with expected Azure OpEx before migrating.

Question 226

AZURE MANAGEMENT & GOVERNANCE

Cost

v5 Q332, Q346

The cost of Azure resources can vary between regions.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Azure prices are different in different regions. The same VM size or storage type can cost more in one region than in another.

✖ **No.**
Microsoft documents different prices per region. You can see them in the Pricing calculator for most services.

EXAM STRATEGY
Prices are set per region, because local costs, taxes and demand are different. 'Can vary between regions' = Yes.

IMPACT
Choosing a cheaper region that still meets your rules can lower compute bills. Region choice really affects cost.

What · Where · When · Why · How

What	Azure resource prices are different in each region.
Where	The Azure Pricing calculator and pricing pages show prices per region.
When	When you choose a region based on cost, as well as speed and legal rules.
Why	Local costs are different, so the same resource type can have a different price in each region.
How	Compare the price in several regions in the Pricing calculator before you deploy.

Exam tip
Cost factors: resource type, usage, region, bandwidth (egress, between regions), subscription or offer type, and Marketplace.

Memory trick
Same coffee, different city, different price.

Real-world example, use case and scenario

EXAMPLE
Region compare
In the Pricing calculator, a D2s v5 VM costs less per hour in East US than in Switzerland North.

USE CASE
Batch rendering
An animation studio runs non-sensitive render jobs in a cheaper region. It has no rules about where the data must stay.

SCENARIO
Data location vs price
A European e-learning company wanted the cheapest region. But GDPR rules said data must stay in the EU. So it chose the cheapest EU region, not the cheapest in the world.

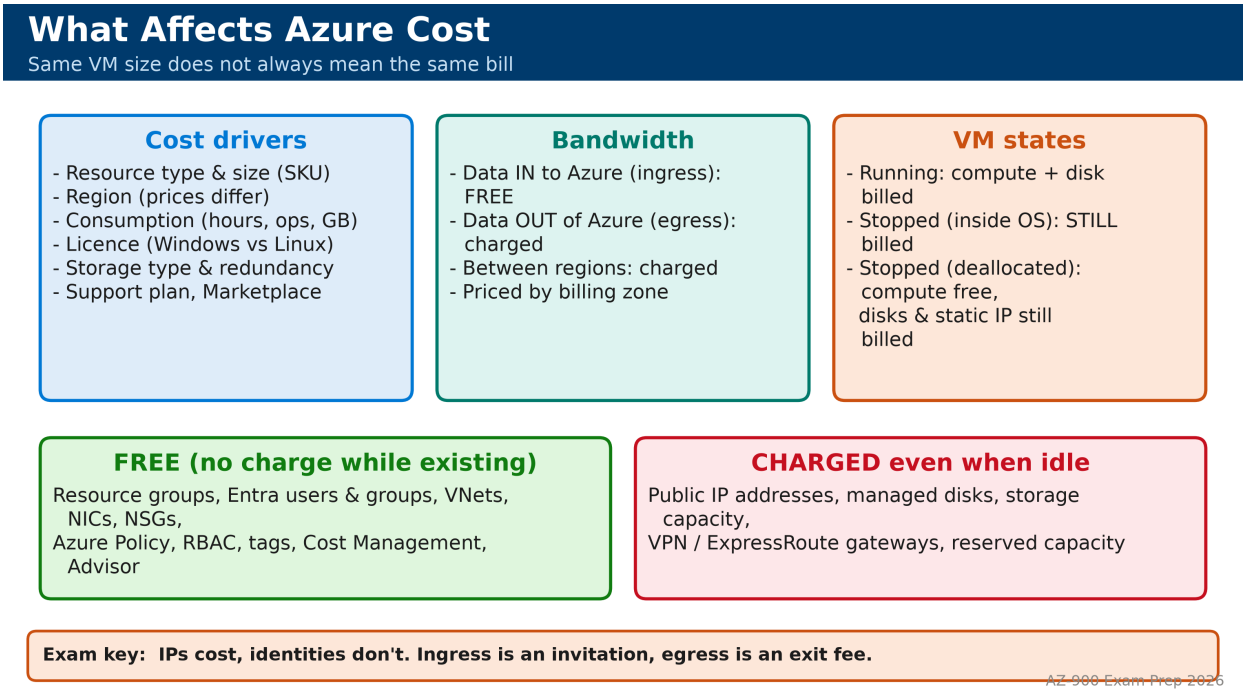


Figure Q226a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

• Bandwidth: ingress is free; the first 100 GB/month of Internet egress is free, then approx. USD 0.087/GB in North America and Europe.

• Inter-region transfer is approx. USD 0.02/GB within a continent; VNet peering is approx. USD 0.01/GB each way in the same region.

• A VM stopped inside the guest OS is still allocated and billed for compute; only Stopped (deallocated) from the portal, CLI or API stops compute charges.

• A Standard static public IP costs approx. USD 3.65/month even unattached; a 128 GB Premium SSD costs approx. USD 20/month even when its VM is deallocated.

• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q226b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 227

AZURE MANAGEMENT & GOVERNANCE

Cost

v5 Q333

An Azure Reservation is used to reserve server capacity at a specific data center.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap words are 'server capacity' and 'specific data center'. A Reservation is a price discount for a region. It does not book servers in a building. You never choose a datacenter in Azure.

Answer corrected in this edition. The previous PDF printed: *Yes.*. The printed answer was Yes. An Azure Reservation is a billing discount for a resource type and region (for VMs it can also give capacity priority in that region). It does not reserve server capacity in a specific datacenter. The answer key also says No.

Answer explained

✔ **No.**
An Azure Reservation is a 1-year or 3-year billing discount for a resource type in a region. It does not reserve server capacity in a specific datacenter.

✖ **Yes.**
You never choose a datacenter. Reservations are billing discounts per region. Capacity reservation is a different feature, set per region or zone.

EXAM STRATEGY
Look for 'specific data center'. Reservations give a price discount for a region and size. They do not book physical capacity in a datacenter. Answer No.

IMPACT
If you need guaranteed capacity, use On-demand Capacity Reservation, which is a different feature. If you mix them up, scaling out may fail when a region is short of capacity.

What · Where · When · Why · How

What	An Azure Reservation is a billing discount for a 1-year or 3-year commitment.
Where	It is applied to a subscription, resource group, management group or shared scope, in a region.
When	When resources such as VMs run all the time for a long period.
Why	It lowers the price in a region. It does not book datacenter capacity.
How	Buy a reservation for a VM size and a region. Matching usage gets the discount automatically.

Exam tip
Reservation = discount for a commitment, applied automatically to matching usage in a region. Never 'a specific datacenter'.

Memory trick
Reservation reserves a price, not a rack.

Real-world example, use case and scenario

EXAMPLE
Automatic discount
A reservation for 5 D4s v5 VMs in West Europe automatically lowers the price of any 5 matching VMs running there.

USE CASE
Shared scope savings
A large group of companies shares its reservations across subscriptions. Unused discounts go to matching VMs in any business unit.

SCENARIO
Datacenter belief
An infrastructure manager at a hospital bought a reservation. He thought it kept his VMs in one building. He learned it was only a billing discount, and he used availability zones for physical protection instead.

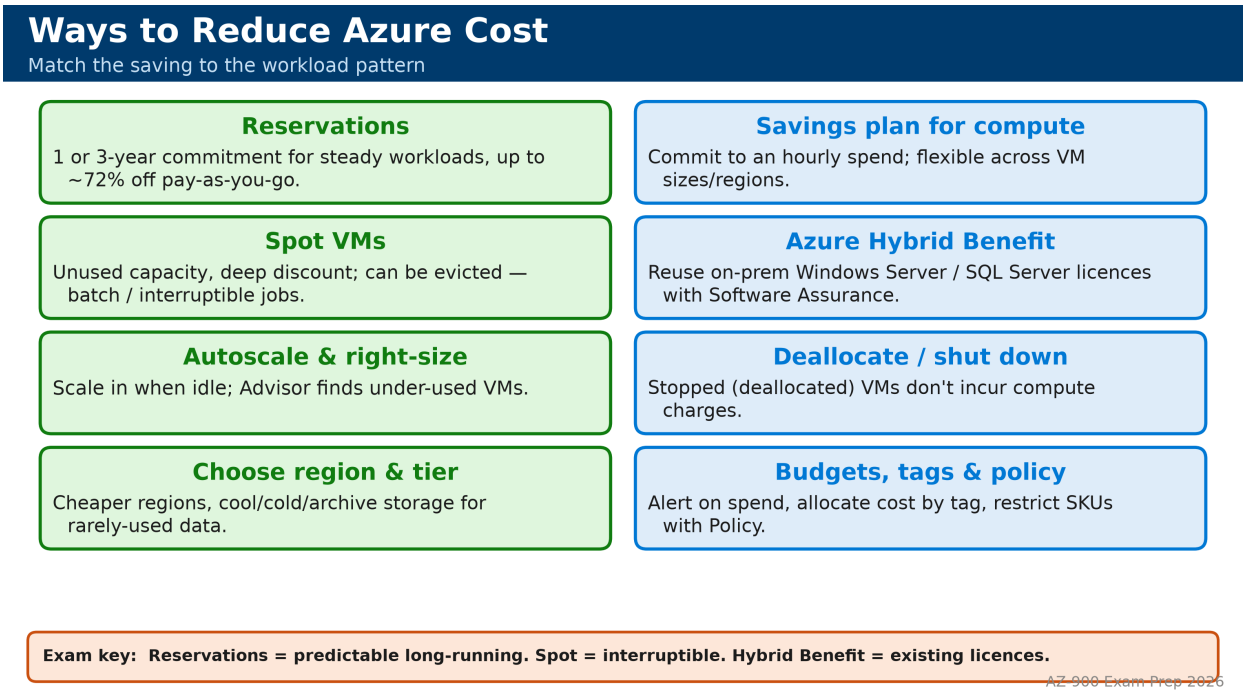


Figure Q227a: Ways to reduce cost

About this figure

• Each box matches a saving to a workload pattern: steady (reservations), flexible (savings plan), interruptible (Spot), existing licences (Hybrid Benefit).

• Reservations save up to approx. 72% versus pay-as-you-go; savings plans for compute save up to approx. 65% and apply across VM families and regions.

• Spot VMs can be up to approx. 90% cheaper, but Azure can evict them with 30 seconds' notice when it needs capacity back.

• Combining Azure Hybrid Benefit with reservations can cut Windows VM costs by up to approx. 80%; Linux subscriptions such as RHEL and SUSE also qualify.

• Example: moving rarely read data from Hot (~USD 0.02/GB-month) to Archive (~USD 0.002 or less) saves storage cost, but reads and early deletion cost more.

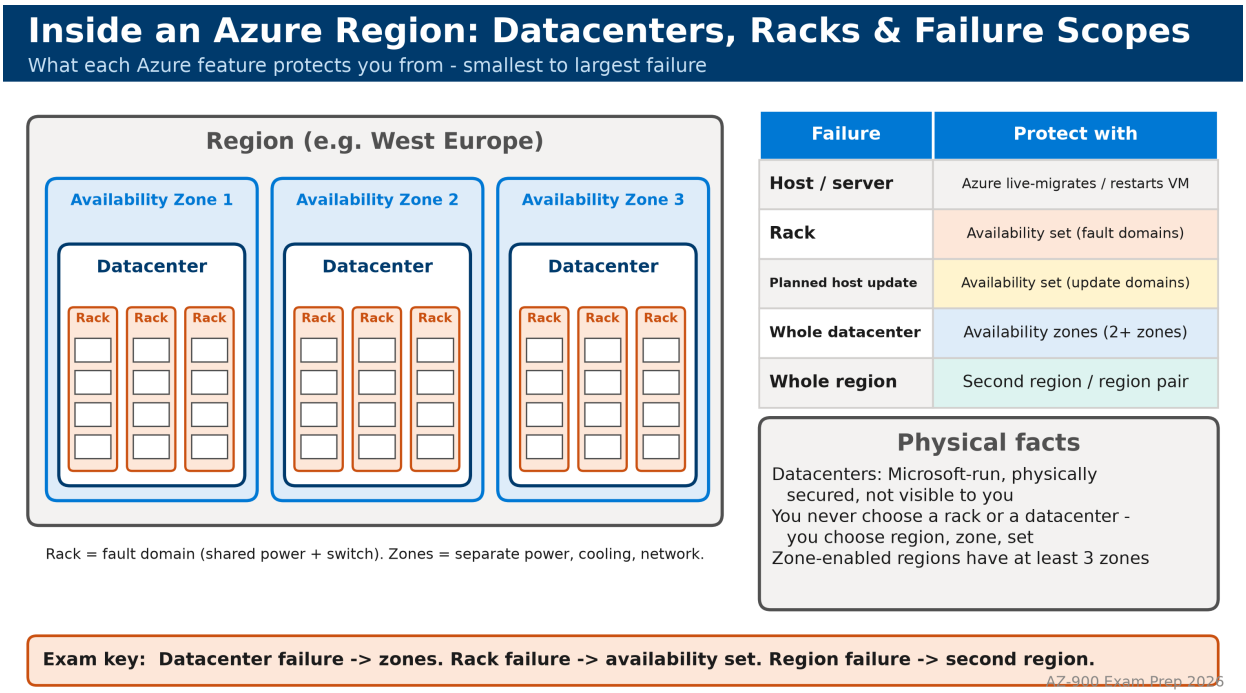


Figure Q227b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q227c: Approximate Azure charges (USD)

About this figure

• Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.

• Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.

• Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.

• Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.

• Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 228

AZURE ARCHITECTURE & SERVICES

Databases
v5 Q334

You can stop an Azure SQL Database instance to decrease costs.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): This is about a normal (provisioned, always-on) Azure SQL Database. It has no Stop button. VMs can be stopped, and SQL Managed Instance can be stopped, but they are different services.

2026 update: Today the serverless tier of Azure SQL Database can auto-pause when idle; then you pay only for storage. Azure SQL Managed Instance can be stopped and started. A provisioned single Azure SQL Database still cannot be stopped, so the answer is still No.

Answer explained

✔ **No.**
A provisioned Azure SQL Database has no stop option. You pay while it exists. To save money, scale it down or use the serverless tier, which can auto-pause.

✖ **Yes.**
VMs and SQL Managed Instance can be stopped. A provisioned Azure SQL Database cannot be stopped to pause billing.

EXAM STRATEGY
A provisioned Azure SQL Database has no Stop button, unlike VMs. If the question says Managed Instance or serverless auto-pause, the answer changes. Plain SQL Database = No.

IMPACT
Teams that expect to stop test databases at night keep paying. Use serverless auto-pause or scale down to really save money.

What · Where · When · Why · How

What	Azure SQL Database is a PaaS (Azure manages the server) database. You pay while it is provisioned.
Where	You set it on the database's Compute + storage page in the Azure portal.
When	When you try to lower database costs while it is not used.
Why	A provisioned database has no stop option, so you cannot stop it.
How	Scale down the tier, or use serverless with auto-pause, to lower the cost.

Exam tip
VMs can be deallocated (stopped and released). A provisioned Azure SQL Database cannot be stopped - scale it down or use serverless auto-pause.

Memory trick
SQL DB has no off switch, only a dimmer (scale down).

Real-world example, use case and scenario

EXAMPLE
No Stop button
In the portal, an Azure SQL Database page shows Scale, Copy and Restore, but no Stop.

USE CASE
Test database
A software company runs test databases on the serverless tier. They pause after 1 idle hour to save money at night.

SCENARIO
Weekend savings
A charity wanted to stop its reporting database at weekends. That is not possible on provisioned tiers. So it moved the database to serverless, which pauses when not used.

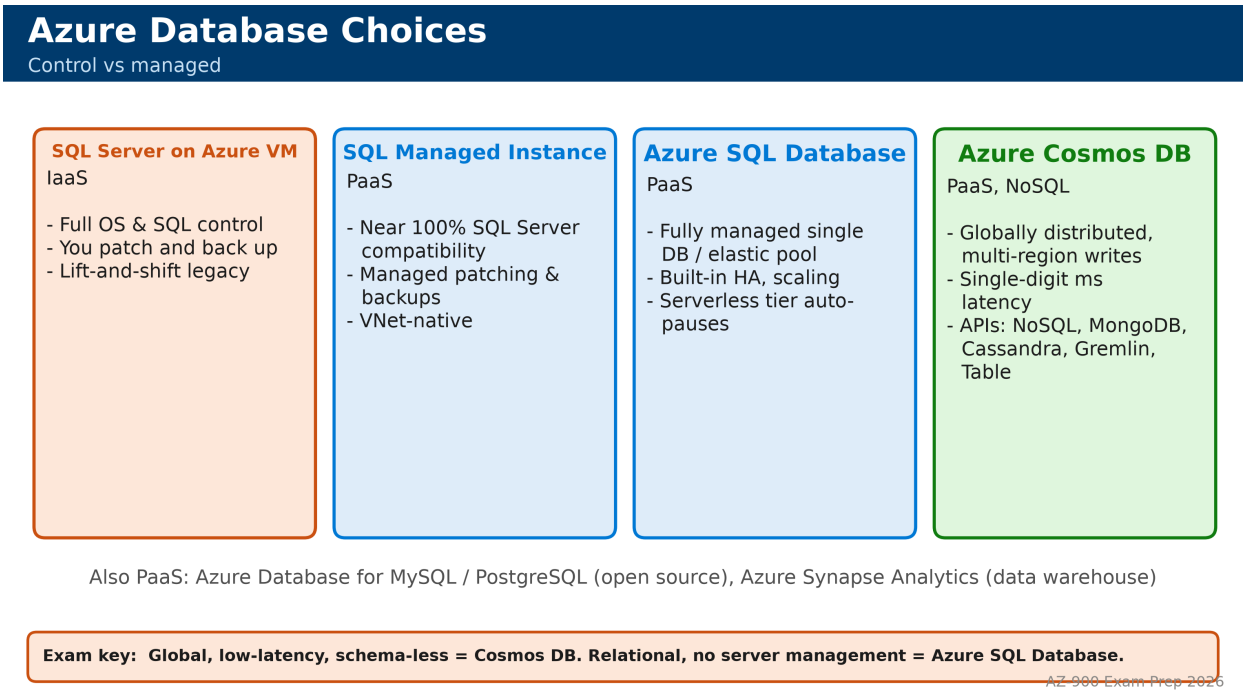


Figure Q228: Azure database choices

About this figure

• Columns run from most control (SQL Server on a VM, IaaS) to fully managed relational PaaS and globally distributed NoSQL Cosmos DB.

• Azure SQL Database Basic costs approx. USD 5/month and S0 approx. USD 15/month; serverless bills per vCore-second and storage only while paused.

• Azure SQL Database SLA is up to 99.99%, and up to 99.995% with zone redundancy in Business Critical; automated backups are kept 7 days by default.

• Cosmos DB free tier gives 1,000 RU/s and 25 GB free per account; it offers five consistency levels and up to 99.999% SLA for multi-region accounts.

• SQL Managed Instance suits migrations needing SQL Agent, cross-database queries or CLR; Azure Hybrid Benefit can reuse SQL Server licences for any of them.

Question 229

CLOUD CONCEPTS

SLA
v5 Q335, Q129

You have an application that is comprised of an Azure web app that has a Service Level Agreement (SLA) of 99.95 percent and an Azure SQL database that has an SLA of 99.99 percent. The composite SLA for the application is [...].

✔ **A.** the product of both SLAS, which equals 99.94 percent.

✖ B. the lowest SLA associated to the application, which is 99.95 percent.

✖ C. the highest SLA associated to the application, which is 99.99 percent.

✖ D. the difference between the two SLAS, which is 0.05 percent.

Correct answer: the product of both SLAS, which equals 99.94 percent.

Clarification (read this first): Composite SLA means the uptime promise of the whole app when it needs BOTH services. You multiply the SLAs, so the result is LOWER than each one. Do not pick the lowest or highest single SLA.

Answer explained

✔ **the product of both SLAS, which equals 99.94 percent.**
When an app needs one service after another, you multiply their SLAs (uptime promises): 0.9995 x 0.9999 = 99.94%. This is lower than each single SLA.

✖ **The lowest SLA associated to the application, which is 99.95 percent.**
The lowest SLA ignores the extra risk from the second service. It is only the highest possible result.

✖ **The highest SLA associated to the application, which is 99.99 percent.**
Adding a service can never make uptime higher than the weaker service's SLA.

✖ **The difference between the two SLAS, which is 0.05 percent.**
Subtracting SLAs has no meaning for uptime. You multiply composite SLAs.

EXAM STRATEGY
Services that depend on each other: multiply the SLAs. 0.9995 x 0.9999 = 0.9994. The result is always lower than the weakest one, so remove highest, lowest and difference.

IMPACT
Each extra dependency lowers the uptime of the whole app. Architects must add backup copies to meet the SLAs they promise customers.

What · Where · When · Why · How

What	A composite SLA is the combined uptime promise of services that depend on each other.
Where	You calculate it across all services that the app needs in a chain.
When	When you estimate the uptime you can promise for an app that uses many services.
Why	Services in a chain multiply: 0.9995 x 0.9999 = 99.94%.
How	Write each SLA as a decimal, multiply them, and change the result back to a percentage.

Exam tip
Services in a chain: multiply, and the result is lower than each one. Backup copies in two regions: 1 - (1-A)(1-B), and the result is higher.

Memory trick
Chain links: multiply, and the chain gets weaker.

Real-world example, use case and scenario

EXAMPLE
Three tiers
Web app 99.95% x SQL 99.99% x Storage 99.9% = about 99.84%.

USE CASE
Customer contract
An HR software company calculates its composite SLA before it promises customers 99.9% in its contract.

SCENARIO
Promise too high
A sales rep at a booking platform promised 99.99% because the database had that SLA. The architect showed that, with the web app, the composite was 99.94%. The contract was changed.

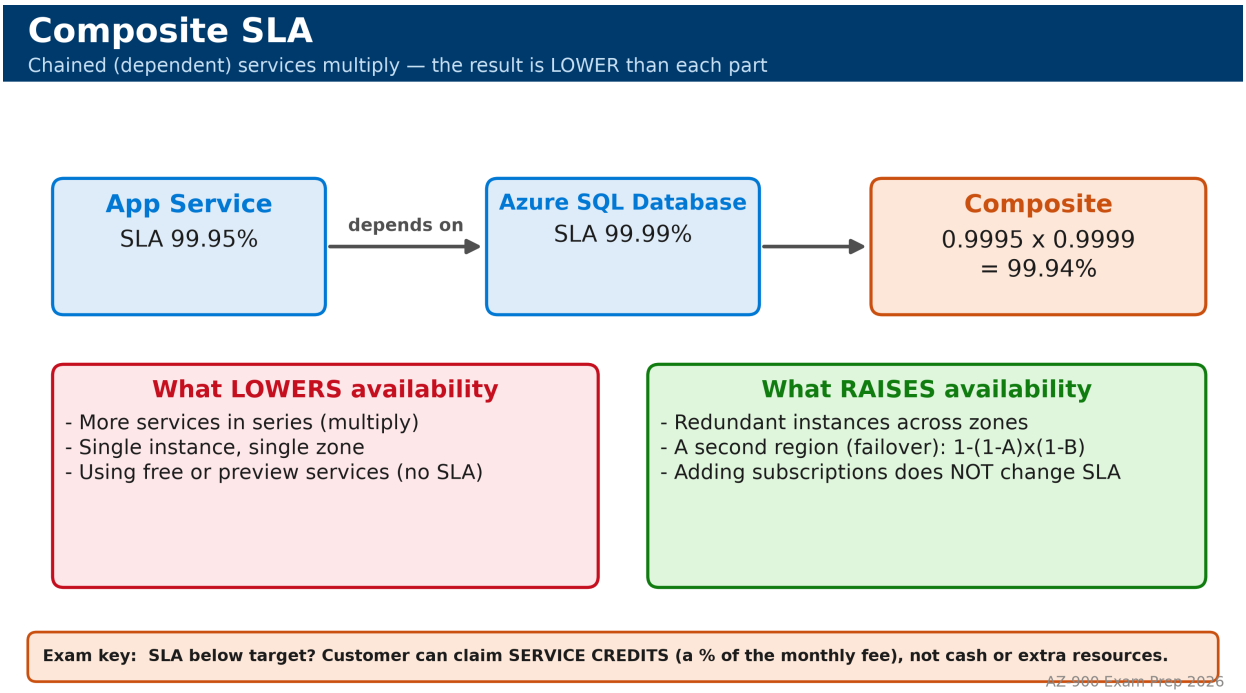


Figure Q229a: Composite SLA

About this figure

• Follow the arrows: when one service depends on another, multiply their SLAs; the result is always lower than the weakest single component.

• Example: 99.95% x 99.99% = 99.94%, which allows roughly 26 minutes of downtime per month instead of about 22 for App Service alone.

• Parallel redundancy raises availability: two independent 99.9% deployments in failover give 1 - (0.001 x 0.001) = 99.9999% in theory.

• Preview services and free tiers (for example App Service F1) carry no SLA, so including them removes any guaranteed availability.

• Credits must be claimed through a support request, usually by the end of the following billing month, and are capped at the service's monthly fee.

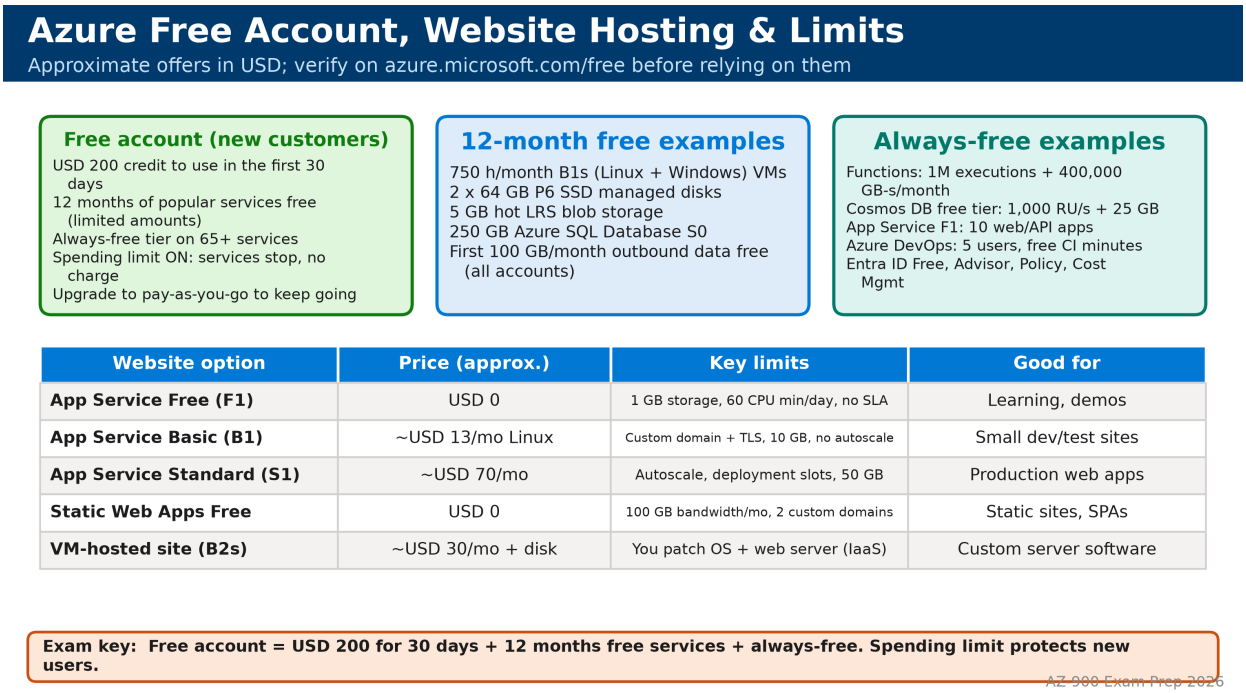


Figure Q229b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 230	AZURE ARCHITECTURE & SERVICES	IoT v5 Q336

Match the Azure Services service to the correct description.

✔ **A.** A managed service that provides bidirectional communication between IoT devices and Azure: IoT Hub. A fully managed software as a service (SaaS) solution to connect, monitor, and manage IoT devices at scale: IoT Central. A software and hardware solution that provides communication and security features for IoT devices: Azure Sphere.

✖ B. A managed service that provides bidirectional communication between IoT devices and Azure: Azure Sphere. A fully managed software as a service (SaaS) solution to connect, monitor, and manage IoT devices at scale: IoT Hub. A software and hardware solution that provides communication and security features for IoT devices: IoT Central.

✖ C. A managed service that provides bidirectional communication between IoT devices and Azure: Azure Sphere. A fully managed software as a service (SaaS) solution to connect, monitor, and manage IoT devices at scale: IoT Central. A software and hardware solution that provides communication and security features for IoT devices: IoT Hub.

✖ D. A managed service that provides bidirectional communication between IoT devices and Azure: IoT Hub. A fully managed software as a service (SaaS) solution to connect, monitor, and manage IoT devices at scale: Azure Sphere. A software and hardware solution that provides communication and security features for IoT devices: IoT Central.

Correct answer: A managed service that provides bidirectional communication between IoT devices and Azure: IoT Hub. A fully managed software as a service (SaaS) solution to connect, monitor, and manage IoT devices at scale: IoT Central. A software and hardware solution that provides communication and security features for IoT devices: Azure Sphere.

Clarification (read this first): Match each description to one service. Key words: 'bidirectional communication' (messages both ways) = IoT Hub; 'SaaS' (ready-made app) = IoT Central; 'hardware and software security' = Azure Sphere.

Answer explained

✔ **A managed service that provides bidirectional communication between IoT devices and Azure: IoT Hub. A fully managed software as a service (SaaS) solution to connect, monitor, and manage IoT devices at scale: IoT Central. A software and hardware solution that provides communication and security features for IoT devices: Azure Sphere.**
IoT Hub is the PaaS (Azure runs the platform) service for two-way messages. IoT Central is the SaaS (ready-made app) for managing many devices. Azure Sphere is the secured chip, OS and security service.

✖ **A managed service that provides bidirectional communication between IoT devices and Azure: Azure Sphere. A fully managed software as a service (SaaS) solution to connect, monitor, and manage IoT devices at scale: IoT Hub. A software and hardware solution that provides communication and security features for IoT devices: IoT Central.**
All three are in the wrong place. Sphere is not a messaging service. IoT Hub is PaaS, not SaaS. IoT Central has no hardware.

✖ **A managed service that provides bidirectional communication between IoT devices and Azure: Azure Sphere. A fully managed software as a service (SaaS) solution to connect, monitor, and manage IoT devices at scale: IoT Central. A software and hardware solution that provides communication and security features for IoT devices: IoT Hub.**
IoT Hub and Sphere are swapped. Sphere is the hardware security solution. IoT Hub is the two-way messaging service.

✖ **A managed service that provides bidirectional communication between IoT devices and Azure: IoT Hub. A fully managed software as a service (SaaS) solution to connect, monitor, and manage IoT devices at scale: Azure Sphere. A software and hardware solution that provides communication and security features for IoT devices: IoT Central.**
IoT Central and Sphere are swapped. IoT Central is the SaaS platform. Sphere is the hardware and OS security solution.

EXAM STRATEGY
Key words: 'bidirectional communication' = IoT Hub, 'SaaS' = IoT Central, 'hardware and software security' = Azure Sphere. Match the clearest key word first.

IMPACT
Choosing SaaS IoT Central or PaaS IoT Hub decides how much you must build. Securing devices at chip level stops hackers from taking over many devices.

What · Where · When · Why · How

What	IoT Hub, IoT Central and Azure Sphere are Azure's main IoT (Internet of Things) services.
Where	IoT Hub is PaaS. IoT Central is SaaS. Sphere covers the device chip, OS and cloud.
When	For messaging, use IoT Hub. For a ready-made device app, use IoT Central. For device security, use Sphere.
Why	Each one matches its description: two-way messaging, SaaS management, hardware security.
How	Devices connect to IoT Hub. IoT Central is built on top of it. Sphere protects the device chip and OS.

Exam tip
Hardware or chip in the description = Azure Sphere. SaaS or dashboards with no coding = IoT Central. Two-way messaging = IoT Hub.

Memory trick
Hub = Highway, Central = Control room (SaaS), Sphere = Secure chip.

Real-world example, use case and scenario

EXAMPLE
Cloud-to-device command
IoT Hub sends a firmware update command to 50,000 smart meters. It also receives their data.

USE CASE
Vending machines
A drinks company uses IoT Central's templates and dashboards to watch vending machine stock. It writes no backend code.

SCENARIO
Hacked ovens
Someone changed the firmware of a company's smart ovens. The company rebuilt them on Azure Sphere chips, with the secured OS and automatic security updates. Device data still went through IoT Hub.

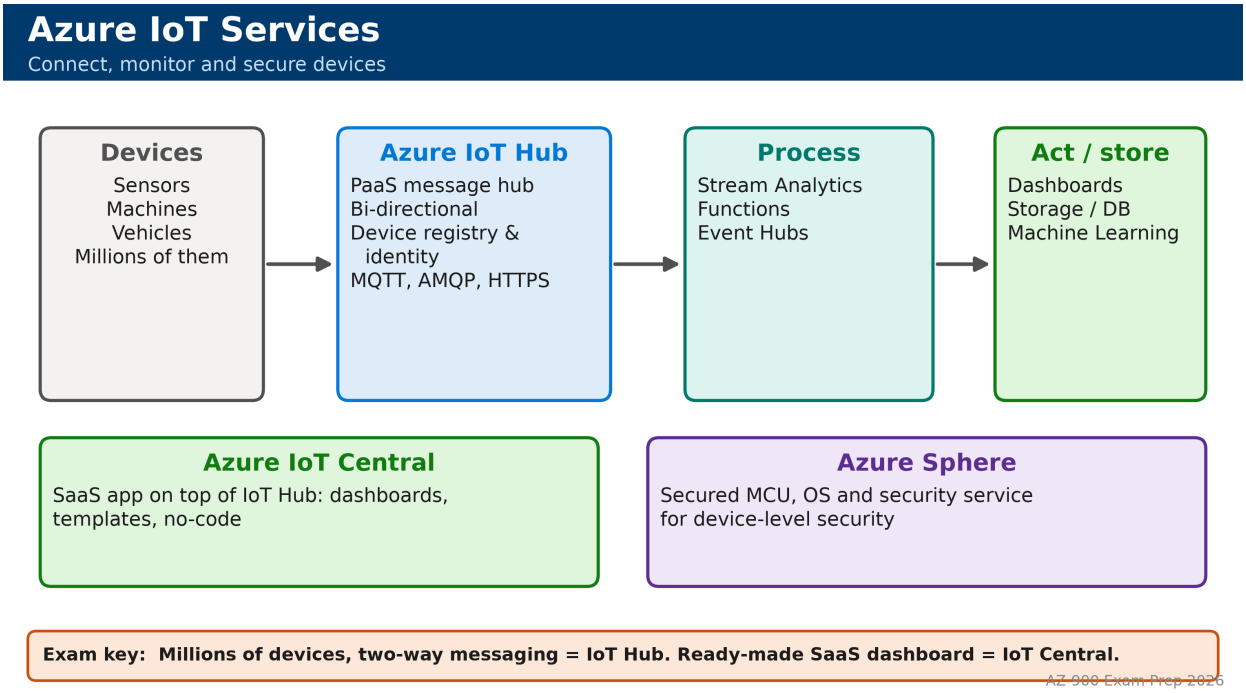


Figure Q230: Azure IoT services

About this figure

- Flow left to right: devices send telemetry to IoT Hub, data is processed, then stored or acted on. Bottom boxes are the SaaS and device-security options.
- IoT Hub has a free tier of 8,000 messages/day; Standard S1 costs approx. USD 25/month per unit for 400,000 messages/day.
- IoT Hub supports cloud-to-device commands and device twins; Event Hubs only ingests events one way, at very high throughput, without per-device identity.
- Azure Sphere combines a certified microcontroller, a Linux-based OS and a cloud security service that delivers updates and certificate-based authentication.
- Use IoT Central for quick, managed dashboards with device templates; use IoT Hub when you build a custom solution and need full control.

Question 231

CLOUD CONCEPTS

Lifecycle
v5 Q338

From the Azure portal, you can distinguish between services that are generally available and services that are in public preview.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
The Azure portal shows a (Preview) label on preview services and features. Anything without the label is generally available (GA).

✖ **No.**
The portal clearly marks previews with a (Preview) label. So you can tell GA and preview services apart.

EXAM STRATEGY
The portal labels preview services and features with a (Preview) tag. 'Distinguish' (tell apart) = Yes.

IMPACT
If you notice preview labels, you will not build production systems on features that have no SLA (uptime promise) by mistake.

What · Where · When · Why · How

What	Public preview is a test stage of an Azure service or feature, before general availability.
Where	In Azure portal menus, pages and Create a resource, and on the Azure updates page.
When	When you decide if a service is ready for production work.
Why	The portal labels previews with (Preview), so you can tell GA and preview apart.
How	Look for the (Preview) label next to the name. Items with no label are GA.

Exam tip
Look for the word (Preview) in the portal. Preview = no SLA, limited support. GA = SLA and full support. The portal shows this on purpose.

Memory trick
Preview wears a name tag - the portal pins (Preview) on its badge.

Real-world example, use case and scenario

EXAMPLE
Portal label
In Create a resource, a new feature shows a (Preview) label. Azure SQL Database has no label because it is GA.

USE CASE
Policy for production
A regional bank's cloud team looks for (Preview) labels in the portal. It blocks those services in production subscriptions because previews have no SLA.

SCENARIO
The intern's catch
An intern at an insurance startup built a demo on a feature marked (Preview). Before go-live, the architect saw the label. He checked Azure updates for the GA date and waited for GA, so the system would have an SLA.

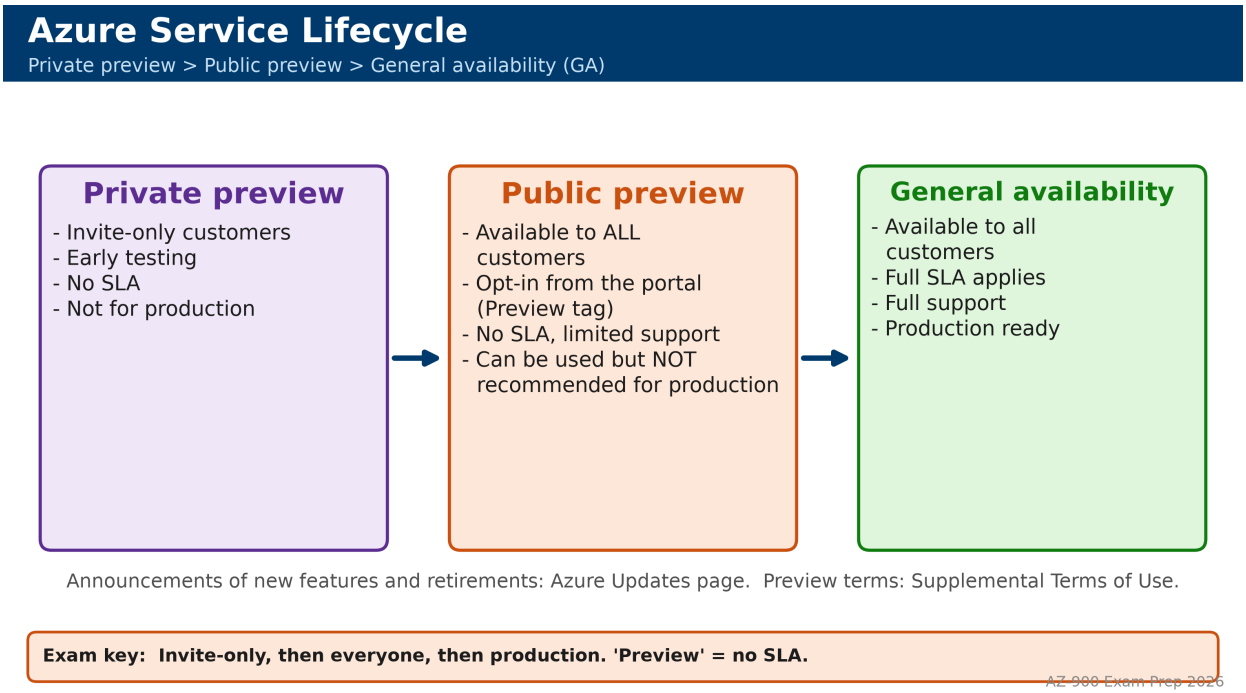


Figure Q231: Service lifecycle: preview to GA

About this figure

- Read the arrows left to right: invite-only private preview, open public preview, then general availability with SLA and full support.
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- Some previews are discounted or free, but many bill at reduced or standard rates; check the pricing page before relying on preview resources.
- Follow Azure Updates for previews, GA announcements and retirements; Service Health also sends retirement notices for services you actually use.
- Microsoft usually gives at least 12 months' notice before retiring a GA service or feature that customers depend on, with migration guidance.

Question 232

CLOUD CONCEPTS

Lifecycle
v5 Q339

After an Azure service becomes generally available, the service is no longer updated with new features.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The statement says GA services get NO new features. That is false. GA means ready for production with an SLA. It does not mean the service stops changing.

Answer explained

✔ **No.**
GA means ready for production, with an SLA (uptime promise) and support. It does not mean frozen. Microsoft keeps adding features to GA services, often as previews inside them.

✖ **Yes.**
GA services keep getting new features, sizes and regions. Only retired services stop getting updates.

EXAM STRATEGY
GA services keep getting updates and new features all the time. 'No longer updated' = No.

IMPACT
Cloud services change all the time. Teams must follow updates and retirements, not think that GA means frozen.

What · Where · When · Why · How

What	General availability (GA) is the stage where an Azure service is ready for production, has an SLA and full support.
Where	Microsoft announces it on the Azure updates page. The portal removes the (Preview) label.
When	When you plan to use a service that may keep changing after release.
Why	GA services keep getting new features, so the statement is false.
How	Microsoft adds new features as previews inside the GA service, then makes them GA later.

Exam tip
GA = SLA + support, not the end of development. A GA service can have preview features at the same time.

Memory trick
GA is graduation, not retirement - the graduate keeps learning.

Real-world example, use case and scenario

EXAMPLE
App Service keeps growing
Azure App Service has been GA for years. It still got new features after GA, such as sidecar containers and new Premium v3/v4 tiers.

USE CASE
Roadmap planning
A logistics company follows the Azure updates RSS feed. It uses new features in GA services, like Azure Kubernetes Service, when they arrive.

SCENARIO
Waiting for the wrong thing
A school district waited to use Azure Virtual Desktop. It thought GA meant no more improvements. A partner showed the many updates after GA, and the district deployed it.

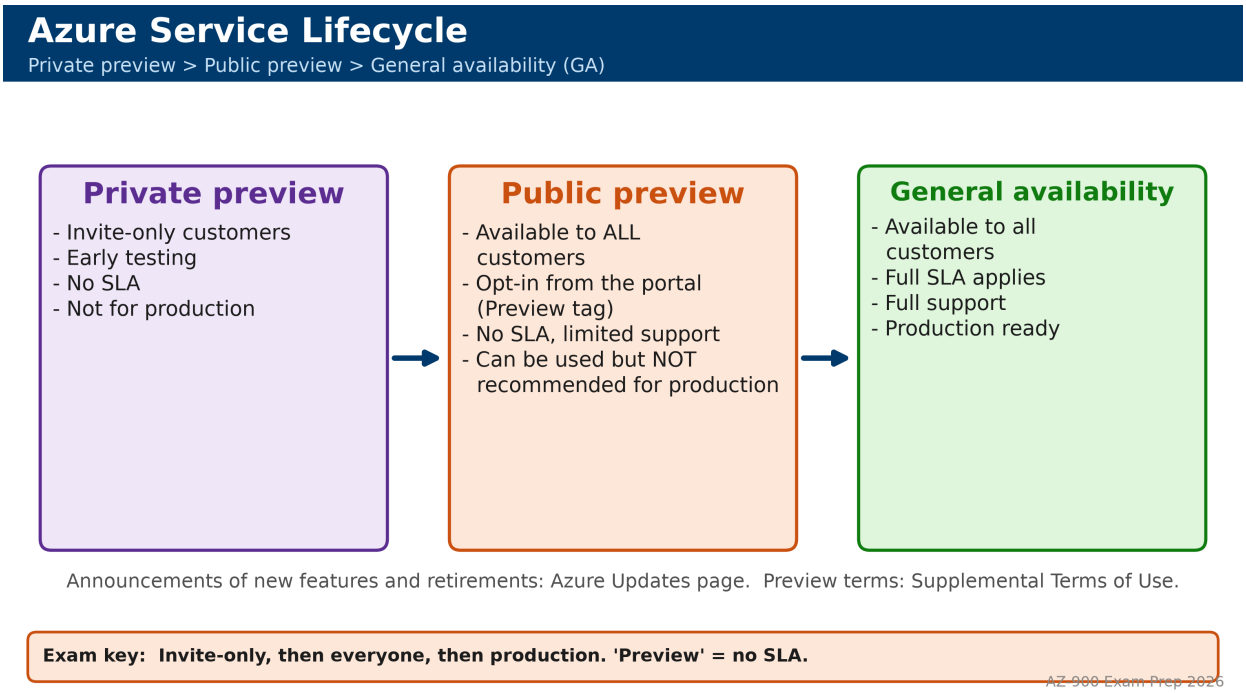


Figure Q232: Service lifecycle: preview to GA

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Question 233

CLOUD CONCEPTS

Lifecycle
v5 Q340

When you create Azure resources for a service in public preview, you must recreate the resources once the service becomes generally available.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap words are 'must recreate'. Resources made in preview normally keep working when the service reaches GA. Only the price and the SLA change.

Answer explained

✔ **No.**
Preview resources normally keep running as they are when a service reaches GA. The price changes to the GA price and the SLA (uptime promise) starts.

✖ **Yes.**
No general rule says you must recreate preview resources at GA. You redeploy only if the release notes say so.

EXAM STRATEGY
Resources made in preview usually continue into GA without redeploying. 'Must recreate' = No.

IMPACT
You avoid needless rebuild work. Still, check the release notes, because a few preview features do need extra steps.

What · Where · When · Why · How

What	The preview-to-GA change is when a service leaves public preview and becomes generally available.
Where	It affects resources you already deployed in your subscription during the preview.
When	When a service you tested in preview reaches general availability.
Why	Preview resources normally keep running, so you do not need to recreate them.
How	Microsoft changes billing to GA prices and starts the SLA. Release notes mention any rare extra steps.

Exam tip
Preview to GA happens in place. The trap words are 'must recreate'.

Memory trick
Preview to GA is a promotion, not a rebuild - same desk, new title.

Real-world example, use case and scenario

EXAMPLE
Same resource ID
A storage feature turned on during preview keeps its resource ID and data after GA. Only the price list and the SLA change.

USE CASE
Early adopter pilot
A medicine lab tests a preview analytics feature in a non-production subscription. It knows the settings will stay when GA arrives.

SCENARIO
Rebuild not needed
A media company planned a weekend to rebuild all preview resources at GA. The architect read the GA announcement and saw no extra steps were needed. He cancelled the work and saved a weekend.

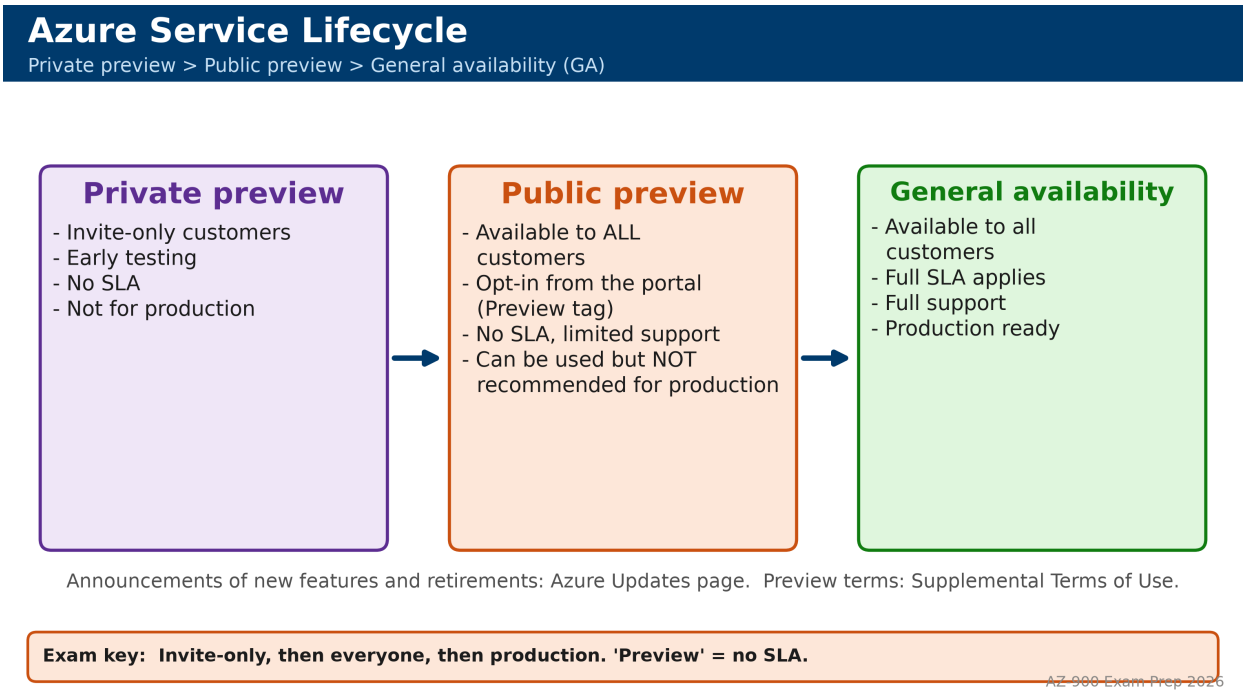


Figure Q233: Service lifecycle: preview to GA

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Question 234

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q343

Data traffic between Azure services within the same Azure region is always free.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The trap word is 'always'. The exam answer is Yes: traffic inside one region is not charged. In real life there are exceptions, for example VNet peering inside a region is charged per GB. Compare Q219 (data into Azure: free) and Q241 (data out of Azure: charged).

2026 update: This is the classic exam answer. In real life, not all traffic inside a region is free. VNet peering inside a region is charged per GB in and out. Some services also charge for traffic through gateways or private endpoints. Other traffic between services in the same region is generally not charged.

Answer explained

✔ **Yes.**
Exam answer: Azure charges for data that leaves a region or leaves Azure. It does not charge for traffic between services in the same region.

✖ **No.**
For AZ-900, traffic between services in the same region is not charged. In real life, some exceptions exist, like VNet peering inside a region, which is charged per GB.

EXAM STRATEGY
Exam rule: same region = free data transfer, so answer Yes. Know that in real life VNet peering is charged even inside one region.

IMPACT
Keeping services that talk a lot in one region keeps network costs low. Spreading them across regions adds charges per GB.

What · Where · When · Why · How

What	Intra-region data transfer is traffic between Azure services in the same region.
Where	Inside one Azure region, for example a VM and a storage account both in West Europe.
When	When you place parts of an app to lower network cost and delay.
Why	Azure charges for data that leaves a region or Azure, not for traffic inside one region.
How	Put services that depend on each other in one region. Note: VNet peering is charged per GB even inside one region.

Exam tip
Same region = free. Different region = paid. Out to the internet or on-premises = paid. Know the peering exception for real life.

Memory trick
Neighbours on the same street don't pay tolls; crossing the region border does.

Real-world example, use case and scenario

EXAMPLE
VM and storage
A VM in West Europe reads 500 GB from a blob container in West Europe. You pay storage transaction charges, but no network charge.

USE CASE
Same region for all parts
An e-learning platform puts its App Service, Azure SQL Database and Redis cache in the same region. This avoids fees and delay between regions.

SCENARIO
Different regions
A marketing agency put its web app in East US and its database in West US 2. The bill showed charges for traffic between regions. Moving the database to East US removed them and made the app faster.

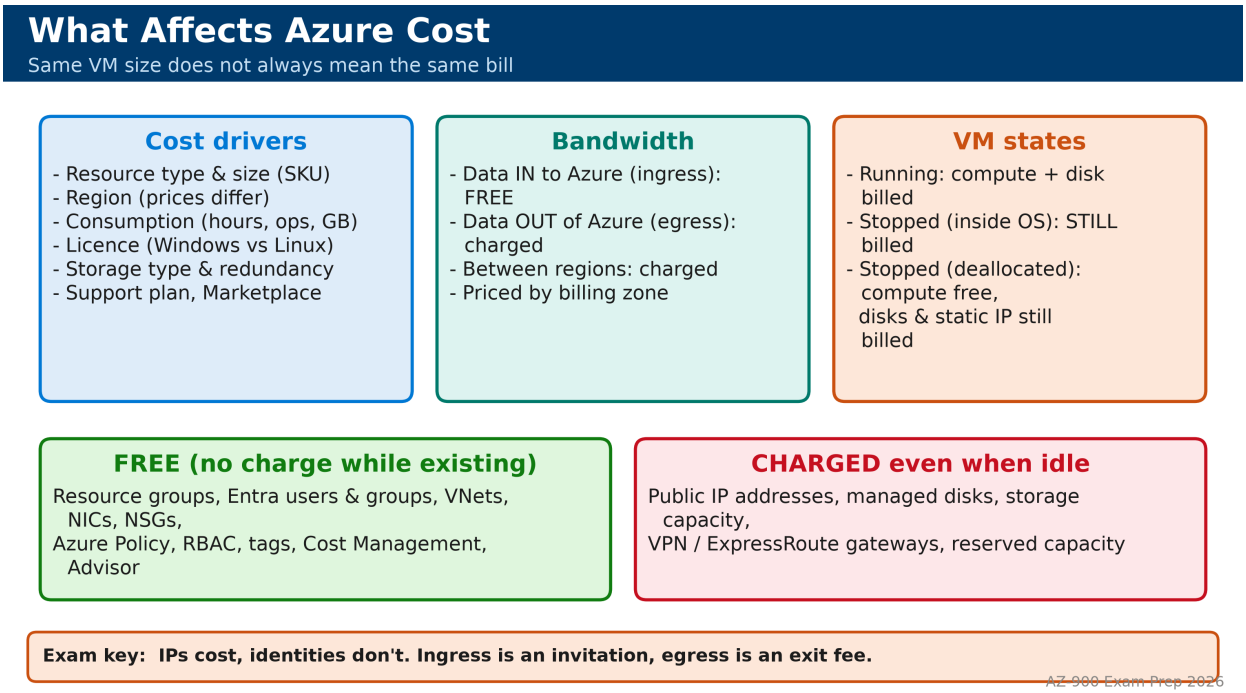


Figure Q234a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

• Bandwidth: ingress is free; the first 100 GB/month of Internet egress is free, then approx. USD 0.087/GB in North America and Europe.

• Inter-region transfer is approx. USD 0.02/GB within a continent; VNet peering is approx. USD 0.01/GB each way in the same region.

• A VM stopped inside the guest OS is still allocated and billed for compute; only Stopped (deallocated) from the portal, CLI or API stops compute charges.

• A Standard static public IP costs approx. USD 3.65/month even unattached; a 128 GB Premium SSD costs approx. USD 20/month even when its VM is deallocated.

• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q234b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 235

AZURE MANAGEMENT & GOVERNANCE

Cost

v5 Q344, Q55, Q146

Your company has an Azure subscription that contains the following unused resources: 20 user accounts in Microsoft Entra ID. Five groups in Microsoft Entra ID. 10 public IP addresses. 10 network interfaces. You need to reduce the Azure costs for the company. Which unused resources should you remove?

✖ A. The network interfaces.

✔ **B.** The public IP addresses.

✖ C. The groups.

✖ D. The user accounts.

Correct answer: The public IP addresses.

Clarification (read this first): The question asks which UNUSED item still COSTS money. Only public IP addresses are charged while unused. Entra ID users, groups and network interfaces are free.

Answer explained

✔ **The public IP addresses.**
Public IP addresses are charged every hour, even when they are not attached to anything. Entra ID users, groups and network interfaces cost nothing.

✖ **The network interfaces.**
Network interfaces have no charge of their own. Deleting them saves nothing.

✖ **The groups.**
Entra ID groups are free. Removing them does not lower the bill.

✖ **The user accounts.**
Entra ID user accounts are free. Only licences, such as P1 or P2, cost money.

EXAM STRATEGY
Look for 'reduce costs' and unused items. Public IPs are charged even when unattached. Entra users, groups and NICs are free. Pick the only charged resource.

IMPACT
Forgotten public IPs are a common waste that Azure Advisor reports. Deleting them also gives attackers fewer ways in.

What · Where · When · Why · How

What	Public IP addresses are Azure resources that the internet can reach. They are charged per hour.
Where	They are network resources in a subscription. They are attached to VMs, load balancers or gateways, or left unattached.
When	When you clean up unused resources to lower Azure costs.
Why	Public IPs are charged even when unattached. Users, groups and NICs are free.
How	Find unused public IPs with Azure Advisor or the portal, and delete them.

Exam tip
Free: resource groups, VNets, NICs, NSGs, users, groups. Charged even when unused: public IPs, disks, reserved capacity.

Memory trick
A public IP is a rented phone number - you pay even if nobody calls.

Real-world example, use case and scenario

EXAMPLE
Forgotten IPs
A team deleted its test VMs. Later it found 10 Standard public IPs still listed and charged every hour. Deleting them stopped the charge.

USE CASE
Advisor cleanup
A hotel chain uses Azure Advisor cost advice every month. It finds unused public IPs and unattached disks in all subscriptions.

SCENARIO
Wrong cleanup target
A charity's admin deleted 50 old user accounts to save money, but the bill did not change. A partner found 12 unused public IPs. Deleting those gave the real saving.

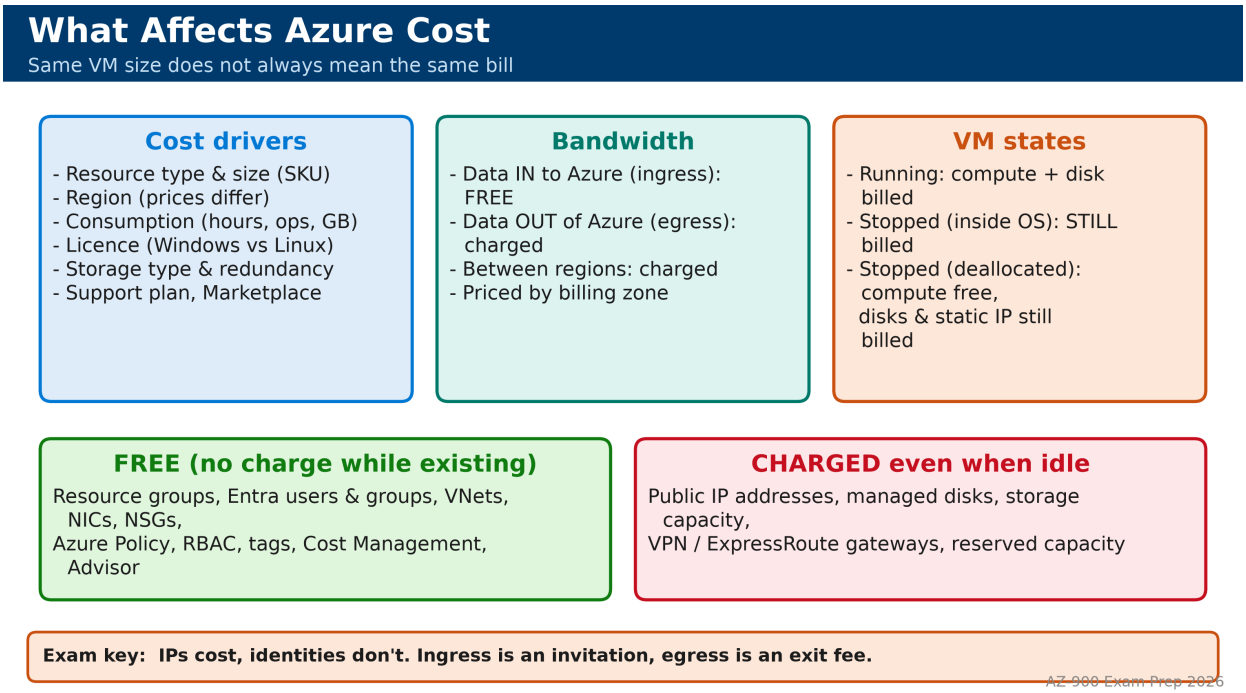


Figure Q235a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

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• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

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Resource	Approx. price	What you pay for / tip
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Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q235b: Approximate Azure charges (USD)

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- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 236

AZURE MANAGEMENT & GOVERNANCE

Cost

v5 Q345, Q16

If an Azure virtual machine has a status of Stopped (deallocated), you will continue to pay for.

✖ A. Compute capacity.

✖ B. I/O operations.

✖ C. Networking.

✔ **D.** Storage.

Correct answer: Storage.

Clarification (read this first): The question asks what you STILL pay for after the VM is Stopped (deallocated - stopped from Azure, so the hardware is released). Answer: storage (the disks). Q283 asks if you still pay for COMPUTE - that answer is No. Q284 asks if the managed disk is still charged - Yes.

Answer explained

✔ **Storage.**
A Stopped (deallocated) VM gives back its compute hardware. But its managed OS disk and data disks still exist, and you pay for them as storage.

✖ **Compute capacity.**
Deallocation releases the vCPUs and memory, so compute charges stop.

✖ **I/O operations.**
A deallocated VM does no disk reads or writes, so there are no transaction charges.

✖ **Networking.**
No traffic flows from a deallocated VM, so there is no network charge. (A static public IP is a separate charge.)

EXAM STRATEGY
'Deallocated' = compute released, disks kept. Only storage (managed disks) is still charged. Remove compute, I/O and networking.

IMPACT
Deallocating test VMs at night saves most of the cost. But disk charges continue. Delete unused disks to stop all charges.

What · Where · When · Why · How

What	Stopped (deallocated) is a VM state where Azure releases the compute hardware.
Where	You do it with the Stop action in the Azure portal, CLI or PowerShell, not from inside the VM's OS.
When	When you do not need a VM for some time, such as nights or weekends.
Why	Compute charges stop, but you still pay storage for the managed OS and data disks.
How	Stop the VM in Azure to deallocate it. The disks stay and are charged until you delete them.

Exam tip
Stopped (deallocated) = no compute charge, but you still pay for disks. Stopped (allocated, shut down from inside the OS) = you still pay for compute.

Memory trick
Deallocate the engine, keep paying for the garage - the disks stay parked.

Real-world example, use case and scenario

EXAMPLE
Weekend shutdown
A D4s_v5 VM is deallocated over the weekend. It has no compute cost, but its 256 GB Premium SSD is charged for the full month.

USE CASE
Test environment schedule
A software consulting company uses auto-shutdown to deallocate test VMs at 7 PM. It pays a small disk cost but saves a lot on compute.

SCENARIO
Shut down but still charged
A university lab shut down its VMs from inside Windows. It was surprised to pay full compute charges. When it deallocated them from the portal, compute charges stopped and only disk storage stayed on the bill.

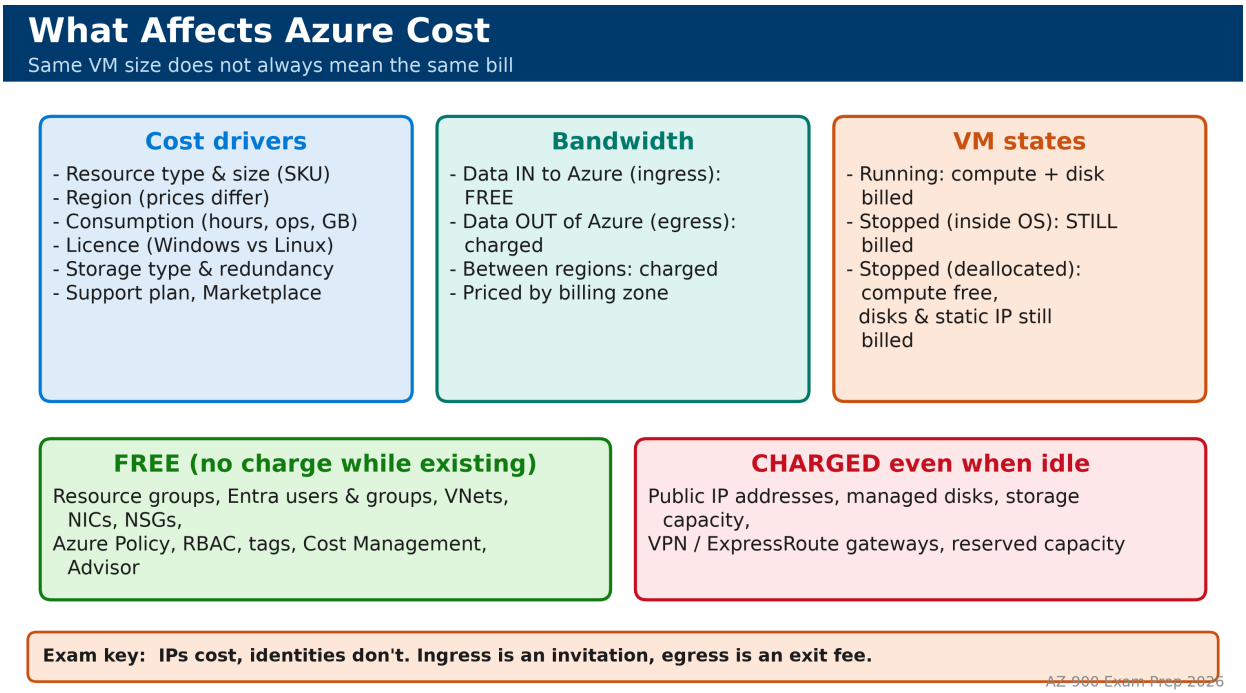


Figure Q236a: What affects Azure cost

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• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

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Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q236b: Approximate Azure charges (USD)

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- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 237

AZURE ARCHITECTURE & SERVICES

Storage
v5 Q347

When you use a general-purpose v2 Azure Storage account, you are only charged for the amount of data that is stored. All read and write operations are free.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap words are 'only' and 'all ... free'. You pay for stored data AND for read and write operations. So the statement is false.

Answer explained

✔ **No.**
GPv2 (general-purpose v2) accounts charge for stored data. They also charge for operations (per 10,000 reads, writes, lists), data retrieval on cooler tiers, geo-replication (copies in another region) and egress (data leaving Azure).

✖ **Yes.**
Reads and writes are charged per 10,000 operations. Stored data is only one part of the storage bill.

EXAM STRATEGY
Look for 'only charged for data stored' and 'operations are free'. GPv2 charges for stored data, operations and egress. Answer No.

IMPACT
Apps that do many operations can pay more for operations than for storage. If you ignore this, you may go over budget.

What · Where · When · Why · How

What	A general-purpose v2 storage account is the standard Azure Storage account for blobs, files, queues and tables.
Where	You create it in a subscription and region, with an access tier and a redundancy setting.
When	When you estimate the full cost of storing and reading data.
Why	Reads and writes are charged per 10,000 operations, so they are not free.
How	Bill = stored data + operations + retrieval on cooler tiers + geo-replication + egress.

Exam tip
Storage cost = stored data + operations + data retrieval (cooler tiers) + egress + replication. Cooler tiers make storing cheaper but reading more expensive.

Memory trick
Storage is a toll road: you pay for the parking space and for every trip in and out.

Real-world example, use case and scenario

EXAMPLE
Busy app
An app stores only 10 GB but does 500 million small reads per month. It pays more for operations than for storage.

USE CASE
Tier choice
A photo-sharing site keeps new images in the Hot tier. The images are read often, so the Cool tier's higher read costs would cost more.

SCENARIO
The logging bill
A finance tech company wrote millions of tiny log entries per hour to blob storage. Operation costs were higher than storage costs. Writing logs in larger batches cut the bill a lot.

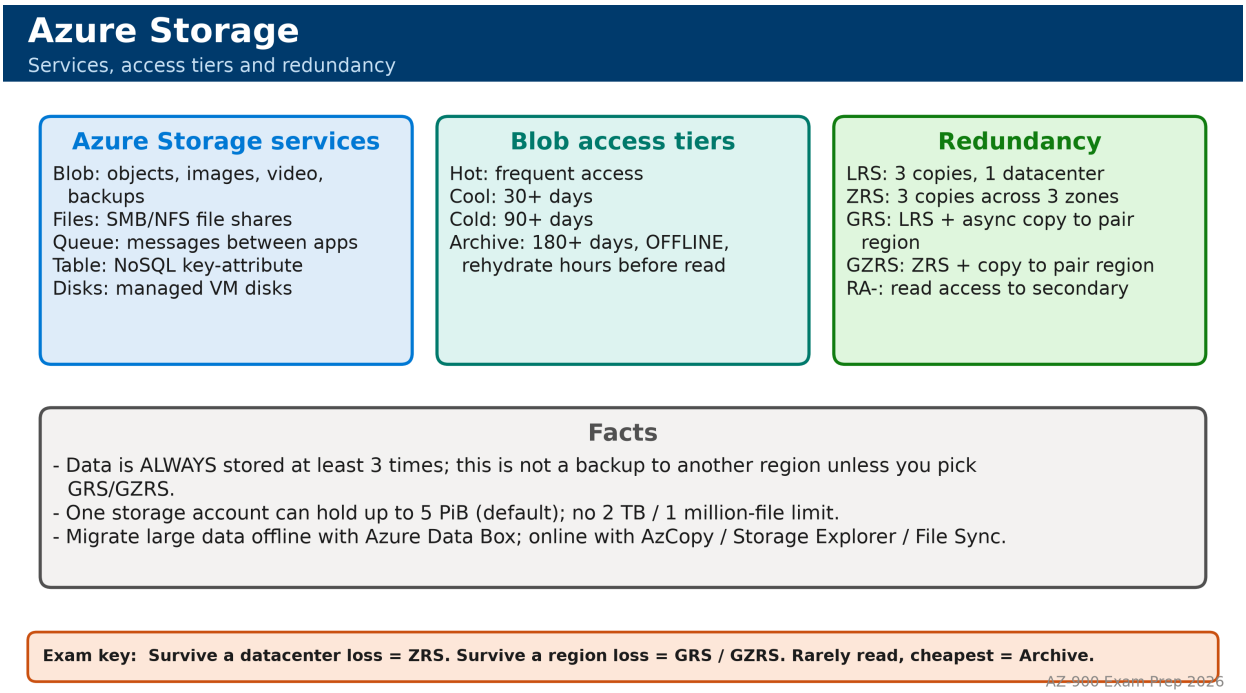


Figure Q237: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 238

CLOUD CONCEPTS

SLA
v5 Q349

In Microsoft Entra ID Premium P2, at least 99.9 percent availability is guaranteed.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): 'At least 99.9 percent' means 99.9% or higher. The paid tiers (P1, P2) promise 99.9%, and Microsoft promises 99.99% for sign-in. So Yes. Q239 asks if P2 has the same SLA as the Free tier - that answer is No, because Free has no SLA.

2026 update: Microsoft Entra ID was called Azure AD before. The paid tiers (P1 and P2) have a 99.9 percent SLA. Microsoft has also promised 99.99 percent for user sign-in. Either way, at least 99.9 percent is guaranteed.

Answer explained

✔ **Yes.**
Entra ID Premium P1 and P2 are paid tiers. They have an SLA (uptime promise) with money back of at least 99.9 percent (99.99 percent for sign-in).

✖ **No.**
P2 is a paid tier with a published SLA of at least 99.9 percent. Only the Free tier has no SLA.

EXAM STRATEGY
Paid Entra ID tiers (P1, P2) have an SLA of 99.9 percent or better, backed by money credits. 'At least 99.9' = Yes.

IMPACT
Identity is the front door to every app. If you know the SLA, you can plan sign-in availability for important systems.

What · Where · When · Why · How

What	Microsoft Entra ID Premium P2 is a paid Entra ID tier. It adds Identity Protection and Privileged Identity Management.
Where	It is licensed per user in the Entra tenant and covered by the Online Services SLA.
When	When a company needs advanced identity security with promised uptime.
Why	Paid Entra ID tiers have an SLA of at least 99.9 percent, backed by money credits.
How	If monthly uptime is below the SLA, the customer can claim a service credit.

Exam tip
Free tier = no SLA. Paid tiers (P1/P2) = 99.9 percent or better. P1 and P2 have different features, not different SLAs.

Memory trick
Pay for Premium, get the Promise.

Real-world example, use case and scenario

EXAMPLE
SLA document
The Online Services SLA lists Microsoft Entra ID Premium with 99.9 percent monthly uptime. It gives service credits if uptime is lower.

USE CASE
Identity backbone
A global airline uses Entra ID P2 for staff sign-in with Conditional Access. It needs an SLA because crews cannot check in without sign-in.

SCENARIO
Auditor question
An auditor asked a pension fund if its identity system had an SLA. The IT lead showed the P2 licences and the 99.9 percent SLA. The auditor was satisfied.

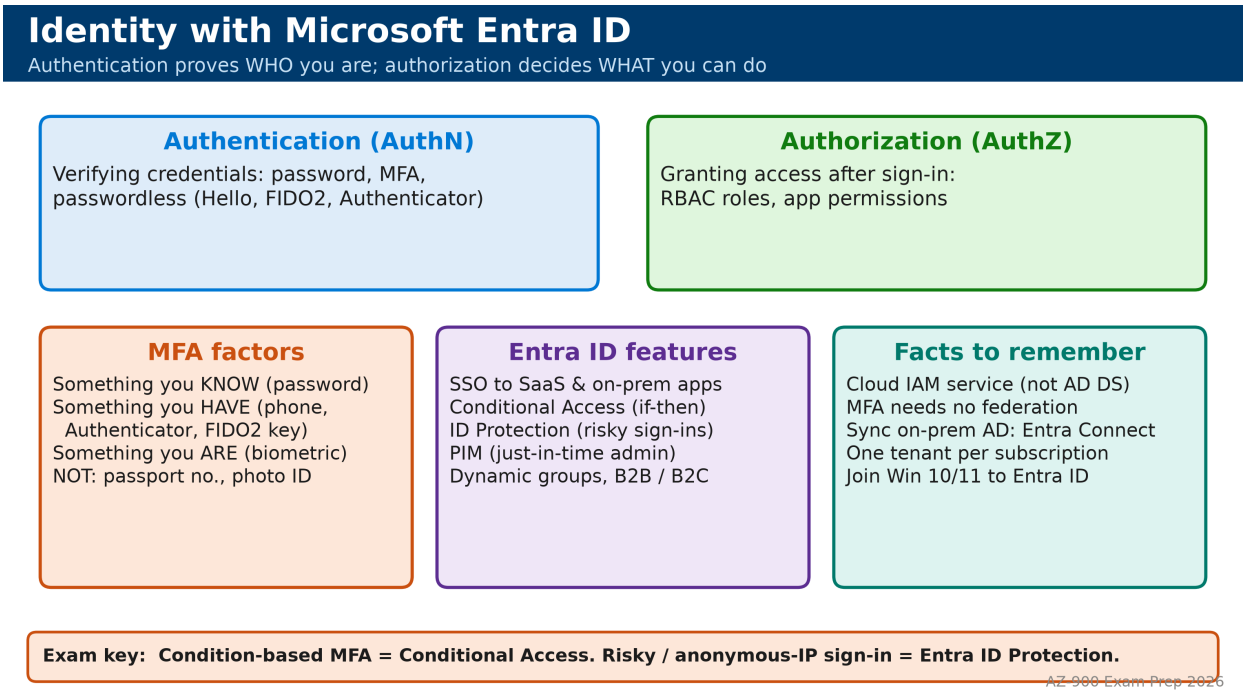


Figure Q238: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 239

CLOUD CONCEPTS

SLA
v5 Q350

The Service Level Agreement (SLA) for Microsoft Entra ID Premium P2 is the same as the SLA for Microsoft Entra ID Free.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The question asks if P2 and Free have the SAME SLA. They do not: Free has no SLA at all. Q238 asks if P2 guarantees at least 99.9% - that answer is Yes.

Answer explained

✔ **No.**
Entra ID Free has no SLA (uptime promise) with money back. Premium P1 and P2 promise at least 99.9 percent. So the SLAs are different.

✖ **Yes.**
The Free tier has no SLA at all, so it cannot match P2's promised uptime.

EXAM STRATEGY
The Free tier has no SLA; paid tiers do. 'Same SLA as Free' = No.

IMPACT
Companies that need promised sign-in uptime must pay for a paid Entra ID tier.

What · Where · When · Why · How

What	Entra ID SLAs are different per tier: Free has none, P1 and P2 have an SLA backed by money credits.
Where	They are written in Microsoft's Online Services SLA document.
When	When you choose an identity tier that must come with an uptime promise.
Why	The Free tier has no SLA, so its SLA cannot equal P2's.
How	Upgrade to P1 or P2 licences to get the SLA and the right to claim service credits.

Exam tip
General AZ-900 rule: free services and free tiers have no SLA. Paid tiers do.

Memory trick
Free comes with no warranty card.

Real-world example, use case and scenario

EXAMPLE
Tier comparison
Entra ID Free gives user management and single sign-on, but no uptime credit. P2 has the 99.9 percent SLA.

USE CASE
Startup upgrade
A fast-growing software startup moved from Entra ID Free to P1. Its large customers wanted an identity provider with an SLA.

SCENARIO
No credit claim
A small charity on Entra ID Free had a sign-in outage and tried to claim a service credit. Support explained that Free has no SLA. Credits are only for paid tiers.

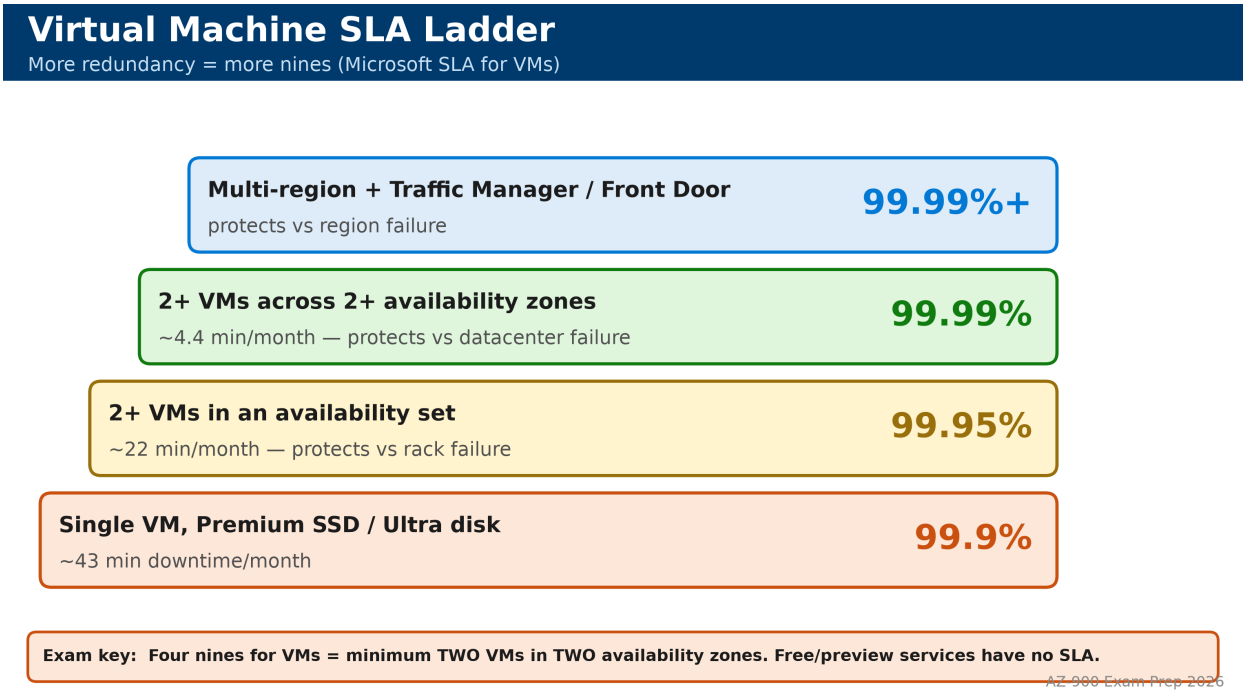


Figure Q239: Virtual machine SLA ladder

About this figure

- Climb the ladder: each step adds redundancy and a higher VM SLA; the downtime figures assume a 30-day month.
- A single VM with Standard SSD gets about 99.5% and with Standard HDD about 95%; all its OS and data disks must meet that tier.
- Availability sets spread VMs across fault domains (racks) and update domains (maintenance batches) in one datacenter; zones spread across datacenters.
- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

Question 240

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q352, Q319

Adding resource groups in an Azure subscription generates additional costs.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Answer explained

✔ **No.**
Resource groups are free containers (folders) in Azure Resource Manager. You pay only for the resources you put inside them.

✖ **Yes.**
Resource groups themselves are never charged. Only the resources you deploy into them cost money.

EXAM STRATEGY
Resource groups are free containers. 'Generates additional costs' = No.

IMPACT
Because they are free, you can split environments clearly. This improves cost reports and access control at no extra cost.

What · Where · When · Why · How

What	A resource group is a free container for Azure resources.
Where	Inside a subscription, managed by Azure Resource Manager.
When	When you organise resources by app, environment or lifetime.
Why	Resource groups cost nothing. Only the resources inside them are charged.
How	Create as many resource groups as you need. Costs appear only after you deploy charged resources.

Exam tip
Free ways to organise: management groups, subscriptions (the container itself), resource groups, tags.

Memory trick
Folders are free; the files inside cost money.

Real-world example, use case and scenario

EXAMPLE
One group per environment
Creating rg-dev, rg-test and rg-prod adds nothing to the bill until you deploy resources into them.

USE CASE
Cost reporting
A retail franchise creates one resource group per store. It sees costs per store in Cost Management at no extra charge.

SCENARIO
Merging myth
A manager asked to merge 40 resource groups into 5 to save money. The architect showed that resource groups are free. The team kept the 40 groups for better access control.

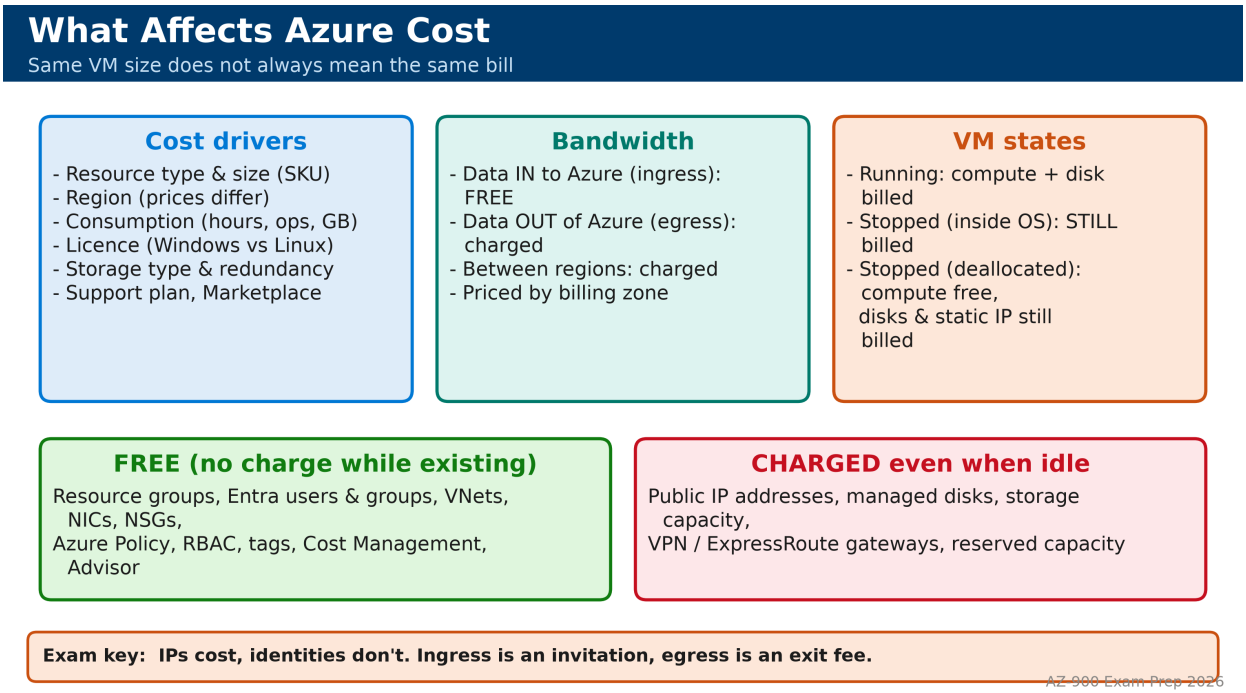


Figure Q240a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

• Bandwidth: ingress is free; the first 100 GB/month of Internet egress is free, then approx. USD 0.087/GB in North America and Europe.

• Inter-region transfer is approx. USD 0.02/GB within a continent; VNet peering is approx. USD 0.01/GB each way in the same region.

• A VM stopped inside the guest OS is still allocated and billed for compute; only Stopped (deallocated) from the portal, CLI or API stops compute charges.

• A Standard static public IP costs approx. USD 3.65/month even unattached; a 128 GB Premium SSD costs approx. USD 20/month even when its VM is deallocated.

• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q240b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 241

AZURE MANAGEMENT & GOVERNANCE

Cost

v5 Q353, Q342

Copying 10 GB of data from Azure to an on-premises network over a VPN generates additional Azure data transfer costs.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): Watch the direction: data goes FROM Azure TO on-premises. Data leaving Azure (egress) is charged. Q219 is the opposite direction (into Azure) - that answer is No, because data coming in is free.

Answer explained

✔ **Yes.**
Data leaving Azure to on-premises over a VPN is charged per GB as outgoing data transfer.

✖ **No.**
Egress (data leaving Azure) is charged. Only the other direction, into Azure, is free.

EXAM STRATEGY
'From Azure to on-premises' = egress = charged. It is the same as the ingress questions, but reversed. The direction decides the answer.

IMPACT
Egress fees grow with the amount of data. So plan a budget for restoring backups or exporting data to on-premises.

What · Where · When · Why · How

What	Egress over VPN is data that leaves Azure through a VPN gateway to on-premises.
Where	Through an Azure VPN gateway on a site-to-site connection.
When	When you download data from Azure to on-premises systems.
Why	Outgoing data is charged per GB, so copying 10 GB creates a transfer cost.
How	It is charged at the normal outgoing data rates, plus the gateway hours.

Exam tip
Same 10 GB, different direction, different answer. Data out = charged.

Memory trick
Coming home from Azure costs the taxi fare.

Real-world example, use case and scenario

EXAMPLE
Log download
You download 10 GB of logs from Azure to an on-premises security system over VPN. This is charged as egress.

USE CASE
Hybrid reporting
A factory copies daily totals from Azure to its on-premises ERP system. It includes egress in its monthly cost plan.

SCENARIO
Copy strategy
A design agency copied all Azure project files back to its office storage every night. After seeing egress costs, it copied only finished projects and cut data transfer by two thirds.

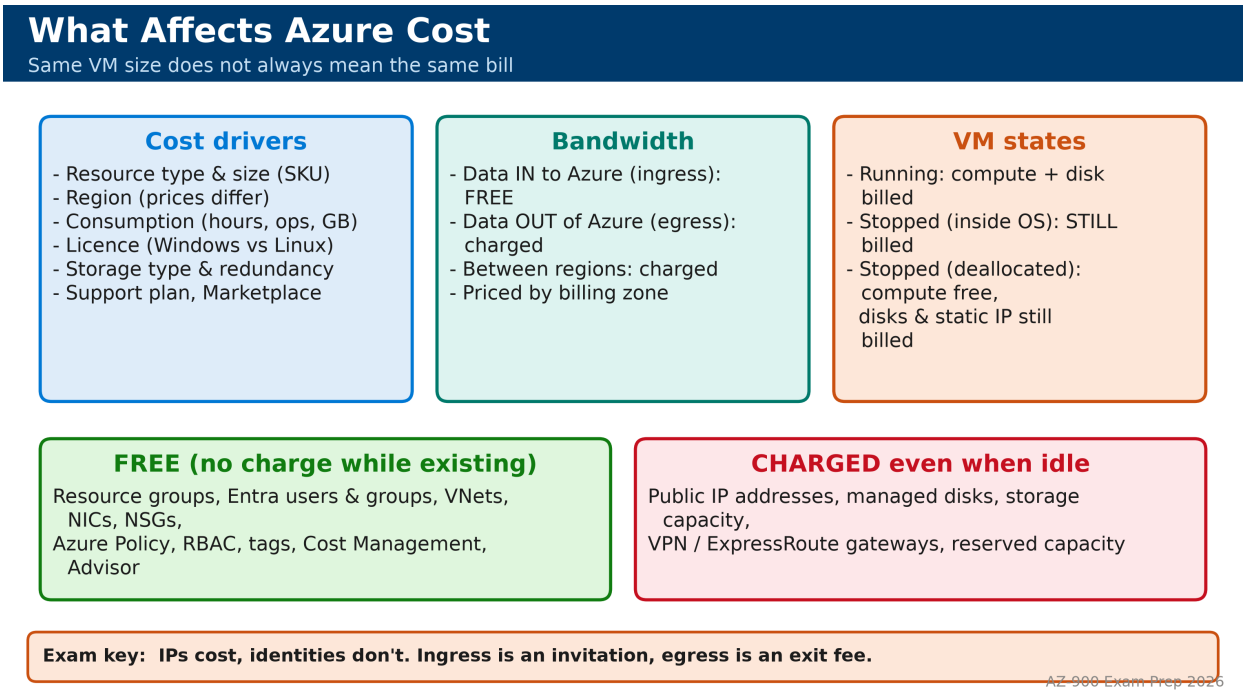


Figure Q241a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

• Bandwidth: ingress is free; the first 100 GB/month of Internet egress is free, then approx. USD 0.087/GB in North America and Europe.

• Inter-region transfer is approx. USD 0.02/GB within a continent; VNet peering is approx. USD 0.01/GB each way in the same region.

• A VM stopped inside the guest OS is still allocated and billed for compute; only Stopped (deallocated) from the portal, CLI or API stops compute charges.

• A Standard static public IP costs approx. USD 3.65/month even unattached; a 128 GB Premium SSD costs approx. USD 20/month even when its VM is deallocated.

• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
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Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q241b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 242

AZURE ARCHITECTURE & SERVICES

Storage
v5 Q354, Q361

Data that is stored in the Archive access tier of an Azure Storage account [...].

✖ A. can be accessed at any time by using azcopy.exe.

✖ B. can only be read by using Azure Backup.

✖ C. must be restored before the data can be accessed.

✔ **D.** must be rehydrated before the data can be accessed.

Correct answer: must be rehydrated before the data can be accessed.

Clarification (read this first): Two options look alike: 'restored' and 'rehydrated'. Azure's exact word for bringing Archive data back online is 'rehydrate'. 'Restore' is a backup word, so it is the trap.

Answer explained

✔ **must be rehydrated before the data can be accessed.**
Archive is an offline tier. You must rehydrate blobs (move them back to Hot, Cool or Cold, by changing the tier or copying) before you can read them. This can take up to 15 hours.

✖ **Can be accessed at any time by using azcopy.exe.**
No tool, not even AzCopy, can read Archive data directly. It is offline until rehydrated.

✖ **Can only be read by using Azure Backup.**
Azure Backup is not the way to read archived blobs. It protects VMs, files and databases.

✖ **Must be restored before the data can be accessed.**
'Restore' is not the official term. Azure calls bringing archived blobs back online 'rehydration'.

EXAM STRATEGY
'Archive tier' = offline storage, and the key word is rehydrate. Remove 'restored' (a backup word), 'any time with azcopy' and 'only Azure Backup'.

IMPACT
Rehydration can take up to 15 hours at standard priority. If you archive data the business needs fast, a quick request can become a one-day wait.

What · Where · When · Why · How

What	The Archive access tier is an offline, lowest-cost blob tier for data you rarely read.
Where	It is set per blob in an Azure Storage account.
When	When you must keep data for a long time and can wait hours to read it.
Why	Archived blobs are offline, so you must rehydrate them before you can read them.
How	Change the tier to Hot, Cool or Cold, or copy the blob to an online tier. Standard priority can take up to 15 hours.

Exam tip
Archive = offline, cheapest storage, highest cost to read, hours to rehydrate, 180-day minimum.

Memory trick
Archive data is freeze-dried - just add water (rehydrate) before use.

Real-world example, use case and scenario

EXAMPLE
Tier change
You set an archived blob's tier to Hot with standard priority. It can be read within about 15 hours.

USE CASE
Compliance records
A broker keeps 7 years of trade recordings in the Archive tier. It rehydrates single files when a regulator asks for them.

SCENARIO
Urgent request
A hospital needed an archived scan within the hour for a court case. It used high-priority rehydration, which brought the small file online in minutes. Later it wrote down archive read times in its service promise.

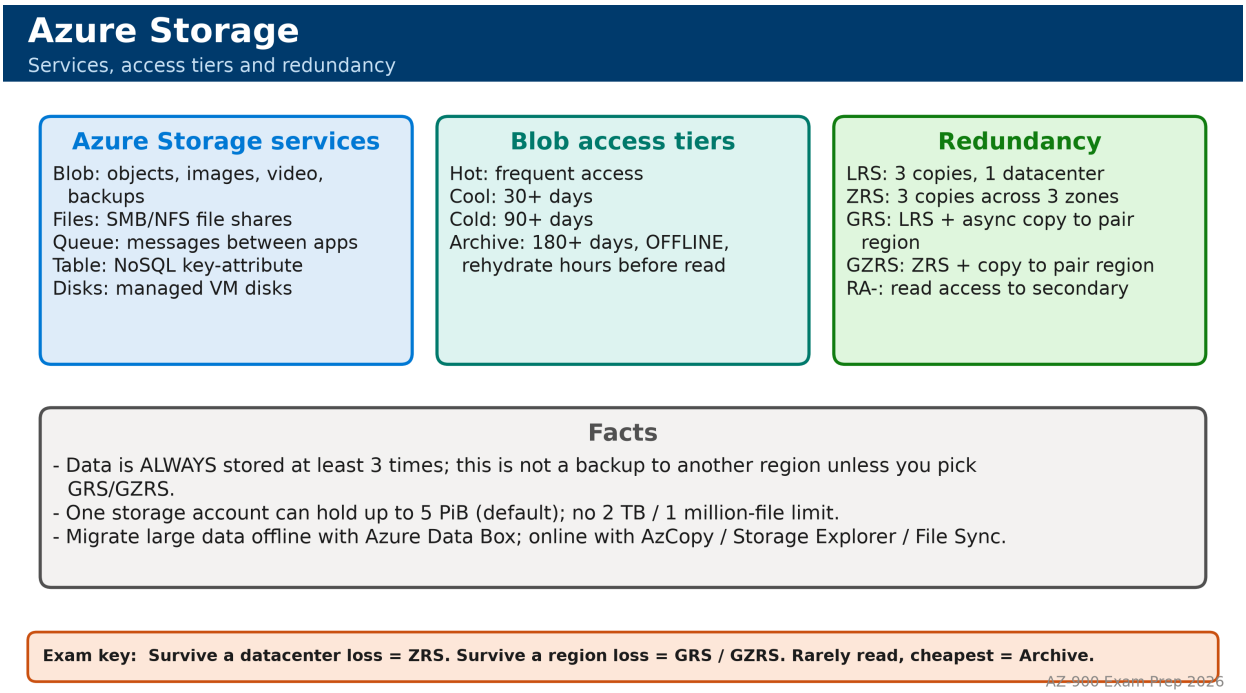


Figure Q242: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 243

AZURE ARCHITECTURE & SERVICES — **Databases** v5 Q356, Q404

Azure Cosmos DB is an example of a [...] offering.

✔ **A.** Platform as a Service (PaaS).

✖ B. Infrastructure as a service (IaaS).

✖ C. Serverless.

✖ D. Software as a service (SaaS).

Correct answer: Platform as a Service (PaaS).

Clarification (read this first): The question asks for the service MODEL (IaaS, PaaS or SaaS). 'Serverless' is a trap: Cosmos DB has a serverless billing mode, but serverless is not a service model.

Answer explained

✔ **Platform as a Service (PaaS).**
Cosmos DB is a fully managed database. Microsoft runs the servers, patching, replication and backups. You manage only the data and throughput (speed you pay for). This is PaaS (Azure runs the platform, you bring the data and code).

✖ **Infrastructure as a service (IaaS).**
Cosmos DB gives you no VMs and no OS access. IaaS (you manage the OS) would be running a database yourself on Azure VMs.

✖ **Serverless.**
Serverless is one of Cosmos DB's capacity and billing modes. It is not its service model.

✖ **Software as a service (SaaS).**
SaaS is a finished app for end users, like Microsoft 365. Cosmos DB is a platform that developers build on.

EXAM STRATEGY
'Fully managed database' means PaaS. Remove SaaS (you build on it) and IaaS (no VM to manage). 'Serverless' is a Cosmos DB capacity mode, not a service model.

IMPACT
The service model decides who patches and backs up a database. Questions about Cosmos DB, SQL Database and PostgreSQL as PaaS come up often in the exam.

What · Where · When · Why · How

What	Azure Cosmos DB is a fully managed NoSQL and vector database that can spread data across the world.
Where	You deploy it as a Cosmos DB account in one or more Azure regions.
When	When apps need fast data in many places in the world without managing servers.
Why	Microsoft runs the infrastructure and you manage the data and throughput, which is PaaS.
How	Create an account and choose an API and a capacity mode. Azure handles copying data and patching.

Exam tip
Managed databases (Azure SQL Database, Cosmos DB, Azure Database for PostgreSQL/MySQL) = PaaS. SQL Server on a VM = IaaS.

Memory trick
Cosmos is a Platform you launch apps from - PaaS to the stars.

Real-world example, use case and scenario

EXAMPLE
Writes in many regions
A Cosmos DB account with multi-region writes copies data across three regions. The customer manages no servers.

USE CASE
Shopping carts
An online fashion shop stores shopping carts in Cosmos DB. Reads take only a few milliseconds anywhere in the world.

SCENARIO
Patching work
A gaming studio ran MongoDB on VMs and spent weekends patching. It moved to Cosmos DB for MongoDB (PaaS), so Microsoft now patches the OS and the database.

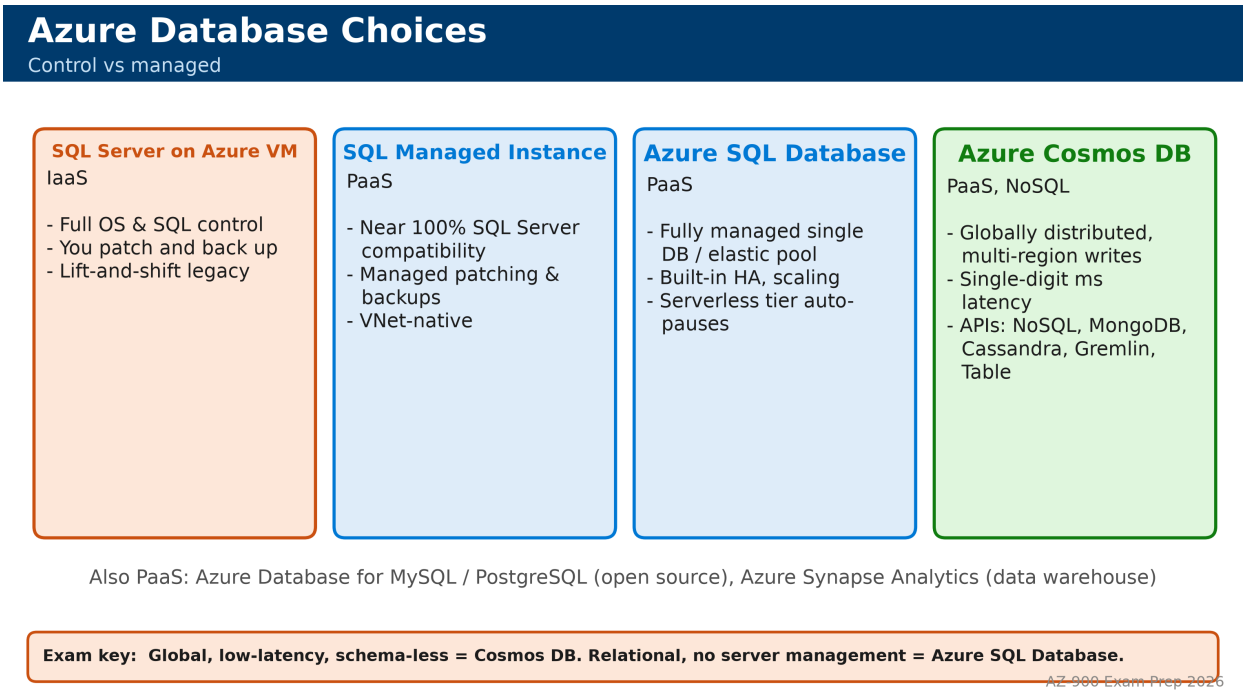


Figure Q243: Azure database choices

About this figure

• Columns run from most control (SQL Server on a VM, IaaS) to fully managed relational PaaS and globally distributed NoSQL Cosmos DB.

• Azure SQL Database Basic costs approx. USD 5/month and S0 approx. USD 15/month; serverless bills per vCore-second and storage only while paused.

• Azure SQL Database SLA is up to 99.99%, and up to 99.995% with zone redundancy in Business Critical; automated backups are kept 7 days by default.

• Cosmos DB free tier gives 1,000 RU/s and 25 GB free per account; it offers five consistency levels and up to 99.999% SLA for multi-region accounts.

• SQL Managed Instance suits migrations needing SQL Agent, cross-database queries or CLR; Azure Hybrid Benefit can reuse SQL Server licences for any of them.

Question 244

AZURE MANAGEMENT & GOVERNANCE — **Compliance** v5 Q357

The Microsoft Service Trust Portal can be accessed by using a Microsoft cloud services account.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
You sign in to the Service Trust Portal with a Microsoft cloud services account (for example a Microsoft 365 or Entra work account). This opens its audit reports.

✖ **No.**
The portal accepts sign-in with a Microsoft cloud services account. You do not need a special separate login.

EXAM STRATEGY
Key fact: you sign in to the Service Trust Portal with a Microsoft cloud services account (an Entra work account). A statement that says 'no account' or 'anonymous only' is the false version.

IMPACT
Auditors often ask for SOC 2 and ISO 27001 reports. You can download them yourself with your work account, which saves weeks in supplier checks.

What · Where · When · Why · How

What	The Service Trust Portal shares Microsoft audit reports, certificates and compliance guides.
Where	At servicetrust.microsoft.com, signed in with a Microsoft cloud services account.
When	When you need Microsoft's SOC, ISO or PCI reports to check a supplier.
Why	Signing in with a Microsoft cloud services account gives full access to the documents.
How	Sign in with a Microsoft 365, Dynamics 365 or Entra work account and download the reports.

Exam tip
Service Trust Portal = Microsoft's audit reports (SOC, ISO, PCI). Sign in with your Microsoft cloud services account to download them.

Memory trick
STP: Show The Paperwork - but show your work badge first.

Real-world example, use case and scenario

EXAMPLE
Downloading SOC 2
A compliance analyst signs in with her company Entra account. She downloads Azure's SOC 2 Type 2 report.

USE CASE
Supplier check
A credit union reads ISO 27001 certificates in the portal before it approves Azure as a supplier.

SCENARIO
Locked report
An auditor at a dental group could see report names but could not download them. She signed in with the group's Microsoft 365 work account, and the documents opened.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

AZ-900 Exam Prep 2026

Figure Q244: Compliance, privacy and trust

About this figure

• Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.

• Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.

• Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.

• Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.

• The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 245

AZURE MANAGEMENT & GOVERNANCE — **Compliance** v5 Q359

The My Library feature can be used to save Microsoft Service Trust Portal documents and resources in a single location.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
My Library in the Service Trust Portal saves documents and resources in one place. It can also tell you when a saved document is updated.

✖ **No.**
My Library is exactly the Service Trust Portal feature that saves documents in one place.

EXAM STRATEGY
'Save documents in a single location' = My Library. Answer Yes. False versions say it stores your own company files, or that it is outside the Service Trust Portal.

IMPACT
My Library sends update alerts when Microsoft publishes a new audit report. So compliance teams do not send old evidence to regulators.

What · Where · When · Why · How

What	My Library is a Service Trust Portal feature that saves documents and resources in one place.
Where	Inside the Service Trust Portal, for users who are signed in.
When	When you need the same Microsoft compliance reports again and again.
Why	It keeps saved documents together and can tell you about updates.
How	Select documents in the portal and add them to My Library.

Exam tip
My Library = bookmarks plus update alerts in the Service Trust Portal.

Memory trick
My Library = my bookshelf of Microsoft audit reports.

Real-world example, use case and scenario

EXAMPLE
Saving reports
A security officer adds the latest ISO 27018 and SOC 1 reports to My Library. Now she can open them with one click.

USE CASE
Update alerts
A government contractor saves FedRAMP documents to My Library. It gets an alert when Microsoft uploads a new version.

SCENARIO
Lost link
A factory's compliance lead kept losing her downloaded PDFs. She saved them to My Library instead, and the team always opened the newest version.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

AZ-900 Exam Prep 2026

Figure Q245: Compliance, privacy and trust

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- Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.
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Question 246

CLOUD CONCEPTS

Deployment models
v5 Q362

A company can extend the computing resources of its internal network by using the public cloud.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): This is about connecting your network to the public cloud and running extra workloads there (hybrid cloud). Q267 is a look-alike about adding your own PHYSICAL servers to the public cloud - that answer is No.

Answer explained

✔ **Yes.**
A company can connect its internal network to Azure with a VPN or ExpressRoute. Then it can run extra workloads in the public cloud. This is the hybrid extension pattern.

✖ **No.**
Extending on-premises capacity into Azure is a normal hybrid cloud case. This includes cloud bursting (using the cloud only at busy times).

EXAM STRATEGY
'Extend internal network using public cloud' is the textbook definition of hybrid, so Yes. Compare 'add physical servers to the public cloud', which is No.

IMPACT
Most large companies use a VPN or ExpressRoute to get extra capacity without buying hardware. Questions about deployment models come up often in AZ-900.

What · Where · When · Why · How

What	Extending on-premises compute into the public cloud is the hybrid cloud pattern.
Where	Azure virtual networks connected to the internal network with VPN Gateway or ExpressRoute.
When	When on-premises capacity runs out or demand is high for a short time.
Why	The public cloud adds capacity when you need it, without buying hardware.
How	Connect a VNet to the internal network and deploy extra workloads there.

Exam tip
Extend the internal network = hybrid cloud. You connect with Site-to-Site VPN or ExpressRoute.

Memory trick
The cloud is your datacenter's extension cord.

Real-world example, use case and scenario

EXAMPLE
Extra VMs
A company adds 50 Azure VMs in a VNet connected by VPN for a short project. Office PCs can reach them.

USE CASE
Tax season
An accounting firm uses Azure for extra processing in tax season. It does not buy servers that would sit unused the rest of the year.

SCENARIO
Full datacenter
A biotech company's datacenter had no more rack space during a project. Instead of waiting six months for hardware, it extended its network into Azure and ran its jobs there within a week.

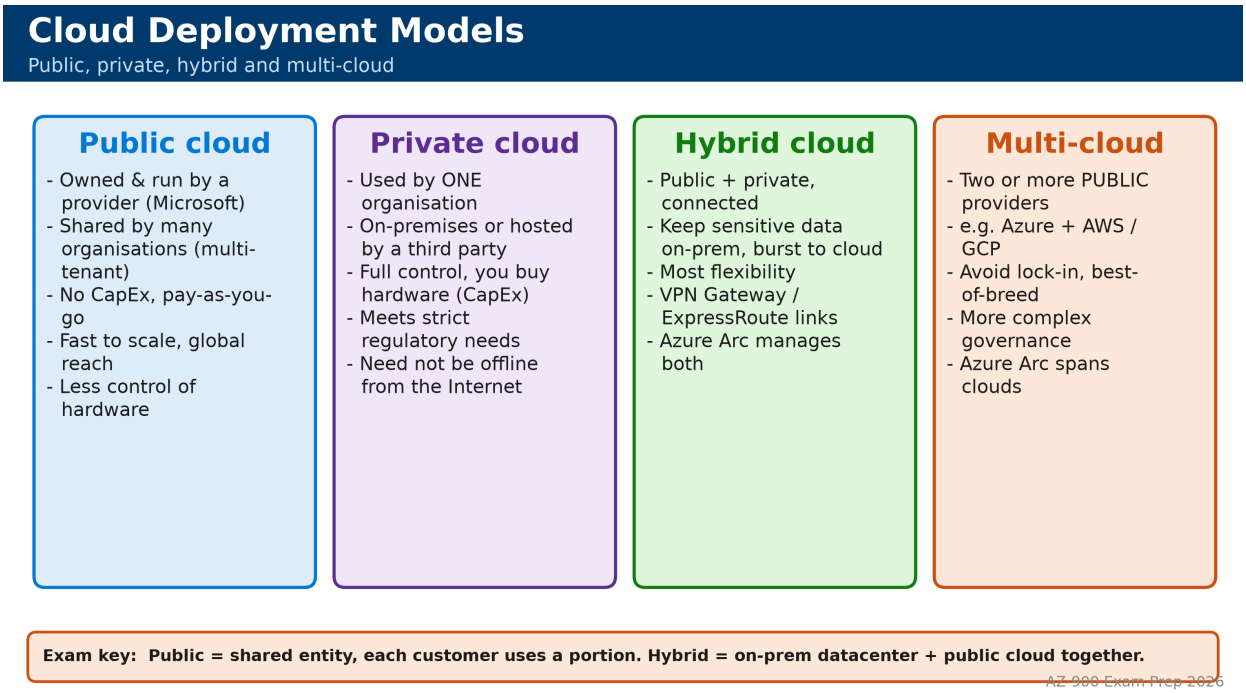


Figure Q246: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 247

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q363, Q26

Azure DevOps Services allows developers to deploy or update applications to Azure using Continuous Integration/Continuous Delivery (CI/CD) pipelines.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The key words are 'deploy or update' with 'CI/CD pipelines' (automatic build, test and release). Azure DevOps does this, so Yes. Q248 asks if Azure DevOps can HOST web apps - that answer is No.

Answer explained

✔ **Yes.**
Azure Pipelines is part of Azure DevOps Services. It gives CI/CD (automatic build, test and release). It builds, tests and deploys apps to Azure targets such as App Service and AKS.

✖ **No.**
Azure Pipelines is the built-in CI/CD part of Azure DevOps Services.

EXAM STRATEGY
'CI/CD pipelines' = Azure Pipelines, a part of Azure DevOps Services, so Yes. In this series, only the 'host web apps' question is No.

IMPACT
Automatic pipelines mean fewer failed manual releases to production. The DevOps Yes/No group (pipelines, Git, hosting) appears together in the exam.

What · Where · When · Why · How

What	Azure Pipelines is the CI/CD (automatic build and release) service in Azure DevOps Services.
Where	In an Azure DevOps organisation and project, deploying to Azure subscriptions.
When	When code changes should be built, tested and deployed automatically.
Why	Pipelines automatically deploy and update apps in Azure.
How	Write a YAML pipeline that starts on each commit. It builds, tests and deploys to targets like App Service.

Exam tip
Azure DevOps = Boards (work tracking), Repos (Git), Pipelines (CI/CD), Test Plans, Artifacts (packages).

Memory trick
Pipelines pipe code from commit to cloud.

Real-world example, use case and scenario

EXAMPLE
Pipeline run
Each merge to main starts an Azure Pipelines build. It runs tests and deploys a container to AKS.

USE CASE
Release speed
A finance tech company moves from monthly manual releases to daily automatic releases to App Service with Azure Pipelines.

SCENARIO
Friday-night releases
A food delivery app's engineers released by hand on Friday nights and often broke production. A pipeline with automatic tests and staging slots ended the late-night fixes.

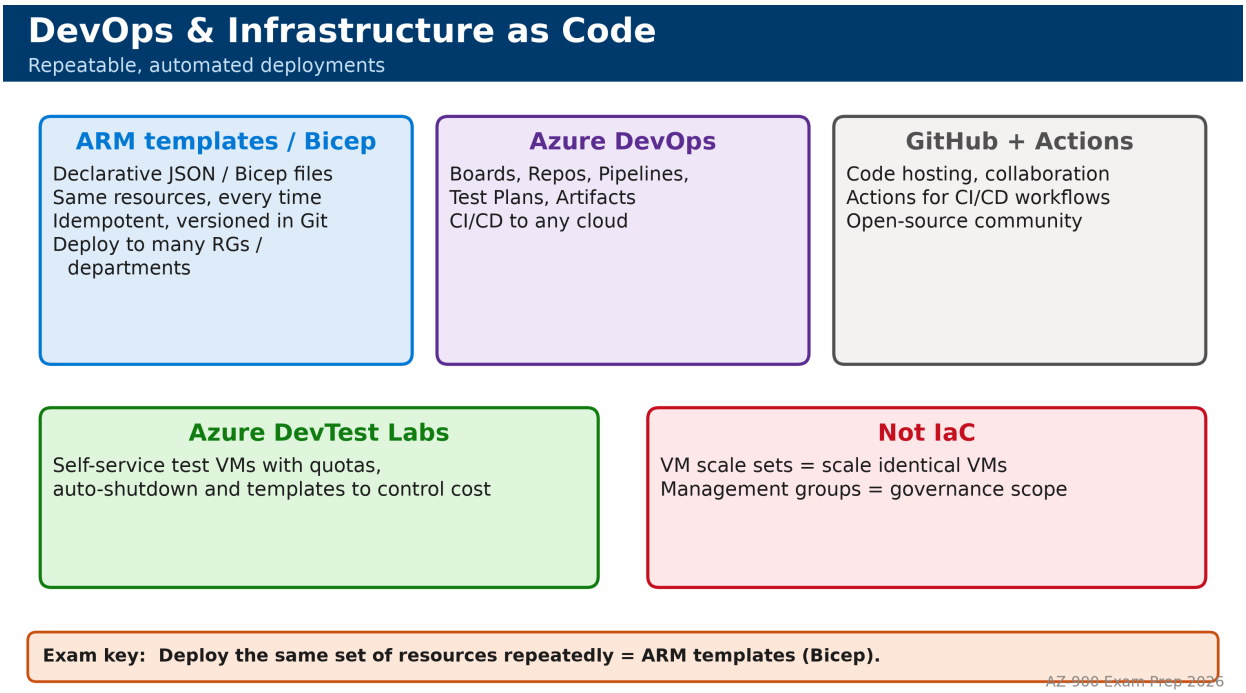


Figure Q247: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 248

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q365

Azure DevOps Services can be used to build and host web apps.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap word is 'host'. Azure DevOps can BUILD and deploy web apps, but it cannot RUN (host) them. Q247 asks if it can deploy with CI/CD - that answer is Yes.

Answer corrected in this edition. The previous PDF printed: *Yes.*. The printed answer Yes is wrong. Azure DevOps Services builds, tests and deploys web apps (with Pipelines), but it does not host them. Hosting is done by services such as Azure App Service, AKS or VMs. The answer key also says No.

Answer explained

✔ **No.**
Azure DevOps builds, tests and deploys web apps, but it has no place to run them. The app runs on a host, such as Azure App Service.

✖ **Yes.**
Azure DevOps can build web apps but cannot host them. Hosting is done by App Service, AKS or VMs.

EXAM STRATEGY
The word 'host' decides it. DevOps builds and deploys, but runs nothing. Answer No. Hosting is done by App Service, AKS or VMs.

IMPACT
If you think DevOps hosts apps, you may forget compute costs in the project budget. 'And host' is a classic one-word exam trap.

What · Where · When · Why · How

What	Azure DevOps Services is a set of developer tools: Boards, Repos, Pipelines, Test Plans and Artifacts.
Where	In an Azure DevOps organisation, separate from hosting services like App Service.
When	When you plan, build, test and deploy software.
Why	It builds and deploys but does not host web apps, so the statement is false.
How	Pipelines deploy the built app to a hosting service, such as Azure App Service.

Exam tip
Watch for 'and host'. DevOps = build + deploy. App Service = host.

Memory trick
DevOps is the delivery truck, App Service is the house.

Real-world example, use case and scenario

EXAMPLE
Two services
An Azure Pipelines release deploys a Node.js site. The site then runs on an App Service plan.

USE CASE
Split of work
At a logistics company, a platform team owns App Service hosting. A dev team owns the Azure DevOps pipelines that deploy to it.

SCENARIO
Missing hosting cost
A startup's budget had only Azure DevOps licences, because it thought DevOps hosted the site. The architect added an App Service plan to the budget, since DevOps only builds and deploys.

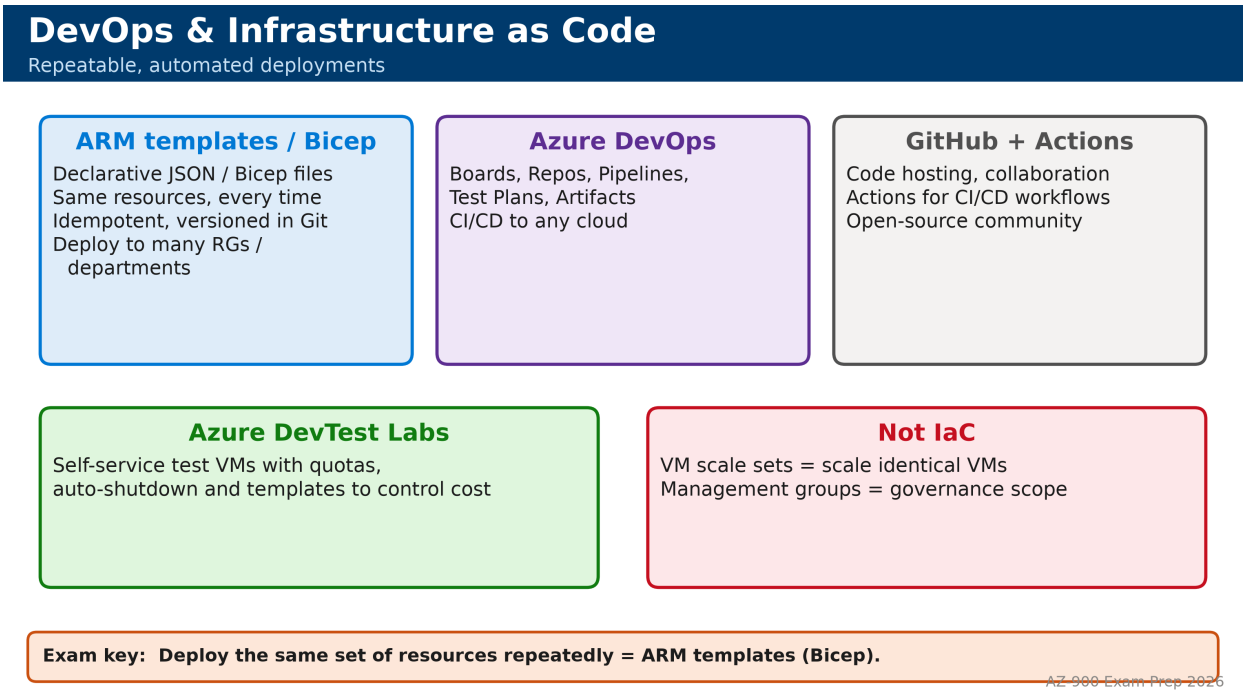


Figure Q248a: DevOps and infrastructure as code

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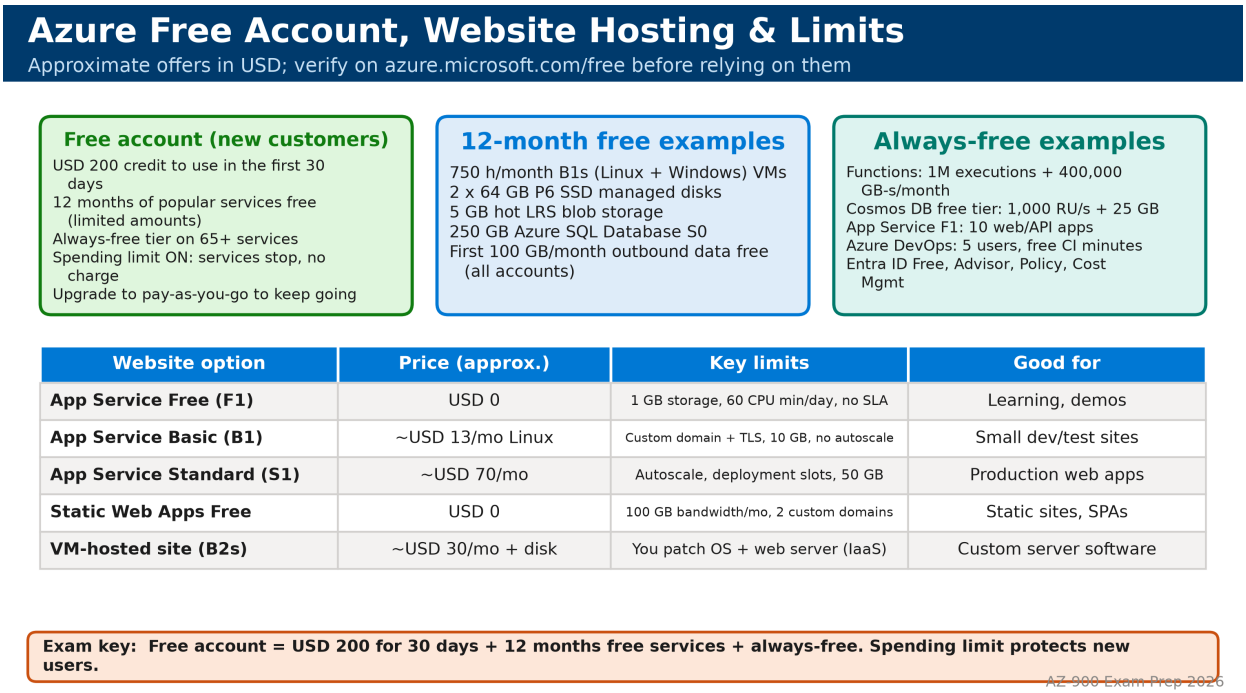


Figure Q248b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 249

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q367

You plan to extend your company network to Azure. The network contains a VPN appliance that uses an IP address of 131.107.200.1. You need to create an Azure resource that identifies the VPN appliance. Which Azure resource should you create?

- ✖ A. Virtual networks.
- ✖ B. Load balancers.
- ✖ C. Virtual networks gateways.
- ✖ D. DNS zones.
- ✖ E. Traffic Manager profiles.
- ✖ F. Network Watcher.
- ✖ G. Virtual networks (classic).
- ✖ H. Application gateways.
- ✔ **I.** Local network gateways.
- ✖ J. CDN profiles.
- ✖ K. ExpressRoute circuits.
- ✖ L. Network security groups.

Correct answer: Local network gateways.

Clarification (read this first): Two names look alike. LOCAL network gateway = the Azure record of YOUR on-premises VPN device (its IP 131.107.200.1). VIRTUAL network gateway = the VPN endpoint on the Azure side. The question asks for the on-premises device.

Answer explained

✔ **Local network gateways.**
A local network gateway represents the on-premises VPN device in Azure. It stores the device's public IP (131.107.200.1) and the on-premises address ranges.

✖ **Virtual networks.**
A virtual network is the private network on the Azure side. It does not represent on-premises equipment.

✖ **Load balancers.**
Load balancers spread traffic across VMs. They do not define VPN partners.

✖ **Virtual networks gateways.**
The virtual network gateway is the VPN endpoint on the Azure side, not the on-premises device.

✖ **DNS zones.**
DNS zones hold domain name records, not VPN device details.

✖ **Traffic Manager profiles.**
Traffic Manager sends users to endpoints using DNS. It has no VPN role.

✖ **Network Watcher.**
Network Watcher monitors and checks networks. It does not define gateways.

✖ **Virtual networks (classic).**
Classic virtual networks are an old, retired model. They do not represent a VPN device.

✖ **Application gateways.**
Application Gateway is a web load balancer (Layer 7) with an optional WAF (web firewall).

✖ **CDN profiles.**
CDN profiles store copies of content near users. They are not related to VPN.

✖ **ExpressRoute circuits.**
ExpressRoute circuits are private connections through a provider. They do not identify a VPN device.

✖ **Network security groups.**
Network security groups filter traffic with rules. They do not identify a remote device.

EXAM STRATEGY
'Identifies the VPN appliance' plus an on-premises IP = local network gateway. Remove the virtual network gateway: it is the Azure side, not your device.

IMPACT
A wrong IP or address range in the local network gateway breaks the Site-to-Site tunnel and the link to on-premises. Gateway pairs are tested often.

What · Where · When · Why · How

What	A local network gateway is the Azure object that represents an on-premises VPN device.
Where	You create it in an Azure resource group and pair it with a virtual network gateway.
When	When you build a Site-to-Site VPN to an on-premises network.
Why	It stores the device's public IP (131.107.200.1) and the on-premises address ranges.
How	Create the local network gateway. Then create a connection to the virtual network gateway.

Exam tip
Local network gateway = on-premises side (your device's IP). Virtual network gateway = Azure side. A Site-to-Site connection needs both.

Memory trick
Local = Location of your Local box; Virtual = in the cloud.

Real-world example, use case and scenario

EXAMPLE
Setup
An admin creates lng-hq with IP 131.107.200.1 and range 10.10.0.0/16. Then he connects it to vgw-prod.

USE CASE
Branch office VPN
A group of animal clinics creates one local network gateway per clinic firewall. This connects each site to Azure.

SCENARIO
Wrong object
A junior engineer entered the office firewall IP into a second virtual network gateway, and the tunnel never worked. Creating a local network gateway for the firewall fixed it.

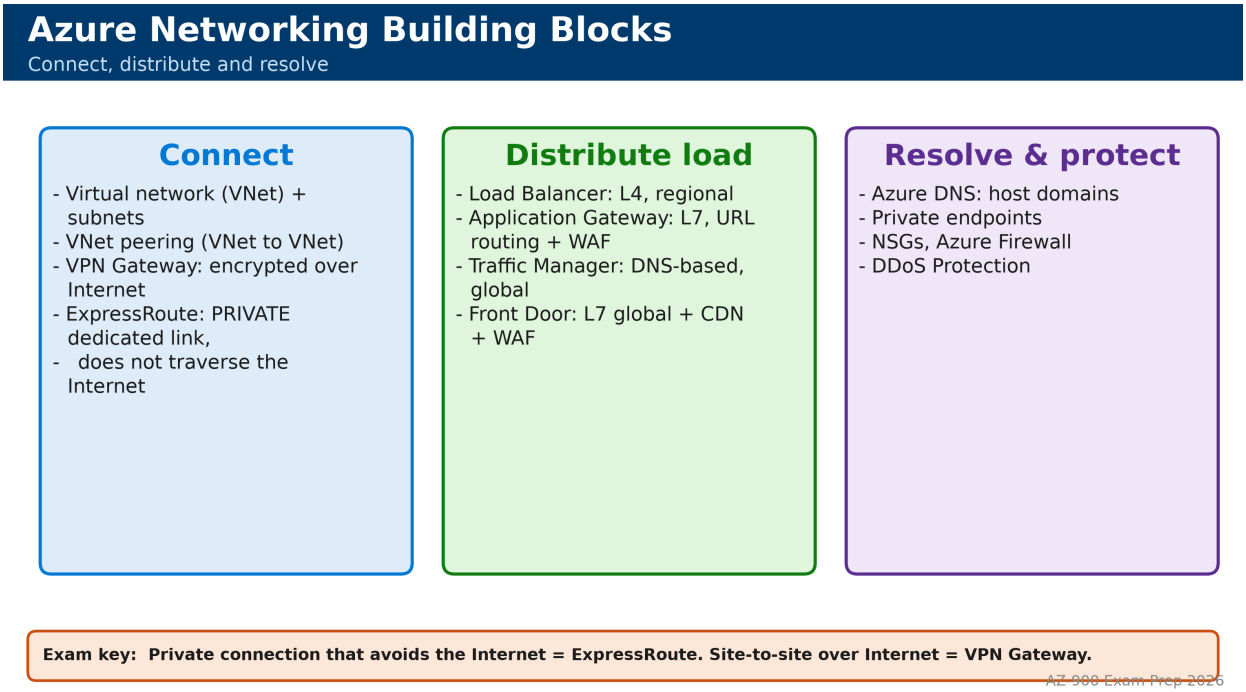


Figure Q249: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 250

AZURE ARCHITECTURE & SERVICES — **Governance**
v5 Q368, Q52

You have several virtual machines in an Azure subscription. You create a new subscription [the virtual machines cannot be moved to the new subscription].

✖ A. No change is needed.

✔ **B.** the virtual machines can be moved to the new subscription.

✖ C. the virtual machines can be moved to the new subscription only if they are all in the same resource group.

✖ D. the virtual machines can be moved to the new subscription only if they run Windows Server 2016.

Correct answer: the virtual machines can be moved to the new subscription.

Clarification (read this first): This is a 'fix the sentence' question. The part in brackets says VMs CANNOT be moved. That is wrong, so 'No change is needed' is wrong too. VMs CAN be moved to a new subscription, with no special conditions.

Answer explained

✔ **the virtual machines can be moved to the new subscription.**
Azure Resource Manager can move VMs, with their disks and network cards, between subscriptions in the same Entra tenant (organisation directory).

✖ **No change is needed.**
VMs can be moved between subscriptions, so 'cannot be moved' is false.

✖ **The virtual machines can be moved to the new subscription only if they are all in the same resource group.**
There is no rule that they must be in the same resource group. You can move VMs from several groups in separate steps.

✖ **The virtual machines can be moved to the new subscription only if they run Windows Server 2016.**
The operating system does not affect whether a VM can be moved.

EXAM STRATEGY
The bracket text 'cannot be moved' is false. Remove the conditions about the same resource group or OS version; they are made up. VMs can move between subscriptions.

IMPACT
Moving to another subscription lets you split billing or change owners without rebuilding VMs. This saves work when a company reorganises.

What · Where · When · Why · How

What	A resource move is an Azure Resource Manager action that moves resources between groups or subscriptions.
Where	Between subscriptions in the same Microsoft Entra tenant, with the portal, CLI or PowerShell.
When	When you reorganise billing or ownership, for example into a new subscription.
Why	VMs with their disks and network cards can be moved, so 'cannot be moved' is false.
How	Select the VM and the resources it needs. Choose Move to another subscription and validate.

Exam tip
Resources can move between resource groups and subscriptions. The resource group itself does not move. The region does not change with a move.

Memory trick
VMs are tenants who can change apartments (subscriptions) in the same building (tenant).

Real-world example, use case and scenario

EXAMPLE
Portal move
In the portal, an admin selects three VMs with their disks and network cards. He chooses Move to another subscription and validates the move.

USE CASE
Billing split
A holding company moves a smaller company's VMs into a new subscription. Now that company gets its own bill.

SCENARIO
Rebuild avoided
A gaming company planned to rebuild 30 VMs in a new production subscription. The architect moved them instead. The VMs stayed the same and nothing had to be reinstalled.

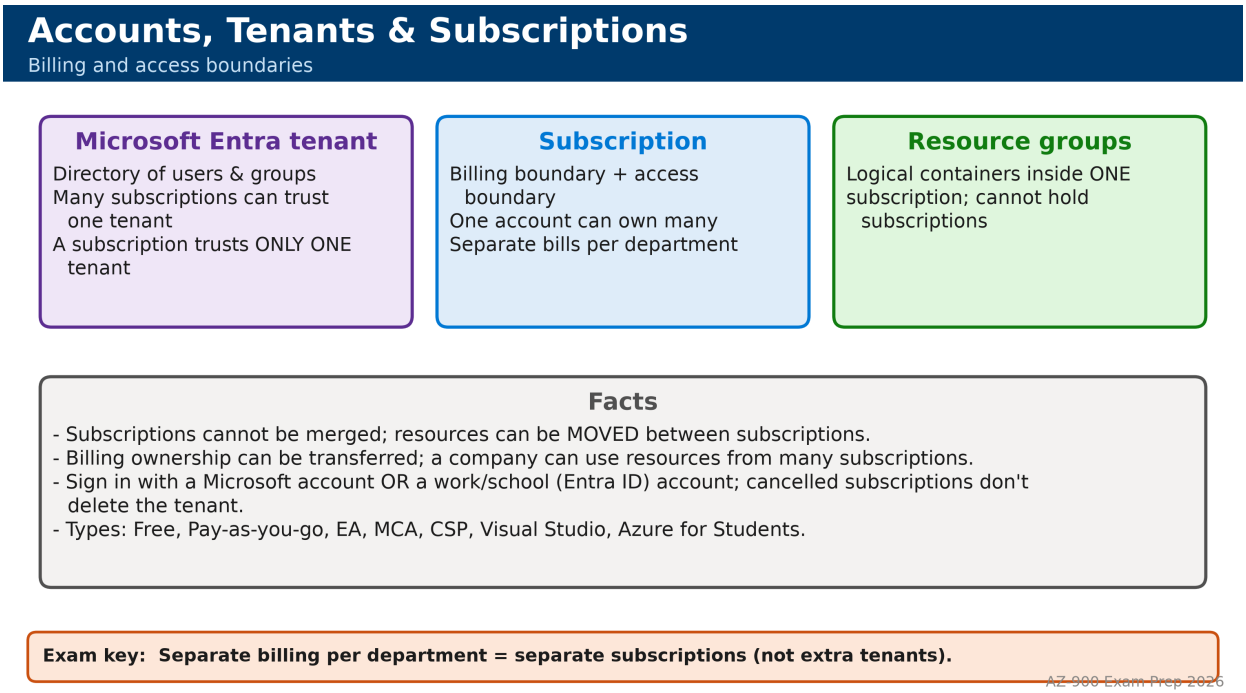


Figure Q250a: Accounts, tenants and subscriptions

About this figure

• Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.

• Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.

• Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.

• Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.

• Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

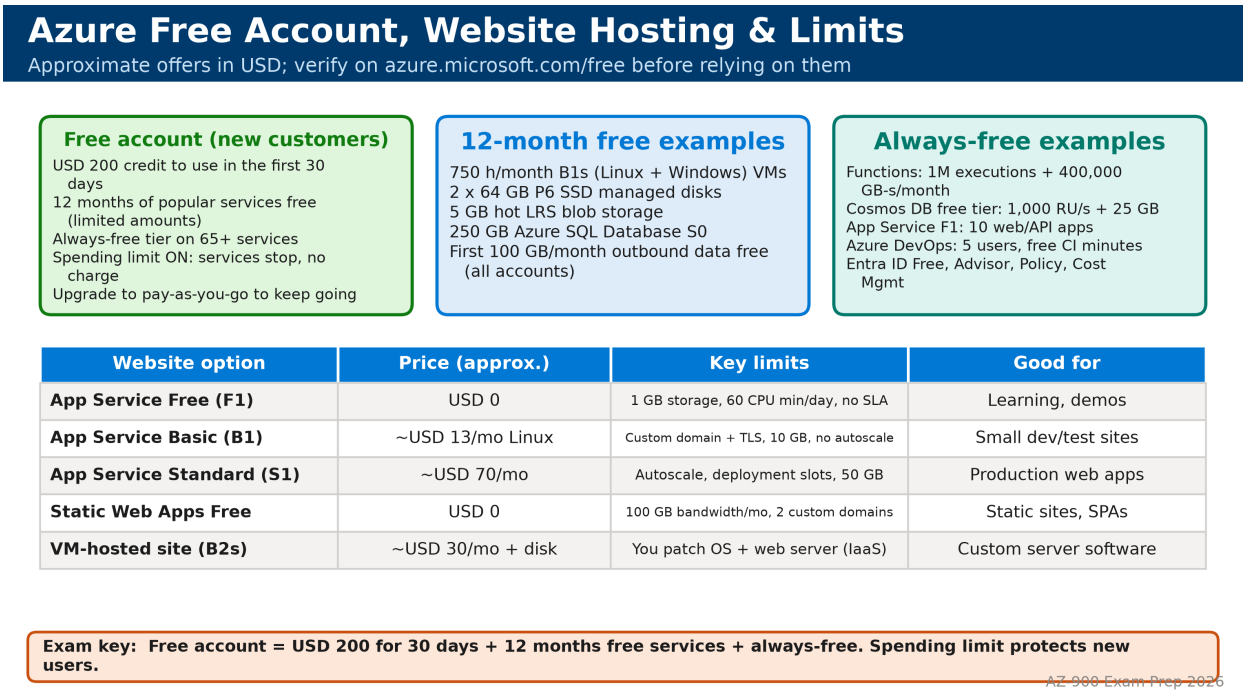


Figure Q250b: Free account, website hosting and limits

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- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 251

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q369

[...] is the process of verifying a user credentials.

✖ A. Authorization.

✔ **B.** Authentication.

✖ C. Federation.

✖ D. Ticketing.

Correct answer: Authentication.

Clarification (read this first): Two words look alike. AuthENTICATION = checking WHO you are (your credentials). AuthORIZATION = deciding WHAT you may do. 'Verifying credentials' = authentication.

Answer explained

✔ **Authentication.**
Authentication checks who a user is. It checks credentials such as passwords, MFA (a second sign-in check) or certificates.

✖ **Authorization.**
Authorization decides what a signed-in user may do, for example with RBAC (role-based access control).

✖ **Federation.**
Federation is a trust between identity providers. It is not the act of checking credentials.

✖ **Ticketing.**
Ticketing is a way, such as Kerberos tickets, to carry proof of an earlier sign-in.

EXAM STRATEGY
'Verifying credentials' = Authentication (who you are). Remove Authorization (what you can do), Federation (trusting another identity provider) and Ticketing (a Kerberos method).

IMPACT
Mixing up authentication and authorization leads to badly designed access control. This difference appears in almost every set of identity questions.

What · Where · When · Why · How

What	Authentication is the process of proving who a user is.
Where	Microsoft Entra ID does it at sign-in.
When	Each time a user or service signs in, before it can use resources.
Why	Checking credentials is the definition of authentication.
How	The user gives credentials, such as a password plus MFA, and Entra ID checks them.

Exam tip
AuthN = who are you (Entra ID sign-in, MFA).
AuthZ = what can you do (RBAC).

Memory trick
AuthN = Name check; AuthZ = Zone access.

Real-world example, use case and scenario

EXAMPLE
Sign-in
You enter a password and approve an Authenticator prompt. This is authentication with Entra ID.

USE CASE
Crew portal
A cruise company's staff portal checks crew with Entra ID (authentication). Then RBAC lets only officers see passenger lists (authorization).

SCENARIO
Right person, wrong rights
A clinic nurse signed in but could not open billing reports. IT explained that authentication worked. Her role (authorization) did not include billing, and this was on purpose.

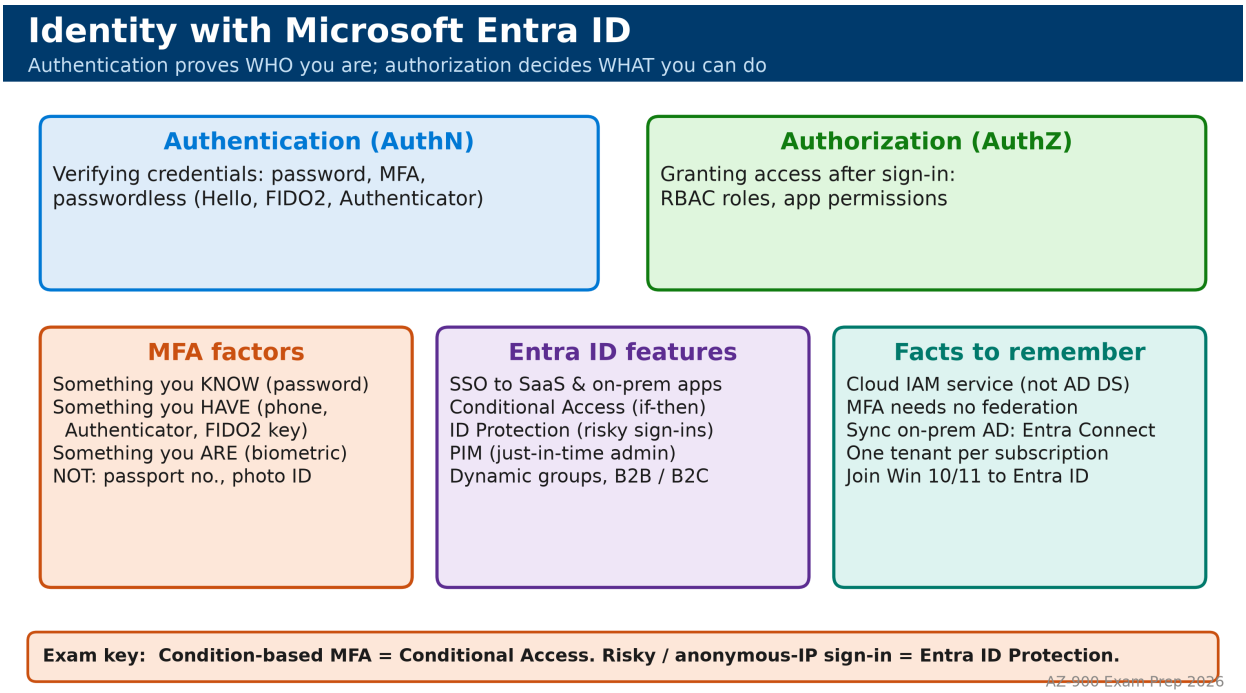


Figure Q251: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 252

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q370

If a resource group named RG1 has a delete lock, [...] can delete RG1.

- ✖ A. only a member of the global administrators group.
- ✔ **B.** the delete lock must be removed before an administrator.
- ✖ C. an Azure policy must be modified before an administrator.
- ✖ D. an Azure tag must be added before an administrator.

Correct answer: the delete lock must be removed before an administrator.

Clarification (read this first): A delete lock stops EVERYONE from deleting, even Owners and Global Administrators. So no person or role can delete RG1 while the lock is on. You must remove the lock first.

Answer explained

✔ **the delete lock must be removed before an administrator.**
A Delete (CanNotDelete) lock blocks deletion for everyone, including Owners. So the lock must be removed before an administrator can delete RG1.

✖ **Only a member of the global administrators group.**
Even Global Administrators and Owners are blocked while a delete lock exists.

✖ **An Azure policy must be modified before an administrator.**
Azure Policy is separate from locks. Changing a policy does not remove a lock.

✖ **An Azure tag must be added before an administrator.**
Tags are only labels. They do not affect locks or permissions.

EXAM STRATEGY
Locks are stronger than RBAC (role-based access control), even for Owner or Global Administrator. Pick 'remove the lock first'. Remove options about policies, tags or admin groups.

IMPACT
A delete lock stops someone from deleting production resource groups by mistake. But if you forget the lock, it also blocks planned cleanup and automatic teardown jobs.

What · Where · When · Why · How

What	A Delete (CanNotDelete) resource lock stops a resource or resource group from being deleted.
Where	You set it on a subscription, resource group or resource. Resources inside inherit it.
When	When important resources must be protected from deletion by mistake.
Why	Locks are stronger than RBAC, so the lock must be removed before anyone can delete RG1.
How	A user with lock permissions removes the lock. Then an administrator deletes RG1.

Exam tip
Locks beat everyone, even Owners. Remove the lock first, then delete. A ReadOnly lock also blocks changes.

Memory trick
A lock is a lock - nobody walks through, you unlock first.

Real-world example, use case and scenario

EXAMPLE
Blocked deletion
An Owner tries to delete rg-core and gets an error that it is locked. It works only after the Delete lock is removed.

USE CASE
Protecting shared services
A bank puts Delete locks on resource groups with ExpressRoute gateways and DNS. Now no one can delete them by mistake.

SCENARIO
Cleanup script fails
A university's cleanup script could not delete an old resource group. The admin found a Delete lock from an old project. He checked with the owner, removed the lock and then deleted the group.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q252: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 253

AZURE ARCHITECTURE & SERVICES — **Compute** v5 Q371

A Platform as a Service (PaaS) solution that hosts web apps in Azure can be provided with additional memory by changing the pricing tier.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
An App Service plan's pricing tier sets the CPU and memory. Moving to a higher tier scales up (gives a bigger machine) and gives the web app more memory.

✖ **No.**
Changing the App Service plan tier is exactly how a PaaS web app gets more memory.

EXAM STRATEGY
'Additional memory' plus 'pricing tier' = scaling up the App Service plan, so Yes. Scaling out would add more copies, not more memory per copy.

IMPACT
Scaling up the plan fixes memory problems without code changes or moving the app, often in minutes. This stops crashes when the app is busy.

What · Where · When · Why · How

What	Scaling up means getting more CPU and memory by moving to a larger App Service pricing tier.
Where	In the App Service plan's Scale up (pricing tier) setting.
When	When a web app needs more memory or CPU per instance (copy).
Why	The tier decides each instance's resources, so changing it adds memory.
How	Choose a higher tier, for example from B1 to P1v3, without redeploying.

Exam tip
Scale up = bigger tier (more CPU and RAM). Scale out = more instances. App Service supports both.

Memory trick
Up = bigger box; Out = more boxes.

Real-world example, use case and scenario

EXAMPLE
Tier change
Moving an App Service plan from B1 (1.75 GB) to P1v3 (8 GB) gives each instance more memory in a few clicks.

USE CASE
Memory-heavy API
A mapping company scales its map API up to a higher tier when large map caches need more RAM.

SCENARIO
Out-of-memory errors
An online exam platform crashed with out-of-memory errors during final exams. Scaling the plan up one tier fixed it in minutes, with no code changes.

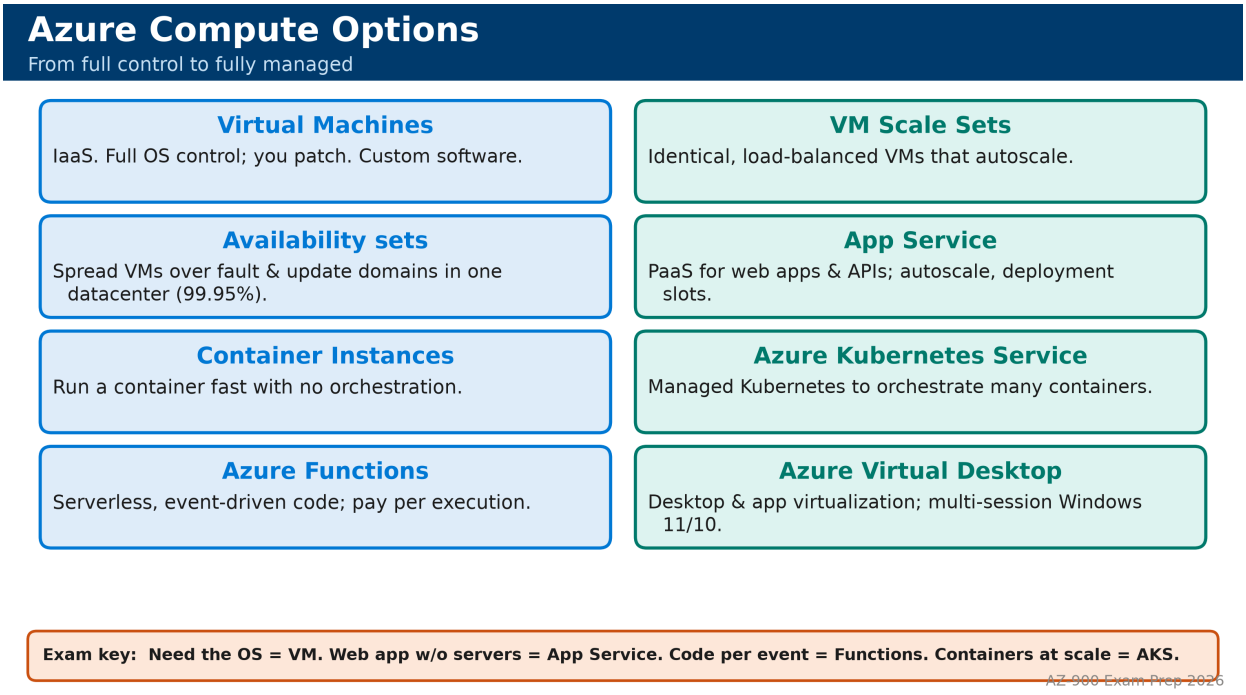


Figure Q253a: Azure compute options

About this figure

- Read left column then right: options run from full control (VMs) to fully managed or serverless (Functions); pick by how much you must manage.
- VM sizes range from tiny B-series (B1s approx. USD 8/month Linux) to large memory and GPU VMs; per-second billing applies while running.
- VM Scale Sets support up to 1,000 instances (600 with custom images); availability sets use up to 3 fault domains and 20 update domains.
- App Service plans set the price: Free F1 USD 0, Basic B1 approx. USD 13/month Linux, Standard S1 approx. USD 70/month with autoscale and slots.
- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

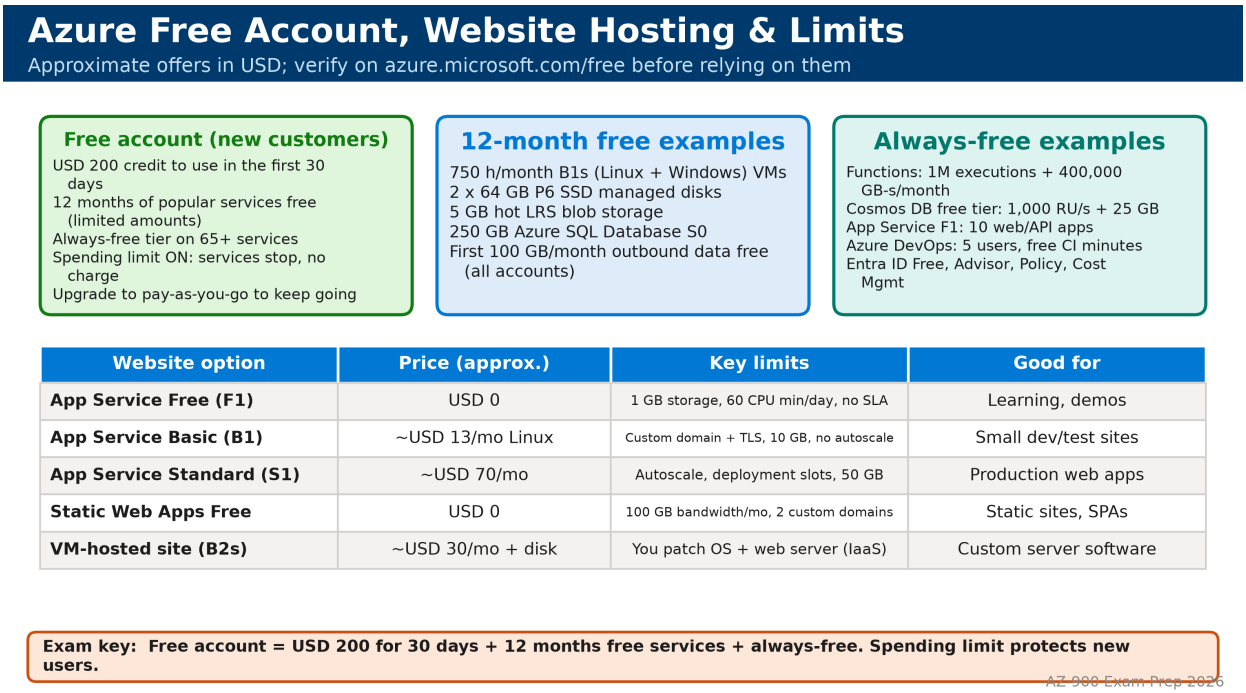


Figure Q253b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 254

AZURE ARCHITECTURE & SERVICES — **Networking**
v5 Q372, Q81, Q366

You plan to deploy 20 virtual machines to an Azure environment. To ensure that a virtual machine named VM1 cannot connect to the other virtual machines, VM1 must [...].

✔ **A.** be deployed to a separate virtual network.

✖ B. run a different operating system than the other virtual machines.

✖ C. be deployed to a separate resource group.

✖ D. have two network interfaces.

Correct answer: be deployed to a separate virtual network.

Clarification (read this first): The goal is that VM1 CANNOT talk to the other VMs. Only a separate virtual network creates a network wall. Resource groups are only folders for management; they do not block traffic.

Answer explained

✔ **be deployed to a separate virtual network.**
VNets (virtual networks) are separate by default. VMs in different VNets cannot talk without peering (linking VNets) or a gateway. So a separate VNet isolates VM1.

✖ **Run a different operating system than the other virtual machines.**
Windows and Linux VMs on the same VNet can talk freely. The OS does not separate them.

✖ **Be deployed to a separate resource group.**
Resource groups do not split networks. A VM in another resource group can use the same VNet.

✖ **Have two network interfaces.**
Adding a network card adds more connections. It does not remove them.

EXAM STRATEGY
Ask which option creates a network wall: only a separate virtual network. Resource groups, OS and the number of network cards do not stop traffic.

IMPACT
The same scenario also appears in a 'No change is needed' version. Learn it once and get both right. A separate VNet is the default way to split networks.

What · Where · When · Why · How

What	A virtual network is a separate, private network for Azure resources.
Where	It is created per subscription and region, with its own address range.
When	When one VM must not reach other VMs.
Why	Separate VNets cannot talk without peering or gateways.
How	Deploy VM1 to a new VNet. Do not peer it or connect it to the others.

Exam tip
The same question appears in a 'No change is needed' format. Both ways, the answer is a separate VNet.

Memory trick
New VNet, new moat around the castle.

Real-world example, use case and scenario

EXAMPLE
Separate address range
VM1 in vnet-quarantine (10.99.0.0/16) cannot reach the 19 VMs in vnet-app (10.1.0.0/16), because there is no peering.

USE CASE
Malware analysis
A cybersecurity company opens suspicious files on a VM in a separate VNet. The malware cannot spread to other systems.

SCENARIO
Supplier VM
A logistics company had to host a supplier's VM, but the VM must not reach internal servers. The company put it in its own VNet and shared only one API through Private Link service.

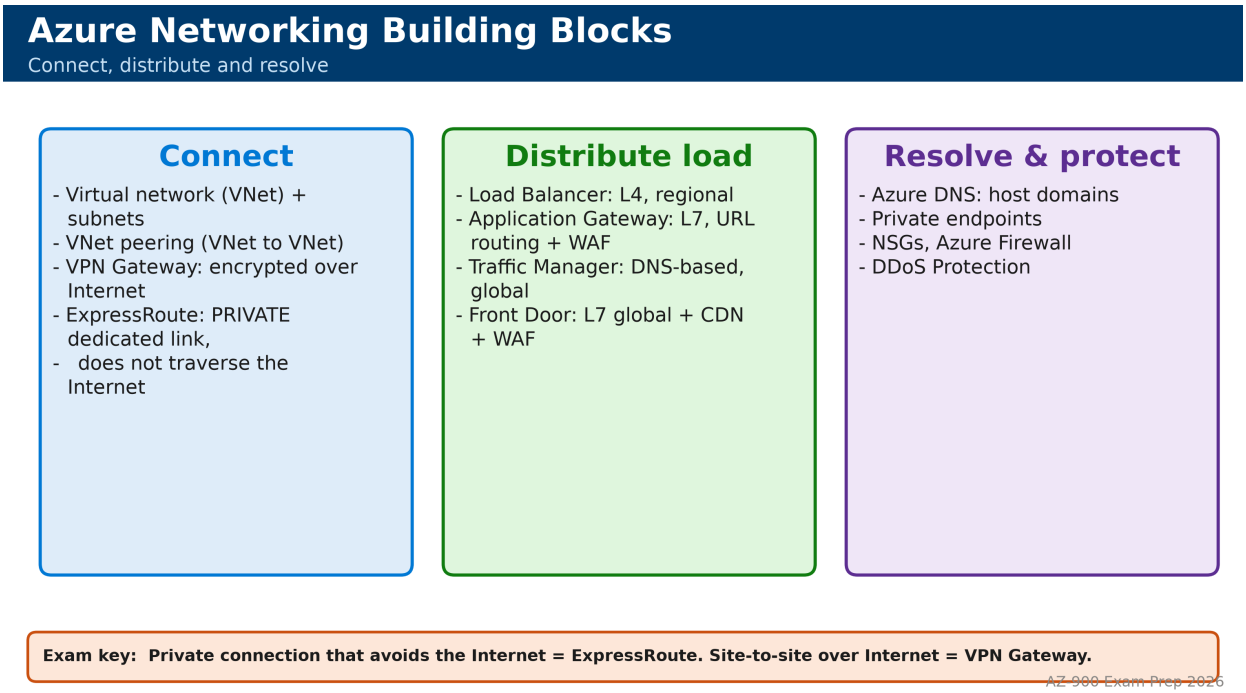


Figure Q254: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 255

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q373, Q237

From Azure Service Health, an administrator can view the health of all the services deployed to an Azure environment and all the other services available in Azure.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The statement has two parts: your own services AND all other Azure services. Service Health covers both, through its Azure status view. So Yes.

Answer explained

✔ **Yes.**
Azure Service Health includes Azure status for all services in all regions. It also shows personal service and resource health for your own deployments.

✖ **No.**
Azure status inside Service Health shows every Azure service in the world, together with a view of your own services.

EXAM STRATEGY
Service Health includes Azure status for all services, plus your own deployments, so Yes. The words 'all the other services' match the Azure status view.

IMPACT
During an outage, check Service Health first. It tells you if the problem is Microsoft's or your own settings, so you find the cause faster.

What · Where · When · Why · How

What	Azure Service Health reports the health of Azure services and of your resources.
Where	In the Azure portal. It combines Azure status, Service health and Resource health.
When	When you check if an outage is caused by Azure or by your own code.
Why	It shows your deployed services and the world status of all Azure services.
How	Open Service Health for your regions and Azure status for all services, and set alerts.

Exam tip
Azure status = all services in the world. Service Health = your services and regions. Resource Health = your single resources.

Memory trick
Status = the world, Service = my neighbourhood, Resource = my house.

Real-world example, use case and scenario

EXAMPLE
Three views
An admin checks Service Health for planned maintenance in West Europe. He checks Azure status for a world-wide DNS problem.

USE CASE
Ops alerting
A payment company sets Service Health alerts on the regions it uses. On-call engineers get a call when there is a problem.

SCENARIO
Is it us or Azure?
A sports streaming site went down during a match. Service Health showed a network problem in its region. So the team switched to the second region instead of checking its own code.

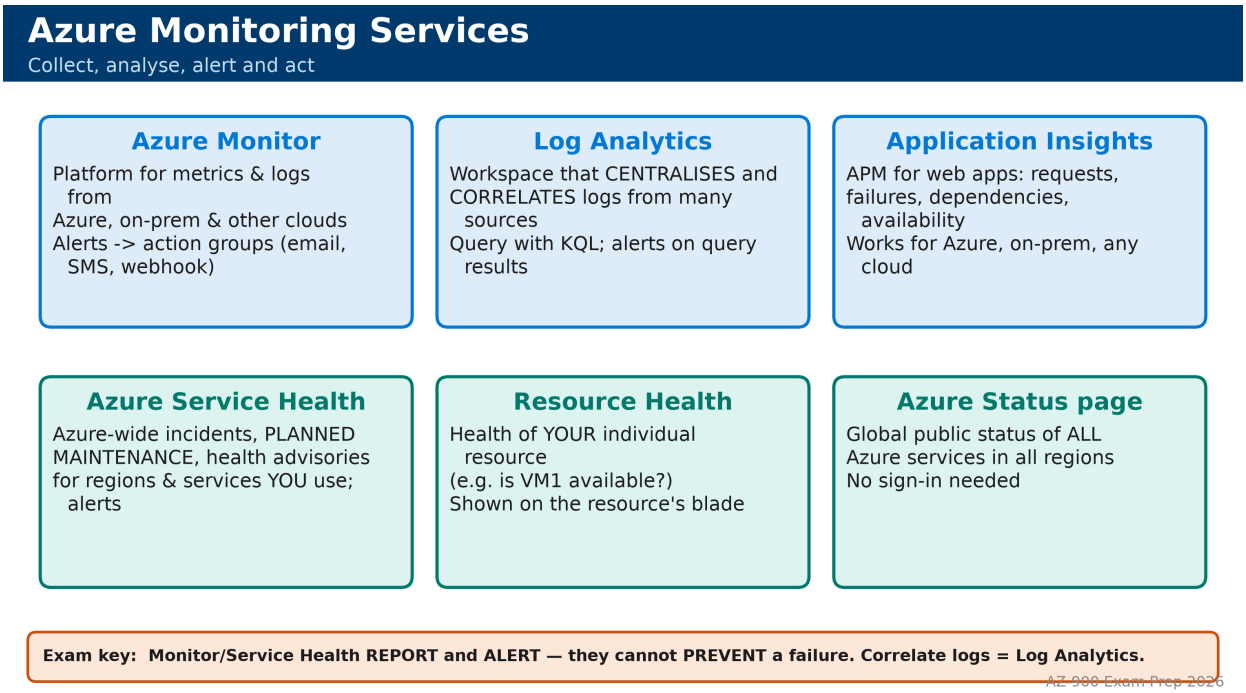


Figure Q255: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 256

AZURE ARCHITECTURE & SERVICES

Security
v5 Q374

Match the Azure Services service to the correct descriptions.

✖ A. Analyze security log files from Azure virtual machines: Azure Security Center (now should be Microsoft Defender for Cloud). Display the secure score for an Azure subscription: Azure Lighthouse. Store passwords for use by Azure Function applications: Azure Key Vault.

✖ B. Analyze security log files from Azure virtual machines: Azure Sentinel. Display the secure score for an Azure subscription: Azure Lighthouse. Store passwords for use by Azure Function applications: Azure Key Vault.

✔ **C.** Analyze security log files from Azure virtual machines: Azure Sentinel. Display the secure score for an Azure subscription: Azure Security Center (now should be Microsoft Defender for Cloud). Store passwords for use by Azure Function applications: Azure Key Vault.

✖ D. Analyze security log files from Azure virtual machines: Azure Sentinel. Display the secure score for an Azure subscription: Azure Security Center (now should be Microsoft Defender for Cloud). Store passwords for use by Azure Function applications: Azure Active Directory (Azure AD) (now should be Microsoft Entra ID).

Correct answer: Analyze security log files from Azure virtual machines: Azure Sentinel. Display the secure score for an Azure subscription: Azure Security Center (now should be Microsoft Defender for Cloud). Store passwords for use by Azure Function applications: Azure Key Vault.

Clarification (read this first): Match three tasks to three services. Old names appear in the options: Azure Sentinel is now Microsoft Sentinel, Azure Security Center is now Microsoft Defender for Cloud, Azure AD is now Microsoft Entra ID. Secure score = Defender for Cloud, not Lighthouse.

2026 update: Azure Sentinel is now Microsoft Sentinel. Azure Security Center is now Microsoft Defender for Cloud. Azure AD is now Microsoft Entra ID.

Answer explained

✔ **Analyze security log files from Azure virtual machines: Azure Sentinel. Display the secure score for an Azure subscription: Azure Security Center (now should be Microsoft Defender for Cloud). Store passwords for use by Azure Function applications: Azure Key Vault.**
Microsoft Sentinel (SIEM - a tool that collects and studies security logs) analyses security logs. Defender for Cloud (formerly Security Center) shows the secure score. Key Vault stores secrets for Functions.

✖ **Analyze security log files from Azure virtual machines: Azure Security Center (now should be Microsoft Defender for Cloud). Display the secure score for an Azure subscription: Azure Lighthouse. Store passwords for use by Azure Function applications: Azure Key Vault.**
Studying security logs from many sources is Sentinel's job. Lighthouse manages many customer tenants and has no secure score.

✖ **Analyze security log files from Azure virtual machines: Azure Sentinel. Display the secure score for an Azure subscription: Azure Lighthouse. Store passwords for use by Azure Function applications: Azure Key Vault.**
Azure Lighthouse lets service providers manage many customer tenants. It does not show a secure score.

✖ **Analyze security log files from Azure virtual machines: Azure Sentinel. Display the secure score for an Azure subscription: Azure Security Center (now should be Microsoft Defender for Cloud). Store passwords for use by Azure Function applications: Azure Active Directory (Azure AD) (now should be Microsoft Entra ID).**
Entra ID stores identities, not app secrets. Functions read passwords from Key Vault.

EXAM STRATEGY
Check one item at a time. 'Secure score' removes every option with Lighthouse. 'Store passwords' removes Entra ID. Only Sentinel, Defender for Cloud and Key Vault is left.

IMPACT
Passwords written directly in Function app settings, instead of Key Vault, are a top cause of breaches. Security matching questions are common in AZ-900.

What · Where · When · Why · How

What	Sentinel is a SIEM (security log analysis tool). Defender for Cloud manages security posture. Key Vault stores secrets.
Where	All three are Azure services in the portal. Function app settings point to Key Vault.
When	When you study security logs, measure security posture and protect app passwords.
Why	Each service matches one task: logs, secure score and secret storage.
How	Send VM logs to Sentinel, view the secure score in Defender for Cloud, and use Key Vault references.

Exam tip
Logs = Sentinel (SIEM). Secure score = Defender for Cloud. Secrets = Key Vault. Lighthouse = service providers managing many tenants.

Memory trick
Sentinel watches, Defender scores, Key Vault keeps.

Real-world example, use case and scenario

EXAMPLE
Function secret
An Azure Function uses a managed identity and a Key Vault reference. It reads its database password while it runs.

USE CASE
Security operations
A utility company sends VM security events to Microsoft Sentinel. Rules find people trying many passwords to break in.

SCENARIO
Board number
A retailer's security chief needed one number for the board. She reported the Defender for Cloud secure score every month. The security team handled incidents in Sentinel.

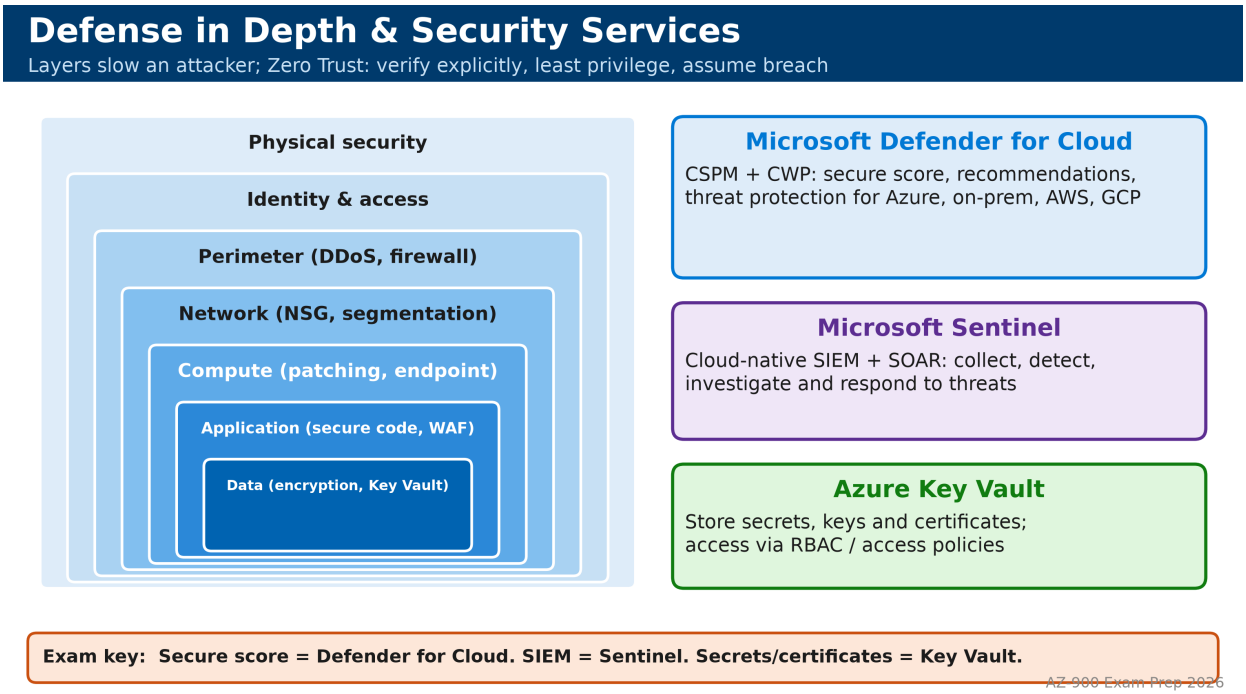


Figure Q256: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 257

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q375, Q388

[Azure policies provide] a common platform for deploying objects to a cloud infrastructure and for implementing consistency across the Azure environment.

- ✖ A. No change is needed.
- ✖ B. Resource groups provide.
- ✔ **C.** Azure Resource Manager provides.
- ✖ D. Management groups provide.

Correct answer: **Azure Resource Manager provides.**

Clarification (read this first): This is a 'fix the sentence' question. Azure Policy ENFORCES rules; it does not deploy. The 'common platform for deploying' is Azure Resource Manager (ARM), so the bracket text must change.

Answer explained

✔ **Azure Resource Manager provides.**
Azure Resource Manager (ARM) is the deployment and management layer. Every request goes through it. It gives the same deployment, templates, RBAC, tags and locks everywhere.

✖ **No change is needed.**
Azure Policy enforces rules on resources, but it does not deploy them.

✖ **Resource groups provide.**
Resource groups are containers for related resources. They are not a deployment platform.

✖ **Management groups provide.**
Management groups organise subscriptions for policy and RBAC (role-based access control). They do not deploy resources.

EXAM STRATEGY
'Common platform for deploying' = Azure Resource Manager. Remove Policy (enforces rules), resource groups (containers) and management groups (organise subscriptions).

IMPACT
Every portal, CLI and API request goes through ARM. So RBAC, tags and locks work the same, whatever tool an admin uses.

What · Where · When · Why · How

What	Azure Resource Manager is the deployment and management service for Azure.
Where	It is the control layer behind the portal, CLI, PowerShell, REST APIs and templates.
When	When you deploy resources the same way across environments.
Why	Every deployment goes through ARM, so it gives one common, consistent platform.
How	Send ARM or Bicep templates. ARM checks who you are, checks your rights, and deploys the resources.

Exam tip
Common platform for deploying = ARM. Enforce rules = Policy. Organise subscriptions = management groups.

Memory trick
ARM is the arm that reaches into Azure for every deployment.

Real-world example, use case and scenario

EXAMPLE
Same API
You create a storage account in the portal or with az CLI. Both send requests to the same ARM endpoint.

USE CASE
Same environments
A restaurant chain deploys the same environment in each country from one Bicep template. ARM processes the template.

SCENARIO
Different setups
Two teams at a telecom company built networks by hand, and the networks did not match. When they used ARM or Bicep templates, every deployment got the same settings.

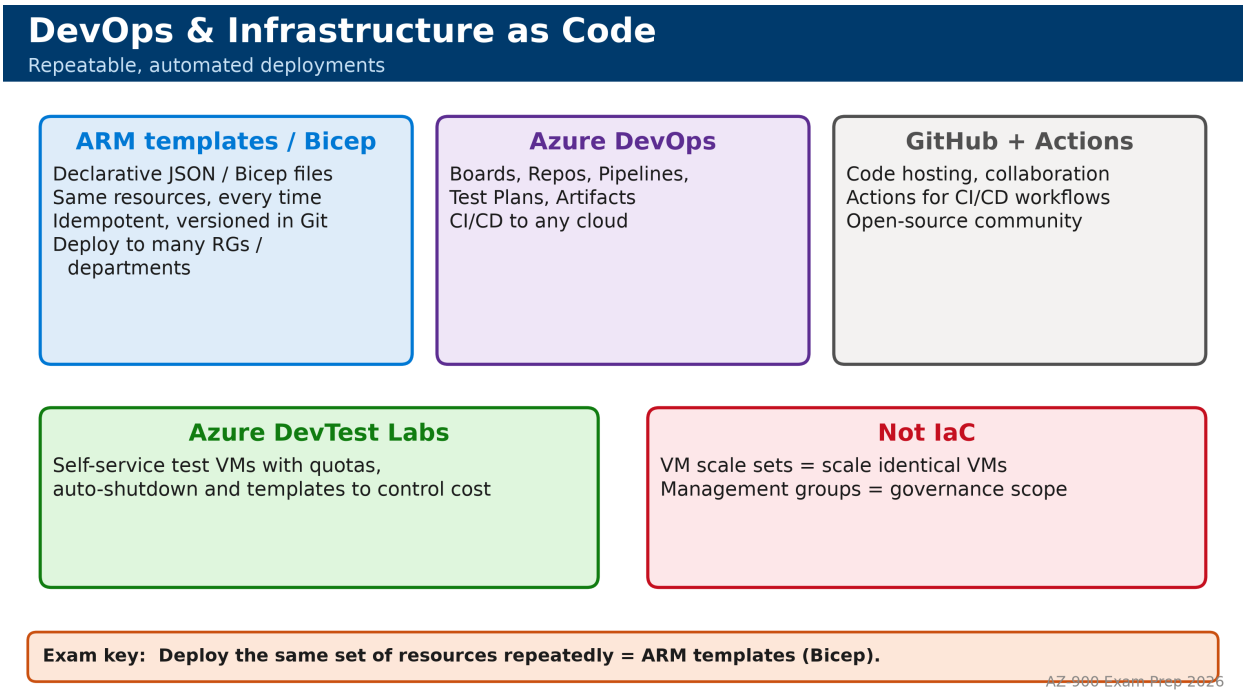


Figure Q257: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 258

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q376, Q34, Q61, Q210

If you set permissions to a resource group, all the Azure resources in that resource group inherit the permissions.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Azure RBAC (role-based access control) role assignments pass down to lower levels. So a role on a resource group applies to every resource in it, also new ones.

✖ **No.**
Role assignments on a resource group pass down to all its resources. Tags are the thing that does not pass down by default.

EXAM STRATEGY
RBAC passes down: management group, then subscription, then resource group, then resource. So Yes. Only tags do not pass down by default.

IMPACT
A role on a resource group quietly gives access to every future resource in it. Too-wide access is a common mistake with 'least privilege' (give only needed rights).

What · Where · When · Why · How

What	Azure RBAC gives access through role assignments at a scope (level).
Where	Scopes: management group, subscription, resource group, resource.
When	When one team needs the same access to all resources in a group.
Why	Role assignments pass down, so resources get the group's permissions.
How	Assign a role, such as Contributor, on the resource group. All its resources inherit it.

Exam tip
RBAC, Policy and locks pass down. Tags do not pass down by default.

Memory trick
Permissions flow downhill like water.

Real-world example, use case and scenario

EXAMPLE
Reader on a resource group
Giving Reader on rg-finance lets an auditor see every VM, database and storage account in that group.

USE CASE
Team ownership
A real estate platform gives the search team Contributor on rg-search. So the team manages all its resources.

SCENARIO
Too wide
A medicine company gave a contractor Contributor on a large resource group. Later it saw the contractor could change a production database there. It moved the contractor's resources to a separate group to limit the access.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

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Figure Q258: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 259

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q377

Your company has a Software Assurance agreement that includes Microsoft SQL Server licenses. You plan to deploy SQL Server on Azure virtual machines. What should you do to minimize licensing costs for the deployment?

✖ A. Use Azure Reservations.

✔ **B.** Use Azure Hybrid Benefit.

✖ C. Deallocate the virtual machines during off hours.

✖ D. Configure Azure Cost Management budgets.

Correct answer: Use Azure Hybrid Benefit.

Clarification (read this first): The key words are 'Software Assurance' and 'SQL Server licenses' you already own. Reusing licences you own = Azure Hybrid Benefit. Reservations lower the compute price, not the licence price.

Answer explained

✔ **Use Azure Hybrid Benefit.**
Azure Hybrid Benefit lets you reuse SQL Server licences that have Software Assurance on Azure VMs. This removes the SQL licence charge from the VM price.

✖ **Use Azure Reservations.**
Reservations lower the compute price for a 1- or 3-year commitment. They do not reuse SQL licences you already own.

✖ **Deallocate the virtual machines during off hours.**
Deallocating (stopping and releasing) VMs cuts compute hours. But while they run, you still pay the price that includes the licence. Your own licences stay unused.

✖ **Configure Azure Cost Management budgets.**
Budgets only track spending and send alerts. They do not lower any price.

EXAM STRATEGY
'Software Assurance' or 'existing licences' = Azure Hybrid Benefit. Remove Reservations (compute discount, not licence) and budgets (tracking only, no savings).

IMPACT
Hybrid Benefit removes the SQL Server licence charge from the VM price, often the biggest cost. This is one of the most valuable cost questions.

What · Where · When · Why · How

What	Azure Hybrid Benefit lets you reuse Windows Server and SQL Server licences that have Software Assurance in Azure.
Where	You set it in the VM's licensing settings in the portal, or when you create the VM.
When	When you already own SQL Server or Windows Server licences with Software Assurance.
Why	It removes the licence part of the VM price, so you pay only for compute.
How	Turn on Hybrid Benefit on each SQL VM that qualifies, and confirm you have the right licences.

Exam tip
The key words 'Software Assurance' or 'existing licences' always point to Azure Hybrid Benefit, not Reservations.

Memory trick
Bring Your Own Licence = Hybrid Benefit: the licence you already paid for crosses the bridge to Azure.

Real-world example, use case and scenario

EXAMPLE
Enterprise SQL VM
A SQL Server Enterprise VM with 16 vCPUs is switched to Hybrid Benefit in the portal. The hourly price drops to the plain Windows VM price, with no SQL Server licence fee.

USE CASE
Insurance migration
An insurer with an Enterprise Agreement and Software Assurance moves 60 SQL Server instances to Azure VMs. It turns on Hybrid Benefit for all of them, so it does not pay for licences twice.

SCENARIO
Using both discounts
A retailer's cloud lead first suggested Reservations for its SQL VMs. The architect explained that Reservations cover only compute, and added Hybrid Benefit for the licence part. Using both cut the VM bill by more than half.

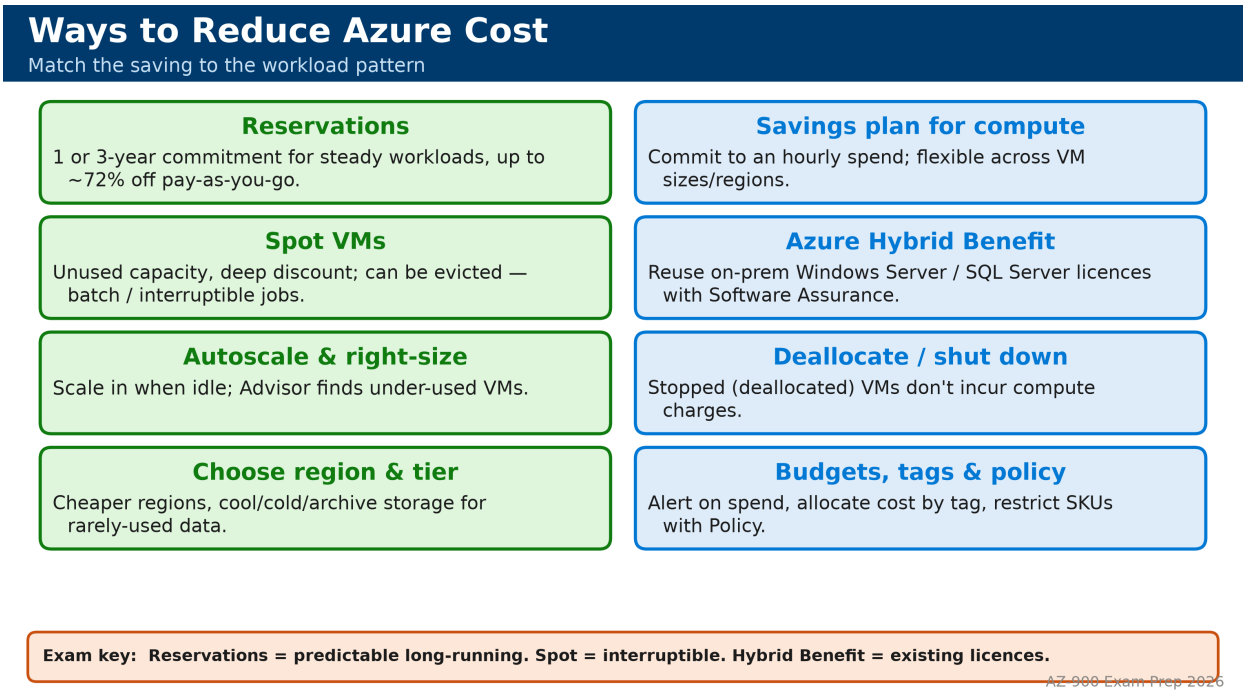


Figure Q259a: Ways to reduce cost

About this figure

• Each box matches a saving to a workload pattern: steady (reservations), flexible (savings plan), interruptible (Spot), existing licences (Hybrid Benefit).

• Reservations save up to approx. 72% versus pay-as-you-go; savings plans for compute save up to approx. 65% and apply across VM families and regions.

• Spot VMs can be up to approx. 90% cheaper, but Azure can evict them with 30 seconds' notice when it needs capacity back.

• Combining Azure Hybrid Benefit with reservations can cut Windows VM costs by up to approx. 80%; Linux subscriptions such as RHEL and SUSE also qualify.

• Example: moving rarely read data from Hot (~USD 0.02/GB-month) to Archive (~USD 0.002 or less) saves storage cost, but reads and early deletion cost more.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q259b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 260

AZURE ARCHITECTURE & SERVICES

Data & AI
v5 Q382, Q4, Q35, Q166

Your company plans to deploy an Artificial Intelligence (AI) solution in Azure. What should the company use to build, test, and deploy predictive analytics solutions?

✖ A. Azure Logic Apps.

✔ **B.** Azure Machine Learning Designer.

✖ C. Azure Batch.

✖ D. Azure Cosmos DB.

Correct answer: Azure Machine Learning Designer.

Clarification (read this first): The key words are 'predictive analytics' (using data to predict the future). That means machine learning. The service is now called Azure Machine Learning designer, inside Azure Machine Learning studio.

2026 update: Today this is the designer (drag-and-drop) inside Azure Machine Learning studio. Azure Machine Learning as a whole (designer, automated ML, notebooks) is the platform to build, train and deploy predictive models.

Answer explained

✔ **Azure Machine Learning Designer.**
Azure Machine Learning designer is a drag-and-drop canvas. You use it to build, train, test and deploy predictive models as endpoints.

✖ **Azure Logic Apps.**
Logic Apps automates workflows between services. It cannot train or deploy predictive models.

✖ **Azure Batch.**
Batch runs large compute jobs in parallel. It has no tools to build, test or deploy ML models.

✖ **Azure Cosmos DB.**
Cosmos DB is a NoSQL database that stores app data. It does not build models.

EXAM STRATEGY
'Build, test, deploy predictive' = Azure Machine Learning designer. Remove Logic Apps (workflow), Batch (large compute jobs) and Cosmos DB (database), because they are not ML services.

IMPACT
Choosing the right ML platform saves you from building your own training systems. AI questions come up often in the Architecture domain.

What · Where · When · Why · How

What	Azure Machine Learning designer is a drag-and-drop tool to build ML (machine learning) pipelines.
Where	Inside Azure Machine Learning studio (ml.azure.com), in a workspace.
When	When you need to build, test and deploy predictive models with little code.
Why	It covers the full cycle for predictive analytics: train, test and deploy.
How	Drag data and algorithm blocks onto the canvas. Train, test, then deploy an endpoint.

Exam tip
'Build, train, test, deploy predictive models' = Azure Machine Learning. Ready-made AI APIs = Azure AI services.

Memory trick
Predict needs a lab: Machine Learning is the lab, designer is the drag-and-drop bench.

Real-world example, use case and scenario

EXAMPLE
Churn pipeline
An analyst drags blocks onto the designer canvas: dataset, Split Data, Two-Class Boosted Decision Tree and Evaluate Model. Then she deploys the best model as a real-time endpoint.

USE CASE
Electricity forecast
An electricity company uses Azure Machine Learning to predict next-day demand from weather and past usage data.

SCENARIO
Workflow vs model
A marketing team first tried to predict which customers would leave with a Logic App and if-rules. Results were poor because nothing was learned. With Azure Machine Learning designer, the team had a trained model in a week.

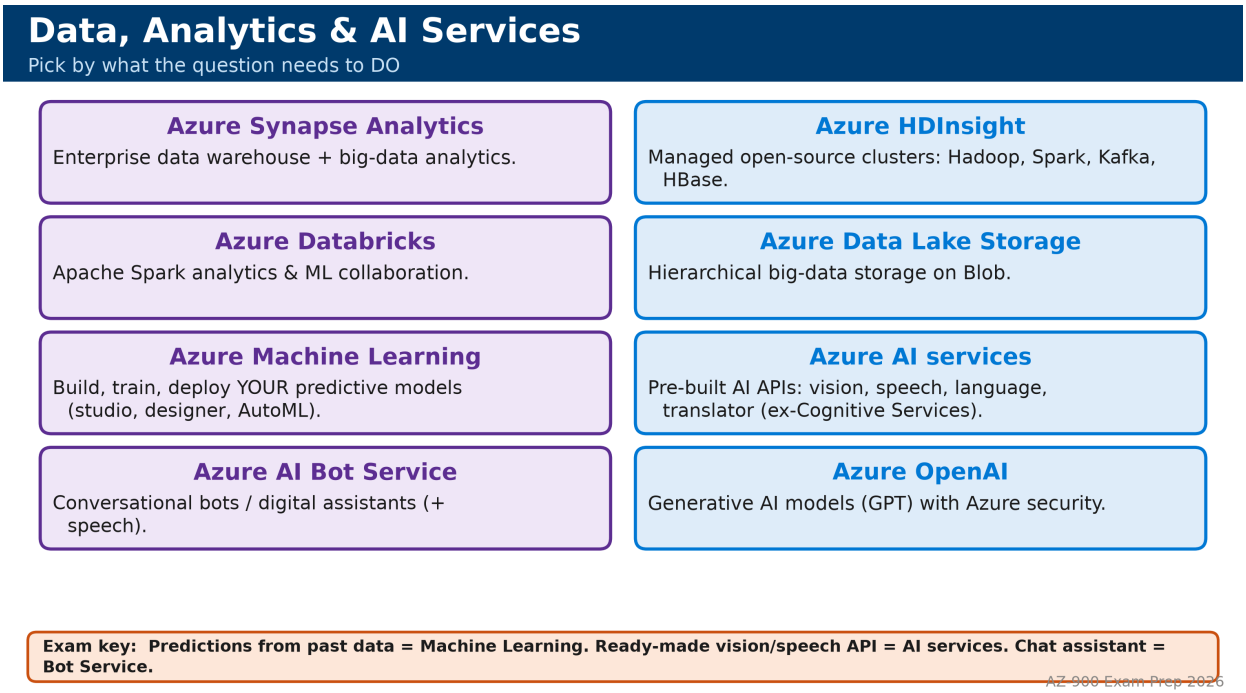


Figure Q260: Data, analytics and AI services

About this figure

- Left column covers data and analytics platforms; right column covers AI. Decide whether the scenario needs storage, processing, custom models or ready-made AI.
- Azure Machine Learning trains your own models from your data; Azure AI services are prebuilt APIs needing no data science, billed per transaction.
- Azure AI services include Vision, Speech, Language, Translator and Document Intelligence; many offer a free F0 tier with limited monthly calls.
- Azure OpenAI models are now offered through Microsoft Foundry (formerly Azure AI Foundry), billed per 1,000 or million tokens, varying widely by model.
- Microsoft Fabric is the newer unified analytics SaaS combining data engineering, warehousing and Power BI, and overlaps much of Synapse.

Question 261

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q385

Azure Multi-Factor Authentication (MFA) can be required for administrative and non administrative user accounts.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

2026 update: Azure MFA is now called Microsoft Entra multifactor authentication.

Answer explained

✔ **Yes.**
Entra MFA (multifactor authentication - a second sign-in check) can be required for any users, groups or roles. You can use Conditional Access, security defaults or per-user MFA. It works for admins and normal users.

✖ **No.**
Nothing limits MFA to one type of account. Policies can target administrators and normal users.

EXAM STRATEGY
Conditional Access can target any users, groups or roles. Admin and non-admin accounts are both covered, so Yes.

IMPACT
MFA blocks most attacks that use stolen passwords. If only admins use MFA, normal users become the easy way in.

What · Where · When · Why · How

What	Microsoft Entra MFA can be required for any user account.
Where	In Conditional Access, for all users, groups or directory roles.
When	When you want to protect both admin and normal user sign-ins.
Why	No account type is left out, so MFA can cover admins and non-admins.
How	Create Conditional Access policies for admin roles and for all users.

Exam tip
MFA can cover any user. Security defaults require MFA for admins at every sign-in, and for users when the sign-in looks risky.

Memory trick
Admin or intern, the second factor knocks for everyone.

Real-world example, use case and scenario

EXAMPLE
Two policies
One Conditional Access policy requires phishing-resistant MFA for Global Administrators. A second policy requires Authenticator MFA for all other users.

USE CASE
Law firm
A law firm requires MFA for every lawyer and assistant who opens case files from outside the office network.

SCENARIO
Phished clerk
A school district used MFA only for IT admins. Then a clerk's password was stolen by phishing, and payroll data was opened. The district added MFA for all staff. A new phishing attack the next month failed at the second check.

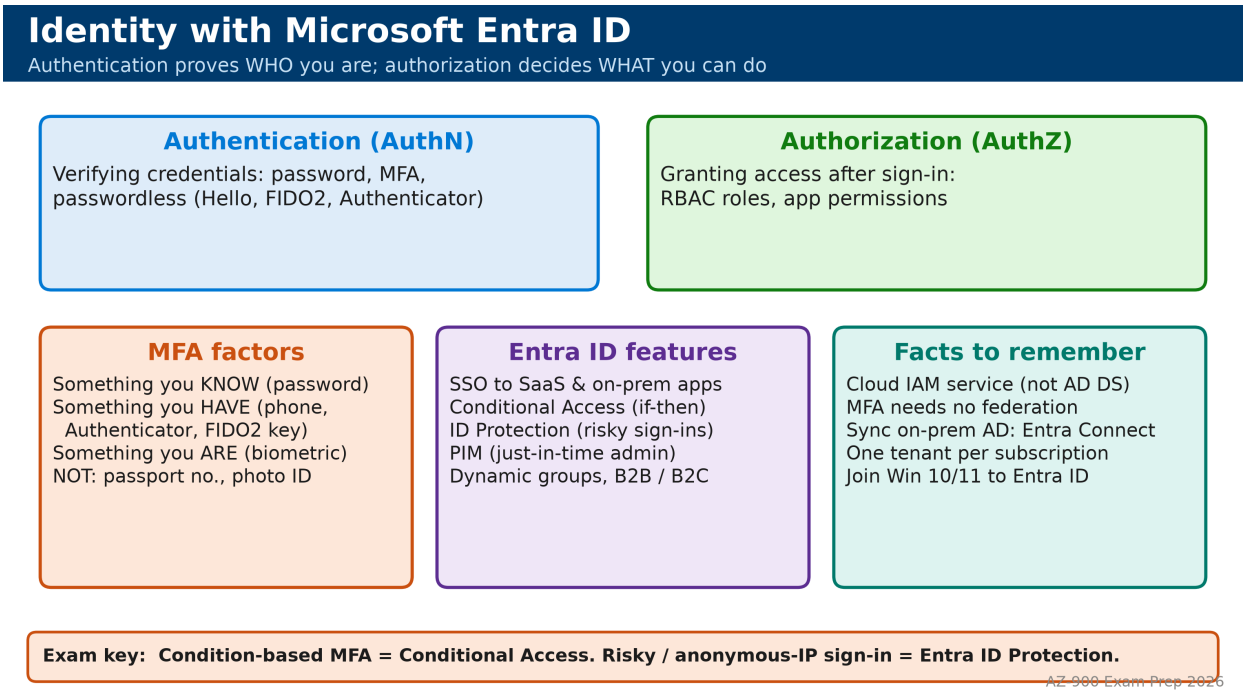


Figure Q261: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 262

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q387

You plan to deploy several Azure virtual machines. You need to ensure that the services running on the virtual machines are available if a single data center fails. Solution: You deploy the virtual machines to two or more resource groups. Does this meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is to survive a DATACENTER failure, which needs physical separation. Resource groups are only logical folders and do not choose the building. Q270 tries a scale set for the same goal - also No. Availability zones are the right answer for this goal.

Answer explained

✔ **No.**
Resource groups are logical containers (folders) for management. They do not control where resources physically run. So they give no protection if a datacenter fails.

✖ **Yes.**
VMs in different resource groups can still be in the same datacenter. Only zones or regions add physical separation.

EXAM STRATEGY
'Single data center fails' needs physical separation. Resource groups are logical containers with no control over location, so No.

IMPACT
High availability built on resource groups gives no real backup. It is an outage waiting to happen. This solution series appears four times in a row.

What · Where · When · Why · How

What	A resource group is a logical container for resources that share a lifetime.
Where	Inside a subscription. It sets the management scope, not the physical location.
When	When you organise resources for RBAC, tags, cost reports or deletion.
Why	It gives no physical separation, so it cannot protect against a datacenter failure.
How	Use availability zones (separate datacenters in one region) or several regions instead.

Exam tip
Datacenter failure = availability zones (or regions). Resource groups and scale sets alone are the No answers.

Memory trick
Folders are not fire walls: an RG is a folder.

Real-world example, use case and scenario

EXAMPLE
Same building possible
VM1 in rg-app-a and VM2 in rg-app-b are both in East US without zones. They may run in the same physical datacenter.

USE CASE
Correct use of resource groups
A software company uses separate resource groups for dev, test and prod. It can delete each one and set permissions on each one separately.

SCENARIO
False feeling of safety
A gaming company split its servers across two resource groups 'for backup'. A cooling failure in one datacenter took both offline. The company redeployed across availability zones 1 and 2 behind a zone-redundant load balancer.

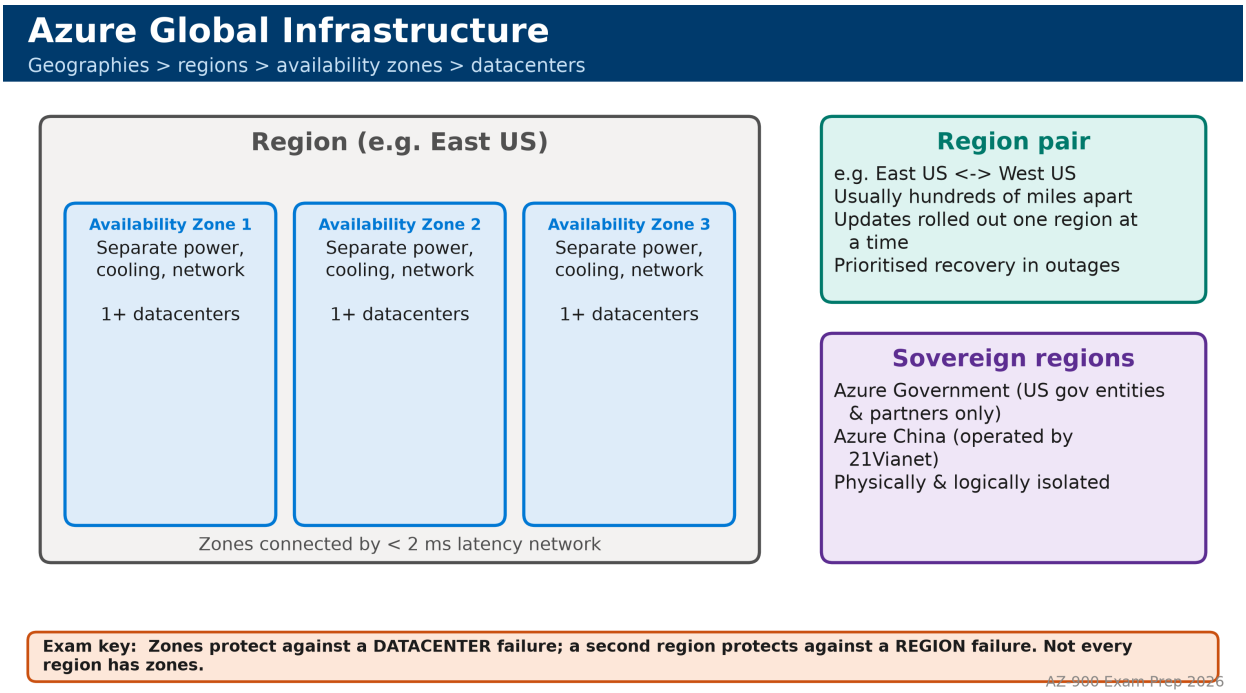


Figure Q262a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

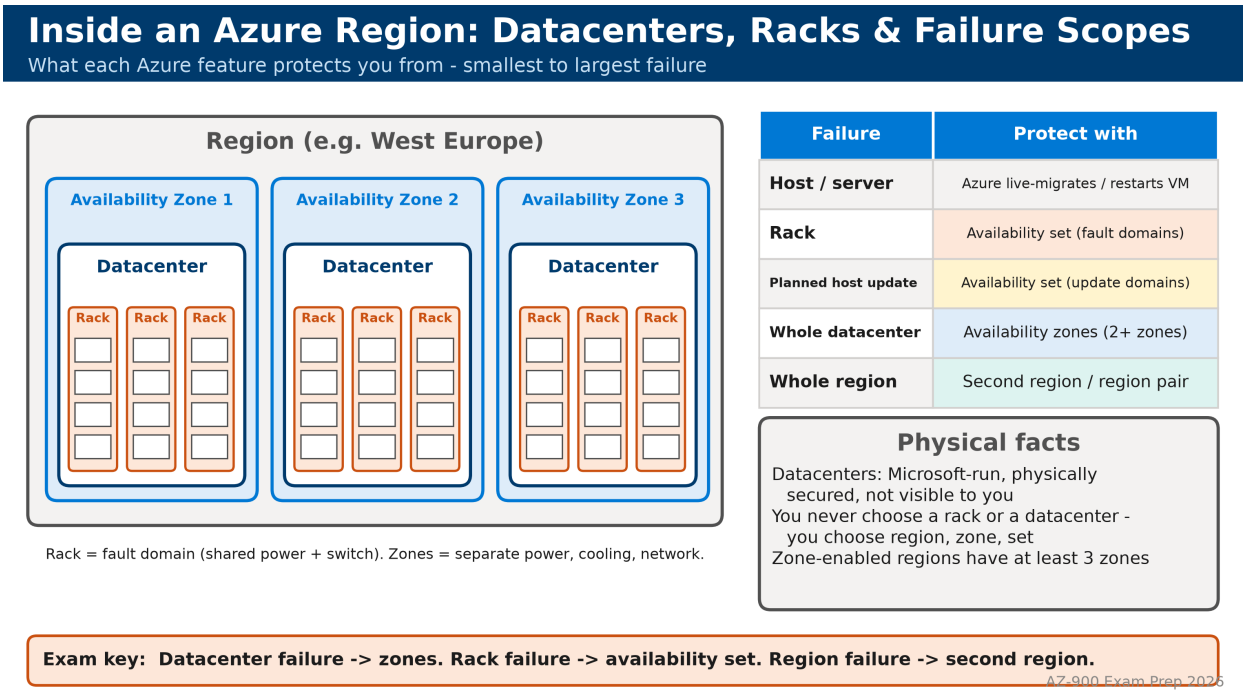


Figure Q262b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 263

CLOUD CONCEPTS

Service models
v5 Q389

Which cloud deployment solution is used for Azure virtual machines and Azure SQL databases?

✔ **A.** Infrastructure as a service (IaaS).

✔ **B.** Platform as a service (PaaS).

✖ C. Software as a service (SaaS).

Correct answer: Infrastructure as a service (IaaS).; Platform as a service (PaaS).

Clarification (read this first): This question has TWO answers, one per service. Azure VMs = IaaS (you manage the OS). Azure SQL Database = PaaS (Microsoft manages the OS and database engine). Careful: SQL Server installed on a VM would be IaaS.

Answer corrected in this edition. The previous PDF printed: *Infrastructure as a service (IaaS).*. In the original, two drop-downs were joined into one. Azure virtual machines are IaaS, but Azure SQL Database is PaaS. So printing only IaaS is wrong for the SQL part. Correct: VMs = IaaS, Azure SQL databases = PaaS.

Answer explained

✔ **Infrastructure as a service (IaaS).; Platform as a service (PaaS).**
Azure VMs are IaaS, because you manage the OS. Azure SQL Database is PaaS, because Microsoft manages the database engine, OS, patching and backups.

✖ **Software as a service (SaaS).**
Neither one is a finished app for end users. Both are building blocks that you set up. SaaS is a finished app, like Microsoft 365.

EXAM STRATEGY
Split the question. VMs: you manage the OS (IaaS). 'SQL databases' means Azure SQL Database (PaaS). Choose both, and remove SaaS.

IMPACT
The right service model shows who patches and backs up. A wrong split leaves SQL VMs unpatched, or doubles the work on managed databases.

What · Where · When · Why · How

What	Service models show who manages which layers: IaaS, PaaS or SaaS.
Where	In the shared responsibility model, Azure VMs are IaaS and Azure SQL Database is PaaS.
When	When you decide how much of the system you want to manage yourself.
Why	With VMs you manage the OS. Microsoft manages SQL Database.
How	Choose VMs for control of the OS. Choose SQL Database for a managed database engine.

Exam tip
SQL Server on an Azure VM = IaaS. Azure SQL Database and SQL Managed Instance = PaaS. Check if the question says 'SQL databases' or 'SQL Server on VMs'.

Memory trick
VM: you patch it (IaaS). SQL Database: Microsoft patches it (PaaS).

Real-world example, use case and scenario

EXAMPLE
Who patches
On a VM, the team plans Windows Update itself. On Azure SQL Database, patching happens automatically, and the customer does nothing.

USE CASE
Logistics app
A shipping company runs an old routing program on VMs (IaaS). It moves its order database to Azure SQL Database (PaaS) so it needs less database maintenance.

SCENARIO
Exam mistake
A candidate answered IaaS for both, because 'SQL' made him think of SQL Server on a VM. When he checked, he saw the question said Azure SQL databases, the managed service. He changed the second answer to PaaS.

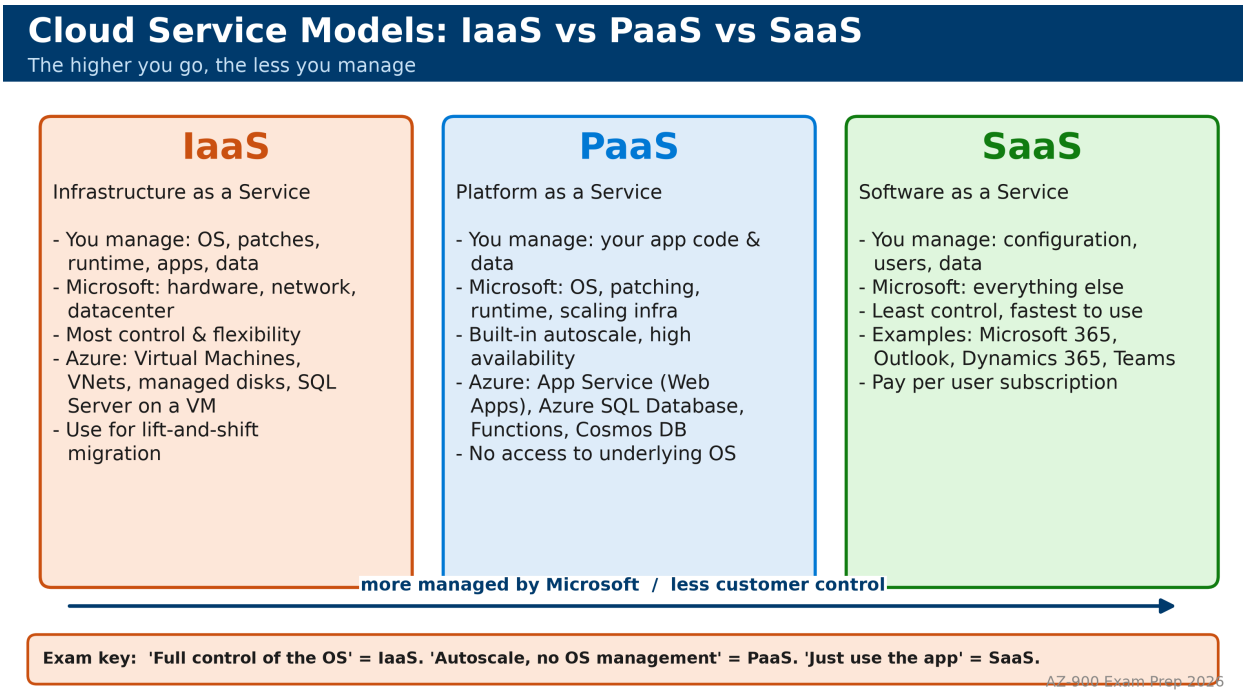


Figure Q263: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

• Serverless options such as Azure Functions are treated as PaaS-style: no OS access, automatic scaling and per-execution billing.

• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

Question 264

AZURE MANAGEMENT & GOVERNANCE

Cost

v5 Q390, Q21, Q379

When planning to migrate a public website to Azure, you must plan to [...].

✖ A. deploy a VPN.

✔ **B.** pay monthly usage costs.

✖ C. pay to transfer all the website data to Azure.

✖ D. reduce the number of connections to the website.

Correct answer: pay monthly usage costs.

Clarification (read this first): The word 'must' asks what is ALWAYS true. After moving to Azure, you will always pay monthly usage costs. A VPN is not needed for a public site, and uploading data into Azure is free (see Q219).

Answer explained

✔ **pay monthly usage costs.**
Azure charges for usage every month. So when you move a public website, you must plan a budget for these repeating usage costs.

✖ **Deploy a VPN.**
Public users reach the site over the internet. No VPN is needed.

✖ **Pay to transfer all the website data to Azure.**
Data coming into Azure (ingress) is free. Only egress (data leaving Azure) is charged.

✖ **Reduce the number of connections to the website.**
Azure services can scale. Moving the site does not mean you must limit connections.

EXAM STRATEGY
The only sure thing after moving is the monthly usage bill. Remove VPN (public site), data transfer fees (ingress is free) and reducing connections.

IMPACT
The same scenario appears in a 'No change is needed' version. Learn it once and get two questions right. It also reminds you to plan for ongoing costs.

What · Where · When · Why · How

What	Azure's pay-as-you-go model charges for used resources every month.
Where	On the subscription's bill in Cost Management + Billing.
When	When you plan the budget to move a website to Azure.
Why	The monthly usage cost is the one sure new cost. Ingress is free and no VPN is needed.
How	Estimate with the Pricing Calculator and set a Cost Management budget.

Exam tip
Same question as the 'No change is needed' version: the right idea is always monthly usage costs.

Memory trick
Moving in is free; the rent comes every month.

Real-world example, use case and scenario

EXAMPLE
Static Web Apps
A small company site moved to the Azure Static Web Apps Standard plan. It shows a fixed monthly charge per app, plus any extra data use.

USE CASE
Charity website
A charity moves its donation website to App Service. It sets a monthly Cost Management budget with alerts at 80 percent.

SCENARIO
VPN not needed
An IT manager planned money for a VPN gateway to 'connect' visitors to the moved site. The partner explained that visitors use the public address. The VPN cost was removed and the monthly App Service cost was added.

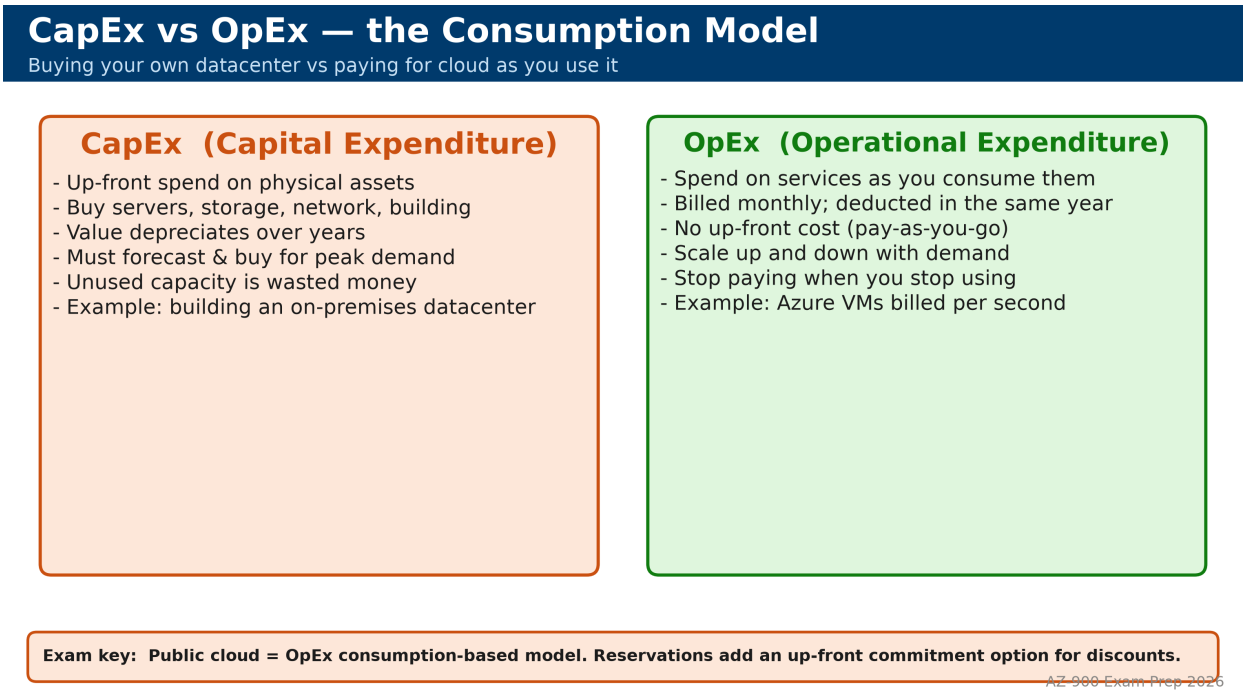


Figure Q264a: CapEx vs OpEx

About this figure

• Left box is buying assets up front (CapEx); right box is paying for consumption (OpEx). Cloud moves spending from left to right.

• CapEx assets are depreciated over several years; OpEx is expensed in the period it is incurred, which suits variable or uncertain demand.

• The consumption model means no up-front cost, pay only for what you use, and stop paying when resources are deleted or deallocated.

• Reservations and savings plans add a 1- or 3-year commitment for discounts, but they remain operational spending, not purchased hardware.

• Use the TCO calculator to compare on-premises CapEx (hardware, power, labour) with expected Azure OpEx before migrating.

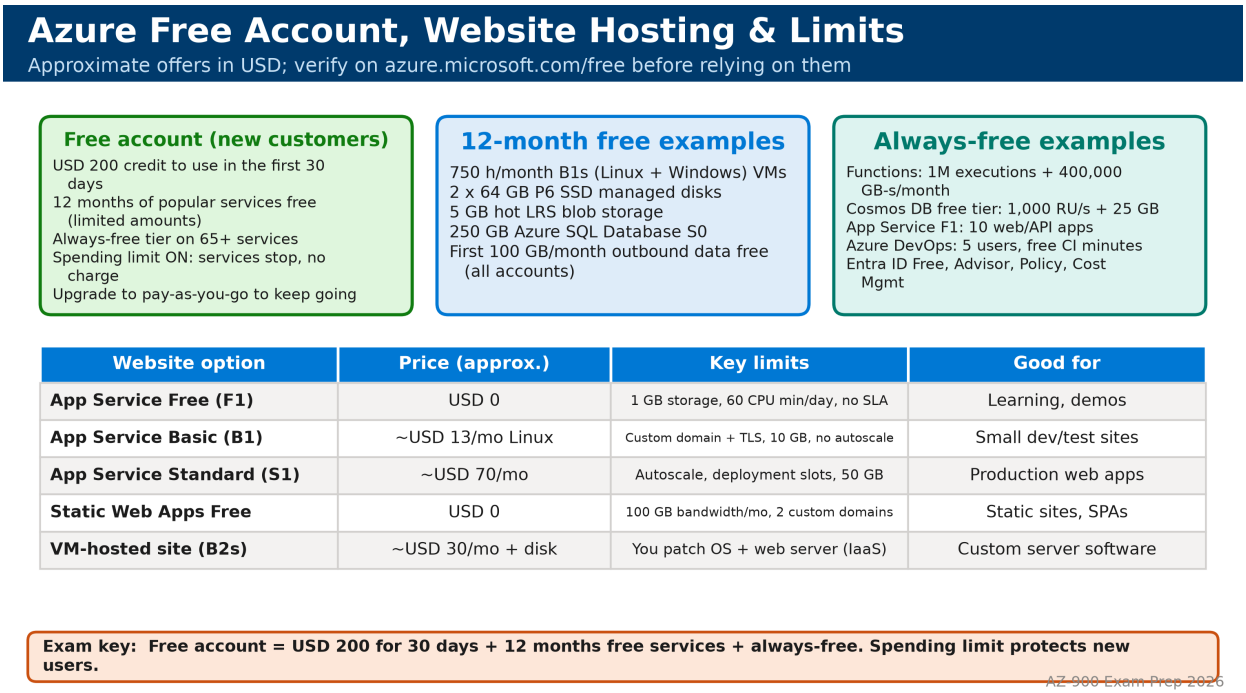


Figure Q264b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 265

CLOUD CONCEPTS

Cloud concepts
v5 Q392

Azure Reserved VM Instances are an example of OpEx.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): OpEx = money spent on renting or running things. CapEx = money spent upfront to OWN physical things. Reserved Instances can be paid upfront, but you still do not own any hardware, so they are OpEx. Q266 (your own datacenter) is the CapEx example.

Answer corrected in this edition. The previous PDF printed: *No.*. The printed answer No is wrong. Reserved VM Instances are a commitment, paid upfront or monthly, for compute you rent. You do not own a physical asset, so they are still operating expenditure (OpEx). This matches the cross-check answer key.

Answer explained

✔ **Yes.**
Reserved Instances are a 1- or 3-year commitment to rented compute. You pay upfront or monthly. You do not own anything, so they are still OpEx.

✖ **No.**
Paying upfront does not make it CapEx by itself. You buy no hardware, and you can pay for reservations monthly.

EXAM STRATEGY
With reservations, you own no asset, so OpEx = Yes. Do not let 'upfront payment' trick you into choosing CapEx.

IMPACT
Reservations are a big way to save money. Knowing they stay OpEx makes approval by finance teams easier.

What · Where · When · Why · How

What	Azure Reservations are 1- or 3-year commitments for cheaper compute.
Where	Bought in the portal under Reservations, for a subscription or resource group.
When	When a VM workload runs all the time for one or three years.
Why	You rent cheaper capacity and own no hardware, so it stays OpEx.
How	Buy a reservation that matches the VM size and region. The discount is applied automatically.

Exam tip
Upfront is not the same as CapEx. CapEx = you own a physical asset. Reservations are still rented capacity, so OpEx.

Memory trick
A prepaid gym membership is still not owning the gym.

Real-world example, use case and scenario

EXAMPLE
3-year reservation
A company buys a 3-year reservation for 20 D8s_v5 VMs and pays monthly. Finance records it as a repeating operating cost.

USE CASE
Software company
A payroll software company reserves its base compute to cut costs. All its cloud spending stays in its OpEx budget.

SCENARIO
Budget type
A project manager put a reservation in the capital budget because it was paid upfront. Finance moved it to OpEx, because no hardware was bought. The capital budget was then free for a new office.

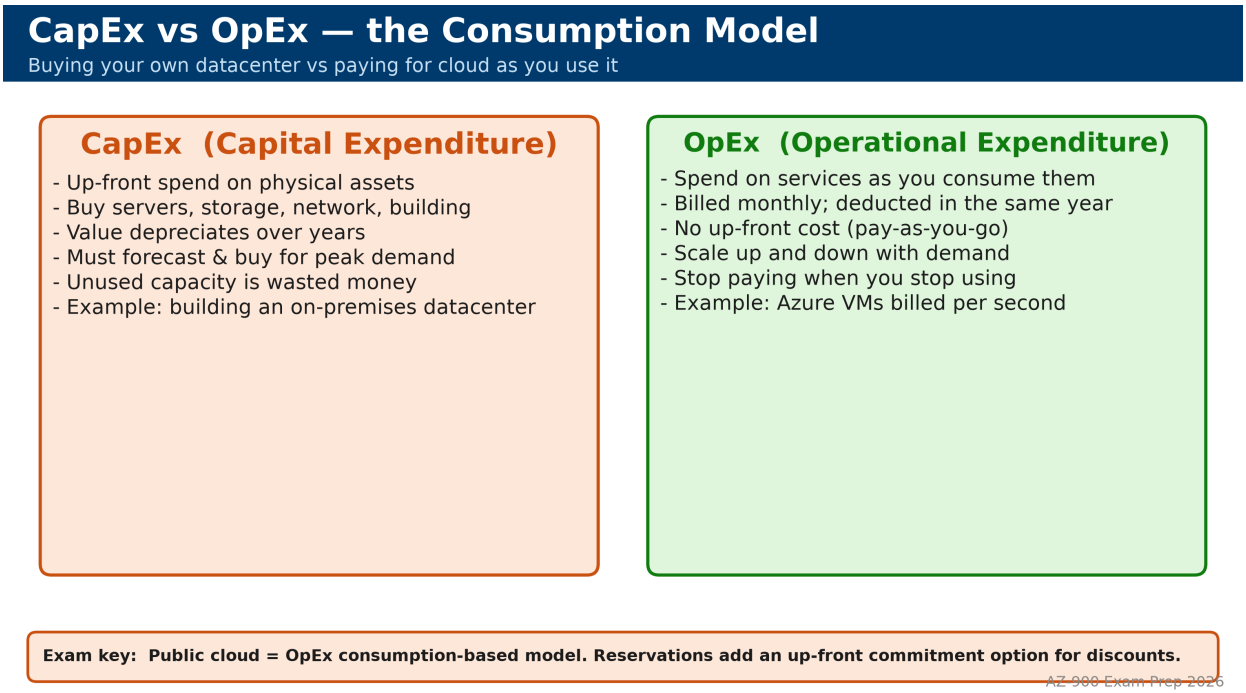


Figure Q265: CapEx vs OpEx

About this figure

- Left box is buying assets up front (CapEx); right box is paying for consumption (OpEx). Cloud moves spending from left to right.
- CapEx assets are depreciated over several years; OpEx is expensed in the period it is incurred, which suits variable or uncertain demand.
- The consumption model means no up-front cost, pay only for what you use, and stop paying when resources are deleted or deallocated.
- Reservations and savings plans add a 1- or 3-year commitment for discounts, but they remain operational spending, not purchased hardware.
- Use the TCO calculator to compare on-premises CapEx (hardware, power, labour) with expected Azure OpEx before migrating.

Question 266

CLOUD CONCEPTS

Cloud concepts
v5 Q393

Deploying your own datacenter is an example of CapEx.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Building your own datacenter means paying upfront for buildings and hardware that you own. Their value is spread over years (depreciation). This is the definition of CapEx.

✖ **No.**
Physical infrastructure that you own and buy upfront is CapEx by definition. Only running costs, like power, are OpEx.

EXAM STRATEGY
Owning buildings and hardware = CapEx, so Yes. Any statement about buying physical infrastructure is CapEx.

IMPACT
Datacenter CapEx ties up money for years. This difference is the main money reason to move to the cloud.

What · Where · When · Why · How

What	CapEx is upfront spending on physical assets whose value is spread over years (depreciation).
Where	It applies to owned on-premises datacenters, servers and network hardware.
When	When a company builds or renews its own datacenter.
Why	Owned infrastructure bought upfront is the textbook CapEx example.
How	Buy and install hardware, then spread its cost over several years.

Exam tip
Own datacenter or buy servers = CapEx. Rent from Azure = OpEx.

Memory trick
Capital = concrete: if you pour concrete, it is CapEx.

Real-world example, use case and scenario

EXAMPLE
Server purchase
A company buys 50 rack servers for USD 400,000. It spreads their cost over five years as a capital asset.

USE CASE
Defence contractor
A defence contractor builds a private datacenter on its own site for secret workloads. It accepts the large upfront CapEx.

SCENARIO
Renewal decision
A university needed a USD 2M datacenter renewal. A study showed Azure would change that CapEx into monthly OpEx. The board approved a step-by-step move to Azure instead of the rebuild.

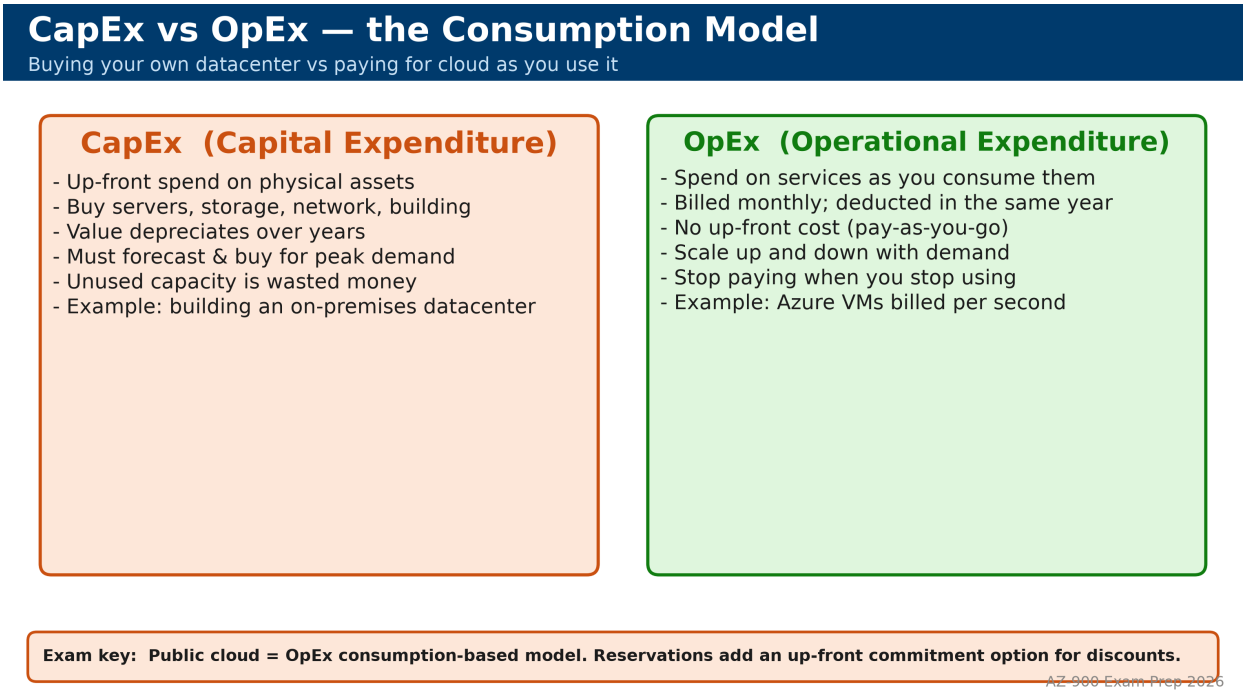


Figure Q266a: CapEx vs OpEx

About this figure

• Left box is buying assets up front (CapEx); right box is paying for consumption (OpEx). Cloud moves spending from left to right.

• CapEx assets are depreciated over several years; OpEx is expensed in the period it is incurred, which suits variable or uncertain demand.

• The consumption model means no up-front cost, pay only for what you use, and stop paying when resources are deleted or deallocated.

• Reservations and savings plans add a 1- or 3-year commitment for discounts, but they remain operational spending, not purchased hardware.

• Use the TCO calculator to compare on-premises CapEx (hardware, power, labour) with expected Azure OpEx before migrating.

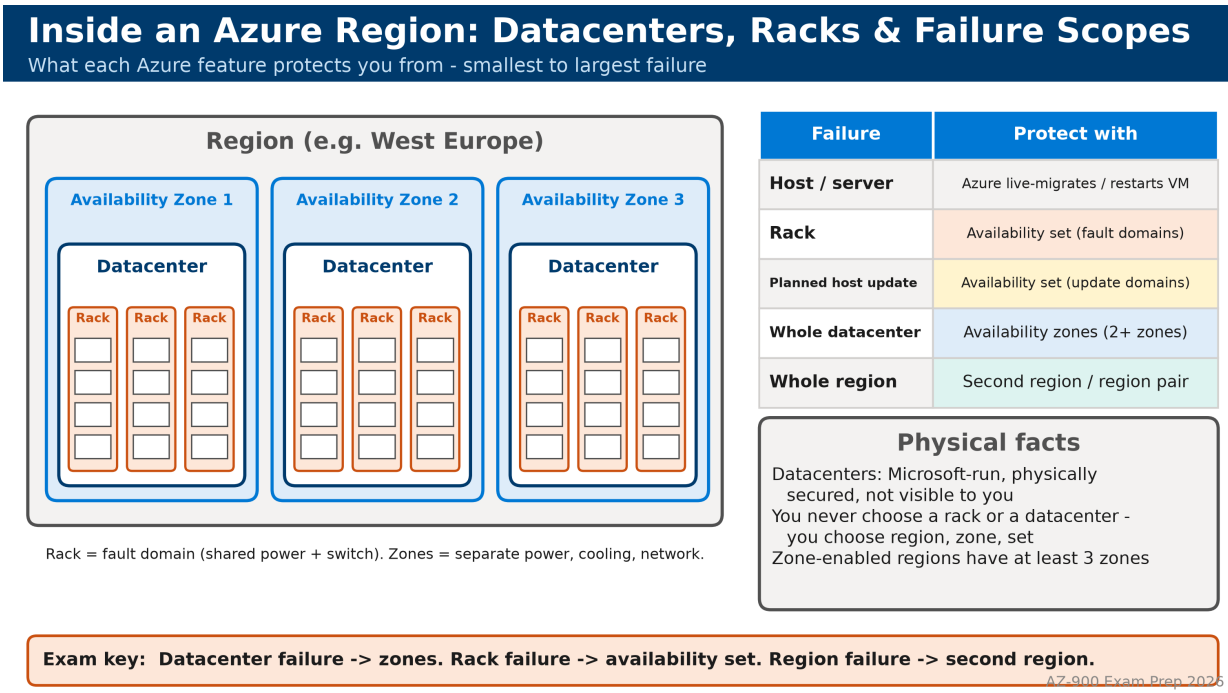


Figure Q266b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 267

CLOUD CONCEPTS

Deployment models
v5 Q394

A company can extend a private cloud by adding its physical servers to the public cloud.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap is 'adding its PHYSICAL servers to the public cloud'. You cannot put your own hardware into Microsoft's public cloud. You extend a private cloud by CONNECTING it to the public cloud. Q246 (extend your network by using the public cloud) is Yes.

Answer explained

✔ **No.**
Customers cannot add their own hardware to the provider's public cloud. You extend a private cloud by connecting it to public cloud resources. This is hybrid cloud.

✖ **Yes.**
Azure's hardware belongs to Microsoft. You extend a private cloud by connecting to Azure, not by adding servers to it.

EXAM STRATEGY
Customers cannot put their own hardware in the provider's public cloud. Extending a private cloud means connecting to public cloud (hybrid), so No.

IMPACT
If you misunderstand hybrid, your migration plans will not work. Azure Local and Azure Arc bring Azure to your hardware, not the other way round.

What · Where · When · Why · How

What	A private cloud runs on infrastructure used by only one company.
Where	In the company's own datacenter, or in a hosting building used only for that company.
When	When you extend private infrastructure with public cloud capacity.
Why	You cannot add your own servers to Azure's public cloud. You connect to it instead.
How	Link the private cloud to Azure with VPN or ExpressRoute, or manage it with Azure Arc.

Exam tip
Extend a private cloud = connect to the public cloud (hybrid). Add your own servers to the public cloud = No.

Memory trick
You can rent a room in the hotel, but you cannot bolt your own bed into its building.

Real-world example, use case and scenario

EXAMPLE
Connect, don't give
A company extends its VMware private cloud by linking it to Azure VNets over ExpressRoute. Its own servers stay in its own building.

USE CASE
Hybrid factory
A car-parts maker keeps its factory servers on-premises and manages them with Azure Arc. New analytics runs on Azure VMs.

SCENARIO
Shipping servers
A CIO asked if the company could ship its spare servers to a Microsoft datacenter to 'join' Azure. The Microsoft team explained this is not possible. Instead, the company connected its private cloud to Azure, making a hybrid cloud.

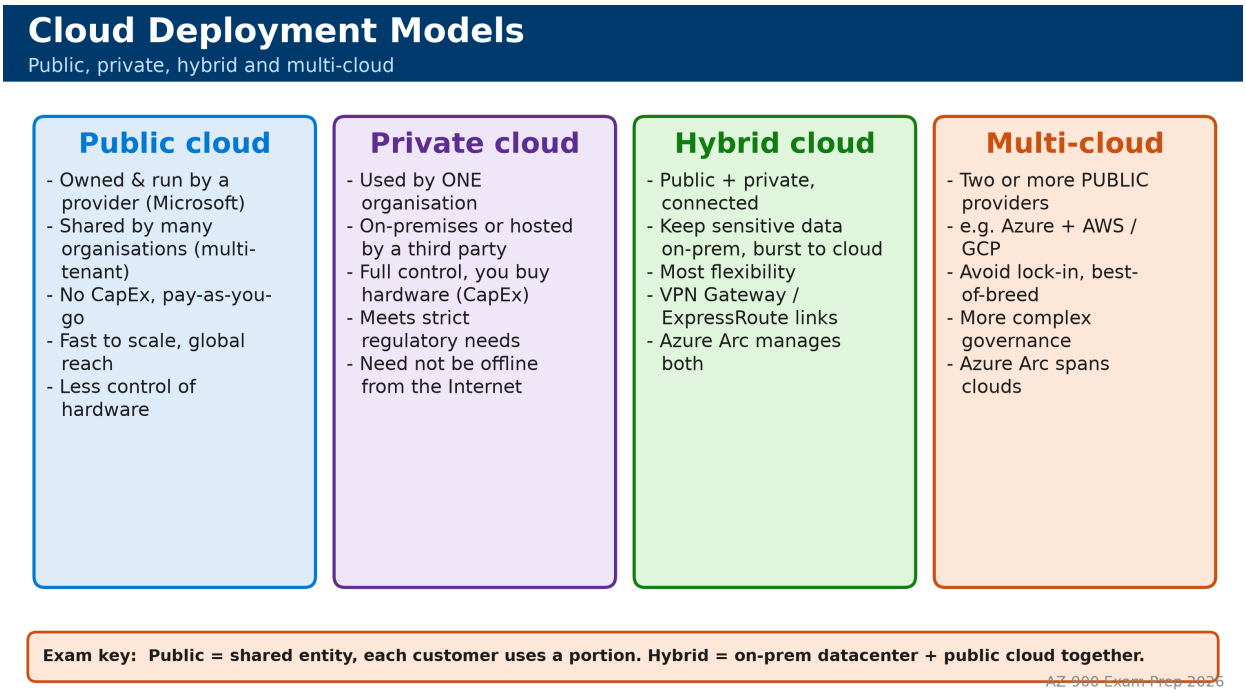


Figure Q267: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 268

CLOUD CONCEPTS

Deployment models
v5 Q395

To build a hybrid cloud, you must deploy resources to the public cloud.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The word 'must' is true here. Hybrid = private (or on-premises) + public. Without any public cloud resources, it is only a private cloud.

Answer explained

✔ **Yes.**
By definition, a hybrid cloud combines private or on-premises resources with public cloud resources. So you need public cloud resources.

✖ **No.**
Without a public cloud part, the environment is only a private cloud, not hybrid.

EXAM STRATEGY
Hybrid by definition = private plus public. Without public cloud resources, it is only a private cloud. So Yes.

IMPACT
Hybrid designs need network, identity and governance (rules) on both sides. Deployment model definitions come up often in the exam.

What · Where · When · Why · How

What	Hybrid cloud combines private or on-premises resources with public cloud resources.
Where	It covers the on-premises datacenter and a public cloud such as Azure.
When	When some workloads stay on-premises and others use the public cloud.
Why	Without public cloud resources, the environment is only private, not hybrid.
How	Deploy resources in Azure and connect them to on-premises networks.

Exam tip
Hybrid = private + public. You need both parts.

Memory trick
A hybrid car needs both the petrol engine and the battery.

Real-world example, use case and scenario

EXAMPLE
Smallest hybrid
An on-premises file server with its backups stored in Azure Backup vaults is already a simple hybrid setup.

USE CASE
Medicine research
A medicine company keeps patient data on-premises. It runs heavy simulations in Azure, linked over ExpressRoute.

SCENARIO
Hybrid in name only
A seller offered a 'hybrid cloud' package that was really only virtual on-premises servers. The buyer's architect saw that nothing ran in a public cloud. The offer was renamed 'private cloud'.

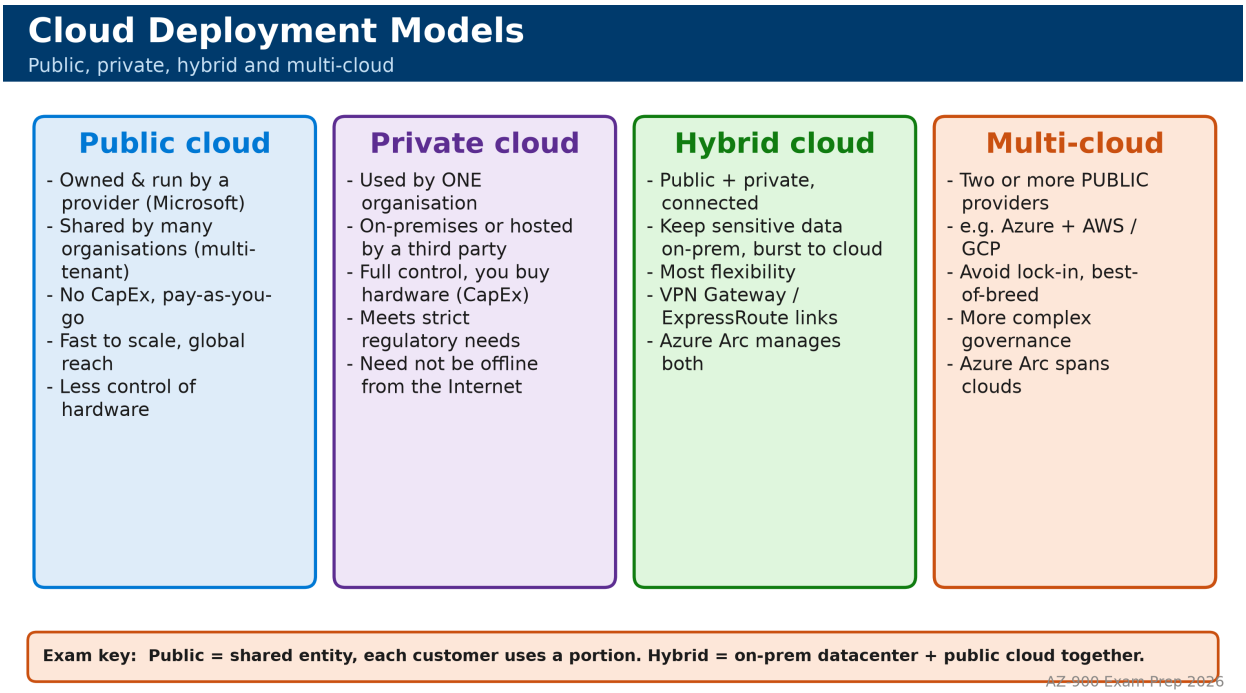


Figure Q268: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 269

CLOUD CONCEPTS

Deployment models
v5 Q396

A private cloud must be disconnected from the internet.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The trap words are 'must' and 'disconnected'. 'Private' means used by only one company. It does not mean cut off from the internet.

Answer explained

✔ **No.**
A private cloud is one used by only one company. It is not defined by being cut off from the internet. Most private clouds are connected to the internet.

✖ **Yes.**
Being disconnected is not part of the private cloud definition. Full isolation (air-gapping) is an optional special case.

EXAM STRATEGY
The words 'must' and 'disconnected' show the statement is false. Private means used by only one company, not cut off from the internet. So No.

IMPACT
If you think private means cut off, you make wrong security choices. Most private clouds face the internet and need full protection at the edge.

What · Where · When · Why · How

What	A private cloud is cloud infrastructure used by only one company.
Where	Hosted on-premises, or by another company only for this one company.
When	When a company needs full control of its infrastructure.
Why	Being used by only one company, not being cut off from the internet, defines a private cloud.
How	Run private cloud services and control internet access with firewalls as needed.

Exam tip
Private = used by only one company. It says nothing about internet access.

Memory trick
A private house still has a front door to the street.

Real-world example, use case and scenario

EXAMPLE
Private cloud on the internet
A bank's private cloud runs its online banking site. Customers reach it over the internet.

USE CASE
University cloud
A university runs an OpenStack private cloud for researchers. It downloads datasets from the internet every day.

SCENARIO
Too much isolation
A new engineer suggested cutting internet access to make the company's cloud 'truly private'. The architect explained that patches, partner APIs and customer access would all stop working. The team kept the connection and made the firewalls stricter.

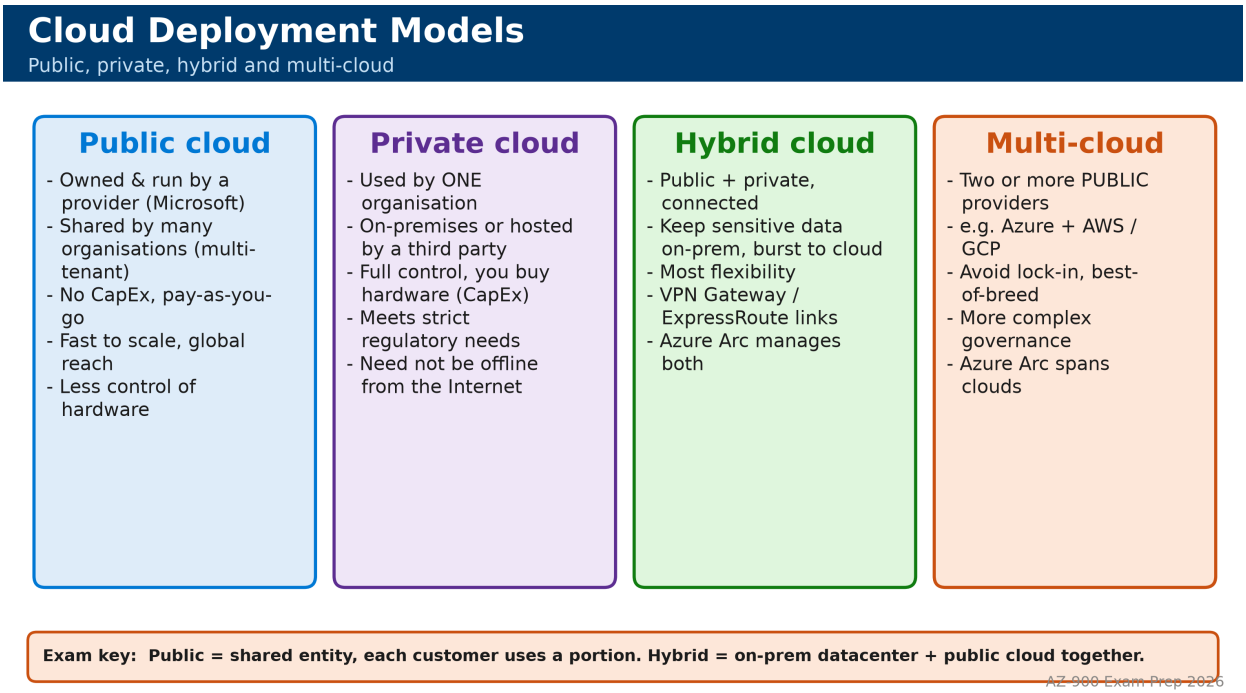


Figure Q269: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

Question 270

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q397, Q126

You plan to deploy several Azure virtual machines. You need to ensure that the services running on the virtual machines are available if a single data center fails. Solution: You deploy the virtual machines to a scale set. Does this meet the goal?

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): The goal is to survive the failure of ONE datacenter (one building). A scale set is about the NUMBER of VMs, not where they are placed. Look-alikes: Q262 (two resource groups) is also No; Q141 and Q148 (availability zones) are Yes.

2026 update: Note: a scale set CAN be set to span availability zones (a zone-redundant scale set). Then it would survive a datacenter failure. The expected answer is still No, because a scale set by itself does not promise this.

Answer explained

✔ **No.**
A scale set adds and removes identical VMs as demand changes. But unless you set it to use availability zones, nothing promises the VMs are in different datacenters.

✖ **Yes.**
A normal scale set does not promise to use more than one datacenter. One datacenter failure could stop all its VMs.

EXAM STRATEGY
Scale set means scalability (handling more load), so the answer is No. Say Yes only if the question clearly says the scale set spans availability zones.

IMPACT
If you think a scale set alone survives a datacenter loss, you may get a full outage. Choose zones on purpose when you need high availability.

What · Where · When · Why · How

What	A virtual machine scale set manages a group of identical VMs that grow and shrink in number.
Where	It is deployed in one region. You can choose to spread it across availability zones (separate datacenters in a region).
When	Use it when a workload must add or remove VMs as demand changes.
Why	Without zones, it does not promise different datacenters, so on its own it fails the goal.
How	Set autoscale rules. For resilience, choose zones 1, 2 and 3 when you create it.

Exam tip
Scale set = scalability. Availability zones = survive a datacenter failure. Regions = survive a region failure.

Memory trick
More copies in one building still burn together.

Real-world example, use case and scenario

EXAMPLE
Set without zones
A scale set of 10 VMs in North Europe has no zones set. Azure spreads the VMs across fault domains (separate racks), but they may all be in one datacenter.

USE CASE
Ticketing site
A concert-ticket site uses a scale set to grow from 4 to 40 web servers when sales open. It also spreads the set across zones 1-3 for resilience.

SCENARIO
Scaling is not high availability
A delivery app trusted a single-zone scale set. A power failure in the datacenter stopped every VM. The team rebuilt the set across three zones.

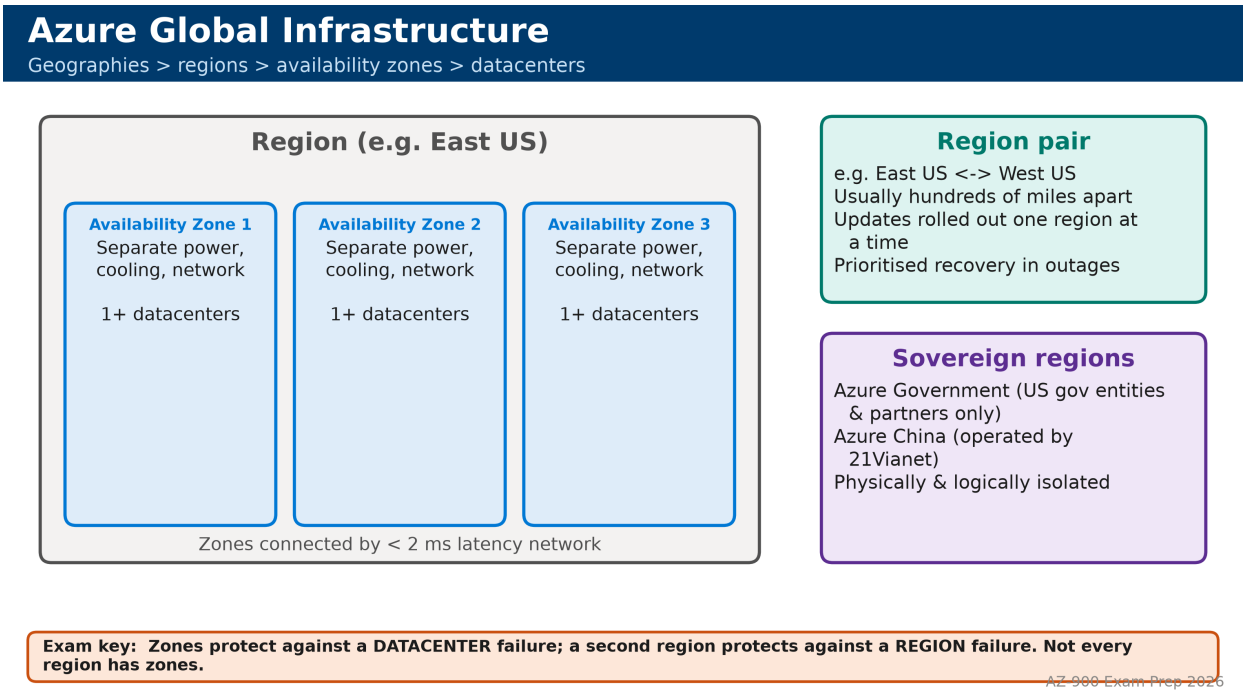


Figure Q270a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

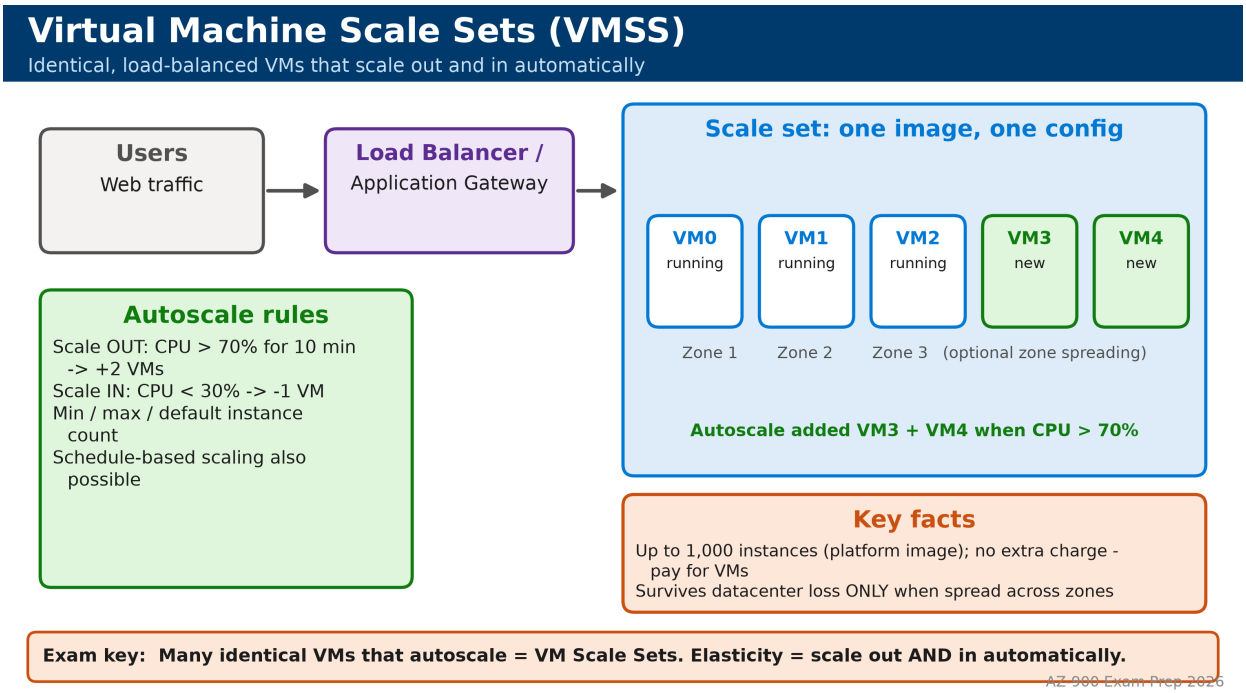


Figure Q270b: Virtual Machine Scale Sets

About this figure

- A scale set runs identical VMs from one image and configuration behind a load balancer or Application Gateway.
- Autoscale adds instances when a metric (e.g. CPU > 70%) is high and removes them when low, within min/max instance counts - this is elasticity.
- Scale sets support up to 1,000 VM instances (platform images); the scale set itself is free - you pay only for the VMs, disks and networking.
- Orchestration modes: Uniform (identical VMs) and Flexible (mixed sizes, recommended for new deployments).
- A scale set survives a datacenter failure only when its instances are spread across availability zones; a non-zonal set does not.
- Scaling out adds more VMs (horizontal); scaling up means a bigger VM size (vertical) and usually needs a restart.

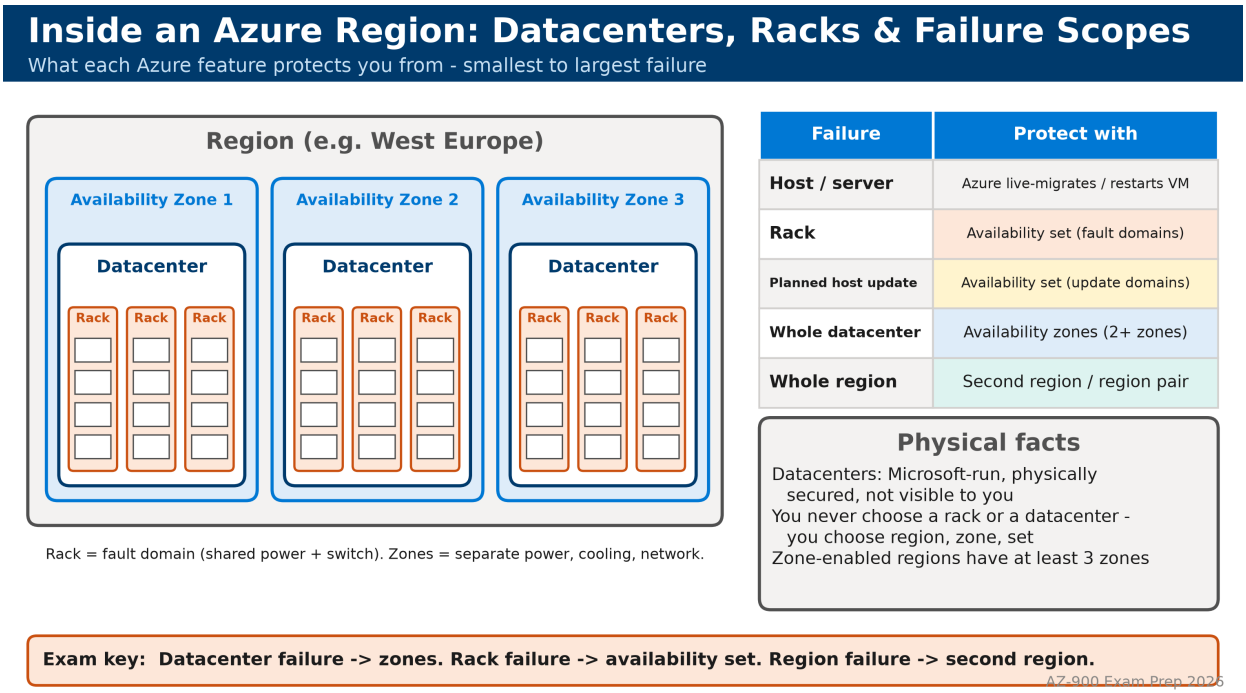


Figure Q270c: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 271

AZURE ARCHITECTURE & SERVICES — Governance — v5 Q400

Azure resources can only access other resources in the same resource group.

✖ A. Yes.

✔ **B.** No.

Correct answer: No.

Clarification (read this first): Watch the word 'only'. A resource group is a folder for management, not a network wall. Resources in different groups can talk to each other.

Answer explained

✔ **No.**
A resource group is a container for managing resources and their lifecycle. It is not a network boundary. Resources can reach resources in other groups, subscriptions or regions.

✖ **Yes.**
Who can connect is controlled by VNets (virtual networks), NSGs (network security groups), firewalls and permissions. Resource group membership does not control it.

EXAM STRATEGY
The word 'only' is a sign the statement is false. Resource groups organize resources; they do not block network traffic. So the answer is No.

IMPACT
Resources often talk across resource groups and subscriptions. If you design around a boundary that does not exist, the system may break or be insecure.

What · Where · When · Why · How

What	A resource group is a logical container for management. It is not a network boundary.
Where	Inside a subscription. Network traffic is controlled by VNets, NSGs and firewalls.
When	When you ask if resources in different groups can communicate.
Why	Being in a group does not limit access between resources.
How	Control connections with VNets, peering, NSGs, private endpoints and RBAC (role-based access control).

Exam tip
Resource groups are for organizing and lifecycle. They never isolate networks.

Memory trick
Different folders, same phone network.

Real-world example, use case and scenario

EXAMPLE
App across groups
A web app in rg-web reads data from an Azure SQL Database in rg-data. It uses a managed identity and a private endpoint.

USE CASE
Shared services
A company keeps Key Vault and Log Analytics in one shared resource group. Apps in many other resource groups use them.

SCENARIO
Wrong idea of isolation
A reviewer thought two client environments were separate because they were in different resource groups. A test showed their VMs could reach each other over a peered VNet. The team added NSG rules to block this.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

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Figure Q271: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 272

AZURE ARCHITECTURE & SERVICES

Governance
v5 Q401

If you delete a resource group, all the resources in the resource group will be deleted.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Answer explained

✔ **Yes.**
Resources share the lifecycle of their resource group. If you delete the group, every resource in it is deleted, unless a delete lock blocks it.

✖ **No.**
A resource cannot exist outside a resource group. So deleting the group removes all of them.

EXAM STRATEGY
Resources live and die with their group. Deleting the group deletes everything inside, so Yes. Only a delete lock can stop it.

IMPACT
One wrong resource group deletion can remove production databases and VMs. This is why critical groups should have delete locks.

What · Where · When · Why · How

What	A resource group shares one lifecycle with the resources inside it.
Where	In a subscription. You delete it from the portal page of the group or with the CLI.
When	When you remove a whole environment, such as a test deployment.
Why	Deleting the group deletes every resource inside it.
How	Run az group delete. Protect production groups with a CanNotDelete lock (a lock that blocks deletion).

Exam tip
Delete the resource group = delete everything inside it. Delete locks protect against this.

Memory trick
Throw away the box, and the contents go with it.

Real-world example, use case and scenario

EXAMPLE
Clean-up
After a hackathon, an engineer runs az group delete --name rg-hack. The VMs, disks, IPs and VNet inside are all removed.

USE CASE
Short-lived test environments
A QA team creates one resource group per code change. It deletes the group after tests pass, which stops all costs at once.

SCENARIO
Wrong group deleted
An intern deleted what he thought was a test resource group. It held a production database. Restoring from backup took hours. The company then added CanNotDelete locks to all production groups.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
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Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
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- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

AZ-900 Exam Prep 2026

Figure Q272: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

A resource group can contain resources from multiple Azure regions.

✔ **A.** Yes.

✖ B. No.

Correct answer: Yes.

Clarification (read this first): The resource group's region only stores its metadata (information about the resources). The resources inside can be in any region. Q128 asks the same idea the other way: 'must all resources share its region?' -> No.

Answer explained

✔ **Yes.**
A resource group's location only stores its metadata (data about the resources). The resources inside it can be in any region.

✖ **No.**
No Azure rule says resources must be in the group's region. Only an optional Azure Policy can add such a limit.

EXAM STRATEGY
The group's region holds metadata only. Resources inside can be in any region, so Yes.

IMPACT
You can keep all parts of a multi-region app in one group. This makes lifecycle and cost tracking simpler without limiting where you deploy.

What · Where · When · Why · How

What	A resource group's location stores only its metadata.
Where	You set it when you create the group. Each resource chooses its own region.
When	When you group a multi-region app under one management scope.
Why	Resources in one group can live in different regions.
How	Create resources in any region and place them in the same resource group.

Exam tip
Resource group region = where metadata is stored. The resources can be in many regions.

Memory trick
One address book, contacts in many cities.

Real-world example, use case and scenario

EXAMPLE
Mixed regions
rg-global-app (location: West Europe) holds a Front Door profile, a web app in West Europe and a copy of the web app in East US.

USE CASE
Travel booking
A travel company keeps all parts of its multi-region booking system in one resource group. This gives one place for access control and cost reports.

SCENARIO
Needless split
A team made one resource group per region because they thought it was required. The architect explained that regions can mix in one group. They regrouped by application lifecycle instead.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

AZ-900 Exam Prep 2026

Figure Q273: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 274

CLOUD CONCEPTS

Service models
v5 Q403

Your company has an on-premises network that contains multiple servers. The company plans to reduce the following administrative responsibilities of network administrators: Backing up application data. Replacing failed server hardware. Managing physical server security. Updating server operating systems. Managing permissions to shared documents. The company plans to migrate several servers to Azure virtual machines. You need to identify which administrative responsibilities will be eliminated after the planned migration. Which two responsibilities should you identify?

✔ **A.** Replacing failed server hardware.

✖ B. Backing up application data.

✔ **C.** Managing physical server security.

✖ D. Updating server operating systems.

✖ E. Managing permissions to shared documents.

Correct answer: Replacing failed server hardware.; Managing physical server security.

Clarification (read this first): The question asks which tasks are REMOVED when servers move to Azure VMs (IaaS - you still manage the OS). Only the physical tasks go to Microsoft. Q277 confirms you still patch the OS on IaaS VMs.

Answer explained

✔ **Replacing failed server hardware.; Managing physical server security.**
With Azure VMs (IaaS), Microsoft owns the physical servers. So replacing failed hardware and physical security become Microsoft's job.

✖ **Backing up application data.**
Backing up data stays your job in every cloud model. You set up Azure Backup yourself.

✖ **Updating server operating systems.**
In IaaS you still patch the guest OS (the OS inside the VM). Microsoft patches it only in PaaS or SaaS.

✖ **Managing permissions to shared documents.**
Access to data and documents is always your job.

EXAM STRATEGY
With IaaS VMs, only the physical layer moves to Microsoft: hardware replacement and physical security. Remove OS updates, backups and permissions, because they stay with you.

IMPACT
If you think Microsoft patches your VMs or backs up your data, systems stay unpatched and data stays unprotected. Shared responsibility questions appear in many exam areas.

What · Where · When · Why · How

What	IaaS moves the physical layer of servers to the cloud provider.
Where	In the shared responsibility model, everything below the hypervisor (the software that runs VMs) is Microsoft's.
When	When you move on-premises servers to Azure VMs.
Why	Microsoft looks after hardware and physical security, so those tasks go away for you.
How	Move to VMs, then keep doing OS updates, backups and permissions yourself.

Exam tip
Moving to VMs removes only the physical layer: hardware, datacenter, physical security and host network.

Memory trick
IaaS: Microsoft owns the metal; you own everything above the hypervisor.

Real-world example, use case and scenario

EXAMPLE
Failed host
When a physical host fails in Azure, the VM restarts on healthy hardware. The customer does not open a hardware ticket.

USE CASE
Law firm move
A law firm moves file servers to Azure VMs. It ends its hardware repair contract but keeps its patching plan and permission checks.

SCENARIO
Patching stopped
After moving to VMs, a retailer stopped patching because it thought Azure did it. A security scan found months of missing updates. They turned on Azure Update Manager and restarted their patch process.

Shared Responsibility Model

Who manages each layer: on-premises vs IaaS vs PaaS vs SaaS

Responsibility	On-premises	IaaS	PaaS	SaaS
Information & data	Customer	Customer	Customer	Customer
Devices (mobile, PCs)	Customer	Customer	Customer	Customer
Accounts & identities	Customer	Customer	Customer	Customer
Identity & directory infra	Customer	Customer	Shared	Shared
Applications	Customer	Customer	Shared	Microsoft
Network controls	Customer	Customer	Shared	Microsoft
Operating system	Customer	Customer	Microsoft	Microsoft
Physical hosts	Customer	Microsoft	Microsoft	Microsoft
Physical network	Customer	Microsoft	Microsoft	Microsoft
Physical datacenter	Customer	Microsoft	Microsoft	Microsoft

Exam key: Data, devices, accounts & identities are ALWAYS the customer's. Physical datacenter is NEVER yours in cloud.

AZ-900 Exam Prep 2026

Figure Q274: Shared responsibility model

About this figure

• Read each row across: orange cells are yours, blue are Microsoft's, yellow are shared. Your share shrinks moving from on-premises to IaaS, PaaS and SaaS.

• Information and data, devices, and accounts and identities stay with the customer in every model, even in SaaS such as Microsoft 365.

• Physical hosts, network and datacenter move to Microsoft in every cloud model, so hardware failures, power and building security are never the customer's task.

• With IaaS you still patch the guest OS and configure network controls; with PaaS the OS becomes Microsoft's job, but your code and access settings remain yours.

• Responsibility does not equal liability: misconfigured access, weak passwords or leaked keys are customer failures even when the service itself is fully managed.

Part 2 - New 2026 Questions (152)

Written for the current AZ-900 skills outline. Same layout as Part 1.

Question 275	CLOUD CONCEPTS	Cloud concepts v5 Q406

Your manager asks you to explain cloud computing to the board. Which statement best describes cloud computing?

- ✔ **A.** The delivery of computing services such as servers, storage, databases, networking and software over the internet, typically with pay-as-you-go pricing
- ✖ B. Running all workloads on servers that the company purchases and installs in its own datacenter
- ✖ C. Installing a hypervisor on existing on-premises servers to run multiple virtual machines
- ✖ D. Leasing a fixed rack in a colocation facility for a flat monthly fee

Correct answer: The delivery of computing services such as servers, storage, databases, networking and software over the internet, typically with pay-as-you-go pricing

Answer explained

✔ **The delivery of computing services such as servers, storage, databases, networking and software over the internet, typically with pay-as-you-go pricing**
Cloud computing means computing services delivered over the internet, when you need them, usually paid for as you use them.

✖ **Running all workloads on servers that the company purchases and installs in its own datacenter**
That is traditional on-premises IT, with a large cost paid up front (CapEx).

✖ **Installing a hypervisor on existing on-premises servers to run multiple virtual machines**
Virtualization alone is not cloud. There is no self-service, on-demand use over the internet.

✖ **Leasing a fixed rack in a colocation facility for a flat monthly fee**
Colocation (renting space for your own servers) is fixed hosting of your own hardware. It is not on-demand, measured service.

EXAM STRATEGY
Look for 'over the internet' plus 'pay-as-you-go'. Remove options about buying servers, installing a hypervisor or renting a rack - these are not cloud.

IMPACT
A clear definition helps you explain cloud spending to a board. This is a very common opening question in Cloud Concepts.

What · Where · When · Why · How

What	On-demand delivery of compute, storage, networking and software over the internet.
Where	From a cloud provider's datacenters around the world, such as Azure regions.
When	Whenever a workload needs resources. You can get them in minutes.
Why	To avoid buying hardware and to pay only for what you use.
How	You rent services through a portal, CLI or API and are billed by usage.

Exam tip
An option with 'over the internet' and 'pay for what you use' is almost always the cloud definition.

Memory trick
Cloud = renting computer power over the internet, like renting a car instead of buying one.

Real-world example, use case and scenario

EXAMPLE
Retail start-up
A start-up runs its online store on Azure App Service instead of buying servers.

USE CASE
Seasonal tax firm
A tax firm rents extra compute only during tax season and gives it back afterward.

SCENARIO
Board briefing
The CIO tells the board that IT will stop owning servers and start renting services over the internet.

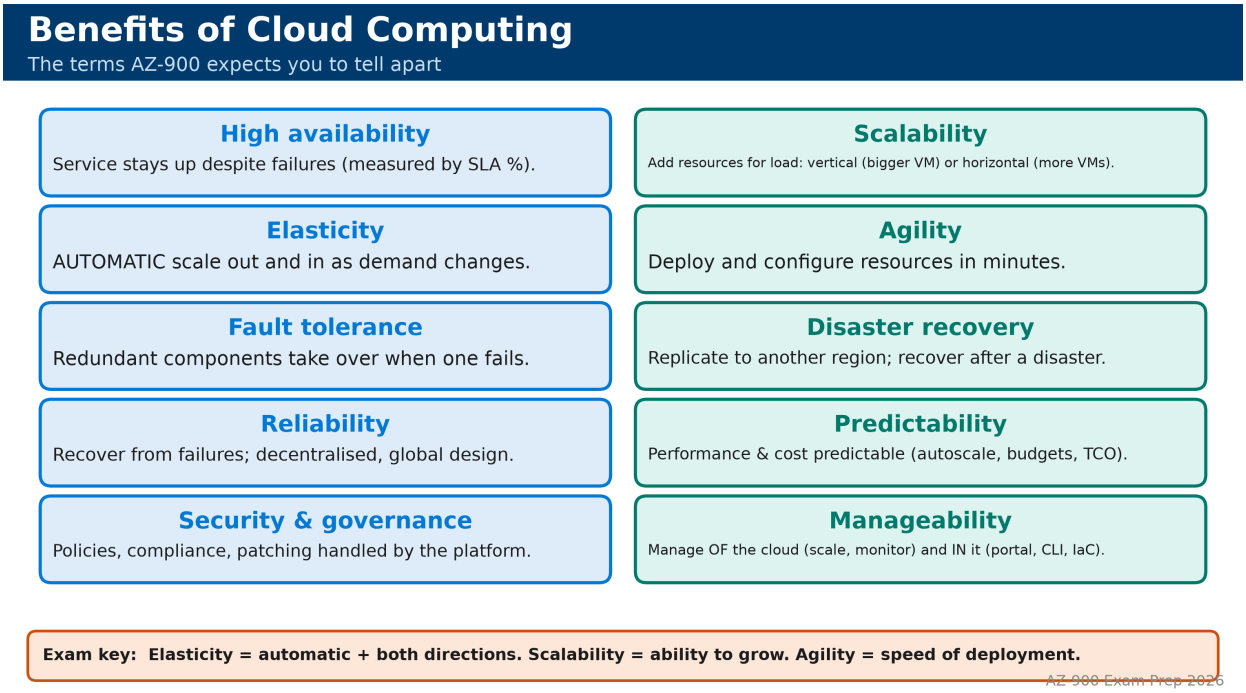


Figure Q275a: Benefits of cloud computing

About this figure

- Each box pairs a term with its meaning; the exam often gives a scenario and asks which benefit it describes, so focus on keywords.
- Scalability is the capacity to grow, manual or automatic; elasticity is automatic scaling out and back in with demand, for example VM Scale Sets autoscale.
- High availability is expressed as an SLA percentage such as 99.9% or 99.99%; fault tolerance and disaster recovery are ways of achieving it.
- Agility means provisioning in minutes instead of weeks of hardware procurement; predictability covers both performance and cost forecasting.
- Governance tools such as Azure Policy, RBAC and locks deliver the security and governance benefit; manageability covers portal, CLI, PowerShell and IaC.

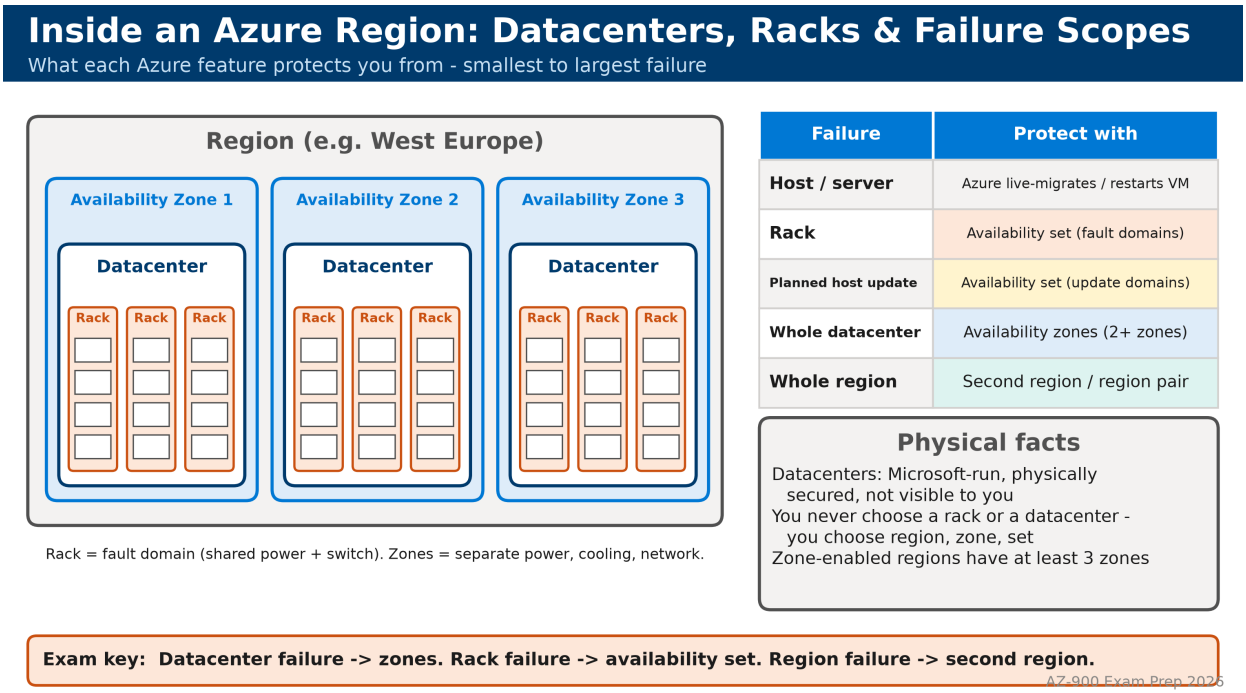


Figure Q275b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 276

CLOUD CONCEPTS

Cloud concepts
v5 Q407

Your company uses a mix of IaaS, PaaS and SaaS services in Azure. Which two responsibilities always remain with the customer, regardless of the service model? Each correct answer presents a complete solution. NOTE: Each correct selection is worth one point. (Choose two.)

✔ **A.** Information and data

✖ B. Physical hosts

✖ C. Physical network

✔ **D.** Accounts and identities

✖ E. Operating system

Correct answer: **Information and data; Accounts and identities**

Clarification (read this first): Watch the word 'always'. The question wants items that stay with the customer in EVERY model (IaaS, PaaS and SaaS). The operating system is yours only in IaaS, so it does not count.

Answer explained

✔ **Information and data; Accounts and identities**
Data, devices (endpoints) and accounts/identities are ALWAYS the customer's job, in IaaS, PaaS and SaaS.

✖ **Physical hosts**
Physical hosts are always the cloud provider's job.

✖ **Physical network**
The physical network inside the datacenter is always Microsoft's job.

✖ **Operating system**
The customer manages the OS only in IaaS. Microsoft manages it in PaaS and SaaS.

EXAM STRATEGY
Remember the three items that are always yours: data, devices, identities. Remove anything physical (always Microsoft) and the OS (it changes by model).

IMPACT
Breaches from wrong permissions or stolen passwords are the customer's fault in every model. This shared responsibility table decides who is accountable for security.

What · Where · When · Why · How

What	The parts of the system the customer can never hand to the provider.
Where	In every Azure, Microsoft 365 and Dynamics 365 service.
When	From the moment the customer stores data or creates users in the cloud.
Why	Only the customer knows who should access which data.
How	Classify data, protect devices and secure identities with MFA (multi-factor sign-in) and RBAC.

Exam tip
Learn the three 'always customer' items: information and data, devices (phones and PCs), accounts and identities.

Memory trick
D-D-A: Data, Devices and Accounts are always yours.

Real-world example, use case and scenario

EXAMPLE
Healthcare
A clinic using Microsoft 365 must still classify patient data and protect staff accounts.

USE CASE
Banking
A bank using Azure SQL Database must still decide who can see account records.

SCENARIO
Stolen password
A user's password is stolen by phishing. Microsoft is not responsible, because identities are the customer's job.

Shared Responsibility Model

Who manages each layer: on-premises vs IaaS vs PaaS vs SaaS

Responsibility	On-premises	IaaS	PaaS	SaaS
Information & data	Customer	Customer	Customer	Customer
Devices (mobile, PCs)	Customer	Customer	Customer	Customer
Accounts & identities	Customer	Customer	Customer	Customer
Identity & directory infra	Customer	Customer	Shared	Shared
Applications	Customer	Customer	Shared	Microsoft
Network controls	Customer	Customer	Shared	Microsoft
Operating system	Customer	Customer	Microsoft	Microsoft
Physical hosts	Customer	Microsoft	Microsoft	Microsoft
Physical network	Customer	Microsoft	Microsoft	Microsoft
Physical datacenter	Customer	Microsoft	Microsoft	Microsoft

Exam key: Data, devices, accounts & identities are ALWAYS the customer's. Physical datacenter is NEVER yours in cloud.

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Figure Q276: Shared responsibility model

About this figure

- Read each row across: orange cells are yours, blue are Microsoft's, yellow are shared. Your share shrinks moving from on-premises to IaaS, PaaS and SaaS.
- Information and data, devices, and accounts and identities stay with the customer in every model, even in SaaS such as Microsoft 365.
- Physical hosts, network and datacenter move to Microsoft in every cloud model, so hardware failures, power and building security are never the customer's task.
- With IaaS you still patch the guest OS and configure network controls; with PaaS the OS becomes Microsoft's job, but your code and access settings remain yours.
- Responsibility does not equal liability: misconfigured access, weak passwords or leaked keys are customer failures even when the service itself is fully managed.

Question 277

CLOUD CONCEPTS

Service models
v5 Q408

Your company deploys Windows Server virtual machines in Azure (IaaS). Statement: The company is responsible for applying operating system updates and security patches to the virtual machines.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): This is IaaS (VMs), so the customer patches the guest OS (the OS inside the VM). Compare Q278: App Service is PaaS, so there the answer is No.

Answer explained

✔ **Yes**
In IaaS the customer owns the guest OS, apps and data. So OS patching is the customer's job.

✖ **No**
In IaaS, Microsoft does not patch the guest OS. That is the customer's task.

EXAM STRATEGY
IaaS means you own the guest OS, so you patch it: Yes. The OS is the line that separates IaaS from PaaS.

IMPACT
Unpatched IaaS VMs are a top cause of breaches. Use Azure Update Manager to automate patching. Do not assume Microsoft does it.

What · Where · When · Why · How

What	Keeping the guest OS of IaaS VMs patched.
Where	On Azure virtual machines the customer has deployed.
When	Whenever Microsoft or Linux vendors release OS updates.
Why	IaaS gives the customer full control of the OS, and control brings responsibility.
How	By hand, or with automation such as Azure Update Manager.

Exam tip
IaaS = most control and most responsibility. The OS is where IaaS and PaaS split.

Memory trick
If you can remote into it (RDP), you patch it.

Real-world example, use case and scenario

EXAMPLE
Manufacturing
A manufacturer plans a monthly patch time for its ERP VMs in Azure.

USE CASE
Old apps
A company moves an old app to a VM without changes and keeps its own patch process.

SCENARIO
Missed patch
Attackers use a missing patch on a VM. The customer, not Microsoft, owned that risk.

Shared Responsibility Model

Who manages each layer: on-premises vs IaaS vs PaaS vs SaaS

Responsibility	On-premises	IaaS	PaaS	SaaS
Information & data	Customer	Customer	Customer	Customer
Devices (mobile, PCs)	Customer	Customer	Customer	Customer
Accounts & identities	Customer	Customer	Customer	Customer
Identity & directory infra	Customer	Customer	Shared	Shared
Applications	Customer	Customer	Shared	Microsoft
Network controls	Customer	Customer	Shared	Microsoft
Operating system	Customer	Customer	Microsoft	Microsoft
Physical hosts	Customer	Microsoft	Microsoft	Microsoft
Physical network	Customer	Microsoft	Microsoft	Microsoft
Physical datacenter	Customer	Microsoft	Microsoft	Microsoft

Exam key: Data, devices, accounts & identities are ALWAYS the customer's. Physical datacenter is NEVER yours in cloud.

AZ-900 Exam Prep 2026

Figure Q277: Shared responsibility model

About this figure

- Read each row across: orange cells are yours, blue are Microsoft's, yellow are shared. Your share shrinks moving from on-premises to IaaS, PaaS and SaaS.
- Information and data, devices, and accounts and identities stay with the customer in every model, even in SaaS such as Microsoft 365.
- Physical hosts, network and datacenter move to Microsoft in every cloud model, so hardware failures, power and building security are never the customer's task.
- With IaaS you still patch the guest OS and configure network controls; with PaaS the OS becomes Microsoft's job, but your code and access settings remain yours.
- Responsibility does not equal liability: misconfigured access, weak passwords or leaked keys are customer failures even when the service itself is fully managed.

Question 278	CLOUD CONCEPTS	Service models v5 Q409, Q15

Your developers host a web app on Azure App Service (PaaS). Statement: The developers are responsible for patching the operating system that runs the App Service instances.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): App Service is PaaS, so Microsoft patches the OS. Compare Q277: for IaaS VMs the customer patches the OS (Yes).

Answer explained

✔ **No**
In PaaS, Microsoft patches the OS and platform. The customer manages only app code, settings and data.

✖ **Yes**
In PaaS, OS patching moves to Microsoft. Developers manage only their code and data.

EXAM STRATEGY
PaaS means Microsoft owns the OS, so No. In App Service, developers manage only code, settings and data.

IMPACT
Teams can stop spending time on OS work and focus on app security. This is a key business benefit of PaaS.

What · Where · When · Why · How

What	Who looks after the OS under a PaaS service.
Where	On Azure App Service and other PaaS services.
When	All the time, as Microsoft rolls out platform updates.
Why	So developers can focus on code, not infrastructure.
How	Microsoft applies OS and runtime patches to App Service automatically.

Exam tip
In PaaS questions, the customer never touches the OS.

Memory trick
PaaS = Platform patched As a Service.

Real-world example, use case and scenario

EXAMPLE
Online shop
A shop's developers deploy code to App Service without ever signing in to a server.

USE CASE
Agency
A marketing agency hosts client websites on App Service with no OS maintenance staff.

SCENARIO
Security audit
An auditor asks who patches the App Service servers. The answer is Microsoft.

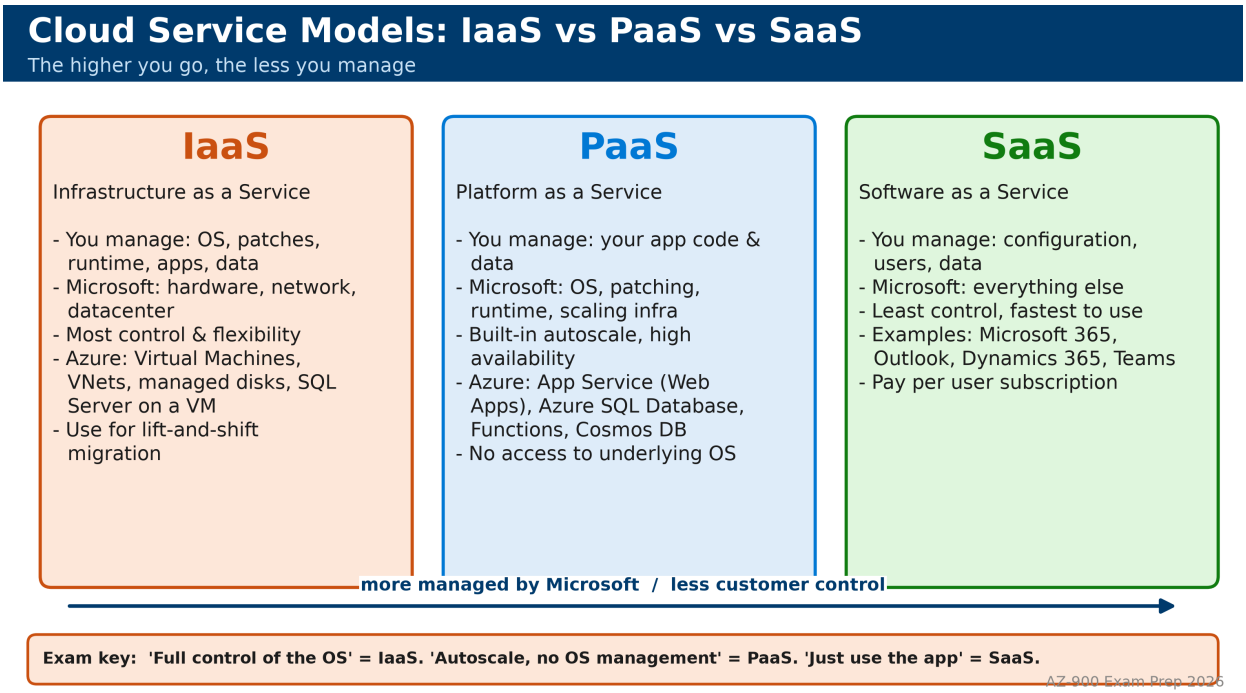


Figure Q278a: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

• Serverless options such as Azure Functions are treated as PaaS-style: no OS access, automatic scaling and per-execution billing.

• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

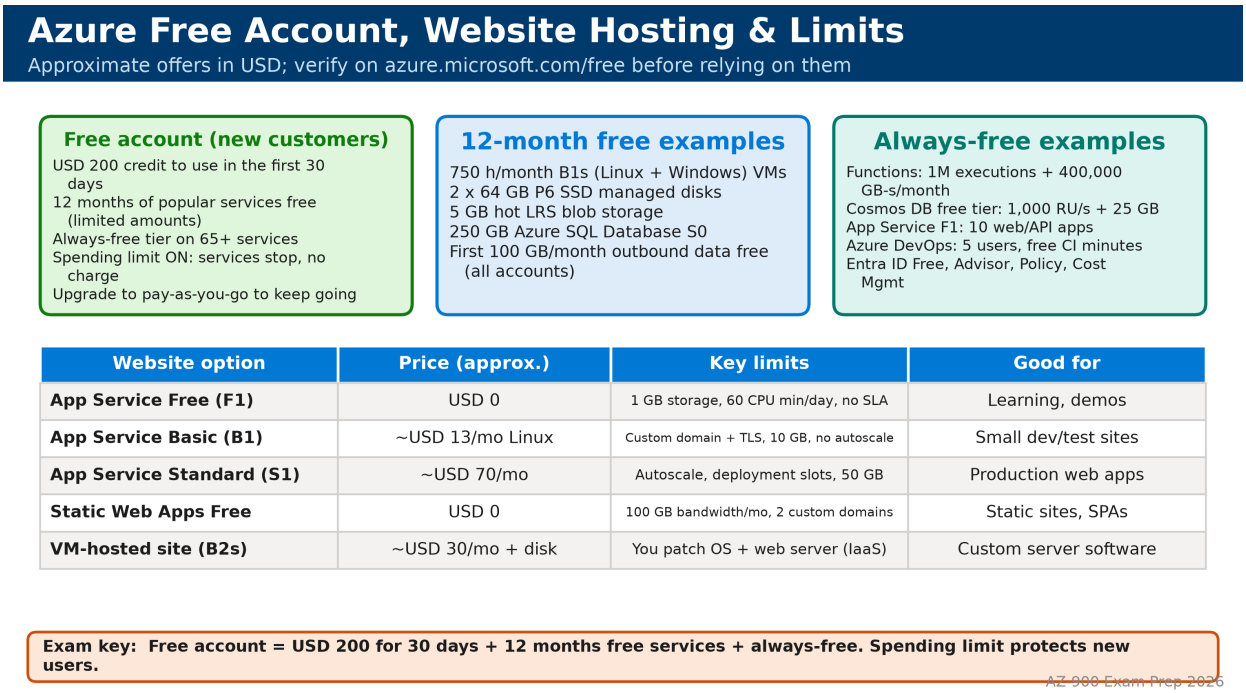


Figure Q278b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 279

CLOUD CONCEPTS

Cloud concepts
v5 Q410

Your company is evaluating Azure. The security officer asks who is responsible for the physical security of the datacenters that host your Azure resources. What should you answer?

✔ **A.** Microsoft, for every service model

✖ B. Your company, for IaaS services only

✖ C. Your company and Microsoft share it equally

✖ D. It depends on the Azure region selected

Correct answer: Microsoft, for every service model

Answer explained

✔ **Microsoft, for every service model**
The physical datacenter, network and hosts are ALWAYS Microsoft's job, in IaaS, PaaS and SaaS.

✖ **Your company, for IaaS services only**
Even in IaaS, customers cannot enter the building. Microsoft secures the facility.

✖ **Your company and Microsoft share it equally**
Physical security is not shared. It belongs fully to Microsoft.

✖ **It depends on the Azure region selected**
The rule is the same in every Azure region.

EXAM STRATEGY
Physical security is always Microsoft's, in every model and every region. Remove any option that gives the customer part of it.

IMPACT
In compliance audits, security officers use Microsoft's physical controls instead of checking datacenters themselves.

What · Where · When · Why · How

What	Protecting the buildings, hardware and physical entry of Azure datacenters.
Where	In every Azure datacenter in every region.
When	At all times, for all service models.
Why	Only the provider runs the buildings and can control them.
How	With guards, biometric entry (fingerprint or face checks), cameras and audited processes run by Microsoft.

Exam tip
The bottom of the shared responsibility chart (physical) is always the provider. The top (data, identities) is always the customer.

Memory trick
Bricks belong to Microsoft; bits of data belong to you.

Real-world example, use case and scenario

EXAMPLE
Insurance
An insurer's auditor uses Microsoft compliance reports for datacenter physical controls.

USE CASE
Government
A city council checks Azure datacenter physical security in the Service Trust Portal.

SCENARIO
Due diligence
A security chief removes 'datacenter guards' from the cloud budget because Microsoft provides them.

Shared Responsibility Model

Who manages each layer: on-premises vs IaaS vs PaaS vs SaaS

Responsibility	On-premises	IaaS	PaaS	SaaS
Information & data	Customer	Customer	Customer	Customer
Devices (mobile, PCs)	Customer	Customer	Customer	Customer
Accounts & identities	Customer	Customer	Customer	Customer
Identity & directory infra	Customer	Customer	Shared	Shared
Applications	Customer	Customer	Shared	Microsoft
Network controls	Customer	Customer	Shared	Microsoft
Operating system	Customer	Customer	Microsoft	Microsoft
Physical hosts	Customer	Microsoft	Microsoft	Microsoft
Physical network	Customer	Microsoft	Microsoft	Microsoft
Physical datacenter	Customer	Microsoft	Microsoft	Microsoft

Exam key: Data, devices, accounts & identities are ALWAYS the customer's. Physical datacenter is NEVER yours in cloud.

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Figure Q279a: Shared responsibility model

About this figure

• Read each row across: orange cells are yours, blue are Microsoft's, yellow are shared. Your share shrinks moving from on-premises to IaaS, PaaS and SaaS.

• Information and data, devices, and accounts and identities stay with the customer in every model, even in SaaS such as Microsoft 365.

• Physical hosts, network and datacenter move to Microsoft in every cloud model, so hardware failures, power and building security are never the customer's task.

• With IaaS you still patch the guest OS and configure network controls; with PaaS the OS becomes Microsoft's job, but your code and access settings remain yours.

• Responsibility does not equal liability: misconfigured access, weak passwords or leaked keys are customer failures even when the service itself is fully managed.

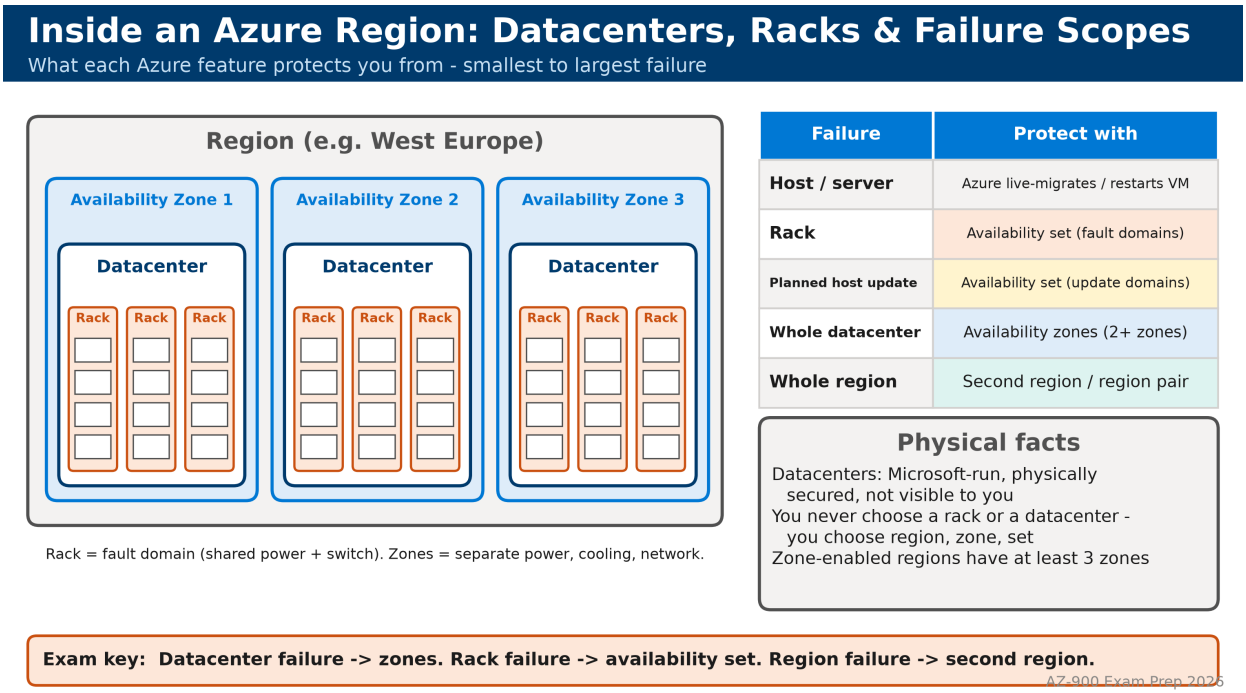


Figure Q279b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 280

CLOUD CONCEPTS

Deployment models
v5 Q411

Your company runs its CRM on Azure and its analytics platform on another public cloud provider, and has no on-premises datacenter. Which cloud model describes this approach?

✔ **A.** Multi-cloud

✖ B. Hybrid cloud

✖ C. Private cloud

✖ D. Single public cloud

Correct answer: Multi-cloud

Clarification (read this first): Hybrid needs an on-premises (or private) part. Here there is NO on-premises datacenter, only two public clouds, so it is multi-cloud, not hybrid.

Answer explained

✔ **Multi-cloud**
Using two or more public cloud providers at the same time is multi-cloud.

✖ **Hybrid cloud**
Hybrid joins private/on-premises with public cloud. There is no on-premises part here.

✖ **Private cloud**
A private cloud is used by one organization only. These are public providers.

✖ **Single public cloud**
The company uses two providers, not one.

EXAM STRATEGY
Two different public providers and no on-premises means multi-cloud. Remove hybrid, which needs a private or on-premises part.

IMPACT
Multi-cloud makes governance harder; Azure Arc helps manage it. Hybrid versus multi-cloud is a common exam trap.

What · Where · When · Why · How

What	Using more than one public cloud provider at the same time.
Where	Across Azure and other providers' datacenters.
When	When workloads need different providers' features, or during a move between clouds.
Why	To use the best service from each provider, or to depend less on one vendor.
How	Run workloads on each provider. You can manage them in one place with Azure Arc.

Exam tip
Hybrid = public + private. Multi-cloud = public + public (different providers).

Memory trick
MULTI = many providers; HYBRID = your own plus public.

Real-world example, use case and scenario

EXAMPLE
Media
A streaming company uses Azure for identity and another provider for video conversion.

USE CASE
Merger
After a merger, a company keeps both companies' different public clouds running.

SCENARIO
Lock-in worry
A fintech spreads workloads across two clouds so it does not depend on one vendor.

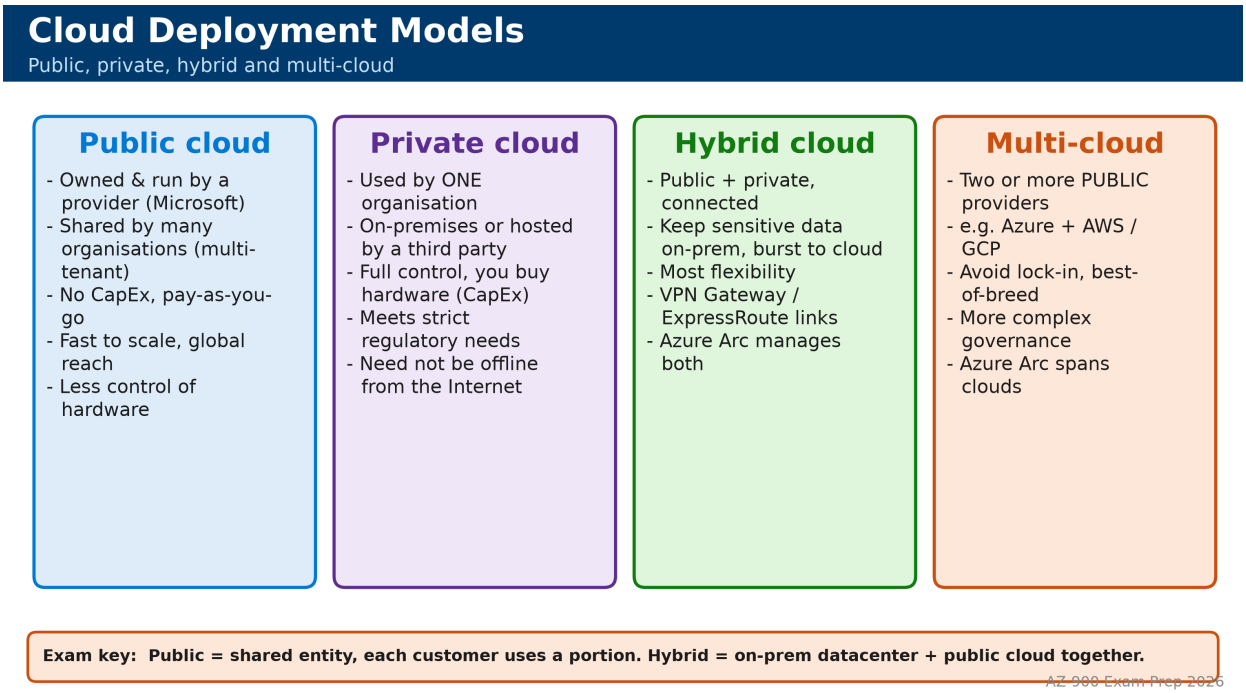


Figure Q280a: Cloud deployment models

About this figure

• Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.

• Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.

• Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.

• Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.

• Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

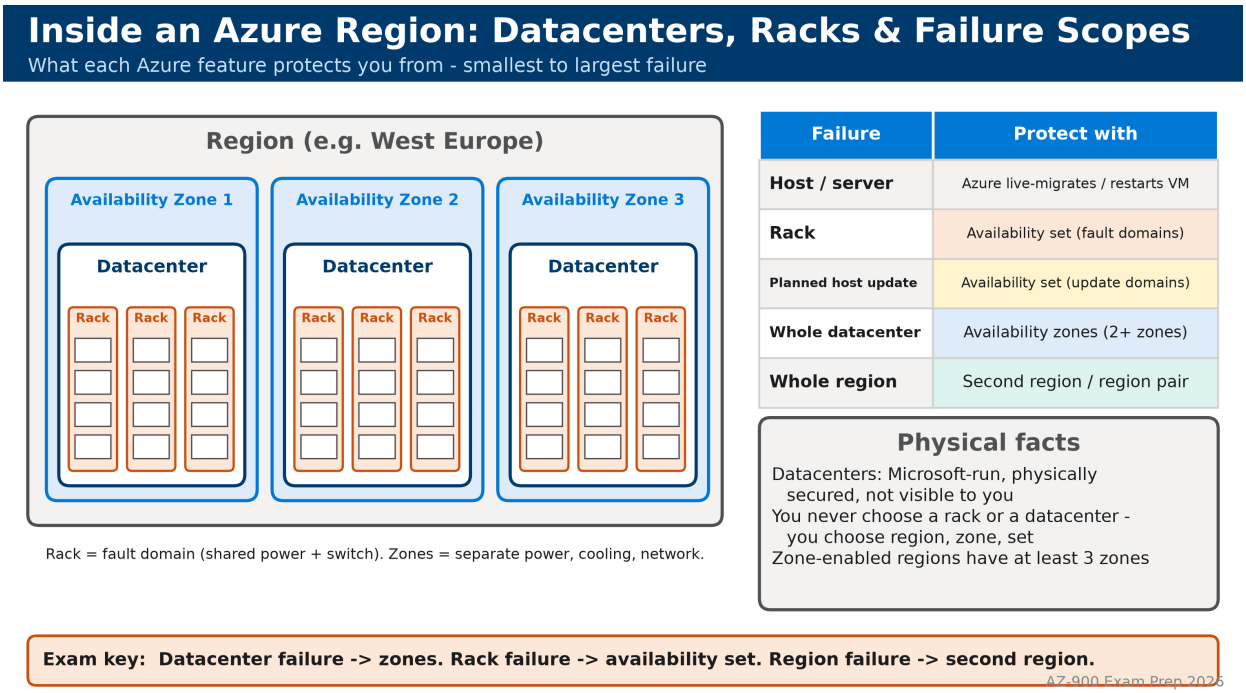


Figure Q280b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 281

CLOUD CONCEPTS

Deployment models
v5 Q412

Statement: A private cloud can be hosted in a datacenter operated by a third-party service provider rather than on the organization's own premises.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): Private cloud is about WHO uses it (one organization only), not WHERE it is. A third party can host it. Q269 is related: a private cloud does not have to be cut off from the internet (No).

Answer explained

✔ **Yes**
A private cloud is used by one organization only. It can be on-premises OR hosted in another company's datacenter.

✖ **No**
Location does not define a private cloud. Use by only one organization does.

EXAM STRATEGY
Private means used by one organization, not a place. A dedicated environment hosted by a third party is still private, so Yes.

IMPACT
Hosted private clouds give isolation without owning a datacenter. Do not treat 'private' as meaning 'on-premises'.

What · Where · When · Why · How

What	A cloud environment used by one organization only.
Where	On-premises, or in a datacenter run by a third party.
When	When an organization needs full control and separation of its infrastructure.
Why	To meet strict control, compliance or custom needs.
How	Give hardware and cloud software to one organization only, wherever it is hosted.

Exam tip
Do not treat 'private cloud' as 'on-premises'. A hosted, dedicated environment is still private.

Memory trick
Private means 'mine alone', not 'in my building'.

Real-world example, use case and scenario

EXAMPLE
Defense contractor
A contractor rents hosted infrastructure that only it can use.

USE CASE
Hospital group
A hospital group runs a private cloud in a partner's secure building.

SCENARIO
Exam trap
A question says private clouds must be in your building. The statement is false.

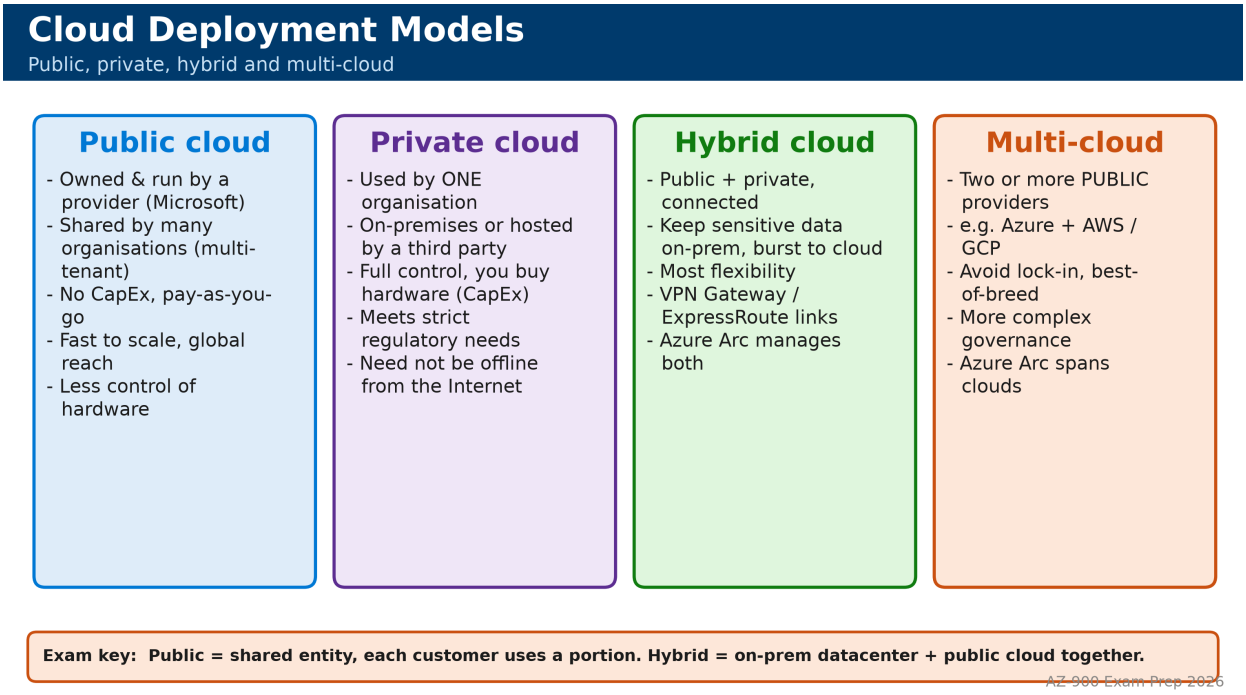


Figure Q281a: Cloud deployment models

About this figure

- Compare the four columns: who owns the hardware, who shares it, and whether workloads span more than one environment or provider.
- Public cloud is multi-tenant but isolated; customers share Microsoft's hardware, not each other's data. No CapEx, consumption pricing.
- Private cloud can be in your datacenter or a hoster's, and can still be Internet-connected; you carry the hardware cost and maintenance.
- Hybrid links private and public, typically via VPN Gateway over the Internet or ExpressRoute private circuits, often to keep regulated data on-premises.
- Azure Arc extends Azure management (Policy, RBAC, Monitor, Defender) to servers and Kubernetes on-premises, at the edge, or in AWS and Google Cloud.

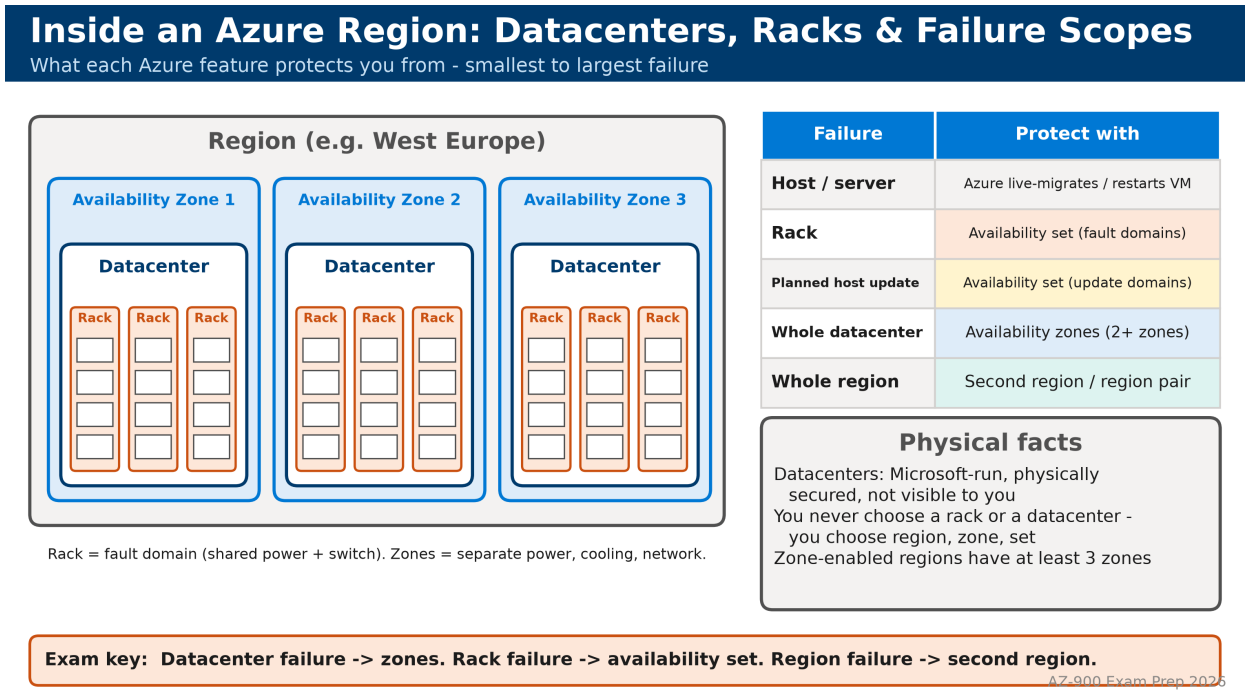


Figure Q281b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
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- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 282

CLOUD CONCEPTS

Cost
v5 Q413, Q330

Your company is moving from on-premises servers to Azure using the consumption-based model. Which statement describes this model?

✔ **A.** There are no up-front costs, and you pay only for the resources you use

✖ B. You pay a fixed annual fee regardless of how many resources you use

✖ C. You must purchase capacity for peak demand before you deploy

✖ D. You pay for hardware up front and recover the cost through depreciation

Correct answer: There are no up-front costs, and you pay only for the resources you use

Answer explained

✔ **There are no up-front costs, and you pay only for the resources you use**
Consumption-based means no cost up front. You pay only for what you use, and you stop paying when you stop using it.

✖ **You pay a fixed annual fee regardless of how many resources you use**
A fixed fee ignores usage. Consumption pricing is measured.

✖ **You must purchase capacity for peak demand before you deploy**
Buying peak capacity in advance is the on-premises way that cloud avoids.

✖ **You pay for hardware up front and recover the cost through depreciation**
Hardware bought up front and written off over time (depreciation) is CapEx, not the consumption model.

EXAM STRATEGY
Look for both signs: no up-front cost and pay only for use. Remove fixed yearly fees, buying for peak in advance, and depreciation.

IMPACT
With consumption billing, resources you leave running still cost money. Turning off unused resources is the main way to save.

What · Where · When · Why · How

What	A pricing model where you pay only for what you use.
Where	In most Azure services, such as VMs, storage and Functions.
When	Every billing period, based on measured use.
Why	To avoid buying too much and to avoid large starting costs.
How	Azure measures use (hours, GB, runs) and bills for it.

Exam tip
Look for 'no up-front cost' and 'pay for what you use'. These are the two signs of the consumption model.

Memory trick
Like an electricity bill: use more, pay more; switch off, pay less.

Real-world example, use case and scenario

EXAMPLE
Event tickets
A ticket site pays for large capacity only during a concert pre-sale.

USE CASE
Research lab
A lab runs a simulation for one week and pays only for that week.

SCENARIO
Budget meeting
The finance chief approves a cloud project with no hardware purchase order.

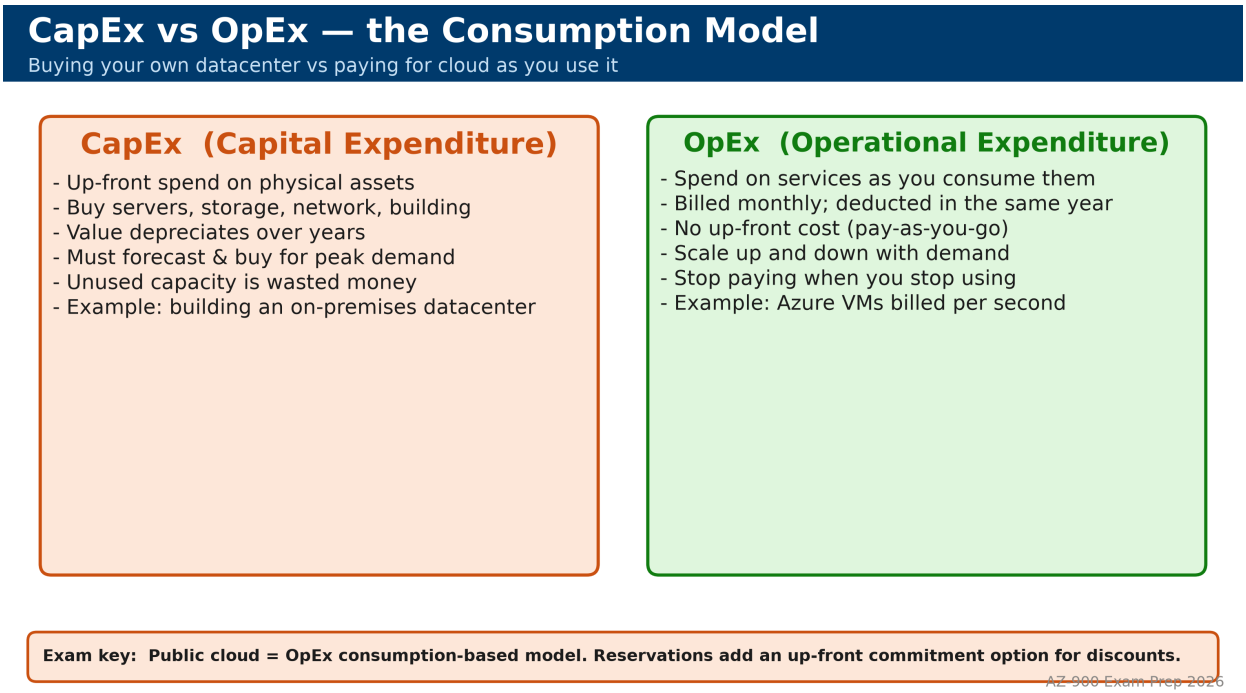


Figure Q282: CapEx vs OpEx

About this figure

• Left box is buying assets up front (CapEx); right box is paying for consumption (OpEx). Cloud moves spending from left to right.

• CapEx assets are depreciated over several years; OpEx is expensed in the period it is incurred, which suits variable or uncertain demand.

• The consumption model means no up-front cost, pay only for what you use, and stop paying when resources are deleted or deallocated.

• Reservations and savings plans add a 1- or 3-year commitment for discounts, but they remain operational spending, not purchased hardware.

• Use the TCO calculator to compare on-premises CapEx (hardware, power, labour) with expected Azure OpEx before migrating.

Question 283

CLOUD CONCEPTS

Cost
v5 Q414

You stop an Azure virtual machine from the Azure portal so that its status is Stopped (deallocated). Statement: The virtual machine continues to incur compute charges while it is deallocated.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): Watch the word 'compute'. A deallocated VM (stopped from Azure, so the hardware is released) does NOT pay for compute. It still pays for its disks (storage) - see Q284 and Q236, where the answer is Yes / Storage.

Answer explained

✔ **No**
Stopped (deallocated) means no compute charges. Disks and static IPs (fixed public IP addresses) are still billed.

✖ **Yes**
Deallocation releases the host hardware, so compute billing stops.

EXAM STRATEGY
Stopped (deallocated) releases the compute, so there are no compute charges: No. Only a shutdown from inside the OS keeps compute billing.

IMPACT
If you shut down inside the OS instead of deallocating, compute charges keep running. This is a common waste in test environments.

What · Where · When · Why · How

What	Whether a deallocated VM is billed for compute.
Where	On Azure VMs stopped from the portal, CLI or PowerShell.
When	For as long as the VM stays Stopped (deallocated).
Why	The consumption model charges only for compute that is assigned to you.
How	Deallocating the VM releases its vCPU (virtual processor) and memory.

Exam tip
Shut down inside the OS = still assigned and billed. Stopped (deallocated) from Azure = no compute charge.

Memory trick
Deallocated = de-billed for compute.

Real-world example, use case and scenario

EXAMPLE
Dev team
Developers deallocate test VMs every evening to save compute costs.

USE CASE
Training lab
A training company deallocates classroom VMs between courses.

SCENARIO
Auto-shutdown
An admin turns on auto-shutdown at 7 PM to stop compute billing overnight.

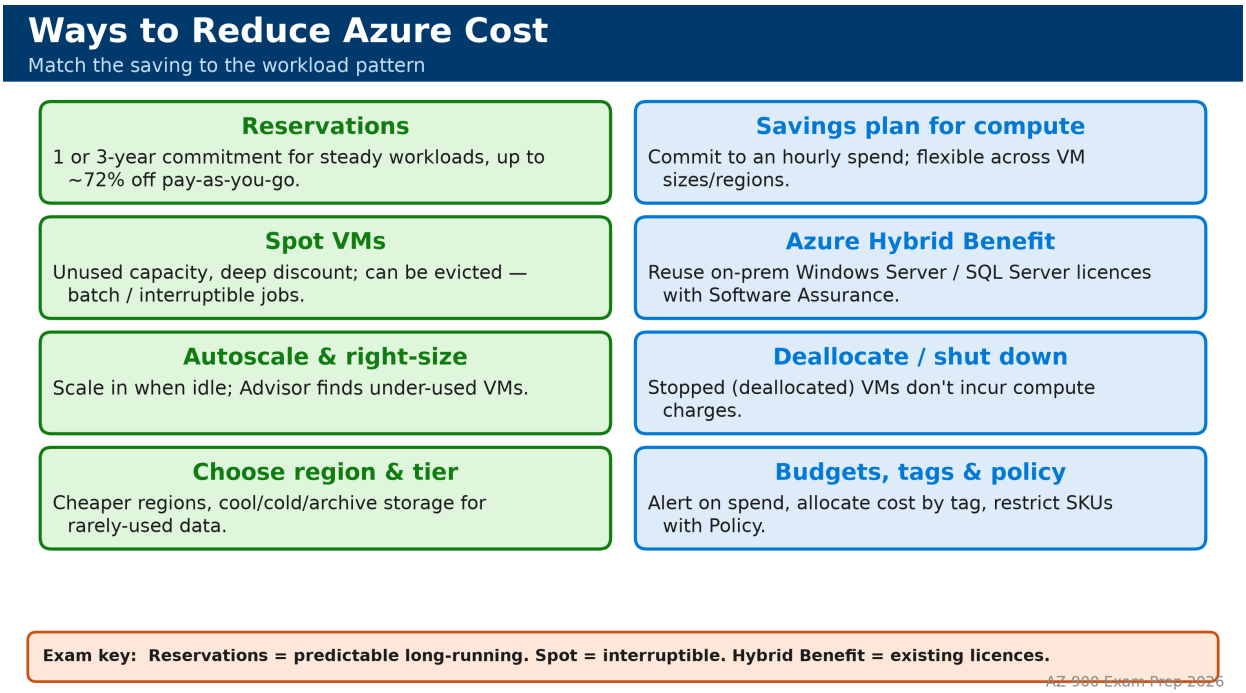


Figure Q283a: Ways to reduce cost

About this figure

• Each box matches a saving to a workload pattern: steady (reservations), flexible (savings plan), interruptible (Spot), existing licences (Hybrid Benefit).

• Reservations save up to approx. 72% versus pay-as-you-go; savings plans for compute save up to approx. 65% and apply across VM families and regions.

• Spot VMs can be up to approx. 90% cheaper, but Azure can evict them with 30 seconds' notice when it needs capacity back.

• Combining Azure Hybrid Benefit with reservations can cut Windows VM costs by up to approx. 80%; Linux subscriptions such as RHEL and SUSE also qualify.

• Example: moving rarely read data from Hot (~USD 0.02/GB-month) to Archive (~USD 0.002 or less) saves storage cost, but reads and early deletion cost more.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q283b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 284

CLOUD CONCEPTS

Cost
v5 Q415

You deallocate an Azure virtual machine that uses a managed OS disk. Statement: You continue to be charged for the managed disk storage while the VM is deallocated.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): This asks about STORAGE (the managed disk), not compute. The disk still exists, so it is still billed: Yes. Compare Q283: compute charges stop (No). Q236 says the same: you still pay for storage.

Answer explained

✔ **Yes**
For a deallocated VM, compute billing stops, but managed disks (storage) are still billed.

✖ **No**
The disk still exists and is billed for storage until you delete it.

EXAM STRATEGY
Disks stay when a VM is deallocated, and storage is billed on its own, so Yes. Compute stops; storage continues.

IMPACT
Forgotten disks from stopped or deleted VMs add charges every month. Azure Advisor points them out for clean-up.

What · Where · When · Why · How

What	Ongoing storage charges for a deallocated VM's disks.
Where	On managed disks attached to stopped Azure VMs.
When	For as long as the disk exists.
Why	The disk data is kept and uses storage space.
How	Billed each month by the disk's size and type.

Exam tip
Remember the split: compute stops, storage continues.

Memory trick
The car is parked (no fuel), but you still pay for the garage.

Real-world example, use case and scenario

EXAMPLE
Old project
A company deallocates old project VMs but still sees disk charges on the bill.

USE CASE
Cost review
A cost analyst finds unattached disks from deleted VMs and removes them.

SCENARIO
Clean-up
An admin deletes a VM's disks after retiring it, to stop all charges.

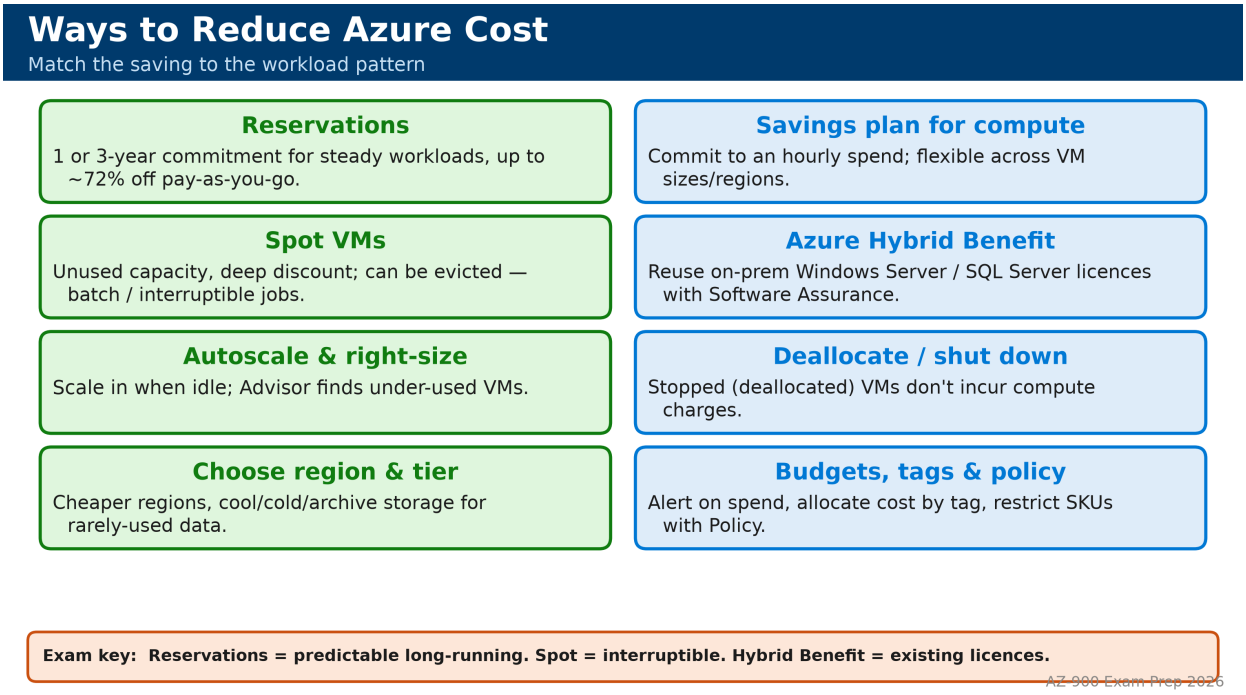


Figure Q284a: Ways to reduce cost

About this figure

- Each box matches a saving to a workload pattern: steady (reservations), flexible (savings plan), interruptible (Spot), existing licences (Hybrid Benefit).
- Reservations save up to approx. 72% versus pay-as-you-go; savings plans for compute save up to approx. 65% and apply across VM families and regions.
- Spot VMs can be up to approx. 90% cheaper, but Azure can evict them with 30 seconds' notice when it needs capacity back.
- Combining Azure Hybrid Benefit with reservations can cut Windows VM costs by up to approx. 80%; Linux subscriptions such as RHEL and SUSE also qualify.
- Example: moving rarely read data from Hot (~USD 0.02/GB-month) to Archive (~USD 0.002 or less) saves storage cost, but reads and early deletion cost more.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
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Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q284b: Approximate Azure charges (USD)

About this figure

• Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.

• Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.

• Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.

• Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.

• Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 285

CLOUD CONCEPTS

Cost

v5 Q416, Q419

Your company runs nightly video-rendering batch jobs that can be restarted if they are interrupted. You need the lowest possible compute cost. Which pricing option should you use?

✔ **A.** Azure Spot Virtual Machines

✖ B. Azure Reservations

✖ C. Azure Hybrid Benefit

✖ D. Pay-as-you-go virtual machines

Correct answer: Azure Spot Virtual Machines

Clarification (read this first): The key words are 'can be restarted if interrupted' and 'lowest cost'. That points to Spot VMs.

Answer explained

✔ **Azure Spot Virtual Machines**
A job that can be interrupted plus lowest cost = Azure Spot VMs. They use spare capacity, can be removed by Azure at any time, and have no SLA (uptime promise).

✖ **Azure Reservations**
Reservations suit steady, always-on workloads with a 1- or 3-year commitment.

✖ **Azure Hybrid Benefit**
Hybrid Benefit lets you reuse Windows/SQL licences you already own. It does not use spare capacity.

✖ **Pay-as-you-go virtual machines**
Pay-as-you-go is flexible but costs much more than Spot for jobs that can be interrupted.

EXAM STRATEGY
'Can be restarted' plus 'lowest cost' = Spot VMs. Remove Reservations (steady 24/7 use), Hybrid Benefit (licences) and pay-as-you-go (most expensive).

IMPACT
Spot gives large discounts for batch jobs that can be interrupted. Using it for critical work risks outages when Azure takes the VMs back (eviction).

What · Where · When · Why · How

What	A discounted VM price option that uses unused Azure capacity.
Where	In Azure regions where spare capacity exists.
When	For work that can handle being stopped and restarted.
Why	To get the lowest compute price for non-critical work.
How	Deploy VMs with Spot priority and an eviction policy (what happens when Azure takes them back).

Exam tip
'Can be interrupted', 'batch', 'lowest cost' point to Spot VMs. 'Runs 24/7 for years' points to reservations.

Memory trick
Spot = Spare capacity, Possibly Taken back.

Real-world example, use case and scenario

EXAMPLE
Animation studio
A studio renders video frames overnight on Spot VMs and continues if a VM is taken back.

USE CASE
Genomics
Researchers run restartable DNA jobs on Spot VMs to make their budget last longer.

SCENARIO
Build servers
A software team runs non-urgent build servers on Spot VMs.

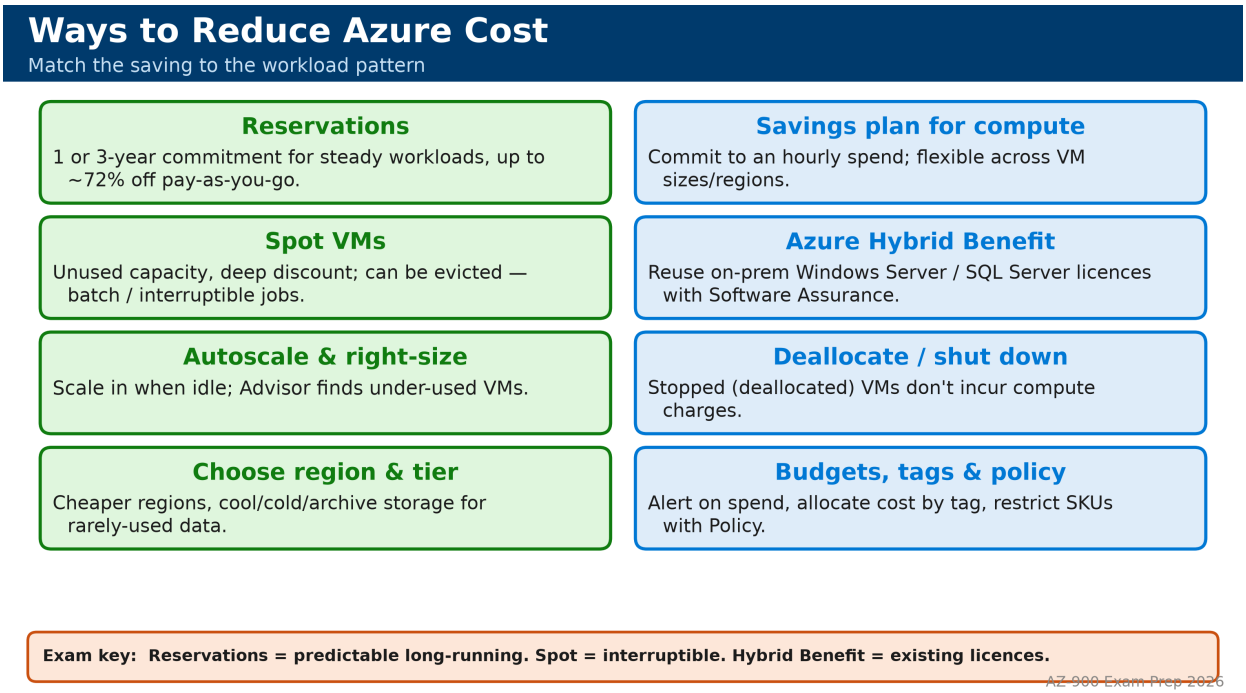


Figure Q285a: Ways to reduce cost

About this figure

• Each box matches a saving to a workload pattern: steady (reservations), flexible (savings plan), interruptible (Spot), existing licences (Hybrid Benefit).

• Reservations save up to approx. 72% versus pay-as-you-go; savings plans for compute save up to approx. 65% and apply across VM families and regions.

• Spot VMs can be up to approx. 90% cheaper, but Azure can evict them with 30 seconds' notice when it needs capacity back.

• Combining Azure Hybrid Benefit with reservations can cut Windows VM costs by up to approx. 80%; Linux subscriptions such as RHEL and SUSE also qualify.

• Example: moving rarely read data from Hot (~USD 0.02/GB-month) to Archive (~USD 0.002 or less) saves storage cost, but reads and early deletion cost more.

Approximate Azure Charges (USD)

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Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q285b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 286

CLOUD CONCEPTS

Cost
v5 Q417

Your company commits to a fixed hourly spend on compute for three years. Workloads frequently change VM families and Azure regions. You need a discount that applies automatically across these changes. What should you use?

✔ **A.** Azure savings plan for compute

✖ B. Azure Reservations for a specific VM size

✖ C. Azure Spot Virtual Machines

✖ D. Azure Dev/Test pricing

Correct answer: Azure savings plan for compute

Clarification (read this first): The clue is that the VM types and regions keep CHANGING. A reservation is locked to one VM size and region. A savings plan commits to an amount of money per hour, so it follows the changes.

Answer explained

✔ **Azure savings plan for compute**
Commit to an hourly spend and keep freedom to change VM families and regions = Azure savings plan for compute.

✖ **Azure Reservations for a specific VM size**
Reservations are tied to a resource type and region, so they do not have this flexibility.

✖ **Azure Spot Virtual Machines**
Spot uses spare capacity that Azure can take back. It has no spending commitment.

✖ **Azure Dev/Test pricing**
Dev/Test pricing is only for non-production work in eligible subscriptions.

EXAM STRATEGY
'Fixed hourly spend' plus 'changes VM families and regions' = savings plan. Remove reservations, which lock a size and region.

IMPACT
Reservations on changing workloads leave discounts unused. Savings plans keep the discount working as the design changes.

What · Where · When · Why · How

What	A pricing model that gives a compute discount in return for a fixed hourly spend.
Where	Across eligible compute services and Azure regions.
When	When spending is steady but VM size or location changes.
Why	To save money without being locked to one VM size or region.
How	Buy a 1-year or 3-year plan with a fixed hourly amount.

Exam tip
Reservation = a specific resource. Savings plan = a specific spend. A flexibility clue means savings plan.

Memory trick
Savings plan = commit dollars, not machines.

Real-world example, use case and scenario

EXAMPLE
SaaS vendor
A SaaS company moves workloads between regions every month but keeps a steady compute budget.

USE CASE
Upgrade
A team moving from older VM series to newer ones uses a savings plan to keep its discount.

SCENARIO
Finance
Finance commits to a fixed hourly compute amount, and Azure applies the discount automatically.

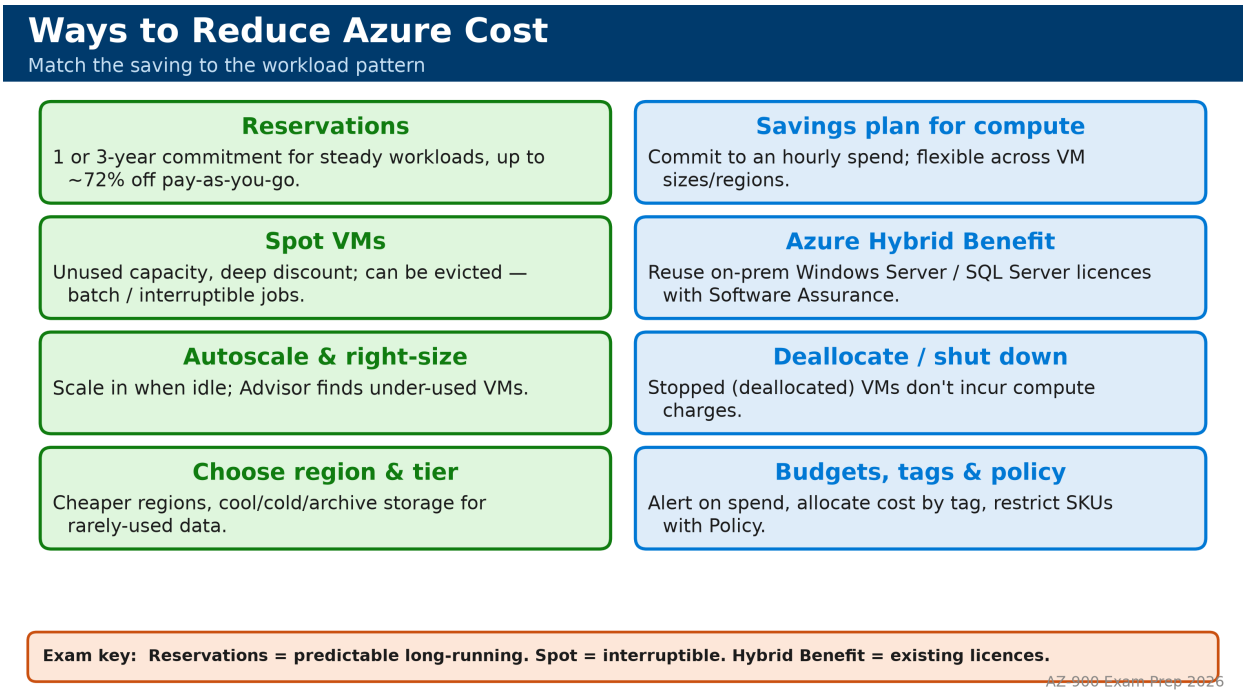


Figure Q286a: Ways to reduce cost

About this figure

• Each box matches a saving to a workload pattern: steady (reservations), flexible (savings plan), interruptible (Spot), existing licences (Hybrid Benefit).

• Reservations save up to approx. 72% versus pay-as-you-go; savings plans for compute save up to approx. 65% and apply across VM families and regions.

• Spot VMs can be up to approx. 90% cheaper, but Azure can evict them with 30 seconds' notice when it needs capacity back.

• Combining Azure Hybrid Benefit with reservations can cut Windows VM costs by up to approx. 80%; Linux subscriptions such as RHEL and SUSE also qualify.

• Example: moving rarely read data from Hot (~USD 0.02/GB-month) to Archive (~USD 0.002 or less) saves storage cost, but reads and early deletion cost more.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
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Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q286b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
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- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 287

CLOUD CONCEPTS

Cost
v5 Q418

You plan to buy Azure Reservations for virtual machines that will run continuously. Which commitment terms can you choose?

✔ **A.** 1 year or 3 years

✖ B. 1 month or 6 months

✖ C. 6 months or 1 year

✖ D. 5 years only

Correct answer: 1 year or 3 years

Answer explained

✔ **1 year or 3 years**
Azure Reservations last 1 year or 3 years. 3 years gives a bigger discount.

✖ **1 month or 6 months**
Reservations are not sold for monthly or 6-month terms.

✖ **6 months or 1 year**
There is no 6-month reservation term.

✖ **5 years only**
There is no 5-year term. The longest is 3 years.

EXAM STRATEGY
Commitment terms are always 1 or 3 years. Remove any monthly, 6-month or 5-year option at once.

IMPACT
3-year terms give the biggest discount but lock you in if workloads change. Match the term to how long the workload will run.

What · Where · When · Why · How

What	How long you can commit to an Azure Reservation.
Where	In the Azure portal under Reservations.
When	When workloads are predictable and run all the time.
Why	To pay less than pay-as-you-go prices.
How	Choose a 1-year or 3-year term. Pay up front or monthly.

Exam tip
Any AZ-900 pricing commitment (reservations, savings plans) uses 1- or 3-year terms.

Memory trick
Reservations come in 1 and 3, like a short and a long lease.

Real-world example, use case and scenario

EXAMPLE
Retail ERP
A retailer reserves VMs for its always-on ERP system for three years.

USE CASE
Database
A hospital reserves database compute for one year while it checks growth.

SCENARIO
Budget planning
Finance compares 1-year and 3-year reservation prices in the Pricing calculator.

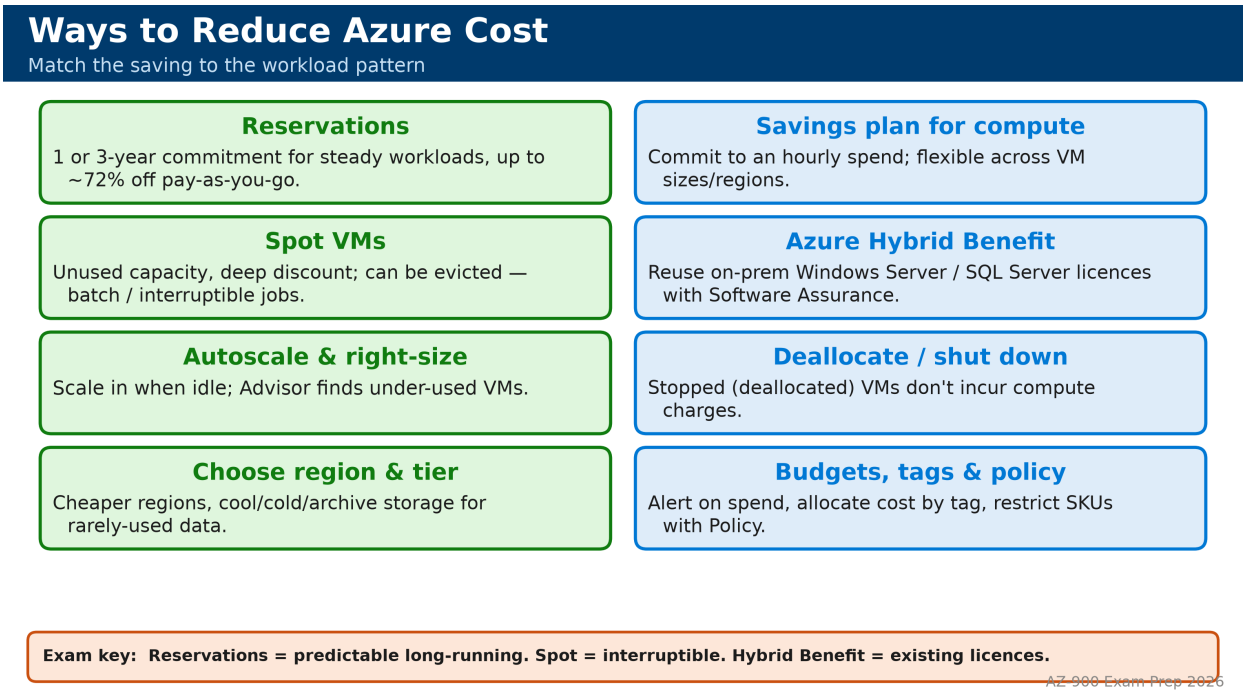


Figure Q287a: Ways to reduce cost

About this figure

• Each box matches a saving to a workload pattern: steady (reservations), flexible (savings plan), interruptible (Spot), existing licences (Hybrid Benefit).

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• Combining Azure Hybrid Benefit with reservations can cut Windows VM costs by up to approx. 80%; Linux subscriptions such as RHEL and SUSE also qualify.

• Example: moving rarely read data from Hot (~USD 0.02/GB-month) to Archive (~USD 0.002 or less) saves storage cost, but reads and early deletion cost more.

Approximate Azure Charges (USD)

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Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q287b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
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- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 288

CLOUD CONCEPTS

Cloud concepts
v5 Q420, Q421

Your developers plan to use a serverless approach for an order-processing component. Which two characteristics apply to serverless computing? Each correct answer presents a complete solution. NOTE: Each correct selection is worth one point. (Choose two.)

✔ **A.** You are billed only when your code runs or for the resources it consumes

✔ **B.** The cloud provider manages the underlying servers and scaling

✖ C. You select and patch the virtual machine operating system

✖ D. You must pre-provision instances for peak load

✖ E. You pay a fixed hourly rate whether or not code runs

Correct answer: You are billed only when your code runs or for the resources it consumes; The cloud provider manages the underlying servers and scaling

Clarification (read this first): Pick TWO. Serverless still uses servers, but you never see or manage them. Remove any option where YOU manage, patch or pre-size machines.

Answer explained

✔ **You are billed only when your code runs or for the resources it consumes; The cloud provider manages the underlying servers and scaling**
Serverless means no server management, automatic scaling when events happen, and paying per run.

✖ **You select and patch the virtual machine operating system**
Choosing and patching the OS is IaaS. Serverless hides the servers.

✖ **You must pre-provision instances for peak load**
Serverless scales by itself, so you do not prepare instances in advance.

✖ **You pay a fixed hourly rate whether or not code runs**
Serverless consumption billing is per run, not a fixed hourly fee.

EXAM STRATEGY
Pick 'provider manages servers' and 'pay per run'. Remove OS patching, pre-sizing and fixed hourly rates, which describe VMs.

IMPACT
Serverless can bring costs close to zero for work that runs now and then. Azure Functions is the standard AZ-900 example.

What · Where · When · Why · How

What	A cloud model where code runs without the customer managing servers.
Where	In services such as Azure Functions and Azure Logic Apps.
When	When an event, such as a web request or a queue message, starts the code.
Why	To reduce operations work and pay only for real use.
How	Deploy functions. The platform scales them and bills per run.

Exam tip
Serverless still uses servers. The customer just never sees or manages them.

Memory trick
Serverless = server-careless, and pay per call.

Real-world example, use case and scenario

EXAMPLE
Online store
An Azure Function sends an order confirmation each time an order message arrives.

USE CASE
IoT
A utility processes meter readings with functions started by incoming events.

SCENARIO
Image processing
A photo app resizes uploads with a function and pays nothing when no one uploads.

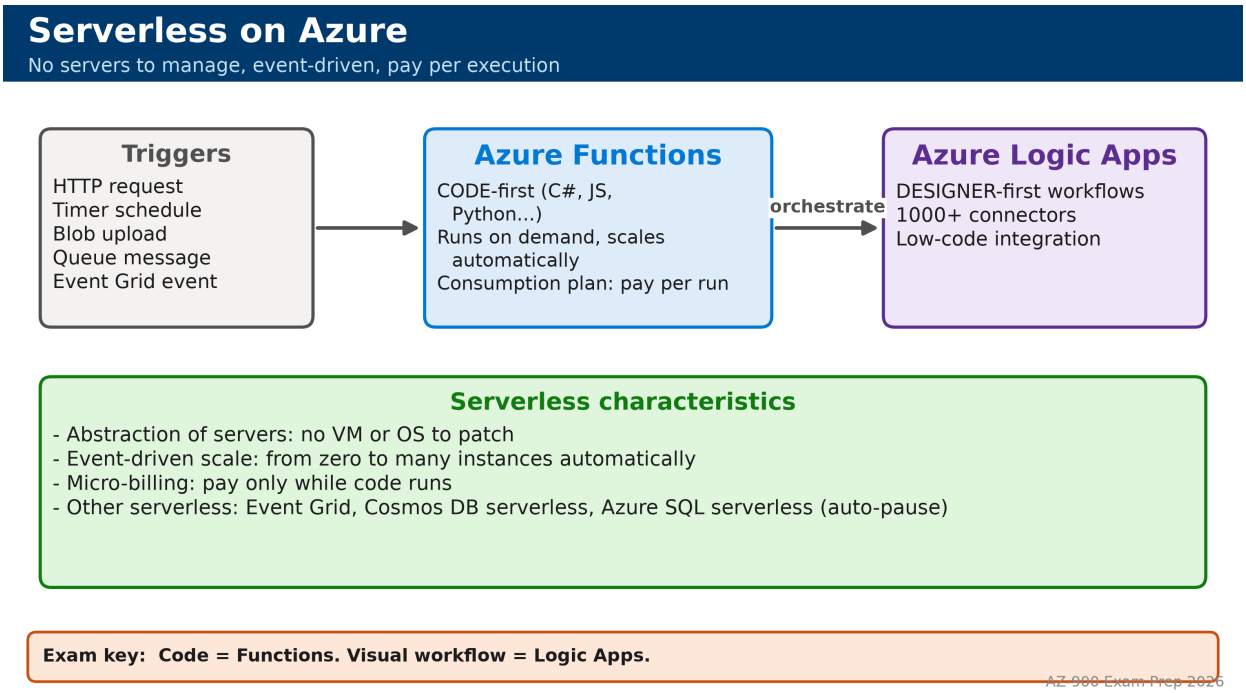


Figure Q288: Serverless on Azure

About this figure

• Flow left to right: an event trigger starts code (Functions) or a workflow (Logic Apps); the green box lists traits shared by serverless services.

• Functions Consumption: approx. USD 0.20 per million executions plus GB-seconds, with 1 million executions and 400,000 GB-s free monthly.

• Consumption-plan functions can cold start and have a default 5-minute, maximum 10-minute timeout; Flex Consumption and Premium plans reduce cold starts.

• Logic Apps Consumption bills per action and connector call, roughly fractions of a cent each; Standard plan bills per hosting plan instead.

• Serverless still runs on servers; Microsoft manages and scales them, and you pay nothing while nothing runs, except storage or plan minimums.

Question 289

CLOUD CONCEPTS

Cloud concepts
v5 Q422

An Azure virtual machine that runs a reporting database is running low on memory. You resize the VM to a size that has more vCPUs and RAM. What is this an example of?

✔ **A.** Vertical scaling (scaling up)

✖ B. Horizontal scaling (scaling out)

✖ C. Load balancing

✖ D. Fault tolerance

Correct answer: Vertical scaling (scaling up)

Clarification (read this first): One VM became BIGGER (more vCPUs and RAM) = scale up (vertical). Adding MORE VMs is scale out (horizontal) - see Q290.

Answer explained

✔ **Vertical scaling (scaling up)**
Making one machine bigger (more CPU and RAM) is vertical scaling, also called scaling up.

✖ **Horizontal scaling (scaling out)**
Scaling out adds more instances. Here one VM got bigger.

✖ **Load balancing**
Load balancing spreads traffic across instances. It does not add power to one VM.

✖ **Fault tolerance**
Fault tolerance means continuing to run after a failure. Resizing is about capacity.

EXAM STRATEGY
More CPU and RAM on the same VM = vertical scaling (scale up). Remove horizontal (more instances), load balancing and fault tolerance.

IMPACT
Resizing a VM usually needs a restart, which causes a short outage. Plan scale-up changes for a maintenance time.

What · Where · When · Why · How

What	Increasing the capacity of one resource.
Where	On Azure VMs, Azure SQL Database tiers and App Service plans.
When	When one instance needs more power for its work.
Why	To improve performance without changing the app to use many instances.
How	Change the VM size or service tier to one with more resources.

Exam tip
Up/down = size of one resource (vertical). Out/in = number of resources (horizontal).

Memory trick
Vertical = taller building; horizontal = more buildings.

Real-world example, use case and scenario

EXAMPLE
Finance reports
A finance team makes its reporting VM bigger before the quarter-end close.

USE CASE
Database
A retailer moves its Azure SQL Database to a higher tier for holiday traffic.

SCENARIO
Scale down
After a project ends, an admin changes a VM to a smaller size to save money.

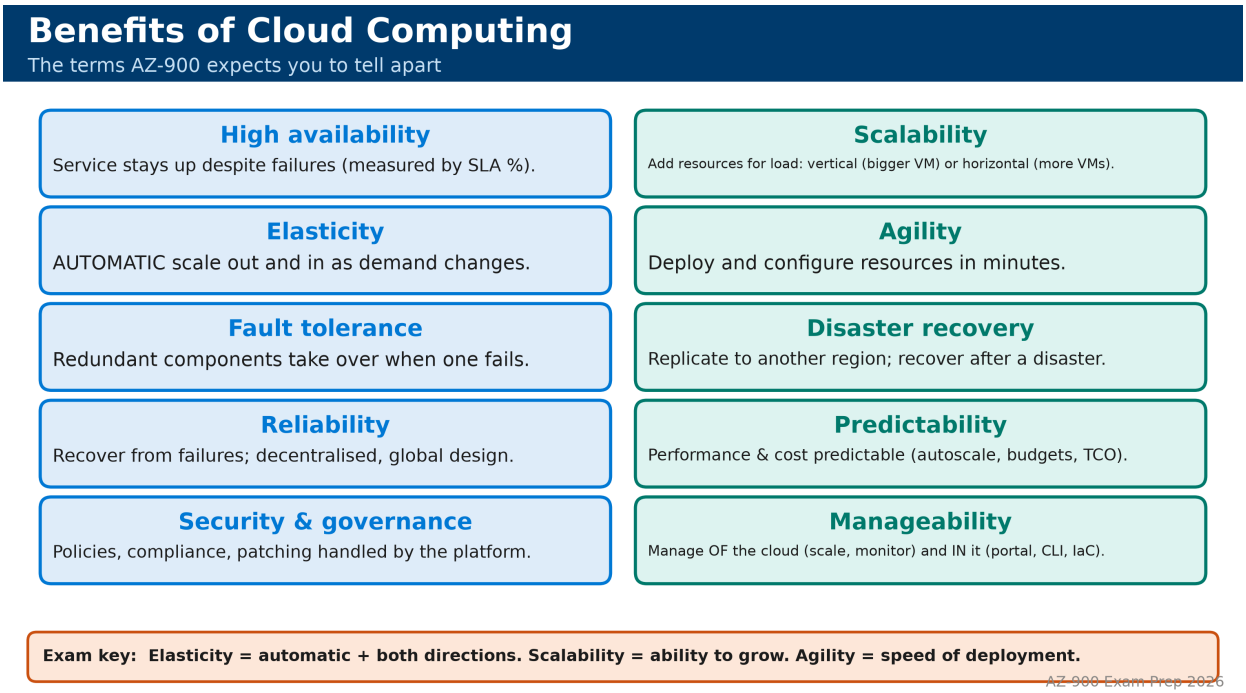


Figure Q289: Benefits of cloud computing

About this figure

- Each box pairs a term with its meaning; the exam often gives a scenario and asks which benefit it describes, so focus on keywords.
- Scalability is the capacity to grow, manual or automatic; elasticity is automatic scaling out and back in with demand, for example VM Scale Sets autoscale.
- High availability is expressed as an SLA percentage such as 99.9% or 99.99%; fault tolerance and disaster recovery are ways of achieving it.
- Agility means provisioning in minutes instead of weeks of hardware procurement; predictability covers both performance and cost forecasting.
- Governance tools such as Azure Policy, RBAC and locks deliver the security and governance benefit; manageability covers portal, CLI, PowerShell and IaC.

Question 290

CLOUD CONCEPTS

Cloud concepts
v5 Q423

Adding more virtual machine instances behind a load balancer to handle increased traffic is an example of [vertical scaling].

✖ A. No change is needed.

✔ **B.** Horizontal scaling

✖ C. Vertical elasticity

✖ D. Disaster recovery

Correct answer: Horizontal scaling

Clarification (read this first): This is a 'fix the bracketed word' question. The bracket says 'vertical scaling', which is wrong, so do NOT pick 'No change is needed'. Adding MORE VMs is horizontal. Compare Q289: one bigger VM is vertical.

Answer explained

✔ **Horizontal scaling**
More instances = horizontal scaling (scale out). A bigger instance = vertical scaling (scale up).

✖ **No change is needed.**
Vertical scaling makes one VM bigger. Adding instances is horizontal, so the sentence must change.

✖ **Vertical elasticity**
This is not the term for adding instances. Adding instances is horizontal scaling.

✖ **Disaster recovery**
Disaster recovery brings service back after a disaster. It is not about adding capacity.

EXAM STRATEGY
'More instances' = horizontal scaling. The bracketed 'vertical' is wrong. Remove vertical elasticity and disaster recovery.

IMPACT
Scaling out adds capacity and resilience without downtime. Autoscale rules on VM scale sets do this in practice.

What · Where · When · Why · How

What	Adding more instances to share the work.
Where	Often with Azure Virtual Machine Scale Sets and App Service.
When	When demand is more than the current instances can handle.
Why	To add capacity and resilience without one bigger machine.
How	Add instances by hand or with autoscale rules.

Exam tip
If the question says 'more instances', 'more VMs' or 'scale set', the answer is horizontal.

Memory trick
Horizontal = add more horses to the team.

Real-world example, use case and scenario

EXAMPLE
Ticketing
A ticket site grows from 4 to 20 web VMs during a sale.

USE CASE
News
A news site adds App Service instances during breaking news.

SCENARIO
Gaming
A game back end adds VMs to its scale set when players log in at night.

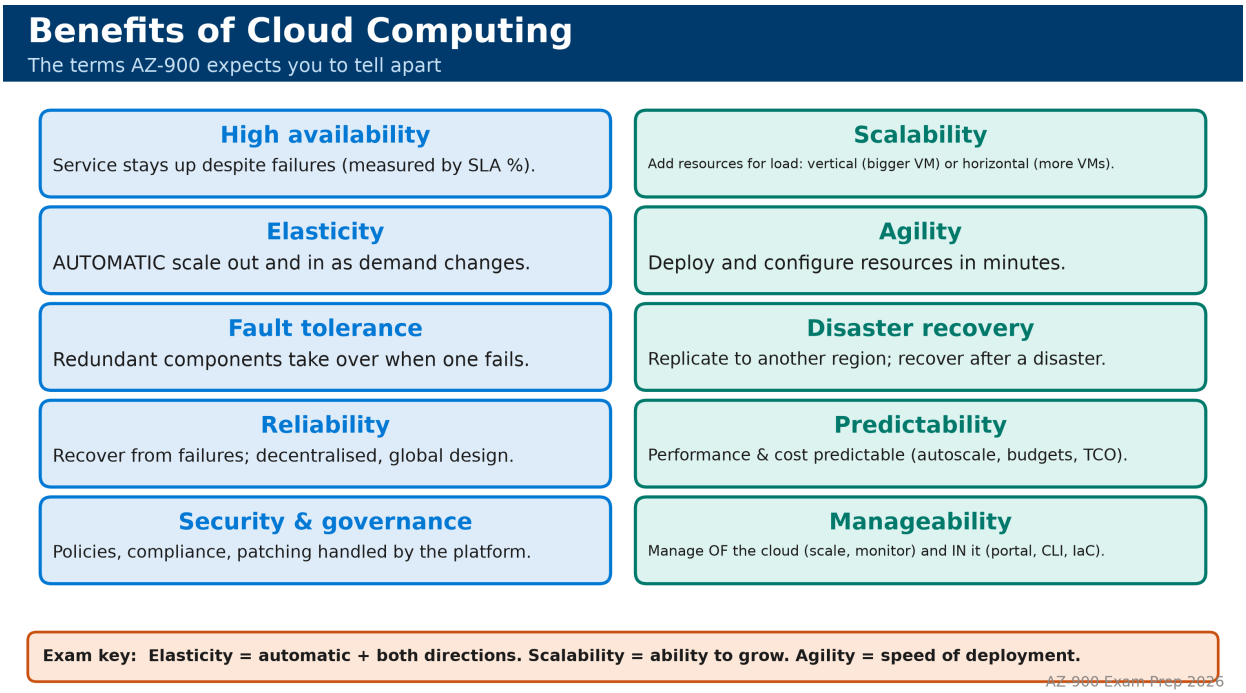


Figure Q290: Benefits of cloud computing

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- High availability is expressed as an SLA percentage such as 99.9% or 99.99%; fault tolerance and disaster recovery are ways of achieving it.
- Agility means provisioning in minutes instead of weeks of hardware procurement; predictability covers both performance and cost forecasting.
- Governance tools such as Azure Policy, RBAC and locks deliver the security and governance benefit; manageability covers portal, CLI, PowerShell and IaC.

Question 291

CLOUD CONCEPTS

Cloud concepts
v5 Q424

Before migrating to Azure, your CFO wants to forecast monthly cloud spending accurately and track it against budgets. Which cloud benefit does this describe?

✔ **A.** Predictability

✖ B. Elasticity

✖ C. Reliability

✖ D. Agility

Correct answer: Predictability

Answer explained

✔ **Predictability**
Forecasting and controlling spend is cost predictability, part of the Predictability benefit.

✖ **Elasticity**
Elasticity means resources change automatically with demand. It is not about forecasting cost.

✖ **Reliability**
Reliability means recovering from failures and continuing to work.

✖ **Agility**
Agility means deploying resources quickly. It does not forecast costs.

EXAM STRATEGY
'Forecast' and 'track against budgets' point to Predictability (cost). Remove elasticity, reliability and agility; they are about capacity, recovery and speed.

IMPACT
Predictable cloud spending helps get approval from finance. Cost Management budgets and the Pricing calculator make it real.

What · Where · When · Why · How

What	Being able to forecast cloud costs and performance with confidence.
Where	With Azure tools such as Microsoft Cost Management and the Pricing calculator.
When	Before the move to cloud and during daily operations.
Why	So the business can plan budgets and avoid surprise bills.
How	Estimate costs first, then track real spending against budgets.

Exam tip
On AZ-900, Predictability has two halves: cost predictability and performance predictability.

Memory trick
Predictability = no surprises, in the bill or in the response time.

Real-world example, use case and scenario

EXAMPLE
Retail chain
A retailer estimates move costs with the Pricing calculator before approving the project.

USE CASE
Public sector
A council sets budgets in Cost Management to stay within its yearly allowance.

SCENARIO
Performance
An e-learning site uses autoscale so pages load at the same speed during exam time.

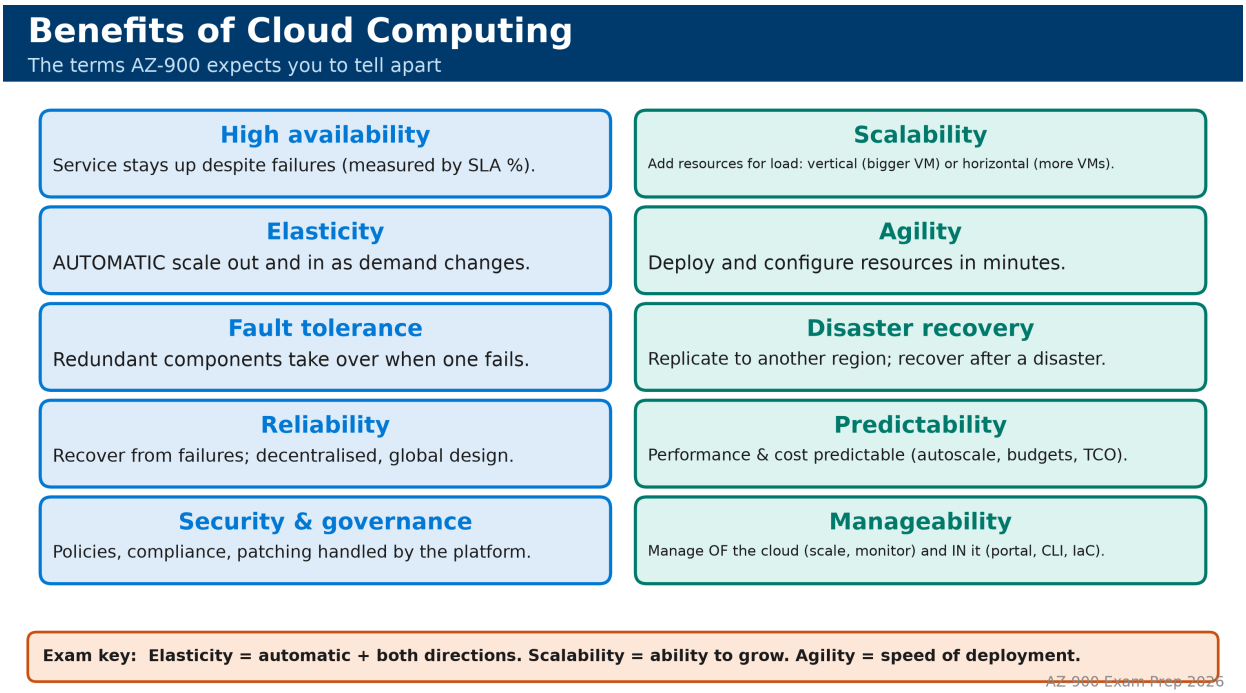


Figure Q291: Benefits of cloud computing

About this figure

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- Agility means provisioning in minutes instead of weeks of hardware procurement; predictability covers both performance and cost forecasting.
- Governance tools such as Azure Policy, RBAC and locks deliver the security and governance benefit; manageability covers portal, CLI, PowerShell and IaC.

Question 292

CLOUD CONCEPTS

Cloud concepts
v5 Q425

Your company deploys its application to multiple Azure regions so that it continues to function and can recover if an entire region fails. Which cloud benefit is the company primarily using?

✔ **A.** Reliability

✖ B. Scalability

✖ C. Manageability

✖ D. Governance

Correct answer: Reliability

Answer explained

✔ **Reliability**
Recovering from failures and continuing to work (for example, by using more than one region) is Reliability.

✖ **Scalability**
Scalability means changing capacity to match demand, not surviving failures.

✖ **Manageability**
Manageability is about how resources are managed and monitored.

✖ **Governance**
Governance enforces standards and compliance, not failure recovery.

EXAM STRATEGY
'Recover if an entire region fails' points to Reliability. Remove scalability (capacity), manageability (operations) and governance (rules).

IMPACT
Using more than one region prevents a regional outage from stopping the business. Cloud benefits are a core Cloud Concepts topic.

What · Where · When · Why · How

What	The ability to recover from failures and keep working.
Where	Across Azure regions and availability zones around the world.
When	When outages, hardware failures or disasters happen.
Why	To protect business continuity and customer trust.
How	Design workloads across several zones or regions.

Exam tip
Know the difference: Reliability = recover and keep working. High availability = keep uptime as high as the SLA (uptime promise) says.

Memory trick
Reliable = it bounces back.

Real-world example, use case and scenario

EXAMPLE
Airline
An airline's booking system runs in two regions so passengers can book during an outage.

USE CASE
Bank
A bank copies core services to a paired region so it can recover from a regional failure.

SCENARIO
Regional outage
When one region has a power problem, traffic moves to another region.

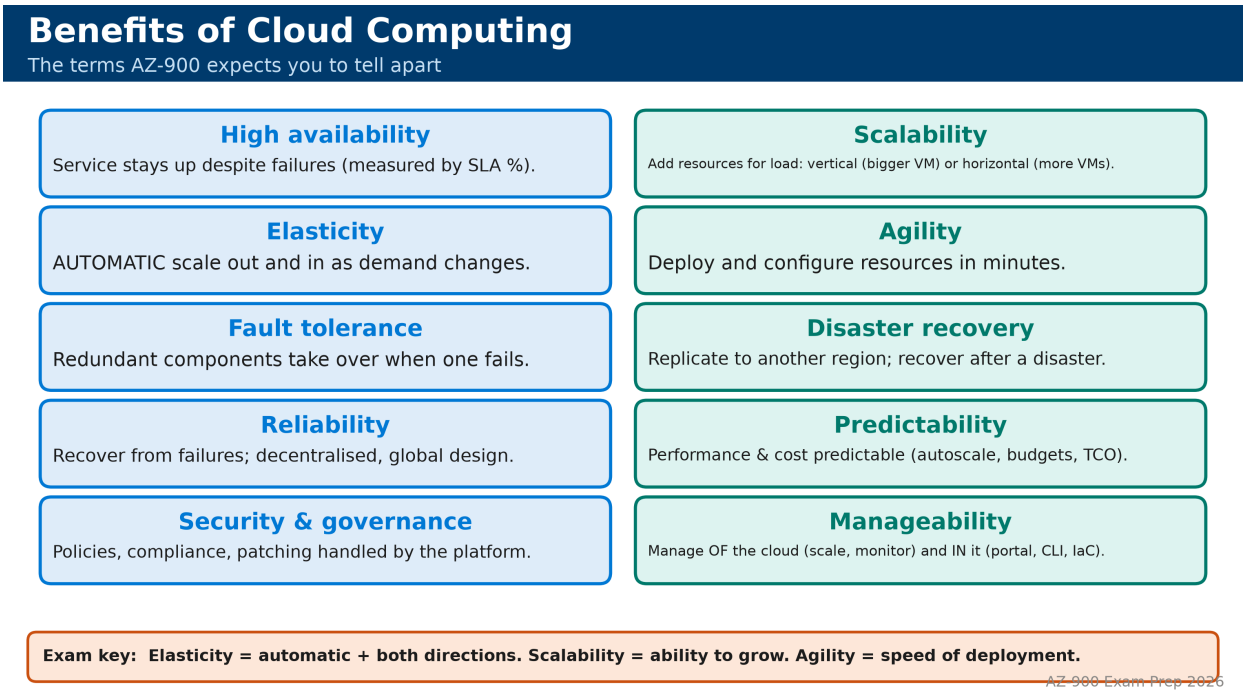


Figure Q292: Benefits of cloud computing

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- High availability is expressed as an SLA percentage such as 99.9% or 99.99%; fault tolerance and disaster recovery are ways of achieving it.
- Agility means provisioning in minutes instead of weeks of hardware procurement; predictability covers both performance and cost forecasting.
- Governance tools such as Azure Policy, RBAC and locks deliver the security and governance benefit; manageability covers portal, CLI, PowerShell and IaC.

Question 293

CLOUD CONCEPTS

Governance
v5 Q426

Your company must ensure that all deployed cloud resources meet corporate standards and regulatory requirements, and that configuration drift is detected and corrected. Which cloud benefit does this describe?

✔ **A.** Governance

✖ B. Elasticity

✖ C. Predictability

✖ D. High availability

Correct answer: Governance

Answer explained

✔ **Governance**
Enforcing company and legal standards and finding drift (settings that changed away from the standard) is Governance.

✖ **Elasticity**
Elasticity changes resources to match demand. It does not enforce standards.

✖ **Predictability**
Predictability is about forecasting cost and performance.

✖ **High availability**
High availability is about uptime, not following standards.

EXAM STRATEGY
The words 'standards', 'regulatory' and 'drift' point to Governance. Remove elasticity, predictability and high availability.

IMPACT
Azure Policy finds and fixes drift automatically. This keeps environments compliant and avoids audit problems.

What · Where · When · Why · How

What	Keeping cloud resources in line with standards and laws.
Where	Across subscriptions, using tools such as Azure Policy.
When	During deployment and all the time afterward.
Why	To lower risk and meet legal duties.
How	Use templates, policies and audits that flag or fix resources that break the rules.

Exam tip
Words like 'standards', 'compliance', 'policies' and 'drift' point to governance.

Memory trick
Governance = the rules of the road for the cloud.

Real-world example, use case and scenario

EXAMPLE
Pharma
A drug company blocks deployments outside approved regions with policies.

USE CASE
Bank
A bank checks that every storage account encrypts data in transit.

SCENARIO
Drift
An admin finds a VM with a public IP that breaks policy and fixes it.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

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Figure Q293: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 294

CLOUD CONCEPTS

Management tools
v5 Q427

Your company configures Azure to automatically add resources when CPU load rises, monitor resource health, and raise alerts based on metrics. Which aspect of cloud manageability does this describe?

✔ **A.** Management of the cloud

✖ B. Management in the cloud

✖ C. Governance of the cloud

✖ D. Security of the cloud

Correct answer: Management of the cloud

Clarification (read this first): Watch the small word: OF vs IN. Management OF the cloud = what Azure does for you automatically (autoscale, monitoring, alerts). Management IN the cloud = the tools you use (portal, CLI, API).

Answer explained

✔ **Management of the cloud**
Autoscale, health monitoring, automatic replacement and metric alerts are management OF the cloud.

✖ **Management in the cloud**
Management IN the cloud means the tools you use: portal, CLI, APIs, PowerShell.

✖ **Governance of the cloud**
Governance is about standards and compliance, not autoscaling and alerts.

✖ **Security of the cloud**
Security is about protecting resources, not automatic scaling and alerts.

EXAM STRATEGY
'Automatically', 'autoscale', 'monitor health' and 'alerts' describe what the cloud does for you, so OF. Remove governance and security. 'IN' needs a tool name like portal or CLI.

IMPACT
The OF versus IN pair appears in several wordings. In real life, autoscale plus alerts helps teams avoid outages and avoid paying for unused peak capacity.

What · Where · When · Why · How

What	Cloud features that manage your resources automatically.
Where	In Azure, through autoscale, Azure Monitor and templates.
When	All the time, in response to metrics and health signals.
Why	To keep workloads healthy with little manual work.
How	Set up autoscale rules, health monitoring and metric alerts.

Exam tip
OF the cloud = what the cloud does for you. IN the cloud = the tools you use (portal, CLI, API).

Memory trick
OF = automatic; IN = interface.

Real-world example, use case and scenario

EXAMPLE
Retail
A shop's web tier adds instances automatically when CPU goes above 70 percent.

USE CASE
Healthcare
A clinic gets an alert when its patient portal reports unhealthy instances.

SCENARIO
Contrast
An admin using Cloud Shell to resize a VM is management IN the cloud.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

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Figure Q294: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 295

CLOUD CONCEPTS

Service models
v5 Q428

Your company wants email, calendars and office productivity apps for all staff. It does not want to manage any servers, platforms or application updates. Which cloud service model should it use?

✔ **A.** Software as a service (SaaS)

✖ B. Platform as a service (PaaS)

✖ C. Infrastructure as a service (IaaS)

✖ D. A private cloud

Correct answer: Software as a service (SaaS)

Clarification (read this first): 'A private cloud' is a deployment model (where it runs), not a service model (what you manage). The question asks for a service model.

Answer explained

✔ **Software as a service (SaaS)**
A ready-made app (email, office, CRM) with no servers or app updates to manage is SaaS.

✖ **Platform as a service (PaaS)**
PaaS is for building your own apps. It does not give you finished email software.

✖ **Infrastructure as a service (IaaS)**
IaaS means you manage the OS and apps yourself.

✖ **A private cloud**
A private cloud is a deployment model, and the company would have to run the infrastructure.

EXAM STRATEGY
'Email, calendars, office apps' plus 'no servers, platforms or updates' = SaaS. Remove IaaS and PaaS. 'Private cloud' is a deployment model, not a service model.

IMPACT
Using SaaS such as Microsoft 365 for common apps removes patching and server costs. Choosing IaaS here means paying staff to run mail servers you do not need.

What · Where · When · Why · How

What	Fully managed, ready-to-use software delivered over the internet.
Where	In services such as Microsoft 365 and Dynamics 365.
When	When a business needs standard apps with no technical management.
Why	To reduce admin work and get updates automatically.
How	Subscribe per user and set up the app.

Exam tip
Email, CRM, office suites and other finished apps = SaaS.

Memory trick
SaaS = Software Already Set up for you.

Real-world example, use case and scenario

EXAMPLE
Law firm
A law firm uses Microsoft 365 for email and documents with no servers to look after.

USE CASE
Sales team
A sales team starts using Dynamics 365 Sales as a subscription CRM.

SCENARIO
Start-up
A start-up signs up for SaaS accounting software instead of installing any.

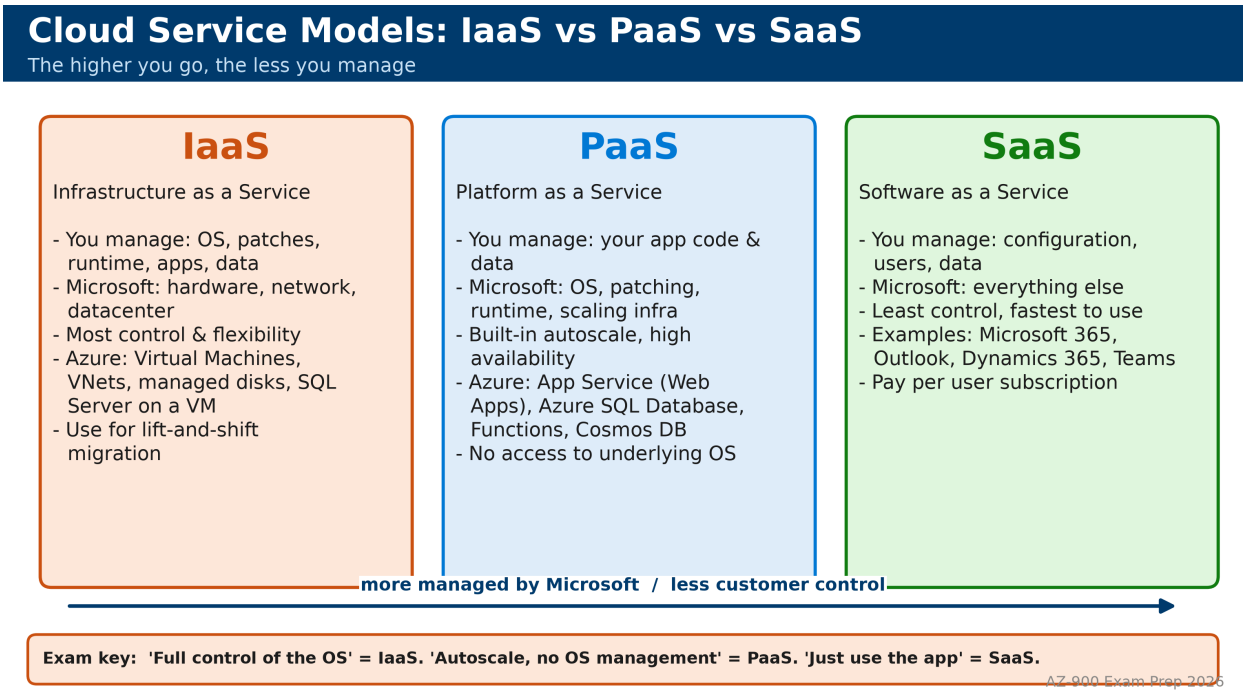


Figure Q295: IaaS vs PaaS vs SaaS

About this figure

• Read left to right: IaaS gives the most control and work, SaaS the least. The arrow shows management shifting from you to Microsoft.

• Exam test: needing to install software or change the OS means IaaS; deploying code without managing servers means PaaS; signing in to a ready app means SaaS.

• Serverless options such as Azure Functions are treated as PaaS-style: no OS access, automatic scaling and per-execution billing.

• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

Question 296

CLOUD CONCEPTS

Service models
v5 Q429

Your company is reviewing scenarios for Platform as a service (PaaS). Which two scenarios are typical PaaS use cases? Each correct answer presents a complete solution. NOTE: Each correct selection is worth one point. (Choose two.)

✔ **A.** Providing a development framework so developers can build and deploy apps without managing servers

✔ **B.** Running analytics or business intelligence on a managed platform without managing infrastructure

✖ C. Lifting and shifting existing virtual machines without changing them

✖ D. Using Microsoft 365 for company email

✖ E. Installing custom software that requires full control of the operating system

Correct answer: Providing a development framework so developers can build and deploy apps without managing servers; Running analytics or business intelligence on a managed platform without managing infrastructure

Clarification (read this first): Pick TWO PaaS cases. Compare Q297, which asks for IaaS cases from a similar list. Here, remove anything that needs full OS control (IaaS) or is a finished app (SaaS).

Answer explained

✔ **Providing a development framework so developers can build and deploy apps without managing servers; Running analytics or business intelligence on a managed platform without managing infrastructure**
Typical PaaS uses: a development platform and analytics/BI (business reports) on a managed platform.

✖ **Lifting and shifting existing virtual machines without changing them**
Moving VMs without changes (lift-and-shift) is a classic IaaS case.

✖ **Using Microsoft 365 for company email**
Microsoft 365 is a finished app, so it is SaaS.

✖ **Installing custom software that requires full control of the operating system**
Full control of the OS needs IaaS.

EXAM STRATEGY
PaaS signs: 'build and deploy without managing servers' or 'managed platform'. Remove 'full OS control' or 'unchanged VMs' (IaaS) and finished apps like Microsoft 365 (SaaS).

IMPACT
PaaS use-case questions appear often. In practice PaaS removes OS patching and helps developers release faster.

What · Where · When · Why · How

What	Managed platforms for building apps or analyzing data.
Where	In services such as Azure App Service, Azure SQL Database and Azure Synapse Analytics.
When	When teams want to build or analyze without managing infrastructure.
Why	To shorten development time and reduce management work.
How	Deploy code or data to a platform that the provider manages.

Exam tip
PaaS = you bring code or data; the platform does the rest.

Memory trick
PaaS = a stage for programmers and analysts.

Real-world example, use case and scenario

EXAMPLE
Insurance
An insurer's developers build a claims website on App Service.

USE CASE
Retail BI
A retailer analyzes sales data with a managed analytics service.

SCENARIO
Mobile app
A start-up uses a managed database and app platform for its mobile back end.

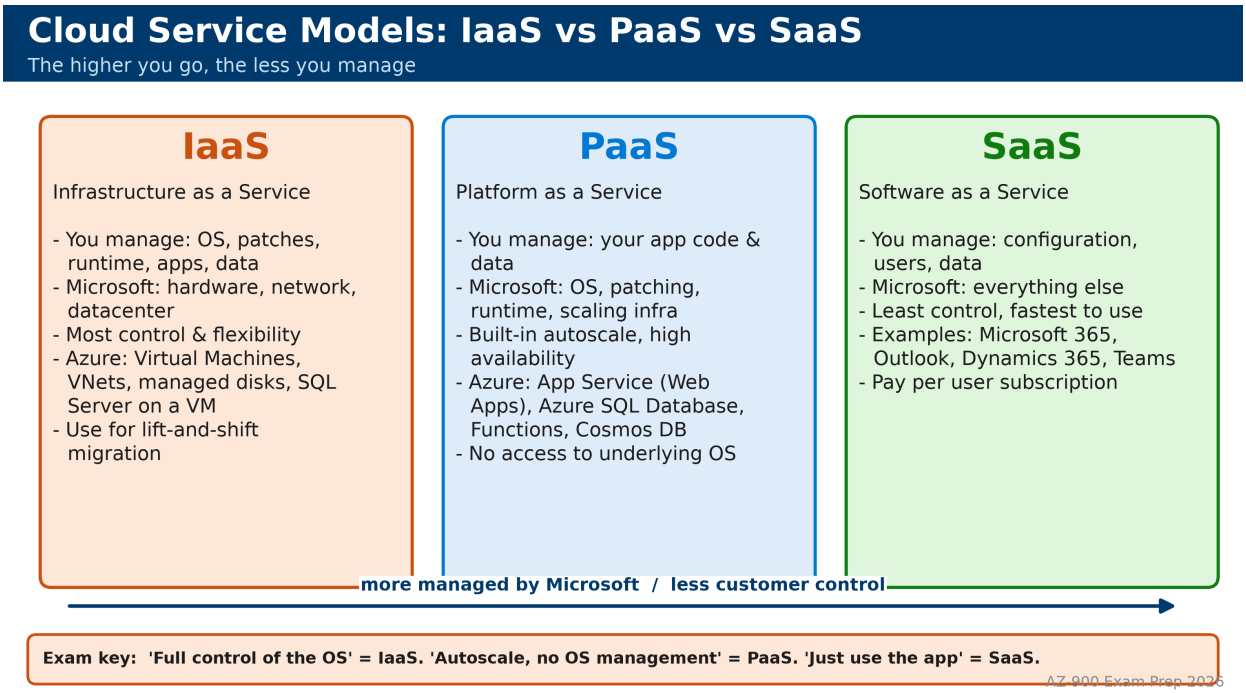


Figure Q296: IaaS vs PaaS vs SaaS

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• IaaS is the usual first step in lift-and-shift migration; PaaS often needs some code or configuration changes but cuts patching and maintenance effort.

• Cost model differs: IaaS VMs bill per second while running, PaaS bills per plan or consumption, and SaaS usually bills per user per month.

Question 297

CLOUD CONCEPTS

Service models
v5 Q430

Your company is reviewing scenarios for Infrastructure as a service (IaaS). Which two scenarios are typical IaaS use cases? Each correct answer presents a complete solution. NOTE: Each correct selection is worth one point. (Choose two.)

✔ **A.** Lift-and-shift migration of on-premises servers to cloud virtual machines

✔ **B.** Test and development environments that need custom operating system configurations

✖ C. Using Microsoft 365 for document collaboration

✖ D. Running event-driven code with Azure Functions

✖ E. Hosting a database on Azure SQL Database without managing the OS

Correct answer: Lift-and-shift migration of on-premises servers to cloud virtual machines; Test and development environments that need custom operating system configurations

Clarification (read this first): Pick TWO IaaS cases. Compare Q296, which asks for PaaS cases. Here, keep the options where YOU control the OS (VMs).

Answer explained

✔ **Lift-and-shift migration of on-premises servers to cloud virtual machines; Test and development environments that need custom operating system configurations**
Typical IaaS uses: lift-and-shift (moving servers without changes) and test/dev environments that need OS control.

✖ **Using Microsoft 365 for document collaboration**
Microsoft 365 is SaaS.

✖ **Running event-driven code with Azure Functions**
Azure Functions is serverless, not IaaS.

✖ **Hosting a database on Azure SQL Database without managing the OS**
Azure SQL Database is PaaS; Microsoft manages the OS.

EXAM STRATEGY
IaaS signs: 'lift-and-shift', 'VMs', 'custom OS settings'. Remove Functions (serverless), Azure SQL Database (PaaS) and Microsoft 365 (SaaS) - in all of these, Microsoft runs the OS.

IMPACT
Lift-and-shift to VMs is the fastest way to move, but you still own OS patching and security. Getting this wrong puts real work on your team.

What · Where · When · Why · How

What	Rented virtual infrastructure that the customer fully controls.
Where	In Azure Virtual Machines, managed disks and virtual networks.
When	When moving existing servers or when you need full OS control.
Why	To move to the cloud quickly without redesigning apps.
How	Deploy and set up VMs, storage and networks yourself.

Exam tip
Lift-and-shift and 'needs OS control' clearly point to IaaS.

Memory trick
IaaS = I administer all the Software on top.

Real-world example, use case and scenario

EXAMPLE
Manufacturing
A manufacturer moves its old factory servers to Azure VMs without changes.

USE CASE
Software house
Developers create custom VMs for testing and delete them afterward.

SCENARIO
Leaving a datacenter
A company's datacenter lease ends, so it moves 50 servers to Azure VMs.

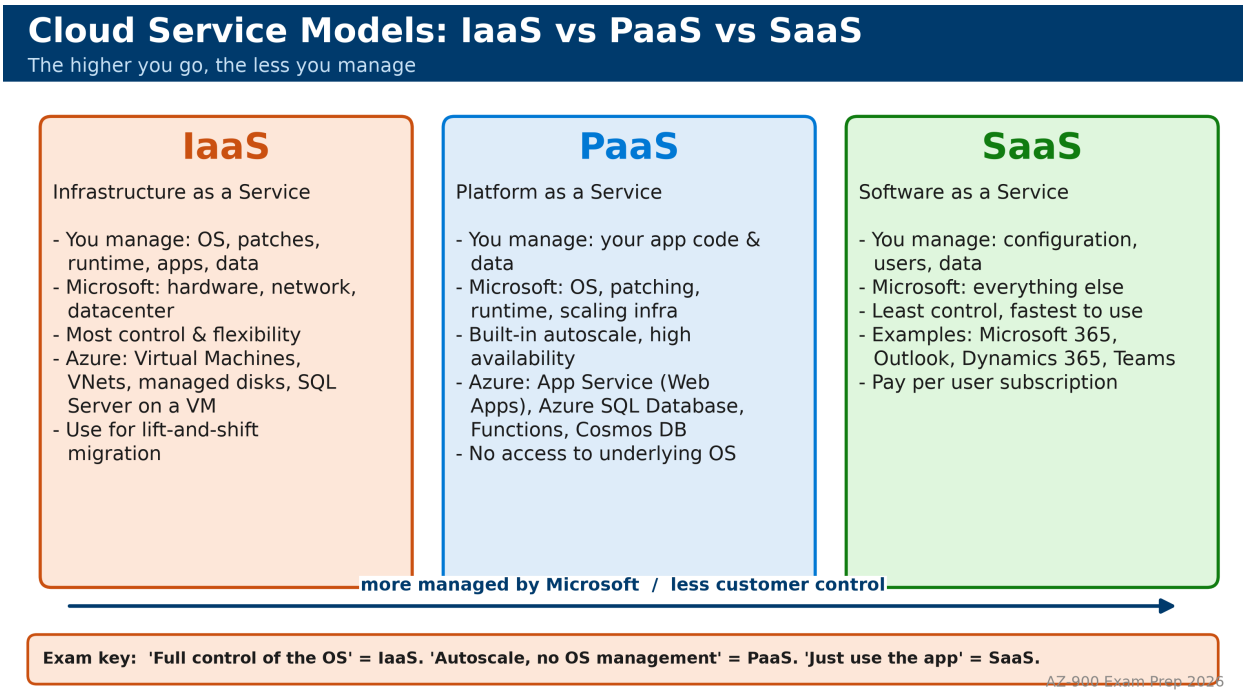


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Question 298

CLOUD CONCEPTS

SLA
v5 Q431

An Azure service has an SLA of 99.9 percent. Approximately how much downtime per month does this SLA allow?

✔ **A.** About 43 minutes

✖ B. About 4 minutes

✖ C. About 7 hours

✖ D. About 1 minute

Correct answer: About 43 minutes

Answer explained

✔ **About 43 minutes**
An SLA (uptime promise) of 99.9% allows about 43 minutes of downtime a month. 99.99% allows about 4 minutes. 99.999% allows under 1 minute.

✖ **About 4 minutes**
About 4 minutes a month matches a 99.99% SLA.

✖ **About 7 hours**
About 7 hours a month is roughly 99% availability.

✖ **About 1 minute**
Under 1 minute a month matches about 99.999%.

EXAM STRATEGY
Learn the ladder: 99.9 = about 43.8 min/month, 99.95 = about 22 min, 99.99 = about 4.4 min, 99.999 = about 26 sec. Each extra nine cuts downtime by ten. Remove hours and seconds.

IMPACT
SLA maths is on almost every AZ-900 exam. It also affects real design: 43 minutes a month may be too much for checkout or payment systems.

What · Where · When · Why · How

What	The most downtime per month that a 99.9% SLA allows.
Where	In the published SLA for each Azure service.
When	Measured over each monthly billing period.
Why	To understand the real uptime promise.
How	Multiply the minutes in a month by 0.1%.

Exam tip
Each extra 9 cuts the allowed downtime by about ten times.

Memory trick
Three nines = 43 minutes ('forty-three for three').

Real-world example, use case and scenario

EXAMPLE
Online shop
A shop checks that 43 minutes of downtime a month is acceptable.

USE CASE
Hospital
A hospital needs higher uptime, so it designs for more nines.

SCENARIO
Design
An architect adds zone redundancy to reach a 99.99% target.

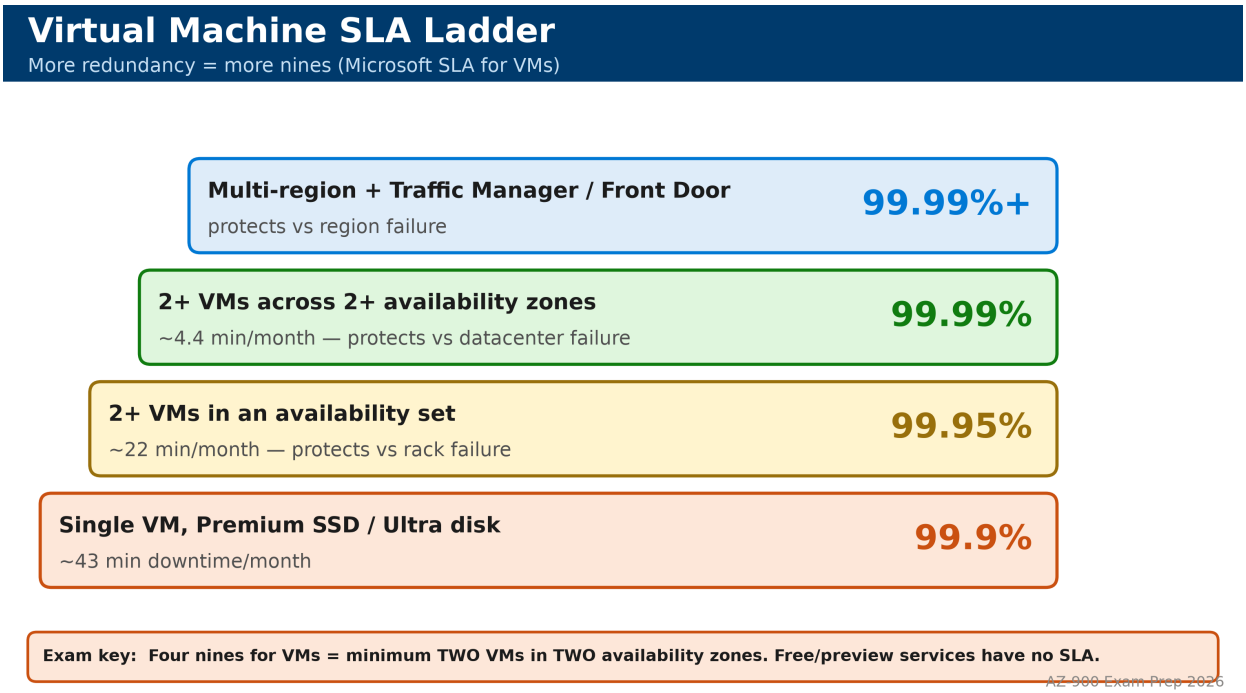


Figure Q298: Virtual machine SLA ladder

About this figure

- Climb the ladder: each step adds redundancy and a higher VM SLA; the downtime figures assume a 30-day month.
- A single VM with Standard SSD gets about 99.5% and with Standard HDD about 95%; all its OS and data disks must meet that tier.
- Availability sets spread VMs across fault domains (racks) and update domains (maintenance batches) in one datacenter; zones spread across datacenters.
- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

Question 299

CLOUD CONCEPTS

SLA
v5 Q432

Statement: Free Azure products and services typically do not have a financially backed SLA.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): 'Financially backed' means you get money back (service credits) if Azure misses the uptime promise. Free services usually give no such promise. Related: Q86 says preview services are also excluded from SLAs.

Answer explained

✔ **Yes**
Free tiers and free products usually have no financially backed SLA (uptime promise with money back). Paid services have SLAs with service credits.

✖ **No**
Free services usually have no SLA. SLAs with credits apply to paid services.

EXAM STRATEGY
Free means no service credit, so no financially backed SLA. 'Free' in the question should lead you to Yes. Free tiers and previews carry no credit promise.

IMPACT
Running production on a Free tier means no uptime promise and no money back. Teams often learn this during their first outage.

What · Where · When · Why · How

What	Whether free Azure offers come with an SLA.
Where	In the SLA documents for Azure services.
When	When choosing between free and paid tiers for a workload.
Why	Service credits are linked to paid use.
How	Check the SLA page for the service and tier before using it in production.

Exam tip
Free tiers and preview services: on the exam, assume no SLA.

Memory trick
Free = no promise, no credit.

Real-world example, use case and scenario

EXAMPLE
Hobby site
A student hosts a portfolio on the App Service Free tier, which has no SLA.

USE CASE
Production
A company moves from a free tier to a paid tier to get an SLA.

SCENARIO
Test project
A team builds a test version on free services and moves to paid tiers for launch.

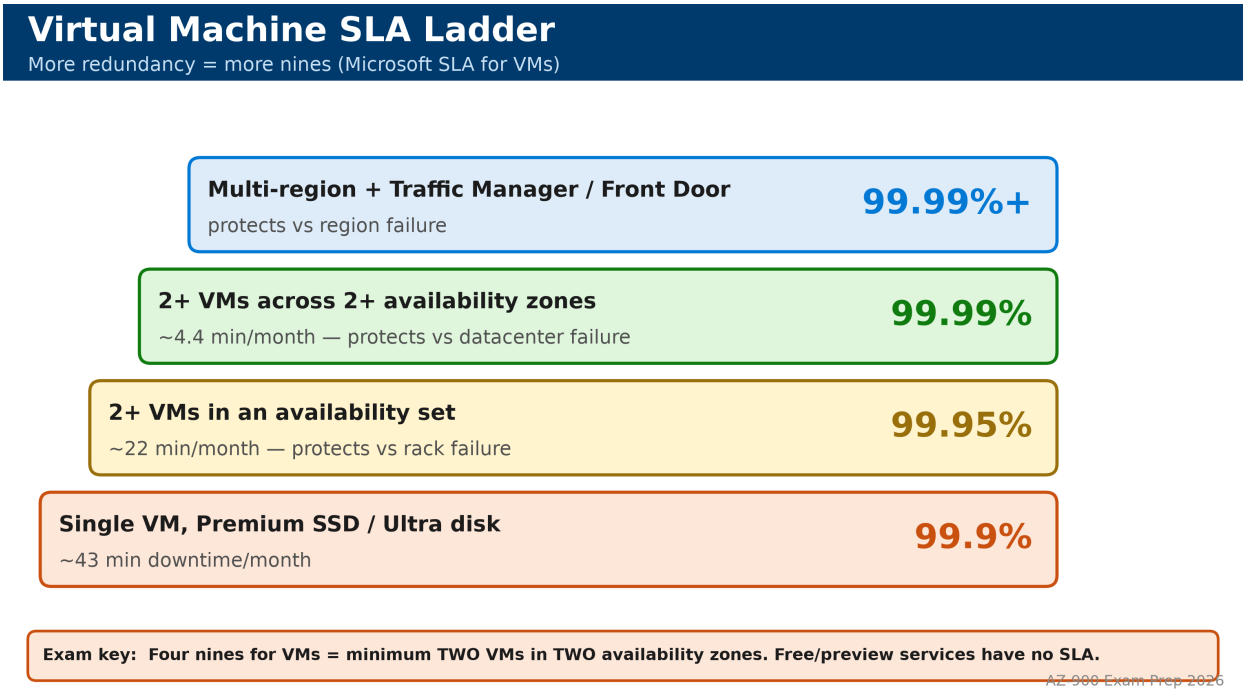


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- Multi-region designs need your own failover via Traffic Manager or Front Door; Microsoft does not publish a single SLA for your combined architecture.
- SLA breaches earn service credits, typically 10% to 100% of the monthly fee depending on how far availability fell; you must file a claim.

Question 300	AZURE ARCHITECTURE & SERVICES	Infrastructure v5 Q433

Your company runs a production app in the East US region and its disaster recovery copy in the paired region West US. Microsoft plans a platform update that affects both regions. How does Azure handle planned system updates across a region pair?

- ✔ **A.** Updates are rolled out to only one region of the pair at a time
- ✖ B. Updates are applied to both regions simultaneously to keep them identical
- ✖ C. Updates are applied only after the customer approves them in Azure Advisor
- ✖ D. Updates are applied only to regions that do not contain availability zones

Correct answer: **Updates are rolled out to only one region of the pair at a time**

Answer explained

✔ **Updates are rolled out to only one region of the pair at a time**
Planned Azure updates go to one region of a pair at a time. If an update goes wrong, the other region is still safe.

✖ **Updates are applied to both regions simultaneously to keep them identical**
Updating both at once would defeat the point of pairing. One bad update could hit both.

✖ **Updates are applied only after the customer approves them in Azure Advisor**
Azure Advisor gives advice. It does not control Microsoft platform updates.

✖ **Updates are applied only to regions that do not contain availability zones**
All regions get updates. Availability zones have nothing to do with update order.

EXAM STRATEGY
Region-pair clue words: 'planned updates', 'disaster recovery'. The answer is one region at a time. Remove 'simultaneously', because it defeats the purpose of pairing.

IMPACT
One-at-a-time updates mean a bad platform patch cannot stop both your main copy and your DR (disaster recovery) copy. This is the main value of using the paired region for DR.

What · Where · When · Why · How

What	Rolling out planned maintenance to paired regions one after the other.
Where	Across the two regions of an Azure region pair, such as East US and West US.
When	Whenever Microsoft deploys planned platform or infrastructure updates.
Why	To lower the risk that a bad update stops both copies of a workload.
How	Microsoft updates one region of the pair, checks it, then updates the other.

Exam tip
Region-pair benefits to learn: updates one at a time, one region of each pair is recovered first in a large outage, and data stays in the same geography (except Brazil South).

Memory trick
Pairs take turns: one dances while the other watches.

Real-world example, use case and scenario

EXAMPLE
Retail DR
A retailer copies its order database from North Europe to West Europe, so a platform update never hits both copies at once.

USE CASE
Bank
A bank uses paired regions for geo-redundant storage, so Microsoft coordinates maintenance and recovery.

SCENARIO
Bad patch
A rare update bug hits Central US. The paired East US 2 is not updated yet, so failover works.

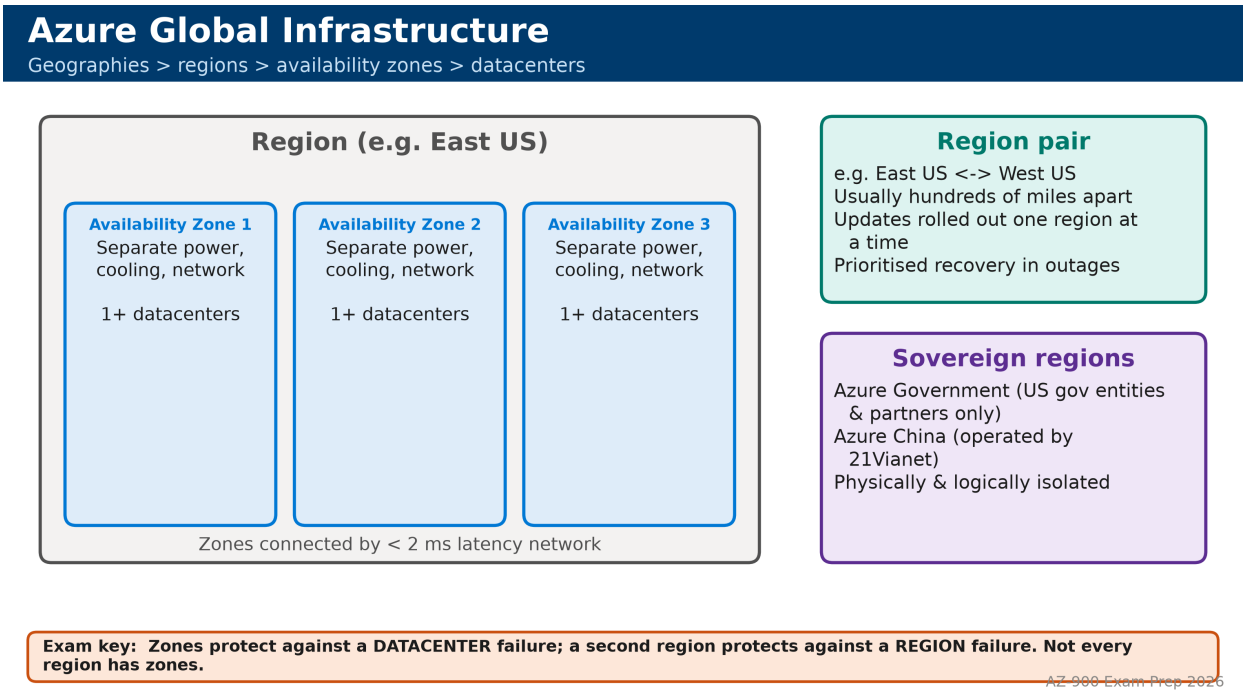


Figure Q300a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

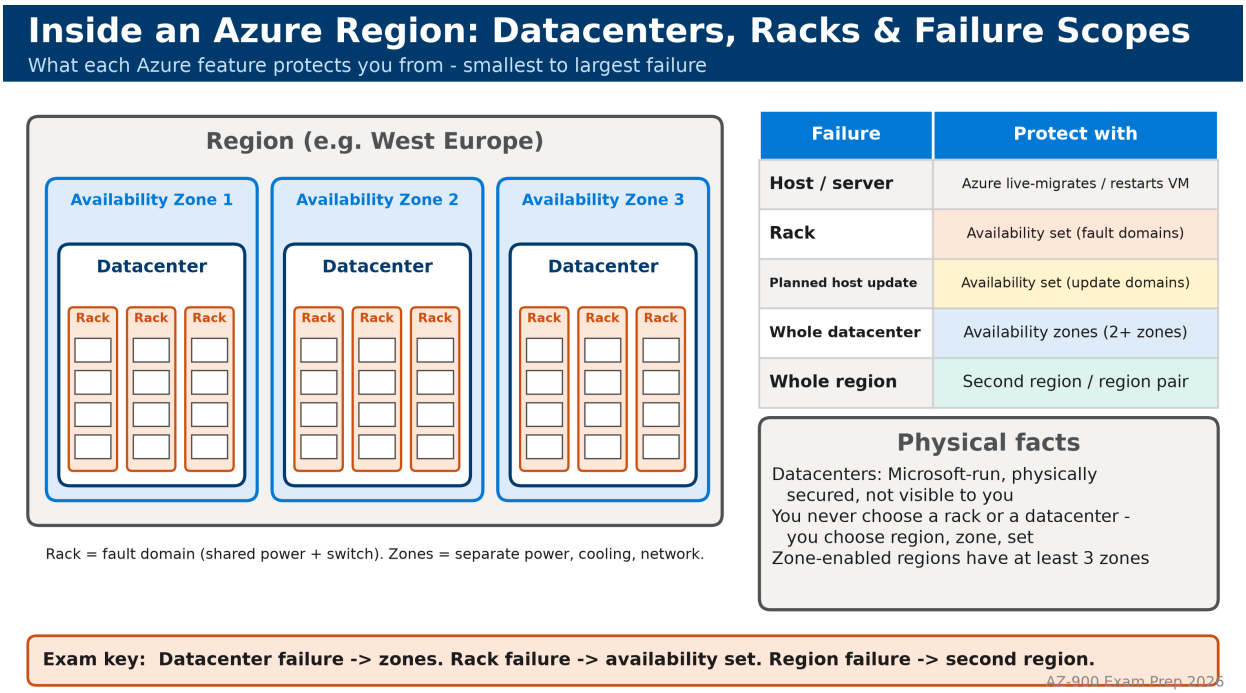


Figure Q300b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 301

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q434

Most Azure region pairs are located within the same geography. Which Azure region is paired with a region OUTSIDE its own geography?

✔ **A.** Brazil South

✖ B. UK South

✖ C. Japan East

✖ D. Australia East

Correct answer: Brazil South

Clarification (read this first): Watch the capital OUTSIDE. Most pairs stay in one geography (country or area). The question asks for the one exception.

Answer explained

✔ **Brazil South**
Brazil South is the exception. It is paired with South Central US, outside Brazil.

✖ **UK South**
UK South pairs with UK West, inside the UK.

✖ **Japan East**
Japan East pairs with Japan West, inside Japan.

✖ **Australia East**
Australia East pairs with Australia Southeast, inside Australia.

EXAM STRATEGY
Remember the exception: Brazil South pairs with South Central US. UK, Japan and Australia regions pair inside their own geography, so remove them.

IMPACT
If you think GRS (geo-redundant storage) keeps Brazilian data in Brazil, you may break data-residency rules (laws about where data is kept). Brazil South copies data to the United States.

What · Where · When · Why · How

What	A region pair that crosses a geography border.
Where	Brazil South in South America, paired with South Central US in Texas.
When	When designing replication that must follow data-residency rules.
Why	Pairing normally keeps copied data in one geography for legal and tax reasons.
How	Check the Azure region pairs list before relying on pairing for residency.

Exam tip
If a question asks which pair breaks the same-geography rule, the answer is Brazil South.

Memory trick
Brazil goes north for its partner: Brazil South pairs with Texas.

Real-world example, use case and scenario

EXAMPLE
Sao Paulo app
A Sao Paulo fintech using GRS in Brazil South learns that its second copy is in South Central US.

USE CASE
Residency check
A legal team checks region pairs to confirm that customer data never leaves its home geography.

SCENARIO
Other design
A Brazilian insurer copies data to Brazil Southeast itself, instead of using the default pair.

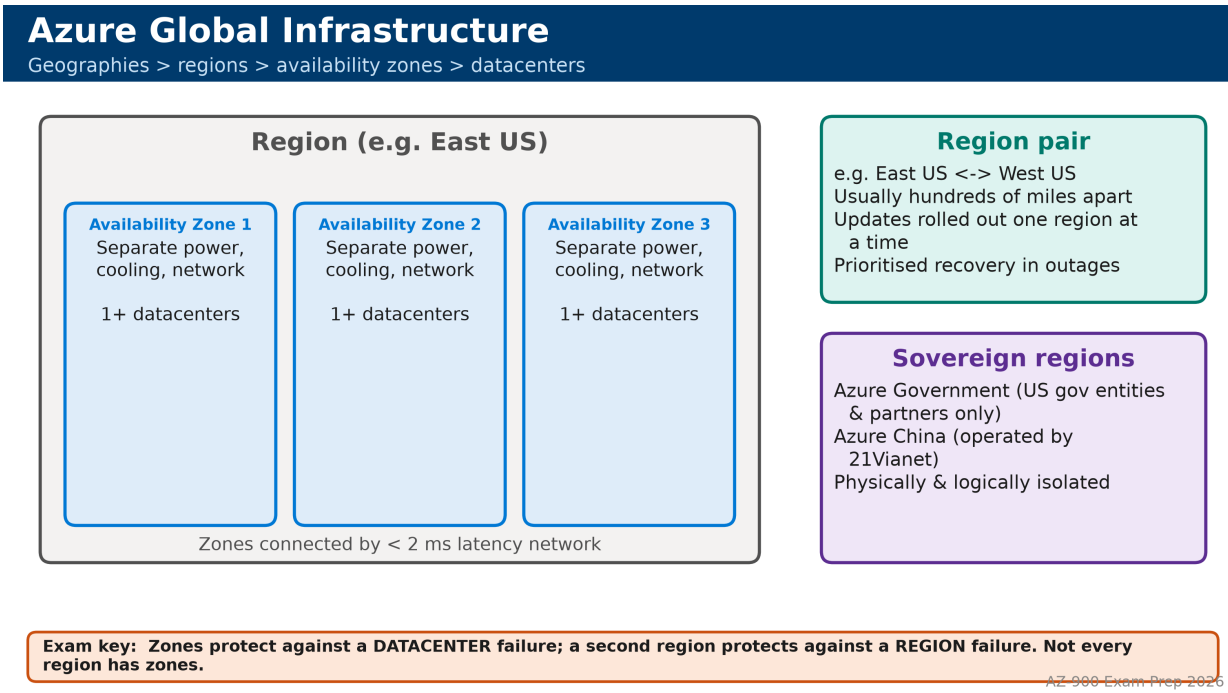


Figure Q301: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

Question 302

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q435

A US federal agency must host workloads in an Azure cloud that is physically isolated from the global public Azure and operated by screened US personnel. What should the agency use?

✔ **A.** Azure Government

✖ B. An Azure region pair in the United States

✖ C. Azure Stack HCI in its own datacenter

✖ D. A dedicated Azure subscription in East US

Correct answer: Azure Government

Answer explained

✔ **Azure Government**
Azure Government is a sovereign cloud (a separate Azure for one country's government). It is isolated from public Azure and run by screened US staff for US government work.

✖ **An Azure region pair in the United States**
Normal US regions are part of public Azure, not a separate sovereign cloud.

✖ **Azure Stack HCI in its own datacenter**
That is on-premises hybrid infrastructure run by the agency, not a sovereign cloud run by Microsoft.

✖ **A dedicated Azure subscription in East US**
A subscription is a billing and management boundary. It does not change the public cloud underneath.

EXAM STRATEGY
'US federal', 'physically isolated' and 'screened US personnel' = Azure Government. Remove region pairs and subscriptions; they are in public Azure.

IMPACT
Federal workloads with strict rules (such as FedRAMP High, CJIS or ITAR) need Azure Government. Putting them in commercial Azure can break contracts and fail audits.

What · Where · When · Why · How

What	A sovereign Azure cloud for the US public sector.
Where	In US government regions such as US Gov Virginia and US Gov Arizona.
When	When federal, state, local or tribal agencies have strict compliance needs.
Why	Some government data must be kept separate and handled only by checked staff.
How	Agencies ask for access and deploy into Azure Government's own portal and endpoints.

Exam tip
Sovereign clouds on AZ-900: Azure Government (US) and Azure China, run by 21Vianet.

Memory trick
Government gets its own gated community of datacenters.

Real-world example, use case and scenario

EXAMPLE
Defense contractor
A defense contractor stores controlled data in Azure Government to meet defense rules.

USE CASE
State agency
A state tax department hosts its tax system in US Gov Virginia.

SCENARIO
Wrong choice
A city tries to meet the rule with a normal East US subscription and fails the audit.

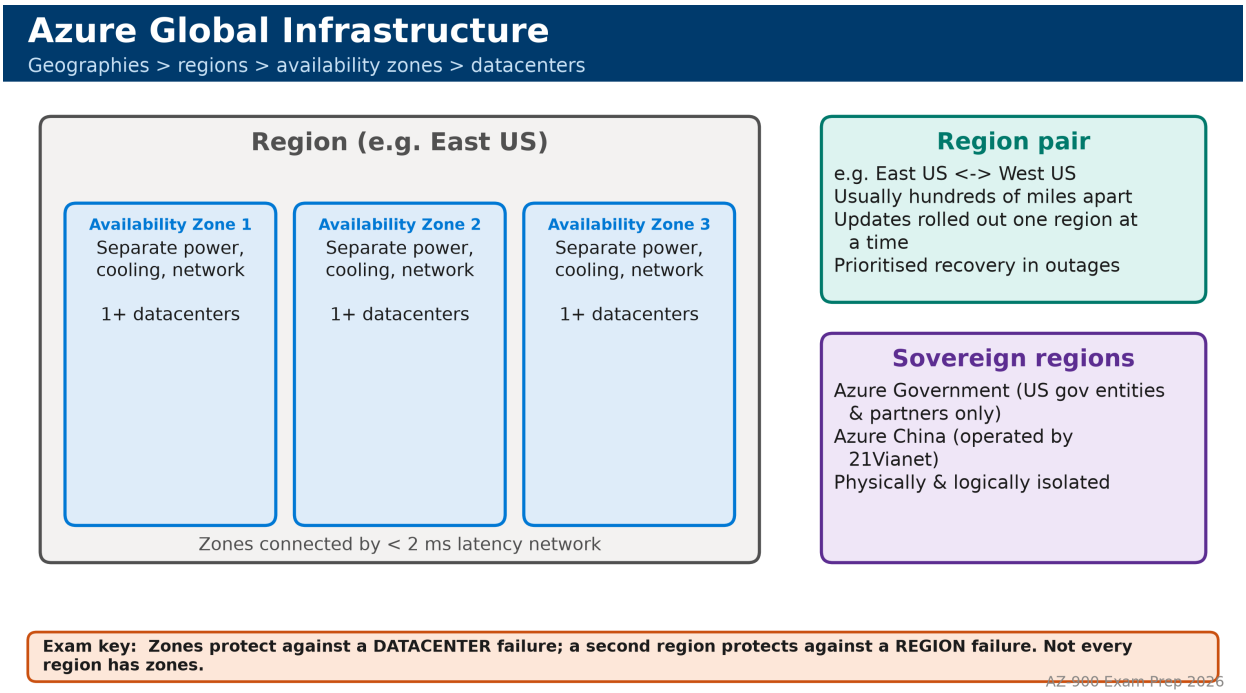


Figure Q302a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

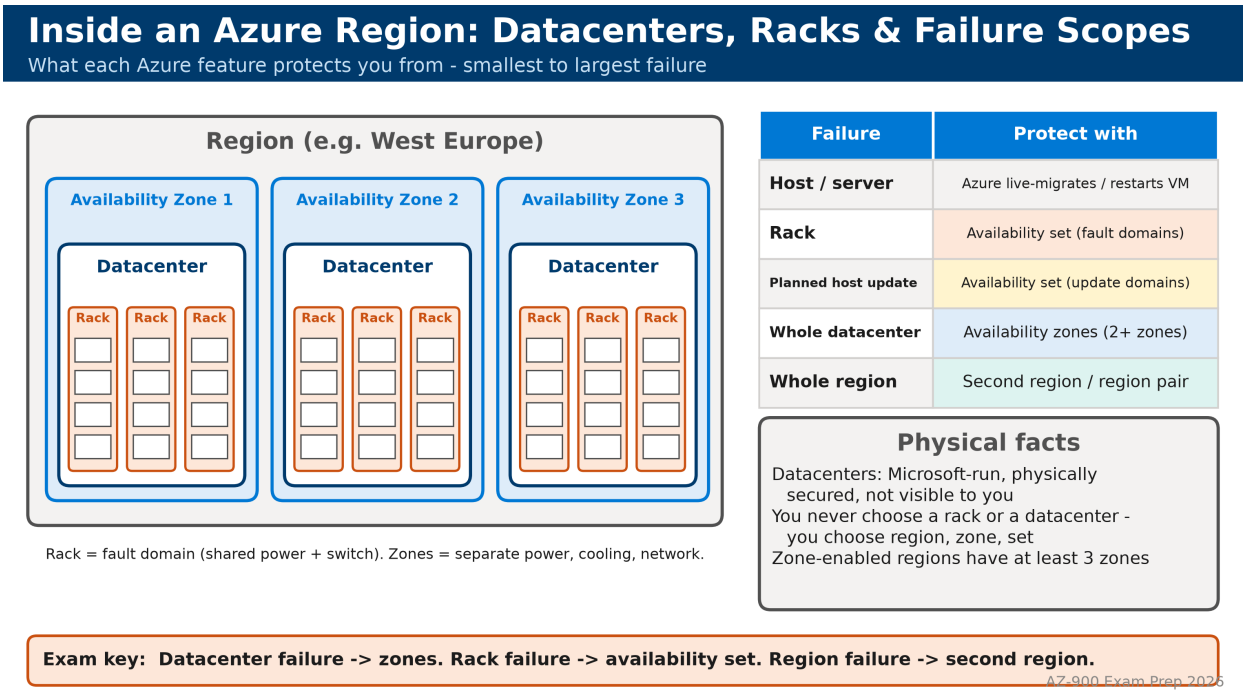


Figure Q302b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 303

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q436, Q282

Azure in mainland China is operated by Microsoft directly from Redmond in the same way as other global Azure regions.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): Compare Q189: Azure Government IS run by Microsoft (Yes). Azure China is NOT run directly by Microsoft; a local company, 21Vianet, runs it.

Answer explained

✔ **No**
No. Azure China is a sovereign cloud run by 21Vianet and kept separate from global Azure.

✖ **Yes**
Microsoft does not run Azure China datacenters directly. 21Vianet does.

EXAM STRATEGY
Azure China is run by 21Vianet, a separate local company, and is isolated from global Azure. Any statement that Microsoft runs it directly is No.

IMPACT
China deployments need separate accounts, endpoints and contracts. If you assume global Azure works there, identity, billing and app connection plans will break.

What · Where · When · Why · How

What	Azure China, a sovereign version of Azure.
Where	In datacenters inside mainland China.
When	When a company must host services for users in mainland China.
Why	Chinese law requires a locally licensed operator.
How	Customers sign up separately and use different portal endpoints for Azure China.

Exam tip
Remember the operator name: 21Vianet. It appears often in sovereign cloud questions.

Memory trick
China's cloud has a local landlord: 21Vianet.

Real-world example, use case and scenario

EXAMPLE
Game launch
A game studio launching in China deploys its servers in Azure China, not global Azure.

USE CASE
Separate account
A car maker finds its global Azure subscription cannot deploy to China regions. It needs a separate Azure China account.

SCENARIO
Data location
An online shop keeps Chinese customer data in China North 3 to follow local law.

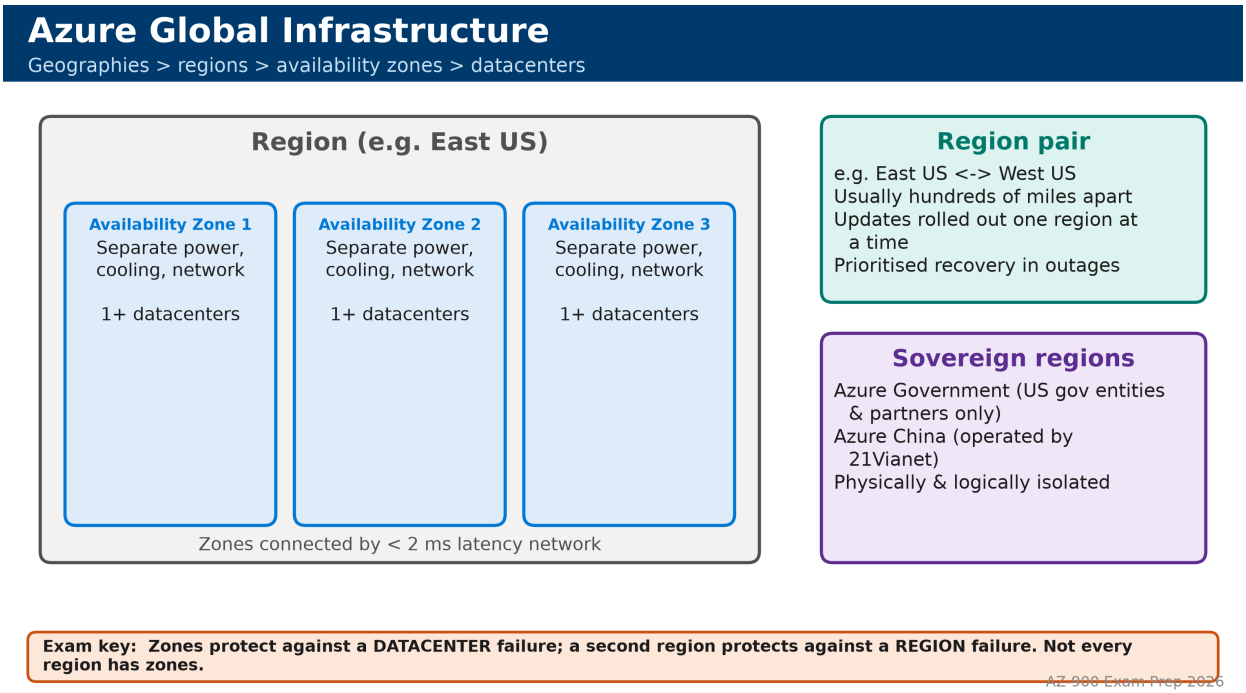


Figure Q303: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

Question 304

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q437

Every Azure region that supports availability zones has a minimum of three separate availability zones.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): Not every region has zones (Q126 -> No). But every region that DOES have zones has at least three. Read 'that supports availability zones' carefully.

Answer explained

✔ **Yes**
Yes. Regions with availability zones always have at least three zones.

✖ **No**
Three is the promised minimum for any region that offers availability zones.

EXAM STRATEGY
Zone-enabled regions have at least three zones. 'Minimum of three' matches the rule, so Yes.

IMPACT
Three zones let zone-redundant services survive the loss of one datacenter and still keep a majority. Architects plan ZRS and zone-redundant databases around this number.

What · Where · When · Why · How

What	The minimum number of availability zones in a zone-enabled region.
Where	Inside one Azure region, such as West Europe or East US 2.
When	When designing zone-redundant deployments.
Why	Three zones let majority-based services survive the loss of one zone.
How	Deploy resources to zones 1, 2 and 3, or choose zone-redundant options.

Exam tip
Not all regions support zones, but those that do have at least three.

Memory trick
Zones come in threes, like a three-legged stool.

Real-world example, use case and scenario

EXAMPLE
Web tier
A ticket site spreads three VMs across zones 1, 2 and 3 in West Europe.

USE CASE
Zone-redundant storage
ZRS copies data to all three zones of a region at the same time.

SCENARIO
Zone outage
A cooling failure stops zone 2. The app keeps working from zones 1 and 3.

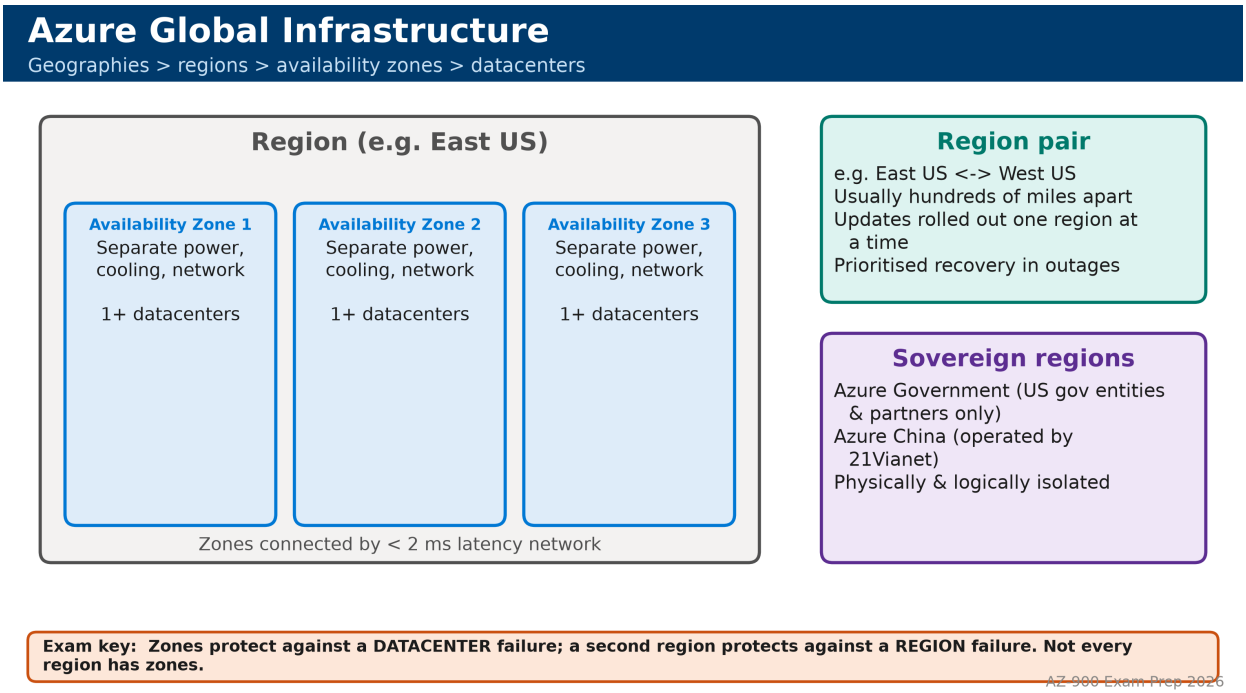


Figure Q304a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

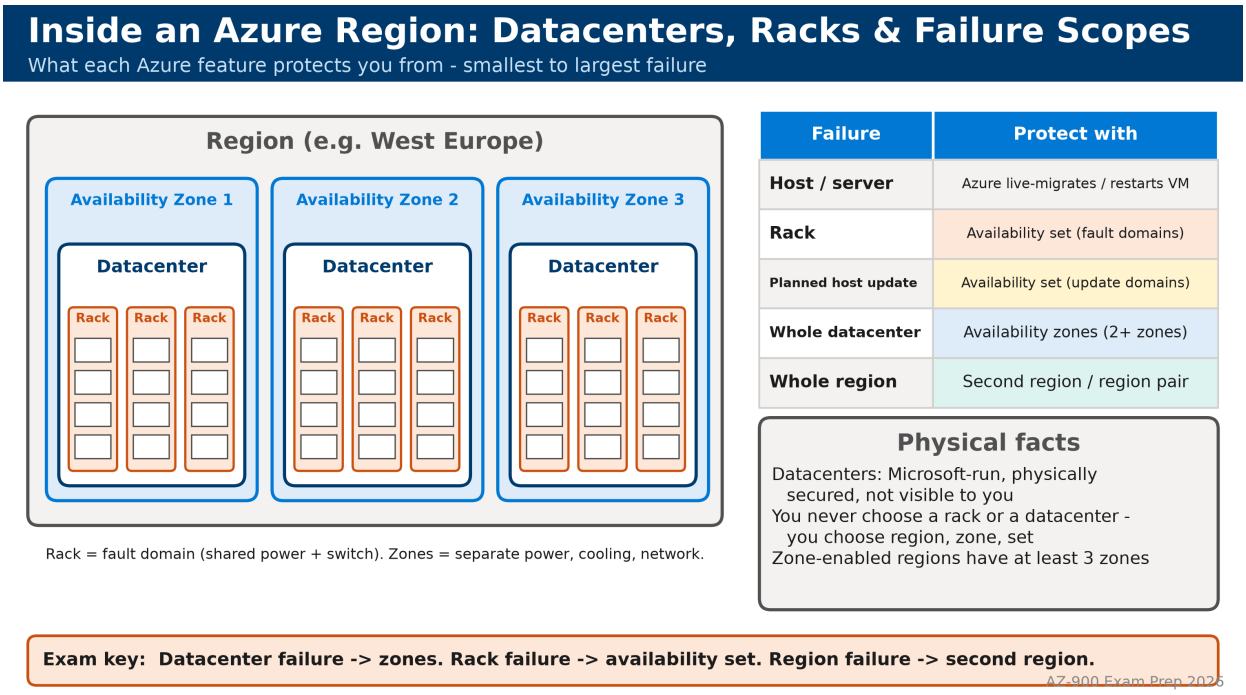


Figure Q304b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 305

AZURE ARCHITECTURE & SERVICES

Infrastructure
v5 Q438

You are reviewing Azure services by how they use availability zones. Azure DNS and Microsoft Entra ID are available across all regions and are resilient to region-wide outages. In which category are these services?

✔ **A.** Non-regional services

✖ B. Zonal services

✖ C. Zone-redundant services

✖ D. Region-pair services

Correct answer: Non-regional services

Clarification (read this first): Three zone categories are easy to mix up. Zonal = pinned to one zone. Zone-redundant = copied across zones in ONE region. Non-regional = not tied to any region, so it survives a whole-region outage.

Answer explained

✔ **Non-regional services**
Non-regional (always-available) services, such as Azure DNS and Entra ID, are not tied to one region. They survive a whole-region outage.

✖ **Zonal services**
Zonal services pin a resource, such as a VM, to one specific zone.

✖ **Zone-redundant services**
These copy across zones but still inside one region, like ZRS.

✖ **Region-pair services**
This is not an official availability-zone service category.

EXAM STRATEGY
'Available across all regions' and 'survives a region-wide outage' = non-regional services. Zonal pins to one zone; zone-redundant stays inside one region.

IMPACT
Knowing that Entra ID and DNS are global stops you from building extra redundancy for them. Spend DR effort on services that really are regional.

What · Where · When · Why · How

What	The non-regional category of Azure services.
Where	Delivered worldwide, not from one region.
When	When deciding which services need extra zone or region design.
Why	Some core services must keep working even if a whole region fails.
How	Microsoft runs them across geographies, so customers set up no zones.

Exam tip
Zonal = pinned to a zone (VM, managed disk). Zone-redundant = spread across zones (ZRS, zone-redundant SQL). Non-regional = global (DNS, Entra ID, Front Door).

Memory trick
Zonal pins, zone-redundant spreads, non-regional floats above.

Real-world example, use case and scenario

EXAMPLE
Name lookup
A media company's Azure DNS zones keep answering during an East US outage.

USE CASE
Sign-in
Staff can still sign in with Entra ID while one region has problems.

SCENARIO
Design review
An architect skips zone planning for DNS but sets up zones for VMs and SQL.

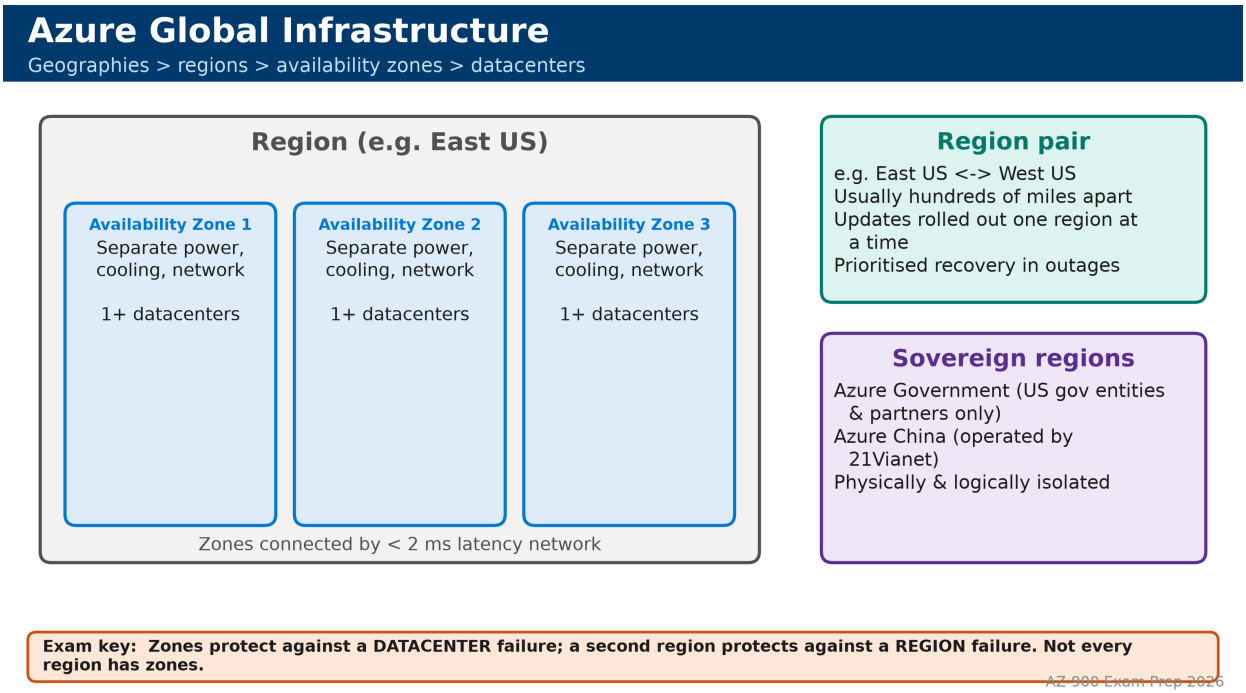


Figure Q305a: Regions, availability zones and region pairs

About this figure

- Nested boxes show the hierarchy: a region holds availability zones, and each zone holds one or more datacenters with independent power, cooling and network.
- Zone-enabled regions have a minimum of three zones. Zonal services pin to one zone; zone-redundant services replicate across zones automatically.
- Azure has 60+ regions worldwide; not all services or VM sizes are available in every region, and prices differ between regions.
- Most region pairs are in the same geography for data residency; some newer regions have no pair, so use another region of your choice for DR.
- Sovereign clouds such as Azure Government and Azure operated by 21Vianet are separate instances with their own portals and accounts.

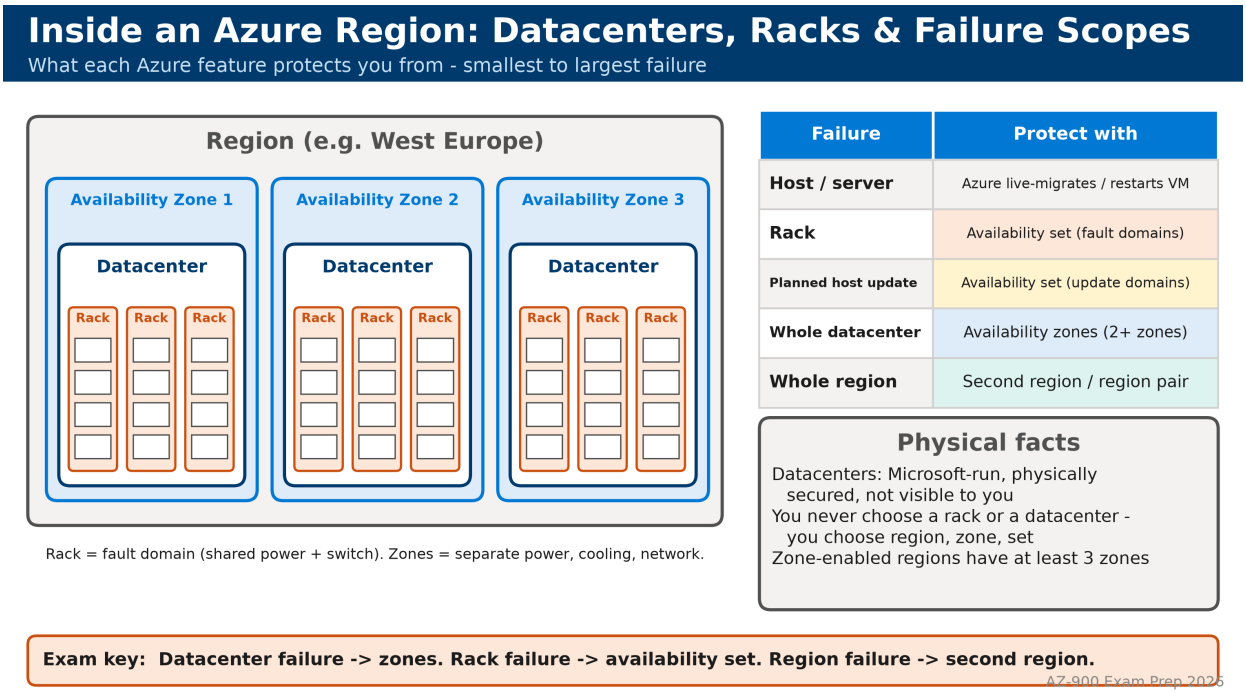


Figure Q305b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 306

AZURE ARCHITECTURE & SERVICES

Subscriptions
v5 Q439

You are designing a management group hierarchy for a large enterprise. Excluding the root level and the subscription level, how many levels deep can a management group hierarchy be?

✔ **A.** 6

✖ B. 3

✖ C. 10

✖ D. 10,000

Correct answer: 6

Clarification (read this first): Two numbers are easy to mix up: 6 = how DEEP the tree can go (levels), 10,000 = how MANY management groups you can have. The question asks about depth.

Answer explained

✔ **6**
Six levels of management groups are allowed below the root. The subscription level is not counted.

✖ **3**
Too low. The documented depth limit is six.

✖ **10**
Not the limit. Depth stops at six levels.

✖ **10,000**
That is the total number of management groups per directory, not the depth.

EXAM STRATEGY
Number questions: 6 levels below root, not counting subscriptions. 10,000 is the limit on how many management groups exist - a classic trap. Pick 6.

IMPACT
Very deep trees hit the six-level limit and make policy inheritance harder. Large companies usually use three or four levels.

What · Where · When · Why · How

What	The depth limit of a management group tree.
Where	Inside one Microsoft Entra tenant, under the tenant root group.
When	When designing governance for many subscriptions.
Why	Limits keep inheritance of policy and RBAC (access roles) easy to manage.
How	Nest management groups up to six levels below root, then add subscriptions.

Exam tip
Numbers to remember: 10,000 management groups per directory, 6 levels deep, one parent for each management group or subscription.

Memory trick
Six floors in the governance tower; 10,000 rooms in the building.

Real-world example, use case and scenario

EXAMPLE
Global company
Root > Geography > Country > Business unit > Environment uses four levels, well within six.

USE CASE
Too deep
A team plans eight nested levels and must make the design flatter.

SCENARIO
Policy inheritance
An allowed-regions policy at level two applies to every subscription below it.

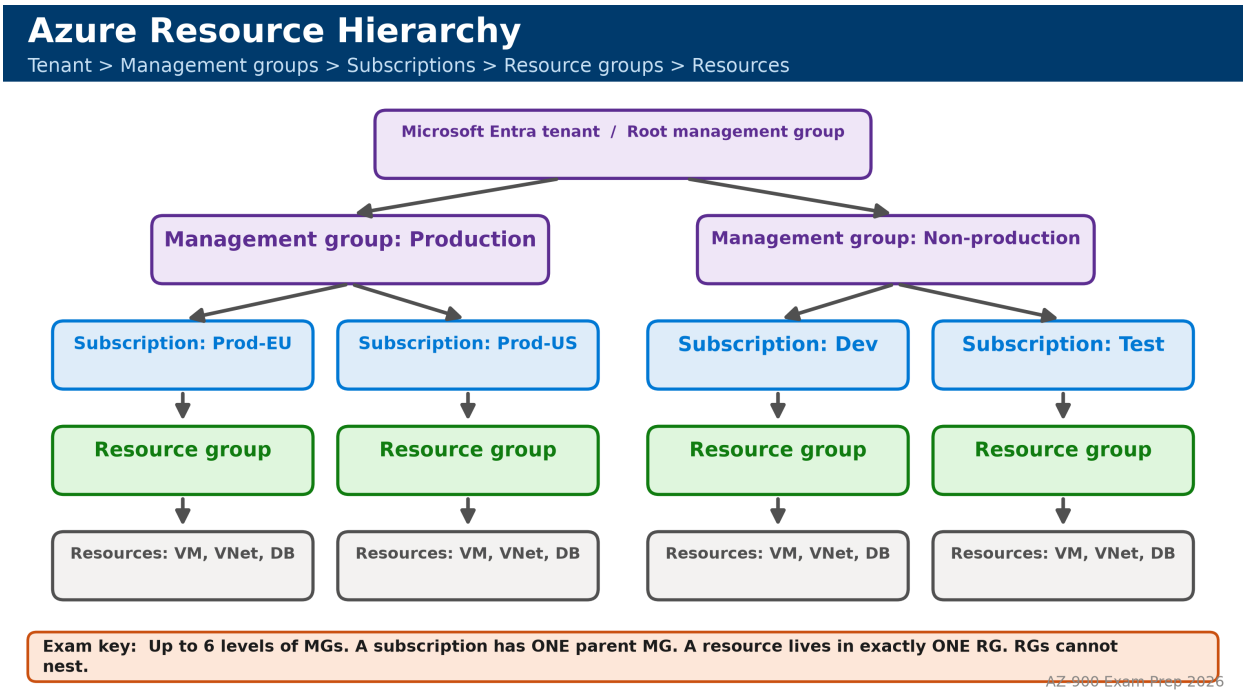


Figure Q306: Azure resource hierarchy

About this figure

• Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.

• A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.

• Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.

• Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.

• Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 307

AZURE ARCHITECTURE & SERVICES

Subscriptions
v5 Q440

An Azure subscription can be placed under two different management groups at the same time so that it inherits policies from both.

✖ A. Yes

✔ **B.** No

Correct answer: No

Answer explained

✔ **No**
No. A subscription has only one parent management group. It still inherits from every group above it in that one chain.

✖ **Yes**
More than one parent is not allowed. The hierarchy is a strict tree.

EXAM STRATEGY
Each management group and subscription has exactly one parent. 'Two at the same time' is impossible, so No.

IMPACT
One parent keeps policy and access inheritance predictable. Place subscriptions carefully, because moving one changes every inherited policy.

What · Where · When · Why · How

What	The one-parent rule for management groups and subscriptions.
Where	In the management group tree under the tenant root group.
When	When organizing or moving subscriptions between management groups.
Why	A strict tree keeps policy and RBAC inheritance predictable.
How	Move a subscription to a new parent; it then inherits from the new chain.

Exam tip
Many children, one parent. Inheritance comes from all the groups above it in that one chain.

Memory trick
Like a folder on disk: it can live in only one parent folder.

Real-world example, use case and scenario

EXAMPLE
Department move
When Marketing joins Sales, its subscription moves to the Sales management group.

USE CASE
Combined rules
To get policies from two groups, place them at different levels of the same chain.

SCENARIO
Audit
An auditor follows a subscription's single path back to the tenant root group.

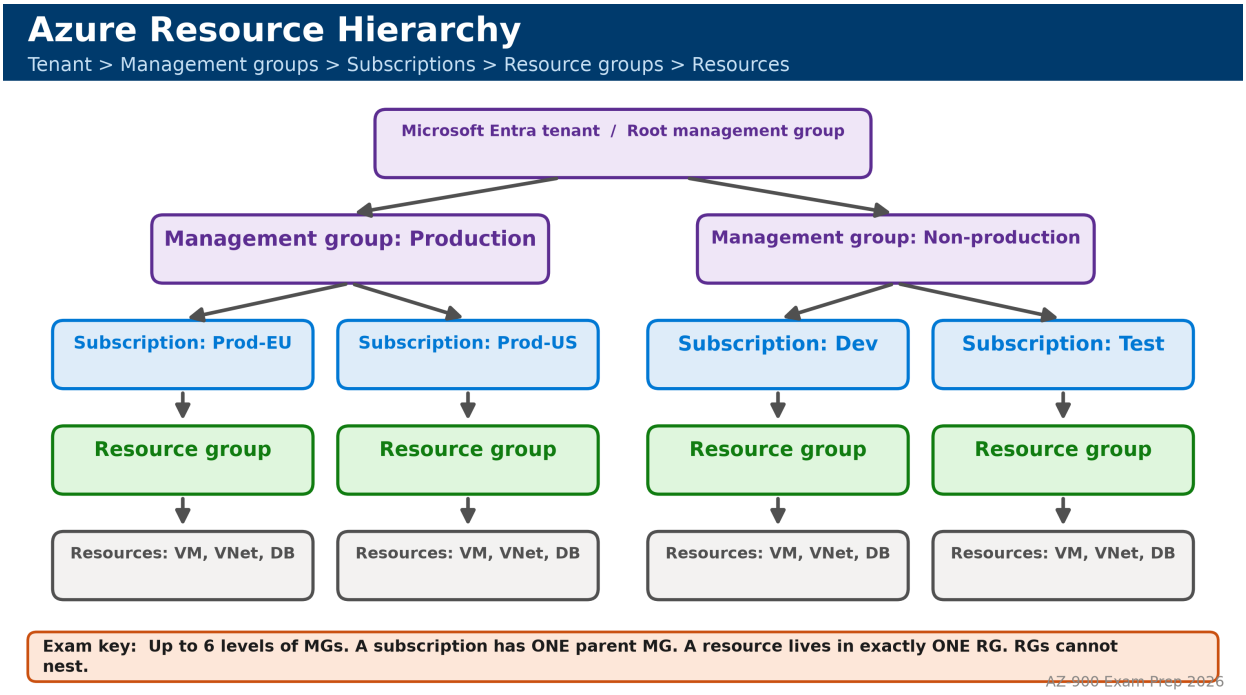


Figure Q307: Azure resource hierarchy

About this figure

- Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.
- A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.
- Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.
- Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.
- Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Your company needs a governance structure in Azure. Which option lists the scopes from the HIGHEST level to the LOWEST level?

- ✔ **A.** Management groups > subscriptions > resource groups > resources
- ✖ B. Subscriptions > management groups > resource groups > resources
- ✖ C. Resource groups > subscriptions > management groups > resources
- ✖ D. Management groups > resource groups > subscriptions > resources

Correct answer: Management groups > subscriptions > resource groups > resources

Answer explained

✔ **Management groups > subscriptions > resource groups > resources**
From top to bottom: management groups, subscriptions, resource groups, resources.

✖ **Subscriptions > management groups > resource groups > resources**
Management groups sit above subscriptions, not below.

✖ **Resource groups > subscriptions > management groups > resources**
Resource groups are inside subscriptions, so this order is upside down.

✖ **Management groups > resource groups > subscriptions > resources**
Subscriptions contain resource groups, not the other way round.

EXAM STRATEGY
Remember the ladder MG > Subscription > RG > Resource. Any option that starts with subscriptions, or puts resource groups above subscriptions, is wrong.

IMPACT
This order is the base for RBAC, Policy, budgets and locks. Almost every governance question expects you to know it.

What · Where · When · Why · How

What	The Azure management hierarchy.
Where	Under one Microsoft Entra tenant's root management group.
When	When planning governance, billing and access boundaries.
Why	Inheritance lets one setting control many resources.
How	Assign policy or RBAC at the highest level that fits, and let it flow down.

Exam tip
Think MG > Sub > RG > R. Settings at a higher level are inherited by lower levels.

Memory trick
My Silly Red Rabbit: Management group, Subscription, Resource group, Resource.

Real-world example, use case and scenario

EXAMPLE
Hospital chain
A hospital group applies an encryption policy at its root management group, so every subscription inherits it.

USE CASE
Dev and prod
Separate subscriptions for dev and prod sit under one Corp management group.

SCENARIO
App grouping
Each app's VMs, disks and network cards are kept together in one resource group.

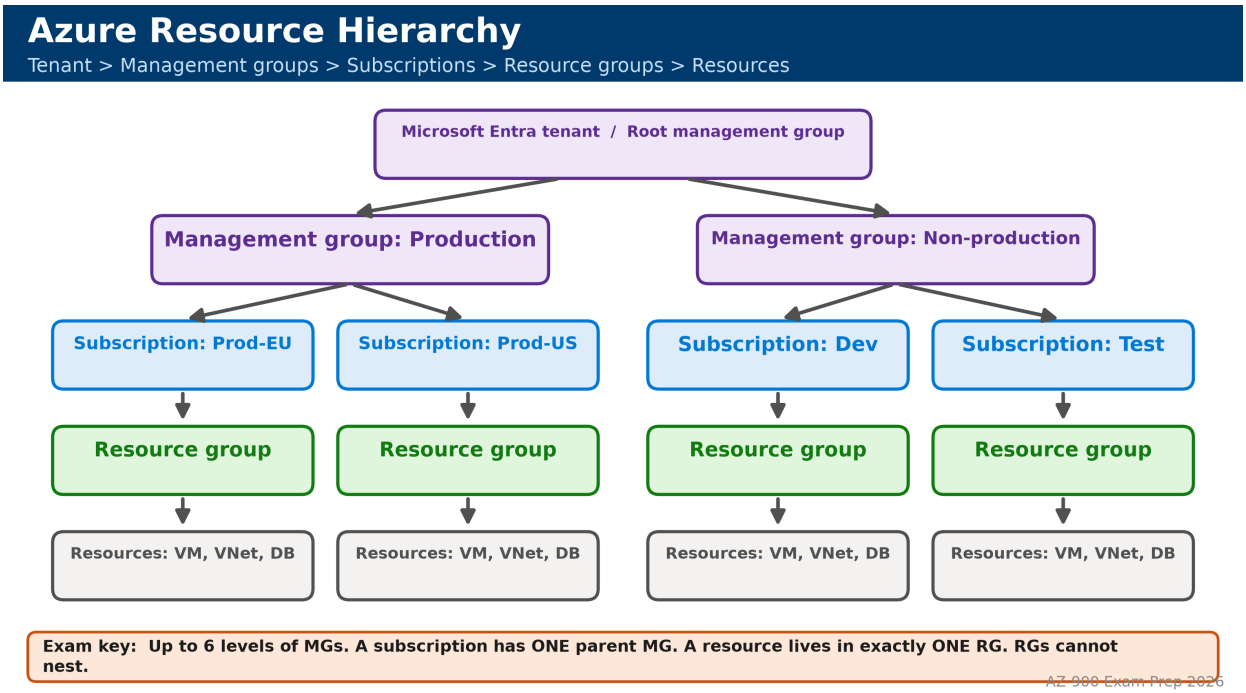


Figure Q308: Azure resource hierarchy

About this figure

• Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.

• A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.

• Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.

• Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.

• Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 309

AZURE ARCHITECTURE & SERVICES

Subscriptions
v5 Q442

Your team wants to organize Azure resources by placing one resource group inside another resource group. Can resource groups be nested?

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): Only MANAGEMENT groups can be nested. RESOURCE groups cannot. Do not mix the two names. Related: Q144 - a resource group cannot contain subscriptions either.

Answer explained

✔ **No**
No. Resource groups are flat and cannot be nested. Use tags or other levels for extra grouping.

✖ **Yes**
Only management groups can be nested. Resource groups have a single level.

EXAM STRATEGY
Resource groups are flat; each resource belongs to exactly one group. Answer No. Use tags or management groups for a hierarchy.

IMPACT
Teams that expect nesting end up with messy resource lists. Tags and naming rules are the real way to organize resources in a subscription.

What · Where · When · Why · How

What	The rule that a resource group cannot hold another resource group.
Where	Inside an Azure subscription.
When	When planning how to group resources for an app or team.
Why	Flat groups keep lifecycle and access control simple.
How	Use tags or management groups instead of nesting resource groups.

Exam tip
Only management groups nest. Resource groups and subscriptions do not.

Memory trick
Resource groups are boxes on one shelf, never boxes inside boxes.

Real-world example, use case and scenario

EXAMPLE
One app per group
An airline puts each booking service in its own resource group instead of nesting them.

USE CASE
Tags instead
A school tags resources with Department=Science instead of nesting groups.

SCENARIO
Wrong plan
A new admin tries to create RG-Child inside RG-Parent and finds no such option.

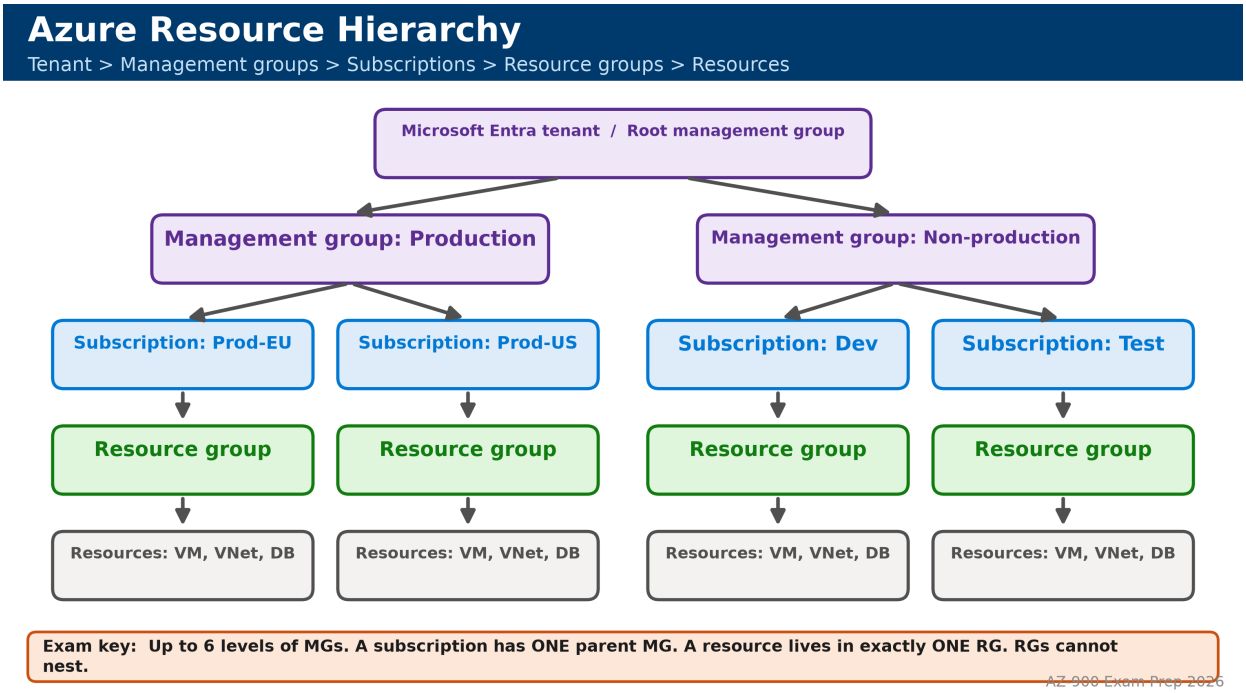


Figure Q309: Azure resource hierarchy

About this figure

- Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.
- A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.
- Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.
- Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.
- Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 310

AZURE ARCHITECTURE & SERVICES

Subscriptions
v5 Q443

A resource group in the West Europe region contains VMs deployed in East US. West Europe experiences a temporary outage. What is the impact on the VMs in East US?

- ✔ **A.** The VMs keep running, but you might be unable to update their configuration until West Europe recovers
- ✖ B. The VMs stop immediately because they are tied to the resource group's region
- ✖ C. The VMs automatically move to the West Europe region pair
- ✖ D. There is no impact of any kind because resource group location is cosmetic

Correct answer: The VMs keep running, but you might be unable to update their configuration until West Europe recovers

Clarification (read this first): The resource group is in West Europe, but the VMs RUN in East US. The group's region only stores metadata (information about the resources). Q273 is related: a group can hold resources from many regions.

Answer explained

✔ **The VMs keep running, but you might be unable to update their configuration until West Europe recovers**
The VMs keep running in their own region. Only management (metadata) changes may be blocked while the group's region is down.

✖ **The VMs stop immediately because they are tied to the resource group's region**
Resources run in their own region. They do not depend on where the group's metadata is stored.

✖ **The VMs automatically move to the West Europe region pair**
Azure never moves resources automatically because of a resource group outage.

✖ **There is no impact of any kind because resource group location is cosmetic**
The location does matter: it stores metadata, so management changes can be affected.

EXAM STRATEGY
Key idea: the group's location stores metadata only. Resources keep running; only management actions may fail. Remove 'stop immediately', 'move automatically' and 'no impact of any kind'.

IMPACT
If the group's metadata is in a far-away region, you may not be able to make urgent changes during an outage. Best practice is to put the group in the same region as its resources.

What · Where · When · Why · How

What	What a resource group's location means.
Where	The region chosen when the resource group is created.
When	When that region has an outage while the resources run somewhere else.
Why	Where metadata is stored matters for compliance and for being able to manage resources.
How	Choose a group region close to, or the same as, its resources.

Exam tip
Resource group location = metadata location. Resources inside can be in any region.

Memory trick
The resource group is the filing cabinet. If the office closes, the workers keep working, but the files cannot be updated.

Real-world example, use case and scenario

EXAMPLE
Compliance
A Swiss bank puts resource groups in Switzerland North so metadata stays in the country.

USE CASE
Outage drill
During a regional problem, admins cannot resize VMs whose resource group is in the affected region.

SCENARIO
Best practice
A start-up keeps resource groups in the same region as their resources to avoid this problem.

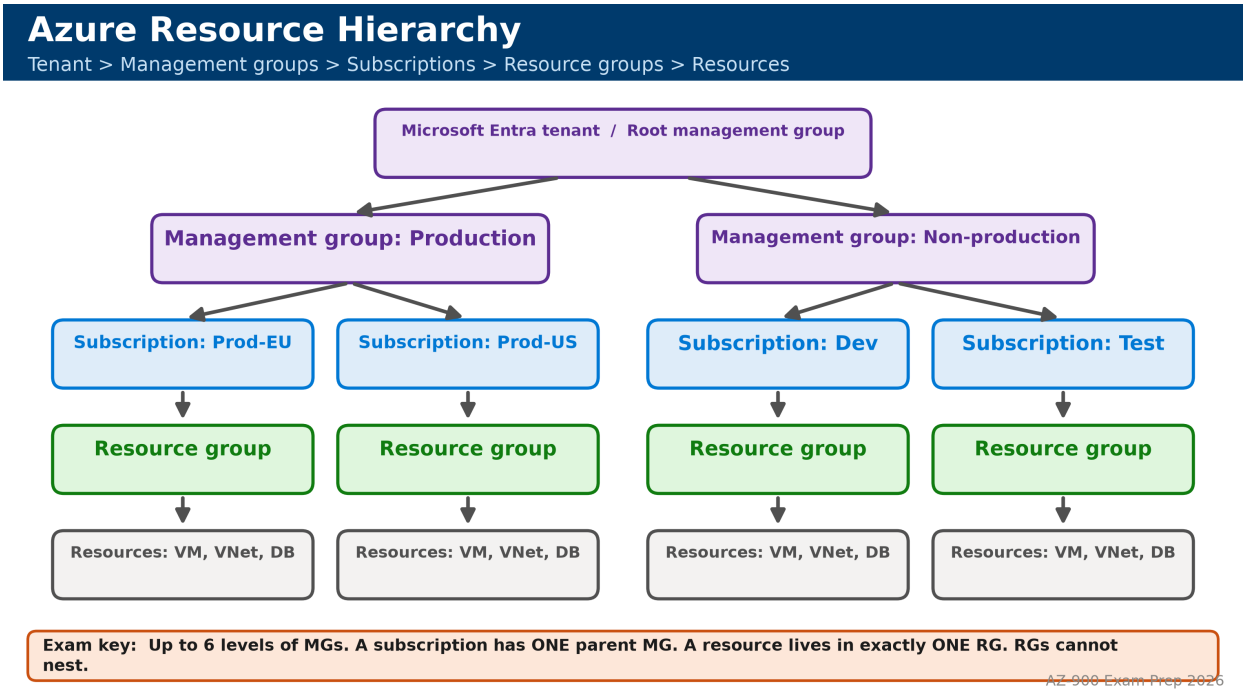


Figure Q310: Azure resource hierarchy

About this figure

- Read top-down: the tenant root management group holds MGs, which hold subscriptions, which hold resource groups, which hold resources.
- A tenant supports up to 10,000 management groups and six levels of depth below the root, not counting the root or subscription level.
- Each management group and subscription has exactly one parent; a resource group belongs to one subscription and cannot contain another resource group.
- Settings applied at a higher level, such as Policy and RBAC, flow down to every child scope, so the root MG governs the whole tenant.
- Subscriptions have service quotas, for example up to 980 resource groups per subscription, so large organisations split workloads across several subscriptions.

Question 311

AZURE ARCHITECTURE & SERVICES

Subscriptions
v5 Q444

Your finance and security teams ask what an Azure subscription provides. Which TWO boundaries does an Azure subscription provide? Each correct answer presents a complete solution.

✔ **A.** A billing boundary

✔ **B.** An access control boundary

✖ C. A network isolation boundary that blocks all cross-subscription traffic

✖ D. A physical datacenter boundary

✖ E. A Microsoft Entra tenant boundary

Correct answer: A billing boundary; An access control boundary

Clarification (read this first): Pick TWO. A subscription separates billing and access. It is not a network wall (VNets in different subscriptions can be peered) and not the tenant (the tenant is the identity boundary).

Answer explained

✔ **A billing boundary; An access control boundary**
A subscription is a billing boundary (separate bills) and an access control boundary (where RBAC and policy apply).

✖ **A network isolation boundary that blocks all cross-subscription traffic**
VNets in different subscriptions can be peered (connected). Subscriptions do not isolate networks.

✖ **A physical datacenter boundary**
A subscription's resources can be in any region and datacenter.

✖ **A Microsoft Entra tenant boundary**
One tenant can be trusted by many subscriptions. The tenant is the identity boundary.

EXAM STRATEGY
Subscription = billing plus access control boundary. Remove 'physical datacenter', 'tenant' (a tenant holds subscriptions) and 'blocks all traffic' (an absolute word).

IMPACT
One subscription per department gives clean bills and separate access. This is the base of charging costs back to teams and of landing-zone design.

What · Where · When · Why · How

What	The two boundaries that a subscription gives.
Where	Between the management group level and resource groups.
When	When deciding whether to create separate subscriptions.
Why	To separate costs and control who can manage which resources.
How	Create subscriptions per environment, department or billing need.

Exam tip
Subscription = billing + access control boundary.
Tenant = identity boundary.

Memory trick
Subscriptions hold the wallet and the keys.

Real-world example, use case and scenario

EXAMPLE
Chargeback
A university gives each faculty its own subscription, so each gets a separate bill.

USE CASE
Locked-down production
Only the operations team has Contributor rights on the production subscription.

SCENARIO
Peering
Two subscriptions' VNets are peered. This shows subscriptions are not network boundaries.

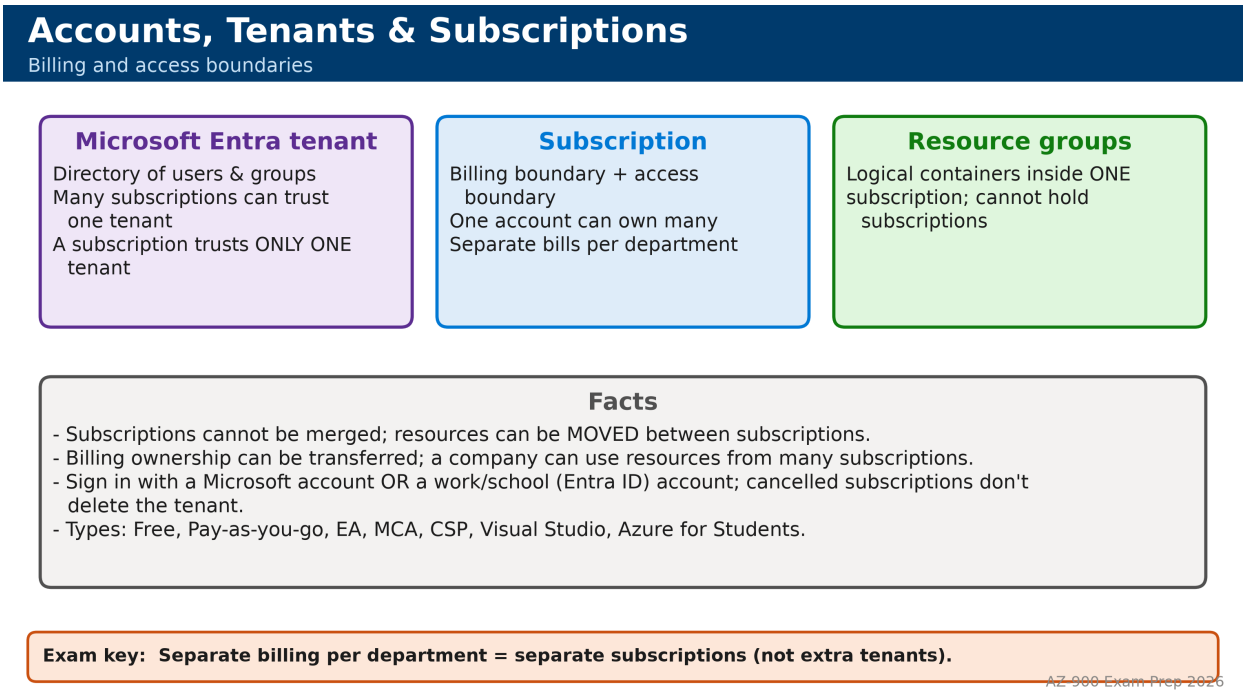


Figure Q311a: Accounts, tenants and subscriptions

About this figure

• Boxes move from identity (tenant) to billing (subscription) to organisation (resource groups); the Facts box lists what can be moved, transferred or merged.

• Key rule: a subscription trusts exactly one Entra tenant, while one tenant can be trusted by many subscriptions; changing the directory can remove RBAC assignments.

• Pay-as-you-go bills monthly by card or invoice; EA suits large organisations with an upfront commitment; CSP is bought through a Microsoft partner.

• Visual Studio subscribers get monthly Azure credit (approx. USD 50 to 150); Azure for Students gives approx. USD 100 credit without a credit card.

• Subscriptions have quotas, such as regional vCPU limits, that can often be raised by a support request at no cost.

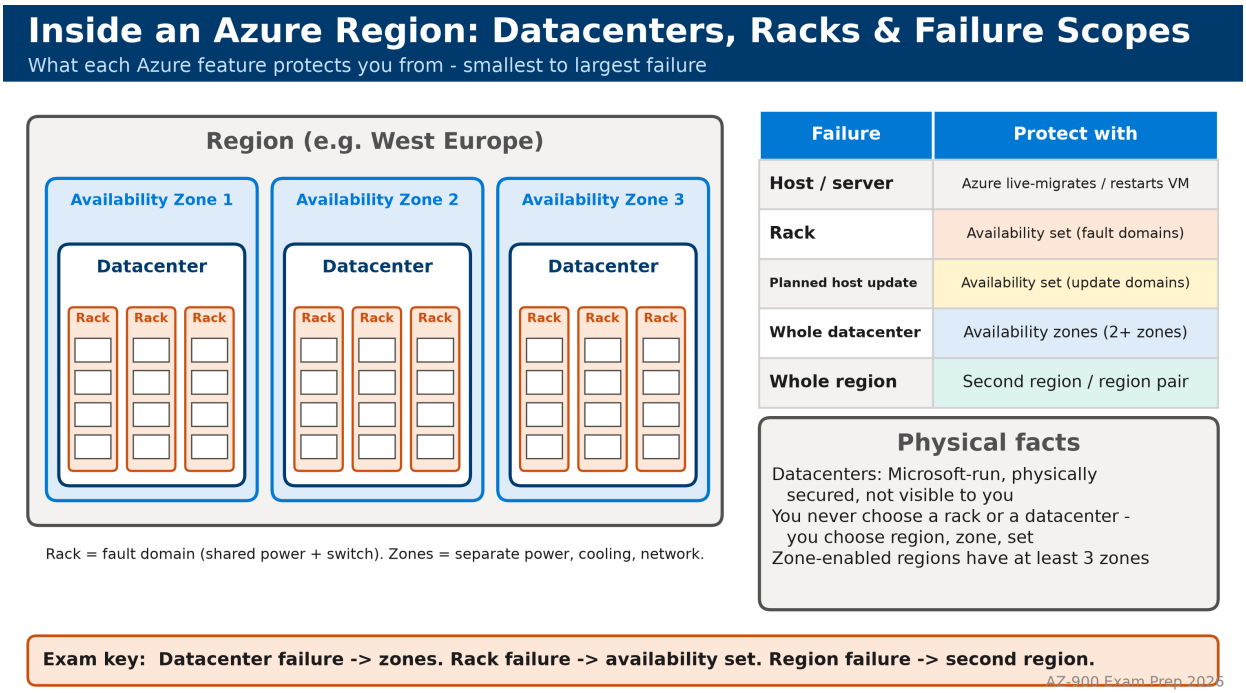


Figure Q311b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

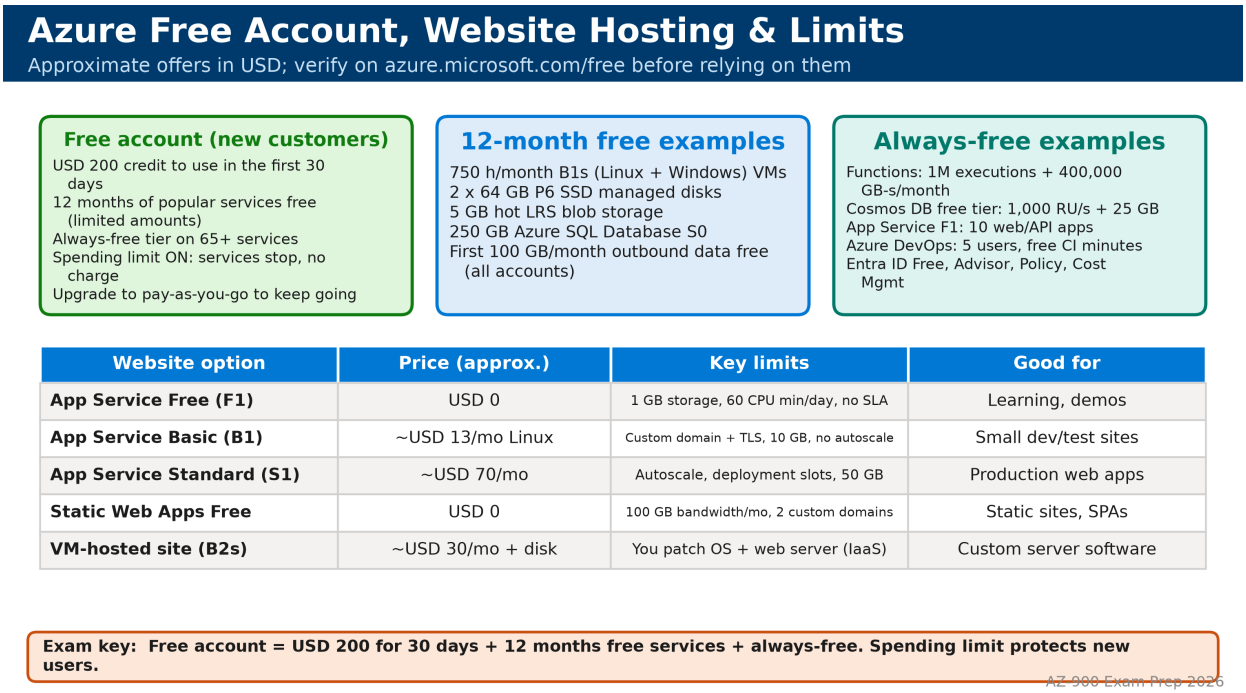


Figure Q311c: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 312

AZURE ARCHITECTURE & SERVICES — **Compute** v5 Q445

Your company runs an online store whose traffic spikes during holiday sales. You need a group of identical, load-balanced VMs that automatically increase or decrease in number based on CPU usage or a schedule. What should you use?

✔ **A.** Azure Virtual Machine Scale Sets

✖ B. An availability set

✖ C. Azure Virtual Desktop

✖ D. Azure Site Recovery

Correct answer: Azure Virtual Machine Scale Sets

Answer explained

✔ **Azure Virtual Machine Scale Sets**
VM Scale Sets give a group of identical, load-balanced VMs that grow or shrink automatically based on metrics or a schedule.

✖ **An availability set**
It spreads VMs across fault and update domains for availability. It does not add or remove VMs.

✖ **Azure Virtual Desktop**
It gives virtual desktops and apps to users, not autoscaling web servers.

✖ **Azure Site Recovery**
It copies workloads for disaster recovery. It does not add VMs when load rises.

EXAM STRATEGY
'Identical VMs' plus 'automatically increase or decrease' = Scale Sets. Availability sets do not autoscale; Site Recovery is DR; Virtual Desktop is user desktops.

IMPACT
Autoscaling handles holiday peaks without paying for peak capacity all year. Fixed VM groups either fail under load or waste money when idle.

What · Where · When · Why · How

What	Virtual Machine Scale Sets.
Where	In an Azure region, optionally spread across availability zones.
When	When demand for a VM-based workload changes over time.
Why	To match capacity to demand automatically and pay only for what runs.
How	Define a VM image and autoscale rules; Azure adds or removes identical VMs.

Exam tip
'Identical VMs' plus 'automatically scale' point to VM Scale Sets.

Memory trick
Scale Sets = a set of clones that grows and shrinks.

Real-world example, use case and scenario

EXAMPLE
Holiday shop
A retailer grows from 4 to 40 web VMs on Black Friday, then goes back down.

USE CASE
Rendering
A film studio adds render VMs overnight and goes down to zero by morning.

SCENARIO
Schedule
A university adds VMs every exam week with a schedule-based rule.

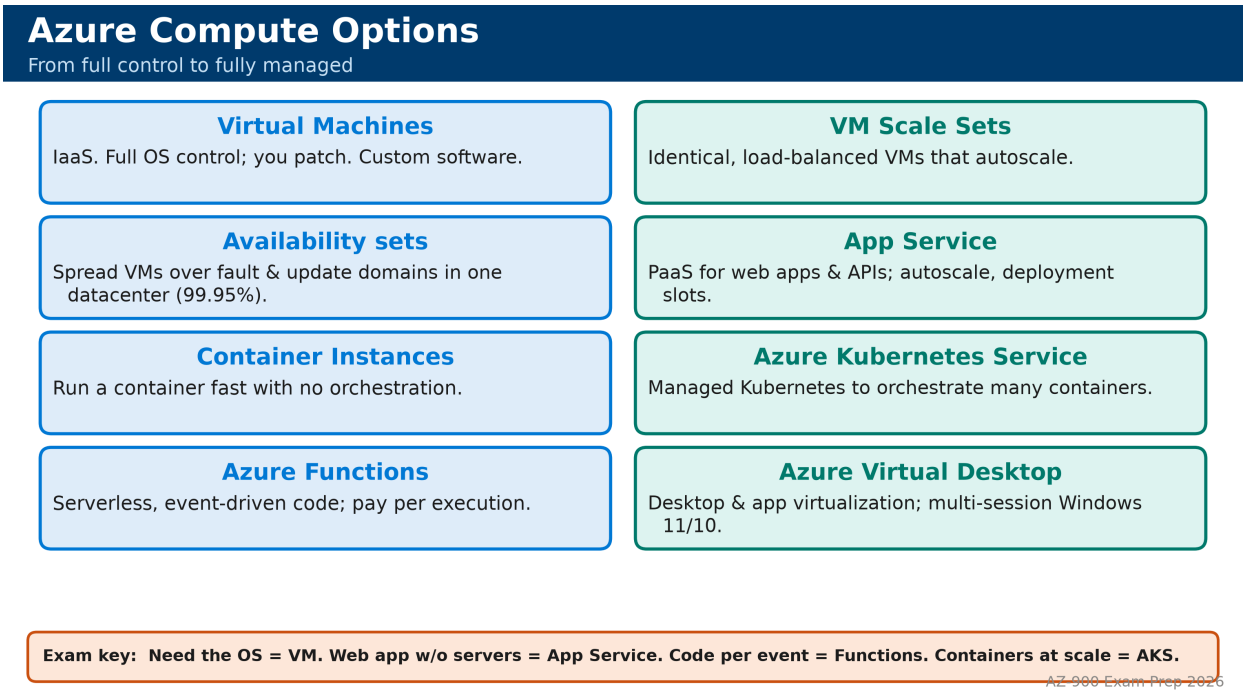


Figure Q312a: Azure compute options

About this figure

- Read left column then right: options run from full control (VMs) to fully managed or serverless (Functions); pick by how much you must manage.
- VM sizes range from tiny B-series (B1s approx. USD 8/month Linux) to large memory and GPU VMs; per-second billing applies while running.
- VM Scale Sets support up to 1,000 instances (600 with custom images); availability sets use up to 3 fault domains and 20 update domains.
- App Service plans set the price: Free F1 USD 0, Basic B1 approx. USD 13/month Linux, Standard S1 approx. USD 70/month with autoscale and slots.
- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

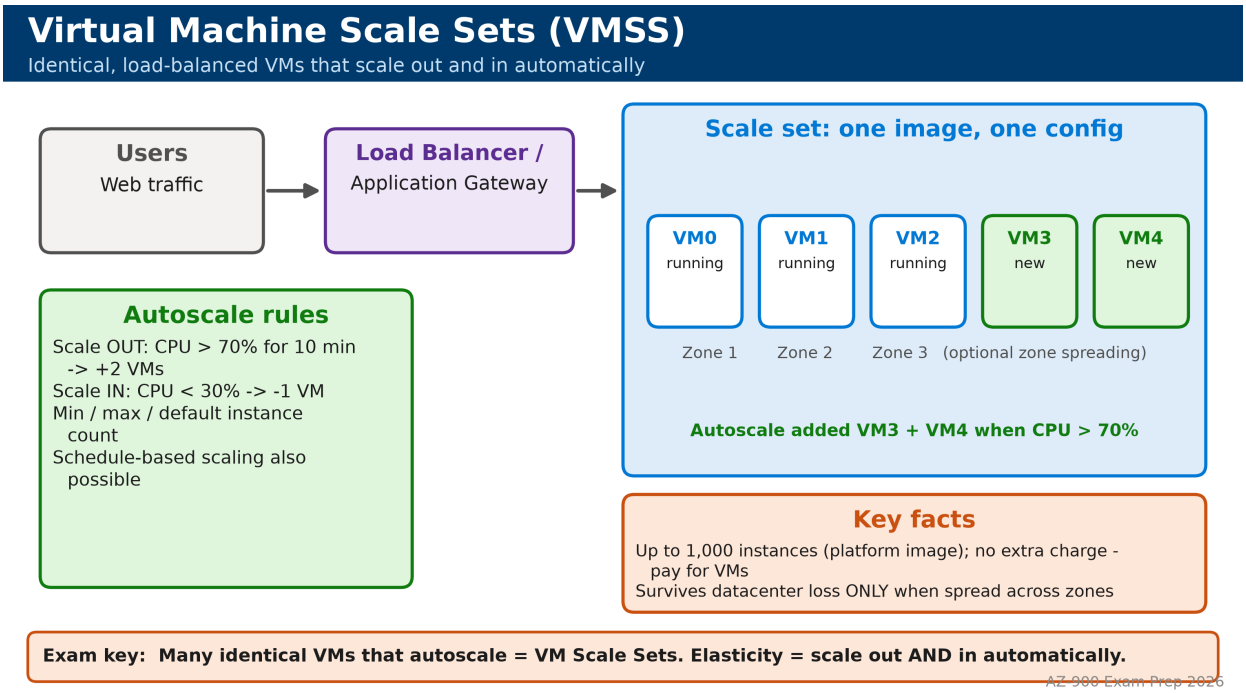


Figure Q312b: Virtual Machine Scale Sets

About this figure

- A scale set runs identical VMs from one image and configuration behind a load balancer or Application Gateway.
- Autoscale adds instances when a metric (e.g. CPU > 70%) is high and removes them when low, within min/max instance counts - this is elasticity.
- Scale sets support up to 1,000 VM instances (platform images); the scale set itself is free - you pay only for the VMs, disks and networking.
- Orchestration modes: Uniform (identical VMs) and Flexible (mixed sizes, recommended for new deployments).
- A scale set survives a datacenter failure only when its instances are spread across availability zones; a non-zonal set does not.
- Scaling out adds more VMs (horizontal); scaling up means a bigger VM size (vertical) and usually needs a restart.

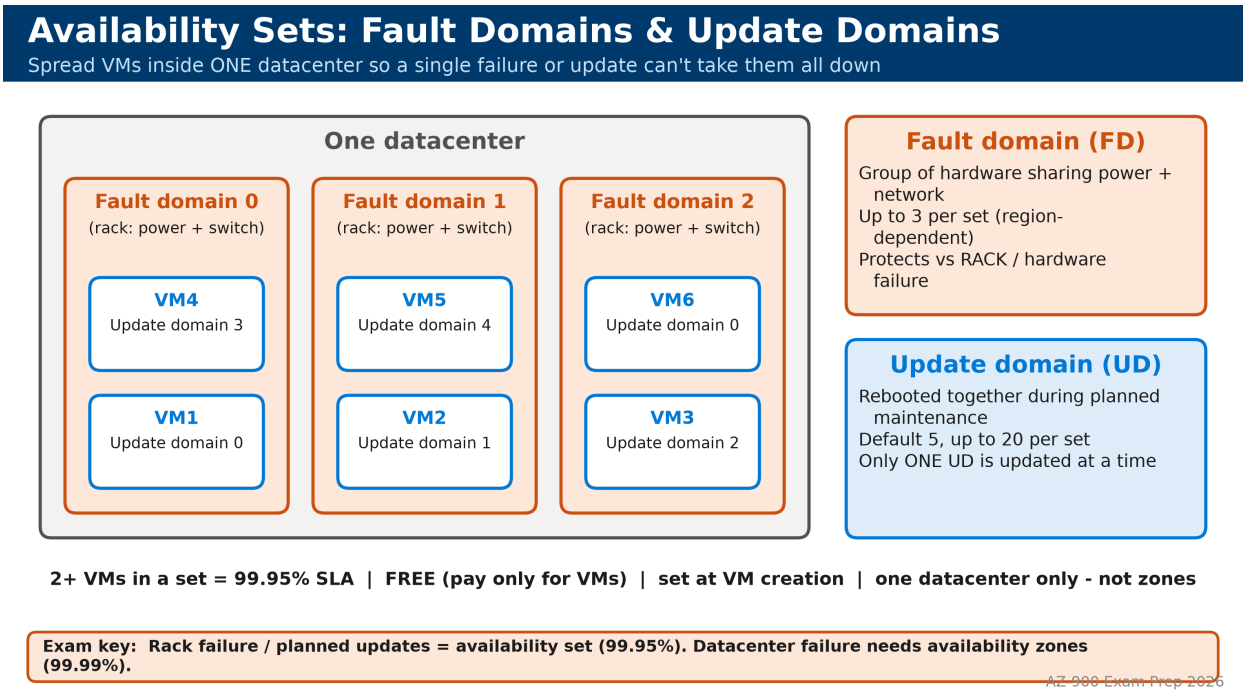


Figure Q312c: Availability sets: fault and update domains

About this figure

- An availability set spreads VMs across fault domains (racks with separate power and switch) and update domains inside ONE datacenter.
- Up to 3 fault domains protect against rack or hardware failure; update domains (default 5, max 20) are rebooted one at a time during planned maintenance.
- With 6 VMs and 5 update domains, the 6th VM wraps back to update domain 0 - as the figure shows.
- Two or more VMs in an availability set get a 99.95% SLA; availability sets are free - you pay only for the VMs.
- A VM is placed in an availability set only at creation; availability sets and availability zones cannot be combined for one VM.
- Availability sets do NOT protect against a datacenter or region failure - use availability zones or a second region for that.

Question 313

AZURE ARCHITECTURE & SERVICES — Compute v5 Q446

You deploy three VMs into an availability set. Which TWO groupings does Azure use to distribute the VMs? Each correct answer presents part of the solution.

- ✔ **A.** Fault domains
- ✔ **B.** Update domains
- ✖ C. Availability zones
- ✖ D. Region pairs
- ✖ E. Management groups

Correct answer: Fault domains; Update domains

Clarification (read this first): Pick TWO. An availability set works INSIDE one datacenter, using fault domains and update domains. Zones and region pairs are bigger, separate features.

Answer explained

✔ **Fault domains; Update domains**
Availability sets use fault domains (separate racks, power and network) and update domains (groups that are rebooted at different times for planned maintenance).

✖ **Availability zones**
Zones are separate datacenters. A VM uses either a zone or an availability set, not both.

✖ **Region pairs**
Region pairs are for cross-region disaster recovery, not an availability set grouping.

✖ **Management groups**
These organize subscriptions for governance, not VM placement.

EXAM STRATEGY
Availability set = fault domains (rack, power, network) plus update domains (maintenance). Remove zones and region pairs; they are separate, larger features.

IMPACT
Without spreading VMs across fault and update domains, one rack failure or planned reboot can stop all of them. That would break the 99.95% VM SLA (uptime promise).

What · Where · When · Why · How

What	Fault domains and update domains in an availability set.
Where	Inside a single Azure datacenter.
When	When you need VM redundancy without availability zones.
Why	To survive a rack failure or a planned host reboot.
How	Put two or more VMs in an availability set; Azure spreads them automatically.

Exam tip
Fault domain = unplanned hardware failure.
Update domain = planned maintenance.

Memory trick
Fault = falls over; Update = upgrades.

Real-world example, use case and scenario

EXAMPLE
Rack failure
A network switch fails in one rack. The VM in another fault domain keeps working.

USE CASE
Host patching
Azure reboots update domain 1 while update domain 2 stays online.

SCENARIO
Region without zones
A company in a region without zones uses an availability set for its SQL VMs.

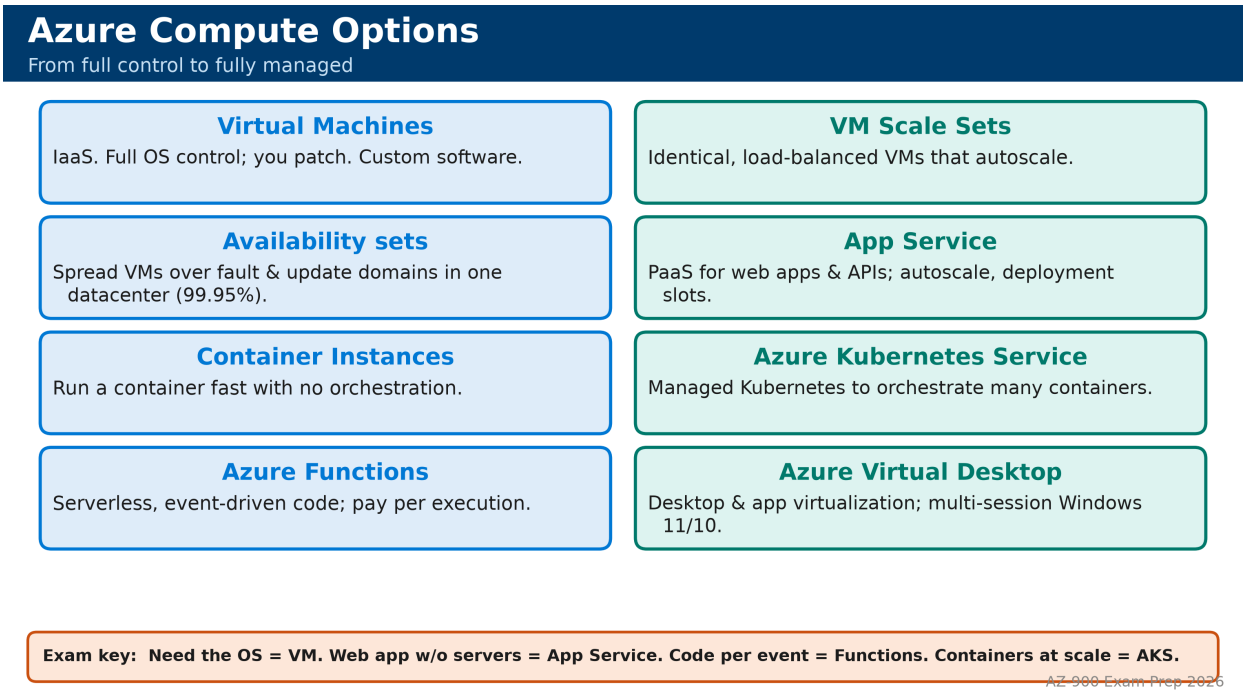


Figure Q313a: Azure compute options

About this figure

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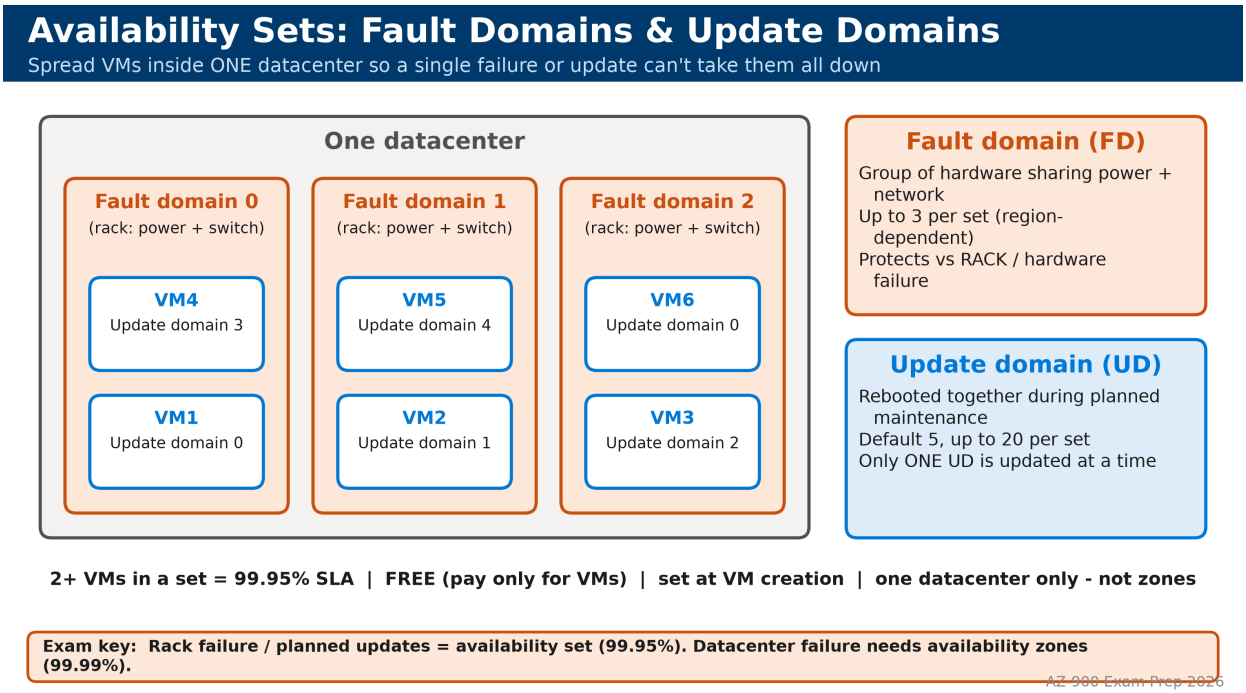


Figure Q313b: Availability sets: fault and update domains

About this figure

- An availability set spreads VMs across fault domains (racks with separate power and switch) and update domains inside ONE datacenter.
- Up to 3 fault domains protect against rack or hardware failure; update domains (default 5, max 20) are rebooted one at a time during planned maintenance.
- With 6 VMs and 5 update domains, the 6th VM wraps back to update domain 0 - as the figure shows.
- Two or more VMs in an availability set get a 99.95% SLA; availability sets are free - you pay only for the VMs.
- A VM is placed in an availability set only at creation; availability sets and availability zones cannot be combined for one VM.
- Availability sets do NOT protect against a datacenter or region failure - use availability zones or a second region for that.

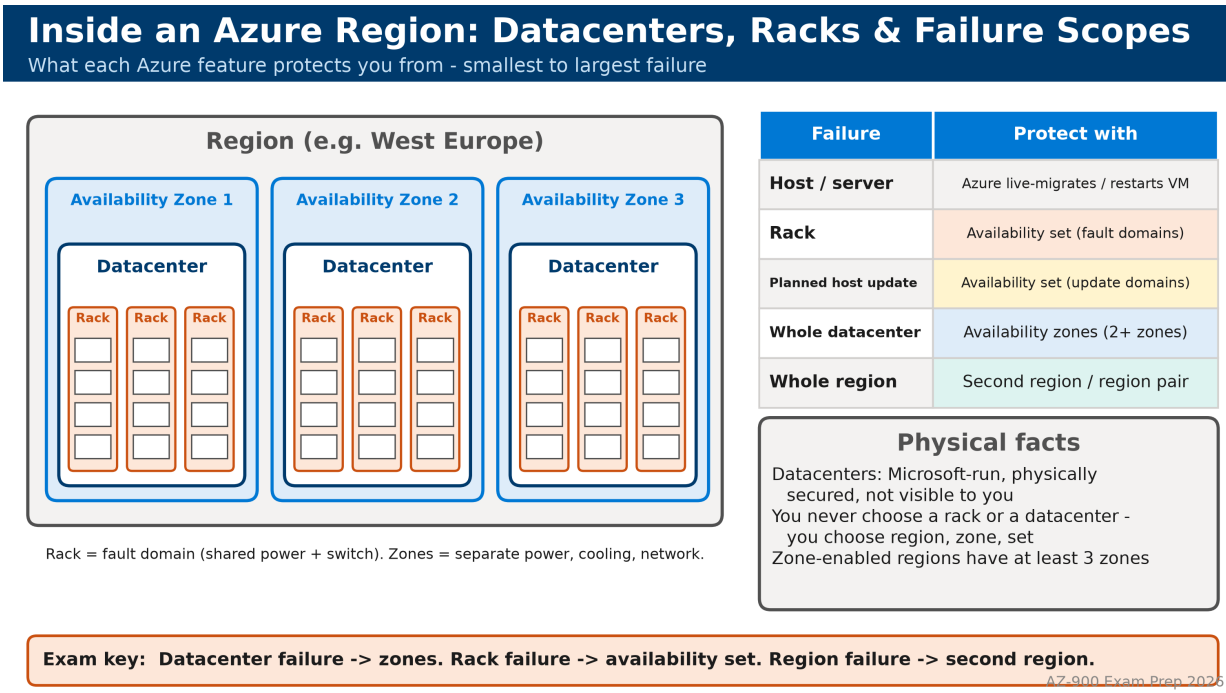


Figure Q313c: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 314

AZURE ARCHITECTURE & SERVICES — Compute — v5 Q447

Placing two VMs in an availability set protects the application if the entire datacenter hosting the availability set fails.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): The question is about losing the WHOLE datacenter. An availability set protects only against rack or maintenance problems inside one datacenter. Q141 (availability zones protect from datacenter failure) is Yes.

Answer explained

✔ **No**
No. An availability set is inside one datacenter. Use availability zones to survive a datacenter outage.

✖ **Yes**
All VMs in an availability set are in the same datacenter, so a datacenter outage hits all of them.

EXAM STRATEGY
An availability set lives inside one datacenter. 'Entire datacenter fails' needs availability zones, so No.

IMPACT
Mixing up sets and zones leaves apps open to datacenter failures. Zones are the right choice for a 99.99% VM SLA.

What · Where · When · Why · How

What	How far an availability set protects you.
Where	Inside a single datacenter in one region.
When	When choosing between availability sets and availability zones.
Why	Rack-level protection is not enough for a whole-building outage.
How	Use zones for datacenter failure and region pairs for region failure.

Exam tip
Protection ladder: availability set (rack) < availability zones (datacenter) < region pairs (region).

Memory trick
Set = same building; Zones = different buildings; Regions = different cities.

Real-world example, use case and scenario

EXAMPLE
Power loss
A datacenter power loss stops both VMs of an availability set at once.

USE CASE
Upgrade
A delivery company moves from an availability set to zones to reach a 99.99% SLA.

SCENARIO
Cost choice
A small blog accepts availability-set protection because it does not need zones.

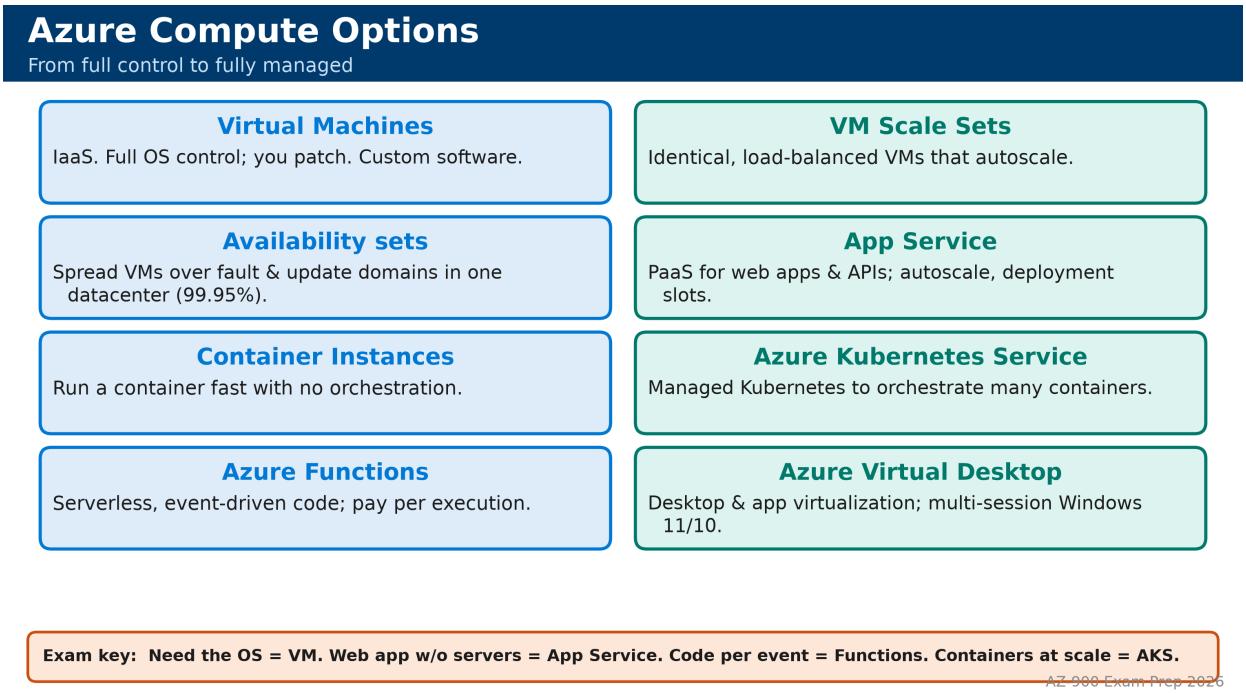


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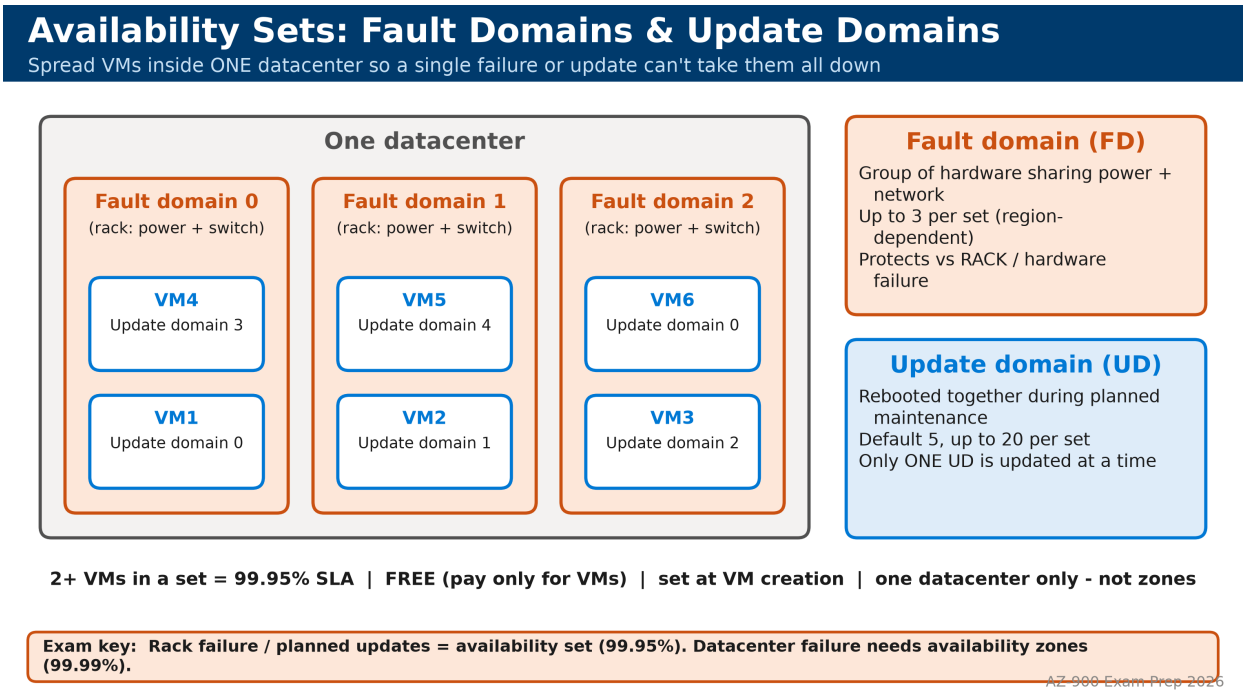


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- With 6 VMs and 5 update domains, the 6th VM wraps back to update domain 0 - as the figure shows.
- Two or more VMs in an availability set get a 99.95% SLA; availability sets are free - you pay only for the VMs.
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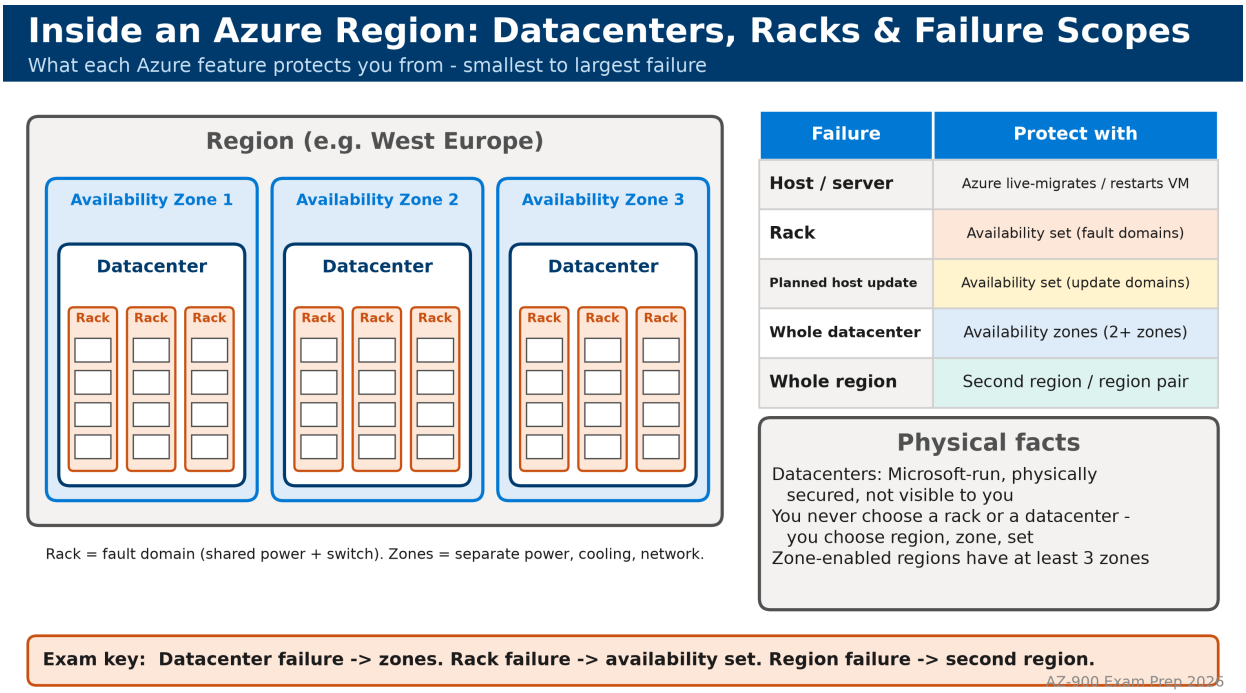


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• Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.

• Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.

• You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.

• Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 315

AZURE ARCHITECTURE & SERVICES — **Compute** v5 Q448

Azure charges an additional fee for creating an availability set, on top of the cost of the VMs placed in it.

✖ A. Yes

✔ **B.** No

Correct answer: No

Answer explained

✔ **No**
No. Availability sets are free. You pay only for the VMs and their resources.

✖ **Yes**
Microsoft does not charge for the availability set itself.

EXAM STRATEGY
Availability sets are free; you pay only for the VMs. Any 'extra fee' for a set is No.

IMPACT
Because sets cost nothing, there is no reason to skip them for multi-VM workloads that cannot use zones. It is free resilience.

What · Where · When · Why · How

What	The price of availability sets.
Where	In any Azure subscription that deploys VMs.
When	When estimating the cost of a highly available VM design.
Why	To see that extra cost comes from extra VMs, not from the set.
How	Create the set for free, then pay normal prices for each VM in it.

Exam tip
Availability sets and resource groups cost nothing themselves; the resources inside them cost money.

Memory trick
The set is a free label; the VMs are what you pay for.

Real-world example, use case and scenario

EXAMPLE
Budget
A clinic budgets for two VMs, not for the availability set they share.

USE CASE
Pricing calculator
An architect finds no line for availability sets in the Pricing calculator.

SCENARIO
Finance question
A finance chief asks why high availability doubled the bill. The answer is the second VM, not the set.

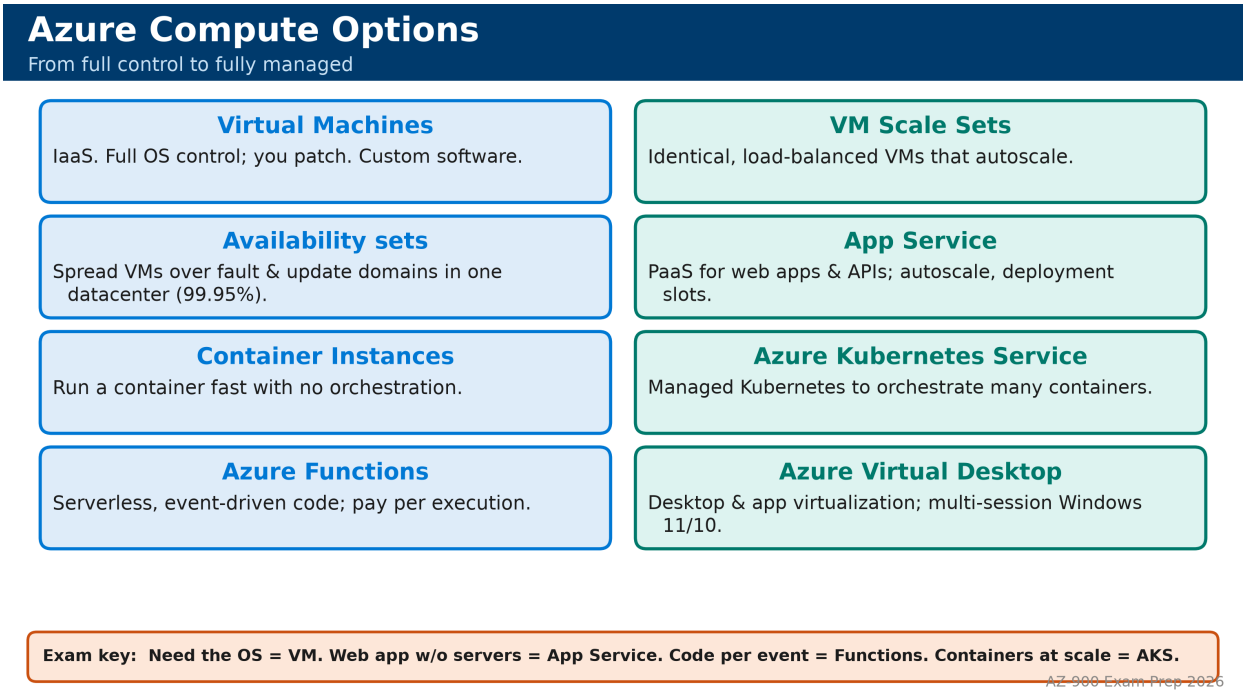


Figure Q315a: Azure compute options

About this figure

- Read left column then right: options run from full control (VMs) to fully managed or serverless (Functions); pick by how much you must manage.
- VM sizes range from tiny B-series (B1s approx. USD 8/month Linux) to large memory and GPU VMs; per-second billing applies while running.
- VM Scale Sets support up to 1,000 instances (600 with custom images); availability sets use up to 3 fault domains and 20 update domains.
- App Service plans set the price: Free F1 USD 0, Basic B1 approx. USD 13/month Linux, Standard S1 approx. USD 70/month with autoscale and slots.
- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

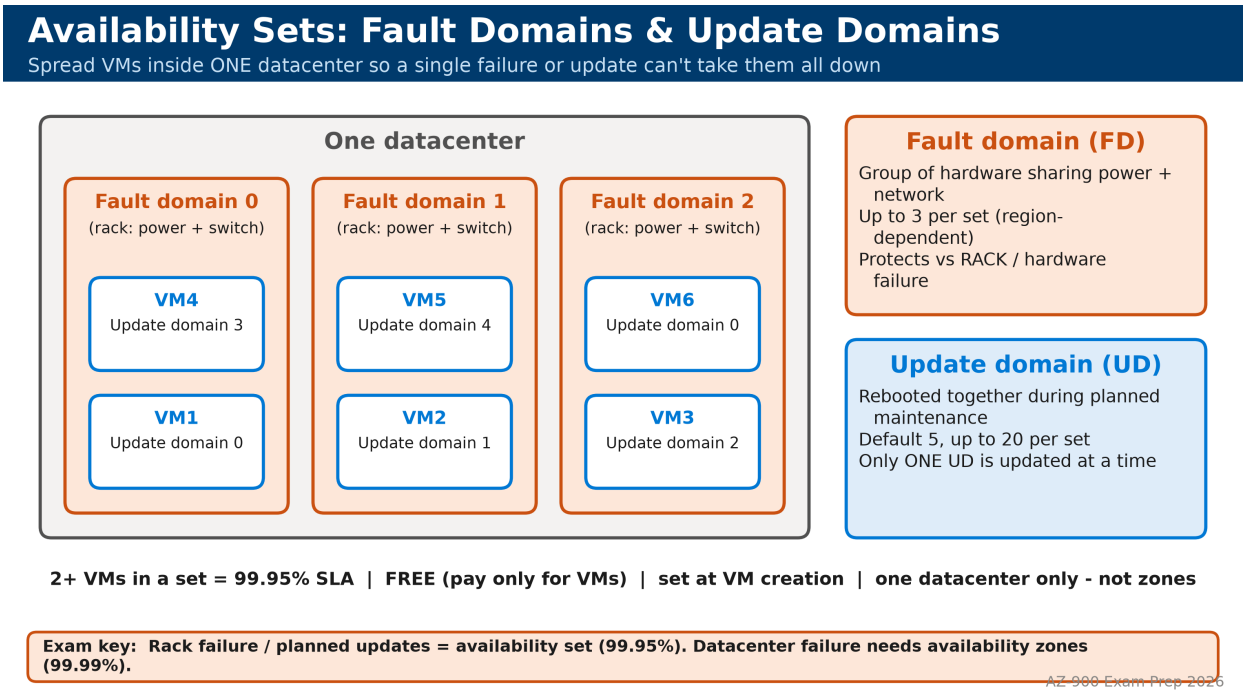


Figure Q315b: Availability sets: fault and update domains

About this figure

- An availability set spreads VMs across fault domains (racks with separate power and switch) and update domains inside ONE datacenter.
- Up to 3 fault domains protect against rack or hardware failure; update domains (default 5, max 20) are rebooted one at a time during planned maintenance.
- With 6 VMs and 5 update domains, the 6th VM wraps back to update domain 0 - as the figure shows.
- Two or more VMs in an availability set get a 99.95% SLA; availability sets are free - you pay only for the VMs.
- A VM is placed in an availability set only at creation; availability sets and availability zones cannot be combined for one VM.
- Availability sets do NOT protect against a datacenter or region failure - use availability zones or a second region for that.

Question 316

AZURE ARCHITECTURE & SERVICES — Compute v5 Q449

Your company wants to give 500 call-center staff a full Windows 11 desktop from Azure. To reduce cost, several users must share each session host VM concurrently using a Windows client OS. What should you use?

✔ **A.** Azure Virtual Desktop with Windows 11 Enterprise multi-session

✖ B. Azure Virtual Machine Scale Sets running Windows Server Core

✖ C. Azure App Service on Windows

✖ D. Azure Container Instances with Windows containers

Correct answer: Azure Virtual Desktop with Windows 11 Enterprise multi-session

Clarification (read this first): Key words: 'several users share each VM' and 'Windows CLIENT OS' (Windows 11, not Windows Server). Only Azure Virtual Desktop offers multi-session Windows 11.

Answer explained

✔ **Azure Virtual Desktop with Windows 11 Enterprise multi-session**
Azure Virtual Desktop offers Windows 11 Enterprise multi-session. Many users can share one VM, each with a full Windows desktop.

✖ **Azure Virtual Machine Scale Sets running Windows Server Core**
Server Core has no desktop, and scale sets do not connect users to desktops.

✖ **Azure App Service on Windows**
App Service hosts web apps and APIs, not user desktops.

✖ **Azure Container Instances with Windows containers**
Containers run apps, not interactive desktops for many users.

EXAM STRATEGY
'Several users share each VM' plus 'Windows client OS' = Windows 11 Enterprise multi-session, which exists only in Azure Virtual Desktop. Remove Server Core and containers.

IMPACT
Sharing VMs between users can greatly cut desktop costs compared with one VM per user. This is why call centers use it.

What · Where · When · Why · How

What	Azure Virtual Desktop multi-session Windows.
Where	On session host VMs (the VMs users connect to) in Azure.
When	When many users need a Windows desktop and cost per user matters.
Why	Multi-session puts more users on fewer VMs and keeps a Windows 11 experience.
How	Create a host pool with Windows 11 Enterprise multi-session images and assign users.

Exam tip
Multi-session Windows 10/11 Enterprise is only available in Azure Virtual Desktop.

Memory trick
AVD = many desks, one machine.

Real-world example, use case and scenario

EXAMPLE
Call center
A phone company puts 15 agents on each AVD session host to use fewer VMs.

USE CASE
Seasonal staff
A tax firm adds AVD host pools for extra accountants every spring.

SCENARIO
Contractors
A drug company gives contractors secure desktops without sending laptops.

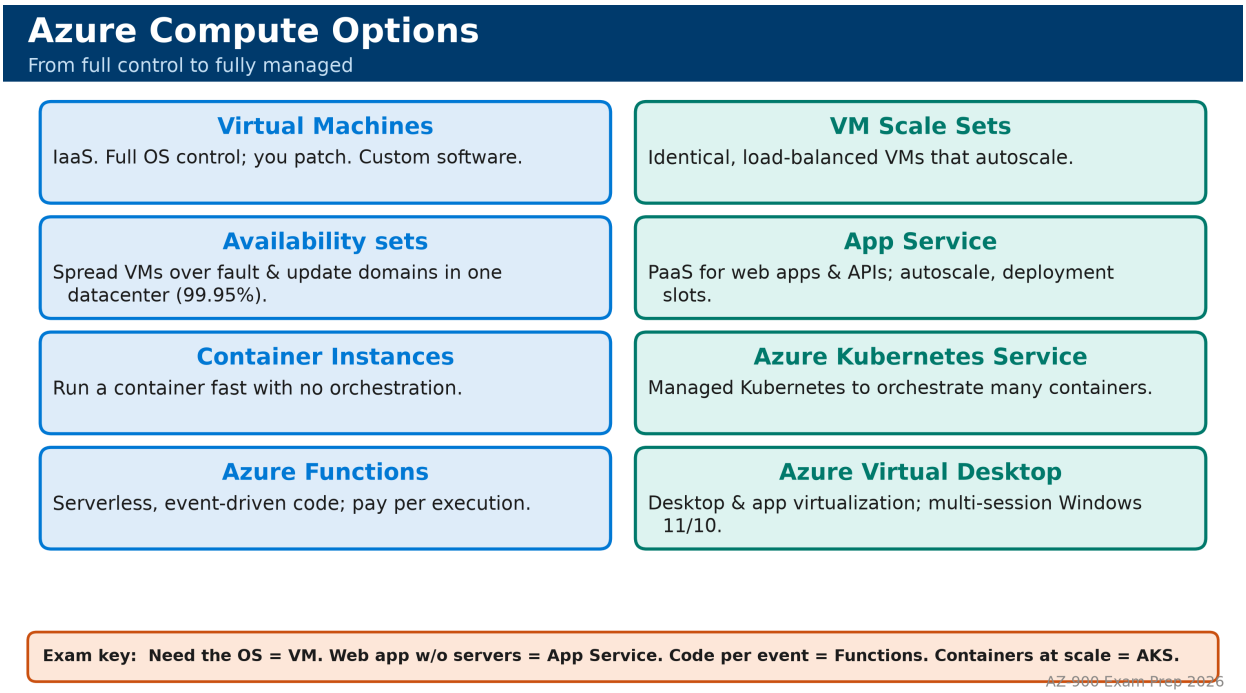


Figure Q316a: Azure compute options

About this figure

- Read left column then right: options run from full control (VMs) to fully managed or serverless (Functions); pick by how much you must manage.
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- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

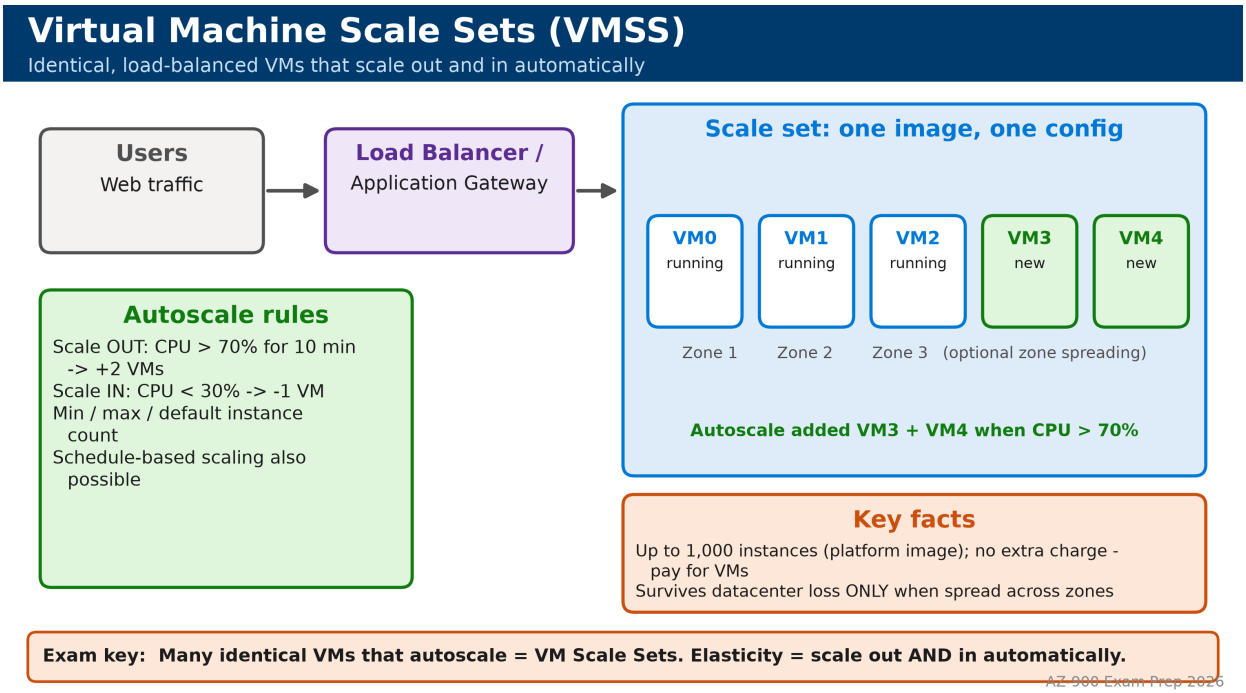


Figure Q316b: Virtual Machine Scale Sets

About this figure

- A scale set runs identical VMs from one image and configuration behind a load balancer or Application Gateway.
- Autoscale adds instances when a metric (e.g. CPU > 70%) is high and removes them when low, within min/max instance counts - this is elasticity.
- Scale sets support up to 1,000 VM instances (platform images); the scale set itself is free - you pay only for the VMs, disks and networking.
- Orchestration modes: Uniform (identical VMs) and Flexible (mixed sizes, recommended for new deployments).
- A scale set survives a datacenter failure only when its instances are spread across availability zones; a non-zonal set does not.
- Scaling out adds more VMs (horizontal); scaling up means a bigger VM size (vertical) and usually needs a restart.

Question 317

AZURE ARCHITECTURE & SERVICES

Compute
v5 Q450

Your company is evaluating Azure Virtual Desktop. Which TWO statements describe benefits of Azure Virtual Desktop? Each correct answer presents a complete solution.

✔ **A.** Users can connect from many devices, including Windows, macOS, iOS, Android and web browsers

✔ **B.** Microsoft manages the service infrastructure, such as the gateway and connection broker

✖ C. It requires you to deploy and maintain on-premises Remote Desktop gateway servers

✖ D. It can deliver only full desktops, not individual published apps

✖ E. It runs only on physical servers in your own datacenter

Correct answer: **Users can connect from many devices, including Windows, macOS, iOS, Android and web browsers; Microsoft manages the service infrastructure, such as the gateway and connection broker**

Clarification (read this first): Pick TWO. Watch the absolute words 'only' and 'requires' in the wrong options - they make those statements false.

Answer explained

✔ **Users can connect from many devices, including Windows, macOS, iOS, Android and web browsers; Microsoft manages the service infrastructure, such as the gateway and connection broker**
AVD works from almost any device, and Microsoft runs the gateway and broker (the parts that connect users to desktops) for you.

✖ **It requires you to deploy and maintain on-premises Remote Desktop gateway servers**
Microsoft runs the gateway and broker roles in Azure. You do not deploy them on-premises.

✖ **It can deliver only full desktops, not individual published apps**
AVD gives both full desktops and single published apps (RemoteApp).

✖ **It runs only on physical servers in your own datacenter**
Session hosts are usually Azure VMs in your subscription.

EXAM STRATEGY
Pick the Microsoft-managed service and any-device options. Remove anything needing on-premises gateways or physical servers, and 'only full desktops', because RemoteApp exists.

IMPACT
Microsoft-managed brokers and gateways remove servers you would otherwise patch and protect. Any-device access supports bring-your-own-device and remote work.

What · Where · When · Why · How

What	Main benefits of Azure Virtual Desktop.
Where	In Azure, used from user devices anywhere.
When	When enabling secure remote or hybrid work.
Why	It cuts infrastructure work and keeps data in Azure, not on devices.
How	Users install a Remote Desktop app or open a browser and sign in with Entra ID.

Exam tip
AVD = desktop and app virtualization, any device, Microsoft manages the service parts.

Memory trick
Your desk follows you to any screen.

Real-world example, use case and scenario

EXAMPLE
Own devices
A design agency lets freelancers use their own Macs to open AVD desktops.

USE CASE
Published app
A hospital publishes only its records app to tablets with RemoteApp.

SCENARIO
Less work
An IT team removes its old remote desktop gateway servers after moving to AVD.

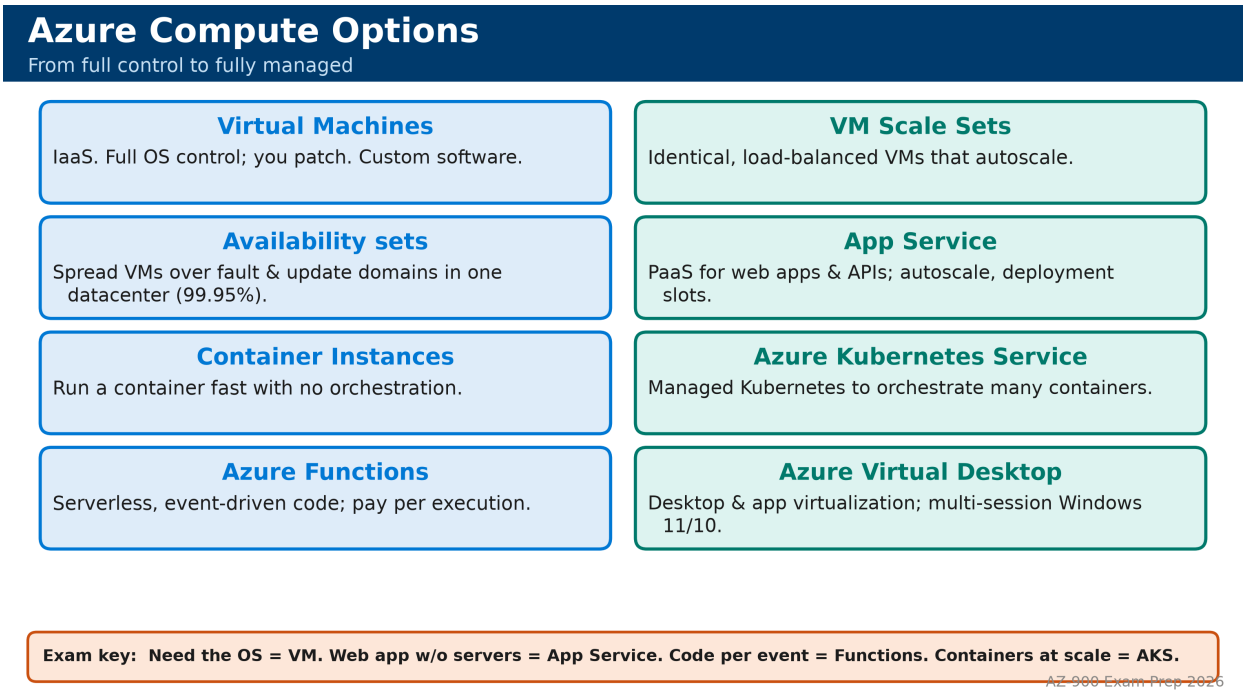


Figure Q317a: Azure compute options

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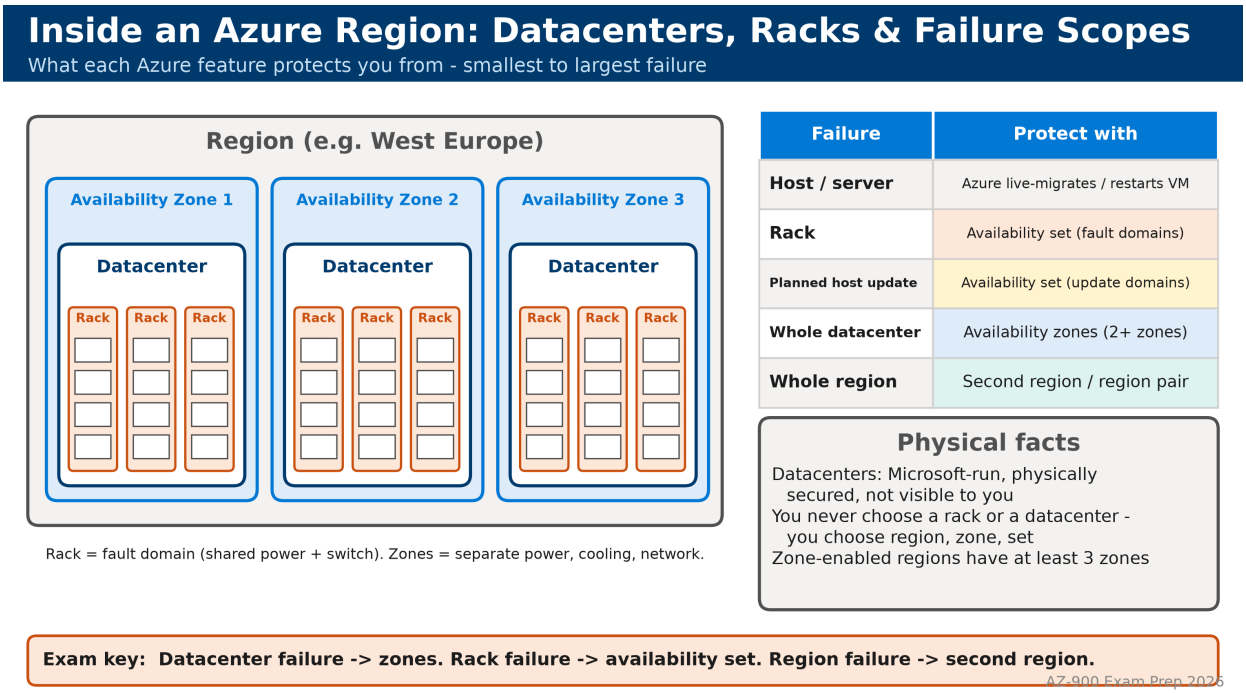


Figure Q317b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 318

AZURE ARCHITECTURE & SERVICES — Compute v5 Q451

Session hosts in an Azure Virtual Desktop host pool are [Azure virtual machines]. Select the answer that correctly completes the sentence. If the sentence is correct, select "No change is needed".

✔ **A.** No change is needed

✖ B. Azure Container Instances

✖ C. Azure Functions apps

✖ D. Azure App Service plans

Correct answer: No change is needed

Clarification (read this first): This is a 'fix the bracketed words' question. Here the bracketed words 'Azure virtual machines' are already correct, so pick 'No change is needed'.

Answer explained

✔ **No change is needed**
AVD session hosts are Azure VMs grouped in host pools, so the sentence is correct.

✖ **Azure Container Instances**
ACI runs containers, not interactive Windows desktops.

✖ **Azure Functions apps**
Functions run event-driven code, not user sessions.

✖ **Azure App Service plans**
App Service plans host web apps, not desktops.

EXAM STRATEGY
Session hosts are VMs; no other option runs full Windows desktops. Choose 'No change is needed'.

IMPACT
Session hosts are billed as VMs. So AVD costs are controlled by VM size, autoscaling and start-on-connect (VMs start only when a user connects).

What · Where · When · Why · How

What	The compute behind Azure Virtual Desktop.
Where	In host pools inside your Azure subscription.
When	Whenever users connect to an AVD desktop or app.
Why	VMs give the full Windows OS that desktops need.
How	Deploy VMs from an image and add them to a host pool.

Exam tip
AVD has VMs underneath, so VM size and VM prices drive AVD cost.

Memory trick
Behind every virtual desk sits a virtual machine.

Real-world example, use case and scenario

EXAMPLE
Sizing
An engineering firm picks GPU VMs as session hosts for design software users.

USE CASE
Cost control
A school uses autoscaling to turn off unused session host VMs at night.

SCENARIO
Image updates
IT patches a master image and redeploys session host VMs every month.

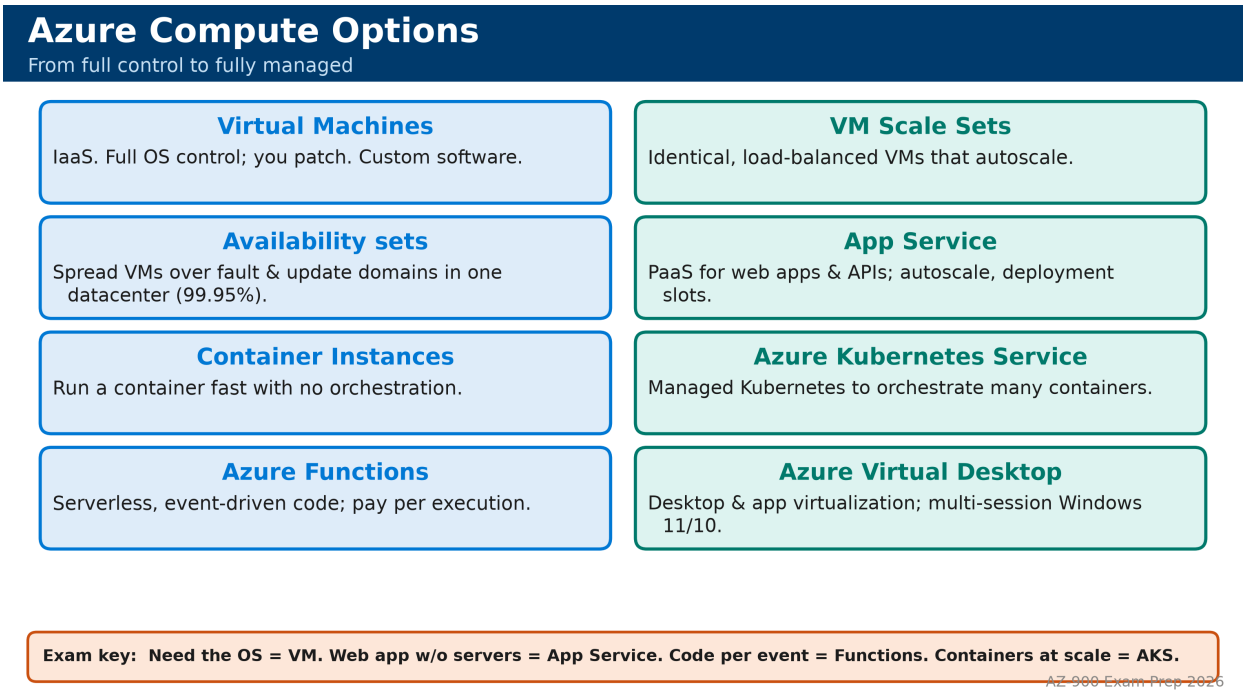


Figure Q318: Azure compute options

About this figure

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- App Service plans set the price: Free F1 USD 0, Basic B1 approx. USD 13/month Linux, Standard S1 approx. USD 70/month with autoscale and slots.
- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

Question 319

AZURE ARCHITECTURE & SERVICES — Compute v5 Q452

A developer needs to run a single container image for a short data-processing task. The solution must start quickly, require no VM or orchestrator management, and be billed per second. What should the developer use?

✔ **A.** Azure Container Instances

✖ B. Azure Kubernetes Service

✖ C. Azure Virtual Machine Scale Sets

✖ D. Azure Virtual Desktop

Correct answer: Azure Container Instances

Clarification (read this first): Three container services look alike. ACI = one simple container, fast, no management. Container Apps (Q320) = microservices with scale to zero. AKS (Q321) = full Kubernetes. Here it is ONE short task, so ACI.

Answer explained

✔ **Azure Container Instances**
Azure Container Instances runs a container directly, with no VMs or orchestrator (a system that manages many containers) to manage. It is billed per second.

✖ **Azure Kubernetes Service**
AKS is a full orchestrator with node pools. It is too much for one short task.

✖ **Azure Virtual Machine Scale Sets**
You would have to manage VMs and install container software.

✖ **Azure Virtual Desktop**
It gives desktops to users, not container workloads.

EXAM STRATEGY
'Single container', 'start quickly', 'no orchestrator' and 'per-second billing' = Container Instances. Remove AKS (orchestration) and Scale Sets (VM management).

IMPACT
Using ACI for short jobs avoids paying for an idle Kubernetes cluster. Using AKS here adds node costs and extra work.

What · Where · When · Why · How

What	Azure Container Instances.
Where	In an Azure region, as a PaaS container group.
When	For simple, short-lived or isolated container workloads.
Why	To avoid managing VMs or Kubernetes.
How	Run az container create, or use the portal with an image name.

Exam tip
One container, fast, no orchestration = ACI. Many microservices with orchestration = AKS or Container Apps.

Memory trick
ACI = Instant Container, Instantly.

Real-world example, use case and scenario

EXAMPLE
Nightly job
An insurer runs a container each night to convert claim files, then it stops.

USE CASE
Build agent
A dev team starts build agents in ACI when needed.

SCENARIO
Burst
A lab runs 50 short simulation containers at once and pays only for the minutes used.

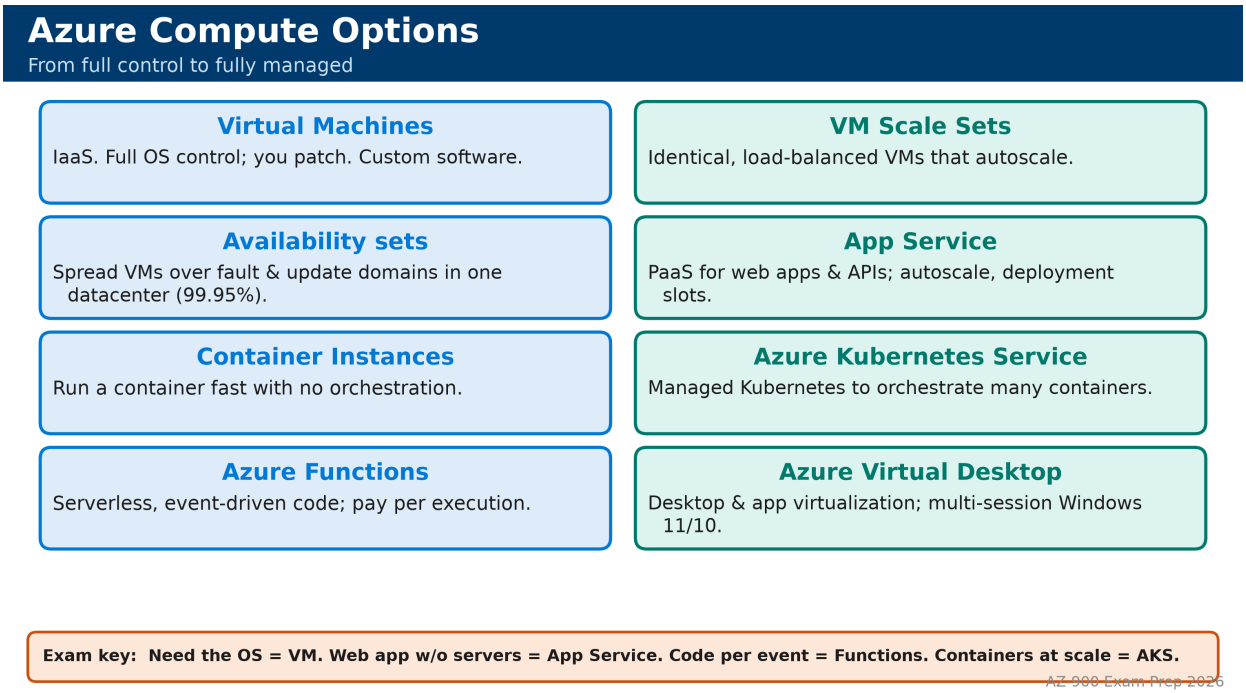


Figure Q319a: Azure compute options

About this figure

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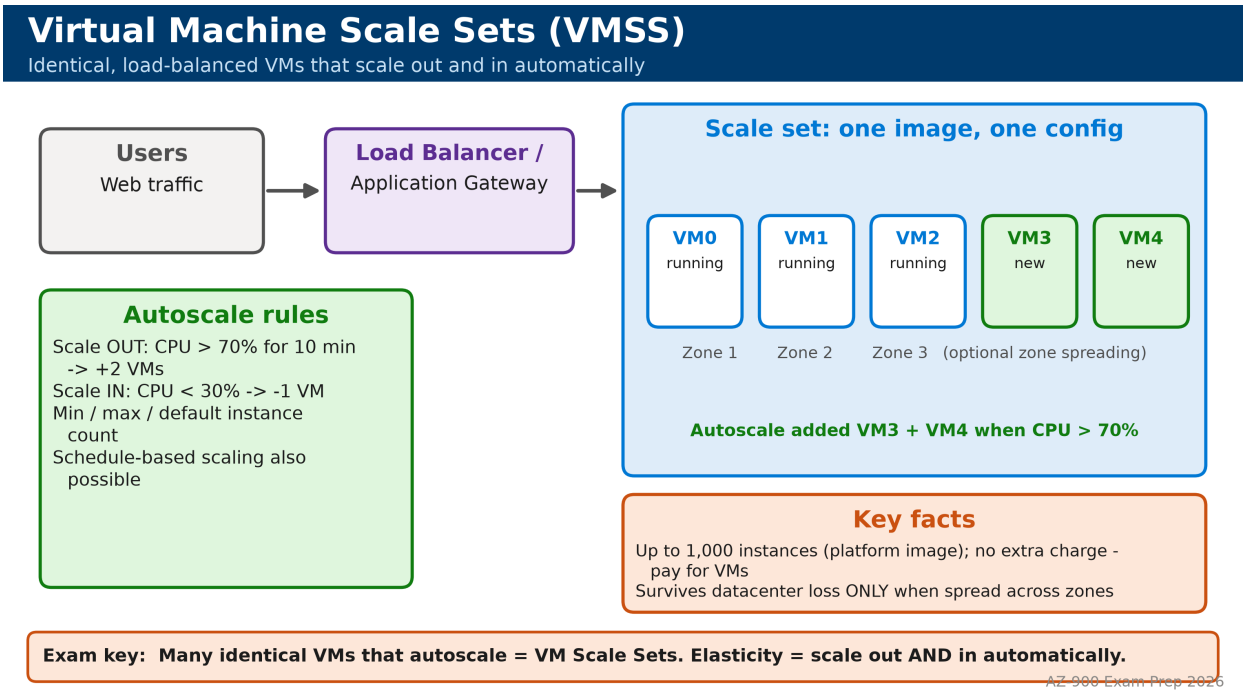


Figure Q319b: Virtual Machine Scale Sets

About this figure

- A scale set runs identical VMs from one image and configuration behind a load balancer or Application Gateway.
- Autoscale adds instances when a metric (e.g. CPU > 70%) is high and removes them when low, within min/max instance counts - this is elasticity.
- Scale sets support up to 1,000 VM instances (platform images); the scale set itself is free - you pay only for the VMs, disks and networking.
- Orchestration modes: Uniform (identical VMs) and Flexible (mixed sizes, recommended for new deployments).
- A scale set survives a datacenter failure only when its instances are spread across availability zones; a non-zonal set does not.
- Scaling out adds more VMs (horizontal); scaling up means a bigger VM size (vertical) and usually needs a restart.

Question 320

AZURE ARCHITECTURE & SERVICES — Compute v5 Q453

Your team is building event-driven microservices in containers. They want automatic scaling, including scaling to zero, built-in support for Dapr and KEDA, and no need to manage Kubernetes clusters or APIs. Which service should they use?

✔ **A.** Azure Container Apps

✖ B. Azure Kubernetes Service

✖ C. Azure Container Instances

✖ D. Azure Batch

Correct answer: Azure Container Apps

Clarification (read this first): The clues are 'scale to zero', Dapr and KEDA (add-ons for microservices and event-based scaling), and 'no Kubernetes to manage'. Compare Q319 (ACI for one simple container) and Q321 (AKS, where you manage the nodes).

Answer explained

✔ **Azure Container Apps**
Azure Container Apps is serverless containers for microservices: KEDA autoscaling, scale to zero, Dapr, and no cluster to manage.

✖ **Azure Kubernetes Service**
AKS gives full Kubernetes access, but you manage cluster and node settings.

✖ **Azure Container Instances**
ACI runs single containers without built-in microservice features like Dapr or event-based scaling.

✖ **Azure Batch**
Batch schedules large parallel compute jobs, not microservice apps.

EXAM STRATEGY
'Scale to zero', 'Dapr', 'KEDA' and 'no Kubernetes management' = Container Apps. AKS exposes Kubernetes; ACI has no built-in autoscaling or Dapr.

IMPACT
Container Apps lets teams run microservices without Kubernetes skills and pay nothing when there is no load. This is a common way to save money on event-driven systems.

What · Where · When · Why · How

What	Azure Container Apps.
Where	In a Container Apps environment in an Azure region.
When	When building microservices or event-driven apps in containers.
Why	To get Kubernetes-style scaling without managing Kubernetes.
How	Deploy container images and set scale rules on web traffic or event triggers.

Exam tip
Containers + microservices + serverless + no Kubernetes management = Container Apps.

Memory trick
Container Apps: Kubernetes power, serverless simplicity.

Real-world example, use case and scenario

EXAMPLE
Order processing
An online shop's order service scales up from zero when messages arrive in a queue.

USE CASE
API back end
A travel start-up hosts several container APIs that scale with web traffic.

SCENARIO
Saving money
A hobby app scales to zero overnight and costs almost nothing when idle.

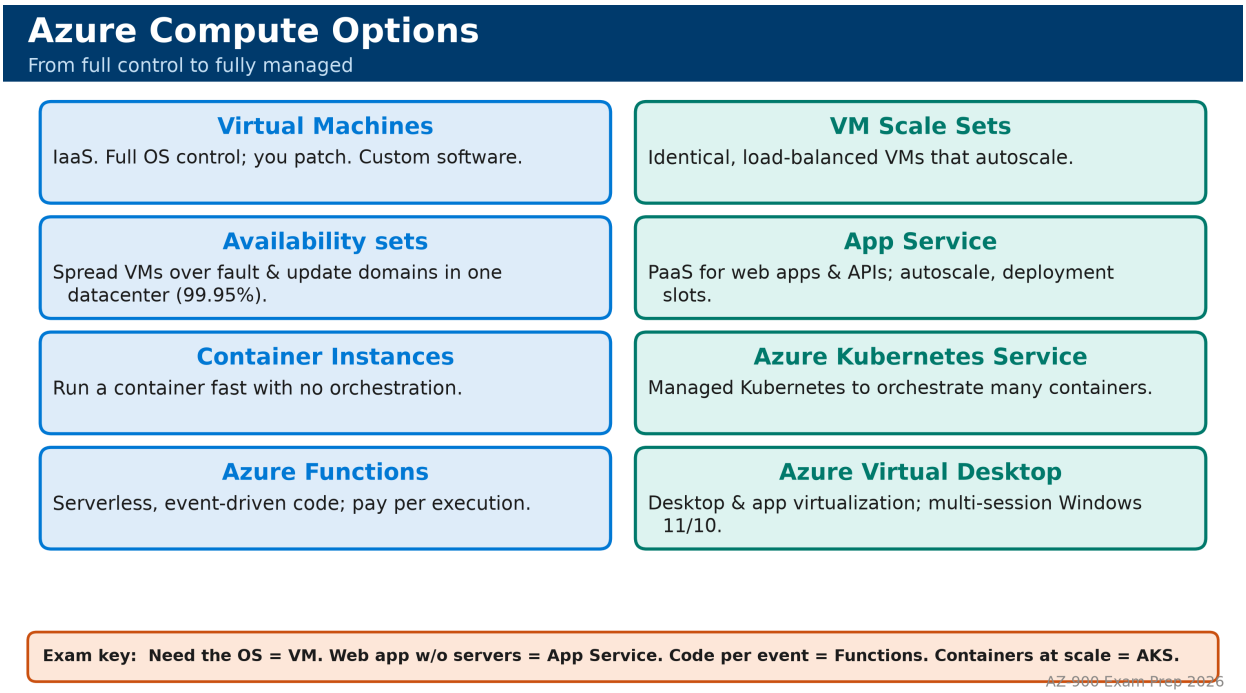


Figure Q320: Azure compute options

About this figure

• Read left column then right: options run from full control (VMs) to fully managed or serverless (Functions); pick by how much you must manage.

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• App Service plans set the price: Free F1 USD 0, Basic B1 approx. USD 13/month Linux, Standard S1 approx. USD 70/month with autoscale and slots.

• Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

Question 321

AZURE ARCHITECTURE & SERVICES — **Compute** v5 Q454

Your company plans to use Azure Kubernetes Service (AKS) to orchestrate containers. Which TWO statements about AKS are true? Each correct answer presents a complete solution.

✔ **A.** Azure manages the Kubernetes control plane for you

✔ **B.** You run your containers on agent (worker) nodes that you pay for as VMs

✖ C. You must install and patch the Kubernetes control plane yourself

✖ D. AKS is a software as a service (SaaS) offering for end users

✖ E. AKS can run only a single container at a time

Correct answer: Azure manages the Kubernetes control plane for you; You run your containers on agent (worker) nodes that you pay for as VMs

Clarification (read this first): Pick TWO. In AKS, Azure manages the control plane (the 'brain' of Kubernetes), but you still pay for the worker node VMs that run your containers.

Answer explained

✔ **Azure manages the Kubernetes control plane for you; You run your containers on agent (worker) nodes that you pay for as VMs**
In AKS, Azure manages the Kubernetes control plane. Your containers run on agent (worker) node VMs that you pay for.

✖ **You must install and patch the Kubernetes control plane yourself**
In AKS, Azure manages the control plane.

✖ **AKS is a software as a service (SaaS) offering for end users**
AKS is a managed container platform for developers and operators, not an app for end users.

✖ **AKS can run only a single container at a time**
AKS runs many containers across many nodes.

EXAM STRATEGY
AKS = managed control plane, and you pay for worker node VMs. Remove 'install the control plane yourself', 'SaaS' and 'only a single container'.

IMPACT
AKS costs come mainly from node pools (groups of worker VMs), not the control plane. If you miss this, you may under-estimate costs or buy too many nodes.

What · Where · When · Why · How

What	Who does what in Azure Kubernetes Service.
Where	In an Azure region, with node pools in your subscription.
When	When you need full container orchestration at large scale.
Why	To use Kubernetes without running its control plane.
How	Create an AKS cluster, add node pools, and deploy with kubectl or Helm.

Exam tip
AKS = orchestration of many containers. You still manage node pools and workloads.

Memory trick
Azure runs the Kubernetes brain; you feed the worker bees.

Real-world example, use case and scenario

EXAMPLE
Video service
A video service runs hundreds of microservices on AKS node pools.

USE CASE
Mixed nodes
A bank uses Linux and Windows node pools in one AKS cluster.

SCENARIO
Upgrades
An operations team upgrades the Kubernetes version with one command while Azure manages the control plane.

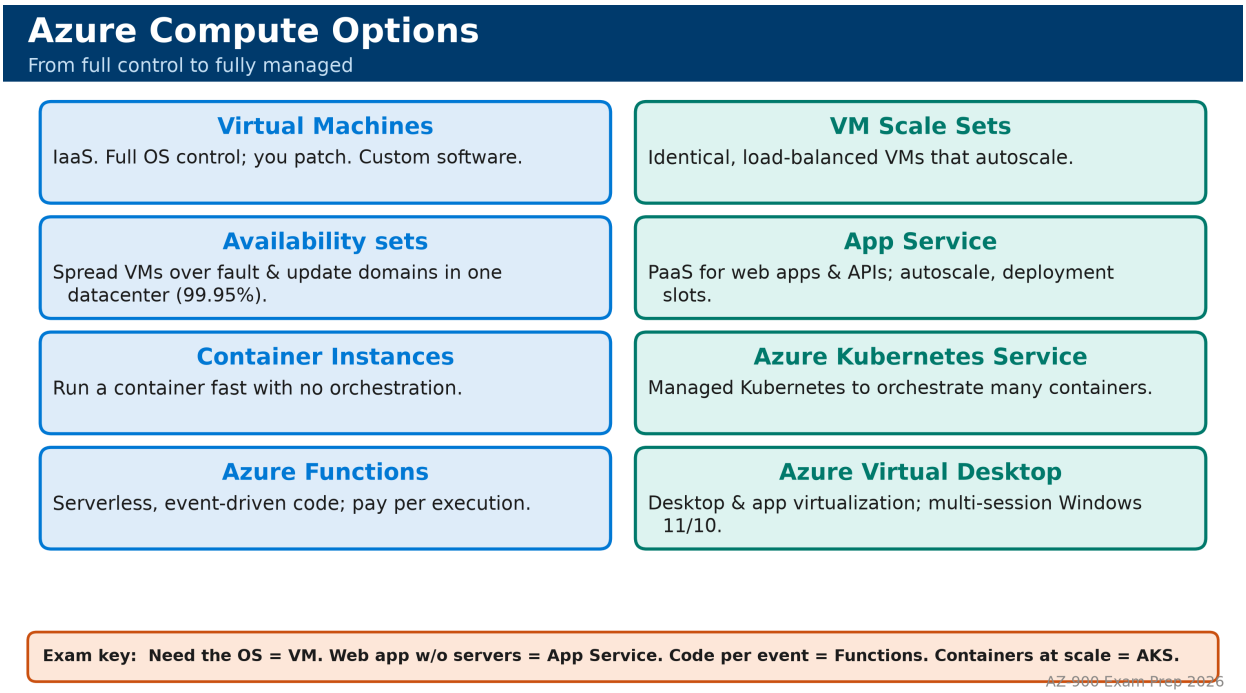


Figure Q321: Azure compute options

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- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

Question 322

AZURE ARCHITECTURE & SERVICES — Compute v5 Q455

Your developers are building a serverless order workflow in Azure Functions that must chain several functions together and keep state between steps, such as waiting for a human approval. Which feature should they use?

✔ **A.** Durable Functions

✖ B. Deployment slots

✖ C. Azure Container Instances

✖ D. Virtual Machine Scale Sets

Correct answer: Durable Functions

Answer explained

✔ **Durable Functions**
Durable Functions adds stateful orchestration (remembering progress, chaining steps, waiting for events) to Azure Functions, which are normally stateless.

✖ **Deployment slots**
Slots let you test a new version before swapping it live. They do not keep workflow state.

✖ **Azure Container Instances**
It runs containers. It is not a stateful serverless workflow feature.

✖ **Virtual Machine Scale Sets**
It is IaaS autoscaling of VMs. It is not serverless and not workflow orchestration.

EXAM STRATEGY
'Chain functions', 'keep state' and 'wait for human approval' = Durable Functions. Deployment slots are for App Service releases. Remove all options that are not Functions.

IMPACT
Durable Functions removes the need for your own state storage and checking code. Long business processes do not lose their progress.

What · Where · When · Why · How

What	Durable Functions, the stateful extension of Azure Functions.
Where	Inside an Azure Functions app.
When	When serverless code must coordinate many steps or wait for events.
Why	Plain functions are stateless (they forget everything between runs) and run for a short time.
How	Write an orchestrator function that calls activity functions; Azure saves the state.

Exam tip
Functions are stateless by default. 'Stateful serverless' means Durable Functions.

Memory trick
Durable = remembers where it left off.

Real-world example, use case and scenario

EXAMPLE
Expense approval
A company's workflow waits until a manager approves an expense, then pays it.

USE CASE
Fan-out
A photo app resizes 1,000 images at the same time and collects the results.

SCENARIO
Order process
A shop reserves stock, charges the card, then ships, and undoes steps if one fails.

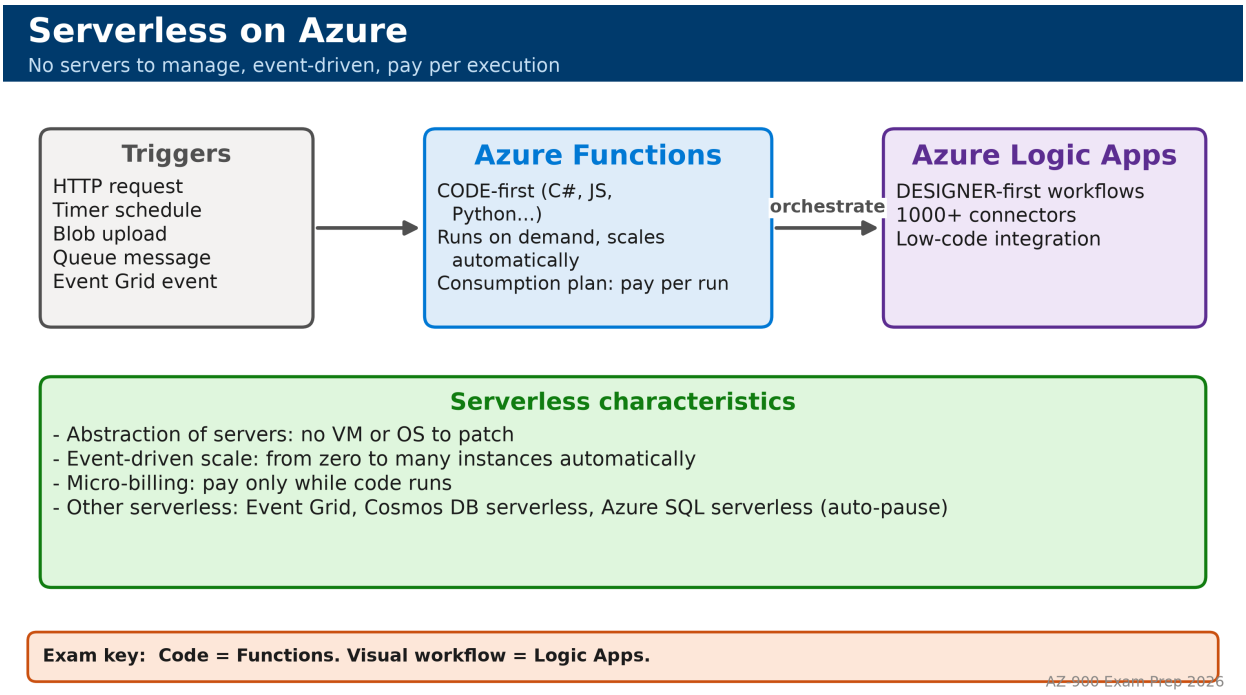


Figure Q322a: Serverless on Azure

About this figure

• Flow left to right: an event trigger starts code (Functions) or a workflow (Logic Apps); the green box lists traits shared by serverless services.

• Functions Consumption: approx. USD 0.20 per million executions plus GB-seconds, with 1 million executions and 400,000 GB-s free monthly.

• Consumption-plan functions can cold start and have a default 5-minute, maximum 10-minute timeout; Flex Consumption and Premium plans reduce cold starts.

• Logic Apps Consumption bills per action and connector call, roughly fractions of a cent each; Standard plan bills per hosting plan instead.

• Serverless still runs on servers; Microsoft manages and scales them, and you pay nothing while nothing runs, except storage or plan minimums.

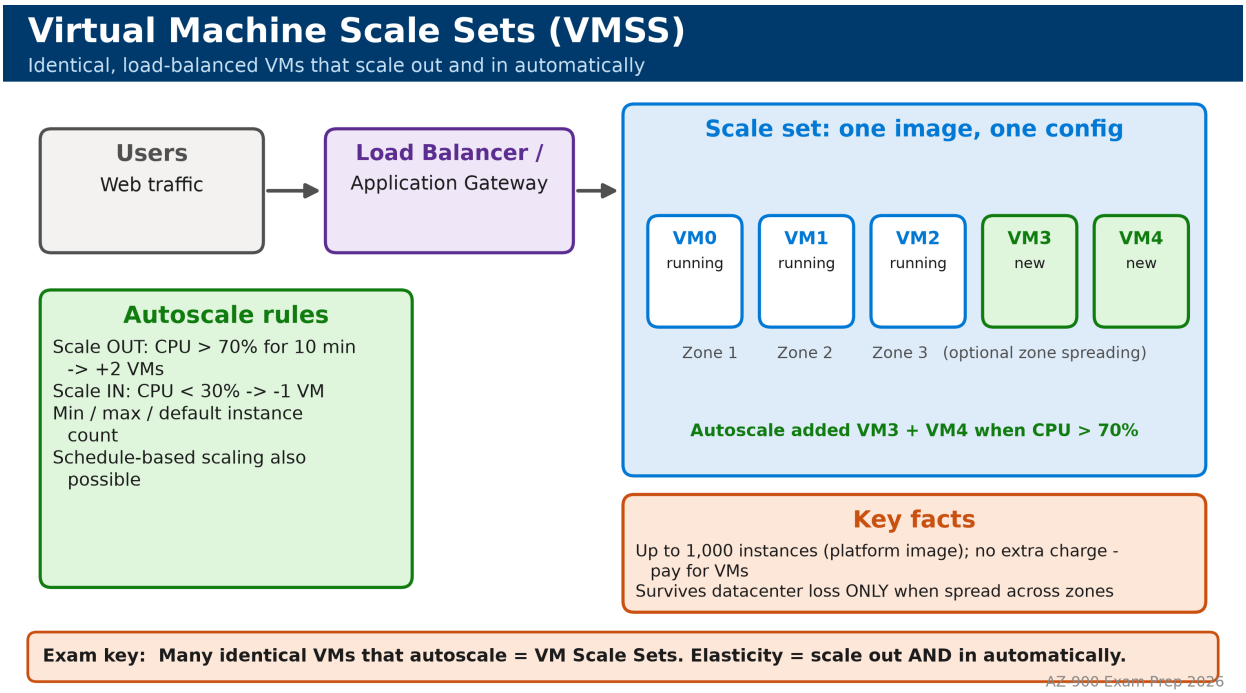


Figure Q322b: Virtual Machine Scale Sets

About this figure

- A scale set runs identical VMs from one image and configuration behind a load balancer or Application Gateway.
- Autoscale adds instances when a metric (e.g. CPU > 70%) is high and removes them when low, within min/max instance counts - this is elasticity.
- Scale sets support up to 1,000 VM instances (platform images); the scale set itself is free - you pay only for the VMs, disks and networking.
- Orchestration modes: Uniform (identical VMs) and Flexible (mixed sizes, recommended for new deployments).
- A scale set survives a datacenter failure only when its instances are spread across availability zones; a non-zonal set does not.
- Scaling out adds more VMs (horizontal); scaling up means a bigger VM size (vertical) and usually needs a restart.

Question 323

AZURE ARCHITECTURE & SERVICES

Compute
v5 Q456, Q331

With the Azure Functions Consumption plan, you are billed [for a fixed number of always-on VMs, even when idle]. Select the answer that correctly completes the sentence. If the sentence is correct, select "No change is needed".

✖ A. No change is needed

✔ **B.** based on executions and resources consumed while your functions run

✖ C. a flat yearly licence fee per function app

✖ D. only for outbound data transfer

Correct answer: based on executions and resources consumed while your functions run

Clarification (read this first): This is a 'fix the bracketed words' question. The bracket says 'always-on VMs, even when idle', which is wrong for the Consumption plan, so do NOT pick 'No change is needed'.

Answer explained

✔ **based on executions and resources consumed while your functions run**
The Consumption plan bills per run and for resources used while code runs. Idle time is not billed.

✖ **No change is needed**
Always-on instances describe a dedicated (App Service) plan, not Consumption.

✖ **A flat yearly licence fee per function app**
Azure Functions has no licence fee per app.

✖ **Only for outbound data transfer**
You pay for runs and resources used, not just for outbound data.

EXAM STRATEGY
Consumption = pay per run and resources used, and it can scale to zero. 'Always-on VMs even when idle' describes dedicated plans, so change the sentence.

IMPACT
Using Consumption for spiky or low-volume work can bring compute bills close to zero. The Premium plan suits apps that need no cold start (startup delay) or heavy VNet use.

What · Where · When · Why · How

What	How the Azure Functions Consumption plan is billed.
Where	In function apps hosted on the Consumption plan.
When	When work is event-driven and happens now and then.
Why	Paying per use matches cost to real demand.
How	Azure measures runs and resources used, and scales instances automatically.

Exam tip
Serverless = no server management + automatic scaling + pay only when code runs.

Memory trick
No knock, no charge: Functions bill only when triggered.

Real-world example, use case and scenario

EXAMPLE
IoT alerts
A farm's sensor alerts start a function a few times a day, costing very little.

USE CASE
Image thumbnails
Each file upload starts a function. No uploads, no cost.

SCENARIO
Wrong plan
A team on a dedicated plan pays for idle instances and moves to Consumption.

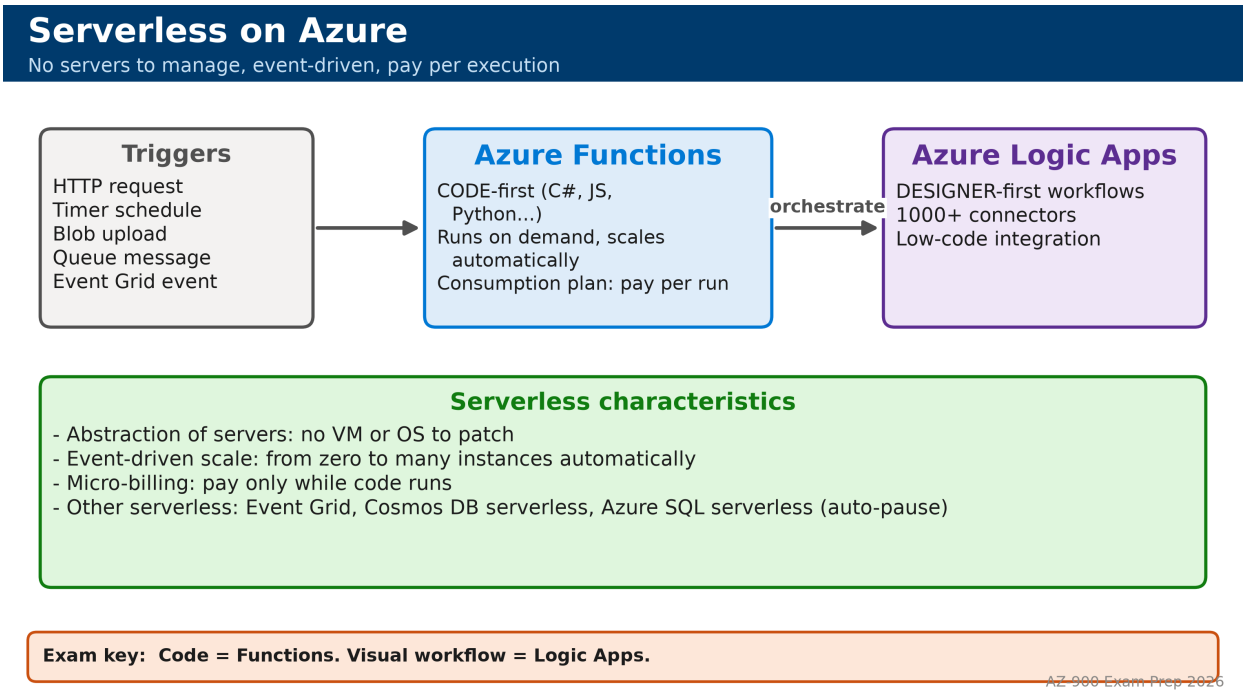


Figure Q323: Serverless on Azure

About this figure

• Flow left to right: an event trigger starts code (Functions) or a workflow (Logic Apps); the green box lists traits shared by serverless services.

• Functions Consumption: approx. USD 0.20 per million executions plus GB-seconds, with 1 million executions and 400,000 GB-s free monthly.

• Consumption-plan functions can cold start and have a default 5-minute, maximum 10-minute timeout; Flex Consumption and Premium plans reduce cold starts.

• Logic Apps Consumption bills per action and connector call, roughly fractions of a cent each; Standard plan bills per hosting plan instead.

• Serverless still runs on servers; Microsoft manages and scales them, and you pay nothing while nothing runs, except storage or plan minimums.

Question 324

AZURE ARCHITECTURE & SERVICES — Compute — v5 Q457

Your team hosts a web app in Azure App Service. They want to deploy a new version to a staging environment, validate it, and then swap it into production with no downtime. Which App Service feature should they use?

✔ **A.** Deployment slots

✖ B. Availability sets

✖ C. Durable Functions

✖ D. Azure Virtual Desktop host pools

Correct answer: Deployment slots

Answer explained

✔ **Deployment slots**
Deployment slots let you put a new version in a staging slot (a separate copy of the app). You then swap it into production with no downtime, and you can swap back if needed.

✖ **Availability sets**
This is a VM feature that spreads VMs over hardware. App Service does not use it.

✖ **Durable Functions**
This runs stateful (remembers progress) workflows in Azure Functions. It does not stage web app releases.

✖ **Azure Virtual Desktop host pools**
These are groups of desktop session hosts. They have nothing to do with web app releases.

EXAM STRATEGY
The words 'staging', 'validate' and 'swap into production with no downtime' point to deployment slots. Remove availability sets and Durable Functions. Neither one swaps releases.

IMPACT
A slot swap gives a release with no downtime. If the new version is bad, you swap back at once. This limits the damage from a bad release.

What · Where · When · Why · How

What	Deployment slots in App Service.
Where	In an App Service plan on the Standard tier or higher.
When	When you release a new version of a web app.
Why	To avoid downtime and to roll back (return to the old version) quickly.
How	Deploy to a staging slot, test it, then swap it with production.

Exam tip
Release with no downtime in App Service = deployment slot swap.

Memory trick
A slot is a stage behind the curtain. Swap it in when the show is ready.

Real-world example, use case and scenario

EXAMPLE
News site
A news company tests a new design in staging, then swaps it live at midnight.

USE CASE
Rollback
A bug appears after a release. The team swaps back in seconds.

SCENARIO
A/B testing
A shop sends 10% of users to a slot to test a new checkout page.

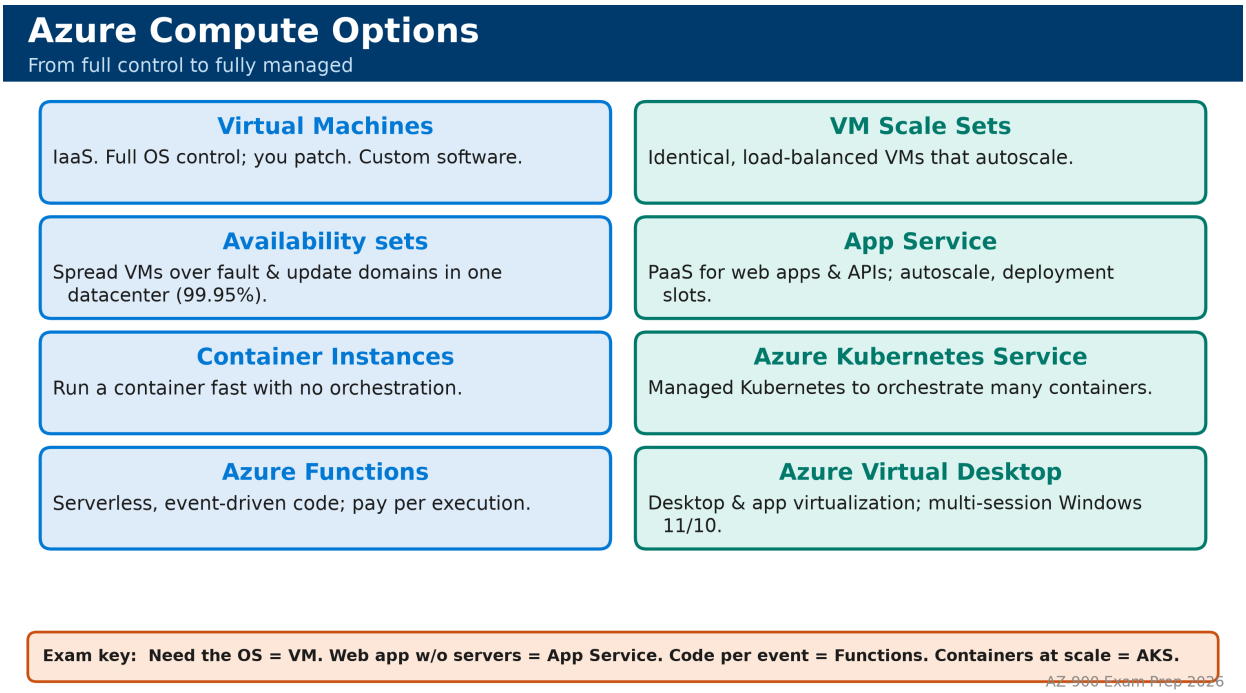


Figure Q324a: Azure compute options

About this figure

- Read left column then right: options run from full control (VMs) to fully managed or serverless (Functions); pick by how much you must manage.
- VM sizes range from tiny B-series (B1s approx. USD 8/month Linux) to large memory and GPU VMs; per-second billing applies while running.
- VM Scale Sets support up to 1,000 instances (600 with custom images); availability sets use up to 3 fault domains and 20 update domains.
- App Service plans set the price: Free F1 USD 0, Basic B1 approx. USD 13/month Linux, Standard S1 approx. USD 70/month with autoscale and slots.
- Container Instances bill per second of vCPU and memory with no cluster; AKS control plane is free tier, you pay for the worker node VMs.

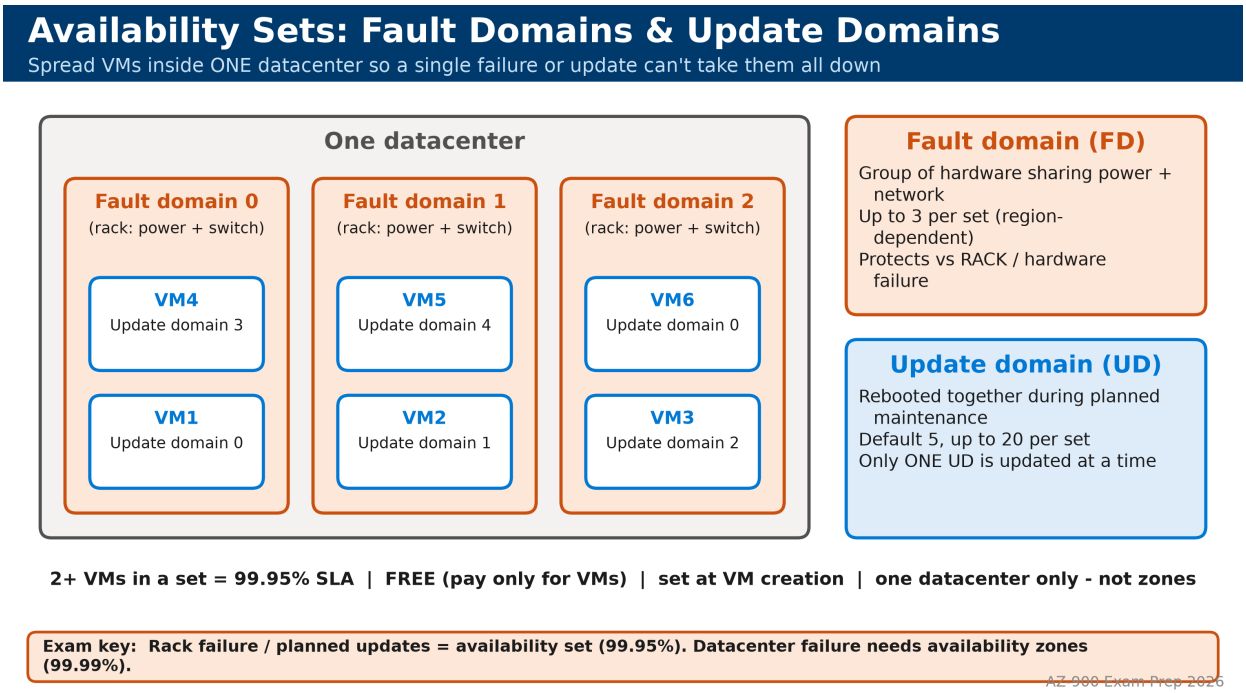


Figure Q324b: Availability sets: fault and update domains

About this figure

- An availability set spreads VMs across fault domains (racks with separate power and switch) and update domains inside ONE datacenter.
- Up to 3 fault domains protect against rack or hardware failure; update domains (default 5, max 20) are rebooted one at a time during planned maintenance.
- With 6 VMs and 5 update domains, the 6th VM wraps back to update domain 0 - as the figure shows.
- Two or more VMs in an availability set get a 99.95% SLA; availability sets are free - you pay only for the VMs.
- A VM is placed in an availability set only at creation; availability sets and availability zones cannot be combined for one VM.
- Availability sets do NOT protect against a datacenter or region failure - use availability zones or a second region for that.

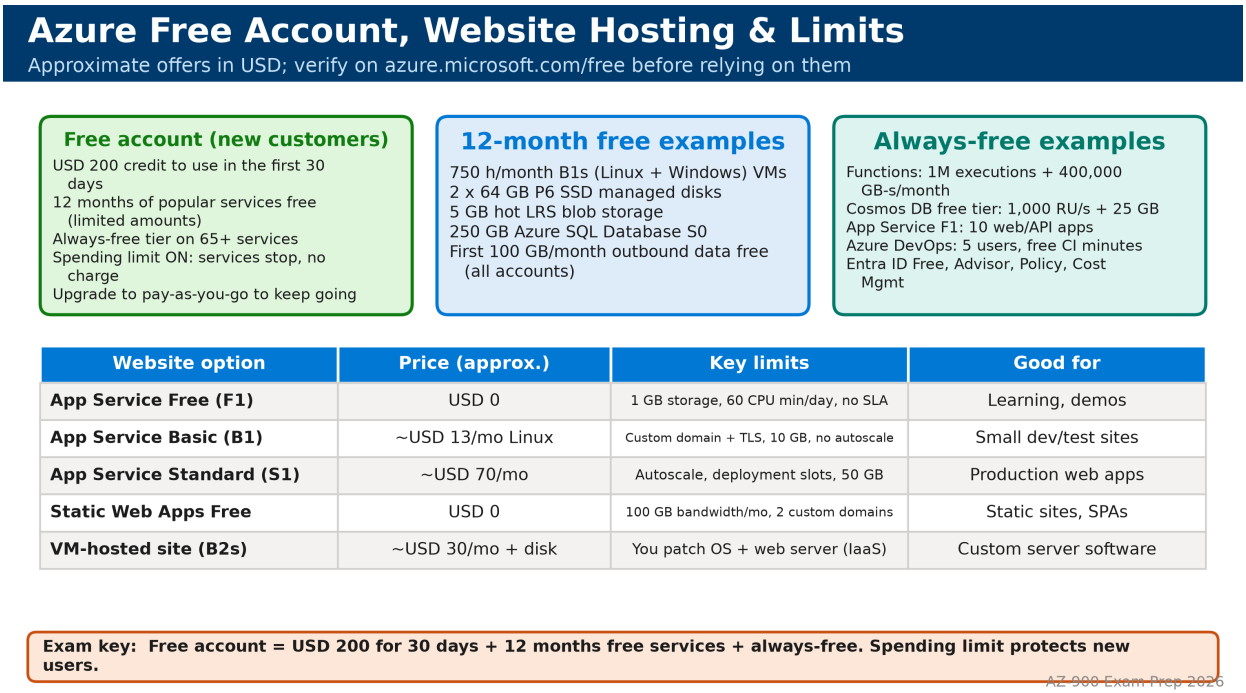


Figure Q324c: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 325

AZURE ARCHITECTURE & SERVICES — Compute v5 Q458

Azure App Service can host web apps on both Windows and Linux, including apps written in .NET, Java, Node.js, Python and PHP.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Answer explained

✔ **Yes**
Yes. App Service runs on Windows or Linux. It supports .NET, Java, Node.js, Python, PHP and custom containers.

✖ **No**
App Service is not limited to Windows or to one language.

EXAM STRATEGY
Key fact: App Service supports Windows and Linux with .NET, Java, Node.js, Python and PHP. The list in the statement is correct, so the answer is Yes.

IMPACT
Teams can move existing web apps to PaaS (Azure manages the servers) without rewriting them. They no longer patch servers for most web apps.

What · Where · When · Why · How

What	Which operating systems and languages Azure App Service supports.
Where	In App Service plans in Azure regions.
When	When you choose a PaaS host for a web app or API.
Why	So teams can move apps without rewriting them.
How	Choose the OS and the language runtime when you create the web app.

Exam tip
App Service = PaaS for web apps, REST APIs and mobile back ends. It runs on Windows or Linux and supports many languages.

Memory trick
App Service speaks many languages and two operating systems.

Real-world example, use case and scenario

EXAMPLE
Python API
A data team hosts a Flask API on App Service for Linux.

USE CASE
Old .NET site
An insurance company moves an ASP.NET site to App Service on Windows.

SCENARIO
Container
A startup deploys its own Docker image to App Service.

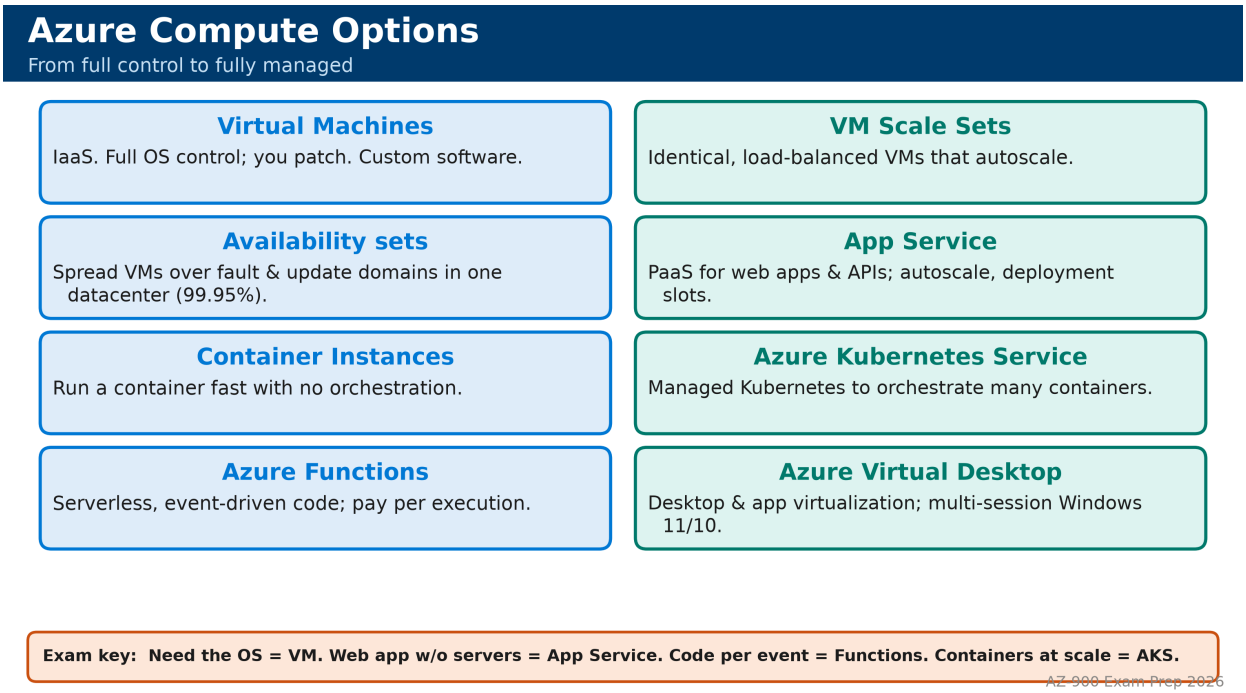


Figure Q325a: Azure compute options

About this figure

- Read left column then right: options run from full control (VMs) to fully managed or serverless (Functions); pick by how much you must manage.
- VM sizes range from tiny B-series (B1s approx. USD 8/month Linux) to large memory and GPU VMs; per-second billing applies while running.
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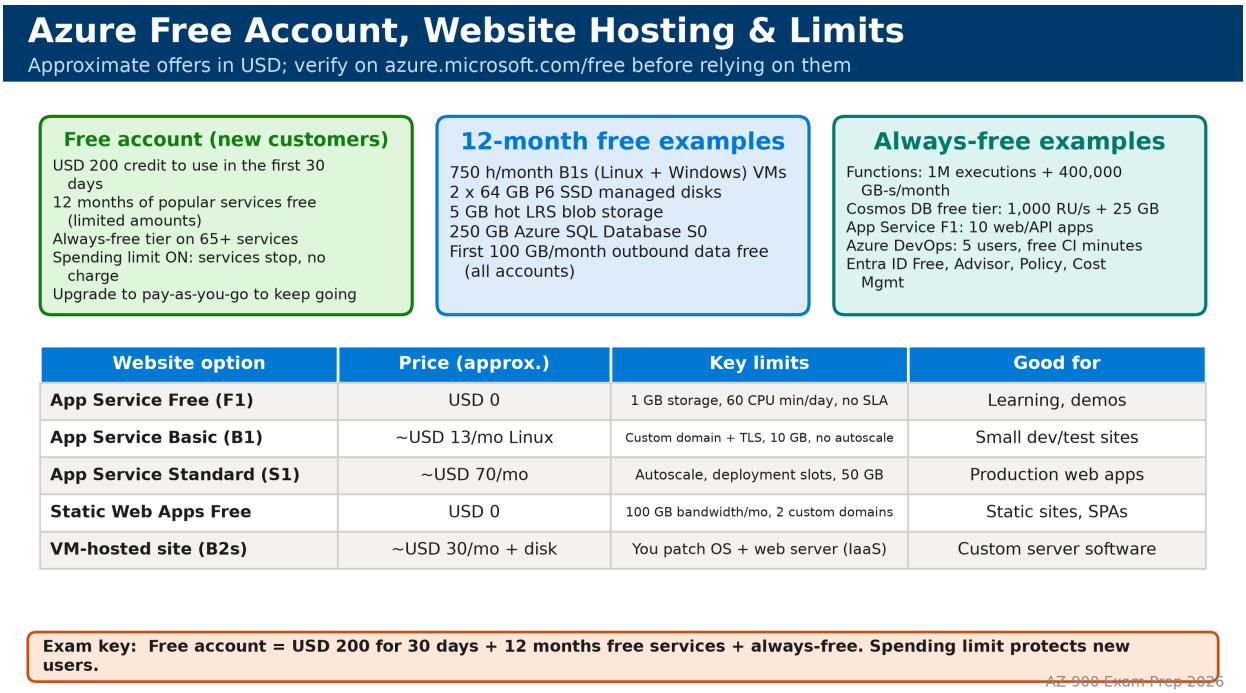


Figure Q325b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
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- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 326

AZURE ARCHITECTURE & SERVICES — **Subscriptions** v5 Q459

A student signs up for an Azure free account to learn Azure and host a small website. Which TWO benefits are included with the Azure free account? Each correct answer presents a complete solution.

- ✔ **A.** A USD 200 credit to use within the first 30 days
- ✔ **B.** Free access to selected popular services for 12 months
- ✖ C. Unlimited free use of every Azure service for 12 months
- ✖ D. A permanent USD 200 credit renewed every month
- ✖ E. Free Premier support for the first year

Correct answer: A USD 200 credit to use within the first 30 days; Free access to selected popular services for 12 months

Clarification (read this first): This question asks for the TWO things you really get. Watch the words 'unlimited', 'permanent' and 'Premier support' - they make options false.

Answer explained

✔ **A USD 200 credit to use within the first 30 days; Free access to selected popular services for 12 months**
The free account gives a USD 200 credit for 30 days, plus 12 months of selected popular services for free, plus some services that are always free.

✖ **Unlimited free use of every Azure service for 12 months**
The free amounts have limits, and only selected services are free.

✖ **A permanent USD 200 credit renewed every month**
The credit is given one time only. It ends after 30 days.

✖ **Free Premier support for the first year**
Paid support plans are not part of the free account.

EXAM STRATEGY
Free account = USD 200 credit for 30 days plus 12 months of selected free services. Remove the options with 'unlimited', 'permanent' and 'Premier support'.

IMPACT
The credit ends after 30 days. If a learner upgrades to pay-as-you-go, they can get a surprise bill. Set budgets and alerts from the first day.

What · Where · When · Why · How

What	What you get with the Azure free account.
Where	You sign up at azure.microsoft.com/free.
When	The credit lasts 30 days. The popular free services last 12 months.
Why	So new users can learn and test Azure at no cost.
How	Sign up with a Microsoft account, a phone check and a card to prove who you are.

Exam tip
USD 200 for 30 days, 12 months of popular services, plus always-free services.

Memory trick
200 for 30, then 12 months, then free basics forever.

Real-world example, use case and scenario

EXAMPLE
Student site
A student hosts a portfolio site on the App Service free tier.

USE CASE
Prototype
A startup tests a VM and SQL Database with the 30-day credit.

SCENARIO
Upgrade
After 30 days, a developer upgrades to pay-as-you-go to keep resources running.

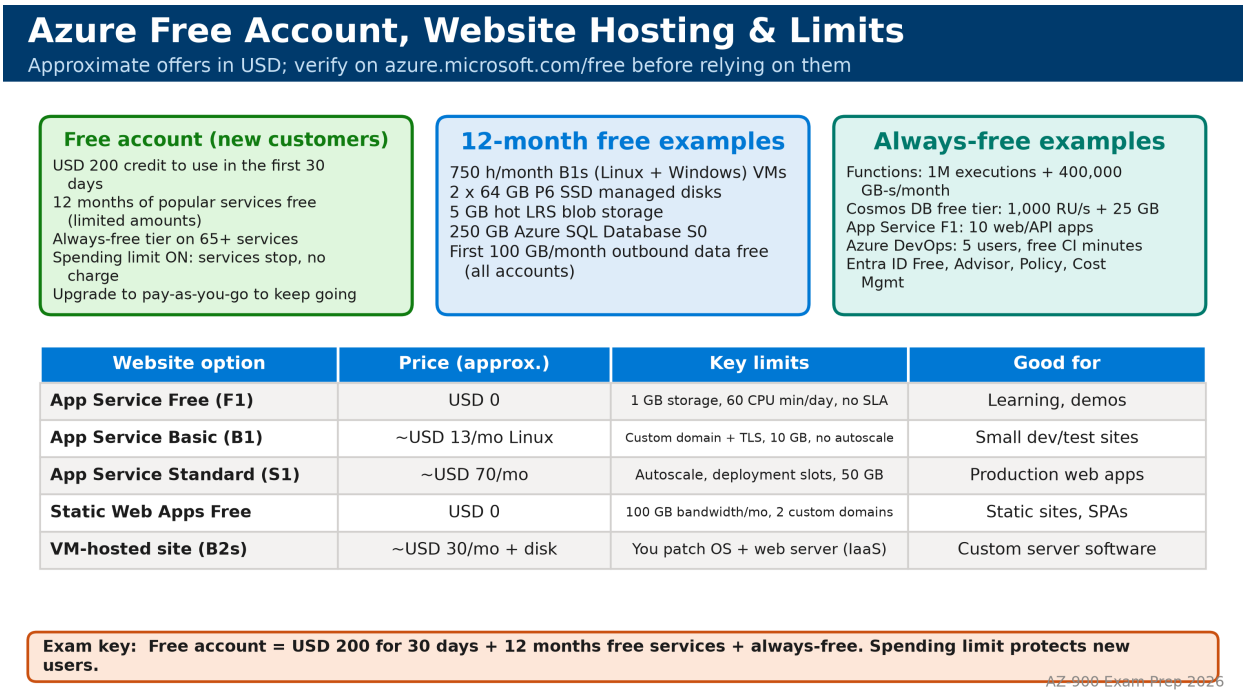


Figure Q326: Free account, website hosting and limits

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- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
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- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 327

AZURE ARCHITECTURE & SERVICES — Networking — v5 Q460

Your company has a storage account and a virtual network named VNet1. Security policy requires that VMs in VNet1 reach the storage account by using a private IP address from VNet1 so that traffic never uses the storage account's public endpoint. What should you create?

✔ **A.** A private endpoint

✖ B. Azure Bastion

✖ C. A NAT gateway

✖ D. A public IP address

Correct answer: A private endpoint

Clarification (read this first): The key words are 'a private IP address from VNet1' and 'never uses the public endpoint (the internet address of the service)'. Only a private endpoint puts the storage account on an IP inside your VNet.

Answer explained

✔ **A private endpoint**
A private endpoint gives a PaaS service (like a storage account) a private IP from your VNet, using Private Link. Traffic then does not use the public endpoint.

✖ **Azure Bastion**
Bastion gives browser-based RDP/SSH (remote login) to VMs. It does not give a storage account a private IP.

✖ **A NAT gateway**
A NAT gateway lets subnets reach the internet outbound (going out). Traffic still uses public endpoints.

✖ **A public IP address**
A public IP makes a resource reachable from the internet. That is the opposite of what is needed.

EXAM STRATEGY
'Private IP address from VNet' plus 'never use the public endpoint' = private endpoint. Bastion is for logging in to VMs. NAT gateway is for outbound internet.

IMPACT
Private endpoints close a path for data theft through public endpoints. Many compliance rules require them for storage and databases.

What · Where · When · Why · How

What	A private IP in your VNet that points to one Azure PaaS resource.
Where	In a subnet of the VNet that needs to reach the service.
When	When a PaaS service must be reached only over a private network.
Why	It keeps data traffic off the public internet.
How	Create a private endpoint for the storage account in a subnet. Update DNS (name lookup) so the name points to the private IP.

Exam tip
Public endpoint = reached through a public IP over the internet. Private endpoint = reached through a private IP from your VNet.

Memory trick
A PRIVATE endpoint gives the service a PRIVATE address inside your house (VNet).

Real-world example, use case and scenario

EXAMPLE
Hospital records
A hospital's VMs read patient files from Blob Storage through a private endpoint. Public access to the account is turned off.

USE CASE
Locked SQL
A bank connects its app subnet to Azure SQL Database through a private endpoint. The database has no path from the internet.

SCENARIO
On-premises access
An on-premises server connected by ExpressRoute uses the private endpoint IP to reach a storage account privately.

Network Security Options

Which control fits which threat?

Service	Layer / scope	Use it to...
Network security group	L3-L4, subnet or NIC	Allow/deny ports & IPs (e.g. open 443 to a VM). Free.
Azure Firewall	L3-L7, VNet-wide (managed)	Central stateful firewall, FQDN rules, threat intel, NAT
DDoS Protection	Public IPs	Absorb volumetric floods; does NOT open ports
Web Application Firewall	L7 (App Gateway / Front Door)	Block SQL injection, XSS on web apps
Azure Bastion	RDP/SSH via portal over TLS	Reach VMs with no public IP
Application security group	Groups NICs for NSG rules	Rule by workload role, not IP list

Exam key: Make a VM reachable on HTTPS = NSG inbound rule for 443. DDoS / Traffic Manager / Firewall alone don't open it.

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Figure Q327: Network security options

About this figure

- Rows compare services by layer and scope; the right column says what problem each solves. Pick the narrowest control that meets the requirement.
- NSG rules are processed by priority (100 to 4096, lowest first); default rules allow VNet and load balancer traffic and deny other inbound Internet traffic.
- Azure Firewall Standard costs approx. USD 900/month plus approx. USD 0.016/GB processed; NSGs and ASGs are free.
- DDoS infrastructure protection is free for all; DDoS Network Protection costs approx. USD 2,900/month for up to 100 public IPs, or IP Protection per IP.
- Azure Bastion removes the need for public IPs on VMs; it bills hourly (Basic approx. USD 0.19/hour) plus outbound data.

Question 328

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q461

Your company owns the domain contoso.com, registered with a third-party registrar. You want to host the DNS zone and manage its records by using the same credentials, billing, and support as your other Azure resources. Which Azure service should you use?

✔ **A.** Azure DNS

✖ B. Azure Traffic Manager

✖ C. Azure Front Door

✖ D. Azure Load Balancer

Correct answer: Azure DNS

Clarification (read this first): The domain is already bought from another company (a registrar). The question asks only where to HOST its DNS records. Q342 asks if Azure DNS can BUY a domain - the answer there is No.

Answer explained

✔ **Azure DNS**
Azure DNS hosts the DNS zone (the list of name records) for your domain on Azure name servers. You manage it like any other Azure resource.

✖ **Azure Traffic Manager**
Traffic Manager sends users to endpoints by using DNS. It does not host your domain's records.

✖ **Azure Front Door**
Front Door is a global layer-7 (HTTP) load balancer and CDN for web apps. It does not host DNS zones.

✖ **Azure Load Balancer**
Load Balancer spreads layer-4 (TCP/UDP) traffic across VMs. It does not turn names into IP addresses.

EXAM STRATEGY
'Host the DNS zone' and 'same credentials and billing as Azure' = Azure DNS. Traffic Manager and Front Door route traffic. Load Balancer spreads connections.

IMPACT
In Azure DNS you can use RBAC (role-based access), locks and activity logs on records. This stops someone from deleting a record by mistake and taking the site offline.

What · Where · When · Why · How

What	A service that hosts DNS zones on Azure's global name servers.
Where	A global service. The zones are kept in a resource group.
When	When you want DNS records managed with your other Azure resources.
Why	One place to manage, with RBAC, logs and high availability.
How	Create a DNS zone, add records, then point the registrar's NS (name server) records to Azure DNS.

Exam tip
Azure DNS hosts domains. You cannot buy or register a domain name in Azure DNS itself.

Memory trick
Azure DNS stores your domain's names; it does not sell the domain.

Real-world example, use case and scenario

EXAMPLE
Shop domain
A shop hosts shop.contoso.com A records in Azure DNS. They point to its App Service app.

USE CASE
Private names
A factory uses Azure DNS private zones so VMs can find db.internal.contoso without their own DNS servers.

SCENARIO
Alias records
A media company uses an alias record so its main domain always follows its load balancer's public IP.

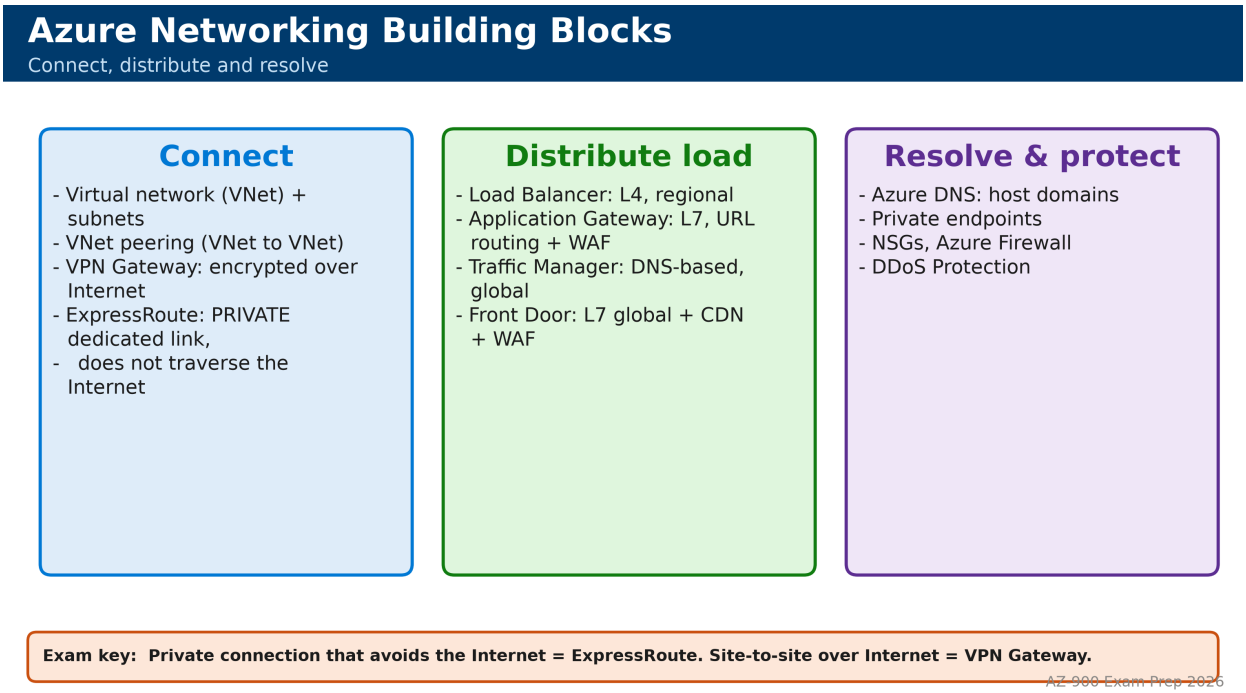


Figure Q328: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 329

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q462

You have four VMs in a single Azure region that run a custom TCP-based application (not HTTP). You need to distribute incoming traffic across the VMs at layer 4 and automatically stop sending traffic to an unhealthy VM. What should you use?

✔ **A.** Azure Load Balancer

✖ B. Azure Application Gateway

✖ C. Azure Front Door

✖ D. Azure DNS

Correct answer: Azure Load Balancer

Clarification (read this first): The key words are 'layer 4' and 'not HTTP'. Application Gateway and Front Door only work with web traffic (HTTP/HTTPS, layer 7). Q350 asks about GLOBAL routing across regions, which has a different answer.

Answer explained

✔ **Azure Load Balancer**
Layer 4 (TCP/UDP), one region, health probes (checks that each VM is working) across VMs = Azure Load Balancer.

✖ **Azure Application Gateway**
Application Gateway is a layer-7 (HTTP/HTTPS) load balancer. It cannot handle a custom TCP app.

✖ **Azure Front Door**
Front Door is a global layer-7 entry point for web apps. It is not for custom TCP apps.

✖ **Azure DNS**
Azure DNS turns names into IPs. It has no health probes and does not spread traffic.

EXAM STRATEGY
'Layer 4', 'TCP, not HTTP' and 'single region' = Azure Load Balancer. Application Gateway and Front Door are layer-7 HTTP. DNS cannot check the health of each VM.

IMPACT
If you match layer 4 and layer 7 correctly, you do not pay for an HTTP gateway that cannot even handle your TCP app.

What · Where · When · Why · How

What	A layer-4 load balancer for TCP and UDP traffic.
Where	In front of VMs or scale sets in one Azure region.
When	When non-web traffic must be shared across many VMs.
Why	It improves uptime and scale because no single server carries all the load.
How	Create a load balancer with a front-end IP, a back-end pool, a health probe and a rule.

Exam tip
Web-aware routing (URL path, WAF) = Application Gateway. Any TCP/UDP = Load Balancer.

Memory trick
Load Balancer = Layer 4. Application Gateway = Application layer 7.

Real-world example, use case and scenario

EXAMPLE
Game servers
A game studio shares UDP traffic across game-server VMs with Azure Load Balancer.

USE CASE
Internal tier
An insurance company uses an internal load balancer for its middle-tier VMs.

SCENARIO
Failed VM
One VM fails its health probe. The load balancer stops sending new connections to it.

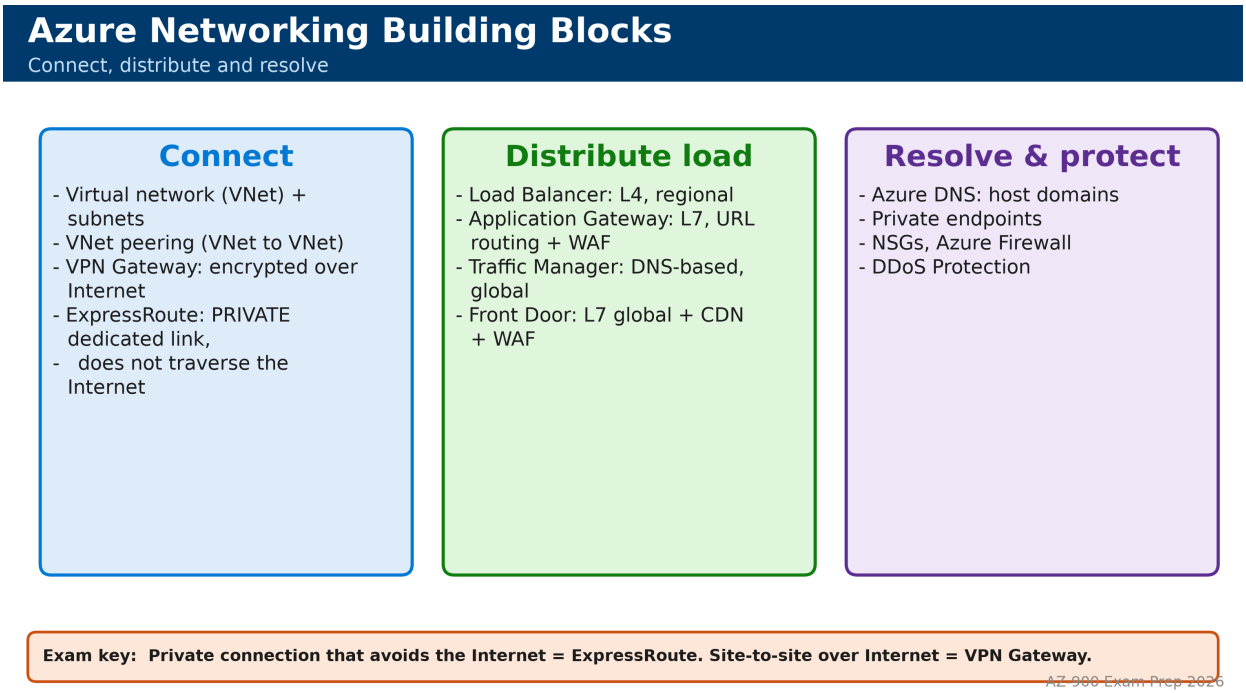


Figure Q329: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 330

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q463

A research lab stores instrument data in Azure Blob Storage. The data is rarely read, will be kept for at least 90 days, and must be readable immediately (no rehydration) when it is needed. You need the lowest storage cost that meets these requirements. Which access tier should you use?

✔ **A.** Cold

✖ B. Hot

✖ C. Cool

✖ D. Archive

Correct answer: Cold

Clarification (read this first): Watch 'readable immediately (no rehydration)'. Archive is the cheapest tier, but it is offline: data must be rehydrated (copied back online, which takes hours) first - see Q242. Cold and Cool are two different tiers; Cold is cheaper to store.

Answer explained

✔ **Cold**
Cold is the cheapest ONLINE tier. Data should stay at least 90 days, and you can read it at once.

✖ **Hot**
Hot is for data you read often. It has the highest storage cost.

✖ **Cool**
Cool is online, but storing data there costs more than Cold. It is for data kept at least 30 days.

✖ **Archive**
Archive is cheapest, but it is offline. Blobs must be rehydrated before you can read them.

EXAM STRATEGY
'Rarely read', '90 days' and 'readable immediately' = Cold. Archive needs rehydration. Cool (30-day minimum) costs more to store. Hot is for frequent use.

IMPACT
The tier you pick changes the storage bill a lot at large scale. Picking Archive by mistake adds hours of delay just when you urgently need the data.

What · Where · When · Why · How

What	An online blob tier for data you rarely read.
Where	On block blobs in general-purpose v2 or Blob Storage accounts.
When	When data is kept at least 90 days and read rarely.
Why	Lower storage cost than Cool, and the data stays online.
How	Set the blob's tier to Cold, or use a lifecycle policy to move it.

Exam tip
Minimum days: Cool 30, Cold 90, Archive 180. Archive is the only offline tier.

Memory trick
Hot, Cool, Cold, Archive: colder = cheaper to store, more costly to read. 30 / 90 / 180.

Real-world example, use case and scenario

EXAMPLE
Lab data
A university moves 6-month-old sensor data to Cold. It is cheap but still instant to read.

USE CASE
Backups
A software company keeps short-term backups in Cold so restores start at once.

SCENARIO
Audit
An auditor sometimes asks for old invoices. They are read straight from the Cold tier.

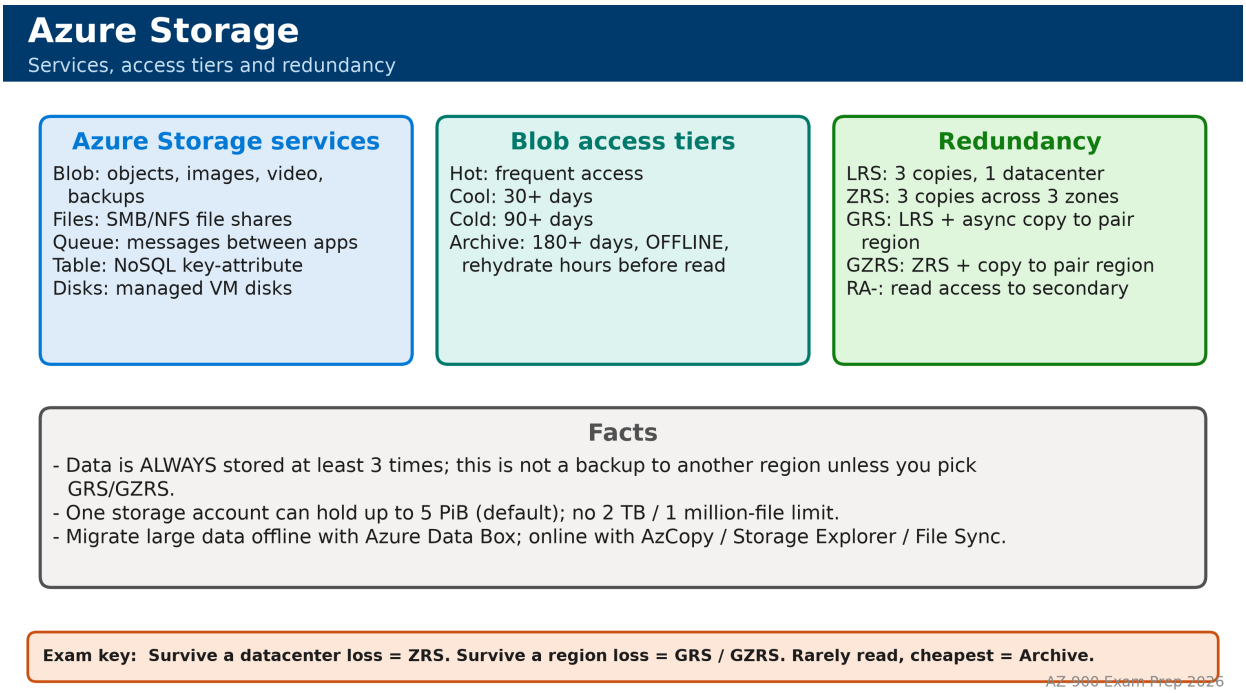


Figure Q330: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 331

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q464

Your company needs a storage redundancy option that synchronously copies data across three availability zones in the primary region, asynchronously copies it to a secondary region, and allows applications to read from the secondary region at any time. Which option should you choose?

✔ **A.** Read-access geo-zone-redundant storage (RA-GZRS)

✖ B. Locally redundant storage (LRS)

✖ C. Zone-redundant storage (ZRS)

✖ D. Geo-redundant storage (GRS)

Correct answer: **Read-access geo-zone-redundant storage (RA-GZRS)**

Clarification (read this first): Three needs in one: 'three zones' (Z), 'secondary region' (G) and 'read from secondary at any time' (RA). Q117 has no zones in its question, so its answer is RA-GRS instead.

Answer explained

✔ **Read-access geo-zone-redundant storage (RA-GZRS)**
Zones in the main region + a copy in a second region + you can read the second copy = RA-GZRS.

✖ **Locally redundant storage (LRS)**
LRS keeps three copies in one datacenter only. It has no zone or region protection.

✖ **Zone-redundant storage (ZRS)**
ZRS spreads copies across zones in one region. It has no copy in a second region.

✖ **Geo-redundant storage (GRS)**
GRS uses LRS (not zones) in the main region. You cannot read the second copy unless a failover happens.

EXAM STRATEGY
Find three clues: 'three zones' = Z, 'secondary region' = G, 'read from secondary any time' = RA. Only RA-GZRS has all three. Remove any option missing one.

IMPACT
RA-GZRS gives the strongest data protection and lets you read during a region outage. It suits critical data with strict recovery rules.

What · Where · When · Why · How

What	The strongest Azure Storage redundancy option, with a readable second copy.
Where	The main region (with availability zones) plus its paired second region.
When	When an app must survive zone and region outages and still read data.
Why	It gives the most durability and read availability.
How	Choose RA-GZRS as the redundancy setting on a general-purpose v2 account.

Exam tip
G = geo (second region), Z = zones, RA = read access to the second copy.

Memory trick
Read the letters: RA-G-Z-RS = Read Access + Geo + Zones.

Real-world example, use case and scenario

EXAMPLE
Bank portal
A bank uses RA-GZRS so statements can still be read if the main region is down.

USE CASE
Global SaaS
A SaaS company stores customer files on RA-GZRS for very high durability.

SCENARIO
Zone outage
One zone fails. The ZRS copies in the other zones keep writes working with no failover.

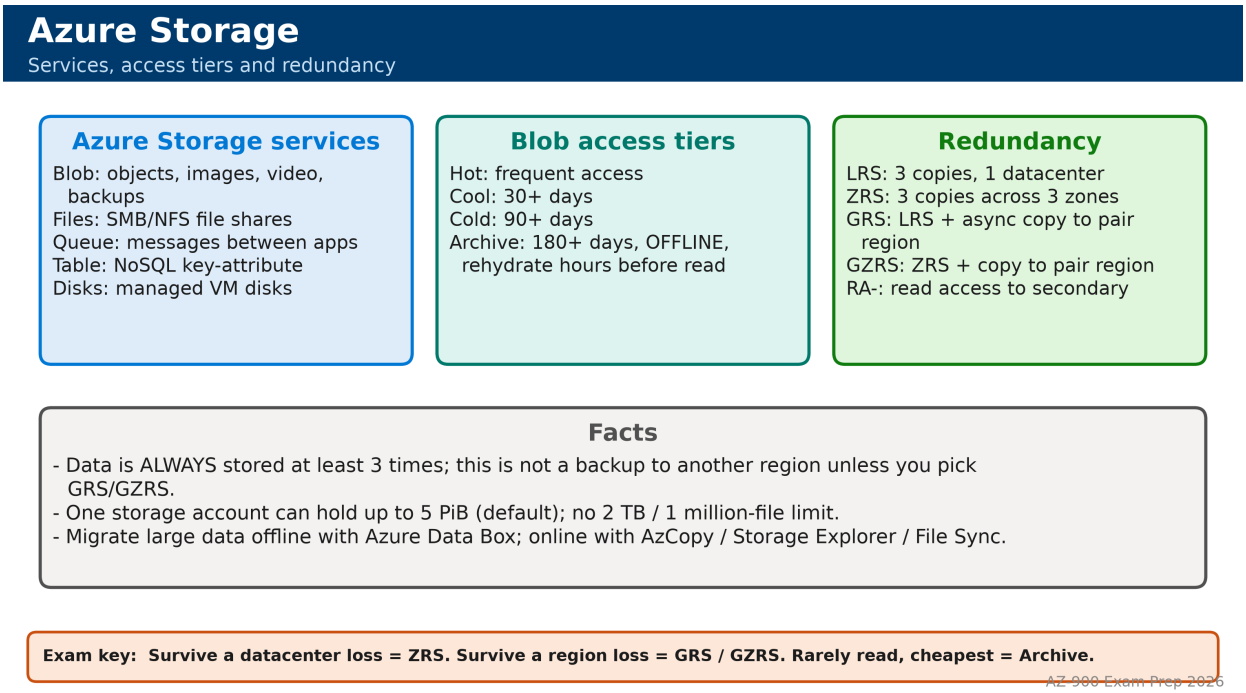


Figure Q331a: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

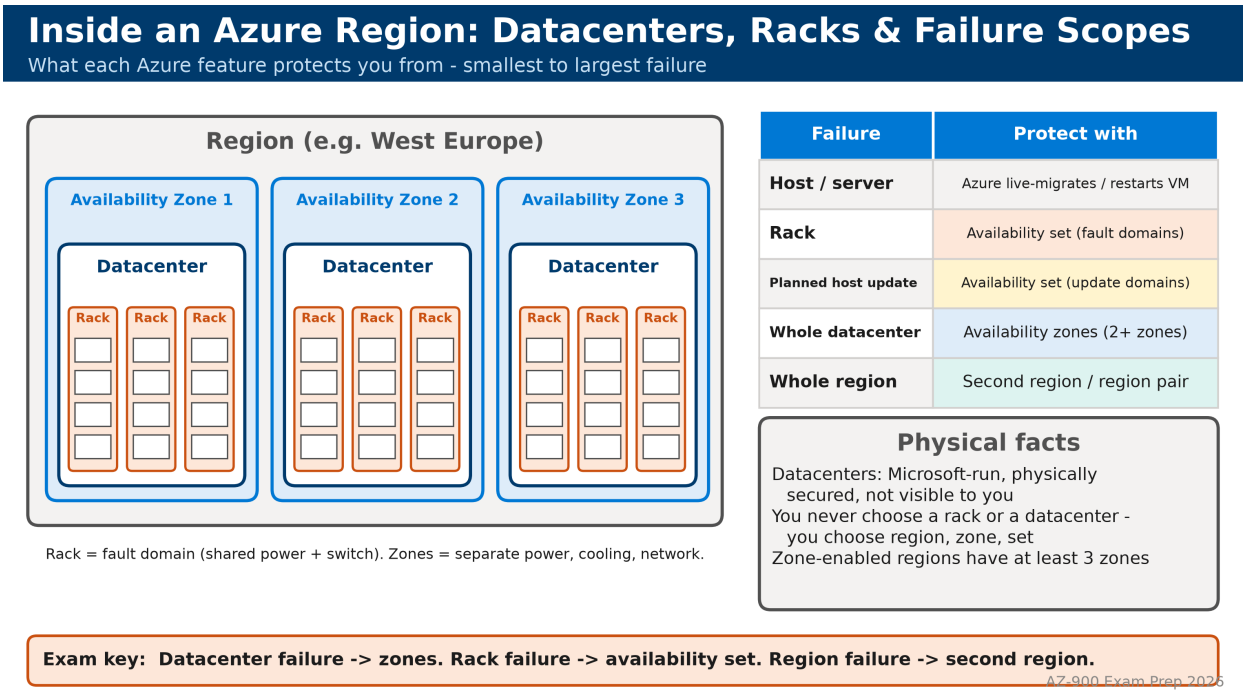


Figure Q331b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 332

AZURE ARCHITECTURE & SERVICES — Storage — v5 Q465

Your company has branch offices with on-premises Windows Server file servers. You want to centralize the files in Azure Files while keeping frequently used files cached locally on each branch server for fast access. What should you use?

✔ **A.** Azure File Sync

✖ B. AzCopy

✖ C. Azure Data Box

✖ D. Azure Migrate

Correct answer: Azure File Sync

Answer explained

✔ **Azure File Sync**
Files kept centrally in Azure Files + a local cache on Windows Server with cloud tiering (rarely used files stay only in the cloud) = Azure File Sync.

✖ **AzCopy**
AzCopy is a command-line tool for one-time copies. It does not keep a synced local cache.

✖ **Azure Data Box**
Data Box is a physical device you ship for large one-time transfers. It does not sync.

✖ **Azure Migrate**
Azure Migrate finds, assesses and moves servers. It does not cache file shares.

EXAM STRATEGY
'Cache locally on branch servers' plus 'centralize in Azure Files' = Azure File Sync. AzCopy and Data Box are one-time transfers. Migrate assesses servers.

IMPACT
Branches keep fast local access, and cloud tiering reduces storage needed on site. This is a common hybrid way to save money.

What · Where · When · Why · How

What	A service that syncs Azure file shares with on-premises Windows Servers.
Where	An agent on Windows Server, linked to an Azure file share.
When	When branch offices need local speed with central cloud storage.
Why	It joins central cloud storage with local speed and multi-site sync.
How	Deploy a Storage Sync Service, install the agent, and create a sync group with cloud tiering.

Exam tip
Azure File Sync needs Windows Server on-premises and an Azure file share (not blob).

Memory trick
File Sync = Files + Sync + Server cache.

Real-world example, use case and scenario

EXAMPLE
Law firm
A law firm caches active case files on each office server. All files live in Azure Files.

USE CASE
Server replacement
A branch server fails. IT installs the agent on a new server and lets the files sync back.

SCENARIO
Multi-site
An engineering company syncs one Azure file share to servers in three countries.

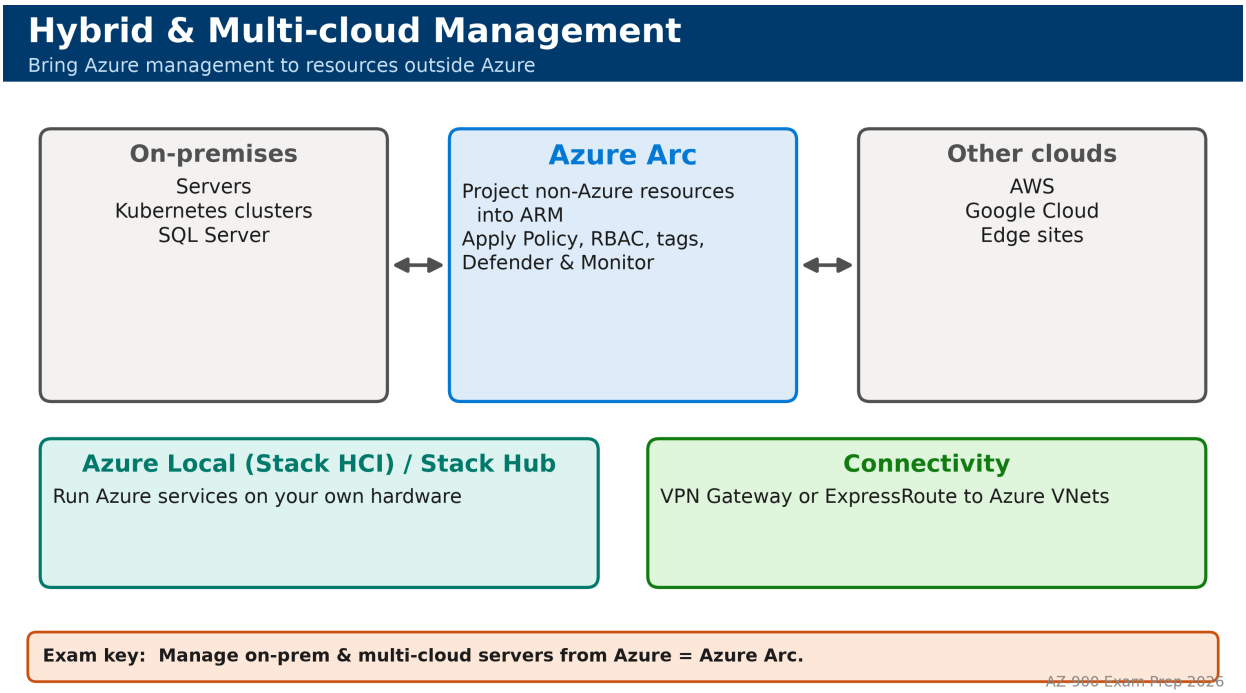


Figure Q332: Hybrid and multi-cloud management

About this figure

- Arrows show Azure Arc linking on-premises and other-cloud resources into Azure Resource Manager; bottom boxes cover local Azure hardware and network links.
- Arc-enabled servers' core control plane (inventory, tags, RBAC, Policy) is free; add-ons such as Defender, Update Manager and Monitor are billed separately.
- Arc can also manage Kubernetes clusters and SQL Server; Arc-enabled SQL Server can be licensed pay-as-you-go through your Azure bill.
- Azure Local (formerly Azure Stack HCI) runs VMs and AKS on validated on-premises hardware, billed per physical core per month in Azure.
- Site-to-site VPN uses the Internet with encryption; ExpressRoute is a private connection through a provider with higher reliability and an SLA up to 99.95%.

Question 333

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q466

Your company must move 70 TB of archived video from its datacenter to Azure Blob Storage. The site has a slow internet connection, and the transfer would take months over the network. What should you use?

✔ **A.** Azure Data Box

✖ B. AzCopy

✖ C. Azure File Sync

✖ D. Azure Storage Explorer

Correct answer: Azure Data Box

Answer explained

✔ **Azure Data Box**
Tens or hundreds of TB + a slow network = Azure Data Box (a device you fill with data and ship to Microsoft).

✖ **AzCopy**
AzCopy sends data over the network, and the network is the problem here.

✖ **Azure File Sync**
File Sync is an ongoing sync over the network. It is still limited by the slow link.

✖ **Azure Storage Explorer**
Storage Explorer is a GUI tool that also copies over the network. It is not suitable for 70 TB on a slow link.

EXAM STRATEGY
'Tens of TB' plus 'slow internet' or 'months over the network' = Data Box offline transfer. Remove AzCopy and Storage Explorer; both depend on the network.

IMPACT
Shipping the data avoids filling the office link for months and helps meet deadlines. Copying 70 TB over the network would slow normal business traffic.

What · Where · When · Why · How

What	A physical device for moving large data into or out of Azure.
Where	Shipped to your datacenter, then sent back to an Azure datacenter.
When	When there is a lot of data and little network bandwidth (speed).
Why	Faster, cheaper and more predictable than months of uploads.
How	Order a Data Box in the Azure portal, copy data onto it, and ship it back.

Exam tip
Data Box can also export data out of Azure. It is not only for import.

Memory trick
Too big for the internet? Ship it in a Data Box.

Real-world example, use case and scenario

EXAMPLE
Media archive
A TV company ships many years of video to Azure on Data Box.

USE CASE
Datacenter exit
A shop closing its datacenter uses Data Box for the first big load, then copies the last changes over the network.

SCENARIO
Disaster recovery
A company exports a large dataset from Azure on a Data Box to rebuild an on-premises site.

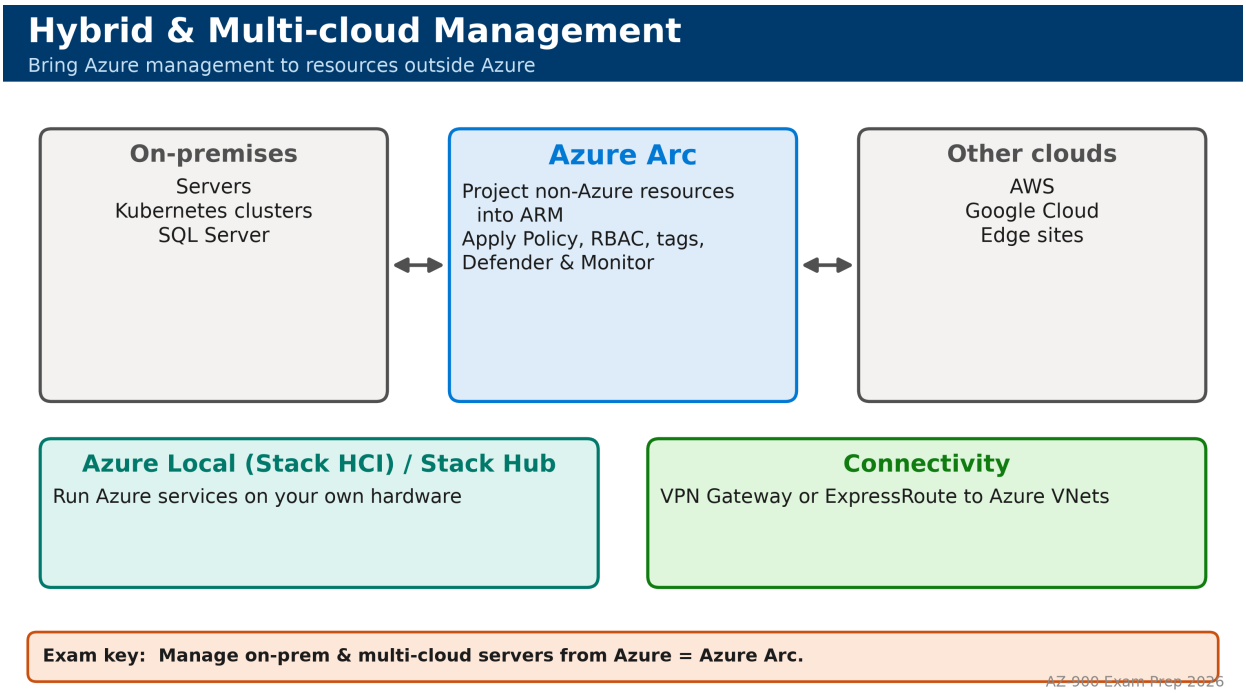


Figure Q333a: Hybrid and multi-cloud management

About this figure

- Arrows show Azure Arc linking on-premises and other-cloud resources into Azure Resource Manager; bottom boxes cover local Azure hardware and network links.
- Arc-enabled servers' core control plane (inventory, tags, RBAC, Policy) is free; add-ons such as Defender, Update Manager and Monitor are billed separately.
- Arc can also manage Kubernetes clusters and SQL Server; Arc-enabled SQL Server can be licensed pay-as-you-go through your Azure bill.
- Azure Local (formerly Azure Stack HCI) runs VMs and AKS on validated on-premises hardware, billed per physical core per month in Azure.
- Site-to-site VPN uses the Internet with encryption; ExpressRoute is a private connection through a provider with higher reliability and an SLA up to 99.95%.

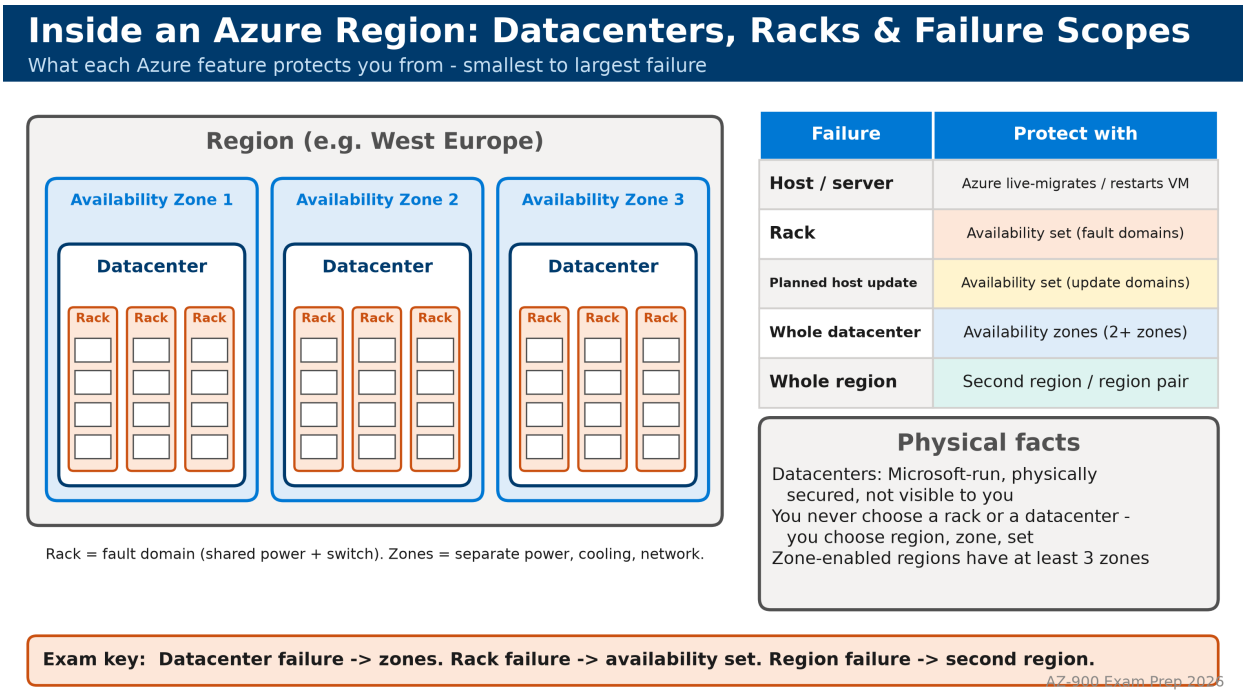


Figure Q333b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 334

AZURE ARCHITECTURE & SERVICES — **Storage** — v5 Q467

Your company plans to run a transaction-heavy SAP HANA database on an Azure VM. The disk must provide the highest IOPS and throughput with sub-millisecond latency, and you want to adjust performance without detaching the disk. Which managed disk type should you choose?

✔ **A.** Ultra Disk

✖ B. Premium SSD

✖ C. Standard SSD

✖ D. Standard HDD

Correct answer: Ultra Disk

Answer explained

✔ **Ultra Disk**
Highest performance, latency (delay) under 1 millisecond, and IOPS you can change while the disk is attached = Ultra Disk.

✖ **Premium SSD**
Premium SSD is fast enough for most production work, but it is below Ultra Disk's top IOPS and latency.

✖ **Standard SSD**
Standard SSD gives steady speed for web servers and light work, not heavy databases.

✖ **Standard HDD**
Standard HDD is the cheapest, for backups and rare use. It has the slowest response.

EXAM STRATEGY
'Highest IOPS (disk reads and writes per second)', 'sub-millisecond', 'SAP HANA' and 'adjust without detaching' = Ultra Disk. Premium SSD is second best.

IMPACT
A slow disk holds back a busy database and every user feels it. With Ultra Disk you can raise performance live for busy periods.

What · Where · When · Why · How

What	A managed disk type for the most disk-heavy workloads.
Where	Attached as a data disk to supported Azure VMs.
When	When databases need very high IOPS and very low latency.
Why	It meets speed needs that other disk types cannot.
How	Create a managed disk with the Ultra SKU and set IOPS and throughput separately.

Exam tip
Disk order by speed: Standard HDD < Standard SSD < Premium SSD < Ultra Disk.

Memory trick
Ultra = the ultimate speed.

Real-world example, use case and scenario

EXAMPLE
SAP HANA
A manufacturer runs SAP HANA log volumes on Ultra Disk.

USE CASE
Trading
A trading platform puts its order database on Ultra Disk for very fast writes.

SCENARIO
Quarter-end
A finance team raises Ultra Disk IOPS for quarter-end work, then lowers it.

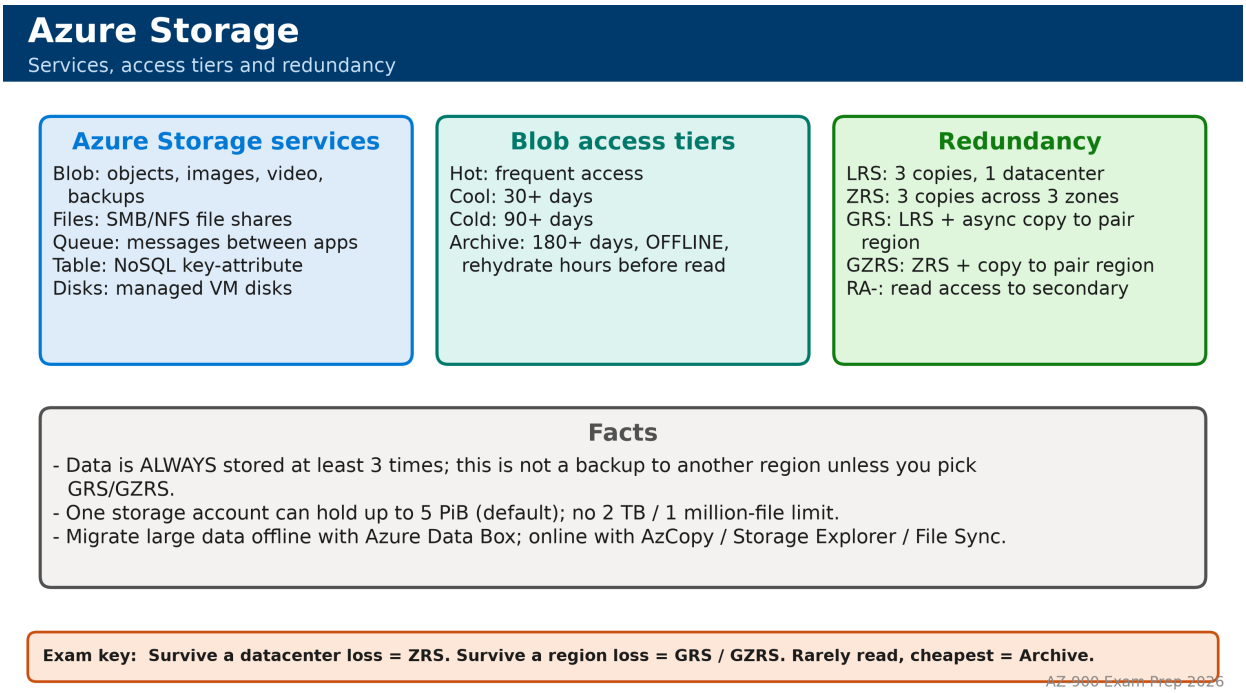


Figure Q334: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 335

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q468

Your support engineers need a free, standalone graphical application that runs on Windows, macOS, and Linux to browse, upload, and download blobs, files, queues, and tables across multiple storage accounts. What should you provide?

✔ **A.** Azure Storage Explorer

✖ B. AzCopy

✖ C. Azure Cloud Shell

✖ D. Azure Data Box

Correct answer: Azure Storage Explorer

Clarification (read this first): The key word is 'graphical' (point-and-click). AzCopy also moves files, but it is command line. Q348 asks only about copying files, so there both AzCopy and Storage Explorer are correct.

Answer explained

✔ **Azure Storage Explorer**
A free desktop app with a GUI (point-and-click screen) for Windows, macOS and Linux to manage storage data = Azure Storage Explorer.

✖ **AzCopy**
AzCopy is a command-line tool, not a graphical app.

✖ **Azure Cloud Shell**
Cloud Shell is a shell (command line) in the browser, not a desktop GUI app.

✖ **Azure Data Box**
Data Box is a physical device for large offline transfers.

EXAM STRATEGY
'Free standalone graphical app' plus 'Windows, macOS, Linux' = Storage Explorer. AzCopy is command line. Cloud Shell is a browser shell. Data Box is hardware.

IMPACT
Support staff can manage data without writing scripts. This means fewer mistakes than typing commands on live accounts.

What · Where · When · Why · How

What	A free desktop GUI app for Azure Storage.
Where	Installed on Windows, macOS or Linux computers.
When	When users prefer to point and click to manage storage data.
Why	It makes work with blobs, files, queues, tables and disks easier.
How	Download it, sign in with Microsoft Entra ID or keys/SAS, and browse your accounts.

Exam tip
GUI = Storage Explorer. Command line = AzCopy.

Memory trick
Storage Explorer = like File Explorer, but for Azure Storage.

Real-world example, use case and scenario

EXAMPLE
Support desk
A help-desk engineer drags a customer log file into a blob container.

USE CASE
Queue check
A developer looks at messages waiting in a storage queue.

SCENARIO
Mac users
A design agency's Mac staff manage files in Blob Storage with Storage Explorer.

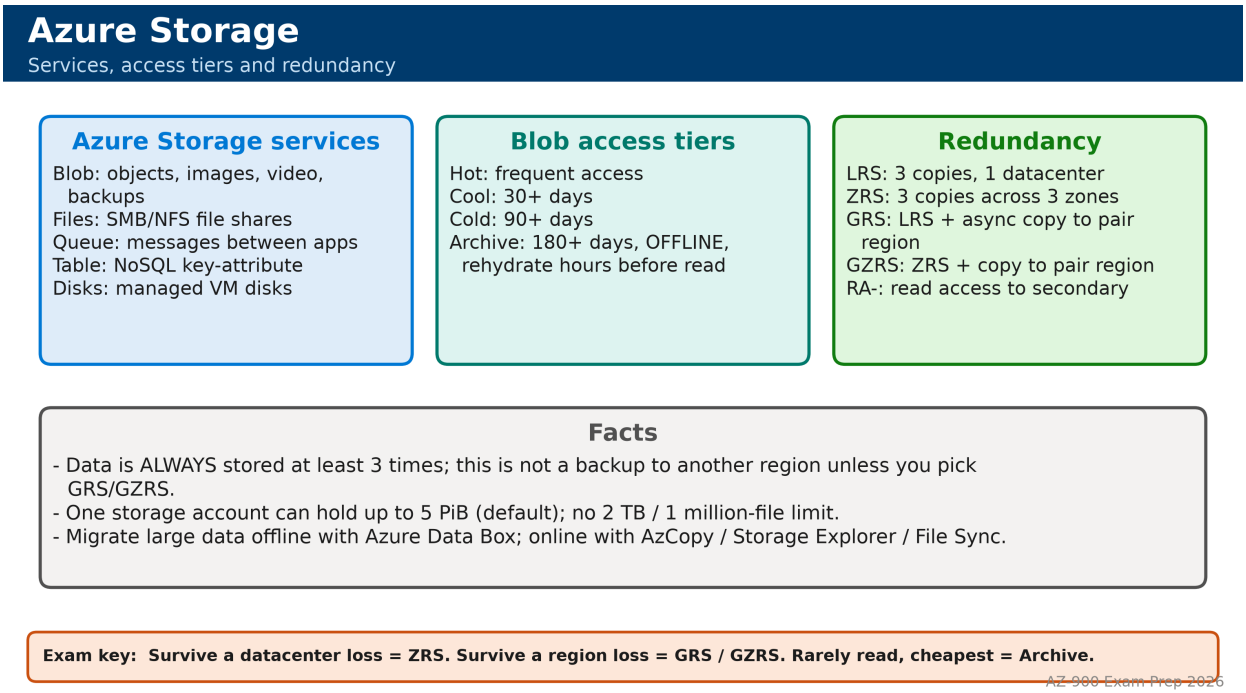


Figure Q335: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

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• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 336

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q469

A web front end must hand off image-processing jobs to a background worker so that the two components are decoupled and jobs are not lost if the worker is busy. You want a simple Azure Storage service that stores millions of messages accessible over HTTP or HTTPS. Which service should you use?

✔ **A.** Azure Queue Storage

✖ B. Azure Table Storage

✖ C. Azure Files

✖ D. Azure Disk Storage

Correct answer: Azure Queue Storage

Clarification (read this first): Do not mix up Queue Storage and Table Storage. 'Messages' and 'hand off jobs' mean Queue. Q337 asks for structured key/attribute data - that is Table Storage.

Answer explained

✔ **Azure Queue Storage**
Passing messages between parts of an app so they work separately (asynchronous) = Queue Storage.

✖ **Azure Table Storage**
Table Storage keeps structured NoSQL records (entities). It is not a message queue.

✖ **Azure Files**
Azure Files gives managed SMB/NFS file shares. It does not pass messages.

✖ **Azure Disk Storage**
Disk Storage gives block-level disks for VMs.

EXAM STRATEGY
'Decouple components (let them work separately)', 'messages', 'HTTP or HTTPS' and 'millions' = Queue Storage. Table stores records, Files is shares, Disk is VM storage.

IMPACT
A queue holds jobs during busy spikes, so no job is lost when workers are busy. This is a basic way to make cloud apps reliable.

What · Where · When · Why · How

What	A message store for work passed between parts of an app.
Where	Inside an Azure storage account.
When	When one part makes work faster than another part can handle it.
Why	It separates the parts and holds the extra load.
How	The front end adds messages. The worker reads each one and deletes it when done.

Exam tip
Each queue message is up to 64 KB. A queue can hold millions of messages, up to the account limit.

Memory trick
Queue = a line of jobs waiting their turn.

Real-world example, use case and scenario

EXAMPLE
Photo app
A photo site queues thumbnail jobs for a background Azure Function.

USE CASE
Orders
An online store queues orders so payment processing can scale on its own.

SCENARIO
Busy sale
During a sale, messages build up in the queue. Workers process them later.

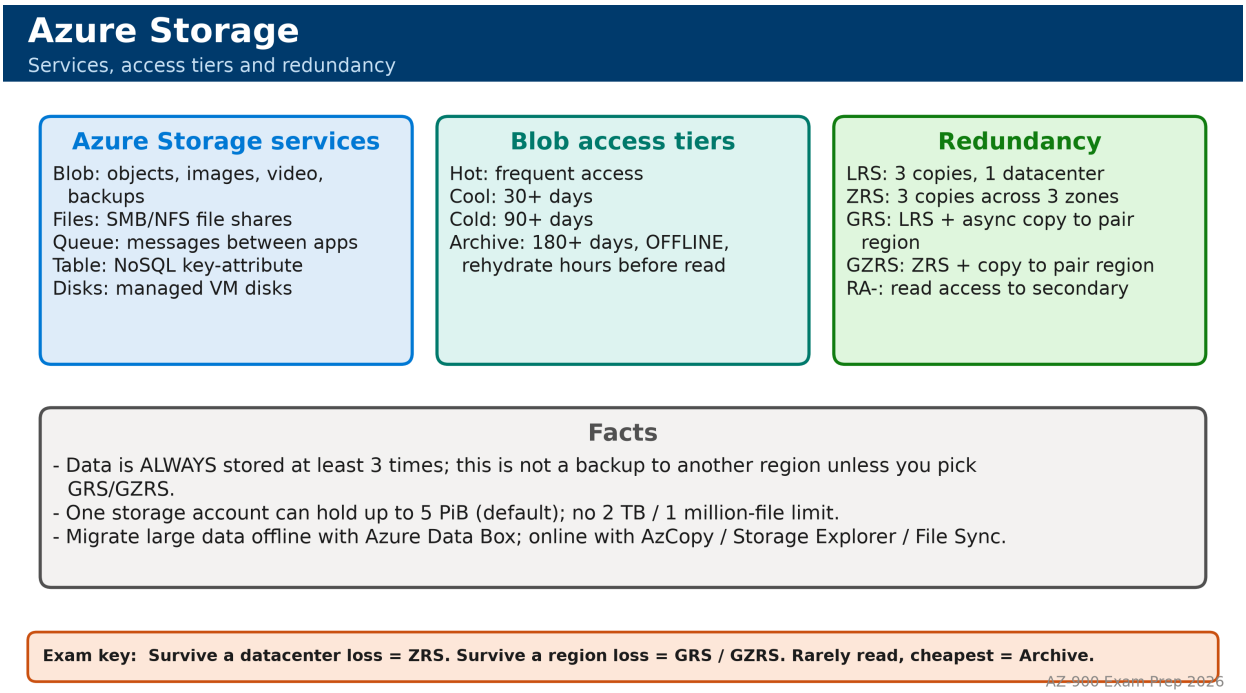


Figure Q336: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 337

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q470

A logistics company needs a low-cost Azure Storage service to store large volumes of structured, non-relational data using a key/attribute model with a schemaless design. Which service should you use?

✔ **A.** Azure Table Storage

✖ B. Azure Queue Storage

✖ C. Azure Blob Storage

✖ D. Azure Files

Correct answer: Azure Table Storage

Clarification (read this first): 'Non-relational' means NoSQL (no fixed tables with joins). Do not confuse Table Storage with Queue Storage (Q336, messages) or with Azure SQL Database (relational).

Answer explained

✔ **Azure Table Storage**
Structured NoSQL data in key/attribute form, with no fixed schema (column list) = Table Storage.

✖ **Azure Queue Storage**
Queue Storage keeps messages for later processing. You cannot query them like records.

✖ **Azure Blob Storage**
Blob Storage keeps unstructured files such as images and video.

✖ **Azure Files**
Azure Files gives SMB/NFS file shares.

EXAM STRATEGY
'Structured, non-relational', 'key/attribute', 'schemaless' and 'low cost' = Table Storage. Queues hold messages. Blob holds files. Files gives shares.

IMPACT
Table Storage is cheap NoSQL storage for big datasets. A relational database would cost more and need schema management.

What · Where · When · Why · How

What	A NoSQL key/attribute data store.
Where	In an Azure storage account (also offered as Azure Cosmos DB for Table).
When	When you store structured data and do not need relational joins.
Why	Low cost and a flexible schema at large scale.
How	Add entities (records) with a PartitionKey and RowKey plus any other properties.

Exam tip
Table Storage = structured NoSQL. Do not confuse it with relational Azure SQL Database.

Memory trick
Table = rows with keys, but no fixed columns.

Real-world example, use case and scenario

EXAMPLE
Shipment tracking
A delivery company stores scan events by shipment ID.

USE CASE
User profiles
A mobile game stores player settings as table entities.

SCENARIO
IoT data
A utility stores device details where each device type has different fields.

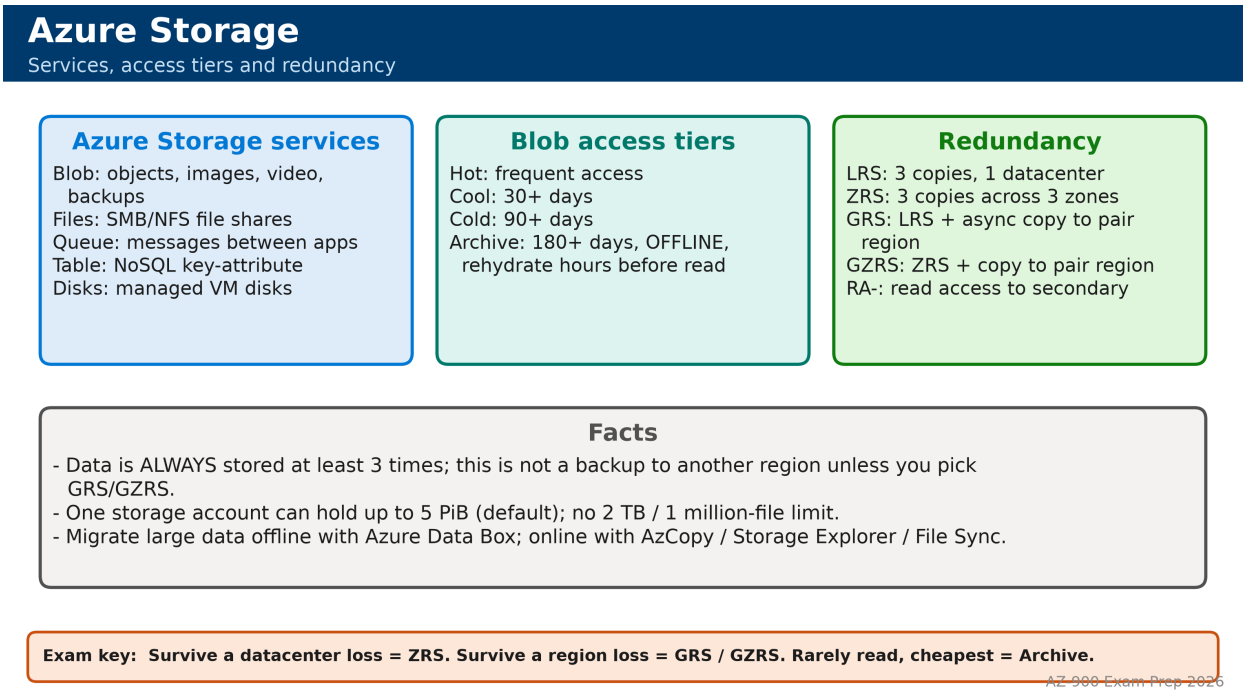


Figure Q337: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

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• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 338

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q471

Your company needs enterprise file shares in Azure that deliver high performance with SSD-backed storage and support both SMB and NFS protocols. Which storage account type should you create?

✔ **A.** Premium file shares

✖ B. Standard general-purpose v2

✖ C. Premium block blobs

✖ D. Premium page blobs

Correct answer: Premium file shares

Clarification (read this first): Watch 'NFS' and 'SSD'. Standard general-purpose v2 also has file shares, but they use HDD and do not support NFS. Only the premium file shares account gives both.

Answer explained

✔ **Premium file shares**
SSD-based file shares with SMB and NFS (file-sharing protocols) = a Premium file shares account.

✖ **Standard general-purpose v2**
It supports file shares, but they are HDD-based and have no NFS.

✖ **Premium block blobs**
This premium account is for block blobs and append blobs, not file shares.

✖ **Premium page blobs**
This premium account is for page blobs only.

EXAM STRATEGY
'SSD-backed', 'file shares' and 'SMB and NFS' = Premium file shares (FileStorage). Standard GPv2 uses HDD for files. Block and page blob types are not file shares.

IMPACT
NFS shares need the premium file shares account. If you pick GPv2, NFS apps cannot run and file server speed is too low.

What · Where · When · Why · How

What	A premium storage account type only for Azure file shares.
Where	Created as a FileStorage account in an Azure region.
When	When file shares need low latency (delay) and high IOPS.
Why	SSD speed plus NFS support for Linux apps.
How	Create a storage account, choose Premium performance and the File shares account type.

Exam tip
Standard GPv2 is the right choice for most cases. Choose premium types when you need low latency.

Memory trick
Premium + Files = fast shares for SMB and NFS.

Real-world example, use case and scenario

EXAMPLE
Render farm
An animation studio uses premium file shares for its render servers.

USE CASE
SAP files
A large company stores SAP shared folders on premium NFS shares.

SCENARIO
Lift and shift
A Linux app that needs NFS moves to Azure on premium file shares.

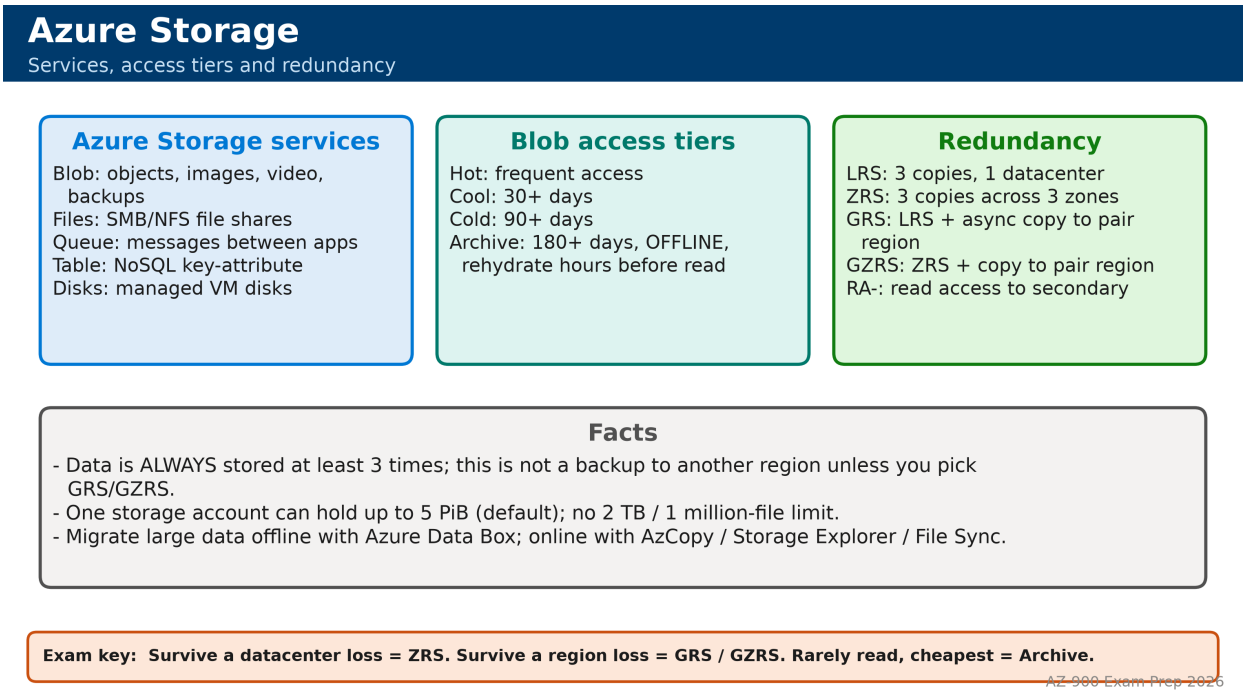


Figure Q338: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

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• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 339

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q472

Your company has a virtual network in East US and another in West Europe. VMs in both must communicate privately over the Microsoft backbone with low latency, without deploying gateways or using the public internet. What should you configure?

✔ **A.** Global virtual network peering

✖ B. A site-to-site VPN connection

✖ C. An ExpressRoute circuit

✖ D. An Azure DNS private zone

Correct answer: Global virtual network peering

Clarification (read this first): The key words are 'two regions' and 'without gateways'. VPN and ExpressRoute need a gateway. Q343 explains why you need this: one VNet cannot span two regions.

Answer explained

✔ **Global virtual network peering**
VNets in different regions + private + no gateways = global VNet peering.

✖ **A site-to-site VPN connection**
Site-to-site VPN links an on-premises network to Azure through a VPN gateway.

✖ **An ExpressRoute circuit**
ExpressRoute is a private link from on-premises through a provider. It is not for VNet-to-VNet.

✖ **An Azure DNS private zone**
A private DNS zone only turns names into IPs. It does not carry traffic.

EXAM STRATEGY
'Different regions', 'Microsoft backbone (Microsoft's own private network)' and 'no gateways' = global VNet peering. VPN and ExpressRoute need gateways. DNS zones only resolve names.

IMPACT
Peering gives fast, private links between regions with no VPN gateway cost or speed limit. It suits apps spread over many regions.

What · Where · When · Why · How

What	A peering link between VNets in different regions.
Where	Set up on both VNets, across Azure regions.
When	When resources in different regions must talk privately.
Why	Fast, private links with no gateways.
How	Create a peering from each VNet to the other in the portal, CLI or ARM/Bicep.

Exam tip
Peering in one region = regional peering. Across regions = global peering. Address ranges must not overlap.

Memory trick
Global peering = neighbours across the world sharing a private hallway.

Real-world example, use case and scenario

EXAMPLE
Multi-region app
A travel site's web tier in East US reads a cache in West Europe through global peering.

USE CASE
Hub and spoke
A bank peers each regional spoke VNet to a central hub VNet.

SCENARIO
DR copy
A database copies data to a disaster recovery region over a globally peered VNet.

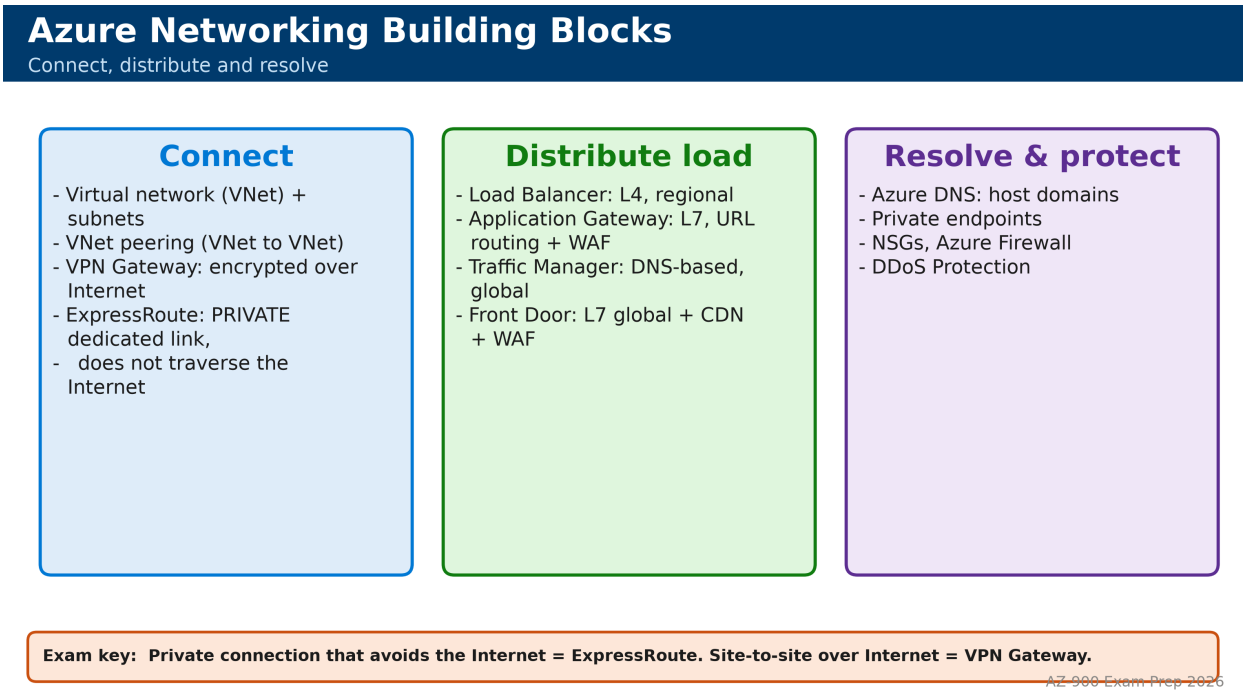


Figure Q339: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 340

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q473

For each statement, select Yes if it is true. Statement: Traffic between two peered Azure virtual networks travels over the Microsoft backbone network and does not traverse the public internet.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Answer explained

✔ **Yes**
Traffic between peered VNets is private. It stays on Microsoft's own backbone network.

✖ **No**
Peering never sends traffic over the public internet.

EXAM STRATEGY
Key fact: peered VNet traffic stays on the Microsoft backbone, never the public internet. The statement matches, so the answer is Yes.

IMPACT
Traffic that never leaves Microsoft's network faces fewer internet threats and has steady speed. This often meets compliance rules without extra encryption tunnels.

What · Where · When · Why · How

What	Private links between peered VNets.
Where	On Microsoft's global backbone network.
When	Whenever resources in peered VNets talk to each other.
Why	It gives low latency, high speed and a private path.
How	After peering is set up on both sides, resources use each other's private IPs.

Exam tip
Peering = private backbone. VPN gateway = encrypted tunnel, often over the internet.

Memory trick
Peers talk in Microsoft's private hallway, not on the public street.

Real-world example, use case and scenario

EXAMPLE
App and data
A shop peers its app VNet with a data VNet so VMs reach databases privately.

USE CASE
Shared services
A company peers spoke VNets to a hub that holds domain controllers.

SCENARIO
Speed check
Engineers see that peered traffic in one region is as fast as traffic inside one VNet.

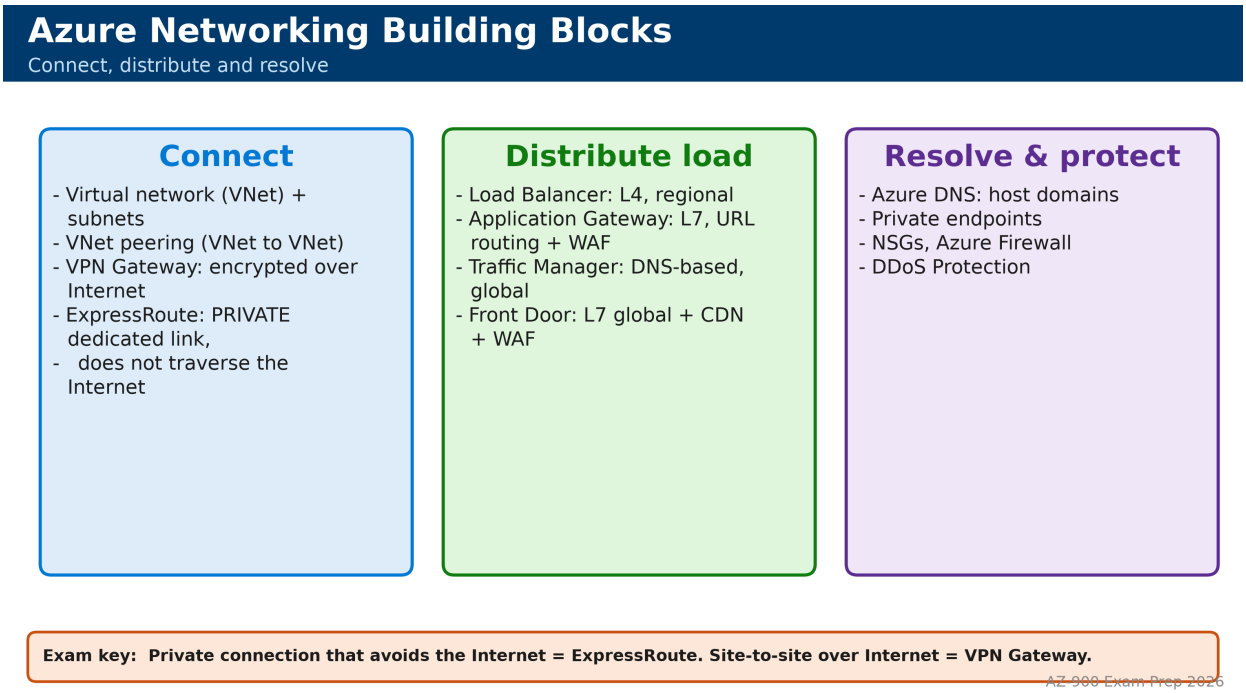


Figure Q340: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 341

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q474

For each statement, select Yes if it is true. Statement: If VNet-A is peered with VNet-B, and VNet-B is peered with VNet-C, then VMs in VNet-A can automatically communicate with VMs in VNet-C.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): The trap word is 'automatically'. Peering is not transitive: it links only the two VNets in that peering. A to B and B to C does NOT give A to C.

Answer explained

✔ **No**
VNet peering is NOT transitive (it does not pass through). A-B plus B-C does not give A-C.

✖ **Yes**
Peering does not pass traffic through a middle VNet by itself.

EXAM STRATEGY
Key fact: peering is not transitive. A to B and B to C does not give A to C, so the answer is No. You need a direct peering or a hub with routing.

IMPACT
If you think peering passes through, hub-and-spoke designs break. Spokes cannot talk unless you add a firewall, Virtual WAN or direct peering. This often surprises teams in production.

What · Where · When · Why · How

What	A rule that peering links only the two VNets involved.
Where	It applies to regional and global VNet peering.
When	When you design hub-and-spoke or many-VNet networks.
Why	It stops network paths you did not plan.
How	Add a direct peering, or route spoke traffic through a firewall or network appliance in the hub.

Exam tip
In hub-and-spoke designs, spoke-to-spoke traffic goes through a hub appliance because peering is not transitive.

Memory trick
A friend of a friend is not your friend in VNet peering.

Real-world example, use case and scenario

EXAMPLE
Spokes apart
Two spoke VNets peered only to a hub cannot talk directly to each other.

USE CASE
Separation
A bank uses this rule to keep its test and production spokes apart.

SCENARIO
Fix
An admin adds Azure Firewall in the hub with route tables so spokes can talk.

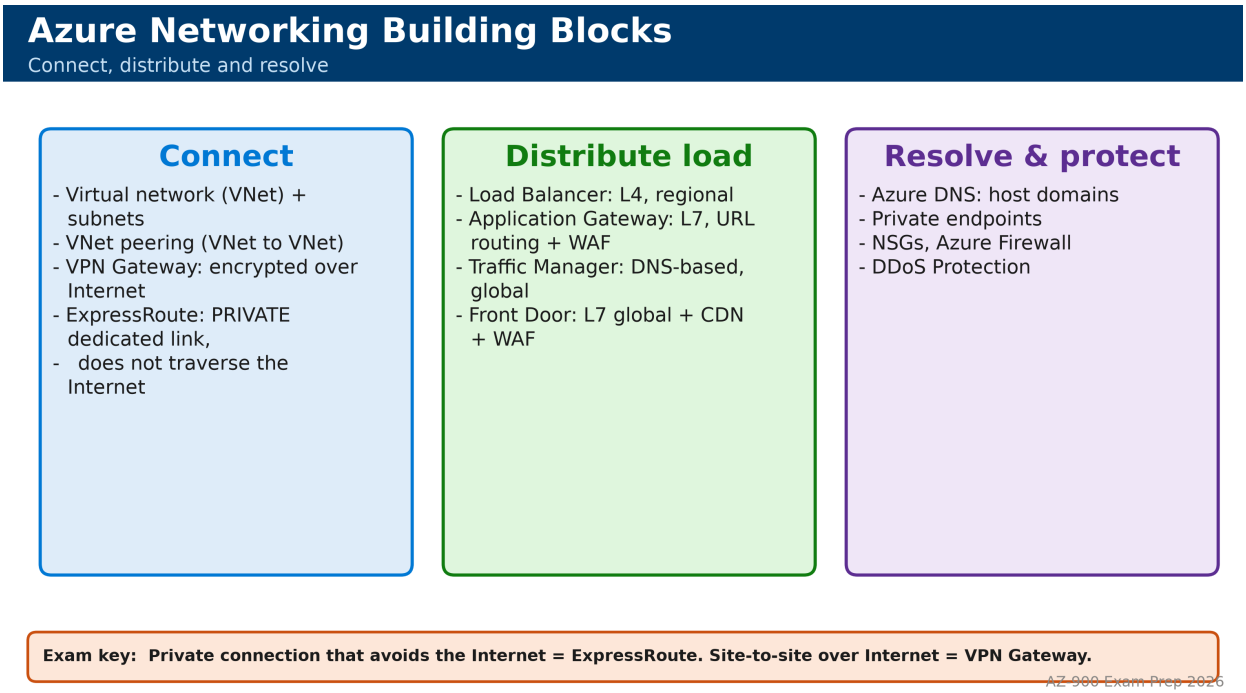


Figure Q341: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 342

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q475

For each statement, select Yes if it is true. Statement: You can use Azure DNS to purchase and register a new public domain name such as fabrikam.com.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): Watch 'purchase and register'. Azure DNS can HOST a domain's records (Q328 - Yes), but it cannot SELL you a new domain name.

Answer explained

✔ **No**
Azure DNS hosts DNS zones. It does not sell or register domain names.

✖ **Yes**
You register a domain with a registrar (a domain seller), not with Azure DNS.

EXAM STRATEGY
Key fact: Azure DNS hosts zones but cannot register domains. 'Purchase and register' means No. App Service domains or a registrar handles the purchase.

IMPACT
Teams who expect Azure DNS to sell domains get delayed at go-live. Buy the domain elsewhere, then point its name servers to Azure DNS.

What · Where · When · Why · How

What	A limit of Azure DNS: it cannot sell domains.
Where	Public DNS zones in Azure DNS.
When	When you plan a new domain for a site hosted in Azure.
Why	It keeps buying (registrar) separate from hosting (Azure DNS).
How	Buy the domain from a registrar, then set its NS records to Azure DNS.

Exam tip
Common exam point: you cannot buy a domain name with Azure DNS.

Memory trick
Azure DNS is the hotel for your domain, not the shop that sells it.

Real-world example, use case and scenario

EXAMPLE
Startup site
A startup buys its domain from a registrar and hosts the records in Azure DNS.

USE CASE
App Service domain
A developer buys a domain through App Service domains, which then uses Azure DNS for hosting.

SCENARIO
Delegation
An admin changes the registrar's name servers to the four Azure DNS name servers.

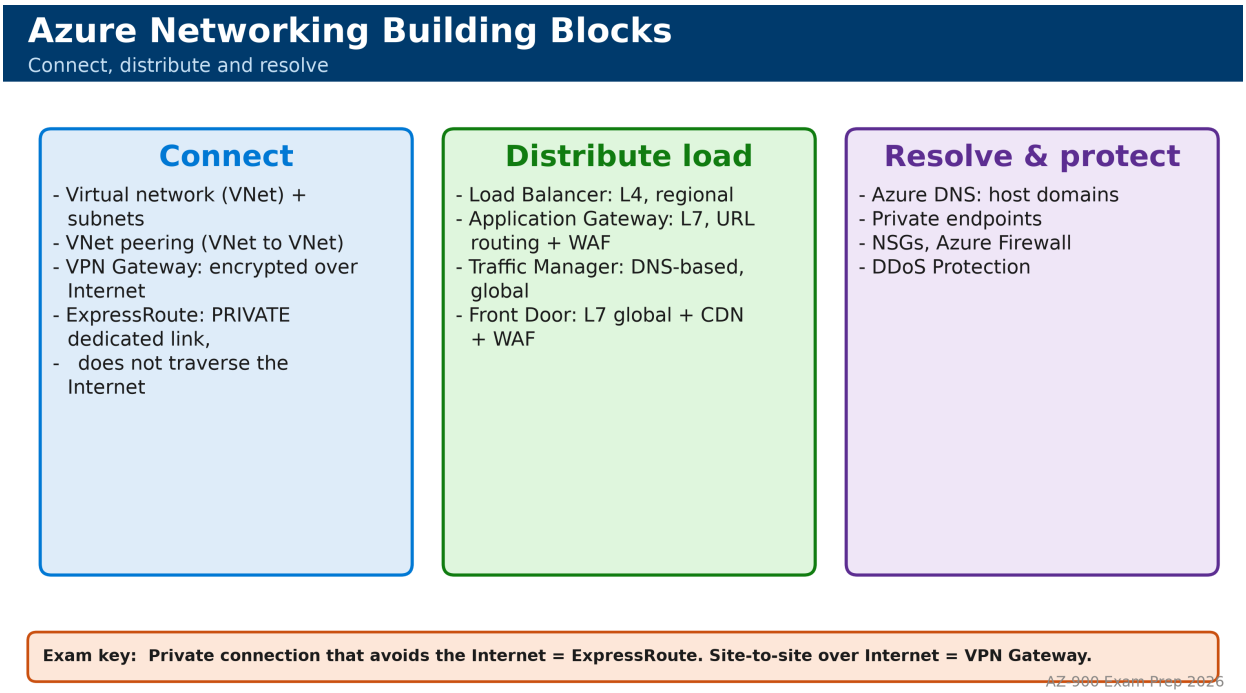


Figure Q342: Azure networking building blocks

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• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 343

AZURE ARCHITECTURE & SERVICES

Networking
v5 Q476

For each statement, select Yes if it is true. Statement: A virtual network can span multiple Azure regions.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): Do not mix up 'region' and 'zone'. A VNet can span the availability zones inside ONE region, but it cannot span two regions. To join regions, use global peering (Q339).

Answer explained

✔ **No**
A VNet lives in one region. Connect regions with global peering.

✖ **Yes**
A VNet cannot span regions. It can span zones inside one region.

EXAM STRATEGY
Key fact: a VNet belongs to one region (it can span zones). 'Multiple regions' means No. Across regions you need peering.

IMPACT
Multi-region designs need a separate VNet per region with IP ranges that do not overlap. Plan IPs before you deploy, or you must change addresses later.

What · Where · When · Why · How

What	The size limit (scope) of an Azure virtual network.
Where	One Azure region per VNet.
When	When you plan networks in many regions.
Why	The region limit decides how you link regions.
How	Create one VNet per region and link them with global peering or VPN.

Exam tip
VNet = one region, many availability zones. A subnet can also span zones.

Memory trick
One VNet, one region: it is a local neighbourhood, not a country.

Real-world example, use case and scenario

EXAMPLE
Two regions
A shop creates VNet-East and VNet-West and links them with global peering.

USE CASE
Zones
A delivery app spreads VMs over three zones inside one VNet.

SCENARIO
Design review
An architect rejects a plan for one VNet across Europe and Asia.

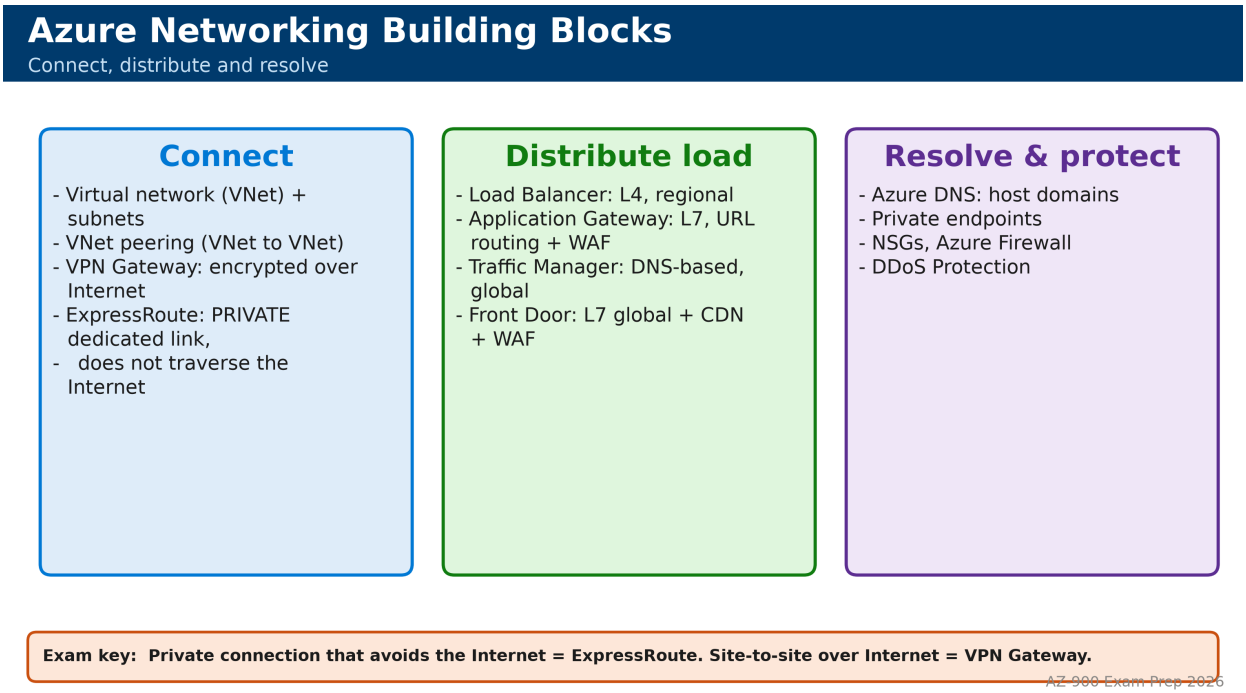


Figure Q343: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

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• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 344

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q477

For each statement, select Yes if it is true. Statement: The address range of a subnet must fall within the address space of the virtual network that contains it.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Answer explained

✔ **Yes**
A subnet's IP range (CIDR) must be inside the VNet's address space and must not overlap other subnets.

✖ **No**
Azure rejects a subnet that is outside the VNet's address space.

EXAM STRATEGY
Key fact: subnets are cut from the VNet's address space. 'Must fall within' is correct, so the answer is Yes.

IMPACT
Good IP planning stops subnets from running out of addresses or overlapping. Overlaps later block peering and hybrid links.

What · Where · When · Why · How

What	A rule for subnet IP addresses.
Where	Inside every Azure virtual network.
When	Every time you create or resize a subnet.
Why	It keeps IP addresses consistent and routable.
How	Choose a subnet range (for example 10.0.1.0/24) inside the VNet range (for example 10.0.0.0/16).

Exam tip
Azure keeps back 5 IP addresses in every subnet.

Memory trick
Subnets are rooms inside the VNet house. A room cannot be outside the house.

Real-world example, use case and scenario

EXAMPLE
Web subnet
VNet 10.0.0.0/16 holds the web subnet 10.0.1.0/24.

USE CASE
Tiers
A hospital puts its web, app and data tiers in three subnets.

SCENARIO
Error
An admin tries to add 192.168.1.0/24 to a 10.0.0.0/16 VNet. It fails until the address space is extended.

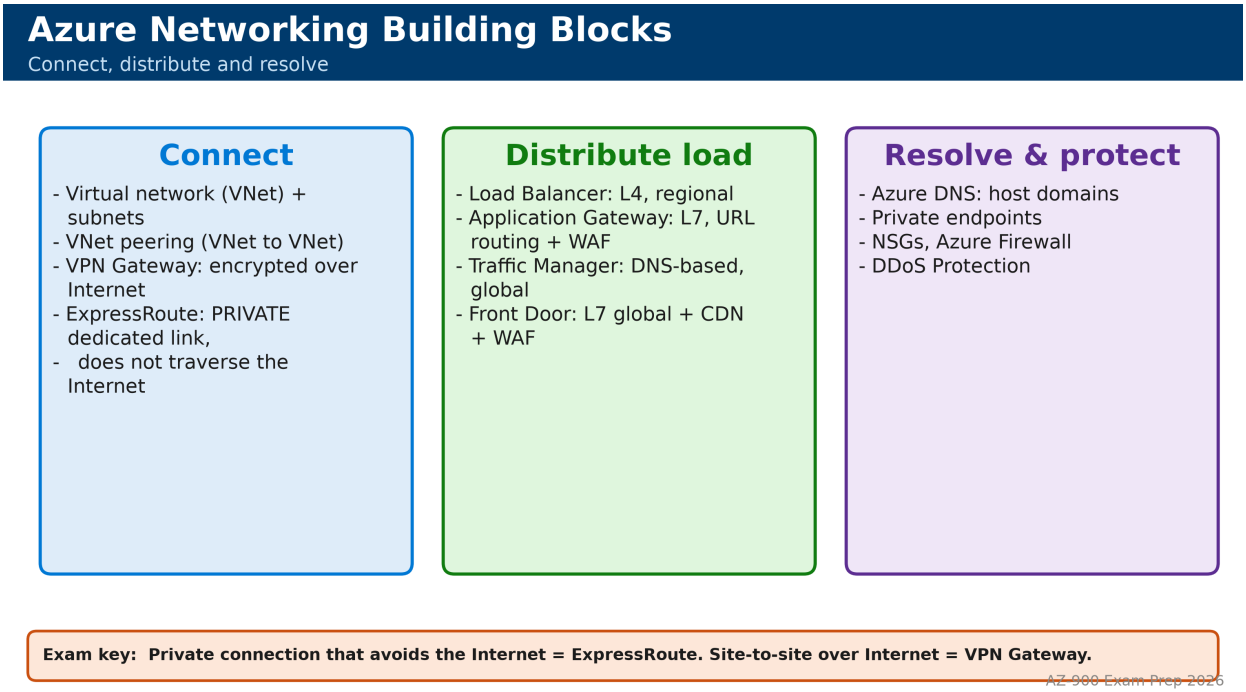


Figure Q344: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

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• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

Question 345

AZURE ARCHITECTURE & SERVICES

Storage
v5 Q478

For each statement, select Yes if it is true. Statement: Zone-redundant storage (ZRS) synchronously replicates your data across three availability zones in the primary region.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): ZRS stays inside ONE region. It protects against a zone failure, not a whole-region failure. Q347 asks for a copy in a SECOND region - ZRS is wrong there.

Answer explained

✔ **Yes**
ZRS = three copies across three availability zones in one region.

✖ **No**
This is exactly how ZRS works. LRS is the option that stays in one datacenter.

EXAM STRATEGY
Key fact: ZRS writes copies at the same time (synchronously) to three zones in one region. The statement matches, so Yes. A second-region copy would be GZRS.

IMPACT
ZRS keeps data available if one datacenter is lost, without the cost of a second region. It is common for production data that needs zone protection.

What · Where · When · Why · How

What	A storage redundancy option across availability zones.
Where	In the main region only.
When	When data must survive a zone (datacenter) failure.
Why	It protects against zone outages with no data loss.
How	Choose ZRS when you create the storage account, or convert later.

Exam tip
ZRS protects against a zone failure but not a full region outage. For that, use GZRS.

Memory trick
Z for Zones: three zones, three copies.

Real-world example, use case and scenario

EXAMPLE
Order API
An online shop uses ZRS so order files survive a zone outage.

USE CASE
Data residency
A government agency uses ZRS to keep data in one country and still survive zone failures.

SCENARIO
Zone down
A zone loses power. Reads and writes continue from the other zones.

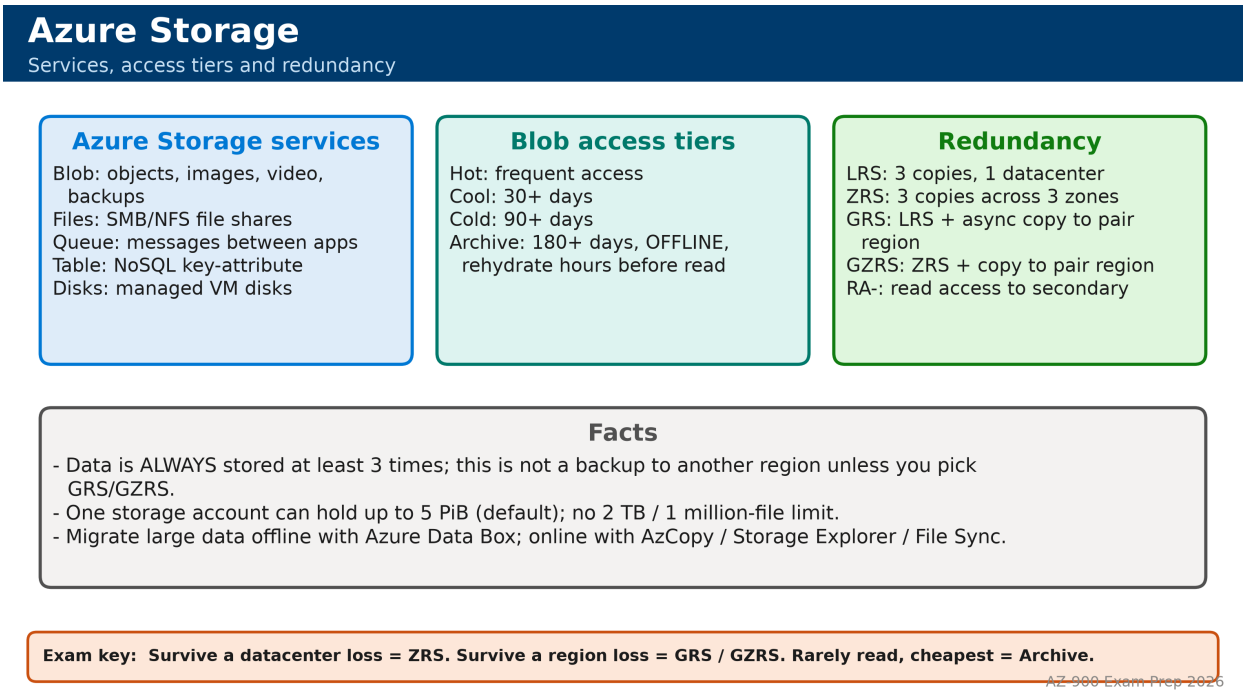


Figure Q345a: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

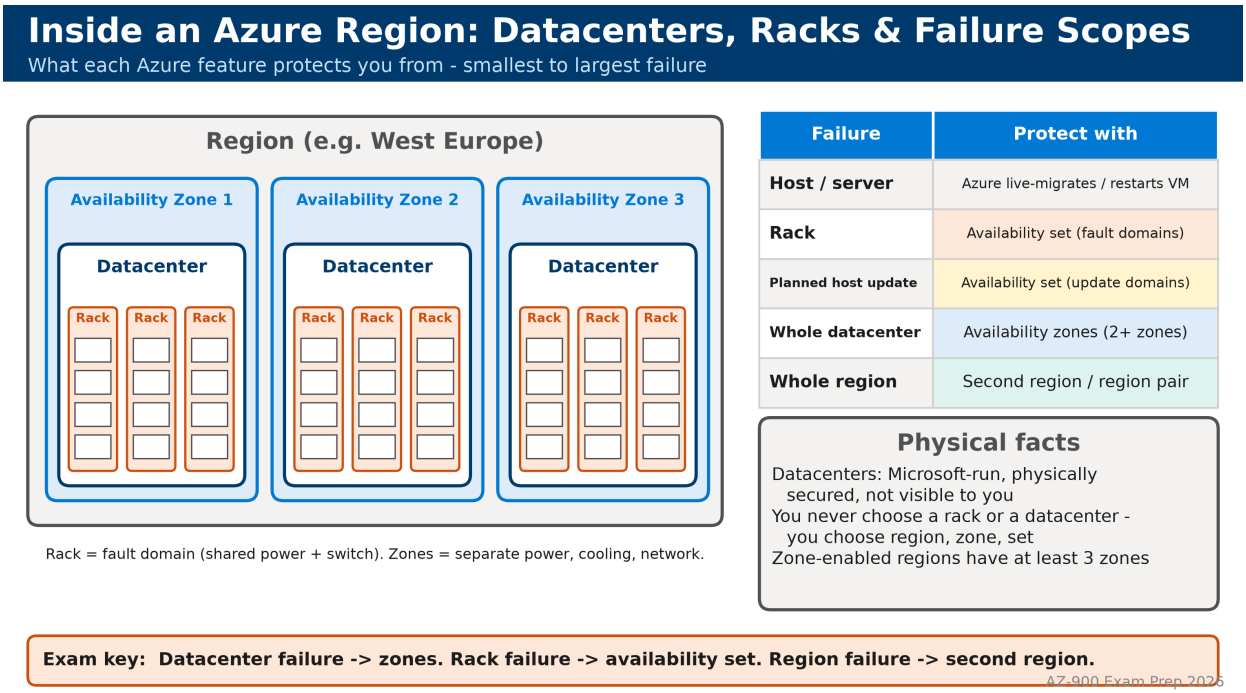


Figure Q345b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 346

AZURE ARCHITECTURE & SERVICES — Storage — v5 Q479

For each statement, select Yes if it is true. Statement: Data written to an Azure Storage account is encrypted at rest automatically, with no configuration required.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Answer explained

✔ **Yes**
Azure Storage always encrypts data at rest (stored data) with AES-256 by default.

✖ **No**
Encryption at rest is on by default and you cannot turn it off.

EXAM STRATEGY
Key fact: Storage Service Encryption is always on, with Microsoft-managed keys by default. 'Automatically, no configuration' means Yes.

IMPACT
Default encryption meets basic compliance rules. You need customer-managed keys only for stricter control of keys.

What · Where · When · Why · How

What	Automatic encryption of stored data by the service.
Where	All Azure Storage accounts, all services and tiers.
When	Every time data is written to the account.
Why	It protects data and helps meet compliance rules.
How	It uses Microsoft-managed keys by default. You can choose your own keys in Key Vault.

Exam tip
Encryption at rest is automatic. Encryption in transit (data moving) is enforced with secure transfer (HTTPS).

Memory trick
Storage arrives locked: AES-256 is the default padlock.

Real-world example, use case and scenario

EXAMPLE
Healthcare
A clinic stores scans in Blob Storage. They are encrypted at rest automatically.

USE CASE
Own keys
A bank adds its own keys in Key Vault to control key rotation.

SCENARIO
Audit
An auditor checks and sees encryption is on by default for the account.

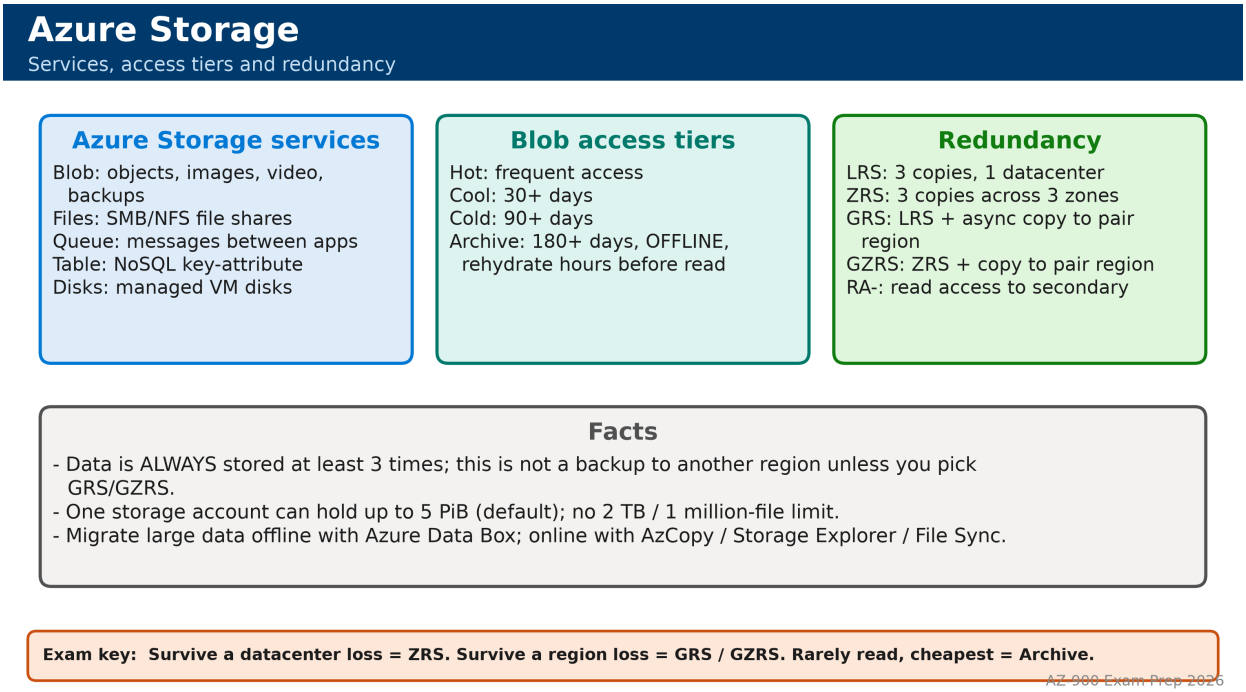


Figure Q346: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 347

AZURE ARCHITECTURE & SERVICES

Storage
v5 Q480

Your company requires that storage account data is replicated to a secondary Azure region to protect against a regional outage. Which two redundancy options meet the requirement? Each correct answer presents a complete solution.

- ✔ **A.** Geo-redundant storage (GRS)
- ✔ **B.** Geo-zone-redundant storage (GZRS)
- ✖ C. Locally redundant storage (LRS)
- ✖ D. Zone-redundant storage (ZRS)

Correct answer: Geo-redundant storage (GRS); Geo-zone-redundant storage (GZRS)

Clarification (read this first): The key words are 'secondary region'. Pick options with the letter G (geo). ZRS sounds strong but stays in one region (Q345). Q331 adds 'read the secondary any time', which needs RA-GZRS.

Answer explained

✔ **Geo-redundant storage (GRS); Geo-zone-redundant storage (GZRS)**
A copy in a second region = any option with G (GRS, GZRS, RA-GRS, RA-GZRS).

✖ **Locally redundant storage (LRS)**
LRS keeps three copies in one datacenter only.

✖ **Zone-redundant storage (ZRS)**
ZRS copies across zones in one region. It has no second region.

EXAM STRATEGY
'Secondary region' = any option with G. Pick GRS and GZRS. Remove LRS and ZRS; they stay in the main region.

IMPACT
If you choose LRS or ZRS for data that must survive a region disaster, the data can be lost. Auditors often check this setting.

What · Where · When · Why · How

What	Geo-replication (copy to another region) options for Azure Storage.
Where	The main region plus a second region, usually the paired region.
When	When a region disaster must not lose more data than your RPO (allowed data loss).
Why	It gives disaster recovery across regions.
How	Choose GRS or GZRS on the account. Start a failover if the main region fails.

Exam tip
Copying to the second region is asynchronous (done a little later), so a small amount of recent data can be lost on failover.

Memory trick
No G, no Geo: LRS and ZRS stay home.

Real-world example, use case and scenario

EXAMPLE
Insurance
An insurer uses GRS so claim files survive a region disaster.

USE CASE
Critical SaaS
A SaaS company picks GZRS for both zone and region protection.

SCENARIO
Failover
After a region outage, an admin starts a failover to the second region.

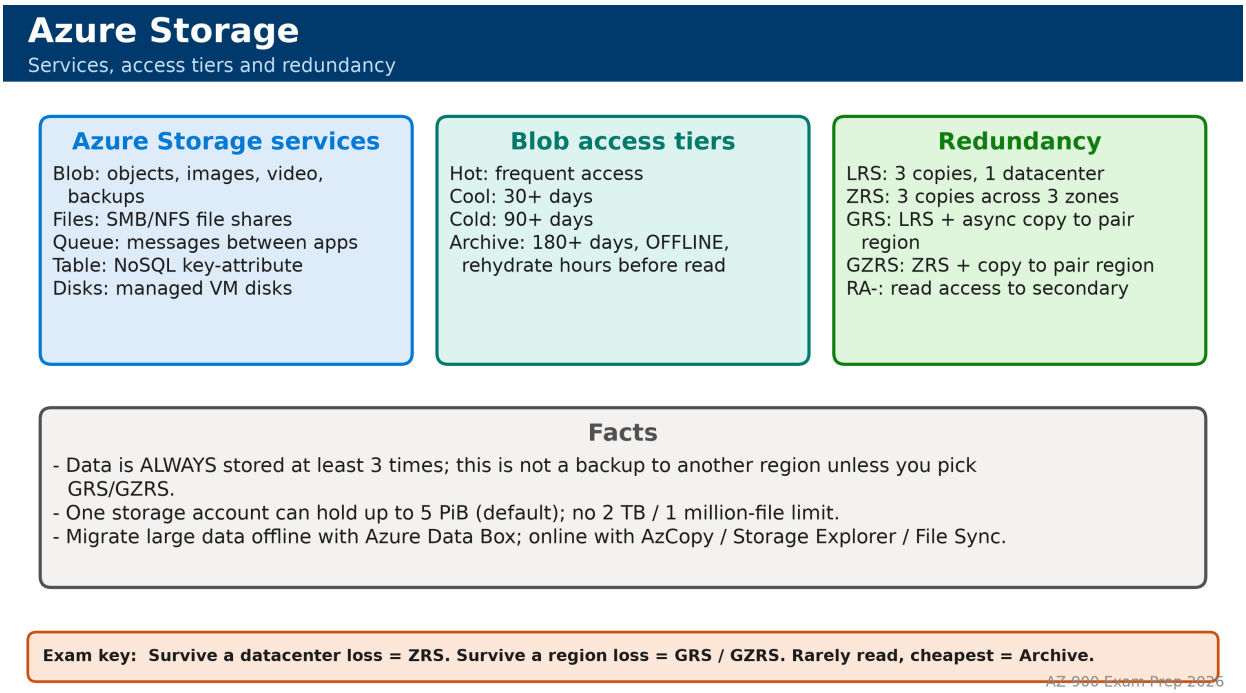


Figure Q347: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 348

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q481

A developer needs to upload thousands of local files to an Azure Blob Storage container. Which two tools can be used to copy the files directly to Blob Storage? Each correct answer presents a complete solution.

✔ **A.** AzCopy

✔ **B.** Azure Storage Explorer

✖ C. Azure File Sync

✖ D. Azure Advisor

Correct answer: AzCopy; Azure Storage Explorer

Clarification (read this first): This question asks only which tools can COPY files to Blob Storage, so both the command-line tool and the GUI tool are correct. Q335 asks for a GRAPHICAL tool only, so there AzCopy is wrong.

Answer explained

✔ **AzCopy; Azure Storage Explorer**
To copy files to blobs: AzCopy (command line) or Storage Explorer (GUI).

✖ **Azure File Sync**
File Sync syncs Windows Server with Azure file shares, not blob containers.

✖ **Azure Advisor**
Advisor gives best-practice advice. It does not move data.

EXAM STRATEGY
'Upload to Blob' = AzCopy (command line) and Storage Explorer (GUI). Remove File Sync (Azure Files only) and Advisor (advice only).

IMPACT
AzCopy handles big transfers fast, in parallel. Storage Explorer suits small, one-off uploads. The right tool saves hours on large moves.

What · Where · When · Why · How

What	Tools to move data into Azure Storage.
Where	On a local computer, a VM, or Cloud Shell (AzCopy).
When	When you move files between local storage and Azure Storage.
Why	Fast, scriptable or point-and-click transfers.
How	Run 'azcopy copy <local> <container-URL> --recursive', or drag files into Storage Explorer.

Exam tip
AzCopy can also copy between storage accounts and sync folders. Each command works in one direction.

Memory trick
AzCopy = the Azure COPY tool.

Real-world example, use case and scenario

EXAMPLE
Website files
A web team uploads images to a blob container with AzCopy in a pipeline.

USE CASE
Analyst upload
A data analyst drags CSV files into a container in Storage Explorer.

SCENARIO
Scripted backup
An admin runs AzCopy every night to copy log files to Blob Storage.

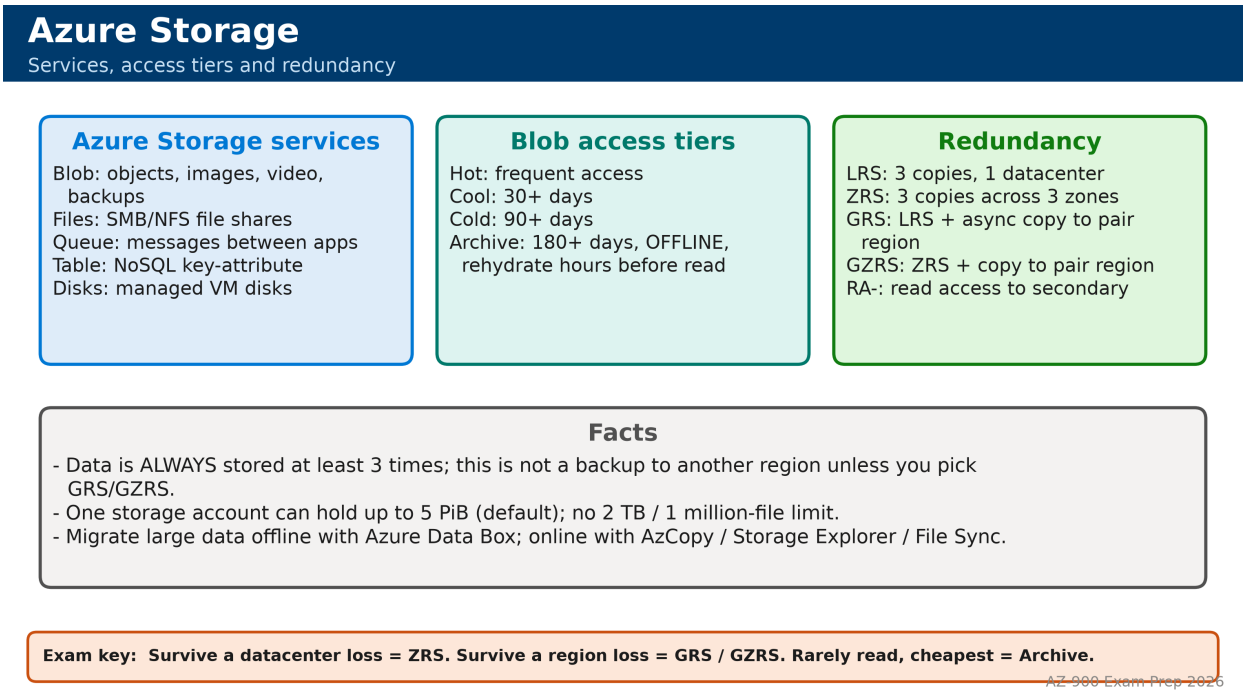


Figure Q348: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 349

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q482

Your company is evaluating Azure ExpressRoute to connect its datacenter to Azure. Which two statements correctly describe ExpressRoute? Each correct answer presents a complete solution.

✔ **A.** Traffic does not traverse the public internet

✔ **B.** Connections are established through a connectivity provider

✖ C. Traffic is encrypted with IPsec by default

✖ D. It connects a single client computer to an Azure virtual network

Correct answer: Traffic does not traverse the public internet; Connections are established through a connectivity provider

Clarification (read this first): Trap: 'private' does NOT mean 'encrypted'. ExpressRoute does not use the internet, but its traffic is not encrypted by default. Also, linking one single computer is point-to-site VPN (Q104), not ExpressRoute.

Answer explained

✔ **Traffic does not traverse the public internet; Connections are established through a connectivity provider**
ExpressRoute = a private line through a connectivity provider (a telecom partner). It does not use the internet.

✖ **Traffic is encrypted with IPsec by default**
ExpressRoute is private but not encrypted by default. Add IPsec or MACsec (encryption) if you need it.

✖ **It connects a single client computer to an Azure virtual network**
That describes a point-to-site VPN.

EXAM STRATEGY
ExpressRoute = private connection through a connectivity provider, not over the internet. Remove 'IPsec by default' (it is not encrypted by default) and 'single client' (point-to-site).

IMPACT
If you assume ExpressRoute is encrypted, you may break security policy. Sensitive traffic may need MACsec or IPsec on top of ExpressRoute.

What · Where · When · Why · How

What	A private, dedicated connection to Microsoft cloud services.
Where	Between an on-premises network (or shared datacenter) and Microsoft edge locations.
When	When you need steady speed and a private connection.
Why	More reliable, faster and lower latency than internet links.
How	Order a circuit from a connectivity provider and link it to VNets with an ExpressRoute gateway.

Exam tip
ExpressRoute is private, but encryption is not automatic. VPN Gateway is encrypted, over the internet.

Memory trick
ExpressRoute = a private express lane, not a public road.

Real-world example, use case and scenario

EXAMPLE
Stock trading
A trading firm uses ExpressRoute for steady, low delay to Azure.

USE CASE
Microsoft 365 and Azure
A company uses ExpressRoute Microsoft peering to reach Microsoft public services.

SCENARIO
Backup path
A company sets up a site-to-site VPN as a backup path if ExpressRoute fails.

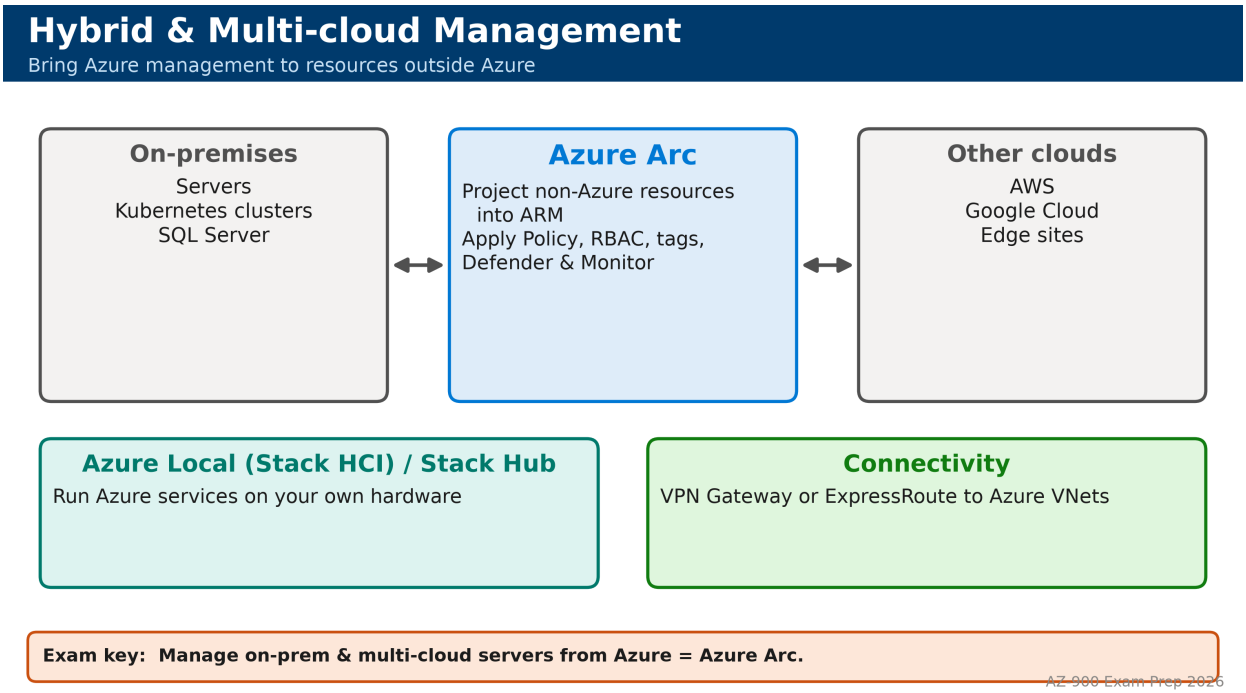


Figure Q349a: Hybrid and multi-cloud management

About this figure

- Arrows show Azure Arc linking on-premises and other-cloud resources into Azure Resource Manager; bottom boxes cover local Azure hardware and network links.
- Arc-enabled servers' core control plane (inventory, tags, RBAC, Policy) is free; add-ons such as Defender, Update Manager and Monitor are billed separately.
- Arc can also manage Kubernetes clusters and SQL Server; Arc-enabled SQL Server can be licensed pay-as-you-go through your Azure bill.
- Azure Local (formerly Azure Stack HCI) runs VMs and AKS on validated on-premises hardware, billed per physical core per month in Azure.
- Site-to-site VPN uses the Internet with encryption; ExpressRoute is a private connection through a provider with higher reliability and an SLA up to 99.95%.

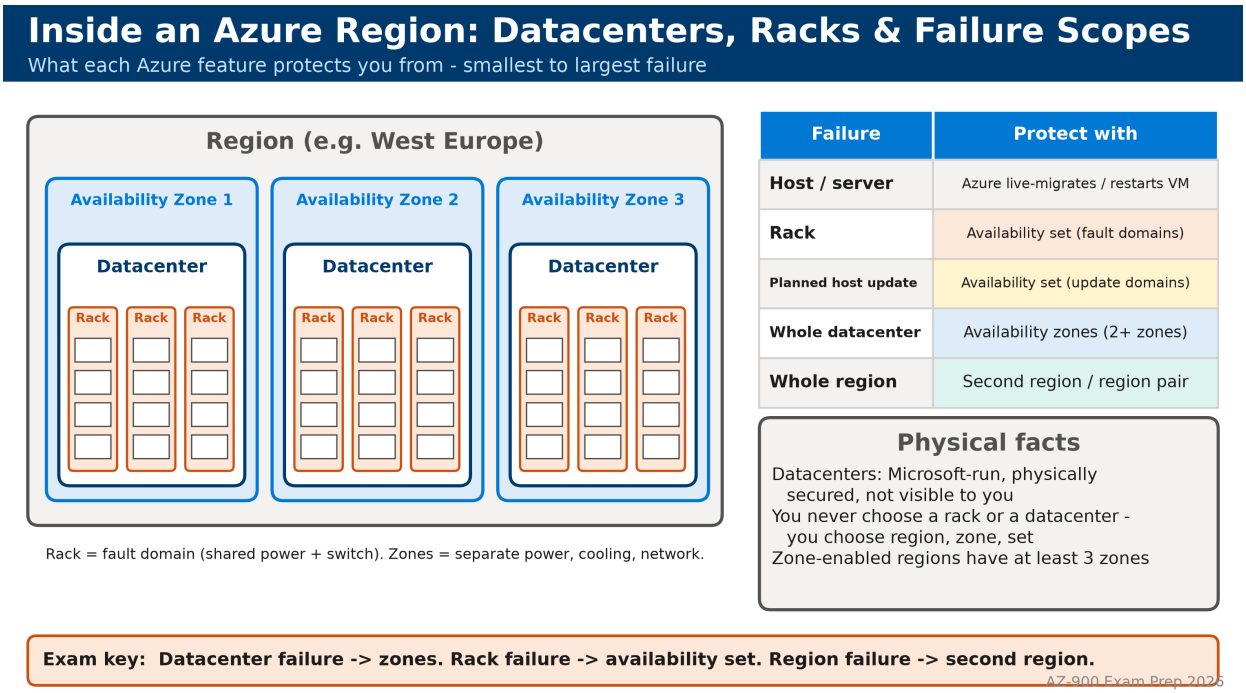


Figure Q349b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 350

AZURE ARCHITECTURE & SERVICES — **Networking** v5 Q483

Your company runs a web application in two Azure regions. You need to direct users globally to the closest healthy region and fail over if a region becomes unavailable. Which two services can provide this global traffic distribution? Each correct answer presents a complete solution.

- ✔ **A.** Azure Front Door
- ✔ **B.** Azure Traffic Manager
- ✖ C. Azure Application Gateway
- ✖ D. Azure VPN Gateway

Correct answer: Azure Front Door; Azure Traffic Manager

Clarification (read this first): The key word is 'globally' (across regions). Application Gateway is similar to Front Door but works in ONE region only. Q329 is a one-region, layer-4 case - Load Balancer.

Answer explained

✔ **Azure Front Door; Azure Traffic Manager**
Routing users across many regions = Front Door (HTTP, at Microsoft's edge) or Traffic Manager (uses DNS).

✖ **Azure Application Gateway**
Application Gateway is a layer-7 load balancer in one region. It does not route across regions.

✖ **Azure VPN Gateway**
VPN Gateway gives encrypted network links. It does not share user traffic.

EXAM STRATEGY
'Globally', 'closest healthy region' and 'fail over between regions' = Front Door and Traffic Manager. Application Gateway is regional. VPN Gateway is for hybrid links.

IMPACT
Global routing keeps users served during a region outage and lowers delay. Regional-only load balancers cannot fail over between regions by themselves.

What · Where · When · Why · How

What	Global traffic routing services across Azure regions.
Where	In front of app copies in many regions.
When	When an app runs in many regions for speed and resilience.
Why	It lowers delay and allows region failover.
How	Add the endpoints in each region, with health probes and a routing method.

Exam tip
Regional: Load Balancer (L4), Application Gateway (L7). Global: Front Door (L7), Traffic Manager (DNS).

Memory trick
Front Door and Traffic Manager both 'travel' around the globe.

Real-world example, use case and scenario

EXAMPLE
Video site
A video platform uses Front Door to send users to the nearest region.

USE CASE
Priority failover
A bank uses Traffic Manager priority routing to fail over to a recovery region.

SCENARIO
Region outage
West Europe goes down. Health probes fail and users are sent to North Europe.

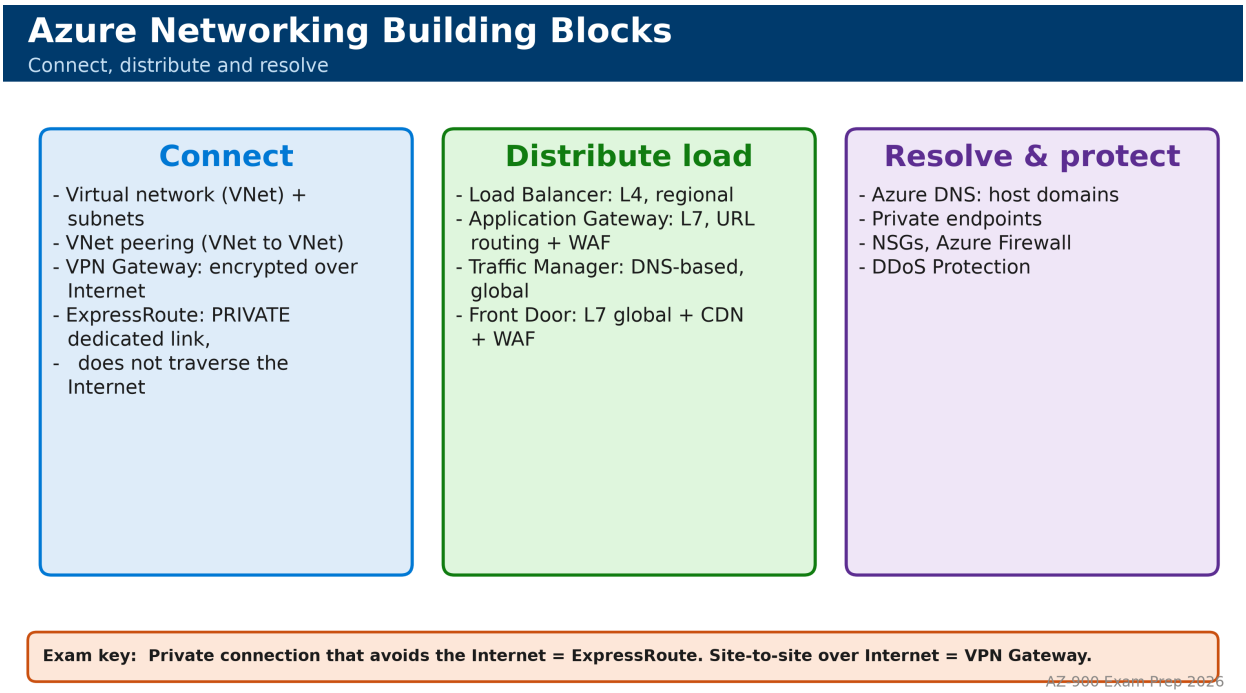


Figure Q350a: Azure networking building blocks

About this figure

• Three columns group services by job: connecting networks, distributing traffic, and resolving names or protecting traffic. Match the scenario keyword to a column.

• VNets and subnets are free; VNet peering costs approx. USD 0.01/GB each way in-region. A VNet can peer with up to about 500 others.

• VPN Gateway VpnGw1 costs approx. USD 140/month and supports up to about 650 Mbps; ExpressRoute circuits start around USD 55/month for 50 Mbps metered.

• Load Balancer and Traffic Manager are not proxies; Application Gateway and Front Door terminate HTTP(S) and can add WAF.

• Azure DNS hosts zones you own but cannot register domain names; private endpoints give a PaaS service a private IP in your VNet.

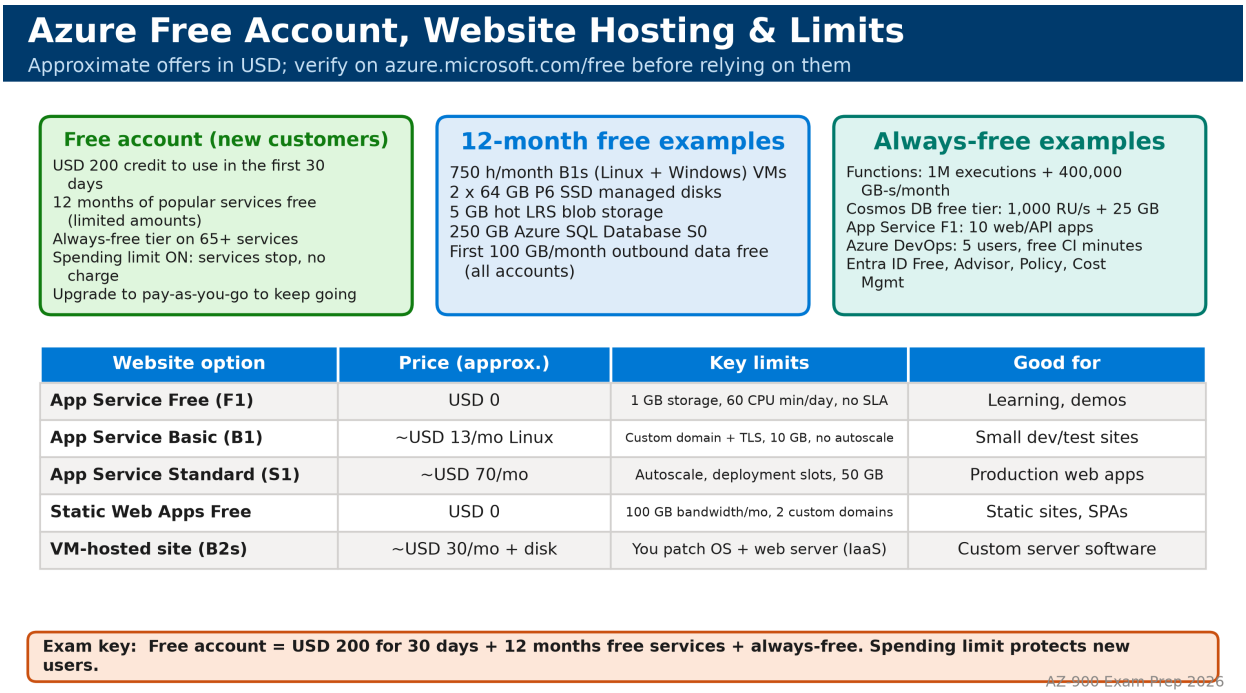


Figure Q350b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 351

AZURE ARCHITECTURE & SERVICES — Storage v5 Q484

Select the answer that correctly completes the sentence, or choose 'No change is needed'. To automatically move block blobs from the Hot tier to the Cool tier when they have not been modified for 30 days, configure [an Azure resource lock].

✖ A. No change is needed

✔ **B.** a lifecycle management policy

✖ C. Azure File Sync

✖ D. a Microsoft Entra Conditional Access policy

Correct answer: a lifecycle management policy

Clarification (read this first): In this question type the [bracketed] text may be wrong. Here it is wrong: a resource lock only stops deletes or changes - it never moves data. 'Automatically move after 30 days' needs a lifecycle management policy.

Answer explained

✔ **a lifecycle management policy**
Moving or deleting blobs automatically by age = a blob lifecycle management policy.

✖ **No change is needed**
Resource locks stop deletes or changes. They never move data between tiers.

✖ **Azure File Sync**
File Sync syncs Azure file shares with Windows Server. It has nothing to do with blob tiers.

✖ **A Microsoft Entra Conditional Access policy**
Conditional Access controls sign-in conditions, not where data is stored.

EXAM STRATEGY
'Automatically move tiers after N days' = lifecycle management policy. Resource locks stop deletion; they never move data, so the sentence must change.

IMPACT
Lifecycle policies cut storage costs by moving old data to cheaper tiers. No manual scripts are needed, and no old data is forgotten in Hot.

What · Where · When · Why · How

What	A rule-based policy for blob data in a storage account.
Where	Set on GPv2 or Blob Storage accounts.
When	When blob use changes in a predictable way as data gets older.
Why	It lowers cost automatically, with no manual moves.
How	Add a rule: if days since last change > 30, set the tier to Cool.

Exam tip
Lifecycle policies can also delete old blobs and snapshots.

Memory trick
Lifecycle = blobs grow old and move to colder rooms by themselves.

Real-world example, use case and scenario

EXAMPLE
Logs
A SaaS company moves logs to Cool after 30 days and deletes them after 365.

USE CASE
Media
A TV company moves raw video to Archive after 180 days.

SCENARIO
Cost cut
A finance team sees storage costs fall after lifecycle rules are turned on.

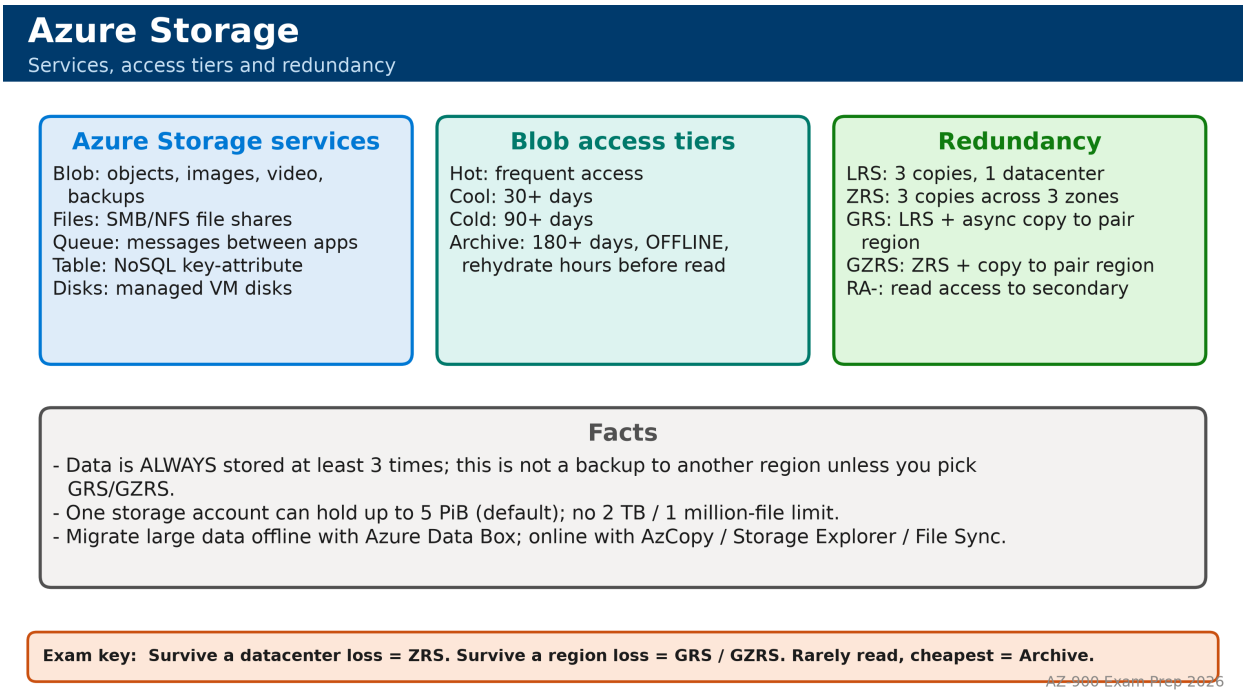


Figure Q351: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 352

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q485

Select the answer that correctly completes the sentence, or choose 'No change is needed'. The Archive access tier can be set at the [storage account] level.

✖ A. No change is needed

✔ **B.** individual blob

✖ C. resource group

✖ D. subscription

Correct answer: individual blob

Clarification (read this first): Watch 'account level' vs 'blob level'. The account's default tier can be only Hot, Cool or Cold. Archive can be set only on each blob.

Answer explained

✔ **individual blob**
Archive is set on each blob. It can never be the account's default tier.

✖ **No change is needed**
The account default tier cannot be Archive.

✖ **Resource group**
Access tiers do not apply to resource groups.

✖ **Subscription**
Access tiers are not set at subscription level.

EXAM STRATEGY
Key fact: Archive is set per blob only. The account default tier offers only Hot, Cool or Cold. Choose 'individual blob'.

IMPACT
Because Archive is per blob, your policies or scripts must target the blobs. If you assume an account setting, data stays in costly tiers.

What · Where · When · Why · How

What	Where you can apply the Archive tier.
Where	On single block blobs in GPv2 or Blob Storage accounts.
When	When certain blobs are rarely read and kept 180 days or more.
Why	It stops a whole account from becoming offline storage by default.
How	Set a blob's tier to Archive in the portal, CLI, AzCopy or a lifecycle policy.

Exam tip
Account default tier: Hot, Cool or Cold. Blob-level tier: Hot, Cool, Cold or Archive.

Memory trick
Archive is a blob-by-blob choice.

Real-world example, use case and scenario

EXAMPLE
Tax records
An accountant archives single year-end statement blobs.

USE CASE
Closed cases
A law firm moves certain case-file blobs to Archive after the case closes.

SCENARIO
Account setup
An admin sets the account default tier and sees no Archive option.

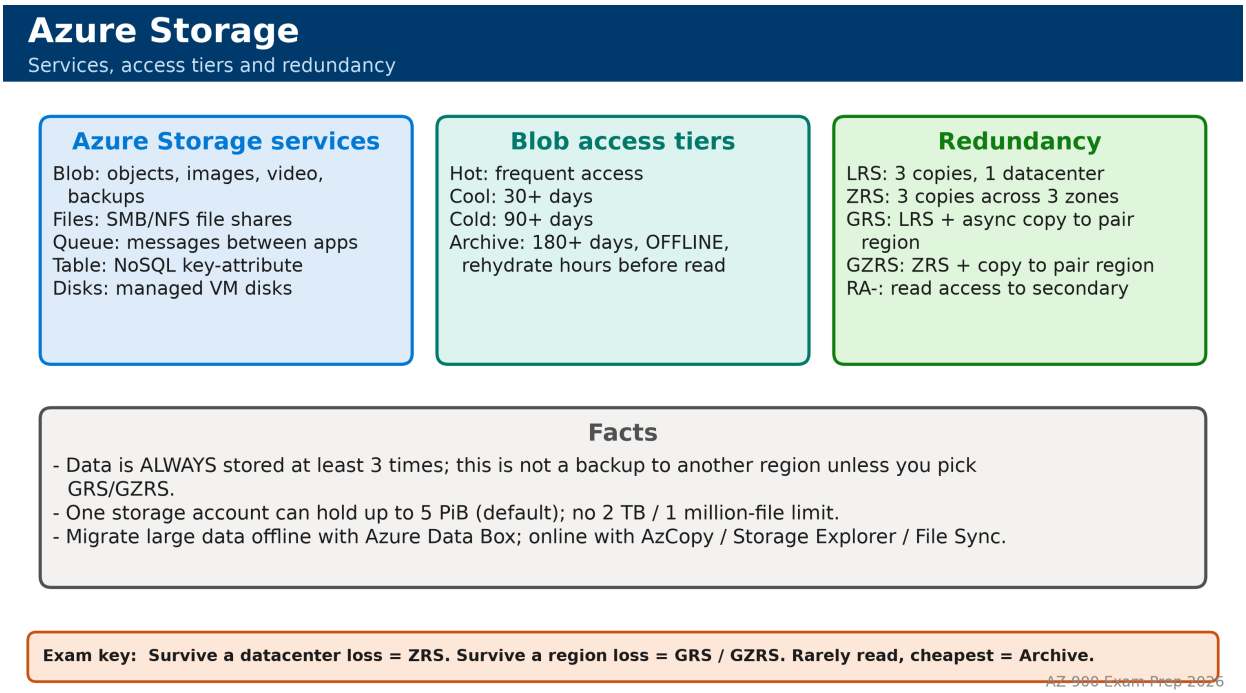


Figure Q352: Azure Storage

About this figure

• Three boxes: the services inside a storage account, the blob access tiers ordered by access frequency, and redundancy options from local to geo-zone.

• Approximate Hot LRS cost is USD 0.02/GB-month, Cool 0.01, Archive 0.002 or less; cooler tiers charge more per read and have early deletion penalties.

• Early deletion minimums: Cool 30 days, Cold 90 days, Archive 180 days. Archive rehydration takes up to 15 hours standard, faster with high priority.

• Durability: LRS at least 11 nines, ZRS 12 nines, GRS and GZRS 16 nines per year. Secondary data is readable only with RA-GRS or RA-GZRS.

• Data Box holds about 80 TB usable per device; Data Box Disk up to 35 TB per order. Use them when network transfer would take too long.

Question 353

AZURE ARCHITECTURE & SERVICES — **Storage** v5 Q486

Select the answer that correctly completes the sentence, or choose 'No change is needed'. To discover on-premises servers, assess their readiness, and estimate the size and cost of running them in Azure before migration, use [Azure Migrate].

✔ **A.** No change is needed

✖ B. Azure Data Box

✖ C. AzCopy

✖ D. Azure File Sync

Correct answer: No change is needed

Clarification (read this first): Here the [bracketed] text is already correct, so the answer is 'No change is needed'. Azure Migrate is the tool that finds servers and estimates their size and cost before a move.

Answer explained

✔ **No change is needed**
Find servers + check if they are ready + choose the right size + move them = Azure Migrate.

✖ **Azure Data Box**
Data Box moves large data offline. It does not assess servers.

✖ **AzCopy**
AzCopy copies files and blobs. It does not find or assess servers.

✖ **Azure File Sync**
File Sync syncs file shares. It does not check if servers are ready to move.

EXAM STRATEGY
'Discover', 'assess readiness' and 'estimate size and cost' before migration = Azure Migrate. The sentence is already correct, so choose No change.

IMPACT
Assessments stop you from picking VMs that are too big and from getting surprise bills after the move. Skipping them often causes cloud overspending.

What · Where · When · Why · How

What	One central place to find, assess and move workloads to Azure.
Where	A project in the Azure portal plus an appliance (helper VM) on-premises.
When	Before and during a move of servers, databases and apps to Azure.
Why	It lowers risk because you size and price workloads first.
How	Create an Azure Migrate project, deploy the appliance, run an assessment, then migrate.

Exam tip
Azure Migrate brings tools like Data Box and Database Migration Service into one hub.

Memory trick
Migrate = Map, Measure, then Move.

Real-world example, use case and scenario

EXAMPLE
VMware servers
A manufacturer assesses 400 VMware VMs before moving them to Azure.

USE CASE
Cost estimate
A finance director reviews Azure Migrate cost estimates before closing a datacenter.

SCENARIO
Dependencies
An architect uses dependency analysis to group servers that must move together.

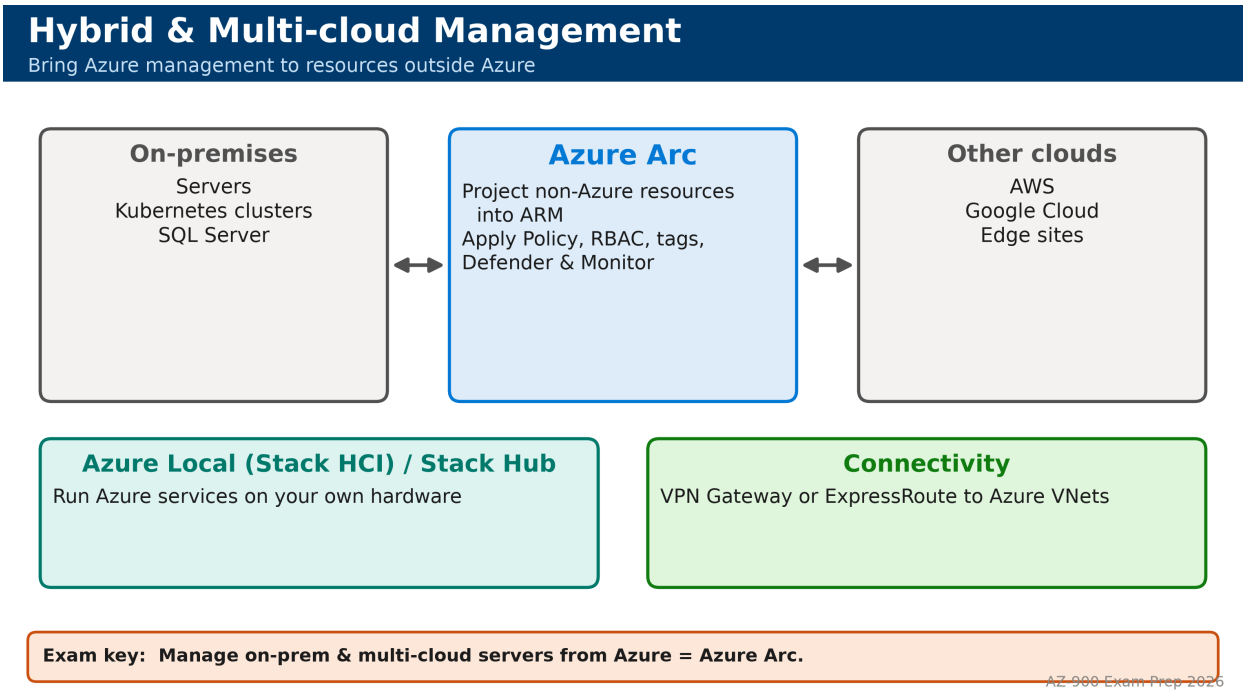


Figure Q353: Hybrid and multi-cloud management

About this figure

- Arrows show Azure Arc linking on-premises and other-cloud resources into Azure Resource Manager; bottom boxes cover local Azure hardware and network links.
- Arc-enabled servers' core control plane (inventory, tags, RBAC, Policy) is free; add-ons such as Defender, Update Manager and Monitor are billed separately.
- Arc can also manage Kubernetes clusters and SQL Server; Arc-enabled SQL Server can be licensed pay-as-you-go through your Azure bill.
- Azure Local (formerly Azure Stack HCI) runs VMs and AKS on validated on-premises hardware, billed per physical core per month in Azure.
- Site-to-site VPN uses the Internet with encryption; ExpressRoute is a private connection through a provider with higher reliability and an SLA up to 99.95%.

Question 354

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q487, Q500

Your company plans to migrate a legacy line-of-business application to Azure VMs. The application requires Kerberos authentication, LDAP queries, and domain-joined servers. You must avoid deploying, patching, and managing domain controllers. What should you use?

✔ **A.** Microsoft Entra Domain Services

✖ B. Microsoft Entra ID Protection

✖ C. Microsoft Entra Connect

✖ D. Azure Key Vault

Correct answer: Microsoft Entra Domain Services

Clarification (read this first): Watch the names. Entra Domain Services (formerly Azure AD Domain Services) gives a managed domain with LDAP and Kerberos. Entra Connect (formerly Azure AD Connect) only SYNCS users; it gives no domain.

Answer explained

✔ **Microsoft Entra Domain Services**
Entra Domain Services = managed domain join, Group Policy, LDAP and Kerberos/NTLM (older sign-in protocols), with no domain controllers for you to manage.

✖ **Microsoft Entra ID Protection**
ID Protection finds and fixes risky sign-ins. It gives no domain join, LDAP or Kerberos.

✖ **Microsoft Entra Connect**
Entra Connect syncs on-premises AD users to Entra ID. It does not host a managed domain.

✖ **Azure Key Vault**
Key Vault stores secrets, keys and certificates. It has no directory or sign-in protocols.

EXAM STRATEGY
'Kerberos', 'LDAP', 'domain-join' plus 'no domain controllers to manage' = Entra Domain Services. Entra Connect only syncs identities and gives no LDAP.

IMPACT
Old apps can move to Azure without you running and patching domain controller VMs. This lowers security risk and admin work.

What · Where · When · Why · How

What	A Microsoft-managed domain in Azure that works like Active Directory.
Where	Deployed into an Azure virtual network in your subscription.
When	When you move old apps that need older AD protocols.
Why	So you do not deploy and patch domain controllers.
How	Turn on a managed domain and join VMs to it. Microsoft runs the domain controllers.

Exam tip
Old protocols (Kerberos, NTLM, LDAP, Group Policy) + no domain controllers to manage = Entra Domain Services.

Memory trick
Domain Services = the 'domain' as a service.

Real-world example, use case and scenario

EXAMPLE
Factory ERP
A manufacturer moves an ERP server that uses LDAP into Azure and joins it to the managed domain.

USE CASE
Old file server
An insurer moves a Windows file server that uses Kerberos and Group Policy, without building domain controller VMs.

SCENARIO
Small IT team
A two-person school IT team needs domain join for lab VMs but cannot run domain controllers.

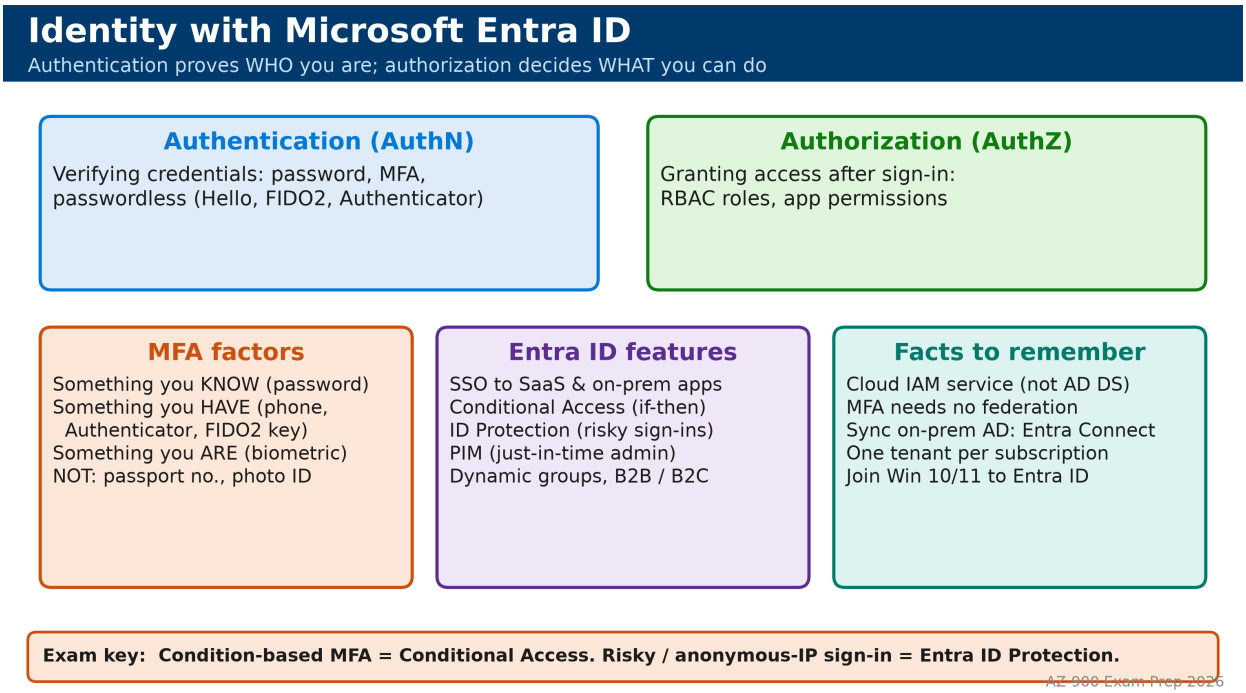


Figure Q354: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 355

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q488

Employees of your company want to sign in to their Windows 11 laptops by using facial recognition or a PIN that is bound to the device instead of a password. Which authentication method should you implement?

✔ **A.** Windows Hello for Business

✖ B. Self-service password reset

✖ C. Microsoft Entra Connect password hash synchronization

✖ D. Conditional Access

Correct answer: Windows Hello for Business

Clarification (read this first): The key words are 'Windows 11 laptops' and 'face or PIN bound to the device'. Q356 uses a USB/NFC hardware key - that is FIDO2, a different answer.

Answer explained

✔ **Windows Hello for Business**
Windows Hello for Business = Windows sign-in with no password, using your face, fingerprint or a PIN tied to the device.

✖ **Self-service password reset**
SSPR lets users reset a forgotten password. It still uses passwords.

✖ **Microsoft Entra Connect password hash synchronization**
This syncs password hashes to the cloud. Users still sign in with passwords.

✖ **Conditional Access**
Conditional Access sets rules for sign-ins. It is not a way to sign in by itself.

EXAM STRATEGY
'Facial recognition' or 'PIN bound to the device' instead of a password = Windows Hello for Business. SSPR resets passwords. Conditional Access is a policy.

IMPACT
Signing in without a password removes the credential most often stolen by phishing (fake login pages). This greatly lowers account takeover risk.

What · Where · When · Why · How

What	A sign-in method with no password, using biometrics (face, fingerprint) or a PIN.
Where	On Windows devices registered or joined to Microsoft Entra ID.
When	When users sign in to their own Windows PC.
Why	To remove passwords that can be phished from daily sign-in.
How	A private key kept in the device's TPM (security chip) is unlocked by the user's face, finger or PIN.

Exam tip
Windows PC + face/fingerprint/PIN = Windows Hello for Business.

Memory trick
Say 'Hello' to your laptop instead of typing a password.

Real-world example, use case and scenario

EXAMPLE
Bank branch PCs
Bank tellers unlock their Windows laptops with a fingerprint instead of a password.

USE CASE
Hospital doctors
Doctors sign in with face recognition on their own workstations to reach patient records faster.

SCENARIO
Law firm
A law firm stops password resets by giving all lawyers device PINs.

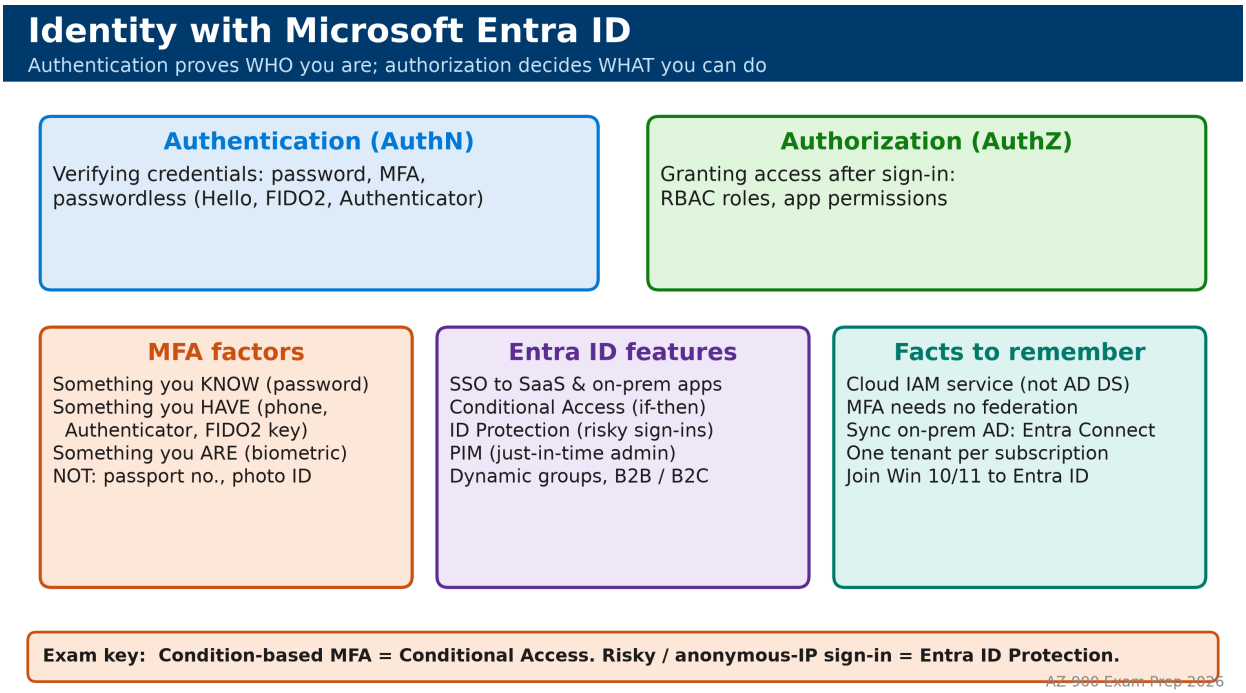


Figure Q355: Identity with Microsoft Entra ID

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- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
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- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 356

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q489

Your security team wants administrators to sign in by inserting a phishing-resistant USB or NFC hardware key, without typing a password. Which Microsoft Entra ID authentication method meets this requirement?

✔ **A.** FIDO2 security keys (passkeys)

✖ B. SMS verification codes

✖ C. Security questions

✖ D. Email one-time passcode

Correct answer: FIDO2 security keys (passkeys)

Clarification (read this first): The key words are 'USB or NFC hardware key' and 'phishing-resistant'. Q355 uses face or PIN on a Windows laptop - that is Windows Hello for Business. SMS and email codes are not phishing-resistant.

Answer explained

✔ **FIDO2 security keys (passkeys)**
FIDO2 security key = a hardware passkey (USB or NFC) for sign-in with no password that resists phishing.

✖ **SMS verification codes**
A code sent by text message. It is not hardware-based and can be phished.

✖ **Security questions**
Used only for password reset. It is based on what you know, and is not a passwordless sign-in.

✖ **Email one-time passcode**
Sends a code by email, mainly for guest users. It is not a hardware key.

EXAM STRATEGY
'Phishing-resistant' and 'USB or NFC hardware key' = FIDO2 security keys or passkeys. Remove SMS, email codes and security questions; they can be phished.

IMPACT
Hardware-key sign-in for admins stops phishing attacks on the most powerful accounts. These accounts cause the worst breaches in the cloud.

What · Where · When · Why · How

What	A hardware passkey based on the FIDO2/WebAuthn standard.
Where	Plugged into USB, or tapped by NFC, on a device.
When	When admins and other powerful users need sign-in that resists phishing.
Why	A key-based credential cannot be phished like a password.
How	The key signs a challenge with a private key that never leaves the key.

Exam tip
Hardware key, USB/NFC, phishing-resistant = FIDO2 security key (passkey).

Memory trick
FIDO = 'Fido the guard dog' on your keyring.

Real-world example, use case and scenario

EXAMPLE
Cloud admins
A SaaS company gives USB-C security keys to all Global Administrators.

USE CASE
Shared tablets
Factory workers tap an NFC key on shared tablets instead of typing passwords.

SCENARIO
Phishing attack
After a phishing attack, a shop requires passkeys, so stolen passwords become useless.

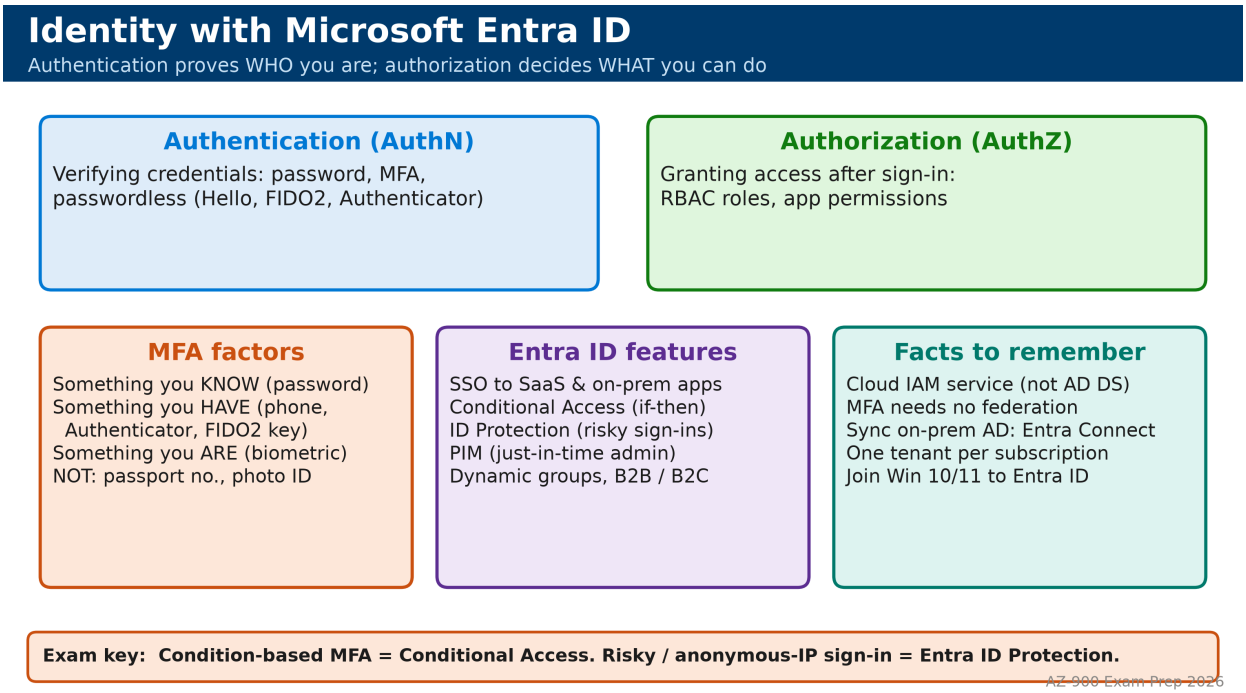


Figure Q356: Identity with Microsoft Entra ID

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- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 357

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q490, Q505

Your company is collaborating with a partner organization. Partner employees must access your SharePoint sites and an internal Azure-hosted app by using their own corporate credentials. You must not manage passwords for them. What should you use?

- ✔ **A.** Microsoft Entra B2B collaboration
- ✖ B. Microsoft Entra External ID for customers
- ✖ C. Microsoft Entra Domain Services
- ✖ D. Microsoft Entra Connect

Correct answer: Microsoft Entra B2B collaboration

Clarification (read this first): Partner EMPLOYEES = B2B (business-to-business). Public CUSTOMERS who sign up themselves = External ID for customers (Q358). Watch which kind of outside user the question names.

Answer explained

✔ **Microsoft Entra B2B collaboration**
B2B collaboration = partner users join your tenant as guests and sign in with their own work accounts.

✖ **Microsoft Entra External ID for customers**
This is for public consumer apps where customers sign up. It is not for partner staff.

✖ **Microsoft Entra Domain Services**
Domain Services gives managed domain join and LDAP/Kerberos. It is not for outside users.

✖ **Microsoft Entra Connect**
Entra Connect syncs your own on-premises AD users. It does not add partners.

EXAM STRATEGY
'Partner employees' plus 'their own corporate credentials' = B2B collaboration. External ID for customers is for consumer sign-up. Domain Services is for old sign-in protocols.

IMPACT
With B2B guest access you do not manage partner passwords. When the partner company disables an account, access ends automatically.

What · Where · When · Why · How

What	Guest access for users from other organizations.
Where	In your own Microsoft Entra workforce tenant (your staff directory).
When	When partners, suppliers or contractors need your apps.
Why	So partners keep using accounts their own company manages.
How	You invite the guest. They accept and sign in with their home account.

Exam tip
Partner/supplier + 'their own credentials' + your apps = B2B collaboration.

Memory trick
B2B = Business-to-Business = guests from another business.

Real-world example, use case and scenario

EXAMPLE
Building project
Two building firms share project files by inviting each other's engineers as B2B guests.

USE CASE
Auditors
Staff from an outside audit firm open a finance app with their own work accounts.

SCENARIO
Ad agency
A shop gives its ad agency access to a Teams channel with no new accounts.

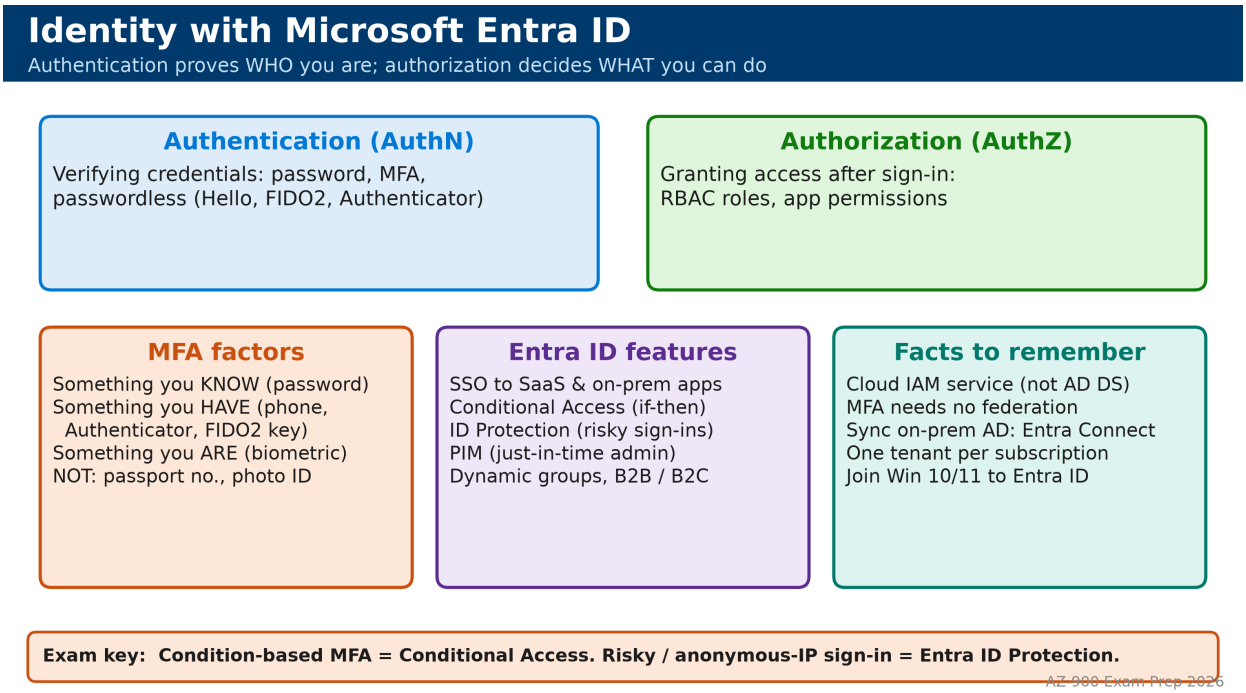


Figure Q357: Identity with Microsoft Entra ID

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- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 358

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q491

Your company is building a public shopping website. Customers must be able to create their own accounts or sign in with Google or Facebook, and the sign-in pages must use your company branding. Which Microsoft Entra capability should you use?

✔ **A.** Microsoft Entra External ID for customers

✖ B. Microsoft Entra B2B collaboration

✖ C. Microsoft Entra Privileged Identity Management

✖ D. Microsoft Entra Domain Services

Correct answer: Microsoft Entra External ID for customers

Clarification (read this first): Entra External ID for customers is the new name for Azure AD B2C. Public CUSTOMERS = External ID for customers. Partner EMPLOYEES = B2B collaboration (Q357).

Answer explained

✔ **Microsoft Entra External ID for customers**
Consumer app + self sign-up + social logins (Google, Facebook) + your branding = Entra External ID for customers (formerly Azure AD B2C).

✖ **Microsoft Entra B2B collaboration**
B2B is for business partners using your company's resources, not for public customer sign-up.

✖ **Microsoft Entra Privileged Identity Management**
PIM gives admins roles just in time (only when needed). It is not for customer sign-in.

✖ **Microsoft Entra Domain Services**
Domain Services is a managed AD domain for old apps. It is not for consumer identity.

EXAM STRATEGY
'Customers', 'create own accounts', 'Google or Facebook' and 'branding' = External ID for customers (the successor to Azure AD B2C). B2B is for partner employees.

IMPACT
Customer identity platforms handle sign-up, social login and branding at large scale. Building this yourself risks security bugs and fewer sign-ups.

What · Where · When · Why · How

What	Customer identity and access management (CIAM).
Where	In a separate external tenant, apart from your staff tenant.
When	When you build apps for the public or consumers.
Why	So customers can safely use their own social accounts.
How	Set up user flows with identity providers such as Google, Facebook or email.

Exam tip
Customers / consumers / social logins = External ID for customers (B2C). Partners = B2B.

Memory trick
B2C = Business-to-Consumer = Customers.

Real-world example, use case and scenario

EXAMPLE
Online grocery
A grocery chain lets shoppers sign in with Google to track orders.

USE CASE
Airline app
An airline's loyalty app uses branded sign-up pages for millions of members.

SCENARIO
Gaming site
A game studio lets players register with Facebook accounts for its community site.

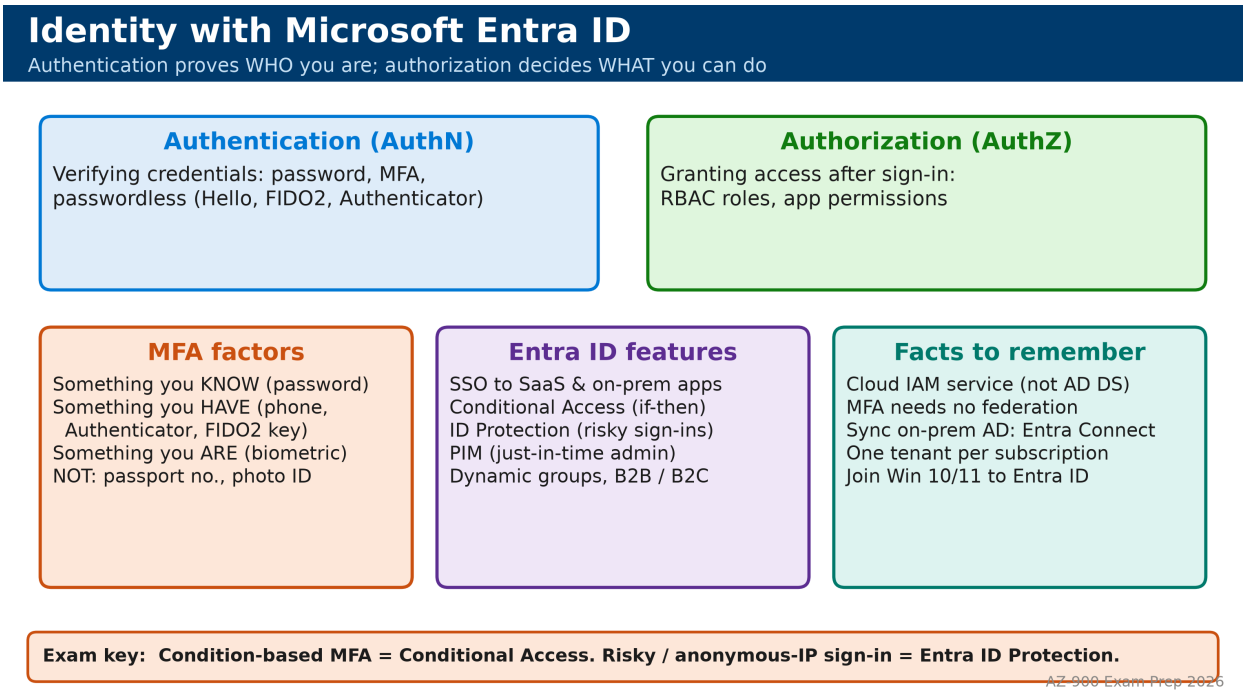


Figure Q358a: Identity with Microsoft Entra ID

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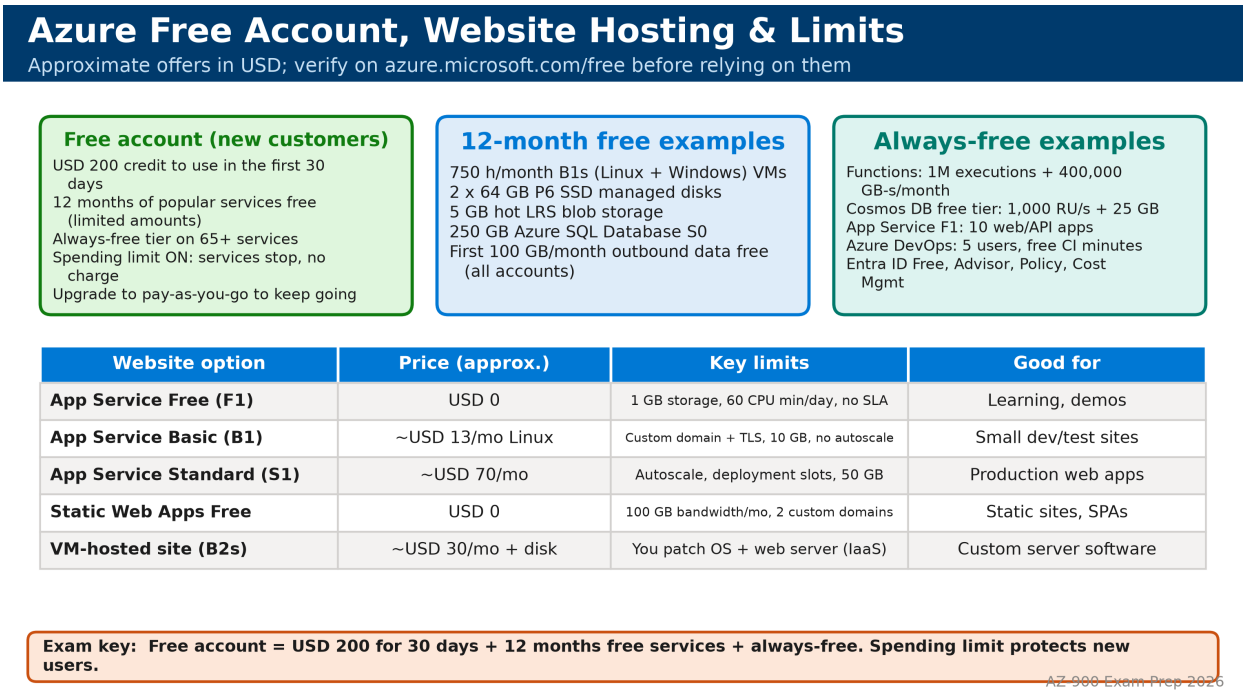


Figure Q358b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 359

AZURE ARCHITECTURE & SERVICES

Security
v5 Q492

Your company adopts a Zero Trust model. You plan to grant administrators only the permissions they need, only for a limited time, by using just-in-time and just-enough-access. Which Zero Trust principle does this implement?

✔ **A.** Use least privilege access

✖ B. Verify explicitly

✖ C. Assume breach

✖ D. Perimeter security

Correct answer: Use least privilege access

Clarification (read this first): Zero Trust has three principles: verify explicitly, use least privilege, assume breach. 'Only the permissions needed, only for a limited time' = least privilege. 'Perimeter security' is not a Zero Trust principle at all. Q373 asks for two of these principles.

Answer explained

✔ **Use least privilege access**
JIT (just-in-time: access only when needed) + JEA (just-enough-access: only the rights needed) = the 'use least privilege access' principle of Zero Trust.

✖ **Verify explicitly**
This means checking every request with all data points. It is not about limiting rights.

✖ **Assume breach**
This means limiting damage, splitting access into small parts, and encrypting end to end.

✖ **Perimeter security**
This is a layer of defense in depth, not a Zero Trust principle.

EXAM STRATEGY
'Just-in-time', 'just-enough-access' and 'only permissions needed' = least privilege. Verify explicitly is about checking sign-in signals. Assume breach is about separation.

IMPACT
Fewer standing admin rights means less damage if an account is stolen. This principle is why companies use PIM in Azure.

What · Where · When · Why · How

What	A Zero Trust principle that limits access to what is needed.
Where	On admin roles, apps and data across the whole company.
When	Every time access is given, especially admin access.
Why	To reduce damage if an account is taken over.
How	With JIT/JEA, risk-based policies and data protection.

Exam tip
Three principles: Verify explicitly, Least privilege, Assume breach. JIT/JEA matches least privilege.

Memory trick
'Just enough, just in time' = least.

Real-world example, use case and scenario

EXAMPLE
Database admins
A bank's database admins turn on a higher role for two hours only after a change is approved.

USE CASE
Contractors
A utility gives contractors read access to one resource group, not the whole subscription.

SCENARIO
Help desk
Help desk staff can reset passwords but cannot give admin roles.

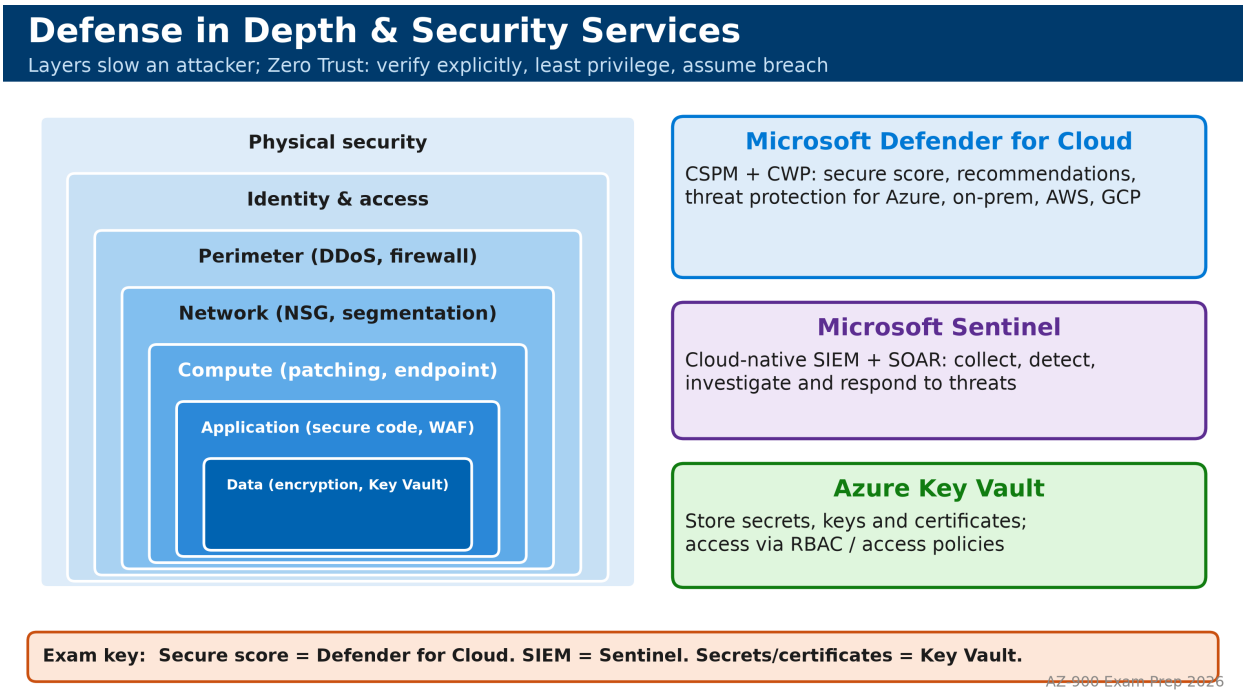


Figure Q359: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 360

AZURE ARCHITECTURE & SERVICES

Security
v5 Q493

In a defense-in-depth strategy, which layer includes distributed denial of service (DDoS) protection and perimeter firewalls that filter large-scale attacks before they reach your network?

✔ **A.** Perimeter

✖ B. Network

✖ C. Compute

✖ D. Physical security

Correct answer: Perimeter

Clarification (read this first): Do not mix up 'Perimeter' and 'Network'. Perimeter is the outer edge facing the internet (DDoS protection, edge firewalls). Network is inside, splitting traffic between resources (for example NSGs).

Answer explained

✔ **Perimeter**
Perimeter layer = DDoS protection + firewalls at the edge.

✖ **Network**
The Network layer splits and limits traffic between resources (for example NSGs), inside the perimeter.

✖ **Compute**
The Compute layer protects VMs and devices with patching and access control.

✖ **Physical security**
Physical security protects datacenter buildings and hardware, not network traffic.

EXAM STRATEGY
'DDoS' and 'perimeter firewalls' = Perimeter layer. The Network layer is splitting traffic and NSGs inside. Physical is datacenter access.

IMPACT
Floods of attack traffic stopped at the perimeter never reach your apps. DDoS Protection costs far less than being offline during an attack.

What · Where · When · Why · How

What	The edge layer of the defense-in-depth model.
Where	Between the internet and your internal networks.
When	When traffic first arrives at your environment.
Why	To stop large attacks before they use up your resources.
How	With Azure DDoS Protection and Azure Firewall at the edge.

Exam tip
Edge / DDoS = Perimeter. Splitting traffic / NSG = Network.

Memory trick
Perimeter is the castle wall. Network is the hallways inside.

Real-world example, use case and scenario

EXAMPLE
Ticket sales
A concert ticket site uses DDoS protection to stay online during a flood of bot traffic.

USE CASE
Government portal
A tax portal puts a firewall at the edge to check incoming traffic.

SCENARIO
Game launch
A game studio expects attacks at launch and turns on DDoS protection for its public IPs.

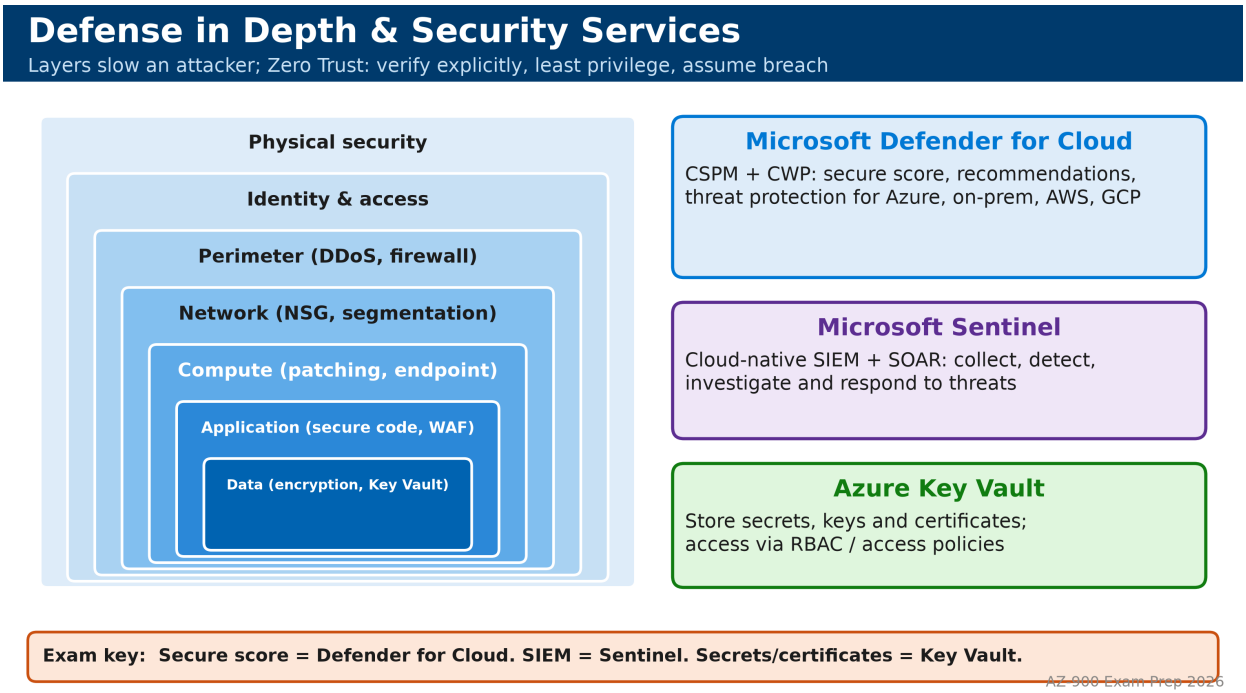


Figure Q360: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
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- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 361

AZURE ARCHITECTURE & SERVICES

Security
v5 Q494

In the defense-in-depth model, each layer protects the layer inside it. Which layer sits at the center and is the ultimate target that all the other layers protect?

✔ **A.** Data

✖ B. Application

✖ C. Identity and access

✖ D. Physical security

Correct answer: Data

Clarification (read this first): The question asks for the CENTER (innermost layer). Data is at the center. Physical security is the outermost layer - the opposite end.

Answer explained

✔ **Data**
Data is the innermost layer. Physical security is the outermost.

✖ **Application**
Application sits just outside Data. It protects apps but is not the main target.

✖ **Identity and access**
Identity and access is a layer that controls access. It is not the center.

✖ **Physical security**
Physical security is the outermost layer. It protects buildings and hardware.

EXAM STRATEGY
The center of the layers is always Data. Physical is the outermost. Identity and Application sit in between, so pick Data.

IMPACT
Every security control in the end protects data. Thinking about data first guides choices on encryption, labels and access.

What · Where · When · Why · How

What	The innermost layer of defense in depth.
Where	In databases, disks, storage accounts and SaaS apps.
When	At all times: at rest (stored) and in transit (moving).
Why	Because data is what attackers really want.
How	By wrapping it in many separate security layers.

Exam tip
Outside to inside: Physical, Identity, Perimeter, Network, Compute, Application, Data.

Memory trick
'People In Perimeters Never Compromise Any Data' = the layer order.

Real-world example, use case and scenario

EXAMPLE
Patient records
A clinic encrypts patient records in Azure SQL as the last layer of defense.

USE CASE
Card payments
A shop puts network rules, app hardening and encryption around card data.

SCENARIO
Stolen laptop
Even if an attacker gets past the network, encrypted data cannot be read.

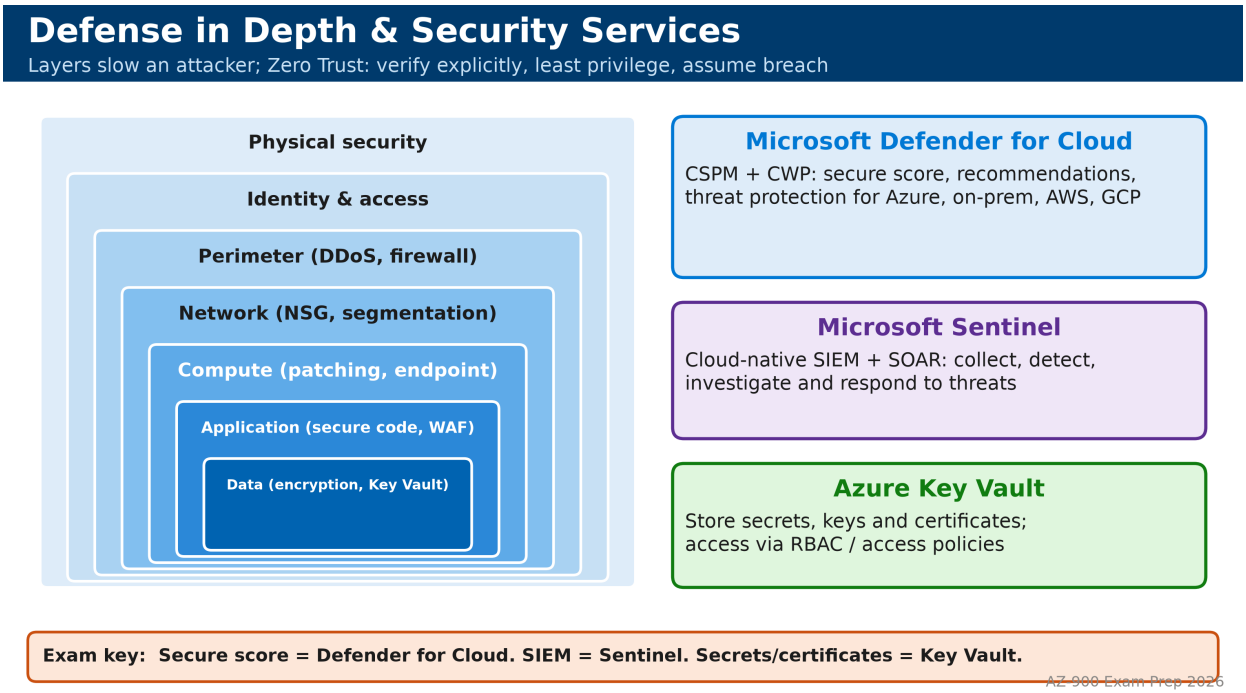


Figure Q361: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 362

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q495

A developer must be able to create, modify, and delete all resources in a resource group, but must NOT be able to grant other users access to those resources. Following the principle of least privilege, which built-in Azure role should you assign?

✔ **A.** Contributor

✖ B. Owner

✖ C. Reader

✖ D. User Access Administrator

Correct answer: Contributor

Clarification (read this first): The trap words are 'must NOT be able to grant other users access'. Owner can do everything, including giving access - too much. Contributor can do everything EXCEPT give access. Q375 asks the opposite: which roles CAN give access.

Answer explained

✔ **Contributor**
Contributor = can manage all resources, but cannot assign roles to others.

✖ **Owner**
Owner can also give access to others. That is more than least privilege allows.

✖ **Reader**
Reader can only view resources. It cannot create or delete.

✖ **User Access Administrator**
User Access Administrator only manages access. It cannot create or change resources.

EXAM STRATEGY
'Create, modify, delete' but 'not grant access' = Contributor. Owner can assign roles. Reader cannot change anything. User Access Administrator manages only access.

IMPACT
Giving Owner instead of Contributor lets developers give access to others. Auditors often find this, and it can lead to privilege escalation (gaining extra rights).

What · Where · When · Why · How

What	A built-in Azure RBAC role for managing resources.
Where	Assigned at management group, subscription, resource group or resource scope.
When	When someone must build and change resources.
Why	To give working rights without the power to give access.
How	Assign the Contributor role to the user at the resource group scope.

Exam tip
Owner = Contributor + access management. Reader = view only.

Memory trick
Contributors build the house. Owners also hand out the keys.

Real-world example, use case and scenario

EXAMPLE
Dev team
A startup's developers get Contributor on the dev resource group.

USE CASE
Agency
A web agency deploys App Service apps for a client but cannot change the client's permissions.

SCENARIO
Audit finding
An audit finds developers with Owner. They are moved down to Contributor.

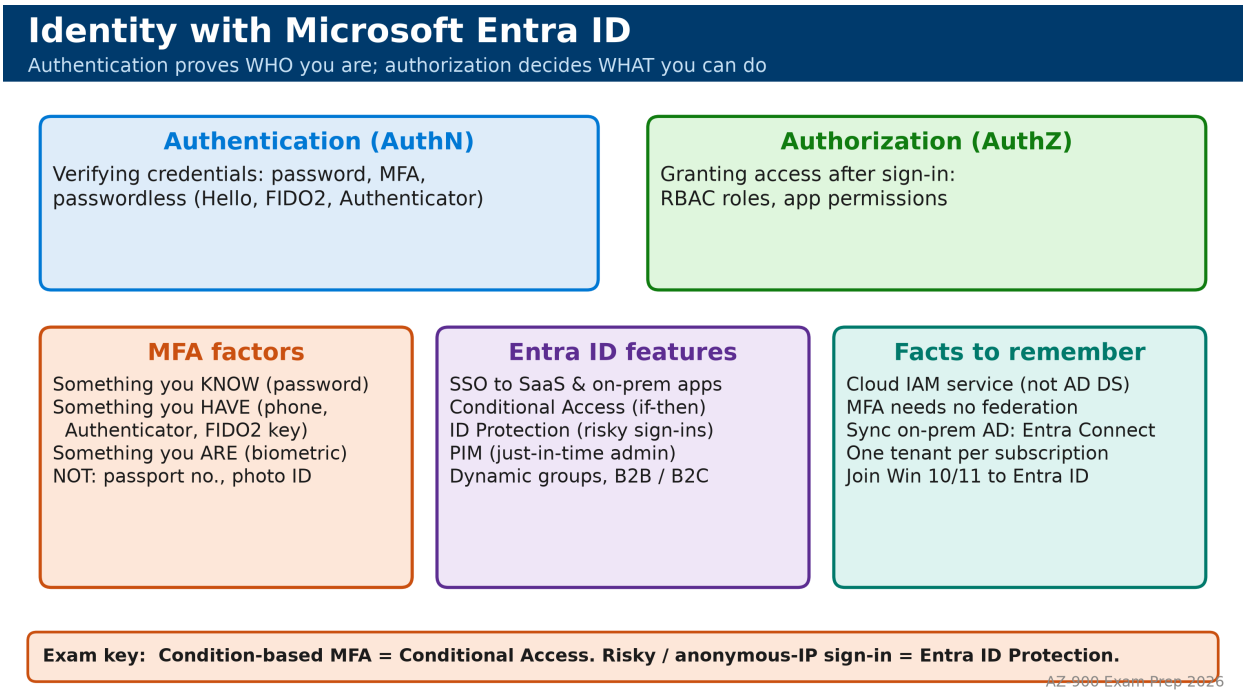


Figure Q362: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 363

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q496

Your company has a division with 12 subscriptions organized under one management group. You need to grant the security team Reader access to all current and future subscriptions in the division by using as few role assignments as possible. At which scope should you assign the role?

✔ **A.** The management group

✖ B. Each subscription

✖ C. Each resource group

✖ D. The individual resources

Correct answer: The management group

Clarification (read this first): Watch the words 'current AND future subscriptions' and 'as few as possible'. One assignment at the management group covers all 12 subscriptions now and any added later. Per-subscription assignments miss new ones.

Answer explained

✔ **The management group**
Assign at the highest scope needed. A role given at a management group is inherited (passed down) by all its subscriptions.

✖ **Each subscription**
This needs 12 assignments and does not cover future subscriptions.

✖ **Each resource group**
This needs many assignments and misses new resource groups.

✖ **The individual resources**
This is the smallest scope and needs the most assignments.

EXAM STRATEGY
'All current and future subscriptions' plus 'fewest assignments' = management group scope. Remove subscription, resource group and resource scopes; they need many assignments.

IMPACT
A management group assignment covers new subscriptions automatically. The security team will not miss any subscription the division adds.

What · Where · When · Why · How

What	The widest RBAC scope above subscriptions.
Where	At the top of the Azure resource hierarchy, below the tenant root group.
When	When the same access applies to many subscriptions.
Why	Inheritance means less admin work.
How	Create one Reader role assignment on the management group.

Exam tip
Four RBAC scopes: management group, subscription, resource group, resource. Permissions flow downward.

Memory trick
Water flows downhill: assign at the top, and it reaches everything below.

Real-world example, use case and scenario

EXAMPLE
Retail divisions
A retailer gives its central security team Reader on the 'Retail' management group.

USE CASE
Auditors
Auditors get read access to all production subscriptions with one assignment.

SCENARIO
New subscription
A new subscription moved into the management group gets the Reader assignment automatically.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

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Figure Q363: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 364

AZURE ARCHITECTURE & SERVICES

Security v5 Q497

Your company runs workloads in Azure, Amazon Web Services (AWS), and Google Cloud Platform (GCP). You need a single Microsoft service that continuously assesses the security posture of resources in all three clouds and provides hardening recommendations. What should you use?

✔ **A.** Microsoft Defender for Cloud

✖ B. Microsoft Entra ID Protection

✖ C. Azure Policy

✖ D. Azure Key Vault

Correct answer: Microsoft Defender for Cloud

Answer explained

✔ **Microsoft Defender for Cloud**
Defender for Cloud = security posture (how securely things are set up) + threat protection across Azure, AWS and GCP.

✖ **Microsoft Entra ID Protection**
ID Protection finds risky sign-ins and users. It does not check how cloud resources are set up.

✖ **Azure Policy**
Azure Policy enforces rules on Azure resources. It is not a security posture tool for many clouds.

✖ **Azure Key Vault**
Key Vault stores secrets and keys. It does not check posture.

EXAM STRATEGY
'Security posture', 'hardening (making more secure) recommendations' and 'Azure, AWS and GCP' = Defender for Cloud. Azure Policy is for Azure. ID Protection is identity risk. Key Vault stores secrets.

IMPACT
One view of all clouds stops bad settings from hiding in another cloud. Bad settings are a main cause of real data breaches.

What · Where · When · Why · How

What	A CNAPP (cloud app protection platform) for posture and workload protection.
Where	Across Azure, AWS, GCP and on-premises (through Azure Arc).
When	All the time, as resources are created and changed.
Why	To find and fix bad settings and to detect threats.
How	Connect your cloud accounts. It checks them against security standards.

Exam tip
Posture across many clouds + recommendations = Defender for Cloud. Identity risk = Entra ID Protection.

Memory trick
Defender 'for Cloud' - any cloud, not just Azure.

Real-world example, use case and scenario

EXAMPLE
Media company
A streaming company sees bad AWS S3 and Azure Storage settings in one Defender for Cloud dashboard.

USE CASE
Merger
A bank buys a startup that uses GCP and checks its security with no new tools.

SCENARIO
Hybrid servers
On-premises servers added with Azure Arc get Defender for Cloud recommendations.

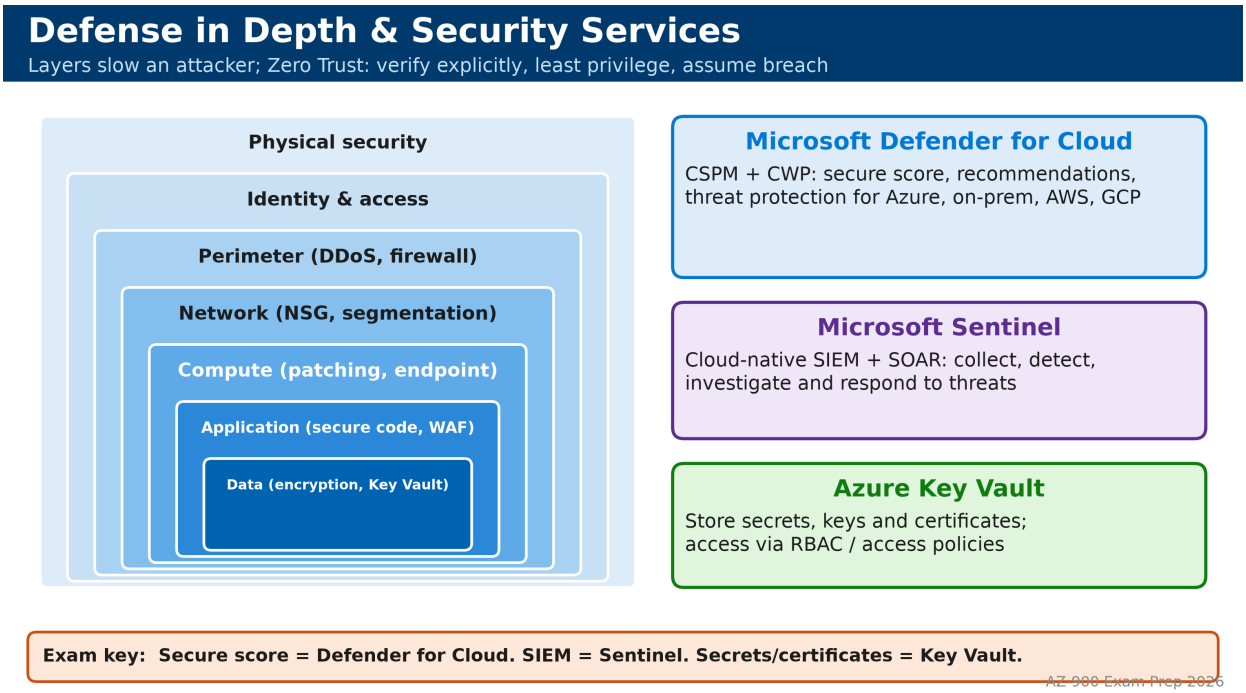


Figure Q364: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 365

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q498

Your company wants to require MFA only when users sign in from outside the corporate network and to block sign-ins from certain countries. What is the minimum Microsoft Entra ID license required to create these Conditional Access policies?

✔ **A.** Microsoft Entra ID P1

✖ B. Microsoft Entra ID Free

✖ C. Microsoft Entra ID P2

✖ D. Microsoft Entra ID Governance

Correct answer: Microsoft Entra ID P1

Clarification (read this first): The key word is 'minimum'. P2 also allows Conditional Access, but it costs more, so it is not the minimum. P2 is needed only for risk-based policies.

Answer explained

✔ **Microsoft Entra ID P1**
Conditional Access needs at least Entra ID P1. Risk-based policies need P2.

✖ **Microsoft Entra ID Free**
Free has security defaults (basic built-in settings) but no custom Conditional Access policies.

✖ **Microsoft Entra ID P2**
P2 works, but it is more than the minimum. P2 adds risk-based features.

✖ **Microsoft Entra ID Governance**
Governance adds identity lifecycle and access review features. It is not the minimum for Conditional Access.

EXAM STRATEGY
'Conditional Access' or location/country rules = P1 minimum. Remove Free (security defaults only). Pick P2 only if the question says risk, ID Protection or PIM.

IMPACT
Buying P2 for every user wastes money. Staying on Free gives no location-based MFA, which auditors often flag for remote workers.

What · Where · When · Why · How

What	The license level that unlocks Conditional Access.
Where	Assigned to users in the Microsoft Entra tenant.
When	When you need custom access policies based on conditions.
Why	Because Free has only basic security defaults.
How	Buy P1 (alone or in Microsoft 365 E3 / Business Premium) and create policies.

Exam tip
Free = security defaults. P1 = Conditional Access. P2 = risk-based (ID Protection, PIM).

Memory trick
P1 = Policies. P2 = Protection from risk.

Real-world example, use case and scenario

EXAMPLE
Remote sales staff
A drug company requires MFA for sales staff who sign in from hotels.

USE CASE
Country block
A local bank blocks sign-ins from countries where it has no staff.

SCENARIO
License planning
An IT manager buys P1, not P2, because no risk-based policies are planned.

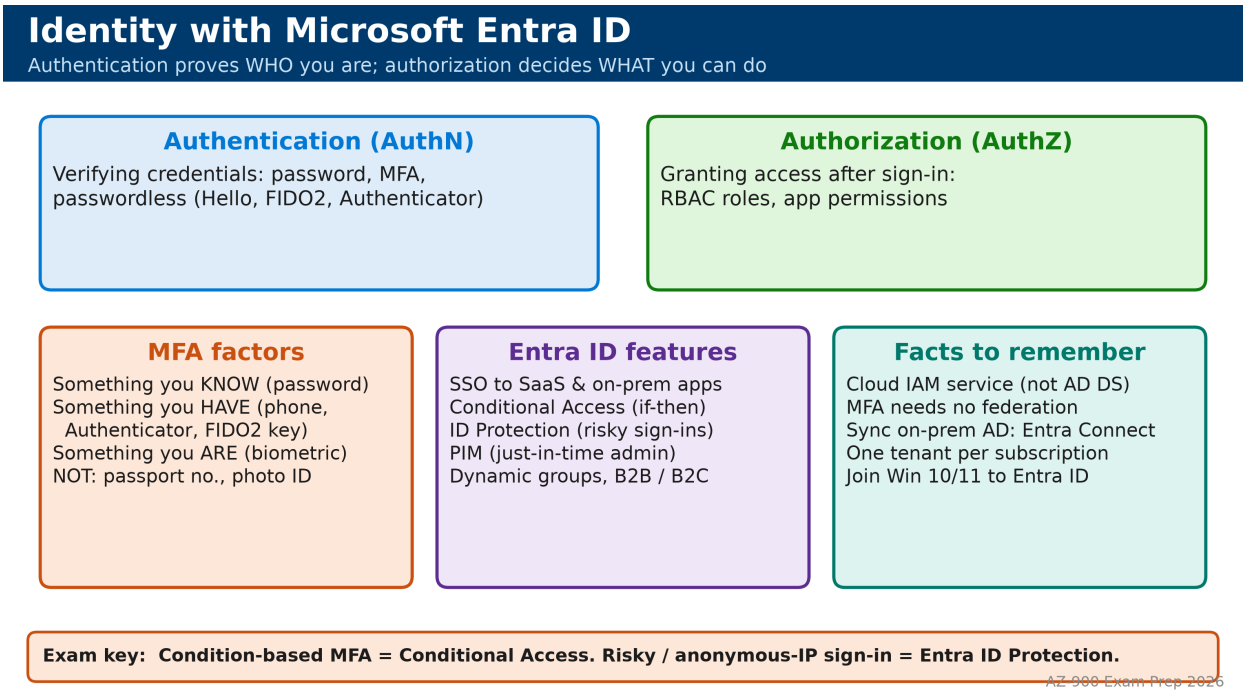


Figure Q365: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 366

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q499

After a user successfully signs in to Microsoft Entra ID, Azure determines whether the user can delete a virtual machine. What is this process called?

✔ **A.** Authorization

✖ B. Authentication

✖ C. Federation

✖ D. Auditing

Correct answer: Authorization

Clarification (read this first): The words 'after a user successfully signs in' mean authentication (proving who you are) is already done. Deciding what the user may DO is authorization. Q251 asks about checking credentials - that is authentication.

Answer explained

✔ **Authorization**
Authorization (AuthZ) = deciding what a signed-in user is allowed to do.

✖ **Authentication**
Authentication checks who you are. It already happened at sign-in.

✖ **Federation**
Federation sets up trust between identity providers. It does not allow actions.

✖ **Auditing**
Auditing records actions after they happen. It does not decide permissions.

EXAM STRATEGY
'Can the user delete/do X' after sign-in = authorization. Authentication is already done. Remove federation (trust between directories) and auditing (logging after the fact).

IMPACT
If you confuse AuthN and AuthZ, you may fix MFA when the real problem is a role that gives too much. Then delete rights stay open.

What · Where · When · Why · How

What	Deciding how much access an identity has.
Where	In Azure, Azure RBAC enforces it through Azure Resource Manager.
When	After authentication, on every action a user asks for.
Why	So users do only what they are allowed to do.
How	Azure checks the user's role assignments at that scope.

Exam tip
AuthN = identity (who you are). AuthZ = permissions (what you can do).

Memory trick
AuthN = your Name. AuthZ = the 'Zones' you may enter.

Real-world example, use case and scenario

EXAMPLE
Hotel key card
Your key card opens your room but not the manager's office. That is authorization.

USE CASE
Read-only finance
A finance analyst can view cost data but cannot delete resources.

SCENARIO
Delete denied
A Reader tries to delete a VM and gets an 'authorization failed' error.

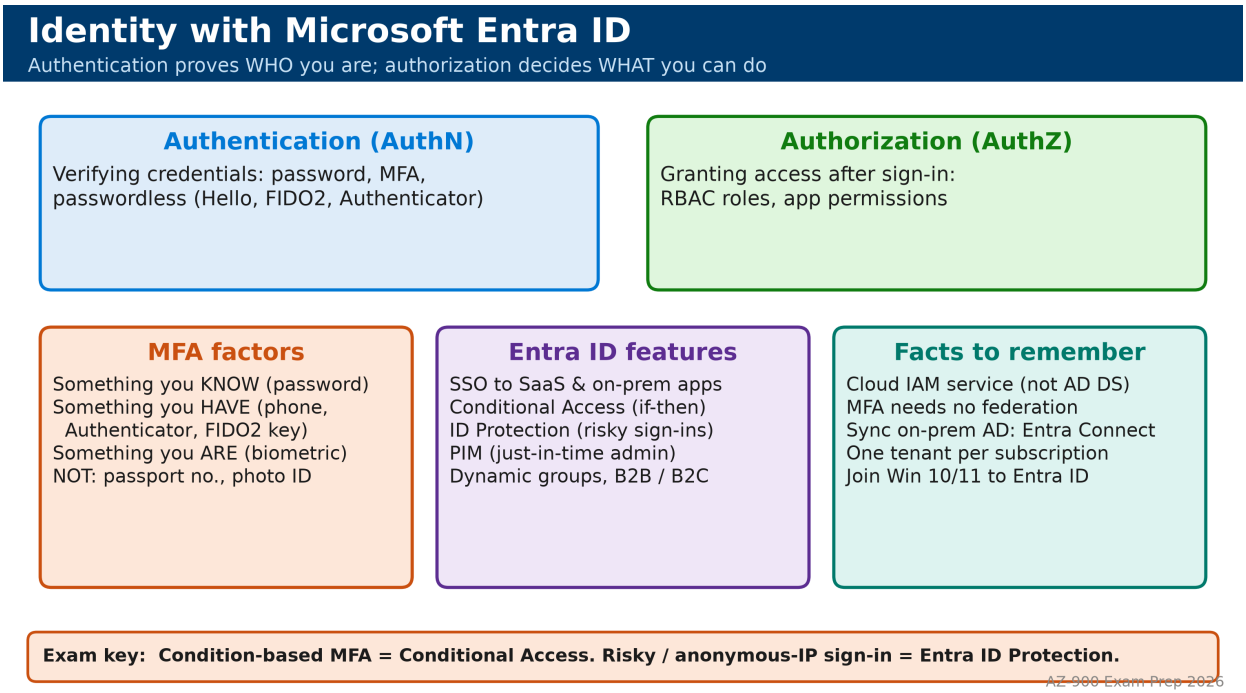


Figure Q366: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
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- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 367

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q501

For the following statement, select Yes if the statement is true. Otherwise, select No. Conditional Access policies are enforced after first-factor authentication is completed.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): The phrase to check is 'after first-factor authentication'. Conditional Access must first know WHO the user is (from the first factor, usually the password). Then it decides to allow, block or ask for MFA.

Answer explained

✔ **Yes**
Yes. Conditional Access is applied after the first factor of sign-in is complete.

✖ **No**
Conditional Access needs sign-in signals such as who the user is. These are known only after the first factor.

EXAM STRATEGY
Key fact: Conditional Access checks signals after the first factor tells it who the user is. 'After first-factor' = Yes. It cannot judge a user it has not identified.

IMPACT
This explains why a correct password can still lead to an MFA prompt or a block. It is the core of fixing sign-in problems for remote users.

What · Where · When · Why · How

What	An if-then rule engine for access decisions.
Where	In Microsoft Entra ID, for cloud apps and actions.
When	After first-factor authentication, before access is given.
Why	To enforce company rules based on signals.
How	It checks user, location, device, app and risk, then allows, blocks or requires MFA.

Exam tip
Conditional Access = signals -> decision -> enforcement, after the first factor.

Memory trick
First prove who you are. Then Conditional Access decides 'if this, then that'.

Real-world example, use case and scenario

EXAMPLE
Unmanaged device
After a password sign-in, a policy asks for MFA because the device is not managed.

USE CASE
Admin portals
A delivery company requires MFA for anyone opening the Azure portal.

SCENARIO
Blocked country
A user signs in with a password from a blocked country and is denied.

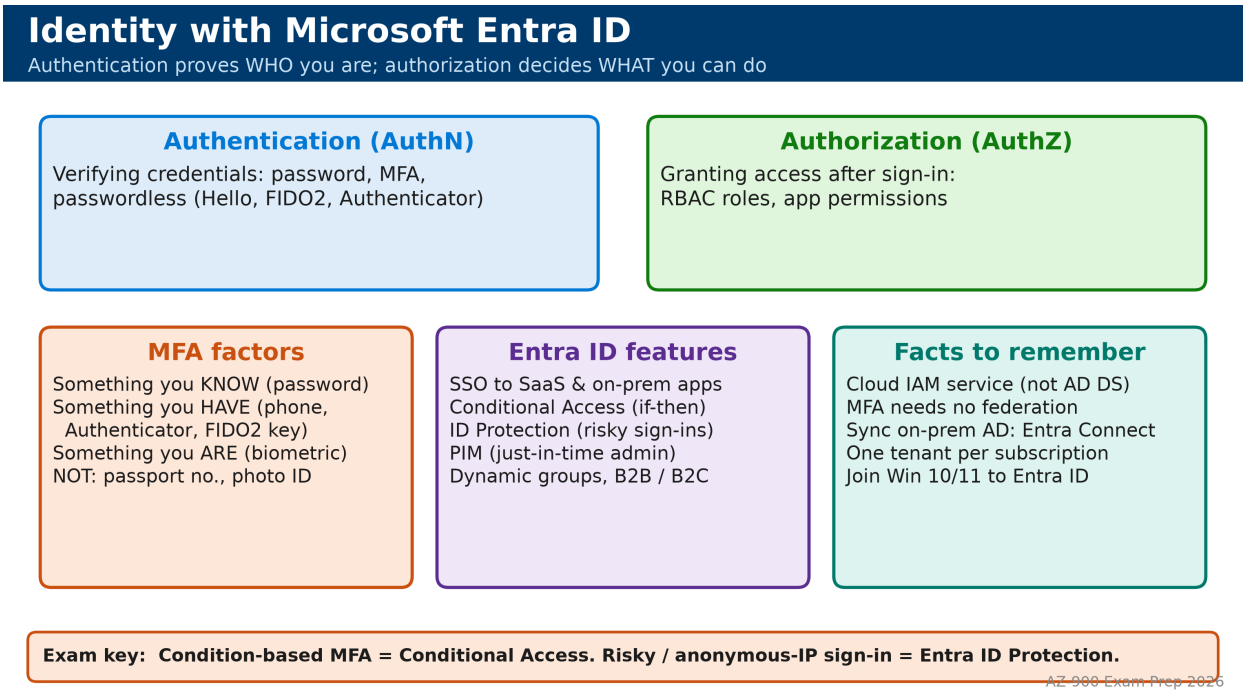


Figure Q367: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 368

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q502

For the following statement, select Yes if the statement is true. Otherwise, select No. If a user is assigned the Reader role at a subscription and the Contributor role on a resource group in that subscription, the user can create resources in that resource group.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): RBAC is additive: roles add up, they do not cancel each other. Reader on the subscription does not take away Contributor on the resource group. Q369 is different: there the user has ONLY Reader, so the answer is No.

Answer explained

✔ **Yes**
Yes. RBAC permissions add up. Contributor on the resource group adds the right to create there.

✖ **No**
Reader does not override Contributor. RBAC combines the permissions from all assignments.

EXAM STRATEGY
Key fact: RBAC is additive, and Reader has no 'deny'. Contributor at the resource group adds create rights there, so the answer is Yes.

IMPACT
If you misunderstand additive RBAC, you may give extra rights elsewhere to 'fix' access. Knowing that scopes combine keeps least privilege.

What · Where · When · Why · How

What	The total permissions from many role assignments.
Where	Across overlapping scopes in the hierarchy.
When	When a user has roles at more than one scope.
Why	Because Azure RBAC is an 'allow' model.
How	Azure Resource Manager adds up the actions from all the user's roles.

Exam tip
RBAC is additive. The widest allowed action wins, unless there is an explicit deny assignment.

Memory trick
Roles stack like keys on a keyring.

Real-world example, use case and scenario

EXAMPLE
Project lead
A project lead can view the whole subscription but builds only in the project's resource group.

USE CASE
Data team
Data engineers get Reader on production and Contributor on the analytics resource group.

SCENARIO
Access review
An admin asks why a Reader could deploy a VM and finds a Contributor role on the resource group.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

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Figure Q368: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 369

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q503

For the following statement, select Yes if the statement is true. Otherwise, select No. A user who has only the Reader role on a resource group can start and stop the virtual machines in that resource group.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): The key word is 'only'. With ONLY Reader, the user can look but cannot start or stop anything. Q368 is different: there the user also has Contributor, so the answer is Yes.

Answer explained

✔ **No**
No. Reader can only view resources. It cannot do actions such as start or stop.

✖ **Yes**
Start and stop are write actions. Reader has only read (*/read) permissions.

EXAM STRATEGY
Key fact: Reader = read-only. Start and stop are actions that change things. 'Only Reader' plus any verb other than 'view' = No.

IMPACT
Operations staff with only Reader cannot restart a stuck production VM during an incident. Give Virtual Machine Contributor or a custom role instead.

What · Where · When · Why · How

What	A built-in, view-only Azure role.
Where	At any RBAC scope, from management group to resource.
When	When users need to see resources but not change them.
Why	To give monitoring access with least privilege.
How	Reader allows */read actions only.

Exam tip
Reader = look, don't touch.

Memory trick
Reader reads; it does not drive.

Real-world example, use case and scenario

EXAMPLE
Auditor
An auditor views VM settings but cannot shut the VMs down.

USE CASE
Operations dashboard
An operations team gets Reader to see resource status without changing anything.

SCENARIO
Failed stop
A Reader clicks Stop on a VM and gets an authorization error.

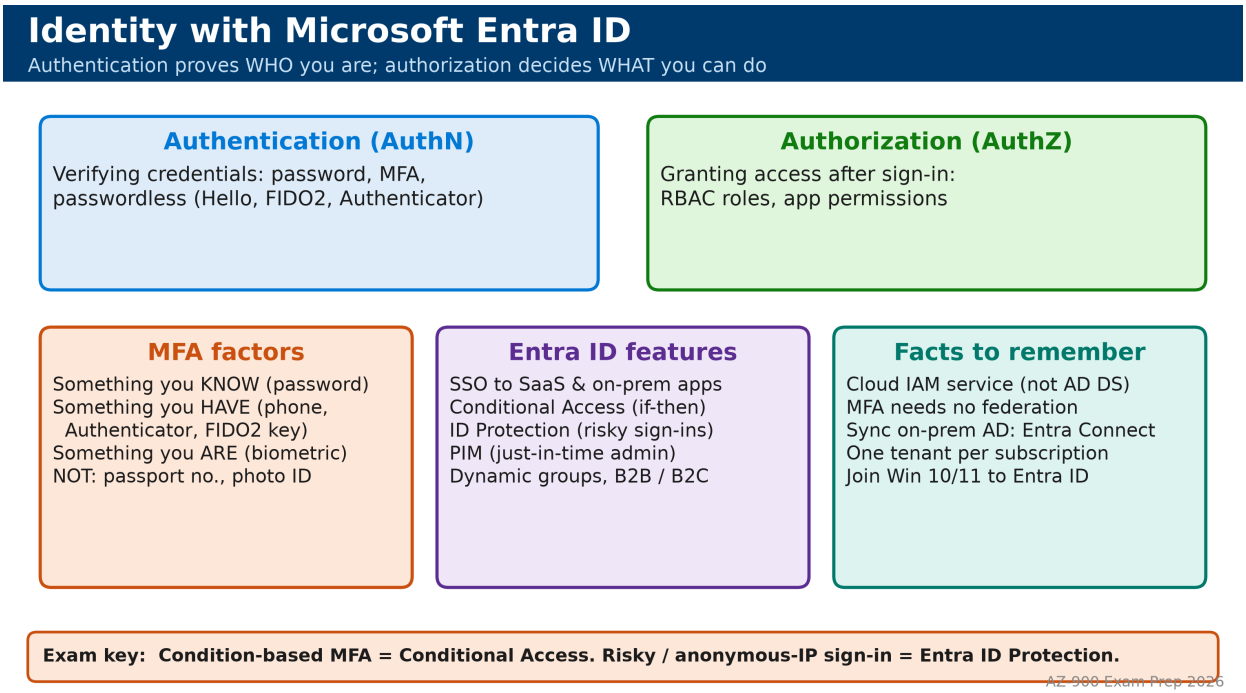


Figure Q369: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 370

AZURE ARCHITECTURE & SERVICES

Security
v5 Q504

For the following statement, select Yes if the statement is true. Otherwise, select No. The Foundational CSPM capabilities of Microsoft Defender for Cloud, including Secure Score and security recommendations, are available at no additional cost.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): The key word is 'Foundational CSPM' (the basic posture features). These are free. Q167 asks if ALL Defender for Cloud features are free - No, because the extra Defender plans cost money.

Answer explained

✔ **Yes**
Yes. Foundational CSPM (Secure Score + security recommendations) is free. Defender plans cost extra.

✖ **No**
Only the extra Defender plans are paid. Foundational CSPM is free.

EXAM STRATEGY
Key fact: 'Foundational CSPM' is the free level. Free = Secure Score and recommendations. Anything named Defender for Servers/Storage/SQL or Defender CSPM costs extra.

IMPACT
Every subscription gets security recommendations at no cost, so every team can measure its Secure Score. Paid plans add workload protection.

What · Where · When · Why · How

What	The free level of Defender for Cloud posture management.
Where	On Azure subscriptions and connected AWS/GCP accounts.
When	Automatically, as soon as you open Defender for Cloud.
Why	To give every customer basic security visibility.
How	Defender for Cloud checks your resources and calculates a Secure Score.

Exam tip
Free = Foundational CSPM. Paid = Defender CSPM and workload plans (Servers, Storage, SQL and more).

Memory trick
The foundation is free; the extra floors cost money.

Real-world example, use case and scenario

EXAMPLE
Startup
A startup uses the free Secure Score to choose what to fix before buying paid plans.

USE CASE
School district
A school district reviews free recommendations to turn on MFA and close open ports.

SCENARIO
Upgrade
A bank turns on Defender for Servers after free CSPM shows unprotected VMs.

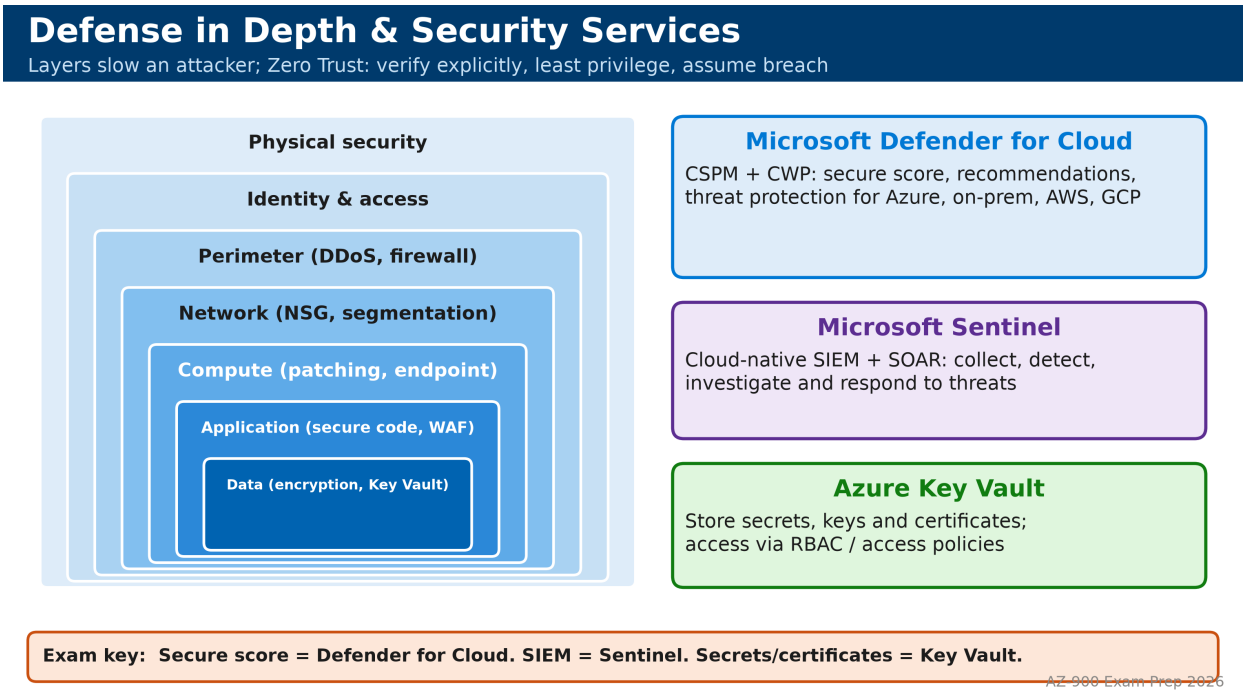


Figure Q370: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
- Microsoft Sentinel bills per GB of data analysed, roughly a few dollars per GB pay-as-you-go, with commitment tiers and some free data sources.
- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 371

AZURE ARCHITECTURE & SERVICES

Security v5 Q506

For the following statement, select Yes if the statement is true. Otherwise, select No. In a Zero Trust model, users and devices located inside the corporate network are trusted by default.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): The trap words are 'trusted by default'. In Zero Trust, being inside the company network never earns trust. Every request is checked.

Answer explained

✔ **No**
No. Zero Trust gives no automatic trust to the internal network. Every request is checked.

✖ **Yes**
That is the old perimeter model (trust everything inside the firewall), which Zero Trust replaces.

EXAM STRATEGY
Signal: 'trusted by default' or 'inside the network' goes against Zero Trust. Any trust based only on location = No.

IMPACT
Automatic inside trust lets ransomware spread from one phished laptop to many machines. Zero Trust checks every request to limit the damage.

What · Where · When · Why · How

What	A security model of 'never trust, always verify'.
Where	Across identities, devices, apps, data, infrastructure and networks.
When	On every access request, from any location.
Why	Because attackers may already be inside the network.
How	Verify explicitly, use least privilege and assume breach.

Exam tip
Zero Trust: location is never a reason to trust.

Memory trick
Zero trust = zero free passes, even at home.

Real-world example, use case and scenario

EXAMPLE
Office Wi-Fi
A law firm requires MFA even for staff in the office.

USE CASE
Hybrid work
A consulting firm uses the same policies for office and home sign-ins.

SCENARIO
Malware inside
Malware on an office PC is blocked because the device fails a compliance check.

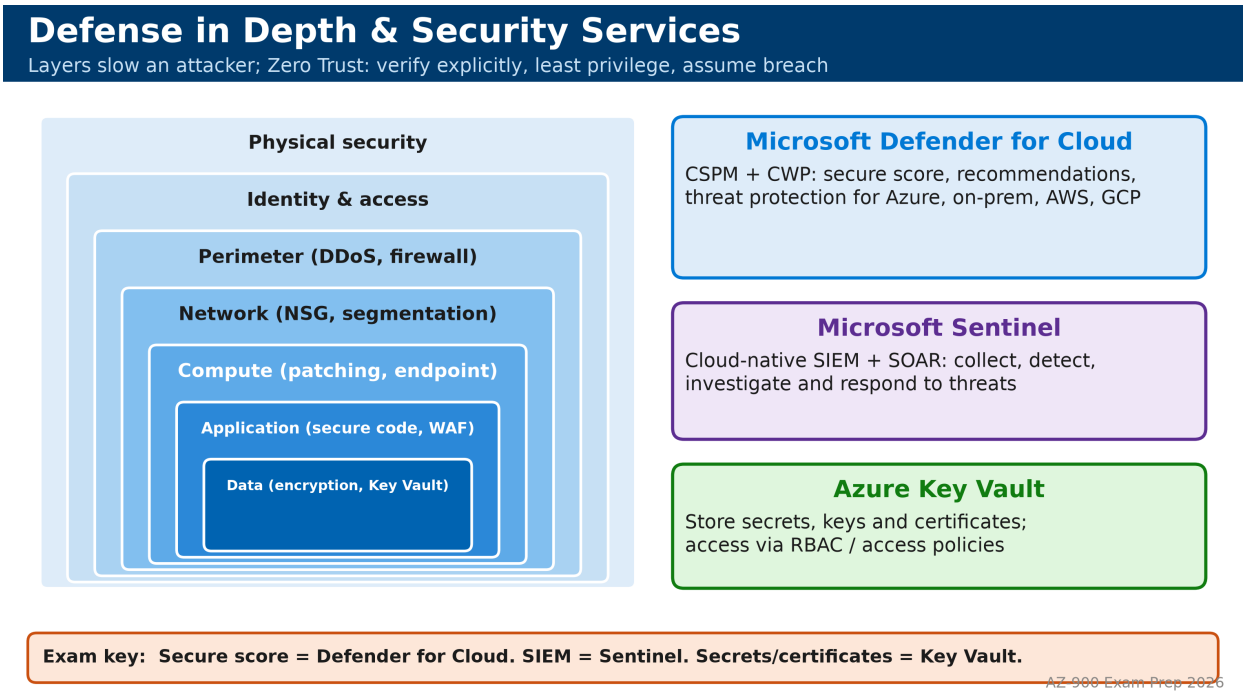


Figure Q371: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
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- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 372

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q507

Your company wants to eliminate passwords for user sign-in to Microsoft Entra ID. Which TWO authentication methods are passwordless? Each correct answer presents a complete solution.

- ✔ **A.** Windows Hello for Business
- ✔ **B.** Microsoft Authenticator passwordless phone sign-in
- ✖ C. Security questions
- ✖ D. Self-service password reset
- ✖ E. Password hash synchronization

Correct answer: **Windows Hello for Business; Microsoft Authenticator passwordless phone sign-in**

Clarification (read this first): Watch options that contain the word 'password'. SSPR and password hash sync still use passwords, and security questions are only for resets. Passwordless means something you have or are, such as a phone or your face. Q355 and Q356 cover other passwordless methods.

Answer explained

✔ **Windows Hello for Business; Microsoft Authenticator passwordless phone sign-in**
Passwordless methods = Windows Hello for Business, Microsoft Authenticator, FIDO2 passkeys and certificate-based authentication.

✖ **Security questions**
Used only for password reset. It is not a sign-in method.

✖ **Self-service password reset**
It resets passwords. It still depends on passwords.

✖ **Password hash synchronization**
A hybrid sync method. Users still sign in with passwords.

EXAM STRATEGY
Passwordless = something you have or are, never something you know. Remove security questions, SSPR (still a password) and password hash sync (it syncs passwords).

IMPACT
Passwordless removes the secret that attackers phish in most account takeovers. It also cuts help-desk password reset tickets.

What · Where · When · Why · How

What	Sign-in methods that do not use a password.
Where	Supported in Microsoft Entra ID for cloud and hybrid users.
When	When you want less phishing risk and fewer password resets.
Why	Passwords are the credential attacked most often.
How	Replace the password with something you have, plus something you are or know (like a PIN).

Exam tip
Passwordless trio: Hello, Authenticator, FIDO2 (plus certificate-based authentication).

Memory trick
Hello, Phone, Key - no password needed.

Real-world example, use case and scenario

EXAMPLE	USE CASE	SCENARIO
Field workers Utility workers approve sign-ins on their phones with Authenticator.	**Office staff** An accounting firm's staff unlock PCs with Windows Hello fingerprint readers.	**Reset tickets** A help desk sees password reset tickets drop after passwordless rollout.

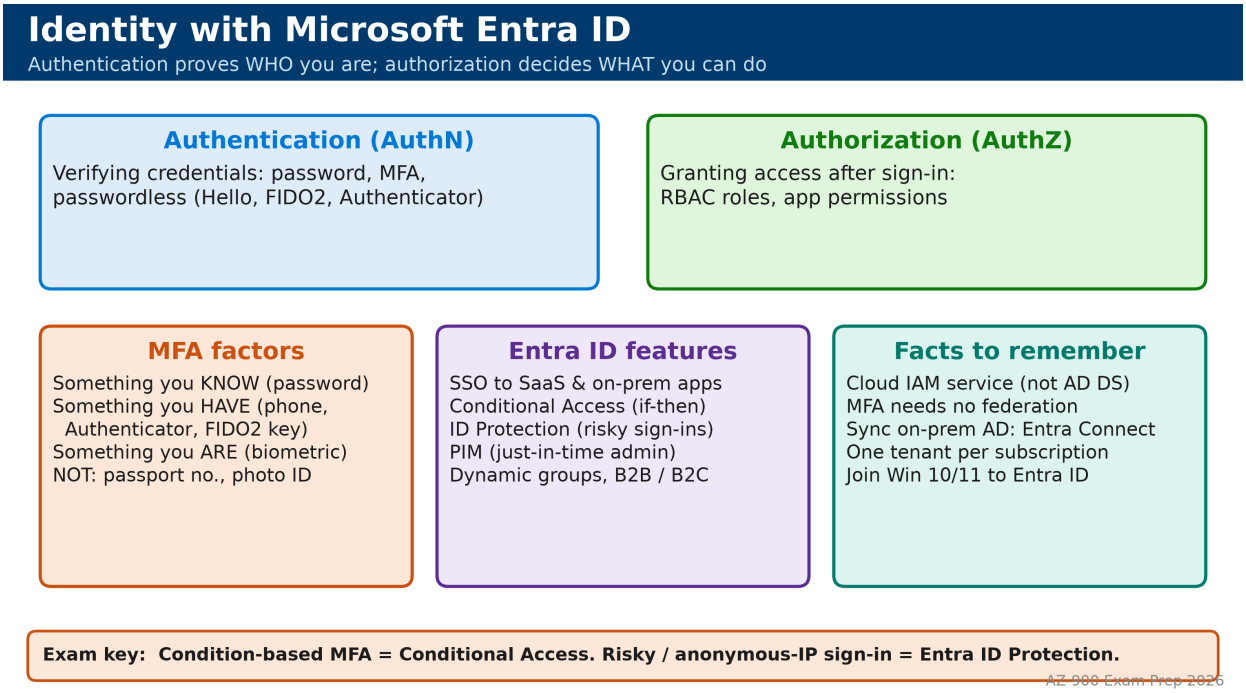


Figure Q372: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 373

AZURE ARCHITECTURE & SERVICES

Security
v5 Q508

Which TWO of the following are guiding principles of the Zero Trust security model? Each correct answer presents part of the solution.

- ✔ **A.** Verify explicitly
- ✔ **B.** Assume breach
- ✖ C. Trust devices on the corporate network
- ✖ D. Grant permanent administrative access
- ✖ E. Rely on the network perimeter firewall

Correct answer: Verify explicitly; Assume breach

Clarification (read this first): Zero Trust has THREE principles, but only two appear here: verify explicitly and assume breach. The third, use least privilege, is the answer in Q359. The wrong options all describe the old 'trust the network' model.

Answer explained

✔ **Verify explicitly; Assume breach**
Zero Trust principles: Verify explicitly, Use least privilege, Assume breach.

✖ **Trust devices on the corporate network**
This goes against Zero Trust. Network location never means trust.

✖ **Grant permanent administrative access**
This goes against least privilege. Zero Trust uses just-in-time access.

✖ **Rely on the network perimeter firewall**
This is the old perimeter model that Zero Trust moves away from.

EXAM STRATEGY
Learn the three: verify explicitly, use least privilege, assume breach. Any option that trusts the network, the perimeter or permanent admin rights is the opposite of Zero Trust.

IMPACT
These principles lead to real designs: MFA everywhere, just-in-time admin with PIM, and small network segments. Zero Trust questions come up often on the exam.

What · Where · When · Why · How

What	The core principles of Microsoft's Zero Trust model.
Where	Across identity, devices, apps, data, infrastructure and network.
When	On every access request.
Why	Because people now work outside one trusted network.
How	Check each request, limit access, and design as if attackers are already inside.

Exam tip
V-L-A: Verify explicitly, Least privilege, Assume breach.

Memory trick
'VLA' - Very Little Allowed.

Real-world example, use case and scenario

EXAMPLE
Split networks
A hospital keeps medical devices apart from office PCs, assuming a breach can happen.

USE CASE
Many signals
A finance app company checks device health, location and risk at every sign-in.

SCENARIO
Ransomware
A manufacturer keeps ransomware inside one segment because it planned for a breach.

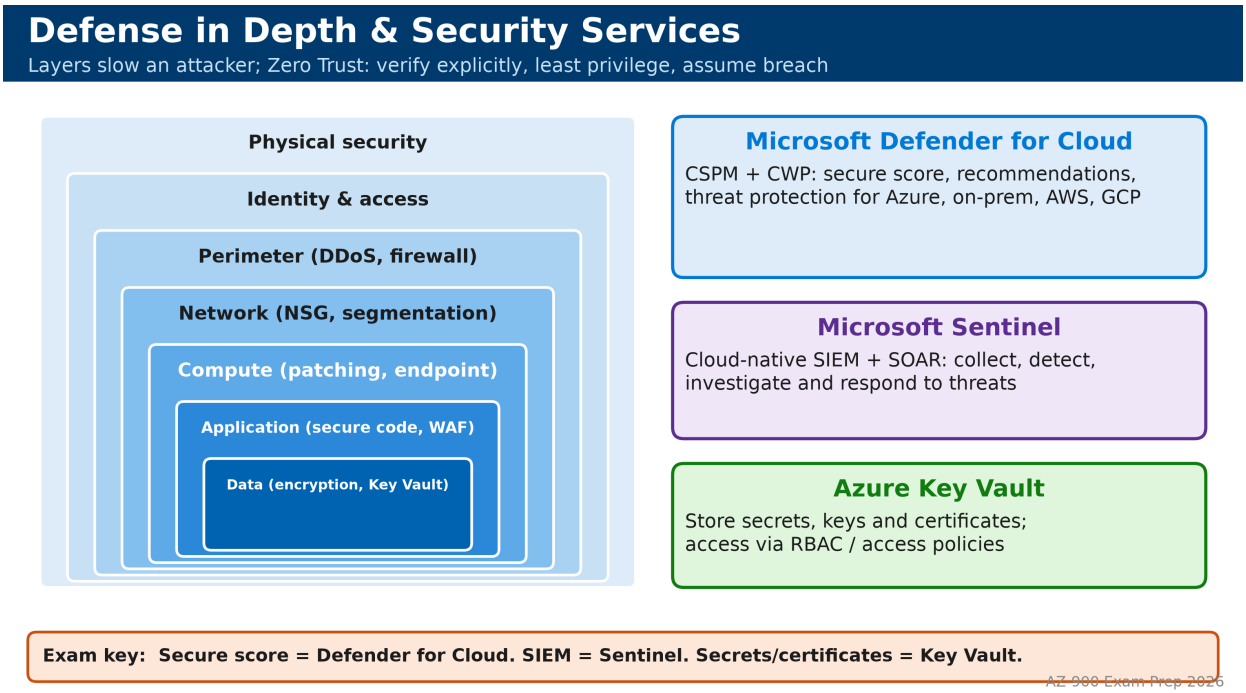


Figure Q373: Defense in depth and security services

About this figure

- Concentric layers run from physical security outside to data at the centre; each layer must be breached in turn, slowing an attacker.
- Zero Trust principles: verify explicitly, use least-privilege access, and assume breach. Never trust a request just because it comes from inside the network.
- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
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- Key Vault Standard bills per operation, approx. USD 0.03 per 10,000 transactions; Premium adds HSM-protected keys. Managed identities avoid storing secrets.

Question 374

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q509

You are configuring a Microsoft Entra Conditional Access policy. Which TWO access controls can the policy apply when its conditions are met? Each correct answer presents a complete solution.

- ✔ **A.** Block access
- ✔ **B.** Require multifactor authentication
- ✖ C. Delete the user account
- ✖ D. Apply a resource lock
- ✖ E. Assign an Azure RBAC role

Correct answer: Block access; Require multifactor authentication

Answer explained

✔ **Block access; Require multifactor authentication**
Conditional Access results: block access, or allow access with conditions such as MFA or a compliant (approved) device.

✖ **Delete the user account**
Conditional Access controls sign-in access. It never deletes accounts.

✖ **Apply a resource lock**
Resource locks are an Azure Resource Manager governance feature.

✖ **Assign an Azure RBAC role**
Role assignments are managed in Azure RBAC, not by Conditional Access.

EXAM STRATEGY
Conditional Access controls are 'block' or 'grant with requirements' (MFA, compliant device). Remove anything that belongs to another tool: deleting accounts, locks, RBAC roles.

IMPACT
The right control lets you block risky countries or force MFA without bothering users who follow the rules. Using the wrong tool leaves access open.

What · Where · When · Why · How

What	The decision a Conditional Access policy enforces.
Where	In Microsoft Entra ID, at sign-in to protected apps.
When	When a sign-in matches the policy's conditions.
Why	To balance security and ease of work using signals.
How	Set the Grant control to Block, or to Grant with requirements.

Exam tip
Conditional Access = IF signals THEN block / require MFA / require compliant device.

Memory trick
A gatekeeper says either 'No entry' or 'Show more ID'.

Real-world example, use case and scenario

EXAMPLE
Old protocols
A university blocks old sign-in protocols that cannot do MFA.

USE CASE
Claims app
An insurer requires MFA whenever users open its claims app.

SCENARIO
Personal phones
A shop requires a compliant device before email works on phones.

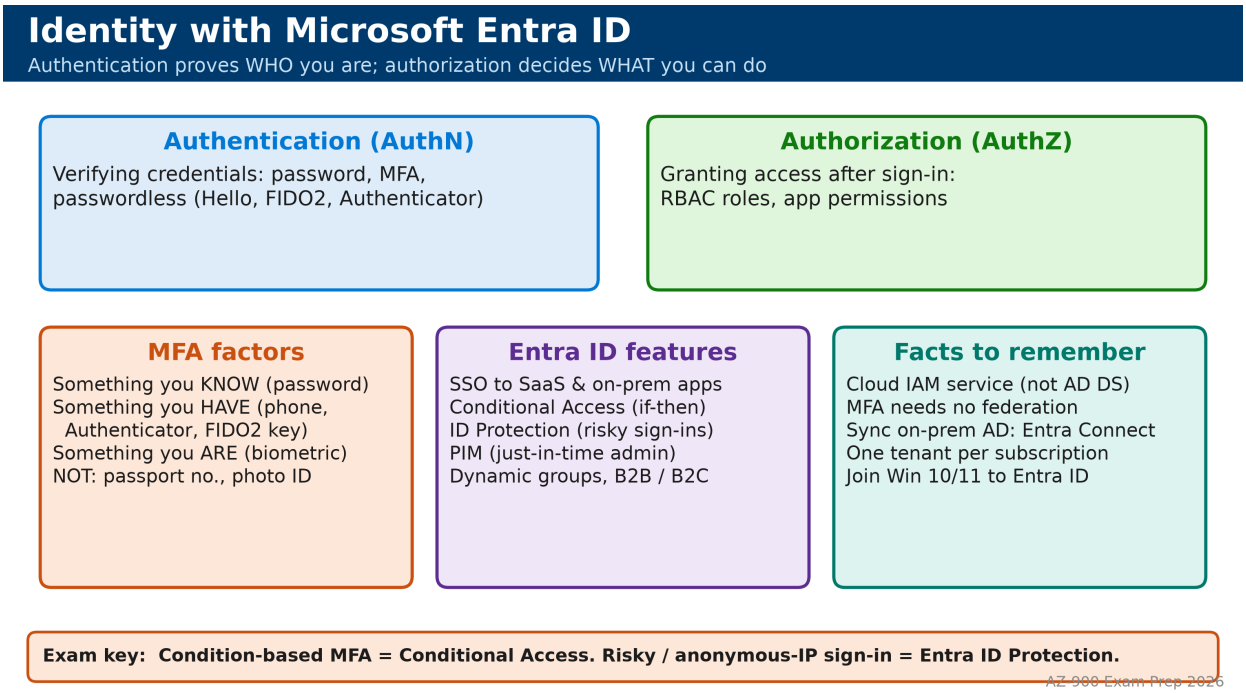


Figure Q374: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
- MFA requires two different factor types; two passwords or a password plus a security question do not count as multi-factor.
- Entra ID Free covers users, groups, SSO and security defaults; Conditional Access needs P1 (~USD 6/user-month), ID Protection and PIM need P2 (~USD 9).
- Entra Connect or the lighter Entra Cloud Sync synchronises on-premises AD users; password hash sync enables cloud sign-in with the same password.
- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 375

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q510

Which TWO built-in Azure roles allow a user to grant other users access to Azure resources by managing role assignments? Each correct answer presents a complete solution.

- ✔ **A.** Owner
- ✔ **B.** User Access Administrator
- ✖ C. Contributor
- ✖ D. Reader

Correct answer: Owner; User Access Administrator

Clarification (read this first): The question asks which roles CAN give access to others. Contributor can change everything, but it cannot give access - see Q362, where that is exactly why Contributor is the answer.

Answer explained

✔ **Owner; User Access Administrator**
Roles that can assign roles: Owner and User Access Administrator. Contributor and Reader cannot.

✖ **Contributor**
Contributor manages resources but cannot give access to others.

✖ **Reader**
Reader is view-only. It cannot manage resources or access.

EXAM STRATEGY
Signal: 'grant others access' = the right to write role assignments. Only Owner and User Access Administrator have it. Contributor can do everything except assign roles.

IMPACT
Giving Owner only so someone can share access gives too many rights. User Access Administrator gives that right without the power to change resources.

What · Where · When · Why · How

What	Roles that can create Azure role assignments.
Where	At any scope where the role is assigned.
When	When someone must give access to others.
Why	To control who can give permissions.
How	Both roles include the Microsoft.Authorization/roleAssignments/write permission.

Exam tip
Main roles: Owner (all + access), Contributor (all but access), Reader (view), User Access Administrator (access only).

Memory trick
The ones who hold the key cabinet: Owner and User Access Administrator.

Real-world example, use case and scenario

EXAMPLE
Team lead
A team lead with Owner on a resource group gives Reader to new staff.

USE CASE
Identity team
A central identity team has User Access Administrator to manage access without touching VMs.

SCENARIO
Delegation
A project manager must add members but not build resources, so gets User Access Administrator.

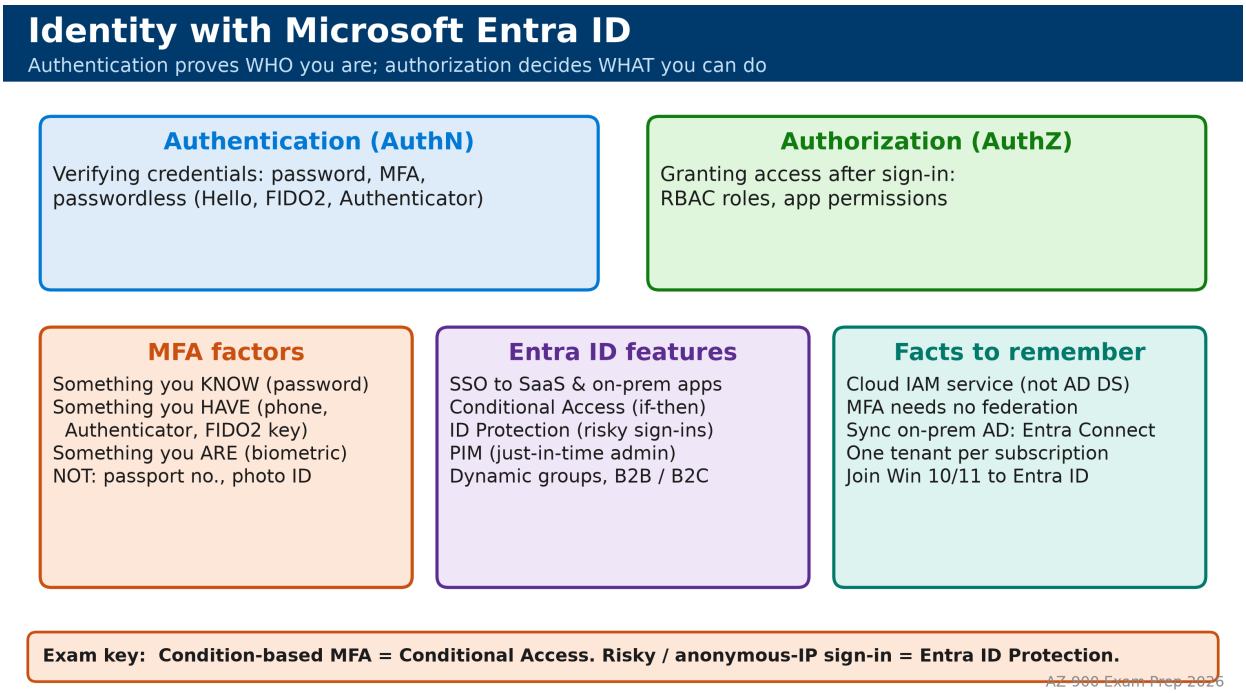


Figure Q375: Identity with Microsoft Entra ID

About this figure

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- Entra Domain Services provides managed LDAP, Kerberos and Group Policy for legacy apps without running your own domain controllers.

Question 376

AZURE ARCHITECTURE & SERVICES

Security v5 Q511

Review the bracketed text. If it makes the statement correct, select 'No change is needed'. Otherwise, select the answer that makes the statement correct. Microsoft Defender for Cloud is a [security information and event management (SIEM) solution] that strengthens security posture and protects workloads across Azure, hybrid and multicloud environments.

✖ A. No change is needed

✔ **B.** cloud-native application protection platform (CNAPP)

✖ C. identity and access management service

✖ D. key and secret management service

Correct answer: cloud-native application protection platform (CNAPP)

Clarification (read this first): Here the [bracketed] text is wrong. A SIEM (a tool that collects and analyzes security logs) is Microsoft Sentinel - see Q178. Defender for Cloud is a CNAPP: it checks security settings and protects workloads.

Answer explained

✔ **cloud-native application protection platform (CNAPP)**
Defender for Cloud = CNAPP (cloud-native application protection platform: posture + workload protection). Microsoft Sentinel = SIEM/SOAR.

✖ **No change is needed**
SIEM describes Microsoft Sentinel, not Defender for Cloud.

✖ **Identity and access management service**
That describes Microsoft Entra ID.

✖ **Key and secret management service**
That describes Azure Key Vault.

EXAM STRATEGY
Signal: 'posture' plus 'protects workloads' = CNAPP (Defender for Cloud). SIEM is Microsoft Sentinel. 'Collect, correlate and hunt through logs' points to Sentinel.

IMPACT
Buying the wrong product leaves a gap. Defender for Cloud makes resources more secure; Sentinel investigates incidents. Most companies need both.

What · Where · When · Why · How

What	A CNAPP that gives CSPM (posture management) and workload protection.
Where	Across Azure, AWS, GCP and on-premises.
When	All the time, for posture checks and threat detection.
Why	To make resources more secure and detect threats to workloads.
How	Turn on foundational CSPM, then turn on Defender plans for each workload type.

Exam tip
Defender for Cloud = protect and harden (CNAPP). Sentinel = collect and investigate (SIEM).

Memory trick
Defender defends workloads. Sentinel stands watch over logs.

Real-world example, use case and scenario

EXAMPLE
Container security
A finance app company uses Defender for Containers to scan AKS images for weak spots.

USE CASE
SQL protection
An online shop gets alerts on SQL injection attempts from Defender for SQL.

SCENARIO
Tool choice
A security team uses Sentinel to link logs and Defender for Cloud for workload alerts.

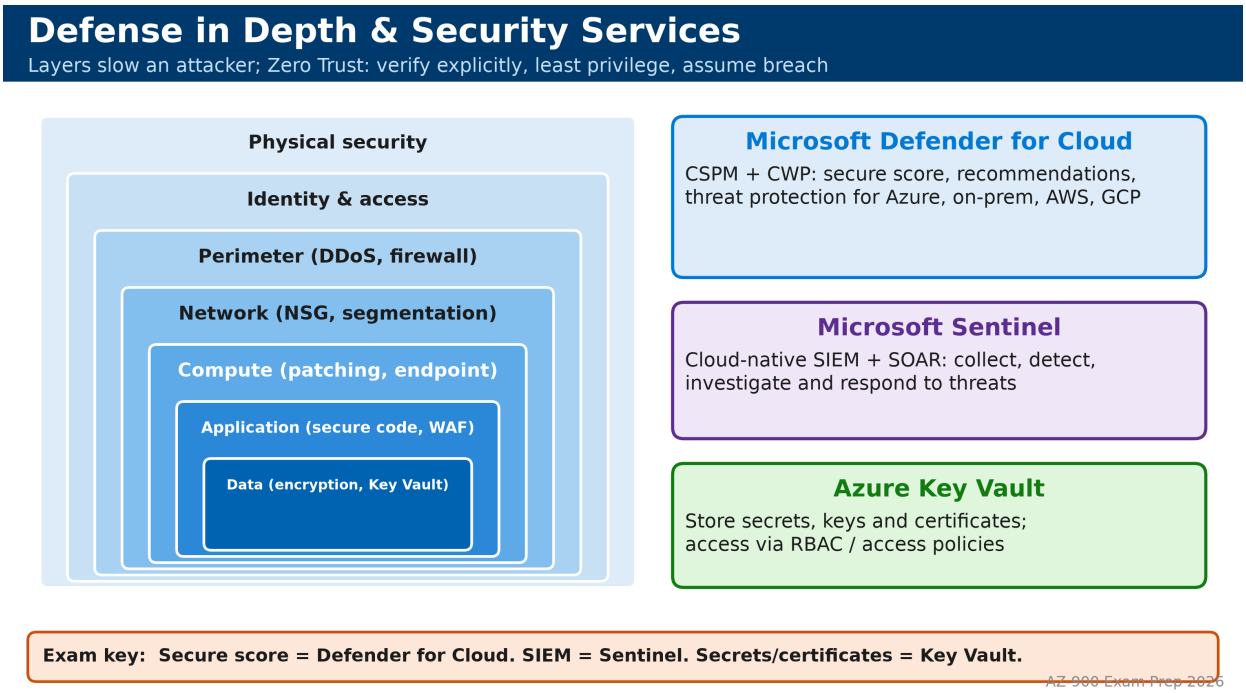


Figure Q376: Defense in depth and security services

About this figure

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- Defender for Cloud's foundational CSPM and secure score are free; paid plans such as Defender for Servers cost approx. USD 5 to 15 per server-month.
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Question 377

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q512

Review the bracketed text. If it makes the statement correct, select 'No change is needed'. Otherwise, select the answer that makes the statement correct. Single sign-on (SSO) enables users to [sign in once with one set of credentials and access multiple applications].

✔ **A.** No change is needed

✖ B. provide two or more forms of verification

✖ C. sign in without any credentials

✖ D. reset their own passwords

Correct answer: No change is needed

Clarification (read this first): Here the [bracketed] text is already correct, so choose 'No change is needed'. Do not confuse SSO (one sign-in, many apps) with MFA (more than one proof at sign-in) - Q378 tests that.

Answer explained

✔ **No change is needed**
SSO = sign in once, with one set of credentials, and use many applications.

✖ **Provide two or more forms of verification**
That describes multifactor authentication (MFA).

✖ **Sign in without any credentials**
SSO still needs a first sign-in with credentials.

✖ **Reset their own passwords**
That describes self-service password reset (SSPR).

EXAM STRATEGY
Key fact: 'sign in once, many apps' is the exact definition of SSO. Remove 'two or more verifications' (MFA) and 'reset passwords' (SSPR).

IMPACT
SSO with Entra ID means fewer passwords. Disabling one account removes access to every connected SaaS app at once.

What · Where · When · Why · How

What	Single sign-on across applications.
Where	Microsoft Entra ID, for Microsoft 365, Azure and thousands of SaaS apps.
When	After the user signs in once in a session.
Why	To reduce password tiredness and make credentials easier to manage.
How	Apps trust tokens (digital proof of sign-in) from Entra ID instead of their own passwords.

Exam tip
SSO = one login, many apps. MFA = many proofs, one login.

Memory trick
Single sign-on: sign once, sail everywhere.

Real-world example, use case and scenario

EXAMPLE
Hospital staff
Nurses sign in once to use scheduling, email and HR apps.

USE CASE
SaaS apps
A marketing firm connects Salesforce and Slack to Entra ID for SSO.

SCENARIO
Leaving staff
Disabling one Entra ID account removes access to every SSO-connected app.

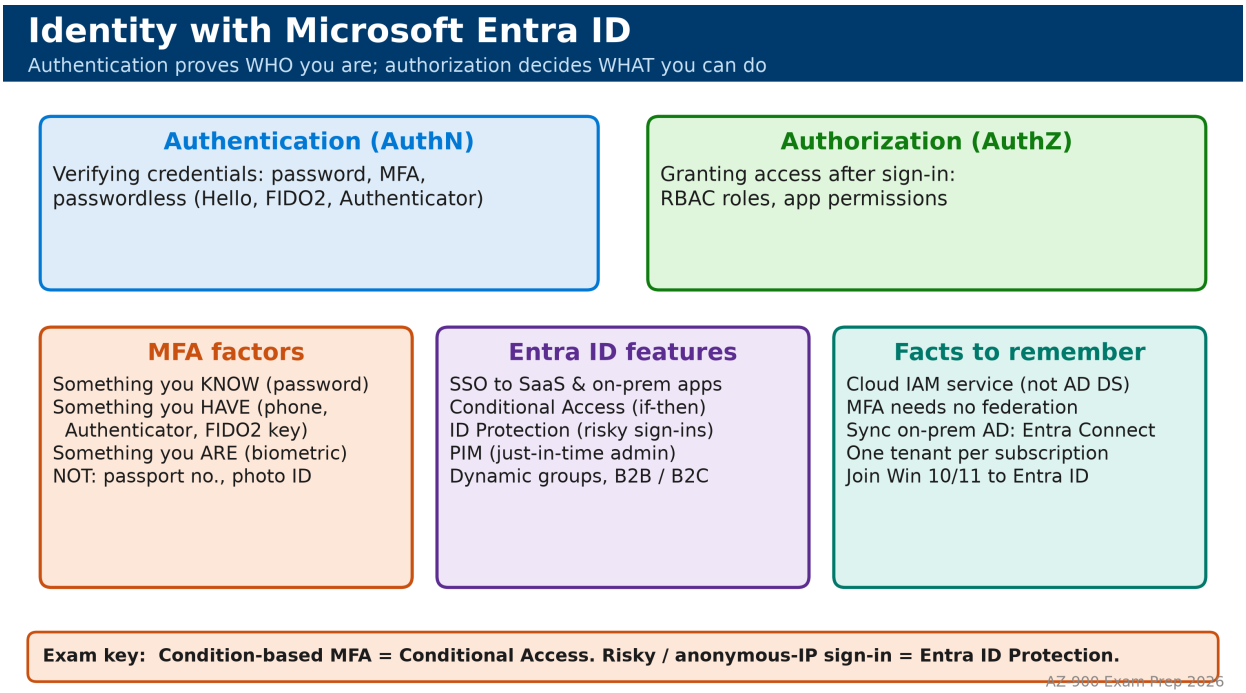


Figure Q377: Identity with Microsoft Entra ID

About this figure

- Top boxes separate authentication (who are you) from authorisation (what may you do); bottom boxes list MFA factors, Entra ID features and key facts.
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Question 378

AZURE ARCHITECTURE & SERVICES

Identity
v5 Q513

Review the bracketed text. If it makes the statement correct, select 'No change is needed'. Otherwise, select the answer that makes the statement correct. Requiring a user to enter a password and then approve a notification in the Microsoft Authenticator app is an example of [single sign-on].

✖ A. No change is needed

✔ **B.** multifactor authentication

✖ C. Conditional Access

✖ D. role-based access control

Correct answer: multifactor authentication

Clarification (read this first): The bracketed words are 'single sign-on'. The question asks what you call a password PLUS a phone approval. That is two checks, so it is MFA, not SSO. Q377 is the look-alike: there SSO (sign in once, use many apps) is correct as written.

Answer explained

✔ **multifactor authentication**
A password plus an Authenticator approval uses two factors (two kinds of proof). Two factors means MFA.

✖ **No change is needed**
SSO (single sign-on) means you sign in once for many apps. It does not add a second check.

✖ **Conditional Access**
Conditional Access (rules that decide when to ask for more checks) can require MFA. But the password plus approval itself is MFA.

✖ **Role-based access control**
RBAC (role-based access control) decides what you may do after sign-in. It does not check who you are.

EXAM STRATEGY
Count the kinds of proof. Password = something you know. Phone approval = something you have. Two kinds = MFA. SSO is about how many times you sign in, not how many checks.

IMPACT
MFA stops most attacks that use stolen passwords. Calling it SSO leads to wrong security rules.

What · Where · When · Why · How

What	Multifactor authentication (MFA) with two kinds of proof.
Where	In Microsoft Entra ID sign-ins with the Authenticator app.
When	At sign-in, often turned on by security defaults or Conditional Access.
Why	To stop attackers who have a stolen password.
How	The user types a password, then taps Approve on a phone notification.

Exam tip
Know + Have + Are (password, phone, fingerprint): any two kinds = MFA.

Memory trick
MFA = More Factors Always.

Real-world example, use case and scenario

EXAMPLE
Online banking
Credit union staff type a password and then approve an Authenticator prompt.

USE CASE
Remote access
A design studio requires MFA for staff who connect by VPN (secure remote connection).

SCENARIO
Stolen password
An attacker has a stolen password, but the user denies the MFA prompt, so the attacker is stopped.

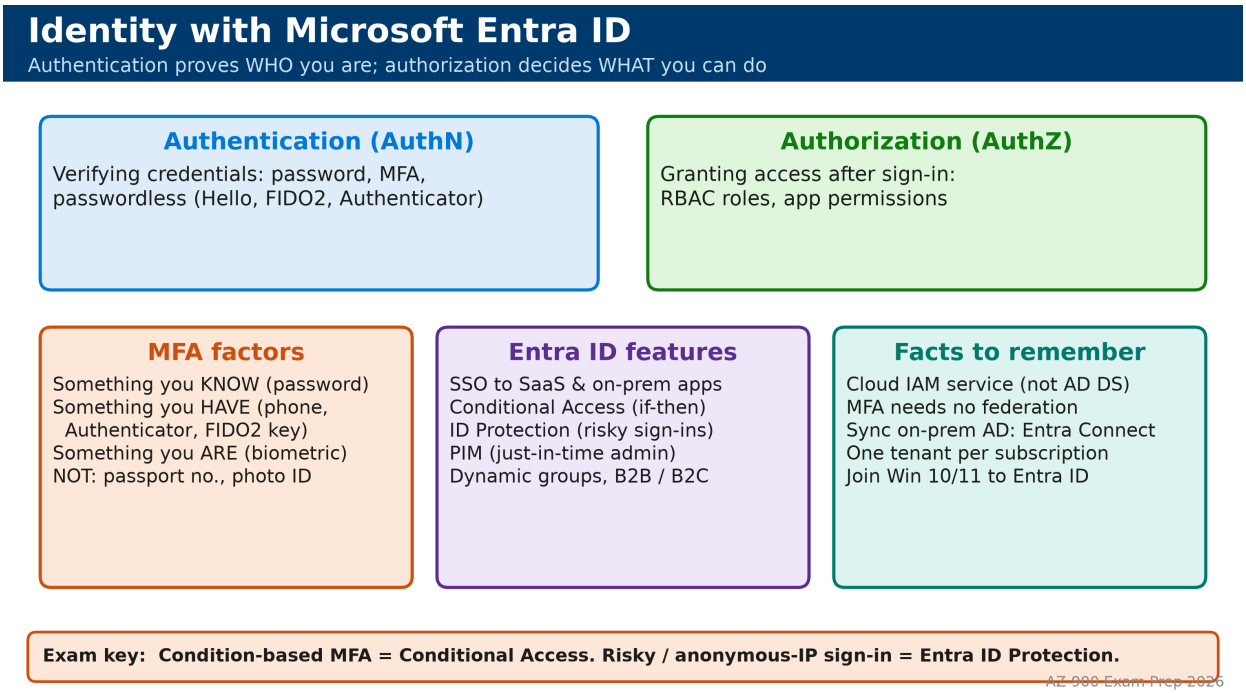


Figure Q378: Identity with Microsoft Entra ID

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Question 379

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q514

Your finance team wants to be notified automatically when daily Azure spending suddenly spikes far above its normal pattern, even if no budget threshold has been reached. What should you use?

✔ **A.** Anomaly alerts in Microsoft Cost Management

✖ B. Azure Service Health alerts

✖ C. Azure Advisor cost recommendations

✖ D. Application Insights smart detection

Correct answer: Anomaly alerts in Microsoft Cost Management

Clarification (read this first): Watch the words 'suddenly spikes above its normal pattern' and 'no budget threshold'. A budget alert needs a fixed amount you set. An anomaly alert needs no amount. Q64 is the look-alike: alert when cost passes a set limit = budget alerts.

Answer explained

✔ **Anomaly alerts in Microsoft Cost Management**
Cost Management anomaly alerts warn you about unusual spending jumps. They compare with past spending, so you set no fixed amount.

✖ **Azure Service Health alerts**
Service Health tells you about Azure outages and planned maintenance. It does not watch your spending.

✖ **Azure Advisor cost recommendations**
Advisor suggests ways to save money, like using smaller VMs. It does not alert on sudden spending jumps.

✖ **Application Insights smart detection**
Smart detection finds unusual app errors and slow responses. It does not look at your bill.

EXAM STRATEGY
Clue words: 'spikes above normal' and 'no budget threshold' = anomaly alerts. Budgets need a fixed amount. Service Health is about outages. Advisor gives savings advice.

IMPACT
You find runaway costs within a day, not at the end of the month. Examples: leaked keys used to mine crypto, or a function stuck in a loop.

What · Where · When · Why · How

What	Anomaly alerts are messages about unusual cost patterns.
Where	You set them up in Microsoft Cost Management in the Azure portal.
When	Use them when you want a warning about surprise jumps without setting a limit.
Why	They catch runaway resources or wrong settings early.
How	Create an anomaly alert rule on a subscription and add email addresses.

Exam tip
Fixed amount = budget alert. Unusual pattern = anomaly alert. Both are in Cost Management.

Memory trick
A budget is a fence you build. An anomaly alert is a smoke detector that notices something odd.

Real-world example, use case and scenario

EXAMPLE
Runaway loop
A dev team's Function App gets stuck in a retry loop overnight. Next morning an anomaly alert emails the cost lead.

USE CASE
Retail peak monitoring
A retailer watches for odd spending outside holiday seasons. A budget set high enough for holidays would not trigger then.

SCENARIO
Forgotten GPU VMs
A research lab forgot to stop GPU VMs after a test. The daily cost jump shows as an anomaly.

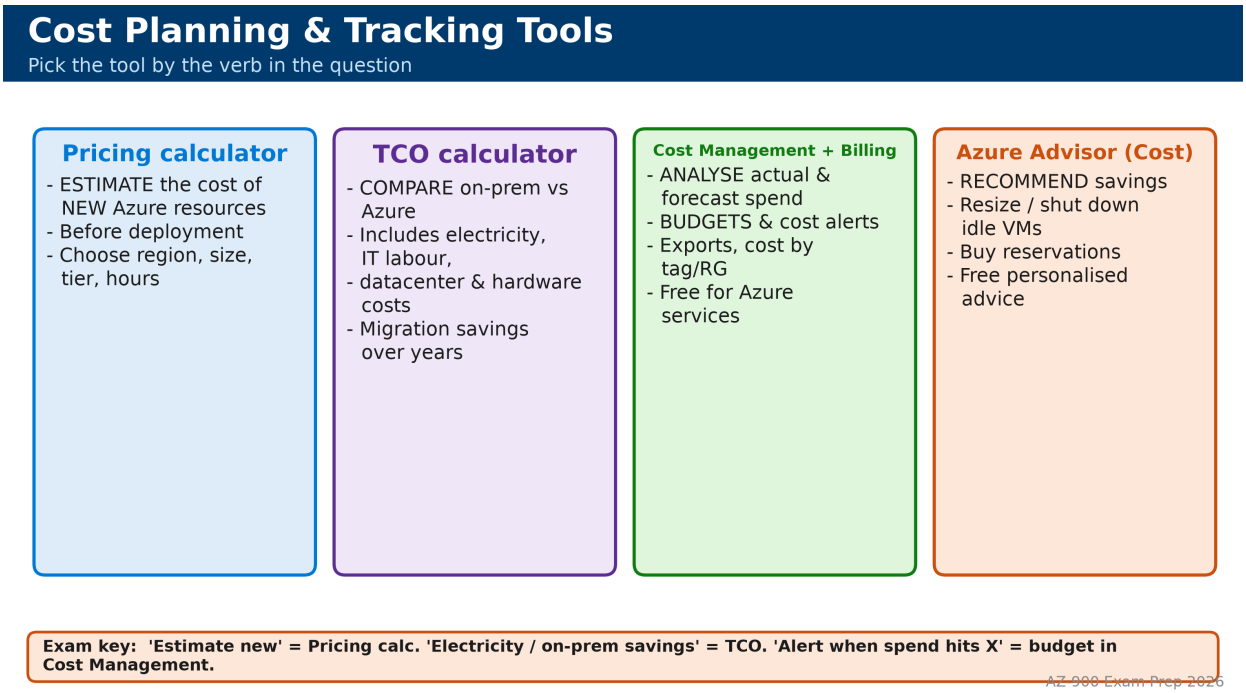


Figure Q379a: Cost planning and tracking tools

About this figure

• Each column is one tool; pick it by the question's verb: estimate (Pricing calculator), compare (TCO), analyse or alert (Cost Management), recommend (Advisor).

• Pricing and TCO calculators are free websites that need no Azure subscription; estimates use list prices and can be saved or exported to Excel.

• Cost Management is free for Azure resources; budgets can be set monthly, quarterly or yearly with alerts at thresholds such as 50%, 80% or 100%.

• Budgets alert but do not stop spending; only the free account and some credit offers have a spending limit that disables resources when exhausted.

• Advisor cost recommendations include resizing or shutting down VMs with low CPU and buying reservations or savings plans based on 30 days of usage.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
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Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
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Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
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Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q379b: Approximate Azure charges (USD)

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- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 380

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q515

A manager wants an interactive chart of the last six months of Azure charges, grouped by service name and resource group, with the ability to filter by tag. Which feature should you use?

✔ **A.** Cost analysis in Microsoft Cost Management

✖ B. Azure Pricing calculator

✖ C. Total Cost of Ownership (TCO) calculator

✖ D. Azure Monitor metrics explorer

Correct answer: Cost analysis in Microsoft Cost Management

Clarification (read this first): Watch 'last six months of Azure charges'. This is money already spent. Cost analysis shows past and forecast spending. The Pricing and TCO calculators only estimate future costs.

Answer explained

✔ **Cost analysis in Microsoft Cost Management**
Cost analysis shows real past costs and forecast costs. You can group and filter them by service, resource group or tag.

✖ **Azure Pricing calculator**
The Pricing calculator estimates costs for things you plan to deploy. It cannot see your real charges.

✖ **Total Cost of Ownership (TCO) calculator**
The TCO (total cost of ownership) calculator compares on-premises costs with Azure before a move. It does not show past spending.

✖ **Azure Monitor metrics explorer**
Metrics explorer charts performance numbers like CPU. It does not show money.

EXAM STRATEGY
Clue words: 'charges', 'last six months', 'group by' and 'filter by tag' = Cost analysis. The calculators only estimate.

IMPACT
Cost analysis is the daily tool for charging costs back to each team. Without it, owners cannot see who spends what.

What · Where · When · Why · How

What	Cost analysis is the report view of Microsoft Cost Management.
Where	It is under Cost Management in the Azure portal, for any billing or resource scope (level).
When	Use it to review past spending or to forecast the month-end cost.
Why	It shows where money is really being spent.
How	Pick a scope, pick a view, then group by service, resource group or tag.

Exam tip
Real past cost = Cost Management. Planned future cost = Pricing calculator. On-premises vs cloud = TCO.

Memory trick
Analysis looks back; calculators look forward.

Real-world example, use case and scenario

EXAMPLE
Monthly review
An IT manager groups last quarter's spending by service and sees storage costs doubled.

USE CASE
Chargeback
A university groups costs by the Department tag to bill each faculty for its cloud use.

SCENARIO
Forecast check
A startup uses the forecast line to see if it will pass its monthly money limit.

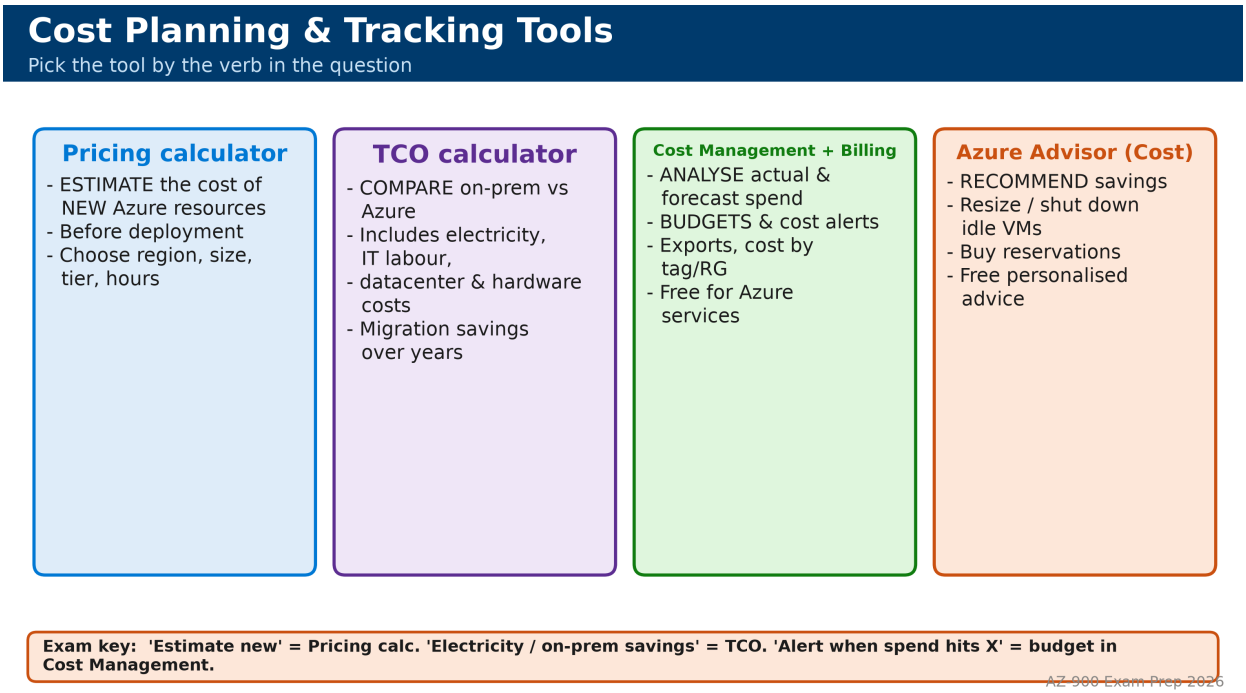


Figure Q380a: Cost planning and tracking tools

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Approximate Azure Charges (USD)

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Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q380b: Approximate Azure charges (USD)

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Question 381

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q516

Your company's BI team needs a daily CSV file of Azure cost and usage data delivered automatically to an Azure Storage account so they can load it into Power BI. What should you configure?

✔ **A.** A scheduled export in Microsoft Cost Management

✖ B. An Azure Policy initiative

✖ C. An Azure Monitor diagnostic setting

✖ D. A budget with an action group

Correct answer: A scheduled export in Microsoft Cost Management

Answer explained

✔ **A scheduled export in Microsoft Cost Management**
Cost Management exports send cost and usage data files to an Azure Storage account on a repeating schedule.

✖ **An Azure Policy initiative**
A Policy initiative (a group of policy rules) checks compliance. It does not export billing data.

✖ **An Azure Monitor diagnostic setting**
A diagnostic setting sends resource logs and metrics. It does not send cost data.

✖ **A budget with an action group**
A budget sends alerts or starts actions at a set amount. It does not create data files.

EXAM STRATEGY
Clue words: 'daily file delivered to Storage' = scheduled export. Policy is governance, diagnostic settings are resource logs, budgets are alerts only.

IMPACT
Exports feed Power BI and cost tools automatically. Manual downloads are slow, error-prone and soon out of date.

What · Where · When · Why · How

What	An export is a scheduled job that writes cost data to files.
Where	The files go into a container in an Azure Storage account you choose.
When	Use it when other tools need regular cost data.
Why	It automates reporting, so nobody downloads files by hand.
How	In Cost Management, open Exports, choose how often, the data set, and the target storage.

Exam tip
Automatic, repeating cost data to storage = Exports.

Memory trick
Export = Express delivery of cost data to storage.

Real-world example, use case and scenario

EXAMPLE
Daily feed
A daily export lands in a blob container. A Power BI report refreshes from it each morning.

USE CASE
Finance data lake
A bank keeps three years of exported cost files in a data lake for audit trends.

SCENARIO
Partner billing
A managed service provider exports monthly costs per subscription to bill its clients.

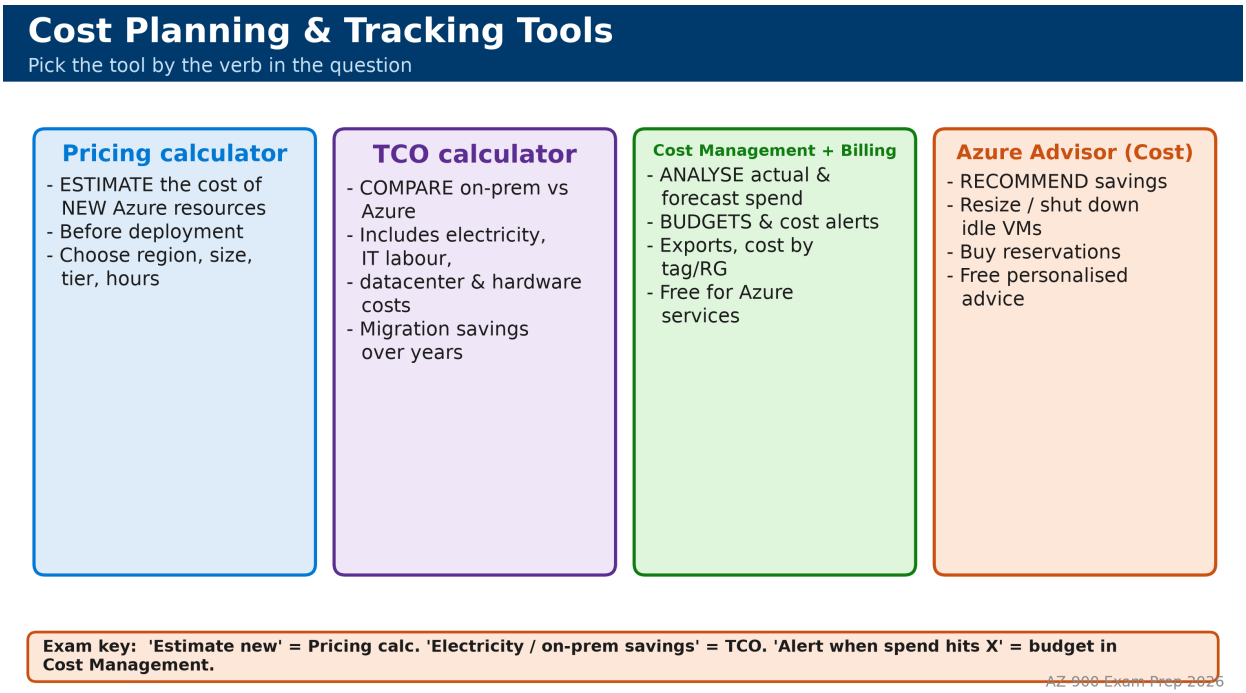


Figure Q381a: Cost planning and tracking tools

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• Advisor cost recommendations include resizing or shutting down VMs with low CPU and buying reservations or savings plans based on 30 days of usage.

Approximate Azure Charges (USD)

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Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q381b: Approximate Azure charges (USD)

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Question 382

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q517

You create a monthly budget of USD 1,000 in Microsoft Cost Management with an alert at 100%. Statement: When spending reaches USD 1,000, the budget automatically stops all resources in the subscription. Is the statement true?

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): Watch 'automatically stops all resources'. A budget only sends alerts. It never stops anything by itself. Resources stop only if you add your own automation through an action group. Q383 (budget by tag) is Yes; this one is No.

Answer explained

✔ **No**
A budget only sends alerts by default. To stop resources, you must add automation through an action group (a list of actions to run).

✖ **Yes**
Budgets never stop resources on their own. They only notify, unless you add your own automation.

EXAM STRATEGY
Key fact: budgets alert; they do not stop resources. 'Automatically stops' = No, unless the question mentions an action group running automation.

IMPACT
Thinking a budget caps spending is costly. Resources keep billing past the limit unless you set up automation yourself.

What · Where · When · Why · How

What	A budget is a spending amount with alert rules.
Where	You create budgets in Microsoft Cost Management at a chosen scope (level).
When	Alerts fire when real or forecast cost passes a set percentage.
Why	They give an early warning so people can act before overspending.
How	Set an amount, a time period and alert levels. You can add an action group.

Exam tip
Budget = alert, not a hard limit. Only the free account and some credit offers have a real spending limit.

Memory trick
A budget barks; it does not bite.

Real-world example, use case and scenario

EXAMPLE
Over budget
A team reaches 100% of its budget. Managers get an email, but the VMs keep running.

USE CASE
Dev shutdown
A software company adds an action group that runs a script to stop dev VMs at 90% of budget.

SCENARIO
Marketing campaign
A marketing team gets a forecast alert at 80% mid-month and pauses a non-essential campaign site.

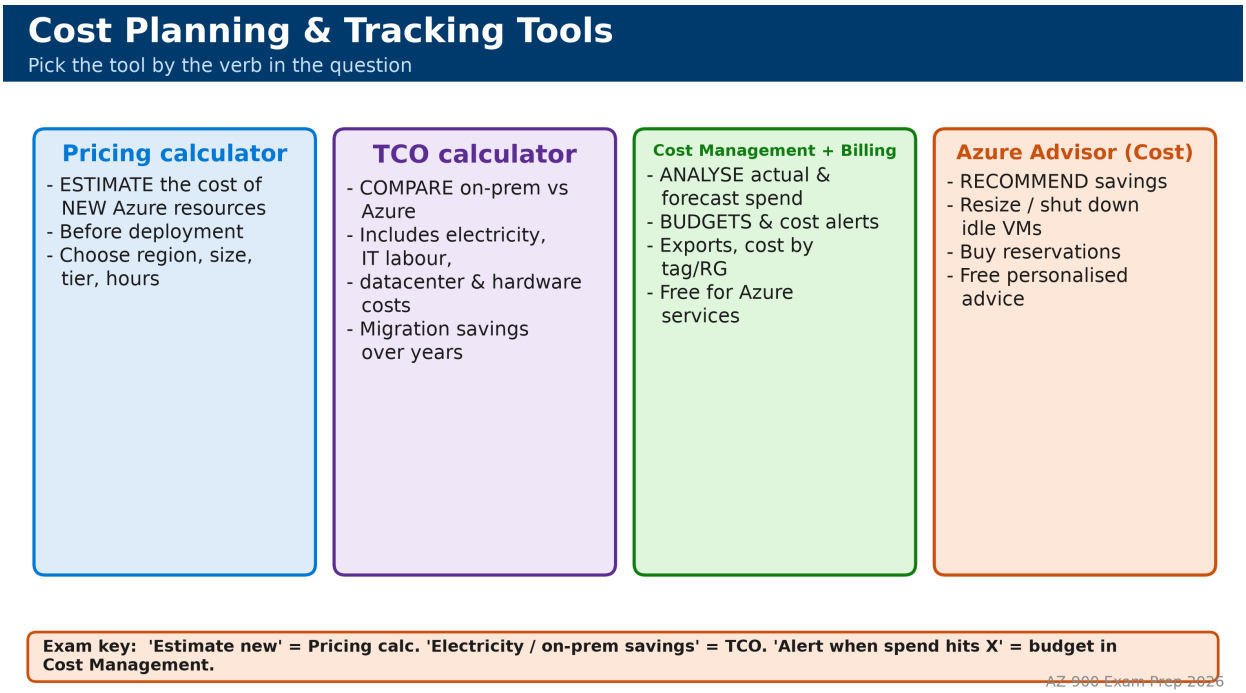


Figure Q382a: Cost planning and tracking tools

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• Advisor cost recommendations include resizing or shutting down VMs with low CPU and buying reservations or savings plans based on 30 days of usage.

Approximate Azure Charges (USD)

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Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q382b: Approximate Azure charges (USD)

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Question 383

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q518

Statement: In Microsoft Cost Management, you can create a budget scoped to a single resource group and filtered to resources with the tag Project=Apollo. Is the statement true?

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): The question asks if a budget can be this narrow: one resource group AND only resources with one tag. Yes, both are allowed. Do not mix it up with Q382 (budget stopping resources), which is No.

Answer explained

✔ **Yes**
Budgets can be set on a resource group and filtered by tags. So a budget for one project is possible.

✖ **No**
Budgets are not limited to whole subscriptions. They can also be filtered by tag values.

EXAM STRATEGY
Key fact: budgets have scopes (management group, subscription, resource group) and filters (tags, resources, services). Resource group plus tag filter is allowed = Yes.

IMPACT
Each project owner gets an early warning. One team's overspending does not hide inside a shared subscription.

What · Where · When · Why · How

What	A tag-filtered budget tracks spending for only some resources.
Where	You define it in Cost Management at the resource group scope (level).
When	Use it when one project shares a resource group or subscription with others.
Why	It gives project owners a cost alert for their own work only.
How	Create a budget, add the filter Tag: Project = Apollo, and set alert levels.

Exam tip
Budgets can be set at management group, subscription and resource group level, and filtered by tags.

Memory trick
Tags turn one big budget into many small ones.

Real-world example, use case and scenario

EXAMPLE
Project budget
Project Apollo's lead gets alerts only for resources tagged Project=Apollo.

USE CASE
Grant tracking
A research institute tracks each grant-funded project's cloud spending with tag-filtered budgets.

SCENARIO
Shared resource group
Two app teams share one resource group. Each team has its own tag-filtered budget and alerts.

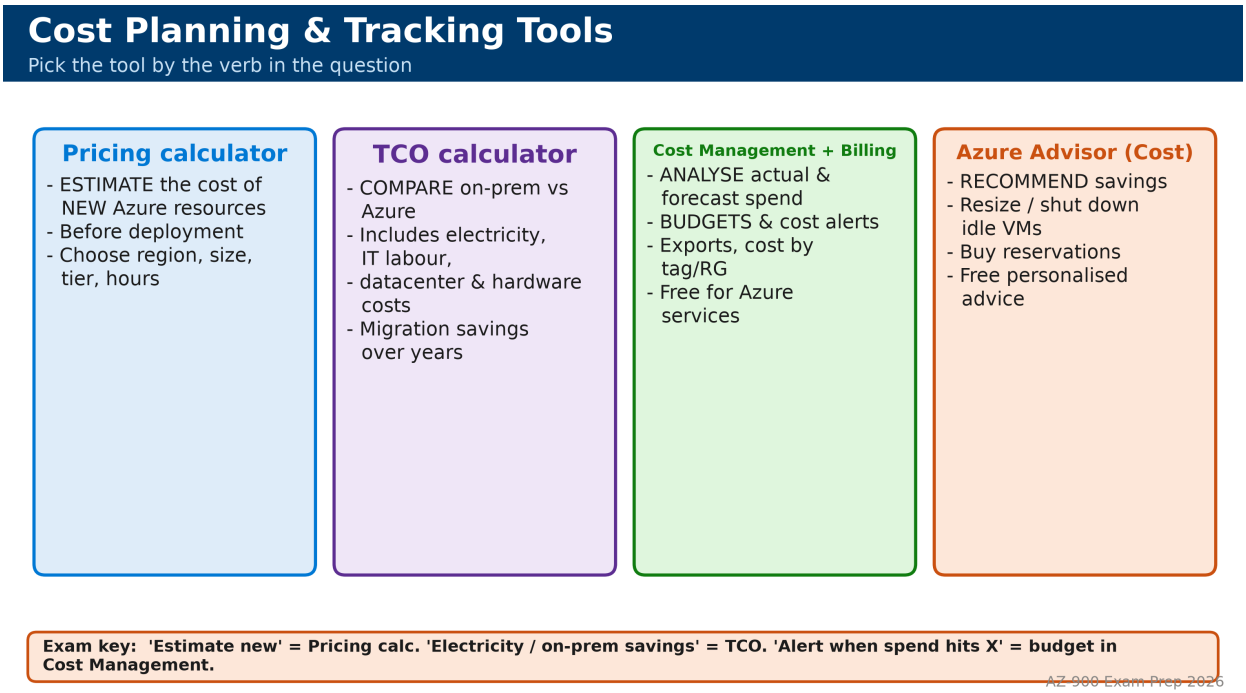


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Approximate Azure Charges (USD)

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Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

AZ-900 Exam Prep 2026

Figure Q383b: Approximate Azure charges (USD)

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- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 384

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q519

Statement: You must sign in with an active Azure subscription before you can build an estimate in the Azure Pricing calculator. Is the statement true?

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): Watch the word 'must'. You do NOT need a subscription or a sign-in to use the Pricing calculator. Signing in is optional; it only lets you save your estimates.

Answer explained

✔ **No**
The Pricing calculator is a public web page. You need no subscription and no sign-in to make estimates.

✖ **Yes**
No subscription is needed. Signing in is optional and only lets you save estimates.

EXAM STRATEGY
Key fact: the Pricing calculator is a public web tool. 'Must sign in' or 'needs a subscription' = No.

IMPACT
Sales staff and architects can price a solution for a customer who has no Azure account yet. This speeds up quotes and approvals.

What · Where · When · Why · How

What	The Pricing calculator estimates the cost of Azure products you plan to use.
Where	It is a public page on the Azure website.
When	Use it before you deploy, to plan budgets.
Why	It shows expected costs before you commit.
How	Add products, choose region, tier and amount, then review or export the estimate.

Exam tip
The Pricing and TCO calculators are both free web tools. No subscription needed.

Memory trick
Calculators are open to everyone, like a price tag in a shop window.

Real-world example, use case and scenario

EXAMPLE
Pre-sales quote
A consultant makes an estimate for a client who has no Azure account yet.

USE CASE
Grant proposal
A nonprofit estimates storage and App Service costs for a funding application.

SCENARIO
Student project
A student compares VM sizes in the calculator before signing up for a free account.

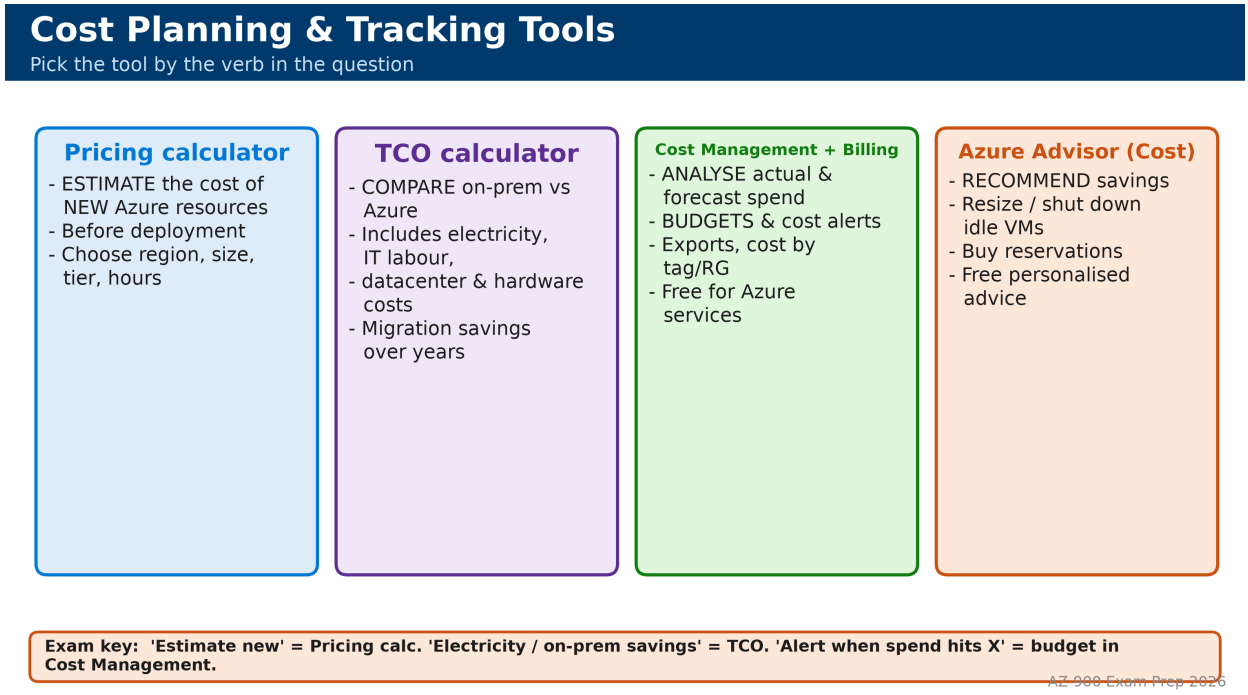


Figure Q384a: Cost planning and tracking tools

About this figure

- Each column is one tool; pick it by the question's verb: estimate (Pricing calculator), compare (TCO), analyse or alert (Cost Management), recommend (Advisor).
- Pricing and TCO calculators are free websites that need no Azure subscription; estimates use list prices and can be saved or exported to Excel.
- Cost Management is free for Azure resources; budgets can be set monthly, quarterly or yearly with alerts at thresholds such as 50%, 80% or 100%.
- Budgets alert but do not stop spending; only the free account and some credit offers have a spending limit that disables resources when exhausted.
- Advisor cost recommendations include resizing or shutting down VMs with low CPU and buying reservations or savings plans based on 30 days of usage.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q384b: Approximate Azure charges (USD)

About this figure

• Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.

• Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.

• Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.

• Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.

• Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 385

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q520

Your company requires every resource to carry the CostCenter tag of its resource group. When a resource is created or updated without the tag, Azure should add it automatically. Which Azure Policy effect should the assigned policy use?

✔ **A.** Modify

✖ B. Deny

✖ C. Audit

✖ D. Disabled

Correct answer: Modify

Clarification (read this first): Watch 'add it automatically'. The policy must CHANGE the resource, so the effect is Modify. Q386 is the look-alike: 'see which ... must NOT block' = Audit. Also note tags are not inherited by default (Q129).

Answer explained

✔ **Modify**
Modify changes resource properties, such as tags, when a resource is created or updated. This lets resources copy the resource group's tag.

✖ **Deny**
Deny blocks the request. It does not add the missing tag.

✖ **Audit**
Audit only records the problem. It does not change the resource.

✖ **Disabled**
Disabled turns the policy off completely.

EXAM STRATEGY
Clue words: 'add it automatically' = Modify (changes tags or settings). Deny blocks, Audit only reports, Disabled does nothing. The built-in 'inherit a tag' policies use Modify.

IMPACT
Automatic tags keep cost reports correct without trusting every engineer to tag by hand. No more 'untagged' costs.

What · Where · When · Why · How

What	Modify is an Azure Policy effect (what the policy does) that changes resource settings.
Where	It is set in the policy definition and assigned to a scope (level).
When	It runs when resources are created or updated, or during remediation (fixing existing resources).
Why	It applies standards without blocking deployments.
How	Assign a Modify policy with a managed identity (an Azure identity for the policy) that may update tags.

Exam tip
Tags are NOT inherited by default. Use Azure Policy with Modify to copy them down.

Memory trick
Modify = Mends the resource. Deny = Door shut. Audit = Just takes notes.

Real-world example, use case and scenario

EXAMPLE
Tag inheritance
A VM created in RG-Finance automatically gets CostCenter=4100 from the group.

USE CASE
Hospital chargeback
A hospital makes sure every resource has a Department tag, so costs map to wards.

SCENARIO
Legacy cleanup
An insurer runs a remediation task to add missing tags to 2,000 existing resources.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

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Figure Q385: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 386

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q521

Your security team wants to see which storage accounts allow public network access, but must NOT block any deployments while they assess the impact. Which Azure Policy effect should they use?

✔ **A.** Audit

✖ B. Deny

✖ C. Modify

✖ D. DeployIfNotExists

Correct answer: Audit

Clarification (read this first): Watch 'must NOT block'. The team only wants to SEE problems, so the answer is Audit. Q385 is the look-alike: 'add it automatically' = Modify. Deny would block deployments.

Answer explained

✔ **Audit**
Audit reports resources that break the rule. It does not stop or change any deployment.

✖ **Deny**
Deny blocks deployments that break the rule. The team does not want that.

✖ **Modify**
Modify changes resource settings automatically. It does more than report.

✖ **DeployIfNotExists**
DeployIfNotExists deploys extra resources or settings when they are missing. It is not report-only.

EXAM STRATEGY
Clue words: 'see which' plus 'must NOT block' = Audit. Deny blocks. Modify and DeployIfNotExists change resources. Teams usually start with Audit, then switch to Deny.

IMPACT
Going straight to Deny can break production pipelines. Auditing first shows the impact, so the rule does not cause an outage.

What · Where · When · Why · How

What	Audit is a report-only Azure Policy effect.
Where	Results show in the Azure Policy compliance view.
When	Use it while you discover problems, before you enforce a rule.
Why	It shows the impact without disturbing teams.
How	Assign the policy with effect Audit and check the compliance percentage.

Exam tip
Start with Audit. Switch to Deny once you know the impact.

Memory trick
Audit = an auditor with a clipboard: watches, writes, never stops you.

Real-world example, use case and scenario

EXAMPLE
Public access check
The dashboard shows 14 of 60 storage accounts allow public access.

USE CASE
Pre-enforcement
A bank audits allowed VM sizes for a month before switching the policy to Deny.

SCENARIO
Merger assessment
After buying another company, IT audits the new subscriptions against company rules.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

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Figure Q386: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 387

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q522

Your company runs a web app on Azure VMs. Which TWO changes would directly affect the monthly Azure bill? Each correct answer presents part of the solution.

✔ **A.** Changing the VM size from D2s v5 to D8s v5

✔ **B.** Increasing the volume of data sent out of Azure to internet users

✖ C. Adding more users to Microsoft Entra ID Free

✖ D. Creating additional resource groups

✖ E. Applying more tags to the VMs

Correct answer: Changing the VM size from D2s v5 to D8s v5; Increasing the volume of data sent out of Azure to internet users

Clarification (read this first): This question asks for TWO answers. Ask of each option: 'Do I pay for this?' VM size and data going out of Azure cost money. Free Entra ID users, resource groups and tags are free.

Answer explained

✔ **Changing the VM size from D2s v5 to D8s v5; Increasing the volume of data sent out of Azure to internet users**
VM size sets the compute price. Egress (data leaving Azure) is charged per GB. Both change the bill.

✖ **Adding more users to Microsoft Entra ID Free**
Microsoft Entra ID Free users have no Azure usage charge.

✖ **Creating additional resource groups**
Resource groups are free folders (logical containers) for resources.

✖ **Applying more tags to the VMs**
Tags are free labels (metadata). They do not change charges.

EXAM STRATEGY
Ask 'is this metered (charged by use)?'. VM size and outbound data are charged. Entra ID Free, resource groups and tags are free, so cross them out.

IMPACT
Egress and too-large VMs are common surprise costs. Right-sizing and CDN caching (copies closer to users) cut the monthly bill.

What · Where · When · Why · How

What	Cost factors are the things that decide Azure charges.
Where	They apply to every resource in every region.
When	Think about them when you design or resize workloads.
Why	Knowing them avoids surprise bills.
How	Choose the right resource sizes and reduce unneeded egress traffic.

Exam tip
Cost factors: resource type and size, usage, region, bandwidth (egress), reservations, licenses. Folders and labels are free.

Memory trick
Bigger box and more data out = bigger bill.

Real-world example, use case and scenario

EXAMPLE
Upsize
Making a VM four times bigger in vCPUs makes its compute price about four times higher.

USE CASE
Video streaming
A media company's bill grows with every terabyte streamed out to viewers.

SCENARIO
Reorganization
An admin splits resources into ten new resource groups and sees no cost change.

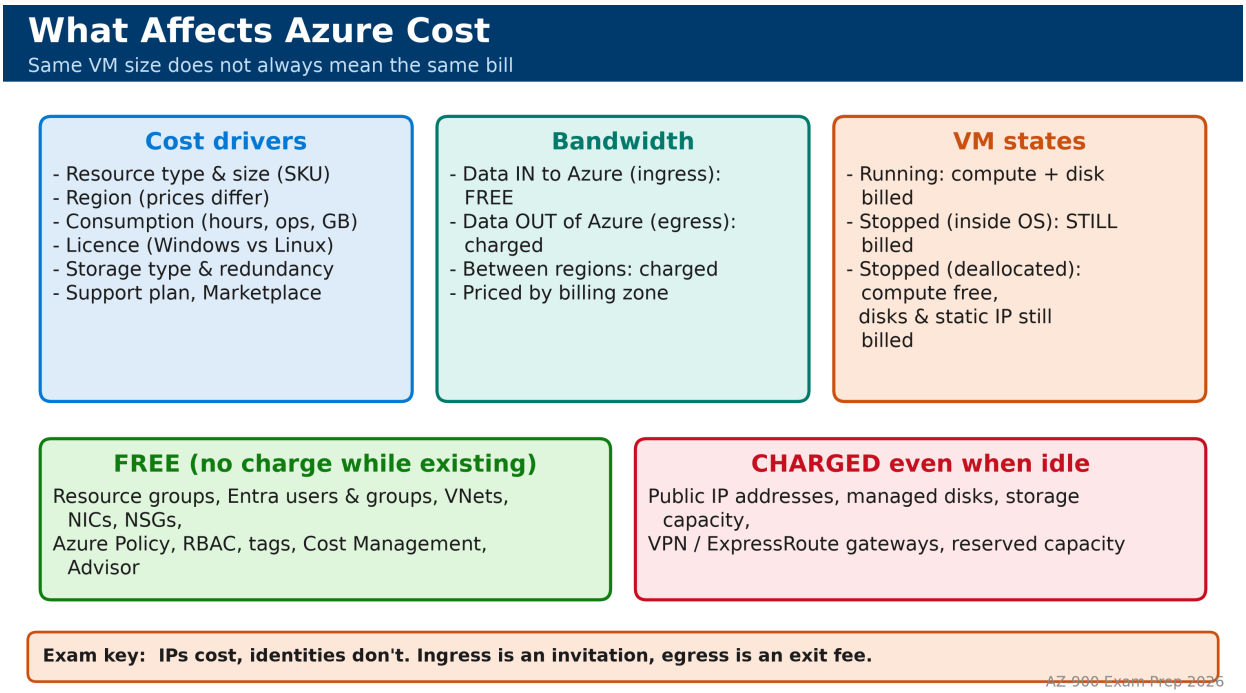


Figure Q387a: What affects Azure cost

About this figure

• Five boxes group what changes a bill: what you deploy, where, how much traffic leaves, the VM state, and resources that cost even when idle.

• Bandwidth: ingress is free; the first 100 GB/month of Internet egress is free, then approx. USD 0.087/GB in North America and Europe.

• Inter-region transfer is approx. USD 0.02/GB within a continent; VNet peering is approx. USD 0.01/GB each way in the same region.

• A VM stopped inside the guest OS is still allocated and billed for compute; only Stopped (deallocated) from the portal, CLI or API stops compute charges.

• A Standard static public IP costs approx. USD 3.65/month even unattached; a 128 GB Premium SSD costs approx. USD 20/month even when its VM is deallocated.

• Windows VMs cost more than Linux because the licence is included unless you apply Azure Hybrid Benefit; Marketplace images may add vendor fees.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q387b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 388

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q523

Statement: Microsoft Cost Management displays cost-saving recommendations, such as shutting down underused VMs, that come from Azure Advisor. Is the statement true?

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Answer explained

✔ **Yes**
Cost Management shows Azure Advisor cost tips, such as using smaller VMs or buying reservations.

✖ **No**
Advisor cost tips are shown directly inside Cost Management.

EXAM STRATEGY
Key fact: Advisor cost recommendations also appear inside Cost Management. So the statement is Yes.

IMPACT
Following these tips (right-size, shut down, reserve) often cuts compute costs with no design change.

What · Where · When · Why · How

What	Recommendations are suggested actions to lower cost.
Where	They appear in Cost Management and in Azure Advisor.
When	Review them often, especially after usage changes.
Why	They find idle or too-large resources.
How	Open Advisor recommendations in Cost Management and apply or dismiss each one.

Exam tip
Advisor's Cost category supplies the recommendations shown in Cost Management.

Memory trick
Advisor gives the advice; Cost Management shows it next to the bill.

Real-world example, use case and scenario

EXAMPLE
Idle VM
A VM at 2% CPU for two weeks is flagged to shut down or resize.

USE CASE
Reservation advice
A logistics firm buys a reservation after Advisor shows steady 24/7 VM use.

SCENARIO
Quarterly review
A cost team works through the list and cuts monthly spending by 15%.

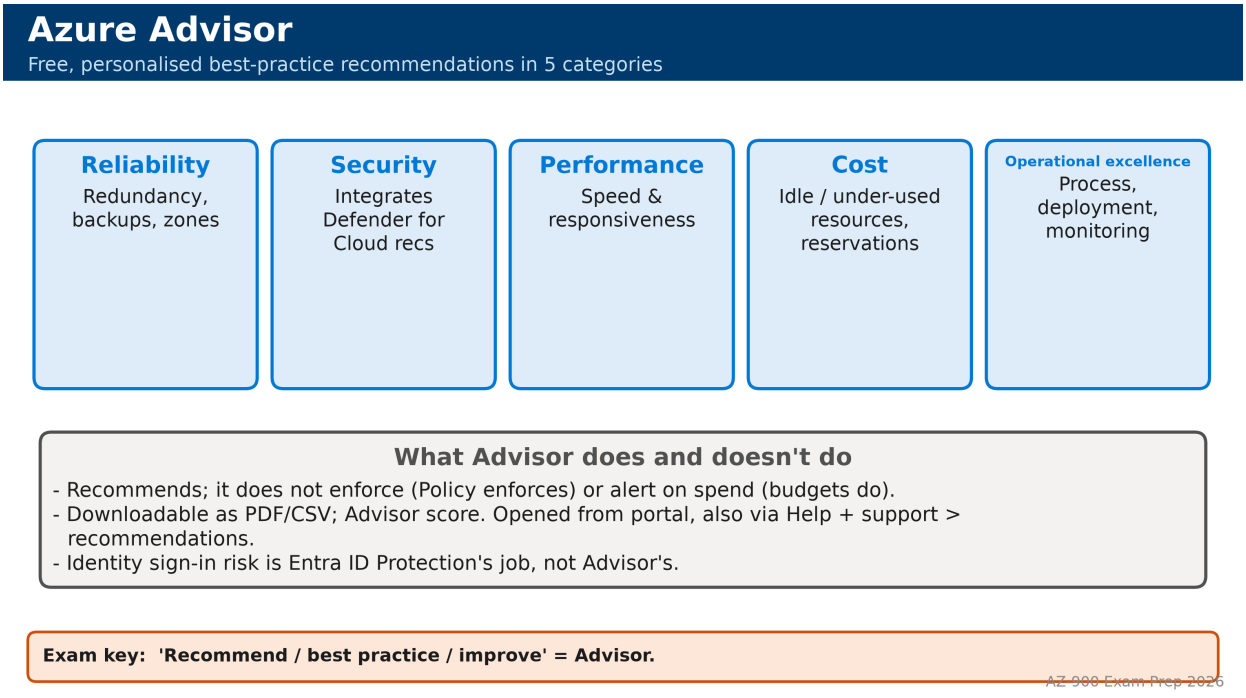


Figure Q388: Azure Advisor

About this figure

• The five boxes are Advisor's categories; each recommendation shows impact (high, medium, low) and suggested actions you can apply or postpone.

• Advisor is free and analyses your deployed resources' configuration and usage telemetry, giving subscription- and resource-group-level recommendations.

• Advisor score rates how well you follow best practices, overall and per category, helping prioritise fixes with the largest impact.

• Recommendations can trigger alerts and digest emails; security recommendations come from Defender for Cloud, so fixing them there also updates Advisor.

• Advisor is aligned with the Azure Well-Architected Framework pillars: reliability, security, cost optimisation, operational excellence and performance efficiency.

Question 389

AZURE MANAGEMENT & GOVERNANCE

Compliance
v5 Q526, Q527

Your company stores data in Azure SQL Database, on-premises SQL Server and Amazon S3. You need a single service to automatically discover and classify this data, build a catalog, and show data lineage. What should you use?

- ✔ **A.** Microsoft Purview
- ✖ B. Microsoft Defender for Cloud
- ✖ C. Azure Policy
- ✖ D. Microsoft Sentinel

Correct answer: Microsoft Purview

Answer explained

✔ **Microsoft Purview**
Purview maps, classifies (labels by type) and catalogs data in Azure, on-premises and other clouds. It also shows lineage (where data came from and where it goes).

✖ **Microsoft Defender for Cloud**
Defender for Cloud is about security posture (how safe your setup is) and threats. It does not catalog data.

✖ **Azure Policy**
Azure Policy enforces rules on Azure resource settings. It does not discover data.

✖ **Microsoft Sentinel**
Sentinel is a SIEM (security event analysis tool). It is not for data governance.

EXAM STRATEGY
Clue words: 'discover, classify, catalog, lineage' across clouds = Microsoft Purview. Defender is security, Policy is resource rules, Sentinel is security events.

IMPACT
One map of data across Azure, on-premises and AWS helps answer privacy requests (like GDPR) and find where sensitive data lives.

What · Where · When · Why · How

What	Microsoft Purview is a platform for data governance, risk and compliance.
Where	It works across Azure, on-premises, other clouds and Microsoft 365.
When	Use it when you need to know what data you have and where it is.
Why	It helps meet governance and privacy rules.
How	Register data sources, run scans, and browse the labelled data in the catalog.

Exam tip
Discover, classify, catalog or trace DATA = Purview. Settings of RESOURCES = Azure Policy.

Memory trick
Purview gives a PURe VIEW of all your data.

Real-world example, use case and scenario

EXAMPLE
PII discovery
Purview scans find columns with national ID numbers in an old SQL Server.

USE CASE
Pharma lineage
A drug company traces clinical trial data from source systems to reports for regulators.

SCENARIO
Data marketplace
Analysts at a retailer search the catalog to find an approved sales data set.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

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Figure Q389: Compliance, privacy and trust

About this figure

• Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.

• Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.

• Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.

• Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.

• The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 390

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q528

Statement: An Azure Policy assignment made at a management group applies to all subscriptions and resources beneath that management group. Is the statement true?

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Answer explained

✔ **Yes**
Policy assignments flow down. An assignment on a management group covers all subscriptions and resources under it.

✖ **No**
Policy assignments are inherited by every level below.

EXAM STRATEGY
Key fact: Policy flows down: management group > subscription > resource group > resource. An assignment at the top covers everything below = Yes.

IMPACT
One assignment on a management group enforces rules, like allowed regions, on hundreds of subscriptions, even ones added later.

What · Where · When · Why · How

What	A policy assignment applies a policy rule to a scope (level).
Where	It can be made at management group, subscription, resource group or resource level.
When	Assign it high in the hierarchy for company-wide rules.
Why	One assignment applies the same rules everywhere.
How	Pick the management group as the scope when you create the assignment. Add exclusions if needed.

Exam tip
Policies, RBAC (access roles) and locks all flow down the hierarchy. Tags do not, by default.

Memory trick
Rules flow downhill.

Real-world example, use case and scenario

EXAMPLE
Region rule
An allowed-locations policy on the root management group applies to all 40 subscriptions.

USE CASE
Government agency
A government agency enforces encryption on every storage account in all departments with one assignment.

SCENARIO
New subscription
A new subscription moved under the Production management group gets its policies at once.

What Inherits Down the Hierarchy?

Scopes: management group > subscription > resource group > resource

Setting	Inherited?	Notes
RBAC role assignment	YES	Role at RG scope = access to every resource in that RG
Azure Policy	YES	Assignment at MG / subscription applies to all children
Resource lock	YES	CanNotDelete / ReadOnly on RG protects all its resources
Tags	NO	Resources do NOT inherit RG tags (use Policy to copy them)

Resource group rules

- A resource group can contain resources from MULTIPLE regions; the RG's own location only stores its metadata.
- Resources in different RGs can talk to each other (e.g. VM in RG-A uses VNet in RG-B).
- Deleting a resource group deletes EVERY resource inside it.
- Resources can be moved between RGs and subscriptions; RGs cannot be nested or renamed.

Exam key: Roles, rules and locks roll down; tags stay put.

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Figure Q390: What inherits down the hierarchy

About this figure

• Green YES rows inherit to child scopes; the red NO row (tags) does not. Scope order: management group, subscription, resource group, resource.

• RBAC is additive: a role at a higher scope cannot be removed at a lower one, though deny assignments and locks can block actions.

• Locks apply to everyone, including Owners; to delete a locked resource you must remove the lock first. ReadOnly can break some management operations.

• Use Azure Policy effects such as 'Inherit a tag from the resource group' to copy tags automatically, since tags are not inherited by default.

• Moving a resource to a new RG changes its inherited policy and RBAC; the resource keeps its region, and some resource types cannot be moved.

Question 391

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q529

You assign an Azure Policy that uses the DeployIfNotExists effect to enable diagnostic settings on key vaults. Existing key vaults created before the assignment show as non-compliant. What should you do to bring them into compliance?

✔ **A.** Create a remediation task

✖ B. Apply a CanNotDelete resource lock

✖ C. Assign the Owner role to the key vaults

✖ D. Change the policy effect to Audit

Correct answer: Create a remediation task

Clarification (read this first): Watch 'existing key vaults created before the assignment'. New resources are fixed automatically. Old ones need a remediation task (a job that applies the fix to existing resources).

Answer explained

✔ **Create a remediation task**
A remediation task applies the Modify or DeployIfNotExists fix to resources that existed before the policy was assigned.

✖ **Apply a CanNotDelete resource lock**
A lock stops deletion. It does not turn on diagnostic settings.

✖ **Assign the Owner role to the key vaults**
RBAC gives permissions to people. It does not fix compliance.

✖ **Change the policy effect to Audit**
Audit only reports. The key vaults would stay non-compliant.

EXAM STRATEGY
Clue words: 'existing resources' plus Modify or DeployIfNotExists = remediation task. New resources are fixed on deploy; old ones need remediation.

IMPACT
Without remediation, older key vaults keep no logs. That leaves gaps that auditors and incident teams need.

What · Where · When · Why · How

What	A remediation task applies a policy's fix to existing resources.
Where	You create it from the Azure Policy compliance or Remediation page.
When	Use it after you assign Modify or DeployIfNotExists policies.
Why	Existing resources are not changed automatically.
How	Pick the assignment, choose the non-compliant resources, and start remediation.

Exam tip
New resources are fixed when deployed. Existing ones need remediation.

Memory trick
Remediation = repair for the past.

Real-world example, use case and scenario

EXAMPLE
Diagnostics
A remediation task turns on logging for 120 existing key vaults in minutes.

USE CASE
Backup enforcement
A healthcare provider remediates existing VMs so they all use Azure Backup.

SCENARIO
Tag backfill
After assigning a tag policy, an admin remediates older resources missing the tag.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q391: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 392

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q530

A user needs to create and delete resource locks on a production resource group. Which built-in role grants the required permissions?

✔ **A.** Owner

✖ B. Contributor

✖ C. Reader

✖ D. Storage Account Contributor

Correct answer: Owner

Clarification (read this first): Trap: Contributor can do almost everything, but it CANNOT manage locks or access. Only Owner or User Access Administrator can. Q375 (who can manage role assignments) gives Owner + User Access Administrator.

Answer explained

✔ **Owner**
Only Owner or User Access Administrator (built-in roles) can create or delete locks. Contributor cannot.

✖ **Contributor**
Contributor cannot do Microsoft.Authorization actions, such as managing locks.

✖ **Reader**
Reader can only view.

✖ **Storage Account Contributor**
Storage Account Contributor manages storage accounts only, not locks.

EXAM STRATEGY
Locks need Microsoft.Authorization/locks permissions: Owner or User Access Administrator. Cross out Contributor and service-specific contributor roles.

IMPACT
Only Owners managing locks stops engineers removing a production delete lock before a risky change.

What · Where · When · Why · How

What	Resource locks stop accidental deletion or changes.
Where	They are set on a subscription, resource group or resource.
When	Use them on important production resources.
Why	Limiting who manages locks keeps the protection real.
How	An Owner adds a CanNotDelete or ReadOnly lock under Settings > Locks.

Exam tip
Contributor can do almost everything EXCEPT manage access and locks.

Memory trick
Only the Owner holds the key to the lock.

Real-world example, use case and scenario

EXAMPLE
Locking prod
The subscription Owner puts a CanNotDelete lock on RG-Prod.

USE CASE
Bank controls
A bank lets only a small Owner group manage locks on payment systems.

SCENARIO
Contributor blocked
A developer with Contributor tries to remove a lock and gets an access error.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q392: Governance tools

About this figure

• Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.

• Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.

• RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.

• A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.

• Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 393

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q531

A storage account has a CanNotDelete (Delete) lock. Statement: Users with the appropriate permissions can still change the storage account's settings, such as its access tier. Is the statement true?

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): The lock name tells you what it stops. CanNotDelete (also called Delete lock) stops DELETING only. Changing settings is still allowed. A ReadOnly lock would stop changes. See also Q252: to delete a locked group you must remove the lock first.

Answer explained

✔ **Yes**
CanNotDelete only stops deletion. Users with permission can still read and change the resource.

✖ **No**
To stop changes you need a ReadOnly lock, not CanNotDelete.

EXAM STRATEGY
Key fact: CanNotDelete blocks only deletion; ReadOnly blocks changes too. So changing settings is still allowed = Yes.

IMPACT
A ReadOnly lock can break normal tasks, such as listing storage keys. CanNotDelete protects against deletion and still lets you manage the resource.

What · Where · When · Why · How

What	CanNotDelete is a lock level that stops deletion.
Where	It can be set on a subscription, resource group or resource.
When	Use it when a resource must change but must never be deleted by mistake.
Why	It protects against costly accidental deletes.
How	Add a lock with type Delete in the resource's Locks page.

Exam tip
Delete lock = change OK, delete NO. ReadOnly lock = change NO, delete NO.

Memory trick
CanNotDelete: change it, just don't kill it.

Real-world example, use case and scenario

EXAMPLE
Tier change
An admin switches a locked storage account's default tier from Hot to Cool.

USE CASE
Media archive
A broadcaster locks its archive account against deletion but still changes its settings.

SCENARIO
Cleanup script
A cleanup script's delete call fails on a locked database. Updates to it still work.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
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Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q393: Governance tools

About this figure

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Question 394

AZURE MANAGEMENT & GOVERNANCE — **Compliance** v5 Q532

Your auditors ask for Microsoft's independent SOC 2 Type 2 and ISO/IEC 27001 audit reports covering Azure. Where should you download them?

✔ **A.** Service Trust Portal

✖ B. Azure Advisor

✖ C. Azure Service Health

✖ D. Microsoft Cost Management

Correct answer: Service Trust Portal

Answer explained

✔ **Service Trust Portal**
The Service Trust Portal is where Microsoft publishes independent audit reports, like SOC 2 and ISO 27001.

✖ **Azure Advisor**
Advisor gives best-practice tips, not audit reports.

✖ **Azure Service Health**
Service Health reports Azure outages and maintenance, not compliance documents.

✖ **Microsoft Cost Management**
Cost Management analyses spending, not compliance documents.

EXAM STRATEGY
Clue words: 'audit reports', SOC, ISO, 'download' = Service Trust Portal. The other three have no compliance reports.

IMPACT
Giving auditors Microsoft's reports quickly shortens vendor reviews. Under the shared responsibility model, Microsoft proves its part this way.

What · Where · When · Why · How

What	The Service Trust Portal is Microsoft's library of compliance documents.
Where	It is at servicetrust.microsoft.com.
When	Use it during audits or vendor risk checks.
Why	Customers must check Microsoft's security controls.
How	Sign in, browse certifications, regulations and standards, and download reports.

Exam tip
Audit reports and certificates from Microsoft = Service Trust Portal.

Memory trick
Need proof you can TRUST? Go to the TRUST portal.

Real-world example, use case and scenario

EXAMPLE
SOC report
A compliance officer downloads the latest Azure SOC 2 Type 2 report for an auditor.

USE CASE
Vendor assessment
A hospital's risk team reviews ISO 27001 certificates before approving Azure.

SCENARIO
Government bid
A contractor adds FedRAMP documents from the portal to a government bid.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

AZ-900 Exam Prep 2026

Figure Q394: Compliance, privacy and trust

About this figure

• Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.

• Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.

• Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.

• Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.

• The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 395

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q533

Select the answer that correctly completes the sentence. If the sentence is already correct, select 'No change is needed'. Statement: '[Azure role-based access control (RBAC)] ensures that only approved VM SKUs can be deployed, regardless of who performs the deployment.'

✖ A. No change is needed

✔ **B.** Azure Policy

✖ C. Resource locks

✖ D. Microsoft Purview

Correct answer: Azure Policy

Clarification (read this first): Watch 'regardless of who'. That means the rule is about WHAT may be deployed, not WHO may deploy. WHAT = Azure Policy; WHO = RBAC. Q205 is similar: block only VM creation = Azure Policy.

Answer explained

✔ **Azure Policy**
Azure Policy limits resource settings, like allowed VM sizes, for every user. RBAC decides who can act.

✖ **No change is needed**
RBAC controls who can do actions. It does not decide which VM sizes are allowed.

✖ **Resource locks**
Locks stop deletion or changes. They cannot limit VM sizes.

✖ **Microsoft Purview**
Purview governs data, not resource settings.

EXAM STRATEGY
Clue words: 'regardless of who' = Azure Policy (what is allowed). RBAC = who can act. Locks stop delete or change. Purview is for data.

IMPACT
An allowed-size policy stops anyone, even Owners, deploying expensive GPU or M-series VMs by mistake. This protects the budget.

What · Where · When · Why · How

What	Azure Policy enforces rules on resource settings.
Where	It is assigned at management group, subscription or resource group level.
When	Use it to keep settings standard across the company.
Why	It prevents costly or non-compliant choices.
How	Assign the built-in 'Allowed virtual machine size SKUs' policy with Deny.

Exam tip
RBAC = WHO can do it. Policy = WHAT can be deployed.

Memory trick
RBAC checks the person; Policy checks the product.

Real-world example, use case and scenario

EXAMPLE
SKU limit
An Owner tries to deploy an M-series VM and the policy denies it.

USE CASE
Cost control
A university allows only B-series VMs in student subscriptions.

SCENARIO
Contractor deploys
A contractor with Contributor rights can still deploy only approved sizes.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q395: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 396

AZURE MANAGEMENT & GOVERNANCE

Cost
v5 Q534

Select the answer that correctly completes the sentence. If the sentence is already correct, select 'No change is needed'. Statement: 'To estimate the monthly cost of a new AKS cluster, an Azure SQL Database and a VPN gateway that you plan to deploy, use the [TCO calculator].'

✖ A. No change is needed

✔ **B.** Pricing calculator

✖ C. Cost analysis in Microsoft Cost Management

✖ D. Azure Advisor

Correct answer: Pricing calculator

Clarification (read this first): Bracketed words: 'TCO calculator'. Watch 'plan to deploy' with no on-premises comparison. Pricing new Azure services = Pricing calculator. Q140 is the look-alike: savings from moving on-premises servers = TCO calculator.

Answer explained

✔ **Pricing calculator**
Estimating the monthly cost of Azure services you plan to deploy is the Pricing calculator's job. TCO compares on-premises with Azure.

✖ **No change is needed**
TCO compares on-premises costs with Azure. It does not price specific new services.

✖ **Cost analysis in Microsoft Cost Management**
Cost analysis shows real costs of resources already deployed, not planned ones.

✖ **Azure Advisor**
Advisor gives tips for existing resources. It does not estimate new deployments.

EXAM STRATEGY
Clue words: 'plan to deploy' Azure services = Pricing calculator. Use TCO only when the question compares on-premises costs with Azure. Cost analysis shows real spending.

IMPACT
Good estimates before deployment set correct budgets and avoid surprises when the first bill arrives.

What · Where · When · Why · How

What	The Pricing calculator builds item-by-item cost estimates.
Where	It is a public tool on the Azure website.
When	Use it while designing and budgeting, before you deploy.
Why	It shows expected monthly cost per service.
How	Add AKS, SQL Database and VPN Gateway, choose options, and check the total.

Exam tip
Estimate new Azure services = Pricing. Compare on-premises with Azure = TCO.

Memory trick
Pricing = price list for Azure items. TCO = Total Comparison with On-premises.

Real-world example, use case and scenario

EXAMPLE
Architecture estimate
An architect exports a price estimate to Excel for managers to approve.

USE CASE
E-commerce launch
An online shop estimates costs for a new AKS-based checkout service.

SCENARIO
Region comparison
A team compares the same setup in two regions to pick the cheaper one.

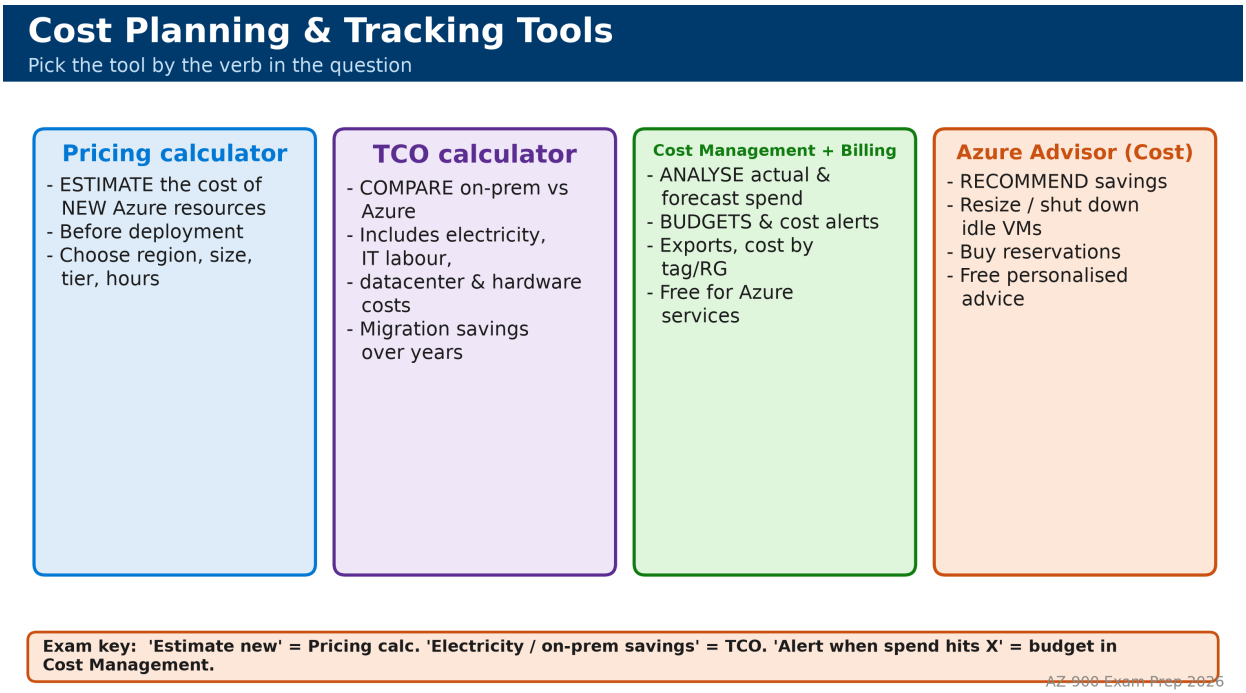


Figure Q396a: Cost planning and tracking tools

About this figure

• Each column is one tool; pick it by the question's verb: estimate (Pricing calculator), compare (TCO), analyse or alert (Cost Management), recommend (Advisor).

• Pricing and TCO calculators are free websites that need no Azure subscription; estimates use list prices and can be saved or exported to Excel.

• Cost Management is free for Azure resources; budgets can be set monthly, quarterly or yearly with alerts at thresholds such as 50%, 80% or 100%.

• Budgets alert but do not stop spending; only the free account and some credit offers have a spending limit that disables resources when exhausted.

• Advisor cost recommendations include resizing or shutting down VMs with low CPU and buying reservations or savings plans based on 30 days of usage.

Approximate Azure Charges (USD)

Pay-as-you-go list prices, East US, Linux unless noted - estimates only; use the Pricing calculator

Resource	Approx. price	What you pay for / tip
VM B1s (1 vCPU, 1 GB)	~USD 8/month	Per second while running; Windows adds licence cost
VM D2s v5 (2 vCPU, 8 GB)	~USD 70/month	Deallocate to stop compute charges; disks still billed
Managed disk Premium SSD 128 GB	~USD 20/month	Billed even when VM is deallocated
Blob storage Hot / Cool / Archive	~0.02 / 0.01 / 0.001 per GB-mo	Cooler = cheaper storage, dearer access
Outbound data (egress)	First 100 GB free, ~0.087/GB	Inbound (ingress) is always free
Inter-region transfer	~0.02/GB	Same-region traffic mostly free (VNet peering ~0.01/GB)
Public IP (Standard, static)	~USD 3.65/month	Charged even when not attached
Azure SQL Database Basic / S0	~USD 5 / 15 per month	Serverless tier auto-pauses (storage only)
Functions (Consumption)	~0.20 per 1M runs	1M runs + 400k GB-s free each month
VPN Gateway VpnGw1	~USD 140/month	Plus egress; ExpressRoute circuits from ~USD 55/mo
Azure Firewall Standard	~USD 900/month	Plus ~0.016 per GB processed
Log Analytics ingestion	~USD 2.30/GB	First 5 GB/month free; Monitor alerts priced per rule
Entra ID P1 / P2	~USD 6 / 9 per user-month	Free tier included; P1 = Conditional Access
Defender for Servers P1 / P2	~USD 5 / 15 per server-mo	Foundational CSPM (secure score) is free

Exam key: Save: Reservations up to ~72%, Spot up to ~90%, Azure Hybrid Benefit reuses Windows/SQL licences.

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Figure Q396b: Approximate Azure charges (USD)

About this figure

- Rows list approximate pay-as-you-go list prices in East US; the right column explains the billing unit and a saving tip. Prices vary by region and change.
- Monthly figures assume about 730 hours of running time; per-second billing means a VM running 8 hours a day costs roughly one third.
- Commit-based discounts: reservations up to approx. 72%, savings plans up to approx. 65%, Spot up to approx. 90%; Azure Hybrid Benefit removes Windows licence cost.
- Idle charges add up: public IPs, disks, gateways and firewalls bill hourly whether used or not, so delete them when no longer needed.
- Use the Pricing calculator for estimates and Cost Management budgets to track real spend; enterprise agreements and CSP partners can offer different prices.

Question 397

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q536

Your organization plans to apply tags to Azure resources. Which TWO tasks can tags help you accomplish? Each correct answer presents a complete solution.

✔ **A.** Group and filter costs by department in Microsoft Cost Management

✔ **B.** Identify resources by environment so an automation script stops only Dev VMs at night

✖ C. Grant users permission to manage specific resources

✖ D. Prevent resources from being deleted

✖ E. Encrypt data stored in the resources

Correct answer: Group and filter costs by department in Microsoft Cost Management; Identify resources by environment so an automation script stops only Dev VMs at night

Clarification (read this first): Choose TWO. Tags are only labels (name-value pairs). They cannot give permissions (RBAC), stop deletion (locks) or encrypt data. Q58 and Q129 also test tags: tags are not inherited by default.

Answer explained

✔ **Group and filter costs by department in Microsoft Cost Management; Identify resources by environment so an automation script stops only Dev VMs at night**
Tags label resources for cost reports and for automation. They have no security or lock function.

✖ **Grant users permission to manage specific resources**
Permissions are given with Azure RBAC, not tags.

✖ **Prevent resources from being deleted**
Resource locks stop deletion, not tags.

✖ **Encrypt data stored in the resources**
Encryption comes from service settings and Key Vault, not tags.

EXAM STRATEGY
Tags are metadata (descriptive labels): use them for cost grouping and automation targeting. Cross out anything about security: RBAC, locks, encryption.

IMPACT
Good tags let you show cost by department and safely automate tasks, like stopping only Dev VMs at night. That is a quick saving.

What · Where · When · Why · How

What	Tags are name-value pairs attached to Azure resources.
Where	They can be added to subscriptions, resource groups and resources.
When	Add them when resources are created, ideally enforced by Azure Policy.
Why	They make chargeback, reports and automation possible.
How	Add tags in the portal, CLI, PowerShell or IaC (infrastructure as code) templates.

Exam tip
Tags = labels only: cost, organisation, automation. Not security, not locks.

Memory trick
Tags are labels on boxes, not locks on doors.

Real-world example, use case and scenario

EXAMPLE
Department costs
Cost analysis grouped by Department shows HR spent USD 2,300 last month.

USE CASE
Night shutdown
A script stops all VMs tagged Environment=Dev at 8 PM to save money.

SCENARIO
Ownership
An Owner=email tag tells the operations team whom to contact about an unused database.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
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Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q397: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 398

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q537

Which TWO Azure Policy effects can change a resource or deploy additional configuration, rather than only blocking or reporting? Each correct answer presents a complete solution.

✔ **A.** Modify

✔ **B.** DeployIfNotExists

✖ C. Audit

✖ D. Deny

✖ E. Disabled

Correct answer: Modify; DeployIfNotExists

Clarification (read this first): Choose TWO. The question asks which effects DO something to the resource. Modify changes it; DeployIfNotExists adds missing settings. Audit only reports and Deny only blocks.

Answer explained

✔ **Modify; DeployIfNotExists**
Modify changes properties. DeployIfNotExists deploys missing settings. Both actively fix resources.

✖ **Audit**
Audit only records the problem. It changes nothing.

✖ **Deny**
Deny blocks the request but makes no change.

✖ **Disabled**
Disabled turns the policy off. Nothing is checked.

EXAM STRATEGY
Clue words: 'change or deploy' = Modify and DeployIfNotExists. Audit reports, Deny blocks, Disabled does nothing.

IMPACT
Fixing effects apply standards (like logging or tags) automatically. Otherwise you get long lists of problems that nobody fixes.

What · Where · When · Why · How

What	Policy effects say what happens when a rule matches.
Where	The effect is set in the policy definition or as a parameter.
When	Use fixing effects when you want automatic correction.
Why	They apply standards without manual work.
How	Assign the policy with a managed identity, and run remediation for existing resources.

Exam tip
Report = Audit. Block = Deny. Fix = Modify or DeployIfNotExists.

Memory trick
Modify and Deploy both DO something.

Real-world example, use case and scenario

EXAMPLE
Add tag
Modify adds Environment=Prod to new resources in a production resource group.

USE CASE
Monitoring agent
A retailer uses DeployIfNotExists to install monitoring agents on every new VM.

SCENARIO
Choosing effects
A governance team uses Audit first, then Modify for tags and Deny for blocked regions.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
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Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

AZ-900 Exam Prep 2026

Figure Q398: Governance tools

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Question 399

AZURE MANAGEMENT & GOVERNANCE — **Compliance** v5 Q538

Your company wants to prevent employees from sharing credit card numbers through email, Teams chats and SharePoint files with people outside the organization. Which solution should you use?

✔ **A.** Microsoft Purview Data Loss Prevention

✖ B. Microsoft Defender for Cloud

✖ C. Azure Policy

✖ D. Microsoft Entra Conditional Access

Correct answer: Microsoft Purview Data Loss Prevention

Clarification (read this first): Watch the word 'sharing' of data. This is about content in email, Teams and SharePoint, so it is Purview DLP. Q179 is a look-alike: auto-watermark (label) documents = Purview Information Protection, not DLP.

Answer explained

✔ **Microsoft Purview Data Loss Prevention**
Purview DLP (Data Loss Prevention) finds sensitive content, like card numbers, and blocks risky sharing in email, Teams and SharePoint.

✖ **Microsoft Defender for Cloud**
Defender for Cloud protects cloud workloads and settings. It does not read email or chat content.

✖ **Azure Policy**
Azure Policy controls Azure resource settings, not what users write or share.

✖ **Microsoft Entra Conditional Access**
Conditional Access controls sign-in conditions, not what data users share.

EXAM STRATEGY
Clue words: 'sharing sensitive information' across email, Teams, SharePoint = Purview DLP. Defender and Policy protect Azure resources, not messages.

IMPACT
One leaked file of card numbers can bring PCI DSS (card industry rules) fines and breach reports. DLP blocks it at the moment of sharing.

What · Where · When · Why · How

What	Data Loss Prevention is a Purview feature that protects sensitive data.
Where	It covers Exchange, Teams, SharePoint, OneDrive and devices.
When	Use it when sensitive data must not leave the organisation.
Why	It reduces data leaks and rule breaks.
How	Create a DLP policy for the credit card data type with block actions.

Exam tip
Stop sensitive data leaving = Purview DLP. Label or encrypt documents = Purview Information Protection.

Memory trick
DLP = Don't Let it Pass.

Real-world example, use case and scenario

EXAMPLE
Blocked email
An email with 16-digit card numbers to an outside address is blocked, with a tip shown to the user.

USE CASE
Healthcare
A clinic stops patient record numbers being posted in external Teams chats.

SCENARIO
Finance file share
A finance user tries to share a payroll spreadsheet outside the company in SharePoint and is stopped.

Compliance, Privacy & Trust

Where to find what

Resource	What you get there
Service Trust Portal	Audit reports (ISO, SOC), compliance documents & whitepapers
Purview Compliance Manager	Track YOUR compliance posture; score & improvement actions
Defender for Cloud (compliance)	Regulatory compliance dashboard for your Azure resources
Microsoft Trust Center	How Microsoft handles security, privacy, compliance
Microsoft Privacy Statement	What personal data Microsoft collects and why
Product Terms + DPA	Licensing terms; Data Protection Addendum obligations
Azure Government	US federal/state/local gov entities & their partners only

Exam key: Download an audit report = Service Trust Portal. Assess your own regulatory posture = Compliance Manager.

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Figure Q399: Compliance, privacy and trust

About this figure

• Each row names a Microsoft resource and what it gives you; match the scenario wording, such as download, assess, read terms or government.

• Service Trust Portal documents require signing in with a Microsoft cloud account; it hosts SOC, ISO, FedRAMP and penetration test reports.

• Compliance Manager in Microsoft Purview gives a compliance score and assessments against templates such as GDPR, ISO 27001 and NIST.

• Microsoft complies with more than 100 offerings; certification of Azure does not make your workload compliant, under the shared responsibility model.

• The Data Protection Addendum is part of the Product Terms and sets Microsoft's obligations as data processor, including GDPR commitments.

Question 400

AZURE MANAGEMENT & GOVERNANCE

Governance
v5 Q539

Select the answer that correctly completes the sentence. If the sentence is already correct, select 'No change is needed'. Statement: 'Azure Policy evaluates resource compliance [only when a resource is first created].'

✖ A. No change is needed

✔ **B.** when resources are created or updated, when assignments change, and on a regular periodic scan

✖ C. only when an administrator manually starts an evaluation

✖ D. only once per month when billing is calculated

Correct answer: when resources are created or updated, when assignments change, and on a regular periodic scan

Clarification (read this first): Bracketed words: 'only when a resource is first created'. Watch the word 'only'. Policy checks many times: on create or update, when assignments change, and in a regular scan (about every 24 hours).

Answer explained

✔ **when resources are created or updated, when assignments change, and on a regular periodic scan**
Policy checks resources when they are created or updated, when assignments change, and in a regular scan about every 24 hours.

✖ **No change is needed**
Existing resources are checked again after creation, including in regular scans.

✖ **Only when an administrator manually starts an evaluation**
You can start a scan by hand, but checks are mostly automatic.

✖ **Only once per month when billing is calculated**
Policy checks have nothing to do with billing.

EXAM STRATEGY
Key fact: 'only' with one trigger is wrong. Policy checks on create or update, on assignment changes, and in a scan about every 24 hours.

IMPACT
Knowing the timing explains why compliance results update a little after a change. This avoids false alarms or early audit sign-off.

What · Where · When · Why · How

What	Compliance evaluation checks resources against assigned policies.
Where	Results show in the Azure Policy compliance dashboard.
When	On create or update, on assignment changes, and about every 24 hours.
Why	It keeps compliance data current as things change.
How	Azure Resource Manager and the policy engine check automatically.

Exam tip
Azure Policy also flags existing non-compliant resources, not just new ones.

Memory trick
Policy never sleeps: on every change and every day.

Real-world example, use case and scenario

EXAMPLE
New assignment
Minutes after a policy is assigned, existing resources show as compliant or non-compliant.

USE CASE
Drift detection
A retailer finds a firewall setting changed outside IaC in the next daily scan.

SCENARIO
Manual change
An engineer turns on public access for a storage account, and it is flagged non-compliant.

Governance Tools — What Each One Does

Read the verb in the question

Tool	Purpose	Exam keyword
Azure Policy	Enforce rules on resource PROPERTIES	'Only allow region X / VM size Y', compliance audit
RBAC	Control WHO can do WHAT at a scope	'Least privilege', 'allow user to manage RG'
Resource locks	Prevent DELETE (CanNotDelete) or change (ReadOnly)	'Prevent accidental deletion' (even by Owners)
Tags	Name:value metadata	'Group costs by department / cost centre'
Management groups	Apply policy/RBAC to many subscriptions	'Govern multiple subscriptions'
Blueprints (retired)	Package policy + RBAC + templates	Replaced by template specs & deployment stacks
Microsoft Purview	Data governance & compliance	'Discover & classify data'

Exam key: Prevent creating = Policy. Prevent deleting = lock. Permissions = RBAC. Cost grouping = tags.

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Figure Q400: Governance tools

About this figure

- Each row pairs a tool with its exam keyword; read the question's verb, such as enforce, permit, prevent deletion, categorise or govern many subscriptions.
- Policy effects include Deny, Audit, Append, Modify and DeployIfNotExists; initiatives group many policies. Policy itself is free for Azure resources.
- RBAC built-in roles include Owner, Contributor, Reader and User Access Administrator; assignments apply at MG, subscription, RG or resource scope.
- A resource supports up to 50 tag name-value pairs; tags help with cost reporting and automation but are not security boundaries.
- Azure Blueprints was retired in July 2026; use template specs, deployment stacks and policy initiatives instead. Purview also catalogues and classifies data.

Question 401

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q540, Q553

Your company runs Windows and Linux servers in its own datacenter and in AWS. Management wants to view these servers in the Azure portal, organize them with tags, and apply Azure Policy to them without migrating them to Azure. What should you use?

✔ **A.** Azure Arc

✖ B. Azure Migrate

✖ C. Azure Lighthouse

✖ D. Azure Bastion

Correct answer: Azure Arc

Clarification (read this first): Watch 'without migrating them'. The servers stay where they are, so the answer is Azure Arc. Azure Migrate is for MOVING servers into Azure.

Answer explained

✔ **Azure Arc**
Azure Arc brings Azure management (portal, tags, Policy, RBAC, Monitor) to servers on-premises and in other clouds, without moving them.

✖ **Azure Migrate**
Azure Migrate checks and moves workloads into Azure. Here the servers must stay where they are.

✖ **Azure Lighthouse**
Lighthouse lets service providers manage other customers' Azure resources. It does not manage non-Azure servers.

✖ **Azure Bastion**
Bastion gives browser-based RDP or SSH (remote login) to Azure VMs. It does not govern outside servers.

EXAM STRATEGY
Clue words: 'without migrating' plus 'view in portal, tags, Policy' = Azure Arc. Migrate moves, Lighthouse manages other tenants, Bastion is secure remote login.

IMPACT
One management place for on-premises, AWS and Azure servers avoids separate tools and uneven security settings.

What · Where · When · Why · How

What	Azure Arc extends Azure's control plane (management layer) to non-Azure machines.
Where	It manages resources on-premises, at the edge, and in other clouds like AWS and Google Cloud.
When	Use it when hybrid or multicloud resources must follow the same rules as Azure resources.
Why	It gives one place for inventory, tags, policy, RBAC and monitoring.
How	Install the Connected Machine agent on each server. This registers it as an Azure resource.

Exam tip
Hybrid or multicloud plus 'manage from Azure' or 'apply Azure Policy' = Azure Arc. 'Move to Azure' = Azure Migrate.

Memory trick
Arc is a rainbow arc reaching from Azure to your other clouds.

Real-world example, use case and scenario

EXAMPLE
Retail stores
A retailer connects Linux servers in 300 stores to Arc, so central IT sees them all in one resource group.

USE CASE
Multicloud policy
A bank applies the same disk encryption policy to Arc-enabled servers running in AWS.

SCENARIO
Factory floor
A manufacturer monitors factory servers in Azure Monitor after connecting them with Azure Arc.

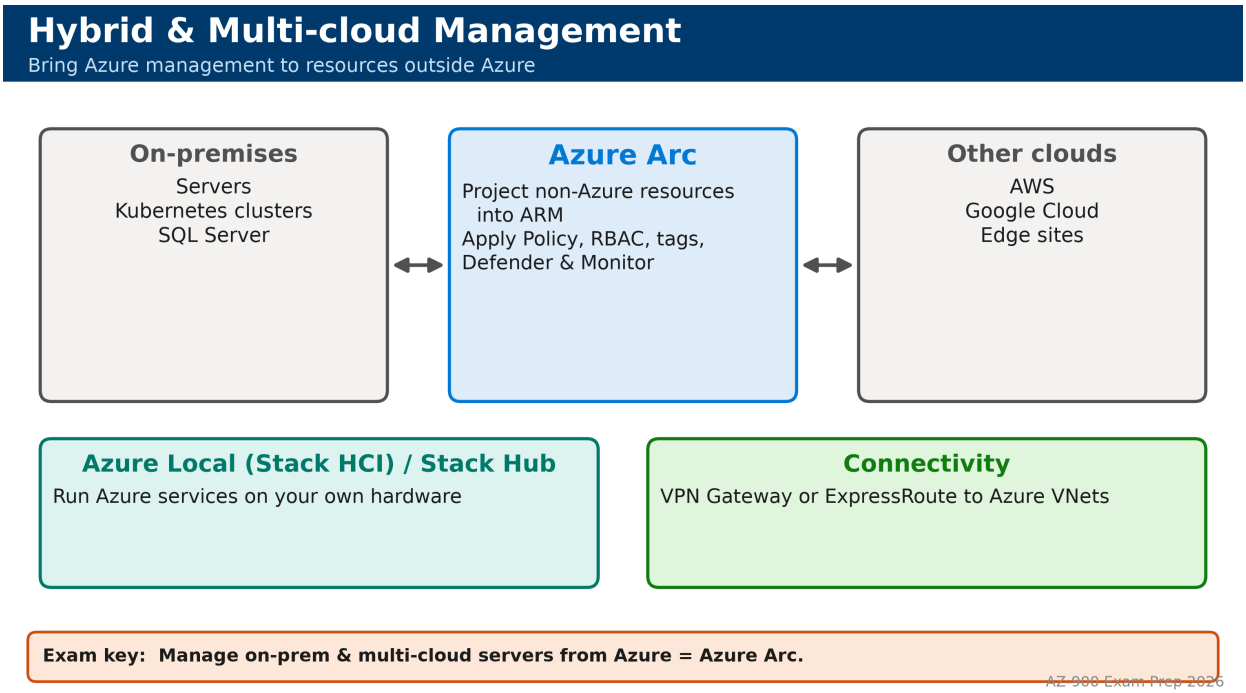


Figure Q401a: Hybrid and multi-cloud management

About this figure

- Arrows show Azure Arc linking on-premises and other-cloud resources into Azure Resource Manager; bottom boxes cover local Azure hardware and network links.
- Arc-enabled servers' core control plane (inventory, tags, RBAC, Policy) is free; add-ons such as Defender, Update Manager and Monitor are billed separately.
- Arc can also manage Kubernetes clusters and SQL Server; Arc-enabled SQL Server can be licensed pay-as-you-go through your Azure bill.
- Azure Local (formerly Azure Stack HCI) runs VMs and AKS on validated on-premises hardware, billed per physical core per month in Azure.
- Site-to-site VPN uses the Internet with encryption; ExpressRoute is a private connection through a provider with higher reliability and an SLA up to 99.95%.

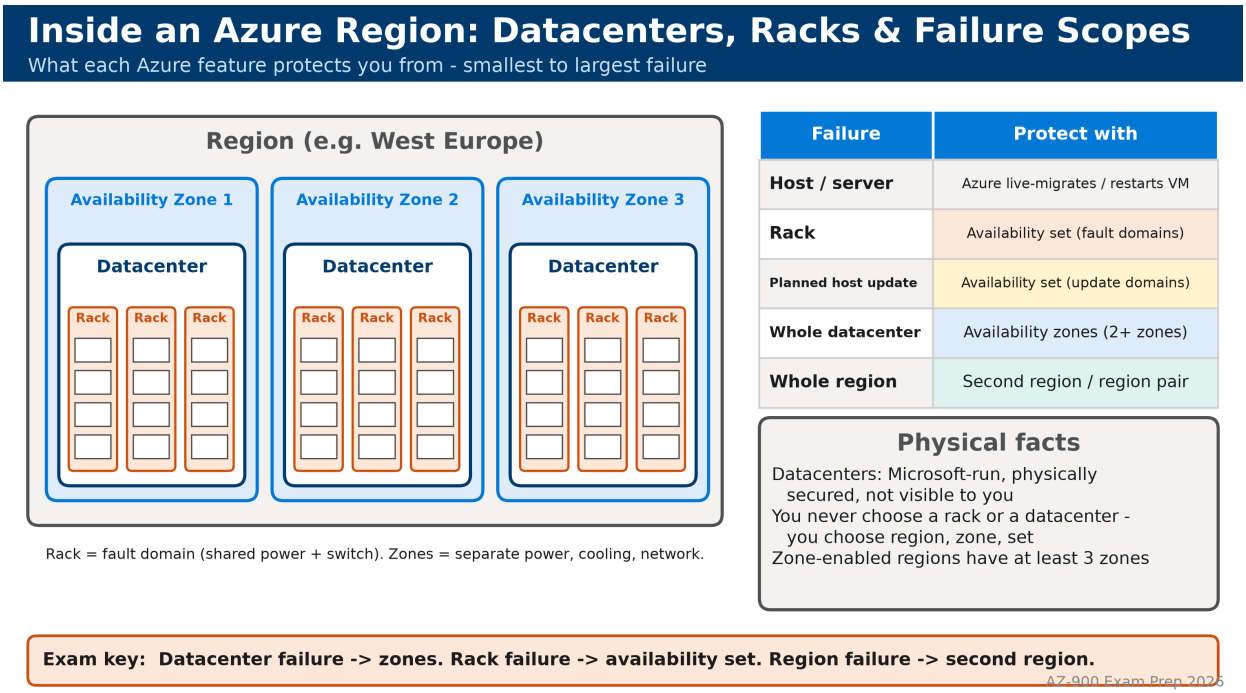


Figure Q401b: Inside a region: datacenters, racks and failure scopes

About this figure

- Read from small to large: a server sits in a rack, racks in a datacenter, datacenters in an availability zone, zones in a region.
- A rack shares power and a network switch, so it is one fault domain; availability sets spread VMs across up to 3 of them.
- Availability zones are physically separate datacenter groups with independent power, cooling and networking, connected by a low-latency network.
- Zones protect against a datacenter failure (99.99% VM SLA with 2+ VMs in 2+ zones); only a second region protects against a whole-region outage.
- You never pick a datacenter or rack - you choose a region, an optional zone, and redundancy options; Microsoft runs physical security.
- Using zones has no separate fee; you pay for the resources you place in them (e.g. two VMs cost twice one VM).

Question 402

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q541

Your team writes ARM templates in JSON and finds them verbose and hard to read. You need a Microsoft domain-specific language with simpler syntax that still deploys through Azure Resource Manager. What should you use?

✔ **A.** Bicep

✖ B. Azure Resource Graph

✖ C. Azure Automation

✖ D. Azure Policy

Correct answer: Bicep

Answer explained

✔ **Bicep**
Bicep is Microsoft's simpler language for infrastructure as code (IaC: infrastructure defined in files). It turns into ARM JSON and deploys through Azure Resource Manager.

✖ **Azure Resource Graph**
Resource Graph searches existing resources at scale with KQL. It does not define deployments.

✖ **Azure Automation**
Automation runs scripts (runbooks) and update tasks. It is not a template language.

✖ **Azure Policy**
Policy checks and enforces rules on resources. It does not describe what to deploy.

EXAM STRATEGY
Clue words: 'simpler than ARM JSON' plus 'domain-specific language' (a language made for one job) = Bicep. Resource Graph searches, Automation runs scripts, Policy enforces rules.

IMPACT
Easy-to-read IaC means fewer deployment errors and faster reviews. This makes repeatable environments practical for new teams.

What · Where · When · Why · How

What	Bicep is a language for describing Azure resources.
Where	Bicep files (.bicep) are kept in source control and deployed to resource groups, subscriptions, management groups or tenants.
When	Use it when you want repeatable Azure deployments with code that is easier to read than JSON.
Why	It makes templates shorter with fewer errors and works fully with ARM.
How	Deploy with az deployment group create or New-AzResourceGroupDeployment. The tools convert it to ARM JSON.

Exam tip
ARM templates = JSON. Bicep = simpler syntax that becomes ARM JSON. Both are declarative (you describe the end result) IaC.

Memory trick
Bicep makes ARM templates stronger and leaner, like flexing an arm muscle.

Real-world example, use case and scenario

EXAMPLE
Smaller file
A 200-line JSON template for a storage account and VNet becomes about 60 lines in Bicep.

USE CASE
Reusable modules
A platform team shares a Bicep module for a standard hub VNet that every project reuses.

SCENARIO
Startup pipeline
A fintech startup deploys its test environment from Bicep files on every pull request in GitHub Actions.

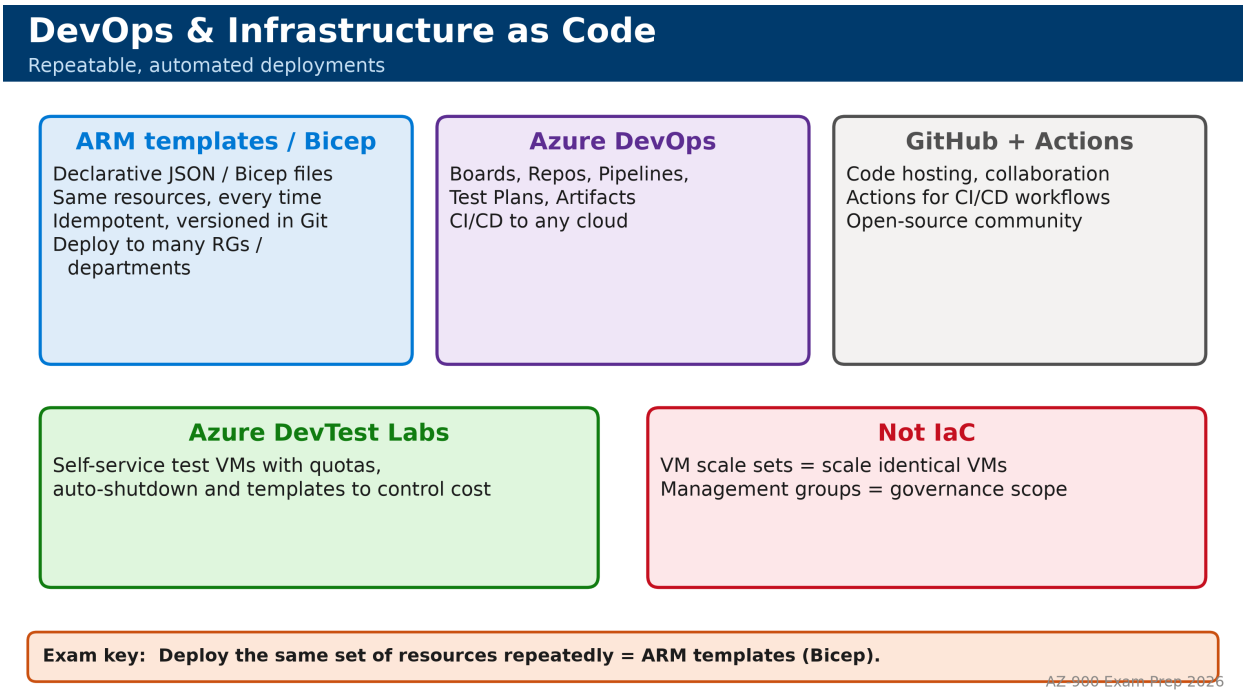


Figure Q402: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 403

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q542

An operations analyst needs to search data collected in a Log Analytics workspace to find all VMs that reported more than 90 percent CPU in the last hour. Which query language should the analyst use?

✔ **A.** Kusto Query Language (KQL)

✖ B. Transact-SQL (T-SQL)

✖ C. Gremlin

✖ D. GraphQL

Correct answer: Kusto Query Language (KQL)

Answer explained

✔ **Kusto Query Language (KQL)**
You search a Log Analytics workspace with KQL (Kusto Query Language). KQL is a read-only language for log and monitoring data.

✖ **Transact-SQL (T-SQL)**
T-SQL is for Azure SQL Database and SQL Server, not Log Analytics.

✖ **Gremlin**
Gremlin is for graph data in Azure Cosmos DB for Apache Gremlin.

✖ **GraphQL**
GraphQL is for web APIs, not Azure Monitor logs.

EXAM STRATEGY
Clue words: 'Log Analytics workspace' = KQL every time. T-SQL is for databases, Gremlin for graphs, GraphQL for APIs.

IMPACT
KQL works in Log Analytics, Sentinel and Resource Graph. One language covers monitoring, security hunting and inventory.

What · Where · When · Why · How

What	KQL is the query language for Azure Monitor Logs.
Where	You run KQL in the Logs page of a Log Analytics workspace or of any resource in the portal.
When	Use it to investigate incidents, build log alerts or make workbook charts.
Why	It quickly filters, joins and sums millions of log records.
How	A query starts with a table name and then uses pipes, for example Perf | where CounterValue > 90.

Exam tip
Log Analytics + query = KQL. Do not confuse it with SQL, even though it looks a bit similar.

Memory trick
KQL: Kusto Queries Logs.

Real-world example, use case and scenario

EXAMPLE
Error search
A KQL query on the AppExceptions table lists the ten most common errors in a web app this week.

USE CASE
Log alert
A hospital runs a KQL log alert every five minutes to find failed sign-ins to a clinical app.

SCENARIO
Capacity review
A gaming company sums average CPU per VM scale set with KQL before a launch event.

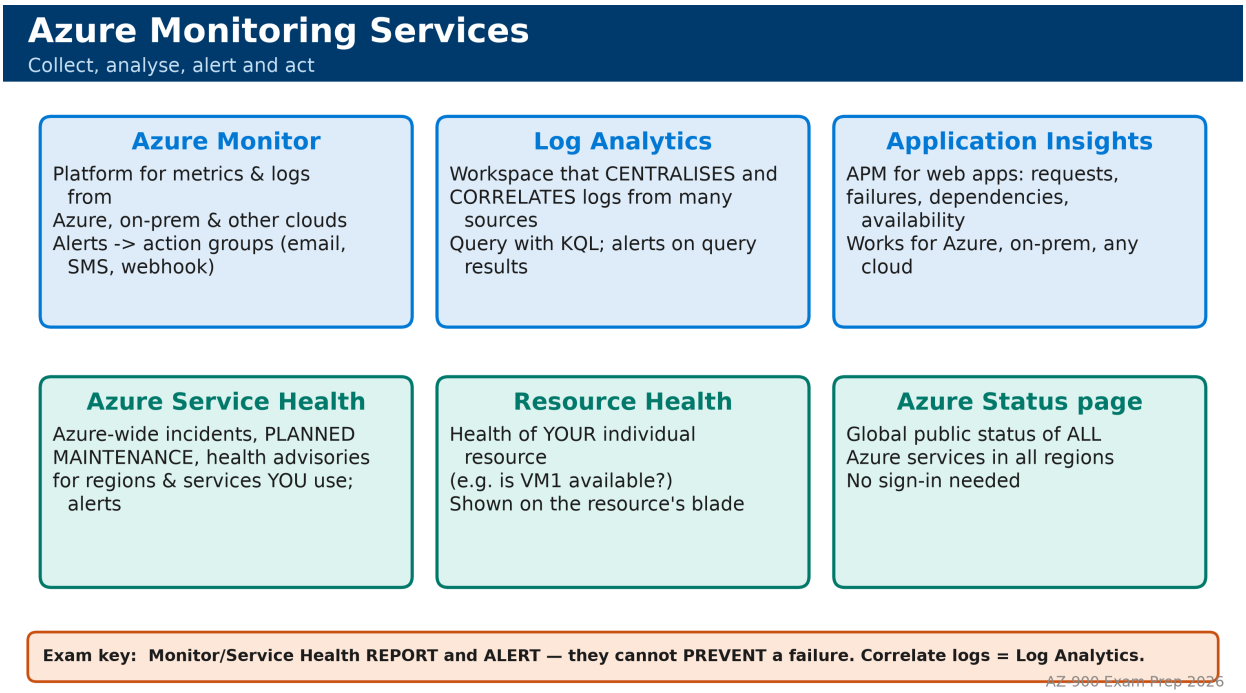


Figure Q403: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 404

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q543

A web server VM named VM1 was unreachable for 15 minutes last night. You need to determine whether VM1 itself was affected by an Azure platform problem and see its availability history. Which service should you use?

✔ **A.** Azure Resource Health

✖ B. Azure Advisor

✖ C. Azure Policy

✖ D. Microsoft Defender for Cloud

Correct answer: Azure Resource Health

Clarification (read this first): Watch 'VM1 itself' - one resource. One resource = Resource Health. Q405 is the look-alike: widespread outage, no account = Azure status page. Q136/Q420: your services in your regions = Service Health.

Answer explained

✔ **Azure Resource Health**
Resource Health shows the current status and past availability of one resource. It also tells you if an Azure platform problem caused the downtime.

✖ **Azure Advisor**
Advisor gives best-practice tips. It does not report past outages of one VM.

✖ **Azure Policy**
Policy checks rules, not whether a resource was up.

✖ **Microsoft Defender for Cloud**
Defender for Cloud is about security and threats, not platform outages.

EXAM STRATEGY
Clue words: one named resource (VM1) and its history = Resource Health. Service Health covers whole services and regions. Advisor gives tips.

IMPACT
It proves whether downtime was Azure's fault. This supports SLA (uptime promise) credit claims and saves hours of debugging your own app.

What · Where · When · Why · How

What	Azure Resource Health is a health view for each single resource, inside Azure Service Health.
Where	It is on each resource's Resource health page and in Service Health.
When	Use it when you need to know if one resource was down because of Azure.
Why	It separates Azure-caused problems from actions by users, so troubleshooting is faster.
How	Open the resource, select Resource health, and review current status and history.

Exam tip
Status page = all of Azure, for everyone. Service Health = Azure services in your regions. Resource Health = one of your resources.

Memory trick
Resource Health = health of one Resource at a time.

Real-world example, use case and scenario

EXAMPLE
Platform reboot
Resource Health shows VM1 was Unavailable because its host failed. Azure moved it to healthy hardware.

USE CASE
SLA claim
An online shop uses Resource Health history as proof for an SLA credit request.

SCENARIO
Database check
A logistics team checks Resource Health on its Azure SQL database before opening a support ticket.

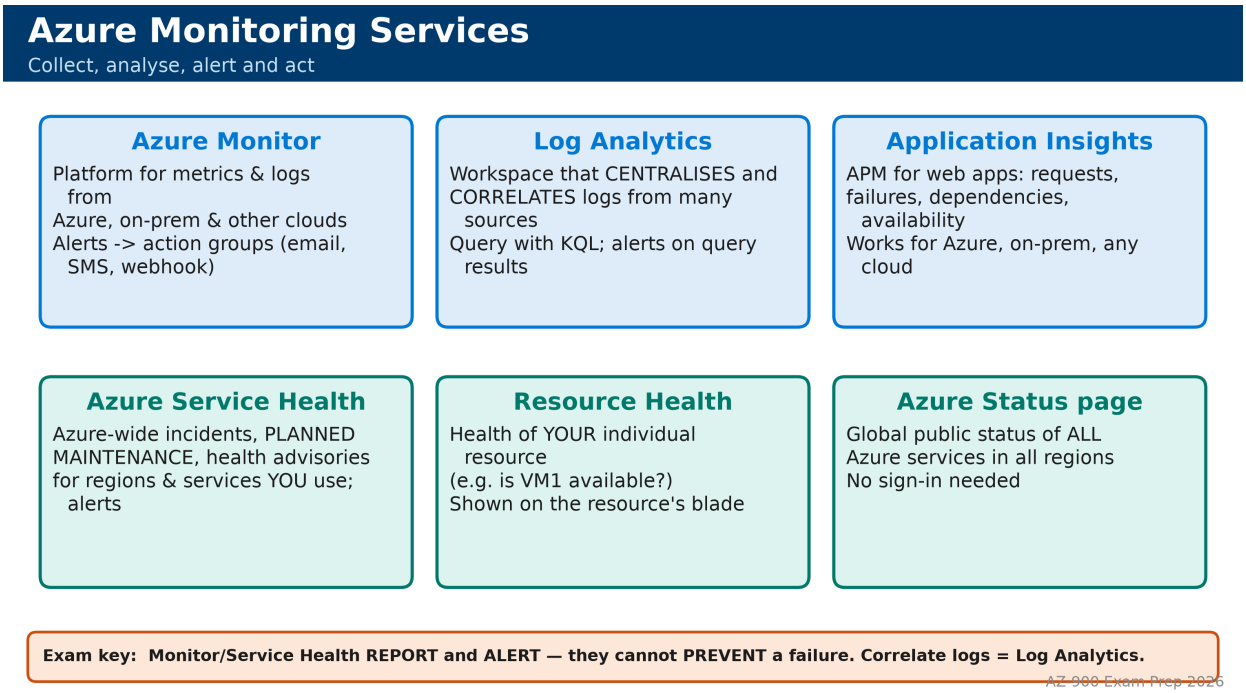


Figure Q404: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 405

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q544

A partner company that does not have an Azure account wants to check whether there is currently a widespread Azure outage affecting many customers. Which resource should they use?

✔ **A.** The Azure status page

✖ B. Azure Service Health

✖ C. Azure Resource Health

✖ D. Azure Monitor alerts

Correct answer: The Azure status page

Clarification (read this first): Watch 'does not have an Azure account'. Service Health and Resource Health need a sign-in. Only the public Azure status page works without one. Q404 (one VM's history) = Resource Health.

Answer explained

✔ **The Azure status page**
The public Azure status page needs no sign-in. It shows large incidents affecting Azure services around the world.

✖ **Azure Service Health**
Service Health is personal to your subscriptions and needs a portal sign-in.

✖ **Azure Resource Health**
Resource Health shows your own single resources and needs subscription access.

✖ **Azure Monitor alerts**
Monitor alerts are set up inside a subscription, about your own resources.

EXAM STRATEGY
Clue words: 'no Azure account' plus 'widespread outage' = public Azure status page. Service Health and Resource Health need a sign-in.

IMPACT
During big incidents, the status page is the first stop for partners and customers who have no portal access.

What · Where · When · Why · How

What	The Azure status page is Microsoft's public view of Azure service health.
Where	It is a public website. Anyone can open it in a browser without an Azure account.
When	Check it when you think there is a large outage affecting many customers.
Why	It gives a broad, non-personal picture of Azure-wide incidents.
How	Open the page and check services by region, plus the history of past incidents.

Exam tip
Public, no sign-in = status page. Personal to your subscription = Service Health.

Memory trick
The status page is a public billboard; Service Health is your personal mailbox.

Real-world example, use case and scenario

EXAMPLE
Vendor check
A SaaS customer checks the status page to see if a reported regional network problem is real.

USE CASE
Help desk
A school help desk links to the status page for staff without portal access.

SCENARIO
Media outage
A news broadcaster reviews the status page history after a streaming problem.

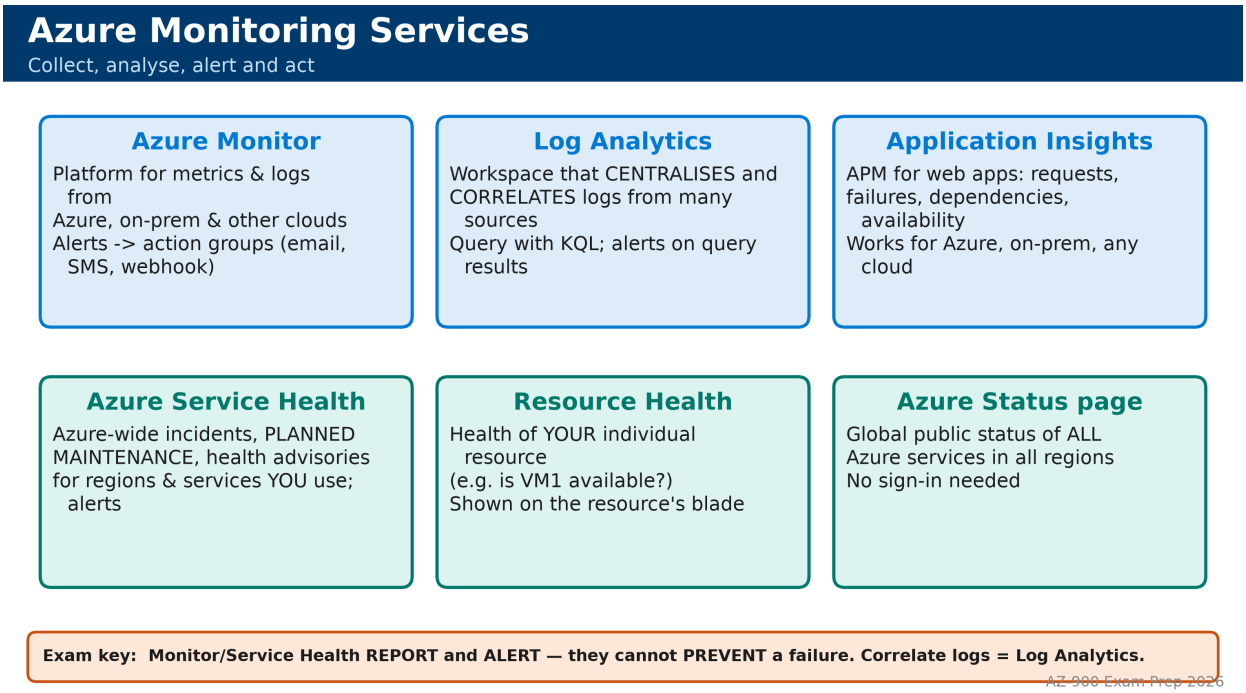


Figure Q405: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 406

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q545

Your operations team wants interactive reports in the Azure portal that combine KQL query results, metrics charts, and explanatory text, and that let users select parameters such as subscription and time range. Which Azure Monitor feature should you use?

✔ **A.** Azure Monitor workbooks

✖ B. Azure Advisor

✖ C. Action groups

✖ D. Azure Activity log

Correct answer: Azure Monitor workbooks

Answer explained

✔ **Azure Monitor workbooks**
Workbooks mix text, KQL queries, metrics and parameters (choices like time range) into interactive reports in the portal.

✖ **Azure Advisor**
Advisor gives recommendations, not custom interactive reports.

✖ **Action groups**
Action groups decide who is notified and what runs when an alert fires.

✖ **Azure Activity log**
The Activity log is a record of operations in a subscription, not a report builder.

EXAM STRATEGY
Clue words: 'interactive reports' mixing text, queries, metrics and parameters = workbooks. Action groups notify, Activity log records operations, Advisor recommends.

IMPACT
Workbooks give teams shared troubleshooting views with filters. This cuts diagnosis time without buying extra dashboard tools.

What · Where · When · Why · How

What	Workbooks are interactive reports built on Azure Monitor data.
Where	They are in Azure Monitor and on many resource pages in the portal.
When	Use them for troubleshooting guides, operations reviews and shared dashboards.
Why	They turn raw logs and metrics into readable charts with filters.
How	Create a workbook, add query, metric, text and parameter parts, then save and share it.

Exam tip
Show or report monitoring data interactively = workbooks (or dashboards). Notify people = action groups.

Memory trick
A workbook is a notebook for your monitoring data.

Real-world example, use case and scenario

EXAMPLE
VM insights
A workbook shows CPU, memory and disk charts for every VM in a resource group chosen from a dropdown list.

USE CASE
Weekly review
An insurance company uses a workbook in weekly operations meetings to review failed requests.

SCENARIO
Troubleshooting guide
An airline builds a workbook that guides on-call engineers through log checks during check-in outages.

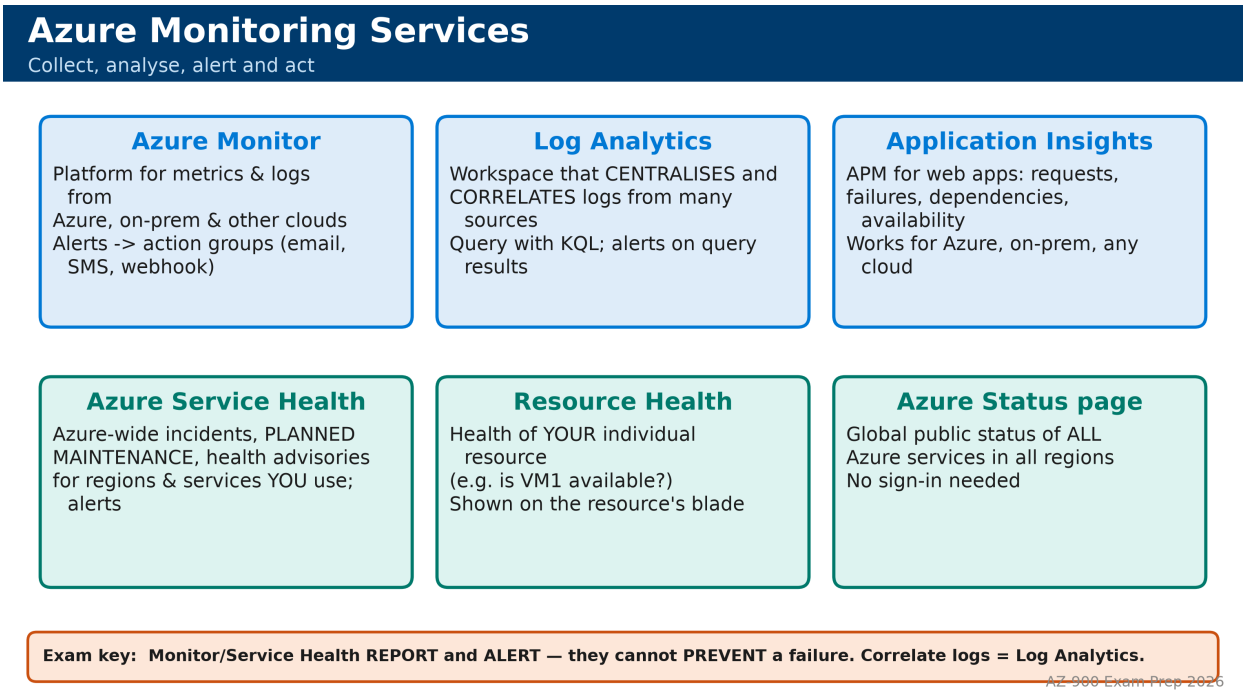


Figure Q406: Azure monitoring services

About this figure

- Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).
- Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.
- Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.
- Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.
- Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 407

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q546

Azure Advisor groups its recommendations into categories. Which of the following is NOT an Azure Advisor recommendation category?

✔ **A.** Compliance

✖ B. Reliability

✖ C. Operational excellence

✖ D. Performance

Correct answer: Compliance

Clarification (read this first): This is a 'which is NOT' question. Pick the option that is NOT an Advisor category. The five real ones are Cost, Reliability, Operational excellence, Performance and Security. Compliance is not one of them.

Answer explained

✔ **Compliance**
Advisor's categories are Reliability, Security, Performance, Cost and Operational excellence. Compliance is not one of them.

✖ **Reliability**
Reliability is a real Advisor category. It covers business continuity and high availability (staying up).

✖ **Operational excellence**
Operational excellence is a real Advisor category for process, workflow and deployment best practices.

✖ **Performance**
Performance is a real Advisor category for making resources faster.

EXAM STRATEGY
Advisor's five: Reliability, Security, Performance, Cost, Operational excellence (same as the Well-Architected Framework). Compliance belongs to Policy or Defender for Cloud.

IMPACT
Knowing Advisor's scope tells you where to look. Cost and reliability advice come from Advisor. Regulatory compliance comes from Defender or Policy.

What · Where · When · Why · How

What	Azure Advisor is a free service that gives best-practice tips for your resources.
Where	Open it from the Advisor page in the Azure portal, or with CLI and API.
When	Review it regularly, for example every month.
Why	It helps improve cost, security, reliability, performance and operations.
How	It studies your resource settings and usage, then lists tips in priority order.

Exam tip
Remember Advisor's five categories: Cost, Security, Reliability, Operational excellence, Performance (same as Well-Architected).

Memory trick
CROPS: Cost, Reliability, Operational excellence, Performance, Security.

Real-world example, use case and scenario

EXAMPLE
Cost
Advisor suggests resizing an underused D8s VM to a smaller size to save money.

USE CASE
Reliability
A healthcare provider follows Advisor's advice to turn on soft delete for blob storage.

SCENARIO
Performance
Advisor recommends Premium SSD disks for a busy shop database VM.

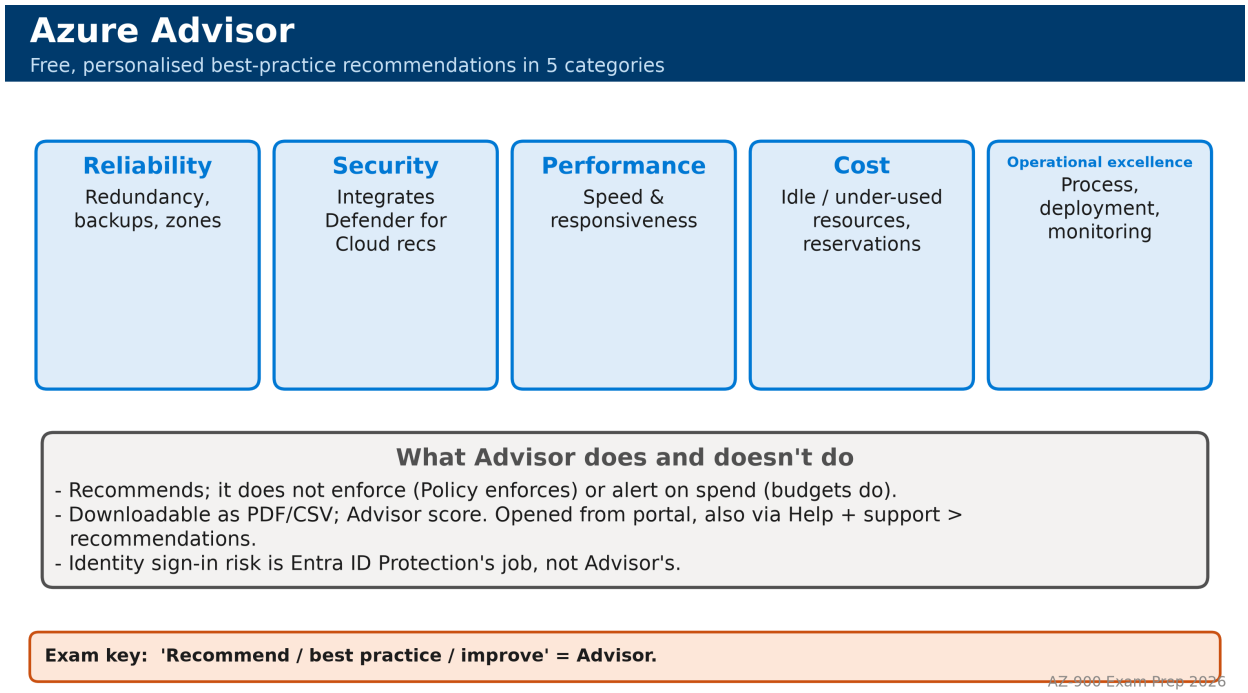


Figure Q407: Azure Advisor

About this figure

• The five boxes are Advisor's categories; each recommendation shows impact (high, medium, low) and suggested actions you can apply or postpone.

• Advisor is free and analyses your deployed resources' configuration and usage telemetry, giving subscription- and resource-group-level recommendations.

• Advisor score rates how well you follow best practices, overall and per category, helping prioritise fixes with the largest impact.

• Recommendations can trigger alerts and digest emails; security recommendations come from Defender for Cloud, so fixing them there also updates Advisor.

• Advisor is aligned with the Azure Well-Architected Framework pillars: reliability, security, cost optimisation, operational excellence and performance efficiency.

Question 408

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q547

An administrator creates a storage account in the Azure portal, a developer creates one with Azure CLI, and a pipeline creates one with Azure PowerShell. Which Azure service receives, authenticates and authorizes all three requests before they reach the storage resource provider?

✔ **A.** Azure Resource Manager

✖ B. Microsoft Entra Connect

✖ C. Azure Front Door

✖ D. Azure API Management

Correct answer: Azure Resource Manager

Answer explained

✔ **Azure Resource Manager**
All tools (portal, CLI, PowerShell, REST, SDKs) send requests to Azure Resource Manager. ARM checks who you are and what you may do.

✖ **Microsoft Entra Connect**
Entra Connect copies on-premises AD identities to Entra ID. It does not handle management requests.

✖ **Azure Front Door**
Front Door is a global load balancer and CDN for user traffic to web apps.

✖ **Azure API Management**
API Management publishes and protects your own APIs. It is not Azure's management layer.

EXAM STRATEGY
Clue words: portal, CLI and PowerShell all meet in one place = Azure Resource Manager, the single control plane (management layer). Front Door and API Management handle app traffic.

IMPACT
Because every tool goes through ARM, RBAC, locks and Policy work the same everywhere. There is no CLI back door around the rules.

What · Where · When · Why · How

What	Azure Resource Manager (ARM) is Azure's deployment and management service.
Where	It sits between every management tool and the Azure resource providers (services that build resources).
When	It is used every time you create, update or delete Azure resources.
Why	It gives the same access control, tags, locks and policy in every tool.
How	Tools call the ARM REST API. ARM checks identity and permissions, then passes the request on.

Exam tip
Same result in the portal, CLI or PowerShell? That is because of Azure Resource Manager.

Memory trick
ARM is the arm every tool must shake before touching Azure.

Real-world example, use case and scenario

EXAMPLE
Denied request
A CLI command fails with AuthorizationFailed. ARM checked RBAC and the user had no Contributor rights.

USE CASE
Locked resource
ARM blocks a PowerShell delete of a production database that has a CanNotDelete lock.

SCENARIO
Mixed tooling
A university IT team uses the portal and Terraform side by side. Both go through ARM.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

AZ-900 Exam Prep 2026

Figure Q408: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 409

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q548

You use Azure Cloud Shell and want scripts saved in your home directory to still be there the next time you start a session. Which Azure service does Cloud Shell use to persist your files?

✔ **A.** An Azure Files share

✖ B. Azure Blob Storage archive tier

✖ C. Azure Managed Disks attached to your VM

✖ D. Azure Queue Storage

Correct answer: An Azure Files share

Answer explained

✔ **An Azure Files share**
Cloud Shell attaches an Azure Files share (called clouddrive) to keep your home folder and files between sessions.

✖ **Azure Blob Storage archive tier**
The Archive tier is offline storage that must be rehydrated (brought back online) first. It cannot hold a live home folder.

✖ **Azure Managed Disks attached to your VM**
Cloud Shell runs on a temporary host managed by Microsoft, not on your VM.

✖ **Azure Queue Storage**
Queue Storage holds messages, not files.

EXAM STRATEGY
Clue words: 'keep files between sessions' = Azure Files share (clouddrive). Archive is offline, queues hold messages, Cloud Shell has no VM disk.

IMPACT
Without it you lose your scripts each session. Note: the file share has a small storage cost in your subscription.

What · Where · When · Why · How

What	Cloud Shell is a browser-based shell with Bash or PowerShell, already signed in.
Where	It runs from the Azure portal, shell.azure.com, the Azure mobile app and VS Code.
When	Use it to run CLI or Az PowerShell commands without installing anything.
Why	A file share keeps your scripts and settings across sessions and devices.
How	On first launch, choose to attach storage. Cloud Shell creates or uses a storage account with a file share.

Exam tip
Cloud Shell saves files in Azure Files. Cloud Shell itself is free; you pay only for the storage.

Memory trick
Cloud Shell keeps its files in Files.

Real-world example, use case and scenario

EXAMPLE
Saved script
An admin saves deploy.sh in clouddrive on Monday and runs it from another laptop on Friday.

USE CASE
Shared tooling
A consulting firm keeps reusable Bicep files in Cloud Shell's file share.

SCENARIO
Ephemeral session
A trainer uses Cloud Shell without storage in a class, knowing files vanish when the session ends.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

AZ-900 Exam Prep 2026

Figure Q409: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 410

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q549

You need to display near real-time, numeric performance values such as CPU percentage and network bytes for an Azure VM on a chart, using data that Azure collects automatically without configuring any agent or workspace. Which Azure Monitor data type should you use?

✔ **A.** Azure Monitor Metrics

✖ B. Azure Monitor Logs

✖ C. Azure Activity log

✖ D. Application Insights traces

Correct answer: Azure Monitor Metrics

Clarification (read this first): Watch 'numeric', 'near real-time' and 'without configuring any agent or workspace'. That is Metrics. Logs need a Log Analytics workspace (Q424).

Answer explained

✔ **Azure Monitor Metrics**
Platform metrics are numbers over time that Azure collects automatically. They are ideal for near real-time charts and alerts.

✖ **Azure Monitor Logs**
Logs need a Log Analytics workspace and data collection. They suit deep searches, not instant number charts.

✖ **Azure Activity log**
The Activity log records management operations (who did what), not performance numbers.

✖ **Application Insights traces**
Traces are messages from app code. They are not VM platform numbers.

EXAM STRATEGY
Clue words: 'numeric', 'near real-time', 'no agent or workspace' = platform Metrics. Logs need a workspace. Activity log tracks operations, not CPU.

IMPACT
Metrics drive fast autoscale and alerts with no setup cost. Using Logs for this adds cost and delay.

What · Where · When · Why · How

What	Metrics are numbers that describe a resource at a point in time.
Where	You see them in Metrics explorer and on a resource's Overview page.
When	Use them for real-time performance charts and fast metric alerts.
Why	They are light, quick and collected by default at no extra cost for platform metrics.
How	Open Metrics explorer, choose the resource and metric, and pin the chart to a dashboard.

Exam tip
Numbers over time, fast = Metrics. Text records searched with KQL = Logs.

Memory trick
Metrics measure; Logs list.

Real-world example, use case and scenario

EXAMPLE
CPU chart
An admin views Percentage CPU for VM1 over the last hour in Metrics explorer.

USE CASE
Autoscale
A ticket site adds VMs to its scale set when average CPU goes above 70 percent.

SCENARIO
Storage latency
A media company charts storage delay metrics during a big video upload.

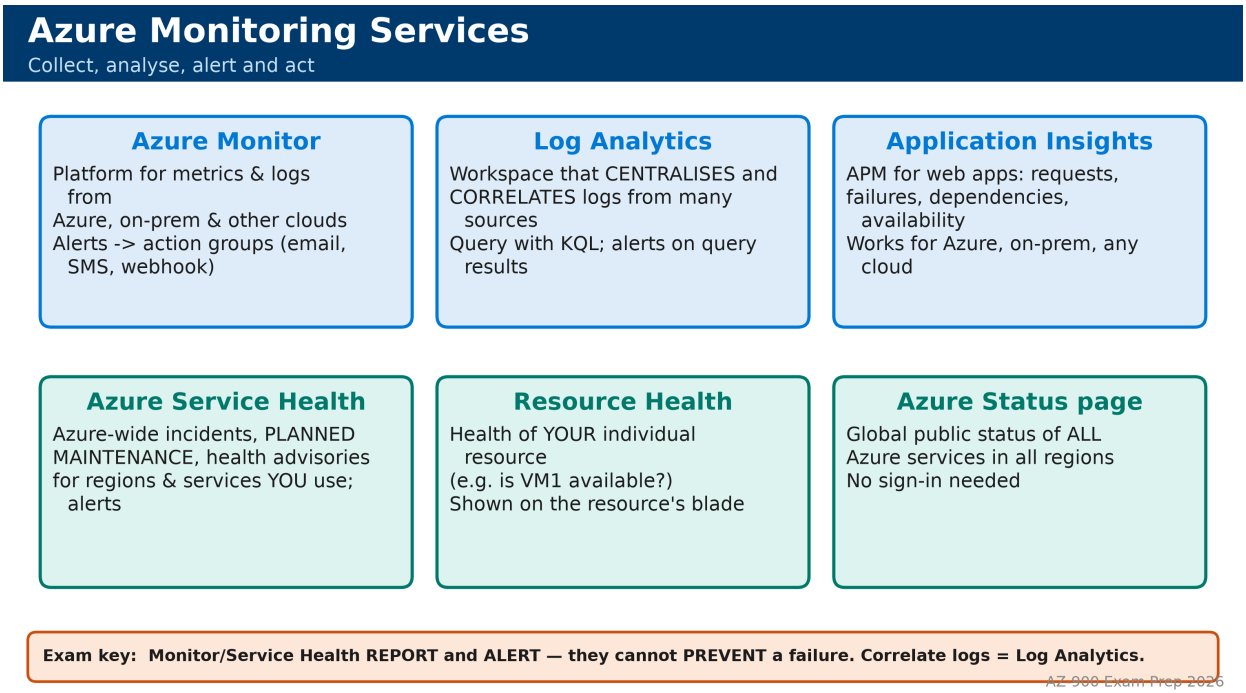


Figure Q410: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 411

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q550

Before deploying an updated Bicep file to a production resource group, you want to preview which resources would be created, modified, or deleted without making any changes. What should you use?

✔ **A.** The what-if operation

✖ B. Azure Advisor

✖ C. A resource lock

✖ D. Azure Resource Health

Correct answer: The what-if operation

Answer explained

✔ **The what-if operation**
What-if shows which resources an ARM or Bicep deployment would create, change or delete. It does not apply anything.

✖ **Azure Advisor**
Advisor gives tips for deployed resources. It does not preview template changes.

✖ **A resource lock**
A lock stops deletion or changes. It does not show what a deployment would change.

✖ **Azure Resource Health**
Resource Health shows if existing resources are up, not planned changes.

EXAM STRATEGY
Clue words: 'preview without making changes' = what-if. Advisor recommends, locks protect, Resource Health shows status. None of them simulate a deployment.

IMPACT
What-if catches accidental deletes or changes before they reach production. It is a cheap guard against outages you cause yourself.

What · Where · When · Why · How

What	What-if is a dry run (test run) of an Azure Resource Manager deployment.
Where	It runs from Azure CLI, Azure PowerShell or deployment pipelines.
When	Use it before deploying template changes, especially to production.
Why	It lowers the risk of accidental changes or deletions.
How	Run az deployment group what-if with the Bicep file and check the colour-coded list of changes.

Exam tip
Preview IaC changes without deploying = what-if.

Memory trick
Ask 'what if I deploy this?' before you really do.

Real-world example, use case and scenario

EXAMPLE
Caught change
What-if shows a template update would change a storage account's SKU (tier). The team did not want that.

USE CASE
Change approval
A bank adds what-if output to every production change request for approval.

SCENARIO
Pipeline gate
A retail DevOps pipeline runs what-if and waits for a review before applying changes.

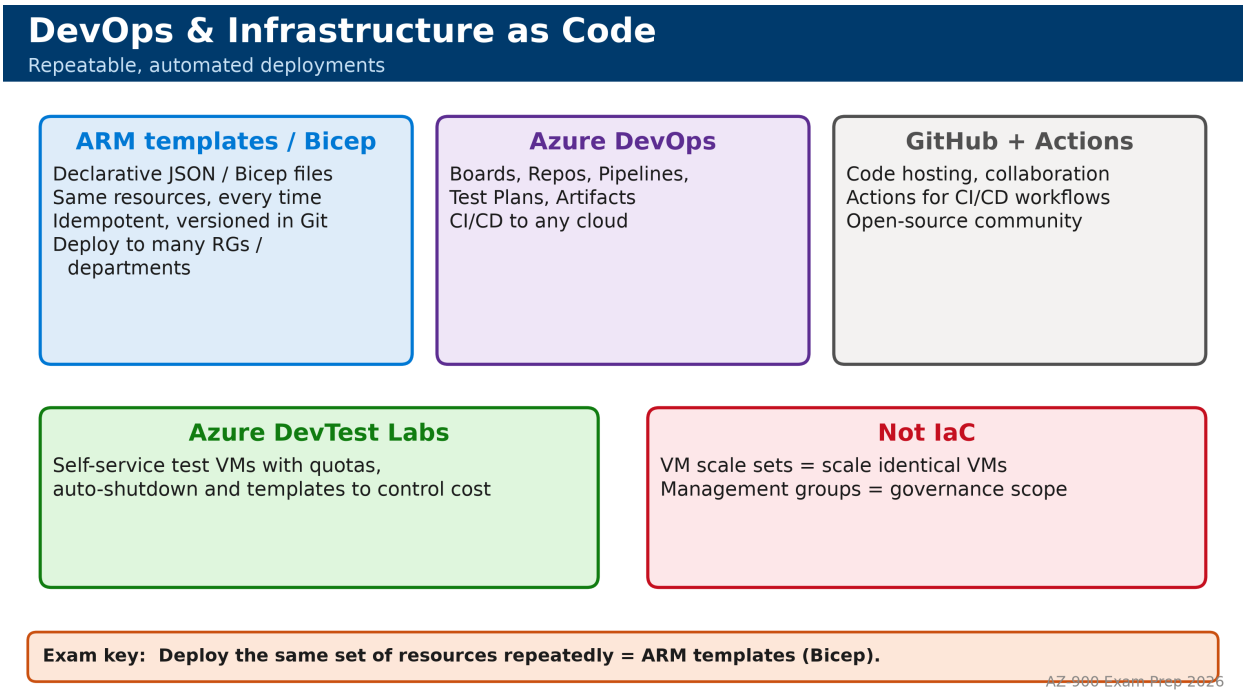


Figure Q411: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 412

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q551, Q106

A production virtual machine was deleted yesterday. You need to identify which user deleted it and when, without having configured any additional data collection. What should you review?

✔ **A.** The Azure Activity log

✖ B. Azure Advisor

✖ C. Application Insights

✖ D. Azure Monitor Metrics

Correct answer: The Azure Activity log

Clarification (read this first): Watch 'which user deleted it and when'. That is a management action, so it is in the Activity log. Metrics only hold numbers like CPU. The Activity log keeps 90 days by default.

Answer explained

✔ **The Azure Activity log**
The Activity log automatically records who created, changed or deleted resources, and when. It keeps this for 90 days.

✖ **Azure Advisor**
Advisor gives tips. It does not record who did what.

✖ **Application Insights**
Application Insights watches app data, not management actions on resources.

✖ **Azure Monitor Metrics**
Metrics store performance numbers, not records of user actions.

EXAM STRATEGY
Clue words: 'who deleted and when' plus 'no extra collection' = Activity log (management events, 90 days). Metrics and Application Insights do not record management actions.

IMPACT
You can quickly find who is responsible after an incident. Send the Activity log to a workspace if audits need more than 90 days.

What · Where · When · Why · How

What	The Activity log is a platform log of subscription-level events.
Where	It is in Azure Monitor and on the Activity log page of each resource and resource group.
When	Use it to check changes or investigate unexpected deletions.
Why	It answers who did what, to which resource, and when.
How	Filter by operation, resource and time. Send it to a workspace to keep it longer.

Exam tip
Who did what and when to an Azure resource = Activity log (90 days by default).

Memory trick
The Activity log logs admin activity.

Real-world example, use case and scenario

EXAMPLE
Deleted VM
The Activity log shows a Delete Virtual Machine action by a contractor's account at 22:14.

USE CASE
Audit evidence
A government agency sends the Activity log to a workspace to keep a year of change history.

SCENARIO
Config drift
A pharmacy chain finds who changed an NSG (network firewall) rule by filtering the Activity log.

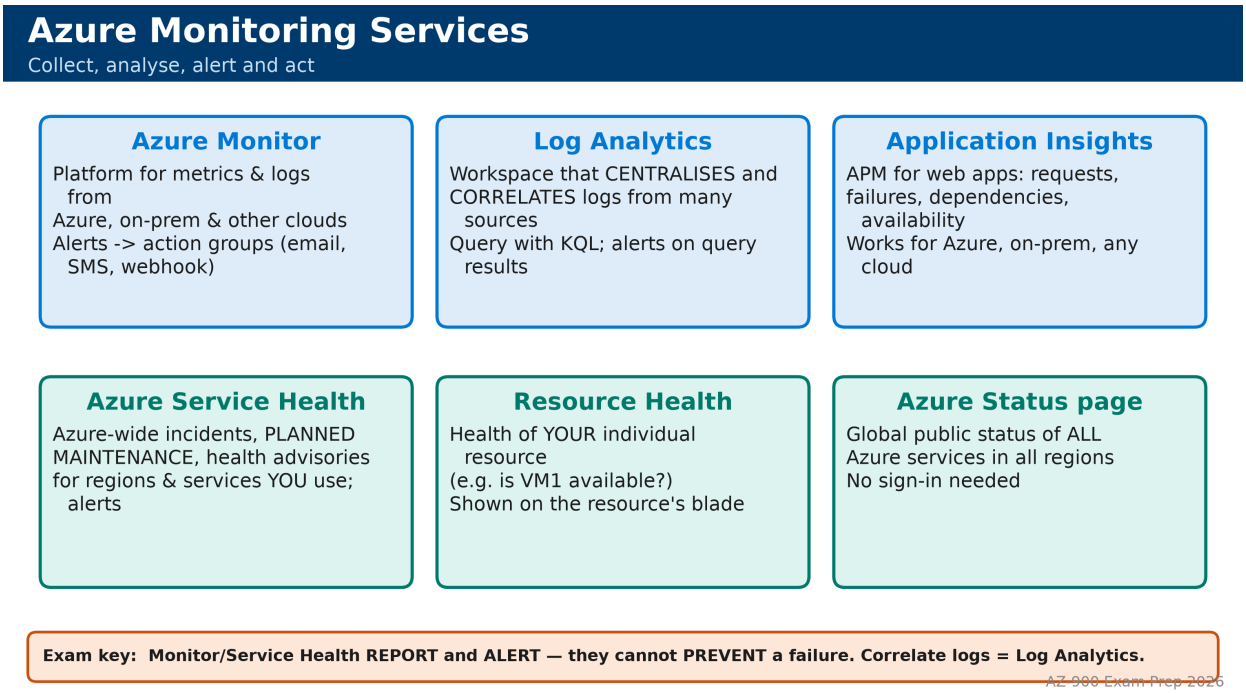


Figure Q412: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 413

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q552

For the following statement, select Yes if the statement is true. Otherwise, select No. Before you can deploy a Bicep file to Azure, you must manually convert it to an ARM JSON template.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): Watch 'must manually convert'. Azure CLI and PowerShell convert Bicep to ARM JSON for you, automatically. So No. You CAN run 'bicep build' by hand, but you do not have to.

Answer explained

✔ **No**
You do not need a manual step. CLI and PowerShell convert Bicep to ARM JSON automatically when you deploy.

✖ **Yes**
You deploy Bicep files directly. The conversion to JSON is automatic and hidden from you.

EXAM STRATEGY
Key fact: Bicep is converted automatically during Azure CLI or PowerShell deployment. 'Must manually convert' = No. 'bicep build' is optional.

IMPACT
It removes a manual pipeline step and a source of errors. This is why Microsoft recommends Bicep for new Azure IaC.

What · Where · When · Why · How

What	Bicep deployment means deploying .bicep files directly.
Where	It works from Azure CLI, Azure PowerShell, Cloud Shell and CI/CD pipelines.
When	Any time you deploy infrastructure written in Bicep.
Why	Automatic conversion keeps Bicep simple and still uses the ARM engine.
How	Run az deployment group create --template-file main.bicep. The CLI converts and sends it.

Exam tip
Bicep becomes ARM JSON automatically. You write in Bicep and deploy in Bicep.

Memory trick
Write Bicep, ship Bicep; ARM JSON is the hidden middleman.

Real-world example, use case and scenario

EXAMPLE
One command
A developer deploys main.bicep to a resource group with one az deployment group create command.

USE CASE
CI pipeline
An insurance firm's Azure DevOps pipeline deploys Bicep files with no separate build step.

SCENARIO
Inspecting output
A curious engineer runs bicep build only to look at the generated JSON.

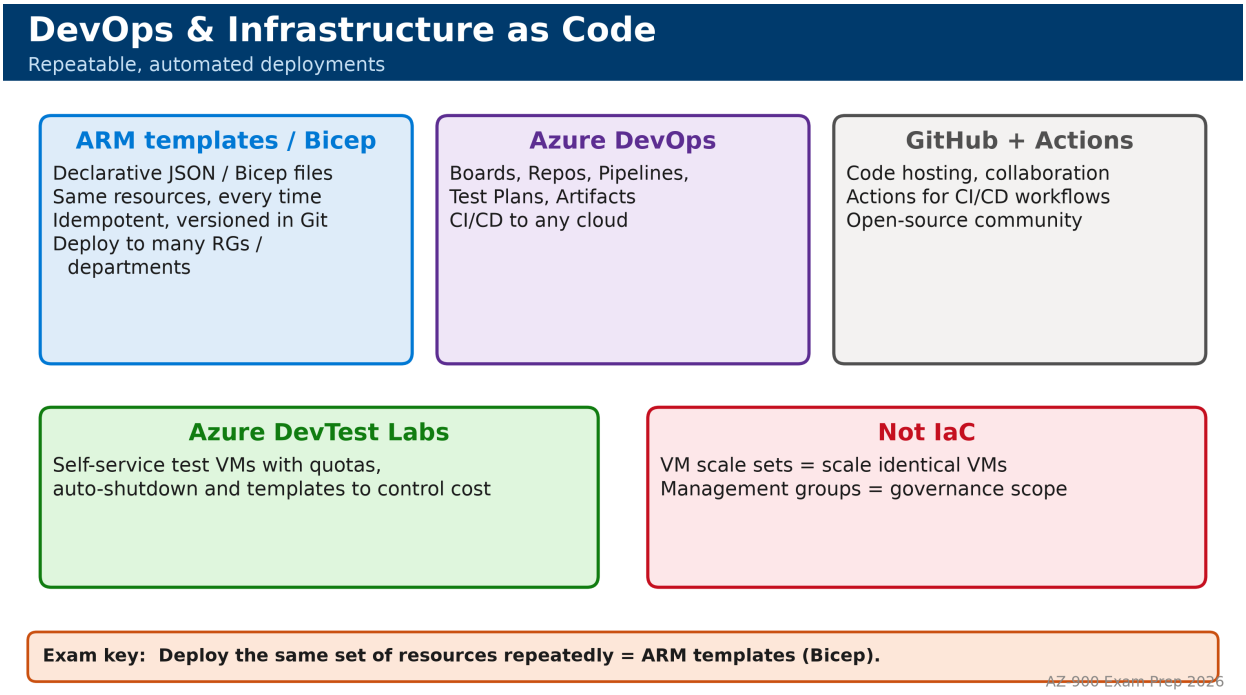


Figure Q413: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
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- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 414

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q554

For the following statement, select Yes if the statement is true. Otherwise, select No. Application Insights can monitor a web application that is hosted on-premises or in another cloud, not only applications hosted in Azure.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): Watch 'not only applications hosted in Azure'. Application Insights works with apps anywhere, as long as they send data to Azure. Q14 asks about apps deployed to Azure; that is also Application Insights.

Answer explained

✔ **Yes**
Application Insights works with any app that can send telemetry (monitoring data) to Azure. The app can be in Azure, on-premises or in another cloud.

✖ **No**
Apps anywhere can send data to Application Insights. Hosting in Azure is not needed.

EXAM STRATEGY
Key fact: Application Insights only needs the app to send data to it, from any host. 'Not only Azure' = Yes.

IMPACT
One APM (app performance monitoring) tool can watch apps in hybrid and multicloud setups, with no blind spots during a migration.

What · Where · When · Why · How

What	Application Insights is app performance monitoring in Azure Monitor.
Where	Apps can run in Azure, on-premises, in other clouds or on mobile devices.
When	Use it to track requests, failures, dependencies (called services) and user behaviour.
Why	It helps find and fix app problems early.
How	Add the Azure Monitor OpenTelemetry distro or SDK with a connection string to the app.

Exam tip
Application Insights = app performance monitoring. The app does not need to be in Azure.

Memory trick
Insights into apps, wherever they live.

Real-world example, use case and scenario

EXAMPLE
On-prem portal
An HR portal in a company datacenter sends request and error data to Application Insights.

USE CASE
Multicloud API
A travel firm monitors an API hosted in AWS with Application Insights.

SCENARIO
Slow pages
A retailer finds a slow checkout page is caused by a slow database call.

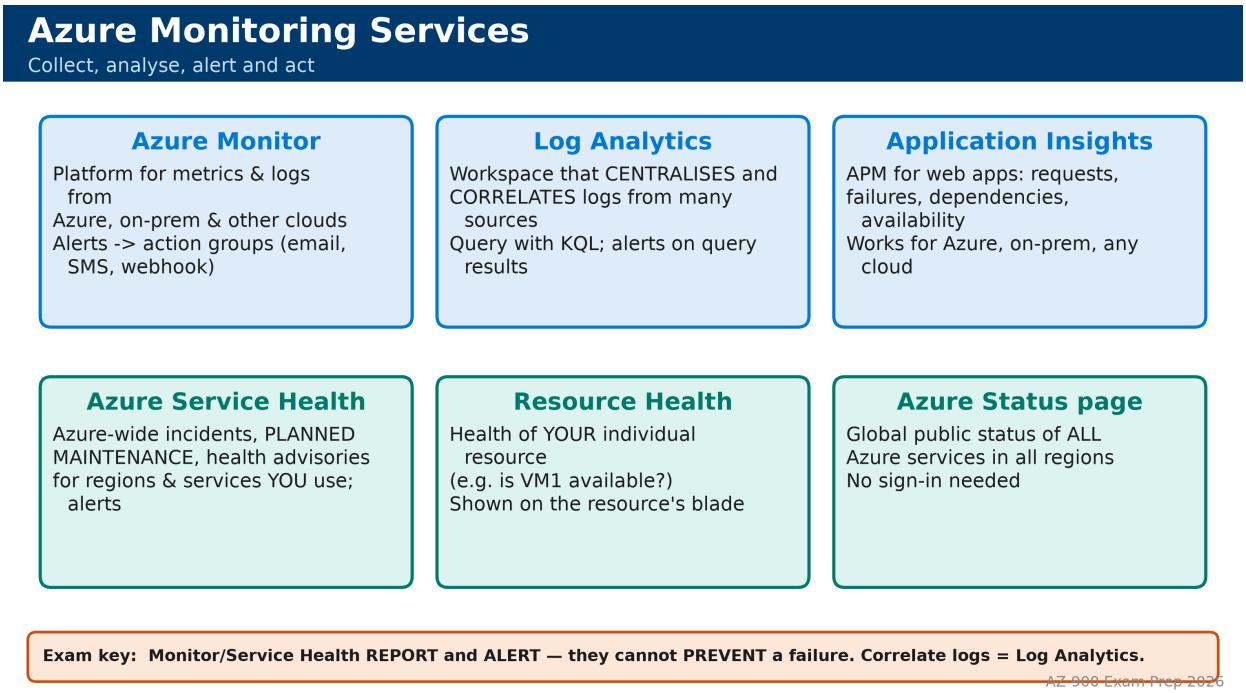


Figure Q414a: Azure monitoring services

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- Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).
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- Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.
- Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.
- Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

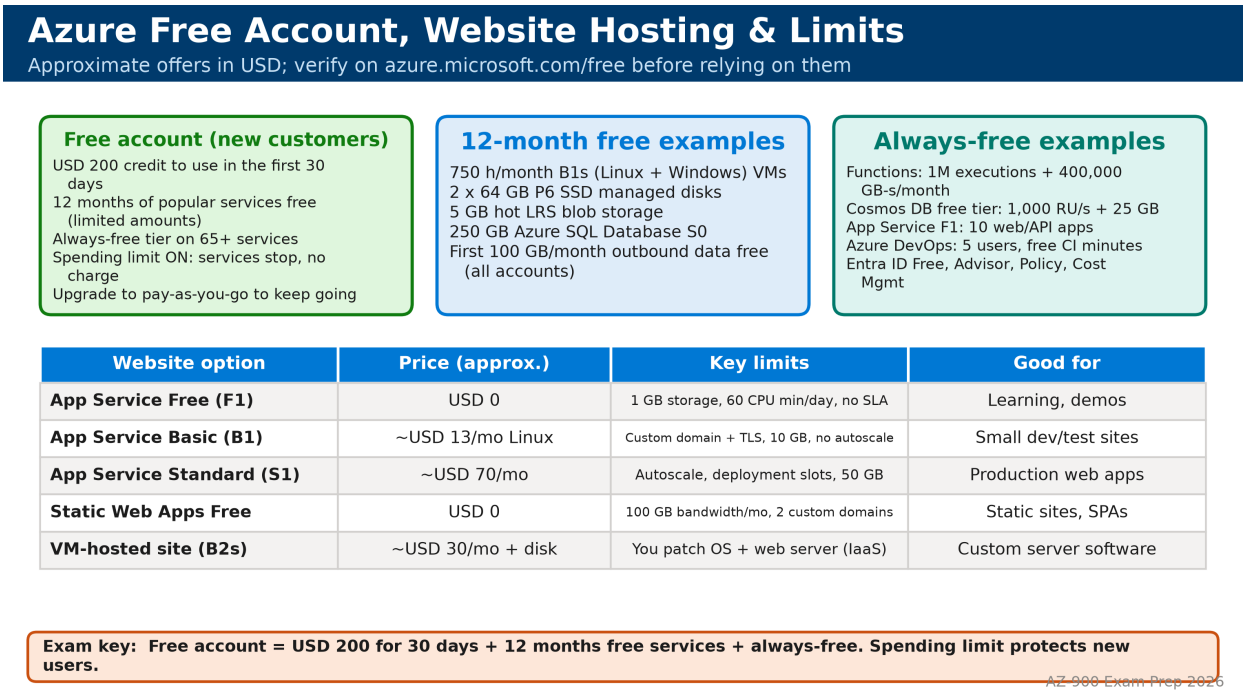


Figure Q414b: Free account, website hosting and limits

About this figure

- Top boxes compare the three free offers: 30-day credit, 12 months of selected services, and always-free tiers; the table compares web hosting choices.
- The approx. USD 200 credit expires after 30 days even if unspent; the spending limit stops services rather than charging your card.
- After upgrading to pay-as-you-go, 12-month free amounts continue, but usage above the free limits is billed at normal rates.
- App Service F1 runs on shared infrastructure with no custom-domain TLS, scale-out or SLA; B1 approx. USD 13/month Linux, S1 approx. USD 70/month.
- Static Web Apps Standard costs approx. USD 9 per app per month and adds SLA, more domains and larger apps than the free tier.
- A small B2s VM is approx. USD 30/month plus disk and public IP, and you must patch the OS and web server yourself (IaaS).

Question 415

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q555

For the following statement, select Yes if the statement is true. Otherwise, select No. An Azure Cloud Shell session ends automatically after 20 minutes without interactive activity.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Answer explained

✔ **Yes**
Cloud Shell sessions end after 20 minutes with no interactive activity. It is built for interactive work.

✖ **No**
Microsoft documents a 20-minute idle timeout for Cloud Shell sessions.

EXAM STRATEGY
Key fact: Cloud Shell times out after 20 idle minutes. It is for interactive sessions, not long-running jobs.

IMPACT
Long scripts can stop mid-run when the session times out. Use Azure Automation or a pipeline for unattended tasks.

What · Where · When · Why · How

What	Cloud Shell is an interactive browser-based shell for managing Azure.
Where	It runs on a temporary host managed by Microsoft, opened from the portal or shell.azure.com.
When	Use it for interactive admin tasks, not long background jobs.
Why	The timeout frees resources, because Cloud Shell is a free per-session service.
How	After a timeout, reconnect. Saved files remain in the attached file share.

Exam tip
Cloud Shell: free, in the browser, Bash or PowerShell, tools ready, 20-minute idle timeout.

Memory trick
Twenty idle minutes and the shell goes to sleep.

Real-world example, use case and scenario

EXAMPLE
Coffee break
An admin comes back from a 30-minute meeting and must reconnect Cloud Shell.

USE CASE
Nightly job
A logistics firm moves a nightly cleanup script from Cloud Shell to Azure Automation.

SCENARIO
Saved work
A student's scripts survive the timeout because they were saved in clouddrive.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

AZ-900 Exam Prep 2026

Figure Q415: Azure management tools

About this figure

- Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.
- Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.
- Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.
- The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.
- All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 416

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q556

For the following statement, select Yes if the statement is true. Otherwise, select No. Azure Advisor is available at no additional cost to view recommendations for your Azure resources.

✔ **A.** Yes

✖ B. No

Correct answer: Yes

Clarification (read this first): Watch 'no additional cost'. Advisor itself is free. You only pay if you act on a tip, for example buy a reservation or a bigger disk.

Answer explained

✔ **Yes**
Advisor is free with every subscription. You pay only for the resources or changes you choose to make.

✖ **No**
Advisor itself costs nothing. Any cost comes only from acting on its tips.

EXAM STRATEGY
Key fact: Advisor is free with every subscription. Cost comes only from changes you make. So 'no additional cost' = Yes.

IMPACT
Free tips on cost, reliability and security are an easy win. Every Azure team should review them often.

What · Where · When · Why · How

What	Azure Advisor is a free recommendation service.
Where	Open it in the Azure portal, with CLI, or through the REST API.
When	Use it any time to review best-practice tips.
Why	Free access encourages every customer to improve cost, security and reliability.
How	Open Advisor, review tips by category, and act on them or postpone them.

Exam tip
Advisor = free, personal tips in five categories.

Memory trick
Free advice from your Azure advisor.

Real-world example, use case and scenario

EXAMPLE
Reservation tip
Advisor recommends a VM reservation for free. The customer pays only if they buy it.

USE CASE
Startup budget
A small startup uses Advisor at no cost to find unused public IP addresses.

SCENARIO
Nonprofit
A charity reviews Advisor every month to cut spending without buying other tools.

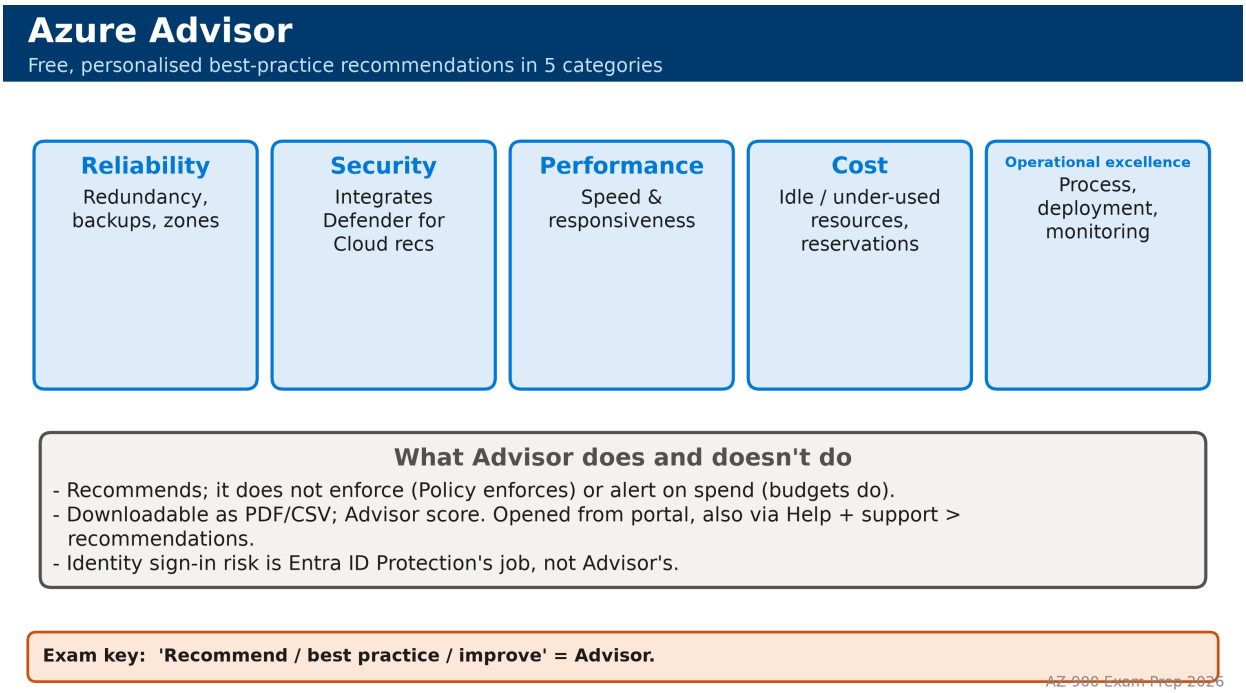


Figure Q416: Azure Advisor

About this figure

• The five boxes are Advisor's categories; each recommendation shows impact (high, medium, low) and suggested actions you can apply or postpone.

• Advisor is free and analyses your deployed resources' configuration and usage telemetry, giving subscription- and resource-group-level recommendations.

• Advisor score rates how well you follow best practices, overall and per category, helping prioritise fixes with the largest impact.

• Recommendations can trigger alerts and digest emails; security recommendations come from Defender for Cloud, so fixing them there also updates Advisor.

• Advisor is aligned with the Azure Well-Architected Framework pillars: reliability, security, cost optimisation, operational excellence and performance efficiency.

Question 417

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q557

For the following statement, select Yes if the statement is true. Otherwise, select No. When you redeploy an ARM template in the default incremental mode, resources in the resource group that are not defined in the template are automatically deleted.

✖ A. Yes

✔ **B.** No

Correct answer: No

Clarification (read this first): Watch 'default incremental mode'. Incremental only adds and updates. It never deletes extra resources. Only Complete mode deletes resources that are not in the template.

Answer explained

✔ **No**
Incremental mode (the default) leaves resources not in the template alone. Only Complete mode deletes them.

✖ **Yes**
That describes Complete mode, not the default Incremental mode.

EXAM STRATEGY
Key fact: Incremental (default) adds and updates only. Complete deletes resources not in the template. 'Default' plus 'deleted' = No.

IMPACT
Running Complete mode by mistake can delete resources not in the template. That is a real production outage risk.

What · Where · When · Why · How

What	Deployment modes control how ARM treats resources that already exist.
Where	They apply to resource group deployments of ARM templates and Bicep files.
When	You choose a mode each time you deploy. Incremental is used unless you pick Complete.
Why	Incremental avoids deleting resources created outside the template.
How	Use --mode Complete in CLI or -Mode Complete in PowerShell to change it.

Exam tip
Incremental = add or update only. Complete = make the resource group match the template exactly.

Memory trick
Incremental increments; Complete cleans.

Real-world example, use case and scenario

EXAMPLE
Manual VM survives
A test VM created by hand stays in the resource group after an incremental redeploy.

USE CASE
Shared group
A marketing team uses incremental mode in a shared resource group so others are not affected.

SCENARIO
Strict environment
A regulated lab uses Complete mode so its resource group always matches source control exactly.

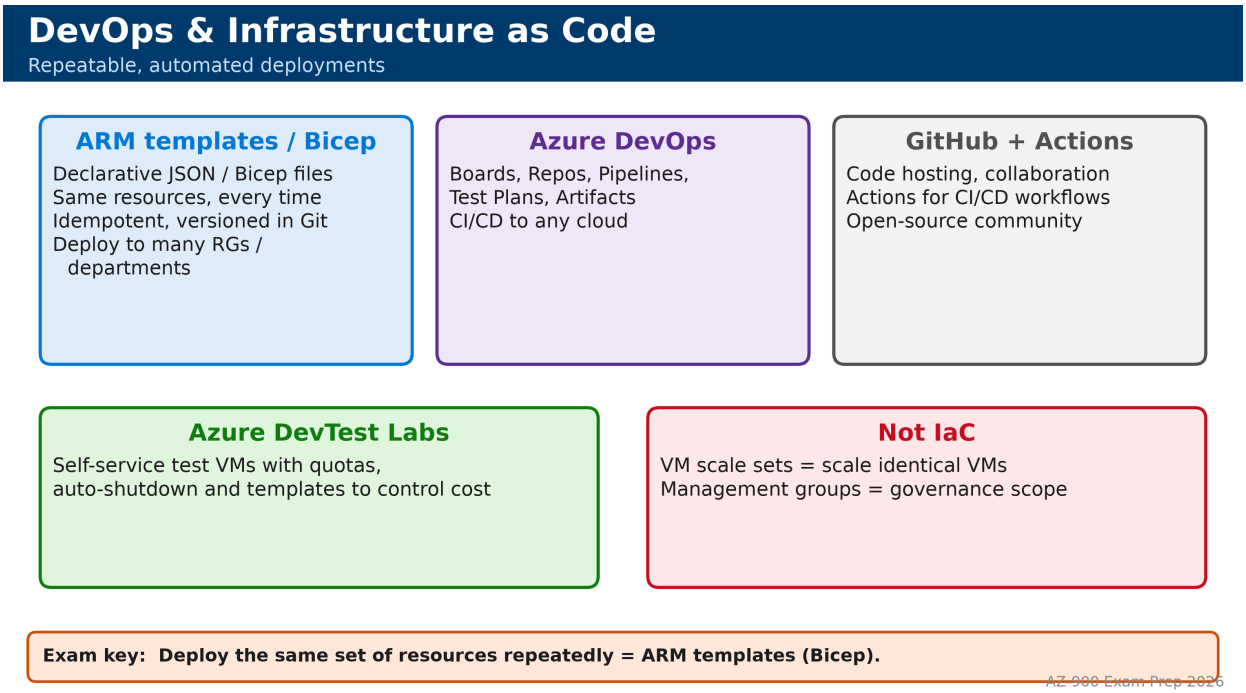


Figure Q417: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 418

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q558

Your company wants to define its Azure environment as infrastructure as code using formats that Azure Resource Manager natively deploys. Which two formats should you use? Each correct answer presents a complete solution.

- ✔ **A.** ARM templates (JSON)
- ✔ **B.** Bicep files
- ✖ C. Azure Monitor workbooks
- ✖ D. Azure Advisor recommendations
- ✖ E. Microsoft Purview data maps

Correct answer: ARM templates (JSON); Bicep files

Clarification (read this first): Choose TWO. The question wants file formats that Azure Resource Manager deploys natively (built in). Those are ARM JSON templates and Bicep. Terraform also works but is third-party and is not an option here.

Answer explained

✔ **ARM templates (JSON); Bicep files**
ARM JSON templates and Bicep are Azure's native IaC (infrastructure as code) formats. Azure Resource Manager deploys them.

✖ **Azure Monitor workbooks**
Workbooks are interactive monitoring reports, not infrastructure definitions.

✖ **Azure Advisor recommendations**
Advisor gives best-practice tips. It does not define or deploy resources.

✖ **Microsoft Purview data maps**
Purview data maps catalog and govern data. They are not a deployment format.

EXAM STRATEGY
Clue words: 'natively deployed by Resource Manager' = ARM JSON and Bicep. Workbooks, Advisor and Purview are not deployment formats.

IMPACT
Native IaC formats support new Azure resource types from day one. They also work with what-if and deployment history.

What · Where · When · Why · How

What	Infrastructure as code means defining infrastructure in files instead of clicking by hand.
Where	The files are kept in Git repositories and deployed to Azure scopes (levels).
When	Use IaC for repeatable environments such as dev, test and production.
Why	It gives consistency, version history, reviews and automation.
How	Write ARM JSON or Bicep, then deploy with CLI, PowerShell or pipelines.

Exam tip
Native Azure IaC = ARM templates and Bicep. Terraform is third-party but also works.

Memory trick
ARM and Bicep: the arm and the muscle of Azure IaC.

Real-world example, use case and scenario

EXAMPLE
Identical environments
A software company deploys the same dev and prod setups from one Bicep file with different parameters.

USE CASE
Disaster recovery
A bank keeps ARM templates to rebuild its network quickly in a second region.

SCENARIO
Code review
A healthcare startup requires a pull request review for every Bicep change before deployment.

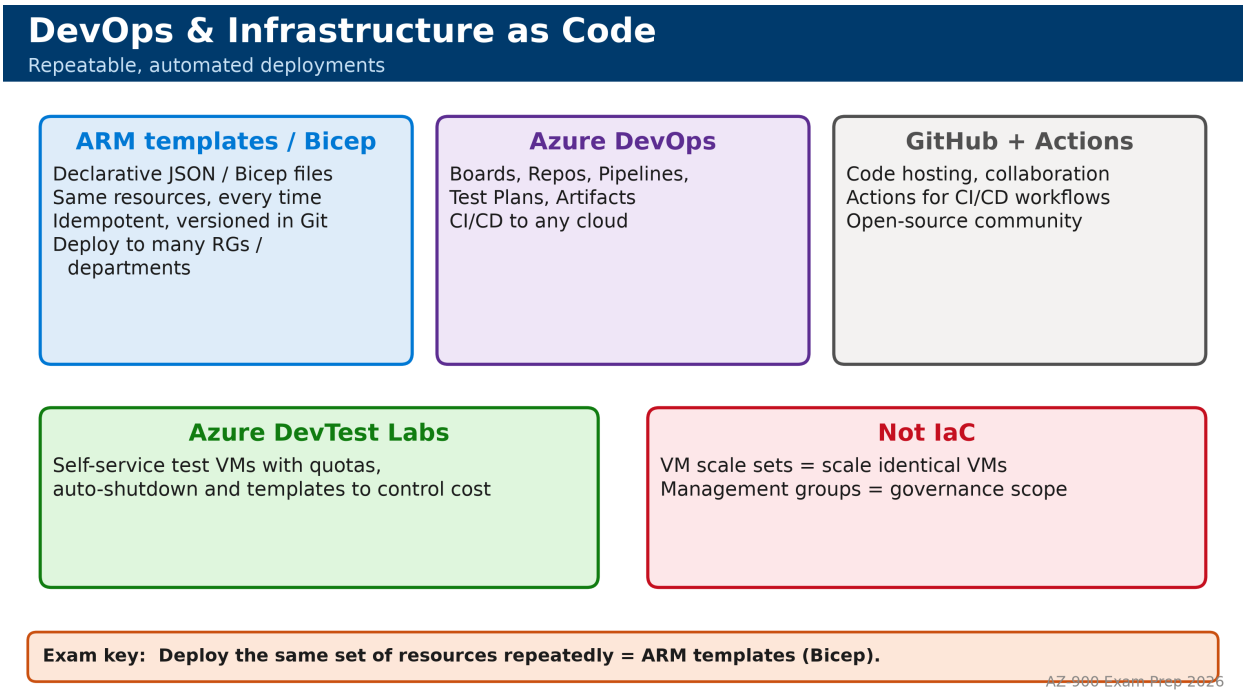


Figure Q418: DevOps and infrastructure as code

About this figure

• Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.

• ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.

• Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.

• GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.

• DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 419

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q559

You create an Azure Monitor alert rule that fires when a VM's CPU exceeds 95 percent. You configure an action group for the alert. Which two actions can the action group perform? Each correct answer presents a complete solution.

✔ **A.** Send an SMS message and email to the on-call engineer

✔ **B.** Trigger an Azure Logic App or webhook

✖ C. Apply a CanNotDelete lock to the VM

✖ D. Assign an Azure Policy initiative to the subscription

Correct answer: **Send an SMS message and email to the on-call engineer; Trigger an Azure Logic App or webhook**

Clarification (read this first): Choose TWO. An action group can only NOTIFY people (email, SMS, voice, push) or RUN something (webhook, Logic App, Function, runbook). It cannot add locks or assign policies.

Answer explained

✔ **Send an SMS message and email to the on-call engineer; Trigger an Azure Logic App or webhook**
Action groups notify people (email, SMS, voice, push) and run actions like webhooks, Logic Apps, Functions or runbooks (automation scripts).

✖ **Apply a CanNotDelete lock to the VM**
Locks are added through ARM management. Adding a lock is not a built-in action group action.

✖ **Assign an Azure Policy initiative to the subscription**
Assigning a policy is a governance task, not an action group action.

EXAM STRATEGY
Action groups = notify (email, SMS, voice, push) or act (webhook, Logic App, Function, runbook). Locks and Policy assignments are not alert actions.

IMPACT
Reliable on-call messages and automatic responses shorten recovery time when a production VM is at full CPU.

What · Where · When · Why · How

What	An action group is a set of notifications and actions that alerts trigger.
Where	You create action groups in Azure Monitor and reuse them in many alert rules.
When	They run whenever a linked alert rule fires.
Why	They make sure the right people know and automatic responses start fast.
How	Create an action group, add notification types and actions, then link it to alert rules.

Exam tip
Alert rule = what condition. Action group = who is told and what runs.

Memory trick
The alert rings the bell; the action group answers the call.

Real-world example, use case and scenario

EXAMPLE
Pager duty
An action group texts and emails the on-call engineer when a payment API fails.

USE CASE
Auto-ticket
A telecom company's action group calls a webhook that opens an incident ticket.

SCENARIO
Self-healing
A gaming studio's action group runs an Automation runbook to restart a stuck service.

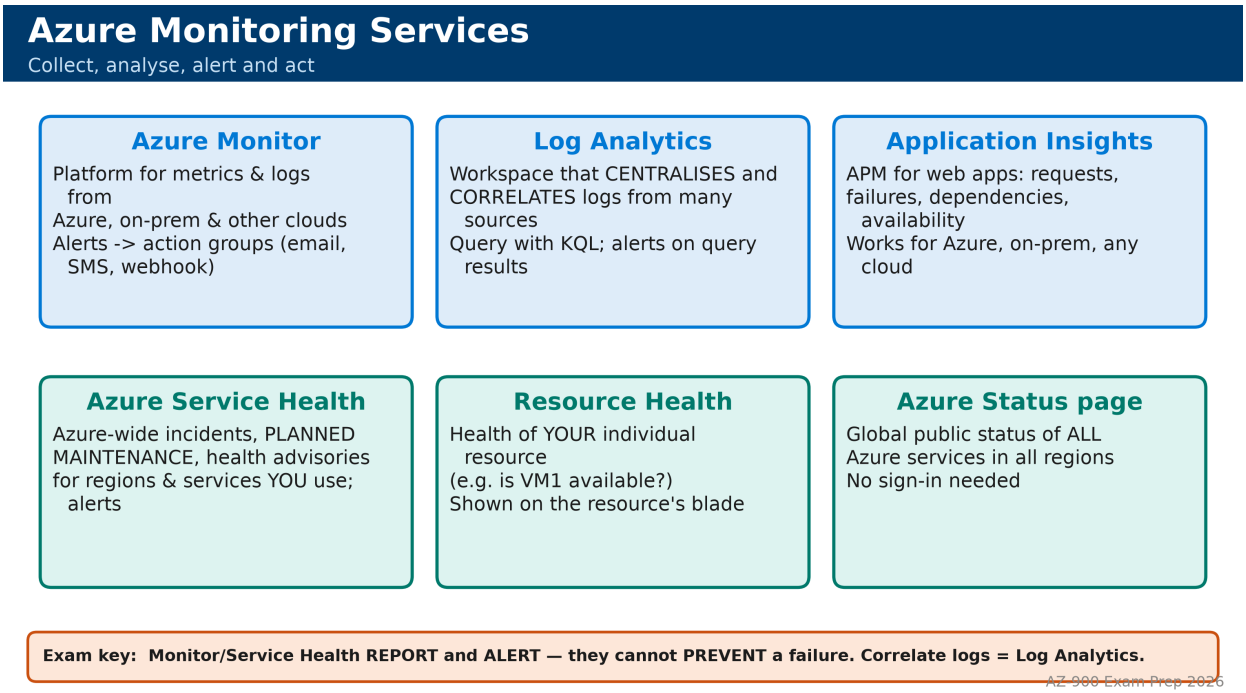


Figure Q419: Azure monitoring services

About this figure

- Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).
- Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.
- Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.
- Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.
- Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 420

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q560

Which two types of information are provided by Azure Service Health in the Azure portal? Each correct answer presents a complete solution.

- ✔ **A.** Health advisories, such as upcoming retirements of features you use
- ✔ **B.** Service issues currently affecting Azure services in your regions
- ✖ C. Recommendations to resize underutilized virtual machines
- ✖ D. Your Microsoft Defender for Cloud secure score
- ✖ E. CPU utilization charts for your virtual machines

Correct answer: **Health advisories, such as upcoming retirements of features you use; Service issues currently affecting Azure services in your regions**

Clarification (read this first): Choose TWO. Service Health tells you about Azure's side: outages, maintenance and advisories (like feature retirements). Tips to resize VMs are from Advisor; secure score is from Defender; CPU charts are from Metrics. Q153: Service Health can alert but cannot PREVENT a failure.

Answer explained

✔ **Health advisories, such as upcoming retirements of features you use; Service issues currently affecting Azure services in your regions**
Service Health reports service issues (outages), planned maintenance, health advisories and security advisories for your subscriptions and regions.

✖ **Recommendations to resize underutilized virtual machines**
Cost tips like resizing VMs come from Azure Advisor.

✖ **Your Microsoft Defender for Cloud secure score**
Secure score is shown in Microsoft Defender for Cloud.

✖ **CPU utilization charts for your virtual machines**
Performance charts come from Azure Monitor Metrics.

EXAM STRATEGY
Service Health = Azure's side: service issues, planned maintenance, health and security advisories for your regions. Cross out Advisor tips, secure score and metrics.

IMPACT
Retirement advisories warn you months ahead, before a feature you use stops working. This avoids rushed emergency moves.

What · Where · When · Why · How

What	Azure Service Health is a personal dashboard of Azure service events.
Where	It is in the Azure portal under Service Health, filtered to your subscriptions and regions.
When	Check it during incidents and to plan for maintenance and retirements.
Why	It tells you how Azure-side events affect your own services.
How	Open Service Health, review the event categories, and create Service Health alerts.

Exam tip
Service Health categories: service issues, planned maintenance, health advisories, security advisories.

Memory trick
Service Health is Azure's news feed about your services.

Real-world example, use case and scenario

EXAMPLE
Retirement notice
A health advisory warns that an API version a company uses will retire next year.

USE CASE
Outage tracking
A food delivery firm follows a service issue in Service Health during a regional storage problem.

SCENARIO
Proactive alert
A school district gets an email when a service issue affects its region.

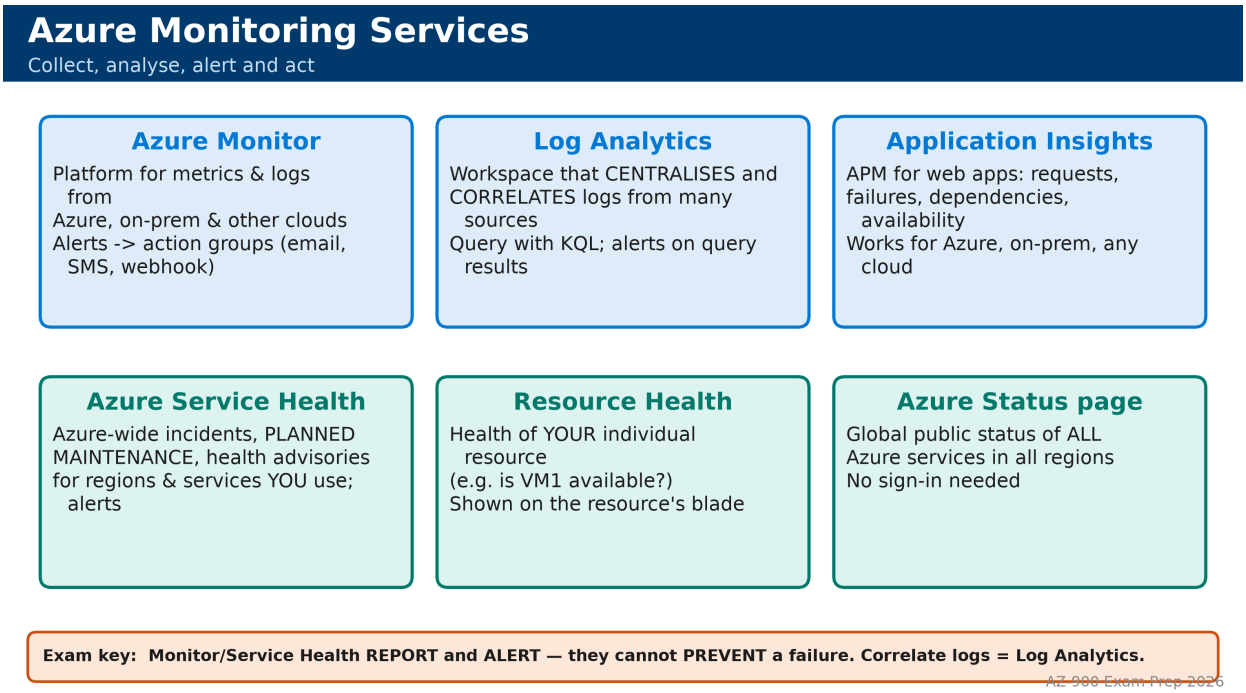


Figure Q420: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.

Question 421

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q561

Your company plans to use Azure Arc. Which two types of resources running outside Azure can be onboarded to Azure Arc? Each correct answer presents a complete solution.

✔ **A.** Windows and Linux physical or virtual servers

✔ **B.** Kubernetes clusters

✖ C. Microsoft 365 user mailboxes

✖ D. Microsoft Entra ID user accounts

Correct answer: **Windows and Linux physical or virtual servers; Kubernetes clusters**

Clarification (read this first): Choose TWO. Arc is for infrastructure (servers, Kubernetes, SQL Server, VMware). It is not for people or mailboxes. Q401 is related: manage outside servers without moving them = Arc.

Answer explained

✔ **Windows and Linux physical or virtual servers; Kubernetes clusters**
Arc can connect servers (Windows or Linux, physical or virtual) and Kubernetes clusters. It also supports SQL Server and VMware environments.

✖ **Microsoft 365 user mailboxes**
Mailboxes are managed in Exchange Online and Microsoft 365 admin tools, not Azure Arc.

✖ **Microsoft Entra ID user accounts**
User accounts already live in Entra ID. Arc connects infrastructure, not users.

EXAM STRATEGY
Arc connects infrastructure: servers, Kubernetes clusters, SQL Server, VMware/SCVMM. Cross out Microsoft 365 mailboxes and Entra users; they are not infrastructure.

IMPACT
Putting on-premises and other-cloud servers and Kubernetes under Arc gives the same policy and security everywhere.

What · Where · When · Why · How

What	Azure Arc supports servers, Kubernetes, SQL Server and virtualization platforms.
Where	These resources can be on-premises, at the edge or in other public clouds.
When	Connect them when you need the same governance and monitoring for all.
Why	One management layer means fewer separate tools in hybrid setups.
How	Servers use the Connected Machine agent. Kubernetes clusters use Arc agents installed with Helm or CLI.

Exam tip
Arc = servers, Kubernetes, data services (SQL), and VMware/SCVMM outside Azure.

Memory trick
Arc reaches servers and clusters, not people and mailboxes.

Real-world example, use case and scenario

EXAMPLE
Edge Kubernetes
A retailer connects in-store Kubernetes clusters to Arc and deploys apps with GitOps (deploy from Git).

USE CASE
SQL inventory
An insurer connects on-premises SQL Server instances to Arc to see them in the Azure portal.

SCENARIO
Cloud servers
A media company connects Google Cloud VMs to Arc to apply Azure Policy guest configuration.

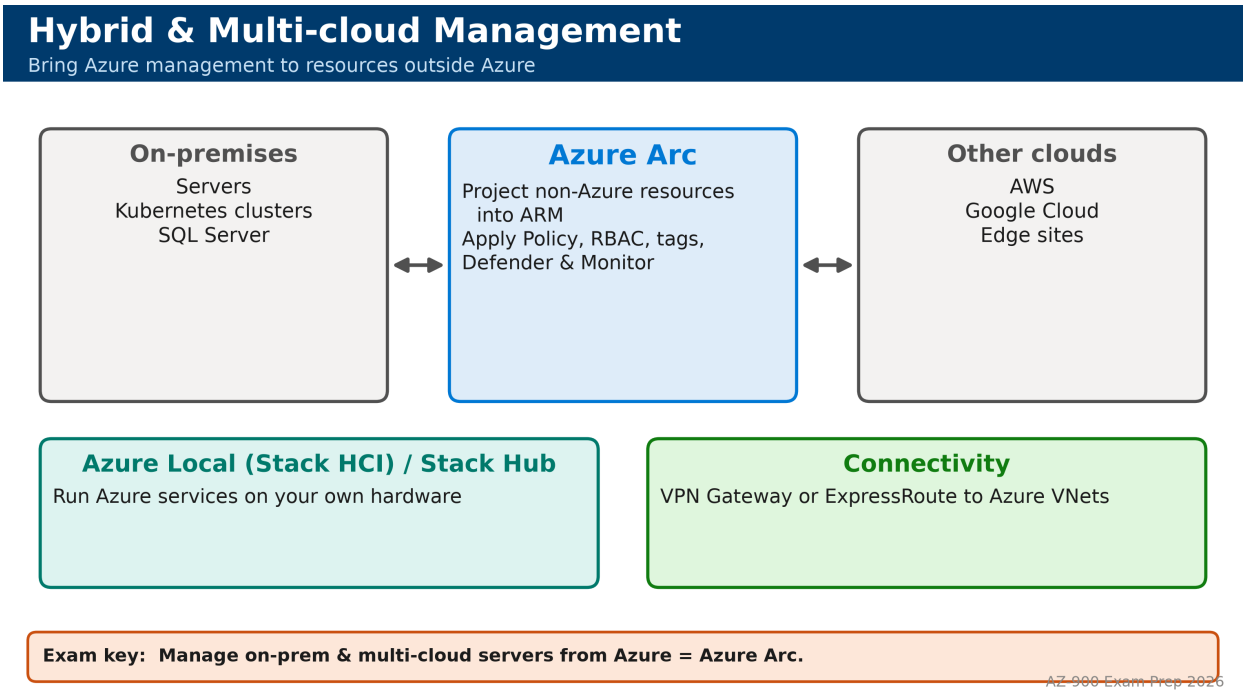


Figure Q421: Hybrid and multi-cloud management

About this figure

- Arrows show Azure Arc linking on-premises and other-cloud resources into Azure Resource Manager; bottom boxes cover local Azure hardware and network links.
- Arc-enabled servers' core control plane (inventory, tags, RBAC, Policy) is free; add-ons such as Defender, Update Manager and Monitor are billed separately.
- Arc can also manage Kubernetes clusters and SQL Server; Arc-enabled SQL Server can be licensed pay-as-you-go through your Azure bill.
- Azure Local (formerly Azure Stack HCI) runs VMs and AKS on validated on-premises hardware, billed per physical core per month in Azure.
- Site-to-site VPN uses the Internet with encryption; ExpressRoute is a private connection through a provider with higher reliability and an SLA up to 99.95%.

Question 422

AZURE MANAGEMENT & GOVERNANCE

DevOps & IaC
v5 Q562

Review the underlined text. If it makes the statement correct, select "No change is needed." If the statement is incorrect, select the answer choice that makes it correct. Azure Resource Manager templates are files written in [YAML] format.

- ✖ A. No change is needed
- ✔ **B.** JSON
- ✖ C. XML
- ✖ D. CSV

Correct answer: JSON

Clarification (read this first): Bracketed word: 'YAML'. ARM templates are JSON. YAML is a trap - it is used for pipelines and Kubernetes files, not ARM templates.

Answer explained

✔ **JSON**
ARM templates are written in JSON. Bicep is a simpler language that becomes that JSON.

✖ **No change is needed**
YAML is used for pipelines like Azure Pipelines, not ARM templates.

✖ **XML**
ARM templates do not use XML. Older classic Azure config files did.

✖ **CSV**
CSV is a flat table format. It cannot describe nested resource definitions.

EXAM STRATEGY
Key fact: ARM templates are JSON. YAML (pipelines and Kubernetes files) is a common trap. Bicep is its own language.

IMPACT
Knowing template formats helps when you review IaC. JSON syntax errors often cause failed deployments.

What · Where · When · Why · How

What	ARM templates are declarative JSON files that describe Azure resources.
Where	They are kept in repositories or template specs, or exported from the portal.
When	Use them to deploy repeatable, consistent infrastructure.
Why	JSON is structured and machine-readable, so it suits nested resource definitions.
How	Templates have $schema, parameters, variables, resources and outputs sections.

Exam tip
ARM templates = JSON. Pipelines = YAML. Bicep = its own language.

Memory trick
ARM wears a JSON jersey.

Real-world example, use case and scenario

EXAMPLE
Export template
An admin exports a resource group's ARM template from the portal and gets a JSON file.

USE CASE
Template spec
A bank stores approved JSON templates as template specs for teams to deploy.

SCENARIO
YAML pipeline
A retail team's YAML pipeline calls a JSON ARM template to deploy a web app.

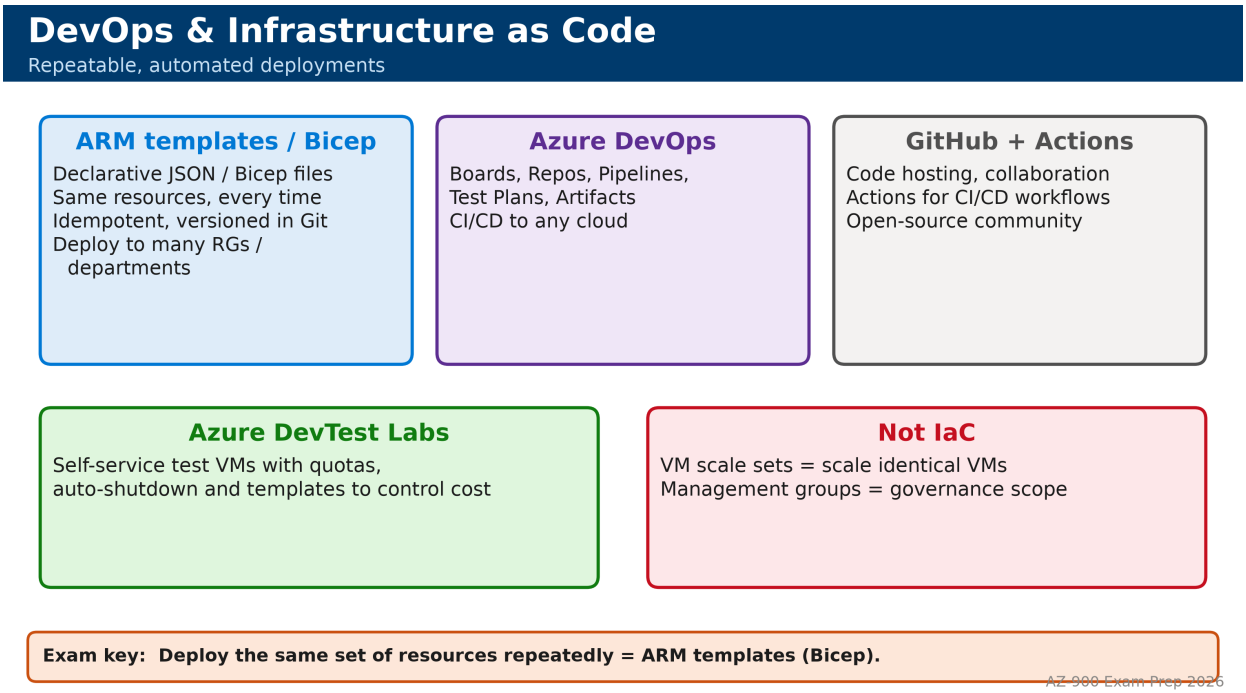


Figure Q422: DevOps and infrastructure as code

About this figure

- Top row is how you define and ship infrastructure: templates, pipelines and Git hosting; bottom boxes separate a dev/test cost tool from things that are not IaC.
- ARM templates and Bicep are free; you pay only for resources deployed. Bicep compiles to ARM JSON and deployments are idempotent.
- Azure DevOps Basic is free for the first 5 users, then approx. USD 6/user-month; it includes 1 free Microsoft-hosted parallel job with 1,800 minutes/month.
- GitHub Actions is free for public repositories and includes monthly free minutes for private ones; Terraform is a third-party IaC alternative.
- DevTest Labs is free itself; you pay for the VMs, and auto-shutdown, quotas and allowed VM sizes keep test spending predictable.

Question 423

AZURE MANAGEMENT & GOVERNANCE

Management tools
v5 Q563

Review the underlined text. If it makes the statement correct, select "No change is needed." If the statement is incorrect, select the answer choice that makes it correct. Azure Cloud Shell [requires you to install the Azure CLI on your local computer before use].

✖ A. No change is needed

✔ **B.** runs in a browser with Azure CLI and Azure PowerShell preinstalled

✖ C. can only run Bash and not PowerShell

✖ D. is available only on Windows computers

Correct answer: runs in a browser with Azure CLI and Azure PowerShell preinstalled

Clarification (read this first): Bracketed words: 'requires you to install the Azure CLI on your local computer'. This is false - Cloud Shell runs in the browser with tools ready. Q47, Q159 and Q162 also test Cloud Shell: it has both Bash and PowerShell and works from any device.

Answer explained

✔ **runs in a browser with Azure CLI and Azure PowerShell preinstalled**
Cloud Shell runs in the browser with Azure CLI, Az PowerShell and other tools already installed. You install nothing on your computer.

✖ **No change is needed**
Cloud Shell needs no local install. The tools are already on its managed host.

✖ **Can only run Bash and not PowerShell**
Cloud Shell offers both Bash and PowerShell.

✖ **Is available only on Windows computers**
It works in any modern browser on Windows, macOS, Linux, ChromeOS and mobile.

EXAM STRATEGY
Key fact: Cloud Shell is in the browser with Azure CLI and Az PowerShell ready. Cross out any option saying local install, Bash-only or Windows-only.

IMPACT
Admins can manage Azure from a locked-down or borrowed computer with no installs. This helps in emergencies.

What · Where · When · Why · How

What	Cloud Shell is a browser-based shell for Azure, already signed in.
Where	It opens from the portal toolbar, shell.azure.com, the Azure mobile app or VS Code.
When	Use it when you cannot or do not want to install tools on your computer.
Why	It gives an up-to-date, ready-to-use setup from any device.
How	Select the Cloud Shell icon, choose Bash or PowerShell, and start running commands.

Exam tip
Cloud Shell: no install, both Bash and PowerShell, both CLI and Az module, any browser.

Memory trick
Cloud Shell = a toolbox in the cloud, carried in your browser.

Real-world example, use case and scenario

EXAMPLE
Locked-down laptop
A contractor without admin rights uses Cloud Shell because they cannot install software.

USE CASE
Tablet admin
A hotel IT manager restarts a web app from a tablet using Cloud Shell.

SCENARIO
Training lab
A training centre uses Cloud Shell so students have the same tools in every browser.

Azure Management Tools

Which tool runs where?

Tool	Windows	macOS	Linux	Browser / tablet / phone
Azure portal	Browser	Browser	Browser	YES (any browser)
Azure Cloud Shell	Browser	Browser	Browser	YES (Bash or PowerShell)
Azure CLI (az)	YES	YES	YES	via Cloud Shell
Azure PowerShell (Az)	YES	YES	YES (PS 7)	via Cloud Shell
Azure mobile app	-	-	-	YES (iOS / Android)

Exam rules

- A PowerShell script needs PowerShell + Az module: Windows, macOS, Linux (PowerShell 7) or Cloud Shell.
- A Linux box with ONLY the Azure CLI cannot run a PowerShell script.
- No local install possible (tablet, kiosk, Chromebook)? Use the portal or Cloud Shell in a browser.
- Everything goes through Azure Resource Manager (ARM).

Exam key: No install? Use a browser. PowerApps is NOT an Azure management tool.

AZ-900 Exam Prep 2026

Figure Q423: Azure management tools

About this figure

• Rows are tools and columns are platforms; green cells show where each runs. Browser-based tools work anywhere, including tablets and Chromebooks.

• Cloud Shell is free to use but needs an Azure file share for persistent storage (a few cents a month); sessions time out after about 20 minutes idle.

• Azure CLI uses 'az' commands and Azure PowerShell uses cmdlets such as New-AzVM; both are cross-platform and produce the same Resource Manager calls.

• The Azure mobile app monitors resources, checks alerts and runs Cloud Shell, but it is not intended for building complex deployments.

• All tools call Azure Resource Manager, so RBAC, Policy and locks apply consistently whichever tool or SDK a user chooses.

Question 424

AZURE MANAGEMENT & GOVERNANCE

Monitoring
v5 Q564

Review the underlined text. If it makes the statement correct, select "No change is needed." If the statement is incorrect, select the answer choice that makes it correct. Data collected by Azure Monitor Logs is stored and queried in [a Log Analytics workspace].

✔ **A.** No change is needed

✖ B. an Azure Advisor score

✖ C. a resource lock

✖ D. an Azure Policy initiative

Correct answer: No change is needed

Clarification (read this first): Here the bracketed text is already correct, so pick 'No change is needed'. Logs live in a Log Analytics workspace. Q410 is related: Metrics need no workspace.

Answer explained

✔ **No change is needed**
Azure Monitor Logs data is stored in a Log Analytics workspace and searched with KQL. So the statement is already correct.

✖ **An Azure Advisor score**
The Advisor score is a percentage that sums up Advisor tips. It does not store logs.

✖ **A resource lock**
A lock stops deletion or changes. It stores no data.

✖ **An Azure Policy initiative**
A Policy initiative is a group of policy rules. It does not store logs.

EXAM STRATEGY
Key fact: Azure Monitor Logs live in a Log Analytics workspace and are searched with KQL. Advisor score, locks and initiatives are unrelated.

IMPACT
Workspace design decides cost, how long logs are kept, and who can see them. Logs spread across places make investigations harder.

What · Where · When · Why · How

What	A Log Analytics workspace is the data store for Azure Monitor Logs.
Where	It is an Azure resource in a chosen region and resource group.
When	Create one when you need to collect and analyse logs from resources or agents.
Why	It keeps log data in one place for searching, alerts and charts.
How	Set up diagnostic settings or data collection rules to send data, then search with KQL.

Exam tip
Logs live in a Log Analytics workspace. Metrics live in the Azure Monitor metrics database.

Memory trick
Log Analytics workspace: where logs go to be analysed.

Real-world example, use case and scenario

EXAMPLE
Diagnostic settings
An admin sends Key Vault audit logs to a Log Analytics workspace using diagnostic settings.

USE CASE
Central security
A bank sends logs from all subscriptions to one workspace used by Microsoft Sentinel.

SCENARIO
Hybrid servers
A manufacturer collects event logs from Arc-enabled servers into a workspace with Azure Monitor Agent.

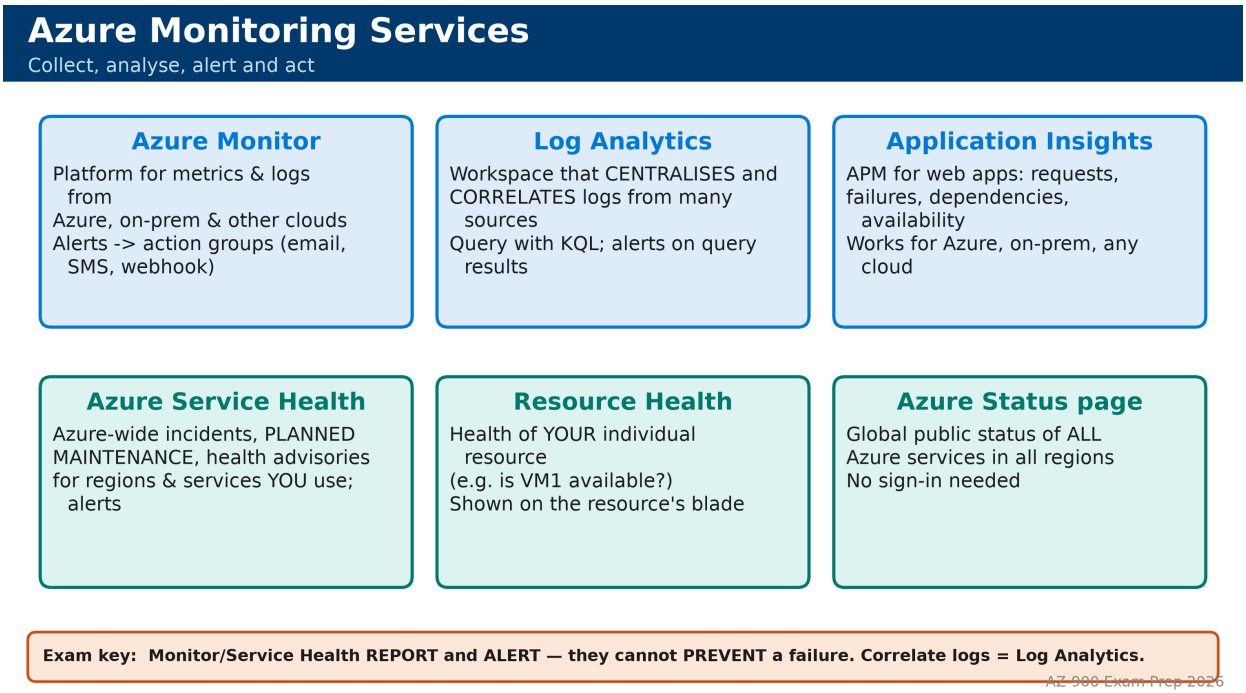


Figure Q424: Azure monitoring services

About this figure

• Top row collects and analyses telemetry; bottom row reports health at three scopes: global (Status), your subscriptions (Service Health), one resource (Resource Health).

• Platform metrics are kept 93 days free; Log Analytics ingestion costs approx. USD 2.30/GB pay-as-you-go, with the first 5 GB/month free per billing account.

• Log Analytics retains data 31 days at no extra charge (90 days with Sentinel); longer retention, up to 12 years in long-term retention, costs extra.

• Application Insights is part of Azure Monitor and stores data in a Log Analytics workspace, so it is billed by the same per-GB ingestion.

• Service Health alerts can notify an action group by email, SMS, push or webhook when an outage or planned maintenance affects your regions.